Dinozorlar

Kapsamlı Wikipedia Derlemesi

31 makale · 6 bölüm · 788 sayfa

Genel bilgi, jeolojik devirler, Saurischia ve Ornithischia soy ağacı, kuşların kökeni ve toplu yok oluşlar.

Kaynak: Wikipedia makaleleri (PDF). Bölümlere göre düzenlenmiştir.

İçindekiler

1. Dinozorlara Genel Bakış 4

Dinozorların tanımı, fizyolojisi ve yumurtaları

Dinosaur 5
Physiology of dinosaurs 77
Dinosaur egg 110

2. Mesozoik Devirler 126

Dinozorların yaşadığı jeolojik dönemler

Triassic 127
Jurassic 152

3. Saurischia (Kertenkele Kalçalılar) 214

Theropod'lar (etçiller) ve Sauropod'lar (uzun boyunlular)

Saurischia 215

Theropoda

Theropoda 219
Ceratosauria 241
Megalosauroidea 255
Allosauridae 261
Spinosauridae 264
Tyrannosauridae 292
Tyrannosaurus 310
Dromaeosauridae 373
Velociraptor 402
Troodontidae 424

Sauropodomorpha

Sauropoda 437

4. Ornithischia (Kuş Kalçalılar) 465

Zırhlı, boynuzlu, plakalı ve ördek gagalı dinozorlar

Ornithischia 466

Thyreophora

Stegosauria 483
Ankylosauria 498

Marginocephalia

Ceratopsidae 509
Triceratops 515

Ornithopoda

Hadrosauridae 545
Maiasaura 565

İçindekiler

5. Kuşların Kökeni 577

Tüylü dinozorlar ve dinozorlardan kuşlara geçiş

- Feathered dinosaur 578
- Archaeopteryx 594
- Origin of birds 625

6. Toplu Yok Oluşlar 656

Triyas–Jura ve Kretase–Paleojen yok oluşları ile nedenleri

- Triassic–Jurassic extinction 657
- Cretaceous–Paleogene extinction event 704
- Chicxulub crater 758
- Deccan Traps 779

1

Dinozorlara Genel Bakış

Dinozorların tanımı, fizyolojisi ve yumurtaları

- Dinosaur
- Physiology of dinosaurs
- Dinosaur egg

WIKIPEDIA 25
25 years of the free encyclopedia

Dinosaur

Dinosaurs are a diverse group of warmblooded reptiles[note 1] of the clade **Dinosauria**. They existed through most of the Mesozoic era, first appearing early in the Triassic period. They became the dominant terrestrial vertebrates after the Triassic–Jurassic extinction event 201.3 mya and their dominance continued throughout the Jurassic and Cretaceous periods. The fossil record shows that birds are feathered dinosaurs, having evolved from earlier theropods during the Late Jurassic epoch, and are the only dinosaur lineage known to have survived the Cretaceous–Paleogene extinction event approximately 66 mya. Dinosaurs can therefore be divided into **avian dinosaurs**—birds—and the extinct **non-avian dinosaurs**, which are all dinosaurs other than birds.

Dinosaurs are varied from taxonomic, morphological and ecological standpoints. Birds, at over 11,000 living species, are among the most diverse groups of vertebrates. Using fossil evidence, paleontologists have identified over 900 distinct genera and more than 1,000 different species of non-avian dinosaurs. Dinosaurs are represented on every continent by both extant species (birds) and fossil remains. Through most of the 20th century, before birds were recognized as dinosaurs, most of the scientific community presumed dinosaurs to have been sluggish and cold-blooded. Most research conducted since the 1970s, however, has indicated that dinosaurs were active animals with elevated metabolisms and numerous adaptations for social interaction. Evidence shows that all dinosaurs had erect legs similar to that of birds and many mammals, as opposed to the sprawling leg stance of cold-blooded tetrapods. Some were herbivorous, others carnivorous. Evidence suggests that all dinosaurs

Dinosaurs

Temporal range: (possible Middle Triassic record)

A collection of dinosaurs (clockwise from top left): *Sinosauropteryx* (a small feathered theropod), *Brontosaurus* (a sauropod), *Tyrannosaurus* (a large theropod), a great egret (an avian theropod), *Triceratops* (a marginocephalian), and *Stegosaurus* (a thyreophoran).

Scientific classification

Kingdom:	Animalia
Phylum:	Chordata
Class:	Reptilia
Clade:	Archosauria
Clade:	Avemetatarsalia
Clade:	Ornithodira
Clade:	Dinosauromorpha
Clade:	Dinosauria
	Owen, 1842

Major groups

- †Ornithischia

were egg-laying, and that nest-building was a trait shared by many dinosaurs, both avian and non-avian.

While dinosaurs were ancestrally bipedal, many extinct groups included quadrupedal species, and some were able to shift between these stances. Elaborate display structures such as horns or crests are common to all dinosaur groups, and some extinct groups developed skeletal modifications such as bony armor and spines. While the dinosaurs' modern-day surviving avian lineage (birds) are generally small due to the constraints of flight, many non-avian dinosaurs were large-bodied—the largest sauropod dinosaurs are estimated to have reached lengths of 39.7 meters (130 feet) and heights of 18 m (59 ft) and were the largest land animals of all time.

The first dinosaur fossils were recognized in the early 19th century, with the name "dinosaur" (meaning "terrible lizard") being coined by Sir Richard Owen in 1842 to refer to these "great fossil lizards". Since then, mounted fossil dinosaur skeletons have been major attractions at museums worldwide, and dinosaurs have become an enduring part of popular culture. The large sizes of some dinosaurs, as well as their seemingly monstrous and fantastic nature, have ensured their regular appearance in best-selling books and films, such as the *Jurassic Park* franchise. Persistent public enthusiasm for the animals has resulted in significant funding for dinosaur science, and new discoveries are regularly covered by the media.

- Saurischia
  - †Sauropodomorpha
  - Theropoda
    - Various extinct groups
    - Aves (birds)

Dinosaurs and possible dinosaurs of uncertain affinity

- †*Agnosphytis*?
- †*Alwalkeria*
- †*Chilesaurus*
- †*Chindesaurus*
- †*Daemonosaurus*
- †*Eodromaeus*
- †*Nhandumirim*
- †*Nyasasaurus*?
- †*Pisanosaurus*?
- †*Smok*?
- †*Tawa*
- †*Thecospondylus*
- †Herrerasauria[1][2]
- †Silesauridae? (paraphyletic?)[3][4][5][6]

Etymology

In 1842 the English paleontologist Sir Richard Owen coined the term "dinosaur", using it to refer to the "distinct tribe or sub-order of Saurian Reptiles" that were then being recognized in England and around the world.[7][8][9][10][11] The term is derived from Ancient Greek δεινός *(deinos)* 'terrible, potent or fearfully great' and σαῦρος *(sauros)* 'lizard or reptile'.[10][12] Though the taxonomic name has often been interpreted as a reference to dinosaurs' teeth, claws, and other fearsome characteristics, Owen intended it also to evoke their size and majesty.[13]

Definition

Birds are avian dinosaurs, and phylogenetic taxonomy includes over 11,000 extant species in the group Dinosauria.

Under phylogenetic nomenclature, dinosaurs are usually defined as the group consisting of the most recent common ancestor (MRCA) of *Triceratops* and modern birds (Neornithes), and all its descendants.[14] It has also been suggested that Dinosauria be defined with respect to the MRCA of *Megalosaurus* and *Iguanodon*, because these were two of the three genera cited by Richard Owen when he recognized the Dinosauria.[15] Both definitions cover the same known genera: Dinosauria = Ornithischia + Saurischia. This includes major groups such as ankylosaurians (armored herbivorous quadrupeds), stegosaurians (plated herbivorous quadrupeds), ceratopsians (bipedal or quadrupedal herbivores with neck frills), pachycephalosaurians (bipedal herbivores with thick skulls), ornithopods (bipedal or quadrupedal herbivores including "duck-bills"), theropods (mostly bipedal carnivores and birds), and sauropodomorphs (mostly large herbivorous quadrupeds with long necks and tails).[16]

Birds are the sole surviving dinosaurs. In traditional taxonomy, birds were considered a separate class that had evolved from dinosaurs. However, most contemporary paleontologists reject the traditional style of classification based on anatomical similarity, in favor of phylogenetic taxonomy based on deduced ancestry, in which each group is defined as all descendants of a given founding genus.[17] Birds belong to the dinosaur subgroup Maniraptora, which are coelurosaurs, which are theropods, which are saurischians.[18]

Research by Matthew G. Baron, David B. Norman, and Paul M. Barrett in 2017 suggested a radical revision of dinosaurian systematics. Phylogenetic analysis by Baron et al. recovered the Ornithischia as being closer to the Theropoda than the Sauropodomorpha, as opposed to the traditional union of theropods with sauropodomorphs. This would cause sauropods and kin to fall outside traditional dinosaurs, so they re-defined Dinosauria as the last common ancestor of *Triceratops horridus*, *Passer domesticus* and *Diplodocus carnegii*, and all of its descendants, to ensure that sauropods and kin remain included as dinosaurs. They also resurrected the clade Ornithoscelida to refer to the group containing Ornithischia and Theropoda.[19][20]

General description

Using one of the above definitions, dinosaurs can be generally described as archosaurs with hind limbs held erect beneath the body.[21] Other prehistoric animals, including pterosaurs, mosasaurs, ichthyosaurs, plesiosaurs, and *Dimetrodon*, while often popularly conceived of as dinosaurs, are not taxonomically classified as dinosaurs. Pterosaurs are distantly related to dinosaurs, being members of the clade Ornithodira. The other groups mentioned are, like dinosaurs and pterosaurs, members of Sauropsida (the reptile and bird clade), except *Dimetrodon* (which is a synapsid). None of them had the erect hind limb posture characteristic of true dinosaurs.[22]

Triceratops skeleton, Natural History Museum of Los Angeles County

Dinosaurs were the dominant terrestrial vertebrates of the Mesozoic Era, especially the Jurassic and Cretaceous periods. Other groups of animals were restricted in size and niches; mammals, for example, rarely exceeded the size of a domestic cat and were generally rodent-sized carnivores of small prey.[23] Dinosaurs have always been recognized as an extremely varied group: over 900 non-avian dinosaur genera have been confidently identified (2018) with 1,124 species (2016). Estimates put the total number of dinosaur genera preserved in the fossil record at 1850, nearly 75% still undiscovered,[24][25][26] and the number that ever existed (in or out of the fossil record) at 3,400.[27] A 2016 estimate put the number of dinosaur species living in the Mesozoic at 1,543–2,468,[25][28] compared to the number of modern-day birds (avian dinosaurs) at 10,806 species.[29]

Extinct dinosaurs, as well as modern birds, include genera that are herbivorous and others carnivorous, including seed-eaters, fish-eaters, insectivores, and omnivores. While dinosaurs were ancestrally bipedal (as are all modern birds), some evolved into quadrupeds, and others, such as *Anchisaurus* and *Iguanodon*, could walk as easily on two or four legs. Cranial modifications like horns and crests are common dinosaurian traits, and some extinct species had bony armor. Although the best-known genera are remarkable for their large size, many Mesozoic dinosaurs were human-sized or smaller, and modern birds are generally small in size. Dinosaurs today inhabit every continent, and fossils show that they had achieved global distribution by the Early Jurassic epoch at latest.[30] Modern birds inhabit most available habitats, from terrestrial to marine, and there is evidence that some non-avian dinosaurs (such as *Microraptor*) could fly or at least glide, and others, such as spinosaurids, had semiaquatic habits.[31]

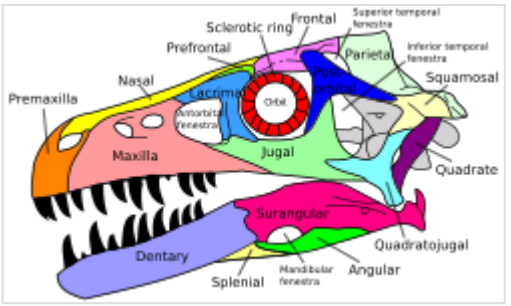
Distinguishing anatomical features

While recent discoveries have made it more difficult to present a universally agreed-upon list of their distinguishing features, nearly all dinosaurs discovered so far share certain modifications to the ancestral archosaurian skeleton, or are clearly descendants of older dinosaurs showing these modifications. Although some later groups of dinosaurs featured further modified versions of these traits, they are considered typical for Dinosauria; the earliest dinosaurs had them and passed them on to their descendants. Such modifications, originating in the most recent common ancestor of a certain taxonomic group, are called the synapomorphies of such a group.[32]

A detailed assessment of archosaur interrelations by Sterling Nesbitt[33] confirmed or found the following twelve unambiguous synapomorphies, some previously known:

- In the skull, a supratemporal fossa (excavation) is present in front of the supratemporal fenestra, the main opening in the rear skull roof
- Epipophyses, obliquely backward-pointing processes on the rear top corners of the anterior (front) neck vertebrae behind the atlas and axis, the first two neck vertebrae
- Apex of a deltopectoral crest (a projection on which the deltopectoral muscles attach) located at or more than 30% down the length of the humerus (upper arm bone)
- Radius, a lower arm bone, shorter than 80% of humerus length

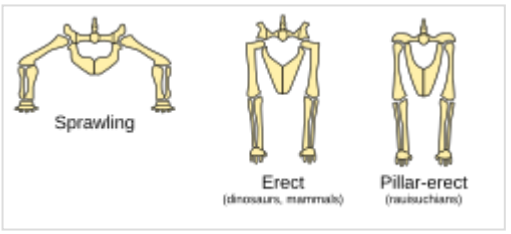
- Fourth trochanter (projection where the caudofemoralis muscle attaches on the inner rear shaft) on the femur (thigh bone) is a sharp flange
- Fourth trochanter asymmetrical, with distal, lower, margin forming a steeper angle to the shaft
- On the astragalus and calcaneum, upper ankle bones, the proximal articular facet, the top connecting surface, for the fibula occupies less than 30% of the transverse width of the element
- Exoccipitals (bones at the back of the skull) do not meet along the midline on the floor of the endocranial cavity, the inner space of the braincase



Labeled diagram of a typical archosaur skull, the skull of *Dromaeosaurus*

- In the pelvis, the proximal articular surfaces of the ischium with the ilium and the pubis are separated by a large concave surface (on the upper side of the ischium a part of the open hip joint is located between the contacts with the pubic bone and the ilium)
- Cnemial crest on the tibia (protruding part of the top surface of the shinbone) arcs anterolaterally (curves to the front and the outer side)
- Distinct proximodistally oriented (vertical) ridge present on the posterior face of the distal end of the tibia (the rear surface of the lower end of the shinbone)
- Concave articular surface for the fibula of the calcaneum (the top surface of the calcaneum, where it touches the fibula, has a hollow profile)

Nesbitt found a number of further potential synapomorphies and discounted a number of synapomorphies previously suggested. Some of these are also present in silesaurids, which Nesbitt recovered as a sister group to Dinosauria, including a large anterior trochanter, metatarsals II and IV of subequal length, reduced contact between ischium and pubis, the presence of a cnemial crest on the tibia and of an ascending process on the astragalus, and many others.[14]



Hip joints and hindlimb postures of: (left to right) typical reptiles (sprawling), dinosaurs and mammals (erect), and rauisuchians (pillar-erect)

A variety of other skeletal features are shared by dinosaurs. However, because they either are common to other groups of archosaurs or were not present in all early dinosaurs, these features are not synapomorphies. For example, as diapsids, dinosaurs ancestrally had two pairs of Infratemporal fenestrae (openings in the skull behind the eyes), and as members of the diapsid group Archosauria, had additional openings in the snout and lower jaw.[34] Additionally, several characteristics once thought to be synapomorphies are now known to have appeared before dinosaurs, or were absent in the earliest dinosaurs and independently evolved by different dinosaur groups. These include an elongated scapula, or shoulder blade; a sacrum composed of three or more fused vertebrae (three are found in some other archosaurs, but only two are found in *Herrerasaurus*);[14] and a perforate acetabulum, or hip socket, with a hole at the center of its inside surface (closed in *Saturnalia tupiniquim*, for example).[35][36] Another difficulty of determining distinctly dinosaurian features is that early dinosaurs and other archosaurs from the Late Triassic epoch are often poorly known and were similar in many ways; these animals have sometimes been misidentified in the literature.[37]

Dinosaurs stand with their hind limbs erect in a manner similar to most modern mammals, but distinct from most other reptiles, whose limbs sprawl out to either side.[38] This posture is due to the development of a laterally facing recess in the pelvis (usually an open socket) and a corresponding inwardly facing

distinct head on the femur.[39] Their erect posture enabled early dinosaurs to breathe easily while moving, which likely permitted stamina and activity levels that surpassed those of "sprawling" reptiles.[40] Erect limbs probably also helped support the evolution of large size by reducing bending stresses on limbs.[41] Some non-dinosaurian archosaurs, including rauisuchians, also had erect limbs but achieved this by a "pillar-erect" configuration of the hip joint, where instead of having a projection from the femur insert on a socket on the hip, the upper pelvic bone was rotated to form an overhanging shelf.[41]

History of study

Pre-scientific history

Dinosaur fossils have been known for millennia, although their true nature was not recognized. Chinese authors have long recognised remains of prehistoric animals as "dragon bones". *Huayang Guo Zhi* (華陽國志), a gazetteer compiled by Chang Qu (常璩) during the Western Jin Dynasty (265–316), reported the discovery of dragon bones at Wucheng in Sichuan Province.[42] These "dragon bones" were and still are often used as ingredients for remedies in Traditional Chinese Medicine.[43] While these "dragon bones" have often popularly been claimed to be dinosaur bones, all known historical examples of vertebrate "dragon bones" from China are actually from Cenozoic mammals.[44] However, a village in Henan, central China was reported in 2007 to have spent the previous several decades excavating and using real dinosaur bones as "dragon bones" for TCM purposes.[43] In Europe, dinosaur fossils were generally thought to be the remains of giants and other biblical creatures.[45]

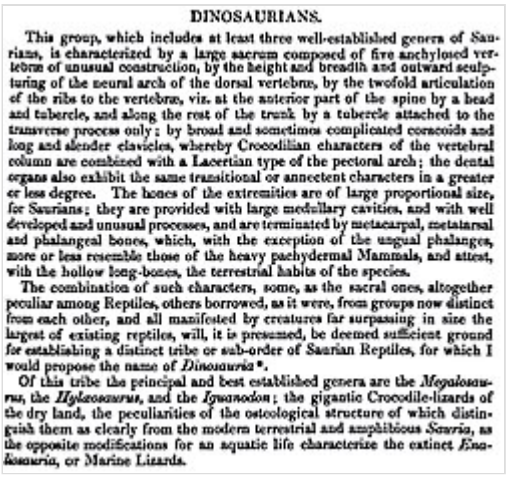
Early dinosaur research

William Buckland

Scholarly descriptions of what would now be recognized as dinosaur bones first appeared in the late 17th century in England. Part of a bone, now known to have been the femur of a *Megalosaurus*,[46] was recovered from a limestone quarry at Cornwell near Chipping Norton, Oxfordshire, in 1676. The fragment was sent to Robert Plot, Professor of Chemistry at the University of Oxford and first curator of the Ashmolean Museum, who published a description in his *The Natural History of Oxford-shire* (1677).[47] He correctly identified the bone as the lower extremity of the femur of a large animal, and recognized that it was too large to belong to any known species. He therefore concluded it to be the femur of a huge human, perhaps a Titan or another type of giant featured in legends.[48][49] Edward Lhuyd, a friend of Sir Isaac Newton, published *Lithophylacii Britannici ichnographia* (1699), the first scientific treatment of what would now be recognized as a dinosaur. In it he described and named a sauropod tooth, "Rutellum impicatum",[50][51] that had been found in Caswell, near Witney, Oxfordshire.[52]

Between 1815 and 1824, the Rev William Buckland, the first Reader of Geology at the University of Oxford, collected more fossilized bones of *Megalosaurus* and became the first person to describe a non-avian dinosaur in a scientific journal.[46][53] The second non-avian dinosaur genus to be identified, *Iguanodon*, was purportedly discovered in 1822 by Mary Ann Mantell, the wife of English geologist

Gideon Mantell, though this is disputed and some historians say Gideon had acquired remains years earlier. Gideon Mantell recognized similarities between his fossils and the bones of modern iguanas and published his findings in 1825.[54][55]



DINOSAURIANS.

This group, which includes at least three well-established genera of Saurians, is characterized by a large sacrum composed of five anchylosed vertebræ of unusual construction, by the height and breadth and outward sculpturing of the neural arch of the dorsal vertebræ, by the twofold articulation of the ribs to the vertebræ, viz. at the anterior part of the spine by a head and tubercle, and along the rest of the trunk by a tubercle attached to the transverse process only; by broad and sometimes complicated coracoids and long and slender clavicles, whereby Crocodilian characters of the vertebral column are combined with a Lacertian type of the pectoral arch; the dental organs also exhibit the same transitional or annectent characters in a greater or less degree. The bones of the extremities are of large proportional size, for Saurians; they are provided with large medullary cavities, and with well developed and unusual processes, and are terminated by metacarpal, metatarsal and phalangeal bones, which, with the exception of the ungual phalanges, more or less resemble those of the heavy pachydermal Mammals, and attest, with the hollow long-bones, the terrestrial habits of the species.

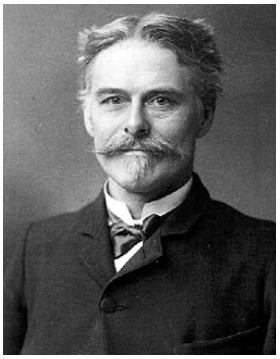
The combination of such characters, some, as the sacral ones, altogether peculiar among Reptiles, others borrowed, as it were, from groups now distinct from each other, and all manifested by creatures far surpassing in size the largest of existing reptiles, will, it is presumed, be deemed sufficient ground for establishing a distinct tribe or sub-order of Saurian Reptiles, for which I would propose the name of *Dinosauria**.

Of this tribe the principal and best established genera are the *Megalosaurus*, the *Hylæosaurus*, and the *Iguanodon*; the gigantic Crocodile-lizards of the dry land, the peculiarities of the osteological structure of which distinguish them as clearly from the modern terrestrial and amphibious *Sauria*, as the opposite modifications for an aquatic life characterize the extinct *Enaliosauria*, or Marine Lizards.

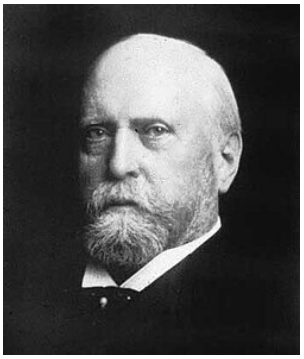
Sir Richard Owen's coining of the word *dinosaur*, in the 1842 revised version of his talk at an 1841 meeting of the British Association for the Advancement of Science.

The study of these "great fossil lizards" soon became of great interest to European and American scientists. Owen recognized that the remains that had been found so far, *Iguanodon*, *Megalosaurus* and *Hylaeosaurus*, shared distinctive features, and so decided to present them as a distinct taxonomic group. As clarified by British geologist and historian Hugh Torrens, Owen had given a presentation about fossil reptiles to the British Association for the Advancement of Science in 1841, but reports of the time show that Owen did not mention the word "dinosaur", nor recognize dinosaurs as a distinct group of reptiles in his address. He introduced the Dinosauria only in the revised text version of his talk published in April 1842.[7][8] With the backing of Prince Albert, the husband of Queen Victoria, Owen established the Natural History Museum, London, to display the national collection of dinosaur fossils and other biological and geological exhibits.[56]

Discoveries in North America



Edward Drinker Cope



Othniel Charles Marsh

In 1858, William Parker Foulke discovered the first known American dinosaur in marl pits in the small town of Haddonfield, New Jersey. (Although fossils had been found before, their nature had not been correctly discerned.) The creature was named *Hadrosaurus foulkii*. It was an extremely important find: *Hadrosaurus* was one of the first nearly complete dinosaur skeletons found (the first was in 1834, in Maidstone, England), and it was clearly a bipedal creature. This was a revolutionary discovery as, until that point, most scientists had assumed dinosaurs walked on four feet, like other lizards. Foulke's discoveries sparked a wave of interests in dinosaurs in the United States, known as Dinosaur Mania.[57]

Dinosaur mania was exemplified by the fierce rivalry between Edward Drinker Cope and Othniel Charles Marsh, both of whom raced to be the first to find new dinosaurs in what came to be known as the Bone Wars. This fight between the two scientists lasted for over 30 years, ending in 1897 when Cope died after spending his entire fortune on the dinosaur hunt. Many valuable dinosaur specimens were damaged or destroyed due to the pair's rough methods: for example, their diggers often used dynamite to unearth bones. Modern paleontologists would find such methods crude and unacceptable, since blasting easily destroys fossil and stratigraphic evidence. Despite their unrefined methods, the contributions of Cope and Marsh to paleontology were vast: Marsh unearthed 86 new species of dinosaur and Cope discovered 56, a total of 142 new species. Cope's collection is now at the American Museum of Natural History in New York City, while Marsh's is at the Peabody Museum of Natural History at Yale University.[58]

"Dinosaur renaissance" and beyond

World War II caused a pause in palaeontological research; after the war, research attention was also diverted increasingly to fossil mammals rather than dinosaurs, which were seen as sluggish and cold-blooded.[59][60] At the end of the 1960s, however, the field of dinosaur research experienced a surge in activity that remains ongoing.[61] Several seminal studies led to this activity. First, John Ostrom discovered the bird-like dromaeosaurid theropod *Deinonychus* and described it in 1969. Its anatomy indicated that it was an active predator that was likely warm-blooded, in marked contrast to the then-prevailing image of dinosaurs.[59] Concurrently, Robert T. Bakker published a series of studies that likewise argued for active lifestyles in dinosaurs based on anatomical and ecological evidence (see § Physiology),[62][63] which were subsequently summarized in his 1986 book *The Dinosaur Heresies*.[64]

Paleontologist Robert T. Bakker with a mounted skeleton of a tyrannosaurid (*Gorgosaurus libratus*)

New revelations were supported by an increase in dinosaur discoveries. Major new dinosaur discoveries have been made by paleontologists working in previously unexplored regions, including India, South America, Madagascar, Antarctica, and most significantly China. Across theropods, sauropodomorphs, and ornithischians, the number of named genera began to increase drastically in the 1990s,[24] with up to 30 new species of dinosaurs named each year by 2008.[65] At least sauropodomorphs experienced a further increase in the number of named species in the 2010s, with an average of 9.3 new species having been named each year between 2009 and 2020. As a consequence, more sauropodomorphs were named between 1990 and 2020 than in all previous years combined.[66] These new localities also led to improvements in overall specimen quality, with new species being increasingly named not on scrappy fossils but on more complete skeletons, sometimes from multiple individuals. Better specimens also led to new species being invalidated less frequently.[65] Asian localities have produced the most complete theropod specimens,[67] while North American localities have produced the most complete sauropodomorph specimens.[66]

Prior to the dinosaur renaissance, dinosaurs were mostly classified using the traditional rank-based system of Linnaean taxonomy. The renaissance was also accompanied by the increasingly widespread application of cladistics, a more objective method of classification based on ancestry and shared traits, which has proved tremendously useful in the study of dinosaur systematics and evolution. Cladistic analysis, among other techniques, helps to compensate for an often incomplete and fragmentary fossil record.[68][69] Reference books summarizing the state of dinosaur research, such as David B. Weishampel and colleagues' *The Dinosauria*, made knowledge more accessible[70] and spurred further interest in dinosaur research. The release of the first and second editions of *The Dinosauria* in 1990 and 2004, and of a review paper by Paul Sereno in 1998, were accompanied by increases in the number of published phylogenetic trees for dinosaurs.[71]

Soft tissue and molecular preservation

Dinosaur fossils are not limited to bones, but also include imprints or mineralized remains of skin coverings, organs, and other tissues. Of these, skin coverings based on keratin proteins are most easily preserved because of their cross-linked, hydrophobic molecular structure.[72] Fossils of keratin-based skin

coverings or bony skin coverings are known from most major groups of dinosaurs. Dinosaur fossils with scaly skin impressions have been found since the 19th century. Samuel Beckles discovered a sauropod forelimb with preserved skin in 1852 that was incorrectly attributed to a crocodile; it was correctly attributed by Marsh in 1888 and subject to further study by Reginald Hooley in 1917.[73] Among ornithischians, in 1884 Jacob Wortman found skin impressions on the first known specimen of *Edmontosaurus annectens*, which were largely destroyed during the specimen's excavation.[74] Owen and Hooley subsequently described skin impressions of *Hypsilophodon* and *Iguanodon* in 1885 and 1917.[73] Since then, scale impressions have been most frequently found among hadrosaurids, where the impressions are known from nearly the entire body across multiple specimens.[75]

An *Edmontosaurus* specimen's skin impressions found in 1999

Color restoration of *Sinosauropteryx*

Color restoration of *Psittacosaurus*

Starting from the 1990s, major discoveries of exceptionally preserved fossils in deposits known as conservation Lagerstätten contributed to research on dinosaur soft tissues.[76][77] Chiefly among these were the rocks that produced the Jehol (Early Cretaceous) and Yanliao (Mid-to-Late Jurassic) biotas of northeastern China, from which hundreds of dinosaur specimens bearing impressions of feather-like structures (both closely related to birds and otherwise, see § Origin of birds) have been described by Xing Xu and colleagues.[78][79] In living reptiles and mammals, pigment-storing cellular structures known as melanosomes are partially responsible for producing colouration.[80][81] Both chemical traces of melanin and characteristically shaped melanosomes have been reported from feathers and scales of Jehol and Yanliao dinosaurs, including both theropods and ornithischians.[82] This has enabled multiple full-body reconstructions of dinosaur colouration, such as for *Sinosauropteryx*[83] and *Psittacosaurus*[84] by Jakob Vinther and colleagues, and similar techniques have also been extended to dinosaur fossils from other localities.[80] (However, some researchers have also suggested that fossilized melanosomes represent bacterial remains.[85][86]) Stomach contents in some Jehol and Yanliao dinosaurs closely related to birds have also provided indirect indications of diet and digestive system anatomy (e.g., crops).[87][88] More concrete evidence of internal anatomy has been reported in *Scipionyx* from the Pietraroja Plattenkalk of Italy. It preserves portions of the intestines, colon, liver, muscles, and windpipe.[89]

Concurrently, a line of work led by Mary Higby Schweitzer, Jack Horner, and colleagues reported various occurrences of preserved soft tissues and proteins within dinosaur bone fossils. Various mineralized structures that likely represented red blood cells and collagen fibres had been found by Schweitzer and others in tyrannosaurid bones as early as 1991.[90][91][92] However, in 2005, Schweitzer and colleagues reported that a femur of *Tyrannosaurus* preserved soft, flexible tissue within, including blood vessels, bone matrix, and connective tissue (bone fibers) that had retained their microscopic structure.[93] This discovery suggested that original soft tissues could be preserved over geological time,[72] with multiple mechanisms having been proposed.[94] Later, in 2009, Schweitzer and colleagues reported that a *Brachylophosaurus* femur preserved similar microstructures, and immunohistochemical techniques (based on antibody binding) demonstrated the presence of proteins such as collagen, elastin, and laminin.[95] Both specimens yielded collagen protein sequences that were viable for molecular

phylogenetic analyses, which grouped them with birds as would be expected.[95][96] The extraction of fragmentary DNA has also been reported for both of these fossils,[97] along with a specimen of *Hypacrosaurus*.[98] In 2015, Sergio Bertazzo and colleagues reported the preservation of collagen fibres and red blood cells in eight Cretaceous dinosaur specimens that did not show any signs of exceptional preservation, indicating that soft tissue may be preserved more commonly than previously thought.[99] Suggestions that these structures represent bacterial biofilms[100] have been rejected,[101] but cross-contamination remains a possibility that is difficult to detect.[102]

Scipionyx fossil with intestines, Natural History Museum of Milan

Evolutionary history

Origins and early evolution

Dinosaurs diverged from their archosaur ancestors during the Middle to Late Triassic epochs, roughly 20 million years after the devastating Permian–Triassic extinction event wiped out an estimated 96% of all marine species and 70% of terrestrial vertebrate species approximately 252 million years ago.[103][104] The oldest dinosaur fossils known from substantial remains date to the Carnian epoch of the Triassic period and have been found primarily in the Ischigualasto and Santa Maria Formations of Argentina and Brazil, and the Pebbly Arkose Formation of Zimbabwe.[105]

The early dinosaurs *Herrerasaurus* (large), *Eoraptor* (small) and a *Plateosaurus* skull, from the Triassic

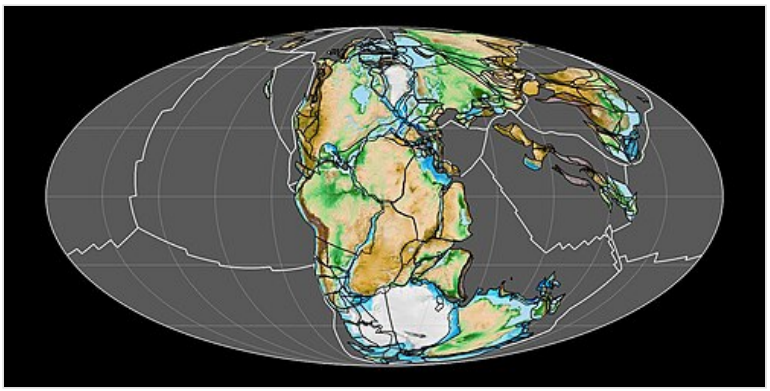
The Ischigualasto Formation (radiometrically dated at 231–230 million years old[106]) has produced the early saurischian *Eoraptor*, originally considered a member of the Herrerasauridae[107] but now considered to be an early sauropodomorph, along with the herrerasaurids *Herrerasaurus* and *Sanjuansaurus*, and the sauropodomorphs *Chromogisaurus*, *Eodromaeus*, and *Panphagia*.[108] *Eoraptor*'s likely resemblance to the common ancestor of all dinosaurs suggests that the first dinosaurs would have been small, bipedal predators.[109][110][111] The Santa Maria Formation (radiometrically dated to be older, at 233.23 million years old[112]) has produced the herrerasaurids *Gnathovorax* and *Staurikosaurus*, along with the sauropodomorphs *Bagualosaurus*, *Buriolestes*, *Guaibasaurus*, *Macrocollum*, *Nhandumirim*, *Pampadromaeus*, *Saturnalia*, and *Unaysaurus*.[108] The Pebbly Arkose Formation, which is of uncertain age but was likely comparable to the other two, has produced the sauropodomorph *Mbiresaurus*, along with an unnamed herrerasaurid.[105]

Less well-preserved remains of the sauropodomorphs *Jaklapallisaurus* and *Nambalia*, along with the early saurischian *Alwalkeria*, are known from the Upper Maleri and Lower Maleri Formations of India.[113] The Carnian-aged Chañares Formation of Argentina preserves primitive, dinosaur-like ornithodirans such as *Lagosuchus* and *Lagerpeton* in Argentina, making it another important site for understanding dinosaur evolution. These ornithodirans support the model of early dinosaurs as small,

bipedal predators.[108][114] Dinosaurs may have appeared as early as the Anisian epoch of the Triassic, approximately 243 million years ago, which is the age of *Nyasasaurus* from the Manda Formation of Tanzania. However, its known fossils are too fragmentary to identify it as a dinosaur or only a close relative.[115] The referral of the Manda Formation to the Anisian is also uncertain. Regardless, dinosaurs existed alongside non-dinosaurian ornithodirans for a period of time, with estimates ranging from 5–10 million years[114] to 21 million years.[112]

When dinosaurs appeared, they were not the dominant terrestrial animals. The terrestrial habitats were occupied by various types of archosauromorphs and therapsids, like cynodonts and rhynchosaurs. Their main competitors were the pseudosuchians, such as aetosaurs, ornithosuchids and rauisuchians, which were more successful than the dinosaurs.[116] Most of these other animals became extinct in the Triassic, in one of two events. First, at about 215 million years ago, a variety of basal archosauromorphs, including the protorosaurs, became extinct. This was followed by the Triassic–Jurassic extinction event (about 201 million years ago), that saw the end of most of the other groups of early archosaurs, like aetosaurs, ornithosuchids, phytosaurs, and rauisuchians. Rhynchosaurs and dicynodonts survived (at least in some areas) at least as late as early–mid Norian and late Norian or earliest Rhaetian stages, respectively,[117][118] and the exact date of their extinction is uncertain. These losses left behind a land fauna of crocodylomorphs, dinosaurs, mammals, pterosaurians, and turtles.[14] The first few lines of early dinosaurs diversified through the Carnian and Norian stages of the Triassic, possibly by occupying the niches of the groups that became extinct.[16] Also notably, there was a heightened rate of extinction during the Carnian pluvial event.[119]

Evolution and paleobiogeography



The supercontinent Pangaea in the early Mesozoic (around 200 million years ago)

Dinosaur evolution after the Triassic followed changes in vegetation and the location of continents. In the Late Triassic and Early Jurassic, the continents were connected as the single landmass Pangaea, and there was a worldwide dinosaur fauna mostly composed of coelophysoid carnivores and early sauropodomorph herbivores.[120] Gymnosperm plants (particularly conifers), a potential food source, radiated in the Late Triassic. Early sauropodomorphs did not have sophisticated mechanisms for processing food in the mouth, and so must have employed other means of breaking down food farther along the digestive tract.[121] The general homogeneity of dinosaurian faunas continued into the Middle and Late Jurassic, where most localities had predators consisting of ceratosaurians, megalosauroids, and allosauroids, and herbivores consisting of stegosaurian ornithischians and large sauropods. Examples of this include the Morrison Formation of North America and Tendaguru Beds of Tanzania. Dinosaurs in China show some differences, with specialized metriacanthosaurid theropods and unusual, long-necked sauropods like *Mamenchisaurus*.[120] Ankylosaurians and ornithopods were also becoming more common, but primitive sauropodomorphs had become extinct. Conifers and pteridophytes were the most common plants. Sauropods, like earlier sauropodomorphs, were not oral processors, but ornithischians were evolving various means of dealing with food in the

mouth, including potential cheek-like organs to keep food in the mouth, and jaw motions to grind food.[121] Another notable evolutionary event of the Jurassic was the appearance of true birds, descended from maniraptoran coelurosaurians.[18]

By the Early Cretaceous and the ongoing breakup of Pangaea, dinosaurs were becoming strongly differentiated by landmass. The earliest part of this time saw the spread of ankylosaurians, iguanodontians, and brachiosaurids through Europe, North America, and northern Africa. These were later supplemented or replaced in Africa by large spinosaurid and carcharodontosaurid theropods, and rebbachisaurid and titanosaurian sauropods, also found in South America. In Asia, maniraptoran coelurosaurians like dromaeosaurids, troodontids, and oviraptorosaurians became the common theropods, and ankylosaurids and early ceratopsians like *Psittacosaurus* became important herbivores. Meanwhile, Australia was home to a fauna of basal ankylosaurians, titanosaurian sauropods and large and small Elasmarian ornithopods.[120] The stegosaurians appear to have gone extinct at some point in the late Early Cretaceous or early Late Cretaceous. A major change in the Early Cretaceous, which would be amplified in the Late Cretaceous, was the evolution of flowering plants. At the same time, several groups of dinosaurian herbivores evolved more sophisticated ways to orally process food. Ceratopsians developed a method of slicing with teeth stacked on each other in batteries, and iguanodontians refined a method of grinding with dental batteries, taken to its extreme in hadrosaurids.[121] Some sauropods also evolved tooth batteries, best exemplified by the rebbachisaurid *Nigersaurus*.[122]

There were three general dinosaur faunas in the Late Cretaceous. In the northern continents of North America and Asia, the major theropods were tyrannosaurids and various types of smaller maniraptoran theropods, with a predominantly ornithischian herbivore assemblage of hadrosaurids, ceratopsians, ankylosaurids, and pachycephalosaurians. In the southern continents that had made up the now-splitting supercontinent Gondwana, abelisaurids were the common theropods, and titanosaurian sauropods the common herbivores. Finally, in Europe, dromaeosaurids, rhabdodontid iguanodontians, nodosaurid ankylosaurians, and titanosaurian sauropods were prevalent.[120] Flowering plants were greatly radiating,[121] with the first grasses appearing by the end of the Cretaceous.[123] Grinding hadrosaurids and shearing ceratopsians became very diverse across North America and Asia. Theropods were also radiating as herbivores or omnivores, with therizinosaurians and ornithomimosaurians becoming common.[121]

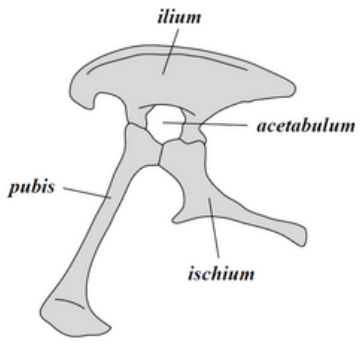
The Cretaceous–Paleogene extinction event, which occurred approximately 66 million years ago at the end of the Cretaceous, caused the extinction of all dinosaur groups except for the neornithine birds. Some other diapsid groups, including crocodilians, dyrosaurs, sebecosuchians, turtles, lizards, snakes, sphenodontians, and choristoderans, also survived the event.[124]

The surviving lineages of neornithine birds, including the ancestors of modern ratites, ducks and chickens, and a variety of waterbirds, diversified rapidly at the beginning of the Paleogene period, entering ecological niches left vacant by the extinction of Mesozoic dinosaur groups such as the arboreal enantiornithines, aquatic hesperornithines, and even the larger terrestrial theropods (in the form of *Gastornis*, eogruiids, bathornithids, ratites, geranoidids, mihirungs, and "terror birds"). It is often stated that mammals out-competed the neornithines for dominance of most terrestrial niches but many of these groups co-existed with rich mammalian faunas for most of the Cenozoic Era.[125] Terror birds and bathornithids occupied carnivorous guilds alongside predatory mammals,[126][127] and ratites are still

fairly successful as midsized herbivores; eogruiids similarly lasted from the Eocene to Pliocene, becoming extinct only very recently after over 20 million years of co-existence with many mammal groups.[128]

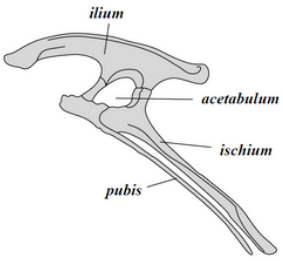
Classification

Dinosaurs belong to a group known as archosaurs, which also includes modern crocodilians. Within the archosaur group, dinosaurs are differentiated most noticeably by their gait. Dinosaur legs extend directly beneath the body, whereas the legs of lizards and crocodilians sprawl out to either side.[32]



Saurischian pelvis structure (left side)

Tyrannosaurus pelvis (showing saurischian structure – left side)



Ornithischian pelvis structure (left side)

Edmontosaurus pelvis (showing ornithischian structure – left side)

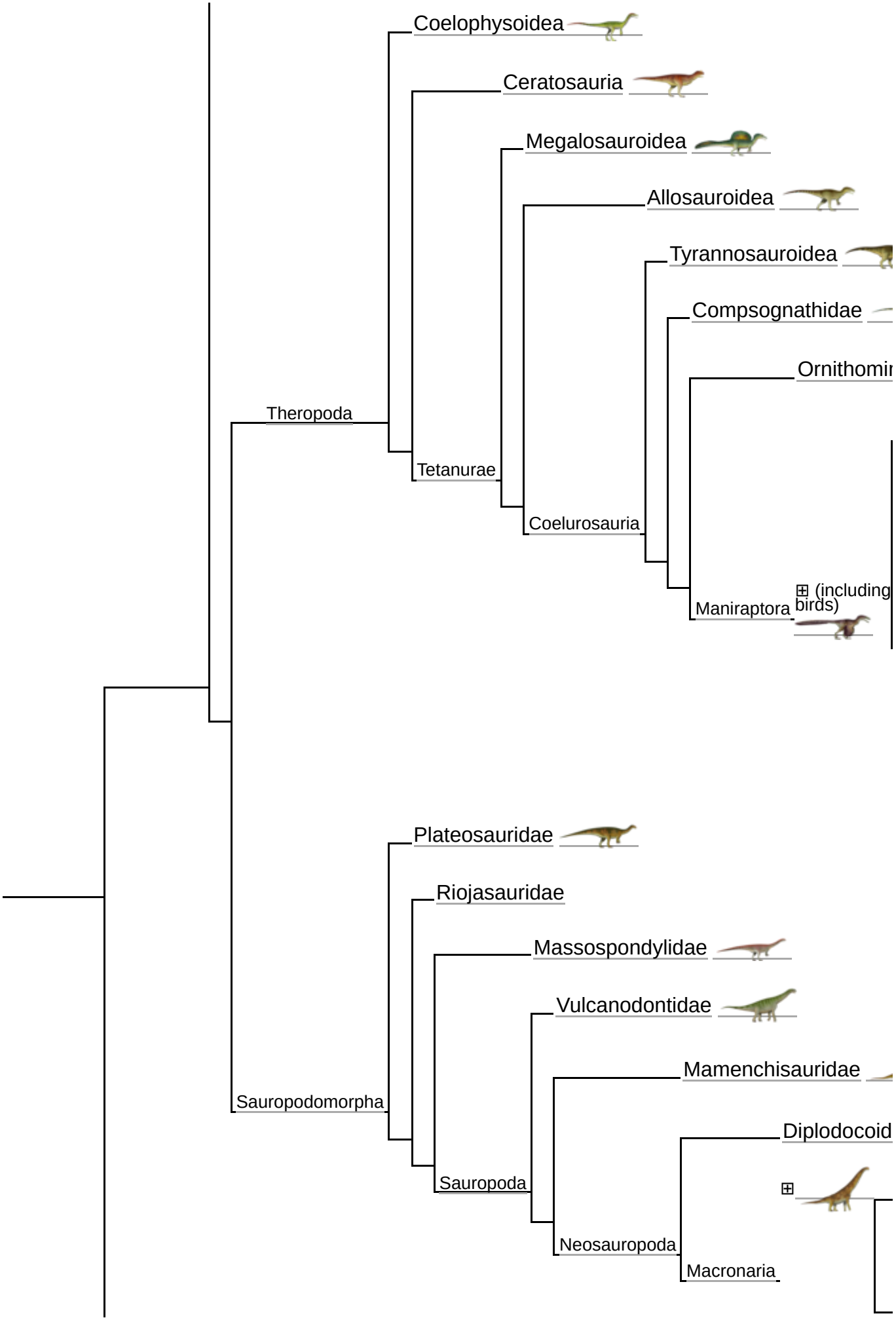
Collectively, dinosaurs as a clade are divided into two primary branches, Saurischia and Ornithischia. Saurischia includes those taxa sharing a more recent common ancestor with birds than with Ornithischia, while Ornithischia includes all taxa sharing a more recent common ancestor with *Triceratops* than with Saurischia. Anatomically, these two groups can be distinguished most noticeably by their pelvic structure. Early saurischians—"lizard-hipped", from the Greek *sauros* (σαῦρος) meaning "lizard" and *ischion* (ἰσχίον) meaning "hip joint"—retained the hip structure of their ancestors, with a pubis bone directed cranially, or forward.[39] This basic form was modified by rotating the pubis backward to varying degrees in several groups (*Herrerasaurus*,[129] therizinosauroids,[130] dromaeosaurids,[131] and birds[18]). Saurischia includes the theropods (exclusively bipedal and with a wide variety of diets) and sauropodomorphs (long-necked herbivores which include advanced, quadrupedal groups).[31][132]

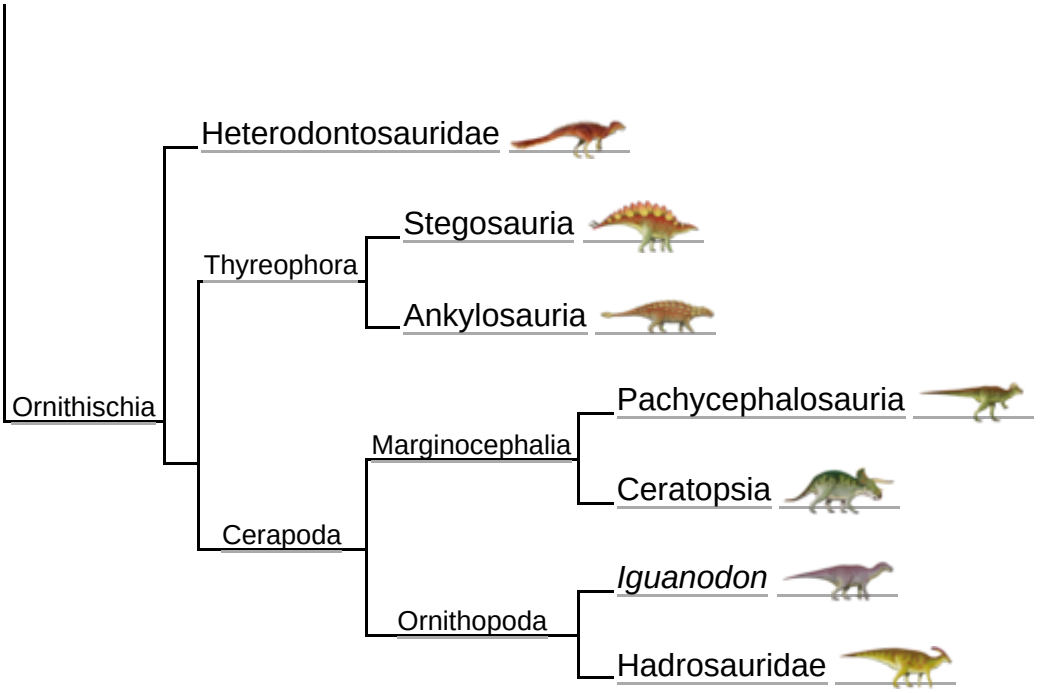
By contrast, ornithischians—"bird-hipped", from the Greek *ornis* (ὀρνίς) meaning "bird" and *ischion* (ἰσχίον) meaning "hip joint"—had a pelvis that superficially resembled a bird's pelvis: the pubic bone was oriented caudally (rear-pointing). Unlike birds, the ornithischian pubis also usually had an additional forward-pointing process. Ornithischia includes a variety of species that were primarily herbivores.

Despite the terms "bird hip" (Ornithischia) and "lizard hip" (Saurischia), birds are not part of Ornithischia. Birds instead belong to Saurischia, the "lizard-hipped" dinosaurs—birds evolved from earlier dinosaurs with "lizard hips".[32]

The following cladogram showing the relationships of dinosaur groups is simplified after Michael Benton, 2015:[39]

Dinosauria Saurischia
- Herrerasauridae

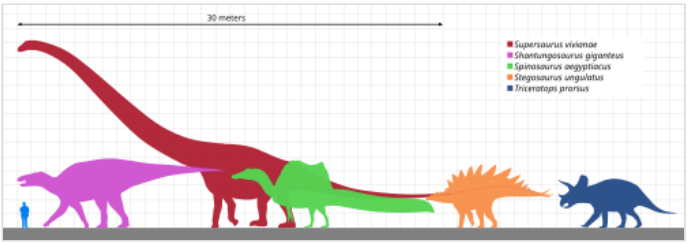

Coelophysoidea
Ceratosauria
Megalosauroidea
Allosauroidea
Tyrannosauroidea
Compsognathidae
Ornithomir
Theropoda
Tetanurae
Coelurosauria
⊞ (including birds)
Maniraptora
Plateosauridae
Riojasauridae
Massospondylidae
Vulcanodontidae
Mamenchisauridae
Diplodocoid
⊞
Sauropodomorpha
Sauropoda
Neosauropoda
Macronaria



Paleobiology

Knowledge about dinosaurs is derived from a variety of fossil and non-fossil records, including fossilized bones, feces, trackways, gastroliths, feathers, impressions of skin, internal organs and other soft tissues.[89][93] Many fields of study contribute to our understanding of dinosaurs, including physics (especially biomechanics), chemistry, biology, and the Earth sciences (of which paleontology is a sub-discipline).[133][134] Two topics of particular interest and study have been dinosaur size and behavior.[135]

Size



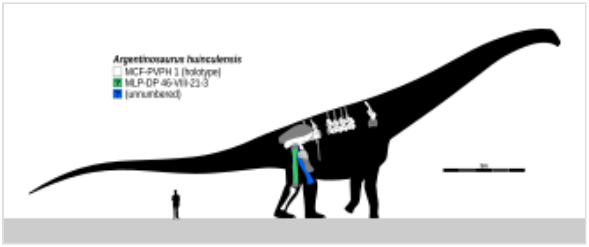
Scale diagram comparing the average human to the longest known dinosaurs in five major clades:
- Sauropoda (*Supersaurus vivianae*)
- Ornithopoda (*Shantungosaurus giganteus*)
- Theropoda (*Spinosaurus aegyptiacus*)
- Thyreophora (*Stegosaurus ungulatus*)
- Marginocephalia (*Triceratops prorsus*)

Current evidence suggests that dinosaur average size varied through the Triassic, Early Jurassic, Late Jurassic and Cretaceous.[110] Predatory theropod dinosaurs, which occupied most terrestrial carnivore niches during the Mesozoic, most often fall into the 100-to-1,000 kg (220-to-2,200 lb) category when sorted by estimated weight into categories based on order of magnitude, whereas recent predatory carnivoran mammals peak in the 10-to-100 kg (22-to-220 lb) category.[136] The mode of Mesozoic dinosaur body masses is between 1 and 10 metric tons (1.1 and 11.0 short tons).[137] This contrasts sharply with the average size of Cenozoic mammals, estimated by the National Museum of Natural History as about 2 to 5 kg (4.4 to 11.0 lb).[138]

The sauropods were the largest and heaviest dinosaurs. For much of the dinosaur era, the smallest sauropods were larger than anything else in their habitat, and the largest was an order of magnitude more massive than anything else that has since walked the Earth. Giant prehistoric mammals such as *Paraceratherium* (the largest land mammal ever) were dwarfed by the giant sauropods, and only modern whales approach or surpass them in size.[139] There are several proposed advantages for the large size of sauropods, including protection from predation, reduction of energy use, and longevity, but it may be that the most important advantage was dietary. Large animals are more efficient at digestion than small animals, because food spends more time in their digestive systems. This also permits them to subsist on food with lower nutritive value than smaller animals. Sauropod remains are mostly found in rock formations interpreted as dry or seasonally dry, and the ability to eat large quantities of low-nutrient browse would have been advantageous in such environments.[140]

Largest and smallest

Scientists will probably never be certain of the largest and smallest dinosaurs to have ever existed. This is because only a tiny percentage of animals were ever fossilized and most of these remain buried in the earth. Few non-avian dinosaur specimens that are recovered are complete skeletons, and impressions of skin and other soft tissues are rare. Rebuilding a complete skeleton by comparing the size and morphology of bones to those of similar, better-known species is an inexact art, and reconstructing the muscles and other organs of the living animal is, at best, a process of educated guesswork.[141]



Comparative size of *Argentinosaurus* to the average human

The tallest and heaviest dinosaur known from good skeletons is *Giraffatitan brancai* (previously classified as a species of *Brachiosaurus*). Its remains were discovered in Tanzania between 1907 and 1912. Bones from several similar-sized individuals were incorporated into the skeleton now mounted and on display at the Museum für Naturkunde in Berlin;[142] this mount is 12 meters (39 ft) tall and 21.8 to 22.5 meters (72 to 74 ft) long,[143][144] and would have belonged to an animal that weighed between 30 000 and 60 000 kilograms (70 000 and 130 000 lb). The longest complete dinosaur is the 27 meters (89 ft) long *Diplodocus*, which was discovered in Wyoming in the United States and displayed in Pittsburgh's Carnegie Museum of Natural History in 1907.[145] The longest dinosaur known from good fossil material is *Patagotitan*: the skeleton mount in the American Museum of Natural History in New York is 37 meters (121 ft) long. The Museo Municipal Carmen Funes in Plaza Huincul, Argentina, has an *Argentinosaurus* reconstructed skeleton mount that is 39.7 meters (130 ft) long.[146]

There were larger dinosaurs, but knowledge of them is based entirely on a small number of fragmentary fossils. Most of the largest herbivorous specimens on record were discovered in the 1970s or later, and include the massive *Argentinosaurus*, which may have weighed 80 000 to 100 000 kilograms (88 to 110 short tons) and reached lengths of 30 to 40 meters (98 to 131 ft); some of the longest were the 33.5-meter (110 ft) long *Diplodocus hallorum*[140] (formerly *Seismosaurus*), the 33-to-34-meter (108 to 112 ft) long *Supersaurus*,[147] and 37-meter (121 ft) long *Patagotitan*; and the tallest, the 18-meter (59 ft) tall *Sauroposeidon*, which could have reached a sixth-floor window. There were a few dinosaurs that were considered either the heaviest or longest. The most famous include *Amphicoelias fragillimus*, known only from a now lost partial vertebral neural arch described in 1878. Extrapolating from the illustration of this

bone, the animal may have been 58 meters (190 ft) long and weighed 122 400 kg (269 800 lb).[140] However, recent research has reclassified *Amphicoelias* from the long, gracile diplodocid to the shorter but much stockier rebbachisaurid. Now renamed as *Maraapunisaurus*, this sauropod measured as much as 40 meters (130 ft) long and weighed as much as 120 000 kg (260 000 lb).[148][149] Another contender of this title includes *Bruhathkayosaurus*, a controversial taxon that was recently confirmed to exist after archived photos were uncovered.[150] *Bruhathkayosaurus* was a titanosaur and would have most likely weighed more than even *Maraapunisaurus*. Recent size estimates in 2023 have this sauropod reaching lengths of up to 44 m (144 ft) long and placed in a colossal weight range of around 110 000–170 000 kg (240 000–370 000 lb). If these upper estimates are true, *Bruhathkayosaurus* would have rivaled the blue whale and *Perucetus colossus* as one of the largest animals to have ever existed.[151]

Bruhathkayosaurus, potentially the largest terrestrial animal to ever exist

The largest carnivorous dinosaur was *Spinosaurus*, reaching a length of 12.6 to 18 meters (41 to 59 ft) and weighing 7 to 20.9 metric tons (7.7 to 23.0 short tons).[152][153] Other large carnivorous theropods included *Giganotosaurus*, *Carcharodontosaurus*, and *Tyrannosaurus*.[153] *Therizinosaurus* and *Deinocheirus* were among the tallest of the theropods. The largest ornithischian dinosaur was probably the hadrosaurid *Shantungosaurus giganteus*, which measured 16.6 meters (54 ft).[154] The largest individuals may have weighed as much as 16 metric tons (18 short tons).[155]

The smallest dinosaur known is the bee hummingbird,[156] with a length of only 5 centimeters (2.0 in) and mass of around 1.8 g (0.063 oz).[157] The smallest known non-avialan dinosaurs were about the size of pigeons and were those theropods most closely related to birds.[158] For example, *Anchiornis huxleyi* is currently the smallest non-avialan dinosaur described from an adult specimen, with an estimated weight of 110 g (3.9 oz)[159] and a total skeletal length of 34 centimeters (1.12 ft).[158][159] The smallest herbivorous non-avialan dinosaurs included *Microceratus* and *Wannanosaurus*, at about 60 centimeters (2.0 ft) long each.[160][161]

An adult bee hummingbird, the smallest known dinosaur

Behavior

Many modern birds are highly social, often found living in flocks. There is general agreement that some behaviors that are common in birds, as well as in crocodilians (closest living relatives of birds), were also common among extinct dinosaur groups. Interpretations of behavior in fossil species are generally based on the pose of skeletons and their habitat, computer simulations of their biomechanics, and comparisons with modern animals in similar ecological niches.[133]

A nesting ground of the hadrosaur *Maiasaura peeblesorum* was discovered in 1978

The first potential evidence for herding or flocking as a widespread behavior common to many dinosaur groups in addition to birds was the 1878 discovery of 31 *Iguanodon*, ornithischians that were then thought to have perished together in Bernissart, Belgium, after they fell into a deep, flooded sinkhole and drowned.[162] Other mass-death sites have been discovered subsequently. Those, along with multiple trackways, suggest that gregarious behavior was common in many early dinosaur species. Trackways of hundreds or even thousands of herbivores indicate that duck-billed (hadrosaurids) may have moved in great herds, like the American bison or the African springbok. Sauropod tracks document that these animals traveled in groups composed of several different species, at least in Oxfordshire, England,[163] although there is no evidence for specific herd structures.[164] Congregating into herds may have evolved for defense, for migratory purposes, or to provide protection for young. There is evidence that many types of slow-growing dinosaurs, including various theropods, sauropods, ankylosaurians, ornithopods, and ceratopsians, formed aggregations of immature individuals. One example is a site in Inner Mongolia that has yielded remains of over 20 *Sinornithomimus*, from one to seven years old. This assemblage is interpreted as a social group that was trapped in mud.[165] The interpretation of dinosaurs as gregarious has also extended to depicting carnivorous theropods as pack hunters working together to bring down large prey.[166][167] However, this lifestyle is uncommon among modern birds, crocodiles, and other reptiles (though it is not unknown as frequently assumed, and is also uncommon in mammals), and the taphonomic evidence suggesting cooperative hunting in such theropods as *Deinonychus* and *Allosaurus* can also be interpreted as the results of fatal disputes between feeding animals.[168]

The crests and frills of some dinosaurs, like the marginocephalians, theropods and lambeosaurines, may have been too fragile to be used for active defense, and so they were likely used for sexual or aggressive displays, though little is known about dinosaur mating and territorialism. Head wounds from bites suggest that theropods, at least, engaged in active aggressive confrontations.[169]

Restoration of two *Centrosaurus apertus* engaged in intra-specific combat

From a behavioral standpoint, one of the most valuable dinosaur fossils was discovered in the Gobi Desert in 1971. It included a *Velociraptor* attacking a *Protoceratops*,[170] providing evidence that dinosaurs did indeed attack each other.[171] Additional evidence for attacking live prey is the partially healed tail of an *Edmontosaurus*, a hadrosaurid dinosaur; the tail is damaged in such a way that shows the animal was bitten by a tyrannosaur but survived.[171] Cannibalism amongst some species of dinosaurs was confirmed by tooth marks found in Madagascar in 2003, involving the theropod *Majungasaurus*.[172]

Comparisons between the scleral rings of dinosaurs and modern birds and reptiles have been used to infer daily activity patterns of dinosaurs. Although it has been suggested that most dinosaurs were active during the day, these comparisons have shown that small predatory dinosaurs such as dromaeosaurids, *Juravenator*, and *Megapnosaurus* were likely nocturnal. Large and medium-sized herbivorous and

Mount of a *Gorgosaurus*, a carnivorous theropod, attacking a herbivorous *Zuul*

omnivorous dinosaurs such as ceratopsians, sauropodomorphs, hadrosaurids, ornithomimosaurs may have been cathemeral, active during short intervals throughout the day, although the small ornithischian *Agilisaurus* was inferred to be diurnal.[173]

Based on fossil evidence from dinosaurs such as *Oryctodromeus*, some ornithischian species seem to have led a partially fossorial (burrowing) lifestyle.[174] Many modern birds are arboreal (tree climbing), and this was also true of many Mesozoic birds, especially the enantiornithines.[175] While some early bird-like species may have already been arboreal as well (including dromaeosaurids) such as *Microraptor*[176]) most non-avialan dinosaurs seem to have relied on land-based locomotion. A good understanding of how dinosaurs moved on the ground is key to models of dinosaur behavior; the science of biomechanics, pioneered by Robert McNeill Alexander, has provided significant insight in this area. For example, studies of the forces exerted by muscles and gravity on dinosaurs' skeletal structure have investigated how fast dinosaurs could run,[133] whether diplodocids could create sonic booms via whip-like tail snapping,[177] and whether sauropods could float.[178]

Communication

Modern birds communicate by visual and auditory signals, and the wide diversity of visual display structures among fossil dinosaur groups, such as horns, frills, crests, sails, and feathers, suggests that visual communication has always been important in dinosaur biology.[179] Reconstruction of the plumage color of *Anchiornis* suggest the importance of color in visual communication in non-avian dinosaurs.[180] A 2009 review indicated that non-avians used visual displays and possibly non-vocal sounds, such as hissing, jaw-grinding or -clapping, splashing, and wing-beating (possible in winged maniraptoran dinosaurs).[179] Vocal communication in non-avian dinosaurs is indicated by the development of advanced hearing in nearly all major groups.[181]

Restoration of a striking and unusual visual display in a *Lambeosaurus magnicristatus*. The crest may also have acted as a resonating chamber for sounds.

Fossilized larynx elements, which often serve as vocal organs in tetrapods, have been identified in two non-avian dinosaurs; the ankylosaurid *Pinacosaurus* and the neornithischian *Pulaosaurus*. Both were ossified (hardened to bone) with flexible joints similar to those of modern birds, which are adapted for opening the throat while the syrinx, a novel organ located at the base of the trachea, produces the calls. This setup suggests that the larynx in dinosaurs did not produce vocalizations but regulated and modified them as they were generated deeper in the respiratory tract.[182][183] Hence, researchers suspect that the syrinx may have been present in non-avian dinosaurs,[183] though the earliest known physical remains are from a specimen of the ancient bird *Vegavis iaai* dated 69–66 million years ago.[184]

A 2016 study concludes that some dinosaurs may have produced closed-mouth vocalizations, such as cooing, hooting, and booming. These occur in both reptiles and birds and involve inflating the esophagus or tracheal pouches. Such vocalizations evolved independently in extant archosaurs numerous times, following increases in body size.[185] The crests of some hadrosaurids and the nasal chambers of ankylosaurids may have been resonators.[186][187]

Reproductive biology

Nest of a plover (*Charadrius*)

All dinosaurs laid amniotic eggs. Dinosaur eggs were usually laid in a nest. Most species create somewhat elaborate nests which can be cups, domes, plates, beds scrapes, mounds, or burrows.[188] Some species of modern bird have no nests; the cliff-nesting common guillemot lays its eggs on bare rock, and male emperor penguins keep eggs between their body and feet. Primitive birds and many non-avialan dinosaurs often lay eggs in communal nests, with males primarily incubating the eggs. While modern birds have only one functional oviduct and lay one egg at a time, more primitive birds and other dinosaurs had two oviducts, like crocodiles. Some non-avialan dinosaurs, such as *Troodon*, exhibited iterative laying, where the adult might lay a pair of eggs every one or two days, and then ensured simultaneous hatching by delaying brooding until all eggs were laid.[189]

When laying eggs, females grow a special type of bone between the hard outer bone and the marrow of their limbs. This medullary bone, which is rich in calcium, is used to make eggshells. A discovery of features in a *Tyrannosaurus* skeleton provided evidence of medullary bone in extinct dinosaurs and, for the first time, allowed paleontologists to establish the sex of a fossil dinosaur specimen. Further research has found medullary bone in the carnosaur *Allosaurus* and the ornithopod *Tenontosaurus*. Because the line of dinosaurs that includes *Allosaurus* and *Tyrannosaurus* diverged from the line that led to *Tenontosaurus* very early in the evolution of dinosaurs, this suggests that the production of medullary tissue is a general characteristic of all dinosaurs.[190]

Another widespread trait among modern birds (but see below in regards to fossil groups and extant megapodes) is parental care for young after hatching. Bob Makela and Jack Horner's 1978 discovery of a *Maiasaura* ("good mother lizard") nesting ground in Montana demonstrated that parental care continued long after birth among ornithopods.[191] A specimen of the oviraptorid *Citipati osmolskae* was discovered in a chicken-like brooding position in 1993,[192] which may indicate that they had begun using an insulating layer of feathers to keep the eggs warm.[193] An embryo of the basal sauropodomorph *Massospondylus* was found without teeth, indicating that some parental care was required to feed the young dinosaurs.[194] Trackways have also confirmed parental behavior among ornithopods from the Isle of Skye in northwestern Scotland.[195]

Fossil interpreted as a nesting oviraptorid *Citipati* at the American Museum of Natural History.

However, there is ample evidence of precociality or superprecociality among many dinosaur species, particularly theropods. For instance, non-ornithuromorph birds have been abundantly demonstrated to have had slow growth rates, megapode-like egg burying behavior and the ability to fly soon after birth.[196][197][198][199] Both *Tyrannosaurus* and *Troodon* had juveniles with clear superprecociality and likely occupying different ecological niches than the adults.[189] Superprecociality has been inferred for sauropods.[200]

Genital structures are unlikely to fossilize as they lack scales that may allow preservation via pigmentation or residual calcium phosphate salts. In 2021, the best preserved specimen of a dinosaur's cloacal vent exterior was described for *Psittacosaurus*, demonstrating lateral swellings similar to crocodylian musk glands used in social displays by both sexes and pigmented regions which could also reflect a signalling function. However, this specimen on its own does not offer enough information to determine whether this dinosaur had sexual signalling functions; it only supports the possibility. Cloacal visual signalling can occur in either males or females in living birds, making it unlikely to be useful to determine sex for extinct dinosaurs.[201] Unlike in some male mammals, there is no evidence of muscle attachments for a mobile penis nor a baculum (penis bone) in dinosaurs. The general sexual position for mating dinosaurs likely involved the male's leg over the female's back as she in positioned her hips upwards and her tail to the side. In some dinosaurs like *Stegosaurus* and *Spinosaurus*, the female may have laid on her side.[202]

Physiology

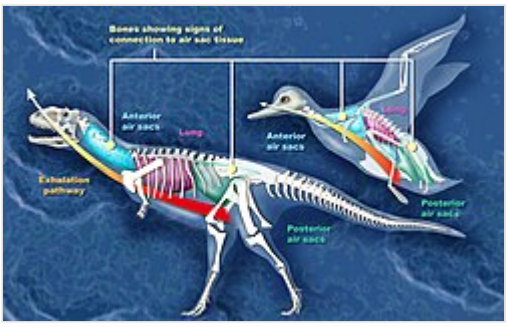
Because both modern crocodilians and birds have four-chambered hearts (albeit modified in crocodilians), it is likely that this is a trait shared by all archosaurs, including all dinosaurs.[203] While all modern birds have high metabolisms and are endothermic ("warm-blooded"), a vigorous debate has been ongoing since the 1960s regarding how far back in the dinosaur lineage this trait extended. Various researchers have supported dinosaurs as being endothermic, ectothermic ("cold-blooded"), or somewhere in between.[204] An emerging consensus among researchers is that, while different lineages of dinosaurs would have had different metabolisms, most of them had higher metabolic rates than other reptiles but lower than living birds and mammals,[205] which is termed mesothermy by some.[206] Evidence from crocodiles and their extinct relatives suggests that such elevated metabolisms could have developed in the earliest archosaurs, which were the common ancestors of dinosaurs and crocodiles.[207][208]

This 1897 restoration of *Brontosaurus* as an aquatic, tail-dragging animal, by Charles R. Knight, typified early views on dinosaur lifestyles.

After non-avian dinosaurs were discovered, paleontologists first posited that they were ectothermic. This was used to imply that the ancient dinosaurs were relatively slow, sluggish organisms, even though many modern reptiles are fast and light-footed despite relying on external sources of heat to regulate their body temperature. The idea of dinosaurs as ectothermic remained a prevalent view until Robert T. Bakker, an early proponent of dinosaur endothermy, published an influential paper on the topic in 1968. Bakker specifically used anatomical and ecological evidence to argue that sauropods, which had hitherto been depicted as sprawling aquatic animals with their tails dragging on the ground, were endotherms that lived vigorous, terrestrial lives. In 1972,

Bakker expanded on his arguments based on energy requirements and predator-prey ratios. This was one of the seminal results that led to the dinosaur renaissance.[62][63][59][209]

One of the greatest contributions to the modern understanding of dinosaur physiology has been paleohistology, the study of microscopic tissue structure in dinosaurs.[210][211] From the 1960s forward, Armand de Ricqlès suggested that the presence of fibrolamellar bone—bony tissue with an irregular, fibrous texture and filled with blood vessels—was indicative of consistently fast growth and therefore endothermy. Fibrolamellar bone was common in both dinosaurs and pterosaurs,[212][213] though not universally present.[214][215] This has led to a significant body of work in reconstructing growth curves and modeling the evolution of growth rates across various dinosaur lineages,[216] which has suggested overall that dinosaurs grew faster than living reptiles.[211] Other lines of evidence suggesting endothermy include the presence of feathers and other types of body coverings in many lineages (see § Feathers); more consistent ratios of the isotope oxygen-18 in bony tissue compared to ectotherms, particularly as latitude and thus air temperature varied, which suggests stable internal temperatures[217][218] (although these ratios can be altered during fossilization[219]); and the discovery of polar dinosaurs, which lived in Australia, Antarctica, and Alaska when these places would have had cool, temperate climates.[220][221][222][223]



Comparison between the air sacs of an abelisaur and a bird

In saurischian dinosaurs, higher metabolisms were supported by the evolution of the avian respiratory system, characterized by an extensive system of air sacs that extended the lungs and invaded many of the bones in the skeleton, making them hollow.[224] Such respiratory systems, which may have appeared in the earliest saurischians,[225] would have provided them with more oxygen compared to a mammal of similar size, while also having a larger resting tidal volume and requiring a lower breathing frequency, which would have allowed them to sustain higher activity levels.[139] The rapid airflow would also have been an effective cooling mechanism, which in conjunction with a lower metabolic rate[226] would have prevented large sauropods from overheating. These traits may have enabled sauropods to grow quickly to gigantic sizes.[227][228] Sauropods may also have benefitted from their size—their small surface area to volume ratio meant that they would have been able to thermoregulate more easily, a phenomenon termed gigantothermy.[139][229]

Like other reptiles, dinosaurs are primarily uricotelic, that is, their kidneys extract nitrogenous wastes from their bloodstream and excrete it as uric acid instead of urea or ammonia via the ureters into the intestine. This would have helped them to conserve water.[205] In most living species, uric acid is excreted along with feces as a semisolid waste.[230][231] However, at least some modern birds (such as hummingbirds) can be facultatively ammonotelic, excreting most of the nitrogenous wastes as ammonia.[232] This material, as well as the output of the intestines, emerges from the cloaca.[233][234] In addition, many species regurgitate pellets,[235] and fossil pellets are known as early as the Jurassic from *Anchiornis*.[236]

The size and shape of the brain can be partly reconstructed based on the surrounding bones. In 1896, Marsh calculated ratios between brain weight and body weight of seven species of dinosaurs, showing that the brain of dinosaurs was proportionally smaller than in today's crocodiles, and that the brain of

Stegosaurus was smaller than in any living land vertebrate. This contributed to the widespread public notion of dinosaurs as being sluggish and extraordinarily stupid. Harry Jerison, in 1973, showed that proportionally smaller brains are expected at larger body sizes, and that brain size in dinosaurs was not smaller than expected when compared to living reptiles.[237] Later research showed that relative brain size progressively increased during the evolution of theropods, with the highest intelligence – comparable to that of modern birds – calculated for the troodontid *Troodon*.[238]

Origin of birds

The possibility that dinosaurs were the ancestors of birds was first suggested in 1868 by Thomas Henry Huxley.[239] After the work of Gerhard Heilmann in the early 20th century, the theory of birds as dinosaur descendants was abandoned in favor of the idea of them being descendants of generalized thecodonts, with the key piece of evidence being the supposed lack of clavicles in dinosaurs.[240] However, as later discoveries showed, clavicles (or a single fused wishbone, which derived from separate clavicles) were not actually absent;[18] they had been found as early as 1924 in *Oviraptor*, but misidentified as an interclavicle.[241] In the 1970s, John Ostrom revived the dinosaur–bird theory,[242] which gained momentum in the coming decades with the advent of cladistic analysis,[243] and a great increase in the discovery of small theropods and early birds.[34] Of particular note have been the fossils of the Jehol Biota, where a variety of theropods and early birds have been found, often with feathers of some type.[18][69] Birds share over a hundred distinct anatomical features with other theropod dinosaurs.[244] They are most closely allied with maniraptoran coelurosaurs.[18] A minority of scientists, most notably Alan Feduccia and Larry Martin, have proposed other evolutionary paths, including revised versions of Heilmann's basal archosaur proposal,[245] or that maniraptoran theropods are the ancestors of birds but themselves are not dinosaurs, only convergent with dinosaurs.[246]

Feathers

Feathers are one of the most recognizable characteristics of modern birds, and a trait that was also shared by several non-avian dinosaurs. Based on the current distribution of fossil evidence, it appears that feathers were an ancestral dinosaurian trait, though one that may have been selectively lost in some species.[247] Direct fossil evidence of feathers or feather-like structures has been discovered in a diverse array of species in many non-avian dinosaur groups,[69] both among saurischians and ornithischians. Simple, branched, feather-like structures are known from heterodontosaurids, primitive neornithischians,[248] and theropods,[249] and primitive ceratopsians. Evidence for true, vaned feathers similar to the flight feathers of modern birds has been found only in the theropod subgroup Maniraptora, which includes oviraptorosaurs, troodontids, dromaeosaurids, and birds.[18][250] Feather-like structures known as pycnofibres have also been found in pterosaurs.[251]

Various feathered non-avian dinosaurs, including *Archaeopteryx*, *Anchiornis*, *Microraptor* and *Zhenyuanlong*

However, researchers do not agree on whether these structures share a common origin between lineages (i.e., they are homologous),[252][253] or if they were the result of widespread experimentation with skin coverings among ornithodirans.[254] If the former is the case, filaments may have been common in the ornithodiran lineage and evolved before the appearance of dinosaurs themselves.[247] Research into the genetics of American alligators has revealed that crocodylian scutes do possess feather-keratins during embryonic development, but these keratins are not expressed by the animals before hatching.[255] The description of feathered dinosaurs has not been without controversy in general; perhaps the most vocal critics have been Alan Feduccia and Theagarten Lingham-Soliar, who have proposed that some purported feather-like fossils are the result of the decomposition of collagenous fiber that underlaid the dinosaurs' skin,[256][257][258] and that maniraptoran dinosaurs with vaned feathers were not actually dinosaurs, but convergent with dinosaurs.[246][257] However, their views have for the most part not been accepted by other researchers, to the point that the scientific nature of Feduccia's proposals has been questioned.[259]

Archaeopteryx was the first fossil found that revealed a potential connection between dinosaurs and birds. It is considered a transitional fossil, in that it displays features of both groups. Brought to light just two years after Charles Darwin's seminal *On the Origin of Species* (1859), its discovery spurred the nascent debate between proponents of evolutionary biology and creationism. This early bird is so dinosaur-like that, without a clear impression of feathers in the surrounding rock, at least one specimen was mistaken for the small theropod *Compsognathus*.[260] Since the 1990s, a number of additional feathered dinosaurs have been found, providing even stronger evidence of the close relationship between dinosaurs and modern birds. Many of these specimens were unearthed in the lagerstätten of the Jehol Biota.[253] If feather-like structures were indeed widely present among non-avian dinosaurs, the lack of abundant fossil evidence for them may be due to the fact that delicate features like skin and feathers are seldom preserved by fossilization and thus often absent from the fossil record.[261]

Skeleton

Because feathers are often associated with birds, feathered dinosaurs are often touted as the missing link between birds and dinosaurs. However, the multiple skeletal features also shared by the two groups represent another important line of evidence for paleontologists. Areas of the skeleton with important similarities include the neck, pubis, wrist (semi-lunate carpal), arm and pectoral girdle, furcula (wishbone), and breast bone. Comparison of bird and dinosaur skeletons through cladistic analysis strengthens the case for the link.[262]

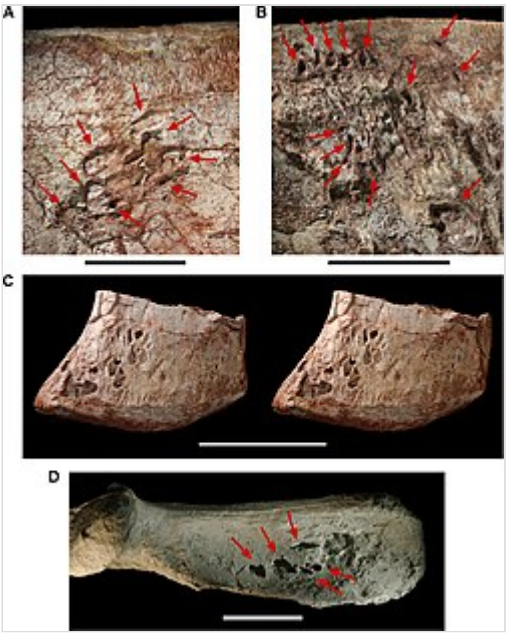
Soft anatomy

Large meat-eating dinosaurs had a complex system of air sacs similar to those found in modern birds, according to a 2005 investigation led by Patrick M. O'Connor. The lungs of theropod dinosaurs (carnivores that walked on two legs and had bird-like feet) likely pumped air into hollow sacs in their skeletons, as is the case in birds. "What was once formally considered unique to birds was present in some form in the ancestors of birds", O'Connor said.[263][264] In 2008, scientists described *Aerosteon riocoloradensis*, the skeleton of which supplies the strongest evidence to date of a dinosaur with a bird-like breathing system. CT scanning of *Aerosteon*'s fossil bones revealed evidence for the existence of air sacs within the animal's body cavity.[224][265]

Behavioral evidence

Fossils of the troodonts *Mei* and *Sinornithoides* demonstrate that some dinosaurs slept with their heads tucked under their arms.[266] This behavior, which may have helped to keep the head warm, is also characteristic of modern birds. Several deinonychosaur and oviraptorosaur specimens have also been found preserved on top of their nests, likely brooding in a bird-like manner.[267] The ratio between egg volume and body mass of adults among these dinosaurs suggest that the eggs were primarily brooded by the male and that the young were highly precocial, similar to many modern ground-dwelling birds.[268]

Some dinosaurs are known to have used gizzard stones like modern birds. These stones are swallowed by animals to aid digestion and break down food and hard fibers once they enter the stomach. When found in association with fossils, gizzard stones are called gastroliths.[269]



Pneumatopores on the left ilium of *Aerosteon riocoloradensis*

Extinction of major groups

All non-avian dinosaurs and most lineages of birds[270] became extinct in a mass extinction event, called the Cretaceous–Paleogene (K-Pg) extinction event, at the end of the Cretaceous period. Above the Cretaceous–Paleogene boundary, which has been dated to 66.038 ± 0.025 million years ago,[271] fossils of non-avian dinosaurs disappear abruptly; the absence of dinosaur fossils was historically used to assign rocks to the ensuing Cenozoic. The nature of the event that caused this mass extinction has been extensively studied since the 1970s, leading to the development of two mechanisms that are thought to have played major roles: an extraterrestrial impact event in the Yucatán Peninsula, along with flood basalt volcanism in India. However, the specific mechanisms of the extinction event and the extent of its effects on dinosaurs are still areas of ongoing research.[272] Alongside dinosaurs, many other groups of animals became extinct: pterosaurs, marine reptiles such as mosasaurs and plesiosaurs, several groups of mammals, ammonites (nautilus-like mollusks), rudists (reef-building bivalves), and various groups of marine plankton.[30][273] In all, approximately 47% of genera and 76% of species on Earth became extinct during the K-Pg extinction event.[274] The relatively large size of most dinosaurs and the low diversity of small-bodied dinosaur species at the end of the Cretaceous may have contributed to their extinction;[275] the extinction of the bird lineages that did not survive may also have been caused by a dependence on forest habitats or a lack of adaptations to eating seeds for survival.[276][277]

Pre-extinction diversity

Just before the K-Pg extinction event, the number of non-avian dinosaur species that existed globally has been estimated at between 628 and 1078.[278] It remains uncertain whether the diversity of dinosaurs was in gradual decline before the K-Pg extinction event, or whether dinosaurs were actually thriving prior to the extinction. Rock formations from the Maastrichtian epoch, which directly preceded the extinction, have been found to have lower diversity than the preceding Campanian epoch, which led to the prevailing

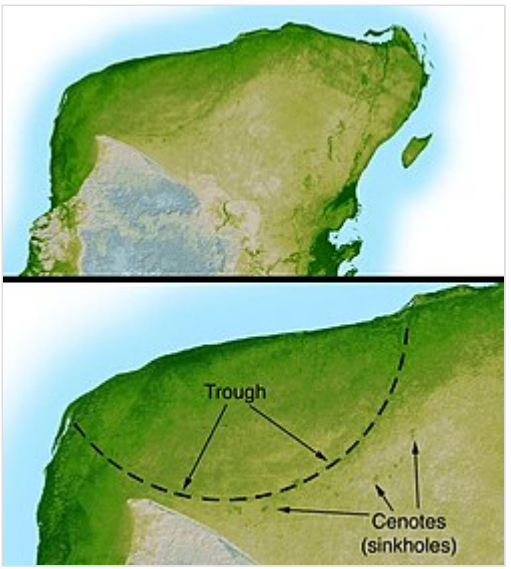
view of a long-term decline in diversity.[30][273][279] However, these comparisons did not account either for varying preservation potential between rock units or for different extents of exploration and excavation.[272] In 1984, Dale Russell carried out an analysis to account for these biases, and found no evidence of a decline;[280] another analysis by David Fastovsky and colleagues in 2004 even showed that dinosaur diversity continually increased until the extinction,[281] but this analysis has been rebutted.[282] Since then, different approaches based on statistics and mathematical models have variously supported either a sudden extinction[272][278][283] or a gradual decline.[284][285] End-Cretaceous trends in diversity may have varied between dinosaur lineages: it has been suggested that sauropods were not in decline, while ornithischians and theropods were in decline.[286][287]

Impact event

The bolide impact hypothesis, first brought to wide attention in 1980 by Walter Alvarez, Luis Alvarez, and colleagues, attributes the K-Pg extinction event to a bolide (extraterrestrial projectile) impact.[288] Alvarez and colleagues proposed that a sudden increase in iridium levels, recorded around the world in rock deposits at the Cretaceous–Paleogene boundary, was direct evidence of the impact.[289] Shocked quartz, indicative of a strong shockwave emanating from an impact, was also found worldwide.[290] The actual impact site remained elusive until a crater measuring 180 km (110 mi) wide was discovered in the Yucatán Peninsula of southeastern Mexico, and was publicized in a 1991 paper by Alan Hildebrand and colleagues.[291] Now, the bulk of the evidence suggests that a bolide 5 to 15 kilometers (3 to 9½ miles) wide impacted the Yucatán Peninsula 66 million years ago, forming this crater[292] and creating a "kill mechanism" that triggered the extinction event.[293][294][295]

Luis (left) and his son Walter Alvarez (right) at the K-T Boundary in Gubbio, Italy, 1981

Within hours, the Chicxulub impact would have created immediate effects such as earthquakes,[296] tsunamis,[297] and a global firestorm that likely killed unsheltered animals and started wildfires.[298][299] However, it would also have had longer-term consequences for the environment. Within days, sulfate aerosols released from rocks at the impact site would have contributed to acid rain and ocean acidification.[300][301] Soot aerosols are thought to have spread around the world over the ensuing months and years; they would have cooled the surface of the Earth by reflecting thermal radiation, and greatly slowed photosynthesis by blocking out sunlight, thus creating an impact winter.[272][302][303] (This role was ascribed to sulfate aerosols until experiments demonstrated otherwise.[301]) The cessation of photosynthesis would have led to the collapse of food webs depending on leafy plants, which included all dinosaurs save for grain-eating birds.[277]



The Chicxulub Crater at the tip of the Yucatán Peninsula; the impactor that formed this crater may have caused the dinosaur extinction.

Deccan Traps

At the time of the K-Pg extinction, the Deccan Traps flood basalts of India were actively erupting. The eruptions can be separated into three phases around the K-Pg boundary, two prior to the boundary and one after. The second phase, which occurred very close to the boundary, would have extruded 70 to 80% of the volume of these eruptions in intermittent pulses that occurred around 100,000 years apart.[304][305] Greenhouse gases such as carbon dioxide and sulfur dioxide would have been released by this volcanic activity,[306][307] resulting in climate change through temperature perturbations of roughly 3 °C (5.4 °F) but possibly as high as 7 °C (13 °F).[308] Like the Chicxulub impact, the eruptions may also have released sulfate aerosols, which would have caused acid rain and global cooling.[309] However, due to large error margins in the dating of the eruptions, the role of the Deccan Traps in the K-Pg extinction remains unclear.[271][272][310]

Before 2000, arguments that the Deccan Traps eruptions—as opposed to the Chicxulub impact—caused the extinction were usually linked to the view that the extinction was gradual. Prior to the discovery of the Chicxulub crater, the Deccan Traps were used to explain the global iridium layer;[306][311] even after the crater's discovery, the impact was still thought to only have had a regional, not global, effect on the extinction event.[312] In response, Luis Alvarez rejected volcanic activity as an explanation for the iridium layer and the extinction as a whole.[313] Since then, however, most researchers have adopted a more moderate position, which identifies the Chicxulub impact as the primary progenitor of the extinction while also recognizing that the Deccan Traps may also have played a role. Walter Alvarez himself has acknowledged that the Deccan Traps and other ecological factors may have contributed to the extinctions in addition to the Chicxulub impact.[314] Some estimates have placed the start of the second phase in the Deccan Traps eruptions within 50,000 years after the Chicxulub impact.[315] Combined with mathematical modelling of the seismic waves that would have been generated by the impact, this has led to the suggestion that the Chicxulub impact may have triggered these eruptions by increasing the permeability of the mantle plume underlying the Deccan Traps.[316][317]

Whether the Deccan Traps were a major cause of the extinction, on par with the Chicxulub impact, remains uncertain. Proponents consider the climatic impact of the sulfur dioxide released to have been on par with the Chicxulub impact, and also note the role of flood basalt volcanism in other mass extinctions like the Permian-Triassic extinction event.[318][319] They consider the Chicxulub impact to have worsened the ongoing climate change caused by the eruptions.[320] Meanwhile, detractors point out the sudden nature of the extinction and that other pulses in Deccan Traps activity of comparable magnitude did not appear to have caused extinctions. They also contend that the causes of different mass extinctions should be assessed separately.[321] In 2020, Alfio Chiarenza and colleagues suggested that the Deccan Traps may even have had the opposite effect: they suggested that the long-term warming caused by its carbon dioxide emissions may have dampened the impact winter from the Chicxulub impact.[295]

Possible Paleocene survivors

Non-avian dinosaur remains have occasionally been found above the K-Pg boundary. In 2000, Spencer Lucas and colleagues reported the discovery of a single hadrosaur right femur in the San Juan Basin of New Mexico, and described it as evidence of Paleocene dinosaurs. The rock unit in which the bone was discovered has been dated to the early Paleocene epoch, approximately 64.8 million years ago.[322] If the bone was not re-deposited by weathering action, it would provide evidence that some dinosaur populations survived at least half a million years into the Cenozoic.[323] Other evidence includes the presence of dinosaur remains in the Hell Creek Formation up to 1.3 m (4.3 ft) above the Cretaceous–Paleogene boundary, representing 40,000 years of elapsed time. This has been used to support the view

that the K-Pg extinction was gradual.[324] However, these supposed Paleocene dinosaurs are considered by many other researchers to be reworked, that is, washed out of their original locations and then reburied in younger sediments.[325][326][327] The age estimates have also been considered unreliable.[328]

Cultural depictions

By human standards, dinosaurs were creatures of fantastic appearance and often enormous size. As such, they have captured the popular imagination and become an enduring part of human culture. The entry of the word "dinosaur" into the common vernacular reflects the animals' cultural importance: in English, "dinosaur" is commonly used to describe anything that is impractically large, obsolete, or bound for extinction.[329]

Outdated *Iguanodon* statues created by Benjamin Waterhouse Hawkins for the Crystal Palace Park in 1853

Public enthusiasm for dinosaurs first developed in Victorian England, where in 1854, three decades after the first scientific descriptions of dinosaur remains, a menagerie of lifelike dinosaur sculptures was unveiled in London's Crystal Palace Park. The Crystal Palace dinosaurs proved so popular that a strong market in smaller replicas soon developed. In subsequent decades, dinosaur exhibits opened at parks and museums around the world, ensuring that successive generations would be introduced to the animals in an immersive and exciting way.[330] The enduring popularity of dinosaurs, in its turn, has resulted in significant public funding for dinosaur science, and has frequently spurred new discoveries. In the United States, for example, the competition between museums for public attention led directly to the Bone Wars of the 1880s and 1890s, during which a pair of feuding paleontologists made enormous scientific contributions.[331]

Gertie the Dinosaur (1914) by Winsor McCay, featuring the first animated dinosaur

The popular preoccupation with dinosaurs has ensured their appearance in literature, film, and other media. Beginning in 1852 with a passing mention in Charles Dickens' *Bleak House*,[332] dinosaurs have been featured in large numbers of fictional works. Jules Verne's 1864 novel *Journey to the Center of the Earth*, Sir Arthur Conan Doyle's 1912 book *The Lost World*, the 1914 animated film *Gertie the Dinosaur* (featuring the first animated dinosaur), the iconic 1933 film *King Kong*, the 1954 *Godzilla* and its many sequels, the best-selling 1990 novel *Jurassic Park* by Michael Crichton and its 1993 film adaptation are just a few notable examples of dinosaur appearances in fiction. Authors of general-interest non-fiction works about dinosaurs, including some prominent paleontologists, have often sought to use the animals as a way to educate readers about science in general. Dinosaurs are ubiquitous in advertising; numerous companies have referenced dinosaurs in printed or televised advertisements, either in order to sell their own products or in order to characterize their rivals as slow-moving, dim-witted, or obsolete.[333][334]

See also

Dinosaurs portal

Paleontology portal

- Dinosaur diet and feeding
- Evolutionary history of life
- Lists of dinosaur-bearing stratigraphic units
- List of dinosaur genera
- List of bird genera
- List of birds
- List of informally named dinosaurs
- List of films featuring dinosaurs

Notes

1. Dinosaurs (including birds) are members of the natural group Reptilia. Their biology does not precisely correspond to the antiquated class Reptilia of Linnaean taxonomy, consisting of cold-blooded amniotes without fur or feathers. As Linnaean taxonomy was formulated for modern animals prior to the study of evolution and paleontology, it fails to account for extinct animals with intermediate traits between traditional classes.

References

1. Matthew G. Baron; Megan E. Williams (2018). "A re-evaluation of the enigmatic dinosauriform Caseosaurus crosbyensis from the Late Triassic of Texas, USA and its implications for early dinosaur evolution". Acta Palaeontologica Polonica. 63. doi:10.4202/app.00372.2017 (https://doi.org/10.4202%2Fapp.00372.2017).
2. Andrea Cau (2018). "The assembly of the avian body plan: a 160-million-year long process" (https://www.paleoitalia.it/wp-content/uploads/2024/03/01_Cau_2018_BSPI_571.pdf) Archived (https://web.archive.org/web/20241203102744/https://www.paleoitalia.it/wp-content/uploads/2024/03/01_Cau_2018_BSPI_571.pdf) December 3, 2024, at the Wayback Machine (PDF). Bollettino della Società Paleontologica Italiana. 57 (1): 1–25.
3. Ferigolo, Jorge; Langer, Max C. (2007). "A Late Triassic dinosauriform from south Brazil and the origin of the ornithischian predentary bone". *Historical Biology*. **19** (1): 23–33. Bibcode:2007HBio...19...23F (https://ui.adsabs.harvard.edu/abs/2007HBio...19...23F). doi:10.1080/08912960600845767 (https://doi.org/10.1080%2F08912960600845767). ISSN 0891-2963 (https://search.worldcat.org/issn/0891-2963). S2CID 85819339 (https://api.semanticscholar.org/CorpusID:85819339).

4. Langer, Max C.; Ferigolo, Jorge (2013). "The Late Triassic dinosauromorph *Sacisaurus agudoensis* (Caturrita Formation; Rio Grande do Sul, Brazil): anatomy and affinities" (https://sp.lyellcollection.org/content/379/1/353). *Geological Society, London, Special Publications*. **379** (1): 353–392. Bibcode:2013GSLSP.379..353L (https://ui.adsabs.harvard.edu/abs/2013GSLSP.379..353L). doi:10.1144/SP379.16 (https://doi.org/10.1144%2FSP379.16). ISSN 0305-8719 (https://search.worldcat.org/issn/0305-8719). S2CID 131414332 (https://api.semanticscholar.org/CorpusID:131414332). Archived (https://web.archive.org/web/20210814015927/https://sp.lyellcollection.org/content/379/1/353) from the original on August 14, 2021. Retrieved March 26, 2021.
5. Cabreira, S.F.; Kellner, A.W.A.; Dias-da-Silva, S.; da Silva, L.R.; Bronzati, M.; de Almeida Marsola, J.C.; Müller, R.T.; de Souza Bittencourt, J.; Batista, B.J.; Raugust, T.; Carrilho, R.; Brodt, A.; Langer, M.C. (2016). "A Unique Late Triassic Dinosauromorph Assemblage Reveals Dinosaur Ancestral Anatomy and Diet" (https://doi.org/10.1016%2Fj.cub.2016.09.040). *Current Biology*. **26** (22): 3090–3095. Bibcode:2016CBio...26.3090C (https://ui.adsabs.harvard.edu/abs/2016CBio...26.3090C). doi:10.1016/j.cub.2016.09.040 (https://doi.org/10.1016%2Fj.cub.2016.09.040). ISSN 0960-9822 (https://search.worldcat.org/issn/0960-9822). PMID 27839975 (https://pubmed.ncbi.nlm.nih.gov/27839975).
6. Müller, Rodrigo Temp; Garcia, Maurício Silva (August 26, 2020). "A paraphyletic 'Silesauridae' as an alternative hypothesis for the initial radiation of ornithischian dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7480155). *Biology Letters*. **16** (8) 20200417. doi:10.1098/rsbl.2020.0417 (https://doi.org/10.1098%2Frsbl.2020.0417). PMC 7480155 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7480155). PMID 32842895 (https://pubmed.ncbi.nlm.nih.gov/32842895).
7. "The 'birth' of dinosaurs" (https://morethanadodo.com/2017/04/28/the-birth-of-dinosaurs/). *More Than A Dodo*. April 28, 2017. Archived (https://web.archive.org/web/20230314201255/https://morethanadodo.com/2017/04/28/the-birth-of-dinosaurs/) from the original on March 14, 2023. Retrieved March 15, 2023.
8. "The Birth of Dinosaurs: Richard Owen and Dinosauria" (https://blog.biodiversitylibrary.org/2015/10/the-birth-of-dinosaurs-richard-owen-and-dinosauria.html). *Biodiversity Heritage Library*. October 16, 2015. Archived (https://web.archive.org/web/20230314201310/https://blog.biodiversitylibrary.org/2015/10/the-birth-of-dinosaurs-richard-owen-and-dinosauria.html) from the original on March 14, 2023. Retrieved March 15, 2023.
9. Brett-Surman, M. K.; Holtz, Thomas R.; Farlow, James O. (June 27, 2012). *The Complete Dinosaur* (https://books.google.com/books?id=pX_I24sDARwC&dq=Hugh+Torrens+Dinosauria+Owen&pg=PA25). Indiana University Press. p. 25. ISBN 978-0-253-00849-7.
10. Owen 1842, p.103 (https://archive.org/details/reportofeleventh42lond/page/n141): "The combination of such characters ... will, it is presumed, be deemed sufficient ground for establishing a distinct tribe or sub-order of Saurian Reptiles, for which I would propose the name of *Dinosauria**. (*Gr. *δεινός*, fearfully great; *σαύρος*, a lizard. ...)
11. "Dinosauria" (https://www.merriam-webster.com/dictionary/Dinosauria). *Merriam-Webster.com Dictionary*. Merriam-Webster. OCLC 1032680871 (https://search.worldcat.org/oclc/1032680871). Retrieved November 10, 2019.
12. Crane, George R. (ed.). "Greek Dictionary Headword Search Results" (https://www.perseus.tufts.edu/hopper/resolveform?type=start&lookup=deino%2Fs&lang=greek). *Perseus 4.0*. Medford and Somerville, MA: Tufts University. Archived (https://web.archive.org/web/20191210013213/https://www.perseus.tufts.edu/hopper/resolveform?type=start&lookup=deino%2Fs&lang=greek) from the original on December 10, 2019. Retrieved October 13, 2019. Lemma for 'δεινός (https://www.perseus.tufts.edu/hopper/text?doc=Perseus%3Atext%3A1999.04.0057%3Aentry%3Ddeino%2Fs1) Archived (https://web.archive.org/web/20191210013242/https://www.perseus.tufts.edu/hopper/text?doc=Perseus%3Atext%3A1999.04.0057%3Aentry%3Ddeino%2Fs1) December 10, 2019, at the Wayback Machine' from Henry George Liddell, Robert Scott, *A Greek-English Lexicon* (1940): 'fearful, terrible'.

13. Farlow & Brett-Surman 1997, pp. ix–xi (https://archive.org/details/isbn_9780253333490/page/n12), Preface, "Dinosaurs: The Terrestrial Superlative" by James O. Farlow and M.K. Brett-Surman.
14. Weishampel, Dodson & Osmólska 2004, pp. 7–19, chpt. 1: "Origin and Relationships of Dinosauria" by Michael J. Benton.
15. Olshevsky 2000
16. Langer, Max C.; Ezcurra, Martin D.; Bittencourt, Jonathas S.; Novas, Fernando E. (February 2010). "The origin and early evolution of dinosaurs". *Biological Reviews*. **85** (1). Cambridge: Cambridge Philosophical Society: 65–66, 82. doi:10.1111/j.1469-185x.2009.00094.x (https://doi.org/10.1111%2Fj.1469-185x.2009.00094.x). hdl:11336/103412 (https://hdl.handle.net/11336%2F103412). ISSN 1464-7931 (https://search.worldcat.org/issn/1464-7931). PMID 19895605 (https://pubmed.ncbi.nlm.nih.gov/19895605). S2CID 34530296 (https://api.semanticscholar.org/CorpusID:34530296).
17. "Using the tree for classification" (https://evolution.berkeley.edu/evolibrary/article/evo_10). *Understanding Evolution*. Berkeley: University of California, Berkeley. Archived (https://web.archive.org/web/20190831003846/https://evolution.berkeley.edu/evolibrary/article/evo_10) from the original on August 31, 2019. Retrieved October 14, 2019.
18. Weishampel, Dodson & Osmólska 2004, pp. 210–231, chpt. 11: "Basal Avialae" by Kevin Padian.
19. Wade, Nicholas (March 22, 2017). "Shaking Up the Dinosaur Family Tree" (https://www.nytimes.com/2017/03/22/science/dinosaur-family-tree.html). *The New York Times*. New York. ISSN 0362-4331 (https://search.worldcat.org/issn/0362-4331). Archived (https://web.archive.org/web/20180407234942/https://www.nytimes.com/2017/03/22/science/dinosaur-family-tree.html) from the original on April 7, 2018. Retrieved October 30, 2019. "A version of this article appears in print on March 28, 2017, on Page D6 of the New York edition with the headline: Shaking Up the Dinosaur Family Tree."
20. Baron, Matthew G.; Norman, David B.; Barrett, Paul M. (2017). "A new hypothesis of dinosaur relationships and early dinosaur evolution". *Nature*. **543** (7646). London: Nature Research: 501–506. Bibcode:2017Natur.543..501B (https://ui.adsabs.harvard.edu/abs/2017Natur.543..501B). doi:10.1038/nature21700 (https://doi.org/10.1038%2Fnature21700). ISSN 0028-0836 (https://search.worldcat.org/issn/0028-0836). PMID 28332513 (https://pubmed.ncbi.nlm.nih.gov/28332513). S2CID 205254710 (https://api.semanticscholar.org/CorpusID:205254710).
21. Glut 1997, p. 40
22. Lambert & The Diagram Group 1990, p. 288 (https://archive.org/details/dinosaurdatabook00lamb/page/288)
23. Farlow & Brett-Surman 1997, pp. 607–624 (https://archive.org/details/isbn_9780253333490/page/607), chpt. 39: "Major Groups of Non-Dinosaurian Vertebrates of the Mesozoic Era" by Michael Morales.
24. Tennant, Jonathan P.; Chiarenza, Alfio Alessandro; Baron, Matthew (February 19, 2018). "How has our knowledge of dinosaur diversity through geologic time changed through research history?" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5822849). *PeerJ*. **6** e4417. doi:10.7717/peerj.4417 (https://doi.org/10.7717%2Fpeerj.4417). PMC 5822849 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5822849). PMID 29479504 (https://pubmed.ncbi.nlm.nih.gov/29479504).

25. Starrfelt, Jostein; Liow, Lee Hsiang (2016). "How many dinosaur species were there? Fossil bias and true richness estimated using a Poisson sampling model" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4810813). *Philosophical Transactions of the Royal Society B*. **371** (1691) 20150219. London: Royal Society. doi:10.1098/rstb.2015.0219 (https://doi.org/10.1098%2Frstb.2015.0219). eISSN 1471-2970 (https://search.worldcat.org/issn/1471-2970). ISSN 0962-8436 (https://search.worldcat.org/issn/0962-8436). PMC 4810813 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4810813). PMID 26977060 (https://pubmed.ncbi.nlm.nih.gov/26977060). S2CID 736657 (https://api.semanticscholar.org/CorpusID:736657).
26. Wang, Steve C.; Dodson, Peter (2006). "Estimating the diversity of dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1564218). *Proc. Natl. Acad. Sci. U.S.A.* **103** (37). Washington, D.C.: National Academy of Sciences: 13601–13605. Bibcode:2006PNAS..10313601W (https://ui.adsabs.harvard.edu/abs/2006PNAS..10313601W). doi:10.1073/pnas.0606028103 (https://doi.org/10.1073%2Fpnas.0606028103). ISSN 0027-8424 (https://search.worldcat.org/issn/0027-8424). PMC 1564218 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1564218). PMID 16954187 (https://pubmed.ncbi.nlm.nih.gov/16954187).
27. Russell, Dale A. (1995). "China and the lost worlds of the dinosaurian era". *Historical Biology*. **10** (1). Milton Park, Oxfordshire: Taylor & Francis: 3–12. Bibcode:1995HBio...10....3R (https://ui.adsabs.harvard.edu/abs/1995HBio...10....3R). doi:10.1080/10292389509380510 (https://doi.org/10.1080%2F10292389509380510). ISSN 0891-2963 (https://search.worldcat.org/issn/0891-2963).
28. Black, Riley (March 23, 2016). "Most Dinosaur Species Are Still Undiscovered" (https://www.nationalgeographic.com/science/article/most-dinosaur-species-are-still-undiscovered/). *National Geographic News*. Archived (https://web.archive.org/web/20210306214843/https://www.nationalgeographic.com/science/article/most-dinosaur-species-are-still-undiscovered/) from the original on March 6, 2021. Retrieved June 6, 2021.
29. Gill, F.; Donsker, D.; Rasmussen, P. (2021). "Welcome" (https://www.worldbirdnames.org/new/). *IOC World Bird List 11.1*. Archived (https://web.archive.org/web/20230730175525/https://www.worldbirdnames.org/new/) from the original on July 30, 2023. Retrieved January 25, 2021.
30. MacLeod, N.; Rawson, P.F.; Forey, P.L.; et al. (1997). "The Cretaceous–Tertiary biotic transition". *Journal of the Geological Society*. **154** (2): 265–292. Bibcode:1997JGSoc.154..265M (https://ui.adsabs.harvard.edu/abs/1997JGSoc.154..265M). doi:10.1144/gsjgs.154.2.0265 (https://doi.org/10.1144%2Fgsjgs.154.2.0265). ISSN 0016-7649 (https://search.worldcat.org/issn/0016-7649). S2CID 129654916 (https://api.semanticscholar.org/CorpusID:129654916).
31. Amiot, Romain; Buffetaut, Éric; Lécuyer, Christophe; et al. (2010). "Oxygen isotope evidence for semi-aquatic habits among spinosaurid theropods". *Geology*. **38** (2). Boulder, CO: Geological Society of America: 139–142. Bibcode:2010Geo....38..139A (https://ui.adsabs.harvard.edu/abs/2010Geo....38..139A). doi:10.1130/G30402.1 (https://doi.org/10.1130%2FG30402.1). ISSN 0091-7613 (https://search.worldcat.org/issn/0091-7613).
32. Brusatte 2012, pp. 9–20, 21
33. Nesbitt, Sterling J. (2011). "The Early Evolution of Archosaurs: Relationships and the Origin of Major Clades" (https://doi.org/10.1206%2F352.1). *Bulletin of the American Museum of Natural History*. **2011** (352). New York: American Museum of Natural History: 1–292. doi:10.1206/352.1 (https://doi.org/10.1206%2F352.1). hdl:2246/6112 (https://hdl.handle.net/2246%2F6112). ISSN 0003-0090 (https://search.worldcat.org/issn/0003-0090). S2CID 83493714 (https://api.semanticscholar.org/CorpusID:83493714).
34. Paul 2000, pp. 140–168, chpt. 3: "Classification and Evolution of the Dinosaur Groups" by Thomas R. Holtz Jr.

35. Smith, Dave; et al. "Dinosauria: Morphology" (https://ucmp.berkeley.edu/diapsids/dinomm.html). Berkeley: University of California Museum of Paleontology. Archived (https://web.archive.org/web/20201112035340/https://ucmp.berkeley.edu/diapsids/dinomm.html) from the original on November 12, 2020. Retrieved October 16, 2019.
36. Langer, Max C.; Abdala, Fernando; Richter, Martha; Benton, Michael J. (1999). "Un dinosaure sauropodomorphe dans le Trias supérieur (Carnien) du Sud du Brésil" [A sauropodomorph dinosaur from the Upper Triassic (Carman) of southern Brazil]. *Comptes Rendus de l'Académie des Sciences, Série IIA*. **329** (7). Amsterdam: Elsevier on behalf of the French Academy of Sciences: 511–517. Bibcode:1999CRASE.329..511L (https://ui.adsabs.harvard.edu/abs/1999CRASE.329..511L). doi:10.1016/S1251-8050(00)80025-7 (https://doi.org/10.1016%2FS1251-8050%2800%2980025-7). ISSN 1251-8050 (https://search.worldcat.org/issn/1251-8050).
37. Nesbitt, Sterling J.; Irmis, Randall B.; Parker, William G. (2007). "A critical re-evaluation of the Late Triassic dinosaur taxa of North America". *Journal of Systematic Palaeontology*. **5** (2). Milton Park, Oxfordshire: Taylor & Francis on behalf of the Natural History Museum, London: 209–243. Bibcode:2007JSPal...5..209N (https://ui.adsabs.harvard.edu/abs/2007JSPal...5..209N). doi:10.1017/S1477201907002040 (https://doi.org/10.1017%2FS1477201907002040). ISSN 1477-2019 (https://search.worldcat.org/issn/1477-2019). S2CID 28782207 (https://api.semanticscholar.org/CorpusID:28782207).
38. This was recognized not later than 1909: Celeskey, Matt (2005). "Dr. W. J. Holland and the Sprawling Sauropods" (https://web.archive.org/web/20110612180650/http://www.hmnh.org/library/diplodocus/holland1910.html). *The Hairy Museum of Natural History*. Archived from the original (http://www.hmnh.org/library/diplodocus/holland1910.html) on June 12, 2011. Retrieved October 18, 2019.
    - Holland, William J. (May 1910). "A Review of Some Recent Criticisms of the Restorations of Sauropod Dinosaurs Existing in the Museums of the United States, with Special Reference to that of *Diplodocus Carnegiei* in the Carnegie Museum" (https://archive.org/details/jstor-2455581). *The American Naturalist*. **44** (521). American Society of Naturalists: 259–283. Bibcode:1910ANat...44..258H (https://ui.adsabs.harvard.edu/abs/1910ANat...44..258H). doi:10.1086/279138 (https://doi.org/10.1086%2F279138). ISSN 0003-0147 (https://search.worldcat.org/issn/0003-0147). S2CID 84424110 (https://api.semanticscholar.org/CorpusID:84424110). Retrieved October 18, 2019.
    - The arguments and many of the images are also presented in Desmond 1975.
39. Benton 2015, p. 205–210
40. Cowen 2005, pp. 151–175, chpt. 12: "Dinosaurs".
41. Kubo, Tai; Benton, Michael J. (November 2007). "Evolution of hindlimb posture in archosaurs: limb stresses in extinct vertebrates" (http://doc.rero.ch/record/14855/files/PAL_E1993.pdf) (PDF). *Palaeontology*. **50** (6). Hoboken, NJ: Wiley-Blackwell: 1519–1529. Bibcode:2007Palgy..50.1519K (https://ui.adsabs.harvard.edu/abs/2007Palgy..50.1519K). doi:10.1111/j.1475-4983.2007.00723.x (https://doi.org/10.1111%2Fj.1475-4983.2007.00723.x). ISSN 0031-0239 (https://search.worldcat.org/issn/0031-0239). S2CID 140698705 (https://api.semanticscholar.org/CorpusID:140698705).
42. Dong 1992
43. "Dinosaur bones 'used as medicine' " (https://news.bbc.co.uk/2/hi/asia-pacific/6276948.stm). *BBC News*. London: BBC. July 6, 2007. Archived (https://web.archive.org/web/20190827184635/https://news.bbc.co.uk/2/hi/asia-pacific/6276948.stm) from the original on August 27, 2019. Retrieved November 4, 2019.
44. Witton, Mark (March 26, 2021). "Mark P. Witton's Blog: Dinosaur fossils and Chinese dragons: ancient association or modern wishful thinking?" (https://markwitton-com.blogspot.com/2021/03/dinosaur-fossils-and-chinese-dragons.html). *Mark P. Witton's Blog*. Retrieved April 12, 2026.

45. Paul 2000, pp. 10–44, chpt. 1: "A Brief History of Dinosaur Paleontology" by Michael J. Benton.
46. Farlow & Brett-Surman 1997, pp. 3–11 (https://archive.org/details/isbn_9780253333490/page/3), chpt. 1: "The Earliest Discoveries" by William A.S. Sarjeant.
47. Plot 1677, pp. 131–139, illus. opp. p. 142, fig. 4
48. Plot 1677, p. (https://archive.org/details/naturalhistoryo00plot/page/131/mode/1up)
49. "Robert Plot" (https://web.archive.org/web/20061001094736/http://www.oum.ox.ac.uk/learning/pdfs/plot.pdf) (PDF). *Learning more*. Oxford: Oxford University Museum of Natural History. 2006. Archived from the original (http://www.oum.ox.ac.uk/learning/pdfs/plot.pdf) (PDF) on October 1, 2006. Retrieved November 14, 2019.
50. Lhuyd 1699, p. 67
51. Delair, Justin B.; Sarjeant, William A.S. (2002). "The earliest discoveries of dinosaurs: the records re-examined". *Proceedings of the Geologists' Association*. **113** (3). Amsterdam: Elsevier on behalf of the Geologists' Association: 185–197. Bibcode:2002PrGA..113..185D (https://ui.adsabs.harvard.edu/abs/2002PrGA..113..185D). doi:10.1016/S0016-7878(02)80022-0 (https://doi.org/10.1016%2FS0016-7878%2802%2980022-0). ISSN 0016-7878 (https://search.worldcat.org/issn/0016-7878).
52. Gunther 1968
53. Buckland, William (1824). "Notice on the Megalosaurus or great Fossil Lizard of Stonesfield" (https://zenodo.org/record/1448577). *Transactions of the Geological Society of London*. **1** (2). London: Geological Society of London: 390–396. doi:10.1144/transgslb.1.2.390 (https://doi.org/10.1144%2Ftransgslb.1.2.390). ISSN 2042-5295 (https://search.worldcat.org/issn/2042-5295). S2CID 129920045 (https://api.semanticscholar.org/CorpusID:129920045). Archived (https://web.archive.org/web/20191021214509/https://zenodo.org/record/1448577/files/article.pdf) (PDF) from the original on October 21, 2019. Retrieved November 5, 2019.
54. Mantell, Gideon A. (1825). "Notice on the *Iguanodon*, a newly discovered fossil reptile, from the sandstone of Tilgate forest, in Sussex" (https://doi.org/10.1098%2Frstl.1825.0010). *Philosophical Transactions of the Royal Society of London*. **115**. London: Royal Society: 179–186. Bibcode:1825RSPT..115..179M (https://ui.adsabs.harvard.edu/abs/1825RSPT..115..179M). doi:10.1098/rstl.1825.0010 (https://doi.org/10.1098%2Frstl.1825.0010). ISSN 0261-0523 (https://search.worldcat.org/issn/0261-0523). JSTOR 107739 (https://www.jstor.org/stable/107739).
55. Farlow & Brett-Surman 1997, pp. 14 (https://archive.org/details/isbn_9780253333490/page/14), chpt. 2: "European Dinosaur Hunters" by Hans-Dieter Sues.
56. Rupke 1994
57. Prieto-Marquez, Albert; Weishampel, David B.; Horner, John R. (March 2006). "The dinosaur *Hadrosaurus foulkii*, from the Campanian of the East Coast of North America, with a reevaluation of the genus" (https://app.pan.pl/archive/published/app51/app51-077.pdf) (PDF). *Acta Palaeontologica Polonica*. **51** (1). Warsaw: Institute of Paleobiology, Polish Academy of Sciences: 77–98. ISSN 0567-7920 (https://search.worldcat.org/issn/0567-7920). Archived (https://web.archive.org/web/20190622113255/https://app.pan.pl/archive/published/app51/app51-077.pdf) (PDF) from the original on June 22, 2019. Retrieved November 5, 2019.
58. Holmes 1998
59. Taylor, M.P. (2010). "Sauropod dinosaur research: a historical review". *Geological Society, London, Special Publications*. **343** (1): 361–386. Bibcode:2010GSLSP.343..361T (https://ui.adsabs.harvard.edu/abs/2010GSLSP.343..361T). doi:10.1144/SP343.22 (https://doi.org/10.1144%2FSP343.22). S2CID 910635 (https://api.semanticscholar.org/CorpusID:910635).
60. Naish, D. (2009). *The Great Dinosaur Discoveries*. London, UK: A & C Black Publishers Ltd. pp. 89–93. ISBN 978-1-4081-1906-8.

61. Arbour, V. (2018). "Results roll in from the dinosaur renaissance". *Science*. **360** (6389): 611. Bibcode:2018Sci...360..611A (https://ui.adsabs.harvard.edu/abs/2018Sci...360..611A). doi:10.1126/science.aat0451 (https://doi.org/10.1126%2Fscience.aat0451). S2CID 46887409 (https://api.semanticscholar.org/CorpusID:46887409).
62. Bakker, R.T. (1968). "The Superiority of Dinosaurs". *Discovery: Magazine of the Peabody Museum of Natural History*. **3** (2): 11–22. ISSN 0012-3625 (https://search.worldcat.org/issn/0012-3625). OCLC 297237777 (https://search.worldcat.org/oclc/297237777).
63. Bakker, R.T. (1972). "Anatomical and Ecological Evidence of Endothermy in Dinosaurs". *Nature*. **238** (5359): 81–85. Bibcode:1972Natur.238...81B (https://ui.adsabs.harvard.edu/abs/1972Natur.238...81B). doi:10.1038/238081a0 (https://doi.org/10.1038%2F238081a0). S2CID 4176132 (https://api.semanticscholar.org/CorpusID:4176132).
64. Bakker 1986
65. Benton, M.J. (2008). "Fossil quality and naming dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2614166). *Biology Letters*. **4** (6): 729–732. Bibcode:2008BiLet...4..729B (https://ui.adsabs.harvard.edu/abs/2008BiLet...4..729B). doi:10.1098/rsbl.2008.0402 (https://doi.org/10.1098%2Frsbl.2008.0402). PMC 2614166 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2614166). PMID 18796391 (https://pubmed.ncbi.nlm.nih.gov/18796391).
66. Cashmore, D.D.; Mannion, P.D.; Upchurch, P.; Butler, R.J. (2020). "Ten more years of discovery: revisiting the quality of the sauropodomorph dinosaur fossil record" (https://doi.org/10.1111%2Fpala.12496). *Palaeontology*. **63** (6): 951–978. Bibcode:2020Palgy..63..951C (https://ui.adsabs.harvard.edu/abs/2020Palgy..63..951C). doi:10.1111/pala.12496 (https://doi.org/10.1111%2Fpala.12496). S2CID 219090716 (https://api.semanticscholar.org/CorpusID:219090716).
67. Cashmore, D.D.; Butler, R.J. (2019). "Skeletal completeness of the non-avian theropod dinosaur fossil record" (https://doi.org/10.1111%2Fpala.12436). *Palaeontology*. **62** (6): 951–981. Bibcode:2019Palgy..62..951C (https://ui.adsabs.harvard.edu/abs/2019Palgy..62..951C). doi:10.1111/pala.12436 (https://doi.org/10.1111%2Fpala.12436). S2CID 197571209 (https://api.semanticscholar.org/CorpusID:197571209).
68. Holtz, T.R. Jr.; Brett-Surman, M.K. (1997). "The Taxonomy and Systematics of Dinosaurs" (https://books.google.com/books?id=Hk5ecvEv0GcC&pg=PA209). *The Complete Dinosaur*. Bloomington: Indiana University Press. pp. 209–223. ISBN 978-0-253-33349-0.
69. St. Fleur, Nicholas (December 8, 2016). "That Thing With Feathers Trapped in Amber? It Was a Dinosaur Tail" (https://www.nytimes.com/2016/12/08/science/dinosaur-feathers-amber.html). Trilobites. *The New York Times*. New York. ISSN 0362-4331 (https://search.worldcat.org/issn/0362-4331). Archived (https://web.archive.org/web/20170831181949/https://www.nytimes.com/2016/12/08/science/dinosaur-feathers-amber.html) from the original on August 31, 2017. Retrieved December 8, 2016.
70. Lockley, M.G.; Wright, J.L. (2000). "Reading About Dinosaurs – An Annotated Bibliography of Books". *Journal of Geoscience Education*. **48** (2): 167–178. Bibcode:2000JGeEd..48..167L (https://ui.adsabs.harvard.edu/abs/2000JGeEd..48..167L). doi:10.5408/1089-9995-48.2.167 (https://doi.org/10.5408%2F1089-9995-48.2.167). S2CID 151426669 (https://api.semanticscholar.org/CorpusID:151426669).
71. Lloyd, G.T.; Davis, K.E.; Pisani, D.; Tarver, J.E.; Ruta, R.; Sakamoto, M.; Hone, D.W.E.; Jennings, R.; Benton, M.J. (2008). "Dinosaurs and the Cretaceous Terrestrial Revolution" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2603200). *Proceedings of the Royal Society B*. **275** (1650): 2483–2490. doi:10.1098/rspb.2008.0715 (https://doi.org/10.1098%2Frspb.2008.0715). PMC 2603200 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2603200). PMID 18647715 (https://pubmed.ncbi.nlm.nih.gov/18647715).
72. Schweitzer, M.H. (2011). "Soft Tissue Preservation in Terrestrial Mesozoic Vertebrates". *Annual Review of Earth and Planetary Sciences*. **39**: 187–216. Bibcode:2011AREPS..39..187S (https://ui.adsabs.harvard.edu/abs/2011AREPS..39..187S). doi:10.1146/annurev-earth-040610-133502 (https://doi.org/10.1146%2Fannurev-earth-040610-133502).

73. Hooley, R.W. (1917). "II—On the Integument of *Iguanodon bernissartensis*, Boulenger, and of *Morosaurus becklesii*, Mantell" (https://zenodo.org/record/1537494). *Geological Magazine*. **4** (4): 148–150. Bibcode:1917GeoM....4..148H (https://ui.adsabs.harvard.edu/abs/1917GeoM....4..148H). doi:10.1017/s0016756800192386 (https://doi.org/10.1017%2Fs0016756800192386). S2CID 129640665 (https://api.semanticscholar.org/CorpusID:129640665). Archived (https://web.archive.org/web/20230415133707/https://zenodo.org/record/1537494) from the original on April 15, 2023. Retrieved February 20, 2023.
74. Osborn, H.F. (1912). "Integument of the iguanodont dinosaur *Trachodon*". *Memoirs of the American Museum of Natural History*. **1**: 33–54.
75. Bell, P.R. (2014). "A review of hadrosaur skin impressions". In Eberth, D.; Evans, D. (eds.). *The Hadrosaurs: Proceedings of the International Hadrosaur Symposium*. Bloomington: Princeton University Press. pp. 572–590.
76. Eliason, C.M.; Hudson, L.; Watts, T.; Garza, H.; Clarke, J.A. (2017). "Exceptional preservation and the fossil record of tetrapod integument" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5597822). *Proceedings of the Royal Society B*. **284** (1862): 1–10. Bibcode:2017RSPSB.28470556E (https://ui.adsabs.harvard.edu/abs/2017RSPSB.28470556E). doi:10.1098/rspb.2017.0556 (https://doi.org/10.1098%2Frspb.2017.0556). PMC 5597822 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5597822). PMID 28878057 (https://pubmed.ncbi.nlm.nih.gov/28878057).
77. Benton, M.J. (1998). "Dinosaur fossils with soft parts" (http://doc.rero.ch/record/14715/files/PAL_E1646.pdf) (PDF). *Trends in Ecology & Evolution*. **13** (8): 303–304. Bibcode:1998TEcoE..13..303B (https://ui.adsabs.harvard.edu/abs/1998TEcoE..13..303B). doi:10.1016/s0169-5347(98)01420-7 (https://doi.org/10.1016%2Fs0169-5347%2898%2901420-7). PMID 21238317 (https://pubmed.ncbi.nlm.nih.gov/21238317).
78. Zhou, Z.-H.; Wang, Y. (2017). "Vertebrate assemblages of the Jurassic Yanliao Biota and the Early Cretaceous Jehol Biota: Comparisons and implications". *Palaeoworld*. **26** (2): 241–252. Bibcode:2017Palae..26..241Z (https://ui.adsabs.harvard.edu/abs/2017Palae..26..241Z). doi:10.1016/j.palwor.2017.01.002 (https://doi.org/10.1016%2Fj.palwor.2017.01.002).
79. Norell, M.A.; Xu, X. (2005). "Feathered Dinosaurs". *Annual Review of Earth and Planetary Sciences*. **33**: 277–299. Bibcode:2005AREPS..33..277N (https://ui.adsabs.harvard.edu/abs/2005AREPS..33..277N). doi:10.1146/annurev.earth.33.092203.122511 (https://doi.org/10.1146%2Fannurev.earth.33.092203.122511).
80. Roy, A.; Pittman, M.; Saitta, E.T.; Kaye, T.G.; Xu, X. (2020). "Recent advances in amniote palaeocolour reconstruction and a framework for future research" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7004074). *Biological Reviews*. **95** (1): 22–50. doi:10.1111/brv.12552 (https://doi.org/10.1111%2Fbrv.12552). PMC 7004074 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7004074). PMID 31538399 (https://pubmed.ncbi.nlm.nih.gov/31538399).
81. Vinther, J. (2020). "Reconstructing Vertebrate Paleocolor" (https://doi.org/10.1146%2Fannurev-earth-073019-045641). *Annual Review of Earth and Planetary Sciences*. **48**: 345–375. Bibcode:2020AREPS..48..345V (https://ui.adsabs.harvard.edu/abs/2020AREPS..48..345V). doi:10.1146/annurev-earth-073019-045641 (https://doi.org/10.1146%2Fannurev-earth-073019-045641). S2CID 219768255 (https://api.semanticscholar.org/CorpusID:219768255).
82. Zhang, F.; Kearns, S.L.; Orr, P.J.; Benton, M.J.; Zhou, Z.; Johnson, D.; Xu, X.; Wang, X. (2010). "Fossilized melanosomes and the colour of Cretaceous dinosaurs and birds" (http://oro.open.ac.uk/22432/2/41064696.pdf) (PDF). *Nature*. **463** (7284): 1075–1078. Bibcode:2010Natur.463.1075Z (https://ui.adsabs.harvard.edu/abs/2010Natur.463.1075Z). doi:10.1038/nature08740 (https://doi.org/10.1038%2Fnature08740). PMID 20107440 (https://pubmed.ncbi.nlm.nih.gov/20107440). S2CID 205219587 (https://api.semanticscholar.org/CorpusID:205219587). Archived (https://web.archive.org/web/20140202214645/http://oro.open.ac.uk/22432/2/41064696.pdf) (PDF) from the original on February 2, 2014. Retrieved February 20, 2023.

83. Smithwick, F.M.; Nicholls, R.; Cuthill, I.C.; Vinther, J. (2017). "Countershading and Stripes in the Theropod Dinosaur *Sinosauropteryx* Reveal Heterogeneous Habitats in the Early Cretaceous Jehol Biota" (https://doi.org/10.1016%2Fj.cub.2017.09.032). *Current Biology*. **27** (21): 3337–3343.e2. Bibcode:2017CBio...27E3337S (https://ui.adsabs.harvard.edu/abs/2017CBio...27E3337S). doi:10.1016/j.cub.2017.09.032 (https://doi.org/10.1016%2Fj.cub.2017.09.032). hdl:1983/8ee95b15-5793-42ad-8e57-da6524635349 (https://hdl.handle.net/1983%2F8ee95b15-5793-42ad-8e57-da6524635349). PMID 29107548 (https://pubmed.ncbi.nlm.nih.gov/29107548).
84. Vinther, J.; Nicholls, R.; Lautenschlager, S.; Pittman, M.; Kaye, T.G.; Rayfield, E.; Mayr, G.; Cuthill, I.C. (2016). "3D Camouflage in an Ornithischian Dinosaur" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5049543). *Current Biology*. **26** (18): 2456–2462. Bibcode:2016CBio...26.2456V (https://ui.adsabs.harvard.edu/abs/2016CBio...26.2456V). doi:10.1016/j.cub.2016.06.065 (https://doi.org/10.1016%2Fj.cub.2016.06.065). PMC 5049543 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5049543). PMID 27641767 (https://pubmed.ncbi.nlm.nih.gov/27641767).
85. Lindgren, J.; Moyer, A.; Schweitzer, M.H.; Sjövall, P.; Uvdal, P.; Nilsson, D.E.; Heimdal, J.; Engdahl, A.; Gren, J.A.; Schultz, B.P.; Kear, B.P. (2015). "Interpreting melanin-based coloration through deep time: a critical review" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4632609). *Proceedings of the Royal Society B*. **282** (1813) 20150614. Bibcode:2015PBioS.28250614L (https://ui.adsabs.harvard.edu/abs/2015PBioS.28250614L). doi:10.1098/rspb.2015.0614 (https://doi.org/10.1098%2Frspb.2015.0614). PMC 4632609 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4632609). PMID 26290071 (https://pubmed.ncbi.nlm.nih.gov/26290071).
86. Schweitzer, M.H.; Lindgren, J.; Moyer, A.E. (2015). "Melanosomes and ancient coloration re-examined: a response to Vinther 2015 (DOI 10.1002/bies.201500018)". *BioEssays*. **37** (11): 1174–1183. doi:10.1002/bies.201500061 (https://doi.org/10.1002%2Fbies.201500061). PMID 26434749 (https://pubmed.ncbi.nlm.nih.gov/26434749). S2CID 45178498 (https://api.semanticscholar.org/CorpusID:45178498).
87. Zhou, Z. (2014). "The Jehol Biota, an Early Cretaceous terrestrial Lagerstätte: new discoveries and implications" (https://doi.org/10.1093%2Fnsr%2Fnwu055). *National Science Review*. **1** (4): 543–559. doi:10.1093/nsr/nwu055 (https://doi.org/10.1093%2Fnsr%2Fnwu055).
88. O'Connor, J.K.; Zhou, Z. (2019). "The evolution of the modern avian digestive system: insights from paravian fossils from the Yanliao and Jehol biotas". *Palaeontology*. **63** (1): 13–27. doi:10.1111/pala.12453 (https://doi.org/10.1111%2Fpala.12453). S2CID 210265348 (https://api.semanticscholar.org/CorpusID:210265348).
89. Dal Sasso, Cristiano; Signore, Marco (March 26, 1998). "Exceptional soft-tissue preservation in a theropod dinosaur from Italy" (http://doc.rero.ch/record/14901/files/PAL_E2043.pdf) (PDF). *Nature*. **392** (6674). London: Nature Research: 383–387. Bibcode:1998Natur.392..383D (https://ui.adsabs.harvard.edu/abs/1998Natur.392..383D). doi:10.1038/32884 (https://doi.org/10.1038%2F32884). ISSN 0028-0836 (https://search.worldcat.org/issn/0028-0836). S2CID 4325093 (https://api.semanticscholar.org/CorpusID:43250093). Archived (https://web.archive.org/web/20160920170653/https://doc.rero.ch/record/14901/files/PAL_E2043.pdf) (PDF) from the original on September 20, 2016.
90. Morell, V. (1993). "Dino DNA: the Hunt and the Hype". *Science*. **261** (5118): 160–162. Bibcode:1993Sci...261..160M (https://ui.adsabs.harvard.edu/abs/1993Sci...261..160M). doi:10.1126/science.8327889 (https://doi.org/10.1126%2Fscience.8327889). PMID 8327889 (https://pubmed.ncbi.nlm.nih.gov/8327889).
91. Pawlicki, R.; Korbel, A.; Kubiak, H. (1996). "Cells, Collagen Fibrils and Vessels in Dinosaur Bone". *Nature*. **211** (5049): 655–657. doi:10.1038/211655a0 (https://doi.org/10.1038%2F211655a0). PMID 5968744 (https://pubmed.ncbi.nlm.nih.gov/5968744). S2CID 4181847 (https://api.semanticscholar.org/CorpusID:4181847).

92. Pawlicki, R.; Nowogrodzka-Zagórska, M. (1998). "Blood vessels and red blood cells preserved in dinosaur bones". *Annals of Anatomy - Anatomischer Anzeiger*. **180** (1): 73–77. doi:10.1016/S0940-9602(98)80140-4 (https://doi.org/10.1016%2FS0940-9602%2898%2980140-4). PMID 9488909 (https://pubmed.ncbi.nlm.nih.gov/9488909).
93. Schweitzer, Mary H.; Wittmeyer, Jennifer L.; Horner, John R.; Toporski, Jan K. (2005). "Soft-Tissue Vessels and Cellular Preservation in *Tyrannosaurus rex*". *Science*. **307** (5717). Washington, D.C.: American Association for the Advancement of Science: 1952–1955. Bibcode:2005Sci...307.1952S (https://ui.adsabs.harvard.edu/abs/2005Sci...307.1952S). doi:10.1126/science.1108397 (https://doi.org/10.1126%2Fscience.1108397). ISSN 0036-8075 (https://search.worldcat.org/issn/0036-8075). PMID 15790853 (https://pubmed.ncbi.nlm.nih.gov/15790853). S2CID 30456613 (https://api.semanticscholar.org/CorpusID:30456613).
94. Anderson, L.A. (2023). "A chemical framework for the preservation of fossil vertebrate cells and soft tissues" (https://doi.org/10.1016%2Fj.earscirev.2023.104367). *Earth-Science Reviews*. **240** 104367. Bibcode:2023ESRv..24004367A (https://ui.adsabs.harvard.edu/abs/2023ESRv..24004367A). doi:10.1016/j.earscirev.2023.104367 (https://doi.org/10.1016%2Fj.earscirev.2023.104367). S2CID 257326012 (https://api.semanticscholar.org/CorpusID:257326012).
95. Schweitzer, M.H.; Zheng, W.; Organ, C.L.; Avci, R.; Suo, Z.; Freimark, L.M.; LeBleu, V.S.; Duncan, M.B.; van der Heiden, M.G.; Neveu, J.M.; Lane, W.S.; Cottrell, J.S.; Horner, J.R.; Cantley, L.C.; Kalluri, R.; Asara, J.M. (2009). "Biomolecular characterization and protein sequences of the Campanian hadrosaur *B. canadensis*". *Science*. **324** (5927): 626–631. Bibcode:2009Sci...324..626S (https://ui.adsabs.harvard.edu/abs/2009Sci...324..626S). doi:10.1126/science.1165069 (https://doi.org/10.1126%2Fscience.1165069). PMID 19407199 (https://pubmed.ncbi.nlm.nih.gov/19407199). S2CID 5358680 (https://api.semanticscholar.org/CorpusID:5358680).
96. Organ, C.L.; Schweitzer, M.H.; Zheng, W.; Freimark, L.M.; Cantley, L.C.; Asara, J.M. (2008). "Molecular Phylogenetics of Mastodon and *Tyrannosaurus rex*". *Science*. **320** (5875): 499. Bibcode:2008Sci...320..499O (https://ui.adsabs.harvard.edu/abs/2008Sci...320..499O). doi:10.1126/science.1154284 (https://doi.org/10.1126%2Fscience.1154284). PMID 18436782 (https://pubmed.ncbi.nlm.nih.gov/18436782). S2CID 24971064 (https://api.semanticscholar.org/CorpusID:24971064).
97. Schweitzer, M.H.; Zheng, W.; Cleland, T.P.; Bern, M. (2013). "Molecular analyses of dinosaur osteocytes support the presence of endogenous molecules". *Bone*. **52** (1). Amsterdam: Elsevier: 414–423. doi:10.1016/j.bone.2012.10.010 (https://doi.org/10.1016%2Fj.bone.2012.10.010). ISSN 8756-3282 (https://search.worldcat.org/issn/8756-3282). PMID 23085295 (https://pubmed.ncbi.nlm.nih.gov/23085295).
98. Bailleul, A.M.; Zheng, W.; Horner, J.R.; Hall, B.K.; Holliday, C.M.; Schweitzer, M.H. (2020). "Evidence of proteins, chromosomes and chemical markers of DNA in exceptionally preserved dinosaur cartilage" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8289162). *National Science Review*. **7** (4): 815–822. doi:10.1093/nsr/nwz206 (https://doi.org/10.1093%2Fnsr%2Fnwz206). PMC 8289162 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8289162). PMID 34692099 (https://pubmed.ncbi.nlm.nih.gov/34692099).
99. Bertazzo, S.; Maidment, S.C.R.; Kallepitis, C.; et al. (2015). "Fibres and cellular structures preserved in 75-million-year-old dinosaur specimens" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4468865). *Nature Communications*. **6** 7352. Bibcode:2015NatCo...6.7352B (https://ui.adsabs.harvard.edu/abs/2015NatCo...6.7352B). doi:10.1038/ncomms8352 (https://doi.org/10.1038%2Fncomms8352). ISSN 2041-1723 (https://search.worldcat.org/issn/2041-1723). PMC 4468865 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4468865). PMID 26056764 (https://pubmed.ncbi.nlm.nih.gov/26056764).

100. Kaye, T.G.; Gaugler, G.; Sawlowicz, Z. (2008). "Dinosaurian Soft Tissues Interpreted as Bacterial Biofilms" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2483347). *PLOS ONE*. **3** (7) e2808. Bibcode:2008PLoSO...3.2808K (https://ui.adsabs.harvard.edu/abs/2008PLoSO...3.2808K). doi:10.1371/journal.pone.0002808 (https://doi.org/10.1371%2Fjournal.pone.0002808). PMC 2483347 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2483347). PMID 18665236 (https://pubmed.ncbi.nlm.nih.gov/18665236).
101. Peterson, J.E.; Lenczewski, M.E.; Scherer, R.P. (2010). "Influence of Microbial Biofilms on the Preservation of Primary Soft Tissue in Fossil and Extant Archosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2953520). *PLOS ONE*. **5** (10) e13334. Bibcode:2010PLoSO...513334P (https://ui.adsabs.harvard.edu/abs/2010PLoSO...513334P). doi:10.1371/journal.pone.0013334 (https://doi.org/10.1371%2Fjournal.pone.0013334). ISSN 1932-6203 (https://search.worldcat.org/issn/1932-6203). PMC 2953520 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2953520). PMID 20967227 (https://pubmed.ncbi.nlm.nih.gov/20967227).
102. Buckley, M.; Warwood, S.; van Dongen, B.; Kitchener, A.C.; Manning, P.L. (2017). "A fossil protein chimera; difficulties in discriminating dinosaur peptide sequences from modern cross-contamination" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5454271). *Proceedings of the Royal Society B*. **284** (1855). Bibcode:2017PBioS.28470544B (https://ui.adsabs.harvard.edu/abs/2017PBioS.28470544B). doi:10.1098/rspb.2017.0544 (https://doi.org/10.1098%2Frspb.2017.0544). PMC 5454271 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5454271). PMID 28566488 (https://pubmed.ncbi.nlm.nih.gov/28566488).
103. Kump, Lee R.; Pavlov, Alexander; Arthur, Michael A. (2005). "Massive release of hydrogen sulfide to the surface ocean and atmosphere during intervals of oceanic anoxia". *Geology*. **33** (5). Boulder, CO: Geological Society of America: 397–400. Bibcode:2005Geo....33..397K (https://ui.adsabs.harvard.edu/abs/2005Geo....33..397K). doi:10.1130/G21295.1 (https://doi.org/10.1130%2FG21295.1). ISSN 0091-7613 (https://search.worldcat.org/issn/0091-7613). S2CID 34821866 (https://api.semanticscholar.org/CorpusID:34821866).
104. Tanner, Lawrence H.; Lucas, Spencer G.; Chapman, Mary G. (March 2004). "Assessing the record and causes of Late Triassic extinctions" (https://web.archive.org/web/20071025225841/http://nmnaturalhistory.org/pdf_files/TJB.pdf) (PDF). *Earth-Science Reviews*. **65** (1–2). Amsterdam: Elsevier: 103–139. Bibcode:2004ESRv...65..103T (https://ui.adsabs.harvard.edu/abs/2004ESRv...65..103T). doi:10.1016/S0012-8252(03)00082-5 (https://doi.org/10.1016%2FS0012-8252%2803%2900082-5). ISSN 0012-8252 (https://search.worldcat.org/issn/0012-8252). Archived from the original (http://nmnaturalhistory.org/pdf_files/TJB.pdf) (PDF) on October 25, 2007. Retrieved October 22, 2007.
105. Griffin, C.T.; Wynd, B.M.; Munyikwa, D.; Broderick, T.J.; Zondo, M.; Tolan, S.; Langer, M.C.; Nesbitt, S.J.; Taruvinga, H.R. (2022). "Africa's oldest dinosaurs reveal early suppression of dinosaur distribution". *Nature*. **609** (7926): 313–319. Bibcode:2022Natur.609..313G (https://ui.adsabs.harvard.edu/abs/2022Natur.609..313G). doi:10.1038/s41586-022-05133-x (https://doi.org/10.1038%2Fs41586-022-05133-x). ISSN 0028-0836 (https://search.worldcat.org/issn/0028-0836). PMID 36045297 (https://pubmed.ncbi.nlm.nih.gov/36045297). S2CID 251977824 (https://api.semanticscholar.org/CorpusID:251977824).
106. Desojo, J.B.; Fiorelli, L.E.; Ezcurra, M.D.; Martinelli, A.G.; Ramezani, J.; Da Rosa, A.A.S.; Belén von Baczko, M.; Jimena Trotteyn, M.; Montefeltro, F.C.; Ezpeleta, M.; Langer, M.C. (2020). "The Late Triassic Ischigualasto Formation at Cerro Las Lajas (La Rioja, Argentina): fossil tetrapods, high-resolution chronostratigraphy, and faunal correlations" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7391656). *Scientific Reports*. **10** (1): 12782. Bibcode:2020NatSR..1012782D (https://ui.adsabs.harvard.edu/abs/2020NatSR..1012782D). doi:10.1038/s41598-020-67854-1 (https://doi.org/10.1038%2Fs41598-020-67854-1). PMC 7391656 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7391656). PMID 32728077 (https://pubmed.ncbi.nlm.nih.gov/32728077).

107. Alcober, Oscar A.; Martinez, Ricardo N. (2010). "A new herrerasaurid (Dinosauria, Saurischia) from the Upper Triassic Ischigualasto Formation of northwestern Argentina" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3088398). *ZooKeys* (63). Sofia: Pensoft Publishers: 55–81. Bibcode:2010ZooK...63...55A (https://ui.adsabs.harvard.edu/abs/2010ZooK...63...55A). doi:10.3897/zookeys.63.550 (https://doi.org/10.3897%2Fzookeys.63.550). ISSN 1313-2989 (https://search.worldcat.org/issn/1313-2989). PMC 3088398 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3088398). PMID 21594020 (https://pubmed.ncbi.nlm.nih.gov/21594020).
108. Novas, F.E.; Agnolin, F.L.; Ezcurra, M.D.; Müller, R.T.; Martinelli, A.; Langer, M. (2021). "Review of the fossil record of early dinosaurs from South America, and its phylogenetic implications". *Journal of South American Earth Sciences*. **110** 103341. Bibcode:2021JSAES.11003341N (https://ui.adsabs.harvard.edu/abs/2021JSAES.11003341N). doi:10.1016/j.jsames.2021.103341 (https://doi.org/10.1016%2Fj.jsames.2021.103341). hdl:11336/167918 (https://hdl.handle.net/11336%2F167918). ISSN 0895-9811 (https://search.worldcat.org/issn/0895-9811).
109. Nesbitt, Sterling J; Sues, Hans-Dieter (2021). "The osteology of the early-diverging dinosaur *Daemonosaurus chauliodus* (Archosauria: Dinosauria) from the Coelophysis Quarry (Triassic: Rhaetian) of New Mexico and its relationships to other early dinosaurs" (https://doi.org/10.1093%2Fzoolinnean%2Fzlaa080). *Zoological Journal of the Linnean Society*. **191** (1): 150–179. doi:10.1093/zoolinnean/zlaa080 (https://doi.org/10.1093%2Fzoolinnean%2Fzlaa080).
110. Sereno, Paul C. (1999). "The Evolution of Dinosaurs" (https://www.researchgate.net/publication/12917068). *Science*. **284** (5423). Washington, D.C.: American Association for the Advancement of Science: 2137–2147. doi:10.1126/science.284.5423.2137 (https://doi.org/10.1126%2Fscience.284.5423.2137). ISSN 0036-8075 (https://search.worldcat.org/issn/0036-8075). PMID 10381873 (https://pubmed.ncbi.nlm.nih.gov/10381873). Archived (https://web.archive.org/web/20180105180335/https://www.researchgate.net/profile/Paul_Sereno/publication/12917068_The_Evolution_of_Dinosaurs/links/56b0c66d08ae8e372151f17e/The-Evolution-of-Dinosaurs.pdf) (PDF) from the original on January 5, 2018. Retrieved November 8, 2019.
111. Sereno, Paul C.; Forster, Catherine A.; Rogers, Raymond R.; Monetta, Alfredo M. (1993). "Primitive dinosaur skeleton from Argentina and the early evolution of Dinosauria". *Nature*. **361** (6407). London: Nature Research: 64–66. Bibcode:1993Natur.361...64S (https://ui.adsabs.harvard.edu/abs/1993Natur.361...64S). doi:10.1038/361064a0 (https://doi.org/10.1038%2F361064a0). ISSN 0028-0836 (https://search.worldcat.org/issn/0028-0836). S2CID 4270484 (https://api.semanticscholar.org/CorpusID:4270484).
112. Langer, Max C.; Ramezani, Jahandar; Da Rosa, Átila A.S. (May 2018). "U-Pb age constraints on dinosaur rise from south Brazil". *Gondwana Research*. **57**. Amsterdam: Elsevier: 133–140. Bibcode:2018GondR..57..133L (https://ui.adsabs.harvard.edu/abs/2018GondR..57..133L). doi:10.1016/j.gr.2018.01.005 (https://doi.org/10.1016%2Fj.gr.2018.01.005). ISSN 1342-937X (https://search.worldcat.org/issn/1342-937X).
113. Novas, F.E.; Ezcurra, M.D.; Chatterjee, S.; Kutty, T.S. (2011). "New dinosaur species from the Upper Triassic Upper Maleri and Lower Dharmaram formations of central India". *Earth and Environmental Science Transactions of the Royal Society of Edinburgh*. **101** (3–4): 333–349. Bibcode:2010EESTR.101..333N (https://ui.adsabs.harvard.edu/abs/2010EESTR.101..333N). doi:10.1017/S1755691011020093 (https://doi.org/10.1017%2FS1755691011020093). S2CID 128620874 (https://api.semanticscholar.org/CorpusID:128620874).

114. Marsicano, C.A.; Irmis, R.B.; Mancuso, A.C.; Mundil, R.; Chemale, F. (2015). "The precise temporal calibration of dinosaur origins" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4725541). *Proceedings of the National Academy of Sciences*. **113** (3): 509–513. Bibcode:2016PNAS..113..509M (https://ui.adsabs.harvard.edu/abs/2016PNAS..113..509M). doi:10.1073/pnas.1512541112 (https://doi.org/10.1073%2Fpnas.1512541112). ISSN 0027-8424 (https://search.worldcat.org/issn/0027-8424). PMC 4725541 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4725541). PMID 26644579 (https://pubmed.ncbi.nlm.nih.gov/26644579).
115. Nesbitt, Sterling J.; Barrett, Paul M.; Werning, Sarah; et al. (2012). "The oldest dinosaur? A Middle Triassic dinosauriform from Tanzania" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3565515). *Biology Letters*. **9** (1) 20120949. London: Royal Society. doi:10.1098/rsbl.2012.0949 (https://doi.org/10.1098%2Frsbl.2012.0949). ISSN 1744-9561 (https://search.worldcat.org/issn/1744-9561). PMC 3565515 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3565515). PMID 23221875 (https://pubmed.ncbi.nlm.nih.gov/23221875).
116. Brusatte, Stephen L.; Benton, Michael J.; Ruta, Marcello; Lloyd, Graeme T. (2008). "Superiority, Competition, and Opportunism in the Evolutionary Radiation of Dinosaurs" (https://www.pure.ed.ac.uk/ws/files/8232088/PDF_Brusatteetal2008SuperiorityCompetition.pdf) (PDF). *Science*. **321** (5895). Washington, D.C.: American Association for the Advancement of Science: 1485–1488. Bibcode:2008Sci...321.1485B (https://ui.adsabs.harvard.edu/abs/2008Sci...321.1485B). doi:10.1126/science.1161833 (https://doi.org/10.1126%2Fscience.1161833). hdl:20.500.11820/00556baf-6575-44d9-af39-bdd0b072ad2b (https://hdl.handle.net/20.500.11820%2F00556baf-6575-44d9-af39-bdd0b072ad2b). ISSN 0036-8075 (https://search.worldcat.org/issn/0036-8075). PMID 18787166 (https://pubmed.ncbi.nlm.nih.gov/18787166). S2CID 13393888 (https://api.semanticscholar.org/CorpusID:13393888). Archived (https://web.archive.org/web/20180719005836/https://www.pure.ed.ac.uk/ws/files/8232088/PDF_Brusatteetal2008SuperiorityCompetition.pdf) (PDF) from the original on July 19, 2018. Retrieved October 22, 2019.
117. Tanner, Spielmann & Lucas 2013, pp. 562–566 (https://econtent.unm.edu/digital/collection/bulletins/id/1688), "The first Norian (Revueltian) rhynchosaur: Bull Canyon Formation, New Mexico, U.S.A." by Justin A. Spielmann, Spencer G. Lucas and Adrian P. Hunt.
118. Sulej, Tomasz; Niedźwiedzki, Grzegorz (2019). "An elephant-sized Late Triassic synapsid with erect limbs" (https://doi.org/10.1126%2Fscience.aal4853). *Science*. **363** (6422). Washington, D.C.: American Association for the Advancement of Science: 78–80. Bibcode:2019Sci...363...78S (https://ui.adsabs.harvard.edu/abs/2019Sci...363...78S). doi:10.1126/science.aal4853 (https://doi.org/10.1126%2Fscience.aal4853). ISSN 0036-8075 (https://search.worldcat.org/issn/0036-8075). PMID 30467179 (https://pubmed.ncbi.nlm.nih.gov/30467179). S2CID 53716186 (https://api.semanticscholar.org/CorpusID:53716186).
119. "Fossil tracks in the Alps help explain dinosaur evolution" (https://www.economist.com/science-and-technology/2018/04/19/fossil-tracks-in-the-alps-help-explain-dinosaur-evolution). Science and Technology. *The Economist*. London. April 19, 2018. ISSN 0013-0613 (https://search.worldcat.org/issn/0013-0613). Retrieved May 24, 2018.
120. Weishampel, Dodson & Osmólska 2004, pp. 627–642, chpt. 27: "Mesozoic Biogeography of Dinosauria" by Thomas R. Holtz Jr., Ralph E. Chapman, and Matthew C. Lamanna.
121. Weishampel, Dodson & Osmólska 2004, pp. 614–626, chpt. 26: "Dinosaur Paleoecology" by David E. Fastovsky and Joshua B. Smith.
122. Sereno, Paul C.; Wilson, Jeffrey A.; Witmer, Lawrence M.; et al. (2007). Kemp, Tom (ed.). "Structural Extremes in a Cretaceous Dinosaur" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2077925). *PLOS ONE*. **2** (11) e1230. San Francisco, CA: PLOS. Bibcode:2007PLoSO...2.1230S (https://ui.adsabs.harvard.edu/abs/2007PLoSO...2.1230S). doi:10.1371/journal.pone.0001230 (https://doi.org/10.1371%2Fjournal.pone.0001230). ISSN 1932-6203 (https://search.worldcat.org/issn/1932-6203). PMC 2077925 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2077925). PMID 18030355 (https://pubmed.ncbi.nlm.nih.gov/18030355).

123. Prasad, Vandana; Strömberg, Caroline A. E.; Alimohammadian, Habib; et al. (2005). "Dinosaur Coprolites and the Early Evolution of Grasses and Grazers". *Science*. **310** (5751). Washington, D.C.: American Association for the Advancement of Science: 1170–1180. Bibcode:2005Sci...310.1177P (https://ui.adsabs.harvard.edu/abs/2005Sci...310.1177P). doi:10.1126/science.1118806 (https://doi.org/10.1126%2Fscience.1118806). ISSN 0036-8075 (https://search.worldcat.org/issn/0036-8075). PMID 16293759 (https://pubmed.ncbi.nlm.nih.gov/16293759). S2CID 1816461 (https://api.semanticscholar.org/CorpusID:1816461).
124. Weishampel, Dodson & Osmólska 2004, pp. 672–684, chpt. 30: "Dinosaur Extinction" by J. David Archibald and David E. Fastovsky.
125. Dyke & Kaiser 2011, chpt. 14: "Bird Evolution Across the K–Pg Boundary and the Basal Neornithine Diversification" by Bent E. K. Lindow. doi:10.1002/9781119990475.ch14 (https://doi.org/10.1002%2F9781119990475.ch14)
126. Cracraft, Joel (1968). "A Review of the Bathornithidae (Aves, Gruiformes), with Remarks on the Relationships of the Suborder Cariamae" (https://www.biodiversitylibrary.org/item/328644#page/1/mode/1up). *American Museum Novitates* (2326). New York: American Museum of Natural History: 1–46. hdl:2246/2536 (https://hdl.handle.net/2246%2F2536). ISSN 0003-0082 (https://search.worldcat.org/issn/0003-0082). Archived (https://archive.org/details/reviewbathornit2326crac/reviewbathornit2326crac/page/6/mode/2up) from the original on November 13, 2023. Retrieved January 30, 2026.
127. Alvarenga, Herculano; Jones, Washington W.; Rinderknecht, Andrés (May 2010). "The youngest record of phorusrhacid birds (Aves, Phorusrhacidae) from the late Pleistocene of Uruguay" (https://www.researchgate.net/publication/233512584). *Neues Jahrbuch für Geologie und Paläontologie*. **256** (2). Stuttgart: E. Schweizerbart: 229–234. Bibcode:2010NJGPA.256..229A (https://ui.adsabs.harvard.edu/abs/2010NJGPA.256..229A). doi:10.1127/0077-7749/2010/0052 (https://doi.org/10.1127%2F0077-7749%2F2010%2F0052). ISSN 0077-7749 (https://search.worldcat.org/issn/0077-7749). Retrieved October 22, 2019.
128. Mayr 2009
129. Paul 1988, pp. 248–250 (https://archive.org/details/predatorydinosaursoftheworld1988/page/n245)
130. Weishampel, Dodson & Osmólska 2004, pp. 151–164, chpt. 7: "Therizinosauroidea" by James M. Clark, Teresa Maryańska, and Rinchen Barsbold.
131. Weishampel, Dodson & Osmólska 2004, pp. 196–210, chpt. 10: "Dromaeosauridae" by Peter J. Makovicky and Mark A. Norell.
132. Taylor, Michael P.; Wedel, Mathew J. (2013). "Why sauropods had long necks; and why giraffes have short necks" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3628838). *PeerJ*. **1** e36. Corte Madera, CA; London. doi:10.7717/peerj.36 (https://doi.org/10.7717%2Fpeerj.36). ISSN 2167-8359 (https://search.worldcat.org/issn/2167-8359). PMC 3628838 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3628838). PMID 23638372 (https://pubmed.ncbi.nlm.nih.gov/23638372).
133. Alexander, R. McNeill (2006). "Dinosaur biomechanics" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1634776). *Proceedings of the Royal Society B*. **273** (1596). London: Royal Society: 1849–1855. Bibcode:2006PBioS.273.1849A (https://ui.adsabs.harvard.edu/abs/2006PBioS.273.1849A). doi:10.1098/rspb.2006.3532 (https://doi.org/10.1098%2Frspb.2006.3532). ISSN 0962-8452 (https://search.worldcat.org/issn/0962-8452). PMC 1634776 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1634776). PMID 16822743 (https://pubmed.ncbi.nlm.nih.gov/16822743).
134. Farlow, James O.; Dodson, Peter; Chinsamy, Anusuya (November 1995). "Dinosaur Biology". *Annual Review of Ecology and Systematics*. **26** (1). Palo Alto, CA: Annual Reviews: 445–471. Bibcode:1995AnRES..26..445F (https://ui.adsabs.harvard.edu/abs/1995AnRES..26..445F). doi:10.1146/annurev.es.26.110195.002305 (https://doi.org/10.1146%2Fannurev.es.26.110195.002305). ISSN 1545-2069 (https://search.worldcat.org/issn/1545-2069).

135. Weishampel, Dodson & Osmólska 2004
136. Dodson & Gingerich 1993, pp. 167–199 (http://earth.geology.yale.edu/~ajs/1993/11.1993.06Farlow.pdf), "On the rareness of big, fierce animals: speculations about the body sizes, population densities, and geographic ranges of predatory mammals and large carnivorous dinosaurs" by James O. Farlow.
137. Peczkis, Jan (1995). "Implications of body-mass estimates for dinosaurs". *Journal of Vertebrate Paleontology*. **14** (4). Milton Park, Oxfordshire: Taylor & Francis for the Society of Vertebrate Paleontology: 520–533. Bibcode:1995JVPal..14..520P (https://ui.adsabs.harvard.edu/abs/1995JVPal..14..520P). doi:10.1080/02724634.1995.10011575 (https://doi.org/10.1080%2F02724634.1995.10011575). ISSN 0272-4634 (https://search.worldcat.org/issn/0272-4634). JSTOR 4523591 (https://www.jstor.org/stable/4523591).
138. "Dinosaur Evolution" (https://web.archive.org/web/20071111204903/http://paleobiology.si.edu/dinosaurs/info/everything/evo_1.html). *Department of Paleobiology*. Dinosaurs. Washington, D.C.: National Museum of Natural History. 2007. Archived from the original (http://paleobiology.si.edu/dinosaurs/info/everything/evo_1.html) on November 11, 2007. Retrieved November 21, 2007.
139. Sander, P. Martin; Christian, Andreas; Clauss, Marcus; et al. (February 2011). "Biology of the sauropod dinosaurs: the evolution of gigantism" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3045712). *Biological Reviews*. **86** (1). Cambridge: Cambridge Philosophical Society: 117–155. Bibcode:2011BioRv..86..117S (https://ui.adsabs.harvard.edu/abs/2011BioRv..86..117S). doi:10.1111/j.1469-185X.2010.00137.x (https://doi.org/10.1111%2Fj.1469-185X.2010.00137.x). ISSN 1464-7931 (https://search.worldcat.org/issn/1464-7931). PMC 3045712 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3045712). PMID 21251189 (https://pubmed.ncbi.nlm.nih.gov/21251189).
140. Foster & Lucas 2006, pp. 131–138 (https://econtent.unm.edu/digital/collection/bulletins/id/790), "Biggest of the big: a critical re-evaluation of the mega-sauropod *Amphicoelias fragillimus* Cope, 1878" by Kenneth Carpenter.
141. Paul 2010
142. Colbert 1971
143. Mazzetta, Gerardo V.; Christiansenb, Per; Fariñaa, Richard A. (2004). "Giants and Bizarres: Body Size of Some Southern South American Cretaceous Dinosaurs" (http://www.miketaylor.org.uk/tmp/papers/Mazzetta-et-al_04_SA-dino-body-size.pdf) (PDF). *Historical Biology*. **16** (2–4). Milton Park, Oxfordshire: Taylor & Francis: 71–83. Bibcode:2004HBio...16...71M (https://ui.adsabs.harvard.edu/abs/2004HBio...16...71M). doi:10.1080/08912960410001715132 (https://doi.org/10.1080%2F08912960410001715132). ISSN 0891-2963 (https://search.worldcat.org/issn/0891-2963). S2CID 56028251 (https://api.semanticscholar.org/CorpusID:56028251). Archived (https://web.archive.org/web/20090225155217/http://www.miketaylor.org.uk/tmp/papers/Mazzetta-et-al_04_SA-dino-body-size.pdf) (PDF) from the original on February 25, 2009.
144. Janensch, Werner (1950). "Die Skelettrekonstruktion von *Brachiosaurus brancai*" (https://paleoglot.org/files/Janensch1950b.pdf) [The Skeleton Reconstruction of Brachiosaurus brancai] (PDF). *Palaeontographica*. Supplement VII (1. Reihe, Teil 3, Lieferung 2). Translation by Gerhard Maier. Stuttgart: E. Schweizerbart: 97–103. OCLC 45923346 (https://search.worldcat.org/oclc/45923346). Archived (https://web.archive.org/web/20170711052046/https://paleoglot.org/files/Janensch1950b.pdf) (PDF) from the original on July 11, 2017. Retrieved October 24, 2019.

145. Lucas, Spencer G.; Herne, Matthew C.; Hecket, Andrew B.; et al. (2004). *Reappraisal of* Seismosaurus*, a Late Jurassic Sauropod Dinosaur From New Mexico* (https://gsa.confex.com/gsa/2004AM/finalprogram/abstract_77727.htm). 2004 Denver Annual Meeting (November 7–10, 2004) (https://gsa.confex.com/gsa/2004AM/webprogram/start.html). Vol. 36. Boulder, CO: Geological Society of America. p. 422. OCLC 62334058 (https://search.worldcat.org/oclc/62334058). Paper No. 181-4. Archived (https://web.archive.org/web/20191008110318/https://gsa.confex.com/gsa/2004AM/finalprogram/abstract_77727.htm) from the original on October 8, 2019. Retrieved October 25, 2019.
146. Sellers, William Irvin.; Margetts, Lee; Coria, Rodolfo Aníbal; Manning, Phillip Lars (2013). Carrier, David (ed.). "March of the Titans: The Locomotor Capabilities of Sauropod Dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3864407). *PLOS ONE*. **8** (10) e78733. San Francisco, CA: PLOS. Bibcode:2013PLoSO...878733S (https://ui.adsabs.harvard.edu/abs/2013PLoSO...878733S). doi:10.1371/journal.pone.0078733 (https://doi.org/10.1371%2Fjournal.pone.0078733). ISSN 1932-6203 (https://search.worldcat.org/issn/1932-6203). PMC 3864407 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3864407). PMID 24348896 (https://pubmed.ncbi.nlm.nih.gov/24348896).
147. Lovelace, David M.; Hartman, Scott A.; Wahl, William R. (October–December 2007). "Morphology of a specimen of *Supersaurus* (Dinosauria, Sauropoda) from the Morrison Formation of Wyoming, and a re-evaluation of diplodocid phylogeny" (https://archive.org/details/biostor-248729). *Arquivos do Museu Nacional*. **65** (4). Rio de Janeiro: National Museum of Brazil; Federal University of Rio de Janeiro: 527–544. ISSN 0365-4508 (https://search.worldcat.org/issn/0365-4508). Retrieved October 26, 2019.
148. Carpenter, Kenneth (2018). "Maraapunisaurus fragillimus, N.G. (formerly Amphicoelias fragillimus), a basal Rebbachisaurid from the Morrison Formation (Upper Jurassic) of Colorado" (https://doi.org/10.31711%2Fgiw.v5i0.28). *Geology of the Intermountain West*. **5**: 227–244. doi:10.31711/giw.v5i0.28 (https://doi.org/10.31711%2Fgiw.v5i0.28).
149. Paul, Gregory S. (2019). "Determining the largest known land animal: A critical comparison of differing methods for restoring the volume and mass of extinct animals" (http://www.gspauldino.com/Titanomass.pdf) (PDF). *Annals of the Carnegie Museum*. **85** (4): 335–358. Bibcode:2019AnCM...85..335P (https://ui.adsabs.harvard.edu/abs/2019AnCM...85..335P). doi:10.2992/007.085.0403 (https://doi.org/10.2992%2F007.085.0403). S2CID 210840060 (https://api.semanticscholar.org/CorpusID:210840060). Archived (https://web.archive.org/web/20200128092205/http://www.gspauldino.com/Titanomass.pdf) (PDF) from the original on January 28, 2020. Retrieved December 2, 2023.
150. Pal, Saurabh; Ayyasami, Krishnan (June 27, 2022). "The lost titan of Cauvery". *Geology Today*. **38** (3): 112–116. Bibcode:2022GeolT..38..112P (https://ui.adsabs.harvard.edu/abs/2022GeolT..38..112P). doi:10.1111/gto.12390 (https://doi.org/10.1111%2Fgto.12390). ISSN 0266-6979 (https://search.worldcat.org/issn/0266-6979). S2CID 250056201 (https://api.semanticscholar.org/CorpusID:250056201).
151. Paul, Gregory S.; Larramendi, Asier (April 11, 2023). "Body mass estimate of *Bruhathkayosaurus* and other fragmentary sauropod remains suggest the largest land animals were about as big as the greatest whales" (https://doi.org/10.18261%2Flet.56.2.5). *Lethaia*. **56** (2): 1–11. Bibcode:2023Letha..56..2.5P (https://ui.adsabs.harvard.edu/abs/2023Letha..56..2.5P). doi:10.18261/let.56.2.5 (https://doi.org/10.18261%2Flet.56.2.5). ISSN 0024-1164 (https://search.worldcat.org/issn/0024-1164). S2CID 259782734 (https://api.semanticscholar.org/CorpusID:259782734).

152. Dal Sasso, Cristiano; Maganuco, Simone; Buffetaut, Éric; et al. (2005). "New information on the skull of the enigmatic theropod *Spinosaurus*, with remarks on its sizes and affinities" (https://web.archive.org/web/20110429015542/http://reocities.com/Athens/bridge/4602/spinoskull.pdf) (PDF). *Journal of Vertebrate Paleontology*. **25** (4). Milton Park, Oxfordshire: Taylor & Francis for the Society of Vertebrate Paleontology: 888–896. doi:10.1671/0272-4634(2005)025[0888:NIOTSO]2.0.CO;2 (https://doi.org/10.1671%2F0272-4634%282005%29025%5B0888%3ANIOTSO%5D2.0.CO%3B2). ISSN 0272-4634 (https://search.worldcat.org/issn/0272-4634). S2CID 85702490 (https://api.semanticscholar.org/CorpusID:85702490). Archived from the original (http://www.reocities.com/Athens/bridge/4602/spinoskull.pdf) (PDF) on April 29, 2011. Retrieved May 5, 2011.
153. Therrien, François; Henderson, Donald M. (2007). "My theropod is bigger than yours ... or not: estimating body size from skull length in theropods". *Journal of Vertebrate Paleontology*. **27** (1). Milton Park, Oxfordshire: Taylor & Francis for the Society of Vertebrate Paleontology: 108–115. doi:10.1671/0272-4634(2007)27[108:MTIBTY]2.0.CO;2 (https://doi.org/10.1671%2F0272-4634%282007%2927%5B108%3AMTIBTY%5D2.0.CO%3B2). ISSN 0272-4634 (https://search.worldcat.org/issn/0272-4634). S2CID 86025320 (https://api.semanticscholar.org/CorpusID:86025320).
154. Zhao, Xijin; Li, Dunjing; Han, Gang; et al. (2007). "Zhuchengosaurus maximus from Shandong Province". *Acta Geoscientia Sinica*. **28** (2). Beijing: Chinese Academy of Geological Sciences: 111–122. ISSN 1006-3021 (https://search.worldcat.org/issn/1006-3021).
155. Weishampel, Dodson & Osmólska 2004, pp. 438–463, chpt. 20: "Hadrosauridae" by John R. Horner David B. Weishampel, and Catherine A. Forster.
156. Norell, Gaffney & Dingus 2000
157. "Bee Hummingbird (*Mellisuga helenae*)" (https://www.birds.com/species/a-b/bee-hummingbird/). *Birds.com*. Paley Media. Archived (https://web.archive.org/web/20150403005328/https://www.birds.com/species/a-b/bee-hummingbird/) from the original on April 3, 2015. Retrieved October 27, 2019.
158. Zhang, Fucheng; Zhou, Zhonghe; Xu, Xing; et al. (2008). "A bizarre Jurassic maniraptoran from China with elongate ribbon-like feathers". *Nature*. **455** (7216). London: Nature Research: 1105–1108. Bibcode:2008Natur.455.1105Z (https://ui.adsabs.harvard.edu/abs/2008Natur.455.1105Z). doi:10.1038/nature07447 (https://doi.org/10.1038%2Fnature07447). ISSN 0028-0836 (https://search.worldcat.org/issn/0028-0836). PMID 18948955 (https://pubmed.ncbi.nlm.nih.gov/18948955). S2CID 4362560 (https://api.semanticscholar.org/CorpusID:4362560).
159. Xu, Xing; Zhao, Qi; Norell, Mark; et al. (February 2008). "A new feathered maniraptoran dinosaur fossil that fills a morphological gap in avian origin" (https://doi.org/10.1007%2Fs11434-009-0009-6). *Chinese Science Bulletin*. **54** (3). Amsterdam: Elsevier on behalf of Science in China Press: 430–435. Bibcode:2009SciBu..54..430X (https://ui.adsabs.harvard.edu/abs/2009SciBu..54..430X). doi:10.1007/s11434-009-0009-6 (https://doi.org/10.1007%2Fs11434-009-0009-6). ISSN 1001-6538 (https://search.worldcat.org/issn/1001-6538). S2CID 53445386 (https://api.semanticscholar.org/CorpusID:53445386).
160. Holtz 2007
161. Butler, Richard J.; Zhao, Qi (February 2009). "The small-bodied ornithischian dinosaurs *Micropachycephalosaurus hongtuyanensis* and *Wannanosaurus yansiensis* from the Late Cretaceous of China". *Cretaceous Research*. **30** (1). Amsterdam: Elsevier: 63–77. Bibcode:2009CrRes..30...63B (https://ui.adsabs.harvard.edu/abs/2009CrRes..30...63B). doi:10.1016/j.cretres.2008.03.002 (https://doi.org/10.1016%2Fj.cretres.2008.03.002). ISSN 0195-6671 (https://search.worldcat.org/issn/0195-6671).

162. Yans, Johan; Dejax, Jean; Pons, Denise; et al. (January–February 2005). "Implications paléontologiques et géodynamiques de la datation palynologique des sédiments à faciès wealdien de Bernissart (bassin de Mons, Belgique)" [Palaeontological and geodynamical implications of the palynological dating of the wealden facies sediments of Bernissart (Mons Basin, Belgium)]. *Comptes Rendus Palevol* (in French). **4** (1–2). Amsterdam: Elsevier of behalf of the French Academy of Sciences: 135–150. Bibcode:2005CRPal...4..135Y (https://ui.adsabs.harvard.edu/abs/2005CRPal...4..135Y). doi:10.1016/j.crpv.2004.12.003 (https://doi.org/10.1016%2Fj.crpv.2004.12.003). ISSN 1631-0683 (https://search.worldcat.org/issn/1631-0683).
163. Day, Julia J.; Upchurch, Paul; Norman, David B.; et al. (2002). "Sauropod Trackways, Evolution, and Behavior" (http://doc.rero.ch/record/15083/files/PAL_E2344.pdf) (PDF). *Science*. **296** (5573). Washington, D.C.: American Association for the Advancement of Science: 1659. doi:10.1126/science.1070167 (https://doi.org/10.1126%2Fscience.1070167). ISSN 0036-8075 (https://search.worldcat.org/issn/0036-8075). PMID 12040187 (https://pubmed.ncbi.nlm.nih.gov/12040187). S2CID 36530770 (https://api.semanticscholar.org/CorpusID:36530770).
164. Curry Rogers & Wilson 2005, pp. 252–284, chpt. 9: "Steps in Understanding Sauropod Biology: The Importance of Sauropods Tracks" by Joanna L. Wright.
165. Varricchio, David J.; Sereno, Paul C.; Zhao, Xijin; et al. (2008). "Mud-trapped herd captures evidence of distinctive dinosaur sociality" (https://www.app.pan.pl/archive/published/app53/APP53-567.pdf) (PDF). *Acta Palaeontologica Polonica*. **53** (4). Warsaw: Institute of Paleobiology, Polish Academy of Sciences: 567–578. Bibcode:2008AcPaP..53..567V (https://ui.adsabs.harvard.edu/abs/2008AcPaP..53..567V). doi:10.4202/app.2008.0402 (https://doi.org/10.4202%2Fapp.2008.0402). ISSN 0567-7920 (https://search.worldcat.org/issn/0567-7920). S2CID 21736244 (https://api.semanticscholar.org/CorpusID:21736244). Archived (https://web.archive.org/web/20190330032513/https://www.app.pan.pl/archive/published/app53/APP53-567.pdf) (PDF) from the original on March 30, 2019. Retrieved May 6, 2011.
166. Lessem & Glut 1993, pp. 19–20 (https://archive.org/details/dinosaursocietys00less/page/19), "Allosaurus"
167. Maxwell, W. Desmond; Ostrom, John H. (1995). "Taphonomy and paleobiological implications of *Tenontosaurus–Deinonychus* associations". *Journal of Vertebrate Paleontology*. **15** (4). Milton Park, Oxfordshire: Taylor & Francis for the Society of Vertebrate Paleontology: 707–712. Bibcode:1995JVPal..15..707M (https://ui.adsabs.harvard.edu/abs/1995JVPal..15..707M). doi:10.1080/02724634.1995.10011256 (https://doi.org/10.1080%2F02724634.1995.10011256). ISSN 0272-4634 (https://search.worldcat.org/issn/0272-4634).
168. Roach, Brian T.; Brinkman, Daniel L. (April 2007). "A Reevaluation of Cooperative Pack Hunting and Gregariousness in *Deinonychus antirrhopus* and Other Nonavian Theropod Dinosaurs". *Bulletin of the Peabody Museum of Natural History*. **48** (1). New Haven, CT: Peabody Museum of Natural History: 103–138. doi:10.3374/0079-032X(2007)48[103:AROCPH]2.0.CO;2 (https://doi.org/10.3374%2F0079-032X%282007%2948%5B103%3AAROCPH%5D2.0.CO%3B2). ISSN 0079-032X (https://search.worldcat.org/issn/0079-032X). S2CID 84175628 (https://api.semanticscholar.org/CorpusID:84175628).
169. Tanke, Darren H. (1998). "Head-biting behavior in theropod dinosaurs: paleopathological evidence" (https://web.archive.org/web/20080227134632/http://www.mnhn.ul.pt/geologia/gaia/12.pdf) (PDF). *Gaia: Revista de Geociências* (15). Lisbon: National Museum of Natural History and Science: 167–184. doi:10.7939/R34T6FJ1P (https://doi.org/10.7939%2FR34T6FJ1P). ISSN 0871-5424 (https://search.worldcat.org/issn/0871-5424). S2CID 90552600 (https://api.semanticscholar.org/CorpusID:90552600). Archived from the original (http://www.mnhn.ul.pt/geologia/gaia/12.pdf) (PDF) on February 27, 2008.
170. "The Fighting Dinosaurs" (https://web.archive.org/web/20120118062252/http://www.amnh.org/exhibitions/fightingdinos/ex-fd.html). New York: American Museum of Natural History. Archived from the original (http://www.amnh.org/exhibitions/fightingdinos/ex-fd.html) on January 18, 2012. Retrieved December 5, 2007.

171. Carpenter, Kenneth (1998). "Evidence of predatory behavior by theropod dinosaurs" (http://www.arca.museus.ul.pt/ArcaSite/obj/gaia/MNHNL-0000778-MG-DOC-web.PDF) (PDF). *Gaia: Revista de Geociências*. **15**. Lisbon: National Museum of Natural History and Science: 135–144. ISSN 0871-5424 (https://search.worldcat.org/issn/0871-5424). Archived (https://web.archive.org/web/20130926134333/http://www.arca.museus.ul.pt/ArcaSite/obj/gaia/MNHNL-0000778-MG-DOC-web.PDF) (PDF) from the original on September 26, 2013.
172. Rogers, Raymond R.; Krause, David W.; Curry Rogers, Kristina (2007). "Cannibalism in the Madagascan dinosaur *Majungatholus atopus*". *Nature*. **422** (6931). London: Nature Research: 515–518. Bibcode:2003Natur.422..515R (https://ui.adsabs.harvard.edu/abs/2003Natur.422..515R). doi:10.1038/nature01532 (https://doi.org/10.1038%2Fnature01532). ISSN 0028-0836 (https://search.worldcat.org/issn/0028-0836). PMID 12673249 (https://pubmed.ncbi.nlm.nih.gov/12673249). S2CID 4389583 (https://api.semanticscholar.org/CorpusID:4389583).
173. Schmitz, Lars; Motani, Ryosuke (2011). "Nocturnality in Dinosaurs Inferred from Scleral Ring and Orbit Morphology". *Science*. **332** (6030). Washington, D.C.: American Association for the Advancement of Science: 705–708. Bibcode:2011Sci...332..705S (https://ui.adsabs.harvard.edu/abs/2011Sci...332..705S). doi:10.1126/science.1200043 (https://doi.org/10.1126%2Fscience.1200043). ISSN 0036-8075 (https://search.worldcat.org/issn/0036-8075). PMID 21493820 (https://pubmed.ncbi.nlm.nih.gov/21493820). S2CID 33253407 (https://api.semanticscholar.org/CorpusID:33253407).
174. Varricchio, David J.; Martin, Anthony J.; Katsura, Yoshihiro (2007). "First trace and body fossil evidence of a burrowing, denning dinosaur" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2176205). *Proceedings of the Royal Society B*. **274** (1616). London: Royal Society: 1361–1368. doi:10.1098/rspb.2006.0443 (https://doi.org/10.1098%2Frspb.2006.0443). ISSN 0962-8452 (https://search.worldcat.org/issn/0962-8452). PMC 2176205 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2176205). PMID 17374596 (https://pubmed.ncbi.nlm.nih.gov/17374596).
175. Chiappe & Witmer 2002
176. Chatterjee, Sankar; Templin, R. Jack (2007). "Biplane wing planform and flight performance of the feathered dinosaur *Microraptor gui*" (https://www.pnas.org/content/pnas/104/5/1576.full.pdf) (PDF). *Proc. Natl. Acad. Sci. U.S.A*. **104** (5). Washington, D.C.: National Academy of Sciences: 1576–1580. Bibcode:2007PNAS..104.1576C (https://ui.adsabs.harvard.edu/abs/2007PNAS..104.1576C). doi:10.1073/pnas.0609975104 (https://doi.org/10.1073%2Fpnas.0609975104). ISSN 0027-8424 (https://search.worldcat.org/issn/0027-8424). PMC 1780066 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1780066). PMID 17242354 (https://pubmed.ncbi.nlm.nih.gov/17242354). Archived (https://web.archive.org/web/20190818174156/https://www.pnas.org/content/pnas/104/5/1576.full.pdf) (PDF) from the original on August 18, 2019. Retrieved October 29, 2019.
177. Goriely, Alain; McMillen, Tyler (2002). "Shape of a Cracking Whip". *Physical Review Letters*. **88** (24) 244301. Ridge, NY: American Physical Society. Bibcode:2002PhRvL..88x4301G (https://ui.adsabs.harvard.edu/abs/2002PhRvL..88x4301G). doi:10.1103/PhysRevLett.88.244301 (https://doi.org/10.1103%2FPhysRevLett.88.244301). ISSN 0031-9007 (https://search.worldcat.org/issn/0031-9007). PMID 12059302 (https://pubmed.ncbi.nlm.nih.gov/12059302).
178. Henderson, Donald M. (2003). "Effects of stomach stones on the buoyancy and equilibrium of a floating crocodilian: a computational analysis". *Canadian Journal of Zoology*. **81** (8). Ottawa: NRC Research Press: 1346–1357. Bibcode:2003CaJZ...81.1346H (https://ui.adsabs.harvard.edu/abs/2003CaJZ...81.1346H). doi:10.1139/z03-122 (https://doi.org/10.1139%2Fz03-122). ISSN 0008-4301 (https://search.worldcat.org/issn/0008-4301).

179. Senter, Phil (June 30, 2009). "Voices of the past: a review of Paleozoic and Mesozoic animal sounds" (https://doi.org/10.1080%2F08912960903033327). *Historical Biology*. **20** (4): 255–287. Bibcode:2008HBio...20..255S (https://ui.adsabs.harvard.edu/abs/2008HBio...20..255S). doi:10.1080/08912960903033327 (https://doi.org/10.1080%2F08912960903033327). ISSN 0891-2963 (https://search.worldcat.org/issn/0891-2963). S2CID 84473967 (https://api.semanticscholar.org/CorpusID:84473967).
180. Li, Quanguo; Gao, Ke-Qin; Vinther, Jakob; et al. (2010). "Plumage Color Patterns of an Extinct Dinosaur" (https://doc.rero.ch/record/210394/files/PAL_E4402.pdf) (PDF). *Science*. **327** (5971). Washington, D.C.: American Association for the Advancement of Science: 1369–1372. Bibcode:2010Sci...327.1369L (https://ui.adsabs.harvard.edu/abs/2010Sci...327.1369L). doi:10.1126/science.1186290 (https://doi.org/10.1126%2Fscience.1186290). ISSN 0036-8075 (https://search.worldcat.org/issn/0036-8075). PMID 20133521 (https://pubmed.ncbi.nlm.nih.gov/20133521). S2CID 206525132 (https://api.semanticscholar.org/CorpusID:206525132). Archived (https://web.archive.org/web/20190330023836/https://doc.rero.ch/record/210394/files/PAL_E4402.pdf) (PDF) from the original on March 30, 2019. Retrieved November 7, 2019.
181. Riede, T. (2019). "The evolution of the syrinx: an acoustic theory" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6366696). *PLOS ONE*. **17** (2) e2006507. doi:10.1371/journal.pbio.2006507 (https://doi.org/10.1371%2Fjournal.pbio.2006507). PMC 6366696 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6366696). PMID 30730882 (https://pubmed.ncbi.nlm.nih.gov/30730882).
182. Yoshida, Junki; Kobayashi, Yoshitsugu; Norell, Mark A. (February 15, 2023). "An ankylosaur larynx provides insights for bird-like vocalization in non-avian dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9932143). *Communications Biology*. **6** (1): 152. doi:10.1038/s42003-023-04513-x (https://doi.org/10.1038%2Fs42003-023-04513-x). ISSN 2399-3642 (https://search.worldcat.org/issn/2399-3642). PMC 9932143 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9932143). PMID 36792659 (https://pubmed.ncbi.nlm.nih.gov/36792659).
183. Yang, Yunfeng; King, James L.; Xu, Xing (2025). "A new neornithischian dinosaur from the Upper Jurassic Tiaojishan Formation of northern China" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12258161). *PeerJ*. **13** e19664. doi:10.7717/peerj.19664 (https://doi.org/10.7717%2Fpeerj.19664). PMC 12258161 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12258161). PMID 40661901 (https://pubmed.ncbi.nlm.nih.gov/40661901).
184. Clarke, Julia A.; Chatterjee, Sankar; Zhiheng, Li; et al. (2016). "Fossil evidence of the avian vocal organ from the Mesozoic". *Nature*. **538** (7626). London: Nature Research: 502–505. Bibcode:2016Natur.538..502C (https://ui.adsabs.harvard.edu/abs/2016Natur.538..502C). doi:10.1038/nature19852 (https://doi.org/10.1038%2Fnature19852). hdl:11336/64786 (https://hdl.handle.net/11336%2F64786). ISSN 0028-0836 (https://search.worldcat.org/issn/0028-0836). PMID 27732575 (https://pubmed.ncbi.nlm.nih.gov/27732575). S2CID 4389926 (https://api.semanticscholar.org/CorpusID:4389926).
185. Riede, Tobias; Eliason, Chad M.; Miller, Edward H.; et al. (2016). "Coos, booms, and hoots: the evolution of closed-mouth vocal behavior in birds" (https://doi.org/10.1111%2Fevo.12988). *Evolution*. **70** (8). Hoboken, NJ: John Wiley & Sons for the Society for the Study of Evolution: 1734–1746. Bibcode:2016Evolu..70.1734R (https://ui.adsabs.harvard.edu/abs/2016Evolu..70.1734R). doi:10.1111/evo.12988 (https://doi.org/10.1111%2Fevo.12988). ISSN 0014-3820 (https://search.worldcat.org/issn/0014-3820). PMID 27345722 (https://pubmed.ncbi.nlm.nih.gov/27345722). S2CID 11986423 (https://api.semanticscholar.org/CorpusID:11986423).

186. Weishampel, David B. (Spring 1981). "Acoustic Analysis of Vocalization of Lambeosaurine Dinosaurs (Reptilia: Ornithischia)" (https://web.archive.org/web/20141006113229/http://www.hopkinsmedicine.org/FAE/DBWpdf/R3_1981aWeishampel.pdf) (PDF). *Paleobiology*. **7** (2). Bethesda, MD: Paleontological Society: 252–261. doi:10.1017/S0094837300004036 (https://doi.org/10.1017%2FS0094837300004036). ISSN 0094-8373 (https://search.worldcat.org/issn/0094-8373). JSTOR 2400478 (https://www.jstor.org/stable/2400478). S2CID 89109302 (https://api.semanticscholar.org/CorpusID:89109302). Archived from the original (http://www.hopkinsmedicine.org/FAE/DBWpdf/R3_1981aWeishampel.pdf) (PDF) on October 6, 2014. Retrieved October 30, 2019.
187. Miyashita, Tetsuto; Arbour, Victoria M.; Witmer, Lawrence M.; et al. (December 2011). "The internal cranial morphology of an armoured dinosaur *Euoplocephalus* corroborated by X-ray computed tomographic reconstruction" (https://web.archive.org/web/20150924062640/http://www.oucom.ohiou.edu/dbms-witmer/Downloads/2011_Miyashita_et_al._Euoplocephalus_head_anatomy.pdf) (PDF). *Journal of Anatomy*. **219** (6). Hoboken, NJ: John Wiley & Sons: 661–675. doi:10.1111/j.1469-7580.2011.01427.x (https://doi.org/10.1111%2Fj.1469-7580.2011.01427.x). ISSN 1469-7580 (https://search.worldcat.org/issn/1469-7580). PMC 3237876 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3237876). PMID 21954840 (https://pubmed.ncbi.nlm.nih.gov/21954840). Archived from the original (http://www.oucom.ohiou.edu/dbms-witmer/Downloads/2011_Miyashita_et_al._Euoplocephalus_head_anatomy.pdf) (PDF) on September 24, 2015. Retrieved October 30, 2019.
188. Hansell 2000
189. Varricchio, David J.; Horner, John R.; Jackson, Frankie D. (2002). "Embryos and eggs for the Cretaceous theropod dinosaur *Troodon formosus*". *Journal of Vertebrate Paleontology*. **22** (3). Milton Park, Oxfordshire: Taylor & Francis for the Society of Vertebrate Paleontology: 564–576. doi:10.1671/0272-4634(2002)022[0564:EAEFTC]2.0.CO;2 (https://doi.org/10.1671%2F0272-4634%282002%29022%5B0564%3AEAEFTC%5D2.0.CO%3B2). ISSN 0272-4634 (https://search.worldcat.org/issn/0272-4634). S2CID 85728452 (https://api.semanticscholar.org/CorpusID:85728452).
190. Lee, Andrew H.; Werning, Sarah (2008). "Sexual maturity in growing dinosaurs does not fit reptilian growth models" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2206579). *Proc. Natl. Acad. Sci. U.S.A.* **105** (2). Washington, D.C.: National Academy of Sciences: 582–587. Bibcode:2008PNAS..105..582L (https://ui.adsabs.harvard.edu/abs/2008PNAS..105..582L). doi:10.1073/pnas.0708903105 (https://doi.org/10.1073%2Fpnas.0708903105). ISSN 0027-8424 (https://search.worldcat.org/issn/0027-8424). PMC 2206579 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2206579). PMID 18195356 (https://pubmed.ncbi.nlm.nih.gov/18195356).
191. Horner, John R.; Makela, Robert (1979). "Nest of juveniles provides evidence of family structure among dinosaurs". *Nature*. **282** (5736). London: Nature Research: 296–298. Bibcode:1979Natur.282..296H (https://ui.adsabs.harvard.edu/abs/1979Natur.282..296H). doi:10.1038/282296a0 (https://doi.org/10.1038%2F282296a0). ISSN 0028-0836 (https://search.worldcat.org/issn/0028-0836). S2CID 4370793 (https://api.semanticscholar.org/CorpusID:4370793).
192. "Discovering Dinosaur Behavior: 1960–present view" (https://web.archive.org/web/20131213042521/https://www.britannica.com/dinosaurs/dinosaurs/BRa.html). *Encyclopædia Britannica*. Chicago, IL: Encyclopædia Britannica, Inc. Archived from the original (https://www.britannica.com/dinosaurs/dinosaurs/BRa.html) on December 13, 2013. Retrieved October 30, 2019.
193. Currie et al. 2004, pp. 234–250 (http://thomas-hopp.com/pdf/DinoBrooding.pdf), chpt. 11: "Dinosaur Brooding Behavior and the Origin of Flight Feathers" by Thomas P. Hopp and Mark J. Orsen.

194. Reisz, Robert R.; Scott, Diane; Sues, Hans-Dieter; et al. (2005). "Embryos of an Early Jurassic Prosauropod Dinosaur and Their Evolutionary Significance" (https://repository.si.edu/bitstream/handle/10088/7530/paleo_REISZ_ET_AL.2005.pdf) (PDF). *Science*. **309** (5735). Washington, D.C.: American Association for the Advancement of Science: 761–764. Bibcode:2005Sci...309..761R (https://ui.adsabs.harvard.edu/abs/2005Sci...309..761R). doi:10.1126/science.1114942 (https://doi.org/10.1126%2Fscience.1114942). ISSN 0036-8075 (https://search.worldcat.org/issn/0036-8075). PMID 16051793 (https://pubmed.ncbi.nlm.nih.gov/16051793). S2CID 37548361 (https://api.semanticscholar.org/CorpusID:37548361). Archived (https://web.archive.org/web/20180722004548/https://repository.si.edu/bitstream/handle/10088/7530/paleo_REISZ_ET_AL.2005.pdf) (PDF) from the original on July 22, 2018.
195. Clark, Neil D. L.; Booth, Paul; Booth, Claire L.; et al. (2004). "Dinosaur footprints from the Duntulm Formation (Bathonian, Jurassic) of the Isle of Skye" (http://eprints.gla.ac.uk/4496/1/4496.pdf) (PDF). *Scottish Journal of Geology*. **40** (1). London: Geological Society of London: 13–21. Bibcode:2004ScJG...40...13C (https://ui.adsabs.harvard.edu/abs/2004ScJG...40...13C). doi:10.1144/sjg40010013 (https://doi.org/10.1144%2Fsjg40010013). ISSN 0036-9276 (https://search.worldcat.org/issn/0036-9276). S2CID 128544813 (https://api.semanticscholar.org/CorpusID:128544813). Archived (https://web.archive.org/web/20130722081936/http://eprints.gla.ac.uk/4496/1/4496.pdf) (PDF) from the original on July 22, 2013. Retrieved December 12, 2019.
196. Zhou, Zhonghe; Zhang, Fucheng (2004). "A Precocial Avian Embryo from the Lower Cretaceous of China". *Science*. **306** (5696). Washington, D.C.: American Association for the Advancement of Science: 653. Bibcode:2004Sci...306..653Z (https://ui.adsabs.harvard.edu/abs/2004Sci...306..653Z). doi:10.1126/science.1100000 (https://doi.org/10.1126%2Fscience.1100000). ISSN 0036-8075 (https://search.worldcat.org/issn/0036-8075). PMID 15499011 (https://pubmed.ncbi.nlm.nih.gov/15499011). S2CID 34504916 (https://api.semanticscholar.org/CorpusID:34504916).
197. Naish, Darren (May 15, 2012). "A drowned nesting colony of Late Cretaceous birds" (https://blogs.scientificamerican.com/tetrapod-zoology/drowned-cretaceous-bird-colony/). *Science*. **306** (5696). Scientific American: 653. Bibcode:2004Sci...306..653Z (https://ui.adsabs.harvard.edu/abs/2004Sci...306..653Z). doi:10.1126/science.1100000 (https://doi.org/10.1126%2Fscience.1100000). PMID 15499011 (https://pubmed.ncbi.nlm.nih.gov/15499011). S2CID 34504916 (https://api.semanticscholar.org/CorpusID:34504916). Archived (https://web.archive.org/web/20180925031243/https://blogs.scientificamerican.com/tetrapod-zoology/drowned-cretaceous-bird-colony/) from the original on September 25, 2018. Retrieved November 16, 2019.
198. Fernández, Mariela S.; García, Rodolfo A.; Fiorelli, Lucas; et al. (2013). "A Large Accumulation of Avian Eggs from the Late Cretaceous of Patagonia (Argentina) Reveals a Novel Nesting Strategy in Mesozoic Birds" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3629076). *PLOS ONE*. **8** (4) e61030. San Francisco, CA: PLOS. Bibcode:2013PLoSO...861030F (https://ui.adsabs.harvard.edu/abs/2013PLoSO...861030F). doi:10.1371/journal.pone.0061030 (https://doi.org/10.1371%2Fjournal.pone.0061030). ISSN 1932-6203 (https://search.worldcat.org/issn/1932-6203). PMC 3629076 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3629076). PMID 23613776 (https://pubmed.ncbi.nlm.nih.gov/23613776).

199. Deeming, Denis Charles; Mayr, Gerald (May 2018). "Pelvis morphology suggests that early Mesozoic birds were too heavy to contact incubate their eggs" (http://eprints.lincoln.ac.uk/id/eprint/31436/13/31436%2031291%20Deeming_et_al-2018-Journal_of_Evolutionary_Biology.pdf) (PDF). *Journal of Evolutionary Biology*. **31** (5). Hoboken, NJ: Wiley-Blackwell on behalf of the European Society for Evolutionary Biology: 701–709. doi:10.1111/jeb.13256 (https://doi.org/10.1111%2Fjeb.13256). ISSN 1010-061X (https://search.worldcat.org/issn/1010-061X). PMID 29485191 (https://pubmed.ncbi.nlm.nih.gov/29485191). S2CID 3588317 (https://api.semanticscholar.org/CorpusID:3588317). Archived (https://web.archive.org/web/20200602094641/http://eprints.lincoln.ac.uk/id/eprint/31436/13/31436%2031291%20Deeming_et_al-2018-Journal_of_Evolutionary_Biology.pdf) (PDF) from the original on June 2, 2020.
200. Myers, Timothy S.; Fiorillo, Anthony R. (2009). "Evidence for gregarious behavior and age segregation in sauropod dinosaurs" (http://doc.rero.ch/record/16633/files/PAL_E962.pdf) (PDF). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **274** (1–2). Amsterdam: Elsevier: 96–104. Bibcode:2009PPP...274...96M (https://ui.adsabs.harvard.edu/abs/2009PPP...274...96M). doi:10.1016/j.palaeo.2009.01.002 (https://doi.org/10.1016%2Fj.palaeo.2009.01.002). ISSN 0031-0182 (https://search.worldcat.org/issn/0031-0182). Archived (https://web.archive.org/web/20200529011847/http://doc.rero.ch/record/16633/files/PAL_E962.pdf) (PDF) from the original on May 29, 2020.
201. Vinther, Jakob; Nicholls, Robert; Kelly, Diane A. (February 22, 2021). "A cloacal opening in a non-avian dinosaur" (https://doi.org/10.1016%2Fj.cub.2020.12.039). *Current Biology*. **31** (4). Elsevier: R1–R3. Bibcode:2021CBio...31.R182V (https://ui.adsabs.harvard.edu/abs/2021CBio...31.R182V). doi:10.1016/j.cub.2020.12.039 (https://doi.org/10.1016%2Fj.cub.2020.12.039). PMID 33472049 (https://pubmed.ncbi.nlm.nih.gov/33472049). S2CID 231644183 (https://api.semanticscholar.org/CorpusID:231644183).
202. Isles, T. (2009). "The socio-sexual behaviour of extant archosaurs: Implications for understanding dinosaur behaviour" (http://doc.rero.ch/record/232923/files/PAL_E4545.pdf) (PDF). *Historical Biology*. **21** (3–4): 139–214. Bibcode:2009HBio...21..139I (https://ui.adsabs.harvard.edu/abs/2009HBio...21..139I). doi:10.1080/08912960903450505 (https://doi.org/10.1080%2F08912960903450505). S2CID 85417112 (https://api.semanticscholar.org/CorpusID:85417112). Archived (https://web.archive.org/web/20210502213306/http://doc.rero.ch/record/232923/files/PAL_E4545.pdf) (PDF) from the original on May 2, 2021. Retrieved October 29, 2019.
203. Weishampel, Dodson & Osmólska 2004, pp. 643–659, chpt. 28: "Physiology of Nonavian Dinosaurs" by Anusuya Chinsamy and Willem J. Hillenius.
204. Pontzer, H.; Allen, V.; Hutchinson, J.R. (2009). "Biomechanics of running indicates endothermy in bipedal dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2772121). *PLOS ONE*. **4** (11) e7783. Bibcode:2009PLoSO...4.7783P (https://ui.adsabs.harvard.edu/abs/2009PLoSO...4.7783P). doi:10.1371/journal.pone.0007783 (https://doi.org/10.1371%2Fjournal.pone.0007783). ISSN 1932-6203 (https://search.worldcat.org/issn/1932-6203). PMC 2772121 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2772121). PMID 19911059 (https://pubmed.ncbi.nlm.nih.gov/19911059).
205. Benson, R.B.J. (2018). "Dinosaur Macroevolution and Macroecology" (https://doi.org/10.1146%2Fannurev-ecolsys-110617-062231). *Annual Review of Ecology, Evolution, and Systematics*. **49**: 379–408. doi:10.1146/annurev-ecolsys-110617-062231 (https://doi.org/10.1146%2Fannurev-ecolsys-110617-062231). S2CID 92837486 (https://api.semanticscholar.org/CorpusID:92837486).
206. Grady, J.M.; Enquist, B.J.; Dettweiler-Robinson, E.; Wright, N.A.; Smith, F.A. (2014). "Evidence for mesothermy in dinosaurs". *Science*. **344** (6189): 1268–1272. Bibcode:2014Sci...344.1268G (https://ui.adsabs.harvard.edu/abs/2014Sci...344.1268G). doi:10.1126/science.1253143 (https://doi.org/10.1126%2Fscience.1253143). PMID 24926017 (https://pubmed.ncbi.nlm.nih.gov/24926017). S2CID 9806780 (https://api.semanticscholar.org/CorpusID:9806780).

207. Legendre, L.J.; Guénard, G.; Botha-Brink, J.; Cubo, J. (2016). "Palaeohistological Evidence for Ancestral High Metabolic Rate in Archosaurs" (https://doi.org/10.1093%2Fsysbio%2Fsyw033). *Systematic Biology*. **65** (6): 989–996. doi:10.1093/sysbio/syw033 (https://doi.org/10.1093%2Fsysbio%2Fsyw033). PMID 27073251 (https://pubmed.ncbi.nlm.nih.gov/27073251).
208. Seymour, R.S.; Bennett-Stamper, C.L.; Johnston, S.D.; Carrier, D.R.; Grigg, G.C. (2004). "Evidence for endothermic ancestors of crocodiles at the stem of archosaur evolution" (https://doi.org/10.1093%2Fsysbio%2Fsyw033). *Physiological and Biochemical Zoology*. **77** (6): 1051–1067. doi:10.1093/sysbio/syw033 (https://doi.org/10.1093%2Fsysbio%2Fsyw033). PMID 27073251 (https://pubmed.ncbi.nlm.nih.gov/27073251).
209. Parsons 2001, pp. 22–48, "The Heresies of Dr. Bakker".
210. Erickson, G.M. (2014). "On dinosaur growth". *Annual Review of Earth and Planetary Sciences*. **42** (1): 675–697. Bibcode:2014AREPS..42..675E (https://ui.adsabs.harvard.edu/abs/2014AREPS..42..675E). doi:10.1146/annurev-earth-060313-054858 (https://doi.org/10.1146%2Fannurev-earth-060313-054858).
211. Bailleul, A.M.; O'Connor, J.; Schweitzer, M.H. (2019). "Dinosaur paleohistology: review, trends and new avenues of investigation" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6768056). *PeerJ*. **7** e7764. Bibcode:2019PeerJ...7e7764B (https://ui.adsabs.harvard.edu/abs/2019PeerJ...7e7764B). doi:10.7717/peerj.7764 (https://doi.org/10.7717%2Fpeerj.7764). PMC 6768056 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6768056). PMID 31579624 (https://pubmed.ncbi.nlm.nih.gov/31579624).
212. De Ricqlès, A. (1974). "Evolution of endothermy: histological evidence" (http://www.stuartsumida.com/BIOL622/Ricqles1974.pdf) (PDF). *Evolutionary Theory*. **1** (2): 51–80. Archived (https://web.archive.org/web/20210417055907/http://www.stuartsumida.com/BIOL622/Ricqles1974.pdf) (PDF) from the original on April 17, 2021.
213. De Ricqlès, A. (1980). "Tissue structures of dinosaur bone, functional significance and possible relation to dinosaur physiology". In Thomas, R.D.K.; Olson, E.C. (eds.). *A Cold Look at the Warm-Blooded Dinosaurs*. New York: American Association for the Advancement of Science. pp. 103–139.
214. Padian, K.; Horner, J.R.; de Ricqlès, A. (2004). "Growth in small dinosaurs and pterosaurs: the evolution of archosaurian growth strategies" (http://doc.rero.ch/record/15191/files/PAL_E2467.pdf) (PDF). *Journal of Vertebrate Paleontology*. **24** (3): 555–571. doi:10.1671/0272-4634(2004)024[0555:GISDAP]2.0.CO;2 (https://doi.org/10.1671%2F0272-4634%282004%2924%5B0555%3AGISDAP%5D2.0.CO%3B2). S2CID 86019906 (https://api.semanticscholar.org/CorpusID:86019906). Archived (https://web.archive.org/web/20230204094637/https://doc.rero.ch/record/15191/files/PAL_E2467.pdf) (PDF) from the original on February 4, 2023. Retrieved January 3, 2023.
215. de Souza, G.A.; Bento Soares, M.; Souza Brum, A.; Zucolotto, M.; Sayão, J.M.; Carlos Weinschütz, L.; Kellner, A.W.A. (2020). "Osteohistology and growth dynamics of the Brazilian noasaurid *Vespersaurus paranaensis* Langer et al., 2019 (Theropoda: Abelisauroidea)" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7500327). *PeerJ*. **8** e9771. doi:10.7717/peerj.9771 (https://doi.org/10.7717%2Fpeerj.9771). PMC 7500327 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7500327). PMID 32983636 (https://pubmed.ncbi.nlm.nih.gov/32983636). S2CID 221906765 (https://api.semanticscholar.org/CorpusID:221906765).

216. For examples of this work conducted on different dinosaur lineages, see

- Erickson, G.M.; Tumanova, T.A. (2000). "Growth curve of *Psittacosaurus mongoliensis* Osborn (Ceratopsia: Psittacosauridae) inferred from long bone histology". *Zoological Journal of the Linnean Society*. **130** (4): 551–566. doi:10.1111/j.1096-3642.2000.tb02201.x (https://doi.org/10.1111%2Fj.1096-3642.2000.tb02201.x). S2CID 84241148 (https://api.semanticscholar.org/CorpusID:84241148).
- Erickson, G.; Rogers, K.; Yerby, S. (2001). "Dinosaurian growth patterns and rapid avian growth rates". *Nature*. **412** (429–433): 429–433. Bibcode:2001Natur.412..429E (https://ui.adsabs.harvard.edu/abs/2001Natur.412..429E). doi:10.1038/35086558 (https://doi.org/10.1038%2F35086558). PMID 11473315 (https://pubmed.ncbi.nlm.nih.gov/11473315). S2CID 4319534 (https://api.semanticscholar.org/CorpusID:4319534). **(Erratum: doi:10.1038/nature16488 (https://doi.org/10.1038%2Fnature16488), PMID 26675731 (https://pubmed.ncbi.nlm.nih.gov/26675731), *Retraction Watch* (http://retractionwatch.com/2016/03/01/high-profile-critic-slams-nature-letters-about-dinosaur-growth-following-corrections/))**
- Erickson, G.; Makovicky, P.; Currie, P.; Norell, M.A.; Yerby, S.A.; Brochu, C.A. (2004). "Gigantism and comparative life-history parameters of tyrannosaurid dinosaurs" (http://doc.rero.ch/record/15279/files/PAL_E2578.pdf) (PDF). *Nature*. **430** (7001): 772–775. Bibcode:2004Natur.430..772E (https://ui.adsabs.harvard.edu/abs/2004Natur.430..772E). doi:10.1038/nature02699 (https://doi.org/10.1038%2Fnature02699). PMID 15306807 (https://pubmed.ncbi.nlm.nih.gov/15306807). S2CID 4404887 (https://api.semanticscholar.org/CorpusID:4404887). Archived (https://web.archive.org/web/20200714024211/http://doc.rero.ch/record/15279/files/PAL_E2578.pdf) (PDF) from the original on July 14, 2020. **(Erratum: doi:10.1038/nature16487 (https://doi.org/10.1038%2Fnature16487), PMID 26675726 (https://pubmed.ncbi.nlm.nih.gov/26675726), *Retraction Watch* (http://retractionwatch.com/2016/03/01/high-profile-critic-slams-nature-letters-about-dinosaur-growth-following-corrections/))**
- Lehman, T.M.; Woodward, H.N. (2008). "Modeling growth rates for sauropod dinosaurs" (http://doc.rero.ch/record/16723/files/PAL_E3766.pdf) (PDF). *Paleobiology*. **34** (2): 264–281. doi:10.1666/0094-8373(2008)034[0264:MGRFSD]2.0.CO;2 (https://doi.org/10.1666%2F0094-8373%282008%29034%5B0264%3AMGRFSD%5D2.0.CO%3B2). S2CID 84163725 (https://api.semanticscholar.org/CorpusID:84163725). Archived (https://web.archive.org/web/20230204092325/https://doc.rero.ch/record/16723/files/PAL_E3766.pdf) (PDF) from the original on February 4, 2023. Retrieved January 3, 2023.
- Horner, J.R.; de Ricqles, A.; Padian, K.; Scheetz, R.D. (2009). "Comparative long bone histology and growth of the "hypsilophodontid" dinosaurs *Orodromeus makelai*, *Dryosaurus altus*, and *Tenontosaurus tillettii* (Ornithischia: Euornithopoda)". *Journal of Vertebrate Paleontology*. **29** (3): 734–747. Bibcode:2009JVPal..29..734H (https://ui.adsabs.harvard.edu/abs/2009JVPal..29..734H). doi:10.1671/039.029.0312 (https://doi.org/10.1671%2F039.029.0312). S2CID 86277619 (https://api.semanticscholar.org/CorpusID:86277619).
- Woodward, H.; Freedman Fowler, E.; Farlow, J.; Horner, J. (2015). "*Maiasaura*, a model organism for extinct vertebrate population biology: A large sample statistical assessment of growth dynamics and survivorship". *Paleobiology*. **41** (4): 503–527. Bibcode:2015Pbio...41..503W (https://ui.adsabs.harvard.edu/abs/2015Pbio...41..503W). doi:10.1017/pab.2015.19 (https://doi.org/10.1017%2Fpab.2015.19). S2CID 85902880 (https://api.semanticscholar.org/CorpusID:85902880).

217. Amiot, R.; Lécuyer, C.; Buffetaut, E.; Escarguel, G.; Fluteau, F.; Martineau, F. (2006). "Oxygen isotopes from biogenic apatites suggest widespread endothermy in Cretaceous dinosaurs" (http://doc.rero.ch/record/14262/files/PAL_E2159.pdf) (PDF). *Earth and Planetary Science Letters*. **246** (1–2): 41–54. Bibcode:2006E&PSL.246...41A (https://ui.adsabs.harvard.edu/abs/2006E&PSL.246...41A). doi:10.1016/j.epsl.2006.04.018 (https://doi.org/10.1016%2Fj.epsl.2006.04.018).

218. Amiot, R.; Wang, X.; Lécuyer, C.; Buffetaut, E.; Boudad, L.; Cavin, L.; Ding, Z.; Fluteau, F.; Kellner, A.W.A.; Tong, H.; Zhang, F. (2010). "Oxygen and carbon isotope compositions of middle Cretaceous vertebrates from North Africa and Brazil: ecological and environmental significance". *Palaeogeography, Palaeoclimatology, Palaeoecology*. **297** (2): 439–451. Bibcode:2010PPP...297..439A (https://ui.adsabs.harvard.edu/abs/2010PPP...297..439A). doi:10.1016/j.palaeo.2010.08.027 (https://doi.org/10.1016%2Fj.palaeo.2010.08.027).
219. Kolodny, Y.; Luz, B.; Sander, M.; Clemens, W.A. (1996). "Dinosaur bones: fossils or pseudomorphs? The pitfalls of physiology reconstruction from apatitic fossils" (http://doc.rero.ch/record/14443/files/PAL_E1627.pdf) (PDF). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **126** (1–2): 161–171. Bibcode:1996PPP...126..161K (https://ui.adsabs.harvard.edu/abs/1996PPP...126..161K). doi:10.1016/S0031-0182(96)00112-5 (https://doi.org/10.1016%2FS0031-0182%2896%2900112-5). Archived (https://web.archive.org/web/20230204094527/https://doc.rero.ch/record/14443/files/PAL_E1627.pdf) (PDF) from the original on February 4, 2023. Retrieved January 3, 2023.
220. Paul, G.S. (1988). "Physiological, migratorial, climatological, geophysical, survival, and evolutionary implications of Cretaceous polar dinosaurs". *Journal of Paleontology*. **62** (4): 640–652. JSTOR 1305468 (https://www.jstor.org/stable/1305468).
221. Clemens, W.A.; Nelms, L.G. (1993). "Paleoecological implications of Alaskan terrestrial vertebrate fauna in latest Cretaceous time at high paleolatitudes". *Geology*. **21** (6): 503–506. Bibcode:1993Geo....21..503C (https://ui.adsabs.harvard.edu/abs/1993Geo....21..503C). doi:10.1130/0091-7613(1993)021<0503:PIOATV>2.3.CO;2 (https://doi.org/10.1130%2F0091-7613%281993%29021%3C0503%3APIOATV%3E2.3.CO%3B2).
222. Rich, T.H.; Vickers-Rich, P.; Gangloff, R.A. (2002). "Polar dinosaurs". *Science*. **295** (5557): 979–980. doi:10.1126/science.1068920 (https://doi.org/10.1126%2Fscience.1068920). PMID 11834803 (https://pubmed.ncbi.nlm.nih.gov/11834803). S2CID 28065814 (https://api.semanticscholar.org/CorpusID:28065814).
223. Buffetaut, E. (2004). "Polar dinosaurs and the question of dinosaur extinction: a brief review" (http://doc.rero.ch/record/15383/files/PAL_E2734.pdf) (PDF). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **214** (3): 225–231. Bibcode:2004PPP...214..225B (https://ui.adsabs.harvard.edu/abs/2004PPP...214..225B). doi:10.1016/j.palaeo.2004.02.050 (https://doi.org/10.1016%2Fj.palaeo.2004.02.050). Archived (https://web.archive.org/web/20200608120307/http://doc.rero.ch/record/15383/files/PAL_E2734.pdf) (PDF) from the original on June 8, 2020.
224. Sereno, Paul C.; Martinez, Ricardo N.; Wilson, Jeffrey A.; et al. (September 2008). Kemp, Tom (ed.). "Evidence for Avian Intrathoracic Air Sacs in a New Predatory Dinosaur from Argentina" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2553519). *PLOS ONE*. **3** (9) e3303. San Francisco, CA: PLOS. Bibcode:2008PLoSO...3.3303S (https://ui.adsabs.harvard.edu/abs/2008PLoSO...3.3303S). doi:10.1371/journal.pone.0003303 (https://doi.org/10.1371%2Fjournal.pone.0003303). ISSN 1932-6203 (https://search.worldcat.org/issn/1932-6203). PMC 2553519 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2553519). PMID 18825273 (https://pubmed.ncbi.nlm.nih.gov/18825273).
225. O'Connor, P.M. (2009). "Evolution of archosaurian body plans: skeletal adaptations of an air-sac-based breathing apparatus in birds and other archosaurs". *Journal of Experimental Zoology Part A: Ecological Genetics and Physiology*. **311** (8): 629–646. Bibcode:2009JEZA..311..629O (https://ui.adsabs.harvard.edu/abs/2009JEZA..311..629O). doi:10.1002/jez.548 (https://doi.org/10.1002%2Fjez.548). PMID 19492308 (https://pubmed.ncbi.nlm.nih.gov/19492308).
226. Eagle, R.A.; Tütken, T.; Martin, T.S.; Tripati, A.K.; Fricke, H.C.; Connely, M.; Cifelli, R.L.; Eiler, J.M. (2011). "Dinosaur body temperatures determined from isotopic (^{13}C-^{18}O) ordering in fossil biominerals". *Science*. **333** (6041): 443–445. Bibcode:2011Sci...333..443E (https://ui.adsabs.harvard.edu/abs/2011Sci...333..443E). doi:10.1126/science.1206196 (https://doi.org/10.1126%2Fscience.1206196). PMID 21700837 (https://pubmed.ncbi.nlm.nih.gov/21700837). S2CID 206534244 (https://api.semanticscholar.org/CorpusID:206534244).

227. Wedel, M.J. (2003). "Vertebral pneumaticity, air sacs, and the physiology of sauropod dinosaurs" (http://doc.rero.ch/record/14872/files/PAL_E2010.pdf) (PDF). *Paleobiology*. **29** (2): 243–255. Bibcode:2003Pbio...29..243W (https://ui.adsabs.harvard.edu/abs/2003Pbio...29..243W). doi:10.1017/S0094837300018091 (https://doi.org/10.1017%2FS0094837300018091). Archived (https://web.archive.org/web/20230518003816/https://doc.rero.ch/record/14872/files/PAL_E2010.pdf) (PDF) from the original on May 18, 2023. Retrieved January 3, 2024.
228. Perry, S.F.; Christian, A.; Breuer, T.; Pajor, N.; Codd, J.R. (2009). "Implications of an avian-style respiratory system for gigantism in sauropod dinosaurs". *Journal of Experimental Zoology Part A: Ecological Genetics and Physiology*. **311** (8): 600–610. Bibcode:2009JEZA..311..600P (https://ui.adsabs.harvard.edu/abs/2009JEZA..311..600P). doi:10.1002/jez.517 (https://doi.org/10.1002%2Fjez.517). PMID 19189317 (https://pubmed.ncbi.nlm.nih.gov/19189317).
229. Alexander, R.M. (1998). "All-time giants: the largest animals and their problems" (https://www.palass.org/publications/palaeontology-journal/archive/41/6/article_pp1231-1245). *Palaeontology*. **41**: 1231–1245. Archived (https://web.archive.org/web/20160927060250/https://www.palass.org/publications/palaeontology-journal/archive/41/6/article_pp1231-1245) from the original on September 27, 2016. Retrieved January 2, 2021.
230. Tsahar, E.; Martínez del Rio, C.; Izhaki, I.; Arad, Z. (2005). "Can birds be ammonotelic? Nitrogen balance and excretion in two frugivores" (https://jeb.biologists.org/content/jexbio/208/6/1025.full.pdf) (PDF). *The Journal of Experimental Biology*. **208** (6): 1025–1034. Bibcode:2005JExpB.208.1025T (https://ui.adsabs.harvard.edu/abs/2005JExpB.208.1025T). doi:10.1242/jeb.01495 (https://doi.org/10.1242%2Fjeb.01495). ISSN 0022-0949 (https://search.worldcat.org/issn/0022-0949). PMID 15767304 (https://pubmed.ncbi.nlm.nih.gov/15767304). S2CID 18540594 (https://api.semanticscholar.org/CorpusID:18540594). Archived (https://web.archive.org/web/20191017110333/https://jeb.biologists.org/content/jexbio/208/6/1025.full.pdf) (PDF) from the original on October 17, 2019. Retrieved October 31, 2019.
231. Skadhauge, E.; Erlwanger, K.H.; Ruziwa, S.D.; Dantzer, V.; Elbrønd, V.S.; Chamunorwa, J.P. (2003). "Does the ostrich (*Struthio camelus*) coprodeum have the electrophysiological properties and microstructure of other birds?". *Comparative Biochemistry and Physiology Part A: Molecular & Integrative Physiology*. **134** (4): 749–755. doi:10.1016/S1095-6433(03)00006-0 (https://doi.org/10.1016%2FS1095-6433%2803%2900006-0). ISSN 1095-6433 (https://search.worldcat.org/issn/1095-6433). PMID 12814783 (https://pubmed.ncbi.nlm.nih.gov/12814783).
232. Preest, M.R.; Beuchat, C.A. (1997). "Ammonia excretion by hummingbirds". *Nature*. **386** (6625): 561–562. Bibcode:1997Natur.386..561P (https://ui.adsabs.harvard.edu/abs/1997Natur.386..561P). doi:10.1038/386561a0 (https://doi.org/10.1038%2F386561a0). ISSN 0028-0836 (https://search.worldcat.org/issn/0028-0836). S2CID 4372695 (https://api.semanticscholar.org/CorpusID:4372695).
233. Mora, J.; Martuscelli, J.; Ortiz Pineda, J.; Soberon, G. (1965). "The Regulation of Urea-Biosynthesis Enzymes in Vertebrates" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1206904). *Biochemical Journal*. **96** (1): 28–35. doi:10.1042/bj0960028 (https://doi.org/10.1042%2Fbj0960028). ISSN 0264-6021 (https://search.worldcat.org/issn/0264-6021). PMC 1206904 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1206904). PMID 14343146 (https://pubmed.ncbi.nlm.nih.gov/14343146).
234. Packard, G.C. (1966). "The Influence of Ambient Temperature and Aridity on Modes of Reproduction and Excretion of Amniote Vertebrates". *The American Naturalist*. **100** (916): 667–682. Bibcode:1966ANat..100..667P (https://ui.adsabs.harvard.edu/abs/1966ANat..100..667P). doi:10.1086/282459 (https://doi.org/10.1086%2F282459). ISSN 0003-0147 (https://search.worldcat.org/issn/0003-0147). JSTOR 2459303 (https://www.jstor.org/stable/2459303). S2CID 85424175 (https://api.semanticscholar.org/CorpusID:85424175).

235. Balgooyen, T.G. (1971). "Pellet Regurgitation by Captive Sparrow Hawks (*Falco sparverius*)" (https://web.archive.org/web/20190404001957/https://sora.unm.edu/sites/default/files/journals/condor/v073n03/p0382-p0385.pdf) (PDF). *Condor*. **73** (3): 382–385. doi:10.2307/1365774 (https://doi.org/10.2307%2F1365774). JSTOR 1365774 (https://www.jstor.org/stable/1365774). Archived from the original (https://sora.unm.edu/sites/default/files/journals/condor/v073n03/p0382-p0385.pdf) (PDF) on April 4, 2019. Retrieved October 30, 2019.
236. Xu, X.; Li, F.; Wang, Y.; Sullivan, C.; Zhang, F.; Zhang, X.; Sullivan, C.; Wang, X.; Zheng, X. (2018). "Exceptional dinosaur fossils reveal early origin of avian-style digestion" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6155034). *Scientific Reports*. **8** (1): 14217. Bibcode:2018NatSR...814217Z (https://ui.adsabs.harvard.edu/abs/2018NatSR...814217Z). doi:10.1038/s41598-018-32202-x (https://doi.org/10.1038%2Fs41598-018-32202-x). ISSN 2045-2322 (https://search.worldcat.org/issn/2045-2322). PMC 6155034 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6155034). PMID 30242170 (https://pubmed.ncbi.nlm.nih.gov/30242170).
237. Russell, Dale A. (1997). "Intelligence". In Kevin Padian; Philip J. Currie (eds.). *Encyclopedia of dinosaurs*. San Diego: Academic Press. pp. 370–372. ISBN 978-0-12-226810-6.
238. Brusatte 2012, p. 83
239. Huxley, Thomas H. (1868). "On the Animals which are most nearly intermediate between Birds and Reptiles" (https://archive.org/details/cbarchive_51934_ontheanimalswhicharemostnearly1840/page/n1). *The Annals and Magazine of Natural History*. **4** (2). London: Taylor & Francis: 66–75. Retrieved October 31, 2019.
240. Heilmann 1926
241. Osborn, Henry Fairfield (1924). "Three new Theropoda, *Protoceratops* zone, central Mongolia" (https://digitallibrary.amnh.org/dspace/bitstream/2246/3223/1/N0144.pdf) (PDF). *American Museum Novitates* (144). New York: American Museum of Natural History: 1–12. ISSN 0003-0082 (https://search.worldcat.org/issn/0003-0082). Archived (https://web.archive.org/web/20070612091741/http://digitallibrary.amnh.org/dspace/bitstream/2246/3223/1/N0144.pdf) (PDF) from the original on June 12, 2007.
242. Ostrom, John H. (1973). "The ancestry of birds" (https://doi.org/10.1038%2F242136a0). *Nature*. **242** (5393). London: Nature Research: 136. Bibcode:1973NPhS..242..136O (https://ui.adsabs.harvard.edu/abs/1973NPhS..242..136O). doi:10.1038/242136a0 (https://doi.org/10.1038%2F242136a0). ISSN 0028-0836 (https://search.worldcat.org/issn/0028-0836). S2CID 29873831 (https://api.semanticscholar.org/CorpusID:29873831).
243. Padian 1986, pp. 1–55 (https://archive.org/details/cbarchive_48104_saurischianmonophylyandtheorig1868), "Saurischian Monophyly and the Origin of Birds" by Jacques Gauthier.
244. Mayr, Gerald; Pohl, Burkhard; Peters, D. Stefan (2005). "A Well-Preserved *Archaeopteryx* Specimen with Theropod Features" (http://doc.rero.ch/record/15488/files/PAL_E2876.pdf) (PDF). *Science*. **310** (5753). Washington, D.C.: American Association for the Advancement of Science: 1483–1486. Bibcode:2005Sci...310.1483M (https://ui.adsabs.harvard.edu/abs/2005Sci...310.1483M). doi:10.1126/science.1120331 (https://doi.org/10.1126%2Fscience.1120331). ISSN 0036-8075 (https://search.worldcat.org/issn/0036-8075). PMID 16322455 (https://pubmed.ncbi.nlm.nih.gov/16322455). S2CID 28611454 (https://api.semanticscholar.org/CorpusID:28611454). Archived (https://web.archive.org/web/20221225090505/https://doc.rero.ch/record/15488/files/PAL_E2876.pdf) (PDF) from the original on December 25, 2022. Retrieved January 3, 2023.
245. Martin, Larry D. (2006). "A basal archosaurian origin for birds". *Acta Zoologica Sinica*. **50** (6): 977–990. ISSN 1674-5507 (https://search.worldcat.org/issn/1674-5507).

246. Feduccia, Alan (October 1, 2002). "Birds are Dinosaurs: Simple Answer to a Complex Problem" (https://academic.oup.com/auk/article/119/4/1187/5562157). *The Auk*. **119** (4). Washington, D.C.: American Ornithologists' Union: 1187–1201. doi:10.1642/0004-8038(2002)119[1187:BADSAT]2.0.CO;2 (https://doi.org/10.1642%2F0004-8038%282002%29119%5B1187%3ABADSAT%5D2.0.CO%3B2) (inactive June 17, 2026). ISSN 0004-8038 (https://search.worldcat.org/issn/0004-8038). JSTOR 4090252 (https://www.jstor.org/stable/4090252). S2CID 86096746 (https://api.semanticscholar.org/CorpusID:86096746). Archived (https://web.archive.org/web/20200729224702/https://academic.oup.com/auk/article/119/4/1187/5562157) from the original on July 29, 2020. Retrieved November 3, 2019.
247. Switek, Brian (July 2, 2012). "Rise of the fuzzy dinosaurs" (https://www.nature.com/articles/nature.2012.10933). News. *Nature*. London: Nature Research. doi:10.1038/nature.2012.10933 (https://doi.org/10.1038%2Fnature.2012.10933). ISSN 0028-0836 (https://search.worldcat.org/issn/0028-0836). S2CID 123219913 (https://api.semanticscholar.org/CorpusID:123219913). Archived (https://web.archive.org/web/20200413155956/https://www.nature.com/articles/nature.2012.10933) from the original on April 13, 2020. Retrieved January 1, 2019.
248. Godefroit, P.; Sinitsa, S.M.; Dhouailly, D.; Bolotsky, Y.L.; Sizov, A.V.; McNamara, M.E.; Benton, M.J.; Spagna, P. (2014). "A Jurassic ornithischian dinosaur from Siberia with both feathers and scales" (https://web.archive.org/web/20190209232112/http://palaeo.gly.bris.ac.uk/Benton/reprints/2014Kulinda.pdf) (PDF). *Science*. **345** (6195): 451–455. Bibcode:2014Sci...345..451G (https://ui.adsabs.harvard.edu/abs/2014Sci...345..451G). doi:10.1126/science.1253351 (https://doi.org/10.1126%2Fscience.1253351). hdl:1983/a7ae6dfb-55bf-4ca4-bd8b-a5ea5f323103 (https://hdl.handle.net/1983%2Fa7ae6dfb-55bf-4ca4-bd8b-a5ea5f323103). PMID 25061209 (https://pubmed.ncbi.nlm.nih.gov/25061209). S2CID 206556907 (https://api.semanticscholar.org/CorpusID:206556907). Archived from the original (http://palaeo.gly.bris.ac.uk/Benton/reprints/2014Kulinda.pdf) (PDF) on February 9, 2019. Retrieved July 27, 2016.
249. Xu, Xing; Norell, Mark A.; Kuang, Xuewen; et al. (2004). "Basal tyrannosauroids from China and evidence for protofeathers in tyrannosauroids" (http://doc.rero.ch/record/15283/files/PAL_E2582.pdf) (PDF). *Nature*. **431** (7009). London: Nature Research: 680–684. Bibcode:2004Natur.431..680X (https://ui.adsabs.harvard.edu/abs/2004Natur.431..680X). doi:10.1038/nature02855 (https://doi.org/10.1038%2Fnature02855). ISSN 0028-0836 (https://search.worldcat.org/issn/0028-0836). PMID 15470426 (https://pubmed.ncbi.nlm.nih.gov/15470426). S2CID 4381777 (https://api.semanticscholar.org/CorpusID:4381777).
250. Göhlich, Ursula B.; Chiappe, Luis M. (2006). "A new carnivorous dinosaur from the Late Jurassic Solnhofen archipelago" (https://web.archive.org/web/20190426005126/https://doc.rero.ch/record/232914/files/PAL_E4515.pdf) (PDF). *Nature*. **440** (7082). London: Nature Research: 329–332. Bibcode:2006Natur.440..329G (https://ui.adsabs.harvard.edu/abs/2006Natur.440..329G). doi:10.1038/nature04579 (https://doi.org/10.1038%2Fnature04579). ISSN 0028-0836 (https://search.worldcat.org/issn/0028-0836). PMID 16541071 (https://pubmed.ncbi.nlm.nih.gov/16541071). S2CID 4427002 (https://api.semanticscholar.org/CorpusID:4427002). Archived from the original (https://doc.rero.ch/record/232914/files/PAL_E4515.pdf) (PDF) on April 26, 2019. Retrieved November 1, 2019.
251. Kellner, Alexander W. A.; Wang, Xiaolin; Tischlinger, Helmut; et al. (2010). "The soft tissue of *Jeholopterus* (Pterosauria, Anurognathidae, Batrachognathinae) and the structure of the pterosaur wing membrane" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2842671). *Proceedings of the Royal Society B*. **277** (1679). London: Royal Society: 321–329. doi:10.1098/rspb.2009.0846 (https://doi.org/10.1098%2Frspb.2009.0846). ISSN 0962-8452 (https://search.worldcat.org/issn/0962-8452). PMC 2842671 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2842671). PMID 19656798 (https://pubmed.ncbi.nlm.nih.gov/19656798).

252. Mayr, G.; Pittman, M.; Saitta, E.; Kaye, T.G.; Vinther, J. (2016). "Structure and homology of *Psittacosaurus* tail bristles" (https://research-information.bris.ac.uk/en/publications/029c668f-08b9-45f6-a0c5-30ce9256e593). *Palaeontology*. **59** (6): 793–802. Bibcode:2016Palgy..59..793M (https://ui.adsabs.harvard.edu/abs/2016Palgy..59..793M). doi:10.1111/pala.12257 (https://doi.org/10.1111%2Fpala.12257). hdl:1983/029c668f-08b9-45f6-a0c5-30ce9256e593 (https://hdl.handle.net/1983%2F029c668f-08b9-45f6-a0c5-30ce9256e593). S2CID 89156313 (https://api.semanticscholar.org/CorpusID:89156313).
253. Benton, M.J.; Dhouailly, D.; Jiang, B.; McNamara, M. (2019). "The Early Origin of Feathers". *Trends in Ecology & Evolution*. **34** (9): 856–869. Bibcode:2019TEcoE..34..856B (https://ui.adsabs.harvard.edu/abs/2019TEcoE..34..856B). doi:10.1016/j.tree.2019.04.018 (https://doi.org/10.1016%2Fj.tree.2019.04.018). hdl:10468/8068 (https://hdl.handle.net/10468%2F8068). PMID 31164250 (https://pubmed.ncbi.nlm.nih.gov/31164250). S2CID 174811556 (https://api.semanticscholar.org/CorpusID:174811556).
254. Barrett, P.M.; Evans, D.C.; Campione, N.E. (2015). "Evolution of dinosaur epidermal structures" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4528472). *Biology Letters*. **11** (6) 20150229. doi:10.1098/rsbl.2015.0229 (https://doi.org/10.1098%2Frsbl.2015.0229). PMC 4528472 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4528472). PMID 26041865 (https://pubmed.ncbi.nlm.nih.gov/26041865).
255. Alibardi, Lorenzo; Knapp, Loren W.; Sawyer, Roger H. (2006). "Beta-keratin localization in developing alligator scales and feathers in relation to the development and evolution of feathers". *Journal of Submicroscopic Cytology and Pathology*. **38** (2–3). Siena: Nuova Immagine Editrice: 175–192. ISSN 1122-9497 (https://search.worldcat.org/issn/1122-9497). PMID 17784647 (https://pubmed.ncbi.nlm.nih.gov/17784647).
256. Lingham-Soliar, Theagarten (December 2003). "The dinosaurian origin of feathers: perspectives from dolphin (Cetacea) collagen fibers". *Naturwissenschaften*. **90** (12). Berlin: Springer Science+Business Media: 563–567. Bibcode:2003NW.....90..563L (https://ui.adsabs.harvard.edu/abs/2003NW.....90..563L). doi:10.1007/s00114-003-0483-7 (https://doi.org/10.1007%2Fs00114-003-0483-7). ISSN 0028-1042 (https://search.worldcat.org/issn/0028-1042). PMID 14676953 (https://pubmed.ncbi.nlm.nih.gov/14676953). S2CID 43677545 (https://api.semanticscholar.org/CorpusID:43677545).
257. Feduccia, Alan; Lingham-Soliar, Theagarten; Hinchliffe, J. Richard (November 2005). "Do feathered dinosaurs exist? Testing the hypothesis on neontological and paleontological evidence". *Journal of Morphology*. **266** (2). Hoboken, NJ: John Wiley & Sons: 125–166. Bibcode:2005JMorp.266..125F (https://ui.adsabs.harvard.edu/abs/2005JMorp.266..125F). doi:10.1002/jmor.10382 (https://doi.org/10.1002%2Fjmor.10382). ISSN 0362-2525 (https://search.worldcat.org/issn/0362-2525). PMID 16217748 (https://pubmed.ncbi.nlm.nih.gov/16217748). S2CID 15079072 (https://api.semanticscholar.org/CorpusID:15079072).
258. Lingham-Soliar, Theagarten; Feduccia, Alan; Wang, Xiaolin (2007). "A new Chinese specimen indicates that 'protofeathers' in the Early Cretaceous theropod dinosaur *Sinosauropteryx* are degraded collagen fibres" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2270928). *Proceedings of the Royal Society B*. **274** (1620). London: Royal Society: 1823–1829. Bibcode:2007PBioS.274.1823L (https://ui.adsabs.harvard.edu/abs/2007PBioS.274.1823L). doi:10.1098/rspb.2007.0352 (https://doi.org/10.1098%2Frspb.2007.0352). ISSN 0962-8452 (https://search.worldcat.org/issn/0962-8452). PMC 2270928 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2270928). PMID 17521978 (https://pubmed.ncbi.nlm.nih.gov/17521978).
259. Prum, Richard O. (2003). "Are Current Critiques Of The Theropod Origin Of Birds Science? Rebuttal To Feduccia 2002". *The Auk*. **120** (2). Washington, D.C.: American Ornithologists' Union: 550–561. doi:10.1642/0004-8038(2003)120[0550:ACCOTT]2.0.CO;2 (https://doi.org/10.1642%2F0004-8038%282003%29120%5B0550%3AACCOTT%5D2.0.CO%3B2) (inactive June 17, 2026). ISSN 0004-8038 (https://search.worldcat.org/issn/0004-8038). JSTOR 4090212 (https://www.jstor.org/stable/4090212).

260. Wellnhofer, Peter (1988). "A New Specimen of *Archaeopteryx*". *Science*. **240** (4860). Washington, D.C.: American Association for the Advancement of Science: 1790–1792. Bibcode:1988Sci...240.1790W (https://ui.adsabs.harvard.edu/abs/1988Sci...240.1790W). doi:10.1126/science.240.4860.1790 (https://doi.org/10.1126%2Fscience.240.4860.1790). ISSN 0036-8075 (https://search.worldcat.org/issn/0036-8075). JSTOR 1701652 (https://www.jstor.org/stable/1701652). PMID 17842432 (https://pubmed.ncbi.nlm.nih.gov/17842432). S2CID 32015255 (https://api.semanticscholar.org/CorpusID:32015255).
    - —— (1988). "Ein neuer Exemplar von *Archaeopteryx*". *Archaeopteryx*. **6**: 1–30.
261. Schweitzer, Mary H.; Watt, J.A.; Avci, R.; et al. (1999). "Beta-keratin specific immunological reactivity in feather-like structures of the Cretaceous Alvarezsaurid, *Shuvuuia deserti*". *Journal of Experimental Zoology Part B*. **285** (2). Hoboken, NJ: Wiley-Blackwell: 146–157. Bibcode:1999JEZ...285..146S (https://ui.adsabs.harvard.edu/abs/1999JEZ...285..146S). doi:10.1002/(SICI)1097-010X(19990815)285:2<146::AID-JEZ7>3.0.CO;2-A (https://doi.org/10.1002%2F%28SICI%291097-010X%2819990815%29285%3A2%3C146%3A%3AAID-JEZ7%3E3.0.CO%3B2-A). ISSN 1552-5007 (https://search.worldcat.org/issn/1552-5007). PMID 10440726 (https://pubmed.ncbi.nlm.nih.gov/10440726).
262. "*Archaeopteryx*: An Early Bird" (https://ucmp.berkeley.edu/diapsids/birds/archaeopteryx.html). Berkeley: University of California Museum of Paleontology. Archived (https://web.archive.org/web/20210203232245/https://ucmp.berkeley.edu/diapsids/birds/archaeopteryx.html) from the original on February 3, 2021. Retrieved October 30, 2019.
263. O'Connor, Patrick M.; Claessens, Leon P. A. M. (2005). "Basic avian pulmonary design and flow-through ventilation in non-avian theropod dinosaurs". *Nature*. **436** (7048). London: Nature Research: 253–256. Bibcode:2005Natur.436..253O (https://ui.adsabs.harvard.edu/abs/2005Natur.436..253O). doi:10.1038/nature03716 (https://doi.org/10.1038%2Fnature03716). ISSN 0028-0836 (https://search.worldcat.org/issn/0028-0836). PMID 16015329 (https://pubmed.ncbi.nlm.nih.gov/16015329). S2CID 4390587 (https://api.semanticscholar.org/CorpusID:4390587).
264. O'Connor, Patrick M.; Claessens, Leon P. A. M. (July 2005). "Basic avian pulmonary design and flow-through ventilation in non-avian theropod dinosaurs" (https://www.nature.com/articles/nature03716). *Nature*. **436** (7048): 253–256. Bibcode:2005Natur.436..253O (https://ui.adsabs.harvard.edu/abs/2005Natur.436..253O). doi:10.1038/nature03716 (https://doi.org/10.1038%2Fnature03716). ISSN 0028-0836 (https://search.worldcat.org/issn/0028-0836). PMID 16015329 (https://pubmed.ncbi.nlm.nih.gov/16015329). Archived (https://web.archive.org/web/20241225213025/https://www.nature.com/articles/nature03716) from the original on December 25, 2024. Retrieved December 25, 2024.
265. "Meat-eating dinosaur from Argentina had bird-like breathing system" (https://news.umich.edu/meat-eating-dinosaur-from-argentina-had-bird-like-breathing-system/). *University of Michigan News*. Ann Arbor, MI: Office of the Vice President for Communications; Regents of the University of Michigan. October 2, 2008. Archived (https://web.archive.org/web/20200730014852/https://news.umich.edu/meat-eating-dinosaur-from-argentina-had-bird-like-breathing-system/) from the original on July 30, 2020. Retrieved November 2, 2019.
266. Xu, Xing; Norell, Mark A. (2004). "A new troodontid dinosaur from China with avian-like sleeping posture" (http://doc.rero.ch/record/15284/files/PAL_E2583.pdf) (PDF). *Nature*. **431** (7010). London: Nature Research: 838–841. Bibcode:2004Natur.431..838X (https://ui.adsabs.harvard.edu/abs/2004Natur.431..838X). doi:10.1038/nature02898 (https://doi.org/10.1038%2Fnature02898). ISSN 0028-0836 (https://search.worldcat.org/issn/0028-0836). PMID 15483610 (https://pubmed.ncbi.nlm.nih.gov/15483610). S2CID 4362745 (https://api.semanticscholar.org/CorpusID:4362745).
267. Norell, Mark A.; Clark, James M.; Chiappe, Luis M.; et al. (1995). "A nesting dinosaur". *Nature*. **378** (6559). London: Nature Research: 774–776. Bibcode:1995Natur.378..774N (https://ui.adsabs.harvard.edu/abs/1995Natur.378..774N). doi:10.1038/378774a0 (https://doi.org/10.1038%2F378774a0). ISSN 0028-0836 (https://search.worldcat.org/issn/0028-0836). S2CID 4245228 (https://api.semanticscholar.org/CorpusID:4245228).

268. Varricchio, David J.; Moore, Jason R.; Erickson, Gregory M.; et al. (2008). "Avian Paternal Care Had Dinosaur Origin" (https://doi.org/10.1126%2Fscience.1163245). *Science*. **322** (5909). Washington, D.C.: American Association for the Advancement of Science: 1826–1828. Bibcode:2008Sci...322.1826V (https://ui.adsabs.harvard.edu/abs/2008Sci...322.1826V). doi:10.1126/science.1163245 (https://doi.org/10.1126%2Fscience.1163245). ISSN 0036-8075 (https://search.worldcat.org/issn/0036-8075). PMID 19095938 (https://pubmed.ncbi.nlm.nih.gov/19095938). S2CID 8718747 (https://api.semanticscholar.org/CorpusID:8718747).
269. Wings, Oliver (2007). "A review of gastrolith function with implications for fossil vertebrates and a revised classification" (https://www.app.pan.pl/archive/published/app52/app52-001.pdf) (PDF). *Palaeontologica Polonica*. **52** (1). Warsaw: Institute of Paleobiology, Polish Academy of Sciences: 1–16. ISSN 0567-7920 (https://search.worldcat.org/issn/0567-7920). Archived (https://web.archive.org/web/20081217091538/http://app.pan.pl/archive/published/app52/app52-001.pdf) (PDF) from the original on December 17, 2008. Retrieved November 2, 2019.
270. Longrich, N.R.; Tokaryk, T.; Field, D.J. (2011). "Mass extinction of birds at the Cretaceous–Paleogene (K–Pg) boundary" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3174646). *Proceedings of the National Academy of Sciences*. **108** (37): 15253–15257. Bibcode:2011PNAS..10815253L (https://ui.adsabs.harvard.edu/abs/2011PNAS..10815253L). doi:10.1073/pnas.1110395108 (https://doi.org/10.1073%2Fpnas.1110395108). PMC 3174646 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3174646). PMID 21914849 (https://pubmed.ncbi.nlm.nih.gov/21914849).
271. Renne, P.R.; Deino, A.L.; Hilgen, F.J.; Kuiper, K.F.; Mark, D.F.; Mitchell, W.S.; Morgan, L.E.; Mundil, R.; Smit, J. (2013). "Time scales of critical events around the Cretaceous-Paleogene boundary". *Science*. **339** (6120): 684–687. Bibcode:2013Sci...339..684R (https://ui.adsabs.harvard.edu/abs/2013Sci...339..684R). doi:10.1126/science.1230492 (https://doi.org/10.1126%2Fscience.1230492). PMID 23393261 (https://pubmed.ncbi.nlm.nih.gov/23393261). S2CID 6112274 (https://api.semanticscholar.org/CorpusID:6112274).
272. Brusatte, S.L.; Butler, R.J.; Barrett, P.M.; Carrano, M.T.; Evans, D.C.; Lloyd, G.T.; Mannion, P.D.; Norell, M.A.; Peppe, D.J.; Upchurch, P.; Williamson, T.E. (2015). "The extinction of the dinosaurs" (https://www.research.ed.ac.uk/en/publications/176e5907-26ec-4959-867f-0f2e52335f88). *Biological Reviews*. **90** (2): 628–642. doi:10.1111/brv.12128 (https://doi.org/10.1111%2Fbrv.12128). hdl:20.500.11820/176e5907-26ec-4959-867f-0f2e52335f88 (https://hdl.handle.net/20.500.11820%2F176e5907-26ec-4959-867f-0f2e52335f88). PMID 25065505 (https://pubmed.ncbi.nlm.nih.gov/25065505). S2CID 115134484 (https://api.semanticscholar.org/CorpusID:115134484).
273. Archibald, J.D.; Clemens, W.A. (1982). "Late Cretaceous Extinctions". *American Scientist*. **70** (4): 377–385. Bibcode:1982AmSci..70..377A (https://ui.adsabs.harvard.edu/abs/1982AmSci..70..377A). JSTOR 27851545 (https://www.jstor.org/stable/27851545).
274. Jablonski, D. (1991). "Extinctions: a paleontological perspective". *Science*. **253** (5021): 754–757. Bibcode:1991Sci...253..754J (https://ui.adsabs.harvard.edu/abs/1991Sci...253..754J). doi:10.1126/science.253.5021.754 (https://doi.org/10.1126%2Fscience.253.5021.754). PMID 17835491 (https://pubmed.ncbi.nlm.nih.gov/17835491).
275. Longrich, N.R.; Bhullar, B.-A. S.; Gauthier, J.A. (2012). "Mass extinction of lizards and snakes at the Cretaceous–Paleogene boundary" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3535637). *Proceedings of the National Academy of Sciences*. **109** (52): 21396–21401. Bibcode:2012PNAS..10921396L (https://ui.adsabs.harvard.edu/abs/2012PNAS..10921396L). doi:10.1073/pnas.1211526110 (https://doi.org/10.1073%2Fpnas.1211526110). ISSN 0027-8424 (https://search.worldcat.org/issn/0027-8424). PMC 3535637 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3535637). PMID 23236177 (https://pubmed.ncbi.nlm.nih.gov/23236177).

276. Field, D.J.; Bercovici, A.; Berv, J.S.; Dunn, R.; Fastovsky, D.E.; Lyson, T.R.; Vajda, V.; Gauthier, J.A. (2018). "Early evolution of modern birds structured by global forest collapse at the end-Cretaceous mass extinction" (https://doi.org/10.1016%2Fj.cub.2018.04.062). *Current Biology*. **28** (11): 1825–1831. Bibcode:2018CBio...28E1825F (https://ui.adsabs.harvard.edu/abs/2018CBio...28E1825F). doi:10.1016/j.cub.2018.04.062 (https://doi.org/10.1016%2Fj.cub.2018.04.062). PMID 29804807 (https://pubmed.ncbi.nlm.nih.gov/29804807). S2CID 44075214 (https://api.semanticscholar.org/CorpusID:44075214).
277. Larson, D.W.; Brown, C.M.; Evans, D.C. (2016). "Dental disparity and ecological stability in bird-like dinosaurs prior to the end-Cretaceous mass extinction" (https://doi.org/10.1016%2Fj.cub.2016.03.039). *Current Biology*. **26** (10): 1325–1333. Bibcode:2016CBio...26.1325L (https://ui.adsabs.harvard.edu/abs/2016CBio...26.1325L). doi:10.1016/j.cub.2016.03.039 (https://doi.org/10.1016%2Fj.cub.2016.03.039). PMID 27112293 (https://pubmed.ncbi.nlm.nih.gov/27112293). S2CID 3937001 (https://api.semanticscholar.org/CorpusID:3937001).
278. Le Loeuff, J. (2012). "Paleobiogeography and biodiversity of Late Maastrichtian dinosaurs: how many dinosaur species went extinct at the Cretaceous-Tertiary boundary?". *Bulletin de la Société Géologique de France*. **183** (6): 547–559. doi:10.2113/gssgfbull.183.6.547 (https://doi.org/10.2113%2Fgssgfbull.183.6.547). ISSN 0037-9409 (https://search.worldcat.org/issn/0037-9409).
279. Carpenter, K. (1983). "Evidence suggesting gradual extinction of latest Cretaceous dinosaurs". *Naturwissenschaften*. **70** (12): 611–612. Bibcode:1983NW.....70..611C (https://ui.adsabs.harvard.edu/abs/1983NW.....70..611C). doi:10.1007/BF00377404 (https://doi.org/10.1007%2FBF00377404). S2CID 20078285 (https://api.semanticscholar.org/CorpusID:20078285).
280. Russell, D.A. (1984). "The gradual decline of the dinosaurs—fact or fallacy?". *Nature*. **307** (5949): 360–361. Bibcode:1984Natur.307..360R (https://ui.adsabs.harvard.edu/abs/1984Natur.307..360R). doi:10.1038/307360a0 (https://doi.org/10.1038%2F307360a0). S2CID 4269426 (https://api.semanticscholar.org/CorpusID:4269426).
281. Fastovsky, D.E.; Huang, Y.; Hsu, J.; Martin-McNaughton, J.; Sheehan, P.M.; Weishampel, D.B. (2004). "Shape of Mesozoic dinosaur richness" (http://doc.rero.ch/record/14984/files/PAL_E2134.pdf) (PDF). *Geology*. **32** (10): 877–880. Bibcode:2004Geo....32..877F (https://ui.adsabs.harvard.edu/abs/2004Geo....32..877F). doi:10.1130/G20695.1 (https://doi.org/10.1130%2FG20695.1). Archived (https://web.archive.org/web/20230204093851/https://doc.rero.ch/record/14984/files/PAL_E2134.pdf) (PDF) from the original on February 4, 2023. Retrieved January 26, 2023.
282. Sullivan, R.M. (2006). "The shape of Mesozoic dinosaur richness: a reassessment" (https://books.google.com/books?id=rCDYCQAAQBAJ&pg=PA403). In Lucas, S.G.; Sullivan, R.M. (eds.). *Late Cretaceous vertebrates from the Western Interior*. New Mexico Museum of Natural History and Science Bulletin. Vol. 35. pp. 403–405.
283. Chiarenza, A.A.; Mannion, P.D.; Lunt, D.J.; Farnsworth, A.; Jones, L.A.; Kelland, S.J.; Allison, P.A. (2019). "Ecological niche modelling does not support climatically-driven dinosaur diversity decline before the Cretaceous/Paleogene mass extinction" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6403247). *Nature Communications*. **10** (1): 1–14. Bibcode:2019NatCo..10.1091C (https://ui.adsabs.harvard.edu/abs/2019NatCo..10.1091C). doi:10.1038/s41467-019-08997-2 (https://doi.org/10.1038%2Fs41467-019-08997-2). PMC 6403247 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6403247). PMID 30842410 (https://pubmed.ncbi.nlm.nih.gov/30842410).
284. Lloyd, G.T. (2012). "A refined modelling approach to assess the influence of sampling on palaeobiodiversity curves: new support for declining Cretaceous dinosaur richness" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3259943). *Biology Letters*. **8** (1): 123–126. doi:10.1098/rsbl.2011.0210 (https://doi.org/10.1098%2Frsbl.2011.0210). PMC 3259943 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3259943). PMID 21508029 (https://pubmed.ncbi.nlm.nih.gov/21508029). S2CID 1376734 (https://api.semanticscholar.org/CorpusID:1376734).

285. Sakamoto, M.; Benton, M.J.; Venditti, C. (2016). "Dinosaurs in decline tens of millions of years before their final extinction" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4983840). *Proceedings of the National Academy of Sciences*. **113** (18): 5036–5040. Bibcode:2016PNAS..113.5036S (https://ui.adsabs.harvard.edu/abs/2016PNAS..113.5036S). doi:10.1073/pnas.1521478113 (https://doi.org/10.1073%2Fpnas.1521478113). PMC 4983840 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4983840). PMID 27092007 (https://pubmed.ncbi.nlm.nih.gov/27092007).
286. Barrett, P.M.; McGowan, A.J.; Page, V. (2009). "Dinosaur diversity and the rock record" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2686664). *Proceedings of the Royal Society B: Biological Sciences*. **276** (1667): 2667–2674. Bibcode:2009RSPSB.276.2667B (https://ui.adsabs.harvard.edu/abs/2009RSPSB.276.2667B). doi:10.1098/rspb.2009.0352 (https://doi.org/10.1098%2Frspb.2009.0352). PMC 2686664 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2686664). PMID 19403535 (https://pubmed.ncbi.nlm.nih.gov/19403535).
287. Upchurch, P.; Mannion, P.D.; Benson, R.B.; Butler, R.J.; Carrano, M.T. (2011). "Geological and anthropogenic controls on the sampling of the terrestrial fossil record: a case study from the Dinosauria". *Geological Society, London, Special Publications*. **358** (1): 209–240. Bibcode:2011GSLSP.358..209U (https://ui.adsabs.harvard.edu/abs/2011GSLSP.358..209U). doi:10.1144/SP358.14 (https://doi.org/10.1144%2FSP358.14). S2CID 130777837 (https://api.semanticscholar.org/CorpusID:130777837).
288. Randall 2015
289. Alvarez, L.W.; Alvarez, W.; Asaro, F.; Michel, H.V. (1980). "Extraterrestrial Cause for the Cretaceous-Tertiary Extinction" (https://web.archive.org/web/20100708202457/http://chaos.swarthmore.edu/courses/soc26/bak-sneppan/13_alverez.pdf) (PDF). *Science*. **208** (4448): 1095–1108. Bibcode:1980Sci...208.1095A (https://ui.adsabs.harvard.edu/abs/1980Sci...208.1095A). doi:10.1126/science.208.4448.1095 (https://doi.org/10.1126%2Fscience.208.4448.1095). ISSN 0036-8075 (https://search.worldcat.org/issn/0036-8075). PMID 17783054 (https://pubmed.ncbi.nlm.nih.gov/17783054). S2CID 16017767 (https://api.semanticscholar.org/CorpusID:16017767). Archived from the original (http://chaos.swarthmore.edu/courses/soc26/bak-sneppan/13_alverez.pdf) (PDF) on July 8, 2010. Retrieved October 30, 2019.
290. Bohor, B.F.; Modreski, P.J.; Foord, E.E. (1987). "Shocked quartz in the Cretaceous-Tertiary boundary clays: Evidence for a global distribution" (https://zenodo.org/record/1230978). *Science*. **236** (4802): 705–709. Bibcode:1987Sci...236..705B (https://ui.adsabs.harvard.edu/abs/1987Sci...236..705B). doi:10.1126/science.236.4802.705 (https://doi.org/10.1126%2Fscience.236.4802.705). PMID 17748309 (https://pubmed.ncbi.nlm.nih.gov/17748309). S2CID 31383614 (https://api.semanticscholar.org/CorpusID:31383614). Archived (https://web.archive.org/web/20210226190627/https://zenodo.org/record/1230978) from the original on February 26, 2021. Retrieved February 8, 2021.
291. Hildebrand, A.R.; Penfield, G.T.; Kring, D.A.; Pilkington, M.; Camargo, Z.A.; Jacobsen, S.B.; Boynton, W.V. (1991). "Chicxulub crater: a possible Cretaceous/Tertiary boundary impact crater on the Yucatan Peninsula, Mexico". *Geology*. **19** (9): 867–871. Bibcode:1991Geo....19..867H (https://ui.adsabs.harvard.edu/abs/1991Geo....19..867H). doi:10.1130/0091-7613(1991)019<0867:CCAPCT>2.3.CO;2 (https://doi.org/10.1130%2F0091-7613%281991%29019%3C0867%3ACCAPCT%3E2.3.CO%3B2).
292. Pope, K.O.; Ocampo, A.C.; Kinsland, G.L.; et al. (1996). "Surface expression of the Chicxulub crater". *Geology*. **24** (6). Boulder, CO: Geological Society of America: 527–530. Bibcode:1996Geo....24..527P (https://ui.adsabs.harvard.edu/abs/1996Geo....24..527P). doi:10.1130/0091-7613(1996)024<0527:SEOTCC>2.3.CO;2 (https://doi.org/10.1130%2F0091-7613%281996%29024%3C0527%3ASEOTCC%3E2.3.CO%3B2). ISSN 0091-7613 (https://search.worldcat.org/issn/0091-7613). PMID 11539331 (https://pubmed.ncbi.nlm.nih.gov/11539331).

293. Schulte, P.; Alegret, L.; Arenillas, I.; Arz, J.A.; Barton, P.J.; Bown, P.R.; Bralower, T.J.; Christeson, G.L.; Claeys, P.; Cockell, C.S.; Collins, G.S.; Deutsch, A.; Goldin, T.J.; Goto, K.; Grajales-Nishimura, J.M.; Grieve, R.A.F.; Gulick, S.P.S.; Johnson, K.R.; Kiessling, W.; Koeberl, C.; Kring, D.A.; MacLeod, K.G.; Matsui, T.; Melosh, J.; Montanari, A.; Morgan, J.V.; Neal, C.R.; Nichols, D.J.; Norris, R.D.; Pierazzo, E.; Ravizza, G.; Rebolledo-Vieyra, M.; Uwe Reimold, W.; Robin, E.; Salge, T.; Speijer, R.P.; Sweet, A.R.; Urrutia-Fucugauchi, J.; Vajda, V.; Whalen, M.T.; Willumsen, P.S. (2010). "The Chicxulub asteroid impact and mass extinction at the Cretaceous-Paleogene boundary" (https://lirias.kuleuven.be/handle/123456789/264213). *Science*. **327** (5970): 1214–1218. Bibcode:2010Sci...327.1214S (https://ui.adsabs.harvard.edu/abs/2010Sci...327.1214S). doi:10.1126/science.1177265 (https://doi.org/10.1126%2Fscience.1177265). PMID 20203042 (https://pubmed.ncbi.nlm.nih.gov/20203042). S2CID 2659741 (https://api.semanticscholar.org/CorpusID:2659741).

294. Kring, D. A. (2007). "The Chicxulub impact event and its environmental consequences at the Cretaceous–Tertiary boundary". *Palaeogeography, Palaeoclimatology, Palaeoecology*. **255** (1–2): 4–21. Bibcode:2007PPP...255....4K (https://ui.adsabs.harvard.edu/abs/2007PPP...255....4K). doi:10.1016/j.palaeo.2007.02.037 (https://doi.org/10.1016%2Fj.palaeo.2007.02.037).

295. Chiarenza, A.A.; Farnsworth, A.; Mannion, P.D.; Lunt, D.J.; Valdes, P.J.; Morgan, J.V.; Allison, P.A. (2020). "Asteroid impact, not volcanism, caused the end-Cretaceous dinosaur extinction" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7382232). *Proceedings of the National Academy of Sciences*. **117** (29): 17084–17093. Bibcode:2020PNAS..11717084C (https://ui.adsabs.harvard.edu/abs/2020PNAS..11717084C). doi:10.1073/pnas.2006087117 (https://doi.org/10.1073%2Fpnas.2006087117). PMC 7382232 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7382232). PMID 32601204 (https://pubmed.ncbi.nlm.nih.gov/32601204).

296. Ivanov, B.A. (2005). "Numerical Modeling of the Largest Terrestrial Meteorite Craters". *Solar System Research*. **39** (5): 381–409. Bibcode:2005SoSyR..39..381I (https://ui.adsabs.harvard.edu/abs/2005SoSyR..39..381I). doi:10.1007/s11208-005-0051-0 (https://doi.org/10.1007%2Fs11208-005-0051-0). S2CID 120305483 (https://api.semanticscholar.org/CorpusID:120305483).

297. Matsui, T.; Imamura, F.; Tajika, E.; Nakano, Y.; Fujisawa, Y. (2002). "Generation and propagation of a tsunami from the Cretaceous-Tertiary impact event". *Geological Society of America Special Papers*. **356**: 69–78. doi:10.1130/0-8137-2356-6.69 (https://doi.org/10.1130%2F0-8137-2356-6.69). ISBN 978-0-8137-2356-3.

298. Robertson, D.S.; McKenna, M.C.; Toon, O.B.; et al. (2004). "Survival in the first hours of the Cenozoic" (https://web.archive.org/web/20120918141759/http://webh01.ua.ac.be/funmorph/raoul/macroevolutie/Robertson2004.pdf) (PDF). *Geological Society of America Bulletin*. **116** (5–6): 760–768. Bibcode:2004GSAB..116..760R (https://ui.adsabs.harvard.edu/abs/2004GSAB..116..760R). doi:10.1130/B25402.1 (https://doi.org/10.1130%2FB25402.1). ISSN 0016-7606 (https://search.worldcat.org/issn/0016-7606). Archived from the original (http://webh01.ua.ac.be/funmorph/raoul/macroevolutie/Robertson2004.pdf) (PDF) on September 18, 2012. Retrieved June 15, 2011.

299. Robertson, D.S.; Lewis, W.M.; Sheehan, P.M.; Toon, O.B. (2013). "K-Pg extinction: Reevaluation of the heat-fire hypothesis" (https://doi.org/10.1002%2Fjgrg.20018). *Journal of Geophysical Research: Biogeosciences*. **118** (1): 329–336. Bibcode:2013JGRG..118..329R (https://ui.adsabs.harvard.edu/abs/2013JGRG..118..329R). doi:10.1002/jgrg.20018 (https://doi.org/10.1002%2Fjgrg.20018). S2CID 17015462 (https://api.semanticscholar.org/CorpusID:17015462).

300. Pope, K.O.; Baines, K.H.; Ocampo, A.C.; Ivanov, B.A. (1997). "Energy, volatile production, and climatic effects of the Chicxulub Cretaceous/Tertiary impact" (https://doi.org/10.1029%2F97JE01743). *Journal of Geophysical Research: Planets*. **102** (E9): 21645–21664. Bibcode:1997JGR...10221645P (https://ui.adsabs.harvard.edu/abs/1997JGR...10221645P). doi:10.1029/97JE01743 (https://doi.org/10.1029%2F97JE01743). PMID 11541145 (https://pubmed.ncbi.nlm.nih.gov/11541145). S2CID 8447773 (https://api.semanticscholar.org/CorpusID:8447773).
301. Ohno, S.; Kadono, T.; Kurosawa, K.; Hamura, T.; Sakaiya, T.; Shigemori, K.; Hironaka, Y.; Sano, T.; Watari, T.; Otani, K.; Matsui, T.; Sugita, S. (2014). "Production of sulphate-rich vapour during the Chicxulub impact and implications for ocean acidification". *Nature Geoscience*. **7** (4): 279–282. Bibcode:2014NatGe...7..279O (https://ui.adsabs.harvard.edu/abs/2014NatGe...7..279O). doi:10.1038/ngeo2095 (https://doi.org/10.1038%2Fngeo2095).
302. Kaiho, K.; Oshima, N.; Adachi, K.; Adachi, Y.; Mizukami, T.; Fujibayashi, M.; Saito, R. (2016). "Global climate change driven by soot at the K-Pg boundary as the cause of the mass extinction" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4944614). *Scientific Reports*. **6** (1): 1–13. Bibcode:2016NatSR...628427K (https://ui.adsabs.harvard.edu/abs/2016NatSR...628427K). doi:10.1038/srep28427 (https://doi.org/10.1038%2Fsrep28427). PMC 4944614 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4944614). PMID 27414998 (https://pubmed.ncbi.nlm.nih.gov/27414998).
303. Lyons, S.L.; Karp, A.T.; Bralower, T.J.; Grice, K.; Schaefer, B.; Gulick, S.P.; Morgan, J.V.; Freeman, K.H. (2020). "Organic matter from the Chicxulub crater exacerbated the K–Pg impact winter" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7568312). *Proceedings of the National Academy of Sciences*. **117** (41): 25327–25334. Bibcode:2020PNAS..11725327L (https://ui.adsabs.harvard.edu/abs/2020PNAS..11725327L). doi:10.1073/pnas.2004596117 (https://doi.org/10.1073%2Fpnas.2004596117). PMC 7568312 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7568312). PMID 32989138 (https://pubmed.ncbi.nlm.nih.gov/32989138).
304. Chenet, A.L.; Courtillot, V.; Fluteau, F.; Gérard, M.; Quidelleur, X.; Khadri, S.F.R.; Subbarao, K.V.; Thordarson, T. (2009). "Determination of rapid Deccan eruptions across the Cretaceous-Tertiary boundary using paleomagnetic secular variation: 2. Constraints from analysis of eight new sections and synthesis for a 3500-m-thick composite section" (https://hal.archives-ouvertes.fr/hal-00406544/file/2008JB005644.pdf) (PDF). *Journal of Geophysical Research: Solid Earth*. **114** (B6): B06103. Bibcode:2009JGRB..114.6103C (https://ui.adsabs.harvard.edu/abs/2009JGRB..114.6103C). doi:10.1029/2008JB005644 (https://doi.org/10.1029%2F2008JB005644). S2CID 140541003 (https://api.semanticscholar.org/CorpusID:140541003).
305. Schoene, B.; Eddy, M.P.; Samperton, K.M.; Keller, C.B.; Keller, G.; Adatte, T.; Khadri, S.F. (2019). "U-Pb constraints on pulsed eruption of the Deccan Traps across the end-Cretaceous mass extinction" (https://doi.org/10.1126%2Fscience.aau2422). *Science*. **363** (6429): 862–866. Bibcode:2019Sci...363..862S (https://ui.adsabs.harvard.edu/abs/2019Sci...363..862S). doi:10.1126/science.aau2422 (https://doi.org/10.1126%2Fscience.aau2422). OSTI 1497969 (https://www.osti.gov/biblio/1497969). PMID 30792300 (https://pubmed.ncbi.nlm.nih.gov/30792300). S2CID 67876950 (https://api.semanticscholar.org/CorpusID:67876950).
306. McLean, D.M. (1985). "Deccan Traps mantle degassing in the terminal Cretaceous marine extinctions". *Cretaceous Research*. **6** (3): 235–259. Bibcode:1985CrRes...6..235M (https://ui.adsabs.harvard.edu/abs/1985CrRes...6..235M). doi:10.1016/0195-6671(85)90048-5 (https://doi.org/10.1016%2F0195-6671%2885%2990048-5).
307. Self, S.; Widdowson, M.; Thordarson, T.; Jay, A.E. (2006). "Volatile fluxes during flood basalt eruptions and potential effects on the global environment: A Deccan perspective" (https://hull-repository.worktribe.com/output/3566573). *Earth and Planetary Science Letters*. **248** (1–2): 518–532. Bibcode:2006E&PSL.248..518S (https://ui.adsabs.harvard.edu/abs/2006E&PSL.248..518S). doi:10.1016/j.epsl.2006.05.041 (https://doi.org/10.1016%2Fj.epsl.2006.05.041).

308. Tobin, T.S.; Bitz, C.M.; Archer, D. (2017). "Modeling climatic effects of carbon dioxide emissions from Deccan Traps volcanic eruptions around the Cretaceous–Paleogene boundary". *Palaeogeography, Palaeoclimatology, Palaeoecology*. **478**: 139–148. Bibcode:2017PPP...478..139T (https://ui.adsabs.harvard.edu/abs/2017PPP...478..139T). doi:10.1016/j.palaeo.2016.05.028 (https://doi.org/10.1016%2Fj.palaeo.2016.05.028).
309. Schmidt, A.; Skeffington, R.A.; Thordarson, T.; Self, S.; Forster, P.M.; Rap, A.; Ridgwell, A.; Fowler, D.; Wilson, M.; Mann, G.W.; Wignall, P.B.; Carslaw, K.S. (2016). "Selective environmental stress from sulphur emitted by continental flood basalt eruptions" (http://eprints.whiterose.ac.uk/92239/1/Schmidt_et_al_NGS_accepted.pdf) (PDF). *Nature Geoscience*. **9** (1): 77–82. Bibcode:2016NatGe...9...77S (https://ui.adsabs.harvard.edu/abs/2016NatGe...9...77S). doi:10.1038/ngeo2588 (https://doi.org/10.1038%2Fngeo2588). S2CID 59518452 (https://api.semanticscholar.org/CorpusID:59518452). Archived (https://web.archive.org/web/20170922012438/http://eprints.whiterose.ac.uk/92239/1/Schmidt_et_al_NGS_accepted.pdf) (PDF) from the original on September 22, 2017.
310. Hofman, C.; Féraud, G.; Courtillot, V. (2000). "$^{40}Ar/^{39}Ar$ dating of mineral separates and whole rocks from the Western Ghats lava pile: further constraints on duration and age of the Deccan traps". *Earth and Planetary Science Letters*. **180** (1–2): 13–27. Bibcode:2000E&PSL.180...13H (https://ui.adsabs.harvard.edu/abs/2000E&PSL.180...13H). doi:10.1016/S0012-821X(00)00159-X (https://doi.org/10.1016%2FS0012-821X%2800%2900159-X). ISSN 0012-821X (https://search.worldcat.org/issn/0012-821X).
311. Sahni, A. (1988). "Cretaceous-Tertiary boundary events: Mass extinctions, iridium enrichment and Deccan volcanism". *Current Science*. **57** (10): 513–519. JSTOR 24090754 (https://www.jstor.org/stable/24090754).
312. Glasby, G.P.; Kunzendorf, H. (1996). "Multiple factors in the origin of the Cretaceous/Tertiary boundary: the role of environmental stress and Deccan Trap volcanism". *Geologische Rundschau*. **85** (2): 191–210. Bibcode:1996IJEaS..85..191G (https://ui.adsabs.harvard.edu/abs/1996IJEaS..85..191G). doi:10.1007/BF02422228 (https://doi.org/10.1007%2FBF02422228). PMID 11543126 (https://pubmed.ncbi.nlm.nih.gov/11543126). S2CID 19155384 (https://api.semanticscholar.org/CorpusID:19155384).
313. Alvarez, L.W. (1987). Mass Extinctions Caused by Large Bolide Impacts (https://www.osti.gov/servlets/purl/875729) (Report). Lawrence Berkeley Laboratory. p. 39. LBL-22786. Retrieved January 27, 2021.
314. Alvarez 1997, pp. 130–146 (https://archive.org/details/trexcraterofdoo000alva/page/130), chpt. 7: "The World after Chicxulub".
315. Renne, P.R.; Sprain, C.J.; Richards, M.A.; Self, S.; Vanderkluysen, L.; Pande, K. (2015). "State shift in Deccan volcanism at the Cretaceous-Paleogene boundary, possibly induced by impact" (https://escholarship.org/uc/item/5fk799n8). *Science*. **350** (6256): 76–78. Bibcode:2015Sci...350...76R (https://ui.adsabs.harvard.edu/abs/2015Sci...350...76R). doi:10.1126/science.aac7549 (https://doi.org/10.1126%2Fscience.aac7549). PMID 26430116 (https://pubmed.ncbi.nlm.nih.gov/26430116). S2CID 30612906 (https://api.semanticscholar.org/CorpusID:30612906). Archived (https://web.archive.org/web/20230323005826/https://escholarship.org/uc/item/5fk799n8) from the original on March 23, 2023. Retrieved October 28, 2022.
316. Richards, M.A.; Alvarez, W.; Self, S.; Karlstrom, L.; Renne, P.R.; Manga, M.; Sprain, C.J.; Smit, J.; Vanderkluysen, L.; Gibson, S.A. (2015). "Triggering of the largest Deccan eruptions by the Chicxulub impact" (https://escholarship.org/uc/item/86f3521g). *Geological Society of America Bulletin*. **127** (11–12): 1507–1520. Bibcode:2015GSAB..127.1507R (https://ui.adsabs.harvard.edu/abs/2015GSAB..127.1507R). doi:10.1130/B31167.1 (https://doi.org/10.1130%2FB31167.1). hdl:1871.1/cc9361fe-f586-44a0-90ac-f5c513e9920b (https://hdl.handle.net/1871.1%2Fcc9361fe-f586-44a0-90ac-f5c513e9920b). OSTI 1512141 (https://www.osti.gov/biblio/1512141). S2CID 3463018 (https://api.semanticscholar.org/CorpusID:3463018). Archived (https://web.archive.org/web/20201022102243/https://escholarship.org/uc/item/86f3521g) from the original on October 22, 2020. Retrieved February 8, 2021.

317. Khazins, V.; Shuvalov, V. (2019). "Chicxulub Impact as a Trigger of One of Deccan Volcanism Phases: Threshold of Seismic Energy Density". In Kocharyan, G.; Lyakhov, A. (eds.). *Trigger Effects in Geosystems*. Springer Proceedings in Earth and Environmental Sciences. Cham: Springer. pp. 523–530. doi:10.1007/978-3-030-31970-0_55 (https://doi.org/10.1007%2F978-3-030-31970-0_55). ISBN 978-3-030-31969-4. S2CID 210277965 (https://api.semanticscholar.org/CorpusID:210277965).
318. Archibald, J.D.; Clemens, W.A.; Padian, K.; Rowe, T.; Macleod, N.; Barrett, P.M.; Gale, A.; Holroyd, P.; Sues, H.-D.; Arens, N.C.; Horner, J.R.; Wilson, G.P.; Goodwin, M.B.; Brochu, C.A.; Lofgren, D.L.; Hurlbert, S.H.; Hartman, J.H.; Eberth, D.A.; Wignall, P.B.; Currie, P.J.; Weil, A.; Prasad, G.V.R.; Dingus, L.; Courtillot, V.; Milner, A.; Milner, A.; Bajpai, S.; Ward, D.J.; Sahni, A. (2010). "Cretaceous extinctions: multiple causes". *Science*. **328** (5981): 973, author reply 975–6. doi:10.1126/science.328.5981.973-a (https://doi.org/10.1126%2Fscience.328.5981.973-a). PMID 20489004 (https://pubmed.ncbi.nlm.nih.gov/20489004).
319. Courtillot, V.; Fluteau, F. (2010). "Cretaceous extinctions: the volcanic hypothesis". *Science*. **328** (5981): 973–974. Bibcode:2010Sci...328..973C (https://ui.adsabs.harvard.edu/abs/2010Sci...328..973C). doi:10.1126/science.328.5981.973-b (https://doi.org/10.1126%2Fscience.328.5981.973-b). PMID 20489003 (https://pubmed.ncbi.nlm.nih.gov/20489003).
320. Keller, G. (2014). "Deccan volcanism, the Chicxulub impact, and the end-Cretaceous mass extinction: Coincidence? Cause and effect". *Geological Society of America Special Papers*. **505**: 57–89. doi:10.1130/2014.2505(03) (https://doi.org/10.1130%2F2014.2505%2803%29). ISBN 978-0-8137-2505-5.
321. Schulte, P.; Alegret, L.; Arenillas, I.; Arz, J.A.; Barton, P.J.; Bown, P.R.; Bralower, T.J.; Christeson, G.L.; Claeys, P.; Cockell, C.S.; Collins, G.S.; Deutsch, A.; Goldin, T.J.; Goto, K.; Grajales-Nishimura, J.M.; Grieve, R.A.F.; Gulick, S.P.S.; Johnson, K.R.; Kiessling, W.; Koeberl, C.; Kring, D.A.; MacLeod, K.G.; Matsui, T.; Melosh, J.; Montanari, A.; Morgan, J.V.; Neal, C.R.; Nichols, D.J.; Norris, R.D.; Pierazzo, E.; Ravizza, G.; Rebolledo-Vieyra, M.; Uwe Reimold, W.; Robin, E.; Salge, T.; Speijer, R.P.; Sweet, A.R.; Urrutia-Fucugauchi, J.; Vajda, V.; Whalen, M.T.; Willumsen, P.S. (2010). "Response—Cretaceous extinctions" (https://lirias.kuleuven.be/handle/123456789/269158). *Science*. **328** (5981): 975–976. Bibcode:2010Sci...328..975S (https://ui.adsabs.harvard.edu/abs/2010Sci...328..975S). doi:10.1126/science.328.5981.975 (https://doi.org/10.1126%2Fscience.328.5981.975).
322. Fassett, J.E.; Heaman, L.M.; Simonetti, A. (2011). "Direct U–Pb dating of Cretaceous and Paleocene dinosaur bones, San Juan Basin, New Mexico". *Geology*. **39** (2): 159–162. Bibcode:2011Geo....39..159F (https://ui.adsabs.harvard.edu/abs/2011Geo....39..159F). doi:10.1130/G31466.1 (https://doi.org/10.1130%2FG31466.1). ISSN 0091-7613 (https://search.worldcat.org/issn/0091-7613).
323. Fassett, J.E.; Heaman, L.M.; Simonetti, A. (2009). "New geochronologic and stratigraphic evidence confirms the Paleocene age of the dinosaur-bearing Ojo Alamo Sandstone and Animas Formation in the San Juan Basin, New Mexico and Colorado" (https://palaeo-electronica.org/2009_1/149/index.html). *Palaeontologia Electronica*. **12** (1): 3A. Archived (https://web.archive.org/web/20210318001522/https://palaeo-electronica.org/2009_1/149/index.html) from the original on March 18, 2021. Retrieved January 27, 2021.
324. Sloan, R.E.; Rigby, J.K. Jr.; Van Valen, L.M.; et al. (1986). "Gradual Dinosaur Extinction and Simultaneous Ungulate Radiation in the Hell Creek Formation". *Science*. **232** (4750): 629–633. Bibcode:1986Sci...232..629S (https://ui.adsabs.harvard.edu/abs/1986Sci...232..629S). doi:10.1126/science.232.4750.629 (https://doi.org/10.1126%2Fscience.232.4750.629). ISSN 0036-8075 (https://search.worldcat.org/issn/0036-8075). PMID 17781415 (https://pubmed.ncbi.nlm.nih.gov/17781415). S2CID 31638639 (https://api.semanticscholar.org/CorpusID:31638639).
325. Lucas, S.G.; Sullivan, R.M.; Cather, S.M.; Jasinski, S.E.; Fowler, D.W.; Heckert, A.B.; Spielmann, J.A.; Hunt, A.P. (2009). "No definitive evidence of Paleocene dinosaurs in the San Juan Basin". *Palaeontologia Electronica*. **12** (2): 8A.

326. Renne, P.R.; Goodwin, M.B. (2012). "Direct U-Pb dating of Cretaceous and Paleocene dinosaur bones, San Juan Basin, New Mexico: COMMENT" (https://doi.org/10.1130%2FG32521C.1). *Geology*. **40** (4): e259. Bibcode:2012Geo....40E.259R (https://ui.adsabs.harvard.edu/abs/2012Geo....40E.259R). doi:10.1130/G32521C.1 (https://doi.org/10.1130%2FG32521C.1).
327. Lofgren, D.L.; Hotton, C.L.; Runkel, A.C. (1990). "Reworking of Cretaceous dinosaurs into Paleocene channel, deposits, upper Hell Creek Formation, Montana". *Geology*. **18** (9): 874–877. Bibcode:1990Geo....18..874L (https://ui.adsabs.harvard.edu/abs/1990Geo....18..874L). doi:10.1130/0091-7613(1990)018<0874:ROCDIP>2.3.CO;2 (https://doi.org/10.1130%2F0091-7613%281990%29018%3C0874%3AROCDIP%3E2.3.CO%3B2).
328. Koenig, A.E.; Lucas, S.G.; Neymark, L.A.; Heckert, A.B.; Sullivan, R.M.; Jasinski, S.E.; Fowler, D.W. (2012). "Direct U-Pb dating of Cretaceous and Paleocene dinosaur bones, San Juan Basin, New Mexico: COMMENT" (https://doi.org/10.1130%2FG32154C.1). *Geology*. **40** (4): e262. Bibcode:2012Geo....40E.262K (https://ui.adsabs.harvard.edu/abs/2012Geo....40E.262K). doi:10.1130/G32154C.1 (https://doi.org/10.1130%2FG32154C.1).
329. "Dinosaur" (https://www.merriam-webster.com/dictionary/Dinosaur). *Merriam-Webster.com Dictionary*. Merriam-Webster. OCLC 1032680871 (https://search.worldcat.org/oclc/1032680871). Retrieved November 7, 2019.
330. Sarjeant 1995, pp. 255–284, chpt. 15: "The Dinosaurs and Dinomania over 150 Years" by Hugh S. Torrens.
331. Currie & Padian 1997, pp. 347–350, "History of Dinosaur Discoveries: First Golden Period" by Brent H. Breithaupt.
332. Dickens 1853, p. 1 (https://archive.org/details/bleakhouse00dick/page/n37), chpt. I: "*London. Michaelmas Term lately over, and the Lord Chancellor sitting in Lincoln's Inn Hall. Implacable November weather. As much mud in the streets, as if the waters had but newly retired from the face of the earth, and it would not be wonderful to meet a Megalosaurus, forty feet long or so, waddling like an elephantine lizard up Holborn Hill.*"
333. Farlow & Brett-Surman 1997, pp. 675–697 (https://archive.org/details/isbn_9780253333490/page/675), chpt. 43: "Dinosaurs and the Media" by Donald F. Glut and M.K. Brett-Surman.
334. Lee, Newton; Madej, Krystina (2012). "Early Animation: Gags and Situations". *Disney Stories*. pp. 17–24. doi:10.1007/978-1-4614-2101-6_3 (https://doi.org/10.1007%2F978-1-4614-2101-6_3). ISBN 978-1-4614-2100-9. S2CID 192335675 (https://api.semanticscholar.org/CorpusID:192335675).

Bibliography

- Alvarez, Walter (1997). *T. rex and the Crater of Doom* (https://archive.org/details/trexcraterofdoo000alva). Princeton, NJ: Princeton University Press. ISBN 978-0-691-01630-6. LCCN 96049208 (https://lccn.loc.gov/96049208). OCLC 1007846558 (https://search.worldcat.org/oclc/1007846558). Retrieved November 4, 2019.
- Bakker, Robert T. (1986). *The Dinosaur Heresies: New Theories Unlocking the Mystery of the Dinosaurs and Their Extinction* (https://archive.org/details/dinosaurheresies00robe). New York: William Morrow and Company. ISBN 978-0-688-04287-5. LCCN 86012643 (https://lccn.loc.gov/86012643). OCLC 13699558 (https://search.worldcat.org/oclc/13699558). Retrieved November 6, 2019.
- Benton, Michael J. (2015). *Vertebrate Palaeontology* (4th [completely revised] ed.). Oxford: Wiley Blackwell. ISBN 978-1-118-40684-7.
- Benton, Michael J. (2005). *Vertebrate Palaeontology* (https://archive.org/details/VertebratePalaeontology) (3rd ed.). Malden, MA: Blackwell Publishing. ISBN 978-0-632-05637-8. LCCN 2003028152 (https://lccn.loc.gov/2003028152). OCLC 53970617 (https://search.worldcat.org/oclc/53970617). Retrieved October 30, 2019.

- Brusatte, Stephen L. (2012). Benton, Michael J. (ed.). *Dinosaur Paleobiology*. Topics in Paleobiology. Foreword by Michael J. Benton. Hoboken, NJ: Wiley-Blackwell. Bibcode:2012dipa.book.....B (https://ui.adsabs.harvard.edu/abs/2012dipa.book.....B). doi:10.1002/9781118274071 (https://doi.org/10.1002%2F9781118274071). ISBN 978-0-470-65658-7. LCCN 2011050466 (https://lccn.loc.gov/2011050466). OCLC 781864955 (https://search.worldcat.org/oclc/781864955).
- Chiappe, Luis M.; Witmer, Lawrence M., eds. (2002). *Mesozoic Birds: Above the Heads of Dinosaurs*. Berkeley: University of California Press. ISBN 978-0-520-20094-4. LCCN 2001044600 (https://lccn.loc.gov/2001044600). OCLC 901747962 (https://search.worldcat.org/oclc/901747962).
- Colbert, Edwin H. (1971) [Originally published, New York: E. P. Dutton, 1968; London: Evans Brothers Ltd, 1969]. *Men and Dinosaurs: The Search in Field and Laboratory* (https://archive.org/details/mendinosaurssear00colb). Harmondsworth: Penguin. ISBN 978-0-14-021288-4. OCLC 16208760 (https://search.worldcat.org/oclc/16208760). Retrieved October 31, 2019.
- Cowen, Richard (2005). *History of Life* (4th ed.). Malden, MA: Blackwell Publishing. ISBN 978-1-4051-1756-2. LCCN 2003027993 (https://lccn.loc.gov/2003027993). OCLC 53970577 (https://search.worldcat.org/oclc/53970577). The 5th edition of the book is available from the Internet Archive (https://archive.org/details/History_of_Life_5th_Edition_by_Richard_Cowen). Retrieved 2019-10-19.
- Currie, Philip J.; Padian, Kevin, eds. (1997). *Encyclopedia of Dinosaurs* (https://archive.org/details/EncyclopediaOfDinosaurs). San Diego, CA: Academic Press. ISBN 978-0-12-226810-6. LCCN 97023430 (https://lccn.loc.gov/97023430). OCLC 436848919 (https://search.worldcat.org/oclc/436848919). Retrieved October 30, 2019.
- Currie, Philip J.; Koppelhus, Eva B.; Shugar, Martin A.; Wright, Joanna L., eds. (2004). *Feathered Dragons: Studies on the Transition from Dinosaurs to Birds*. Life of the Past. Bloomington, IN: Indiana University Press. ISBN 978-0-253-34373-4. LCCN 2003019035 (https://lccn.loc.gov/2003019035). OCLC 52942941 (https://search.worldcat.org/oclc/52942941).
- Curry Rogers, Kristina A.; Wilson, Jeffrey A., eds. (2005). *The Sauropods: Evolution and Paleobiology*. Berkeley: University of California Press. ISBN 978-0-520-24623-2. LCCN 2005010624 (https://lccn.loc.gov/2005010624). OCLC 879179542 (https://search.worldcat.org/oclc/879179542).
- Desmond, Adrian J. (1975). *The Hot-Blooded Dinosaurs: A Revolution in Palaeontology* (https://archive.org/details/hotbloodeddinosa00desm). London: Blond & Briggs. ISBN 978-0-8037-3755-6. LCCN 76359907 (https://lccn.loc.gov/76359907). OL 4933052M (https://openlibrary.org/books/OL4933052M). Retrieved October 30, 2019.
- Dickens, Charles (1853). *Bleak House* (https://archive.org/details/bleakhouse00dick/page/n7). London: Bradbury and Evans. Retrieved November 7, 2019.
- Dodson, Peter; Gingerich, Philip D., eds. (1993). *Functional Morphology and Evolution*. American Journal of Science. Vol. 293A. New Haven, CT: Kline Geology Laboratory, Yale University. ISSN 0002-9599 (https://search.worldcat.org/issn/0002-9599). OCLC 27781160 (https://search.worldcat.org/oclc/27781160).
- Dong, Zhiming (1992). *Dinosaurian Faunas of China* (English ed.). Beijing; Berlin; New York: China Ocean Press; Springer-Verlag. ISBN 978-3-540-52084-9. LCCN 92207835 (https://lccn.loc.gov/92207835). OCLC 26522845 (https://search.worldcat.org/oclc/26522845).
- Dyke, Gareth; Kaiser, Gary, eds. (2011). *Living Dinosaurs: The Evolutionary History of Modern Birds*. Chichester; Hoboken, NJ: Wiley-Blackwell. ISBN 978-0-470-65666-2. LCCN 2010043277 (https://lccn.loc.gov/2010043277). OCLC 729724640 (https://search.worldcat.org/oclc/729724640).

- Farlow, James O.; Brett-Surman, M.K., eds. (1997). *The Complete Dinosaur* (https://archive.org/details/isbn_9780253333490). Bloomington, IN: Indiana University Press. ISBN 978-0-253-33349-0. LCCN 97-23698 (https://lccn.loc.gov/97-23698). OCLC 924985811 (https://search.worldcat.org/oclc/924985811). Retrieved October 14, 2019.
- Foster, John R.; Lucas, Spencer G., eds. (2006). "Paleontology and Geology of the Upper Jurassic Morrison Formation" (https://econtent.unm.edu/digital/collection/bulletins/id/803). *Bulletin of the New Mexico Museum of Natural History and Science*. New Mexico Museum of Natural History and Science Bulletin. **36**. Albuquerque, NM: New Mexico Museum of Natural History and Science. ISSN 1524-4156 (https://search.worldcat.org/issn/1524-4156). OCLC 77520577 (https://search.worldcat.org/oclc/77520577). Retrieved October 21, 2019.
- Glut, Donald F. (1997). *Dinosaurs: The Encyclopedia*. Foreword by Michael K. Brett-Surman. Jefferson, NC: McFarland & Company. ISBN 978-0-89950-917-4. LCCN 95047668 (https://lccn.loc.gov/95047668). OCLC 33665881 (https://search.worldcat.org/oclc/33665881).
- Gunther, Robert Theodore, ed. (1968) [First printed in Oxford 1945]. *Life and Letters of Edward Lhwyd* (https://archive.org/details/earlyscienceinox14gunt/page/n3). Early Science in Oxford. Vol. XIV. Preface by Albert Everard Gunther (Reprint ed.). London: Dawsons of Pall Mall. ISBN 978-0-7129-0292-2. LCCN 22005926 (https://lccn.loc.gov/22005926). OCLC 43529321 (https://search.worldcat.org/oclc/43529321). Retrieved November 4, 2019.
- Hansell, Mike (2000). *Bird Nests and Construction Behaviour* (https://archive.org/details/birdnestsconstru0000hans). Pen and ink illustration by Raith Overhill. Cambridge: University of Cambridge Press. ISBN 978-0-521-46038-5. LCCN 99087681 (https://lccn.loc.gov/99087681). OCLC 876286627 (https://search.worldcat.org/oclc/876286627). Retrieved October 30, 2019.
- Heilmann, Gerhard (1926). *The Origin of Birds*. London; New York: H. F. & G. Witherby; D. Appleton & Company. LCCN 27001127 (https://lccn.loc.gov/27001127). OCLC 606021642 (https://search.worldcat.org/oclc/606021642).
- Holmes, Thom (1998). *Fossil Feud: The Rivalry of the First American Dinosaur Hunters* (https://archive.org/details/isbn_9790382391483_c8k9). Parsippany, NJ: Julian Messner. ISBN 978-0-382-39149-1. LCCN 96013610 (https://lccn.loc.gov/96013610). OCLC 34472600 (https://search.worldcat.org/oclc/34472600).
- Holtz, Thomas R. Jr. (2007). *Dinosaurs: The Most Complete, Up-to-Date Encyclopedia for Dinosaur Lovers of All Ages* (https://archive.org/details/dinosaursmostcom00holt). Illustrated by Luis V. Rey. New York: Random House. ISBN 978-0-375-82419-7. LCCN 2006102491 (https://lccn.loc.gov/2006102491). OCLC 77486015 (https://search.worldcat.org/oclc/77486015). Retrieved October 22, 2019.
- Lambert, David; The Diagram Group (1990). *The Dinosaur Data Book: The Definitive, Fully Illustrated Encyclopedia of Dinosaurs* (https://archive.org/details/dinosaurdatabook00lamb). New York: Avon Books. ISBN 978-0-380-75896-8. LCCN 89092487 (https://lccn.loc.gov/89092487). OCLC 21833417 (https://search.worldcat.org/oclc/21833417). Retrieved October 14, 2019.
- Lessem, Don; Glut, Donald F. (1993). *The Dinosaur Society's Dinosaur Encyclopedia* (https://archive.org/details/dinosaursocietys00less). Illustrations by Tracy Lee Ford; scientific advisors, Peter Dodson, et al. New York: Random House. ISBN 978-0-679-41770-5. LCCN 94117716 (https://lccn.loc.gov/94117716). OCLC 30361459 (https://search.worldcat.org/oclc/30361459). Retrieved October 30, 2019.
- Lhuyd, Edward (1699). *Lithophylacii Britannici ichnographia* (https://web.archive.org/web/20200814045952/http://lhldigital.lindahall.org/cdm/ref/collection/earththeory/id/1913) [*British figured stones*]. London: Ex Officina M.C. Archived from the original (http://lhldigital.lindahall.org/cdm/ref/collection/earththeory/id/1913) on August 14, 2020. Retrieved November 4, 2019.

- Mayr, Gerald (2009). *Paleogene Fossil Birds* (https://archive.org/details/PaleogeneFossilBirds). Berlin: Springer-Verlag. doi:10.1007/978-3-540-89628-9 (https://doi.org/10.1007%2F978-3-540-89628-9). ISBN 978-3-540-89627-2. LCCN 2008940962 (https://lccn.loc.gov/2008940962). OCLC 916182693 (https://search.worldcat.org/oclc/916182693). S2CID 88941254 (https://api.semanticscholar.org/CorpusID:88941254). Retrieved October 30, 2019.
- Norell, Mark; Gaffney, Eugene S.; Dingus, Lowell (2000) [Originally published as *Discovering Dinosaurs in the American Museum of Natural History*. New York: Knopf, 1995]. *Discovering Dinosaurs: Evolution, Extinction, and the Lessons of Prehistory* (https://archive.org/details/discoveringdinos00nore) (Revised ed.). Berkeley: University of California Press. ISBN 978-0-520-22501-5. LCCN 99053335 (https://lccn.loc.gov/99053335). OCLC 977125867 (https://search.worldcat.org/oclc/977125867). Retrieved October 30, 2019.
- Olshevsky, George (2000). *An Annotated Checklist of Dinosaur Species by Continent*. Mesozoic Meanderings. Vol. 3. Illustrated by Tracy Lee Ford. San Diego, CA: Publications Requiring Research. ISSN 0271-9428 (https://search.worldcat.org/issn/0271-9428). LCCN 00708700 (https://lccn.loc.gov/00708700). OCLC 44433611 (https://search.worldcat.org/oclc/44433611).
- Owen, Richard (1842). "Report on British Fossil Reptiles. Part II" (https://archive.org/details/reportofeleventh42lond/page/n99). *Report of the Eleventh Meeting of the British Association for the Advancement of Science; Held at Plymouth in July 1841* (https://archive.org/details/reportofeleventh42lond/page/n3/mode/2up). London: John Murray. pp. 60 (https://archive.org/details/reportofeleventh42lond/page/60)–204. LCCN 99030427 (https://lccn.loc.gov/99030427). OCLC 1015526268 (https://search.worldcat.org/oclc/1015526268). Retrieved October 13, 2019.
- Padian, Kevin, ed. (1986). *The Origin of Birds and the Evolution of Flight*. Memoirs of the California Academy of Sciences. Vol. 8. San Francisco, CA: California Academy of Sciences. ISBN 978-0-940228-14-6. OCLC 946083441 (https://search.worldcat.org/oclc/946083441). OL 9826926M (https://openlibrary.org/books/OL9826926M).
- Parsons, Keith M. (2001). *Drawing out Leviathan: Dinosaurs and the Science Wars* (https://archive.org/details/drawingoutleviat0000pars). Life in the Past. Bloomington, IN: Indiana University Press. ISBN 978-0-253-33937-9. LCCN 2001016803 (https://lccn.loc.gov/2001016803). OCLC 50174737 (https://search.worldcat.org/oclc/50174737). Retrieved October 30, 2019.
- Paul, Gregory S. (1988). *Predatory Dinosaurs of the World: A Complete Illustrated Guide* (https://archive.org/details/predatorydinosaursoftheworld1988). New York: Simon & Schuster. ISBN 978-0-671-61946-6. LCCN 88023052 (https://lccn.loc.gov/88023052). OCLC 859819093 (https://search.worldcat.org/oclc/859819093). Retrieved October 30, 2019.
- Paul, Gregory S., ed. (2000). *The Scientific American Book of Dinosaurs* (1st ed.). New York: St. Martin's Press. ISBN 978-0-312-26226-6. LCCN 2001269051 (https://lccn.loc.gov/2001269051). OCLC 45256074 (https://search.worldcat.org/oclc/45256074).
- Paul, Gregory S. (2010). *The Princeton Field Guide to Dinosaurs*. Princeton Field Guides. Princeton, NJ: Princeton University Press. ISBN 978-0-691-13720-9. LCCN 2010014916 (https://lccn.loc.gov/2010014916). OCLC 907619291 (https://search.worldcat.org/oclc/907619291).
- Plot, Robert (1677). *The Natural History of Oxford-shire: Being an Essay toward the Natural History of England* (https://archive.org/details/naturalhistoryo00plot/page/n9). Oxford; London: S. Millers. LCCN 11004267 (https://lccn.loc.gov/11004267). OCLC 933062622 (https://search.worldcat.org/oclc/933062622). Retrieved November 13, 2019.
- Randall, Lisa (2015). *Dark Matter and the Dinosaurs: The Astounding Interconnectedness of the Universe*. New York: HarperCollins. ISBN 978-0-06-232847-2. LCCN 2016427646 (https://lccn.loc.gov/2016427646). OCLC 962371431 (https://search.worldcat.org/oclc/962371431).

- Rupke, Nicolaas A. (1994). *Richard Owen: Victorian Naturalist* (https://archive.org/details/richardowenvicto00rupk). New Haven: Yale University Press. ISBN 978-0-300-05820-8. LCCN 93005739 (https://lccn.loc.gov/93005739). OCLC 844183804 (https://search.worldcat.org/oclc/844183804). Retrieved November 5, 2019.
- Sarjeant, William A.S., ed. (1995). *Vertebrate Fossils and the Evolution of Scientific Concepts: Writings in Tribute to Beverly Halstead, by Some of His Many Friends*. Modern Geology, vol. 18. Reading: Gordon and Breach Publishers. ISBN 978-2-88124-996-9. ISSN 0026-7775 (https://search.worldcat.org/issn/0026-7775). LCCN 00500382 (https://lccn.loc.gov/00500382). OCLC 34672546 (https://search.worldcat.org/oclc/34672546). "Reprint of papers published in a special volume of Modern geology [v. 18 (Halstead memorial volume), 1993], with five additional contributions.--Pref."
- Tanner, Lawrence H.; Spielmann, Justin A.; Lucas, Spencer G., eds. (2013). "The Triassic System: New Developments in Stratigraphy and Paleontology" (https://econtent.unm.edu/digital/collection/bulletins/id/1645/rec/1). *Bulletin of the New Mexico Museum of Natural History and Science*. New Mexico Museum of Natural History and Science Bulletin. **61**. Albuquerque, NM: New Mexico Museum of Natural History and Science. ISSN 1524-4156 (https://search.worldcat.org/issn/1524-4156). OCLC 852432407 (https://search.worldcat.org/oclc/852432407). Retrieved October 21, 2019.
- Weishampel, David B.; Dodson, Peter; Osmólska, Halszka, eds. (2004). *The Dinosauria* (2nd ed.). Berkeley: University of California Press. ISBN 978-0-520-25408-4. LCCN 2004049804 (https://lccn.loc.gov/2004049804). OCLC 154697781 (https://search.worldcat.org/oclc/154697781).

Further reading

- Paul, Gregory S. (2002). *Dinosaurs of the Air: The Evolution and Loss of Flight in Dinosaurs and Birds* (https://archive.org/details/dinosaursofairev0000paul). Baltimore; London: Johns Hopkins University Press. ISBN 978-0-8018-6763-7. LCCN 2001000242 (https://lccn.loc.gov/2001000242). OCLC 1088130487 (https://search.worldcat.org/oclc/1088130487)..
- Sternberg, Charles Mortram (1966) [Original edition published by E. Cloutier, printer to the King, 1946]. *Canadian Dinosaurs*. Geological Series. Vol. 54 (2nd ed.). Ottawa: National Museum of Canada. LCCN gs46000214 (https://lccn.loc.gov/gs46000214). OCLC 1032865683 (https://search.worldcat.org/oclc/1032865683).
- Stewart, Tabori & Chang (1997). *The Humongous Book of Dinosaurs*. New York: Stewart, Tabori & Chang. ISBN 978-1-55670-596-0. LCCN 97000398 (https://lccn.loc.gov/97000398). OCLC 1037269801 (https://search.worldcat.org/oclc/1037269801).
- University of Southampton (September 29, 2021). "Two New Species of Large Predatory Dinosaur with Crocodile-Like Skulls Discovered on Isle of Wight" (https://scitechdaily.com/two-new-species-of-large-predatory-dinosaur-with-crocodile-like-skulls-discovered-on-isle-of-wight/). *SciTechDaily*.
- Zhou, Zhonghe (October 2004). "The origin and early evolution of birds: discoveries, disputes, and perspectives from fossil evidence" (https://web.archive.org/web/20110721144552/http://www.cisneros-heredia.org/infotrans/usfq/ornitofauna/pdfs/zhou2004.pdf) (PDF). *Naturwissenschaften*. **91** (10). Berlin: Springer Science+Business Media: 455–471. Bibcode:2004NW.....91..455Z (https://ui.adsabs.harvard.edu/abs/2004NW.....91..455Z). doi:10.1007/s00114-004-0570-4 (https://doi.org/10.1007%2Fs00114-004-0570-4). ISSN 0028-1042 (https://search.worldcat.org/issn/0028-1042). PMID 15365634 (https://pubmed.ncbi.nlm.nih.gov/15365634). S2CID 3329625 (https://api.semanticscholar.org/CorpusID:3329625). Archived from the original (http://www.cisneros-heredia.org/infotrans/usfq/ornitofauna/pdfs/zhou2004.pdf) (PDF) on July 21, 2011. Retrieved November 6, 2019.

Retrieved from "https://en.wikipedia.org/w/index.php?title=Dinosaur&oldid=1381030563"

WIKIPEDIA 25
25 years of the free encyclopedia

Physiology of dinosaurs

The **physiology of non-avian dinosaurs** has historically been a controversial subject, particularly their thermoregulation. Recently, many new lines of evidence have been brought to bear on dinosaur physiology generally, including not only metabolic systems and thermoregulation, but on respiratory and cardiovascular systems as well.

During the early years of dinosaur paleontology, it was widely considered that they were sluggish, cumbersome, and sprawling cold-blooded lizards. However, with the discovery of much more complete skeletons in the western United States, starting in the 1870s, scientists made more informed interpretations of dinosaur biology and physiology. Edward Drinker Cope, opponent of Othniel Charles Marsh in the Bone Wars, propounded at least some dinosaurs as active and agile, as seen in the painting of two fighting *Laelaps* produced under his direction by Charles R. Knight.[1]

In parallel, the development of Darwinian evolution, and the discoveries of *Archaeopteryx* and *Compsognathus*, led Thomas Henry Huxley to propose that dinosaurs were closely related to birds.[2] Despite these considerations, the image of dinosaurs as large reptiles had already taken root,[1] and most aspects of their paleobiology were interpreted as being typically reptilian for the first half of the twentieth century.[3] Beginning in the 1960s and with the advent of the dinosaur renaissance, views of dinosaurs and their physiology have changed dramatically, including the discovery of feathered dinosaurs in Early Cretaceous age deposits in China, indicating that birds evolved from highly agile maniraptoran dinosaurs.

History

Early interpretations

The study of dinosaurs began in the 1820s in England. Pioneers in the field, such as William Buckland, Gideon Mantell, and Richard Owen, interpreted the first, very fragmentary remains as belonging to large quadrupedal beasts.[4] Their early work can be seen today in the Crystal Palace Dinosaurs, constructed in the 1850s, which present known dinosaurs as elephantine lizard-like reptiles.[5] Despite these reptilian appearances, Owen speculated that dinosaur heart and respiratory systems were more similar to that of a mammal than a reptile.[4]

Reconstruction of *Megalosaurus* from 1854, in Crystal Palace, London

Changing views and the dinosaur renaissance

In the late 1960s, similar ideas reappeared, beginning with John Ostrom's work on *Deinonychus* and bird evolution.[6] His student, Bob Bakker, popularized the changing thought in a series of papers beginning with *The superiority of dinosaurs* in 1968.[7] In these publications, he argued strenuously that dinosaurs

were warm-blooded and active animals, capable of sustained periods of high activity. In most of his writings Bakker framed his arguments as new evidence leading to a revival of ideas popular in the late 19th century, frequently referring to an ongoing *dinosaur renaissance*. He used a variety of anatomical and statistical arguments to defend his case,[8][9] the methodology of which was fiercely debated among scientists.[10]

The 1897 painting of "*Laelaps*" (now *Dryptosaurus*) by Charles R. Knight

These debates sparked interest in new methods for ascertaining the palaeobiology of extinct animals, such as bone histology, which have been successfully applied to determining the growth-rates of many dinosaurs.

Today, it is generally thought that many or perhaps all dinosaurs had higher metabolic rates than living reptiles, but also that the situation is more complex and varied than Bakker originally proposed. For example, while smaller dinosaurs may have been true endotherms, the larger forms could have been inertial homeotherms,[11][12] or that many dinosaurs could have had intermediate metabolic rates.[13]

Feeding and digestion

The earliest dinosaurs were almost certainly predators, and shared several predatory features with their nearest non-dinosaur relatives like *Lagosuchus*, including: relatively large, curved, blade-like teeth in large, wide-opening jaws that closed like scissors; relatively small abdomens, as carnivores do not require large digestive systems. Later dinosaurs regarded as predators sometimes grew much larger, but retained the same set of features. Instead of chewing their food, these predators swallowed it whole.[14]

The feeding habits of ornithomimosaurs and oviraptorosaurs are a mystery: although they evolved from a predatory theropod lineage, they have small jaws and lack the blade-like teeth of typical predators, but there is no evidence of their diet or how they ate and digested it.[14]

Features of other groups of dinosaurs indicate they were herbivores. These features include:

- Jaws that only slightly opened and closed so that all the teeth met at the same time
- Large abdomens that could accommodate large amounts of vegetation and store it for the long time it takes to digest vegetation
- Guts that likely contained endosymbiotic micro-organisms that digest cellulose, as no known animal can digest this tough material directly[14]

Sauropods, which were herbivores, did not chew their food, as their teeth and jaws appear suitable only for stripping leaves off plants. Ornithischians, also herbivores, show a variety of approaches. The armored ankylosaurs and stegosaurs had small heads and weak jaws and teeth, and are thought to have fed in much the same way as sauropods. The pachycephalosaurs had small heads and weak jaws and teeth, but their lack of large digestive systems suggests a different diet, possibly fruits, seeds, or young shoots, which would have been more nutritious to them than leaves.[14]

On the other hand, ornithopods such as *Hypsilophodon*, *Iguanodon* and various hadrosaurs had horny beaks for snipping off vegetation and jaws and teeth that were well-adapted for chewing. The horned ceratopsians had similar mechanisms.[14]

It has often been suggested that at least some non-avian dinosaurs used swallowed stones, known as gastroliths, to aid digestion by grinding their food in muscular gizzards, and that this was a feature they shared with avian dinosaurs (birds). In 2007 Oliver Wings reviewed references to gastroliths in scientific literature and found considerable confusion, starting with the lack of an agreed and objective definition of "gastrolith". He found that swallowed hard stones or grit can assist digestion in birds that mainly feed on grain but may not be essential—and that birds that eat insects in summer and grain in winter usually get rid of the stones and grit in summer. Gastroliths have often been described as important for sauropod dinosaurs, whose diet of vegetation required very thorough digestion, but Wings concluded that this idea was incorrect: gastroliths are found with only a small percentage of sauropod fossils; where they have been found, the amounts are too small and in many cases the stones are too soft to have been effective in grinding food; most of these gastroliths are highly polished, but gastroliths used by modern animals to grind food are roughened by wear and corroded by stomach acids; hence the sauropod gastroliths were probably swallowed accidentally. On the other hand, he concluded that gastroliths found with fossils of advanced theropod dinosaurs such as *Sinornithomimus* and *Caudipteryx* resemble those of birds, and that the use of gastroliths for grinding food may have appeared early in the group of dinosaurs from which these dinosaurs and birds both evolved.[15][16]

Reproductive biology

When laying eggs, female birds grow a special type of bone in their limbs between the hard outer bone and the marrow.[17] This medullary bone, which is rich in calcium, is used to make eggshells, and the birds that produced it absorb it when they have finished laying eggs.[18] Medullary bone has been found in fossils of the theropods *Tyrannosaurus* and *Allosaurus* and of the ornithopod *Tenontosaurus*.[18][19]

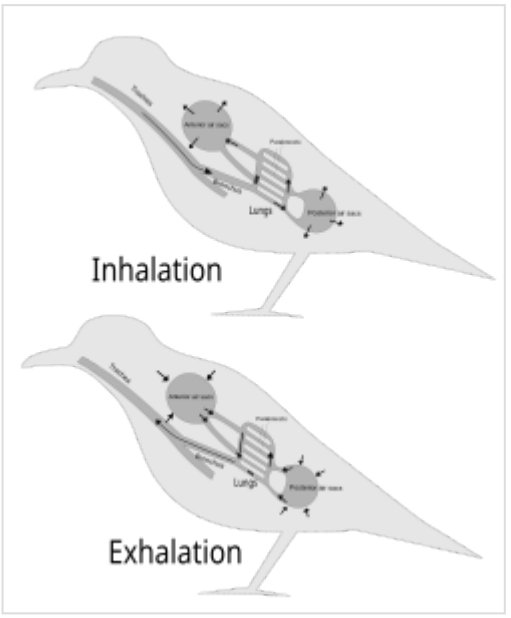
Because the line of dinosaurs that includes *Allosaurus* and *Tyrannosaurus* diverged from the line that led to *Tenontosaurus* very early in the evolution of dinosaurs, the presence of medullary bone in both groups suggests that dinosaurs in general produced medullary tissue. On the other hand, crocodilians, which are dinosaurs' second closest extant relatives after birds, do not produce medullary bone. This tissue may have first appeared in ornithodires, the Triassic archosaur group from which dinosaurs are thought to have evolved.[18]

Medullary bone has been found in specimens of sub-adult size, which suggests that dinosaurs reached sexual maturity before they were full-grown. Sexual maturity at sub-adult size is also found in reptiles and in medium- to large-sized mammals, but birds and small mammals reach sexual maturity only after they are full-grown—which happens within their first year. Early sexual maturity is also associated with specific features of animals' life cycles: the young are born relatively well-developed rather than helpless; and the death-rate among adults is high.[18]

Respiratory system

Air sacs

From about 1870 onwards scientists have generally agreed that the post-cranial skeletons of many dinosaurs contained many air-filled cavities (postcranial skeletal pneumaticity), especially in the vertebrae. Pneumatization of the skull (such as paranasal sinuses) is found in both synapsids and archosaurs, but postcranial pneumatization is found only in birds, non-avian saurischian dinosaurs, and pterosaurs.



Birds' lungs obtain fresh air during both exhalation and inhalation, because the air sacs do all the "pumping" and the lungs simply absorb oxygen.

For a long time these cavities were regarded simply as weight-saving devices, but Bakker proposed that they were connected to air sacs like those that make birds' respiratory systems the most efficient of all animals'.[9]

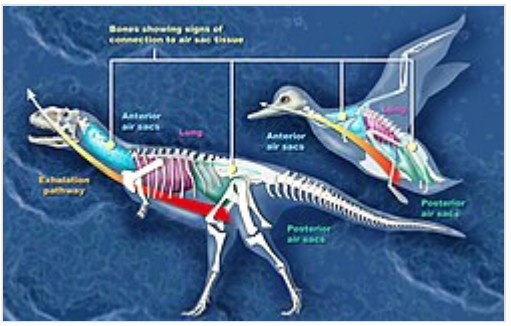
John Ruben *et al.* (1997, 1999, 2003, 2004) disputed this and suggested that dinosaurs had a "tidal" respiratory system (in and out) powered by a crocodile-like hepatic piston mechanism – muscles attached mainly to the pubis pull the liver backwards, which makes the lungs expand to inhale; when these muscles relax, the lungs return to their previous size and shape, and the animal exhales. They also presented this as a reason for doubting that birds descended from dinosaurs.[20][21][22][23][24]

Critics have claimed that, without avian air sacs, modest improvements in a few aspects of a modern reptile's circulatory and respiratory systems would enable the reptile to achieve 50% to 70% of the oxygen flow of a mammal of similar size,[25] and that lack of avian air sacs would not prevent the development of endothermy.[26] Very few formal rebuttals have been published in scientific journals of Ruben et al.'s claim that dinosaurs could not have had avian-style air sacs; but one points out that the *Sinosauropteryx* fossil on which they based much of their argument was severely flattened and therefore it was impossible to tell whether the liver was the right shape to act as part of a hepatic piston mechanism.[27] Some recent papers simply note without further comment that Ruben et al. argued against the presence of air sacs in dinosaurs.[28]

Researchers have presented evidence and arguments for air sacs in sauropods, "prosauropods", coelurosaurs, ceratosaurs, and the theropods *Aerosteon* and *Coelophysis*.

In advanced sauropods ("neosauropods") the vertebrae of the lower back and hip regions show signs of air sacs. In early sauropods only the cervical (neck) vertebrae show these features. If the developmental sequence found in bird embryos is a guide, air sacs actually evolved before the channels in the skeleton that accommodate them in later forms.[29][30]

Evidence of air sacs has also been found in theropods. Studies indicate that fossils of coelurosaurs,[31] ceratosaurs,[28] and the theropods *Coelophysis* and *Aerosteon* exhibit evidence of air sacs. *Coelophysis*, from the late Triassic, is one of the earliest dinosaurs whose fossils show evidence of channels for air sacs.[30] *Aerosteon*, a Late Cretaceous allosaur, had the most bird-like air sacs found so far.[32]



Comparison between the air sacs of *Majungasaurus* and a bird

Early sauropodomorphs, including the group traditionally called "prosauropods", may also have had air sacs. Although possible pneumatic indentations have been found in *Plateosaurus* and *Thecodontosaurus*, the indentations are very small. One study in 2007 concluded that prosauropods likely had abdominal and cervical air sacs, based on the evidence for them in sister taxa (theropods and sauropods). The study concluded that it was impossible to determine whether prosauropods had a bird-like flow-through lung, but that the air sacs were almost certainly present.[33] A further indication for the presence of air sacs and their use in lung ventilation comes from a reconstruction of the air exchange volume (the volume of air exchanged with each breath) of *Plateosaurus*, which when expressed as a ratio of air volume per body weight at 29 ml/kg is similar to values of geese and other birds, and much higher than typical mammalian values.[34]

So far no evidence of air sacs has been found in ornithischian dinosaurs. But this does not imply that ornithischians could not have had metabolic rates comparable to those of mammals, since mammals also do not have air sacs.[35]

Three explanations have been suggested for the development of air sacs in dinosaurs:[32]

- Increase in respiratory capacity. This is probably the most common hypothesis, and fits well with the idea that many dinosaurs had fairly high metabolic rates.
- Improving balance and maneuvrability by lowering the center of gravity and reducing rotational inertia. However this does not explain the expansion of air sacs in the quadrupedal sauropods.
- As a cooling mechanism. It seems that air sacs and feathers evolved at about the same time in coelurosaurs. If feathers retained heat, their owners would have required a means of dissipating excess heat. This idea is plausible but needs further empirical support.

Calculations of the volumes of various parts of the sauropod *Apatosaurus*' respiratory system support the evidence of bird-like air sacs in sauropods:

- Assuming that *Apatosaurus*, like dinosaurs' nearest surviving relatives crocodilians and birds, did not have a diaphragm, the dead-space volume of a 30-ton specimen would be about 184 liters. This is the total volume of the mouth, trachea and air tubes. If the animal exhales less than this, stale air is not expelled and is sucked back into the lungs on the following inhalation.
- Estimates of its tidal volume – the amount of air moved into or out of the lungs in a single breath – depend on the type of respiratory system the animal had: 904 liters if avian; 225 liters if mammalian; 19 liters if reptilian.

On this basis, *Apatosaurus* could not have had a reptilian respiratory system, as its tidal volume would have been less than its dead-space volume, so that stale air was not expelled but was sucked back into the lungs. Likewise, a mammalian system would only provide to the lungs about 225 − 184 = 41 liters of fresh, oxygenated air on each breath. *Apatosaurus* must therefore have had either a system unknown in

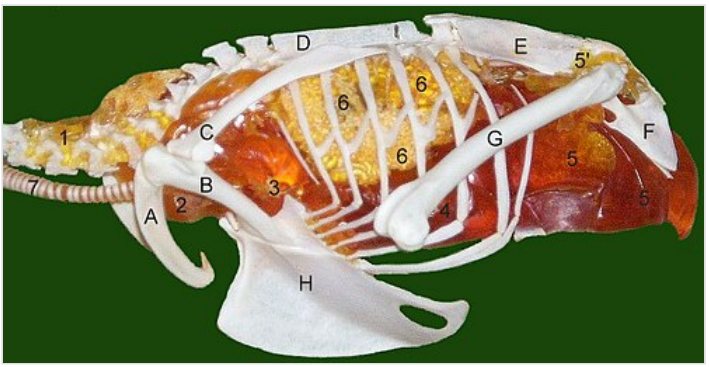
the modern world or one like birds', with multiple air sacs and a flow-through lung. Furthermore, an avian system would only need a lung volume of about 600 liters while a mammalian one would have required about 2,950 liters, which would exceed the estimated 1,700 liters of space available in a 30-ton *Apatosaurus*' chest.[36]

Dinosaur respiratory systems with bird-like air sacs may have been capable of sustaining higher activity levels than mammals of similar size and build can sustain. In addition to providing a very efficient supply of oxygen, the rapid airflow would have been an effective cooling mechanism, which is essential for animals that are active but too large to get rid of all the excess heat through their skins.[35]

The palaeontologist Peter Ward has argued that the evolution of the air sac system, which first appears in the very earliest dinosaurs, may have been in response to the very low (11%) atmospheric oxygen of the Carnian and Norian ages of the Triassic Period.[37]

Uncinate processes on the ribs

Birds have spurs called "uncinate processes" on the rear edges of their ribs, and these give the chest muscles more leverage when pumping the chest to improve oxygen supply. The size of the uncinate processes is related to the bird's lifestyle and oxygen requirements: they are shortest in walking birds and longest in diving birds, which need to replenish their oxygen reserves quickly when they surface. Non-avian maniraptoran dinosaurs also had these uncinate processes, and they were proportionately as long as in modern diving birds, which indicates that maniraptorans needed a high-capacity oxygen supply.[38][39]



The uncinate processes are the small white spurs about halfway along the ribs. The rest of this diagram shows the air sacs and other parts of a bird's respiratory system:1 cervical air sac, 2 clavicular air sac, 3 cranial thoracal air sac, 4 caudal thoracal air sac, 5 abdominal air sac (5' diverticulus into pelvic girdle), 6 lung, 7 trachea

Plates that may have functioned the same way as uncinate processes have been observed in fossils of the ornithischian dinosaur *Thescelosaurus*, and have been interpreted as evidence of high oxygen consumption and therefore high metabolic rate.[40]

Nasal turbinates

Nasal turbinates are convoluted structures of thin bone in the nasal cavity. In most mammals and birds these are present and lined with mucous membranes that perform two functions. They improve the sense of smell by increasing the area available to absorb airborne chemicals, and they warm and moisten inhaled air, and extract heat and moisture from exhaled air to prevent desiccation of the lungs.

John Ruben and others have argued that no evidence of nasal turbinates has been found in dinosaurs. All the dinosaurs they examined had nasal passages that were too narrow and short to accommodate nasal turbinates, so dinosaurs could not have sustained the breathing rate required for a mammal-like or bird-like metabolic rate while at rest, because their lungs would have dried out.[20][21][41][42] However, objections have been raised against this argument. Nasal turbinates are absent or very small in some birds (e.g. ratites, Procellariiformes and Falconiformes) and mammals (e.g. whales, anteaters, bats, elephants,

and most primates), although these animals are fully endothermic and in some cases very active.[43][44][45][46] Other studies conclude that nasal turbinates are fragile and seldom found in fossils. In particular none have been found in fossil birds.[47]

In 2014 Jason Bourke and others in *Anatomical Record* reported finding nasal turbinates in pachycephalosaurs.[48]

Cardiovascular system

The possible heart of "Willo" the thescelosaur (center).

In principle one would expect dinosaurs to have had two-part circulations driven by four-chambered hearts, since many would have needed high blood pressure to deliver blood to their heads, which were high off the ground, but vertebrate lungs can only tolerate fairly low blood pressure.[35] In 2000, a skeleton of *Thescelosaurus*, now on display at the North Carolina Museum of Natural Sciences, was described as including the remnants of a four-chambered heart and an aorta. The authors interpreted the structure of the heart as indicating an elevated metabolic rate for *Thescelosaurus*, not reptilian cold-bloodedness.[49] Their conclusions have been disputed; other researchers published a paper where they assert that the heart is really a concretion of entirely mineral "cement". As they note: the anatomy given for the object is incorrect, for example the alleged "aorta" is narrowest where it meets the "heart" and lacks arteries branching from it; the "heart" partially engulfs one of the ribs and has an internal structure of concentric layers in some places; and another concretion is preserved behind the right leg.[50] The original authors defended their position; they agreed that the chest did contain a type of concretion, but one that had formed around and partially preserved the more muscular portions of the heart and aorta.[51]

Regardless of the object's identity, it may have little relevance to dinosaurs' internal anatomy and metabolic rate. Both modern crocodilians and birds, the closest living relatives of dinosaurs, have four-chambered hearts, although modified in crocodilians, and so dinosaurs probably had them as well. However such hearts are not necessarily tied to metabolic rate.[52]

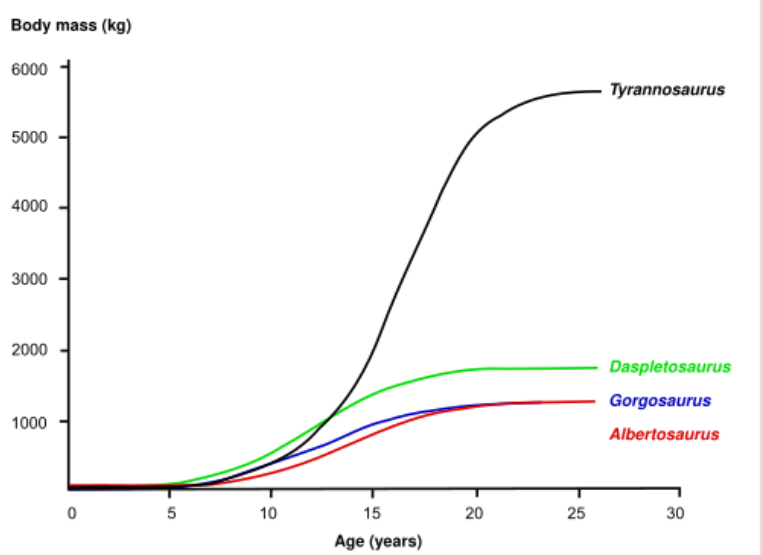
Growth and lifecycle

No dinosaur egg has been found that is larger than a basketball and embryos of large dinosaurs have been found in relatively small eggs, e.g. Maiasaura.[53] Like mammals, dinosaurs stopped growing when they reached the typical adult size of their species, while mature reptiles continued to grow slowly if they had enough food. Dinosaurs of all sizes grew faster than similarly sized modern reptiles; but the results of comparisons with similarly sized "warm-blooded" modern animals depend on their sizes:[54][55]

Weight (kg)	Comparative growth rate of dinosaurs	Modern animals in this size range
0.22	Slower than marsupials	Rat
1 – 20	Similar to marsupials, slower than precocial birds (those that are born capable of running)	From guinea pig to Andean condor
100 – 1000	Faster than marsupials, similar to precocial birds, slower than placental mammals	From red kangaroo to polar bear
1500 – 3500	Similar to most placental mammals	From American bison to rhinoceros
25000 and over	Very fast, similar to modern whales; but about half that of a scaled-up altricial bird (one that is born helpless) – if one could scale up a bird to 25,000 kilograms (25 long tons; 28 short tons)	Whales

Tyrannosaurus rex showed a "teenage growth spurt":[56][57]

- ½ ton at age 10
- very rapid growth to around two tons in the mid-teens (about ½ ton per year).
- negligible growth after the second decade.



A graph showing the hypothesized growth curves (body mass versus age) of four tyrannosaurids. *Tyrannosaurus rex* is drawn in black. Based on Erickson et al. 2004.

A 2008 study of one skeleton of the hadrosaur *Hypacrosaurus* concluded that this dinosaur grew even faster, reaching its full size at the age of about 15; the main evidence was the number and spacing of growth rings in its bones. The authors found this consistent with a life-cycle theory that prey species should grow faster than their predators if they lose a lot of juveniles to predators and the local environment provides enough resources for rapid growth.[58]

It appears that individual dinosaurs were rather short-lived, e.g. the oldest (at death) *Tyrannosaurus* found so far was 28 and the oldest sauropod was 38.[56] Predation was probably responsible for the high death rate of very young dinosaurs and sexual competition for the high death rate of sexually mature dinosaurs.[59]

Metabolism

Scientific opinion about the life-style, metabolism and temperature regulation of dinosaurs has varied over time since the discovery of dinosaurs in the mid-19th century. The activity of metabolic enzymes varies with temperature, so temperature control is vital for any organism, whether endothermic or ectothermic. Organisms can be categorized as poikilotherms (poikilo – changing), which are tolerant of

internal temperature fluctuations, and homeotherms (homeo – same), which must maintain a constant core temperature. Animals can be further categorized as endotherms, which regulate their temperature internally, and ectotherms, which regulate temperature by the use of external heat sources.

"Warm-bloodedness" is a complex and rather ambiguous term, because it includes some or all of:

- Homeothermy, i.e. maintaining a fairly constant body temperature. Modern endotherms maintain a variety of temperatures: 28 °C (82 °F) to 30 °C (86 °F) in monotremes and sloths; 33 °C (91 °F) to 36 °C (97 °F) in marsupials; 36 °C (97 °F) to 38 °C (100 °F) in most placentals; and around 41 °C (106 °F) in birds.[35]
- Tachymetabolism, i.e. maintaining a high metabolic rate, particularly when at rest. This requires a fairly high and stable body temperature, since biochemical processes run about half as fast if an animal's temperature drops by 10 °C; most enzymes have an optimum operating temperature and their efficiency drops rapidly outside the preferred range.[60]
- Endothermy, i.e. the ability to generate heat internally, for example by "burning" fat, rather than via behaviors such as basking or muscular activity. Although endothermy is in principle the most reliable way to maintain a fairly constant temperature, it is expensive; for example modern mammals need 10 to 13 times as much food as modern reptiles.[35]

Large dinosaurs may also have maintained their temperatures by inertial homeothermy, also known as "bulk homeothermy" or "mass homeothermy". In other words, the thermal capacity of such large animals was so high that it would take two days or more for their temperatures to change significantly, and this would have smoothed out variations caused by daily temperature cycles. This smoothing effect has been observed in large turtles and crocodilians, but *Plateosaurus*, which weighed about 700 kilograms (1,500 lb), may have been the smallest dinosaur in which it would have been effective. Inertial homeothermy would not have been possible for small species nor for the young of larger species.[35] Vegetation fermenting in the guts of large herbivores can also produce considerable heat, but this method of maintaining a high and stable temperature would not have been possible for carnivores or for small herbivores or the young of larger herbivores.[61]

Since the internal mechanisms of extinct creatures are unknowable, most discussion focuses on homeothermy and tachymetabolism.

Assessment of metabolic rates is complicated by the distinction between the rates while resting and while active. In all modern reptiles and most mammals and birds the maximum rates during all-out activity are 10 to 20 times higher than minimum rates while at rest. However, in a few mammals these rates differ by a factor of 70. Theoretically it would be possible for a land vertebrate to have a reptilian metabolic rate at rest and a bird-like rate while working flat out. However, an animal with such a low resting rate would be unable to grow quickly. The huge herbivorous sauropods may have been on the move so constantly in search of food that their energy expenditure would have been much the same irrespective of whether their resting metabolic rates were high or low.[62]

Theories

The main possibilities are that:[35]

- Non-avian dinosaurs were cold-blooded, like modern reptiles, except that the large size of many would have stabilized their body temperatures.
- They were warm-blooded, more like birds or modern mammals than modern reptiles.

- They were neither cold-blooded nor warm-blooded in modern terms, but had metabolisms that were different from and in some ways intermediate between those of modern cold-blooded and warm-blooded animals.
- They included animals with two or three of these types of metabolism.

Non-avian dinosaurs were around for about 150 million years, so it is very likely that different groups evolved different metabolisms and thermoregulatory regimes, and that some developed different physiologies from the first dinosaurs.

If all or some non-avian dinosaurs had intermediate metabolisms, they may have had the following features:[35]

- Low resting metabolic rates—which would reduce the amount of food they needed and allow them to use more of that food for growth than do animals with high resting metabolic rates.
- Inertial homeothermy
- The ability to control heat loss by expanding and contracting blood vessels just under the skin, as many modern reptiles do.
- Two-part circulations driven by four-chambered hearts.
- High aerobic capacity, allowing sustained activity.

Robert Reid has suggested that such animals could be regarded as "failed endotherms". He envisaged both dinosaurs and the Triassic ancestors of mammals passing through a stage with these features. Mammals were forced to become smaller as archosaurs came to dominate ecological niches for medium to large animals. Their decreasing size made them more vulnerable to heat loss because it increased their ratios of surface area to mass, and thus forced them to increase internal heat generation and thus become full endotherms. On the other hand, dinosaurs became medium to very large animals and thus were able to retain the "intermediate" type of metabolism.[35]

Bone structure

Armand de Ricqlès discovered Haversian canals in dinosaur bones, and argued that there was evidence of endothermy in dinosaurs. These canals are common in "warm-blooded" animals and are associated with fast growth and an active life style because they help to recycle bone to facilitate rapid growth and repair damage caused by stress or injuries.[63] Dense secondary Haversian bone, which is formed during remodeling, is found in many living endotherms as well as dinosaurs, pterosaurs and therapsids. Secondary Haversian canals are correlated with size and age, mechanical stress and nutrient turnover. The presence of secondary Haversian canals suggests comparable bone growth and lifespans in mammals and dinosaurs.[64] Bakker argued that the presence of fibrolamellar bone (produced quickly and having a fibrous, woven appearance) in dinosaur fossils was evidence of endothermy.[9]

However, as a result of other, mainly later research, bone structure is not considered a reliable indicator of metabolism in dinosaurs, mammals or reptiles:

- Dinosaur bones often contain lines of arrested growth (LAGs), formed by alternating periods of slow and fast growth; in fact many studies count growth rings to estimate the ages of dinosaurs.[55][56] The formation of growth rings is usually driven by seasonal changes in temperature, and this seasonal influence has sometimes been regarded as a sign of slow metabolism and ectothermy. But growth rings are found in polar bears and in mammals that

hibernate.[65][66] The relationship between LAGs and seasonal growth dependency remains unresolved.[67]

- Fibrolamellar bone is fairly common in young crocodilians and sometimes found in adults.[68][69]
- Haversian bone has been found in turtles, crocodilians and tortoises,[70] but is often absent in small birds, bats, shrews and rodents.[69]

Nevertheless, de Ricqlès persevered with studies of the bone structure of dinosaurs and archosaurs. In mid-2008 he co-authored a paper that examined bone samples from a wide range of archosaurs, including early dinosaurs, and concluded that:[71]

- Even the earliest archosauriforms may have been capable of very fast growth, which suggests they had fairly high metabolic rates. Although drawing conclusions about the earliest archosauriformes from later forms is tricky, because species-specific variations in bone structure and growth rate are very likely, there are research strategies that can minimize the risk that such factors will cause errors in the analysis.
- Archosaurs split into three main groups in the Triassic: ornithodirans, from which dinosaurs evolved, remained committed to rapid growth; crocodilians' ancestors adopted more typical "reptilian" slow growth rates; and most other Triassic archosaurs had intermediate growth rates.

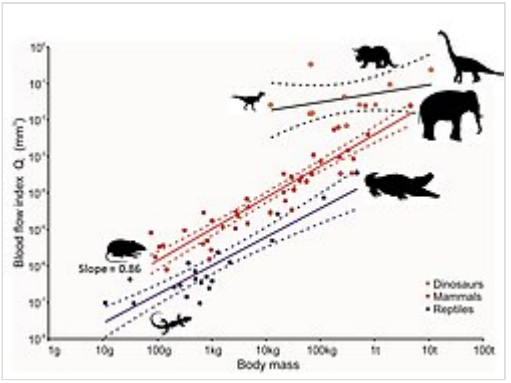
An osteohistological analysis of vascular density and density, shape and area of osteocytes concluded non-avian dinosaurs and the majority of archosauriforms (except *Proterosuchus*, crocodilians and phytosaurs) retained heat and had resting metabolic rates similar to those of extant mammals and birds.[72]

Metabolic rate, blood pressure and flow

Endotherms rely highly on aerobic metabolism and have high rates of oxygen consumption during activity and rest. The oxygen required by the tissues is carried by the blood, and consequently blood flow rates and blood pressures at the heart of warm-blooded endotherms are considerably higher than those of cold-blooded ectotherms.[73] It is possible to measure the minimum blood pressures of dinosaurs by estimating the vertical distance between the heart and the top of the head, because this column of blood must have a pressure at the bottom equal to the hydrostatic pressure derived from the density of blood and gravity. Added to this pressure is that required to move the blood through the circulatory system. It was pointed out in 1976 that, because of their height, many dinosaurs had minimum blood pressures within the endothermic range, and that they must have had four-chambered hearts to separate the high pressure circuit to the body from the low pressure circuit to the lungs.[74] It was not clear whether these dinosaurs had high blood pressure simply to support the blood column or to support the high blood flow rates required by endothermy or both.

However, recent analysis of the tiny holes in fossil leg bones of dinosaurs provides a gauge for blood flow rate and hence metabolic rate.[75] The holes are called nutrient foramina, and the nutrient artery is the major blood vessel passing through to the interior of the bone, where it branches into tiny vessels of the Haversian canal system. This system is responsible for replacing old bone with new bone, thereby repairing microbreaks that occur naturally during locomotion. Without this repair, microbreaks would build up, leading to stress fractures and ultimately catastrophic bone failure. The size of the nutrient foramen provides an index of blood flow through it, according to the Hagen-Poiseuille equation. The size is also related to the body size of animal, of course, so this effect is removed by analysis of allometry. Blood flow index of the nutrient foramen of the femurs in living mammals increases in direct proportion

to the animals' maximum metabolic rates, as measured during maximum sustained locomotion. Mammalian blood flow index is about 10 times greater than in ectothermic reptiles. Ten species of fossil dinosaurs from five taxonomic groups reveal indices even higher than in mammals, when body size is accounted for, indicating that they were highly active, aerobic animals. Thus high blood flow rate, high blood pressure, a four-chambered heart and sustained aerobic metabolism are all consistent with endothermy.



Foramen blood flow index, derived from the size of the nutrient foramen of the femurs of mammals, reptiles and dinosaurs

Growth rates

Dinosaurs grew from small eggs to several tons in weight relatively quickly. A natural interpretation of this is that dinosaurs converted food into body weight very quickly, which requires a fairly fast metabolism both to forage actively and to assimilate the food quickly.[76] Developing bone found in juveniles is distinctly porous, which has been linked to vascularization and bone deposition rate, all suggesting growth rates close to those observed in modern birds.

But a preliminary study of the relationship between adult size, growth rate, and body temperature concluded that larger dinosaurs had higher body temperatures than smaller ones had; *Apatosaurus*, the largest dinosaur in the sample, was estimated to have a body temperature exceeding 41 °C (106 °F), whereas smaller dinosaurs were estimated to have body temperatures around 25 °C (77 °F)[77] – for comparison, normal human body temperature is about 37 °C (99 °F).[78][79] Based on these estimations, the study concluded that large dinosaurs were inertial homeotherms (their temperatures were stabilized by their sheer bulk) and that dinosaurs were ectothermic (in colloquial terms, "cold-blooded", because they did not generate as much heat as mammals when not moving or digesting food).[77] These results are consistent with the relationship between dinosaurs' sizes and growth rates (described above).[80][81] Studies of the sauropodomorph *Massospondylus* and early theropod *Syntarsus* (*Megapnosaurus*) reveal growth rates of 3 kg/year and 17 kg/year, respectively, much slower than those estimated of *Maiasaura* and observed in modern birds.[82]

Oxygen isotope ratios in bone

The ratio of the isotopes ^{16}O and ^{18}O in bone depends on the temperature the bone formed at: the higher the temperature, the more ^{16}O. Barrick and Showers (1999) analyzed the isotope ratios in two theropods that lived in temperate regions with seasonal variation in temperature, *Tyrannosaurus* (USA) and *Giganotosaurus* (Argentina):[83]

- dorsal vertebrae from both dinosaurs showed no sign of seasonal variation, indicating that both maintained a constant core temperature despite seasonal variations in air temperature.
- ribs and leg bones from both dinosaurs showed greater variability in temperature and a lower average temperature as the distance from the vertebrae increased.

Barrick and Showers concluded that both dinosaurs were endothermic but at lower metabolic levels than modern mammals, and that inertial homeothermy was an important part of their temperature regulation as adults. Their similar analysis of some Late Cretaceous ornithischians in 1996 concluded that these animals showed a similar pattern.[84]

However this view has been challenged. The evidence indicates homeothermy, but by itself cannot prove endothermy. Secondly, the production of bone may not have been continuous in areas near the extremities of limbs – in allosaur skeletons lines of arrested growth ("LAGs"; rather like growth rings) are sparse or absent in large limb bones but common in the fingers and toes. While there is no absolute proof that LAGs are temperature-related, they could mark times when the extremities were so cool that the bones ceased to grow. If so, the data about oxygen isotope ratios would be incomplete, especially for times when the extremities were coolest. Oxygen isotope ratios may be an unreliable method of estimating temperatures if it cannot be shown that bone growth was equally continuous in all parts of the animal.[35]

Predator–prey ratios

Bakker argued that:[85]

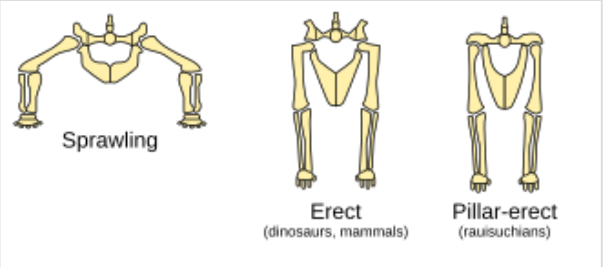
- cold-blooded predators need much less food than warm-blooded ones, so a given mass of prey can support far more cold-blooded predators than warm-blooded ones.
- the ratio of the total mass of predators to prey in dinosaur communities was much more like that of modern and recent warm-blooded communities than that of recent or fossil cold-blooded communities.
- hence predatory dinosaurs were warm-blooded. And since the earliest dinosaurs (e.g. *Staurikosaurus*, *Herrerasaurus*) were predators, all dinosaurs must have been warm-blooded.

This argument was criticized on several grounds and is no longer taken seriously (the following list of criticisms is far from exhaustive):[86][87]

- Estimates of dinosaur weights vary widely, and even a small variation can make a large difference to the calculated predator–prey ratio.
- His sample may not have been representative. Bakker obtained his numbers by counting museum specimens, but these have a bias towards rare or especially well-preserved specimens, and do not represent what exists in fossil beds. Even fossil beds may not accurately represent the actual populations, for example smaller and younger animals have less robust bones and are therefore less likely to be preserved.
- There are no published predator–prey ratios for large ectothermic predators, because such predators are very rare and mostly occur only on fairly small islands. Large ectothermic herbivores are equally rare. So Bakker was forced to compare mammalian predator–prey ratios with those of fish and invertebrate communities, where life expectancies are much shorter and other differences also distort the comparison.
- The concept assumes that predator populations are limited only by the availability of prey. However other factors such as shortage of nesting sites, cannibalism or predation of one predator on another can hold predator populations below the limit imposed by prey biomass, and this would misleadingly reduce the predator–prey ratio.
- Ecological factors can misleadingly reduce the predator–prey ratio, for example: a predator might prey on only some of the "prey" species present; disease, parasites and starvation might kill some of the prey animals before the predators get a chance to hunt them.
- It is very difficult to state precisely what preys on what. For example, the young of herbivores may be preyed upon by lizards and snakes while the adults are preyed on by mammals. Conversely the young of many predators live largely on invertebrates and switch to vertebrates as they grow.

Posture and gait

Dinosaurs' limbs were erect and held under their bodies, rather than sprawling out to the sides like those of lizards and newts. The evidence for this is the angles of the joint surfaces and the locations of muscle and tendon attachments on the bones. Attempts to represent dinosaurs with sprawling limbs result in creatures with dislocated hips, knees, shoulders and elbows.[88]



Hip joints and limb postures.

Carrier's constraint states that air-breathing vertebrates with two lungs that flex their bodies sideways during locomotion find it difficult to move and breathe at the same time. This severely limits stamina, and forces them to spend more time resting than moving.[89]

Sprawling limbs require sideways flexing during locomotion (except for tortoises and turtles, which are very slow and whose armor keeps their bodies fairly rigid). However, despite Carrier's constraint, sprawling limbs are efficient for creatures that spend most of their time resting on their bellies and only move for a few seconds at a time—because this arrangement minimizes the energy costs of getting up and lying down.

Erect limbs increase the costs of getting up and lying down, but avoid Carrier's constraint. This indicates that dinosaurs were active animals because natural selection would have favored the retention of sprawling limbs if dinosaurs had been sluggish and spent most of their waking time resting. An active lifestyle requires a metabolism that quickly regenerates energy supplies and breaks down waste products which cause fatigue, i.e., it requires a fairly fast metabolism and a considerable degree of homeothermy.

Additionally, an erect posture demands precise balance, the result of a rapidly functioning neuromuscular system. This suggests endothermic metabolism, because an ectothermic animal would be unable to walk or run, and thus to evade predators, when its core temperature was lowered. Other evidence for endothermy includes limb length (many dinosaurs possessed comparatively long limbs) and bipedalism, both found today only in endotherms.[90] Many bipedal dinosaurs possessed gracile leg bones with a short thigh relative to calf length. This is generally an adaptation to frequent sustained running, characteristic of endotherms which, unlike ectotherms, are capable of producing sufficient energy to stave off the onset of anaerobic metabolism in the muscle.[91]

Bakker and Ostrom both pointed out that all dinosaurs had erect hindlimbs and that all quadrupedal dinosaurs had erect forelimbs; and that among living animals only the endothermic ("warm-blooded") mammals and birds have erect limbs (Ostrom acknowledged that crocodilians' occasional "high walk" was a partial exception). Bakker claimed this was clear evidence of endothermy in dinosaurs, while Ostrom regarded it as persuasive but not conclusive.[9][92]

A 2009 study supported the hypothesis that endothermy was widespread in at least larger non-avian dinosaurs, and that It was plausibly ancestral for all dinosauriforms, based on the biomechanics of running,[93] though it has also been suggested that endothermy appeared much earlier in archosauromorph evolution, perhaps even preceding the origin of Archosauriformes.[72]

Feathers and filaments

There is now no doubt that many theropod dinosaur species had feathers, including *Shuvuuia*, *Sinosauropteryx* and *Dilong* (an early tyrannosaur).[94][27][95] These have been interpreted as insulation and therefore evidence of warm-bloodedness.

Skin impression of the hadrosaur *Edmontosaurus*

But direct, unambiguous impressions of feathers have only been found in coelurosaurs (which include the birds and tyrannosaurs, among others), so at present feathers give us no information about the metabolisms of the other major dinosaur groups, e.g. coelophysids, ceratosaurs, carnosaurs, or sauropods. Filamentous integument was also present in at least some ornithischians, such as *Tianyulong*, *Kulindadromeus* and *Psittacosaurus*, not only indicating endothermy in this group, but also that feathers were already present in the first ornithodiran (the last common ancestor of dinosaurs and pterosaurs). Their absence in certain groups like Ankylosauria could be the result of suppression of feather genes.[96] Though filaments only first appeared in Coelurosauria according to maximum likelihood reconstructions and that the integumentary structures of *Psittacosaurus*, *Tianyulong*, and *Kulindadromeus* independently evolved from filaments but this was by assuming primitive pterosaur ancestors were scaly.[97]

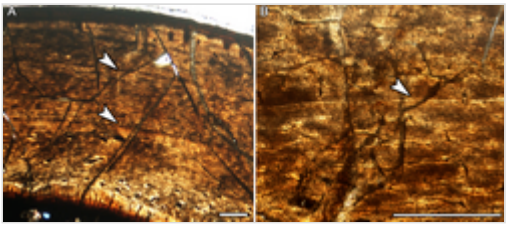
The fossilised skin of *Carnotaurus* (an abelisaurid and therefore not a coelurosaur) shows an unfeathered, reptile-like skin with rows of bumps,[98] but the conclusion that *Carnotaurus* was necessarily featherless has been criticized as the impressions do not cover the whole body, being found only in the lateral region but not the dorsum.[99] An adult *Carnotaurus* weighed about 2 tonnes,[100] and mammals of this size and larger have either very short, sparse hair or naked skins, so perhaps the skin of *Carnotaurus* tells us nothing about whether smaller non-coelurosaurian theropods had feathers. The tyrannosauroid *Yutyrannus* is known to have possessed feathers and weighed 1.1 tonne.[100]

Skin-impressions of *Pelorosaurus* and other sauropods (dinosaurs with elephantine bodies and long necks) reveal large hexagonal scales, and some sauropods, such as *Saltasaurus*, had bony plates in their skin.[101] The skin of ceratopsians consisted of large polygonal scales, sometimes with scattered circular plates.[102] "Mummified" remains and skin impressions of hadrosaurids reveal pebbly scales. It is unlikely that the ankylosaurids, such as *Euoplocephalus*, had insulation, as most of their surface area was covered in bony knobs and plates.[103] Likewise there is no evidence of insulation in the stegosaurs. Thus insulation, and the elevated metabolic rate behind evolving them, may have been limited to the theropods, or even just a subset of theropods. Lack of feathers or other sort of insulation does not indicate ectothermy or low metabolisms, as observed in the relative hairlessness of mammalian megafauna, pigs, human children and the hairless bat being compatible with endothermy.[100]

Polar dinosaurs

Dinosaur fossils have been found in regions that were close to the poles at the relevant times, notably in southeastern Australia, Antarctica and the North Slope of Alaska. There is no evidence of major changes in the angle of the Earth's axis, so polar dinosaurs and the rest of these ecosystems would have had to cope with the same extreme variation of day length through the year that occurs at similar latitudes today (up to a full day with no darkness in summer, and a full day with no sunlight in winter).[104]

Studies of fossilized vegetation suggest that the Alaska North Slope had a maximum temperature of 13 °C (55 °F) and a minimum temperature of 2 °C (36 °F) to 8 °C (46 °F) in the last 35 million years of the Cretaceous (slightly cooler than Portland, Oregon but slightly warmer than Calgary, Alberta). Even so, the Alaska North Slope has no fossils of large cold-blooded animals such as lizards and crocodilians, which were common at the same time in Alberta, Montana, and Wyoming. This suggests that at least some non-avian dinosaurs were warm-blooded.[104] It has been proposed that North American polar dinosaurs may have migrated to warmer regions as winter approached, which would allow them to inhabit Alaska during the summers even if they were cold-blooded.[105] But a round trip between there and Montana would probably have used more energy than a cold-blooded land vertebrate produces in a year; in other words the Alaskan dinosaurs would have to be warm-blooded, irrespective of whether they migrated or stayed for the winter.[106] A 2008 paper on dinosaur migration by Phil R. Bell and Eric Snively proposed that most polar dinosaurs, including theropods, sauropods, ankylosaurians, and hypsilophodonts, probably overwintered, although hadrosaurids like *Edmontosaurus* were probably capable of annual 2,600 km (1,600 mi) round trips.[107][108]



Bone microstructure of polar hypsilophodontid approaching skeletal maturity

It is more difficult to determine the climate of southeastern Australia when the dinosaur fossil beds were laid down 115 to 105 million years ago, towards the end of the Early Cretaceous: these deposits contain evidence of permafrost, ice wedges, and hummocky ground formed by the movement of subterranean ice, which suggests mean annual temperatures ranged between −6 °C (21 °F) and 5 °C (41 °F); oxygen isotope studies of these deposits give a mean annual temperature of 1.5 °C (34.7 °F) to 2.5 °C (36.5 °F). However the diversity of fossil vegetation and the large size of some of fossil trees exceed what is found in such cold environments today, and no-one has explained how such vegetation could have survived in the cold temperatures suggested by the physical indicators – for comparison Fairbanks, Alaska presently has a mean annual temperature of 2.9 °C (37.2 °F).[104] An annual migration from and to southeastern Australia would have been very difficult for fairly small dinosaurs in such as *Leaellynasaura*, a herbivore about 60 centimetres (2.0 ft) to 90 centimetres (3.0 ft) long, because seaways to the north blocked the passage to warmer latitudes.[104] Bone samples from *Leaellynasaura* and *Timimus*, an ornithomimid about 3.5 metres (11 ft) long and 1.5 metres (4.9 ft) high at the hip, suggested these two dinosaurs had different ways of surviving the cold, dark winters: the *Timimus* sample had lines of arrested growth (LAGs for short; similar to growth rings), and it may have hibernated; but the *Leaellynasaura* sample showed no signs of LAGs, so it may have remained active throughout the winter.[109] A 2011 study focusing on hypsilophodont and theropod bones also concluded that these dinosaurs did not hibernate through the winter, but stayed active.[110]

Evidence for behavioral thermoregulation

Some dinosaurs, e.g. *Spinosaurus* and *Ouranosaurus*, had on their backs "sails" supported by spines growing up from the vertebrae. (This was also true, incidentally, for the synapsid *Dimetrodon*.) Such dinosaurs could have used these sails to:

- take in heat by basking with the "sails" at right angles to the sun's rays.
- to lose heat by using the "sails" as radiators while standing in the shade or while facing directly towards or away from the sun.

But these were a very small minority of known dinosaur species. One common interpretation of the plates on stegosaurs' backs is as heat exchangers for thermoregulation, as the plates are filled with blood vessels, which, theoretically, could absorb and dissipate heat.[111]

This might have worked for a stegosaur with large plates, such as *Stegosaurus*, but other stegosaurs, such as *Wuerhosaurus*, *Tuojiangosaurus* and *Kentrosaurus* possessed much smaller plates with a surface area of doubtful value for thermo-regulation. However, the idea of stegosaurian plates as heat exchangers has recently been questioned.[112]

Other evidence

Endothermy demands frequent respiration, which can result in water loss. In living birds and mammals, water loss is limited by pulling moisture out of exhaled air with mucus-covered respiratory turbinates, tissue-covered bony sheets in the nasal cavity. Several dinosaurs have olfactory turbinates, used for smell, but none have yet been identified with respiratory turbinates.[113]

Because endothermy allows refined neuromuscular control, and because brain matter requires large amounts of energy to sustain, some speculate that increased brain size indicates increased activity and, thus, endothermy. The encephalization quotient (EQ) of dinosaurs, a measure of brain size calculated using brain endocasts, varies on a spectrum from bird-like to reptile-like. Using EQ alone, coelurosaurs appear to have been as active as living mammals, while theropods and ornithopods fall somewhere between mammals and reptiles, and other dinosaurs resemble reptiles.[113]

A study published by Roger Seymour in 2013 added more support to the idea that dinosaurs were endothermic. After studying saltwater crocodiles, Seymour found that even if their large sizes could provide stable and high body temperatures, during activity the crocodile's ectothermic metabolism provided less aerobic abilities and generate only 14% of the total muscle power of a similar sized endothermic mammal before full fatigue. Seymour reasoned that dinosaurs would have needed to be endothermic since they would have needed better aerobic abilities and higher power generation to compete with and dominate over mammals as active land animals throughout the Mesozoic era.[114]

Early archosaur metabolism

It appears that the earliest dinosaurs had the features that form the basis for arguments for warm-blooded dinosaurs—especially erect limbs. This raises the question "How did dinosaurs become warm-blooded?" The most obvious possible answers are:

- "Their immediate ancestors (archosaurs) were cold-blooded, and dinosaurs began developing warm-bloodedness very early in their evolution." This implies that dinosaurs developed a significant degree of warm-bloodedness in a very short time, possibly less than 20M years. But in mammals' ancestors the evolution of warm-bloodedness seems to have taken much longer, starting with the beginnings of a secondary palate around the beginning of the mid-Permian[115] and going on possibly until the appearance of hair about 164M years ago in the mid Jurassic[116]).
- "Dinosaurs' immediate ancestors (archosaurs) were at least fairly warm-blooded, and dinosaurs evolved further in that direction." This answer raises 2 problems: (A) The early evolution of archosaurs is still very poorly understood – large numbers of individuals and species are found from the start of the Triassic but only 2 species are known from the very

late Permian (*Archosaurus rossicus* and *Protorosaurus speneri*); (B) Crocodilians evolved shortly before dinosaurs and are closely related to them, but are cold-blooded (see below).

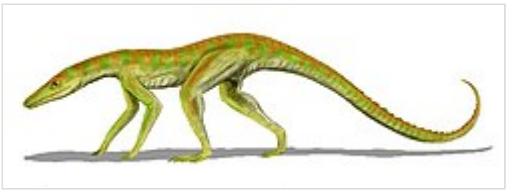
Crocodilians present some puzzles if one regards dinosaurs as active animals with fairly constant body temperatures. Crocodilians evolved shortly before dinosaurs and, second to birds, are dinosaurs' closest living relatives – but modern crocodilians are cold-blooded. This raises some questions:

- If dinosaurs were to a large extent "warm-blooded", when and how fast did warm-bloodedness evolve in their lineage?
- Modern crocodilians are cold-blooded but have several features associated with warm-bloodedness. How did they acquire these features?

Modern crocodilians are cold-blooded but can move with their limbs erect, and have several features normally associated with warm-bloodedness because they improve the animal's oxygen supply:[117]

- 4-chambered hearts. Mammals and birds have four-chambered hearts. Non-crocodilian reptiles have three-chambered hearts, which are less efficient because they allow oxygenated and de-oxygenated blood to mix and therefore send some de-oxygenated blood out to the body instead of to the lungs. Modern crocodilians' hearts are four-chambered, but are smaller relative to body size and run at lower pressure than those of modern mammals and birds. They also have a bypass that makes them functionally three-chambered when under water, conserving oxygen.[118]
- a diaphragm, which aids breathing.
- a secondary palate, which allows the animal to eat and breathe at the same time.
- a hepatic piston mechanism for pumping the lungs. This is different from the lung-pumping mechanisms of mammals and birds but similar to what some researchers claim to have found in some dinosaurs.[20][22]

So why did natural selection favor these features, which are important for active warm-blooded creatures but of little apparent use to cold-blooded aquatic ambush predators that spend most of their time floating in water or lying on river banks?



Reconstruction of *Terrestrisuchus*, a very slim, leggy Triassic crocodylomorph.

It was suggested in the late 1980s that crocodilians were originally active, warm-blooded predators and that their archosaur ancestors were warm-blooded.[89][117] More recently, developmental studies indicate that crocodilian embryos develop fully four-chambered hearts first—then develop the modifications that make their hearts function as three-chambered under water. Using the principle that ontogeny recapitulates phylogeny, the researchers concluded that the original crocodilians had fully 4-chambered hearts and were therefore warm-blooded and that later crocodilians developed the bypass as they reverted to being cold-blooded aquatic ambush predators.[119][120]

More recent research on archosaur bone structures and their implications for growth rates also suggests that early archosaurs had fairly high metabolic rates and that the Triassic ancestors of crocodilians dropped back to more typically "reptilian" metabolic rates.[71]

If this view is correct, the development of warm-bloodedness in archosaurs (reaching its peak in dinosaurs) and in mammals would have taken more similar amounts of time. It would also be consistent with the fossil evidence:

- The earliest crocodylomorphs, e.g. *Terrestrisuchus*, were slim, leggy terrestrial predators.
- Erect limbs appeared quite early in archosaurs' evolution, and those of rauisuchians are very poorly adapted for any other posture.[121]

See also

- Dinosaur classification
- Dinosaur renaissance
- Evolution of dinosaurs
- Evolutionary physiology
- List of dinosaurs
- Origin of birds
- Argentine black and white tegu#Warm-bloodedness

References

1. Lucas, Spencer G. (1999). *Dinosaurs: The Textbook* (https://archive.org/details/dinosaurstextboo00luca/page/3) (3rd ed.). McGraw-Hill Companies, Inc. pp. 3–9 (https://archive.org/details/dinosaurstextboo00luca/page/3). ISBN 978-0-07-303642-7.
2. Fastovsky DE, Weishampel DB (2005). "Theropoda I:Nature red in tooth and claw". In Fastovsky DE, Weishampel DB (eds.). *The Evolution and Extinction of the Dinosaurs* (2nd ed.). Cambridge University Press. pp. 265–299. ISBN 978-0-521-81172-9.
3. Colbert, E.H.; Cowles, R.B.; Bogert, C.M. (June 1947). "Temperature Tolerances in the American Alligator and Their Bearing on the Habits, Evolution, and Extinction of the Dinosaurs". *The Quarterly Review of Biology*. **22** (2): 145–146. doi:10.1086/395712 (https://doi.org/10.1086%2F395712).
4. Lucas, Spencer G. (2000). *Dinosaurs: The Textbook* (https://archive.org/details/dinosaurstextboo00luca/page/1) (3rd ed.). McGraw-Hill Companies, Inc. pp. 1–3 (https://archive.org/details/dinosaurstextboo00luca/page/1). ISBN 978-0-07-303642-7.
5. Torrens, Hugh (1997). "Politics and Paleontology". In Farlow, James O.; Brett-Surman, Michael K. (eds.). *The Complete Dinosaur* (https://archive.org/details/isbn_9780253333490/page/175). Bloomington: Indiana University Press. pp. 175–190 (https://archive.org/details/isbn_9780253333490/page/175). ISBN 978-0-253-33349-0.
6. Benton, Michael J. (2000). "A brief history of dinosaur paleontology". In Paul, Gregory S. (ed.). *The Scientific American Book of Dinosaurs*. New York: St. Martin's Press. pp. 10–44. ISBN 978-0-312-26226-6.
7. Bakker, R.T. (1968). "The superiority of dinosaurs". *Discovery*. **3** (2): 11–22.
8. Bakker, R. T. (1987). "The Return of the Dancing Dinosaurs" (https://archive.org/details/dinosaurspastpre0000unse). In Czerkas, S. J.; Olson, E. C. (eds.). *Dinosaurs Past and Present, vol. I*. University of Washington Press. ISBN 978-0-295-96541-3.
9. Bakker, Robert T. (1972). "Anatomical and ecological evidence of endothermy in dinosaurs". *Nature*. **238** (5359): 81–85. Bibcode:1972Natur.238...81B (https://ui.adsabs.harvard.edu/abs/1972Natur.238...81B). doi:10.1038/238081a0 (https://doi.org/10.1038%2F238081a0). S2CID 4176132 (https://api.semanticscholar.org/CorpusID:4176132).
10. Thomas, R.D.K.; Olson, E.C. (1980). *A Cold Look at the Warm-Blooded Dinosaurs*. Westview Press. ISBN 978-0-89158-464-3.
11. Benton, M.J. (2005). *Vertebrate Palaeontology* (3rd ed.). Wiley-Blackwell. pp. 221–223. ISBN 978-0-632-05637-8.

12. Paladino, F.V.; O'Connor, M.P.; Spotila, J.R. (1990). "Metabolism of leatherback turtles, gigantothermy, and thermoregulation of dinosaurs". *Nature*. **344** (6269): 858–860. Bibcode:1990Natur.344..858P (https://ui.adsabs.harvard.edu/abs/1990Natur.344..858P). doi:10.1038/344858a0 (https://doi.org/10.1038%2F344858a0). S2CID 4321764 (https://api.semanticscholar.org/CorpusID:4321764).
13. Barrick, R.E.; Showers, W.J.; Fischer, A.G. (1996). "Comparison of Thermoregulation of Four Ornithischian Dinosaurs and a Varanid Lizard from the Cretaceous Two Medicine Formation: Evidence from Oxygen Isotopes". *PALAIOS*. **11** (4): 295–305. Bibcode:1996Palai..11..295B (https://ui.adsabs.harvard.edu/abs/1996Palai..11..295B). doi:10.2307/3515240 (https://doi.org/10.2307%2F3515240). JSTOR 3515240 (https://www.jstor.org/stable/3515240).
14. Norman, D.B. (April 2001). "Dinosaur Feeding" (http://mrw.interscience.wiley.com/emrw/9780470015902/els/article/a0003321/current/abstract). *Encyclopedia of Life Sciences*. John Wiley & Sons. doi:10.1038/npg.els.0003321 (https://doi.org/10.1038%2Fnpg.els.0003321). ISBN 978-0-470-01617-6. Retrieved 10 September 2009.
15. Wings, O. (2007). "A review of gastrolith function with implications for fossil vertebrates and a revised classification" (http://app.pan.pl/archive/published/app52/app52-001.pdf) (PDF). *Acta Palaeontologica Polonica*. **52** (1): 1–16. Archived (https://web.archive.org/web/20110523033418/http://www.app.pan.pl/archive/published/app52/app52-001.pdf) (PDF) from the original on 23 May 2011. Retrieved 10 September 2008.
16. Wings, O. & Sander. P.M. (March 2007). "No gastric mill in sauropod dinosaurs: new evidence from analysis of gastrolith mass and function in ostriches" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2197205). *Proceedings of the Royal Society B*. **274** (1610): 635–640. doi:10.1098/rspb.2006.3763 (https://doi.org/10.1098%2Frspb.2006.3763). PMC 2197205 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2197205). PMID 17254987 (https://pubmed.ncbi.nlm.nih.gov/17254987).
17. Dacke, C. G.; Arkle, S.; Cook, D. J.; Wormstone, I. M.; Jones, S.; Zaidi, M.; Bascal, Z. A. (1 November 1993). "Medullary Bone and Avian Calcium Regulation" (https://web.archive.org/web/20150426093845/http://jeb.biologists.org/content/184/1/63.full.pdf) (PDF). *Journal of Experimental Biology*. **184** (1): 63–88. Bibcode:1993JExpB.184...63D (https://ui.adsabs.harvard.edu/abs/1993JExpB.184...63D). doi:10.1242/jeb.184.1.63 (https://doi.org/10.1242%2Fjeb.184.1.63). Archived from the original (https://journals.biologists.com/jeb/article-abstract/184/1/63/6483/Medullary-Bone-and-Avian-Calcium-Regulation?redirectedFrom=PDF) on 26 April 2015 – via Silverchair.
18. Lee, Andrew H.; Werning, Sarah (2008). "Sexual maturity in growing dinosaurs does not fit reptilian growth models" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2206579). *Proceedings of the National Academy of Sciences*. **105** (2): 582–587. Bibcode:2008PNAS..105..582L (https://ui.adsabs.harvard.edu/abs/2008PNAS..105..582L). doi:10.1073/pnas.0708903105 (https://doi.org/10.1073%2Fpnas.0708903105). PMC 2206579 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2206579). PMID 18195356 (https://pubmed.ncbi.nlm.nih.gov/18195356).
19. Schweitzer, M.H.; Wittmeyer, J.L.; Horner, J.R. (2005). "Gender-specific reproductive tissue in ratites and *Tyrannosaurus rex*" (http://doc.rero.ch/record/14767/files/PAL_E1895.pdf) (PDF). *Science*. **308** (5727): 1456–1460. Bibcode:2005Sci...308.1456S (https://ui.adsabs.harvard.edu/abs/2005Sci...308.1456S). doi:10.1126/science.1112158 (https://doi.org/10.1126%2Fscience.1112158). PMID 15933198 (https://pubmed.ncbi.nlm.nih.gov/15933198). S2CID 30264554 (https://api.semanticscholar.org/CorpusID:30264554). Archived (https://web.archive.org/web/20230312052627/https://doc.rero.ch/record/14767/files/PAL_E1895.pdf) (PDF) from the original on 12 March 2023. Retrieved 3 January 2023.
20. Ruben, J.A.; Jones, T.D.; Geist, N.R.; Hillenius, W. J. (November 1997). "Lung structure and ventilation in theropod dinosaurs and early birds". *Science*. **278** (5341): 1267–1270. Bibcode:1997Sci...278.1267R (https://ui.adsabs.harvard.edu/abs/1997Sci...278.1267R). doi:10.1126/science.278.5341.1267 (https://doi.org/10.1126%2Fscience.278.5341.1267).

21. Ruben, J.A.; Jones, T.D.; Geist, N.R.; Leitch, A.; Hillenius, W.J. (1997). "Lung ventilation and gas exchange in theropod dinosaurs". *Science*. **278** (5341): 1267–1270. Bibcode:1997Sci...278.1267R (https://ui.adsabs.harvard.edu/abs/1997Sci...278.1267R). doi:10.1126/science.278.5341.1267 (https://doi.org/10.1126%2Fscience.278.5341.1267).
22. Ruben, J.A.; Dal Sasso, C.; Geist, N.R.; Hillenius, W. J.; Jones, T.D.; Signore, M. (January 1999). "Pulmonary function and metabolic physiology of theropod dinosaurs" (http://doc.rero.ch/record/13886/files/PAL_E818.pdf) (PDF). *Science*. **283** (5401): 514–516. Bibcode:1999Sci...283..514R (https://ui.adsabs.harvard.edu/abs/1999Sci...283..514R). doi:10.1126/science.283.5401.514 (https://doi.org/10.1126%2Fscience.283.5401.514). PMID 9915693 (https://pubmed.ncbi.nlm.nih.gov/9915693). Archived (https://web.archive.org/web/20230122224748/https://doc.rero.ch/record/13886/files/PAL_E818.pdf) (PDF) from the original on 22 January 2023. Retrieved 3 January 2023.
23. Ruben, J. A.; Jones, T. D.; Geist, N. R. (2003). "Respiration and reproductive paleophysiology of dinosaurs and early birds" (http://sonoma-dspace.calstate.edu/bitstream/10211.1/1472/1/Geist_Respiratory.pdf) (PDF). *Physiol. Biochem. Zool*. **76** (2): 141–164. doi:10.1086/375425 (https://doi.org/10.1086%2F375425). hdl:10211.1/1472 (https://hdl.handle.net/10211.1%2F1472). PMID 12794669 (https://pubmed.ncbi.nlm.nih.gov/12794669). S2CID 222819 (https://api.semanticscholar.org/CorpusID:222819).
24. Hillenius, W. J. & Ruben, J.A. (November–December 2004). "The Evolution of Endothermy in Terrestrial Vertebrates: Who? When? Why?". *Physiological and Biochemical Zoology*. **77** (6): 1019–1042. doi:10.1086/425185 (https://doi.org/10.1086%2F425185). PMID 15674773 (https://pubmed.ncbi.nlm.nih.gov/15674773). S2CID 29300018 (https://api.semanticscholar.org/CorpusID:29300018).
25. Hicks, J.W. & Farmer, C.G. (November 1997). "Lung Ventilation and Gas Exchange in Theropod Dinosaurs" (http://www.sciencemag.org/cgi/content/summary/281/5373/45a). *Science*. **278** (5341): 1267–1270. Bibcode:1997Sci...278.1267R (https://ui.adsabs.harvard.edu/abs/1997Sci...278.1267R). doi:10.1126/science.278.5341.1267 (https://doi.org/10.1126%2Fscience.278.5341.1267). Archived (https://web.archive.org/web/20051113021056/http://www.sciencemag.org/cgi/content/summary/281/5373/45a) from the original on 13 November 2005. Retrieved 17 April 2008.
26. Hicks, J.W. & Farmer, C.G. (September 1999). "Gas exchange potential in reptilian lungs: implications for the dinosaur–avian connection". *Respiration Physiology*. **117** (2–3): 73–83. doi:10.1016/S0034-5687(99)00060-2 (https://doi.org/10.1016%2FS0034-5687%2899%2900060-2). PMID 10563436 (https://pubmed.ncbi.nlm.nih.gov/10563436).
27. Currie, P.J. & Chen, P-j. (December 2001). "Anatomy of *Sinosauropteryx prima* from Liaoning, northeastern China" (http://doc.rero.ch/record/14314/files/PAL_E1480.pdf) (PDF). *Canadian Journal of Earth Sciences*. **38** (12): 1705–1727. Bibcode:2001CaJES..38.1705C (https://ui.adsabs.harvard.edu/abs/2001CaJES..38.1705C). doi:10.1139/cjes-38-12-1705 (https://doi.org/10.1139%2Fcjes-38-12-1705). Archived (https://web.archive.org/web/20230122224657/https://doc.rero.ch/record/14314/files/PAL_E1480.pdf) (PDF) from the original on 22 January 2023. Retrieved 3 January 2023.
28. O'Connor, P. & Claessens, L. (July 2005). "Basic avian pulmonary design and flow-through ventilation in non-avian theropod dinosaurs". *Nature*. **436** (7048): 253–256. Bibcode:2005Natur.436..253O (https://ui.adsabs.harvard.edu/abs/2005Natur.436..253O). doi:10.1038/nature03716 (https://doi.org/10.1038%2Fnature03716). PMID 16015329 (https://pubmed.ncbi.nlm.nih.gov/16015329). S2CID 4390587 (https://api.semanticscholar.org/CorpusID:4390587).

29. Wedel, M.J. (2003). "Vertebral pneumaticity, air sacs, and the physiology of sauropod dinosaurs" (http://paleobiol.geoscienceworld.org/cgi/content/abstract/29/2/243). *Paleobiology*. **29** (2): 243–255. Bibcode:2003Pbio...29..243W (https://ui.adsabs.harvard.edu/abs/2003Pbio...29..243W). doi:10.1666/0094-8373(2003)029<0243:VPASAT>2.0.CO;2 (https://doi.org/10.1666%2F0094-8373%282003%29029%3C0243%3AVPASAT%3E2.0.CO%3B2). S2CID 46619244 (https://api.semanticscholar.org/CorpusID:46619244). Archived (https://web.archive.org/web/20110726102155/http://paleobiol.geoscienceworld.org/cgi/content/abstract/29/2/243) from the original on 26 July 2011. Retrieved 17 April 2008. Full text currently online at "Findarticles.com: Vertebral pneumaticity, air sacs, and the physiology of sauropod dinosaurs" (https://web.archive.org/web/20110411160617/http://findarticles.com/p/articles/mi_qa4067/is_200304/ai_n9202774/). *Paleobiology*. 2003. Archived from the original on 11 April 2011. Retrieved 9 November 2006. and "Vertebral pneumaticity, air sacs, and the physiology of sauropod dinosaurs" (https://web.archive.org/web/20081217091539/http://sauroposeidon.net/Wedel_2003a_sauropod-pneumaticity.pdf) (PDF). Archived from the original on 17 December 2008. Detailed anatomical analyses can be found at Wedel, M.J. (2003). "The Evolution of Vertebral Pneumaticity in Sauropod Dinosaurs" (http://doc.rero.ch/record/15186/files/PAL_E2462.pdf) (PDF). *Journal of Vertebrate Paleontology*. **23** (2): 344–357. doi:10.1671/0272-4634(2003)023[0344:TEOVPI]2.0.CO;2 (https://doi.org/10.1671%2F0272-4634%282003%29023%5B0344%3ATEOVPI%5D2.0.CO%3B2). S2CID 55884062 (https://api.semanticscholar.org/CorpusID:55884062). Archived (https://web.archive.org/web/20230122224730/https://doc.rero.ch/record/15186/files/PAL_E2462.pdf) (PDF) from the original on 22 January 2023. Retrieved 3 January 2023.
30. Wedel, M.J. (June 2006). "Origin of postcranial skeletal pneumaticity in dinosaurs" (http://doc.rero.ch/record/16156/files/PAL_E3107.pdf) (PDF). *Integrative Zoology*. **1** (2): 80–85. doi:10.1111/j.1749-4877.2006.00019.x (https://doi.org/10.1111%2Fj.1749-4877.2006.00019.x). PMID 21395998 (https://pubmed.ncbi.nlm.nih.gov/21395998). Archived (https://web.archive.org/web/20230122224720/https://doc.rero.ch/record/16156/files/PAL_E3107.pdf) (PDF) from the original on 22 January 2023. Retrieved 3 January 2023.
31. Naish, D.; Martill, D. M.; Frey, E. (June 2004). "Ecology, systematics and biogeographical relationships of dinosaurs, including a new theropod, from the Santana Formation (?Albian, Early Cretaceous) of Brazil". *Historical Biology*. **16** (2–4): 57–70. Bibcode:2004HBio...16...57N (https://ui.adsabs.harvard.edu/abs/2004HBio...16...57N). CiteSeerX 10.1.1.394.9219 (https://citeseerx.ist.psu.edu/viewdoc/summary?doi=10.1.1.394.9219). doi:10.1080/08912960410001674200 (https://doi.org/10.1080%2F08912960410001674200). S2CID 18592288 (https://api.semanticscholar.org/CorpusID:18592288). {{cite journal}}: Cite uses deprecated parameter |citeseerx= (help) This is also one of several topics featured in a post on Naish's blog, "Basal tyrant dinosaurs and my pet *Mirischia*" (https://darrennaish.blogspot.com/2006/06/basal-tyrant-dinosaurs-and-my-pet.html). Archived (https://web.archive.org/web/20101019211637/http://darrennaish.blogspot.com/2006/06/basal-tyrant-dinosaurs-and-my-pet.html) from the original on 19 October 2010. Retrieved 17 April 2008. – note *Mirischia* was a coelurosaur, which Naish believes was closely related to *Compsognathus*.
32. Sereno PC, Martinez RN, Wilson JA, Varricchio DJ, Alcober OA, et al. (September 2008). Kemp T (ed.). "Evidence for Avian Intrathoracic Air Sacs in a New Predatory Dinosaur from Argentina" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2553519). *PLOS ONE*. **3** (9) e3303. Bibcode:2008PLoSO...3.3303S (https://ui.adsabs.harvard.edu/abs/2008PLoSO...3.3303S). doi:10.1371/journal.pone.0003303 (https://doi.org/10.1371%2Fjournal.pone.0003303). PMC 2553519 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2553519). PMID 18825273 (https://pubmed.ncbi.nlm.nih.gov/18825273).
33. Wedel, Mathew (2007). "What pneumaticity tells us about 'prosauropods', and vice versa" (https://web.archive.org/web/20080705145519/http://sauroposeidon.net/Wedel_2007_pneumaticity-and-prosauropods.pdf) (PDF). *Special Papers in Palaeontology*. **77**: 207–222. Archived from the original on 5 July 2008. Retrieved 31 October 2007.

34. Mallison, H. (2010). "The digital *Plateosaurus* II: an assessment of the range of motion of the limbs and vertebral column and of previous reconstructions using a digital skeletal mount" (https://doi.org/10.4202%2Fapp.2009.0075). *Acta Palaeontologica Polonica*. **55** (3): 433–458. doi:10.4202/app.2009.0075 (https://doi.org/10.4202%2Fapp.2009.0075).
35. Reid, R.E.H. (1997). "Dinosaurian Physiology: the Case for "Intermediate" Dinosaurs". In Farlow, J.O.; Brett-Surman, M.K. (eds.). *The Complete Dinosaur* (https://books.google.com/books?id=FOViD-lDPy0C&q=%22The+Complete+Dinosaur%22+%22intermediate+dinosaurs%22&pg=PA449). Bloomington: Indiana University Press. pp. 449–473. ISBN 978-0-253-33349-0. Retrieved 12 September 2008.
36. Paladino, F.V.; Spotila, J.R.; Dodson, P. (1997). "A Blueprint for Giants: Modeling the Physiology of Large Dinosaurs". In Farlow, J.O.; Brett-Surman, M.K. (eds.). *The Complete Dinosaur*. Bloomington, Ind.: Indiana University Press. pp. 491–504. ISBN 978-0-253-21313-6.
37. Ward, Peter (2006) *Out of thin air: Dinosaurs, birds, and earth's ancient atmosphere* (https://books.google.com/books?id=z6iG67mkfZoC&q=%22Out+of+Thin+Air%22) Pages 159–198, National Academies Press. ISBN 9780309141239.
38. Codd, J.R.; Manning, P.L.; Norell, M.A.; Perry, S.F. (January 2008). "Avian-like breathing mechanics in maniraptoran dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2596187). *Proceedings of the Royal Society B*. **275** (1631): 157–161. doi:10.1098/rspb.2007.1233 (https://doi.org/10.1098%2Frspb.2007.1233). PMC 2596187 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2596187). PMID 17986432 (https://pubmed.ncbi.nlm.nih.gov/17986432). News summary at "Why Dinosaurs Had 'Fowl' Breath" (https://www.sciencedaily.com/releases/2007/11/071107074326.htm). 7 November 2007. Archived (https://web.archive.org/web/20160827171625/https://www.sciencedaily.com/releases/2007/11/071107074326.htm) from the original on 27 August 2016. Retrieved 28 February 2018.
39. Tickle, P.G.; Ennos, A.R.; Lennox, L.E.; Perry, S.F.; Codd, J.R. (November 2007). "Functional significance of the uncinate processes in birds" (https://doi.org/10.1242%2Fjeb.008953). *Journal of Experimental Biology*. **210** (Pt 22): 3955–3961. Bibcode:2007JExpB.210.3955T (https://ui.adsabs.harvard.edu/abs/2007JExpB.210.3955T). doi:10.1242/jeb.008953 (https://doi.org/10.1242%2Fjeb.008953). PMID 17981863 (https://pubmed.ncbi.nlm.nih.gov/17981863).
40. Fisher, P.E.; Russell, D.A.; Stoskopf, M.K.; Barrick, R.E.; Hammer, M.; Kuzmitz A.A. (April 2000). "Cardiovascular Evidence for an Intermediate or Higher Metabolic Rate in an Ornithischian Dinosaur". *Science*. **288** (5465): 503–505. Bibcode:2000Sci...288..503F (https://ui.adsabs.harvard.edu/abs/2000Sci...288..503F). doi:10.1126/science.288.5465.503 (https://doi.org/10.1126%2Fscience.288.5465.503). PMID 10775107 (https://pubmed.ncbi.nlm.nih.gov/10775107). But note that this paper's main subject is that the fossil provided strong evidence of a 4-chambered heart, which is not widely accepted.
41. Ruben, J.A.; Hillenius, W.J.; Geist, N.R.; Leitch, A.; Jones, T.D.; Currie, P.J.; Horner, J.R.; Espe, G. (August 1996). "The metabolic status of some Late Cretaceous dinosaurs" (http://doc.rero.ch/record/14694/files/PAL_E1414.pdf) (PDF). *Science*. **273** (5279): 1204–1207. Bibcode:1996Sci...273.1204R (https://ui.adsabs.harvard.edu/abs/1996Sci...273.1204R). doi:10.1126/science.273.5279.1204 (https://doi.org/10.1126%2Fscience.273.5279.1204). S2CID 84693210 (https://api.semanticscholar.org/CorpusID:84693210). Archived (https://web.archive.org/web/20230416181502/https://doc.rero.ch/record/14694/files/PAL_E1414.pdf) (PDF) from the original on 16 April 2023. Retrieved 3 January 2023.
42. Ruben, J.A. & Jones, T.D. (2000). "Selective Factors Associated with the Origin of Fur and Feathers" (https://doi.org/10.1093%2Ficb%2F40.4.585). *American Zoologist*. **40** (4): 585–596. doi:10.1093/icb/40.4.585 (https://doi.org/10.1093%2Ficb%2F40.4.585).
43. Bang, B.G. (1966). "The olfactory apparatus of Procellariiformes". *Acta Anatomica*. **65** (1): 391–415. doi:10.1159/000142884 (https://doi.org/10.1159%2F000142884). PMID 5965973 (https://pubmed.ncbi.nlm.nih.gov/5965973).

44. Bang, B.G. (1971). "Functional anatomy of the olfactory system in 23 orders of birds". *Acta Anatomica*. 79. **79**: 1–76. doi:10.1159/isbn.978-3-318-01866-0 (https://doi.org/10.1159%2Fisbn.978-3-318-01866-0). ISBN 978-3-8055-1193-3. PMID 5133493 (https://pubmed.ncbi.nlm.nih.gov/5133493).
45. Scott, J.H. (1954). "Heat regulating function of the nasal mucous membrane". *Journal of Larynology and Otology*. **68** (5): 308–317. doi:10.1017/S0022215100049707 (https://doi.org/10.1017%2FS0022215100049707). PMID 13163588 (https://pubmed.ncbi.nlm.nih.gov/13163588). S2CID 32082759 (https://api.semanticscholar.org/CorpusID:32082759).
46. Coulombe, H.N.; Sam H. Ridgway; S.H.; Evans, W.E. (1965). "Respiratory water exchange in two species of porpoise". *Science*. **149** (3679): 86–88. Bibcode:1965Sci...149...86C (https://ui.adsabs.harvard.edu/abs/1965Sci...149...86C). doi:10.1126/science.149.3679.86 (https://doi.org/10.1126%2Fscience.149.3679.86). PMID 17737801 (https://pubmed.ncbi.nlm.nih.gov/17737801). S2CID 38947951 (https://api.semanticscholar.org/CorpusID:38947951).
47. Witmer, L.M. (August 2001). "Nostril Position in Dinosaurs and Other Vertebrates and Its Significance for Nasal Function". *Science*. **293** (5531): 850–853. CiteSeerX 10.1.1.629.1744 (https://citeseerx.ist.psu.edu/viewdoc/summary?doi=10.1.1.629.1744). doi:10.1126/science.1062681 (https://doi.org/10.1126%2Fscience.1062681). PMID 11486085 (https://pubmed.ncbi.nlm.nih.gov/11486085). S2CID 7328047 (https://api.semanticscholar.org/CorpusID:7328047). {{cite journal}}: Cite uses deprecated parameter |citeseerx= (help)
48. "Dinosaur breathing study shows that noses enhanced smelling and cooled brain" (http://phys.org/news/2014-10-dinosaur-noses-cooled-brain.html). Archived (https://web.archive.org/web/20141021200417/http://phys.org/news/2014-10-dinosaur-noses-cooled-brain.html) from the original on 21 October 2014. Retrieved 16 October 2014.
49. Fisher, Paul E.; Russell, Dale A.; Stoskopf, Michael K.; Barrick, Reese E.; Hammer, Michael; Kuzmitz, Andrew A. (April 2000). "Cardiovascular evidence for an intermediate or higher metabolic rate in an ornithischian dinosaur". *Science*. **288** (5465): 503–505. Bibcode:2000Sci...288..503F (https://ui.adsabs.harvard.edu/abs/2000Sci...288..503F). doi:10.1126/science.288.5465.503 (https://doi.org/10.1126%2Fscience.288.5465.503). PMID 10775107 (https://pubmed.ncbi.nlm.nih.gov/10775107).
50. Rowe, Timothy; McBride, Earle F.; Sereno, Paul C. (February 2001). "Technical comment: dinosaur with a heart of stone" (https://doi.org/10.1126%2Fscience.291.5505.783a). *Science*. **291** (5505): 783a. doi:10.1126/science.291.5505.783a (https://doi.org/10.1126%2Fscience.291.5505.783a). PMID 11157158 (https://pubmed.ncbi.nlm.nih.gov/11157158).
51. Russell, Dale A.; Fisher, Paul E.; Barrick, Reese E.; Stoskopf, Michael K. (February 2001). "Reply: dinosaur with a heart of stone" (https://doi.org/10.1126%2Fscience.291.5505.783a). *Science*. **291** (5505): 783a. doi:10.1126/science.291.5505.783a (https://doi.org/10.1126%2Fscience.291.5505.783a). PMID 11157158 (https://pubmed.ncbi.nlm.nih.gov/11157158).
52. Chinsamy, Anusuya; and Hillenius, Willem J. (2004). "Physiology of nonavian dinosaurs". *The Dinosauria*, 2nd. 643–659.
53. Carpenter, K.; Hirsch, K.F.; Horner, J.R. (1994). "Introduction". In Carpenter, K.; Hirsch, K.F.; Horner, J.R. (eds.). *Dinosaur Eggs and Babies* (https://books.google.com/books?id=9BPIRx-OpRMC&q=dinsaur+egg+size&pg=PA256). Cambridge University Press. ISBN 978-0-521-56723-7.

54. Erickson, G.M.; Curry Rogers, K.; Yerby, S.A. (July 2001). "Dinosaurian growth patterns and rapid avian growth rates". *Nature*. **412** (6845): 429–433. Bibcode:2001Natur.412..429E (https://ui.adsabs.harvard.edu/abs/2001Natur.412..429E). doi:10.1038/35086558 (https://doi.org/10.1038%2F35086558). PMID 11473315 (https://pubmed.ncbi.nlm.nih.gov/11473315). S2CID 4319534 (https://api.semanticscholar.org/CorpusID:4319534). **(Erratum: doi:10.1038/nature16488 (https://doi.org/10.1038%2Fnature16488), PMID 26675731 (https://pubmed.ncbi.nlm.nih.gov/26675731), *Retraction Watch* (http://retractionwatch.com/2016/03/01/high-profile-critic-slams-nature-letters-about-dinosaur-growth-following-corrections/))** Note Kristina Rogers also published papers under her maiden name, Kristina Curry.
55. Curry, K.A. (1999). "Ontogenetic Histology of *Apatosaurus* (Dinosauria: Sauropoda): New Insights on Growth Rates and Longevity" (https://web.archive.org/web/20100715065946/http://www.vertpaleo.org/publications/jvp/19-654-665.cfm). *Journal of Vertebrate Paleontology*. **19** (4): 654–665. Bibcode:1999JVPal..19..654C (https://ui.adsabs.harvard.edu/abs/1999JVPal..19..654C). doi:10.1080/02724634.1999.10011179 (https://doi.org/10.1080%2F02724634.1999.10011179). Archived from the original (http://www.vertpaleo.org/publications/jvp/19-654-665.cfm) on 15 July 2010.
56. Erickson, G.M.; Makovicky, P.J.; Currie, P.J.; Norell, M.A.; Yerby, S.A.; Brochu, C.A. (August 2004). "Gigantism and comparative life-history parameters of tyrannosaurid dinosaurs" (http://doc.rero.ch/record/15279/files/PAL_E2578.pdf) (PDF). *Nature*. **430** (7001): 772–775. Bibcode:2004Natur.430..772E (https://ui.adsabs.harvard.edu/abs/2004Natur.430..772E). doi:10.1038/nature02699 (https://doi.org/10.1038%2Fnature02699). PMID 15306807 (https://pubmed.ncbi.nlm.nih.gov/15306807). S2CID 4404887 (https://api.semanticscholar.org/CorpusID:4404887). Archived (https://web.archive.org/web/20200714024211/http://doc.rero.ch/record/15279/files/PAL_E2578.pdf) (PDF) from the original on 14 July 2020. Retrieved 22 November 2022. **(Erratum: doi:10.1038/nature16487 (https://doi.org/10.1038%2Fnature16487), PMID 26675726 (https://pubmed.ncbi.nlm.nih.gov/26675726), *Retraction Watch* (http://retractionwatch.com/2016/03/01/high-profile-critic-slams-nature-letters-about-dinosaur-growth-following-corrections/))**
57. Horner, J. R.; Padian, K. (September 2004). "Age and growth dynamics of Tyrannosaurus rex" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1691809). *Proceedings of the Royal Society B*. **271** (1551): 1875–1880. doi:10.1098/rspb.2004.2829 (https://doi.org/10.1098%2Frspb.2004.2829). PMC 1691809 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1691809). PMID 15347508 (https://pubmed.ncbi.nlm.nih.gov/15347508).
58. Cooper, L.N.; Lee, A.H.; Taper, M.L.; Horner, J.R. (August 2008). "Relative growth rates of predator and prey dinosaurs reflect effects of predation" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2605812). *Proceedings of the Royal Society B*. **275** (1651): 2609–2615. doi:10.1098/rspb.2008.0912 (https://doi.org/10.1098%2Frspb.2008.0912). PMC 2605812 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2605812). PMID 18682367 (https://pubmed.ncbi.nlm.nih.gov/18682367).
59. Erickson, G.M.; Currie, P.J.; Inouye, B.D.; Winn, A.A. (July 2006). "Tyrannosaur Life Tables: An Example of Nonavian Dinosaur Population Biology". *Science*. **313** (5784): 213–217. Bibcode:2006Sci...313..213E (https://ui.adsabs.harvard.edu/abs/2006Sci...313..213E). doi:10.1126/science.1125721 (https://doi.org/10.1126%2Fscience.1125721). PMID 16840697 (https://pubmed.ncbi.nlm.nih.gov/16840697). S2CID 34191607 (https://api.semanticscholar.org/CorpusID:34191607).
60. McGowan, C. (1991). *Dinosaurs, Spitfires and Sea Dragons* (https://archive.org/details/dinosaursspitfir00mcgo/page/133). Harvard University Press. pp. 133–135 (https://archive.org/details/dinosaursspitfir00mcgo/page/133). ISBN 978-0-674-20769-1.
61. Mackie, R.I. (2002). "Mutualistic Fermentative Digestion in the Gastrointestinal Tract: Diversity and Evolution" (https://doi.org/10.1093%2Ficb%2F42.2.319). *Integrative and Comparative Biology*. **42** (2): 319–326. doi:10.1093/icb/42.2.319 (https://doi.org/10.1093%2Ficb%2F42.2.319). PMID 21708724 (https://pubmed.ncbi.nlm.nih.gov/21708724).

62. Paul, G.S. (1998). "Cope's rule" (https://books.google.com/books?id=CLEPeg_SjTcC&q=sprawling+limbs+efficient). In Carpenter, K.; Fraser, N.; Chure, D.; Kirkland, J.I. (eds.). *Morrison Symposium Proceedings: A Special Issue Of The Journal Modern Geology*. Taylor & Francis. p. 211. ISBN 978-90-5699-183-8. Retrieved 12 September 2008.
63. Ricqles, A. J. de. (1974). *Evolution of endothermy: histological evidence*. Evolutionary Theory 1: 51–80
64. Fastovsky & Weishampel 2009, p.258.
65. Chinsamy, A.; Rich, T.; Vickers-Rich, P. (1998). "Polar dinosaur bone histology" (https://web.archive.org/web/20100715110202/http://www.vertpaleo.org/publications/jvp/18-385-390.cfm). *Journal of Vertebrate Paleontology*. **18** (2): 385–390. Bibcode:1998JVPal..18..385C (https://ui.adsabs.harvard.edu/abs/1998JVPal..18..385C). doi:10.1080/02724634.1998.10011066 (https://doi.org/10.1080%2F02724634.1998.10011066). Archived from the original (http://www.vertpaleo.org/publications/jvp/18-385-390.cfm) on 15 July 2010.
66. Klevezal, G.A.; Mina, M.V.; Oreshkin, A.V. (1996). *Recording structures of mammals. Determination of age and reconstruction of life history*. CRC Press. ISBN 978-90-5410-621-0.
67. Fastovsky & Weishampel 2009, p.260.
68. Enlow, D.H. (1963). *Principles of Bone Remodeling. An account of post-natal growth and remodeling processes in long bones and the mandible*. Springfield, IL: C.C. Thomas.
69. Reid, R.E.H. (1984). "Primary bone and dinosaurian physiology". *Geological Magazine*. **121** (6): 589–598. Bibcode:1984GeoM..121..589R (https://ui.adsabs.harvard.edu/abs/1984GeoM..121..589R). doi:10.1017/S0016756800030739 (https://doi.org/10.1017%2FS0016756800030739). S2CID 128981835 (https://api.semanticscholar.org/CorpusID:128981835).
70. Reid, R.E.H (1997). "How dinosaurs grew". In Farlow, J.O.; Brett-Surman, M.K. (eds.). *The Complete Dinosaur* (https://archive.org/details/isbn_9780253333490/page/403). Bloomington: Indiana University Press. pp. 403–413 (https://archive.org/details/isbn_9780253333490/page/403). ISBN 978-0-253-33349-0.
71. de Ricqlès, A.; Padian, K.; Knoll, F.; Horner, J.R. (April–June 2008). "On the origin of high growth rates in archosaurs and their ancient relatives: Complementary histological studies on Triassic archosauriforms and the problem of a "phylogenetic signal" in bone histology". *Annales de Paléontologie*. **94** (2): 57–76. Bibcode:2008AnPal..94...57D (https://ui.adsabs.harvard.edu/abs/2008AnPal..94...57D). doi:10.1016/j.annpal.2008.03.002 (https://doi.org/10.1016%2Fj.annpal.2008.03.002). Abstract also online at "The Origin of High Growth Rates in Archosaurs" (https://palaeoblog.blogspot.com/2008/05/origin-of-high-growth-rates-in.html). 31 May 2008. Archived (https://web.archive.org/web/20210318164521/http://palaeoblog.blogspot.com/2008/05/origin-of-high-growth-rates-in.html) from the original on 18 March 2021. Retrieved 3 June 2008.
72. Legendre, Lucas J.; Guénard, Guillaume; Botha-Brink, Jennifer; Cubo, Jorge (1 November 2016). "Palaeohistological Evidence for Ancestral High Metabolic Rate in Archosaurs" (https://academic.oup.com/sysbio/article/65/6/989/2281633). *Systematic Biology*. **65** (6): 989–996. doi:10.1093/sysbio/syw033 (https://doi.org/10.1093%2Fsysbio%2Fsyw033). ISSN 1063-5157 (https://search.worldcat.org/issn/1063-5157). PMID 27073251 (https://pubmed.ncbi.nlm.nih.gov/27073251). Archived (https://web.archive.org/web/20190128202826/https://academic.oup.com/sysbio/article/65/6/989/2281633) from the original on 28 January 2019. Retrieved 2 May 2020.

73. Seymour, Roger S.; Bennett-Stamper, Christina L.; Johnston, Sonya D.; Carrier, David R.; Grigg, Gordon C. (2004). "Evidence for endothermic ancestors of crocodiles at the stem of archosaur evolution" (http://digital.library.adelaide.edu.au/dspace/bitstream/2440/1933/1/hdl1933.pdf) (PDF). *Physiological and Biochemical Zoology*. **77** (6): 1051–1067. doi:10.1086/422766 (https://doi.org/10.1086%2F422766). hdl:2440/1933 (https://hdl.handle.net/2440%2F1933). PMID 15674775 (https://pubmed.ncbi.nlm.nih.gov/15674775). S2CID 10111065 (https://api.semanticscholar.org/CorpusID:10111065). Archived (https://web.archive.org/web/20120904072242/http://digital.library.adelaide.edu.au/dspace/bitstream/2440/1933/1/hdl1933.pdf) (PDF) from the original on 4 September 2012. Retrieved 1 October 2012.
74. Seymour, Roger S. (1976). "Dinosaurs, endothermy, and blood pressure". *Nature*. **262** (5565): 207–208. Bibcode:1976Natur.262..207S (https://ui.adsabs.harvard.edu/abs/1976Natur.262..207S). doi:10.1038/262207a0 (https://doi.org/10.1038%2F262207a0). S2CID 4157084 (https://api.semanticscholar.org/CorpusID:4157084).
75. Seymour, Roger S.; Smith, Sarah L; White, Craig R.; Henderson, Donald M.; Schwarz-Wings, Daniela (2012). "Blood flow to long bones indicates activity metabolism in mammals, reptiles and dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3234558). *Proceedings of the Royal Society B*. **279** (1728): 451–456. doi:10.1098/rspb.2011.0968 (https://doi.org/10.1098%2Frspb.2011.0968). PMC 3234558 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3234558). PMID 21733896 (https://pubmed.ncbi.nlm.nih.gov/21733896).
76. Paul, G.S. (1988). *Predatory Dinosaurs of the World* (https://archive.org/details/predatorydinosau00paul/page/155). New York: Simon and Schuster. p. 155 (https://archive.org/details/predatorydinosau00paul/page/155). ISBN 978-0-671-61946-6.
77. Gillooly, J.F.; Allen, A.P.; Charnov, E.L. (August 2006). "Dinosaur Fossils Predict Body Temperatures" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1489189). *PLOS Biology*. **4** (8) e248. doi:10.1371/journal.pbio.0040248 (https://doi.org/10.1371%2Fjournal.pbio.0040248). PMC 1489189 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1489189). PMID 16817695 (https://pubmed.ncbi.nlm.nih.gov/16817695). There is a less technical summary at Gross, L. (August 2006). "Math and Fossils Resolve a Debate on Dinosaur Metabolism" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1489188). *PLOS Biology*. **4** (8) e255. doi:10.1371/journal.pbio.0040255 (https://doi.org/10.1371%2Fjournal.pbio.0040255). PMC 1489188 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1489188). PMID 20076616 (https://pubmed.ncbi.nlm.nih.gov/20076616).
78. Mackowiak, P. A.; S. S. Wasserman; M. M. Levine (23 September 1992). "A critical appraisal of 98.6 °F (37.0 °C), the upper limit of the normal body temperature, and other legacies of Carl Reinhold August Wunderlich". *JAMA*. **268** (12): 1578–1580. doi:10.1001/jama.1992.03490120092034 (https://doi.org/10.1001%2Fjama.1992.03490120092034). PMID 1302471 (https://pubmed.ncbi.nlm.nih.gov/1302471).
79. Sund-Levander, Märtha; Forsberg, Christina; Wahren, Lis Karin (2002). "Normal oral, rectal, tympanic and axillary body temperature in adult men and women: a systematic literature review". *Scandinavian Journal of Caring Sciences*. **16** (2): 122–128. doi:10.1046/j.1471-6712.2002.00069.x (https://doi.org/10.1046%2Fj.1471-6712.2002.00069.x). PMID 12000664 (https://pubmed.ncbi.nlm.nih.gov/12000664).
80. Erickson, Gregory M.; Rogers, Kristina Curry; Yerby, Scott A. (2001). "Dinosaurian growth patterns and rapid avian growth rates". *Nature*. **412** (6845): 429–433. Bibcode:2001Natur.412..429E (https://ui.adsabs.harvard.edu/abs/2001Natur.412..429E). doi:10.1038/35086558 (https://doi.org/10.1038%2F35086558). PMID 11473315 (https://pubmed.ncbi.nlm.nih.gov/11473315). S2CID 4319534 (https://api.semanticscholar.org/CorpusID:4319534). **(Erratum: doi:10.1038/nature16488 (https://doi.org/10.1038%2Fnature16488), PMID 26675731 (https://pubmed.ncbi.nlm.nih.gov/26675731), *Retraction Watch* (http://retractionwatch.com/2016/03/01/high-profile-critic-slams-nature-letters-about-dinosaur-growth-following-corrections/))**

81. Curry, Kristina A. (1999). "Ontogenetic histology of Apatosaurus (Dinosauria: Sauropoda): New insights on growth rates and longevity". *Journal of Vertebrate Paleontology*. **19** (4): 654–665. Bibcode:1999JVPal..19..654C (https://ui.adsabs.harvard.edu/abs/1999JVPal..19..654C). doi:10.1080/02724634.1999.10011179 (https://doi.org/10.1080%2F02724634.1999.10011179).
82. Fastovsky, David E.; Weishampel, Dvid B. (2009). *Dinosaurs: A Concise Natural History* (http://www.evolbiol.ru/docs/docs/large_files/dinosaurs2009.pdf) (PDF). p. 258. Archived (https://web.archive.org/web/20180706192139/http://www.evolbiol.ru/docs/docs/large_files/dinosaurs2009.pdf) (PDF) from the original on 6 July 2018. Retrieved 6 July 2018.
83. Barrick, R.E. & Showers, W.J. (1999). "Thermophysiology and biology of *Giganotosaurus*: comparison with *Tyrannosaurus*" (https://web.archive.org/web/20110517220816/http://palaeo-electronica.org/1999_2/gigan/issue2_99.htm). *Palaeontologia Electronica*. **2** (2). Archived from the original (http://palaeo-electronica.org/1999_2/gigan/issue2_99.htm) on 17 May 2011. Retrieved 10 September 2008.
84. Barrick, R.E; Showers, W.J.; Fischer, A.G. (August 1996). "Comparison of Thermoregulation of Four Ornithischian Dinosaurs and a Varanid Lizard from the Cretaceous Two Medicine Formation: Evidence from Oxygen Isotopes". *PALAIOS*. **11** (4): 295–305. Bibcode:1996Palai..11..295B (https://ui.adsabs.harvard.edu/abs/1996Palai..11..295B). doi:10.2307/3515240 (https://doi.org/10.2307%2F3515240). JSTOR 3515240 (https://www.jstor.org/stable/3515240).
85. Bakker, R.T. (September 1974). "Dinosaur Bioenergetics – A Reply to Bennett and Dalzell, and Feduccia". *Evolution*. **28** (3): 497–503. doi:10.2307/2407178 (https://doi.org/10.2307%2F2407178). JSTOR 2407178 (https://www.jstor.org/stable/2407178). PMID 28564852 (https://pubmed.ncbi.nlm.nih.gov/28564852).
86. Fastovsky, D.E.; Weishampel, D.B.; Sibbick, J. (2005). *The Evolution and Extinction of the Dinosaurs* (https://books.google.com/books?id=M8nt10bkovIC&q=dinosaur+predator+prey+ratio&pg=PA371). Cambridge University Press. ISBN 978-0-521-81172-9.
87. Farlow, J.O. (1980). "Predator/prey biomass ratios, community food webs and dinosaur physiology" (https://web.archive.org/web/20081217091538/http://bio.fsu.edu/~amarquez/Evolutionary%20Morphology%20fall%202004/Farlow/Predator-Prey%20biomass%20ratios%2C%20community%20food%20webs%20and%20dinosaur%20physiology.pdf) (PDF). In Thomas, R.D.K.; Olson, E.C. (eds.). *A cold look at the warm-blooded dinosaurs*. Boulder, CO: American Association for the Advancement of Science. pp. 55–83. Archived from the original (http://bio.fsu.edu/~amarquez/Evolutionary%20Morphology%20fall%202004/Farlow/Predator-Prey%20biomass%20ratios,%20community%20food%20webs%20and%20dinosaur%20physiology.pdf) (PDF) on 17 December 2008.
88. This was recognized not later than 1909: "Dr. Holland and the Sprawling Sauropods" (https://web.archive.org/web/20110612180650/http://www.hmnh.org/library/diplodocus/holland1910.html). Archived from the original (http://www.hmnh.org/library/diplodocus/holland1910.html) on 12 June 2011. The arguments and many of the images are also presented in Desmond, A. (1976). *Hot Blooded Dinosaurs*. DoubleDay. ISBN 978-0-385-27063-2.
89. Carrier, D.R. (1987). "The evolution of locomotor stamina in tetrapods: circumventing a mechanical constraint". *Paleobiology*. **13** (3): 326–341. Bibcode:1987Pbio...13..326C (https://ui.adsabs.harvard.edu/abs/1987Pbio...13..326C). doi:10.1017/S0094837300008903 (https://doi.org/10.1017%2FS0094837300008903). JSTOR 2400734 (https://www.jstor.org/stable/2400734). S2CID 83722453 (https://api.semanticscholar.org/CorpusID:83722453).
90. Fastovsky & Weishampel 2009, p.251.
91. Fastovsky & Weishampel 2009, p.252.

92. Ostrom, J.H. (1980). "The evidence of endothermy in dinosaurs" (https://web.archive.org/web/20080527192758/http://bio.fsu.edu/~amarquez/Evolutionary%20Morphology%20fall%202004/Ostrom/The%20evidence%20for%20endothermy%20in%20dinosaurs%20-%20Ostrom.pdf) (PDF). In Thomas, R.D.K.; Olson, E.C. (eds.). *A cold look at the warm-blooded dinosaurs*. Boulder, CO: American Association for the Advancement of Science. pp. 82–105. Archived from the original (http://bio.fsu.edu/~amarquez/Evolutionary%20Morphology%20fall%202004/Ostrom/The%20evidence%20for%20endothermy%20in%20dinosaurs%20-%20Ostrom.pdf) (PDF) on 27 May 2008.
93. Pontzer, H. .; Allen, V. .; Hutchinson, J. .; Farke, A. A. (2009). Farke, Andrew Allen (ed.). "Biomechanics of running indicates endothermy in bipedal dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2772121). *PLOS ONE*. **4** (11) e7783. Bibcode:2009PLoSO...4.7783P (https://ui.adsabs.harvard.edu/abs/2009PLoSO...4.7783P). doi:10.1371/journal.pone.0007783 (https://doi.org/10.1371%2Fjournal.pone.0007783). PMC 2772121 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2772121). PMID 19911059 (https://pubmed.ncbi.nlm.nih.gov/19911059).
94. Schweitzer, M.H.; J.A. Watt; R. Avci; L. Knapp; L. Chiappe; M. Norell; M. Marshall. (July 1999). "Beta-keratin specific immunological reactivity in feather-like structures of the Cretaceous alvarezsaurid, *Shuvuuia deserti*" (http://www3.interscience.wiley.com/cgi-bin/abstract/63000744/ABSTRACT). *Journal of Experimental Zoology*. **285** (2): 146–157. Bibcode:1999JEZ...285..146S (https://ui.adsabs.harvard.edu/abs/1999JEZ...285..146S). doi:10.1002/(SICI)1097-010X(19990815)285:2<146::AID-JEZ7>3.0.CO;2-A (https://doi.org/10.1002%2F%28SICI%291097-010X%2819990815%29285%3A2%3C146%3A%3AAID-JEZ7%3E3.0.CO%3B2-A). PMID 10440726 (https://pubmed.ncbi.nlm.nih.gov/10440726).
95. Xu X.; Norell, M.A.; Kuang X.; Wang X.; Zhao Q. & Jia C. (2004). "Basal tyrannosauroids from China and evidence for protofeathers in tyrannosauroids" (http://doc.rero.ch/record/15283/files/PAL_E2582.pdf) (PDF). *Nature*. **431** (7009): 680–684. Bibcode:2004Natur.431..680X (https://ui.adsabs.harvard.edu/abs/2004Natur.431..680X). doi:10.1038/nature02855 (https://doi.org/10.1038%2Fnature02855). PMID 15470426 (https://pubmed.ncbi.nlm.nih.gov/15470426). S2CID 4381777 (https://api.semanticscholar.org/CorpusID:4381777).
96. Benton, Michael J.; Dhouailly, Danielle; Jiang, Baoyu; McNamara, Maria (1 September 2019). "The Early Origin of Feathers" (http://www.sciencedirect.com/science/article/pii/S0169534719301405). *Trends in Ecology & Evolution*. **34** (9): 856–869. Bibcode:2019TEcoE..34..856B (https://ui.adsabs.harvard.edu/abs/2019TEcoE..34..856B). doi:10.1016/j.tree.2019.04.018 (https://doi.org/10.1016%2Fj.tree.2019.04.018). hdl:10468/8068 (https://hdl.handle.net/10468%2F8068). ISSN 0169-5347 (https://search.worldcat.org/issn/0169-5347). PMID 31164250 (https://pubmed.ncbi.nlm.nih.gov/31164250). S2CID 174811556 (https://api.semanticscholar.org/CorpusID:174811556). Archived (https://web.archive.org/web/20190603115654/https://www.sciencedirect.com/science/article/pii/S0169534719301405) from the original on 3 June 2019. Retrieved 25 April 2020.
97. Barrett, Paul M.; Evans, David C.; Campione, Nicolás E. (30 June 2015). "Evolution of dinosaur epidermal structures" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4528472). *Biology Letters*. **11** (6) 20150229. doi:10.1098/rsbl.2015.0229 (https://doi.org/10.1098%2Frsbl.2015.0229). PMC 4528472 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4528472). PMID 26041865 (https://pubmed.ncbi.nlm.nih.gov/26041865).
98. Bonaparte, J.F.; Novas, E.E.; Coria, R.A. (1990). "*Carnotaurus sastrei* Bonaparte, the horned, lightly built carnosaur from the Middle Cretaceous of Patagonia" (https://web.archive.org/web/20080510084320/http://dinosaurs.nhm.org/staff/coriaworks.htm). *Natural History Museum of Los Angeles County Contributions in Science*. **416**: 1–42. Archived from the original (http://dinosaurs.nhm.org/staff/coriaworks.htm) on 10 May 2008.

99. Phd, Andrea Cau (1 March 2016). "Theropoda: Miti e leggende post-moderne sui dinosauri mesozoici: Le squame di Carnotaurus" (https://theropoda.blogspot.com/2016/03/miti-e-leggende-post-moderne-sui.html). *Theropoda*. Archived (https://web.archive.org/web/20220519092654/https://theropoda.blogspot.com/2016/03/miti-e-leggende-post-moderne-sui.html) from the original on 19 May 2022. Retrieved 25 April 2020.
100. Paul, Gregory S. (2016). *The Princeton Field Guide to Dinosaurs - 2nd Edition*. Princeton University Press. ISBN 978-0-691-16766-4.
101. Czerkas, S. A. (1994). Lockley, M.G.; dos Santos, V.F.; Meyer, C.A.; Hunt, A.P. (eds.). "The history and interpretation of sauropod skin impressions". *Gaia*. **10**. Lisbon, Portugal.
102. Dodson, P. & Barlowe, W.D. (1996). *The Horned Dinosaurs: A Natural History* (https://books.google.com/books?id=NsBhg_H5FEwC&q=%22ceratopsian%22+skin&pg=PA31). Princeton University Press. ISBN 978-0-691-05900-6. See also image at "Ceratopsian skin" (https://web.archive.org/web/20140203051057/http://piclib.nhm.ac.uk/piclib/www/image.php?img=96447&frm=ser&search=cat_4). Archived from the original (http://piclib.nhm.ac.uk/piclib/www/image.php?img=96447&frm=ser&search=cat_4) on 3 February 2014.
103. Gosline, A. (16 November 2004). "Dinosaurs' "bulletproof" armour revealed" (https://www.newscientist.com/article/dn6674-dinosaurs-bulletproof-armour-revealed.html). New Scientist.com news service. Archived (https://web.archive.org/web/20150122032507/http://www.newscientist.com/article/dn6674-dinosaurs-bulletproof-armour-revealed.html) from the original on 22 January 2015. Retrieved 8 September 2017.
104. Rich, T.H.; Vickers-Rich, P.; Gangloff, R.A. (February 2002). "Polar Dinosaurs" (http://www.sciencemag.org/cgi/content/summary/295/5557/979). *Science*. **295** (5557): 979–980. doi:10.1126/science.1068920 (https://doi.org/10.1126%2Fscience.1068920). PMID 11834803 (https://pubmed.ncbi.nlm.nih.gov/11834803). S2CID 28065814 (https://api.semanticscholar.org/CorpusID:28065814). Archived (https://web.archive.org/web/20081011202804/http://www.sciencemag.org/cgi/content/summary/295/5557/979) from the original on 11 October 2008. Retrieved 19 April 2008. See also Vickers-Rich, P. & Rich, T.H. (2004). "Dinosaurs of the Antarctic" (https://web.archive.org/web/20080517100516/http://aprn.org/wp-content/uploads/2007/08/dinosaurs-antarctic.pdf) (PDF). Scientific American. Archived from the original (http://aprn.org/wp-content/uploads/2007/08/dinosaurs-antarctic.pdf) (PDF) on 17 May 2008. and Fiorillo, A.R. (2004). "Dinosaurs of Arctic Alaska" (https://web.archive.org/web/20080517100549/http://aprn.org/wp-content/uploads/2007/08/dinosaurs-arctic-alaska.pdf) (PDF). *Scientific American*. **291** (6): 84–91. Bibcode:2004SciAm.291f..84F (https://ui.adsabs.harvard.edu/abs/2004SciAm.291f..84F). doi:10.1038/scientificamerican1204-84 (https://doi.org/10.1038%2Fscientificamerican1204-84). PMID 15597984 (https://pubmed.ncbi.nlm.nih.gov/15597984). Archived from the original (http://aprn.org/wp-content/uploads/2007/08/dinosaurs-arctic-alaska.pdf) (PDF) on 17 May 2008.
105. Hotton, N. (1980). "An alternative to dinosaur endothermy: The happy wanderers". In Thomas, R.D.K.; Olson, E.C. (eds.). *A cold look at the warm-blooded dinosaurs*. Boulder, CO: American Association for the Advancement of Science. pp. 311–350. ISBN 978-0-89158-464-3.
106. Paul, G.S. (1988). *Predatory Dinosaurs of the World* (https://archive.org/details/predatorydinosau00paul/page/160). New York: Simon and Schuster. pp. 160–161 (https://archive.org/details/predatorydinosau00paul/page/160). ISBN 978-0-671-61946-6.
107. Bell, Phil R.; Snively, E. (2008). "Polar dinosaurs on parade: a review of dinosaur migration" (https://doi.org/10.1080%2F03115510802096101). *Alcheringa*. **32** (3): 271–284. Bibcode:2008Alch...32..271B (https://ui.adsabs.harvard.edu/abs/2008Alch...32..271B). doi:10.1080/03115510802096101 (https://doi.org/10.1080%2F03115510802096101).
108. Lloyd, Robin (4 December 2008). "Polar Dinosaurs Endured Cold Dark Winters" (http://www.livescience.com/animals/081204-polar-dinosaurs.html). *LiveScience.com*. Imaginova. Retrieved 11 December 2008.

109. Chinsamy, A.; Rich, T.H.; Vickers-Rich, P. (1998). "Polar dinosaur bone histology". *Journal of Vertebrate Paleontology*. **18** (2): 385–390. Bibcode:1998JVPal..18..385C (https://ui.adsabs.harvard.edu/abs/1998JVPal..18..385C). doi:10.1080/02724634.1998.10011066 (https://doi.org/10.1080%2F02724634.1998.10011066). JSTOR 4523908 (https://www.jstor.org/stable/4523908). See also Leslie, M. (December 2007). "Taking a Dinosaur's Temperature" (https://web.archive.org/web/20081210205003/http://www.smithsonianmag.com/history-archaeology/dinosaur-temperature-200712.html). Smithsonian magazine. Archived from the original (http://www.smithsonianmag.com/history-archaeology/dinosaur-temperature-200712.html) on 10 December 2008. Retrieved 19 April 2008.
110. Woodward, Holly N.; Rich, Thomas H.; Chinsamy, Anusuya; Vickers-Rich, Patricia (2011). "Growth Dynamics of Australia's Polar Dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3149654). *PLOS ONE*. **6** (8) e23339. Bibcode:2011PLoSO...623339W (https://ui.adsabs.harvard.edu/abs/2011PLoSO...623339W). doi:10.1371/journal.pone.0023339 (https://doi.org/10.1371%2Fjournal.pone.0023339). PMC 3149654 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3149654). PMID 21826250 (https://pubmed.ncbi.nlm.nih.gov/21826250).
111. de Bufrenil, V.; Farlow, J.O.; de Riqles, A. (1986). "Growth and function of *Stegosaurus* plates: evidence from bone histology". *Paleobiology*. **12** (4): 459–473. Bibcode:1986Pbio...12..459D (https://ui.adsabs.harvard.edu/abs/1986Pbio...12..459D). doi:10.1017/S0094837300003171 (https://doi.org/10.1017%2FS0094837300003171). JSTOR 2400518 (https://www.jstor.org/stable/2400518). S2CID 89139852 (https://api.semanticscholar.org/CorpusID:89139852).
112. "Stegosaur plates and spikes for looks only" (https://web.archive.org/web/20110927124153/http://www.innovations-report.de/html/berichte/biowissenschaften_chemie/bericht-44440.html). 18 May 2005. Archived from the original (http://www.innovations-report.de/html/berichte/biowissenschaften_chemie/bericht-44440.html) on 27 September 2011.
113. Fastovsky & Weishampel 2009, p.255.
114. Seymour Roger S. (5 July 2013). "Maximal Aerobic and Anaerobic Power Generation in Large Crocodiles versus Mammals: Implications for Dinosaur Gigantothermy" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3702618). *PLOS ONE*. **8** (7) e69361. Bibcode:2013PLoSO...869361S (https://ui.adsabs.harvard.edu/abs/2013PLoSO...869361S). doi:10.1371/journal.pone.0069361 (https://doi.org/10.1371%2Fjournal.pone.0069361). PMC 3702618 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3702618). PMID 23861968 (https://pubmed.ncbi.nlm.nih.gov/23861968).
115. Kermack, D.M. & Kermack, K.A. (1984). *The evolution of mammalian characters*. London: Croom Helm Kapitan Szabo Publishers. p. 149. ISBN 978-0-7099-1534-8.

116. The earliest clear evidence of hair or fur is in fossils of *Castorocauda*, from 164M years ago in the mid Jurassic, Ji, Q.; Luo, Z-X; Yuan, C-X; Tabrum, A.R. (February 2006). "A Swimming Mammaliaform from the Middle Jurassic and Ecomorphological Diversification of Early Mammals" (http://www.sciencemag.org/cgi/content/abstract/311/5764/1123). *Science*. **311** (5764): 1123–7. Bibcode:2006Sci...311.1123J (https://ui.adsabs.harvard.edu/abs/2006Sci...311.1123J). doi:10.1126/science.1123026 (https://doi.org/10.1126%2Fscience.1123026). PMID 16497926 (https://pubmed.ncbi.nlm.nih.gov/16497926). S2CID 46067702 (https://api.semanticscholar.org/CorpusID:46067702). Archived (https://web.archive.org/web/20091021130220/http://sciencemag.org/cgi/content/abstract/311/5764/1123) from the original on 21 October 2009. Retrieved 19 April 2008. See also the news item at "Jurassic "Beaver" Found; Rewrites History of Mammals" (https://web.archive.org/web/20060303071809/http://news.nationalgeographic.com/news/2006/02/0223_060223_beaver.html). Archived from the original (http://news.nationalgeographic.com/news/2006/02/0223_060223_beaver.html) on 3 March 2006. It has been argued since the 1950s at the latest that there may be evidence of hair in early-Triassic cynodonts such as *Thrinaxodon*: Brink, A.S. (1955). "A study on the skeleton of *Diademodon*". *Palaeontologia Africana*. **3**: 3–39. and Kemp, T.S. (1982). *Mammal-like reptiles and the origin of mammals* (https://archive.org/details/mammallikereptil00kemp). London: Academic Press. p. 363 (https://archive.org/details/mammallikereptil00kemp/page/n191). ISBN 978-0-12-404120-2. but the foramina (small tunnels) in cynodont snout bones are ambiguous evidence at best, since similar foramina are found in a few living reptiles: Estes, R. (1961). "Cranial anatomy of the cynodont reptile *Thrinaxodon liorhinus*". *Bulletin of the Museum of Comparative Zoology* (1253): 165–180. and Ruben, J.A. & Jones, T.D. (2000). "Selective Factors Associated with the Origin of Fur and Feathers" (https://doi.org/10.1093%2Ficb%2F40.4.585). *American Zoologist*. **40** (4): 585–596. doi:10.1093/icb/40.4.585 (https://doi.org/10.1093%2Ficb%2F40.4.585).

117. McGowan, C. (1991). *Dinosaurs, Spitfires and Sea Dragons* (https://archive.org/details/dinosaursspitfir00mcgo/page/151). Harvard University Press. pp. 151–152 (https://archive.org/details/dinosaursspitfir00mcgo/page/151). ISBN 978-0-674-20769-1.

118. "Secret of the crocodile heart" (http://www.abc.net.au/science/news/stories/s167223.htm). *Australian Broadcasting Corporation*. Archived (https://web.archive.org/web/20210318164751/https://www.abc.net.au/science/news/stories/s167223.htm) from the original on 18 March 2021. Retrieved 31 March 2008.

119. Summers, A.P. (2005). "Evolution: Warm-hearted crocs" (https://doi.org/10.1038%2F434833a). *Nature*. **434** (7035): 833–834. Bibcode:2005Natur.434..833S (https://ui.adsabs.harvard.edu/abs/2005Natur.434..833S). doi:10.1038/434833a (https://doi.org/10.1038%2F434833a). PMID 15829945 (https://pubmed.ncbi.nlm.nih.gov/15829945). S2CID 4399224 (https://api.semanticscholar.org/CorpusID:4399224).

120. Seymour, R. S.; Bennett-Stamper, C. L.; Johnston, S. D.; Carrier, D. R.; Grigg, G. C. (2004). "Evidence for endothermic ancestors of crocodiles at the stem of archosaur evolution" (https://espace.library.uq.edu.au/view/UQ:71147/UQ71147_OA.pdf) (PDF). *Physiological and Biochemical Zoology*. **77** (6): 1051–1067. doi:10.1086/422766 (https://doi.org/10.1086%2F422766). hdl:2440/1933 (https://hdl.handle.net/2440%2F1933). PMID 15674775 (https://pubmed.ncbi.nlm.nih.gov/15674775). S2CID 10111065 (https://api.semanticscholar.org/CorpusID:10111065). See also the explanation of this, with useful illustrations, at Myers, P.Z. (19 April 2005). "Hot-blooded crocodiles?" (https://web.archive.org/web/20080216115144/http://pharyngula.org/index/weblog/comments/hot_blooded_crocodiles/). Archived from the original (http://pharyngula.org/index/weblog/comments/hot_blooded_crocodiles/) on 16 February 2008.

121. Hutchinson, J.R. (March–April 2006). "The evolution of locomotion in archosaurs" (http://doc.rero.ch/record/15518/files/PAL_E2922.pdf) (PDF). *Comptes Rendus Palevol*. **5** (3–4): 519–530. Bibcode:2006CRPal...5..519H (https://ui.adsabs.harvard.edu/abs/2006CRPal...5..519H). doi:10.1016/j.crpv.2005.09.002 (https://doi.org/10.1016%2Fj.crpv.2005.09.002). Archived (https://web.archive.org/web/20230214150906/https://doc.rero.ch/record/15518/files/PAL_E2922.pdf) (PDF) from the original on 14 February 2023. Retrieved 3 January 2023.

External links

- Thermophysiology and Biology of Giganotosaurus: Comparison with Tyrannosaurus (http://palaeo-electronica.org/1999_2/gigan/issue2_99.htm) Archived (https://web.archive.org/web/20110517220816/http://palaeo-electronica.org/1999_2/gigan/issue2_99.htm) 17 May 2011 at the Wayback Machine by RE Barrick and WJ Showers (1999)
- Heart of a Dinosaur Is Reported Found (https://www.nytimes.com/library/national/science/042100sci-paleo-dinosaur.html)
- Crocodile evolution no heart-warmer (http://www.adelaide.edu.au/adelaidean/issues/5501/news5550.html)

Retrieved from "https://en.wikipedia.org/w/index.php?title=Physiology_of_dinosaurs&oldid=1373334387"

WIKIPEDIA
25 years of the free encyclopedia

Dinosaur egg

Dinosaur eggs are the organic vessels in which a dinosaur embryo develops. When the first scientifically documented remains of non-avian dinosaurs were being described in England during the 1820s, it was presumed that dinosaurs had laid eggs because they were reptiles.[1] In 1859, the first scientifically documented dinosaur egg fossils were discovered in France by Jean-Jacques Pouech, although they were mistaken for giant bird eggs (birds were not yet recognized as dinosaurs at the time).

The first scientifically recognized non-avian dinosaur egg fossils were discovered in 1923 by an American Museum of Natural History crew in Mongolia. Dinosaur eggshell can be studied in thin section and viewed under a microscope. The interior of a dinosaur egg can be studied using CAT scans or by gradually dissolving away the shell with acid. Sometimes the egg preserves the remains of the developing embryo inside. The oldest known dinosaur eggs and embryos are from *Massospondylus*, which lived during the Early Jurassic, about 190 million years ago.[2][3]

Fossilized sauropod eggs displayed at Indroda Dinosaur and Fossil Park

History

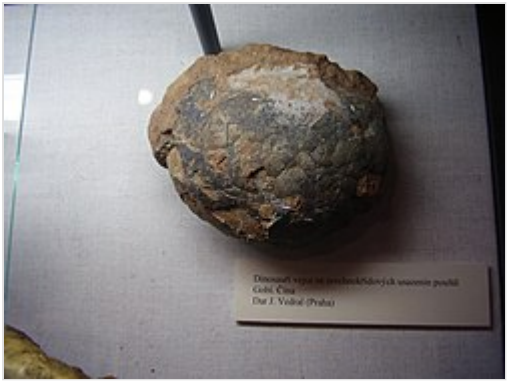
In 1859, the first scientifically documented dinosaur egg fossils were discovered in southern France by a Catholic priest and amateur naturalist named Father Jean-Jacques Pouech; he thought, however, that they were laid by giant birds.[4] The first scientifically recognized dinosaur egg fossils were discovered serendipitously in 1923 by an American Museum of Natural History crew while looking for evidence of early humans in Mongolia.[5] These eggs were mistakenly attributed to the locally abundant herbivore *Protoceratops*, but are now known to be *Oviraptor* eggs. Egg discoveries continued to mount all over the world, leading to the development of multiple competing classification schemes.

A *Citipati osmolskae* egg with preserved embryo, at the AMNH.

Identification

Fossil dinosaur eggshell fragments can be recognized based on three important traits. Their thickness should be roughly uniform, they are usually slightly curved, and their surface is covered in tiny pores. Less frequently, the concave underside of the eggshell fragment will preserve bumps known as

mammillae. Sometimes the embryo had absorbed so much of the calcium that the mammillae need a magnifying glass or microscope to be seen.[6] However, there are many kinds of naturally occurring objects which can resemble fossil eggs. These can fool even professional paleontologists.[7]



Fossilized dinosaur egg from the Gobi Desert, National Museum in Prague

False eggs

Calculus: Calculi are egg-like objects formed in the stomachs of ruminants such as cattle, deer, elk, and goats. Calculus formation is a defense mechanism protecting the ruminant's stomach from damage if it swallows a foreign object while grazing. After ingestion, the object is covered by the same material composing bone, calcium phosphate, and eventually vomited out of the animal's system. These "stomach stones" tend to range in size from 1 to 6 centimeters. Larger sizes are known but very rare.[8] Sometimes tiny dimples cover the surface of a stomach stone, which can fool observers into thinking they are the pores of an egg.[9] Fossil egg expert Ken Carpenter has described stomach stones as the most egg-like natural objects, noting that they are "the trickiest [egg-like] objects to correctly identify".[10] Calculi are so egg-like that on one occasion a detailed description of a stomach stone misidentified as a fossil egg was published in the scientific literature.[9] Calculi can be distinguished from real egg fossils because when they are broken open, they show the layers of calcium phosphate and the foreign object at the core.[9] Multiple layers of eggshell are known in pathological eggs, but these layers don't go all the way down to its core the way a stomach stone's do. Calculi are often suspiciously intact, unlike fossil eggs, which are usually damaged.[8] Stomach stones also lack distinct shells with their attending structural components like continuous or prismatic layers, mammillae, and pores.[9]

Concretions: Concretions are formed when decaying organisms change the chemistry of their immediate surroundings in a manner that is conducive to minerals precipitating out of solution. These minerals accumulate in a mass roughly shaped like the region of altered chemistry. Sometimes the mass produced is egg-shaped.[11] Most egg-shaped concretions have uniform interiors, however some form through the accumulation of mineral in layers.[12] These layered concretions can be even harder to recognize than those with uniform interiors because the layers can resemble egg white and yolk. The yellow of the false yolk comes from minerals like limonite, siderite, and sulfur.[9]

Concretions also generally lack distinct shells, although sometimes they can appear to have them if their outside surfaces have been case-hardened. Since their interiors are softer, erosion can separate the two, creating eggshell pseudofossils. Real egg fossils should preserve eggshell structures like pores, mammillae, and prismatic or continuous layers, which are not present in concretions. Any given concretion is unlikely to be exactly the same size as any other, so associations of egg-like objects of different sizes are probably not real eggs at all. Concretions can also be far larger than any real egg so an apparently unnaturally large "egg" has probably been misidentified.[9]

Insect trace fossils: Sometimes the living or breeding chambers of an insect burrow are so perfectly egg-shaped that even a paleontologist can mistake a natural cast of these chambers for a fossil egg. Insect burrow fossils can sometimes be distinguished from real egg fossils by the presence of "scratch marks" on their surface left by the insect during the burrow's original excavation. Fossil insect pupae can also

resemble eggs. After death and burial, the decomposition of a deceased pupa would leave a gap in the sediment that could be filled with minerals carried by groundwater, forming an egg-like cast. These pseudo-eggs can be recognized by their small size (usually not much longer than a centimeter or two) and lack of an eggshell with its typical anatomy.[9]

Stones: The erosive effects of water can sometimes round rocks into egg-like shapes.[11]

Structure

Paleontologists' knowledge of the structure of dinosaur eggs is limited to the hard shell. However, it can be inferred that dinosaur eggs had an amnion, chorion, and an allantois, the three major membranes in modern bird and reptile eggs. Dinosaur eggs vary greatly in size and shape, but even the largest non-avian dinosaur eggs (*Megaloolithus*) are smaller than the largest known bird eggs, which were laid by the extinct elephant bird. Dinosaur eggs range in shape from spherical to highly elongated (some specimens three times longer than they are wide). Some elongated eggs are symmetrical, whereas others have one rounded end and one pointed end (similar to bird eggs). Most elongated eggs were laid by theropods and have an avian-like eggshell, whereas the spherical eggs typically represent non-theropod dinosaurs.[13]

Fossil dinosaur eggshells, like modern bird and reptile eggshells, are made up of calcium carbonate crystal units. The basic arrangement and structure of these eggshell units (called the ultrastructure) is used to divide fossil eggs into several basic types, including the spherulitic, prismatic, and ornithoid basic types, which contain dinosaurs.[14] Dinosaur eggs further classified by the microstructural aspects of the crystalline structure of the eggshell units and by the type of their pores and their shell ornamentation.[15]

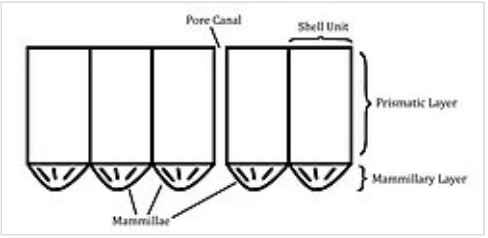


Diagram of a two-layered eggshell.

Layers

Dinosaur eggshells are divided into one, two, or three layers of distinct ultrastructure.[15][16][17][18]

The innermost layer, known as the mammillary layer or the cone layer, is only found in theropod eggs (the prismatic and ornithoid basic types). It is composed of cone-shaped structures called mammillae at the base of each shell unit. Mammillae are the first part of the eggshell to form. Each mammilla forms from crystals radiating outward from an organic core until they touch neighboring mammillae and grow upwards into the next layer.[13][15] In spherulitic eggs, the eggs of non-theropod dinosaurs, the eggshell units grow upward from their organic cores; the base of each eggshell unit is rounded, but is not a true mammilla because it does not have a distinct ultrastructure from the top of the unit.[13][14]

The second layer is alternately called the prismatic layer, the columnar layer, the continuous layer, the crystalline layer,[13] the cryptoprismatic layer,[19] the palisade layer,[15] the spongy layer,[20] or the single layer.[21] In this layer, the shell units can be distinct, partially fused together, or entirely continuous.[14] In some dinosaur eggs, the prismatic layer exhibits squamatic ultrastructure, where the prismatic structure is obscured by a rough texture resembling lizard skin.[15][14]

Though rare in non-avian dinosaurs, some theropod eggs and most bird eggs have a third layer (known as the external layer) made up of vertical calcite crystals.[15][13]

Pore canals

In all eggs, the embryo must breathe. In egg-laying amniotes (including dinosaurs), pore canals cutting through the eggshell allow gas exchange between the embryo and the outside world. Dinosaur eggshells exhibit a lot of diversity in pore size, density, and shape. One early attempt at classification of dinosaurian eggs, proposed by the Soviet paleontologist A. Sochava, was based on grouping eggs by their pore systems.[22] This system was abandoned when it was discovered that different eggs could have very similar pores, but pore systems continue to play an important role in modern eggshell parataxonomy.[14] The density and width of the pores, combined with the eggshell's thickness can be used to predict the gas conductance of a dinosaur's egg.[15] This can provide both information about nesting behavior and about the climate: eggs buried in sediment have higher rates of gas conductance than those laid in the open, and eggs laid in arid environments have lower gas conductance (to prevent water loss) than those laid in more humid conditions.[23]

Paleontologist and fossil egg expert Kenneth Carpenter catalogued six types of pore systems:[14]

1. Angusticanaliculate - Long, narrow, straight pores with low pore density. These eggs would have a low gas exchange rate, and therefore they were typically laid in dry areas.[14]
2. Tubocanaliculate - Large diameter pores with funnel-shaped openings on both inner and outer surfaces of the shell. These eggs would have a high gas exchange rate, and therefore were probably buried in humid mounds.[14]
3. Multicanaliculate - Numerous large, branching, and closely spaced pore canals. They have a high gas exchange rate, so like tubocanaliculate eggs they were probably also buried humid mounds.[14]
4. Prolatocanaliculate - Pores vary in width throughout their length. Gas exchange water loss rates are variable, so these eggs could have been laid in many different environments. This type is subdivided into foveocanaliculate with larger pore openings, and lagenocanaliculate with narrower pore openings.[14]
5. Rimocanaliculate - Very narrow slitlike pore canals. This pore system is seen in modern ostriches, so these eggs were laid in open nests, similar to how ostriches do today.[14]
6. Obliquicanaliculate - These canals cut diagonally through multiple eggshell units instead of going between them like in other pore systems. Obliquicanaliculate pores are only found in a single oogenus: *Preprismatoolithus*.[14]

Ornamentation

Unlike most modern eggs, many non-avian dinosaur eggs had a rough texture formed by nodes and ridges ornamenting the surface of their shell.[15] This is predominant in Cretaceous dinosaur eggs, but very rare in eggs from the Jurassic or Triassic.[24] Because of the lack of modern analogues, the purpose of eggshell ornamentation is unknown,[15] but many functions have been proposed.[24] Possibly, they provided extra strength to the eggshell without having pore canals too long for adequate gas exchange. They could also have helped keep substrate away from the pore openings of eggs that were buried, but modern turtles and crocodylians which bury their eggs have smooth eggshells, so this adaptation is not necessary for animals which bury their eggs. Another hypothesis, proposed by R. M. Mellon in 1982 in

his senior thesis at Princeton University, is that the ridges and nodes would have formed pathways for gas to flow across the surface of the eggshell, preventing accumulation of too much CO_2 and aiding the flow of oxygen and water vapor.[24]

The dinosaur nest, a reconstruction from the National History Museum in London.

Since it varies from egg to egg, the texture of an eggshell's ornamentation is useful for classification. Six types of ornamentation were catalogued by Carpenter in 1999:[14]

1. Compactituberculate - The dome-shaped tops of the shell units form a dense covering of nodes on the surface of the eggshell. This type of ornamentation is most commonly seen in megaloolithids.[25]
2. Sagenotuberculate - The nodes and ridges form a netlike pattern interspersed with pits and grooves.
3. Dispersituberculate - Scattered nodes. This ornamentation is seen on the poles of elongated eggs, which may have allowed accumulations CO_2 at the poles to escape between the nodes.[24]
4. Lineartuberculate - Ridges, and chains of ridges and nodes form lines parallel to the long axis of the egg.
5. Ramotuberculate - Irregular chains of nodes, typically found as a transition between the lineartuberculate midsection and dispersituberculate ends of elongated eggs.
6. Anastomotuberculate - Ridges similar to lineartuberculate, but instead form wavy, branching, or anastomosing patterns resembling the water ripple marks in sand.

Classification

The classification of dinosaur eggs is based on the structure of the egg shells viewed in thin section via microscope, although new techniques such as electron backscatter diffraction have been used.[26] There are three main categories of dinosaur eggs: spherulitic (sauropods and hadrosaurs),[27] prismatic,[28] and ornithoid (theropods, including modern birds).[29]

Oogenera

Oogenera are taxonomic names for types of eggshell. Nearly three dozen oogenera have been named for dinosaur eggs:

- *Ageroolithus*
- *Apheloolithus*[30]
- *Boletuoolithus*[31]
- *Cairanoolithus*[32]
- *Continuoolithus*[30]
- *Dendroolithus*[30]
- *Dictyoolithus*[33]
- *Dispersituberoolithus*
- *Ellipsoolithus*[34]
- *Elongatoolithus*[35]
- *Faveoolithus*[30]
- *Heishanoolithus*[36]
- *Laevisoolithus*
- *Macroolithus*[37]
- *Macroelongatoolithus*[38]
- *Megaloolithus*[39]
- *Nanshiungoolithus*[40]
- *Oblongoolithus*
- *Ovaloolithus*[41]
- *Pachycorioolithus*[42]

- *Paraspheroolithus*[43]
- *Phaceloolithus*[30]
- *Placoolithus*[44]
- *Porituberoolithus*
- *Polyclonoolithus*[45]
- *Preprismatoolithus*[30]
- *Prismatoolithus*[46]
- *Protoceratopsidovum*[47]
- *Pseudogeckoolithus*
- *Shixingoolithus*[48]
- *Sphaerovum*[30]
- *Spheroolithus*[49]
- *Spheruprismatoolithus*[47]
- *Stromatoolithus*[30]
- *Subtiliolithus*[30]
- *Tacuarembovum*[30]
- *Trachoolithus*[50]
- *Tristraguloolithus*
- *Youngoolithus*[30]

Therizinosaur nest and eggs (*cf. Segnosaurus*) in from *Dinosaurland* in Lyme Regis, England.

Embryos

Non-avian dinosaur embryos, the animal inside the eggs, are very rare but useful to understand ontogeny, heterochrony, and dinosaur systematics. Embryo fossils are known from:

"Baby Yingliang", an especially well-preserved specimen of embryo-bearing Oviraptorid egg

- *Beibeilong*
- *Citipati*[51]
- *Heyuannia*
- *Hypacrosaurus*
- *Lufengosaurus*
- *Lourinhanosaurus*[52][53]
- *Massospondylus*[54]
- *Maiasaura*
- *Qianlong shouhu*
- *Troodon*[55]

Taphonomy

The formation of fossil eggs begins with the original egg itself. Not all eggs that end up fossilizing experience the death of their embryo beforehand. Fossil eggs with open tops are common and could result from the preservation of eggs that hatched successfully.[56] Dinosaur eggs whose embryos died were

likely victims of similar causes to those that kill embryos in modern reptile and bird eggs. Typical causes of death include congenital problems, diseases, suffocation from being buried too deep, inimical temperatures, or too much or too little water.[57]

Whether or not hatching was successful, burial would begin with sediments gradually entering any large openings in the shell.[56] Even intact eggs are likely to fill with sediment once they crack under the strain of deep burial. Sometimes, though, fossilization can begin fast enough to prevent the eggs from being cracked. If the water table is high enough dissolved minerals like calcite can percolate through the pores of the eggshell. When the egg is completely filled it can become sturdy enough to withstand the weight of the overlying sediments.[57] Not all fossil egg specimens are of complete specimens, however. Individual pieces of eggshell are much more robust than the entire egg and can be transported intact long distances from where they were originally laid.[58]

When the egg is buried deeply enough, the bacteria decomposing it no longer have access to oxygen and need to power their metabolisms with different substances. These physiological changes in the decomposers also alter the local environment in a way that allows certain minerals to be deposited, while others remain in solution.[57] Generally, however, a fossilizing egg's shell keeps the same calcite it had in life, which allows scientists to study its original structure millions of years after the developing dinosaur hatched or died.[59] However, eggs can also sometimes be altered after burial. This process is called diagenesis.[59] One form of diagenesis is a microscopic cross-hatched pattern imposed on the eggshell by the pressure of being buried deeply.[60] If the pressure gets severe enough, sometimes the eggshell's internal microscopic structure can be completely destroyed. Diagenesis can also happen chemically in addition to physically. The chemical conditions of a decomposing egg can make it easy for silica to be incorporated into eggshell and damage its structure. When iron-bearing substances alter eggshell it can be obvious because compounds like hematite, pyrite, and iron sulfide can turn the shell blackish or rusty colors.[61]

Depositional environments

Dinosaur eggs are known from a variety of depositional environments.

Beach sands: Beach sands were a good place for dinosaurs to lay their eggs because the sand would be effective at absorbing and holding enough heat to incubate the eggs. One ancient beach deposit in northeastern Spain actually preserves about 300,000 fossil dinosaur eggs.[62]

Floodplains: Dinosaurs often laid their eggs on ancient floodplains. The mudstones deposited at these sites are therefore excellent sources of dinosaur egg fossils.[58]

Sand dunes: Many dinosaur eggs have been recovered from sandstone deposits that formed in the ancient dune fields of what are now northern China and Mongolia.[63] The presence of *Oviraptor* preserved in their life brooding position suggests that the eggs, nests, and parents may have been rapidly buried by sandstorms.[62]

Excavation and preparation

Usually the first evidence of fossil dinosaur eggs to be discovered are shell fragments that have eroded away from the original eggs and been transported downhill by the elements.[6] If the source eggs can be found the area must be examined for more unexposed eggs. If the paleontologists are fortunate enough to have found a nest, the number and arrangement of the eggs must be estimated. Excavation must proceed to significant depth since many dinosaur nests include multiple layers of eggs. As the underside of the nest is excavated, it would be covered by material like newspaper, tin foil, or tissue. Afterwards, the entire block is covered in multiple layers of plaster-soaked strips of burlap. When the plaster is dried, the block is undercut the rest of the way and turned over.[64]

The fine work of cleaning the egg fossils is performed in a laboratory. Preparation usually begins from the underside of the block, which tends to be the best preserved.[64] Because of their fragility, cleaning fossil eggs requires patience and skill.[65] Scientists use delicate instruments like dental picks, needles, small pneumatic engraving tools, and X-Acto knives.[64] Scientists must determine at what point to stop cleaning based on their own criteria. If eggs are fully extracted they can be more fully studied individually at the cost of information regarding the spatial relationships between eggs or if the eggs had hatched. Commercial fossil dealers tend to expose only the bottom of the eggs since the topsides might be damaged by hatching and therefore less visually appealing to potential customers.[66]

Research techniques

Acid dissolution

Acids can be used to learn more about fossil eggs. Diluted acetic acid or EDTA can be used to expose the microstructure of shell that has been damaged by weathering. Acids are also used to extract embryo skeletons from the egg encasing them.[67] Even fossilized soft tissue like muscle and cartilage as well as fat globules from the original egg yolk can be uncovered using this method.[68] Amateur paleontologist Terry Manning has been credited with groundbreaking work developing this technique.[69] First, the paleontologist must submerge the egg in a very dilute phosphoric acid bath. Since the acid solution can penetrate the egg, every few days the specimen must be soaked in distilled water to prevent the acid from damaging the embryo before it is even exposed. If embryonic fossil bone is revealed after drying from the water bath, the exposed fossils must be delicately cleaned with fine instruments like needles and paint brushes. The exposed bone is then coated with plastic preservatives like Acryloid B67, Paraloid B72, or Vinac B15 to protect it from the acid when submerged for another round. The complete process can take months before the whole embryo is revealed.[67] Even then only about 20% of the eggs subjected to the process reveal any embryo fossils at all.[70]

CAT scans

CAT scans can be used to infer the 3D structure of a fossil egg's interior by compiling images taken of slices through the egg in small regular increments. Scientists have tried to use CAT scans to look for embryo fossils contained inside the egg without having to damage the egg itself by physically extracting them. However, as of Ken Carpenter's 1999 book on dinosaur eggs, *Eggs, Nests, and Baby Dinosaurs*, all

alleged embryos discovered using this method were actually false alarms. Variations in the type of infilling mineral or cement binding the infilling sediment into rock sometimes resemble bones in CAT scan images. Sometimes eggshell fragments that fell back into the egg when it hatched have been mistaken for embryonic bones.[67][71] The use of CAT scans to search for embryonic remains is actually conceptually flawed since embryonic bones have not yet mineralized. Since the infilling sediment is their only source of minerals they will be preserved at basically the same density and therefore have poor visibility in the scan. The validity of this issue has been confirmed by performing Cat scans on fossil eggs known to have embryos inside and noting their poor visibility in the scan images. The only truly reliable way to discover a dinosaur embryo is to cut the egg open or dissolve some of its eggshell away.[67]

Cathodoluminescence

Cathodoluminescence is the most important tool paleontologists have for revealing whether or not the calcium in fossil eggshell has been altered.[72] Calcite in eggshell is either pure or rich in calcium carbonate. However, the calcite composing the egg can be altered after burial to include significant calcium content. Cathodoluminescence causes calcite altered in this fashion to glow orange.[73]

Gel electrophoresis

Gel electrophoresis has been used in attempts to identify the amino acids present in the organic components of dinosaur eggshell. Contact with human skin can contaminate eggs with foreign amino acids, so only untouched eggs can be investigated using this technique. EDTA can be used to dissolve the calcite of the eggshell while leaving the shell's organic content intact. The resultant organic residue would be blended and then implanted into gel. Electricity would then be run through the sample, causing the amino acids to migrate through the gel until they stop at levels determined by their physical properties. Protein silver stain is then used to dye the amino acids and make them visible.[72] The bands of amino acids from the dinosaur eggs can then be compared with the banding of samples with known composition for identification.[72]

Gel electrophoresis is not necessarily a perfect means of discovering the amino acid composition of dinosaur eggshell because sometimes the amount or type of amino acids present could be altered during or after preservation. One potential confounding factor would be the heating of deeply buried egg fossils, which can break down amino acids. Another potential source of error is groundwater, which can leach away amino acids. These issues cast doubt as to whether the results these sorts of studies give are reliable as the actual composition of the eggshell's organic material in life. However, studies applying these techniques have made suggestive findings, including amino acid profiles in dinosaur eggs similar to those in modern birds.[72]

Geneva lens measure

The Geneva Lens Measure is a device used to measure curved surfaces. It is most commonly used by opticians to measure lenses but can also be used by paleontologists to estimate the life size of dinosaur eggs from shell fragments. The instrument can be used to help estimate the size of fossil eggshells by measuring their curved surfaces. Since most eggs aren't perfectly round measurements from multiple parts of the egg with varying shell curvatures may be needed to get a full idea of the egg's size. Ideally an eggshell fragment being used to estimate the full size of an egg should be more than 3 cm long. Smaller eggshell fragments are better suited to other methods of study, like the Obrig radius dial gauge. The

Geneva Lens measure gives units in diopters which must be converted to the radius in millimeters. Use of the Geneva Lens Measure to estimate the size of a fossil egg was first done by Sauer on fossil ostrich eggs.[73]

Light microscopy

Light microscopy can be used to magnify the structure of dinosaur eggshell for scientific research. To do so an eggshell fragment must be embedded in epoxy resin and sliced into a thin section with a thin-bladed rock saw. This basic method was invented by French paleontologist Paul Gervais and has remained almost unchanged ever since. Horizontally cut thin sections are called tangential thin sections while vertically cut thin sections are called radial sections. Regardless of direction, the sample must be abraded by fine-grit sand or emery paper until it is translucent. Then the structure of the shell's calcite crystals or pores can be examined under a petrographic microscope.[74] The calcite crystal structure of dinosaur eggshell can be classified by their effect on polarized light. Calcite is capable of acting as a polarizing light filter.[75] When a microscopic thin section sample is rotated relative to polarized light it can eventually block all the light and seem opaque. This phenomenon is called extinction. Different varieties of dinosaur eggs with their different calcite crystal structures have different light extinction properties that can be used to identify and distinguish even eggs that seem very similar on the surface.[76] To reconstruct the three-dimensional structures of the shell's pore channels scientists require a series of multiple radial sections.[74]

Scanning electron microscopy

Scanning electron microscopy is used to view dinosaur eggshell under even greater magnification than is possible with light microscopy. However, this does not mean that scanning electron microscopy is necessarily the superior research method. Since both techniques provide differing amounts and types of information they can be used together synergistically to provide a more complete understanding of the specimen under scrutiny. Eggshell specimens best suited for scanning electron microscopy are those recently broken because such a break will usually occur along the plane of the eggshell's calcite crystal lattice. First, a small specimen would be covered with a very thin layer of gold or platinum. The specimen would then be bombarded with electrons. The electrons bounce back off the metal and due to their small size, can be used to form a detailed image of the specimen.[76]

Mass spectrometry

Mass spectrometry is a method for determining eggshell composition that uses a device called a mass spectrometer. First, the eggshell sample must be powdered and placed in the mass spectrometer's vacuum chamber.[68] The powder is vaporized by the heat of an intense laser beam. A stream of electrons then bombard the gaseous eggshell molecules, which breaks down the molecules in the eggshell and imbues them with a positive charge. A magnetic field then sorts them by mass before they are detected by the spectrometer.[77] One application of mass spectrometry has been to study the isotope ratios of dinosaur eggshell in order to ascertain their diets and living conditions. However this research is complicated by the fact that isotope ratios can be altered post mortem before or during fossilization. Bacterial decomposition can alter carbon isotope ratios in eggs and groundwater can alter the oxygen isotope ratios of eggshell.[78] More recently, uranium–lead (U–Pb) mass spectrometry has been applied directly to carbonate within dinosaur eggshells, providing absolute age estimates of egg-bearing strata.[79]

X rays

X-ray equipment, like CAT scans, are used to study the interior of fossil eggs. Unlike CAT scans, x-ray imaging condenses the entire interior of the egg into a single two-dimensional image rather than a series of images documenting the interior in three dimensions. X-ray imaging in the context of dinosaur research has generally been used to look for evidence of embryonic fossils contained inside the egg. However, as of Kenneth Carpenter's 1999 book *Eggs, Nests, and Baby Dinosaurs*, all putative embryos discovered using x-rays have been misidentifications. This is because the use of x-rays to find embryos is conceptually flawed. Embryo bones are incompletely developed and will generally lack their own mineral content, as such the only source of minerals for these bones is the sediment that fills the egg after burial. The fossilized bones will therefore have the same density as the sediment filling the interior of the egg which served as the source for their mineral content and will be poorly visible in an x-ray image. So far the only reliable method for examining embryonic fossils preserved in dinosaur eggs is to physically extract them through means such as acid dissolution.[67]

X-rays can be used to chemically analyze dinosaur eggshell. This technique requires pure shell samples, so the fossil must be completely free of its surrounding rock matrix. The shell must then be further cleaned by an ultrasonic bath. The sample can then be bombarded by electrons emitted by the same sort of probe used by scanning electron microscopes. Upon impact with the samples x-rays are emitted that can be used to identify the composition of the shell.[68]

X-ray diffraction is a method for determining eggshell composition that uses X-rays to directly bombard powdered eggshell. Upon impact some of the x-rays will be diffracted at different angles and intensities depending on the specific elements present in the eggshell.[68]

Allosterics

In order to test out how allosterics played a part in dinosaur egg size, scientists used modern day animal species such as birds, crocodiles, and tortoises in their experiment. They set the bird group as representing the theropods with the reptiles representing the sauropod group. The laid eggs of each species where compared with one another over the course of the study as well as against the fossilized eggs. The results that was retrieved from the experiment was that while sauropods laid smaller eggs in greater amounts each year, dinosaur of the theropod group was revealed to lay larger eggs less frequently over the years, similar to modern birds today.

See also

- Baby Yingliang

Footnotes

1. "First Discoveries," Carpenter (1999); page 1.
2. Skinner, Justin."ROM Puts Oldest Dinosaur Eggs Ever Discovered on Display" (http://www.insidetoronto.com/news/local/article/812684--rom-puts-oldest-dinosaur-eggs-ever-discovered-on-display) Archived (https://web.archive.org/web/20100510154418/http://www.insidetoronto.com/news/local/article/812684--rom-puts-oldest-dinosaur-eggs-ever-discovered-on-display) May 10, 2010, at the Wayback Machine. insidetoronto.com. May 6, 2010.

3. Moskvitch, Katia. "Eggs with the Oldest Known Embryos of a Dinosaur Found" (https://www.bbc.co.uk/news/science-environment-11734616). BBC News. November 12, 2010.
4. "First Discoveries," Carpenter (1999); page 5.
5. "First Discoveries," Carpenter (1999); page 4.
6. "Collecting Eggs," Carpenter (1999); page 115.
7. "Fake Eggs," Carpenter (1999); page 118.
8. "Fake Eggs," Carpenter (1999); page 121.
9. "Fake Eggs," Carpenter (1999); page 120.
10. "Fake Eggs," Carpenter (1999); pages 120–121.
11. "Fake Eggs," Carpenter (1999); page 119.
12. "Fake Eggs," Carpenter (1999); pages 119–120.
13. Carpenter, Kenneth (1999). "Making an Egg". *Eggs, Nests, and Baby Dinosaurs: A Look at Dinosaur Reproduction (Life of the Past)* (https://archive.org/details/isbn_9780253334978/page/85). Bloomington, Indiana: Indiana University Press. pp. 85–107 (https://archive.org/details/isbn_9780253334978/page/85). ISBN 978-0-253-33497-8.
14. Carpenter, Kenneth (1999). "How to Study a Fossil Egg". *Eggs, Nests, and Baby Dinosaurs: A Look at Dinosaur Reproduction (Life of the Past)* (https://archive.org/details/isbn_9780253334978/page/122). Bloomington, Indiana: Indiana University Press. pp. 122–144 (https://archive.org/details/isbn_9780253334978/page/122). ISBN 978-0-253-33497-8.
15. Laura E. Wilson, Karen Chin, Frankie D. Jackson, and Emily S. Bray. II. Eggshell morphology and structure (http://www.ucmp.berkeley.edu/science/eggshell/eggshell2.php). *UCMP Online Exhibits: Fossil Eggshell*
16. Dauphin, Y. (1990). "Comparative microstructural studies of eggshells. 1. Dinosaurs of the Southern France". *Revue de Paléobiologie*. **9**: 127–133.
17. Dauphin, Y. (1990). "Incidence de l'état diagénétique des coquilles d'oeufs de dinosaures sur la reconnaissance des morphotypes - exemple du Bassin d'Aix en Provence". *C. R. Acad. Sci. Paris*. sér/ II, 310: 849–954.
18. Dauphin, Y.; Jaeger, J.J. (1991). "Implications de l'analyse microstructurale et géochimique des œufs de dinosaures de la cairanne (Bassin d'Aix en Provence, France, Rognacien inférieur)". *Paläontologische Zeitschrift*. **65** (3–4): 391–404. Bibcode:1991PalZ...65..391D (https://ui.adsabs.harvard.edu/abs/1991PalZ...65..391D). doi:10.1007/bf02989853 (https://doi.org/10.1007%2Fbf02989853). ISSN 0031-0220 (https://search.worldcat.org/issn/0031-0220). S2CID 129041143 (https://api.semanticscholar.org/CorpusID:129041143).
19. Simon, D. J. (2014). "Giant Dinosaur (theropod) Eggs of the Oogenus Macroelongatoolithus (Elongatoolithidae) from Southeastern Idaho: Taxonomic, Paleobiogeographic, and Reproductive Implications. (http://scholarworks.montana.edu/xmlui/bitstream/handle/1/8693/SimonD0814.pdf?sequence=1)" (Doctoral dissertation, Montana State University, Bozeman).
20. Mikhailov, Konstantin (1996). "Bird Eggs in the Upper Cretaceous of Mongolia". *Paleontological Journal*. **30** (1): 114–116.
21. Vianey-Liaud, Monique; Lopez-Martinez, Nieves (1997). "Late Cretaceous Dinosaur Eggshells from the Tremp Basin, Southern Pyrenees, Lleida, Spain" (https://eprints.ucm.es/id/eprint/69803/1/3Vianey%26LopezM97-1.pdf) (PDF). *Journal of Paleontology*. **71** (6): 1157–1171. Bibcode:1997JPal...71.1157V (https://ui.adsabs.harvard.edu/abs/1997JPal...71.1157V). doi:10.1017/s002233600003609x (https://doi.org/10.1017%2Fs002233600003609x). S2CID 131405598 (https://api.semanticscholar.org/CorpusID:131405598).
22. Carpenter, Kenneth; Hirsch, Karl; Horner, John (1994). "Introduction". In Carpenter, Kenneth; Hirsch, Karl; Horner, John (eds.). *Dinosaur Eggs and Babies*. Trumpington Street, Cambridge: University of Cambridge. pp. 1–11. ISBN 978-0-521-44342-5.

23. Laura E. Wilson, Karen Chin, Frankie D. Jackson, and Emily S. Bray. V. Paleobiology and eggs (http://www.ucmp.berkeley.edu/science/eggshell/eggshell2.php). *UCMP Online Exhibits: Fossil Eggshell*
24. Moratalla, J.J.; Powell, J.E. (1994). "Dinosaur Nesting Patterns". In Carpenter, Kenneth; Hirsch, Karl; Horner, John (eds.). *Dinosaur Eggs and Babies*. The Pitt Building, Trumpington Street, Cambridge: Cambridge University Press. pp. 37–46.
25. Soto, M.; Perea, D.; Cambiaso, A.V. (2012). "First sauropod (Dinosauria: Saurischia) remains from the Guichón Formation, Late Cretaceous of Uruguay" (https://www.researchgate.net/publication/232399931). *Journal of South American Earth Sciences*. **33** (1): 68–79. Bibcode:2012JSAES..33...68S (https://ui.adsabs.harvard.edu/abs/2012JSAES..33...68S). doi:10.1016/j.jsames.2011.08.003 (https://doi.org/10.1016%2Fj.jsames.2011.08.003).
26. Moreno-Azanza, Miguel; Bauluz, Blanca; Canudo, José Ignacio; Gasca, José Manuel; Fernández-Baldor, Fidel Torcida (2016). "Combined Use of Electron and Light Microscopy Techniques Reveals False Secondary Shell Units in Megaloolithidae Eggshells" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4856302). *PLOS ONE*. **11** (5) e0153026 (published May 4, 2016). Bibcode:2016PLoSO..1153026M (https://ui.adsabs.harvard.edu/abs/2016PLoSO..1153026M). doi:10.1371/journal.pone.0153026 (https://doi.org/10.1371%2Fjournal.pone.0153026). PMC 4856302 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4856302). PMID 27144767 (https://pubmed.ncbi.nlm.nih.gov/27144767).
27. "Basic Types Eggshell: Spherulitic Basic Type," Carpenter (1999); pages 136-137.
28. "Basic Types Eggshell: Prismatic Basic Type," Carpenter (1999); page 137.
29. *What are dinosaur eggs?* (https://web.archive.org/web/20140201213751/http://palaeo.gly.bris.ac.uk/palaeofiles/eggs/default.html), archived from the original (http://palaeo.gly.bris.ac.uk/palaeofiles/eggs/default.html) on February 1, 2014
30. Glut (2003).
31. The Palaeobiology Database (https://paleobiodb.org/classic/checkTaxonInfo?taxon_no=82906)
32. The Palaeobiology Database (https://paleobiodb.org/classic/checkTaxonInfo?taxon_no=67205)
33. The Palaeobiology Database (https://paleobiodb.org/classic/checkTaxonInfo?taxon_no=67198)
34. The Palaeobiology Database (https://paleobiodb.org/classic/checkTaxonInfo?taxon_no=81056)
35. The Palaeobiology Database (https://paleobiodb.org/classic/checkTaxonInfo?taxon_no=67185)
36. The Palaeobiology Database (https://paleobiodb.org/classic/checkTaxonInfo?taxon_no=81709)
37. The Palaeobiology Database (https://paleobiodb.org/classic/checkTaxonInfo?taxon_no=67183)
38. The Palaeobiology Database (https://paleobiodb.org/classic/checkTaxonInfo?taxon_no=98798)
39. The Palaeobiology Database (https://paleobiodb.org/classic/checkTaxonInfo?taxon_no=65584)
40. The Palaeobiology Database (https://paleobiodb.org/classic/checkTaxonInfo?taxon_no=67189)
41. The Palaeobiology Database (https://paleobiodb.org/classic/checkTaxonInfo?taxon_no=56741)

42. Lawver, Daniel R.; Jin, Xingsheng; Jackson, Frankie D.; Wang, Qiongying (2016). "An Avian Egg from the Lower Cretaceous (Albian) Liangtoutang Formation of Zhejiang Province, China". *Journal of Vertebrate Paleontology*. **36** (3) e1100631. Bibcode:2016JVPal..36E0631L (https://ui.adsabs.harvard.edu/abs/2016JVPal..36E0631L). doi:10.1080/02724634.2016.1100631 (https://doi.org/10.1080%2F02724634.2016.1100631). S2CID 88169746 (https://api.semanticscholar.org/CorpusID:88169746).
43. The Palaeobiology Database (https://paleobiodb.org/classic/checkTaxonInfo?taxon_no=67196)
44. The Palaeobiology Database (https://paleobiodb.org/classic/checkTaxonInfo?taxon_no=68519)
45. Xie, J.-F.; Zhang, S.-K.; Jin, X.-S.; Li, D.-Q.; Zhou, L.-Q. (2016). "A new type of dinosaur eggs from Early Cretaceous of Gansu Province, China" (https://web.archive.org/web/20160129003559/http://www.ivpp.ac.cn/cbw/gjzdwxb/pressonline/201512/P020151203499501638411.pdf) (PDF). *Vertebrata PalAsiatica*. **54** (1): 1–10. Archived from the original (http://www.ivpp.ac.cn/cbw/gjzdwxb/pressonline/201512/P020151203499501638411.pdf) (PDF) on January 29, 2016. Retrieved January 11, 2016.
46. The Palaeobiology Database (https://paleobiodb.org/classic/checkTaxonInfo?taxon_no=65562)
47. The Palaeobiology Database (https://paleobiodb.org/classic/checkTaxonInfo?taxon_no=84312)
48. The Palaeobiology Database (https://paleobiodb.org/classic/checkTaxonInfo?taxon_no=67195)
49. The Palaeobiology Database (https://paleobiodb.org/classic/checkTaxonInfo?taxon_no=67290)
50. The Palaeobiology Database (https://paleobiodb.org/classic/checkTaxonInfo?taxon_no=81708)
51. Norell, M. A.; Clark, J. M.; Dashzeveg, D.; Barsbold, T.; Chiappe, L. M.; Davidson, A. R.; McKenna, M. C.; Novacek, M. J. (1994). "A theropod dinosaur embryo, and the affinities of the Flaming Cliffs Dinosaur eggs". *Science*. **266** (5186): 779–782. Bibcode:1994Sci...266..779N (https://ui.adsabs.harvard.edu/abs/1994Sci...266..779N). doi:10.1126/science.266.5186.779 (https://doi.org/10.1126%2Fscience.266.5186.779). PMID 17730398 (https://pubmed.ncbi.nlm.nih.gov/17730398). S2CID 22333224 (https://api.semanticscholar.org/CorpusID:22333224).
52. Mateus et al. (1998).
53. de Ricqles et al. (2001).
54. "Abstract," Reisz et al. (2005); page 761.
55. "Correction: A comparative embryological study of two ornithischian dinosaurs," Horner and Weishampel (1996); page 103.
56. "How to Fossilize an Egg," Carpenter (1999); page 112.
57. "How to Fossilize an Egg," Carpenter (1999); page 113.
58. "How to Fossilize an Egg," Carpenter (1999); page 108.
59. "How to Fossilize an Egg," Carpenter (1999); page 114.
60. "How to Fossilize an Egg," Carpenter (1999); pages 114–115.
61. "How to Fossilize an Egg," Carpenter (1999); page 115.
62. "How to Fossilize an Egg," Carpenter (1999); page 111.
63. "How to Fossilize an Egg," Carpenter (1999); page 110.
64. "Collecting Eggs," Carpenter (1999); page 117.
65. "Collecting Eggs," Carpenter (1999); pages 117–118.
66. "Collecting Eggs," Carpenter (1999); page 118.

67. "Tools of the Trade," Carpenter (1999); page 128.
68. "Tools of the Trade," Carpenter (1999); page 130.
69. https://embryo.asu.edu/pages/acid-dissolution-fossil-dinosaur-eggs Last paragraph
70. "Tools of the Trade," Carpenter (1999); pages 128–130.
71. "Fig 7.11," Carpenter (1999); page 118.
72. "Tools of the Trade," Carpenter (1999); page 133.
73. "Tools of the Trade," Carpenter (1999); page 134.
74. "Tools of the Trade," Carpenter (1999); page 122.
75. "Tools of the Trade," Carpenter (1999); page 124.
76. "Tools of the Trade," Carpenter (1999); page 125.
77. "Tools of the Trade," Carpenter (1999); page 131.
78. "Tools of the Trade," Carpenter (1999); page 132.
79. Chen, Qingmin; Cheng, Xing; Wang, Jian; Zhao, Bi; Zhang, Shukang; Ning, Youfeng; Wang, Gaohong; He, Kaikai; Zhang, Wenshuo; Yu, Dongxiang; Li, Jiangli; Zou, Yarui; Chen, Gang; Li, Min; Cheng, Hai (September 11, 2025). "Geological age of the Yunyang dinosaur eggs revealed by in-situ carbonate U-Pb dating and its scientific implications" (https://doi.org/10.3389%2Ffeart.2025.1638838). *Frontiers in Earth Science*. **13** 1638838. Bibcode:2025FrEaS..1338838C (https://ui.adsabs.harvard.edu/abs/2025FrEaS..1338838C). doi:10.3389/feart.2025.1638838 (https://doi.org/10.3389%2Ffeart.2025.1638838). ISSN 2296-6463 (https://search.worldcat.org/issn/2296-6463).

References

- Carpenter, Kenneth (1999). *Eggs, Nests, and Baby Dinosaurs: A Look at Dinosaur Reproduction (Life of the Past)*, Indiana University Press; ISBN 0-253-33497-7.
- Deeming, D. C. and M. W. J. Ferguson (eds.) 1991. Egg incubation: its effect on embryonic development in birds and reptiles. Cambridge University Press, UK. 448pp.
- Glut, Donald F. (2003), "Appendix: Dinosaur Tracks and Eggs" (https://archive.org/details/dinosaursencyclo00glut_2/page/613), *Dinosaurs: The Encyclopedia. 3rd Supplement*, Jefferson, North Carolina: McFarland & Company, Inc., pp. 613–652 (https://archive.org/details/dinosaursencyclo00glut_2/page/613), ISBN 978-0-7864-1166-5
- Horner, John R.; Weishampel, David B. (1996). "A comparative embryological study of two ornithischian dinosaurs - a correction" (https://doi.org/10.1038%2F383103b0). *Nature*. **383** (6595): 256–257. Bibcode:1996Natur.383..103H (https://ui.adsabs.harvard.edu/abs/1996Natur.383..103H). doi:10.1038/383103b0 (https://doi.org/10.1038%2F383103b0).
- Mateus, I; Mateus, H; Antunes, MT; Mateus, O; Taquet, P; Ribeiro, V; Manuppella, G (1998). "Upper Jurassic theropod dinosaur embryos from Lourinhã (Portugal)". *Memórias da Academia das Ciências de Lisboa*. **37**: 101–110.
- Moskvitch, Katia. "Eggs with the Oldest Known Embryos of a Dinosaur Found" (https://www.bbc.co.uk/news/science-environment-11734616). BBC News. November 12, 2010.
- de Ricqlès, A.; Mateus, O.; Antunes, M. T.; Taquet, P. (2001). "Histomorphogenesis of embryos of Upper Jurassic theropods from Lourinhã (Portugal)". *Comptes Rendus de l'Académie des Sciences, Série IIA*. **332** (10): 647–656. Bibcode:2001CRASE.332..647D (https://ui.adsabs.harvard.edu/abs/2001CRASE.332..647D). doi:10.1016/s1251-8050(01)01580-4 (https://doi.org/10.1016%2Fs1251-8050%2801%2901580-4).
- Reisz, Robert R.; Scott, Diane; Sues, Hans-Dieter; Evans, David C.; Raath, Michael A. (2005). "Embryos of an Early Jurassic prosauropod dinosaur and their evolutionary significance" (http://doc.rero.ch/record/14975/files/PAL_E2124.pdf) (PDF). *Science*. **309** (5735): 761–764. Bibcode:2005Sci...309..761R (https://ui.adsabs.harvard.edu/abs/2005Sc

i...309..761R). doi:10.1126/science.1114942 (https://doi.org/10.1126%2Fscience.1114942). PMID 16051793 (https://pubmed.ncbi.nlm.nih.gov/16051793). S2CID 37548361 (https://api.semanticscholar.org/CorpusID:37548361).

- Skinner, Justin."ROM Puts Oldest Dinosaur Eggs Ever Discovered on Display" (http://www.insidetoronto.com/news/local/article/812684--rom-puts-oldest-dinosaur-eggs-ever-discovered-on-display) Archived (https://web.archive.org/web/20100510154418/http://www.insidetoronto.com/news/local/article/812684--rom-puts-oldest-dinosaur-eggs-ever-discovered-on-display) May 10, 2010, at the Wayback Machine. insidetoronto.com. May 6, 2010.
- "What are dinosaur eggs?" (https://web.archive.org/web/20140201213751/http://palaeo.gly.bris.ac.uk/palaeofiles/eggs/default.html), *University of Bristol Earth Sciences*, archived from the original (http://palaeo.gly.bris.ac.uk/palaeofiles/eggs/default.html) on February 1, 2014, retrieved June 20, 2013

External links

- Media related to Dinosauria eggs at Wikimedia Commons

Dinosaurs portal

Retrieved from "https://en.wikipedia.org/w/index.php?title=Dinosaur_egg&oldid=1370426901"

2

Mesozoik Devirler

Dinozorların yaşadığı jeolojik dönemler

- Triassic
- Jurassic

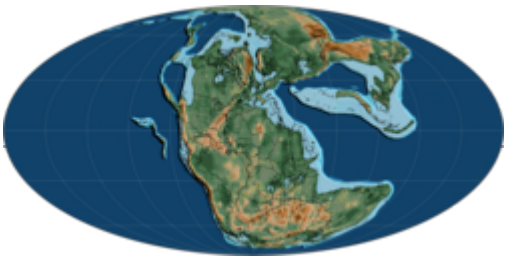
WIKIPEDIA
25 years of the free encyclopedia

Triassic

The **Triassic** (/traɪˈæsɪk/ *try-A-sik*) is a geologic period and a stratigraphic system that spans 50.5 million years from the end of the Permian Period 251.902 Ma (million years ago) to the beginning of the Jurassic Period 201.4 Ma.[8][9] The Triassic Period is the first and shortest geologic period of the Mesozoic Era, and the seventh period of the Phanerozoic Eon. The start and the end of the Triassic Period featured major extinction events.[10]

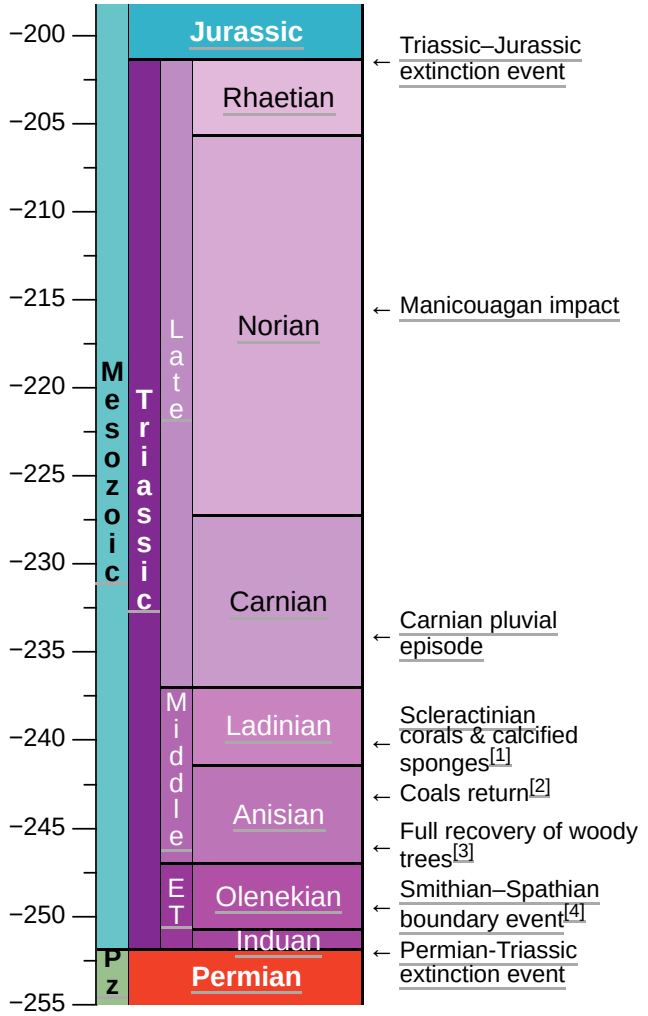
Chronologically, the Triassic Period is divided into three epochs: (i) the Early Triassic, (ii) the Middle Triassic, and (iii) the Late Triassic. The Triassic Period began after the Permian–Triassic extinction event that much reduced the biosphere of planet Earth. The fossil record of the Triassic Period presents three categories of organisms: (i) animals that survived the Permian–Triassic extinction event, (ii) new animals that briefly flourished in the Triassic biosphere, and (iii) new animals that evolved and dominated the Mesozoic Era. Reptiles, especially archosaurs, were the chief terrestrial vertebrates during this time. A specialized group of archosaurs, called dinosaurs, first appeared in the Late Triassic but did not become dominant until the succeeding Jurassic Period.[11] Archosaurs that became dominant in this period were primarily pseudosuchians, relatives and ancestors of modern crocodilians, while some archosaurs specialized in flight, the first time among vertebrates, becoming the pterosaurs. Therapsids, the dominant vertebrates of the preceding Permian period, saw a brief surge in diversification in the Triassic, with dicynodonts and cynodonts quickly becoming dominant, but they declined throughout the period with the majority

Triassic



A paleomap of Earth as it appeared 225 Ma during the Norian Age of the Late Triassic Epoch

Chronology



Subdivision of the Triassic according to the ICS, as of 2024.[5]

Vertical axis scale: Millions of years ago

Etymology

becoming extinct by the end. However, the first stem-group mammals (mammaliamorphs), themselves a specialized subgroup of cynodonts, appeared during the Triassic and would survive the extinction event, allowing them to radiate during the Jurassic. Amphibians were primarily represented by the temnospondyls, giant aquatic predators that had survived the end-Permian extinction and saw a new burst of diversification in the Triassic, before going extinct by the end; however, early crown-group lissamphibians (including stem-group frogs, salamanders and caecilians) also became more common during the Triassic and survived the extinction event. The earliest known neopterygian fish, including early holosteans and teleosts, appeared near the beginning of the Triassic, and quickly diversified to become among the dominant groups of fish in both freshwater and marine habitats.

Name formality	Formal
Usage information	
Celestial body	Earth
Regional usage	Global (ICS)
Time scale(s) used	ICS Time Scale
Definition	
Chronological unit	Period
Stratigraphic unit	System
Time span formality	Formal
Lower boundary definition	First appearance of the conodont *Hindeodus parvus*
Lower boundary GSSP	Meishan, Zhejiang, China 31.0798°N 119.7058°E
Lower GSSP ratified	2001[6]
Upper boundary definition	First appearance of the ammonite *Psiloceras spelae tirolicum*
Upper boundary GSSP	Kuhjoch section, Karwendel mountains, Northern Calcareous Alps, Austria 47.4839°N 11.5306°E
Upper GSSP ratified	2010[7]

The vast supercontinent of Pangaea dominated the globe during the Triassic, but in the latest Triassic (Rhaetian) and Early Jurassic it began to gradually rift into two separate landmasses: Laurasia to the north and Gondwana to the south. The global climate during the Triassic was mostly hot and dry,[12] with deserts spanning much of Pangaea's interior. However, the climate shifted and became more humid as Pangaea began to drift apart. The end of the period was marked by yet another major mass extinction, the Triassic–Jurassic extinction event, that wiped out many groups, including most pseudosuchians, and allowed dinosaurs to assume dominance in the Jurassic.

Etymology

The Triassic was named in 1834 by Friedrich Alberti, after a succession of three distinct rock layers (Greek *triás* meaning 'triad') that are widespread in southern Germany: the lower Buntsandstein (colourful sandstone), the middle Muschelkalk (shell-bearing limestone) and the upper Keuper (coloured clay).[13]

Early Triassic sandstone (Buntsandstein) near Stadtroda, Germany

Middle Triassic Muschelkalk (shell-bearing limestone) near Dörzbach, Germany

Late Triassic Steigerwald Formation and overlying Hassberge Formation in Schönbuch, Germany

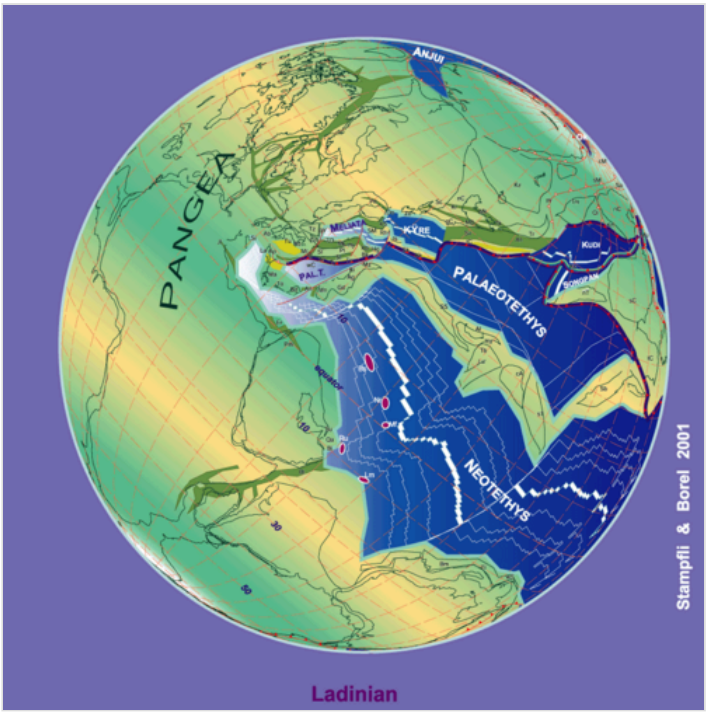
Dating and subdivisions

On the geologic time scale, the Triassic is usually divided into Early, Middle, and Late Triassic Epochs, and the corresponding rocks are referred to as Lower, Middle, or Upper Triassic. The faunal stages from the youngest to oldest are:

Series/epoch	Stage/age	Lower boundary
Lower/Early Jurassic	Hettangian	201.4 ± 0.2 Ma
	Rhaetian	205.7 Ma
	Norian	227.3 Ma
	Carnian	237 Ma
		241.464 ± 0.28 Ma
		247 Ma
		250.8 Ma
		251.902 ± 0.024 Ma

Paleogeography

At the beginning of the Triassic, all the major continents were amalgamated into the supercontinent of Pangea. Centred on the equator, this stretched in an arc from the north to south polar regions with Laurussia in the north and Gondwana in the south. The Paleo- and Neo-Tethys oceans lay within the arc of the supercontinent with the vast Panthalassa Ocean beyond.[14] North China and Amuria, and South China were separated from Pangea by the Paleoasian Ocean, but this closed by the Late Triassic.[15]



View of the Tethys area during the Ladinian stage (230 Ma)

Pangea was surrounded by subduction zones that dipped beneath the supercontinent. The great mountain ranges that marked the Late Paleozoic continental collisions were largely eroded and were being replaced by regions of thinned crust that lay along the lines of the future Atlantic, Indian and Southern oceans.[14][16]

The supercontinent changed motion from drifting westward to rotating counterclockwise during late Permian. This continued until the Carnian (c. 230 Ma), after which it resumed the westward motion. These changes in motion were triggered by the opening of the Neo-Tethys, and closing of the Paleo-Tethys respectively, and affected tectonic regimes particularly along the southern and western margins.[17]

230 Ma continental reconstruction

The narrow Cimmerian terranes that had rifted from the northern margin of Gondwana in the Permian continued to drift northwards; the Paleo-Tethys Ocean closing in front of them and the Neo-Tethys opening behind.[15]

Eruptions of the Siberian Traps Large Igneous Province (LIP) persisted into the Early Triassic[14] and the Central Atlantic Magmatic Province (CAMP) were active by the Late Triassic as a prelude to seafloor spreading in the Central Atlantic at the boundary of the Triassic and Jurassic.[18]

Northern Pangea (Laurussia)

In northern-eastern Pangea, the Siberian Traps LIP continued to erupt into the Middle Triassic. Lower Triassic major deltaic systems (including the Triassic Boreal Ocean delta plain)[19] advanced across the shallow Arctic Ocean. Their catchment areas the high ground of the Urals, Fennoscandinavia, the Canada-Greenland Shield and the Lomonosov High.[14] During the Late Triassic, in response to the opening of the Central Atlantic to the south, tectonic movements between North America and Baltica led to convergence in the High Arctic with uplift, folding and thrusting in the Barents Sea and North Siberian margin.[14]

Major extensional tectonic movements, that began in the late Permian, continued across Laurussia from North/Central Europe in the east to eastern Canada in the west, with north-south trending graben systems developing across Europe, including further subsidence in the Northern and Southern Permian basins.[14][16] Lower Triassic sediments deposited in these basins are continental to brackish marine in nature. They are overlain by shallow marine carbonates and mudstones and evaporites of the Zechstein Sea.[14]

Further south, during the Norian, the opening of the Central Atlantic led to the formation of narrow, deep water basins in the Eastern Mediterranean area.[14] Corsica, Sardinia, Calabria, and the Balearic terranes were attached to Europe, whilst Apulia, Adria, and the terranes of southern Turkey remained attached to the African plate.[16]

Along the western margin of Laurussia, a continental magmatic arc extended from the southwestern United States to the Arctic with an associated continental-scale foreland basin system. Localised back-arc basins developed within this, in regions of extension.[20] Along the northern section of the margin, the Permian to Early Triassic counterclockwise rotation of Pangea resulted in the closure of the narrow Slide Mountain Ocean and the Sonoma orogeny.[14][17] In south, it led to a reduction in the dip of the subducting slab and thickening of the continental crust.[17]

Southern Pangea (Gondwana)

Much of Africa was stable and above sea level, with only a few Triassic-aged lake sediments known, although along the northern coast marine sediments were deposited during periods of higher sea levels.[14]

Northeast-southwest trending rifting along the eastern edge of Africa between Madagascar (Indian plate) and eastern Africa (Somali plate) continued from the Late Carboniferous into the Triassic, with substantial mountains rising along the edge of the rift and the formation of a series of pull-apart basins. Gradual marine incursions from the Neo-Tethys resulted in the deposition of Lower and Middle Triassic marine sediments in these basins. Upper Triassic sediments are continental in nature and this rift system ultimately failed and a new north-south trending rift system developed in the Jurassic.[14][21]

The opening of the Neo-Tethys created passive margins along the Arabian and Indian margins.[14][22] Rifting in the Neo-Tethys extended westwards between the Pontides and Taurides terranes of Turkey during the Late Triassic.[14] The Tethyan Himalayan block remained attached to India but was separated by the thinned crust of Greater India, the northern margin of which supplied sediments to the passive margin.[23]

The counterclockwise rotation of Pangea precipitated dextral transpression across the NNE-SSW trending, west-dipping subduction zone, along the eastern Australian margin, which culminated in the Hunter-Bowen orogeny (c. 260-230 Ma).[17][24] Following this, the magmatic arc rotated to north-south and compression gave way to extension. This was accompanied by subduction rollback and back-arc basin formation.[17][24]

Along the southwestern margin of South America, low plate convergence and subduction rates, triggered by the assembly and rotation of Pangea, resulted in subduction rollback and extension across the back-arc region. This generated large amounts of felsic magmatism.[25][17] These extensional forces stretched across the continent with the formation of large northwest-trending basins with thick sedimentary deposits and the extension related magmatism.[14]

Closure of the Paleo-Tethys

The Paleo-Tethys ocean formed as the continents surrounding it assembled to form Pangea in the Late Palaeozoic.[26] The Eurasian sector of Pangea lay along its north and northwestern margin.[14] To the northeast, the narrow Paleoasian Ocean (a branch of the Paleo-Tethys) lay between Eurasia, and North China and Tarim, and to the east, South China and Annamia (Southeast Asia). To the south were the Cimmerian terranes (Central Iran, Qiangtang(north Tibet), Lhasa (south Tibet), and Sibumasu (eastern Myanmar, Thailand, Malay peninsula and Sumatra). These terranes had rifted from northeastern Gondwana during the Permian. As they drifted northwards through the Triassic, the Paleo-Tethys closed in front of them, and the Neotethys opened behind.[14][15]

The Paleo-Tethys was being consumed by subduction zones along the southern margin of North China, much of the Eurasian margin, and along the northern margin of the Qiangtang-Annamia and Lhasa-Sibumasu blocks.[14][15]

Collisions between Annamia and South China (c. 246-230 Ma); between Sibumasu and South China–Annamia (c. 240-230 Ma); and, between Qiangtang and Lhasa (c. 250–230 Ma) resulted in the Indosinian orogeny and the formation of a single large Eastern Asian continent.[15] At about the same time (c. 240-230 Ma), the final closure of the Paleoasian Ocean led to the collision of Tarim and North China with the Kazakhstan and Siberian regions of Pangea, to form the Central Asian orogenic belt.[15] South China collided with North China (c. 220 Ma), forming the Central China orogenic belt.[15] The segment of the Paleo-Tethys between North China and Qiangtang may never have fully closed, but was filled with Permo-Triassic turbidites preserved in the West Kunlun and Bayanhar belts of the Central China orogenic belt.[27]

The amalgamation of these East Asian blocks with Pangea in the Late Triassic maximised the land area of the supercontinent. It coincided with a period of dramatic climate change and the development of the megamonsoon, although the relationship between these is the subject of ongoing research.[15]

The western Paleo-Tethys remained open until about 205 Ma, when the Iranian blocks collided with the Turan platform, on the southern margin of Eurasia, resulting in the Cimmerian orogeny. This extended from the Anatolian Plateau in the northwest, and merged with the Indosinian orogenic belt in the east.[14][22] Late Triassic deformation across the Eastern Mediterranean area and much of the Middle East was complex, with regional scale strike-slip faulting and continued subduction below the Iranian margin.[14]

Opening of the Central Atlantic

Beginning in the latest Permian, a broad zone of lithospheric extension developed across Pangea along the line of the future Central Atlantic Ocean.[18][28] The location of this rifting followed the pre-existing structures of the Variscan orogeny, and began immediately after the orogenic collapse of the Variscan belt.[18][21] Extension began in the northern Central Atlantic region in the Anisian, and in the southern Central Atlantic in the Carnian.[18][28]

Major rift basins formed along the present-day eastern North American margin from Florida to Newfoundland (Newark Supergroup basins), and along the Europe/African margin (Moroccan and Iberian basins). The Moroccan basins are the equivalent of Nova Scotian basins, and the Iberian the equivalent of the Newfoundland basins.[28] These basins formed broad depressions on the continental crust that extended for hundreds of kilometres across central Pangea, with localised faulting formed sub-basins.[21] The basins were filled by mainly continental deposits from regional-scale river systems and lakes, with only minor, late marine incursions in some areas.[28][21]

The period of rifting came to an end with the emplacement of the Central Atlantic Magmatic Province (CAMP) around 201 Ma. This was followed by seafloor spreading and the opening of the Central Atlantic Ocean.[18] The CAMP is one of the largest LIPs and covered a region of about 10 million km^2 across North America, northeastern South America, northwestern Africa, and southwestern Europe.[14][18] The magmatism produced dense dyke swarms, with individual dykes up to 800 km long, massive sill complexes, and lava fields that covered several hundred kilometres.[29] Despite its size, the period of magmatism was brief, lasting only about 1 million years. Such intense igneous activity indicates widespread mantle melting, rather than a simple plume within the mantle. The varied petrological composition of the CAMP magmatism reflects local contamination of the upper mantle by continental lithosphere, including partial melting of previously subducted slabs.[18][29] The magmatism, with its large scale injection of carbon and sulphur into the atmosphere, precipitated volcanic winters. This was followed by longer-term climate warming and ocean acidification, which caused the end-Triassic mass extinction.[18][30][29]

Panthalassic Ocean

Although no direct evidence remains, Panthalassa is thought to have been divided into three major tectonic plates: the Farallon; Izanagi; and, Phoenix. These were separated by oceanic spreading ridges. In the northeast, the smaller Cache Creek plate was being subducted beneath the western margin of North America, and beneath the Farallon plate to the south.[14]

Paleooceanography

Eustatic sea level in the Triassic was consistently low compared to the other geological periods. The beginning of the Triassic was around present sea level, rising to about 10–20 metres (33–66 ft) above present-day sea level during the Early and Middle Triassic. Sea level rise accelerated in the Ladinian, culminating with a sea level up to 50 metres (164 ft) above present-day levels during the Carnian. Sea level began to decline in the Norian, reaching a low of 50 metres (164 ft) below present sea level during the mid-Rhaetian. Low global sea levels persisted into the earliest Jurassic. The long-term sea level trend is superimposed by 22 sea level drop events widespread in the geologic record, mostly of minor (less than 25-metre (82 ft)) and medium (25–75-metre (82–246 ft)) magnitudes. A lack of evidence for Triassic continental ice sheets suggest that glacial eustasy is unlikely to be the cause of these changes.[32] It has

Sydney, Australia lies on Triassic shales and sandstones. Almost all of the exposed rocks around Sydney belong to the Triassic Sydney sandstone.[31]

generally been assumed that the cause was changes in volume of the global ocean basin due to variations in oceanic volcanism, with largest volumes occurring in volcanism's absence when the ocean basins were subsiding.[33] Variation in water and sediment delivery to the oceans, with higher sea levels during pluvial eras lasting up to four million years, is also hypothesised to be behind these sea level variations.[32]

Climate

The Triassic continental interior climate was generally hot and dry, so that typical deposits are red bed sandstones and evaporites. There is no evidence of glaciation at or near either pole; in fact, the polar regions were apparently moist and temperate, providing a climate suitable for forests and vertebrates, including reptiles. Pangaea's large size limited the moderating effect of the global ocean; its continental climate was highly seasonal, with very hot summers and cool winters.[34] The strong contrast between the Pangea supercontinent and the global ocean triggered intense cross-equatorial monsoons,[34] sometimes referred to as the Pangean megamonsoons.[35]

The Triassic may have mostly been a dry period, but evidence exists that it was punctuated by several episodes of increased rainfall in tropical and subtropical latitudes of the Tethys Sea and its surrounding land.[36] Sediments and fossils suggestive of a more humid climate are known from the Anisian to Ladinian of the Tethysian domain, and from the Carnian and Rhaetian of a larger area that includes also the Boreal domain (e.g., Svalbard Islands), the North American continent, the South China block and Argentina. The best-studied of such episodes of humid climate, and probably the most intense and widespread, was the Carnian Pluvial Event.

Early Triassic

The Early Triassic was the hottest portion of the entire Phanerozoic, seeing as it occurred during and immediately after the discharge of titanic volumes of greenhouse gases from the Siberian Traps. The Early Triassic began with the Permian-Triassic Thermal Maximum (PTTM) and was followed by the brief Dienerian Cooling (DC) from 251 to 249 Ma, which was in turn followed by the Latest Smithian Thermal Maximum (LSTT) around 249 to 248 Ma. During the Latest Olenekian Cooling (LOC), from 248 to 247 Ma, temperatures cooled by about 6 °C.[37]

Middle Triassic

The Middle Triassic was cooler than the Early Triassic, with temperatures falling over most of the Anisian, with the exception of a warming spike in the latter portion of the stage.[38] From 242 to 233 Ma, the Ladinian-Carnian Cooling (LCC) ensued.[37]

Late Triassic

At the beginning of the Carnian, global temperatures continued to be relatively cool.[39] The eruption of the Wrangellia Large Igneous Province around 234 Ma caused abrupt global warming, terminating the cooling trend of the LCC.[40] This warming was responsible for the Carnian Pluvial Event and resulted in an episode of widespread global humidity.[41] The CPE ushered in the Mid-Carnian Warm Interval (MCWI), which lasted from 234 to 227 Ma.[37] At the Carnian-Norian boundary occurred a positive $\delta^{13}C$ excursion believed to signify an increase in organic carbon burial.[42] From 227 to 217 Ma, there was a relatively cool period known as the Early Norian Cool Interval (ENCI), after which occurred the Mid-Norian Warm Interval (MNWI) from 217 to 209 Ma. The MNWI was briefly interrupted around 214 Ma by a cooling possibly related to the Manicouagan impact.[37] Around 212 Ma, a 10 Myr eccentricity maximum caused a paludification of Pangaea and a reduction in the size of arid climatic zones.[43] The Rhaetian Cool Interval (RCI) lasted from 209 to 201 Ma.[37] At the terminus of the Triassic, there was an extreme warming event referred to as the End-Triassic Thermal Event (ETTE), which was responsible for the Triassic-Jurassic mass extinction.[37] Bubbles of carbon dioxide in basaltic rocks dating back to the end of the Triassic indicate that volcanic activity from the Central Atlantic Magmatic Province helped trigger climate change in the ETTE.[44]

Flora

Land plants

During the Early Triassic, lycophytes, particularly those of the order Isoetales (which contains living quillworts), rose to prominence due to the environmental instability following the Permian-Triassic extinction, with one particularly notable example being the genus *Pleuromeia*, which grew in columnar like fashion, sometimes reaching a height of 2 metres (6.6 ft). The relevance of lycophytes declined from the Middle Triassic onwards, following the return of more stable environmental conditions.[45]

While having first appeared during the Permian, the extinct seed plant group Bennettitales first became a prominent element in global floras during the Late Triassic, a position they would hold for much of the Mesozoic.[46] In the Southern Hemisphere landmasses of Gondwana, the tree *Dicroidium*, an extinct "seed fern" belong to the order Corystospermales was a dominant element in forest habitats across the region during the Middle-Late Triassic.[47] During the Late Triassic, the Ginkgoales (which today are represented by only a single species, *Ginkgo biloba*) underwent considerable diversification.[48] Conifers were abundant during the Triassic, and included the Voltziales (which contains various lineages, probably including those ancestral to modern conifers),[49] as well as the extinct family Cheirolepidiaceae, which first appeared in the Late Triassic, and would be prominent throughout most of the rest of the Mesozoic.[50] The Middle Triassic also included herbaceous ruderal conifers, a growth form which no longer exists in modern conifers.[51]

Coal

Immediately above the Permian–Triassic boundary the glossopteris flora was suddenly largely displaced by an Australia-wide coniferous flora.[52]

No known coal deposits date from the start of the Triassic Period. This is known as the Early Triassic "coal gap" and can be seen as part of the Permian–Triassic extinction event.[53] Possible explanations for the coal gap include sharp drops in sea level at the time of the Permo-Triassic boundary;[54] acid rain from the Siberian Traps eruptions or from an impact event that overwhelmed acidic swamps; climate shift to a greenhouse climate that was too hot and dry for peat accumulation; evolution of fungi or herbivores that were more destructive of wetlands; the extinction of all plants adapted to peat swamps, with a hiatus of several million years before new plant species evolved that were adapted to peat swamps;[53] or soil anoxia as oxygen levels plummeted.[55]

An Early Triassic Australian landscape including *Pleuromeia* (central foreground) as well as horsetails, *Dicroidium* trees (prominently on right), as well as the temnospondyl amphibian *Bulgosuchus* (foreground) and an erythrosuchid reptile (background left). Art by Michael Rothman

Phytoplankton

Before the Permian extinction, Archaeplastida (red and green algae) had been the major marine phytoplanktons since about 659–645 million years ago,[56] when they replaced marine planktonic cyanobacteria, which first appeared about 800 million years ago, as the dominant phytoplankton in the oceans.[57] In the Triassic, secondary endosymbiotic algae became the most important plankton.[58]

Fauna

Marine invertebrates

Middle Triassic marginal marine sequence, southwestern Utah

In marine environments, new modern types of corals appeared in the Early Triassic, forming small patches of reefs of modest extent compared to the great reef systems of Devonian or modern times. At the end of the Carnian, a reef crisis occurred in South China.[59] Serpulids appeared in the Middle Triassic.[60] Microconchids were abundant. The shelled cephalopods called ammonites recovered, diversifying from a single line that survived the Permian extinction. Bivalves began to rapidly diversify during the Middle Triassic, becoming highly abundant in the oceans.[61]

Insects

Aquatic insects rapidly diversified during the Middle Triassic, with this time interval representing a crucial diversification for Holometabola, the clade containing the majority of modern insect species.[62]

Fish

Birgeria

In the wake of the Permian-Triassic mass extinction event, the fish fauna was remarkably uniform, with many families and genera exhibiting a cosmopolitan distribution.[63] Coelacanths show their highest post-Devonian diversity in the Early Triassic.[64] Ray-finned fishes (actinopterygians) went through a remarkable diversification in the beginning of the Triassic, leading to peak diversity during the Middle Triassic; however, the pattern of this diversification is still not well understood due to a taphonomic megabias.[65] The first stem-group teleosts appeared during the Triassic (teleosts are by far the most diverse group of fish today).[63] Predatory actinopterygians such as saurichthyids and birgeriids, some of which grew over 1.2 m (3.9 ft) in length, appeared in the Early Triassic and became widespread and successful during the period as a whole.[66] Lakes and rivers were populated by lungfish (Dipnoi), such as *Ceratodus*, which are mainly known from the dental plates, abundant in the fossils record.[67] Hybodonts, a group of shark-like cartilaginous fish, were dominant in both freshwater and marine environments throughout the Triassic.[68] Last survivors of the mainly Palaeozoic Eugeneodontida are known from the Early Triassic.[69]

Amphibians

Reconstruction of the Triassic amphibian *Mastodonsaurus*

Temnospondyl amphibians were among those groups that survived the Permian–Triassic extinction. Once abundant in both terrestrial and aquatic environments, the terrestrial species had mostly died out during the extinction event. The Triassic survivors were aquatic or semi-aquatic, and were represented by *Tupilakosaurus*, *Thabanchuia*, Branchiosauridae and *Micropholis*, all of which died out in Early Triassic, and the successful Stereospondyli, with survivors into the Cretaceous Period. The largest Triassic stereospondyls, such as *Mastodonsaurus*, were up to 4 to 6 metres (13 to 20 ft) in length.[70][71] Some lineages (e.g. trematosaurs) flourished briefly in the Early Triassic, while others (e.g. capitosaurs) remained successful throughout the whole period, or only came to prominence in the Late Triassic (e.g. *Plagiosaurus*, metoposaurs).

The first Lissamphibians (modern amphibians) appear in the Triassic, with the progenitors of the first frogs already present by the Early Triassic. However, the group as a whole did not become common until the Jurassic, when the temnospondyls had become very rare.

Most of the Reptiliomorpha, stem-amniotes that gave rise to the amniotes, disappeared in the Triassic, but two water-dwelling groups survived: Embolomeri that only survived into the early part of the period, and the Chroniosuchia, which survived until the end of the Triassic.

The Ashfield Shale of western Sydney, formed during the Middle Triassic, features amphibian fossils from that period.[72]

Reptiles

Archosauromorphs

The Permian–Triassic extinction devastated terrestrial life. Biodiversity rebounded as the surviving species repopulated empty terrain, but these were short-lived. Diverse communities with complex food-web structures took 30 million years to reestablish.[10][73] Archosauromorph reptiles, which had already appeared and diversified to an extent in the Permian Period, exploded in diversity as an adaptive radiation in response to the Permian-Triassic mass extinction. By the Early Triassic, several major archosauromorph groups had appeared. Long-necked, lizard-like early archosauromorphs were known as protorosaurs, which is likely a paraphyletic group rather than a true clade. Tanystropheids were a family of protorosaurs which elevated their neck size to extremes, with the largest genus *Tanystropheus* having a neck longer than its body. The protorosaur family Sharovipterygidae used their elongated hindlimbs for gliding. Other archosauromorphs, such as rhynchosaurs and allokotosaurs, were mostly stocky-bodied herbivores with specialized jaw structures.

Allokotosaurs were iguana-like reptiles, including *Trilophosaurus* (a common Late Triassic reptile with three-crowned teeth), *Teraterpeton* (which had a long beak-like snout), and *Shringasaurus* (a horned herbivore which reached a body length of 3–4 metres (9.8–13.1 ft)).

One group of archosauromorphs, the archosauriforms, were distinguished by their active predatory lifestyle, with serrated teeth and upright limb postures. Archosauriforms were diverse in the Triassic, including various terrestrial and semiaquatic predators of all shapes and sizes. The large-headed and robust erythrosuchids were among the dominant carnivores in the early Triassic.

True archosaurs appeared in the early Triassic, splitting into two branches: Avemetatarsalia (the ancestors to birds) and Pseudosuchia (the ancestors to crocodilians). Avemetatarsalians were a minor component of their ecosystems, but eventually produced the earliest pterosaurs and dinosaurs in the Late Triassic. Early long-tailed pterosaurs appeared in the Norian and quickly spread worldwide. Triassic dinosaurs evolved in the Carnian and include early sauropodomorphs and theropods. Most Triassic dinosaurs were small predators and only a few were common, such as *Coelophysis*, which was 1 to 2 metres (3.3 to 6.6 ft) long. Triassic sauropodomorphs primarily inhabited cooler regions of the world.[74]

The large predator *Smok* was most likely also an archosaur, but it is uncertain if it was a primitive dinosaur or a pseudosuchian.

Pseudosuchians were far more ecologically dominant in the Triassic, including large herbivores (such as aetosaurs), large carnivores ("rauisuchians"), and the first crocodylomorphs ("sphenosuchians"). "Rauisuchians" were ancestral to small, lightly-built crocodylomorphs, the only pseudosuchians which survived into the Jurassic.

Tanystropheus, a long-necked tanystropheid

Proterosuchus, a crocodile-like early archosauriform from the Early Triassic

Staurikosaurus, one of the earliest dinosaurs, a member of the Triassic family Herrerasauridae

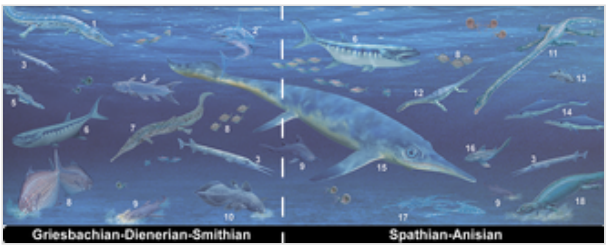
Postosuchus, a rauisuchid which was an apex predator in parts of Late Triassic North America

Plateosaurus was one of the largest of early sauropodomorphs, or "prosauropods", of the Late Triassic

Coelophysis was one of the most abundant theropod dinosaurs in the Late Triassic

Marine reptiles

There were many types of marine reptiles. These included the Sauropterygia, which featured pachypleurosaurs and nothosaurs (both common during the Middle Triassic, especially in the Tethys region), placodonts, the earliest known herbivorous marine reptile *Atopodentatus*, and the first plesiosaurs. The first of the lizard-like Thalattosauria (*Askeptosaurus*) and the highly successful ichthyopterygians, which appeared in Early Triassic seas, soon diversified. By the Middle Triassic, some ichthyopterygians were achieving very large body masses.[76]



Marine vertebrate apex predators of the Early Triassic and Anisian (Middle Triassic)[75]

Other reptiles

Among other reptiles, the earliest turtles, like *Proganochelys* and *Proterochersis*, appeared during the Norian Age (Stage) of the Late Triassic Period. The Lepidosauromorpha, specifically the Sphenodontia, are first found in the fossil record of the earlier Carnian Age, though the earliest lepidosauromorphs likely occurred in the Permian. The Procolophonidae, the last surviving parareptiles, were an important group of small lizard-like herbivores. The drepanosaurs were a clade of unusual, chameleon-like arboreal reptiles with birdlike heads and specialised claws.

Synapsids

Three therapsid groups survived into the Triassic: dicynodonts, therocephalians, and cynodonts. The cynodont *Cynognathus* was a characteristic top predator in the Olenekian and Anisian of Gondwana. Both kannemeyeriiform dicynodonts and gomphodont cynodonts remained important herbivores during much of the period. Therocephalians included both large predators (*Moschorhinus*) and herbivorous forms (bauriids) until their extinction midway through the period. Ecteniniid cynodonts played a role as large-sized, cursorial predators in the Late Triassic. During the Carnian (early part of the Late Triassic), some advanced cynodonts gave rise to the first mammals.

During the Triassic, archosaurs displaced therapsids as the largest and most ecologically prolific terrestrial amniotes. This "Triassic Takeover" may have contributed to the evolution of mammals by forcing the surviving therapsids and their mammaliaform successors to live as small, mainly nocturnal insectivores. Nocturnal life may have forced the mammaliaforms to develop fur and a higher metabolic rate.[77]

Lystrosaurus was a widespread dicynodont and the most common land vertebrate during the Early Triassic, after animal life had been greatly diminished

Cynognathus was a carnivorous mammal-like cynodont from the Early Triassic.

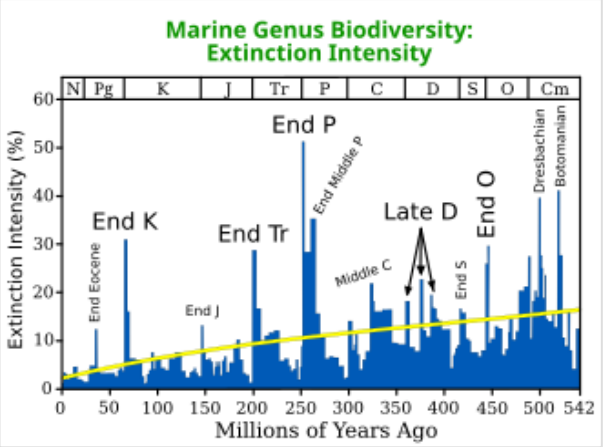
Lagerstätten

Two Early Triassic lagerstätten (high-quality fossil beds), the Dienerian aged Guiyang biota[78] and the earliest Spathian aged Paris biota[79] stand out due to their exceptional preservation and diversity. They represent the earliest lagerstätten of the Mesozoic era and provide insight into the biotic recovery from the Permian-Triassic mass extinction event.

The Monte San Giorgio lagerstätte, now in the Lake Lugano region of northern Italy and southern Switzerland, was in Middle Triassic times a lagoon behind reefs with an anoxic bottom layer, so there were no scavengers and little turbulence to disturb fossilization, a situation that can be compared to the better-known Jurassic Solnhofen Limestone lagerstätte. The remains of fish and various marine reptiles (including the common pachypleurosaur *Neusticosaurus*, and the bizarre long-necked archosauromorph *Tanystropheus*), along with some terrestrial forms like *Ticinosuchus* and *Macrocnemus*, have been recovered from this locality. All these fossils date from the Anisian and Ladinian ages (about 242 Ma ago).

Triassic–Jurassic extinction event

The Triassic Period ended with a mass extinction, which was particularly severe in the oceans; the conodonts disappeared, as did all the marine reptiles except ichthyosaurs and plesiosaurs. Invertebrates like brachiopods and molluscs (such as gastropods) were severely affected. In the oceans, 22% of marine families and possibly about half of marine genera went missing.



The mass extinction event is marked by 'End Tr'

Though the end-Triassic extinction event was not equally devastating in all terrestrial ecosystems, several important clades of crurotarsans (large archosaurian reptiles previously grouped together as the thecodonts) disappeared, as did most of the large labyrinthodont amphibians, groups of small reptiles, and most synapsids. Some of the early, primitive dinosaurs also became extinct, but more adaptive ones survived to evolve into the Jurassic. Surviving plants that went on to dominate the Mesozoic world included modern conifers and cycadeoids.

The cause of the Late Triassic extinction is uncertain. It was accompanied by huge volcanic eruptions that occurred as the supercontinent Pangaea began to break apart about 202 to 191 million years ago (40Ar/39Ar dates),[80] forming the Central Atlantic Magmatic Province (CAMP),[81] one of the largest known inland volcanic events since the planet had first cooled and stabilized. Other possible but less likely causes for the extinction events include global cooling or even a bolide impact, for which an impact crater containing Manicouagan Reservoir in Quebec, Canada, has been singled out. However, the Manicouagan impact melt has been dated to 214±1 Ma. The date of the Triassic-Jurassic boundary has also been more accurately fixed recently, at 201.4 Ma. Both dates are gaining accuracy by using more accurate forms of radiometric dating, in particular the decay of uranium to lead in zircons formed at time of the impact. So, the evidence suggests the Manicouagan impact preceded the end of the Triassic by approximately 10±2 Ma. It could not therefore be the immediate cause of the observed mass extinction.[82]

The number of Late Triassic extinctions is disputed. Some studies suggest that there are at least two periods of extinction towards the end of the Triassic, separated by 12 to 17 million years. But arguing against this is a recent study of North American faunas. In the Petrified Forest of northeast Arizona there is a unique sequence of late Carnian-early Norian terrestrial sediments. An analysis in 2002 found no significant change in the paleoenvironment.[83] Phytosaurs, the most common fossils there, experienced a change-over only at the genus level, and the number of species remained the same. Some aetosaurs, the next most common tetrapods, and early

Skull of a Triassic Period phytosaur found in the Petrified Forest National Park

dinosaurs, passed through unchanged. However, both phytosaurs and aetosaurs were among the groups of archosaur reptiles completely wiped out by the end-Triassic extinction event.

It seems likely then that there was some sort of end-Carnian extinction, when several herbivorous archosauromorph groups died out, while the large herbivorous therapsids—the kannemeyeriid dicynodonts and the traversodont cynodonts—were much reduced in the northern half of Pangaea (Laurasia).

These extinctions within the Triassic and at its end allowed the dinosaurs to expand into many niches that had become unoccupied. Dinosaurs became increasingly dominant, abundant and diverse, and remained that way for the next 150 million years. The true "Age of Dinosaurs" is during the following Jurassic and Cretaceous periods, rather than the Triassic.

See also

- Geologic time scale – System that relates geologic strata to time
- List of fossil sites *(with link directory)*
- Triassic land vertebrate faunachrons – Subdivisions of geological time
- Phylloceratina – Extinct suborder of molluscs
- Dinosaur – Diverse clade of specialized reptiles

Notes

1. Payne, Johnathan L.; Lehrmann, Daniel J.; Wei, Jiayong; Orchard, Michael J.; Schrag, Daniel P.; Knoll, Andrew H. (July 23, 2004). "Large Perturbations of the Carbon Cycle During Recovery from the End-Permian Extinction" (http://www.sciencemag.org/cgi/content/abstract/305/5683/506). *Science*. **305** (5683): 506–509. Bibcode:2004Sci...305..506P (https://ui.adsabs.harvard.edu/abs/2004Sci...305..506P). doi:10.1126/science.1097023 (https://doi.org/10.1126%2Fscience.1097023). PMID 15273391 (https://pubmed.ncbi.nlm.nih.gov/15273391). S2CID 35498132 (https://api.semanticscholar.org/CorpusID:35498132).
2. Retallack, Gregory J.; Veevers, John J.; Morante, Ric (February 1996). "Global coal gap between Permian–Triassic extinctions and middle Triassic recovery of peat forming plants" (http://bulletin.geoscienceworld.org/cgi/content/abstract/108/2/195). *Geological Society of America Bulletin*. **108** (2): 195–207. Bibcode:1996GSAB..108..195R (https://ui.adsabs.harvard.edu/abs/1996GSAB..108..195R). doi:10.1130/0016-7606(1996)108<0195:GCGBPT>2.3.CO;2 (https://doi.org/10.1130%2F0016-7606%281996%29108%3C0195%3AGCGBPT%3E2.3.CO%3B2). Retrieved September 29, 2007.
3. McElwain, Jennifer C.; Punyasena, Surangi W. (October 2007). "Mass extinction events and the plant fossil record" (https://www.cell.com/trends/ecology-evolution/abstract/S0169-5347(07)00256-X). *Trends in Ecology & Evolution*. **22** (10): 548–557. doi:10.1016/j.tree.2007.09.003 (https://doi.org/10.1016%2Fj.tree.2007.09.003). PMID 17919771 (https://pubmed.ncbi.nlm.nih.gov/17919771). Retrieved October 4, 2025.
4. Widmann, Philipp; Bucher, Hugo; Leu, Marc; Vennemann, Torsten; Bagherpour, Borhan; Schneebeli-Hermann, Elke; Goudemand, Nicolas; Schaltegger, Urs (2020). "Dynamics of the Largest Carbon Isotope Excursion During the Early Triassic Biotic Recovery" (https://doi.org/10.3389%2Ffeart.2020.00196). *Frontiers in Earth Science*. **8** (196): 196. Bibcode:2020FrEaS...8..196W (https://ui.adsabs.harvard.edu/abs/2020FrEaS...8..196W). doi:10.3389/feart.2020.00196 (https://doi.org/10.3389%2Ffeart.2020.00196).

5. "International Chronostratigraphic Chart" (https://stratigraphy.org/ICSchart/ChronostratChart2024-12.pdf) (PDF). *International Commission on Stratigraphy*. December 2024. Retrieved October 23, 2025.
6. Hongfu, Yin; Kexin, Zhang; Jinnan, Tong; Zunyi, Yang; Shunbao, Wu (June 2001). "The Global Stratotype Section and Point (GSSP) of the Permian-Triassic Boundary" (https://doi.org/10.18814%2Fepiiugs%2F2001%2Fv24i2%2F004). *Episodes*. **24** (2): 102–114. doi:10.18814/epiiugs/2001/v24i2/004 (https://doi.org/10.18814%2Fepiiugs%2F2001%2Fv24i2%2F004).
7. Hillebrandt, A.v.; Krystyn, L.; Kürschner, W. M.; et al. (September 2013). "The Global Stratotype Sections and Point (GSSP) for the base of the Jurassic System at Kuhjoch (Karwendel Mountains, Northern Calcareous Alps, Tyrol, Austria)" (https://doi.org/10.18814%2Fepiiugs%2F2013%2Fv36i3%2F001). *Episodes*. **36** (3): 162–198. doi:10.18814/epiiugs/2013/v36i3/001 (https://doi.org/10.18814%2Fepiiugs%2F2013%2Fv36i3%2F001).
8. "International Stratigraphic Chart" (https://stratigraphy.org/ICSchart/ChronostratChart2023-06.pdf) (PDF). International Commission on Stratigraphy. June 2023. Retrieved 19 July 2023.
9. "Triassic" (https://www.dictionary.com/browse/Triassic). *Dictionary.com Unabridged* (Online). n.d.
10. Sahney, Sarda; Benton, Michael J (7 April 2008). "Recovery from the most profound mass extinction of all time" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2596898). *Proceedings of the Royal Society B: Biological Sciences*. **275** (1636): 759–765. doi:10.1098/rspb.2007.1370 (https://doi.org/10.1098%2Frspb.2007.1370). PMC 2596898 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2596898). PMID 18198148 (https://pubmed.ncbi.nlm.nih.gov/18198148).
11. Brusatte, S. L.; Benton, M. J.; Ruta, M.; Lloyd, G. T. (12 September 2008). "Superiority, Competition, and Opportunism in the Evolutionary Radiation of Dinosaurs". *Science*. **321** (5895): 1485–1488. Bibcode:2008Sci...321.1485B (https://ui.adsabs.harvard.edu/abs/2008Sci...321.1485B). doi:10.1126/science.1161833 (https://doi.org/10.1126%2Fscience.1161833). hdl:20.500.11820/00556baf-6575-44d9-af39-bdd0b072ad2b (https://hdl.handle.net/20.500.11820%2F00556baf-6575-44d9-af39-bdd0b072ad2b). PMID 18787166 (https://pubmed.ncbi.nlm.nih.gov/18787166).
12. Dell'Amore, Christine (19 October 2012). " 'Lethally Hot' Earth Was Devoid of Life—Could It Happen Again?" (https://www.nationalgeographic.com/history/article/121018-triassic-extinctions-hot-global-warming-science-environment). *National Geographic*.
13. Friedrich von Alberti, *Beitrag zu einer Monographie des bunten Sandsteins, Muschelkalks und Keupers, und die Verbindung dieser Gebilde zu einer Formation* [Contribution to a monograph on the colored sandstone, shell limestone and mudstone, and the joining of these structures into one formation] (Stuttgart and Tübingen, (Germany): J. G. Cotta, 1834). Alberti coined the term "Trias" on page 324 (https://archive.org/details/bub_gb_Ie27AAAAIAAJ/page/n347) :
"… bunter Sandstein, Muschelkalk und Keuper das Resultat einer Periode, ihre Versteinerungen, um mich der Worte E. de Beaumont's zu bedeinen, die Thermometer einer geologischen Epoche seyen, … also die bis jezt beobachtete Trennung dieser Gebilde in 3 Formationen nicht angemessen, und es mehr dem Begriffe Formation entsprechend sey, sie zu einer Formation, welche ich vorläufig *Trias* nennen will, zu verbinden."
(… colored sandstone, shell limestone, and mudstone are the result of a period; their fossils are, to avail myself of the words of E. de Beaumont, the thermometer of a geologic epoch; … thus the separation of these structures into 3 formations, which has been maintained until now, isn't appropriate, and it is more consistent with the concept of "formation" to join them into one formation, which for now I will name "trias".)
14. Torsvik, Trond H.; Cocks, Leonard Robert Morrison (2017). *Earth history and palaeogeography*. Cambridge: Cambridge university press. ISBN 978-1-107-10532-4.

15. Song, Dongfang; Xiao, Wenjiao; Ao, Songjian; Mao, Qigui; Wan, Bo; Zeng, Hao (June 2024). "Contemporaneous closure of the Paleo-Asian Ocean in the Middle-Late Triassic: A synthesis of new evidence and tectonic implications for the final assembly of Pangea". *Earth-Science Reviews*. **253** 104771. Bibcode:2024ESRv..25304771S (https://ui.adsabs.harvard.edu/abs/2024ESRv..25304771S). doi:10.1016/j.earscirev.2024.104771 (https://doi.org/10.1016%2Fj.earscirev.2024.104771).
16. Scotese, C.R.; Schettino, A. (2017). "Late Permian-Early Jurassic Paleogeography of Western Tethys and the World". *Permo-Triassic Salt Provinces of Europe, North Africa and the Atlantic Margins*. pp. 57–95. doi:10.1016/B978-0-12-809417-4.00004-5 (https://doi.org/10.1016%2FB978-0-12-809417-4.00004-5). ISBN 978-0-12-809417-4.
17. Riel, Nicolas; Jaillard, Etienne; Martelat, Jean-Emmanuel; Guillot, Stéphane; Braun, Jean (January 2018). "Permian-Triassic Tethyan realm reorganization: Implications for the outward Pangea margin". *Journal of South American Earth Sciences*. **81**: 78–86. Bibcode:2018JSAES..81...78R (https://ui.adsabs.harvard.edu/abs/2018JSAES..81...78R). doi:10.1016/j.jsames.2017.11.007 (https://doi.org/10.1016%2Fj.jsames.2017.11.007).
18. Peace, Alexander L.; Phethean, J.J.J.; Franke, D.; Foulger, G.R.; Schiffer, C.; Welford, J.K.; McHone, G.; Rocchi, S.; Schnabel, M.; Doré, A.G. (July 2020). "A review of Pangaea dispersal and Large Igneous Provinces – In search of a causative mechanism". *Earth-Science Reviews*. **206** 102902. Bibcode:2020ESRv..20602902P (https://ui.adsabs.harvard.edu/abs/2020ESRv..20602902P). doi:10.1016/j.earscirev.2019.102902 (https://doi.org/10.1016%2Fj.earscirev.2019.102902).
19. Klausen, Tore Grane; Nyberg, Björn; Helland-Hansen, William (May 2019). "The largest delta plain in Earth's history" (https://doi.org/10.1130%2FG45507.1). *Geology*. **47** (5): 470–474. Bibcode:2019Geo....47..470K (https://ui.adsabs.harvard.edu/abs/2019Geo....47..470K). doi:10.1130/G45507.1 (https://doi.org/10.1130%2FG45507.1). hdl:1956/22168 (https://hdl.handle.net/1956%2F22168).
20. Hadlari, Thomas; Midwinter, Derrick; Poulton, Terry P.; Matthews, William A. (13 April 2017). "A Pangean rim of fire: Reviewing the Triassic of western Laurentia" (https://doi.org/10.1130%2FL643.1). *Lithosphere* L643.1. doi:10.1130/L643.1 (https://doi.org/10.1130%2FL643.1).
21. Frizon de Lamotte, Dominique; Fourdan, Brendan; Leleu, Sophie; Leparmentier, François; de Clarens, Philippe (2015). "Style of rifting and the stages of Pangea breakup". *Tectonics*. **34** (5): 1009–1029. Bibcode:2015Tecto..34.1009F (https://ui.adsabs.harvard.edu/abs/2015Tecto..34.1009F). doi:10.1002/2014TC003760 (https://doi.org/10.1002%2F2014TC003760).
22. Madanipour, Saeed; Najafi, Mahdi; Nozaem, Reza; Vergés, Jaume; Yassaghi, Ali; Heydari, Iraj; Khodaparast, Sedigheh; Soudmand, Zahra; Aghajari, Lotfollah (2024). "The Arabia – Eurasia Collision Zone in Iran: Tectonostratigraphic and Structural Synthesis". *Journal of Petroleum Geology*. **47** (2): 123–171. Bibcode:2024JPetG..47..123M (https://ui.adsabs.harvard.edu/abs/2024JPetG..47..123M). doi:10.1111/jpg.12854 (https://doi.org/10.1111%2Fjpg.12854).
23. Cao, Huawen; Huang, Yong; Li, Guangming; Zhang, Linkui; Wu, Jianyang; Dong, Lei; Dai, Zuowen; Lu, Liu (January 2018). "Late Triassic sedimentary records in the northern Tethyan Himalaya: Tectonic link with Greater India" (https://doi.org/10.1016%2Fj.gsf.2017.04.001). *Geoscience Frontiers*. **9** (1): 273–291. Bibcode:2018GeoFr...9..273C (https://ui.adsabs.harvard.edu/abs/2018GeoFr...9..273C). doi:10.1016/j.gsf.2017.04.001 (https://doi.org/10.1016%2Fj.gsf.2017.04.001).
24. Rosenbaum, Gideon (30 May 2018). "The Tasmanides: Phanerozoic Tectonic Evolution of Eastern Australia". *Annual Review of Earth and Planetary Sciences*. **46** (1): 291–325. Bibcode:2018AREPS..46..291R (https://ui.adsabs.harvard.edu/abs/2018AREPS..46..291R). doi:10.1146/annurev-earth-082517-010146 (https://doi.org/10.1146%2Fannurev-earth-082517-010146).

25. del Rey, Álvaro; Deckart, Katja; Planavsky, Noah; Arriagada, César; Martínez, Fernando (December 2019). "Tectonic evolution of the southwestern margin of Pangea and its global implications: Evidence from the mid Permian–Triassic magmatism along the Chilean-Argentine border". *Gondwana Research*. **76**: 303–321. Bibcode:2019GondR..76..303D (https://ui.adsabs.harvard.edu/abs/2019GondR..76..303D). doi:10.1016/j.gr.2019.05.007 (https://doi.org/10.1016%2Fj.gr.2019.05.007).
26. Şengör, A.M. Celâl; Altıner, Demir; Zabcı, Cengiz; Sunal, Gürsel; Lom, Nalan; Aylan, Eşref; Öner, Tayfun (December 2023). "On the nature of the Cimmerian Continent" (https://doi.org/10.1016%2Fj.earscirev.2023.104520). *Earth-Science Reviews*. **247** 104520. Bibcode:2023ESRv..24704520S (https://ui.adsabs.harvard.edu/abs/2023ESRv..24704520S). doi:10.1016/j.earscirev.2023.104520 (https://doi.org/10.1016%2Fj.earscirev.2023.104520).
27. Dong, Yunpeng; Sun, Shengsi; Santosh, M.; Zhao, Jie; Sun, Jiaopeng; He, Dengfeng; Shi, Xiaohui; Hui, Bo; Cheng, Chao; Zhang, Guowei (December 2021). "Central China Orogenic Belt and amalgamation of East Asian continents". *Gondwana Research*. **100**: 131–194. Bibcode:2021GondR.100..131D (https://ui.adsabs.harvard.edu/abs/2021GondR.100..131D). doi:10.1016/j.gr.2021.03.006 (https://doi.org/10.1016%2Fj.gr.2021.03.006).
28. Leleu, Sophie; Hartley, Adrian J.; van Oosterhout, Cees; Kennan, Lorcan; Ruckwied, Katrin; Gerdes, Keith (July 2016). "Structural, stratigraphic and sedimentological characterisation of a wide rift system: The Triassic rift system of the Central Atlantic Domain". *Earth-Science Reviews*. **158**: 89–124. Bibcode:2016ESRv..158...89L (https://ui.adsabs.harvard.edu/abs/2016ESRv..158...89L). doi:10.1016/j.earscirev.2016.03.008 (https://doi.org/10.1016%2Fj.earscirev.2016.03.008). hdl:2164/8901 (https://hdl.handle.net/2164%2F8901).
29. Marzoli, Andrea; Callegaro, Sara; Dal Corso, Jacopo; Davies, Joshua H. F. L.; Chiaradia, Massimo; Youbi, Nassrrdine; Bertrand, Hervé; Reisberg, Laurie; Merle, Renaud; Jourdan, Fred (2018). "The Central Atlantic Magmatic Province (CAMP): A Review". *The Late Triassic World*. Topics in Geobiology. Vol. 46. pp. 91–125. doi:10.1007/978-3-319-68009-5_4 (https://doi.org/10.1007%2F978-3-319-68009-5_4). ISBN 978-3-319-68008-8.
30. Kent, Dennis V.; Olsen, Paul E.; Wang, Huapei; Schaller, Morgan F.; Et-Touhami, Mohammed (12 November 2024). "Correlation of sub-centennial-scale pulses of initial Central Atlantic Magmatic Province lavas and the end-Triassic extinctions" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11573653). *Proceedings of the National Academy of Sciences*. **121** (46) e2415486121. Bibcode:2024PNAS..12115486K (https://ui.adsabs.harvard.edu/abs/2024PNAS..12115486K). doi:10.1073/pnas.2415486121 (https://doi.org/10.1073%2Fpnas.2415486121). PMC 11573653 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11573653). PMID 39467154 (https://pubmed.ncbi.nlm.nih.gov/39467154).
31. Herbert, Chris; Helby, Robin (1980). *A Guide to the Sydney basin*. Maitland, NSW: Geological Survey of NSW. p. 582. ISBN 978-0-7240-1250-3.
32. Haq, Bilal U. (December 2018). "Triassic eustatic variations reexamined" (https://doi.org/10.1130%2FGSATG381A.1). *GSA Today*. **28** (12). Geological Society of America: 4–9. Bibcode:2018GSAT...28l...4H (https://ui.adsabs.harvard.edu/abs/2018GSAT...28l...4H). doi:10.1130/GSATG381A.1 (https://doi.org/10.1130%2FGSATG381A.1).
33. Embry, Ashton F. (1988). "Triassic Sea-Level Changes: Evidence from the Canadian Arctic Archipelago". *Sea-Level Changes*. pp. 249–259. doi:10.2110/pec.88.01.0249 (https://doi.org/10.2110%2Fpec.88.01.0249). ISBN 978-1-56576-089-9.
34. Stanley, 452–53.
35. Zeng, Zhiwei; Zhu, Hongtao; Yang, Xianghua; Zeng, Hongliu; Hu, Xiaolin; Xia, Chenchen (May 2019). "The Pangaea Megamonsoon records: Evidence from the Triassic Mungaroo Formation, Northwest Shelf of Australia". *Gondwana Research*. **69**: 1–24. Bibcode:2019GondR..69....1Z (https://ui.adsabs.harvard.edu/abs/2019GondR..69....1Z). doi:10.1016/j.gr.2018.11.015 (https://doi.org/10.1016%2Fj.gr.2018.11.015).

36. Preto, N.; Kustatscher, E.; Wignall, P. B. (2010). "Triassic climates – State of the art and perspectives". *Palaeogeography, Palaeoclimatology, Palaeoecology*. **290** (1–4): 1–10. Bibcode:2010PPP...290....1P (https://ui.adsabs.harvard.edu/abs/2010PPP...290....1P). doi:10.1016/j.palaeo.2010.03.015 (https://doi.org/10.1016%2Fj.palaeo.2010.03.015).
37. Scotese, Christopher Robert; Song, Haijun; Mills, Benjamin J. W.; van der Meer, Douwe G. (April 2021). "Phanerozoic paleotemperatures: The earth's changing climate during the last 540 million years". *Earth-Science Reviews*. **215** 103503. Bibcode:2021ESRv..21503503S (https://ui.adsabs.harvard.edu/abs/2021ESRv..21503503S). doi:10.1016/j.earscirev.2021.103503 (https://doi.org/10.1016%2Fj.earscirev.2021.103503).
38. Trotter, Julie A.; Williams, Ian S.; Nicora, Alda; Mazza, Michele; Rigo, Manuel (April 2015). "Long-term cycles of Triassic climate change: a new δ18O record from conodont apatite". *Earth and Planetary Science Letters*. **415**: 165–174. Bibcode:2015E&PSL.415..165T (https://ui.adsabs.harvard.edu/abs/2015E&PSL.415..165T). doi:10.1016/j.epsl.2015.01.038 (https://doi.org/10.1016%2Fj.epsl.2015.01.038).
39. Dal Corso, Jacopo; Mills, Benjamin J.W.; Chu, Daoliang; Newton, Robert J.; Song, Haijun (15 January 2022). "Background Earth system state amplified Carnian (Late Triassic) environmental changes". *Earth and Planetary Science Letters*. **578** 117321. Bibcode:2022E&PSL.57817321D (https://ui.adsabs.harvard.edu/abs/2022E&PSL.57817321D). doi:10.1016/j.epsl.2021.117321 (https://doi.org/10.1016%2Fj.epsl.2021.117321).
40. Dal Corso, J.; Mietto, P.; Newton, R. J.; Pancost, R. D.; Preto, N.; Roghi, G.; Wignall, P. B. (2012). "Discovery of a major negative 13C spike in the Carnian (Late Triassic) linked to the eruption of Wrangellia flood basalts". *Geology*. **40** (1): 79–82. Bibcode:2012Geo....40...79D (https://ui.adsabs.harvard.edu/abs/2012Geo....40...79D). doi:10.1130/G32473.1 (https://doi.org/10.1130%2FG32473.1).
41. Li, Liqin; Kürschner, Wolfram M.; Lu, Ning; Chen, Hongyu; An, Pengcheng; Wang, Yongdong (September 2022). "Palynological record of the Carnian Pluvial Episode from the northwestern Sichuan Basin, SW China". *Review of Palaeobotany and Palynology*. **304** 104704. Bibcode:2022RPaPa.30404704L (https://ui.adsabs.harvard.edu/abs/2022RPaPa.30404704L). doi:10.1016/j.revpalbo.2022.104704 (https://doi.org/10.1016%2Fj.revpalbo.2022.104704). hdl:10852/99190 (https://hdl.handle.net/10852%2F99190).
42. Muttoni, Giovanni; Mazza, Michele; Mosher, David; Katz, Miriam E.; Kent, Dennis V.; Balini, Marco (April 2014). "A Middle–Late Triassic (Ladinian–Rhaetian) carbon and oxygen isotope record from the Tethyan Ocean". *Palaeogeography, Palaeoclimatology, Palaeoecology*. **399**: 246–259. Bibcode:2014PPP...399..246M (https://ui.adsabs.harvard.edu/abs/2014PPP...399..246M). doi:10.1016/j.palaeo.2014.01.018 (https://doi.org/10.1016%2Fj.palaeo.2014.01.018).
43. Ikeda, Masayuki; Ozaki, Kazumi; Legrand, Julien (23 July 2020). "Impact of 10-Myr scale monsoon dynamics on Mesozoic climate and ecosystems" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7378230). *Scientific Reports*. **10** (1): 11984. Bibcode:2020NatSR..1011984I (https://ui.adsabs.harvard.edu/abs/2020NatSR..1011984I). doi:10.1038/s41598-020-68542-w (https://doi.org/10.1038%2Fs41598-020-68542-w). PMC 7378230 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7378230). PMID 32704030 (https://pubmed.ncbi.nlm.nih.gov/32704030).
44. Capriolo, Manfredo; Marzoli, Andrea; Aradi, László E.; Callegaro, Sara; Dal Corso, Jacopo; Newton, Robert J.; Mills, Benjamin J. W.; Wignall, Paul B.; Bartoli, Omar; Baker, Don R.; Youbi, Nasrrddine; Remusat, Laurent; Spiess, Richard; Szabó, Csaba (7 April 2020). "Deep CO2 in the end-Triassic Central Atlantic Magmatic Province" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7138847). *Nature Communications*. **11** (1): 1670. Bibcode:2020NatCo..11.1670C (https://ui.adsabs.harvard.edu/abs/2020NatCo..11.1670C). doi:10.1038/s41467-020-15325-6 (https://doi.org/10.1038%2Fs41467-020-15325-6). PMC 7138847 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7138847). PMID 32265448 (https://pubmed.ncbi.nlm.nih.gov/32265448).

45. Looy, Cindy V.; van Konijnenburg-van Cittert, Johanna H. A.; Duijnstee, Ivo A. P. (2 March 2021). "Proliferation of Isoëtalean Lycophytes During the Permo-Triassic Biotic Crises: A Proxy for the State of the Terrestrial Biosphere" (https://doi.org/10.3389%2Ffeart.2021.615370). *Frontiers in Earth Science*. **9** 615370. Bibcode:2021FrEaS...9...55L (https://ui.adsabs.harvard.edu/abs/2021FrEaS...9...55L). doi:10.3389/feart.2021.615370 (https://doi.org/10.3389%2Ffeart.2021.615370).
46. Blomenkemper, Patrick; Bäumer, Robert; Backer, Malte; Abu Hamad, Abdalla; Wang, Jun; Kerp, Hans; Bomfleur, Benjamin (2021). "Bennettitalean Leaves From the Permian of Equatorial Pangea—The Early Radiation of an Iconic Mesozoic Gymnosperm Group" (https://doi.org/10.3389%2Ffeart.2021.652699). *Frontiers in Earth Science*. **9** 652699: 162. Bibcode:2021FrEaS...9..162B (https://ui.adsabs.harvard.edu/abs/2021FrEaS...9..162B). doi:10.3389/feart.2021.652699 (https://doi.org/10.3389%2Ffeart.2021.652699).
47. Mays, Chris; McLoughlin, Stephen (2020-02-25). "Caught between two mass extinctions: The rise and fall of Dicroidium" (https://depositsmag.com/2020/02/25/caught-between-two-mass-extinctions-the-rise-and-fall-of-dicroidium/). *Deposits Mag*. Retrieved 2023-09-23.
48. Zhou, Zhi-Yan (March 2009). "An overview of fossil Ginkgoales". *Palaeoworld*. **18** (1): 1–22. doi:10.1016/j.palwor.2009.01.001 (https://doi.org/10.1016%2Fj.palwor.2009.01.001).
49. Forte, Giuseppa; Kustatscher, Evelyn; Van Konijnenburg-van Cittert, Johanna H. A. (July 2021). "Conifer Diversity in the Middle Triassic: New Data from the Fossillagerstätte Kühwiesenkopf/Monte Prà della Vacca (Pelsonian, Anisian) in the Dolomites (Northeastern Italy)". *International Journal of Plant Sciences*. **182** (6): 445–467. Bibcode:2021IJPlS.182..445F (https://ui.adsabs.harvard.edu/abs/2021IJPlS.182..445F). doi:10.1086/714280 (https://doi.org/10.1086%2F714280).
50. Escapa, Ignacio; Leslie, Andrew (February 2017). "A new Cheirolepidiaceae (Coniferales) from the Early Jurassic of Patagonia (Argentina): Reconciling the records of impression and permineralized fossils". *American Journal of Botany*. **104** (2): 322–334. doi:10.3732/ajb.1600321 (https://doi.org/10.3732%2Fajb.1600321). hdl:11336/40738 (https://hdl.handle.net/11336%2F40738). PMID 28213347 (https://pubmed.ncbi.nlm.nih.gov/28213347).
51. Rothwell, Gar W.; Grauvogel-Stamm, Léa; Mapes, Gene (2000). "An herbaceous fossil conifer: Gymnospermous ruderals in the evolution of Mesozoic vegetation" (https://linkinghub.elsevier.com/retrieve/pii/S0031018299001364). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **156** (1–2): 139–145. doi:10.1016/S0031-0182(99)00136-4 (https://doi.org/10.1016%2FS0031-0182%2899%2900136-4). Retrieved 2026-09-08.
52. Hosher, WT Magaritz M Clark D (1987). "Events near the time of the Permian-Triassic boundary". *Mod. Geol*. **11**: 155–80 [173–74].
53. Retallack, G. J.; Veevers, J. J.; Morante, R. (1996). "Global coal gap between Permian-Triassic extinction and Middle Triassic recovery of peat-forming plants". *Bulletin of the Geological Society of America*. **108** (2): 195–207. Bibcode:1996GSAB..108..195R (https://ui.adsabs.harvard.edu/abs/1996GSAB..108..195R). doi:10.1130/0016-7606(1996)108<0195:GCGBPT>2.3.CO;2 (https://doi.org/10.1130%2F0016-7606%281996%29108%3C0195%3AGCGBPT%3E2.3.CO%3B2).
54. Holser, WT; Schoenlaub, H-P; Klein, P; Attrep, M; Boeckelmann, Klaus; et al. (1989). "A unique geochemical record at the Permian/Triassic boundary". *Nature*. **337** (6202): 39 [42]. Bibcode:1989Natur.337...39H (https://ui.adsabs.harvard.edu/abs/1989Natur.337...39H). doi:10.1038/337039a0 (https://doi.org/10.1038%2F337039a0).
55. Retallack, Gregory J.; Krull, Evelyn S. (2006). "Carbon isotopic evidence for terminal-Permian methane outbursts and their role in extinctions of animals, plants, coral reefs, and peat swamps". *Wetlands through Time*. doi:10.1130/2006.2399(12) (https://doi.org/10.1130%2F2006.2399%2812%29). ISBN 978-0-8137-2399-0.
56. "How snowball Earth gave rise to complex life – Cosmos Magazine" (https://cosmosmagazine.com/geoscience/how-snowball-earth-gave-rise-to-complex-life). 16 August 2017.

57. "December: Phytoplankton | News | University of Bristol" (https://www.bristol.ac.uk/news/2015/december/bacteria-bloom-evolution.html). December 2015.
58. Brocks, Jochen J.; Jarrett, Amber J. M.; Sirantoine, Eva; Hallmann, Christian; Hoshino, Yosuke; Liyanage, Tharika (August 2017). "The rise of algae in Cryogenian oceans and the emergence of animals". *Nature*. **548** (7669): 578–581. doi:10.1038/nature23457 (https://doi.org/10.1038%2Fnature23457). PMID 28813409 (https://pubmed.ncbi.nlm.nih.gov/28813409).
59. Jin, Xin; Shi, Zhiqiang; Rigo, Manuel; Franceschi, Marco; Preto, Nereo (15 September 2018). "Carbonate platform crisis in the Carnian (Late Triassic) of Hanwang (Sichuan Basin, South China): Insights from conodonts and stable isotope data". *Journal of Asian Earth Sciences*. **164**: 104–124. Bibcode:2018JAESc.164..104J (https://ui.adsabs.harvard.edu/abs/2018JAESc.164..104J). doi:10.1016/j.jseaes.2018.06.021 (https://doi.org/10.1016%2Fj.jseaes.2018.06.021).
60. Vinn, O.; Mutvei, H. (2009). "Calcareous tubeworms of the Phanerozoic" (https://doi.org/10.3176%2Fearth.2009.4.07). *Estonian Journal of Earth Sciences*. **58** (4): 286–96. Bibcode:2009EsJES..58..286V (https://ui.adsabs.harvard.edu/abs/2009EsJES..58..286V). doi:10.3176/earth.2009.4.07 (https://doi.org/10.3176%2Fearth.2009.4.07).
61. Komatsu, Toshifumi; Chen, Jin-hua; Cao, Mei-zhen; Stiller, Frank; Naruse, Hajime (10 June 2004). "Middle Triassic (Anisian) diversified bivalves: depositional environments and bivalve assemblages in the Leidapo Member of the Qingyan Formation, southern China". *Palaeogeography, Palaeoclimatology, Palaeoecology*. **208** (3–4): 227–223. Bibcode:2004PPP...208..207K (https://ui.adsabs.harvard.edu/abs/2004PPP...208..207K). doi:10.1016/j.palaeo.2004.03.005 (https://doi.org/10.1016%2Fj.palaeo.2004.03.005).
62. Zheng, Daran; Chang, Su-Chin; Wang, He; Fang, Yan; Wang, Jun; Feng, Chongqing; Xie, Guwei; Jarzembowski, Edmund A.; Zhang, Haichun; Wang, Bo (7 September 2018). "Middle-Late Triassic insect radiation revealed by diverse fossils and isotopic ages from China" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6124916). *Science Advances*. **4** (9) eaat1380. Bibcode:2018SciA....4.1380Z (https://ui.adsabs.harvard.edu/abs/2018SciA....4.1380Z). doi:10.1126/sciadv.aat1380 (https://doi.org/10.1126%2Fsciadv.aat1380). PMC 6124916 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6124916). PMID 30191177 (https://pubmed.ncbi.nlm.nih.gov/30191177).
63. Romano, Carlo; Koot, Martha B.; Kogan, Ilja; Brayard, Arnaud; Minikh, Alla V.; Brinkmann, Winand; Bucher, Hugo; Kriwet, Jürgen (February 2016). "Permian-Triassic Osteichthyes (bony fishes): diversity dynamics and body size evolution" (https://hal.science/hal-01253154). *Biological Reviews*. **91** (1): 106–47. doi:10.1111/brv.12161 (https://doi.org/10.1111%2Fbrv.12161). PMID 25431138 (https://pubmed.ncbi.nlm.nih.gov/25431138).
64. Cavin, Lionel; Furrer, Heinz; Obrist, Christian (2013). "New coelacanth material from the Middle Triassic of eastern Switzerland, and comments on the taxic diversity of actinistans" (https://doi.org/10.1007%2Fs00015-013-0143-7). *Swiss Journal of Geoscience*. **106** (2): 161–177. Bibcode:2013SwJG..106..161C (https://ui.adsabs.harvard.edu/abs/2013SwJG..106..161C). doi:10.1007/s00015-013-0143-7 (https://doi.org/10.1007%2Fs00015-013-0143-7).
65. Romano, Carlo (January 2021). "A hiatus obscures the early evolution of Modern lineages of bony fishes" (https://doi.org/10.3389%2Ffeart.2020.618853). *Frontiers in Earth Science*. **8** 618853. Bibcode:2021FrEaS...8.8853R (https://ui.adsabs.harvard.edu/abs/2021FrEaS...8.8853R). doi:10.3389/feart.2020.618853 (https://doi.org/10.3389%2Ffeart.2020.618853).
66. Romano, Carlo; Jenks, James F.; Jattiot, Romain; Scheyer, Torsten M. (2017). "Marine Early Triassic Actinopterygii from Elko County (Nevada, USA): implications for the Smithian equatorial vertebrate eclipse" (https://doi.org/10.1017%2Fjpa.2017.36). *Journal of Paleontology*. **91** (5): 1–22. Bibcode:2017JPal...91.1025R (https://ui.adsabs.harvard.edu/abs/2017JPal...91.1025R). doi:10.1017/jpa.2017.36 (https://doi.org/10.1017%2Fjpa.2017.36).

67. Agnolin, Federico L.; Mateus, Octávio; Milàn, Jesper; Marzola, Marco; Wings, Oliver; Adolfssen, Jan Schulz; Clemmensen, Lars B. (4 March 2018). "Ceratodus tunuensis, sp. nov., a new lungfish (Sarcopterygii, Dipnoi) from the Upper Triassic of central East Greenland". *Journal of Vertebrate Paleontology*. **38** (2) e1439834. doi:10.1080/02724634.2018.1439834 (https://doi.org/10.1080%2F02724634.2018.1439834).
68. Kumar, Krishna; Bajpai, Sunil; Pandey, Pragya; Ghosh, Triparna; Bhattacharya, Debasish (3 June 2022). "Hybodont sharks from the Jurassic of Jaisalmer, western India". *Historical Biology*. **34** (6): 953–963. doi:10.1080/08912963.2021.1954920 (https://doi.org/10.1080%2F08912963.2021.1954920).
69. Mutter, Raoul J.; Neuman, Andrew G. (2008). "New eugeneodontid sharks from the Lower Triassic Sulphur Mountain Formation of Western Canada". In Cavin, L.; Longbottom, A.; Richter, M. (eds.). *Fishes and the Break-up of Pangaea*. Geological Society of London, Special Publications. Vol. 295. London: Geological Society of London. pp. 9–41. doi:10.1144/sp295.3 (https://doi.org/10.1144%2Fsp295.3).
70. Wells, Kentwood D. (2010). *The Ecology and Behavior of Amphibians* (https://books.google.com/books?id=eDKEKy5JJbIC&pg=PA8). University of Chicago Press. p. 8. ISBN 978-0-226-89333-4.
71. Benton, Michael J. (2009). "Temnospondyls and Reptiliomorphs after the Carboniferous" (https://books.google.com/books?id=VThUUUtM8A4C&pg=PA97). *Vertebrate Palaeontology*. John Wiley & Sons. pp. 97–99. ISBN 978-1-4051-4449-0.
72. Farman, Roy M.; Bell, Phil R. (September 2020). "Australia's earliest tetrapod swimming traces from the Hawkesbury Sandstone (Middle Triassic) of the Sydney Basin". *Journal of Paleontology*. **94** (5): 966–978. doi:10.1017/jpa.2020.22 (https://doi.org/10.1017%2Fjpa.2020.22).
73. Douglas Palmer; Peter Barrett (2009). *Evolution: The Story of Life*. London: The Natural History Museum. ISBN 978-1-84533-339-3.
74. Dunne, Emma M.; Farnsworth, Alexander; Benson, Roger B. J.; Godoy, Pedro L.; Greene, Sarah E.; Valdes, Paul J.; Lunt, Daniel J.; Butler, Richard J. (9 January 2023). "Climatic controls on the ecological ascendancy of dinosaurs" (https://doi.org/10.1016%2Fj.cub.2022.11.064). *Current Biology*. **33** (1): 206–214. Bibcode:2023CBio...33E.206D (https://ui.adsabs.harvard.edu/abs/2023CBio...33E.206D). doi:10.1016/j.cub.2022.11.064 (https://doi.org/10.1016%2Fj.cub.2022.11.064). hdl:1983/aea1ae86-2260-4d4d-a9d5-0fe38a0f470e (https://hdl.handle.net/1983%2Faea1ae86-2260-4d4d-a9d5-0fe38a0f470e). PMID 36528026 (https://pubmed.ncbi.nlm.nih.gov/36528026).
75. Scheyer, Torsten M.; Romano, Carlo; Jenks, Jim; Bucher, Hugo (19 March 2014). "Early Triassic Marine Biotic Recovery: The Predators' Perspective" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3960099). *PLOS ONE*. **9** (3) e88987. doi:10.1371/journal.pone.0088987 (https://doi.org/10.1371%2Fjournal.pone.0088987). PMC 3960099 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3960099). PMID 24647136 (https://pubmed.ncbi.nlm.nih.gov/24647136).
76. Zakharov, Y. D.; Nakajima, Y.; Arkhangelsky, M. S.; Popov, A. M.; Bondarenko, L. G.; Smyshlyaeva, O. P.; Pokrovsky, V. K. (15 May 2024). "New Finds of Triassic Marine Reptiles from Eastern Russia: Ammonoid Age Control and Possible Evidence for Ichthyopterygian". *Stratigraphy and Geological Correlation*. **32** (3): 242–264. Bibcode:2024SGC....32..242Z (https://ui.adsabs.harvard.edu/abs/2024SGC....32..242Z). doi:10.1134/S0869593824030080 (https://doi.org/10.1134%2FS0869593824030080).
77. Ruben, John A.; Jones, Terry D. (August 2000). "Selective Factors Associated with the Origin of Fur and Feathers" (https://doi.org/10.1093%2Ficb%2F40.4.585). *American Zoologist*. **40** (4): 585–596. doi:10.1093/icb/40.4.585 (https://doi.org/10.1093%2Ficb%2F40.4.585).

78. Dai, Xu; Davies, Joshua H.F.L.; Yuan, Zhiwei; Brayard, Arnaud; Ovtcharova, Maria; Xu, Guanghui; Liu, Xiaokang; Smith, Christopher P.A.; Schweitzer, Carrie E.; Li, Mingtao; Perrot, Morgann G.; Jiang, Shouyi; Miao, Luyi; Cao, Yiran; Yan, Jia; Bai, Ruoyu; Wang, Fengyu; Guo, Wei; Song, Huyue; Tian, Li; Dal Corso, Jacopo; Liu, Yuting; Chu, Daoliang; Song, Haijun (2023). "A Mesozoic fossil lagerstätte from 250.8 million years ago shows a modern-type marine ecosystem" (https://u-bourgogne.hal.science/hal-04016004). *Science*. **379** (6632): 567–572. Bibcode:2023Sci...379..567D (https://ui.adsabs.harvard.edu/abs/2023Sci...379..567D). doi:10.1126/science.adf1622 (https://doi.org/10.1126%2Fscience.adf1622). PMID 36758082 (https://pubmed.ncbi.nlm.nih.gov/36758082).
79. Brayard, Arnaud; Krumenacker, L. J.; Botting, Joseph P.; Jenks, James F.; Bylund, Kevin G.; Fara, Emmanuel; Vennin, Emmanuelle; Olivier, Nicolas; Goudemand, Nicolas; Saucède, Thomas; Charbonnier, Sylvain; Romano, Carlo; Doguzhaeva, Larisa; Thuy, Ben; Hautmann, Michael; Stephen, Daniel A.; Thomazo, Christophe; Escarguel, Gilles (2017). "Unexpected Early Triassic marine ecosystem and the rise of the Modern evolutionary fauna" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5310825). *Science Advances*. **3** (2) e1602159. Bibcode:2017SciA....3E2159B (https://ui.adsabs.harvard.edu/abs/2017SciA....3E2159B). doi:10.1126/sciadv.1602159 (https://doi.org/10.1126%2Fsciadv.1602159). PMC 5310825 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5310825). PMID 28246643 (https://pubmed.ncbi.nlm.nih.gov/28246643).
80. Nomade, S.; Knight, K.B.; Beutel, E.; Renne, P.R.; Verati, C.; Féraud, G.; Marzoli, A.; Youbi, N.; Bertrand, H. (February 2007). "Chronology of the Central Atlantic Magmatic Province: Implications for the Central Atlantic rifting processes and the Triassic–Jurassic biotic crisis". *Palaeogeography, Palaeoclimatology, Palaeoecology*. **244** (1–4): 326–344. doi:10.1016/j.palaeo.2006.06.034 (https://doi.org/10.1016%2Fj.palaeo.2006.06.034).
81. Marzoli, Andrea; Renne, Paul R.; Piccirillo, Enzo M.; Ernesto, Marcia; Bellieni, Giuliano; Min, Angelo De (23 April 1999). "Extensive 200-Million-Year-Old Continental Flood Basalts of the Central Atlantic Magmatic Province". *Science*. **284** (5414): 616–618. doi:10.1126/science.284.5414.616 (https://doi.org/10.1126%2Fscience.284.5414.616). PMID 10213679 (https://pubmed.ncbi.nlm.nih.gov/10213679).
82. Hodych, J. P.; Dunning, G. R. (1992). "Did the Manicouagan impact trigger end-of-Triassic mass extinction?". *Geology*. **20** (1): 51. doi:10.1130/0091-7613(1992)020<0051:DTMITE>2.3.CO;2 (https://doi.org/10.1130%2F0091-7613%281992%29020%3C0051%3ADTMITE%3E2.3.CO%3B2).
83. "No Significant Nonmarine Carnian-Norian (Late Triassic) Extinction Event: Evidence From Petrified Forest National Park" (https://web.archive.org/web/20031106231251/http://gsa.confex.com/gsa/2002AM/finalprogram/abstract_42936.htm). *gsa.confex.com*. Archived from the original (http://gsa.confex.com/gsa/2002AM/finalprogram/abstract_42936.htm) on 2003-11-06. Retrieved 2003-12-12.

References

- Emiliani, Cesare. (1992). *Planet Earth: Cosmology, Geology, & the Evolution of Life & the Environment*. Cambridge University Press. (Paperback Edition ISBN 0-521-40949-7)
- Ogg, Jim; June, 2004, *Overview of Global Boundary Stratotype Sections and Points (GSSP's)* Stratigraphy.org (http://www.stratigraphy.org/bak/gssp.htm), Accessed April 30, 2006
- Stanley, Steven M. *Earth System History.* New York: W.H. Freeman and Company, 1999. ISBN 0-7167-2882-6
- Sues, Hans-Dieter & Fraser, Nicholas C. *Triassic Life on Land: The Great Transition* New York: Columbia University Press, 2010. Series: Critical Moments and Perspectives in Earth History and Paleobiology. ISBN 978-0-231-13522-1

- van Andel, Tjeerd, (1985) 1994, *New Views on an Old Planet: A History of Global Change*, Cambridge University Press

External links

- Overall introduction (https://web.archive.org/web/20151210044546/http://palaeos.com/mesozoic/triassic/triassic.htm)
- 'The Triassic world' (https://web.archive.org/web/20160222224358/http://rainbow.ldgo.columbia.edu/courses/v1001/9.html)
- Paleofiles page on the Triassic extinctions (https://web.archive.org/web/20170801065542/http://palaeo.gly.bris.ac.uk/palaeofiles/triassic/triextict.htm)
- Examples of Triassic Fossils (http://www.geo-lieven.com/erdzeitalter/trias/trias.htm)
- Triassic (chronostratigraphy scale) (https://web.archive.org/web/20200809123734/https://ghkclass.com/ghkC.html?triassic)

Retrieved from "https://en.wikipedia.org/w/index.php?title=Triassic&oldid=1380461221"

WIKIPEDIA
25 years of the free encyclopedia

Jurassic

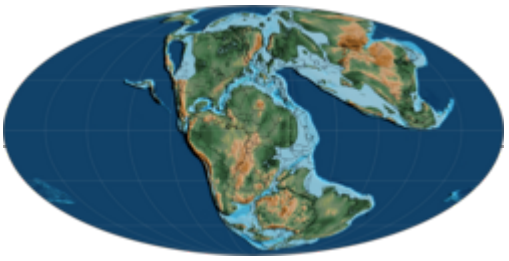
The **Jurassic** (/dʒʊˈræsɪk/ *juurr-ASS-ik*[2]) is a geologic period and stratigraphic system that lasted about 58.3 million years, spanning from the end of the Triassic Period 201.4 Ma (million years ago) to the beginning of the Cretaceous Period, 143.1 Ma. The Jurassic constitutes the second and middle period of the Mesozoic Era as well as the eighth period of the Phanerozoic Eon and is named after the Jura Mountains, where limestone strata from the period were first identified.

The start of the Jurassic was marked by the major Triassic–Jurassic extinction event, associated with the eruption of the Central Atlantic Magmatic Province (CAMP). The beginning of the Toarcian Age started around 183 Ma and is marked by the Toarcian Oceanic Anoxic Event, a global episode of oceanic anoxia, ocean acidification, and elevated global temperatures associated with extinctions, likely caused by the eruption of the Karoo-Ferrar large igneous provinces. The end of the Jurassic, however, has no clear, definitive boundary with the Cretaceous and is the only boundary between geological periods to remain formally undefined.

By the beginning of the Jurassic, the supercontinent Pangaea had begun rifting into two landmasses: Laurasia to the north and Gondwana to the south. The climate of the Jurassic was warmer than the present, and there were no ice caps. Forests grew close to the poles, with large arid expanses in the lower latitudes.

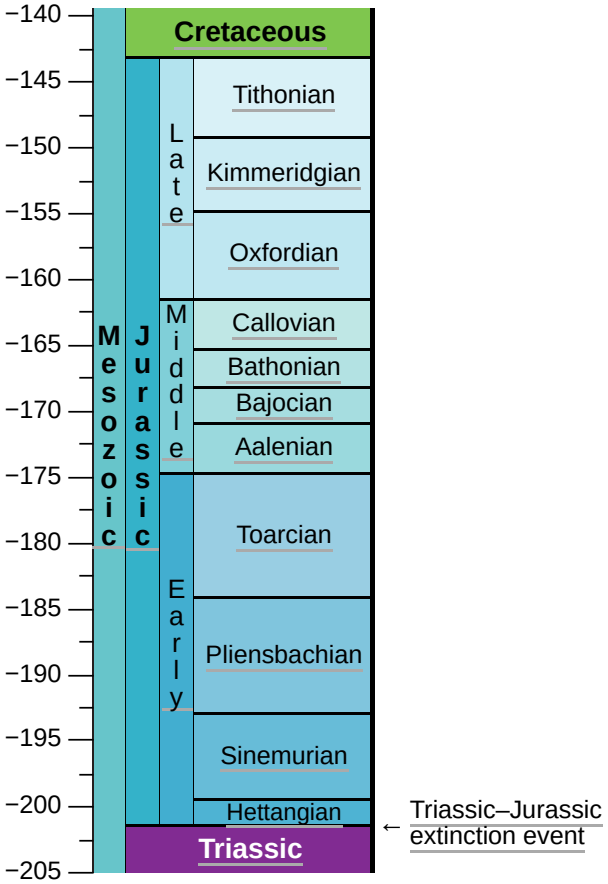
On land, the fauna transitioned from the Triassic fauna, dominated jointly by dinosauromorph and pseudosuchian

Jurassic



A paleomap of Earth as it appeared 170 million years ago during the Middle Jurassic Epoch

Chronology



Subdivision of the Jurassic according to the ICS, as of 2024.[1]

Vertical axis scale: Millions of years ago

Etymology	
Name formality	Formal
Usage information	
Regional usage	Global (ICS)

archosaurs to one dominated by dinosaurs alone. The first stem-group birds appeared during the Jurassic, evolving from a branch of theropod dinosaurs. Other major events include the appearance of the earliest crabs and modern frogs, salamanders and lizards. Mammaliaformes, one of the few cynodont lineages to survive the end of the Triassic, continued to diversify throughout the period, with the Jurassic seeing the emergence of the first crown group mammals. Crocodylomorphs made the transition from a terrestrial to an aquatic life. The oceans were inhabited by marine reptiles such as ichthyosaurs and plesiosaurs, while pterosaurs were the dominant flying vertebrates. Modern sharks and rays first appeared and diversified during the period, while the first known crown-group teleost fish (the dominant group of modern fish) appeared near the end of the period. The flora was dominated by ferns and gymnosperms, including conifers, of which many modern groups made their first appearance during the period, as well as other groups like the extinct Bennettitales.

Time scale(s) used	ICS Time Scale
	Definition
Chronological unit	Period
Stratigraphic unit	System
Time span formality	Formal
Lower boundary definition	First appearance of the ammonite *Psiloceras spelae tirolicum*.
Lower boundary GSSP	Kuhjoch section, Karwendel mountains, Northern Calcareous Alps, Austria 47.4839°N 11.5306°E
Lower GSSP ratified	2010
Upper boundary definition	Not formally defined
Upper boundary definition candidates	▪ Magnetic—base of Chron M18r ▪ Base of Calpionellid zone B ▪ FAD of ammonite *Berriasella jacobi*
Upper boundary GSSP candidate section(s)	None

Etymology and history

The chronostratigraphic term "Jurassic" originates from the Jura Mountains, a forested mountain range adjacent to the Alps that mainly follows the France–Switzerland border. During a tour of the region in 1795, German naturalist Alexander von Humboldt recognized carbonate deposits within the Jura Mountains as geologically distinct from the Triassic aged Muschelkalk of southern Germany, but he erroneously concluded that they were older. He then named them *Jura-Kalkstein* ('Jura limestone') in 1799.[3]

In 1829, the French naturalist Alexandre Brongniart published a book entitled *Tableau des terrains qui composent l'écorce du globe, ou Essai sur la structure de la partie connue de le Terre* [*Description of the Terrains that Constitute the Crust of the Earth or Essay on the Structure of the Known Lands of the Earth*]. In this book, Brongniart used the phrase *terrains jurassiques* when correlating the "Jura-Kalkstein" of Humboldt with similarly aged oolitic limestones in Britain, thus coining and publishing the term "Jurassic".[4][3]

The German geologist Leopold von Buch in 1839 established the three-fold division of the Jurassic, originally named from oldest to the youngest: the Black Jurassic, Brown Jurassic, and White Jurassic.[5] The term "Lias" had previously been used for strata of equivalent age to the Black Jurassic in England by

Portrait of Alexandre Brongniart, who coined the term "Jurassic"

William Conybeare and William Phillips in 1822. William Phillips, the geologist, worked with William Conybeare to find out more about the Black Jurassic in England.

The French palaeontologist Alcide d'Orbigny in papers between 1842 and 1852 divided the Jurassic into ten stages based on ammonite and other fossil assemblages in England and France, of which seven are still used, but none has retained its original definition. The German geologist and palaeontologist Friedrich August von Quenstedt in 1858 divided the three series of von Buch in the Swabian Jura into six subdivisions defined by ammonites and other fossils.

The German palaeontologist Albert Oppel in his studies between 1856 and 1858 altered d'Orbigny's original scheme and further subdivided the stages into biostratigraphic zones, based primarily on ammonites. Most of the modern stages of the Jurassic were formalized at the Colloque du Jurassique à Luxembourg in 1962.[3]

Geology

The Jurassic Period is divided into three epochs: Early, Middle, and Late. Similarly, in stratigraphy, the Jurassic is divided into the Lower Jurassic, Middle Jurassic, and Upper Jurassic series. Geologists divide the rocks of the Jurassic into a stratigraphic set of units called stages, each formed during corresponding time intervals called ages.

Stages can be defined globally or regionally. For global stratigraphic correlation, the International Commission on Stratigraphy (ICS) ratify global stages based on a Global Boundary Stratotype Section and Point (GSSP) from a single formation (a stratotype) identifying the lower boundary of the stage.[3] The ages of the Jurassic from youngest to oldest are as follows:[6]

Series/epoch	Stage/age	Lower boundary
Early Cretaceous	Berriasian	143.1 ± 0.6 Ma
Upper/Late Jurassic	Tithonian	149.2 ± 0.7 Ma
	Kimmeridgian	154.8 ± 0.8 Ma
	Oxfordian	161.5 ± 1.0 Ma
Middle Jurassic	Callovian	165.3 ± 1.1 Ma
	Bathonian	168.2 ± 1.2 Ma
	Bajocian	170.9 ± 0.8 Ma
	Aalenian	174.7 ± 0.8 Ma
Lower/Early Jurassic	Toarcian	184.2 ± 0.3 Ma
	Pliensbachian	192.9 ± 0.3 Ma
	Sinemurian	199.5 ± 0.3 Ma
	Hettangian	201.4 ± 0.2 Ma

Stratigraphy

Jurassic stratigraphy is primarily based on the use of ammonites as index fossils. The first appearance datum of specific ammonite taxa is used to mark the beginnings of stages, as well as smaller timespans within stages, referred to as "ammonite zones"; these, in turn, are also sometimes subdivided further into subzones. Global stratigraphy is based on standard European ammonite zones, with other regions being calibrated to the European successions.[3]

Folded Lower Jurassic limestone layers of the Doldenhorn nappe at Gasteretal, Switzerland

Early Jurassic

The oldest part of the Jurassic Period has historically been referred to as the Lias or Liassic, roughly equivalent in extent to the Early Jurassic, but also including part of the preceding Rhaetian. The Hettangian Stage was named by Swiss palaeontologist Eugène Renevier in 1864 after Hettange-Grande in north-eastern France.[3] The GSSP for the base of the Hettangian is located at the Kuhjoch Pass, Karwendel Mountains, Northern Calcareous Alps, Austria; it was ratified in 2010. The beginning of the Hettangian, and thus the Jurassic as a whole, is marked by the first appearance of the ammonite *Psiloceras spelae tirolicum* in the Kendlbach Formation exposed at Kuhjoch.[7] The base of the Jurassic was previously defined as the first appearance of *Psiloceras planorbis* by Albert Oppel in 1856–58, but this was changed as the appearance was seen as too localised an event for an international boundary.[3]

Middle Jurassic strata in Neuquén Province, Argentina

Tidwell Member of the Morrison Formation (Upper Jurassic), Colorado

The Sinemurian Stage was first defined and introduced into scientific literature by Alcide d'Orbigny in 1842. It takes its name from the French town of Semur-en-Auxois, near Dijon. The original definition of Sinemurian included what is now the Hettangian. The GSSP of the Sinemurian is located at a cliff face north of the hamlet of East Quantoxhead, 6 kilometres east of Watchet, Somerset, England, within the Blue Lias, and was ratified in 2000. The beginning of the Sinemurian is defined by the first appearance of the ammonite *Vermiceras quantoxense*.[3][8]

Base Aalenian GSSP at Fuentelsaz

Albert Oppel in 1858 named the Pliensbachian Stage after the hamlet of Pliensbach in the community of Zell unter Aichelberg in the Swabian Alb, near Stuttgart, Germany. The GSSP for the base of the Pliensbachian is found at the Wine Haven locality in Robin Hood's Bay, Yorkshire, England, in

the Redcar Mudstone Formation, and was ratified in 2005. The beginning of the Pliensbachian is defined by the first appearance of the ammonite *Bifericeras donovani*.[9]

The village Thouars (Latin: *Toarcium*), just south of Saumur in the Loire Valley of France, lends its name to the Toarcian Stage. The Toarcian was named by Alcide d'Orbigny in 1842, with the original locality being Vrines quarry around 2 km northwest of Thouars. The GSSP for the base of the Toarcian is located at Peniche, Portugal, and was ratified in 2014. The boundary is defined by the first appearance of ammonites belonging to the subgenus *Dactylioceras* (*Eodactylites*).[10]

Middle Jurassic

The Aalenian is named after the city of Aalen in Germany. The Aalenian was defined by Swiss geologist Karl Mayer-Eymar in 1864. The lower boundary was originally between the dark clays of the Black Jurassic and the overlying clayey sandstone and ferruginous oolite of the Brown Jurassic sequences of southwestern Germany.[3] The GSSP for the base of the Aalenian is located at Fuentelsaz in the Iberian range near Guadalajara, Spain, and was ratified in 2000. The base of the Aalenian is defined by the first appearance of the ammonite *Leioceras opalinum*.[11]

Alcide d'Orbigny in 1842 named the Bajocian Stage after the town of Bayeux (Latin: *Bajoce*) in Normandy, France. The GSSP for the base of the Bajocian is located in the Murtinheira section at Cabo Mondego, Portugal; it was ratified in 1997. The base of the Bajocian is defined by the first appearance of the ammonite *Hyperlioceras mundum*.[12]

The Bathonian is named after the city of Bath, England, introduced by Belgian geologist d'Omalius d'Halloy in 1843, after an incomplete section of oolitic limestones in several quarries in the region. The GSSP for the base of the Bathonian is Ravin du Bès, Bas-Auran area, Alpes de Haute Provence, France; it was ratified in 2009. The base of the Bathonian is defined by the first appearance of the ammonite *Gonolkites convergens*, at the base of the *Zigzagiceras zigzag* ammonite zone.[13]

The Callovian is derived from the Latinized name of the village of Kellaways in Wiltshire, England, and was named by Alcide d'Orbigny in 1852, originally the base at the contact between the Forest Marble Formation and the Cornbrash Formation. However, this boundary was later found to be within the upper part of the Bathonian.[3] The base of the Callovian does not yet have a certified GSSP. The working definition for the base of the Callovian is the first appearance of ammonites belonging to the genus *Kepplerites*.[14]

Late Jurassic

The Oxfordian is named after the city of Oxford in England and was named by Alcide d'Orbigny in 1844 in reference to the Oxford Clay. The base of the Oxfordian lacks a defined GSSP. W. J. Arkell in studies in 1939 and 1946 placed the lower boundary of the Oxfordian as the first appearance of the ammonite *Quenstedtoceras mariae* (then placed in the genus *Vertumniceras*). Subsequent proposals have suggested the first appearance of *Cardioceras redcliffense* as the lower boundary.[3][14]

The village of Kimmeridge on the coast of Dorset, England, is the origin of the name of the Kimmeridgian. The stage was named by Alcide d'Orbigny in 1842 in reference to the Kimmeridge Clay. The GSSP for the base of the Kimmeridgian is the Flodigarry section at Staffin Bay on the Isle of Skye,

Scotland,[15] which was ratified in 2021. The boundary is defined by the first appearance of ammonites marking the boreal Bauhini Zone and the subboreal Baylei Zone.[14]

Kimmeridge Clay at Egmont Bight in the Jurassic Coast, southern England

The Tithonian was introduced in scientific literature by Albert Oppel in 1865. The name Tithonian is unusual in geological stage names because it is derived from Greek mythology rather than a place name. Tithonus was the son of Laomedon of Troy and fell in love with Eos, the Greek goddess of dawn. His name was chosen by Albert Oppel for this stratigraphical stage because the Tithonian finds itself hand in hand with the dawn of the Cretaceous. The base of the Tithonian currently lacks a GSSP.[3] The working definition for the base of the Tithonian is the first appearance of the ammonite genus *Gravesia*.[14]

The upper boundary of the Jurassic is currently undefined, and the Jurassic–Cretaceous boundary is currently the only system boundary to lack a defined GSSP. Placing a GSSP for this boundary has been difficult because of the strong regionality of most biostratigraphic markers, and lack of any chemostratigraphic events, such as isotope excursions (large sudden changes in ratios of isotopes), that could be used to define or correlate a boundary. Calpionellids, an enigmatic group of planktonic protists with urn-shaped calcitic tests briefly abundant during the latest Jurassic to earliest Cretaceous, have been suggested to represent the most promising candidates for fixing the Jurassic–Cretaceous boundary[16] In particular, the first appearance *Calpionella alpina*, co-inciding with the base of the eponymous Alpina subzone, has been proposed as the definition of the base of the Cretaceous.[17] The working definition for the boundary has often been placed as the first appearance of the ammonite *Strambergella jacobi*, formerly placed in the genus *Berriasella*, but its use as a stratigraphic indicator has been questioned, as its first appearance does not correlate with that of *C. alpina*.[18]

Mineral and hydrocarbon deposits

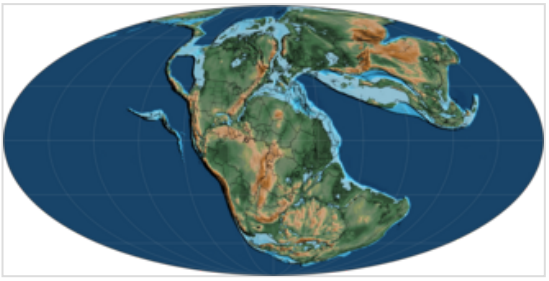
The Kimmeridge Clay and equivalents are the major source rock for the North Sea oil.[19] The Arabian Intrashelf Basin, deposited during the Middle and Late Jurassic, is the setting of the world's largest oil reserves, including the Ghawar Field, the world's largest oil field.[20] The Jurassic-aged Sargelu[21] and Naokelekan formations[22] are major source rocks for oil in Iraq. Over 1500 gigatons of Jurassic coal reserves are found in north-west China, primarily in the Turpan-Hami Basin and the Ordos Basin.[23]

Impact events

Major impact events include the Morokweng impact structure, a 70 km diameter impact structure buried beneath the Kalahari desert in northern South Africa. The impact is dated to the Tithonian, approximately 146.06 ± 0.16 Ma.[24] Another major structure is the Puchezh-Katunki crater, 40 kilometres in diameter, buried beneath Nizhny Novgorod Oblast in western Russia. The impact has been dated to the Sinemurian, 195.9 ± 1.0 Ma.[25]

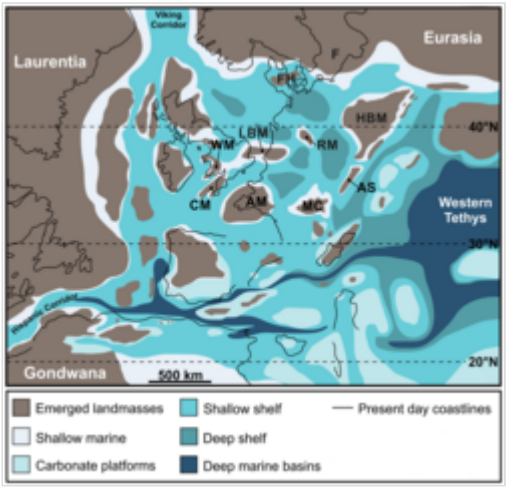
Paleogeography and tectonics

At the beginning of the Jurassic, all of the world's major landmasses were coalesced into the supercontinent Pangaea, which during the Early Jurassic began to break up into northern supercontinent Laurasia and the southern supercontinent Gondwana.[26] The rifting between North America and Africa was the first to initiate, beginning in the early Jurassic, associated with the emplacement of the Central Atlantic Magmatic Province.[27]



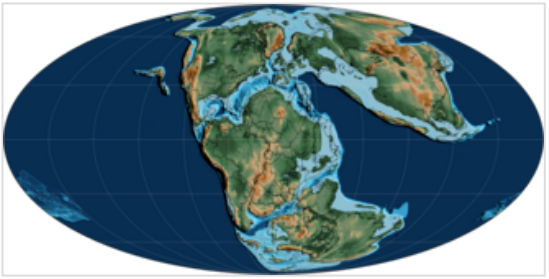
Map of geography during the Early Jurassic, around 190 million years ago

During the Jurassic, the North Atlantic Ocean remained relatively narrow, while the South Atlantic did not open until the Cretaceous.[28][27] The continents were surrounded by Panthalassa, with the Tethys Ocean between Gondwana and Asia. At the end of the Triassic, there was a marine transgression in Europe, flooding most parts of central and western Europe transforming it into an archipelago of islands surrounded by shallow seas.[29] During the Jurassic, both the North and South Pole were covered by oceans.[26] Beginning in the Early Jurassic, the Boreal Ocean was connected to the proto-Atlantic by the "Viking corridor" or Transcontinental Laurasian Seaway, a passage between the Baltic Shield and Greenland several hundred kilometers wide.[30][31][32] During the Callovian, the Turgai Epicontinental Sea formed, creating a marine barrier between Europe and Asia.[33]



Map of Europe during the Toarcian Age

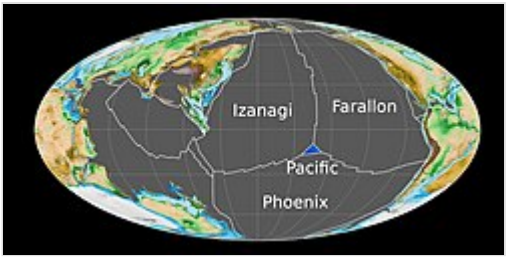
Madagascar and Antarctica began to rift away from Africa during the late Early Jurassic in association with the eruption of the Karoo-Ferrar large igneous provinces, opening the western Indian Ocean and beginning the fragmentation of Gondwana.[34][35] At the beginning of the Jurassic, North and South America remained connected, but by the beginning of the Late Jurassic they had rifted apart to form the Caribbean Seaway, also known as the Hispanic Corridor, which connected the North Atlantic Ocean with eastern Panthalassa. Palaeontological data suggest that the seaway had been open since the Early Jurassic.[36]



Map of global geography during the Late Jurassic, around 155 million years ago

As part of the Nevadan orogeny, which began during the Triassic, the Cache Creek Ocean closed, and various terranes including the large Wrangellia Terrane accreted onto the western margin of North America.[37][38] By the Middle Jurassic the Siberian plate and the North China-Amuria block had collided, resulting in the closure of the Mongol-Okhotsk Ocean.[39]

During the Early Jurassic, around 190 million years ago, the Pacific Plate originated at the triple junction of the Farallon, Phoenix, and Izanagi tectonic plates, the three main oceanic plates of Panthalassa. The previously stable triple junction had converted to an unstable arrangement surrounded on all sides by transform faults because of a kink in one of the plate boundaries, resulting in the formation of the Pacific Plate at the centre of the junction.[40] During the Middle to early Late Jurassic, the Sundance Seaway, a shallow epicontinental sea, covered much of northwest North America.[41]



Formation of the Pacific Plate during the Early Jurassic

The eustatic sea level is estimated to have been close to present levels during the Hettangian and Sinemurian, rising several tens of metres during the late Sinemurian–Pliensbachian before regressing to near present levels by the late Pliensbachian. There seems to have been a gradual rise to a peak of ~75 m above present sea level during the Toarcian. During the latest part of the Toarcian, the sea level again dropped by several tens of metres. It progressively rose from the Aalenian onwards, aside from dips of a few tens of metres in the Bajocian and around the Callovian–Oxfordian boundary, peaking possibly as high as 140 metres above present sea level at the Kimmeridgian–Tithonian boundary. The sea levels falls in the late Tithonian, perhaps to around 100 metres, before rebounding to around 110 metres at the Tithonian–Berriasian boundary.

Grainstone with calcitic ooids and sparry calcite cement; Carmel Formation, Middle Jurassic, of southern Utah, US

The sea level within the long-term trends across the Jurassic was cyclical, with 64 fluctuations, 15 of which were over 75 metres. The most noted cyclicity in Jurassic rocks is fourth order, with a periodicity of approximately 410,000 years.[42]

During the Early Jurassic the world's oceans transitioned from an aragonite sea to a calcite sea chemistry, favouring the dissolution of aragonite and precipitation of calcite.[43] The rise of calcareous plankton during the Middle Jurassic profoundly altered ocean chemistry, with the deposition of biomineralized plankton on the ocean floor acting as a buffer against large CO_2 emissions.[44]

Climate

The climate of the Jurassic was generally warmer than that of present, by around 5–10 °C (9–18 °F), with atmospheric carbon dioxide likely about four times higher. Intermittent "cold snap" intervals are known to have occurred during this time period, however, interrupting the otherwise warm greenhouse climate.[45] Forests likely grew near the poles, where they experienced warm summers and cold, sometimes snowy winters; there were unlikely to have been ice sheets given the high summer temperatures that prevented the accumulation of snow, though there may have been mountain glaciers.[46] Dropstones and glendonites in northeastern Siberia during the Early to Middle Jurassic indicate cold winters.[47] The ocean depths were likely 8 °C (14 °F) warmer than present, and coral reefs grew 10° of

latitude further north and south. The Intertropical Convergence Zone likely existed over the oceans, resulting in large areas of desert and scrubland in the lower latitudes between 40° N and S of the equator. Tropical rainforest and tundra biomes are likely to have been rare or absent.[46] The Jurassic also witnessed the decline of the Pangaean megamonsoon that had characterised the preceding Permian and Triassic periods.[48] Variation in the frequency of wildfire activity in the Jurassic was governed by the 405 kyr eccentricity cycle.[49] Thanks to the breakup of Pangaea, the hydrological cycle during the Jurassic was significantly enhanced.[50]

The beginning of the Jurassic was likely marked by a thermal spike corresponding to the Triassic–Jurassic extinction and eruption of the Central Atlantic magmatic province. The first part of the Jurassic was marked by the Early Jurassic Cool Interval between 199 and 183 million years ago.[47] It has been proposed that glaciation was present in the Northern Hemisphere during both the early Pliensbachian[51] and the latest Pliensbachian.[52][53] There was a spike in global temperatures of around 4–8 °C (7–14 °F) during the early part of the Toarcian corresponding to the Toarcian Oceanic Anoxic Event and the eruption of the Karoo-Ferrar large igneous provinces in southern Gondwana, with the warm interval extending to the end of the Toarcian around 174 million years ago.[47] During the Toarcian Warm Interval, ocean surface temperatures likely exceeded 30 °C (86 °F), and equatorial and subtropical (30°N–30°S) regions are likely to have been extremely arid, with temperatures in the interior of Pangea likely in excess of 40 °C (104 °F).The Toarcian Warm Interval is followed by the Middle Jurassic Cool Interval (MJCI) between 174 and 164 Ma,[47] which may have been punctuated by brief, ephemeral icehouse intervals.[54][55] During the Aalenian, precessionally forced climatic changes dictated peatland wildfire magnitude and frequency.[56] The European climate appears to have become noticeably more humid at the Aalenian-Bajocian boundary but then became more arid during the middle Bajocian.[57] A transient ice age possibly occurred in the late Bajocian.[58] The Callovian-Oxfordian boundary at the end of the MJCI witnessed particularly notable global cooling,[59][60] potentially even an ice age.[61] This is followed by the Kimmeridgian Warm Interval (KWI) between 164 and 150 Ma.[47] Based on fossil wood distribution, this was one of the wettest intervals of the Jurassic.[62] The Pangaean interior had less severe seasonal swings than in previous warm periods as the expansion of the Central Atlantic and Western Indian Ocean provided new sources of moisture.[47] A prominent drop in temperatures occurred during the Tithonian, known as the Early Tithonian Cooling Event (ETCE).[60] The end of the Jurassic was marked by the Tithonian–early Barremian Cool Interval (TBCI), beginning 150 Ma and continuing into the Early Cretaceous.[47]

Climatic events

Toarcian Oceanic Anoxic Event

The Toarcian Oceanic Anoxic Event (TOAE), also known as the Jenkyns Event, was an episode of widespread oceanic anoxia during the early part of the Toarcian Age, c. 183 Ma. It is marked by a globally documented high amplitude negative carbon isotope excursion,[63][64] as well as the deposition of black shales[65] and the extinction and collapse of carbonate-producing marine organisms, associated with a major rise in global temperatures.[66]

The TOAE is often attributed to the eruption of the Karoo-Ferrar large igneous provinces and the associated increase of carbon dioxide concentration in the atmosphere, as well as the possible associated release of methane clathrates.[66] This likely accelerated the hydrological cycle and increased silicate

weathering, as evidenced by an increased amount of organic matter of terrestrial origin found in marine deposits during the TOAE.[67] Groups affected include ammonites,[68] ostracods,[65][69] foraminifera,[70][71] bivalves,[65] cnidarians, and especially brachiopods,[72][73][74] for which the TOAE represented one of the most severe extinctions in their evolutionary history.[75] While the event had significant impact on marine invertebrates, it had little effect on marine reptiles.[76] During the TOAE, the Sichuan Basin was transformed into a giant lake, probably three times the size of modern-day Lake Superior, represented by the Da'anzhai Member of the Ziliujing Formation. The lake likely sequestered ~460 gigatons (Gt) of organic carbon and ~1,200 Gt of inorganic carbon during the event.[77] Seawater pH, which had already substantially decreased prior to the event, increased slightly during the early stages of the TOAE, before dropping to its lowest point around the middle of the event.[78] This ocean acidification is the probable cause of the collapse of carbonate production.[79][80] Additionally, anoxic conditions were exacerbated by enhanced recycling of phosphorus back into ocean water as a result of high ocean acidity and temperature inhibiting its mineralisation into apatite; the abundance of phosphorus in marine environments caused further eutrophication and consequent anoxia in a positive feedback loop.[81]

End-Jurassic transition

The end-Jurassic transition was originally considered one of eight mass extinctions, but is now considered to be a complex interval of faunal turnover, with the increase in diversity of some groups and decline in others, though the evidence for this is primarily European, probably controlled by changes in eustatic sea level.[82]

Flora

End-Triassic extinction

There is no evidence of a mass extinction of plants at the Triassic–Jurassic boundary.[83] At the Triassic–Jurassic boundary in Greenland, the sporomorph (pollen and spores) record suggests a complete floral turnover.[84] An analysis of macrofossil floral communities in Europe suggests that changes were mainly due to local ecological succession.[85] At the end of the Triassic, the Peltaspermaceae became extinct in most parts of the world, with *Lepidopteris* persisting into the Early Jurassic in Patagonia.[86] *Dicroidium*, a corystosperm seed fern that was a dominant part of Gondwanan floral communities during the Triassic, also declined at the Triassic–Jurassic boundary, surviving as a relict in Antarctica into the Early Jurassic.[87]

Floral composition

Conifers

Conifers formed a dominant component of Jurassic floras. The Late Triassic and Jurassic was a major time of diversification of conifers, with most modern conifer groups appearing in the fossil record by the end of the Jurassic, having evolved from voltzialean ancestors.[88][89]

Araucarian conifers have their first unambiguous records during the Early Jurassic, and members of the modern genus *Araucaria* were widespread across both hemispheres by the Middle Jurassic.[89][90][91]

Petrified *Araucaria mirabilis* cone from the Middle Jurassic of Argentina

Also abundant during the Jurassic is the extinct family Cheirolepidiaceae, often recognised through their highly distinctive *Classopolis* pollen. Jurassic representatives include the pollen cone *Classostrobus* and the seed cone *Pararaucaria*. Araucarian and Cheirolepidiaceae conifers often occur in association.[92]

The oldest definitive record of the cypress family (Cupressaceae) is *Austrohamia minuta* from the Early Jurassic (Pliensbachian) of Patagonia, known from many parts of the plant.[93] The reproductive structures of *Austrohamia* have strong similarities to those of the primitive living cypress genera *Taiwania* and *Cunninghamia*. By the Middle to Late Jurassic Cupressaceae were abundant in warm temperate–tropical regions of the Northern Hemisphere, most abundantly represented by the genus *Elatides*.[94] The Jurassic also saw the first appearances of some modern genera of cypresses, such as *Sequoia*.[95]

Members of the extinct genus *Schizolepidopsis* which likely represent a stem-group to the pine family (Pinaceae), were widely distributed across Eurasia during the Jurassic.[96][97] The oldest unambiguous record of Pinaceae is the pine cone *Eathiestrobus*, known from the Late Jurassic (Kimmeridgian) of Scotland, which remains the only known unequivocal fossil of the group before the Cretaceous.[98] Despite being the earliest known member of the Pinaceae, *Eathiestrobus* appears to be a member of the pinoid clade of the family, suggesting that the initial diversification of Pinaceae occurred earlier than has been found in the fossil record.[99][89]

The earliest record of the yew family (Taxaceae) is *Palaeotaxus rediviva*, from the Hettangian of Sweden, suggested to be closely related to the living *Austrotaxus*, while *Marskea jurassica* from the Middle Jurassic of Yorkshire, England and material from the Callovian–Oxfordian Daohugou Bed in China are thought to be closely related to *Amentotaxus*, with the latter material assigned to the modern genus, indicating that Taxaceae had substantially diversified by the end of the Jurassic.[100]

The oldest unambiguous members of Podocarpaceae are known from the Jurassic, found across both hemispheres, including *Scarburgia* and *Harrisiocarpus* from the Middle Jurassic of England, as well as unnamed species from the Middle-Late Jurassic of Patagonia.[101]

During the Early Jurassic, the flora of the mid-latitudes of Eastern Asia were dominated by the extinct deciduous broad leafed conifer *Podozamites*, which appears to not be closely related to any living family of conifer. Its range extended northwards into polar latitudes of Siberia and then contracted northward in the Middle to Late Jurassic, corresponding to the increasing aridity of the region.[102]

Ginkgoales

Ginkgoales, of which the sole living species is *Ginkgo biloba*, were more diverse during the Jurassic: they were among the most important components of Eurasian Jurassic floras and were adapted to a wide variety of climatic conditions.[103] The earliest representatives of the genus *Ginkgo*, represented by

Leaves of *Ginkgo huttonii* from the Middle Jurassic of England

ovulate and pollen organs similar to those of the modern species, are known from the Middle Jurassic in the Northern Hemisphere.[103] Several other lineages of ginkgoaleans are known from Jurassic rocks, including *Yimaia*, *Grenana*, *Nagrenia* and *Karkenia*. These lineages are associated with *Ginkgo*-like leaves, but are distinguished from living and fossil representatives of *Ginkgo* by having differently arranged reproductive structures.[103][104] *Umaltolepis* from the Jurassic of Asia has strap-shaped ginkgo-like leaves with highly distinct reproductive structures with similarities to those of peltasperm and corystosperm seed ferns, has been suggested to be a member of Ginkgoales *sensu lato*.[105]

Bennettitales

Bennettitales, having first become widespread during the preceding Triassic, were diverse and abundant members of Jurassic floras across both hemispheres.[106] The foliage of Bennettitales bears strong similarities to those of cycads, to such a degree that they cannot be reliably distinguished on the basis of morphology alone. Leaves of Bennettitales can be distinguished from those of cycads their different arrangement of stomata, and the two groups are not thought to be closely related.[107] Jurassic Bennettitales predominantly belong to the group Williamsoniaceae,[106] which grew as shrubs and small trees. The Williamsoniaceae are thought to have had a divaricate branching habit, similar to that of living *Banksia*, and adapted to growing in open habitats with poor soil nutrient conditions.[108] Bennettitales exhibit complex, flower-like reproductive structures some of which are thought to have been pollinated by insects. Several groups of insects that bear long proboscis, including extinct families such as kalligrammatid lacewings[109] and extant ones such as acrocerid flies,[110] are suggested to have been pollinators of bennettitales, feeding on nectar produced by bennettitalean cones.

Restoration of a member of Bennettitales belonging to Williamsoniaceae.

Cycads

Cycads reached their apex of diversity during the Jurassic and Cretaceous Periods.[111] Despite the Mesozoic sometimes being called the "Age of Cycads", cycads are thought to have been a relatively minor component of mid-Mesozoic floras, with the Bennettitales and Nilssoniales, which have cycad-like foliage, being dominant.[112] The Nilssoniales have often been considered cycads or cycad relatives, but have been found to be distinct on chemical grounds, and perhaps more closely allied with Bennettitales.[113] The relationships of most Mesozoic cycads to living groups are ambiguous,[112] with no Jurassic cycads belonging to either of the two modern groups of cycads, though some Jurassic cycads possibly represent stem-group relatives of modern Cycadaceae, like the leaf genus *Paracycas* known Europe, and Zamiaceae, like some European species of the leaf genus *Pseudoctenis*. Also widespread

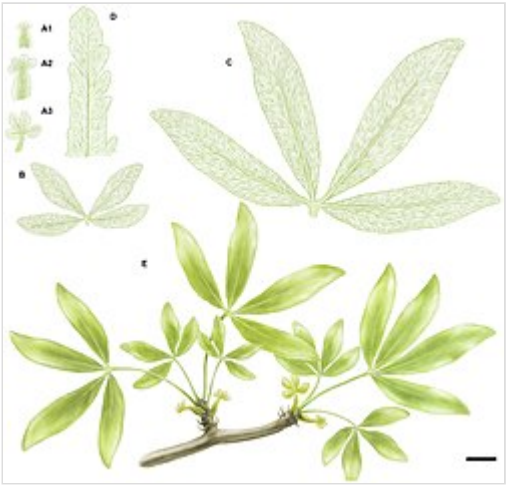
during the Jurassic was the extinct *Ctenis* lineage, which appears to be distantly related to modern cycads.[114] Modern cycads are pollinated by beetles, and such an association is thought to have formed by the Early Jurassic.[111]

Other seed plants

Although there have been several claimed records, there are no widely accepted Jurassic fossil records of flowering plants, which make up 90% of living plant species, and fossil evidence suggests that the group diversified during the following Cretaceous.[115]

The earliest known gnetophytes, one of the four main living groups of gymnosperms, appeared by the end of the Jurassic, with the oldest unequivocal gnetophyte being the seed *Dayvaultia* from the Late Jurassic of North America.[116]

"Seed ferns" (Pteridospermatophyta) is a collective term to refer to disparate lineages of fern like plants that produce seeds but have uncertain affinities to living seed plant groups. A prominent group of Jurassic seed ferns is the Caytoniales, which reached their zenith during the Jurassic, with widespread records in the Northern Hemisphere, though records in the Southern Hemisphere remain rare. Due to their berry-like seed-bearing capsules, they have often been suggested to have been closely related or perhaps ancestral to flowering plants, but the evidence for this is inconclusive.[117] Corystosperm-aligned seed ferns, such as *Pachypteris* and *Komlopteris* were widespread across both hemispheres during the Jurassic.[118]



Life restoration of the stem and leaves of a member of the extinct seed plant order Caytoniales, a group which reached its apex of abundance during the Jurassic, and which may be closely related to living flowering plants.

Czekanowskiales, also known as Leptostrobales, are a group of seed plants uncertain affinities with persistent heavily dissected leaves borne on deciduous short shoots, subtended by scale-like leaves, known from the Late Triassic (possibly Late Permian[119]) to Cretaceous.[120] They are thought to have had a tree- or shrub-like habit and formed a conspicuous component of Northern Hemisphere Mesozoic temperate and warm-temperate floras.[119] The genus *Phoenicopsis* was widespread in Early-Middle Jurassic floras of Eastern Asia and Siberia.[121]

The Pentoxylales, a small but clearly distinct group of liana-like seed plants of obscure affinities, first appeared during the Jurassic. Their distribution appears to have been confined to Eastern Gondwana.[122]

Ferns and allies

Living families of ferns widespread during the Jurassic include Dipteridaceae, Matoniaceae, Gleicheniaceae, Osmundaceae and Marattiaceae.[123][124] Although recent phylogenetic evidence supports the hypothesis that the Jurassic marks the beginning of a long-term decline in fern diversity and subsequent turnover.[125] Polypodiales, which make up 80% of living fern diversity, have no record from

the Jurassic and are thought to have diversified in the Cretaceous,[126] though the widespread Jurassic herbaceous fern genus *Coniopteris*, historically interpreted as a close relative of tree ferns of the family Dicksoniaceae, has recently been reinterpreted as an early relative of the group.[127]

The Cyatheales, the group containing most modern tree ferns, appeared during the Late Jurassic, represented by members of the genus *Cyathocaulis*, which are suggested to be early members of Cyatheaceae on the basis of cladistic analysis.[128] Only a handful of possible records exist of the Hymenophyllaceae from the Jurassic, including *Hymenophyllites macrosporangiatus* from the Russian Jurassic.[129]

The oldest remains of modern horsetails of the genus *Equisetum* first appear in the Early Jurassic, represented by *Equisetum dimorphum* from the Early Jurassic of Patagonia[130] and *Equisetum laterale* from the Early to Middle Jurassic of Australia.[131][132] Silicified remains of *Equisetum thermale* from the Late Jurassic of Argentina exhibit all the morphological characters of modern members of the genus.[133] The estimated split between *Equisetum bogotense* and all other living *Equisetum* is estimated to have occurred no later than the Early Jurassic.[132]

Lower plants

Quillworts virtually identical to modern species are known from the Jurassic onwards. *Isoetites rolandii* from the Middle Jurassic of Oregon is the earliest known species to represent all major morphological features of modern *Isoetes*. More primitive forms such as *Nathorstiana*, which retain an elongated stem, persisted into the Early Cretaceous.[134]

The moss *Kulindobryum* from the Middle Jurassic of Russia, which was found associated with dinosaur bones, is thought to be related to the Splachnaceae, which grow on animal carcasses.[135] *Bryokhutuliinia* from the same region is thought to be related to Dicranales.[135] *Heinrichsiella* from the Jurassic of Patagonia is thought to belong to either Polytrichaceae or Timmiellaceae.[136]

The liverwort *Pellites hamiensis* from the Middle Jurassic Xishanyao Formation of China is the oldest record of the family Pelliaceae.[137] *Pallaviciniites sandaolingensis* from the same deposit is thought to belong to the subclass Pallaviciniineae within the Pallaviciniales.[138] *Ricciopsis sandaolingensis*, also from the same deposit, is the only Jurassic record of Ricciaceae.[139]

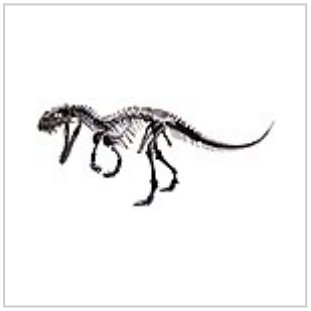
Fauna

Dinosaurs

Dinosaurs, which had morphologically diversified in the Late Triassic, experienced a major increase in diversity and abundance during the Early Jurassic in the aftermath of the end-Triassic extinction and the extinction of other reptile groups, becoming the dominant vertebrates in terrestrial ecosystems.[140][141] *Chilesaurus*, a morphologically aberrant herbivorous dinosaur from the Late Jurassic of South America, has uncertain relationships to the three main groups of dinosaurs, having been recovered as a member of all three in different analyses.[142]

Theropods

Advanced theropods belonging to Neotheropoda first appeared in the Late Triassic. Basal neotheropods, such as coelophysoids and dilophosaurs, persisted into the Early Jurassic, but became extinct by the Middle Jurassic.[143] The earliest averostrans appear during the Early Jurassic, with the earliest known member of Ceratosauria being *Saltriovenator* from the early Sinemurian (199.3–197.5 Ma) of Italy.[144] The unusual ceratosaur *Limusaurus* from the Late Jurassic of China had a herbivorous diet, with adults having edentulous beaked jaws,[145] making it the earliest known theropod to have converted from an ancestrally carnivorous diet.[146] The earliest members of the Tetanurae appeared during the late Early Jurassic or early Middle Jurassic.[147] The Megalosauridae represent the oldest radiation of the Tetanurae, first appearing in Europe during the Bajocian.[148] The oldest member of Allosauroidea has been suggested to be *Asfaltovenator* from the Middle Jurassic of South America.[147] Coelurosaurs first appeared during the Middle Jurassic, including early tyrannosaurs such as *Proceratosaurus* from the Bathonian of Britain.[149] Some coelurosaurs from the Late Jurassic of China including *Shishugounykus* and *Haplocheirus* are suggested to represent early alvarezsaurs,[150] however, this has been questioned.[151] Scansoriopterygids, a group of small feathered coelurosaurs with membraneous, bat-like wings for gliding, are known from the Middle to Late Jurassic of China.[152] The oldest record of troodontids is suggested to be *Hesperornithoides* from the Late Jurassic of North America. Tooth remains suggested to represent those of dromaeosaurs are known from the Jurassic, but no body remains are known until the Cretaceous.[153]



Skeleton of *Ceratosaurus*, a ceratosaurid from the Late Jurassic of North America

Skeleton of *Monolophosaurus*, a basal tetanuran from the Middle Jurassic of China

Restoration of *Yi qi*, a scansoriopterygid from the Middle to Late Jurassic of China

Birds

The earliest avialans, which include birds and their ancestors, appear during the Middle to Late Jurassic, definitively represented by *Archaeopteryx* from the Late Jurassic of Germany. Avialans belong to the clade Paraves within Coelurosauria, which also includes dromaeosaurs and troodontids. The Anchiornithidae from the Middle-Late Jurassic of Eurasia have frequently suggested to be avialans, but have also alternatively found as a separate lineage of paravians.[154]

Ornithischians

The earliest definitive ornithischians appear during the Early Jurassic, represented by basal ornithischians like *Lesothosaurus*, heterodontosaurids, and early members of Thyreophora. The earliest members of Ankylosauria and Stegosauria appear during the Middle Jurassic.[155] The basal neornithischian *Kulindadromeus* from the Middle Jurassic of Russia indicates that at least some ornithischians were

Skeleton of *Heterodontosaurus*, a primitive ornithischian from the Early Jurassic of South Africa

covered in protofeathers.[156] The earliest members of Ankylopollexia, which become prominent in the Cretaceous, appeared during the Late Jurassic, represented by bipedal forms such as *Camptosaurus*.[157] Ceratopsians first appeared in the Late Jurassic of China, represented by members of Chaoyangsauridae.[158]

Archaeopteryx lithographica from the Late Jurassic (Tithonian) of Germany

Sauropodomorphs

Skeleton of *Mamenchisaurus sinocanadorum* from the Middle-Late Jurassic of China

Sauropods became the dominant large herbivores in terrestrial ecosystems during the Jurassic.[159] Some Jurassic sauropods reached gigantic sizes, becoming the largest organisms to have ever lived on land.[160]

Basal bipedal sauropodomorphs, such as massospondylids, continued to exist into the Early Jurassic, but became extinct by the beginning of the Middle Jurassic.[159] Quadrupedal sauropomorphs appeared during the Late Triassic. The quadrupedal *Ledumahadi* from the earliest Jurassic of South Africa reached an estimated weight of 12 tons, far in excess of other known basal sauropodomorphs.[161] Gravisaurian sauropods first appeared during the Early Jurassic, with the oldest definitive record being *Vulcanodon* from Zimbabwe, likely of Sinemurian age.[162] Eusauropods first appeared during the late Early Jurassic (Toarcian) and diversified during the Middle Jurassic;[159] these included cetiosaurids, turiasaurs,[163] and mamenchisaurs.[164] Neosauropods such as macronarians and diplodocoids first appeared during the Middle Jurassic, before becoming abundant and globally distributed during the Late Jurassic.[165]

Other reptiles

Crocodylomorphs

The Triassic–Jurassic extinction decimated pseudosuchian diversity, with crocodylomorphs, which originated during the early Late Triassic, being the only group of pseudosuchians to survive. All other pseudosuchians, including the herbivorous aetosaurs and carnivorous "rauisuchians", became extinct.[166] The morphological diversity of crocodylomorphs during the Early Jurassic was around the same as that of Late Triassic pseudosuchians, but they occupied different areas of morphospace, suggesting that they occupied different ecological niches to their Triassic counterparts and that there was an extensive and rapid radiation of crocodylomorphs during this interval.[167] While living crocodilians are mostly

confined to an aquatic ambush predator lifestyle, Jurassic crocodylomorphs exhibited a wide variety of life habits. An unnamed protosuchid known from teeth from the Early Jurassic of Arizona represents the earliest known herbivorous crocodylomorph, an adaptation that appeared several times during the Mesozoic.[168]

Holotype specimen of *Platysuchus,* a telosaurid thalattosuchian

The Thalattosuchia, a clade of predominantly marine crocodylomorphs, first appeared during the Early Jurassic and became a prominent part of marine ecosystems.[169] Within Thalattosuchia, the Metriorhynchidae became highly adapted for life in the open ocean, including the transformation of limbs into flippers, the development of a tail fluke, and smooth, scaleless skin.[170] The morphological diversity of crocodylomorphs during the Early and Middle Jurassic was relatively low compared to that in later time periods and was dominated by terrestrial small-bodied, long-legged sphenosuchians, early crocodyliforms and thalattosuchians.[171][169] The Neosuchia, a major group of crocodylomorphs, first appeared during the Early to Middle Jurassic. The Neosuchia represents the transition from an ancestrally terrestrial lifestyle to a freshwater aquatic ecology similar to that occupied by modern crocodilians.[172] The timing of the origin of Neosuchia is disputed. The oldest record of Neosuchians has been suggested to be *Calsoyasuchus*, from the Early Jurassic of Arizona, which in many analyses has been recovered as the earliest branching member of the neosuchian family Goniopholididae, which radically alters times of diversification for crocodylomorphs. However, this placement has been disputed, with some analyses finding it outside Neosuchia, which would place the oldest records of Neosuchia in the Middle Jurassic.[172] *Razanandrongobe* from the Middle Jurassic of Madagascar has been suggested to represent the oldest record of Notosuchia, a primarily Gondwanan clade of mostly terrestrial crocodylomorphs, otherwise known from the Cretaceous and Cenozoic.[173]

Turtles

Stem-group turtles (Testudinata) diversified during the Jurassic. Jurassic stem-turtles belong to two progressively more advanced clades, the Mesochelydia and Perichelydia.[174] It is thought that the ancestral condition for mesochelydians is aquatic, as opposed to terrestrial for testudinates.[175] The two modern groups of turtles (Testudines), Pleurodira and Cryptodira, diverged by the beginning of the Late Jurassic.[174] The oldest known pleurodires, the Platychelyidae, are known from the Late Jurassic of Europe and the Americas,[176] while the oldest unambiguous cryptodire, *Sinaspideretes*, an early relative of softshell turtles, is known from the Late Jurassic of China.[177] The Thalassochelydia, a diverse lineage of marine turtles unrelated to modern sea turtles, are known from the Late Jurassic of Europe and South America.[178]

Thalassemys, a thalassochelydian sea turtle known from the Late Jurassic of Germany

Lepidosaurs

Rhynchocephalians (the sole living representative being the tuatara) had achieved a global distribution by the beginning of the Jurassic,[179] and represented the dominant group of small reptiles during the Jurassic globally.[180] Rhynchocephalians reached their highest morphological diversity in their evolutionary history during the Jurassic, occupying a wide range of lifestyles, including the aquatic pleurosaurs with long snake-like bodies and reduced limbs, the specialized herbivorous eilenodontines, as well as the sapheosaurs which had broad tooth plates indicative of durophagy.[181] Rhynchocephalians disappeared from Asia after the Early Jurassic.[179] The last common ancestor of living squamates (which includes lizards and snakes) is estimated to have lived around 190 million years ago during the Early Jurassic, with the major divergences between modern squamate lineages estimated to have occurred during the Early to Middle Jurassic.[182] Squamates first appear in the fossil record during the Middle Jurassic[183] including members of modern clades such as Scincomorpha,[184] though many Jurassic squamates have unclear relationships to living groups.[185] *Eichstaettisaurus* from the Late Jurassic of Germany has been suggested to be an early relative of geckos and displays adaptations for climbing.[186] *Dorsetisaurus* from the Late Jurassic of North America and Europe represents the oldest widely accepted record of Anguimorpha.[187] *Marmoretta* from the Middle Jurassic of Britain has been suggested to represent a late surviving lepidosauromorph outside both Rhynchocephalia and Squamata, though some studies have recovered it as a stem-squamate.[188]

Vadasaurus herzogi, a rynchocephalian from the Upper Jurassic Solnhofen Limestone of Germany

Homeosaurus maximiliani, a rynchocephalian from the Solnhofen Limestone

Pleurosaurus,, an aquatic rhynchocephalian from the Late Jurassic of Europe

Eichstaettisaurus schroederi,, an extinct lizard from the Solnhofen Limestone

Choristoderes

The earliest known remains of Choristodera, a group of freshwater aquatic reptiles with uncertain affinities to other reptile groups, are found in the Middle Jurassic. Only two genera of choristodere are known from the Jurassic. One is the small lizard-like *Cteniogenys*, thought to be the most basal known choristodere; it is known from the Middle to Late Jurassic of Europe and Late Jurassic of North America, with similar remains also known from the upper Middle Jurassic of Kyrgyzstan and western Siberia.[189] The other is *Coeruleodraco* from the Late Jurassic of China, which is a more advanced choristodere, though still small and lizard-like in morphology.[190]

Ichthyosaurs

Ichthyosaurs suffered an evolutionary bottleneck during the end-Triassic extinction, with all non-neoichthyosaurians becoming extinct. Ichthyosaurs reached their apex of species diversity during the Early Jurassic, with an array of morphologies including the huge apex predator *Temnodontosaurus* and

Skeleton of *Coeruleodraco*

swordfish-like *Eurhinosaurus*, though Early Jurassic ichthyosaurs were significantly less morphologically diverse than their Triassic counterparts.[191][192] At the Early–Middle Jurassic boundary, between the end of the Toarcian and the beginning of the Bajocian, most lineages of ichythosaur appear to have become extinct, with the first appearance of the Ophthalmosauridae, the clade that would encompass almost all ichthyosaurs from then on, during the early Bajocian.[193] Ophthalmosaurids were diverse by the Late Jurassic, but failed to fill many of the niches that had been occupied by ichthyosaurs during the Early Jurassic.[193][191][192]

Fossil of *Ichthyosaurus somersetensis* at the Natural History Museum, London

Plesiosaurs

Plesiosaurs originated at the end of the Triassic (Rhaetian). By the end of the Triassic, all other sauropterygians, including placodonts and nothosaurs, had become extinct. At least six lineages of plesiosaur crossed the Triassic–Jurassic boundary.[194] Plesiosaurs were already diverse in the earliest Jurassic, with the majority of plesiosaurs in the Hettangian-aged Blue Lias belonging to the Rhomaleosauridae. Early plesiosaurs were generally small-bodied, with body size increasing into the Toarcian.[195] There appears to have been a strong turnover around the Early–Middle Jurassic boundary, with microcleidids and rhomaleosaurids becoming extinct and nearly extinct respectively after the end of the Toarcian with the first appearance of the dominant clade of plesiosaurs of the latter half of the Jurassic, the Cryptoclididae during the Bajocian.[193] The Middle Jurassic saw the evolution of short-necked and large-headed thalassophonean pliosaurs from ancestrally small-headed, long-necked forms.[196][193] Some thalassophonean pliosaurs, such as some species of *Pliosaurus*, had skulls up to two metres in length with body lengths estimated around 10–12 meters (33–39 ft), making them the apex predators of Late Jurassic oceans.[197][193] Plesiosaurs invaded freshwater environments during the Jurassic, with indeterminate remains of small-bodied pleisosaurs known from freshwater sediments from the Jurassic of China and Australia.[198][199]

Rhomaleosaurus cramptoni at the Natural History Museum, London

Pterosaurs

Pterosaurs first appeared in the Late Triassic. A major radiation of Jurassic pterosaurs is the Rhamphorhynchidae, which first appeared in the late Early Jurassic (Toarcian);[200] they are thought to have been piscivorous.[201] Anurognathids, which first appeared in the Middle Jurassic, possessed short heads and densely furred bodies, and are thought to have been insectivores.[201] Derived monofenestratan pterosaurs such as wukongopterids appeared in the late Middle Jurassic. Advanced short-tailed pterodactyloids first appeared at the Middle–Late Jurassic boundary. Jurassic pterodactyloids include the

Skeleton of *Rhamphorhynchus muensteri* at Teylers Museum, Haarlem

ctenochasmatids, like *Ctenochasma*, which have closely spaced needle-like teeth that were presumably used for filter feeding.[201] The bizarre Late Jurassic ctenochasmatoid *Cycnorhamphus* had a jaw with teeth only at the tips, with bent jaws like those of living openbill storks that may have been used to hold and crush hard invertebrates.[201]

Amphibians

Skeleton of *Karaurus sharovi*, a stem-group salamander from the Middle to Late Jurassic of Kazakhstan

The diversity of temnospondyls had progressively declined through the Late Triassic, with only brachyopoids surviving into the Jurassic and beyond.[202] Members of the family Brachyopidae are known from Jurassic deposits in Asia,[203] while the chigutisaurid *Siderops* is known from the Early Jurassic of Australia.[204] Modern lissamphibians began to diversify during the Jurassic. The Early Jurassic *Prosalirus* thought to represent the first frog relative with a morphology capable of hopping like living frogs.[205] Morphologically recognisable stem-frogs like the South American *Notobatrachus* are known from the Middle Jurassic,[206] with modern crown-group frogs like *Enneabatrachus* and *Rhadinosteus* appearing by the Late Jurassic.[207] While the earliest salamander-line amphibians are known from the Triassic,[208] crown group salamanders first appear during the Middle to Late Jurassic in Eurasia, alongside stem-group relatives. Many Jurassic stem-group salamanders, such as *Marmorerpeton* and *Kokartus*, are thought to have been neotenic.[209] Early representatives of crown group salamanders include *Chunerpeton*, *Pangerpeton* and *Linglongtriton* from the Middle to Late Jurassic Yanliao Biota of China. Some of these are suggested to belong to Cryptobranchoidea, which contains living Asiatic and giant salamanders.[210] *Beiyanerpeton*, and *Qinglongtriton* from the same biota are thought to be early members of Salamandroidea, the group which contains all other living salamanders.[211][212] Salamanders dispersed into North America by the end of the Jurassic, as evidenced by *Iridotriton*, found in the Late Jurassic Morrison Formation.[213] The stem-caecilian *Eocaecilia* is known from the Early Jurassic of Arizona.[214] The fourth group of lissamphibians, the extinct salamander-like albanerpetontids, first appeared in the Middle Jurassic, represented by *Anoualerpeton priscus* from the Bathonian of Britain, as well as indeterminate remains from equivalently aged sediments in France and the Anoual Formation of Morocco.[215]

Mammaliaformes

Mammaliaformes, including mammals, having originated from cynodonts at the end of the Triassic, diversified extensively during the Jurassic.[216] While most Jurassic mammalaliaforms are solely known from isolated teeth and jaw fragments, exceptionally preserved remains have revealed a variety of lifestyles.[216] The docodontan *Castorocauda* was adapted to aquatic life, similarly to the platypus and otters.[217] Some members of Haramiyida[218] and the eutriconodontan tribe Volaticotherini[219] had a

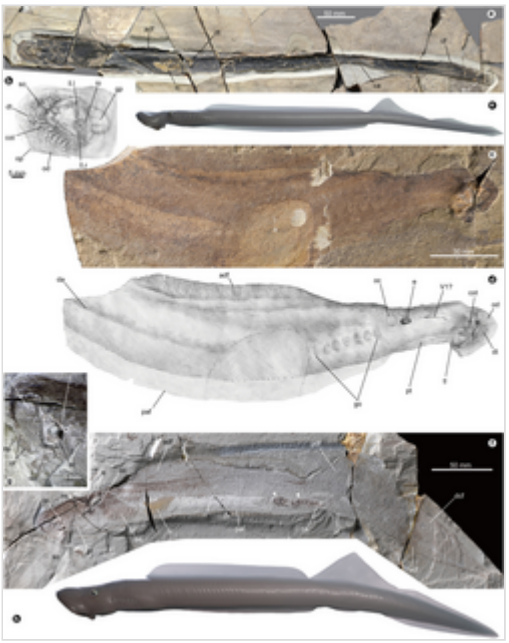
Henkelotherium, a likely arboreal dyolestoid from the Late Jurassic of Portugal

patagium akin to those of flying squirrels, allowing them to glide through the air. The aardvark-like mammal *Fruitafossor*, of uncertain taxonomy, was likely a specialist on colonial insects, similarly to living anteaters.[220] Australosphenida, a group of mammals possibly related to living monotremes, first appeared in the Middle Jurassic of Gondwana.[221] The earliest records of multituberculates, one of the longest lasting and most successful orders of mammals, are known from the Middle Jurassic.[222] Therian mammals, represented today by living placentals and marsupials, diversified meteorically during the Middle Jurassic.[223] They have their earliest records during the early Late Jurassic, represented by *Juramaia*, a eutherian mammal closer to the ancestry of placentals than marsupials.[224] *Juramaia* is much more advanced than expected for its age, as other therian mammals are not known until the Early Cretaceous, and it has been suggested that *Juramaia* may also originate from the Early Cretaceous instead.[225] Two groups of non-mammaliaform cynodonts persisted beyond the end of the Triassic. The insectiviorous Tritheledontidae has a few records from the Early Jurassic. The Tritylodontidae, a herbiviorous group of cynodonts that first appeared during the Rhaetian, has abundant records from the Jurassic, overwhelmingly from the Northern Hemisphere.[226][227]

Fish

Jawless fish

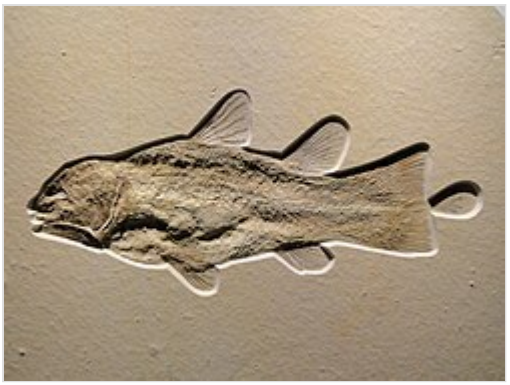
The last known species of conodont, a class of jawless fish whose hard, tooth-like elements are key index fossils, finally became extinct during the earliest Jurassic after over 300 million years of evolutionary history, with an asynchronous extinction occurring first in the Tethys and eastern Panthalassa and survivors persisting into the earliest Hettangian of Hungary and central Panthalassa.[228] End-Triassic conodonts were represented by only a handful of species and had been progressively declining through the Middle and Late Triassic.[229] *Yanliaomyzon* from the Middle Jurassic of China represents the oldest post Paleozoic lamprey, and the oldest lamprey to have the toothed feeding apparatus and likely the three stage life cycle typical of modern members of the group.[230]



Fossils and life restorations of the two species of *Yanliaomyzon* , a lamprey known from the Middle Jurassic of China

Sarcopterygii

Lungfish (Dipnoi) were present in freshwater environments of both hemispheres during the Jurassic.[231] Some studies have proposed that the last common ancestor of all living lungfish



Coelacanth from the Solnhofen Limestone

lived during the Jurassic.[232] Mawsoniids, a marine and freshwater/brackish group of coelacanths, which first appeared in North America during the Triassic, expanded into Europe and South America by the end of the Jurassic.[233] The marine Latimeriidae, which contains the living coelacanths of the genus *Latimeria*, were also present in the Jurassic, having originated in the Triassic, with a number of records from the Jurassic of Europe including *Swenzia*, thought to be the closest known relative of living coelacanths.[234]

Actinopterygii

Fossil of *Thrissops*, an ichthyodectid stem-group teleost from the Late Jurassic Solnhofen Limestone of Germany, showing preserved colouration

Ray-finned fish (Actinopterygii) were major components of Jurassic freshwater and marine ecosystems. Archaic "palaeoniscoid" fish, which were common in both marine and freshwater habitats during the preceding Triassic declined during the Jurassic, being largely replaced by more derived actinopterygian lineages.[235] The oldest known Acipenseriformes, the group that contains living sturgeon and paddlefish, are from the Early Jurassic.[236] Amiiform fish (which today only includes the bowfin) first appeared during the Early Jurassic, represented by *Caturus* from the Pliensbachian of Britain; after their appearance in the western Tethys, they expanded to Africa, North America and Southeast and East Asia by the end of the Jurassic,[237] with the modern family Amiidae appearing during the Late Jurassic.[238] Pycnodontiformes, which first appeared in the western Tethys during the Late Triassic, expanded to South America and Southeast Asia by the end of the Jurassic, having a high diversity in Europe during the Late Jurassic.[237] During the Jurassic, the Ginglymodi, the only living representatives being gars (Lepisosteidae) were diverse in both freshwater and marine environments. The oldest known representatives of anatomically modern gars appeared during the Late Jurassic.[239] Stem-group teleosts, which make up over 99% of living Actinopterygii, had first appeared during the Triassic in the western Tethys; they underwent a major diversification beginning in the Late Jurassic, with early representatives of modern teleost clades such as Elopomorpha and Osteoglossoidei appearing during this time.[240][241] The Pachycormiformes, a group of marine stem-teleosts, first appeared in the Early Jurassic and included both tuna-like predatory and filter-feeding forms, the latter included the largest bony fish known to have existed: *Leedsichthys*, with an estimated maximum length of over 15 metres, known from the late Middle to Late Jurassic.[242]

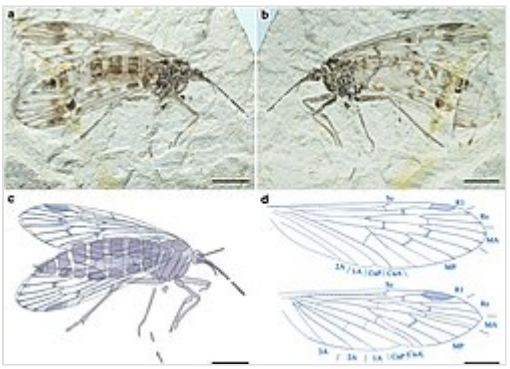
Chondrichthyes

During the Early Jurassic, the shark-like hybodonts, which represented the dominant group of chondrichthyans during the preceding Triassic, were common in both marine and freshwater settings; however, by the Late Jurassic, hybodonts had become minor components of most marine communities, having been largely replaced by modern neoselachians, but remained common in freshwater and restricted marine environments.[243][244] The Neoselachii, which contains all living sharks and rays, radiated beginning in the Early Jurassic.[245] The oldest known ray (Batoidea) is *Antiquaobatis* from the Pliensbachian of Germany.[246] Jurassic batoids known from complete remains retain a conservative,

guitarfish-like morphology.[247] The oldest known Hexanchiformes and carpet sharks (Orectolobiformes) are from the Early Jurassic (Pliensbachian and Toarcian, respectively) of Europe.[248][249] The oldest known members of the Heterodontiformes, the only living members of which are the bullhead shark (*Heterodontus*), first appeared in the Early Jurassic, with representatives of the living genus appearing during the Late Jurassic.[250] The oldest record of angelsharks (Squatiniformes) is *Pseudorhina* from the Late Jurassic (Oxfordian–Tithonian) of Europe, which already has a bodyform similar to members of the only living genus of the order, *Squatina*.[251] The oldest known remains of Carcharhiniformes, the largest order of living sharks, first appear in the late Middle Jurassic (Bathonian) of the western Tethys (England and Morocco). Known dental and exceptionally preserved body remains of Jurassic Carchariniformes are similar to those of living catsharks.[252] Synechodontiformes, an extinct group of sharks closely related to Neoselachii, were also widespread during the Jurassic.[253] The oldest remains of modern chimaeras are from the Early Jurassic of Europe, with members of the living family Callorhinchidae appearing during the Middle Jurassic. Unlike most living chimaeras, Jurassic chimeras are often found in shallow water environments.[254] The closely related myriacanthids and the flattened *Squaloraja* are also known from the Jurassic of Europe.[255]

Fossil of *Pseudorhina* from the Late Jurassic of Germany, a close relative of modern angelsharks.

Insects and arachnids



Lichnomesopsyche daohugouensis, an extinct mesopsychid scorpionfly from the Late Jurassic of China

There appears to have been no major extinction of insects at the Triassic–Jurassic boundary.[83] Many important insect fossil localities are known from the Jurassic of Eurasia, the most important being the Karabastau Formation of Kazakhstan and the various Yanliao Biota deposits in Inner Mongolia, China, such as the Daohugou Bed, dating to the Callovian–Oxfordian. The diversity of insects stagnated throughout the Early and Middle Jurassic, but during the latter third of the Jurassic origination rates increased substantially while extinction rates remained flat.[256] The increasing diversity of insects in the Middle–Late Jurassic corresponds with a substantial increase in the diversity of insect mouthparts.[257] The Middle to Late Jurassic was a time of major diversification for beetles,[258] particularly for the suborder Polyphaga, which represents 90% of living beetle species but which was rare during the preceding Triassic.[259] Weevils first appear in the fossil record during the Middle to Late Jurassic, but are suspected to have originated during the Late Triassic to Early Jurassic.[260] Orthopteran diversity had declined during the Late Triassic, but recovered during the Early Jurassic,[261] with the Hagloidea, a superfamily of ensiferan orthopterans today confined to a few living species, being particularly diverse during the Jurassic.[262] The oldest known lepidopterans (the group containing butterflies and moths) are known from the Triassic–Jurassic boundary, with wing scales belonging to the suborder Glossata and Micropterigidae-grade moths from the deposits of this age in Germany.[263] Modern representatives of both dragonflies and damselflies also first appeared during

the Jurassic.[264] Although modern representatives are not known until the Cenozoic, ectoparasitic insects thought to represent primitive fleas, belonging to the family Pseudopulicidae, are known from the Middle Jurassic of Asia. These insects are substantially different from modern fleas, lacking the specialised morphology of the latter and being larger.[265][266] Parasitoid wasps (Apocrita) first appeared during the Early Jurassic and subsequently became widespread, reshaping terrestrial food webs.[267] Several other groups of insects, including Phasmatodea (stick insects),[268] Mantophasmatidae (gladiators),[269] Embioptera (webspinners),[270] and Raphidioptera (snakeflies)[271] also first appeared in the Jurassic. The earliest scale insect (Coccomorpha) **is** known from amber dating to the Late Jurassic, though the group probably originated earlier during the Triassic.[272]

Only a handful of records of mites are known from the Jurassic, including *Jureremus*, an oribatid mite belonging to the family Cymbaeremaeidae known from the Late Jurassic of Britain and Russia,[273] and a member of the still living orbatid genus *Hydrozetes* from the Early Jurassic of Sweden.[274] Spiders diversified through the Jurassic.[275] The Early Jurassic *Seppo koponeni* may represent a stem group to Palpimanoidea.[276] *Eoplectreurys* from the Middle Jurassic of China is considered a stem lineage of Synspermiata. The oldest member of the family Archaeidae, *Patarchaea*, is known from the Middle Jurassic of China.[275] *Mongolarachne* from the Middle Jurassic of China is among the largest known fossil spiders, with legs over 5 centimetres long.[277] The only scorpion known from the Jurassic is *Liassoscorpionides* from the Early Jurassic of Germany, of uncertain placement.[278] Eupnoi harvestmen (Opiliones) are known from the Middle Jurassic of China, including members of the family Sclerosomatidae.[279][280]

Mongolarachne from the Late Jurassic of China

Marine invertebrates

End-Triassic extinction

During the end-Triassic extinction, 46%–72% of all marine genera became extinct. The effects of the end Triassic extinction were greatest at tropical latitudes and were more severe in Panthalassa than the Tethys or Boreal oceans. Tropical reef ecosystems collapsed during the event, and would not fully recover until much later in the Jurassic. Sessile filter feeders and photosymbiotic organisms were among those most severely affected.[281]

Marine ecosystems

Having declined at the Triassic–Jurassic boundary, reefs substantially expanded during the Late Jurassic, including both sponge reefs and scleractinian coral reefs. Late Jurassic reefs were similar in form to modern reefs but had more microbial carbonates and hypercalcified sponges, and had weak biogenic binding. Reefs sharply declined at the close of the Jurassic,[282] which caused an associated drop in diversity in decapod crustaceans.[283] The earliest planktonic foraminifera, which constitute the suborder Globigerinina, are known from the late Early Jurassic (mid-Toarcian) of the western Tethys, expanding across the whole Tethys by the Middle Jurassic and becoming globally distributed in tropical latitudes by

the Late Jurassic.[284] Coccolithophores and dinoflagellates, which had first appeared during the Triassic, radiated during the Early to Middle Jurassic, becoming prominent members of the phytoplankton.[285] Microconchid tube worms, the last remaining order of Tentaculita, a group of animals of uncertain affinities that were convergent on *Spirorbis* tube worms, were rare after the Triassic and had become reduced to the single genus *Punctaconchus,* which became extinct in the late Bathonian.[286] The oldest known diatom is from Late Jurassic–aged amber from Thailand, assigned to the living genus *Hemiaulus*.[287]

Echinoderms

Crinoids diversified throughout the Jurassic, reaching their peak Mesozoic diversity during the Late Jurassic, primarily due to the radiation of sessile forms belonging to the orders Cyrtocrinida and Millericrinida.[288] Echinoids (sea urchins) underwent substantial diversification beginning in the Early Jurassic, primarily driven by the radiation of irregular (asymmetrical) forms, which were adapting to deposit feeding. Rates of diversification sharply dropped during the Late Jurassic.[289]

Crustaceans

The Jurassic was a significant time for the evolution of decapods.[283] The first true crabs (Brachyura) are known from the Early Jurassic, with the earliest being *Eocarcinus praecursor* from the early Pliensbachian of England, which lacked the crab-like morphology (carcinisation) of modern crabs,[290] and *Eoprosopon klugi* from the late Pliensbachian of Germany, which may belong to the living family Homolodromiidae.[291] Most Jurassic crabs are known only from carapace pieces, which makes it difficult to determine their relationships.[292] While rare in the Early and Middle Jurassic, crabs became abundant during the Late Jurassic as they expanded from their ancestral silty sea floor habitat into hard substrate habitats like reefs, with crevices in reefs providing refuge from predators.[292][283] Hermit crabs also first appeared during the Jurassic, with the earliest known being *Schobertella hoelderi* from the late Hettangian of Germany.[293] Early hermit crabs are associated with ammonite shells rather than those of gastropods.[294] Glypheids, which today are only known from two species, reached their peak diversity during the Jurassic, with around 150 species out of a total fossil record of 250 known from the period.[295] Jurassic barnacles were of low diversity compared to present,[296] but several important evolutionary innovations are known, including the first appearances of calcite shelled forms and species with an epiplanktonic mode of life.[297]

Eryon, a polychelidan decapod crustacean from the Late Jurassic of Germany.

Brachiopods

Brachiopod diversity declined during the Triassic–Jurassic extinction. Spire-bearing brachiopods (Spiriferinida and Athyridida) did not recover their biodiversity, becoming extinct in the TOAE.[298] Rhynchonellida and Terebratulida also declined during the Triassic–Jurassic extinction but rebounded

during the Early Jurassic; neither clade underwent much morphological variation.[299] Brachiopods substantially declined in the Late Jurassic; the causes are poorly understood. Proposed reasons include increased predation, competition with bivalves, enhanced bioturbation or increased grazing pressure.[300]

Bryozoans

Like the preceding Triassic, bryozoan diversity was relatively low compared to the Paleozoic. The vast majority of Jurassic bryozoans are members of Cyclostomatida, which experienced a radiation during the Middle Jurassic, with all Jurassic representatives belonging to the suborders Tubuliporina and Cerioporina. Cheilostomata, the dominant group of modern bryozoans, first appeared during the Late Jurassic.[301]

Molluscs

Gastropods

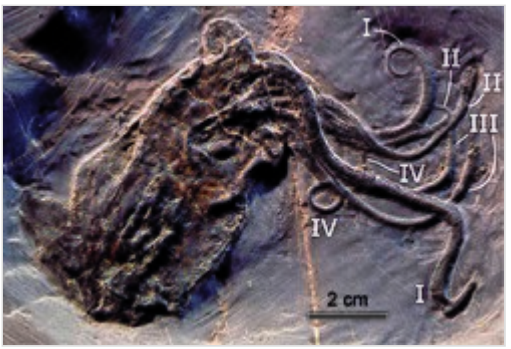
Marine gastropods were significantly affected by the T-J extinction, with around 56% of genera going extinct, with Neritimorpha being particularly strongly effected, while Heterobranchia suffered much lower losses than other groups.[302] While present, the diversity of freshwater and land snails was much lower during the Jurassic than in contemporary ecosystems, with the diversity of these groups not reaching levels comparable to modern times until the following Cretaceous.[303]

Bivalves

The end-Triassic extinction had a severe impact on bivalve diversity, though it had little impact on bivalve ecological diversity. The extinction was selective, having less of an impact on deep burrowers, but there is no evidence of a differential impact between surface-living (epifaunal) and burrowing (infaunal) bivalves.[304] Bivalve family level diversity after the Early Jurassic was static, though genus diversity experienced a gradual increase throughout the period.[305] Rudists, the dominant reef-building organisms of the Cretaceous, first appeared in the Late Jurassic (mid-Oxfordian) in the northern margin of the western Tethys, expanding to the eastern Tethys by the end of the Jurassic.[306]

Cephalopods

Ammonites were devastated by the end-Triassic extinction, with only a handful of genera belonging to the family Psiloceratidae of the suborder Phylloceratina surviving and becoming ancestral to all later Jurassic and Cretaceous ammonites. Ammonites explosively diversified during the Early Jurassic, with the orders Psiloceratina, Ammonitina, Lytoceratina, Haploceratina, Perisphinctina and Ancyloceratina all appearing during the Jurassic. Ammonite faunas during the Jurassic were regional, being divided into around 20 distinguishable provinces and subprovinces in two realms, the northern high latitude Pan-Boreal realm, consisting of the Arctic, northern Panthalassa and northern Atlantic regions, and the equatorial–southern Pan-Tethyan



Fossil specimen of *Proteroctopus* from the Middle Jurassic of France, formerly thought to be world's oldest known octopus

realm, which included the Tethys and most of Panthalassa.[307] Ammonite diversifications occurred coevally with marine transgressions, while their diversity nadirs occurred during marine regressions.[308]

The oldest definitive records of the squid-like belemnites are from the earliest Jurassic (Hettangian–Sinemurian) of Europe and Japan; they expanded worldwide during the Jurassic.[309] Belemnites were shallow-water dwellers, inhabiting the upper 200 metres of the water column on the continental shelves and in the littoral zone. They were key components of Jurassic ecosystems, both as predators and prey, as evidenced by the abundance of belemnite guards in Jurassic rocks.[310]

The earliest vampyromorphs, of which the only living member is the vampire squid, first appeared during the Early Jurassic.[311] The earliest octopuses appeared during the Middle Jurassic, having split from their closest living relatives, the vampyromorphs, during the Triassic to Early Jurassic.[312] All Jurassic octopuses are solely known from the hard gladius.[312][313] Octopuses likely originated from bottom-dwelling (benthic) ancestors which lived in shallow environments.[312] *Proteroctopus* from the late Middle Jurassic La Voulte-sur-Rhône lagerstätte, previously interpreted as an early octopus, is now thought to be a basal taxon outside the clade containing vampyromorphs and octopuses.[314]

References

Citations

1. "International Chronostratigraphic Chart" (https://stratigraphy.org/ICSchart/ChronostratChart2024-12.pdf) (PDF). *International Commission on Stratigraphy*. December 2024. Retrieved October 23, 2025.
2. "Dictionary.com | Meanings & Definitions of English Words" (https://www.dictionary.com/browse/jurassic). *Dictionary.com*. 2024-04-17. Retrieved 2024-04-18.
3. Ogg, J.G.; Hinnov, L.A.; Huang, C. (2012). "Jurassic". *The Geologic Time Scale*. pp. 731–791. doi:10.1016/b978-0-444-59425-9.00026-3 (https://doi.org/10.1016%2Fb978-0-444-59425-9.00026-3). ISBN 978-0-444-59425-9.
4. Brongniart, Alexandre (1829). *Tableau des terrains qui composent l'écorce du globe ou essai sur la structure de la partie connue de la terre* (https://gallica.bnf.fr/ark:/12148/bpt6k255061/f1.image) [*Description of the Terrains that Constitute the Crust of the Earth or Essay on the Structure of the Known Lands of the Earth*] (in French). Strasbourg, France: F.G. Levrault – via Gallica. From p. 221, footnote 2: *"Souvent aussi calcaire oolithique moyen ou principal (great oolithe); mais le nom de terrains jurassiques nous paroît préférable […] analogue à celle de la chaîne du Jura."* (Often also middle or principal oolitic limestone (great oolithe); but the name of "Jurassic terrains" seems to us preferable, because it is more general, because it indicates a terrain composed of different rocks, being in a geognostic position analogous to that of the Jura chain.)
5. von Buch, L., 1839. Über den Jura in Deutschland. Der Königlich Preussischen Akademie der Wissenschaften, Berlin, p. 87.
6. Cohen, K.M., Finney, S.C., Gibbard, P.L. & Fan, J.-X. (2013; updated) The ICS International Chronostratigraphic Chart (https://stratigraphy.org/ICSchart/ChronostratChart2024-12.pdf). Episodes 36: 199–204.

7. Hillebrandt, A.v.; Krystyn, L.; Kürschner, W.M.; Bonis, N.R.; Ruhl, M.; Richoz, S.; Schobben, M. A. N.; Urlichs, M.; Bown, P.R.; Kment, K.; McRoberts, C.A. (2013-09-01). "The Global Stratotype Sections and Point (GSSP) for the base of the Jurassic System at Kuhjoch (Karwendel Mountains, Northern Calcareous Alps, Tyrol, Austria)" (https://doi.org/10.18814%2Fepiiugs%2F2013%2Fv36i3%2F001). *Episodes*. **36** (3): 162–198. Bibcode:2013Episo..36..162H (https://ui.adsabs.harvard.edu/abs/2013Episo..36..162H). doi:10.18814/epiiugs/2013/v36i3/001 (https://doi.org/10.18814%2Fepiiugs%2F2013%2Fv36i3%2F001).
8. Bloos, Gert; Page, Kevin N. (2002-03-01). "Global Stratotype Section and Point for base of the Sinemurian Stage (Lower Jurassic)" (https://doi.org/10.18814%2Fepiiugs%2F2002%2Fv25i1%2F003). *Episodes*. **25** (1): 22–28. Bibcode:2002Episo..25...22B (https://ui.adsabs.harvard.edu/abs/2002Episo..25...22B). doi:10.18814/epiiugs/2002/v25i1/003 (https://doi.org/10.18814%2Fepiiugs%2F2002%2Fv25i1%2F003).
9. Meister, Christian; Aberhan, Martin; Blau, Joachim; Dommergues, Jean-Louis; Feist-Burkhardt, Susanne; Hailwood, Ernie A.; Hart, Malcom; Hesselbo, Stephen P.; Hounslow, Mark W.; Hylton, Mark; Morton, Nicol (2006-06-01). "The Global Boundary Stratotype Section and Point (GSSP) for the base of the Pliensbachian Stage (Lower Jurassic), Wine Haven, Yorkshire, UK" (https://doi.org/10.18814%2Fepiiugs%2F2006%2Fv29i2%2F003). *Episodes*. **29** (2): 93–106. Bibcode:2006Episo..29...93M (https://ui.adsabs.harvard.edu/abs/2006Episo..29...93M). doi:10.18814/epiiugs/2006/v29i2/003 (https://doi.org/10.18814%2Fepiiugs%2F2006%2Fv29i2%2F003).
10. Rocha, Rogério Bordalo da; Mattioli, Emanuela; Duarte, Luís Vítor; Pittet, Bernard; Elmi, Serge; Mouterde, René; Cabral, Maria Cristina; Comas-Rengifo, Maria José; Gómez, Juan José; Goy, António; Hesselbo, Stephen P. (2016-09-01). "Base of the Toarcian Stage of the Lower Jurassic defined by the Global Boundary Stratotype Section and Point (GSSP) at the Peniche section (Portugal)" (https://doi.org/10.18814%2Fepiiugs%2F2016%2Fv39i3%2F99741). *Episodes*. **39** (3): 460–481. Bibcode:2016Episo..39..460R (https://ui.adsabs.harvard.edu/abs/2016Episo..39..460R). doi:10.18814/epiiugs/2016/v39i3/99741 (https://doi.org/10.18814%2Fepiiugs%2F2016%2Fv39i3%2F99741). hdl:10261/140775 (https://hdl.handle.net/10261%2F140775).
11. Barrón, Eduardo; Ureta, Soledad; Goy, Antonio; Lassaletta, Luis (August 2010). "Palynology of the Toarcian–Aalenian Global Boundary Stratotype Section and Point (GSSP) at Fuentelsaz (Lower–Middle Jurassic, Iberian Range, Spain)" (https://eprints.ucm.es/16616/1/1-s3.pdf) (PDF). *Review of Palaeobotany and Palynology*. **162** (1): 11–28. Bibcode:2010RPaPa.162...11B (https://ui.adsabs.harvard.edu/abs/2010RPaPa.162...11B). doi:10.1016/j.revpalbo.2010.04.003 (https://doi.org/10.1016%2Fj.revpalbo.2010.04.003).
12. Pavia, G.; Enay, R. (1997-03-01). "Definition of the Aalenian-Bajocian Stage boundary" (https://doi.org/10.18814%2Fepiiugs%2F1997%2Fv20i1%2F004). *Episodes*. **20** (1): 16–22. Bibcode:1997Episo..20...16P (https://ui.adsabs.harvard.edu/abs/1997Episo..20...16P). doi:10.18814/epiiugs/1997/v20i1/004 (https://doi.org/10.18814%2Fepiiugs%2F1997%2Fv20i1%2F004).
13. López, Fernández; Rafael, Sixto; Pavia, Giulio; Erba, Elisabetta; Guiomar, Myette; Paiva Henriques, María Helena; Lanza, Roberto; Mangold, Charles; Morton, Nicol; Olivero, Davide; Tiraboschi, Daniele (2009). "The Global Boundary Stratotype Section and Point (GSSP) for base of the Bathonian Stage (Middle Jurassic), Ravin du Bès Section, SE France" (https://doi.org/10.18814%2Fepiiugs%2F2009%2Fv32i4%2F001). *Episodes*. **32** (4): 222–248. Bibcode:2009Episo..32..222F (https://ui.adsabs.harvard.edu/abs/2009Episo..32..222F). doi:10.18814/epiiugs/2009/v32i4/001 (https://doi.org/10.18814%2Fepiiugs%2F2009%2Fv32i4%2F001).
14. "International Commission on Stratigraphy-Subcommission on Jurassic Stratigraphy" (https://jurassic.stratigraphy.org/our-work). *jurassic.stratigraphy.org*. Retrieved 2021-04-09.

15. Barski, Marcin (6 September 2018). "Dinoflagellate cyst assemblages across the Oxfordian/Kimmeridgian boundary (Upper Jurassic) at Flodigarry, Staffin Bay, Isle of Skye, Scotland – a proposed GSSP for the base of the Kimmeridgian" (https://bibliotekanauki.pl/articles/2060961.pdf) (PDF). *Volumina Jurassica*. **15** (1): 51–62.
16. Wimbledon, William A.P. (2017). "News and Reviews – Developments with fixing a Tithonian/Berriasian (J/K) boundary" (https://vjs.pgi.gov.pl/article/view/26340). *Volumina Jurassica*. **15** (1).
17. Wimbledon, William A.P.; Rehakova, Daniela; Svobodová, Andrea; Schnabl, Petr; Pruner, Petr; Elbra, Tiiu; Šifnerová, Kristýna; Kdýr, Šimon; Frau, Camille; Schnyder, Johann; Galbrun, Bruno (11 February 2020). "Fixing a J/K boundary: A comparative account of key Tithonian–Berriasian profiles in the departments of Drôme and Hautes-Alpes, France" (https://doi.org/10.31577%2FGeolCarp.71.1.3). *Geologica Carpathica*. **71** (1). Bibcode:2020GCarp..71..1.3W (https://ui.adsabs.harvard.edu/abs/2020GCarp..71..1.3W). doi:10.31577/GeolCarp.71.1.3 (https://doi.org/10.31577%2FGeolCarp.71.1.3).
18. Frau, Camille; Bulot, Luc G.; Reháková, Daniela; Wimbledon, William A.P.; Ifrim, Christina (November 2016). "Revision of the ammonite index species Berriasella jacobi Mazenot, 1939 and its consequences for the biostratigraphy of the Berriasian Stage". *Cretaceous Research*. **66**: 94–114. Bibcode:2016CrRes..66...94F (https://ui.adsabs.harvard.edu/abs/2016CrRes..66...94F). doi:10.1016/j.cretres.2016.05.007 (https://doi.org/10.1016%2Fj.cretres.2016.05.007).
19. Gautier D.L. (2005). "Kimmeridgian Shales Total Petroleum System of the North Sea Graben Province" (https://pubs.usgs.gov/bul/2204/c/pdf/B2204C.pdf) (PDF). United States Geological Survey. Retrieved 2 November 2018.
20. Wilson, A. O. (2020). "Chapter 1 Introduction to the Jurassic Arabian Intrashelf Basin" (https://doi.org/10.1144%2FM53.1). *Geological Society, London, Memoirs*. **53**: 1–19. doi:10.1144/M53.1 (https://doi.org/10.1144%2FM53.1).
21. Abdula, Rzger A. (August 2015). "Hydrocarbon potential of Sargelu Formation and oil-source correlation, Iraqi Kurdistan". *Arabian Journal of Geosciences*. **8** (8): 5845–5868. Bibcode:2015ArJG....8.5845A (https://ui.adsabs.harvard.edu/abs/2015ArJG....8.5845A). doi:10.1007/s12517-014-1651-0 (https://doi.org/10.1007%2Fs12517-014-1651-0).
22. Soran University; Abdula, Rzger A. (2016-10-16). "Source Rock Assessment of Naokelekan Formation in Iraqi Kurdistan" (https://doi.org/10.17656%2Fjzs.10589). *Journal of Zankoy Sulaimani – Part A*. **19** (1): 103–124. doi:10.17656/jzs.10589 (https://doi.org/10.17656%2Fjzs.10589).
23. Ao, Weihua; Huang, Wenhui; Weng, Chengmin; Xiao, Xiuling; Liu, Dameng; Tang, Xiuyi; Chen, Ping; Zhao, Zhigen; Wan, Huan; Finkelman, Robert B. (January 2012). "Coal petrology and genesis of Jurassic coal in the Ordos Basin, China" (https://doi.org/10.1016%2Fj.gsf.2011.09.004). *Geoscience Frontiers*. **3** (1): 85–95. Bibcode:2012GeoFr...3...85A (https://ui.adsabs.harvard.edu/abs/2012GeoFr...3...85A). doi:10.1016/j.gsf.2011.09.004 (https://doi.org/10.1016%2Fj.gsf.2011.09.004).
24. Kenny, Gavin G.; Harrigan, Claire O.; Schmitz, Mark D.; Crowley, James L.; Wall, Corey J.; Andreoli, Marco A. G.; Gibson, Roger L.; Maier, Wolfgang D. (2021-08-01). "Timescales of impact melt sheet crystallization and the precise age of the Morokweng impact structure, South Africa" (https://doi.org/10.1016%2Fj.epsl.2021.117013). *Earth and Planetary Science Letters*. **567** 117013. Bibcode:2021E&PSL.56717013K (https://ui.adsabs.harvard.edu/abs/2021E&PSL.56717013K). doi:10.1016/j.epsl.2021.117013 (https://doi.org/10.1016%2Fj.epsl.2021.117013).
25. Holm-Alwmark, Sanna; Jourdan, Fred; Ferrière, Ludovic; Alwmark, Carl; Koeberl, Christian (15 May 2021). "Resolving the age of the Puchezh-Katunki impact structure (Russia) against alteration and inherited 40Ar* – No link with extinctions" (https://doi.org/10.1016%2Fj.gca.2021.03.001). *Geochimica et Cosmochimica Acta*. **301**: 116–140. Bibcode:2021GeCoA.301..116H (https://ui.adsabs.harvard.edu/abs/2021GeCoA.301..116H). doi:10.1016/j.gca.2021.03.001 (https://doi.org/10.1016%2Fj.gca.2021.03.001).

26. Scotese, Christopher R. (2021-05-30). "An Atlas of Phanerozoic Paleogeographic Maps: The Seas Come In and the Seas Go Out" (https://doi.org/10.1146%2Fannurev-earth-081320-064052). *Annual Review of Earth and Planetary Sciences*. **49** (1): 679–728. Bibcode:2021AREPS..49..679S (https://ui.adsabs.harvard.edu/abs/2021AREPS..49..679S). doi:10.1146/annurev-earth-081320-064052 (https://doi.org/10.1146%2Fannurev-earth-081320-064052).
27. Frizon de Lamotte, Dominique; Fourdan, Brendan; Leleu, Sophie; Leparmentier, François; de Clarens, Philippe (24 April 2015). "Style of rifting and the stages of Pangea breakup" (https://doi.org/10.1002%2F2014TC003760). *Tectonics*. **34** (5): 1009–1029. Bibcode:2015Tecto..34.1009F (https://ui.adsabs.harvard.edu/abs/2015Tecto..34.1009F). doi:10.1002/2014TC003760 (https://doi.org/10.1002%2F2014TC003760).
28. Hosseinpour, Maral; Williams, Simon; Seton, Maria; Barnett-Moore, Nicholas; Müller, R. Dietmar (2016-10-02). "Tectonic evolution of Western Tethys from Jurassic to present day: coupling geological and geophysical data with seismic tomography models". *International Geology Review*. **58** (13): 1616–1645. Bibcode:2016IGRv...58.1616H (https://ui.adsabs.harvard.edu/abs/2016IGRv...58.1616H). doi:10.1080/00206814.2016.1183146 (https://doi.org/10.1080%2F00206814.2016.1183146). hdl:2123/20835 (https://hdl.handle.net/2123%2F20835).
29. Barth, G.; Franz, M.; Heunisch, C.; Ernst, W.; Zimmermann, J.; Wolfgramm, M. (2018-01-01). "Marine and terrestrial sedimentation across the T–J transition in the North German Basin". *Palaeogeography, Palaeoclimatology, Palaeoecology*. **489**: 74–94. Bibcode:2018PPP...489...74B (https://ui.adsabs.harvard.edu/abs/2018PPP...489...74B). doi:10.1016/j.palaeo.2017.09.029 (https://doi.org/10.1016%2Fj.palaeo.2017.09.029).
30. Korte, Christoph; Hesselbo, Stephen P.; Ullmann, Clemens V.; Dietl, Gerd; Ruhl, Micha; Schweigert, Günter; Thibault, Nicolas (December 2015). "Jurassic climate mode governed by ocean gateway" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4682040). *Nature Communications*. **6** (1) 10015. Bibcode:2015NatCo...610015K (https://ui.adsabs.harvard.edu/abs/2015NatCo...610015K). doi:10.1038/ncomms10015 (https://doi.org/10.1038%2Fncomms10015). PMC 4682040 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4682040). PMID 26658694 (https://pubmed.ncbi.nlm.nih.gov/26658694).
31. Bjerrum, Christian J.; Surlyk, Finn; Callomon, John H.; Slingerland, Rudy L. (August 2001). "Numerical paleoceanographic study of the Early Jurassic Transcontinental Laurasian Seaway" (https://discovery.ucl.ac.uk/id/eprint/188170/). *Paleoceanography and Paleoclimatology*. **16** (4): 390–404. Bibcode:2001PalOc..16..390B (https://ui.adsabs.harvard.edu/abs/2001PalOc..16..390B). doi:10.1029/2000PA000512 (https://doi.org/10.1029%2F2000PA000512).
32. Mitchell, Andrew J.; Allison, Peter A.; Gorman, Gerald J.; Piggott, Matthew D.; Pain, Christopher C. (1 March 2011). "Tidal circulation in an ancient epicontinental sea: The Early Jurassic Laurasian Seaway". *Geology*. **39** (3): 207–210. Bibcode:2011Geo....39..207M (https://ui.adsabs.harvard.edu/abs/2011Geo....39..207M). doi:10.1130/G31496.1 (https://doi.org/10.1130%2FG31496.1).
33. Upchurch, Paul; Hunn, Craig A.; Norman, David B. (22 March 2002). "An analysis of dinosaurian biogeography: evidence for the existence of vicariance and dispersal patterns caused by geological events" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1690931). *Proceedings of the Royal Society B: Biological Sciences*. **269** (1491): 613–621. doi:10.1098/rspb.2001.1921 (https://doi.org/10.1098%2Frspb.2001.1921). PMC 1690931 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1690931). PMID 11916478 (https://pubmed.ncbi.nlm.nih.gov/11916478).
34. Geiger, Markus; Clark, David Norman; Mette, Wolfgang (March 2004). "Reappraisal of the timing of the breakup of Gondwana based on sedimentological and seismic evidence from the Morondava Basin, Madagascar". *Journal of African Earth Sciences*. **38** (4): 363–381. Bibcode:2004JAfES..38..363G (https://ui.adsabs.harvard.edu/abs/2004JAfES..38..363G). doi:10.1016/j.jafrearsci.2004.02.003 (https://doi.org/10.1016%2Fj.jafrearsci.2004.02.003).

35. Nguyen, Luan C.; Hall, Stuart A.; Bird, Dale E.; Ball, Philip J. (June 2016). "Reconstruction of the East Africa and Antarctica continental margins" (https://doi.org/10.1002%2F2015JB012776). *Journal of Geophysical Research: Solid Earth*. **121** (6): 4156–4179. Bibcode:2016JGRB..121.4156N (https://ui.adsabs.harvard.edu/abs/2016JGRB..121.4156N). doi:10.1002/2015JB012776 (https://doi.org/10.1002%2F2015JB012776).
36. Bartolini, Claudio; Buffler, Richard T.; Blickwede, Jon F., eds. (2003). *The Circum-Gulf of Mexico and the Caribbean*. doi:10.1306/M79877 (https://doi.org/10.1306%2FM79877). ISBN 0-89181-360-8.
37. Blakey, Ronald C.; Ranney, Wayne D. (2018). "The Arrival of Wrangellia and the Nevadan Orogeny: Late Triassic to Late Jurassic: Ca. 240–145 Ma". *Ancient Landscapes of Western North America*. pp. 89–101. doi:10.1007/978-3-319-59636-5_7 (https://doi.org/10.1007%2F978-3-319-59636-5_7). ISBN 978-3-319-59634-1.
38. Clennett, Edward J.; Sigloch, Karin; Mihalynuk, Mitchell G.; Seton, Maria; Henderson, Martha A.; Hosseini, Kasra; Mohammadzaheri, Afsaneh; Johnston, Stephen T.; Müller, R. Dietmar (August 2020). "A Quantitative Tomotectonic Plate Reconstruction of Western North America and the Eastern Pacific Basin". *Geochemistry, Geophysics, Geosystems*. **21** (8) e09117. Bibcode:2020GGG....2109117C (https://ui.adsabs.harvard.edu/abs/2020GGG....2109117C). doi:10.1029/2020GC009117 (https://doi.org/10.1029%2F2020GC009117).
39. Yi, Zhiyu; Meert, Joseph G. (2020-08-16). "A Closure of the Mongol-Okhotsk Ocean by the Middle Jurassic: Reconciliation of Paleomagnetic and Geological Evidence". *Geophysical Research Letters*. **47** (15) e2020GL088235. Bibcode:2020GeoRL..4788235Y (https://ui.adsabs.harvard.edu/abs/2020GeoRL..4788235Y). doi:10.1029/2020GL088235 (https://doi.org/10.1029%2F2020GL088235).
40. Boschman, Lydian M.; van Hinsbergen, Douwe J. J. (July 2016). "On the enigmatic birth of the Pacific Plate within the Panthalassa Ocean" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5919776). *Science Advances*. **2** (7) e1600022. Bibcode:2016SciA....2E0022B (https://ui.adsabs.harvard.edu/abs/2016SciA....2E0022B). doi:10.1126/sciadv.1600022 (https://doi.org/10.1126%2Fsciadv.1600022). PMC 5919776 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5919776). PMID 29713683 (https://pubmed.ncbi.nlm.nih.gov/29713683).
41. Danise, Silvia; Holland, Steven M. (July 2018). "A Sequence Stratigraphic Framework for the Middle to Late Jurassic of the Sundance Seaway, Wyoming: Implications for Correlation, Basin Evolution, and Climate Change". *The Journal of Geology*. **126** (4): 371–405. Bibcode:2018JG....126..371D (https://ui.adsabs.harvard.edu/abs/2018JG....126..371D). doi:10.1086/697692 (https://doi.org/10.1086%2F697692).
42. Haq, Bilal U. (2018). "Jurassic Sea-Level Variations: A Reappraisal" (https://doi.org/10.1130%2FGSATG359A.1). *GSA Today*: 4–10. doi:10.1130/GSATG359A.1 (https://doi.org/10.1130%2FGSATG359A.1).
43. Vulpius, Sara; Kiessling, Wolfgang (January 2018). "New constraints on the last aragonite–calcite sea transition from early Jurassic ooids". *Facies*. **64** (1) 3. Bibcode:2018Faci...64....3V (https://ui.adsabs.harvard.edu/abs/2018Faci...64....3V). doi:10.1007/s10347-017-0516-x (https://doi.org/10.1007%2Fs10347-017-0516-x).
44. Eichenseer, Kilian; Balthasar, Uwe; Smart, Christopher W.; Stander, Julian; Haaga, Kristian A.; Kiessling, Wolfgang (August 2019). "Jurassic shift from abiotic to biotic control on marine ecological success". *Nature Geoscience*. **12** (8): 638–642. Bibcode:2019NatGe..12..638E (https://ui.adsabs.harvard.edu/abs/2019NatGe..12..638E). doi:10.1038/s41561-019-0392-9 (https://doi.org/10.1038%2Fs41561-019-0392-9). hdl:10026.1/14472 (https://hdl.handle.net/10026.1%2F14472).

45. Dai, Xianduo; Du, Yuansheng; Ziegler, Martin; Wang, Chaowen; Ma, Qianli; Chai, Rong; Guo, Hua (1 January 2022). "Middle Triassic to Late Jurassic climate change on the northern margin of the South China Plate: Insights from chemical weathering indices and clay mineralogy". *Palaeogeography, Palaeoclimatology, Palaeoecology*. **585** 110744. Bibcode:2022PPP...58510744D (https://ui.adsabs.harvard.edu/abs/2022PPP...58510744D). doi:10.1016/j.palaeo.2021.110744 (https://doi.org/10.1016%2Fj.palaeo.2021.110744). hdl:1874/419504 (https://hdl.handle.net/1874%2F419504).
46. Sellwood, Bruce W.; Valdes, Paul J. (January 2008). "Jurassic climates". *Proceedings of the Geologists' Association*. **119** (1): 5–17. Bibcode:2008PrGA..119....5S (https://ui.adsabs.harvard.edu/abs/2008PrGA..119....5S). doi:10.1016/S0016-7878(59)80068-7 (https://doi.org/10.1016%2FS0016-7878%2859%2980068-7).
47. Scotese, Christopher R.; Song, Haijun; Mills, Benjamin J.W.; van der Meer, Douwe G. (April 2021). "Phanerozoic paleotemperatures: The earth's changing climate during the last 540 million years". *Earth-Science Reviews*. **215** 103503. Bibcode:2021ESRv..21503503S (https://ui.adsabs.harvard.edu/abs/2021ESRv..21503503S). doi:10.1016/j.earscirev.2021.103503 (https://doi.org/10.1016%2Fj.earscirev.2021.103503).
48. Landwehrs, Jan; Feulner, Georg; Petri, Stefan; Sames, Benjamin; Wagreich, Michael (20 May 2021). "Investigating Mesozoic Climate Trends and Sensitivities With a Large Ensemble of Climate Model Simulations" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8251552). *Paleoceanography and Paleoclimatology*. **36** (6) e2020PA004134. Bibcode:2021PaPa...36.4134L (https://ui.adsabs.harvard.edu/abs/2021PaPa...36.4134L). doi:10.1029/2020PA004134 (https://doi.org/10.1029%2F2020PA004134). PMC 8251552 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8251552). PMID 34240008 (https://pubmed.ncbi.nlm.nih.gov/34240008).
49. Zhang, Zhihui; Lv, Dawei; Lu, Man; Yu, Zicheng; Gao, Yuan; Wang, Tiantian; Gao, Jie; Wang, Chengshan (March 2023). "Wildfire activity driven by the 405-kyr orbital climate cycles in the Middle Jurassic". *Global and Planetary Change*. **222** 104069. Bibcode:2023GPC...22204069Z (https://ui.adsabs.harvard.edu/abs/2023GPC...22204069Z). doi:10.1016/j.gloplacha.2023.104069 (https://doi.org/10.1016%2Fj.gloplacha.2023.104069).
50. Allmon, Warren D.; Martin, Ronald E. (2014). "Seafood through time revisited: the Phanerozoic increase in marine trophic resources and its macroevolutionary consequences". *Paleobiology*. **40** (2): 256–287. doi:10.1666/13065 (https://doi.org/10.1666%2F13065).
51. Bougeault, Cédric; Pellenard, Pierre; Deconinck, Jean-François; Hesselbo, Stephen P.; Dommergues, Jean-Louis; Bruneau, Ludovic; Cocquerez, Théophile; Laffont, Rémi; Huret, Emilia; Thibault, Nicholas (February 2017). "Climatic and palaeoceanographic changes during the Pliensbachian (Early Jurassic) inferred from clay mineralogy and stable isotope (C-O) geochemistry (NW Europe)". *Global and Planetary Change*. **149**: 139–152. Bibcode:2017GPC...149..139B (https://ui.adsabs.harvard.edu/abs/2017GPC...149..139B). doi:10.1016/j.gloplacha.2017.01.005 (https://doi.org/10.1016%2Fj.gloplacha.2017.01.005). hdl:10871/25335 (https://hdl.handle.net/10871%2F25335).
52. Ruebsam, Wolfgang; Mayer, Bernhard; Schwark, Lorenz (January 2019). "Cryosphere carbon dynamics control early Toarcian global warming and sea level evolution". *Global and Planetary Change*. **172**: 440–453. Bibcode:2019GPC...172..440R (https://ui.adsabs.harvard.edu/abs/2019GPC...172..440R). doi:10.1016/j.gloplacha.2018.11.003 (https://doi.org/10.1016%2Fj.gloplacha.2018.11.003).
53. Ruebsam, Wolfgang; Schwark, Lorenz (2021-05-11). "Impact of a northern-hemispherical cryosphere on late Pliensbachian–early Toarcian climate and environment evolution". *Geological Society, London, Special Publications*. **514** (1): SP514–2021–11. Bibcode:2021GSLSP.514..359R (https://ui.adsabs.harvard.edu/abs/2021GSLSP.514..359R). doi:10.1144/SP514-2021-11 (https://doi.org/10.1144%2FSP514-2021-11).

54. Ikeda, Masayuki; Bôle, Maximilien; Baumgartner, Peter O. (1 September 2016). "Orbital-scale changes in redox condition and biogenic silica/detrital fluxes of the Middle Jurassic Radiolarite in Tethys (Sogno, Lombardy, N-Italy): Possible link with glaciation?". *Palaeogeography, Palaeoclimatology, Palaeoecology*. **457**: 247–257. Bibcode:2016PPP...457..247I (https://ui.adsabs.harvard.edu/abs/2016PPP...457..247I). doi:10.1016/j.palaeo.2016.06.009 (https://doi.org/10.1016%2Fj.palaeo.2016.06.009).
55. Li, Jun; Huang, Cheng-Min; Yang, Guo-Lin; Pan, Yuan-Yuan; Wen, Xing-Yue (January 2022). "Middle Jurassic climate oscillations from paleosol records of the Sichuan Basin, SW China" (https://doi.org/10.1016%2Fj.jop.2022.01.003). *Journal of Palaeogeography*. **11** (1): 97–122. Bibcode:2022JPalG..11...97L (https://ui.adsabs.harvard.edu/abs/2022JPalG..11...97L). doi:10.1016/j.jop.2022.01.003 (https://doi.org/10.1016%2Fj.jop.2022.01.003).
56. Zhang, Zhihui; Wang, Chengshan; Lv, Dawei; Hay, William W.; Wang, Tiantian; Cao, Shuo (1 January 2020). "Precession-scale climate forcing of peatland wildfires during the early middle Jurassic greenhouse period". *Global and Planetary Change*. **184** 103051. Bibcode:2020GPC...18403051Z (https://ui.adsabs.harvard.edu/abs/2020GPC...18403051Z). doi:10.1016/j.gloplacha.2019.103051 (https://doi.org/10.1016%2Fj.gloplacha.2019.103051).
57. Hesselbo, Stephen P.; Morgans-Bell, Helen S.; McElwain, Jennifer C.; Rees, P. McAllister; Robinson, Stuart A.; Ross, C. Elizabeth (May 2003). "Carbon-Cycle Perturbation in the Middle Jurassic and Accompanying Changes in the Terrestrial Paleoenvironment". *The Journal of Geology*. **111** (3): 259–276. Bibcode:2003JG....111..259H (https://ui.adsabs.harvard.edu/abs/2003JG....111..259H). doi:10.1086/373968 (https://doi.org/10.1086%2F373968).
58. Wierzbowski, H.; Joachimski, M. (22 October 2007). "Reconstruction of late Bajocian–Bathonian marine palaeoenvironments using carbon and oxygen isotope ratios of calcareous fossils from the Polish Jura Chain (central Poland)". *Palaeogeography, Palaeoclimatology, Palaeoecology*. **254** (3–4): 523–540. Bibcode:2007PPP...254..523W (https://ui.adsabs.harvard.edu/abs/2007PPP...254..523W). doi:10.1016/j.palaeo.2007.07.010 (https://doi.org/10.1016%2Fj.palaeo.2007.07.010).
59. Jenkyns, Hugh C.; Schouten-Huibers, L.; Schouten, S.; Sinninghe Damsté, Jaap S. (2 February 2012). "Warm Middle Jurassic–Early Cretaceous high-latitude sea-surface temperatures from the Southern Ocean" (https://doi.org/10.5194%2Fcp-8-215-2012). *Climate of the Past*. **8** (1): 215–226. Bibcode:2012CliPa...8..215J (https://ui.adsabs.harvard.edu/abs/2012CliPa...8..215J). doi:10.5194/cp-8-215-2012 (https://doi.org/10.5194%2Fcp-8-215-2012).
60. Li, Gaojie; Xia, Guoqing; Yi, Haisheng; Wu, Chihua; Wagreich, Michael (15 September 2022). "Climate changes as recorded in stable carbon isotopic compositions of the Late Jurassic marine sedimentary succession in the Qiangtang Basin, Northern Tibet". *Journal of Asian Earth Sciences*. **236** 105317. Bibcode:2022JAESc.23605317L (https://ui.adsabs.harvard.edu/abs/2022JAESc.23605317L). doi:10.1016/j.jseaes.2022.105317 (https://doi.org/10.1016%2Fj.jseaes.2022.105317).
61. Dromart, G.; Garcia, J.-P.; Picard, S.; Atrops, F.; Lécuyer, C.; Sheppard, S. M. F. (25 August 2003). "Ice age at the Middle–Late Jurassic transition?". *Earth and Planetary Science Letters*. **213** (3–4): 205–220. Bibcode:2003E&PSL.213..205D (https://ui.adsabs.harvard.edu/abs/2003E&PSL.213..205D). doi:10.1016/S0012-821X(03)00287-5 (https://doi.org/10.1016%2FS0012-821X%2803%2900287-5).
62. Philippe, Marc; Puijalon, Sara; Suan, Guillaume; Mousset, Sylvain; Thévenard, Frédéric; Mattioli, Emanuela (15 January 2017). "The palaeolatitudinal distribution of fossil wood genera as a proxy for European Jurassic terrestrial climate". *Palaeogeography, Palaeoclimatology, Palaeoecology*. **466**: 373–381. Bibcode:2017PPP...466..373P (https://ui.adsabs.harvard.edu/abs/2017PPP...466..373P). doi:10.1016/j.palaeo.2016.11.029 (https://doi.org/10.1016%2Fj.palaeo.2016.11.029).

63. Them, T.R.; Gill, B.C.; Caruthers, A.H.; Gröcke, D.R.; Tulsky, E.T.; Martindale, R.C.; Poulton, T.P.; Smith, P.L. (February 2017). "High-resolution carbon isotope records of the Toarcian Oceanic Anoxic Event (Early Jurassic) from North America and implications for the global drivers of the Toarcian carbon cycle" (https://doi.org/10.1016%2Fj.epsl.2016.11.021). *Earth and Planetary Science Letters*. **459**: 118–126. Bibcode:2017E&PSL.459..118T (https://ui.adsabs.harvard.edu/abs/2017E&PSL.459..118T). doi:10.1016/j.epsl.2016.11.021 (https://doi.org/10.1016%2Fj.epsl.2016.11.021).
64. Ros-Franch, Sonia; Echevarría, Javier; Damborenea, Susana E.; Manceñido, Miguel O.; Jenkyns, Hugh C.; Al-Suwaidi, Aisha; Hesselbo, Stephen P.; Riccardi, Alberto C. (1 July 2019). "Population response during an Oceanic Anoxic Event: The case of Posidonotis (Bivalvia) from the Lower Jurassic of the Neuquén Basin, Argentina". *Palaeogeography, Palaeoclimatology, Palaeoecology*. **525**: 57–67. Bibcode:2019PPP...525...57R (https://ui.adsabs.harvard.edu/abs/2019PPP...525...57R). doi:10.1016/j.palaeo.2019.04.009 (https://doi.org/10.1016%2Fj.palaeo.2019.04.009). hdl:11336/128130 (https://hdl.handle.net/11336%2F128130).
65. Wignall, Paul B.; Bond, David P. G. (2008). "The end-Triassic and Early Jurassic mass extinction records in the British Isles". *Proceedings of the Geologists' Association*. **119** (1): 73–84. Bibcode:2008PrGA..119...73W (https://ui.adsabs.harvard.edu/abs/2008PrGA..119...73W). doi:10.1016/S0016-7878(08)80259-3 (https://doi.org/10.1016%2FS0016-7878%2808%2980259-3).
66. Reolid, Matías; Mattioli, Emanuela; Duarte, Luís V.; Ruebsam, Wolfgang (2021-09-22). "The Toarcian Oceanic Anoxic Event: where do we stand?". *Geological Society, London, Special Publications*. **514** (1): 1–11. Bibcode:2021GSLSP.514....1R (https://ui.adsabs.harvard.edu/abs/2021GSLSP.514....1R). doi:10.1144/SP514-2021-74 (https://doi.org/10.1144%2FSP514-2021-74).
67. Rodrigues, Bruno; Duarte, Luís V.; Silva, Ricardo L.; Mendonça Filho, João Graciano (15 September 2020). "Sedimentary organic matter and early Toarcian environmental changes in the Lusitanian Basin (Portugal)". *Palaeogeography, Palaeoclimatology, Palaeoecology*. **554** 109781. Bibcode:2020PPP...55409781R (https://ui.adsabs.harvard.edu/abs/2020PPP...55409781R). doi:10.1016/j.palaeo.2020.109781 (https://doi.org/10.1016%2Fj.palaeo.2020.109781).
68. Dera, Guillaume; Neige, Pascal; Dommergues, Jean-Louis; Fara, Emmanuel; Laffont, Rémi; Pellenard, Pierre (January 2010). "High-resolution dynamics of Early Jurassic marine extinctions: the case of Pliensbachian–Toarcian ammonites (Cephalopoda)" (https://figshare.com/articles/dataset/High-resolution_dynamics_of_Early_Jurassic_marine_extinctions_the_case_of_Pliensbachian_Toarcian_ammonites_Cephalopoda_/3454760). *Journal of the Geological Society*. **167** (1): 21–33. Bibcode:2010JGSoc.167...21D (https://ui.adsabs.harvard.edu/abs/2010JGSoc.167...21D). doi:10.1144/0016-76492009-068 (https://doi.org/10.1144%2F0016-76492009-068).
69. Arias, Carmen (1 October 2013). "The early Toarcian (early Jurassic) ostracod extinction events in the Iberian Range: The effect of temperature changes and prolonged exposure to low dissolved oxygen concentrations" (https://eprints.ucm.es/id/eprint/26708/1/1-s2.0-S0031018213003258-_01.pdf) (PDF). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **387**: 40–55. Bibcode:2013PPP...387...40A (https://ui.adsabs.harvard.edu/abs/2013PPP...387...40A). doi:10.1016/j.palaeo.2013.07.004 (https://doi.org/10.1016%2Fj.palaeo.2013.07.004).
70. Hess, Silvia; Nagy, Jenő; Laursen, Gitte Vestergaard (28 January 2014). "Benthic foraminifera from the Lower Jurassic transgressive mudstones of the south-western Barents Sea—a possible high-latitude expression of the global Pliensbachian–Toarcian turnover?" (https://doi.org/10.3402%2Fpolar.v33.20206). *Polar Research*. **33** (1) 20206. Bibcode:2014PolRe..3320206H (https://ui.adsabs.harvard.edu/abs/2014PolRe..3320206H). doi:10.3402/polar.v33.20206 (https://doi.org/10.3402%2Fpolar.v33.20206).

71. Reolid, Matías; Copestake, Philip; Johnson, Ben (15 October 2019). "Foraminiferal assemblages, extinctions and appearances associated with the Early Toarcian Oceanic Anoxic Event in the Llanbedr (Mochras Farm) Borehole, Cardigan Bay Basin, United Kingdom". *Palaeogeography, Palaeoclimatology, Palaeoecology*. **532** 109277. Bibcode:2019PPP...53209277R (https://ui.adsabs.harvard.edu/abs/2019PPP...53209277R). doi:10.1016/j.palaeo.2019.109277 (https://doi.org/10.1016%2Fj.palaeo.2019.109277).
72. Danise, Silvia; Clémence, Marie-Emilie; Price, Gregory D.; Murphy, Daniel P.; Gómez, Juan J.; Twitchett, Richard J. (15 June 2019). "Stratigraphic and environmental control on marine benthic community change through the early Toarcian extinction event (Iberian Range, Spain)". *Palaeogeography, Palaeoclimatology, Palaeoecology*. **524**: 183–200. Bibcode:2019PPP...524..183D (https://ui.adsabs.harvard.edu/abs/2019PPP...524..183D). doi:10.1016/j.palaeo.2019.03.039 (https://doi.org/10.1016%2Fj.palaeo.2019.03.039). hdl:10026.1/13668 (https://hdl.handle.net/10026.1%2F13668).
73. Caruthers, Andrew H.; Smith, Paul L.; Gröcke, Darren R. (September 2013). "The Pliensbachian–Toarcian (Early Jurassic) extinction, a global multi-phased" (https://durham-repository.worktribe.com/output/1405383). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **386**: 104–118. Bibcode:2013PPP...386..104C (https://ui.adsabs.harvard.edu/abs/2013PPP...386..104C). doi:10.1016/j.palaeo.2013.05.010 (https://doi.org/10.1016%2Fj.palaeo.2013.05.010).
74. Vörös, Attila; Kocsis, Ádám; Pálfy, József (1 September 2016). "Demise of the last two spire-bearing brachiopod orders (Spiriferinida and Athyridida) at the Toarcian (Early Jurassic) extinction event". *Palaeogeography, Palaeoclimatology, Palaeoecology*. **457**: 233–241. Bibcode:2016PPP...457..233V (https://ui.adsabs.harvard.edu/abs/2016PPP...457..233V). doi:10.1016/j.palaeo.2016.06.022 (https://doi.org/10.1016%2Fj.palaeo.2016.06.022).
75. Joran, Fernando García; Baeza-Carratalá, José Francisco; Goy, Antonio (1 October 2018). "Changes in brachiopod body size prior to the Early Toarcian (Jurassic) Mass Extinction". *Palaeogeography, Palaeoclimatology, Palaeoecology*. **506**: 242–249. Bibcode:2018PPP...506..242G (https://ui.adsabs.harvard.edu/abs/2018PPP...506..242G). doi:10.1016/j.palaeo.2018.06.045 (https://doi.org/10.1016%2Fj.palaeo.2018.06.045). hdl:10045/77781 (https://hdl.handle.net/10045%2F77781).
76. Maxwell, Erin E.; Vincent, Peggy (2015-11-06). "Effects of the early Toarcian Oceanic Anoxic Event on ichthyosaur body size and faunal composition in the Southwest German Basin". *Paleobiology*. **42** (1): 117–126. doi:10.1017/pab.2015.34 (https://doi.org/10.1017%2Fpab.2015.34).
77. Xu, Weimu; Ruhl, Micha; Jenkyns, Hugh C.; Hesselbo, Stephen P.; Riding, James B.; Selby, David; Naafs, B. David A.; Weijers, Johan W. H.; Pancost, Richard D.; Tegelaar, Erik W.; Idiz, Erdem F. (February 2017). "Carbon sequestration in an expanded lake system during the Toarcian oceanic anoxic event". *Nature Geoscience*. **10** (2): 129–134. Bibcode:2017NatGe..10..129X (https://ui.adsabs.harvard.edu/abs/2017NatGe..10..129X). doi:10.1038/ngeo2871 (https://doi.org/10.1038%2Fngeo2871). hdl:10871/24965 (https://hdl.handle.net/10871%2F24965).
78. Müller, Tamás; Jurikova, Hana; Gutjahr, Marcus; Tomašových, Adam; Schlögl, Jan; Liebetrau, Volker; Duarte, Luís v.; Milovský, Rastislav; Suan, Guillaume; Mattioli, Emanuela; Pittet, Bernard (2020-12-01). "Ocean acidification during the early Toarcian extinction event: Evidence from boron isotopes in brachiopods" (https://doi.org/10.1130%2FG47781.1). *Geology*. **48** (12): 1184–1188. Bibcode:2020Geo....48.1184M (https://ui.adsabs.harvard.edu/abs/2020Geo....48.1184M). doi:10.1130/G47781.1 (https://doi.org/10.1130%2FG47781.1). hdl:10023/20595 (https://hdl.handle.net/10023%2F20595).
79. Trecalli, Alberto; Spangenberg, Jorge; Adatte, Thierry; Föllmi, Karl B.; Parente, Mariano (December 2012). "Carbonate platform evidence of ocean acidification at the onset of the early Toarcian oceanic anoxic event". *Earth and Planetary Science Letters*. 357–358: 214–225. Bibcode:2012E&PSL.357..214T (https://ui.adsabs.harvard.edu/abs/2012E&PSL.357..214T). doi:10.1016/j.epsl.2012.09.043 (https://doi.org/10.1016%2Fj.epsl.2012.09.043).

80. Ettinger, Nicholas P.; Larson, Toti E.; Kerans, Charles; Thibodeau, Alyson M.; Hattori, Kelly E.; Kacur, Sean M.; Martindale, Rowan C. (2020-09-23). Eberli, Gregor (ed.). "Ocean acidification and photic-zone anoxia at the Toarcian Oceanic Anoxic Event: Insights from the Adriatic Carbonate Platform". *Sedimentology*. **68**: 63–107. doi:10.1111/sed.12786 (https://doi.org/10.1111%2Fsed.12786).
81. Papadomanolaki, Nina M.; Lenstra, Wytze K.; Wolthers, Mariette; Slomp, Caroline P. (1 July 2022). "Enhanced phosphorus recycling during past oceanic anoxia amplified by low rates of apatite authigenesis" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10883373). *Science Advances*. **8** (26) eabn2370. Bibcode:2022SciA....8N2370P (https://ui.adsabs.harvard.edu/abs/2022SciA....8N2370P). doi:10.1126/sciadv.abn2370 (https://doi.org/10.1126%2Fsciadv.abn2370). hdl:1874/421467 (https://hdl.handle.net/1874%2F421467). PMC 10883373 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10883373). PMID 35776794 (https://pubmed.ncbi.nlm.nih.gov/35776794).
82. Tennant, Jonathan P.; Mannion, Philip D.; Upchurch, Paul (2016-09-02). "Sea level regulated tetrapod diversity dynamics through the Jurassic/Cretaceous interval" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5025807). *Nature Communications*. **7** (1) 12737. Bibcode:2016NatCo...712737T (https://ui.adsabs.harvard.edu/abs/2016NatCo...712737T). doi:10.1038/ncomms12737 (https://doi.org/10.1038%2Fncomms12737). PMC 5025807 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5025807). PMID 27587285 (https://pubmed.ncbi.nlm.nih.gov/27587285).
83. Lucas, Spencer G.; Tanner, Lawrence H. (October 2015). "End-Triassic nonmarine biotic events" (https://doi.org/10.1016%2Fj.jop.2015.08.010). *Journal of Palaeogeography*. **4** (4): 331–348. Bibcode:2015JPalG...4..331L (https://ui.adsabs.harvard.edu/abs/2015JPalG...4..331L). doi:10.1016/j.jop.2015.08.010 (https://doi.org/10.1016%2Fj.jop.2015.08.010).
84. Mander, Luke; Kürschner, Wolfram M.; McElwain, Jennifer C. (2010-08-31). "An explanation for conflicting records of Triassic–Jurassic plant diversity" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2932585). *Proceedings of the National Academy of Sciences*. **107** (35): 15351–15356. Bibcode:2010PNAS..10715351M (https://ui.adsabs.harvard.edu/abs/2010PNAS..10715351M). doi:10.1073/pnas.1004207107 (https://doi.org/10.1073%2Fpnas.1004207107). PMC 2932585 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2932585). PMID 20713737 (https://pubmed.ncbi.nlm.nih.gov/20713737).
85. Barbacka, Maria; Pacyna, Grzegorz; Kocsis, Ádam T.; Jarzynka, Agata; Ziaja, Jadwiga; Bodor, Emese (August 2017). "Changes in terrestrial floras at the Triassic-Jurassic Boundary in Europe" (http://ruj.uj.edu.pl/xmlui/handle/item/41636). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **480**: 80–93. Bibcode:2017PPP...480...80B (https://ui.adsabs.harvard.edu/abs/2017PPP...480...80B). doi:10.1016/j.palaeo.2017.05.024 (https://doi.org/10.1016%2Fj.palaeo.2017.05.024).
86. Elgorriaga, Andrés; Escapa, Ignacio H.; Cúneo, N. Rubén (July 2019). "Relictual Lepidopteris (Peltaspermales) from the Early Jurassic Cañadón Asfalto Formation, Patagonia, Argentina". *International Journal of Plant Sciences*. **180** (6): 578–596. Bibcode:2019IJPlS.180..578E (https://ui.adsabs.harvard.edu/abs/2019IJPlS.180..578E). doi:10.1086/703461 (https://doi.org/10.1086%2F703461).
87. Bomfleur, Benjamin; Blomenkemper, Patrick; Kerp, Hans; McLoughlin, Stephen (2018). "Polar Regions of the Mesozoic–Paleogene Greenhouse World as Refugia for Relict Plant Groups" (http://urn.kb.se/resolve?urn=urn:nbn:se:nrm:diva-2905). *Transformative Paleobotany*. pp. 593–611. doi:10.1016/b978-0-12-813012-4.00024-3 (https://doi.org/10.1016%2Fb978-0-12-813012-4.00024-3). ISBN 978-0-12-813012-4.

88. Atkinson, Brian A.; Serbet, Rudolph; Hieger, Timothy J.; Taylor, Edith L. (October 2018). "Additional evidence for the Mesozoic diversification of conifers: Pollen cone of Chimaerostrobus minutus gen. et sp. nov. (Coniferales), from the Lower Jurassic of Antarctica" (https://doi.org/10.1016%2Fj.revpalbo.2018.06.013). *Review of Palaeobotany and Palynology*. **257**: 77–84. Bibcode:2018RPaPa.257...77A (https://ui.adsabs.harvard.edu/abs/2018RPaPa.257...77A). doi:10.1016/j.revpalbo.2018.06.013 (https://doi.org/10.1016%2Fj.revpalbo.2018.06.013).
89. Leslie, Andrew B.; Beaulieu, Jeremy; Holman, Garth; Campbell, Christopher S.; Mei, Wenbin; Raubeson, Linda R.; Mathews, Sarah (September 2018). "An overview of extant conifer evolution from the perspective of the fossil record" (https://digitalcommons.lsu.edu/biosci_pubs/2619). *American Journal of Botany*. **105** (9): 1531–1544. Bibcode:2018AmJB..105.1531L (https://ui.adsabs.harvard.edu/abs/2018AmJB..105.1531L). doi:10.1002/ajb2.1143 (https://doi.org/10.1002%2Fajb2.1143). PMID 30157290 (https://pubmed.ncbi.nlm.nih.gov/30157290).
90. Stockey, Ruth A.; Rothwell, Gar W. (July 2020). "Diversification of crown group Araucaria: the role of Araucaria famii sp. nov. in the Late Cretaceous (Campanian) radiation of Araucariaceae in the Northern Hemisphere" (https://doi.org/10.1002%2Fajb2.1505). *American Journal of Botany*. **107** (7): 1072–1093. doi:10.1002/ajb2.1505 (https://doi.org/10.1002%2Fajb2.1505). PMID 32705687 (https://pubmed.ncbi.nlm.nih.gov/32705687).
91. Escapa, Ignacio H.; Catalano, Santiago A. (October 2013). "Phylogenetic Analysis of Araucariaceae: Integrating Molecules, Morphology, and Fossils". *International Journal of Plant Sciences*. **174** (8): 1153–1170. Bibcode:2013IJPlS.174.1153E (https://ui.adsabs.harvard.edu/abs/2013IJPlS.174.1153E). doi:10.1086/672369 (https://doi.org/10.1086%2F672369). hdl:11336/3583 (https://hdl.handle.net/11336%2F3583).
92. Stockey, Ruth A.; Rothwell, Gar W. (March 2013). "Pararaucaria carrii sp. nov., Anatomically Preserved Evidence for the Conifer Family Cheirolepidiaceae in the Northern Hemisphere". *International Journal of Plant Sciences*. **174** (3): 445–457. Bibcode:2013IJPlS.174..445S (https://ui.adsabs.harvard.edu/abs/2013IJPlS.174..445S). doi:10.1086/668614 (https://doi.org/10.1086%2F668614).
93. Escapa, Ignacio; Cúneo, Rubén; Axsmith, Brian (September 2008). "A new genus of the Cupressaceae (sensu lato) from the Jurassic of Patagonia: Implications for conifer megasporangiate cone homologies". *Review of Palaeobotany and Palynology*. **151** (3–4): 110–122. Bibcode:2008RPaPa.151..110E (https://ui.adsabs.harvard.edu/abs/2008RPaPa.151..110E). doi:10.1016/j.revpalbo.2008.03.002 (https://doi.org/10.1016%2Fj.revpalbo.2008.03.002).
94. Contreras, Dori L.; Escapa, Ignacio H.; Iribarren, Rocio C.; Cúneo, N. Rubén (October 2019). "Reconstructing the Early Evolution of the Cupressaceae: A Whole-Plant Description of a New Austrohamia Species from the Cañadón Asfalto Formation (Early Jurassic), Argentina". *International Journal of Plant Sciences*. **180** (8): 834–868. Bibcode:2019IJPlS.180..834C (https://ui.adsabs.harvard.edu/abs/2019IJPlS.180..834C). doi:10.1086/704831 (https://doi.org/10.1086%2F704831).
95. Ma, Qing-Wen; K. Ferguson, David; Liu, Hai-Ming; Xu, Jing-Xian (2020). "Compressions of Sequoia (Cupressaceae sensu lato) from the Middle Jurassic of Daohugou, Ningcheng, Inner Mongolia, China". *Palaeobiodiversity and Palaeoenvironments*. **1** (9): 1. doi:10.1007/s12549-020-00454-z (https://doi.org/10.1007%2Fs12549-020-00454-z).
96. Domogatskaya, Ksenia V.; Herman, Alexei B. (May 2019). "New species of the genus Schizolepidopsis (conifers) from the Albian of the Russian high Arctic and geological history of the genus". *Cretaceous Research*. **97**: 73–93. Bibcode:2019CrRes..97...73D (https://ui.adsabs.harvard.edu/abs/2019CrRes..97...73D). doi:10.1016/j.cretres.2019.01.012 (https://doi.org/10.1016%2Fj.cretres.2019.01.012).

97. Matsunaga, Kelly K. S.; Herendeen, Patrick S.; Herrera, Fabiany; Ichinnorov, Niiden; Crane, Peter R.; Shi, Gongle (2021-05-10). "Ovulate Cones of Schizolepidopsis ediae sp. nov. Provide Insights into the Evolution of Pinaceae" (https://doi.org/10.1086%2F714281). *International Journal of Plant Sciences*. **182** (6): 490–507. Bibcode:2021IJPlS.182..490M (https://ui.adsabs.harvard.edu/abs/2021IJPlS.182..490M). doi:10.1086/714281 (https://doi.org/10.1086%2F714281).
98. Rothwell, Gar W.; Mapes, Gene; Stockey, Ruth A.; Hilton, Jason (April 2012). "The seed cone Eathiestrobus gen. nov.: Fossil evidence for a Jurassic origin of Pinaceae". *American Journal of Botany*. **99** (4): 708–720. Bibcode:2012AmJB...99..708R (https://ui.adsabs.harvard.edu/abs/2012AmJB...99..708R). doi:10.3732/ajb.1100595 (https://doi.org/10.3732%2Fajb.1100595). PMID 22491001 (https://pubmed.ncbi.nlm.nih.gov/22491001).
99. Smith, Selena Y.; Stockey, Ruth A.; Rothwell, Gar W.; Little, Stefan A. (2017-01-02). "A new species of Pityostrobus (Pinaceae) from the Cretaceous of California: moving towards understanding the Cretaceous radiation of Pinaceae". *Journal of Systematic Palaeontology*. **15** (1): 69–81. Bibcode:2017JSPal..15...69S (https://ui.adsabs.harvard.edu/abs/2017JSPal..15...69S). doi:10.1080/14772019.2016.1143885 (https://doi.org/10.1080%2F14772019.2016.1143885).
100. Dong, Chong; Shi, Gongle; Herrera, Fabiany; Wang, Yongdong; Herendeen, Patrick S; Crane, Peter R (2020-06-18). "Middle–Late Jurassic fossils from northeastern China reveal morphological stasis in the catkin-yew" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8288717). *National Science Review*. **7** (11): 1765–1767. doi:10.1093/nsr/nwaa138 (https://doi.org/10.1093%2Fnsr%2Fnwaa138). PMC 8288717 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8288717). PMID 34691509 (https://pubmed.ncbi.nlm.nih.gov/34691509).
101. Andruchow-Colombo, Ana; Escapa, Ignacio H; Aagesen, Lone; Matsunaga, Kelly K S (2023-08-04). "In search of lost time: tracing the fossil diversity of Podocarpaceae through the ages". *Botanical Journal of the Linnean Society*. **203** (4): 315–336. doi:10.1093/botlinnean/boad027 (https://doi.org/10.1093%2Fbotlinnean%2Fboad027). hdl:11336/227952 (https://hdl.handle.net/11336%2F227952).
102. Pole, Mike; Wang, Yongdong; Bugdaeva, Eugenia V.; Dong, Chong; Tian, Ning; Li, Liqin; Zhou, Ning (2016-12-15). "The rise and demise of Podozamites in east Asia—An extinct conifer life style". *Palaeogeography, Palaeoclimatology, Palaeoecology*. Mesozoic ecosystems – Climate and Biota. **464**: 97–109. Bibcode:2016PPP...464...97P (https://ui.adsabs.harvard.edu/abs/2016PPP...464...97P). doi:10.1016/j.palaeo.2016.02.037 (https://doi.org/10.1016%2Fj.palaeo.2016.02.037).
103. Zhou, Zhi-Yan (March 2009). "An overview of fossil Ginkgoales". *Palaeoworld*. **18** (1): 1–22. doi:10.1016/j.palwor.2009.01.001 (https://doi.org/10.1016%2Fj.palwor.2009.01.001).
104. Nosova, Natalya (October 2013). "Revision of the genus Grenana Samylina from the Middle Jurassic of Angren, Uzbekistan". *Review of Palaeobotany and Palynology*. **197**: 226–252. Bibcode:2013RPaPa.197..226N (https://ui.adsabs.harvard.edu/abs/2013RPaPa.197..226N). doi:10.1016/j.revpalbo.2013.06.005 (https://doi.org/10.1016%2Fj.revpalbo.2013.06.005).
105. Dong, Chong; Shi, Gongle; Zhang, Xiaoqing; Wang, Zixi; Wang, Yongdong (November 2022). "Middle-Late Jurassic fossils from Northeast China confirm the affiliation of Umaltolepis seed-bearing structures and Pseudotorellia leaves". *Review of Palaeobotany and Palynology*. **306** 104763. Bibcode:2022RPaPa.30604763D (https://ui.adsabs.harvard.edu/abs/2022RPaPa.30604763D). doi:10.1016/j.revpalbo.2022.104763 (https://doi.org/10.1016%2Fj.revpalbo.2022.104763).
106. Popa, Mihai E. (June 2014). "Early Jurassic bennettitalean reproductive structures of Romania". *Palaeobiodiversity and Palaeoenvironments*. **94** (2): 327–362. Bibcode:2014PdPe...94..327P (https://ui.adsabs.harvard.edu/abs/2014PdPe...94..327P). doi:10.1007/s12549-014-0165-9 (https://doi.org/10.1007%2Fs12549-014-0165-9).
107. Taylor, T. (2009). "Cycadophytes". *Biology and Evolution of Fossil Plants*. pp. 703–741. doi:10.1016/b978-0-12-373972-8.00017-6 (https://doi.org/10.1016%2Fb978-0-12-373972-8.00017-6). ISBN 978-0-12-373972-8.

108. Pott, Christian; McLoughlin, Stephen (2014-06-01). "Divaricate growth habit in Williamsoniaceae (Bennettitales): unravelling the ecology of a key Mesozoic plant group". *Palaeobiodiversity and Palaeoenvironments*. **94** (2): 307–325. Bibcode:2014PdPe...94..307P (https://ui.adsabs.harvard.edu/abs/2014PdPe...94..307P). doi:10.1007/s12549-014-0157-9 (https://doi.org/10.1007%2Fs12549-014-0157-9).
109. Labandeira, Conrad C.; Yang, Qiang; Santiago-Blay, Jorge A.; Hotton, Carol L.; Monteiro, Antónia; Wang, Yong-Jie; Goreva, Yulia; Shih, ChungKun; Siljeström, Sandra; Rose, Tim R.; Dilcher, David L. (2016-02-10). "The evolutionary convergence of mid-Mesozoic lacewings and Cenozoic butterflies" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4760178). *Proceedings of the Royal Society B: Biological Sciences*. **283** (1824) 20152893. Bibcode:2016PBioS.28352893L (https://ui.adsabs.harvard.edu/abs/2016PBioS.28352893L). doi:10.1098/rspb.2015.2893 (https://doi.org/10.1098%2Frspb.2015.2893). PMC 4760178 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4760178). PMID 26842570 (https://pubmed.ncbi.nlm.nih.gov/26842570).
110. Khramov, Alexander V.; Lukashevich, Elena D. (July 2019). "A Jurassic dipteran pollinator with an extremely long proboscis". *Gondwana Research*. **71**: 210–215. Bibcode:2019GondR..71..210K (https://ui.adsabs.harvard.edu/abs/2019GondR..71..210K). doi:10.1016/j.gr.2019.02.004 (https://doi.org/10.1016%2Fj.gr.2019.02.004).
111. Cai, Chenyang; Escalona, Hermes E.; Li, Liqin; Yin, Ziwei; Huang, Diying; Engel, Michael S. (September 2018). "Beetle Pollination of Cycads in the Mesozoic" (https://doi.org/10.1016%2Fj.cub.2018.06.036). *Current Biology*. **28** (17): 2806–2812.e1. Bibcode:2018CBio...28E2806C (https://ui.adsabs.harvard.edu/abs/2018CBio...28E2806C). doi:10.1016/j.cub.2018.06.036 (https://doi.org/10.1016%2Fj.cub.2018.06.036). PMID 30122529 (https://pubmed.ncbi.nlm.nih.gov/30122529).
112. Coiro, Mario; Pott, Christian (December 2017). "Eobowenia gen. nov. from the Early Cretaceous of Patagonia: indication for an early divergence of Bowenia?" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5383990). *BMC Evolutionary Biology*. **17** (1): 97. Bibcode:2017BMCEE..17...97C (https://ui.adsabs.harvard.edu/abs/2017BMCEE..17...97C). doi:10.1186/s12862-017-0943-x (https://doi.org/10.1186%2Fs12862-017-0943-x). PMC 5383990 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5383990). PMID 28388891 (https://pubmed.ncbi.nlm.nih.gov/28388891).
113. Vajda, Vivi; Pucetaite, Milda; McLoughlin, Stephen; Engdahl, Anders; Heimdal, Jimmy; Uvdal, Per (August 2017). "Molecular signatures of fossil leaves provide unexpected new evidence for extinct plant relationships". *Nature Ecology & Evolution*. **1** (8): 1093–1099. Bibcode:2017NatEE...1.1093V (https://ui.adsabs.harvard.edu/abs/2017NatEE...1.1093V). doi:10.1038/s41559-017-0224-5 (https://doi.org/10.1038%2Fs41559-017-0224-5). PMID 29046567 (https://pubmed.ncbi.nlm.nih.gov/29046567).
114. Coiro, Mario; Allio, Rémi; Mazet, Nathan; Seyfullah, Leyla J.; Condamine, Fabien L. (2023-06-11). "Reconciling fossils with phylogenies reveals the origin and macroevolutionary processes explaining the global cycad biodiversity" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10953041). *New Phytologist*. **240** (4): 1616–1635. Bibcode:2023NewPh.240.1616C (https://ui.adsabs.harvard.edu/abs/2023NewPh.240.1616C). doi:10.1111/nph.19010 (https://doi.org/10.1111%2Fnph.19010). PMC 10953041 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10953041). PMID 37302411 (https://pubmed.ncbi.nlm.nih.gov/37302411).
115. Bateman, Richard M (2020). "Hunting the Snark: the flawed search for mythical Jurassic angiosperms". *Journal of Experimental Botany*. **71** (1): 22–35. doi:10.1093/jxb/erz411 (https://doi.org/10.1093%2Fjxb%2Ferz411). PMID 31538196 (https://pubmed.ncbi.nlm.nih.gov/31538196).

116. Coiro, Mario; Roberts, Emily A.; Hofmann, Christa-Ch.; Seyfullah, Leyla J. (2022-12-14). "Cutting the long branches: Consilience as a path to unearth the evolutionary history of Gnetales" (https://doi.org/10.3389%2Ffevo.2022.1082639). *Frontiers in Ecology and Evolution*. **10** 1082639. Bibcode:2022FrEEv..1082639C (https://ui.adsabs.harvard.edu/abs/2022FrEEv..1082639C). doi:10.3389/fevo.2022.1082639 (https://doi.org/10.3389%2Ffevo.2022.1082639).
117. Elgorriaga, Andrés; Escapa, Ignacio H.; Cúneo, N. Rubén (2019-09-02). "Southern Hemisphere Caytoniales: vegetative and reproductive remains from the Lonco Trapial Formation (Lower Jurassic), Patagonia". *Journal of Systematic Palaeontology*. **17** (17): 1477–1495. Bibcode:2019JSPal..17.1477E (https://ui.adsabs.harvard.edu/abs/2019JSPal..17.1477E). doi:10.1080/14772019.2018.1535456 (https://doi.org/10.1080%2F14772019.2018.1535456).
118. Slodownik, Miriam; Hill, Robert S.; McLoughlin, Stephen (October 2023). "Komlopteris: A persistent lineage of post-Triassic corystosperms in Gondwana" (https://doi.org/10.1016%2Fj.revpalbo.2023.104950). *Review of Palaeobotany and Palynology*. **317** 104950. Bibcode:2023RPaPa.31704950S (https://ui.adsabs.harvard.edu/abs/2023RPaPa.31704950S). doi:10.1016/j.revpalbo.2023.104950 (https://doi.org/10.1016%2Fj.revpalbo.2023.104950).
119. Kustatscher, Evelyn; Visscher, Henk; van Konijnenburg-van Cittert, Johanna H. A. (2019-09-01). "Did the Czekanowskiales already exist in the late Permian?" (https://doi.org/10.1007%2Fs12542-019-00468-9). *PalZ*. **93** (3): 465–477. Bibcode:2019PalZ...93..465K (https://ui.adsabs.harvard.edu/abs/2019PalZ...93..465K). doi:10.1007/s12542-019-00468-9 (https://doi.org/10.1007%2Fs12542-019-00468-9).
120. Taylor, T. (2009). "Gymnosperms with obscure affinities". *Biology and Evolution of Fossil Plants*. pp. 757–785. doi:10.1016/b978-0-12-373972-8.00019-x (https://doi.org/10.1016%2Fb978-0-12-373972-8.00019-x). ISBN 978-0-12-373972-8.
121. Sun, Chunlin; Li, Yunfeng; Dilcher, David L.; Wang, Hongshan; Li, Tao; Na, Yuling; Wang, Anping (November 2015). "An introductory report on the biodiversity of Middle Jurassic Phoenicopsis (Czekanowskiales) from the Ordos Basin, China". *Science Bulletin*. **60** (21): 1858–1865. Bibcode:2015SciBu..60.1858S (https://ui.adsabs.harvard.edu/abs/2015SciBu..60.1858S). doi:10.1007/s11434-015-0904-y (https://doi.org/10.1007%2Fs11434-015-0904-y).
122. Pattemore, G.A., Rigby, J.F. and Playford, G., 2015. Triassic-Jurassic pteridosperms of Australasia: speciation, diversity and decline. Boletín Geológico y Minero, 126 (4): 689–722
123. Skog, Judith E. (April 2001). "Biogeography of Mesozoic leptosporangiate ferns related to extant ferns". *Brittonia*. **53** (2): 236–269. Bibcode:2001Britt..53..236S (https://ui.adsabs.harvard.edu/abs/2001Britt..53..236S). doi:10.1007/bf02812701 (https://doi.org/10.1007%2Fbf02812701).
124. Tian, Ning; Wang, Yong-Dong; Zhang, Wu; Zheng, Shao-Lin; Zhu, Zhi-Peng; Liu, Zhong-Jian (2018-03-01). "Permineralized osmundaceous and gleicheniaceous ferns from the Jurassic of Inner Mongolia, NE China". *Palaeobiodiversity and Palaeoenvironments*. **98** (1): 165–176. Bibcode:2018PdPe...98..165T (https://ui.adsabs.harvard.edu/abs/2018PdPe...98..165T). doi:10.1007/s12549-017-0313-0 (https://doi.org/10.1007%2Fs12549-017-0313-0).
125. Urrea, María José; Yañez, Agustina; Flores, Jorge R (2026-08-28). "Total-evidence dating analysis of Osmundaceae reveals long-term extinction amid global climatic and palaeogeographical changes" (https://doi.org/10.1093/aob/mcag269). *Annals of Botany*: mcag269. doi:10.1093/aob/mcag269 (https://doi.org/10.1093%2Faob%2Fmcag269). ISSN 1095-8290 (https://search.worldcat.org/issn/1095-8290).
126. Regalado, Ledis; Schmidt, Alexander R.; Müller, Patrick; Niedermeier, Lisa; Krings, Michael; Schneider, Harald (July 2019). "Heinrichsia cheilanthoides gen. et sp. nov., a fossil fern in the family Pteridaceae (Polypodiales) from the Cretaceous amber forests of Myanmar" (https://doi.org/10.1111%2Fjse.12514). *Journal of Systematics and Evolution*. **57** (4): 329–338. Bibcode:2019JSyEv..57..329R (https://ui.adsabs.harvard.edu/abs/2019JSyEv..57..329R). doi:10.1111/jse.12514 (https://doi.org/10.1111%2Fjse.12514).

127. Li, Chunxiang; Miao, Xinyuan; Zhang, Li-Bing; Ma, Junye; Hao, Jiasheng (January 2020). "Re-evaluation of the systematic position of the Jurassic–Early Cretaceous fern genus Coniopteris". *Cretaceous Research*. **105** 104136. Bibcode:2020CrRes.10504136L (https://ui.adsabs.harvard.edu/abs/2020CrRes.10504136L). doi:10.1016/j.cretres.2019.04.007 (https://doi.org/10.1016%2Fj.cretres.2019.04.007).
128. Korall, Petra; Pryer, Kathleen M. (February 2014). Parmakelis, Aristeidis (ed.). "Global biogeography of scaly tree ferns (Cyatheaceae): evidence for Gondwanan vicariance and limited transoceanic dispersal" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4238398). *Journal of Biogeography*. **41** (2): 402–413. Bibcode:2014JBiog..41..402K (https://ui.adsabs.harvard.edu/abs/2014JBiog..41..402K). doi:10.1111/jbi.12222 (https://doi.org/10.1111%2Fjbi.12222). PMC 4238398 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4238398). PMID 25435648 (https://pubmed.ncbi.nlm.nih.gov/25435648).
129. Axsmith, Brian J.; Krings, Michael; Taylor, Thomas N. (September 2001). "A filmy fern from the Upper Triassic of North Carolina (USA)" (https://doi.org/10.2307%2F3558399). *American Journal of Botany*. **88** (9): 1558–1567. Bibcode:2001AmJB...88.1558A (https://ui.adsabs.harvard.edu/abs/2001AmJB...88.1558A). doi:10.2307/3558399 (https://doi.org/10.2307%2F3558399). JSTOR 3558399 (https://www.jstor.org/stable/3558399). PMID 21669688 (https://pubmed.ncbi.nlm.nih.gov/21669688).
130. Elgorriaga, Andrés; Escapa, Ignacio H.; Bomfleur, Benjamin; Cúneo, Rubén; Ottone, Eduardo G. (February 2015). "Reconstruction and Phylogenetic Significance of a New Equisetum Linnaeus Species from the Lower Jurassic of Cerro Bayo (Chubut Province, Argentina)". *Ameghiniana*. **52** (1): 135–152. Bibcode:2015Amegh..52..135E (https://ui.adsabs.harvard.edu/abs/2015Amegh..52..135E). doi:10.5710/AMGH.15.09.2014.2758 (https://doi.org/10.5710%2FAMGH.15.09.2014.2758). hdl:11336/66623 (https://hdl.handle.net/11336%2F66623).
131. Gould, Re (April 1968). "Morphology of Equisetum laterale Phillips, 1829, and E. bryanii sp. nov. from the Mesozoic of South-Eastern Queensland". *Australian Journal of Botany*. **16** (1): 153–176. Bibcode:1968AuJB...16..153G (https://ui.adsabs.harvard.edu/abs/1968AuJB...16..153G). doi:10.1071/BT9680153 (https://doi.org/10.1071%2FBT9680153).
132. Elgorriaga, Andrés; Escapa, Ignacio H.; Rothwell, Gar W.; Tomescu, Alexandru M. F.; Rubén Cúneo, N. (August 2018). "Origin of Equisetum: Evolution of horsetails (Equisetales) within the major euphyllophyte clade Sphenopsida" (https://doi.org/10.1002%2Fajb2.1125). *American Journal of Botany*. **105** (8): 1286–1303. Bibcode:2018AmJB..105.1286E (https://ui.adsabs.harvard.edu/abs/2018AmJB..105.1286E). doi:10.1002/ajb2.1125 (https://doi.org/10.1002%2Fajb2.1125). PMID 30025163 (https://pubmed.ncbi.nlm.nih.gov/30025163).
133. Channing, Alan; Zamuner, Alba; Edwards, Dianne; Guido, Diego (2011). "Equisetum thermale sp. nov. (Equisetales) from the Jurassic San Agustín hot spring deposit, Patagonia: Anatomy, paleoecology, and inferred paleoecophysiology" (https://doi.org/10.3732%2Fajb.1000211). *American Journal of Botany*. **98** (4): 680–697. Bibcode:2011AmJB...98..680C (https://ui.adsabs.harvard.edu/abs/2011AmJB...98..680C). doi:10.3732/ajb.1000211 (https://doi.org/10.3732%2Fajb.1000211). hdl:11336/95234 (https://hdl.handle.net/11336%2F95234). PMID 21613167 (https://pubmed.ncbi.nlm.nih.gov/21613167).
134. Wood, Daniel; Besnard, Guillaume; Beerling, David J.; Osborne, Colin P.; Christin, Pascal-Antoine (2020-06-18). "Phylogenomics indicates the "living fossil" Isoetes diversified in the Cenozoic" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7302493). *PLOS ONE*. **15** (6) e0227525. Bibcode:2020PLoSO..1527525W (https://ui.adsabs.harvard.edu/abs/2020PLoSO..1527525W). doi:10.1371/journal.pone.0227525 (https://doi.org/10.1371%2Fjournal.pone.0227525). PMC 7302493 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7302493). PMID 32555586 (https://pubmed.ncbi.nlm.nih.gov/32555586).

135. Mamontov, Yuriy S.; Ignatov, Michael S. (July 2019). "How to rely on the unreliable: Examples from Mesozoic bryophytes of Transbaikalia" (https://doi.org/10.1111%2Fjse.12483). *Journal of Systematics and Evolution*. **57** (4): 339–360. Bibcode:2019JSyEv..57..339M (https://ui.adsabs.harvard.edu/abs/2019JSyEv..57..339M). doi:10.1111/jse.12483 (https://doi.org/10.1111%2Fjse.12483).
136. Bippus, Alexander C.; Savoretti, Adolfina; Escapa, Ignacio H.; Garcia-Massini, Juan; Guido, Diego (October 2019). "Heinrichsiella patagonica gen. et sp. nov.: A Permineralized Acrocarpous Moss from the Jurassic of Patagonia". *International Journal of Plant Sciences*. **180** (8): 882–891. Bibcode:2019IJPlS.180..882B (https://ui.adsabs.harvard.edu/abs/2019IJPlS.180..882B). doi:10.1086/704832 (https://doi.org/10.1086%2F704832).
137. Li, Ruiyun; Li, Xiaoqiang; Deng, Shenghui; Sun, Bainian (August 2020). "Morphology and microstructure of Pellites hamiensis nov. sp., a Middle Jurassic liverwort from northwestern China and its evolutionary significance". *Geobios*. **62**: 23–29. Bibcode:2020Geobi..62...23L (https://ui.adsabs.harvard.edu/abs/2020Geobi..62...23L). doi:10.1016/j.geobios.2020.07.003 (https://doi.org/10.1016%2Fj.geobios.2020.07.003).
138. Li, Rui-Yun; Wang, Xue-lian; Chen, Jing-Wei; Deng, Sheng-Hui; Wang, Zi-Xi; Dong, Jun-Ling; Sun, Bai-Nian (June 2016). "A new thalloid liverwort: Pallaviciniites sandaolingensis sp. nov. from the Middle Jurassic of Turpan–Hami Basin, NW China". *PalZ*. **90** (2): 389–397. Bibcode:2016PalZ...90..389L (https://ui.adsabs.harvard.edu/abs/2016PalZ...90..389L). doi:10.1007/s12542-016-0299-3 (https://doi.org/10.1007%2Fs12542-016-0299-3).
139. Li, Ruiyun; Li, Xiaoqiang; Wang, Hongshan; Sun, Bainian (2019). "Ricciopsis sandaolingensis sp. nov., a new fossil bryophyte from the Middle Jurassic Xishanyao Formation in the Turpan-Hami Basin, Xinjiang, Northwest China" (https://doi.org/10.26879%2F917). *Palaeontologia Electronica*. **22** (2): 917. Bibcode:2019PalEl..22..917L (https://ui.adsabs.harvard.edu/abs/2019PalEl..22..917L). doi:10.26879/917 (https://doi.org/10.26879%2F917).
140. Brusatte, Stephen L; Benton, Michael J; Ruta, Marcello; Lloyd, Graeme T (2008-12-23). "The first 50 Myr of dinosaur evolution: macroevolutionary pattern and morphological disparity" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2614175). *Biology Letters*. **4** (6): 733–736. Bibcode:2008BiLet...4..733B (https://ui.adsabs.harvard.edu/abs/2008BiLet...4..733B). doi:10.1098/rsbl.2008.0441 (https://doi.org/10.1098%2Frsbl.2008.0441). PMC 2614175 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2614175). PMID 18812311 (https://pubmed.ncbi.nlm.nih.gov/18812311).
141. Brusatte, S. L.; Benton, M. J.; Ruta, M.; Lloyd, G. T. (2008-09-12). "Superiority, Competition, and Opportunism in the Evolutionary Radiation of Dinosaurs" (https://www.research.ed.ac.uk/en/publications/00556baf-6575-44d9-af39-bdd0b072ad2b). *Science*. **321** (5895): 1485–88. Bibcode:2008Sci...321.1485B (https://ui.adsabs.harvard.edu/abs/2008Sci...321.1485B). doi:10.1126/science.1161833 (https://doi.org/10.1126%2Fscience.1161833). hdl:20.500.11820/00556baf-6575-44d9-af39-bdd0b072ad2b (https://hdl.handle.net/20.500.11820%2F00556baf-6575-44d9-af39-bdd0b072ad2b). PMID 18787166 (https://pubmed.ncbi.nlm.nih.gov/18787166).
142. Temp Müller, Rodrigo; Augusto Pretto, Flávio; Kerber, Leonardo; Silva-Neves, Eduardo; Dias-da-Silva, Sérgio (28 March 2018). "Comment on 'A dinosaur missing-link? Chilesaurus and the early evolution of ornithischian dinosaurs' " (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5897605). *Biology Letters*. **14** (3) 20170581. doi:10.1098/rsbl.2017.0581 (https://doi.org/10.1098%2Frsbl.2017.0581). PMC 5897605 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5897605). PMID 29593074 (https://pubmed.ncbi.nlm.nih.gov/29593074).

143. Zahner, Marion; Brinkmann, Winand (August 2019). "A Triassic averostran-line theropod from Switzerland and the early evolution of dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6669044). *Nature Ecology & Evolution*. **3** (8): 1146–1152. Bibcode:2019NatEE...3.1146Z (https://ui.adsabs.harvard.edu/abs/2019NatEE...3.1146Z). doi:10.1038/s41559-019-0941-z (https://doi.org/10.1038%2Fs41559-019-0941-z). PMC 6669044 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6669044). PMID 31285577 (https://pubmed.ncbi.nlm.nih.gov/31285577).
144. Sasso, Cristiano Dal; Maganuco, Simone; Cau, Andrea (2018-12-19). "The oldest ceratosaurian (Dinosauria: Theropoda), from the Lower Jurassic of Italy, sheds light on the evolution of the three-fingered hand of birds" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6304160). *PeerJ*. **6** e5976. Bibcode:2018PeerJ...6e5976D (https://ui.adsabs.harvard.edu/abs/2018PeerJ...6e5976D). doi:10.7717/peerj.5976 (https://doi.org/10.7717%2Fpeerj.5976). PMC 6304160 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6304160). PMID 30588396 (https://pubmed.ncbi.nlm.nih.gov/30588396).
145. Wang, Shuo; Stiegler, Josef; Amiot, Romain; Wang, Xu; Du, Guo-hao; Clark, James M.; Xu, Xing (January 2017). "Extreme Ontogenetic Changes in a Ceratosaurian Theropod" (https://doi.org/10.1016%2Fj.cub.2016.10.043). *Current Biology*. **27** (1): 144–148. Bibcode:2017CBio...27..144W (https://ui.adsabs.harvard.edu/abs/2017CBio...27..144W). doi:10.1016/j.cub.2016.10.043 (https://doi.org/10.1016%2Fj.cub.2016.10.043). PMID 28017609 (https://pubmed.ncbi.nlm.nih.gov/28017609).
146. Zanno, Lindsay E.; Makovicky, Peter J. (2011-01-04). "Herbivorous ecomorphology and specialization patterns in theropod dinosaur evolution" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3017133). *Proceedings of the National Academy of Sciences*. **108** (1): 232–237. Bibcode:2011PNAS..108..232Z (https://ui.adsabs.harvard.edu/abs/2011PNAS..108..232Z). doi:10.1073/pnas.1011924108 (https://doi.org/10.1073%2Fpnas.1011924108). PMC 3017133 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3017133). PMID 21173263 (https://pubmed.ncbi.nlm.nih.gov/21173263).
147. Rauhut, Oliver W. M.; Pol, Diego (2019-12-11). "Probable basal allosauroid from the early Middle Jurassic Cañadón Asfalto Formation of Argentina highlights phylogenetic uncertainty in tetanuran theropod dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6906444). *Scientific Reports*. **9** (1): 18826. Bibcode:2019NatSR...918826R (https://ui.adsabs.harvard.edu/abs/2019NatSR...918826R). doi:10.1038/s41598-019-53672-7 (https://doi.org/10.1038%2Fs41598-019-53672-7). PMC 6906444 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6906444). PMID 31827108 (https://pubmed.ncbi.nlm.nih.gov/31827108).
148. Benson, R.B.J (2010). "A description of Megalosaurus bucklandii (Dinosauria: Theropoda) from the Bathonian of the UK and the relationships of Middle Jurassic theropods" (https://doi.org/10.1111%2Fj.1096-3642.2009.00569.x). *Zoological Journal of the Linnean Society*. **158** (4): 882–935. doi:10.1111/j.1096-3642.2009.00569.x (https://doi.org/10.1111%2Fj.1096-3642.2009.00569.x).
149. Rauhut, Oliver W. M.; Milner, Angela C.; Moore-Fay, Scott (2010). "Cranial osteology and phylogenetic position of the theropod dinosaur *Proceratosaurus* bradleyi(Woodward, 1910) from the Middle Jurassic of England" (https://doi.org/10.1111%2Fj.1096-3642.2009.00591.x). *Zoological Journal of the Linnean Society*. **158**: 155–195. doi:10.1111/j.1096-3642.2009.00591.x (https://doi.org/10.1111%2Fj.1096-3642.2009.00591.x).
150. Qin, Zichuan; Clark, James; Choiniere, Jonah; Xu, Xing (13 August 2019). "A new alvarezsaurian theropod from the Upper Jurassic Shishugou Formation of western China" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6692367). *Scientific Reports*. **9** (1) 11727. Bibcode:2019NatSR...911727Q (https://ui.adsabs.harvard.edu/abs/2019NatSR...911727Q). doi:10.1038/s41598-019-48148-7 (https://doi.org/10.1038%2Fs41598-019-48148-7). PMC 6692367 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6692367). PMID 31409823 (https://pubmed.ncbi.nlm.nih.gov/31409823).

151. Agnolín, Federico L.; Lu, Jun-Chang; Kundrát, Martin; Xu, Li (2021-06-02). "Alvarezsaurid osteology: new data on cranial anatomy". *Historical Biology*. **34** (3): 443–452. doi:10.1080/08912963.2021.1929203 (https://doi.org/10.1080%2F08912963.2021.1929203).
152. Wang, Min; O'Connor, Jingmai K.; Xu, Xing; Zhou, Zhonghe (May 2019). "A new Jurassic scansoriopterygid and the loss of membranous wings in theropod dinosaurs" (http://publication.plazi.org/id/A235FFC976436969FFAEFFCAFF8FDF1B). *Nature*. **569** (7755): 256–259. Bibcode:2019Natur.569..256W (https://ui.adsabs.harvard.edu/abs/2019Natur.569..256W). doi:10.1038/s41586-019-1137-z (https://doi.org/10.1038%2Fs41586-019-1137-z). PMID 31068719 (https://pubmed.ncbi.nlm.nih.gov/31068719).
153. Hartman, Scott; Mortimer, Mickey; Wahl, William R.; Lomax, Dean R.; Lippincott, Jessica; Lovelace, David M. (2019-07-10). "A new paravian dinosaur from the Late Jurassic of North America supports a late acquisition of avian flight" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6626525). *PeerJ*. **7** e7247. Bibcode:2019PeerJ...7e7247H (https://ui.adsabs.harvard.edu/abs/2019PeerJ...7e7247H). doi:10.7717/peerj.7247 (https://doi.org/10.7717%2Fpeerj.7247). PMC 6626525 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6626525). PMID 31333906 (https://pubmed.ncbi.nlm.nih.gov/31333906).
154. Rauhut, Oliver W. M.; Foth, Christian (2020), "The Origin of Birds: Current Consensus, Controversy, and the Occurrence of Feathers", in Foth, Christian; Rauhut, Oliver W. M. (eds.), *The Evolution of Feathers: From Their Origin to the Present*, Fascinating Life Sciences, Cham: Springer International Publishing, pp. 27–45, doi:10.1007/978-3-030-27223-4_3 (https://doi.org/10.1007%2F978-3-030-27223-4_3), ISBN 978-3-030-27223-4
155. Norman, David B (2021-01-01). "Scelidosaurus harrisonii (Dinosauria: Ornithischia) from the Early Jurassic of Dorset, England: biology and phylogenetic relationships". *Zoological Journal of the Linnean Society*. **191** (1): 1–86. doi:10.1093/zoolinnean/zlaa061 (https://doi.org/10.1093%2Fzoolinnean%2Fzlaa061).
156. Godefroit, Pascal; Sinitsa, Sofia M.; Cincotta, Aude; McNamara, Maria E.; Reshetova, Svetlana A.; Dhouailly, Danielle (2020), "Integumentary Structures in Kulindadromeus zabaikalicus, a Basal Neornithischian Dinosaur from the Jurassic of Siberia", in Foth, Christian; Rauhut, Oliver W. M. (eds.), *The Evolution of Feathers: From Their Origin to the Present*, Fascinating Life Sciences, Cham: Springer International Publishing, pp. 47–65, doi:10.1007/978-3-030-27223-4_4 (https://doi.org/10.1007%2F978-3-030-27223-4_4), ISBN 978-3-030-27223-4
157. McDonald, Andrew T. (2012-05-22). Farke, Andrew A. (ed.). "Phylogeny of Basal Iguanodonts (Dinosauria: Ornithischia): An Update" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3358318). *PLOS ONE*. **7** (5) e36745. Bibcode:2012PLoSO...736745M (https://ui.adsabs.harvard.edu/abs/2012PLoSO...736745M). doi:10.1371/journal.pone.0036745 (https://doi.org/10.1371%2Fjournal.pone.0036745). PMC 3358318 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3358318). PMID 22629328 (https://pubmed.ncbi.nlm.nih.gov/22629328).
158. Han, Fenglu; Forster, Catherine A.; Clark, James M.; Xu, Xing (2015-12-09). "A New Taxon of Basal Ceratopsian from China and the Early Evolution of Ceratopsia" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4674058). *PLOS ONE*. **10** (12) e0143369. Bibcode:2015PLoSO..1043369H (https://ui.adsabs.harvard.edu/abs/2015PLoSO..1043369H). doi:10.1371/journal.pone.0143369 (https://doi.org/10.1371%2Fjournal.pone.0143369). PMC 4674058 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4674058). PMID 26649770 (https://pubmed.ncbi.nlm.nih.gov/26649770).

159. Pol, D.; Ramezani, J.; Gomez, K.; Carballido, J. L.; Carabajal, A. Paulina; Rauhut, O. W. M.; Escapa, I. H.; Cúneo, N. R. (2020-11-25). "Extinction of herbivorous dinosaurs linked to Early Jurassic global warming event" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7739499). *Proceedings of the Royal Society B: Biological Sciences*. **287** (1939) 20202310. Bibcode:2020PBioS.28702310P (https://ui.adsabs.harvard.edu/abs/2020PBioS.28702310P). doi:10.1098/rspb.2020.2310 (https://doi.org/10.1098%2Frspb.2020.2310). PMC 7739499 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7739499). PMID 33203331 (https://pubmed.ncbi.nlm.nih.gov/33203331).
160. Sander, P. Martin; Christian, Andreas; Clauss, Marcus; Fechner, Regina; Gee, Carole T.; Griebeler, Eva-Maria; Gunga, Hanns-Christian; Hummel, Jürgen; Mallison, Heinrich; Perry, Steven F.; Preuschoft, Holger (February 2011). "Biology of the sauropod dinosaurs: the evolution of gigantism" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3045712). *Biological Reviews*. **86** (1): 117–155. Bibcode:2011BioRv..86..117S (https://ui.adsabs.harvard.edu/abs/2011BioRv..86..117S). doi:10.1111/j.1469-185X.2010.00137.x (https://doi.org/10.1111%2Fj.1469-185X.2010.00137.x). PMC 3045712 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3045712). PMID 21251189 (https://pubmed.ncbi.nlm.nih.gov/21251189).
161. McPhee, Blair W.; Benson, Roger B.J.; Botha-Brink, Jennifer; Bordy, Emese M.; Choiniere, Jonah N. (8 October 2018). "A Giant Dinosaur from the Earliest Jurassic of South Africa and the Transition to Quadrupedality in Early Sauropodomorphs" (https://doi.org/10.1016%2Fj.cub.2018.07.063). *Current Biology*. **28** (19): 3143–3151.e7. Bibcode:2018CBio...28E3143M (https://ui.adsabs.harvard.edu/abs/2018CBio...28E3143M). doi:10.1016/j.cub.2018.07.063 (https://doi.org/10.1016%2Fj.cub.2018.07.063). PMID 30270189 (https://pubmed.ncbi.nlm.nih.gov/30270189).
162. Viglietti, Pia A.; Barrett, Paul M.; Broderick, Tim J.; Munyikwa, Darlington; MacNiven, Rowan; Broderick, Lucy; Chapelle, Kimberley; Glynn, Dave; Edwards, Steve; Zondo, Michel; Broderick, Patricia (January 2018). "Stratigraphy of the Vulcanodon type locality and its implications for regional correlations within the Karoo Supergroup". *Journal of African Earth Sciences*. **137**: 149–156. Bibcode:2018JAfES.137..149V (https://ui.adsabs.harvard.edu/abs/2018JAfES.137..149V). doi:10.1016/j.jafrearsci.2017.10.015 (https://doi.org/10.1016%2Fj.jafrearsci.2017.10.015).
163. Royo-Torres, Rafael; Cobos, Alberto; Mocho, Pedro; Alcalá, Luis (2021-01-01). "Origin and evolution of turiasaur dinosaurs set by means of a new 'rosetta' specimen from Spain" (https://doi.org/10.1093%2Fzoolinnean%2Fzlaa091). *Zoological Journal of the Linnean Society*. **191** (1): 201–227. doi:10.1093/zoolinnean/zlaa091 (https://doi.org/10.1093%2Fzoolinnean%2Fzlaa091).
164. Ren, Xin-Xin; Huang, Jian-Dong; You, Hai-Lu (2020-05-27). "The second mamenchisaurid dinosaur from the Middle Jurassic of Eastern China". *Historical Biology*. **32** (5): 602–610. Bibcode:2020HBio...32..602R (https://ui.adsabs.harvard.edu/abs/2020HBio...32..602R). doi:10.1080/08912963.2018.1515935 (https://doi.org/10.1080%2F08912963.2018.1515935).
165. Ren, Xin-Xin; Jiang, Shan; Wang, Xu-Ri; Peng, Guang-Zhao; Ye, Yong; Jia, Lei; You, Hai-Lu (2022-11-14). "Re-examination of Dashanpusaurus dongi (Sauropoda: Macronaria) supports an early Middle Jurassic global distribution of neosauropod dinosaurs". *Palaeogeography, Palaeoclimatology, Palaeoecology*. **610** 111318. doi:10.1016/j.palaeo.2022.111318 (https://doi.org/10.1016%2Fj.palaeo.2022.111318).
166. Allen, Bethany J.; Stubbs, Thomas L.; Benton, Michael J.; Puttick, Mark N. (March 2019). Mannion, Philip (ed.). "Archosauromorph extinction selectivity during the Triassic-Jurassic mass extinction" (https://doi.org/10.1111%2Fpala.12399). *Palaeontology*. **62** (2): 211–224. Bibcode:2019Palgy..62..211A (https://ui.adsabs.harvard.edu/abs/2019Palgy..62..211A). doi:10.1111/pala.12399 (https://doi.org/10.1111%2Fpala.12399). hdl:1983/e3fc2e40-c849-42ed-99fe-ea17fc26b2ec (https://hdl.handle.net/1983%2Fe3fc2e40-c849-42ed-99fe-ea17fc26b2ec).

167. Toljagić, Olja; Butler, Richard J. (2013-06-23). "Triassic–Jurassic mass extinction as trigger for the Mesozoic radiation of crocodylomorphs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3645043). *Biology Letters*. **9** (3) 20130095. Bibcode:2013BiLet...930095T (https://ui.adsabs.harvard.edu/abs/2013BiLet...930095T). doi:10.1098/rsbl.2013.0095 (https://doi.org/10.1098%2Frsbl.2013.0095). PMC 3645043 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3645043). PMID 23536443 (https://pubmed.ncbi.nlm.nih.gov/23536443).
168. Melstrom, Keegan M.; Irmis, Randall B. (July 2019). "Repeated Evolution of Herbivorous Crocodyliforms during the Age of Dinosaurs" (https://doi.org/10.1016%2Fj.cub.2019.05.076). *Current Biology*. **29** (14): 2389–2395.e3. Bibcode:2019CBio...29E2389M (https://ui.adsabs.harvard.edu/abs/2019CBio...29E2389M). doi:10.1016/j.cub.2019.05.076 (https://doi.org/10.1016%2Fj.cub.2019.05.076). PMID 31257139 (https://pubmed.ncbi.nlm.nih.gov/31257139).
169. Stubbs, Thomas L.; Pierce, Stephanie E.; Elsler, Armin; Anderson, Philip S. L.; Rayfield, Emily J.; Benton, Michael J. (2021-03-31). "Ecological opportunity and the rise and fall of crocodylomorph evolutionary innovation" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8059953). *Proceedings of the Royal Society B: Biological Sciences*. **288** (1947) 20210069. doi:10.1098/rspb.2021.0069 (https://doi.org/10.1098%2Frspb.2021.0069). PMC 8059953 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8059953). PMID 33757349 (https://pubmed.ncbi.nlm.nih.gov/33757349).
170. Spindler, Frederik; Lauer, René; Tischlinger, Helmut; Mäuser, Matthias (2021-07-05). "The integument of pelagic crocodylomorphs (Thalattosuchia: Metriorhynchidae)" (https://doi.org/10.26879%2F1099). *Palaeontologia Electronica*. **24** (2): 1–41. doi:10.26879/1099 (https://doi.org/10.26879%2F1099).
171. Irmis, Randall B.; Nesbitt, Sterling J.; Sues, Hans-Dieter (2013). "Early Crocodylomorpha". *Geological Society, London, Special Publications*. **379** (1): 275–302. Bibcode:2013GSLSP.379..275I (https://ui.adsabs.harvard.edu/abs/2013GSLSP.379..275I). doi:10.1144/SP379.24 (https://doi.org/10.1144%2FSP379.24).
172. Wilberg, Eric W.; Turner, Alan H.; Brochu, Christopher A. (2019-01-24). "Evolutionary structure and timing of major habitat shifts in Crocodylomorpha" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6346023). *Scientific Reports*. **9** (1): 514. Bibcode:2019NatSR...9..514W (https://ui.adsabs.harvard.edu/abs/2019NatSR...9..514W). doi:10.1038/s41598-018-36795-1 (https://doi.org/10.1038%2Fs41598-018-36795-1). PMC 6346023 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6346023). PMID 30679529 (https://pubmed.ncbi.nlm.nih.gov/30679529).
173. Dal Sasso, C.; Pasini, G.; Fleury, G.; Maganuco, S. (2017). "*Razanandrongobe sakalavae*, a gigantic mesoeucrocodylian from the Middle Jurassic of Madagascar, is the oldest known notosuchian" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5499610). *PeerJ*. **5** e3481. doi:10.7717/peerj.3481 (https://doi.org/10.7717%2Fpeerj.3481). PMC 5499610 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5499610). PMID 28690926 (https://pubmed.ncbi.nlm.nih.gov/28690926).
174. Joyce, Walter G. (April 2017). "A Review of the Fossil Record of Basal Mesozoic Turtles". *Bulletin of the Peabody Museum of Natural History*. **58** (1): 65–113. Bibcode:2017BPMNH..58...65J (https://ui.adsabs.harvard.edu/abs/2017BPMNH..58...65J). doi:10.3374/014.058.0105 (https://doi.org/10.3374%2F014.058.0105).
175. Sterli, Juliana; de la Fuente, Marcelo S.; Rougier, Guillermo W. (4 July 2018). "New remains of Condorchelys antiqua (Testudinata) from the Early-Middle Jurassic of Patagonia: anatomy, phylogeny, and paedomorphosis in the early evolution of turtles". *Journal of Vertebrate Paleontology*. **38** (4): (1)-(17). Bibcode:2018JVPal..38....1S (https://ui.adsabs.harvard.edu/abs/2018JVPal..38....1S). doi:10.1080/02724634.2018.1480112 (https://doi.org/10.1080%2F02724634.2018.1480112). hdl:11336/99525 (https://hdl.handle.net/11336%2F99525).

176. Sullivan, Patrick M.; Joyce, Walter G. (August 2017). "The shell and pelvic anatomy of the Late Jurassic turtle Platychelys oberndorferi based on material from Solothurn, Switzerland". *Swiss Journal of Palaeontology*. **136** (2): 323–343. Bibcode:2017SwJP..136..323S (https://ui.adsabs.harvard.edu/abs/2017SwJP..136..323S). doi:10.1007/s13358-017-0136-7 (https://doi.org/10.1007%2Fs13358-017-0136-7).
177. Evers, Serjoscha W.; Benson, Roger B. J. (January 2019). Smith, Andrew (ed.). "A new phylogenetic hypothesis of turtles with implications for the timing and number of evolutionary transitions to marine lifestyles in the group". *Palaeontology*. **62** (1): 93–134. Bibcode:2019Palgy..62...93E (https://ui.adsabs.harvard.edu/abs/2019Palgy..62...93E). doi:10.1111/pala.12384 (https://doi.org/10.1111%2Fpala.12384).
178. Anquetin, Jérémy; Püntener, Christian; Joyce, Walter G. (October 2017). "A Review of the Fossil Record of Turtles of the Clade Thalassochelydia". *Bulletin of the Peabody Museum of Natural History*. **58** (2): 317–369. Bibcode:2017BPMNH..58..317A (https://ui.adsabs.harvard.edu/abs/2017BPMNH..58..317A). doi:10.3374/014.058.0205 (https://doi.org/10.3374%2F014.058.0205).
179. Evans, Susan E.; Jones, Marc E.H. (2010). "The Origin, Early History and Diversification of Lepidosauromorph Reptiles". *New Aspects of Mesozoic Biodiversity*. Lecture Notes in Earth Sciences. Vol. 132. pp. 27–44. doi:10.1007/978-3-642-10311-7_2 (https://doi.org/10.1007%2F978-3-642-10311-7_2). ISBN 978-3-642-10310-0.
180. Brownstein, Chase D.; Meyer, Dalton L.; Fabbri, Matteo; Bhullar, Bhart-Anjan S.; Gauthier, Jacques A. (2022-11-29). "Evolutionary origins of the prolonged extant squamate radiation" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9708687). *Nature Communications*. **13** (1): 7087. Bibcode:2022NatCo..13.7087B (https://ui.adsabs.harvard.edu/abs/2022NatCo..13.7087B). doi:10.1038/s41467-022-34217-5 (https://doi.org/10.1038%2Fs41467-022-34217-5). PMC 9708687 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9708687). PMID 36446761 (https://pubmed.ncbi.nlm.nih.gov/36446761).
181. Herrera-Flores, Jorge A.; Stubbs, Thomas L.; Benton, Michael J. (2017). "Macroevolutionary patterns in Rhynchocephalia: is the tuatara (Sphenodon punctatus) a living fossil?" (https://doi.org/10.1111%2Fpala.12284). *Palaeontology*. **60** (3): 319–328. Bibcode:2017Palgy..60..319H (https://ui.adsabs.harvard.edu/abs/2017Palgy..60..319H). doi:10.1111/pala.12284 (https://doi.org/10.1111%2Fpala.12284).
182. Burbrink, Frank T; Grazziotin, Felipe G; Pyron, R Alexander; Cundall, David; Donnellan, Steve; Irish, Frances; Keogh, J Scott; Kraus, Fred; Murphy, Robert W; Noonan, Brice; Raxworthy, Christopher J (2020-05-01). Thomson, Robert (ed.). "Interrogating Genomic-Scale Data for Squamata (Lizards, Snakes, and Amphisbaenians) Shows no Support for Key Traditional Morphological Relationships". *Systematic Biology*. **69** (3): 502–520. doi:10.1093/sysbio/syz062 (https://doi.org/10.1093%2Fsysbio%2Fsyz062). PMID 31550008 (https://pubmed.ncbi.nlm.nih.gov/31550008).
183. Cleary, Terri J.; Benson, Roger B. J.; Evans, Susan E.; Barrett, Paul M. (21 March 2018). "Lepidosaurian diversity in the Mesozoic–Palaeogene: the potential roles of sampling biases and environmental drivers" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5882712). *Royal Society Open Science*. **5** (3) 171830. Bibcode:2018RSOS....571830C (https://ui.adsabs.harvard.edu/abs/2018RSOS....571830C). doi:10.1098/rsos.171830 (https://doi.org/10.1098%2Frsos.171830). PMC 5882712 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5882712). PMID 29657788 (https://pubmed.ncbi.nlm.nih.gov/29657788).
184. Evans, S. E. (1998). "Crown group lizards (Reptilia, Squamata) from the Middle Jurassic of the British Isles". *Palaeontographica, Abteilung A*. **250** (4–6): 123–154. Bibcode:1998PalAA.250..123E (https://ui.adsabs.harvard.edu/abs/1998PalAA.250..123E). doi:10.1127/pala/250/1998/123 (https://doi.org/10.1127%2Fpala%2F250%2F1998%2F123).

185. Dong, Liping; Wang, Yuan; Mou, Lijie; Zhang, Guoze; Evans, Susan E. (2019-09-13). "A new Jurassic lizard from China" (https://doi.org/10.5252%2Fgeodiversitas2019v41a16). *Geodiversitas*. **41** (16): 623. Bibcode:2019Geodv..41..623D (https://ui.adsabs.harvard.edu/abs/2019Geodv..41..623D). doi:10.5252/geodiversitas2019v41a16 (https://doi.org/10.5252%2Fgeodiversitas2019v41a16).
186. Simões, Tiago R.; Caldwell, Michael W.; Nydam, Randall L.; Jiménez-Huidobro, Paulina (September 2016). "Osteology, phylogeny, and functional morphology of two Jurassic lizard species and the early evolution of scansoriality in geckoes". *Zoological Journal of the Linnean Society*. doi:10.1111/zoj.12487 (https://doi.org/10.1111%2Fzoj.12487).
187. Daza, J. D.; Bauer, A. M.; Stanley, E. L.; Bolet, A.; Dickson, B.; Losos, J. B. (November 2018). "An Enigmatic Miniaturized and Attenuate Whole Lizard from the Mid-Cretaceous Amber of Myanmar". *Breviora*. **563** (1): 1. doi:10.3099/MCZ49.1 (https://doi.org/10.3099%2FMCZ49.1). hdl:1983/0955fcf4-a32a-4498-b920-1421dcea67de (https://hdl.handle.net/1983%2F0955fcf4-a32a-4498-b920-1421dcea67de).
188. Griffiths, Elizabeth F.; Ford, David P.; Benson, Roger B.J.; Evans, Susan E. (November 2021). "New information on the Jurassic lepidosauromorph Marmoretta oxoniensis". *Papers in Palaeontology*. **7** (4): 2255–2278. Bibcode:2021PPal....7.2255G (https://ui.adsabs.harvard.edu/abs/2021PPal....7.2255G). doi:10.1002/spp2.1400 (https://doi.org/10.1002%2Fspp2.1400).
189. Matsumoto, R.; Evans, S. E. (2010). "Choristoderes and the freshwater assemblages of Laurasia" (https://doi.org/10.5209%2Frev_JIGE.2010.v36.n2.11). *Journal of Iberian Geology*. **36** (2): 253–274. Bibcode:2010JIbG...36..253M (https://ui.adsabs.harvard.edu/abs/2010JIbG...36..253M). doi:10.5209/rev_JIGE.2010.v36.n2.11 (https://doi.org/10.5209%2Frev_JIGE.2010.v36.n2.11).
190. Matsumoto, Ryoko; Dong, Liping; Wang, Yuan; Evans, Susan E. (2019-06-18). "The first record of a nearly complete choristodere (Reptilia: Diapsida) from the Upper Jurassic of Hebei Province, People's Republic of China" (https://discovery.ucl.ac.uk/id/eprint/10067787/). *Journal of Systematic Palaeontology*. **17** (12): 1031–1048. Bibcode:2019JSPal..17.1031M (https://ui.adsabs.harvard.edu/abs/2019JSPal..17.1031M). doi:10.1080/14772019.2018.1494220 (https://doi.org/10.1080%2F14772019.2018.1494220).
191. Thorne, P. M.; Ruta, M.; Benton, M. J. (2011-05-17). "Resetting the evolution of marine reptiles at the Triassic-Jurassic boundary" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3100925). *Proceedings of the National Academy of Sciences*. **108** (20): 8339–8344. Bibcode:2011PNAS..108.8339T (https://ui.adsabs.harvard.edu/abs/2011PNAS..108.8339T). doi:10.1073/pnas.1018959108 (https://doi.org/10.1073%2Fpnas.1018959108). PMC 3100925 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3100925). PMID 21536898 (https://pubmed.ncbi.nlm.nih.gov/21536898).
192. Moon, Benjamin C.; Stubbs, Thomas L. (2020-02-13). "Early high rates and disparity in the evolution of ichthyosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7018711). *Communications Biology*. **3** (1): 68. Bibcode:2020CmBio...3...68M (https://ui.adsabs.harvard.edu/abs/2020CmBio...3...68M). doi:10.1038/s42003-020-0779-6 (https://doi.org/10.1038%2Fs42003-020-0779-6). PMC 7018711 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7018711). PMID 32054967 (https://pubmed.ncbi.nlm.nih.gov/32054967).
193. Fischer, Valentin; Weis, Robert; Thuy, Ben (2021-02-22). "Refining the marine reptile turnover at the Early–Middle Jurassic transition" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7906043). *PeerJ*. **9** e10647. doi:10.7717/peerj.10647 (https://doi.org/10.7717%2Fpeerj.10647). PMC 7906043 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7906043). PMID 33665003 (https://pubmed.ncbi.nlm.nih.gov/33665003).

194. Wintrich, Tanja; Hayashi, Shoji; Houssaye, Alexandra; Nakajima, Yasuhisa; Sander, P. Martin (2017-12-01). "A Triassic plesiosaurian skeleton and bone histology inform on evolution of a unique body plan" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5729018). *Science Advances*. **3** (12) e1701144. Bibcode:2017SciA....3E1144W (https://ui.adsabs.harvard.edu/abs/2017SciA....3E1144W). doi:10.1126/sciadv.1701144 (https://doi.org/10.1126%2Fsciadv.1701144). PMC 5729018 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5729018). PMID 29242826 (https://pubmed.ncbi.nlm.nih.gov/29242826).
195. Benson, Roger B. J.; Evans, Mark; Druckenmiller, Patrick S. (2012-03-16). "High Diversity, Low Disparity and Small Body Size in Plesiosaurs (Reptilia, Sauropterygia) from the Triassic–Jurassic Boundary" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3306369). *PLOS ONE*. **7** (3) e31838. Bibcode:2012PLoSO...731838B (https://ui.adsabs.harvard.edu/abs/2012PLoSO...731838B). doi:10.1371/journal.pone.0031838 (https://doi.org/10.1371%2Fjournal.pone.0031838). PMC 3306369 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3306369). PMID 22438869 (https://pubmed.ncbi.nlm.nih.gov/22438869).
196. O'Keefe, F. Robin (2002). "The evolution of plesiosaur and pliosaur morphotypes in the Plesiosauria (Reptilia: Sauropterygia)". *Paleobiology*. **28** (1): 101–112. Bibcode:2002Pbio...28..101O (https://ui.adsabs.harvard.edu/abs/2002Pbio...28..101O). doi:10.1666/0094-8373(2002)028<0101:TEOPAP>2.0.CO;2 (https://doi.org/10.1666%2F0094-8373%282002%29028%3C0101%3ATEOPAP%3E2.0.CO%3B2).
197. Benson, Roger B. J.; Evans, Mark; Smith, Adam S.; Sassoon, Judyth; Moore-Faye, Scott; Ketchum, Hilary F.; Forrest, Richard (2013-05-31). "A Giant Pliosaurid Skull from the Late Jurassic of England" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3669260). *PLOS ONE*. **8** (5) e65989. Bibcode:2013PLoSO...865989B (https://ui.adsabs.harvard.edu/abs/2013PLoSO...865989B). doi:10.1371/journal.pone.0065989 (https://doi.org/10.1371%2Fjournal.pone.0065989). PMC 3669260 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3669260). PMID 23741520 (https://pubmed.ncbi.nlm.nih.gov/23741520).
198. Gao, Ting; Li, Da-Qing; Li, Long-Feng; Yang, Jing-Tao (2019-08-13). "The first record of freshwater plesiosaurian from the Middle Jurassic of Gansu, NW China, with its implications to the local palaeobiogeography" (https://doi.org/10.1186%2Fs42501-019-0043-5). *Journal of Palaeogeography*. **8** (1): 27. Bibcode:2019JPalG...8...27G (https://ui.adsabs.harvard.edu/abs/2019JPalG...8...27G). doi:10.1186/s42501-019-0043-5 (https://doi.org/10.1186%2Fs42501-019-0043-5).
199. Kear, Benjamin P. (2 August 2012). "A revision of Australia's Jurassic plesiosaurs" (https://doi.org/10.1111%2Fj.1475-4983.2012.01183.x). *Palaeontology*. **55** (5): 1125–1138. Bibcode:2012Palgy..55.1125K (https://ui.adsabs.harvard.edu/abs/2012Palgy..55.1125K). doi:10.1111/j.1475-4983.2012.01183.x (https://doi.org/10.1111%2Fj.1475-4983.2012.01183.x).
200. O'Sullivan, Michael; Martill, David M. (2017-11-17). "The taxonomy and systematics of Parapsicephalus purdoni (Reptilia: Pterosauria) from the Lower Jurassic Whitby Mudstone Formation, Whitby, U.K". *Historical Biology*. **29** (8): 1009–1018. Bibcode:2017HBio...29.1009O (https://ui.adsabs.harvard.edu/abs/2017HBio...29.1009O). doi:10.1080/08912963.2017.1281919 (https://doi.org/10.1080%2F08912963.2017.1281919).
201. Bestwick, Jordan; Unwin, David M.; Butler, Richard J.; Henderson, Donald M.; Purnell, Mark A. (November 2018). "Pterosaur dietary hypotheses: a review of ideas and approaches: Pterosaur dietary hypotheses" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6849529). *Biological Reviews*. **93** (4): 2021–2048. doi:10.1111/brv.12431 (https://doi.org/10.1111%2Fbrv.12431). PMC 6849529 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6849529). PMID 29877021 (https://pubmed.ncbi.nlm.nih.gov/29877021).
202. Lucas, Spencer G. (2018). "Late Triassic Terrestrial Tetrapods: Biostratigraphy, Biochronology and Biotic Events". *The Late Triassic World*. Topics in Geobiology. Vol. 46. pp. 351–405. doi:10.1007/978-3-319-68009-5_10 (https://doi.org/10.1007%2F978-3-319-68009-5_10). ISBN 978-3-319-68008-8.

203. Averianov, Alexander O.; Martin, Thomas; Skutschas, Pavel P.; Rezvyi, Anton S.; Bakirov, Aizek A. (March 2008). "Amphibians from the Middle Jurassic Balabansai Svita in the Fergana Depression, Kyrgyzstan (Central Asia)" (https://doi.org/10.1111%2Fj.1475-4983.2007.00748.x). *Palaeontology*. **51** (2): 471–485. Bibcode:2008Palgy..51..471A (https://ui.adsabs.harvard.edu/abs/2008Palgy..51..471A). doi:10.1111/j.1475-4983.2007.00748.x (https://doi.org/10.1111%2Fj.1475-4983.2007.00748.x).
204. Warren, A. A.; Hutchinson, M. N.; Hill, Dorothy (1983-09-13). "The last Labyrinthodont? A new brachyopoid (Amphibia, Temnospondyli) from the early Jurassic Evergreen formation of Queensland, Australia". *Philosophical Transactions of the Royal Society of London. B, Biological Sciences*. **303** (1113): 1–62. Bibcode:1983RSPTB.303....1W (https://ui.adsabs.harvard.edu/abs/1983RSPTB.303....1W). doi:10.1098/rstb.1983.0080 (https://doi.org/10.1098%2Frstb.1983.0080).
205. Reilly, Stephen M.; Jorgensen, Michael E. (February 2011). "The evolution of jumping in frogs: Morphological evidence for the basal anuran locomotor condition and the radiation of locomotor systems in crown group anurans". *Journal of Morphology*. **272** (2): 149–168. Bibcode:2011JMorp.272..149R (https://ui.adsabs.harvard.edu/abs/2011JMorp.272..149R). doi:10.1002/jmor.10902 (https://doi.org/10.1002%2Fjmor.10902). PMID 21210487 (https://pubmed.ncbi.nlm.nih.gov/21210487).
206. Báez, Ana Maria; Nicoli, Laura (March 2008). "A new species of Notobatrachus (Amphibia, Salientia) from the Middle Jurassic of northwestern Patagonia". *Journal of Paleontology*. **82** (2): 372–376. Bibcode:2008JPal...82..372B (https://ui.adsabs.harvard.edu/abs/2008JPal...82..372B). doi:10.1666/06-117.1 (https://doi.org/10.1666%2F06-117.1). hdl:11336/135748 (https://hdl.handle.net/11336%2F135748).
207. Marjanović, David; Laurin, Michel (2014-07-04). "An updated paleontological timetree of lissamphibians, with comments on the anatomy of Jurassic crown-group salamanders (Urodela)". *Historical Biology*. **26** (4): 535–550. Bibcode:2014HBio...26..535M (https://ui.adsabs.harvard.edu/abs/2014HBio...26..535M). doi:10.1080/08912963.2013.797972 (https://doi.org/10.1080%2F08912963.2013.797972).
208. Schoch, Rainer R.; Werneburg, Ralf; Voigt, Sebastian (2020-05-26). "A Triassic stem-salamander from Kyrgyzstan and the origin of salamanders" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7261083). *Proceedings of the National Academy of Sciences*. **117** (21): 11584–11588. Bibcode:2020PNAS..11711584S (https://ui.adsabs.harvard.edu/abs/2020PNAS..11711584S). doi:10.1073/pnas.2001424117 (https://doi.org/10.1073%2Fpnas.2001424117). PMC 7261083 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7261083). PMID 32393623 (https://pubmed.ncbi.nlm.nih.gov/32393623).
209. Skutschas, Pavel; Stein, Koen (April 2015). "Long bone histology of the stem salamander Kokartus honorarius (Amphibia: Caudata) from the Middle Jurassic of Kyrgyzstan" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4386933). *Journal of Anatomy*. **226** (4): 334–347. doi:10.1111/joa.12281 (https://doi.org/10.1111%2Fjoa.12281). PMC 4386933 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4386933). PMID 25682890 (https://pubmed.ncbi.nlm.nih.gov/25682890).
210. Jia, Jia; Gao, Ke-Qin (2019-03-04). "A new stem hynobiid salamander (Urodela, Cryptobranchoidea) from the Upper Jurassic (Oxfordian) of Liaoning Province, China". *Journal of Vertebrate Paleontology*. **39** (2) e1588285. Bibcode:2019JVPal..39E8285J (https://ui.adsabs.harvard.edu/abs/2019JVPal..39E8285J). doi:10.1080/02724634.2019.1588285 (https://doi.org/10.1080%2F02724634.2019.1588285).
211. Gao, K.-Q.; Shubin, N. H. (2012-04-10). "Late Jurassic salamandroid from western Liaoning, China" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3326464). *Proceedings of the National Academy of Sciences*. **109** (15): 5767–5772. Bibcode:2012PNAS..109.5767G (https://ui.adsabs.harvard.edu/abs/2012PNAS..109.5767G). doi:10.1073/pnas.1009828109 (https://doi.org/10.1073%2Fpnas.1009828109). PMC 3326464 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3326464). PMID 22411790 (https://pubmed.ncbi.nlm.nih.gov/22411790).

212. Jia, Jia; Gao, Ke-Qin (2016-05-04). "A New Basal Salamandroid (Amphibia, Urodela) from the Late Jurassic of Qinglong, Hebei Province, China" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4856324). *PLOS ONE*. **11** (5) e0153834. Bibcode:2016PLoSO..1153834J (https://ui.adsabs.harvard.edu/abs/2016PLoSO..1153834J). doi:10.1371/journal.pone.0153834 (https://doi.org/10.1371%2Fjournal.pone.0153834). PMC 4856324 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4856324). PMID 27144770 (https://pubmed.ncbi.nlm.nih.gov/27144770).
213. Evans, S. E.; Lally, C.; Chure, D. C.; Elder, A.; Maisano, J. A. (2005). "A Late Jurassic salamander (Amphibia: Caudata) from the Morrison Formation of North America" (https://doi.org/10.1111%2Fj.1096-3642.2005.00159.x). *Zoological Journal of the Linnean Society*. **143** (4): 599–616. doi:10.1111/j.1096-3642.2005.00159.x (https://doi.org/10.1111%2Fj.1096-3642.2005.00159.x).
214. Santos, Rodolfo Otávio; Laurin, Michel; Zaher, Hussam (2020-11-03). "A review of the fossil record of caecilians (Lissamphibia: Gymnophionomorpha) with comments on its use to calibrate molecular timetrees" (https://doi.org/10.1093%2Fbiolinnean%2Fblaa148). *Biological Journal of the Linnean Society*. **131** (4): 737–755. doi:10.1093/biolinnean/blaa148 (https://doi.org/10.1093%2Fbiolinnean%2Fblaa148).
215. Haddoumi, Hamid; Allain, Ronan; Meslouh, Said; Metais, Grégoire; Monbaron, Michel; Pons, Denise; Rage, Jean-Claude; Vullo, Romain; Zouhri, Samir; Gheerbrant, Emmanuel (January 2016). "Guelb el Ahmar (Bathonian, Anoual Syncline, eastern Morocco): First continental flora and fauna including mammals from the Middle Jurassic of Africa" (https://insu.hal.science/insu-01163723). *Gondwana Research*. **29** (1): 290–319. Bibcode:2016GondR..29..290H (https://ui.adsabs.harvard.edu/abs/2016GondR..29..290H). doi:10.1016/j.gr.2014.12.004 (https://doi.org/10.1016%2Fj.gr.2014.12.004).
216. Lee, Michael S.Y.; Beck, Robin M.D. (31 August 2015). "Mammalian Evolution: A Jurassic Spark" (https://doi.org/10.1016%2Fj.cub.2015.07.008). *Current Biology*. **25** (17): R759–R761. Bibcode:2015CBio...25.R759L (https://ui.adsabs.harvard.edu/abs/2015CBio...25.R759L). doi:10.1016/j.cub.2015.07.008 (https://doi.org/10.1016%2Fj.cub.2015.07.008). PMID 26325137 (https://pubmed.ncbi.nlm.nih.gov/26325137).
217. Ji, Qiang; Luo, Zhe-Xi; Yuan, Chong-Xi; Tabrum, Alan R. (24 February 2006). "A Swimming Mammaliaform from the Middle Jurassic and Ecomorphological Diversification of Early Mammals". *Science*. **311** (5764): 1123–1127. Bibcode:2006Sci...311.1123J (https://ui.adsabs.harvard.edu/abs/2006Sci...311.1123J). doi:10.1126/science.1123026 (https://doi.org/10.1126%2Fscience.1123026). PMID 16497926 (https://pubmed.ncbi.nlm.nih.gov/16497926).
218. Meng, Qing-Jin; Grossnickle, David M.; Liu, Di; Zhang, Yu-Guang; Neander, April I.; Ji, Qiang; Luo, Zhe-Xi (August 2017). "New gliding mammaliaforms from the Jurassic". *Nature*. **548** (7667): 291–296. Bibcode:2017Natur.548..291M (https://ui.adsabs.harvard.edu/abs/2017Natur.548..291M). doi:10.1038/nature23476 (https://doi.org/10.1038%2Fnature23476). PMID 28792929 (https://pubmed.ncbi.nlm.nih.gov/28792929).
219. Meng, Jin; Hu, Yaoming; Wang, Yuanqing; Wang, Xiaolin; Li, Chuankui (March 2007). "Erratum: A Mesozoic gliding mammal from northeastern China". *Nature*. **446** (7131): 102. doi:10.1038/nature05639 (https://doi.org/10.1038%2Fnature05639).
220. Luo, Z.-X.; Wible, J.R. (2005). "A Late Jurassic Digging Mammal and Early Mammalian Diversification". *Science*. **308** (5718): 103–107. Bibcode:2005Sci...308..103L (https://ui.adsabs.harvard.edu/abs/2005Sci...308..103L). doi:10.1126/science.1108875 (https://doi.org/10.1126%2Fscience.1108875). PMID 15802602 (https://pubmed.ncbi.nlm.nih.gov/15802602).
221. Luo, Zhe-Xi; Cifelli, Richard L.; Kielan-Jaworowska, Zofia (January 2001). "Dual origin of tribosphenic mammals". *Nature*. **409** (6816): 53–57. Bibcode:2001Natur.409...53L (https://ui.adsabs.harvard.edu/abs/2001Natur.409...53L). doi:10.1038/35051023 (https://doi.org/10.1038%2F35051023). PMID 11343108 (https://pubmed.ncbi.nlm.nih.gov/11343108).

222. Mao, Fangyuan; Brewer, Philippa; Hooker, Jerry J.; Meng, Jin (2022-12-31). "New allotherian specimens from the Middle Jurassic Woodeaton Quarry (Oxfordshire) and implications for haramiyidan diversity and phylogeny". *Journal of Systematic Palaeontology*. **20** (1): 1–37. Bibcode:2022JSPal..2097021M (https://ui.adsabs.harvard.edu/abs/2022JSPal..2097021M). doi:10.1080/14772019.2022.2097021 (https://doi.org/10.1080%2F14772019.2022.2097021).
223. Close, Roger A.; Friedman, Matt; Lloyd, Graeme T.; Benson, Roger B. J. (17 August 2015). "Evidence for a Mid-Jurassic Adaptive Radiation in Mammals". *Current Biology*. **25** (16): 2137–2142. Bibcode:2015CBio...25.2137C (https://ui.adsabs.harvard.edu/abs/2015CBio...25.2137C). doi:10.1016/j.cub.2015.06.047 (https://doi.org/10.1016%2Fj.cub.2015.06.047). PMID 26190074 (https://pubmed.ncbi.nlm.nih.gov/26190074).
224. Zhe-Xi Luo; Chong-Xi Yuan; Qing-Jin Meng; Qiang Ji (25 August 2011). "A Jurassic eutherian mammal and divergence of marsupials and placentals". *Nature*. **476** (7361): 442–445. Bibcode:2011Natur.476..442L (https://ui.adsabs.harvard.edu/abs/2011Natur.476..442L). doi:10.1038/nature10291 (https://doi.org/10.1038%2Fnature10291). PMID 21866158 (https://pubmed.ncbi.nlm.nih.gov/21866158).
225. King, Benedict; Beck, Robin M. D. (2020-06-10). "Tip dating supports novel resolutions of controversial relationships among early mammals" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7341916). *Proceedings of the Royal Society B: Biological Sciences*. **287** (1928) 20200943. doi:10.1098/rspb.2020.0943 (https://doi.org/10.1098%2Frspb.2020.0943). PMC 7341916 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7341916). PMID 32517606 (https://pubmed.ncbi.nlm.nih.gov/32517606).
226. Ruta, Marcello; Botha-Brink, Jennifer; Mitchell, Stephen A.; Benton, Michael J. (2013-10-22). "The radiation of cynodonts and the ground plan of mammalian morphological diversity" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3768321). *Proceedings of the Royal Society B: Biological Sciences*. **280** (1769) 20131865. Bibcode:2013PBioS.28031865R (https://ui.adsabs.harvard.edu/abs/2013PBioS.28031865R). doi:10.1098/rspb.2013.1865 (https://doi.org/10.1098%2Frspb.2013.1865). PMC 3768321 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3768321). PMID 23986112 (https://pubmed.ncbi.nlm.nih.gov/23986112).
227. Abdala, Fernando; Gaetano, Leandro C. (2018). "The Late Triassic Record of Cynodonts: Time of Innovations in the Mammalian Lineage". *The Late Triassic World*. Topics in Geobiology. Vol. 46. pp. 407–445. doi:10.1007/978-3-319-68009-5_11 (https://doi.org/10.1007%2F978-3-319-68009-5_11). ISBN 978-3-319-68008-8.
228. Du, Yixing; Chiari, Marco; Karádi, Viktor; Nicora, Alda; Onoue, Tetsuji; Pálfy, József; Roghi, Guido; Tomimatsu, Yuki; Rigo, Manuel (April 2020). "The asynchronous disappearance of conodonts: New constraints from Triassic-Jurassic boundary sections in the Tethys and Panthalassa". *Earth-Science Reviews*. **203** 103176. Bibcode:2020ESRv..20303176D (https://ui.adsabs.harvard.edu/abs/2020ESRv..20303176D). doi:10.1016/j.earscirev.2020.103176 (https://doi.org/10.1016%2Fj.earscirev.2020.103176). hdl:11577/3338908 (https://hdl.handle.net/11577%2F3338908).
229. Ginot, Samuel; Goudemand, Nicolas (December 2020). "Global climate changes account for the main trends of conodont diversity but not for their final demise". *Global and Planetary Change*. **195** 103325. Bibcode:2020GPC...19503325G (https://ui.adsabs.harvard.edu/abs/2020GPC...19503325G). doi:10.1016/j.gloplacha.2020.103325 (https://doi.org/10.1016%2Fj.gloplacha.2020.103325).
230. Wu, Feixiang; Janvier, Philippe; Zhang, Chi (2023-10-31). "The rise of predation in Jurassic lampreys" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10618186). *Nature Communications*. **14** (1): 6652. Bibcode:2023NatCo..14.6652W (https://ui.adsabs.harvard.edu/abs/2023NatCo..14.6652W). doi:10.1038/s41467-023-42251-0 (https://doi.org/10.1038%2Fs41467-023-42251-0). PMC 10618186 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10618186). PMID 37907522 (https://pubmed.ncbi.nlm.nih.gov/37907522).

231. Kemp, Anne; Cavin, Lionel; Guinot, Guillaume (April 2017). "Evolutionary history of lungfishes with a new phylogeny of post-Devonian genera" (https://doi.org/10.1016%2Fj.palaeo.2016.12.051). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **471**: 209–219. Bibcode:2017PPP...471..209K (https://ui.adsabs.harvard.edu/abs/2017PPP...471..209K). doi:10.1016/j.palaeo.2016.12.051 (https://doi.org/10.1016%2Fj.palaeo.2016.12.051).
232. Brownstein, Chase Doran; Harrington, Richard C; Near, Thomas J. (July 2023). "The biogeography of extant lungfishes traces the breakup of Gondwana". *Journal of Biogeography*. **50** (7): 1191–1198. Bibcode:2023JBiog..50.1191B (https://ui.adsabs.harvard.edu/abs/2023JBiog..50.1191B). doi:10.1111/jbi.14609 (https://doi.org/10.1111%2Fjbi.14609).
233. Cavin, Lionel; Cupello, Camila; Yabumoto, Yoshitaka; Léo, Fragoso; Deersi, Uthumporn; Brito, Paul M. (2019). "Phylogeny and evolutionary history of mawsoniid coelacanths" (http://www.kmnh.jp/wp-content/uploads/2019/05/A17-3-Cavin.pdf) (PDF). *Bulletin of the Kitakyushu Museum of Natural History and Human History, Series A*. **17**: 3–13.
234. Clement, Gaël (2005-09-30). "A new coelacanth (Actinistia, Sarcopterygii) from the Jurassic of France, and the question of the closest relative fossil to Latimeria". *Journal of Vertebrate Paleontology*. **25** (3): 481–491. doi:10.1671/0272-4634(2005)025[0481:ANCASF]2.0.CO;2 (https://doi.org/10.1671%2F0272-4634%282005%29025%5B0481%3AANCASF%5D2.0.CO%3B2).
235. Skrzycka, Roksana (2014-07-03). "Revision of two relic actinopterygians from the Middle or Upper Jurassic Karabastau Formation, Karatau Range, Kazakhstan". *Alcheringa: An Australasian Journal of Palaeontology*. **38** (3): 364–390. Bibcode:2014Alch...38..364S (https://ui.adsabs.harvard.edu/abs/2014Alch...38..364S). doi:10.1080/03115518.2014.880267 (https://doi.org/10.1080%2F03115518.2014.880267).
236. Hilton, Eric J.; Grande, Lance; Jin, Fan (January 2021). "Redescription of † Yanosteus longidorsalis Jin et al., (Chondrostei, Acipenseriformes, †Peipiaosteidae) from the Early Cretaceous of China". *Journal of Paleontology*. **95** (1): 170–183. Bibcode:2021JPal...95..170H (https://ui.adsabs.harvard.edu/abs/2021JPal...95..170H). doi:10.1017/jpa.2020.80 (https://doi.org/10.1017%2Fjpa.2020.80).
237. Poyato-Ariza, Francisco José; Martín-Abad, Hugo (2020-07-19). "History of two lineages: Comparative analysis of the fossil record in Amiiformes and Pycnodontiformes (Osteischtyes, Actinopterygii)" (https://doi.org/10.7203%2Fsjp.28.1.17833). *Spanish Journal of Palaeontology*. **28** (1): 79. doi:10.7203/sjp.28.1.17833 (https://doi.org/10.7203%2Fsjp.28.1.17833). hdl:10486/710030 (https://hdl.handle.net/10486%2F710030).
238. Deesri, Uthumporn; Naksri, Wilailuck; Jintasakul, Pratueng; Noda, Yoshikazu; Yukawa, Hirokazu; Hossny, Tamara El; Cavin, Lionel (2023-03-27). "A New Sinamiin Fish (Actinopterygii) from the Early Cretaceous of Thailand: Implications on the Evolutionary History of the Amiid Lineage" (https://doi.org/10.3390%2Fd15040491). *Diversity*. **15** (4): 491. Bibcode:2023Diver..15..491D (https://ui.adsabs.harvard.edu/abs/2023Diver..15..491D). doi:10.3390/d15040491 (https://doi.org/10.3390%2Fd15040491).
239. Brito, Paulo M.; Alvarado-Ortega, Jésus; Meunier, François J. (December 2017). "Earliest known lepisosteoid extends the range of anatomically modern gars to the Late Jurassic" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5736718). *Scientific Reports*. **7** (1): 17830. Bibcode:2017NatSR...717830B (https://ui.adsabs.harvard.edu/abs/2017NatSR...717830B). doi:10.1038/s41598-017-17984-w (https://doi.org/10.1038%2Fs41598-017-17984-w). PMC 5736718 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5736718). PMID 29259200 (https://pubmed.ncbi.nlm.nih.gov/29259200).
240. Arratia G. Mesozoic halecostomes and the early radiation of teleosts. In: Arratia G, Tintori A, editors. Mesozoic Fishes 3 – Systematics, Paleoenvironments and Biodiversity. München: Verlag Dr. Friedrich Pfeil; 2004. p. 279–315.

241. Tse, Tze-Kei; Pittman, Michael; Chang, Mee-mann (2015-03-26). "A specimen of Paralycoptera Chang & Chou 1977 (Teleostei: Osteoglossoidei) from Hong Kong (China) with a potential Late Jurassic age that extends the temporal and geographical range of the genus" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4380157). *PeerJ*. **3** e865. doi:10.7717/peerj.865 (https://doi.org/10.7717%2Fpeerj.865). PMC 4380157 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4380157). PMID 25834774 (https://pubmed.ncbi.nlm.nih.gov/25834774).
242. Liston, J., Newbrey, M., Challands, T., and Adams, C., 2013, "Growth, age and size of the Jurassic pachycormid *Leedsichthys problematicus* (Osteichthyes: Actinopterygii) in: Arratia, G., Schultze, H. and Wilson, M. (eds.) *Mesozoic Fishes 5 – Global Diversity and Evolution*. Verlag Dr. Friedrich Pfeil, München, Germany, pp. 145–175
243. Rees, Jan; Underwood, Charlie J. (2008-01-17). "Hybodont sharks of the English Bathonian and Callovian (Middle Jurassic)" (https://doi.org/10.1111%2Fj.1475-4983.2007.00737.x). *Palaeontology*. **51** (1): 117–147. Bibcode:2008Palgy..51..117R (https://ui.adsabs.harvard.edu/abs/2008Palgy..51..117R). doi:10.1111/j.1475-4983.2007.00737.x (https://doi.org/10.1111%2Fj.1475-4983.2007.00737.x).
244. Corso, Jacopo Dal; Bernardi, Massimo; Sun, Yadong; Song, Haijun; Seyfullah, Leyla J.; Preto, Nereo; Gianolla, Piero; Ruffell, Alastair; Kustatscher, Evelyn; Roghi, Guido; Merico, Agostino (September 2020). "Extinction and dawn of the modern world in the Carnian (Late Triassic)" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7494334). *Science Advances*. **6** (38) eaba0099. Bibcode:2020SciA....6...99D (https://ui.adsabs.harvard.edu/abs/2020SciA....6...99D). doi:10.1126/sciadv.aba0099 (https://doi.org/10.1126%2Fsciadv.aba0099). PMC 7494334 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7494334). PMID 32938682 (https://pubmed.ncbi.nlm.nih.gov/32938682).
245. Underwood, Charlie J. (March 2006). "Diversification of the Neoselachii (Chondrichthyes) during the Jurassic and Cretaceous" (https://eprints.bbk.ac.uk/id/eprint/467/1/Underwood_Paleobiology.pdf) (PDF). *Paleobiology*. **32** (2): 215–235. Bibcode:2006Pbio...32..215U (https://ui.adsabs.harvard.edu/abs/2006Pbio...32..215U). doi:10.1666/04069.1 (https://doi.org/10.1666%2F04069.1).
246. Stumpf, Sebastian; Kriwet, Jürgen (2019-12-01). "A new Pliensbachian elasmobranch (Vertebrata, Chondrichthyes) assemblage from Europe, and its contribution to the understanding of late Early Jurassic elasmobranch diversity and distributional patterns" (https://doi.org/10.1007%2Fs12542-019-00451-4). *PalZ*. **93** (4): 637–658. Bibcode:2019PalZ...93..637S (https://ui.adsabs.harvard.edu/abs/2019PalZ...93..637S). doi:10.1007/s12542-019-00451-4 (https://doi.org/10.1007%2Fs12542-019-00451-4).
247. Underwood, Charlie J.; Claeson, Kerin M. (June 2019). "The Late Jurassic ray Kimmerobatis etchesi gen. et sp. nov. and the Jurassic radiation of the Batoidea". *Proceedings of the Geologists' Association*. **130** (3–4): 345–354. Bibcode:2019PrGA..130..345U (https://ui.adsabs.harvard.edu/abs/2019PrGA..130..345U). doi:10.1016/j.pgeola.2017.06.009 (https://doi.org/10.1016%2Fj.pgeola.2017.06.009).
248. Kriwet, Jürgen; Klug, Stefanie (December 2011). "A new Jurassic cow shark (Chondrichthyes, Hexanchiformes) with comments on Jurassic hexanchiform systematics" (https://doi.org/10.1007%2Fs00015-011-0075-z). *Swiss Journal of Geosciences*. **104** (S1): 107–114. Bibcode:2011SwJG..104..107K (https://ui.adsabs.harvard.edu/abs/2011SwJG..104..107K). doi:10.1007/s00015-011-0075-z (https://doi.org/10.1007%2Fs00015-011-0075-z).
249. Srdic, Alex; Duffin, Christopher J.; Martill, David M. (August 2016). "First occurrence of the orectolobiform shark Akaimia in the Oxford Clay Formation (Jurassic, Callovian) of England" (https://researchportal.port.ac.uk/portal/en/publications/first-occurrence-of-the-orectolobiform-shark-akaimia-in-the-oxford-clay-formation-jurassic-callovian-of-england(dfc43ced-c714-4737-9a73-83277c51be7a).html). *Proceedings of the Geologists' Association*. **127** (4): 506–513. Bibcode:2016PrGA..127..506S (https://ui.adsabs.harvard.edu/abs/2016PrGA..127..506S). doi:10.1016/j.pgeola.2016.07.002 (https://doi.org/10.1016%2Fj.pgeola.2016.07.002).

250. Slater, Tiffany S.; Ashbrook, Kate; Kriwet, Jürgen (August 2020). Cavin, Lionel (ed.). "Evolutionary relationships among bullhead sharks (Chondrichthyes, Heterodontiformes)". *Papers in Palaeontology*. **6** (3): 425–437. Bibcode:2020PPal....6..425S (https://ui.adsabs.harvard.edu/abs/2020PPal....6..425S). doi:10.1002/spp2.1299 (https://doi.org/10.1002%2Fspp2.1299). hdl:10468/10339 (https://hdl.handle.net/10468%2F10339).
251. López-Romero, Faviel A.; Stumpf, Sebastian; Pfaff, Cathrin; Marramà, Giuseppe; Johanson, Zerina; Kriwet, Jürgen (2020-07-28). "Evolutionary trends of the conserved neurocranium shape in angel sharks (Squatiniformes, Elasmobranchii)" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7387474). *Scientific Reports*. **10** (1): 12582. Bibcode:2020NatSR..1012582L (https://ui.adsabs.harvard.edu/abs/2020NatSR..1012582L). doi:10.1038/s41598-020-69525-7 (https://doi.org/10.1038%2Fs41598-020-69525-7). PMC 7387474 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7387474). PMID 32724124 (https://pubmed.ncbi.nlm.nih.gov/32724124).
252. Stumpf, Sebastian; Scheer, Udo; Kriwet, Jürgen (2019-03-04). "A new genus and species of extinct ground shark, †Diprosopovenator hilperti, gen. et sp. nov. (Carcharhiniformes, †Pseudoscyliorhinidae, fam. nov.), from the Upper Cretaceous of Germany" (https://doi.org/10.1080%2F02724634.2019.1593185). *Journal of Vertebrate Paleontology*. **39** (2) e1593185. Bibcode:2019JVPal..39E3185S (https://ui.adsabs.harvard.edu/abs/2019JVPal..39E3185S). doi:10.1080/02724634.2019.1593185 (https://doi.org/10.1080%2F02724634.2019.1593185).
253. Klug, Stefanie; Tütken, Thomas; Wings, Oliver; Pfretzschner, Hans-Ulrich; Martin, Thomas (September 2010). "A Late Jurassic freshwater shark assemblage (Chondrichthyes, Hybodontiformes) from the southern Junggar Basin, Xinjiang, Northwest China". *Palaeobiodiversity and Palaeoenvironments*. **90** (3): 241–257. Bibcode:2010PdPe...90..241K (https://ui.adsabs.harvard.edu/abs/2010PdPe...90..241K). doi:10.1007/s12549-010-0032-2 (https://doi.org/10.1007%2Fs12549-010-0032-2).
254. Popov, Evgeny V.; Delsate, Dominique; Felten, Roland (2019-07-02). "A New Callorhinchid Genus (Holocephali, Chimaeroidei) from the Early Bajocian of Ottange-Rumelange, on the Luxembourg-French Border". *Paleontological Research*. **23** (3): 220. Bibcode:2019PalRe..23..220P (https://ui.adsabs.harvard.edu/abs/2019PalRe..23..220P). doi:10.2517/2018PR021 (https://doi.org/10.2517%2F2018PR021).
255. Duffin, Christopher J.; Milàn, Jesper (2017-11-14). "A new myriacanthid holocephalian from the Early Jurassic of Denmark" (https://doi.org/10.37570%2Fbgsd-2017-65-10). *Bulletin of the Geological Society of Denmark*. **65**: 161–170. Bibcode:2017BuGSD..65..161D (https://ui.adsabs.harvard.edu/abs/2017BuGSD..65..161D). doi:10.37570/bgsd-2017-65-10 (https://doi.org/10.37570%2Fbgsd-2017-65-10).
256. Labandeira, Conrad C. (2018-05-23), "The Fossil History of Insect Diversity", *Insect Biodiversity*, Chichester, UK: John Wiley & Sons, Ltd, pp. 723–788, doi:10.1002/9781118945582.ch24 (https://doi.org/10.1002%2F9781118945582.ch24), ISBN 978-1-118-94558-2
257. Nel, Patricia; Bertrand, Sylvain; Nel, André (December 2018). "Diversification of insects since the Devonian: a new approach based on morphological disparity of mouthparts" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5824790). *Scientific Reports*. **8** (1): 3516. Bibcode:2018NatSR...8.3516N (https://ui.adsabs.harvard.edu/abs/2018NatSR...8.3516N). doi:10.1038/s41598-018-21938-1 (https://doi.org/10.1038%2Fs41598-018-21938-1). PMC 5824790 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5824790). PMID 29476087 (https://pubmed.ncbi.nlm.nih.gov/29476087).

258. McKenna, Duane D.; Shin, Seunggwan; Ahrens, Dirk; Balke, Michael; Beza-Beza, Cristian; Clarke, Dave J.; Donath, Alexander; Escalona, Hermes E.; Friedrich, Frank; Letsch, Harald; Liu, Shanlin (2019-12-03). "The evolution and genomic basis of beetle diversity" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6900523). *Proceedings of the National Academy of Sciences*. **116** (49): 24729–24737. Bibcode:2019PNAS..11624729M (https://ui.adsabs.harvard.edu/abs/2019PNAS..11624729M). doi:10.1073/pnas.1909655116 (https://doi.org/10.1073%2Fpnas.1909655116). PMC 6900523 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6900523). PMID 31740605 (https://pubmed.ncbi.nlm.nih.gov/31740605).

259. Beutel, Rolf G.; Xu, Chunpeng; Jarzembowski, Edmund; Kundrata, Robin; Boudinot, Brendon E.; McKenna, Duane D.; Goczał, Jakub (13 February 2024). "The evolutionary history of Coleoptera (Insecta) in the late Palaeozoic and the Mesozoic" (https://doi.org/10.1111%2Fsyen.12623). *Systematic Entomology*. **49** (3): 355–388. Bibcode:2024SysEn..49..355B (https://ui.adsabs.harvard.edu/abs/2024SysEn..49..355B). doi:10.1111/syen.12623 (https://doi.org/10.1111%2Fsyen.12623).

260. Shin, Seunggwan; Clarke, Dave J; Lemmon, Alan R; Moriarty Lemmon, Emily; Aitken, Alexander L; Haddad, Stephanie; Farrell, Brian D; Marvaldi, Adriana E; Oberprieler, Rolf G; McKenna, Duane D (2018-04-01). "Phylogenomic Data Yield New and Robust Insights into the Phylogeny and Evolution of Weevils" (https://doi.org/10.1093%2Fmolbev%2Fmsx324). *Molecular Biology and Evolution*. **35** (4): 823–836. doi:10.1093/molbev/msx324 (https://doi.org/10.1093%2Fmolbev%2Fmsx324). hdl:11336/57287 (https://hdl.handle.net/11336%2F57287). PMID 29294021 (https://pubmed.ncbi.nlm.nih.gov/29294021).

261. Xu, Chunpeng; Fang, Yanan; Fang, Yan; Wang, He; Zhou, Qian; Jiang, Xueying; Zhang, Haichun (14 February 2024). "Early Jurassic orthopteran insects from the southern Junggar Basin, NW China, with discussion of biodiversity changes of Orthoptera across the Triassic–Jurassic boundary". *Geological Society, London, Special Publications*. **538** (1): 147–154. Bibcode:2024GSLSP.538..147X (https://ui.adsabs.harvard.edu/abs/2024GSLSP.538..147X). doi:10.1144/SP538-2021-184 (https://doi.org/10.1144%2FSP538-2021-184).

262. Woodrow, Charlie; Baker, Ed; Jonsson, Thorin; Montealegre-Z, Fernando (2022-08-10). Nityananda, Vivek (ed.). "Reviving the sound of a 150-year-old insect: The bioacoustics of Prophalangopsis obscura (Ensifera: Hagloidea)" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9365155). *PLOS ONE*. **17** (8) e0270498. Bibcode:2022PLoSO..1770498W (https://ui.adsabs.harvard.edu/abs/2022PLoSO..1770498W). doi:10.1371/journal.pone.0270498 (https://doi.org/10.1371%2Fjournal.pone.0270498). PMC 9365155 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9365155). PMID 35947546 (https://pubmed.ncbi.nlm.nih.gov/35947546).

263. van Eldijk, Timo J. B.; Wappler, Torsten; Strother, Paul K.; van der Weijst, Carolien M. H.; Rajaei, Hossein; Visscher, Henk; van de Schootbrugge, Bas (January 2018). "A Triassic-Jurassic window into the evolution of Lepidoptera" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5770165). *Science Advances*. **4** (1) e1701568. Bibcode:2018SciA....4.1568V (https://ui.adsabs.harvard.edu/abs/2018SciA....4.1568V). doi:10.1126/sciadv.1701568 (https://doi.org/10.1126%2Fsciadv.1701568). PMC 5770165 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5770165). PMID 29349295 (https://pubmed.ncbi.nlm.nih.gov/29349295).

264. Kohli, Manpreet Kaur; Ware, Jessica L.; Bechly, Günter (2016). "How to date a dragonfly: Fossil calibrations for odonates" (https://doi.org/10.26879%2F576). *Palaeontologia Electronica*. **19** (1): 576. Bibcode:2016PalEl..19..576K (https://ui.adsabs.harvard.edu/abs/2016PalEl..19..576K). doi:10.26879/576 (https://doi.org/10.26879%2F576).

265. Huang, DiYing; Engel, Michael S.; Cai, ChenYang; Nel, André (May 2013). "Mesozoic giant fleas from northeastern China (Siphonaptera): Taxonomy and implications for palaeodiversity" (https://doi.org/10.1007%2Fs11434-013-5769-3). *Chinese Science Bulletin*. **58** (14): 1682–1690. Bibcode:2013ChSBu..58.1682H (https://ui.adsabs.harvard.edu/abs/2013ChSBu..58.1682H). doi:10.1007/s11434-013-5769-3 (https://doi.org/10.1007%2Fs11434-013-5769-3). hdl:1808/14426 (https://hdl.handle.net/1808%2F14426).

266. Gao, Taiping; Shih, Chungkun; Rasnitsyn, Alexandr P.; Xu, Xing; Wang, Shuo; Ren, Dong (July 2013). "New Transitional Fleas from China Highlighting Diversity of Early Cretaceous Ectoparasitic Insects" (https://doi.org/10.1016%2Fj.cub.2013.05.040). *Current Biology*. **23** (13): 1261–1266. Bibcode:2013CBio...23.1261G (https://ui.adsabs.harvard.edu/abs/2013CBio...23.1261G). doi:10.1016/j.cub.2013.05.040 (https://doi.org/10.1016%2Fj.cub.2013.05.040). PMID 23810530 (https://pubmed.ncbi.nlm.nih.gov/23810530).
267. Labandeira, Conrad C.; Li, Longfeng (2021). "The History of Insect Parasitism and the Mid-Mesozoic Parasitoid Revolution". *The Evolution and Fossil Record of Parasitism*. Topics in Geobiology. Vol. 49. pp. 377–533. doi:10.1007/978-3-030-42484-8_11 (https://doi.org/10.1007%2F978-3-030-42484-8_11). ISBN 978-3-030-42483-1.
268. Yang, Hongru; Shi, Chaofan; Engel, Michael S; Zhao, Zhipeng; Ren, Dong; Gao, Taiping (2020-04-02). "Early specializations for mimicry and defense in a Jurassic stick insect" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8288419). *National Science Review*. **8** (1) nwaa056. doi:10.1093/nsr/nwaa056 (https://doi.org/10.1093%2Fnsr%2Fnwaa056). PMC 8288419 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8288419). PMID 34691548 (https://pubmed.ncbi.nlm.nih.gov/34691548).
269. Huang, Di-ying; Nel, André; Zompro, Oliver; Waller, Alain (October 2008). "Mantophasmatodea now in the Jurassic". *Naturwissenschaften*. **95** (10): 947–952. Bibcode:2008NW.....95..947H (https://ui.adsabs.harvard.edu/abs/2008NW.....95..947H). doi:10.1007/s00114-008-0412-x (https://doi.org/10.1007%2Fs00114-008-0412-x). PMID 18545982 (https://pubmed.ncbi.nlm.nih.gov/18545982).
270. Huang, Di-Ying; Nel, André (August 2009). "Oldest webspinners from the Middle Jurassic of Inner Mongolia, China (Insecta: Embiodea)" (https://doi.org/10.1111%2Fj.1096-3642.2008.00499.x). *Zoological Journal of the Linnean Society*. **156** (4): 889–895. doi:10.1111/j.1096-3642.2008.00499.x (https://doi.org/10.1111%2Fj.1096-3642.2008.00499.x).
271. Engel, Michael S.; Winterton, Shaun L.; Breitkreuz, Laura C.V. (2018-01-07). "Phylogeny and Evolution of Neuropterida: Where Have Wings of Lace Taken Us?". *Annual Review of Entomology*. **63** (1): 531–551. Bibcode:2018AREnt..63..531E (https://ui.adsabs.harvard.edu/abs/2018AREnt..63..531E). doi:10.1146/annurev-ento-020117-043127 (https://doi.org/10.1146%2Fannurev-ento-020117-043127). PMID 29324039 (https://pubmed.ncbi.nlm.nih.gov/29324039).
272. Vršanský, P.; Sendi, H.; Kotulová, J.; Szwedo, J.; Havelcová, M.; Palková, H.; Vršanská, L.; Sakala, J.; Puškelová, L.; Golej, M.; Biroň, A.; Peyrot, D.; Quicke, D.; Néraudeau, D.; Uher, P.; Maksoud, S.; Azar, D. (2024). "Jurassic Park approached: a coccid from Kimmeridgian cheirolepidiacean Aintourine Lebanese amber" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11895504). *National Science Review*. **12** (3). nwae200. doi:10.1093/nsr/nwae200 (https://doi.org/10.1093%2Fnsr%2Fnwae200). PMC 11895504 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11895504). PMID 40070804 (https://pubmed.ncbi.nlm.nih.gov/40070804).
273. Selden, Paul A.; Baker, Anne S.; Phipps, Kenneth J. (2008). "An Oribatid Mite (arachnida: Acari) from the Oxford Clay (jurassic: Upper Callovian) of South Cave Station Quarry, Yorkshire, Uk". *Palaeontology*. **51** (3): 623–633. Bibcode:2008Palgy..51..623S (https://ui.adsabs.harvard.edu/abs/2008Palgy..51..623S). doi:10.1111/j.1475-4983.2008.00769.x (https://doi.org/10.1111%2Fj.1475-4983.2008.00769.x). hdl:1808/8353 (https://hdl.handle.net/1808%2F8353).
274. Sivhed, Ulf; Wallwork, John A. (March 1978). "An Early Jurassic oribatid mite from southern Sweden". *Geologiska Föreningen i Stockholm Förhandlingar*. **100** (1): 65–70. doi:10.1080/11035897809448562 (https://doi.org/10.1080%2F11035897809448562).
275. Magalhaes, Ivan L. F.; Azevedo, Guilherme H. F.; Michalik, Peter; Ramírez, Martín J. (February 2020). "The fossil record of spiders revisited: implications for calibrating trees and evidence for a major faunal turnover since the Mesozoic". *Biological Reviews*. **95** (1): 184–217. doi:10.1111/brv.12559 (https://doi.org/10.1111%2Fbrv.12559). PMID 31713947 (https://pubmed.ncbi.nlm.nih.gov/31713947).

276. Selden, Paul A.; Dunlop, Jason A. (2014). "The first fossil spider (Araneae: Palpimanoidea) from the Lower Jurassic (Grimmen, Germany)". *Zootaxa*. **3894** (1): 161–168. doi:10.11646/zootaxa.3894.1.13 (https://doi.org/10.11646%2Fzootaxa.3894.1.13). PMID 25544628 (https://pubmed.ncbi.nlm.nih.gov/25544628).
277. Selden, P. A.; Shih, C.K.; Ren, D. (2013). "A giant spider from the Jurassic of China reveals greater diversity of the orbicularian stem group" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3889289). *Naturwissenschaften*. **100** (12): 1171–1181. Bibcode:2013NW....100.1171S (https://ui.adsabs.harvard.edu/abs/2013NW....100.1171S). doi:10.1007/s00114-013-1121-7 (https://doi.org/10.1007%2Fs00114-013-1121-7). PMC 3889289 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3889289). PMID 24317464 (https://pubmed.ncbi.nlm.nih.gov/24317464).
278. Dunlop, Jason A.; Kamenz, Carsten; Scholtz, Gerhard (June 2007). "Reinterpreting the morphology of the Jurassic scorpion Liassoscorpionides". *Arthropod Structure & Development*. **36** (2): 245–252. Bibcode:2007ArtSD..36..245D (https://ui.adsabs.harvard.edu/abs/2007ArtSD..36..245D). doi:10.1016/j.asd.2006.09.003 (https://doi.org/10.1016%2Fj.asd.2006.09.003). PMID 18089103 (https://pubmed.ncbi.nlm.nih.gov/18089103).
279. Huang, Diying; Selden, Paul A.; Dunlop, Jason A. (August 2009). "Harvestmen (Arachnida: Opiliones) from the Middle Jurassic of China". *Naturwissenschaften*. **96** (8): 955–962. Bibcode:2009NW.....96..955H (https://ui.adsabs.harvard.edu/abs/2009NW.....96..955H). doi:10.1007/s00114-009-0556-3 (https://doi.org/10.1007%2Fs00114-009-0556-3). PMID 19495718 (https://pubmed.ncbi.nlm.nih.gov/19495718).
280. Giribet, Gonzalo; Tourinho, Ana Lúcia; Shih, ChungKun; Ren, Dong (March 2012). "An exquisitely preserved harvestman (Arthropoda, Arachnida, Opiliones) from the Middle Jurassic of China". *Organisms Diversity & Evolution*. **12** (1): 51–56. Bibcode:2012ODivE..12...51G (https://ui.adsabs.harvard.edu/abs/2012ODivE..12...51G). doi:10.1007/s13127-011-0067-x (https://doi.org/10.1007%2Fs13127-011-0067-x).
281. Dunhill, Alexander M.; Foster, William J.; Sciberras, James; Twitchett, Richard J. (January 2018). Hautmann, Michael (ed.). "Impact of the Late Triassic mass extinction on functional diversity and composition of marine ecosystems" (https://doi.org/10.1111%2Fpala.12332). *Palaeontology*. **61** (1): 133–148. Bibcode:2018Palgy..61..133D (https://ui.adsabs.harvard.edu/abs/2018Palgy..61..133D). doi:10.1111/pala.12332 (https://doi.org/10.1111%2Fpala.12332).
282. Kiessling, Wolfgang (December 2009). "Geologic and Biologic Controls on the Evolution of Reefs". *Annual Review of Ecology, Evolution, and Systematics*. **40** (1): 173–192. doi:10.1146/annurev.ecolsys.110308.120251 (https://doi.org/10.1146%2Fannurev.ecolsys.110308.120251).
283. Klompmaker, A. A.; Schweitzer, C. E.; Feldmann, R. M.; Kowalewski, M. (2013-11-01). "The influence of reefs on the rise of Mesozoic marine crustaceans". *Geology*. **41** (11): 1179–1182. Bibcode:2013Geo....41.1179K (https://ui.adsabs.harvard.edu/abs/2013Geo....41.1179K). doi:10.1130/G34768.1 (https://doi.org/10.1130%2FG34768.1).
284. Hudson, Wendy; Hart, Malcolm B.; Smart, Christopher W. (2009-01-01). "Palaeobiogeography of early planktonic foraminifera". *Bulletin de la Société Géologique de France*. **180** (1): 27–38. Bibcode:2009BSGF..180...27H (https://ui.adsabs.harvard.edu/abs/2009BSGF..180...27H). doi:10.2113/gssgfbull.180.1.27 (https://doi.org/10.2113%2Fgssgfbull.180.1.27).
285. Wiggan, Nickolas J.; Riding, James B.; Fensome, Robert A.; Mattioli, Emanuela (2018-05-01). "The Bajocian (Middle Jurassic): A key interval in the early Mesozoic phytoplankton radiation" (https://univ-lyon1.hal.science/hal-02322995). *Earth-Science Reviews*. **180**: 126–146. Bibcode:2018ESRv..180..126W (https://ui.adsabs.harvard.edu/abs/2018ESRv..180..126W). doi:10.1016/j.earscirev.2018.03.009 (https://doi.org/10.1016%2Fj.earscirev.2018.03.009).
286. Zatoń, M.; Taylor, P.D. (2009-12-31). "Microconchids (Tentaculita) from the Middle Jurassic of Poland" (https://doi.org/10.3140%2Fbull.geosci.1167). *Bulletin of Geosciences*: 653–660. doi:10.3140/bull.geosci.1167 (https://doi.org/10.3140%2Fbull.geosci.1167).

287. Girard, Vincent; Saint Martin, Simona; Buffetaut, Eric; Saint Martin, Jean-Paul; Néraudeau, Didier; Peyrot, Daniel; Roghi, Guido; Ragazzi, Eugenio; Suteethorn, Varavudh (2020). Saint Martin, J.-P.; Saint Martin, S. (eds.). "Thai amber: insights into early diatom history?" (https://doi.org/10.1051%2Fbsgf%2F2020028). *BSGF – Earth Sciences Bulletin*. **191**: 23. doi:10.1051/bsgf/2020028 (https://doi.org/10.1051%2Fbsgf%2F2020028). hdl:11577/3391076 (https://hdl.handle.net/11577%2F3391076).
288. Gorzelak, Przemysław; Salamon, Mariusz A.; Trzęsiok, Dawid; Lach, Rafał; Baumiller, Tomasz K. (April 2016). "Diversity dynamics of post-Palaeozoic crinoids – in quest of the factors affecting crinoid macroevolution". *Lethaia*. **49** (2): 231–244. Bibcode:2016Letha..49..231G (https://ui.adsabs.harvard.edu/abs/2016Letha..49..231G). doi:10.1111/let.12141 (https://doi.org/10.1111%2Flet.12141).
289. Hopkins, Melanie J.; Smith, Andrew B. (2015-03-24). "Dynamic evolutionary change in post-Paleozoic echinoids and the importance of scale when interpreting changes in rates of evolution" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4378421). *Proceedings of the National Academy of Sciences*. **112** (12): 3758–3763. Bibcode:2015PNAS..112.3758H (https://ui.adsabs.harvard.edu/abs/2015PNAS..112.3758H). doi:10.1073/pnas.1418153112 (https://doi.org/10.1073%2Fpnas.1418153112). PMC 4378421 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4378421). PMID 25713369 (https://pubmed.ncbi.nlm.nih.gov/25713369).
290. Scholtz, Gerhard (November 2020). "Eocarcinus praecursor Withers, 1932 (Malacostraca, Decapoda, Meiura) is a stem group brachyuran" (https://doi.org/10.1016%2Fj.asd.2020.100991). *Arthropod Structure & Development*. **59** 100991. Bibcode:2020ArtSD..5900991S (https://ui.adsabs.harvard.edu/abs/2020ArtSD..5900991S). doi:10.1016/j.asd.2020.100991 (https://doi.org/10.1016%2Fj.asd.2020.100991). PMID 32891896 (https://pubmed.ncbi.nlm.nih.gov/32891896).
291. Schweitzer, Carrie E.; Feldmann, Rodney M. (2010-05-01). "The Oldest Brachyura (Decapoda: Homolodromioidea: Glaessneropsoidea) Known to Date (Jurassic)" (https://doi.org/10.1651%2F09-3231.1). *Journal of Crustacean Biology*. **30** (2): 251–256. Bibcode:2010JCBio..30..251F (https://ui.adsabs.harvard.edu/abs/2010JCBio..30..251F). doi:10.1651/09-3231.1 (https://doi.org/10.1651%2F09-3231.1).
292. Guinot, Danièle (2019-11-14). "New hypotheses concerning the earliest brachyurans (Crustacea, Decapoda, Brachyura)" (https://doi.org/10.5252%2Fgeodiversitas2019v41a22). *Geodiversitas*. **41** (1): 747. Bibcode:2019Geodv..41..747G (https://ui.adsabs.harvard.edu/abs/2019Geodv..41..747G). doi:10.5252/geodiversitas2019v41a22 (https://doi.org/10.5252%2Fgeodiversitas2019v41a22).
293. Fraaije, René; Schweigert, Günter; Nützel, Alexander; Havlik, Philipe (2013-01-01). "New Early Jurassic hermit crabs from Germany and France" (https://doi.org/10.1163%2F1937240X-00002191). *Journal of Crustacean Biology*. **33** (6): 802–817. Bibcode:2013JCBio..33..802F (https://ui.adsabs.harvard.edu/abs/2013JCBio..33..802F). doi:10.1163/1937240X-00002191 (https://doi.org/10.1163%2F1937240X-00002191).
294. Mironenko, Aleksandr (January 2020). "A hermit crab preserved inside an ammonite shell from the Upper Jurassic of central Russia: Implications to ammonoid palaeoecology" (https://doi.org/10.1016%2Fj.palaeo.2019.109397). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **537** 109397. Bibcode:2020PPP...53709397M (https://ui.adsabs.harvard.edu/abs/2020PPP...53709397M). doi:10.1016/j.palaeo.2019.109397 (https://doi.org/10.1016%2Fj.palaeo.2019.109397).
295. Bracken-Grissom, Heather D.; Ahyong, Shane T.; Wilkinson, Richard D.; Feldmann, Rodney M.; Schweitzer, Carrie E.; Breinholt, Jesse W.; Bendall, Matthew; Palero, Ferran; Chan, Tin-Yam; Felder, Darryl L.; Robles, Rafael (2014-07-01). "The Emergence of Lobsters: Phylogenetic Relationships, Morphological Evolution and Divergence Time Comparisons of an Ancient Group (Decapoda: Achelata, Astacidea, Glypheidea, Polychelida)" (https://doi.org/10.1093%2Fsysbio%2Fsyu008). *Systematic Biology*. **63** (4): 457–479. doi:10.1093/sysbio/syu008 (https://doi.org/10.1093%2Fsysbio%2Fsyu008). PMID 24562813 (https://pubmed.ncbi.nlm.nih.gov/24562813).

296. Chan, Benny K K; Dreyer, Niklas; Gale, Andy S; Glenner, Henrik; Ewers-Saucedo, Christine; Pérez-Losada, Marcos; Kolbasov, Gregory A; Crandall, Keith A; Høeg, Jens T (2021-02-25). "The evolutionary diversity of barnacles, with an updated classification of fossil and living forms" (https://doi.org/10.1093%2Fzoolinnean%2Fzlaa160). *Zoological Journal of the Linnean Society*. **193** (3): 789–846. doi:10.1093/zoolinnean/zlaa160 (https://doi.org/10.1093%2Fzoolinnean%2Fzlaa160). hdl:11250/2990967 (https://hdl.handle.net/11250%2F2990967).
297. Gale, Andy; Schweigert, Günter (January 2016). Hautmann, Michael (ed.). "A new phosphatic-shelled cirripede (Crustacea, Thoracica) from the Lower Jurassic (Toarcian) of Germany – the oldest epiplanktonic barnacle" (https://doi.org/10.1111%2Fpala.12207). *Palaeontology*. **59** (1): 59–70. Bibcode:2016Palgy..59...59G (https://ui.adsabs.harvard.edu/abs/2016Palgy..59...59G). doi:10.1111/pala.12207 (https://doi.org/10.1111%2Fpala.12207).
298. Vörös, Attila; Kocsis, Ádám T.; Pálfy, József (September 2016). "Demise of the last two spire-bearing brachiopod orders (Spiriferinida and Athyridida) at the Toarcian (Early Jurassic) extinction event". *Palaeogeography, Palaeoclimatology, Palaeoecology*. **457**: 233–241. Bibcode:2016PPP...457..233V (https://ui.adsabs.harvard.edu/abs/2016PPP...457..233V). doi:10.1016/j.palaeo.2016.06.022 (https://doi.org/10.1016%2Fj.palaeo.2016.06.022).
299. Vörös, Attila; Kocsis, Ádám T.; Pálfy, József (2019). "Mass extinctions and clade extinctions in the history of brachiopods: Brief review and a post-Paleozoic case study". *Rivista Italiana di Paleontologia e Stratigrafia*. **125** (3). doi:10.13130/2039-4942/12184 (https://doi.org/10.13130%2F2039-4942%2F12184).
300. Manojlovic, Marko; Clapham, Matthew E. (2020-11-23). "The role of bioturbation-driven substrate disturbance in the Mesozoic brachiopod decline" (https://doi.org/10.1017%2Fpab.2020.50). *Paleobiology*. **47**: 86–100. doi:10.1017/pab.2020.50 (https://doi.org/10.1017%2Fpab.2020.50).
301. Taylor, Paul D.; Ernst, Andrej (June 2008). "Bryozoans in transition: The depauperate and patchy Jurassic biota". *Palaeogeography, Palaeoclimatology, Palaeoecology*. **263** (1–2): 9–23. Bibcode:2008PPP...263....9T (https://ui.adsabs.harvard.edu/abs/2008PPP...263....9T). doi:10.1016/j.palaeo.2008.01.028 (https://doi.org/10.1016%2Fj.palaeo.2008.01.028).
302. Ferrari, Mariel; Hautmann, Michael (2022-11-02). Kiel, Steffen (ed.). "Gastropods underwent a major taxonomic turnover during the end-Triassic marine mass extinction event" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9629647). *PLOS ONE*. **17** (11) e0276329. Bibcode:2022PLoSO..1776329F (https://ui.adsabs.harvard.edu/abs/2022PLoSO..1776329F). doi:10.1371/journal.pone.0276329 (https://doi.org/10.1371%2Fjournal.pone.0276329). PMC 9629647 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9629647). PMID 36322518 (https://pubmed.ncbi.nlm.nih.gov/36322518).
303. Neubauer, Thomas A. (February 2024). "The fossil record of freshwater Gastropoda – a global review" (https://doi.org/10.1111%2Fbrv.13016). *Biological Reviews*. **99** (1): 177–199. doi:10.1111/brv.13016 (https://doi.org/10.1111%2Fbrv.13016). PMID 37698140 (https://pubmed.ncbi.nlm.nih.gov/37698140).
304. Ros, Sonia; De Renzi, Miquel; Damborenea, Susana E.; Márquez-Aliaga, Ana (November 2011). "Coping between crises: Early Triassic–early Jurassic bivalve diversity dynamics". *Palaeogeography, Palaeoclimatology, Palaeoecology*. **311** (3–4): 184–199. Bibcode:2011PPP...311..184R (https://ui.adsabs.harvard.edu/abs/2011PPP...311..184R). doi:10.1016/j.palaeo.2011.08.020 (https://doi.org/10.1016%2Fj.palaeo.2011.08.020). hdl:11336/81358 (https://hdl.handle.net/11336%2F81358).
305. Mondal, Subhronil; Harries, Peter J. (February 2016). "The Effect of Taxonomic Corrections on Phanerozoic Generic Richness Trends in Marine Bivalves with a Discussion on the Clade's Overall History". *Paleobiology*. **42** (1): 157–171. Bibcode:2016Pbio...42..157M (https://ui.adsabs.harvard.edu/abs/2016Pbio...42..157M). doi:10.1017/pab.2015.35 (https://doi.org/10.1017%2Fpab.2015.35).

306. Sha, J.; Cestari, R.; Fabbi, S. (April 2020). "Paleobiogeographic distribution of rudist bivalves (Hippuritida) in the Oxfordian–early Aptian (Late Jurassic–Early Cretaceous)". *Cretaceous Research*. **108** 104289. Bibcode:2020CrRes.10804289S (https://ui.adsabs.harvard.edu/abs/2020CrRes.10804289S). doi:10.1016/j.cretres.2019.104289 (https://doi.org/10.1016%2Fj.cretres.2019.104289).
307. Page, Kevin N. (January 2008). "The evolution and geography of Jurassic ammonoids". *Proceedings of the Geologists' Association*. **119** (1): 35–57. Bibcode:2008PrGA..119...35P (https://ui.adsabs.harvard.edu/abs/2008PrGA..119...35P). doi:10.1016/S0016-7878(08)80257-X (https://doi.org/10.1016%2FS0016-7878%2808%2980257-X).
308. Sandoval, José; O'Dogherty, Jean; Guex, Jean (1 August 2001). "Evolutionary Rates of Jurassic Ammonites in Relation to Sea-level Fluctuations". *PALAIOS*. **16** (4): 311–335. Bibcode:2001Palai..16..311S (https://ui.adsabs.harvard.edu/abs/2001Palai..16..311S). doi:10.1669/0883-1351(2001)016<0311:EROJAI>2.0.CO;2 (https://doi.org/10.1669%2F0883-1351%282001%29016%3C0311%3AEROJAI%3E2.0.CO%3B2).
309. Iba, Yasuhiro; Sano, Shin-ichi; Mutterlose, Jörg (2014-05-02). Samonds, Karen E. (ed.). "The Early Evolutionary History of Belemnites: New Data from Japan" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4008418). *PLOS ONE*. **9** (5) e95632. Bibcode:2014PLoSO...995632I (https://ui.adsabs.harvard.edu/abs/2014PLoSO...995632I). doi:10.1371/journal.pone.0095632 (https://doi.org/10.1371%2Fjournal.pone.0095632). PMC 4008418 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4008418). PMID 24788872 (https://pubmed.ncbi.nlm.nih.gov/24788872).
310. Hoffmann, René; Stevens, Kevin (February 2020). "The palaeobiology of belemnites – foundation for the interpretation of rostrum geochemistry" (https://doi.org/10.1111%2Fbrv.12557). *Biological Reviews*. **95** (1): 94–123. Bibcode:2020BioRv..95...94H (https://ui.adsabs.harvard.edu/abs/2020BioRv..95...94H). doi:10.1111/brv.12557 (https://doi.org/10.1111%2Fbrv.12557). PMID 31729839 (https://pubmed.ncbi.nlm.nih.gov/31729839).
311. Fuchs, Dirk; Weis, Robert (2008-07-11). "Taxonomy, morphology and phylogeny of Lower Jurassic loligosepiid coleoids (Cephalopoda)". *Neues Jahrbuch für Geologie und Paläontologie – Abhandlungen*. **249** (1): 93–112. Bibcode:2008NJGPA.249...93F (https://ui.adsabs.harvard.edu/abs/2008NJGPA.249...93F). doi:10.1127/0077-7749/2008/0249-0093 (https://doi.org/10.1127%2F0077-7749%2F2008%2F0249-0093).
312. Fuchs, Dirk; Iba, Yasuhiro; Heyng, Alexander; Iijima, Masaya; Klug, Christian; Larson, Neal L.; Schweigert, Günter (February 2020). Brayard, Arnaud (ed.). "The Muensterelloidea: phylogeny and character evolution of Mesozoic stem octopods". *Papers in Palaeontology*. **6** (1): 31–92. Bibcode:2020PPal....6...31F (https://ui.adsabs.harvard.edu/abs/2020PPal....6...31F). doi:10.1002/spp2.1254 (https://doi.org/10.1002%2Fspp2.1254).
313. Fuchs, Dirk; Schweigert, Günter (June 2018). "First Middle–Late Jurassic gladius vestiges provide new evidence on the detailed origin of incirrate and cirrate octopuses (Coleoidea)". *PalZ*. **92** (2): 203–217. Bibcode:2018PalZ...92..203F (https://ui.adsabs.harvard.edu/abs/2018PalZ...92..203F). doi:10.1007/s12542-017-0399-8 (https://doi.org/10.1007%2Fs12542-017-0399-8).
314. Kruta, Isabelle; Rouget, Isabelle; Charbonnier, Sylvain; Bardin, Jérémie; Fernandez, Vincent; Germain, Damien; Brayard, Arnaud; Landman, Neil (2016). "Proteroctopus ribeti in coleoid evolution" (https://doi.org/10.1111%2Fpala.12265). *Palaeontology*. **59** (6): 767–773. Bibcode:2016Palgy..59..767K (https://ui.adsabs.harvard.edu/abs/2016Palgy..59..767K). doi:10.1111/pala.12265 (https://doi.org/10.1111%2Fpala.12265).

External links

- Examples of Jurassic Fossils (http://www.geo-lieven.com/erdzeitalter/jura/jura.htm)
- Jurassic (chronostratigraphy scale) (https://ghkclass.com/ghkC.html?jurassic)

- Jurassic fossils in Harbury, Warwickshire (https://web.archive.org/web/20081208011604/http://harbury.villagebuzz.co.uk/viewtopic.php?f=16&t=297)
- Jurassic Microfossils: 65+ images of Foraminifera (http://www.foraminifera.eu/querydb.php?period=Jurassic&aktion=suche)
- "Jurassic" (https://en.wikisource.org/wiki/1911_Encyclop%C3%A6dia_Britannica/Jurassic). *Encyclopædia Britannica*. Vol. 15 (11th ed.). 1911. With map and table.

Retrieved from "https://en.wikipedia.org/w/index.php?title=Jurassic&oldid=1380614854"

3

Saurischia (Kertenkele Kalçalılar)

Theropod'lar (etçiller) ve Sauropod'lar (uzun boyunlular)

- Saurischia
- Theropoda
- Ceratosauria
- Megalosauroidea
- Allosauridae
- Spinosauridae
- Tyrannosauridae
- Tyrannosaurus
- Dromaeosauridae
- Velociraptor
- Troodontidae
- Sauropoda

WIKIPEDIA
25 years of the free encyclopedia

Saurischia

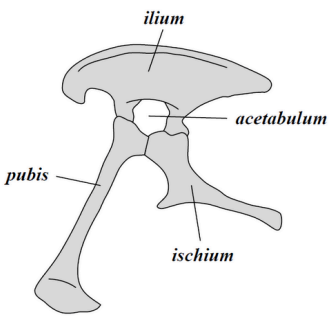
Saurischia (/sɔːˈrɪskiə/ *saw-RIS-kee-ə*, meaning "reptile-hipped" from the Greek *sauros* (σαῦρος) meaning 'lizard' and *ischion* (ἴσχιον) meaning 'hip joint')[3] is one of the two basic divisions of dinosaurs (the other being Ornithischia), classified by their hip structure. Saurischia and Ornithischia were originally called orders by Harry Seeley in 1888,[4] though today most paleontologists classify Saurischia as an unranked clade rather than an order.[5]

Description

All carnivorous dinosaurs (certain types of theropods) are traditionally classified as saurischians, as are all of the birds and one of the two primary lineages of herbivorous dinosaurs, the sauropodomorphs. At the end of the Cretaceous Period, all saurischians except birds became extinct in the course of the Cretaceous–Paleogene extinction event. Birds, as a group of maniraptoran theropod dinosaurs, are a sub-clade of saurischian dinosaurs in phylogenetic classification.[6]

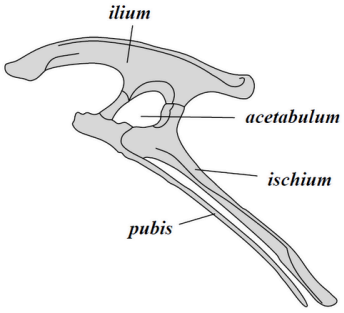
Saurischian dinosaurs are traditionally distinguished from ornithischian dinosaurs by their three-pronged pelvic structure, with the pubis pointed forward. The ornithischians' pelvis is arranged with the pubis rotated backward, parallel with the ischium, often also with a forward-pointing process, giving a four-pronged structure. The saurischian hip structure led Seeley to name them "lizard-hipped" dinosaurs, because they retained the ancestral hip anatomy also found in modern lizards and other reptiles. He named ornithischians "bird-hipped" dinosaurs because their hip arrangement was superficially similar to that of birds, though he did not propose any specific relationship between ornithischians and birds. However, in the view which has long been held, this "bird-hipped" arrangement evolved several times independently in dinosaurs, first in the ornithischians, then in the lineage of saurischians including birds (Avialae), and lastly in the therizinosaurians. This would then be an example of convergent evolution: avialans, therizinosaurians, and ornithischian dinosaurs all developed a similar hip anatomy independently of each other, possibly as an adaptation to their herbivorous or omnivorous diets.[7]

Classification

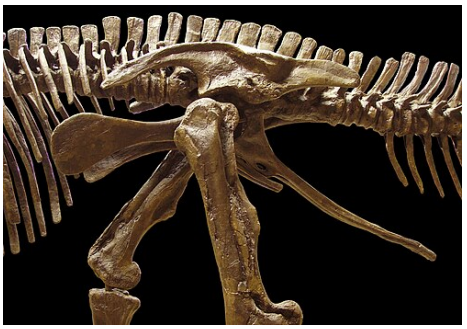


Saurischian pelvis structure (left side)

Tyrannosaurus pelvis (showing saurischian structure – left side)



Ornithischian pelvis structure (left side)



Edmontosaurus pelvis (showing ornithischian structure – left side)

Saurischians

Temporal range:

Late Triassic – Present, Possible Middle Triassic record[2]

Montage of six different representatives of saurischian dinosaurs.

1st row (early saurischians): *Herrerasaurus ischigualastensis* (herrerasaur), *Eodromaeus murphi* (basal theropod);
2nd row (theropods): Brown pelican, *Tyrannosaurus rex*;
3rd row (sauropodomorphs): *Apatosaurus louisae*, *Plateosaurus trossingensis*.

Scientific classification

Kingdom:	Animalia
Phylum:	Chordata
Class:	Reptilia
Clade:	Dinosauria
Clade:	Saurischia Seeley, 1888

Subgroups

- †Herrerasauria
- **Eusaurischia** Padian *et al.*, 1999
  - †Sauropodomorpha
  - Theropoda

Saurischians of uncertain affinity

- †*Alwalkeria*
- †*Eodromaeus*
- †*Saltopus*?
- †*Saurophaganax*

In his paper naming the two groups, Seeley reviewed previous classification schemes put forth by other paleontologists to divide up the traditional order Dinosauria. He preferred one that had been put forward by Othniel Charles Marsh in 1878, which divided dinosaurs into four orders: Sauropoda, Theropoda, Ornithopoda, and Stegosauria (these names are still used today in much the same way to refer to suborders or clades within Saurischia and Ornithischia).[4]

Seeley, however, wanted to formulate a classification that would take into account a single primary difference between major dinosaurian groups based on a characteristic that also differentiated them from other reptiles. He found this in the configuration of the hip bones, and found that all four of Marsh's orders could be divided neatly into two major groups based on this feature. He placed the Stegosauria and Ornithopoda in the Ornithischia, and the Theropoda and Sauropoda in the Saurischia. Furthermore, Seeley used this major difference in the hip bones, along with many other noted differences between the two groups, to argue that "dinosaurs" were not a natural grouping at all, but rather two distinct orders that had arisen independently from more primitive archosaurs.[4] This concept that "dinosaur" was an outdated term for two distinct orders lasted many decades in the scientific and popular literature, and it was not until the 1960s that scientists began to again consider the possibility that saurischians and ornithischians were more closely related to each other than they were to other archosaurs.

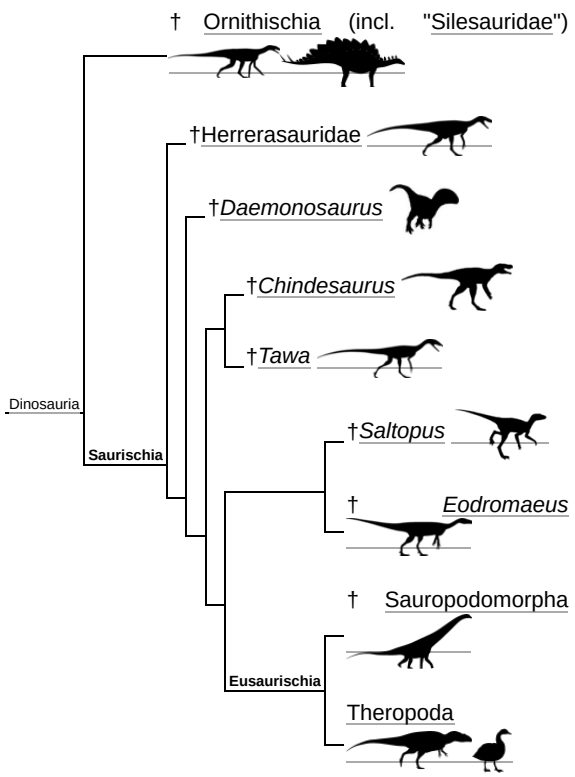
Although his concept of a polyphyletic Dinosauria is no longer accepted by most paleontologists, Seeley's basic division of the two dinosaurian groups has stood the test of time, and has been supported by modern cladistic analysis of relationships among dinosaurs.[8] A node-base clade, **Eusaurischia**, was named for the least inclusive group containing sauropodomorphs (represented by *Cetiosaurus*) and theropods (represented by Neornithes). Any saurischian that diverged before the theropod-sauropodomorph split is therefore outside clade Eusaurischia.[9]

One alternative hypothesis challenging Seeley's classification was proposed by Robert T. Bakker in his 1986 book *The Dinosaur Heresies*. Bakker's classification separated the theropods into their own group and placed the two groups of herbivorous dinosaurs (the sauropodomorphs and ornithischians) together in a separate group he named the Phytodinosauria ("plant dinosaurs").[10] The Phytodinosauria hypothesis was based partly on the supposed link between ornithischians and prosauropods, and the idea that the former had evolved directly from the latter, possibly by way of an enigmatic family that seemed to possess characters of both groups, the segnosaurs.[11] However, it was later found that segnosaurs were an unusual type of herbivorous theropod saurischian closely related to birds, and the Phytodinosauria hypothesis fell out of favor.

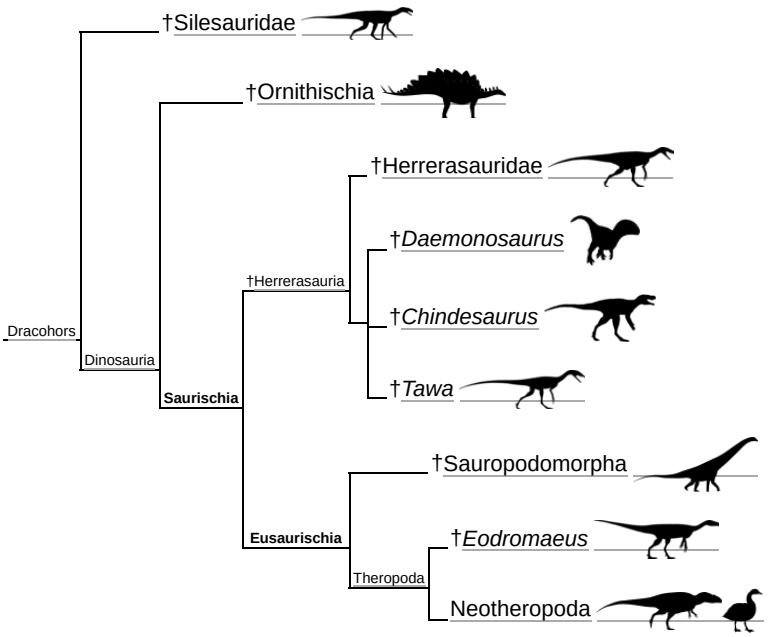
A 2017 study by Matthew Grant Baron, David B. Norman and Paul M. Barrett did not find support for a monophyletic Saurischia, according to its traditional definition. Instead, the group was found to be paraphyletic. As a solution, Theropoda was removed from the group and placed as the sister group to the Ornithischia in the newly defined clade Ornithoscelida. As another result, the authors redefined Saurischia as "the most inclusive clade that contains *Diplodocus carnegii*, but not *Triceratops horridus*, resulting in a clade containing only the Sauropodomorpha and Herrerasauridae.[12][13] Thomas Holtz (2017) recommended using the name Sauropodomorpha to refer to a possible clade that includes traditional sauropodomorphs and herrerasaurids; alternatively, he proposed redefining the long-disused taxon Pachypodosauria to include Sauropodomorpha and Herrerasauridae as subclades.[14] Cau (2018) also supported Ornithoscelida but placed herrerasaurids, *Tawa* and *Daemonosaurus* in a clade (Herrerasauria) outside Dinosauria.[15] Other recent studies support a view closer to the traditional Saurischia hypothesis, with theropods closer to sauropodomorphs than to ornithischians. Novas et al. (2021) support Cau's herrerasaur phylogeny but place this clade in Saurischia.[16]

Phylogenetic position of saurischians in different topologies

Müller & Garcia, 2020[17]

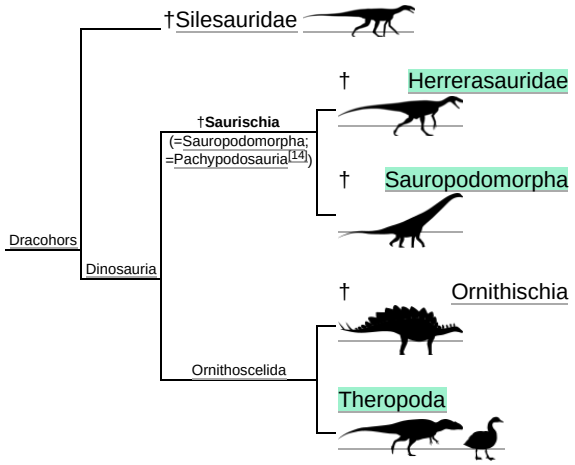


Novas et al., 2021[16]

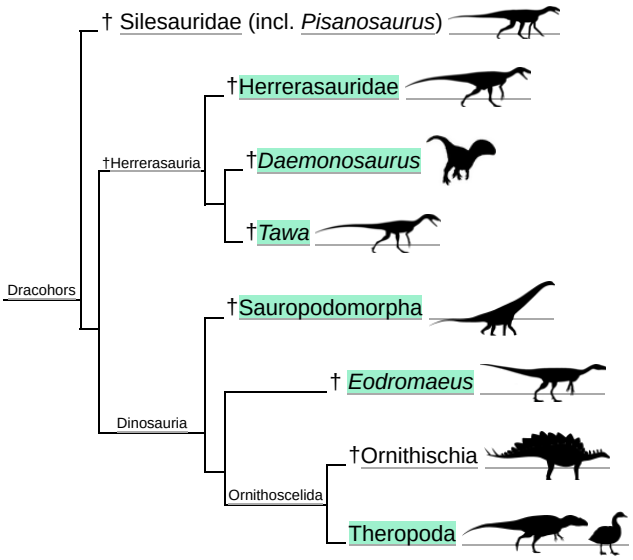


Ornithoscelida hypothesis. Taxa traditionally classified as saurischians are highlighted in green.

Baron et al., 2017[12]



Cau, 2018[15]



See also

- Ornithoscelida
- Phytodinosauria

References

1. Langer, M.C.; Ramezani, J.; Da Rosa, Á.A.S. (2018). "U-Pb age constraints on dinosaur rise from south Brazil". *Gondwana Research*. **X** (18): 133–140. Bibcode:2018GondR..57..133L (https://ui.adsabs.harvard.edu/abs/2018GondR..57..133L). doi:10.1016/j.gr.2018.01.005 (https://doi.org/10.1016%2Fj.gr.2018.01.005).

2. Nesbitt, S. J.; Barrett, P. M.; Werning, S.; Sidor, C. A.; Charig, A. J. (2013). "The oldest dinosaur? A Middle Triassic dinosauriform from Tanzania" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3565515). *Biol. Lett*. **9** (1) 20120949. doi:10.1098/rsbl.2012.0949 (https://doi.org/10.1098%2Frsbl.2012.0949). PMC 3565515 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3565515). PMID 23221875 (https://pubmed.ncbi.nlm.nih.gov/23221875).
3. Oxford English Dictionary
4. "I. On the classification of the fossil animals commonly named Dinosauria" (https://doi.org/10.1098%2Frspl.1887.0117). *Proceedings of the Royal Society of London*. **43** (258–265): 165–171. 31 December 1888. doi:10.1098/rspl.1887.0117 (https://doi.org/10.1098%2Frspl.1887.0117). S2CID 129792059 (https://api.semanticscholar.org/CorpusID:129792059).
5. Weishampel, D.B., Dodson, P., and Osmólska, H. (eds.). (2004). *The Dinosauria. 2nd edition*. University of California Press, Berkeley. 833 pp.
6. Padian, K. (2004). "Basal Avialae". In Weishampel, David B.; Dodson, Peter; Osmólska, Halszka (eds.). *The Dinosauria* (Second ed.). Berkeley: University of California Press. pp. 210–231. ISBN 978-0-520-24209-8.
7. Zanno, Lindsay E.; Gillette, David D.; Albright, L. Barry; Titus, Alan L. (7 October 2009). "A new North American therizinosaurid and the role of herbivory in 'predatory' dinosaur evolution" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2817200). *Proceedings of the Royal Society B: Biological Sciences*. **276** (1672): 3505–3511. doi:10.1098/rspb.2009.1029 (https://doi.org/10.1098%2Frspb.2009.1029). PMC 2817200 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2817200). PMID 19605396 (https://pubmed.ncbi.nlm.nih.gov/19605396).
8. Weishampel, David B.; Dodson, Peter; Osmólska, Halszka (eds.) (2004). *The Dinosauria*, Second Edition. University of California Press., 861 pp.
9. Weishampel, D. B., Dodson, P., & Osmólska, H. (Eds.). (2007). *The dinosauria*. Univ of California Press.
10. Bakker, R. T. (1986). *The Dinosaur Heresies*. New York: William Morrow. p. 203. ISBN 0-14-010055-5.
11. Paul, G.S. (1988). *Predatory Dinosaurs of the World, a Complete Illustrated Guide*. New York: Simon & Schuster. 464 p.
12. Baron, M.G.; Norman, D.B.; Barrett, P.M. (23 March 2017). "A new hypothesis of dinosaur relationships and early dinosaur evolution". *Nature*. **543** (7646): 501–506. Bibcode:2017Natur.543..501B (https://ui.adsabs.harvard.edu/abs/2017Natur.543..501B). doi:10.1038/nature21700 (https://doi.org/10.1038%2Fnature21700). PMID 28332513 (https://pubmed.ncbi.nlm.nih.gov/28332513). S2CID 205254710 (https://api.semanticscholar.org/CorpusID:205254710).
13. "New study shakes the roots of the dinosaur family tree" (https://www.cam.ac.uk/research/news/new-study-shakes-the-roots-of-the-dinosaur-family-tree). 22 March 2017.
14. Holtz, T.R. (2017). "Share names for dinosaur divisions" (https://doi.org/10.1038%2F545030d). *Nature*. **545** (30): 30. doi:10.1038/545030d (https://doi.org/10.1038%2F545030d). PMID 28470208 (https://pubmed.ncbi.nlm.nih.gov/28470208). S2CID 4395996 (https://api.semanticscholar.org/CorpusID:43959596).
15. Cau, A. (2018). "The assembly of the avian body plan: a 160-million-year long process". *Bollettino della Società Paleontologica Italiana* (1): 1–25. doi:10.4435/BSPI.2018.01 (https://doi.org/10.4435%2FBSPI.2018.01) (inactive 12 July 2025). S2CID 44078918 (https://api.semanticscholar.org/CorpusID:44078918).
16. Novas, F.E.; Agnolin, F.L.; Ezcurra, M.D.; Müller, R.T.; Martinelli, A.; Langer, M. (2021). "Review of the fossil record of early dinosaurs from South America, and its phylogenetic implications". *Journal of South American Earth Sciences*. **10** 103341. Bibcode:2021JSAES.11003341N (https://ui.adsabs.harvard.edu/abs/2021JSAES.11003341N). doi:10.1016/j.jsames.2021.103341 (https://doi.org/10.1016%2Fj.jsames.2021.103341).
17. Müller, R.T.; Garcia, M.S. (2020). "A paraphyletic 'Silesauridae' as an alternative hypothesis for the initial radiation of ornithischian dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7480155). *Biology Letters*. **16** (8) 20200417. doi:10.1098/rsbl.2020.0417 (https://doi.org/10.1098%2Frsbl.2020.0417). PMC 7480155 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7480155). PMID 32842895 (https://pubmed.ncbi.nlm.nih.gov/32842895). S2CID 221298572 (https://api.semanticscholar.org/CorpusID:221298572).

Retrieved from "https://en.wikipedia.org/w/index.php?title=Saurischia&oldid=1381503433"

WIKIPEDIA
25 years of the free encyclopedia

Theropoda

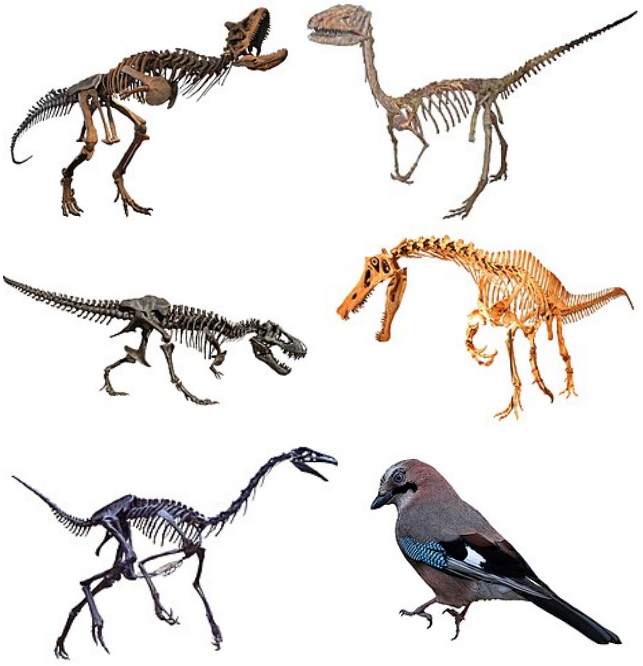
Theropoda (/θɪˈrɒpədə/;[2] from ancient Greek θηρίο- ποδός [*θηρίον*, (*therion*) "wild beast"; *πούς*, *ποδός* (*pous*, *podos*) "foot"][3]) is one of the three major clades of dinosaurs, alongside Ornithischia and Sauropodomorpha. Theropods, both extant and extinct, are characterized by hollow bones and three toes and claws on each limb. They are generally classed as saurischian dinosaurs, placing them closer to sauropodomorphs than to ornithischians. They were ancestrally carnivorous, although a number of theropod groups evolved to become herbivores and omnivores. Many members of the subgroup Coelurosauria were covered with feathers, and it is possible that they were also present in other theropods. In the Jurassic, birds evolved from small specialized coelurosaurian theropods. There are around 11,000 living bird species, making theropods the only known living group of dinosaurs.

Theropods first appeared during the Carnian age of the Late Triassic period 231.4 million years ago (Ma)[4] and included the majority of large terrestrial carnivores from the Early Jurassic until the end of the Cretaceous, about 66 Ma, including the largest terrestrial carnivorous animals ever, such as *Tyrannosaurus* and *Giganotosaurus*, though non-avian theropods exhibited considerable size diversity, with some non-avian theropods like scansoriopterygids being no bigger than small birds.

Theropoda

Temporal range:

Late Triassic – present, including birds



Gallery of theropods (clockwise from top left) *Carnotaurus*, *Coelophysis*, *Irritator*, a Eurasian jay, *Struthiomimus* and *Tyrannosaurus*

Scientific classification

Kingdom:	Animalia
Phylum:	Chordata
Class:	Reptilia
Clade:	Dinosauria
Clade:	Saurischia
Clade:	Theropoda Marsh, 1881

Subgroups[1]

- †*Anteavis*
- †*Eodromaeus*
- †*Erythrovenator*

Biology

Traits

Various synapomorphies for Theropoda have been proposed based on which taxa are included in the group. For example, a 1999 paper by Paul Sereno suggests that theropods are characterized by traits such as an ectopterygoid fossa (a depression around the ectopterygoid bone), an intramandibular joint located within the lower jaw, and extreme internal cavitation within the bones.[5] However, since taxa like *Herrerasaurus* may not be theropods, these traits may have been more widely distributed among early saurischians rather than being unique to theropods.

Instead, taxa with a higher probability of being within the Theropoda may share more specific traits, such as a prominent promaxillary fenestra, cervical vertebrae with pleurocoels in the anterior part of the centrum leading to a more pneumatic neck, five or more sacral vertebrae, enlargement of the carpal bone, and a distally concave portion of the tibia, among a few other traits found throughout the skeleton. Like the early sauropodomorphs, the second digit in a theropod's hand is enlarged. Theropods also have a very well developed ball and socket joint near their neck and head.[6][7]

Most theropods belong to the clade Neotheropoda, characterized by the reduction of several foot bones, thus leaving three toed footprints on the ground when they walk (tridactyl feet). Digit V was reduced to a remnant early in theropod evolution and was gone by the late Triassic. Digit I is reduced and generally do not touch the ground, and greatly reduced in some lineages.[8] They also lack a digit V on their hands and have developed a furcula which is otherwise known as a wishbone.[6] Early neotheropods like the coelophysoids have a noticeable kink in the upper jaw known as a subnarial gap. Averostrans are some of the most derived theropods and contain

- †*Lepidus*
- †*Velocipes*
- **Neotheropoda** Bakker, 1986
  - †**Coelophysoidea**
  - **Averostriformes** Wang et al., 2026
    - †*Chuandongocoelurus*
    - †*Cryolophosaurus*
    - †*Dilophosaurus*
    - †*Dracovenator*
    - †*Gojirasaurus*?
    - †*Notatesseraeraptor*
    - †*Sarcosaurus*
    - †*Sinosaurus*
    - †*Spinostropheus*
    - †*Tachiraptor*
    - †*Zupaysaurus*
    - **Averostra** Paul, 2002
      - †**Ceratosauria**
      - **Tetanurae**

Theropods of uncertain affinity

- †*Austrocheirus*
- †*Bahariasaurus*
- †*Camarillasaurus*
- †*Chilantaisaurus*
- †*Coeluroides*
- †*Compsosuchus*
- †*Dandakosaurus*
- †*Deltadromeus*
- †*Dornraptor*
- †*Gojirasaurus*
- †*Gualicho*
- †*Kakuru*
- †*Labocania*

the Tetanurae and Ceratosauria. While some used to consider coelophysoids and ceratosaurs to be within the same group due to features such as a fused hip, later studies showed that it is more likely that these were features ancestral to neotheropods and were lost in basal tetanurans.[9] Averostrans and their close relatives are united via the complete loss of any digit V remnants, fewer teeth in the maxilla, the movement of the tooth row further down the maxilla and a lacrimal fenestra. Averostrans also share features in their hips and teeth.[10]

- †*Newtonsaurus*
- †*Orthogoniosaurus*
- †*Ostafrikasaurus*
- †*Ozraptor*
- †*Shuangbaisaurus*
- †*Szechuanosaurus*
- †*Wakinosaurus*
- †*Xuanhanosaurus*

Possible theropods

- †*Alwalkeria*
- †*Caseosaurus*
- †*Chilesaurus*
- †*Chindesaurus*
- †*Daemonosaurus*
- †*Nyasasaurus*
- †*Tawa*
- †**Herrerasauria**

Diet and teeth

Theropods exhibit a wide range of diets, from insectivores to herbivores and carnivores. Strict carnivory has always been considered the ancestral diet for theropods as a group, and a wider variety of diets was historically considered a characteristic exclusive to the avian theropods (birds). However, discoveries in the late 20th and early 21st centuries showed that a variety of diets existed even in more basal lineages.[11] All early finds of theropod fossils showed them to be primarily carnivorous. Fossilized specimens of early theropods known to scientists in the 19th and early 20th centuries all possessed sharp teeth with serrated edges for cutting flesh, and some specimens even showed direct evidence of predatory behavior. For example, a *Compsognathus longipes* fossil was found with a lizard in its stomach, and a *Velociraptor mongoliensis* specimen was found locked in combat with a *Protoceratops andrewsi* (a type of ornithischian dinosaur). This was likely caused by a sand dune blowing over the two animals mid combat, resulting in fossilization.

In the modern fauna, theropods are represented by over 11,000 species of birds, which are a group of maniraptoran theropods within the clade Avialae.

The first confirmed non-carnivorous fossil theropods found were the therizinosaurs, originally known as "segnosaurs". First thought to be prosauropods, these enigmatic dinosaurs were later proven to be highly specialized, herbivorous theropods. Therizinosaurs possessed large abdomens for processing plant food, and small heads with beaks and leaf-shaped teeth. Further study of maniraptoran theropods and their relationships showed that therizinosaurs were not the only early members of this group to abandon carnivory. Several other lineages of early maniraptorans show adaptations for an omnivorous diet, including seed-eating (some

troodontids) and insect-eating (many avialans and alvarezsaurs). Oviraptorosaurs, ornithomimosaurs and advanced troodontids were likely omnivorous as well, and some theropods (such as *Masiakasaurus knopfleri* and the spinosaurids) appear to have specialized in catching fish.[12][13]

Specimen of the troodontid *Jinfengopteryx elegans*, with seeds preserved in the stomach region

Diet is largely deduced by the tooth morphology,[14] tooth marks on bones of the prey, and gut contents. Some theropods, such as *Baryonyx*, *Lourinhanosaurus*, ornithomimosaurs, and birds, are known to use gastroliths, or gizzard-stones.

The majority of theropod teeth are blade-like, with serration on the edges,[15] called ziphodont. Others are pachydont or folidont depending on the shape of the tooth or denticles.[15] The morphology of the teeth is distinct enough to tell the major families apart,[14] which indicate different diet strategies. An investigation in July 2015 discovered that what appeared to be "cracks" in their teeth were actually folds that helped to prevent tooth breakage by strengthening individual serrations as they attacked their prey.[16] The folds helped the teeth stay in place longer, especially as theropods evolved into larger sizes and had more force in their bite.[17][18]

As a general rule, predatory theropods above between 16 and 32 kg preyed predominantly on prey as large or larger than themselves, while those below this body mass threshold preyed primarily on prey smaller than themselves. This is similar to the pattern observed in predatory mammals, where taxa above ~21.5 kg predominantly prey on animals larger than themselves, because the energy animals above this body mass threshold gain from catching small prey is insufficient to offset the bioenergetic expenditure required to capture them.[19]

Integument (skin, scales and feathers)

Fossil of an *Anchiornis*, showing large preserved feather imprints

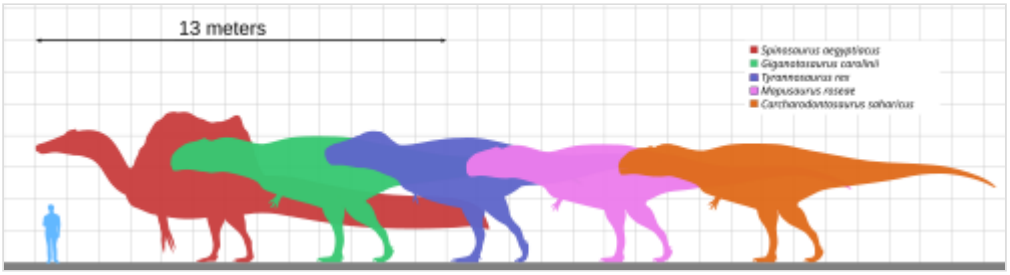
Mesozoic theropods were also very diverse in terms of skin texture and covering. Feathers or feather-like structures (filaments) are attested in most lineages of coelurosaurs (see feathered dinosaur). However, outside the coelurosaurs, feathers may have been confined to the young, smaller species, or limited parts of the animal. Many larger theropods had skin covered in small, bumpy scales. In some species, these were interspersed with larger scales. This type of skin is best known in the ceratosaur *Carnotaurus*, which has been preserved with extensive skin impressions.[20] Osteoderms, scales with a bony core, are known from *Ceratosaurus*, which was discovered with segments of osteoderms on top of its neck and tail, probably forming a continuous row in life.[21] In carnosaurs rectangular scutate scales are known from the top of the feet and the underside of the tail in *Concavenator*, and from the underside of the neck in *Allosaurus*.[22]

There is evidence of some lineages of theropods being ancestrally feathered but losing them in favor of scales in later members. In tyrannosauroids, the early members *Dilong* and *Yutyrannus* are preserved with evidence of feathers, while in the later tyrannosaurids, like *Tyrannosaurus*, *Tarbosaurus*, *Albertosaurus*, *Gorgosaurus*, and *Daspletosaurus*, there is evidence of scales, though it is unknown if they lost all feathers entirely.[23]

The coelurosaur lineages most distant from birds had feathers that were relatively short and composed of simple, possibly branching filaments.[24] Simple filaments are also seen in therizinosaurs, which also possessed large, stiffened "quill"-like feathers. More fully feathered theropods, such as dromaeosaurids, usually retain scales only on the feet. Some species may have mixed feathers elsewhere on the body as well. *Scansoriopteryx* preserved scales near the underside of the tail,[25] and *Juravenator* may have been predominantly scaly with some simple filaments interspersed.[26] On the other hand, some theropods were completely covered with feathers, such as the anchiornithid *Anchiornis*, which even had feathers on the feet and toes.[27]

Based on relationships between tooth size and skull length and also a comparison of the degree of wear of the teeth of non-avian theropods and modern lepidosaurs, it is concluded that theropods had lips that protected their teeth from the outside. Visually, the snouts of such theropods as *Daspletosaurus* had more similarities with lizards than crocodilians, which lack lips.[28]

Size



Size comparison of selected giant theropod dinosaurs – the longest (left) is *Spinosaurus aegyptiacus*, shortest (right) is *Carcharodontosaurus saharicus*.

Tyrannosaurus was for many decades the largest and best-known theropod to the general public. Since its discovery, however, a number of other, comparably-sized giant carnivorous dinosaurs have been described, including *Spinosaurus*, *Carcharodontosaurus*, and *Giganotosaurus*.[29] These large theropod dinosaurs are estimated to exceed *Tyrannosaurus* in length, especially in the case of *Spinosaurus*, though more recent studies and reconstructions show that *Tyrannosaurus*, although shorter, was more heavily built and thus still comparable to them in mass and overall size. The largest known *Tyrannosaurus* specimens such as Sue and Scotty are currently estimated to be the most massive individual theropod specimens known to science, though by only a narrow margin; a considerable bias in sample size resulting from the much greater number of adult specimens in *Tyrannosaurus* compared to all other theropods of similar mass, combined with the majority of individual adult giant theropods (even with *Tyrannosaurus*) never having entered the fossil record, mean it is difficult to judge which, if any, of these theropods was the largest by mass in reality.[30][31] There is still no clear explanation for exactly why giant theropods grew to be so massive compared to any land predators that came before and after them.

The largest extant theropod is the common ostrich, up to 2.74 m (9 ft) tall and weighing between 90 and 130 kg (200 – 290 lb).[32] The smallest non-avian theropod known from adult specimens is the anchiornithid *Anchiornis huxleyi*, at 110 grams in weight and 34 centimeters (1 ft) in length.[27] When modern birds are included, the bee hummingbird (*Mellisuga helenae*) is smallest at 1.9 g and 5.5 cm (2.2 in) long.[33][34]

An adult male bee hummingbird, the smallest known theropod and smallest living dinosaur

Recent theories propose that theropod body size shrank continuously over a period of 50 million years, from an average of 163 kilograms (359 lb) down to 0.8 kilograms (1.8 lb), eventually evolving into over 11,000 species of modern birds. This was based on evidence that theropods were the only dinosaurs to get continuously smaller, and that their skeletons changed four times as fast as those of other dinosaur species.[35][36]

Growth rates

In order to estimate the growth rates of theropods, scientists need to calculate both age and body mass of a dinosaur. Both of these measures can only be calculated through fossilized bone and tissue, so regression analysis and extant animal growth rates as proxies are used to make predictions. Fossilized bones exhibit growth rings that appear as a result of growth or seasonal changes, which can be used to approximate age at the time of death.[37] However, the amount of rings in a skeleton can vary from bone to bone, and old rings can also be lost at advanced age, so scientists need to properly control these two possibly confounding variables.

Body mass is harder to determine as bone mass only represents a small proportion of the total body mass of animals. One method is to measure the circumference of the femur, which in non-avian theropod dinosaurs has been shown to be a relatively proportional to quadrupedal mammals,[38] and use this measurement as a function of body weight, as the proportions of long bones like the femur grow proportionately with body mass.[38] The method of using extant animal bone proportion to body mass ratios to make predictions about extinct animals is known as the extant-scaling (ES) approach.[39] A second method, known as the volumetric-density (VD) approach, uses full-scale models of skeletons to make inferences about potential mass. The ES approach is better for wide-range studies including many specimens and doesn't require as much of a complete skeleton as the VD approach, but the VD approach allows scientists to better answer more physiological questions about the animal, such as locomotion and center of gravity.[39]

The current consensus is that non-avian theropods didn't exhibit a group wide growth rate, but instead had varied rates depending on their size. However, all non-avian theropods had faster growth rates than extant reptiles, even when modern reptiles are scaled up to the large size of some non-avian theropods. As body mass increases, the relative growth rate also increases. This trend may be due to the need to reach the size required for reproductive maturity.[40] For example, one of the smallest known theropods was *Microraptor zhaoianus*, which had a body mass of 200 grams, grew at a rate of approximately 0.33 grams per day.[41] A comparable reptile of the same size grows at half of this rate. The growth rates of medium-

sized non-avian theropods (100–1000 kg) approximated those of precocial birds, which are much slower than altricial birds. Large theropods (1500–3500 kg) grew even faster, similar to rates displayed by eutherian mammals.[41] The largest non-avian theropods, like *Tyrannosaurus rex*, had similar growth dynamics to the largest living land animal today, the African elephant, which is characterized by a rapid period of growth until maturity, subsequently followed by slowing growth in adulthood.[42]

Stance and gait

As a hugely diverse group of animals, the posture adopted by theropods likely varied considerably between various lineages through time.[43] All known theropods are bipedal, with the forelimbs reduced in length and specialized for a wide variety of tasks (see below). In modern birds, the body is typically held in a somewhat upright position, with the upper leg (femur) held parallel to the spine and with the forward force of locomotion generated at the knee. Scientists are not certain how far back in the theropod family tree this type of posture and locomotion extends.[43]

An ostrich walking on a road in Etosha National Park, Namibia

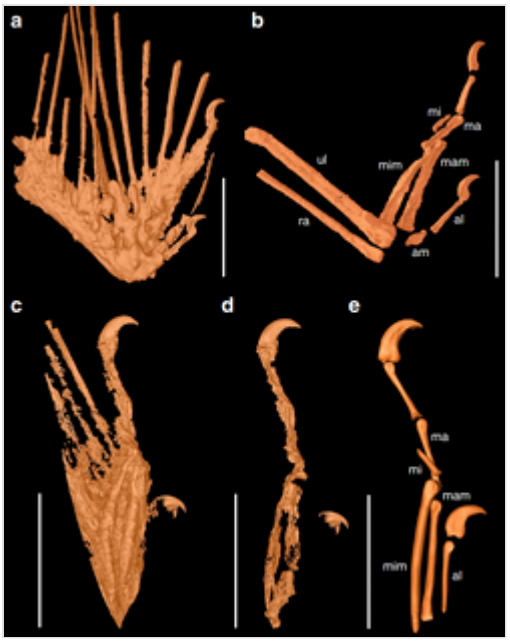
Non-avian theropods were first recognized as bipedal during the 19th century, before their relationship to birds was widely accepted. During this period, theropods such as carnosaurs and tyrannosaurids were thought to have walked with vertical femurs and spines in an upright, nearly erect posture, using their long, muscular tails as additional support in a kangaroo-like tripodal stance.[43] Beginning in the 1970s, biomechanical studies of extinct giant theropods cast doubt on this interpretation. Studies of limb bone articulation and the relative absence of trackway evidence for tail dragging suggested that, when walking, the giant, long-tailed theropods would have adopted a more horizontal posture with the tail held parallel to the ground.[43][44] However, the orientation of the legs in these species while walking remains controversial. Some studies support a traditional vertically oriented femur, at least in the largest long-tailed theropods,[44] while others suggest that the knee was normally strongly flexed in all theropods while walking, even giants like the tyrannosaurids.[45][46] It is likely that a wide range of body postures, stances, and gaits existed in the many extinct theropod groups.[43][47]

Nervous system and senses

Although rare, complete casts of theropod endocrania are known from fossils. Theropod endocrania can also be reconstructed from preserved brain cases without damaging valuable specimens by using a computed tomography scan and 3D reconstruction software. These finds are of evolutionary significance because they help document the emergence of the neurology of modern birds from that of earlier reptiles. An increase in the proportion of the brain occupied by the cerebrum seems to have occurred with the advent of the Coelurosauria and "continued throughout the evolution of maniraptorans and early birds."[48]

Studies show that theropods had very sensitive snouts. It is suggested they might have been used for temperature detection, feeding behavior, and wave detection.[49][50]

Forelimb morphology



Mummified enantiornithean wing (of an unknown genus) from Cenomanian amber from Myanmar

Shortened forelimbs in relation to hind legs were a common trait among theropods, most notably in the abelisaurids (such as *Carnotaurus*) and the tyrannosaurids (such as *Tyrannosaurus*). This trait was, however, not universal: spinosaurids had well developed forelimbs, as did many coelurosaurs. The relatively robust forelimbs of one genus, *Xuanhanosaurus*, led Dong Zhiming to suggest that the animal might have been quadrupedal.[51] However, this is no longer thought to be likely.[52]

The hands are also very different among the different groups. The most common form among non-avian theropods is an appendage consisting of three fingers; the digits I, II and III (or possibly II, III and IV), with sharp claws. Some basal theropods, like most Ceratosaurians, had four digits, and also a reduced metacarpal V (e.g. *Dilophosaurus*). The majority of tetanurans had three,[a][53] but some had even fewer.[54]

The forelimbs' scope of use is also believed to have also been different among different families. The spinosaurids could have used their powerful forelimbs to hold fish. Some small maniraptorans such as scansoriopterygids are believed to have used their forelimbs to climb in trees.[25] The wings of modern birds are used primarily for flight, though they are adapted for other purposes in certain groups. For example, aquatic birds such as penguins use their wings as flippers.

Forelimb movement

Contrary to the way theropods have often been reconstructed in art and the popular media, the range of motion of theropod forelimbs was severely limited, especially compared with the forelimb dexterity of humans and other primates.[55] Most notably, theropods and other bipedal saurischian dinosaurs (including the bipedal prosauropods) could not pronate their hands—that is, they could not rotate the forearm so that the palms faced the ground or backwards towards the legs. In humans, pronation is achieved by motion of the radius relative to the ulna (the two bones of the forearm). In saurischian dinosaurs, however, the end of the radius near the elbow was actually locked into a groove of the ulna, preventing any movement. Movement at the wrist was also limited in many species, forcing the entire forearm and hand to move as a single unit with little flexibility.[56] In theropods and prosauropods, the only way for the palm to face the ground would have been by lateral splaying of the entire forelimb, as in a bird raising its wing.[55]

In carnosaurs like *Acrocanthosaurus*, the hand itself retained a relatively high degree of flexibility, with mobile fingers. This was also true of more basal theropods, such as herrerasaurs. Coelurosaurs showed a shift in the use of the forearm, with greater flexibility at the shoulder allowing the arm to be raised towards the horizontal plane, and to even greater degrees in flying birds. However, in coelurosaurs, such as ornithomimosaurs and especially dromaeosaurids, the hand itself had lost most flexibility, with highly inflexible fingers. Dromaeosaurids and other maniraptorans also showed increased mobility at the wrist

not seen in other theropods, thanks to the presence of a specialized half-moon shaped wrist bone (the semi-lunate carpal) that allowed the whole hand to fold backward towards the forearm in the manner of modern birds.[56]

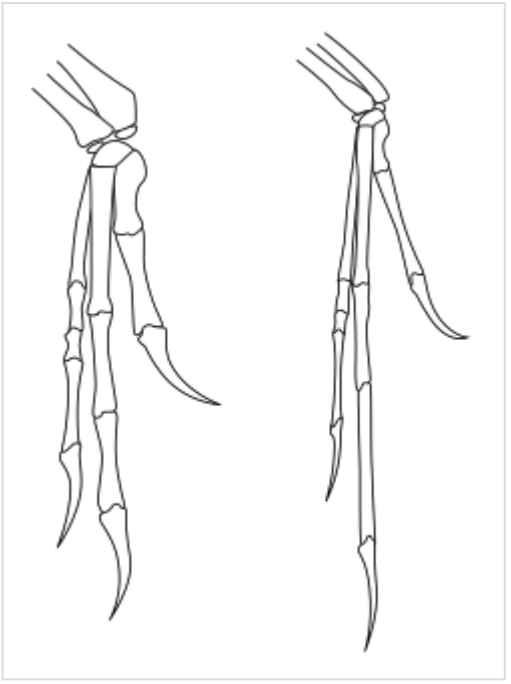


Diagram of *Deinonychus* (left) and *Archaeopteryx* (right) forelimbs illustrating wing-like posture

Paleopathology

In 2001, Ralph E. Molnar published a survey of pathologies in theropod dinosaur bone. He found pathological features in 21 genera from 10 families. Pathologies were found in theropods of all body size although they were less common in fossils of small theropods, although this may be an artifact of preservation. They are very widely represented throughout the different parts of theropod anatomy. The most common sites of preserved injury and disease in theropod dinosaurs are the ribs and tail vertebrae. Despite being abundant in ribs and vertebrae, injuries seem to be "absent... or very rare" on the bodies' primary weight supporting bones like the sacrum, femur, and tibia. The lack of preserved injuries in these bones suggests that they were selected by evolution for resistance to breakage. The least common sites of preserved injury are the cranium and forelimb, with injuries occurring in about equal frequency at each site. Most pathologies preserved in theropod fossils are the remains of injuries like fractures, pits, and punctures, often likely originating with bites. Some theropod paleopathologies seem to be evidence of infections, which tended to be confined only to small regions of the animal's body. Evidence for congenital malformities have also been found in theropod remains. Such discoveries can provide information useful for understanding the evolutionary history of the processes of biological development. Unusual fusions in cranial elements or asymmetries in the same are probably evidence that one is examining the fossils of an extremely old individual rather than a diseased one.[57]

Swimming

The trackway of a swimming theropod, the first in China of the ichnogenus named *Characichnos*, was discovered at the Feitianshan Formation in Sichuan.[58] These swim tracks support the hypothesis that theropods were adapted to swimming and capable of traversing moderately deep water. Dinosaur swim tracks are considered to be rare trace fossils, and are among a class of vertebrate swim tracks that also include those of pterosaurs and crocodylomorphs. The study described and analyzed four complete natural molds of theropod foot prints that are now stored at the Huaxia Dinosaur Tracks Research and Development Center (HDT). These dinosaur footprints were in fact claw marks, which suggest that this theropod was swimming near the surface of a river and just the tips of its toes and claws could touch the bottom. The tracks indicate a coordinated, left-right, left-right progression, which supports the proposition that theropods were well-coordinated swimmers.[58]

Evolutionary history

Possible early forms *Herrerasaurus* (large) and *Eoraptor* (small)

During the late Triassic, a number of primitive proto-theropod and theropod dinosaurs existed and evolved alongside each other.

The earliest and most primitive of the theropod dinosaurs were the carnivorous *Eodromaeus* and, possibly, the herrerasaurids of Argentina. The herrerasaurs existed during the early late Triassic (Late Carnian to Early Norian). They were found in North America and South America and possibly also India and Southern Africa. The herrerasaurs were characterised by a mosaic of primitive and advanced features. Some paleontologists have in the past considered the herrerasaurians to be members of Theropoda, while other theorized the group to be basal saurischians, and may even have evolved prior to the saurischian-ornithischian split. Cladistic analysis following the discovery of *Tawa*, another Triassic dinosaur, suggests the herrerasaurs likely were early theropods.[59]

The earliest and most primitive unambiguous theropods are the Coelophysoidea. The coelophysoids were a group of widely distributed, lightly built and potentially gregarious animals. They included small hunters like *Coelophysis* and *Camposaurus*. These successful animals continued from the Late Carnian (early Late Triassic) through to the Toarcian (late Early Jurassic). Although in the early cladistic classifications they were included under the Ceratosauria and considered a side-branch of more advanced theropods,[60] they may have been ancestral to all other theropods (which would make them a paraphyletic group).[61][62]

Neotheropoda (meaning "new theropods") is a clade that includes coelophysoids and more advanced theropod dinosaurs, and is the only group of theropods that survived the Triassic–Jurassic extinction event. Neotheropoda was named by R.T. Bakker in 1986 as a group including the relatively derived theropod subgroups Ceratosauria and Tetanurae, and excluding coelophysoids.[63] However, most later researchers have used it to denote a broader group. Neotheropoda was first defined as a clade by Paul Sereno in 1998 as *Coelophysis* plus modern birds, which includes almost all theropods except the most primitive species.[64] Dilophosauridae was formerly considered a small clade within Neotheropoda, but was later considered to be paraphyletic. By the Early Jurassic, all non-averostran neotheropods had gone extinct.[65]

Averostra (or "bird snouts") is a clade within Neotheropoda that includes most theropod dinosaurs, namely Ceratosauria and Tetanurae. It represents the only group of post-Early Jurassic theropods. One important diagnostic feature of Averostra is the absence of the fifth metacarpal. Other saurischians retained this bone, albeit in a significantly reduced form.[66]

The somewhat more advanced ceratosaurs (including *Ceratosaurus* and *Carnotaurus*) appeared during the Early Jurassic and continued through to the Late Jurassic in Laurasia. They competed alongside their more anatomically advanced tetanuran relatives and—in the form of the abelisaur lineage—lasted to the end of the Cretaceous in Gondwana.

The Tetanurae are more specialised again than the ceratosaurs. They are subdivided into the basal Megalosauroidea (alternately Spinosauroidea) and the more derived Avetheropoda. Megalosauridae were primarily Middle Jurassic to Early Cretaceous predators, and their spinosaurid relatives' remains are mostly from Early and Middle Cretaceous rocks. Avetheropoda, as their name indicates, were more closely related to birds and are again divided into the Allosauroidea (the diverse carcharodontosaurs) and the Coelurosauria (a very large and diverse dinosaur group including the birds).

Thus, during the late Jurassic, there were no fewer than four distinct lineages of theropods—ceratosaurs, megalosaurs, allosaurs, and coelurosaurs—preying on the abundance of small and large herbivorous dinosaurs. All four groups survived into the Cretaceous, and two of those—the ceratosaurs and coelurosaurs—survived to end of the period, where they were geographically separate, the ceratosaurs in Gondwana and Europe, and the coelurosaurs in Laurasia.

Of all the theropod groups, the coelurosaurs were by far the most diverse. Some coelurosaur groups that flourished during the Cretaceous were the tyrannosaurids (including *Tyrannosaurus*), the dromaeosaurids (including *Velociraptor* and *Deinonychus*, which are remarkably similar in form to one of the oldest known birds, *Archaeopteryx*),[67][68] the bird-like troodontids and oviraptorosaurs, the ornithomimosaurs (or "ostrich Dinosaurs"), the strange giant-clawed herbivorous therizinosaurs, and the avialans, which include modern birds and is the only dinosaur lineage to survive the Cretaceous–Paleogene extinction event.[69] While the roots of these various groups are found in the Middle Jurassic, they only became abundant during the Early Cretaceous. A few palaeontologists, such as Gregory S. Paul, have suggested that some or all of these advanced theropods were actually descended from flying dinosaurs or proto-birds like *Archaeopteryx* that lost the ability to fly and returned to a terrestrial habitat.[70]

The evolution of birds from other theropod dinosaurs has also been reported, with some of the linking features being the furcula (wishbone), pneumatized bones, brooding of the eggs, and (in coelurosaurs, at least) feathers.[35][36][71]

Classification

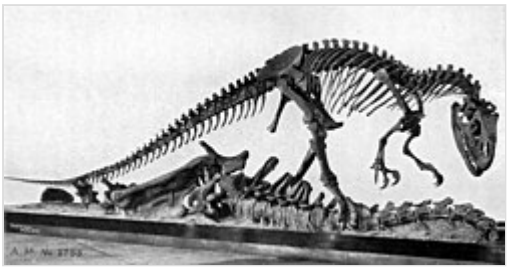
History of classification

O. C. Marsh coined the name Theropoda (meaning "beast feet") in 1881.[72] Marsh initially named Theropoda as a suborder to include the family Allosauridae, but later expanded its scope, reranking it as an order to include a wide array of "carnivorous" dinosaur families, including Megalosauridae, Compsognathidae, Ornithomimidae, Plateosauridae and Anchisauridae (now known to be herbivorous sauropodomorphs) and Hallopodidae (subsequently revealed as relatives of crocodilians). Due to the scope of Marsh's Order Theropoda, it came to replace a previous taxonomic group that Marsh's rival E. D. Cope had created in 1866 for the carnivorous dinosaurs: **Goniopoda** ("angled feet").[52]

By the early 20th century, some palaeontologists, such as Friedrich von Huene, no longer considered carnivorous dinosaurs to have formed a natural group. Huene abandoned the name "Theropoda", instead using Harry Seeley's Order Saurischia, which Huene divided into the suborders Coelurosauria and Pachypodosauria. Huene placed most of the small theropod groups into Coelurosauria, and the large theropods and prosauropods into Pachypodosauria, which he considered ancestral to the Sauropoda (prosauropods were still thought of as carnivorous at that time, owing to the incorrect association of

rauisuchian skulls and teeth with prosauropod bodies, in animals such as *Teratosaurus*).[52] Describing the first known dromaeosaurid (*Dromaeosaurus albertensis*) in 1922,[73] W. D. Matthew and Barnum Brown became the first paleontologists to exclude prosauropods from the carnivorous dinosaurs, and attempted to revive the name "Goniopoda" for that group, but other scientists did not accept either of these suggestions.[52]

Othniel Charles Marsh, who coined the name Theropoda. Photo c. 1870



Allosaurus was one of the first dinosaurs classified as a theropod.

In 1956, "Theropoda" came back into use—as a taxon containing the carnivorous dinosaurs and their descendants—when Alfred Romer reclassified the Order Saurischia into two suborders, Theropoda and Sauropoda. This basic division has survived into modern palaeontology, with the exception of, again, the Prosauropoda, which Romer included as an infraorder of theropods. Romer also maintained a division between Coelurosauria and Carnosauria (which he also ranked as infraorders). This dichotomy was upset by the discovery of *Deinonychus* and *Deinocheirus* in 1969, neither of which could be classified easily as "carnosaurs" or "coelurosaurs". In light of these and other discoveries, by the late 1970s Rinchen Barsbold had created a new series of theropod infraorders: Coelurosauria, Deinonychosauria, Oviraptorosauria, Carnosauria, Ornithomimosauria, and Deinocheirosauria.[52]

With the advent of cladistics and phylogenetic nomenclature in the 1980s, and their development in the 1990s and 2000s, a clearer picture of theropod relationships began to emerge. Jacques Gauthier named several major theropod groups in 1986, including the clade Tetanurae for one branch of a basic theropod split with another group, the Ceratosauria. As more information about the link between dinosaurs and birds came to light, the more bird-like theropods were grouped in the clade Maniraptora (also named by Gauthier in 1986[60]). These new developments also came with a recognition among most scientists that birds arose directly from maniraptoran theropods and, on the abandonment of ranks in cladistic classification, with the re-evaluation of birds as a subset of theropod dinosaurs that survived the Mesozoic extinctions and lived into the present.[52]

Major groups

The following is a simplified classification of theropod groups based on their evolutionary relationships, and organized based on the list of Mesozoic dinosaur species provided by Holtz.[1] A more detailed version can be found at dinosaur classification. The dagger (†) is used to signify groups with no living members.

Ceratosaurus, a ceratosaurid

- †Coelophysoidea (small, early predatory theropods; includes *Coelophysis* and close relatives)
- †Ceratosauria (often elaborately horned, among the dominant southern carnivores of the Late Cretaceous; includes *Ceratosaurus*, the abelisaurids, the noasaurids, and their

relatives)

- Tetanurae ("stiff tails"; includes most theropods)
  - †Megalosauroidea (early group of large carnivores, including the megalosaurids and the semi-aquatic spinosaurids)
  - †Allosauroidea (a highly successful group of large terrestrial carnivores, dominant worldwide during the Late Jurassic, Early Cretaceous and the start of the Late Cretaceous; includes *Allosaurus* and its relatives, like the carcharodontosaurids)
  - †Megaraptora (A group of medium to large, long-armed and large-clawed predatory theropods of controversial affinities, classified variously as being carnosaurs, tyrannosauroids or a separate group of coelurosaurs; among the dominant southern carnivores during the Late Cretaceous)
  - Coelurosauria (feathered theropods, with a range of body sizes and niches)
    - †Compsognathidae (early coelurosaurs with reduced forelimbs)
    - †Tyrannosauroidea (The tyrannosaurids, the dominant northern carnivores for most of the Late Cretaceous, and close relatives; often had reduced forelimbs)
    - †Ornithomimosauria ("ostrich-mimics"; mostly toothless; omnivores to possible herbivores)
    - Maniraptora ("hand snatchers"; had long, slender arms and fingers)
      - †Alvarezsauroidea (small insectivores with reduced forelimbs each bearing one enlarged claw)
      - †Therizinosauria (bipedal herbivores with large hand claws and small heads)
      - †Scansoriopterygidae (small, arboreal maniraptors with long third fingers)
      - †Oviraptorosauria (mostly toothless; their diet and lifestyle are uncertain)
      - Paraves ("near-birds"; generally carnivorous or omnivorous, sickle-claws)
        - †Anchiornithidae (small, winged protobirds)

Irritator, a spinosaurid

Mapusaurus, a carcharodontosaurid

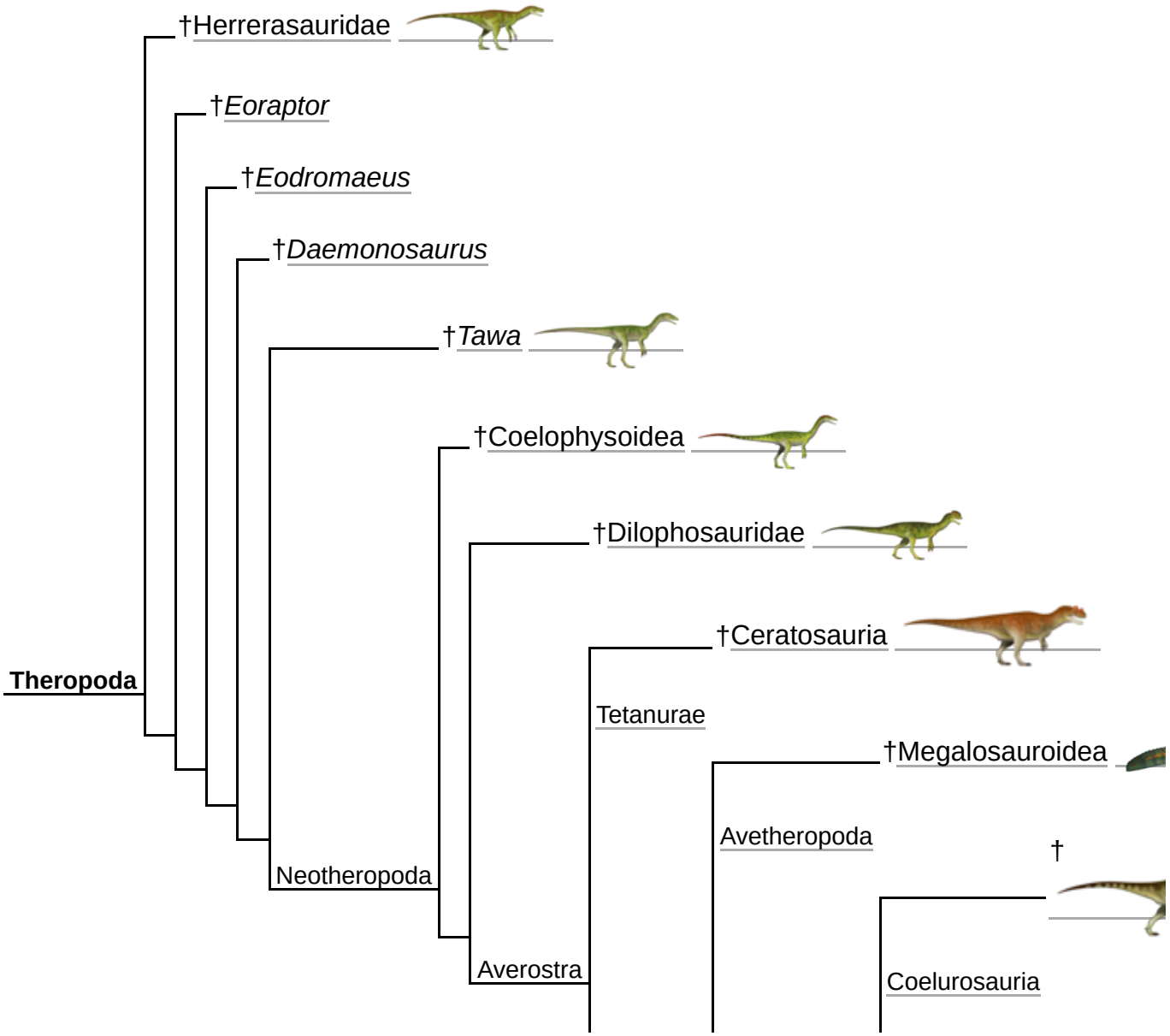
Microraptor, a dromaeosaurid

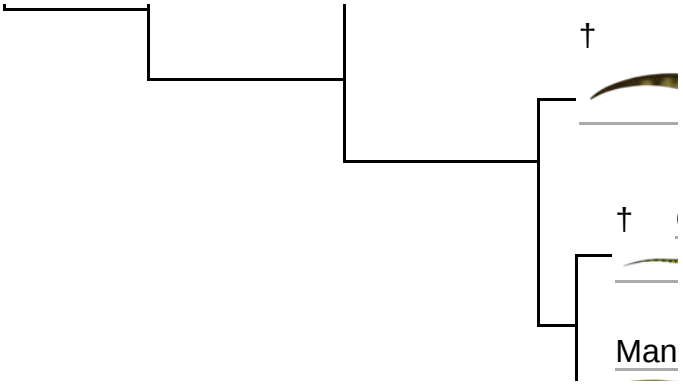
The house sparrow, an avian, is the world's most widespread extant wild theropod.[74]

- †Dromaeosauridae (small to medium-sized predatory theropods with large teeth and sickle claws)
- †Troodontidae (small, gracile predatory or omnivorous theropods with small teeth and sickle claws)
- Avialae (birds and extinct relatives)
    - †Omnivoropterygidae (large, early short-tailed avialans)
    - †Confuciusornithidae (small toothless birds)
    - †Enantiornithes (primitive tree-dwelling, flying birds)
    - Euornithes (advanced flying birds)
        - †Yanornithiformes (toothed Cretaceous Chinese birds)
        - †Hesperornithes (specialized aquatic diving birds)
        - **Aves** (modern, beaked birds and their extinct relatives)

Relationships

The following family tree illustrates a synthesis of the relationships of the major theropod groups based on various studies conducted in the 2010s.[75]





Averostra was named by G.S. Paul in 2002 as an apomorphy-based clade defined as the group including the Dromaeosauridae and other Avepoda with (an ancestor with) a promaxillary fenestra (*fenestra promaxillaris*) which can also be referred to as a maxillary fenestra,[76] an extra opening in the front outer side of the maxilla, the bone that makes up the upper jaw.[77] It was later redefined by Martin Ezcurra and Gilles Cuny in 2007 as a node-based clade containing *Ceratosaurus nasicornis*, *Allosaurus fragilis*, their last common ancestor and all its descendants.[78] Mickey Mortimer commented that Paul's original apomorphy-based definition may make Averostra a much broader clade than the *Ceratosaurus+Allosaurus* node, potentially including all of Avepoda or more.[79]

A large study of early dinosaurs by Dr Matthew G. Baron, David Norman and Paul M. Barrett (2017) published in the journal *Nature* suggested that Theropoda is actually more closely related to Ornithischia, to which it formed the sister group within the clade Ornithoscelida. This new hypothesis also recovered Herrerasauridae as the sister group to Sauropodomorpha in the redefined Saurischia and suggested that the hypercarnivore morphologies that are observed in specimens of theropods and herrerasaurids were acquired convergently.[80][81] However, this phylogeny remains controversial and additional work is being done to clarify these relationships.[82]

Footnotes

a. Some genera within Avetheropoda, however, had four digits, see " "Theropoda I" on *Avetheropoda*" (http://www.geol.umd.edu/~tholtz/G104/10422ther.htm). Department of Geology. University of Maryland. 14 July 2006.

See also

- Feathered dinosaurs
- Origin of birds
- Other major clades of dinosaur
  - Marginocephalia
  - Ornithopoda
  - Sauropodomorpha
  - Thyreophora

Dinosaurs portal
Birds portal
Amphibians portal
Reptiles portal

References

1. Holtz, Thomas R., Jr. (Winter 2011). "Appendix" (http://www.geol.umd.edu/~tholtz/dinoappendix/HoltzappendixWinter2011.pdf) (PDF). *Dinosaurs: The most complete, up-to-date encyclopedia for dinosaur lovers of all ages* (published 2012).
2. "Theropoda" (https://www.merriam-webster.com/dictionary/Theropoda). *Merriam-Webster.com Dictionary*. Merriam-Webster. OCLC 1032680871 (https://search.worldcat.org/oclc/1032680871).
3. Creisler, B. (2003). "Dinosauria translation and pronunciation guide T" (https://web.archive.org/web/20100920042319/http://dinosauria.com/dml/names/dinot.htm). *Dinosauria On-Line*. Archived from the original (http://www.dinosauria.com/dml/names/dinot.htm) on 20 September 2010. Retrieved 28 April 2025.
4. Martinez, Ricardo N.; Sereno, Paul C.; Alcober, Oscar A.; Colombi, Carina E.; Renne, Paul R.; Montañez, Isabel P.; Currie, Brian S. (14 January 2011). "A Basal Dinosaur from the Dawn of the Dinosaur Era in Southwestern Pangaea". *Science*. **331** (6014): 206–210. Bibcode:2011Sci...331..206M (https://ui.adsabs.harvard.edu/abs/2011Sci...331..206M). doi:10.1126/science.1198467 (https://doi.org/10.1126%2Fscience.1198467). hdl:11336/69202 (https://hdl.handle.net/11336%2F69202). PMID 21233386 (https://pubmed.ncbi.nlm.nih.gov/21233386).
5. Sereno, Paul C. (25 June 1999). "The Evolution of Dinosaurs". *Science*. **284** (5423): 2137–2147. doi:10.1126/science.284.5423.2137 (https://doi.org/10.1126%2Fscience.284.5423.2137). PMID 10381873 (https://pubmed.ncbi.nlm.nih.gov/10381873).
6. Novas, Fernando E.; Agnolin, Federico L.; Ezcurra, Martín D.; Temp Müller, Rodrigo; Martinelli, Agustín G.; Langer, Max C. (October 2021). "Review of the fossil record of early dinosaurs from South America, and its phylogenetic implications". *Journal of South American Earth Sciences*. **110** 103341. Bibcode:2021JSAES.11003341N (https://ui.adsabs.harvard.edu/abs/2021JSAES.11003341N). doi:10.1016/j.jsames.2021.103341 (https://doi.org/10.1016%2Fj.jsames.2021.103341). hdl:11336/167918 (https://hdl.handle.net/11336%2F167918).
7. Dingus, Lowell; Rowe, Timothy (1998). *The mistaken extinction: Dinosaur evolution and the origin of birds*. New York: W.H. Freeman. ISBN 978-0-7167-2944-0.
8. Harris, J.D. (1997). "Four-toed theropod footprints and a paleomagnetic age from the Whetstone Falls Member of the Harebell Formation (Upper Cretaceous: Maastrichtian), northwestern Wyoming" (https://cactus.utahtech.edu/jharris/Saurexallopus.pdf) (PDF). *Cretaceous Research*. **18** (1): 139. doi:10.1006/cres.1996.0053 (https://doi.org/10.1006%2Fcres.1996.0053) – via Dixie State University, St. George, UT. "(*Saurexallopus lovei*)"
9. Holtz, Thomas R.; Rey, Luis V. (2007). *Dinosaurs: the most complete, up-to-date encyclopedia for dinosaur lovers of all ages* (1st ed.). New York: Random House. ISBN 978-0-375-82419-7.
10. Smith, Nathan D.; Makovicky, Peter J.; Hammer, William R.; Currie, Philip J. (October 2007). "Osteology of Cryolophosaurus ellioti (Dinosauria: Theropoda) from the Early Jurassic of Antarctica and implications for early theropod evolution" (http://publication.plazi.org/id/FF9DFFDFFF9CFFFFFFDFCD51C053FFA0). *Zoological Journal of the Linnean Society*. **151** (2): 377–421. doi:10.1111/j.1096-3642.2007.00325.x (https://doi.org/10.1111%2Fj.1096-3642.2007.00325.x).
11. Zanno, Lindsay E.; Gillette, David D.; Albright, L. Barry; Titus, Alan L. (25 August 2010). "A new North American therizinosaurid and the role of herbivory in 'predatory' dinosaur evolution" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2817200). *Proceedings of the Royal Society B*. **276** (1672): 3505–3511. doi:10.1098/rspb.2009.1029 (https://doi.org/10.1098%2Frspb.2009.1029). PMC 2817200 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2817200). PMID 19605396 (https://pubmed.ncbi.nlm.nih.gov/19605396).

12. Longrich, Nicholas R.; Currie, Philip J. (February 2009). "*Albertonykus borealis*, a new alvarezsaur (Dinosauria: Theropoda) from the Early Maastrichtian of Alberta, Canada: Implications for the systematics and ecology of the Alvarezsauridae". *Cretaceous Research*. **30** (1): 239–252. Bibcode:2009CrRes..30..239L (https://ui.adsabs.harvard.edu/abs/2009CrRes..30..239L). doi:10.1016/j.cretres.2008.07.005 (https://doi.org/10.1016%2Fj.cretres.2008.07.005).
13. Holtz, T. R. Jr.; Brinkman, D. L.; Chandler, C. L. (1998). "Dental morphometrics and a possibly omnivorous feeding habit for the theropod dinosaur *Troodon*". *GAIA*. **15**: 159–166.
14. Hendrickx, Christophe; Mateus, Octávio (30 January 2014). "Abelisauridae (Dinosauria: Theropoda) from the Late Jurassic of Portugal and dentition-based phylogeny as a contribution for the identification of isolated theropod teeth". *Zootaxa*. **3759** (1): 1–74. Bibcode:2014Zoot.37599.1.1H (https://ui.adsabs.harvard.edu/abs/2014Zoot.37599.1.1H). doi:10.11646/zootaxa.3759.1.1 (https://doi.org/10.11646%2Fzootaxa.3759.1.1). PMID 24869965 (https://pubmed.ncbi.nlm.nih.gov/24869965).
15. Hendrickx, Christophe; Mateus, Octávio; Araújo, Ricardo (3 September 2015). "A proposed terminology of theropod teeth (Dinosauria, Saurischia)" (https://figshare.com/articles/journal_contribution/1568195). *Journal of Vertebrate Paleontology*. **35** (5) e982797. Bibcode:2015JVPal..35E2797H (https://ui.adsabs.harvard.edu/abs/2015JVPal..35E2797H). doi:10.1080/02724634.2015.982797 (https://doi.org/10.1080%2F02724634.2015.982797).
16. Geggel, Laura (28 July 2015). "One tough bite: T. rex's teeth had secret weapon" (https://www.foxnews.com/science/one-tough-bite-t-rexs-teeth-had-secret-weapon/). Fox News. Retrieved 1 August 2015.
17. "Special Serrations Gave Carnivorous Dinosaurs an Evolutionary Edge" (https://web.archive.org/web/20150818052557/http://phenomena.nationalgeographic.com/2015/08/17/special-serrations-gave-carnivorous-dinosaurs-an-evolutionary-edge/). 17 August 2015. Archived from the original (http://phenomena.nationalgeographic.com/2015/08/17/special-serrations-gave-carnivorous-dinosaurs-an-evolutionary-edge) on 18 August 2015.
18. Brink, K. S.; Reisz, Robert Rafael; LeBlanc, Aaron R. H.; Chang, R. S.; Lee, Y. C.; Chiang, C. C.; Huang, T.; Evans, David Christopher (28 July 2015). "Developmental and evolutionary novelty in the serrated teeth of theropod dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4648475). *Scientific Reports*. **5** (1) 12338. Bibcode:2015NatSR...512338B (https://ui.adsabs.harvard.edu/abs/2015NatSR...512338B). doi:10.1038/srep12338 (https://doi.org/10.1038%2Fsrep12338). PMC 4648475 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4648475). PMID 26216577 (https://pubmed.ncbi.nlm.nih.gov/26216577).
19. Codron, Daryl; Carbone, Chris; Clauss, Marcus (30 October 2013). Farke, Andrew A. (ed.). "Ecological Interactions in Dinosaur Communities: Influences of Small Offspring and Complex Ontogenetic Life Histories" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3812983). *PLOS ONE*. **8** (10) e77110. Bibcode:2013PLoSO...877110C (https://ui.adsabs.harvard.edu/abs/2013PLoSO...877110C). doi:10.1371/journal.pone.0077110 (https://doi.org/10.1371%2Fjournal.pone.0077110). ISSN 1932-6203 (https://search.worldcat.org/issn/1932-6203). PMC 3812983 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3812983). PMID 24204749 (https://pubmed.ncbi.nlm.nih.gov/24204749).
20. Hendrickx, Christophe; Bell, Phil R. (December 2021). "The scaly skin of the abelisaurid Carnotaurus sastrei (Theropoda: Ceratosauria) from the Upper Cretaceous of Patagonia". *Cretaceous Research*. **128** 104994. Bibcode:2021CrRes.12804994H (https://ui.adsabs.harvard.edu/abs/2021CrRes.12804994H). doi:10.1016/j.cretres.2021.104994 (https://doi.org/10.1016%2Fj.cretres.2021.104994). hdl:11336/148293 (https://hdl.handle.net/11336%2F148293).
21. Gilmore, Charles W. (1920). "Osteology of the carnivorous Dinosauria in the United States National Museum, with special reference to the genera Antrodemus (Allosaurus) and Ceratosaurus". *Bulletin of the United States National Museum* (110): i–159. doi:10.5479/si.03629236.110.i (https://doi.org/10.5479%2Fsi.03629236.110.i). hdl:10088/10107 (https://hdl.handle.net/10088%2F10107).

22. Hendrickx, Christophe; Bell, Phil R.; Pittman, Michael; Milner, Andrew R. C.; Cuesta, Elena; O'Connor, Jingmai; Loewen, Mark; Currie, Philip J.; Mateus, Octávio; Kaye, Thomas G.; Delcourt, Rafael (June 2022). "Morphology and distribution of scales, dermal ossifications, and other non-feather integumentary structures in non-avialan theropod dinosaurs". *Biological Reviews*. **97** (3): 960–1004. doi:10.1111/brv.12829 (https://doi.org/10.1111%2Fbrv.12829). PMID 34991180 (https://pubmed.ncbi.nlm.nih.gov/34991180).
23. Bell, Phil R.; Campione, Nicolás E.; Persons, W. Scott; Currie, Philip J.; Larson, Peter L.; Tanke, Darren H.; Bakker, Robert T. (7 June 2017). "Tyrannosauroid integument reveals conflicting patterns of gigantism and feather evolution" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5493735). *Biology Letters*. **13** (6) 20170092. doi:10.1098/rsbl.2017.0092 (https://doi.org/10.1098%2Frsbl.2017.0092). PMC 5493735 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5493735). PMID 28592520 (https://pubmed.ncbi.nlm.nih.gov/28592520).
24. Göhlich, Ursula B.; Chiappe, Luis M. (16 March 2006). "A new carnivorous dinosaur from the Late Jurassic Solnhofen archipelago". *Nature*. **440** (7082): 329–332. Bibcode:2006Natur.440..329G (https://ui.adsabs.harvard.edu/abs/2006Natur.440..329G). doi:10.1038/nature04579 (https://doi.org/10.1038%2Fnature04579). PMID 16541071 (https://pubmed.ncbi.nlm.nih.gov/16541071).
25. Czerkas, S.A.; Yuan, C. (2002). "An arboreal maniraptoran from northeast China" (http://www.dinosaur-museum.org/featheredinosaurs/arboreal_maniraptoran.pdf) (PDF). In Czerkas, S.J. (ed.). *Feathered Dinosaurs and the Origin of Flight*. The Dinosaur Museum Journal. Vol. 1. Blanding, UT: The Dinosaur Museum. pp. 63–95.
26. Goehlich, U.B.; Tischlinger, H.; Chiappe, L.M. (2006). "*Juraventaor starki* (Reptilia, Theropoda) ein nuer Raubdinosaurier aus dem Oberjura der Suedlichen Frankenalb (Sueddeutschland): Skelettanatomie und Wiechteilbefunde". *Archaeopteryx*. **24**: 1–26.
27. Xu, Xing; Zhao, Qi; Norell, Mark; Sullivan, Corwin; Hone, David; Erickson, Gregory; Wang, XiaoLin; Han, FengLu; Guo, Yu (February 2009). "A new feathered maniraptoran dinosaur fossil that fills a morphological gap in avian origin". *Science Bulletin*. **54** (3): 430–435. Bibcode:2009SciBu..54..430X (https://ui.adsabs.harvard.edu/abs/2009SciBu..54..430X). doi:10.1007/s11434-009-0009-6 (https://doi.org/10.1007%2Fs11434-009-0009-6).
28. Cullen, Thomas M.; Larson, Derek William; Witton, Mark P.; Scott, Diane; Maho, Tea; Brink, Kirstin S.; et al. (30 March 2023). "Theropod dinosaur facial reconstruction and the importance of soft tissues in paleobiology" (https://doi.org/10.1126%2Fscience.abo7877). *Science*. **379** (6639): 1348–1352. Bibcode:2023Sci...379.1348C (https://ui.adsabs.harvard.edu/abs/2023Sci...379.1348C). doi:10.1126/science.abo7877 (https://doi.org/10.1126%2Fscience.abo7877). PMID 36996202 (https://pubmed.ncbi.nlm.nih.gov/36996202).
29. Therrien, F.; Henderson, D. M. (2007). "My theropod is bigger than yours...or not: estimating body size from skull length in theropods". *Journal of Vertebrate Paleontology*. **27** (1): 108–115. doi:10.1671/0272-4634(2007)27[108:MTIBTY]2.0.CO;2 (https://doi.org/10.1671%2F0272-4634%282007%2927%5B108%3AMTIBTY%5D2.0.CO%3B2). S2CID 86025320 (https://api.semanticscholar.org/CorpusID:86025320).
30. Mallon, Jordan C.; Hone, David W. E. (24 July 2024). "Estimation of maximum body size in fossil species: A case study using *Tyrannosaurus rex*" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11267449). *Ecology and Evolution*. **14** (7) 11658. Bibcode:2024EcoEv..1411658M (https://ui.adsabs.harvard.edu/abs/2024EcoEv..1411658M). doi:10.1002/ece3.11658 (https://doi.org/10.1002%2Fece3.11658). ISSN 2045-7758 (https://search.worldcat.org/issn/2045-7758). PMC 11267449 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11267449). PMID 39050661 (https://pubmed.ncbi.nlm.nih.gov/39050661).
31. Witton, Mark Paul (2025). *King Tyrant: A Natural History of Tyrannosaurus Rex*. Princeton University Press. ISBN 978-0-691-24558-4.
32. "Ostrich" (http://www.awf.org/wildlife-conservation/ostrich). *African Wildlife Foundation*. Retrieved 28 October 2020.

33. Brusatte, Stephen L.; O'Connor, Jingmai K.; Jarvis, Erich D. (October 2015). "The Origin and Diversification of Birds" (https://doi.org/10.1016%2Fj.cub.2015.08.003). *Current Biology*. **25** (19): R888–R898. Bibcode:2015CBio...25.R888B (https://ui.adsabs.harvard.edu/abs/2015CBio...25.R888B). doi:10.1016/j.cub.2015.08.003 (https://doi.org/10.1016%2Fj.cub.2015.08.003). hdl:10161/11144 (https://hdl.handle.net/10161%2F11144). PMID 26439352 (https://pubmed.ncbi.nlm.nih.gov/26439352).
34. Hendry, Lisa (2023). "Are birds the only surviving dinosaurs?" (https://www.nhm.ac.uk/discover/why-are-birds-the-only-surviving-dinosaurs.html). The Trustees of The Natural History Museum, London. Retrieved 14 April 2023.
35. Borenstein, Seth (31 July 2014). "Study traces dinosaur evolution into early birds" (https://web.archive.org/web/20181023110310/http://apnews.excite.com/article/20140731/us-sci-shrinking-dinosaurs-a5c053f221.html). *AP News*. Archived from the original (http://apnews.excite.com/article/20140731/us-sci-shrinking-dinosaurs-a5c053f221.html) on 23 October 2018. Retrieved 3 August 2014.
36. Zoe Gough (31 July 2014). "Dinosaurs 'shrank' regularly to become birds" (https://www.bbc.co.uk/nature/28563682). BBC.
37. Padian, Kevin; de Ricqlès, Armand J.; Horner, John R. (July 2001). "Dinosaurian growth rates and bird origins". *Nature*. **412** (6845): 405–408. Bibcode:2001Natur.412..405P (https://ui.adsabs.harvard.edu/abs/2001Natur.412..405P). doi:10.1038/35086500 (https://doi.org/10.1038%2F35086500). PMID 11473307 (https://pubmed.ncbi.nlm.nih.gov/11473307).
38. Anderson, J. F.; Hall-Martin, A.; Russell, D. A. (September 1985). "Long-bone circumference and weight in mammals, birds and dinosaurs". *Journal of Zoology*. **207** (1): 53–61. doi:10.1111/j.1469-7998.1985.tb04915.x (https://doi.org/10.1111%2Fj.1469-7998.1985.tb04915.x).
39. Campione, Nicolás E.; Evans, David C. (2020). "The accuracy and precision of body mass estimation in non-avian dinosaurs". *Biological Reviews*. **95** (6): 1759–1797. Bibcode:2020BioRv..95.1759C (https://ui.adsabs.harvard.edu/abs/2020BioRv..95.1759C). doi:10.1111/brv.12638 (https://doi.org/10.1111%2Fbrv.12638). PMID 32869488 (https://pubmed.ncbi.nlm.nih.gov/32869488).
40. Lee, Andrew H.; Werning, Sarah (15 January 2008). "Sexual maturity in growing dinosaurs does not fit reptilian growth models" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2206579). *Proceedings of the National Academy of Sciences*. **105** (2): 582–587. Bibcode:2008PNAS..105..582L (https://ui.adsabs.harvard.edu/abs/2008PNAS..105..582L). doi:10.1073/pnas.0708903105 (https://doi.org/10.1073%2Fpnas.0708903105). PMC 2206579 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2206579). PMID 18195356 (https://pubmed.ncbi.nlm.nih.gov/18195356).
41. Erickson, Gregory M.; Rogers, Kristina Curry; Yerby, Scott A. (26 July 2001). "Dinosaurian growth patterns and rapid avian growth rates". *Nature*. **412** (6845): 429–433. Bibcode:2001Natur.412..429E (https://ui.adsabs.harvard.edu/abs/2001Natur.412..429E). doi:10.1038/35086558 (https://doi.org/10.1038%2F35086558). PMID 11473315 (https://pubmed.ncbi.nlm.nih.gov/11473315). **(Erratum: doi:10.1038/nature16488 (https://doi.org/10.1038%2Fnature16488), PMID 26675731 (https://pubmed.ncbi.nlm.nih.gov/26675731), *Retraction Watch* (http://retractionwatch.com/2016/03/01/high-profile-critic-slams-nature-letters-about-dinosaur-growth-following-corrections/))**
42. Horner, John R.; Padian, Kevin (22 September 2004). "Age and growth dynamics of Tyrannosaurus rex" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1691809). *Proceedings of the Royal Society of London. Series B: Biological Sciences*. **271** (1551): 1875–1880. doi:10.1098/rspb.2004.2829 (https://doi.org/10.1098%2Frspb.2004.2829). PMC 1691809 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1691809). PMID 15347508 (https://pubmed.ncbi.nlm.nih.gov/15347508).

43. Hutchinson, J.R. (March–April 2006). "The evolution of locomotion in archosaurs" (http://doc.rero.ch/record/15518/files/PAL_E2922.pdf) (PDF). *Comptes Rendus Palevol*. **5** (3–4): 519–530. Bibcode:2006CRPal...5..519H (https://ui.adsabs.harvard.edu/abs/2006CRPal...5..519H). doi:10.1016/j.crpv.2005.09.002 (https://doi.org/10.1016%2Fj.crpv.2005.09.002).
44. Newman, B.H. (1970). "Stance and gait in the flesh-eating *Tyrannosaurus*" (http://publication.plazi.org/id/FFA0107EFFCFFFE3803CFFE12A45FFEC). *Biological Journal of the Linnean Society*. **2** (2): 119–123. doi:10.1111/j.1095-8312.1970.tb01707.x (https://doi.org/10.1111%2Fj.1095-8312.1970.tb01707.x).
45. Padian, K.; Olsen, P.E. (1989). "Ratite footprints and the stance and gait of Mesozoic theropods". In Gillette, D.D.; Lockley, M.G. (eds.). *Dinosaur Tracks and Traces*. Cambridge, UK: Cambridge University Press. pp. 231–241.
46. Paul, G.S. (1998). "Limb design, function and running performance in ostrich-mimics and tyrannosaurs". *Gaia*. **15**: 257–270.
47. Farlow, J. O.; Gatesy, S. M.; Holtz, T. R. Jr.; Hutchinson, J. R.; Robinson, J. M. (2000). "Theropod locomotion" (https://doi.org/10.1093%2Ficb%2F40.4.640). *Am. Zool*. **40** (4): 640–663. doi:10.1093/icb/40.4.640 (https://doi.org/10.1093%2Ficb%2F40.4.640).
48. "Abstract", in Chure (2001). Pg. 19.
49. Barker, Chris Tijani; Naish, Darren; Newham, Elis; Katsamenis, Orestis L.; Dyke, Gareth (16 June 2017). "Complex neuroanatomy in the rostrum of the Isle of Wight theropod Neovenator salerii" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5473926). *Scientific Reports*. **7** (1) 3749. Bibcode:2017NatSR...7.3749B (https://ui.adsabs.harvard.edu/abs/2017NatSR...7.3749B). doi:10.1038/s41598-017-03671-3 (https://doi.org/10.1038%2Fs41598-017-03671-3). PMC 5473926 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5473926). PMID 28623335 (https://pubmed.ncbi.nlm.nih.gov/28623335).
50. Kawabe, Soichiro; Hattori, Soki (3 July 2022). "Complex neurovascular system in the dentary of Tyrannosaurus". *Historical Biology*. **34** (7): 1137–1145. Bibcode:2022HBio...34.1137K (https://ui.adsabs.harvard.edu/abs/2022HBio...34.1137K). doi:10.1080/08912963.2021.1965137 (https://doi.org/10.1080%2F08912963.2021.1965137).
51. Dong, Z. (1984). "A new theropod dinosaur from the Middle Jurassic of Sichuan Basin". *Vertebrata PalAsiatica*. **22** (3): 213–218.
52. Rauhut, O.W. (2003). *The Interrelationships and Evolution of Basal Theropod Dinosaurs*. Blackwell Publishing. ISBN 0-901702-79-X.
53. Barta, D.E.; Nesbitt, S.J.; Norell, M.A. (2017). "The evolution of the manus of early theropod dinosaurs is characterized by high inter- and intraspecific variation" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5735062). *Journal of Anatomy*. **232** (1): 80–104. doi:10.1111/joa.12719 (https://doi.org/10.1111%2Fjoa.12719). PMC 5735062 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5735062). PMID 29114853 (https://pubmed.ncbi.nlm.nih.gov/29114853).
54. "The first single-fingered dinosaur" (https://phys.org/news/2011-01-single-fingered-dinosaur.html). *phys.org*. January 2011.
55. Carpenter, Kenneth (June 2002). "Forelimb biomechanics of nonavian theropod dinosaurs in predation". *Senckenbergiana Lethaea*. **82** (1): 59–75. doi:10.1007/BF03043773 (https://doi.org/10.1007%2FBF03043773).
56. Senter, Phil; Robins, James H. (July 2005). "Range of motion in the forelimb of the theropod dinosaur *Acrocanthosaurus atokensis*, and implications for predatory behaviour" (https://doi.org/10.1017%2FS0952836905006989). *Journal of Zoology*. **266** (3): 307–318. Bibcode:2005JZoo..266..307S (https://ui.adsabs.harvard.edu/abs/2005JZoo..266..307S). doi:10.1017/S0952836905006989 (https://doi.org/10.1017%2FS0952836905006989).
57. Molnar, R.E. (2001). "Theropod paleopathology: A literature survey". In Tanke, D.H.; Carpenter, K. (eds.). *Mesozoic Vertebrate Life*. Indiana University Press. pp. 337–363.

58. Xing, L.D.; Lockley, M.G.; Zhang, J.P.; et al. (2013). "A new Early Cretaceous dinosaur track assemblage and the first definite non-avian theropod swim trackway from China" (https://doi.org/10.1007%2Fs11434-013-5802-6). *Chin Sci Bull*. **58** (19): 2370–2378. Bibcode:2013ChSBu..58.2370X (https://ui.adsabs.harvard.edu/abs/2013ChSBu..58.2370X). doi:10.1007/s11434-013-5802-6 (https://doi.org/10.1007%2Fs11434-013-5802-6).
59. Nesbitt, S.J.; Smith, N.D.; Irmis, R.B.; Turner, A.H.; Downs, A. & Norell, M.A. (11 December 2009). "A complete skeleton of a Late Triassic saurischian and the early evolution of dinosaurs". *Science*. **326** (5959): 1530–1533. Bibcode:2009Sci...326.1530N (https://ui.adsabs.harvard.edu/abs/2009Sci...326.1530N). doi:10.1126/science.1180350 (https://doi.org/10.1126%2Fscience.1180350). PMID 20007898 (https://pubmed.ncbi.nlm.nih.gov/20007898).
60. Rowe, T. & Gauthier, J. (1990). "Ceratosauria". In Weishampel, D.B.; Dodson, P. & Osmólska, H. (eds.). *The Dinosauria*. Berkeley, CA / Los Angeles, CA / Oxford, UK: University of California Press. pp. 151–168.
61. Mortimer, M. (4 July 2001). "Rauhut's Thesis" (http://dml.cmnh.org/2001Jul/msg00110.html). *dml.cmnh.org*. Dinosaur Mailing List Archives.
62. Carrano, Matthew T.; Sampson, Scott D.; Forster, Catherine A. (19 September 2002). "The osteology of *Masiakasaurus knopfleri*, a small abelisauroid (Dinosauria: Theropoda) from the Late Cretaceous of Madagascar". *Journal of Vertebrate Paleontology*. **22** (3): 510–534. doi:10.1671/0272-4634(2002)022[0510:TOOMKA]2.0.CO;2 (https://doi.org/10.1671%2F0272-4634%282002%29022%5B0510%3ATOOMKA%5D2.0.CO%3B2).
63. Bakker, R.T. (1986). *The Dinosaur Heresies* (http://doc.rero.ch/record/232376/files/PAL_E1363.pdf) (PDF). New York, NY: William Morrow – via doc.rero.ch.
64. Sereno, P. (1998). "A rationale for phylogenetic definitions, with application to the higher-level taxonomy of Dinosauria". Abhandlungen. *Neues Jahrbuch für Geologie und Paläontologie* (annual). **210**: 41–83. doi:10.1127/njgpa/210/1998/41 (https://doi.org/10.1127%2Fnjgpa%2F210%2F1998%2F41).
65. Marsh, Adam D.; Rowe, Timothy B. (2020). "A comprehensive anatomical and phylogenetic evaluation of Dilophosaurus wetherilli (Dinosauria, Theropoda) with descriptions of new specimens from the Kayenta Formation of northern Arizona" (https://doi.org/10.1017%2Fjpa.2020.14). *Journal of Paleontology*. **94** (S78): 1–103. Bibcode:2020JPal...94S...1M (https://ui.adsabs.harvard.edu/abs/2020JPal...94S...1M). doi:10.1017/jpa.2020.14 (https://doi.org/10.1017%2Fjpa.2020.14).
66. Sasso, Cristiano Dal; Maganuco, Simone; Cau, Andrea (19 December 2018). "The oldest ceratosaurian (Dinosauria: Theropoda), from the Lower Jurassic of Italy, sheds light on the evolution of the three-fingered hand of birds" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6304160). *PeerJ*. **6** e5976. Bibcode:2018PeerJ...6e5976D (https://ui.adsabs.harvard.edu/abs/2018PeerJ...6e5976D). doi:10.7717/peerj.5976 (https://doi.org/10.7717%2Fpeerj.5976). PMC 6304160 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6304160). PMID 30588396 (https://pubmed.ncbi.nlm.nih.gov/30588396).
67. Ostrom, J.H. (1969). "Osteology of *Deinonychus antirrhopus*, an unusual theropod from the Lower Cretaceous of Montana". *Peabody Museum Natural History Bulletin*. **30**: 1–165.
68. Paul, G.S. (1988). *Predatory Dinosaurs of the World*. New York, NY: Simon and Schuster Co. ISBN 0-671-61946-2.
69. Dingus, L.; Rowe, T. (1998). *The Mistaken Extinction: Dinosaur evolution and the origin of birds*. Freeman.
70. Paul, G.S. (2002). *Dinosaurs of the Air: The evolution and loss of flight in dinosaurs and birds*. Baltimore, MD: Johns Hopkins University Press. ISBN 0-8018-6763-0.
71. Lee, MichaelS.Y.; Cau, Andrea; Naish, Darren; Dyke, Gareth J. (August 2014). "Sustained miniaturization and anatomical innovation in the dinosaurian ancestors of birds". *Science*. **345** (6196): 562–566. Bibcode:2014Sci...345..562L (https://ui.adsabs.harvard.edu/abs/2014Sci...345..562L). doi:10.1126/science.1252243 (https://doi.org/10.1126%2Fscience.1252243). PMID 25082702 (https://pubmed.ncbi.nlm.nih.gov/25082702).

72. Marsh, O.C. (1881). "Principal characters of American Jurassic dinosaurs. Part V". *The American Journal of Science and Arts*. 3. **21** (125): 417–423. Bibcode:1881AmJS...21..417M (https://ui.adsabs.harvard.edu/abs/1881AmJS...21..417M). doi:10.2475/ajs.s3-21.125.417 (https://doi.org/10.2475%2Fajs.s3-21.125.417).
73. Matthew, W.D.; Brown, B. (1922). "The family Deinodontidae, with notice of a new genus from the Cretaceous of Alberta". *Bulletin of the American Museum of Natural History*. **46**: 367–385.
74. Anderson, Ted R. (2006). Biology of the Ubiquitous House Sparrow: from Genes to Populations. Oxford: Oxford University Press. ISBN 0-19-530411-X.
75. Hendrickx, C.; Hartman, S.A.; Mateus, O. (2015). "An overview of non-avian theropod discoveries and classification". *PalArch's Journal of Vertebrate Palaeontology*. **12** (1): 1–73.
76. Carr, Thomas (22 October 2013). "Osteology VIII: The antorbital cavity" (https://tyrannosauroideacentral.blogspot.com/2013/10/osteology-viii-antorbital-cavity.html). *Tyrannosauroidea central*. Retrieved 2 April 2024.
77. Paul, G.S. (2002). *Dinosaurs of the Air* (https://books.google.com/books?id=OUwXzD3iihAC). Baltimore, MD / London, UK: The Johns Hopkins University Press. ISBN 978-0-8018-6763-7 – via Google books.
78. Ezcurra, M.D.; Cuny, G. (2007). "The coelophysoid *Lophostropheus airelensis*, gen. nov.: A review of the systematics of *"Liliensternus" airelensis* from the Triassic-Jurassic outcrops of Normandy (France)" (https://www.researchgate.net/publication/216007083). *Journal of Vertebrate Paleontology*. **27** (1): 73–86. doi:10.1671/0272-4634(2007)27[73:TCLAGN]2.0.CO;2 (https://doi.org/10.1671%2F0272-4634%282007%2927%5B73%3ATCLAGN%5D2.0.CO%3B2) – via researchgate.net.
79. Mortimer, Mickey (15 November 2010). "Averostra and Avepoda" (https://theropoddatabase.blogspot.com/2010/11/averostra-and-avepoda.html). *The Theropod Database Blog*. Retrieved 6 September 2020.
80. Baron, Matthew G.; Norman, David B.; Barrett, Paul M. (23 March 2017). "A new hypothesis of dinosaur relationships and early dinosaur evolution". *Nature*. **543** (7646): 501–506. Bibcode:2017Natur.543..501B (https://ui.adsabs.harvard.edu/abs/2017Natur.543..501B). doi:10.1038/nature21700 (https://doi.org/10.1038%2Fnature21700). PMID 28332513 (https://pubmed.ncbi.nlm.nih.gov/28332513).
81. "New study shakes the roots of the dinosaur family tree" (https://www.cam.ac.uk/research/news/new-study-shakes-the-roots-of-the-dinosaur-family-tree). *cam.ac.uk*. Cambridge, UK: University of Cambridge. 22 March 2017.
82. Langer, Max C.; Ezcurra, Martín D.; Rauhut, Oliver W. M.; Benton, Michael J.; Knoll, Fabien; McPhee, Blair W.; Novas, Fernando E.; Pol, Diego; Brusatte, Stephen L. (2 November 2017). "Untangling the dinosaur family tree". *Nature*. **551** (7678): E1–E3. Bibcode:2017Natur.551E...1L (https://ui.adsabs.harvard.edu/abs/2017Natur.551E...1L). doi:10.1038/nature24011 (https://doi.org/10.1038%2Fnature24011). hdl:1983/d088dae2-c7fa-4d41-9fa2-aeebbfcd2fa3 (https://hdl.handle.net/1983%2Fd088dae2-c7fa-4d41-9fa2-aeebbfcd2fa3). PMID 29094688 (https://pubmed.ncbi.nlm.nih.gov/29094688).

External links

- Media related to Theropoda at Wikimedia Commons

Retrieved from "https://en.wikipedia.org/w/index.php?title=Theropoda&oldid=1381370558"

WIKIPEDIA 25
25 years of the free encyclopedia

Ceratosauria

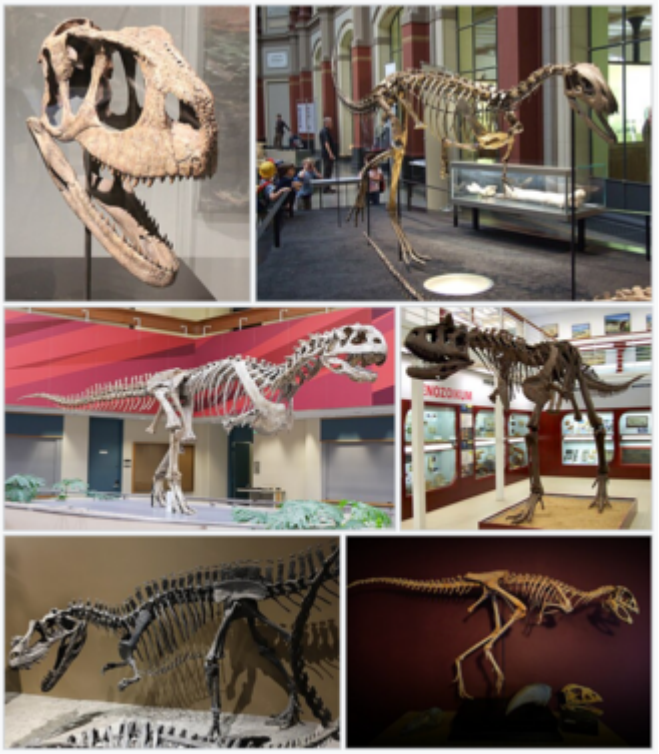
Ceratosaurs are members of the clade **Ceratosauria**, a group of dinosaurs defined as all theropods sharing a more recent common ancestor with *Ceratosaurus* than with birds. The oldest known ceratosaur, *Saltriovenator*, dates to the earliest part of the Jurassic, around 199 million years ago. Ceratosauria includes three major clades: Ceratosauridae, Noasauridae, and Abelisauridae, found primarily (though not exclusively) in the Southern Hemisphere. Originally, Ceratosauria included the above dinosaurs plus the Late Triassic to Early Jurassic Coelophysoidea and Dilophosauridae, implying a much earlier divergence of ceratosaurs from other theropods. However, most recent studies have shown that coelophysoids and dilophosaurids do not form a natural group with other ceratosaurs, and are excluded from this group.[1]

Ceratosauria derives its names from the type species, *Ceratosaurus nasicornis*, described by O.C. Marsh in 1884. A moderately large predator from the Late Jurassic, *Ceratosaurus nasicornis*, was the first ceratosaur to be discovered. Ceratosaurs are generally moderately large in size, with some exceptions like the larger *Carnotaurus* and the significantly smaller noasaurs. The major defining characteristics of Ceratosauria include a robust skull with increased ornamentation or height and a shortening of the arms.[2] Both of these characteristics are generally accentuated in later members of the group, such as the abelisaurs, whereas more basal species such as *C. nasicornis* appear more similar to other basal theropods. The highly fragmented nature of the ceratosaur fossil record means that the characteristics, relationships, and early history of Ceratosauria remain mysterious and highly debated.

Ceratosaurs

Temporal range:

Sinemurian-Maastrichtian,



Six ceratosaurs (top left to bottom right): *Rugops*, *Elaphrosaurus*, *Majungasaurus*, *Carnotaurus*, *Ceratosaurus*, *Berthasaura*

Scientific classification

Kingdom:	Animalia
Phylum:	Chordata
Class:	Reptilia
Clade:	Dinosauria
Clade:	Saurischia
Clade:	Theropoda
Clade:	Averostriformes
Clade:	Averostra
Clade:	†Ceratosauria Marsh, 1884

History of study

An illustration of the skeleton of *Ceratosaurus* by O.C. Marsh

An early life reconstruction of *Ceratosaurus*

Ceratosauria was first described by O.C. Marsh in the *American Journal of Science* in 1884. Writing about the newly discovered *C. nasicornis*, he noted the similarities between the firmly united metatarsals of *C. nasicornis* and those of *Archaeopteryx*. Since *C. nasicornis* was the only other dinosaur discovered at the time to share this trait, Marsh concluded that Ceratosauria must be placed very near *Archaeopteryx* and its related groups.[3] Marsh also named the family **Ceratosauridae** in 1884 to contain *C. nasicornis*. Since then, a number of other species have been referred to this family, mainly from the genus *Ceratosaurus*.

The idea of the Ceratosauria would be contested by Marsh's rival, Edward Drinker Cope. Cope argued that the taxon was invalid.[4] The idea of the Ceratosauria would regain some support more than thirty years later when Gilmore argued in its favor in 1920. Despite Gilmore's support, few species were added to the group following World War I, and little emphasis was placed on it. In fact, the scientific community's most

Type species

†*Ceratosaurus nasicornis*
Marsh, 1884

Subgroups

- †*Berberosaurus*
- †*Fosterovenator*
- †*Saltriovenator*
- †**Neoceratosauria** Novas, 1991
  - †**Ceratosauridae** Marsh, 1884
    - †*Ceratosaurus*
    - †*Genyodectes*
  - †**Berthasauridae** Hendrickx *et al.*, 2024
    - †*Afromimus*?
    - †*Berthasaura*
  - †**Abelisauroidea** Bonaparte and Novas, 1985
    - †*Betasuchus*
    - †*Dahalokely*
    - †*Eoabelisaurus*
    - †*Ligabueino*
    - †**Abelisauridae**
    - †**Noasauridae**

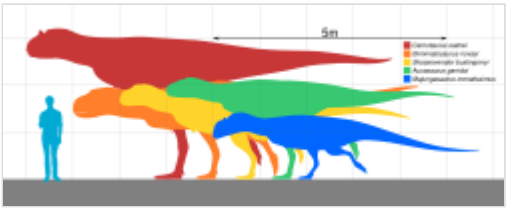
Possible ceratosaurs

- †*Austrocheirus*
- †*Bahariasaurus*
- †*Dandakosaurus*
- †*Deltadromeus*
- †*Gualicho*
- †*Ostafrikasaurus*
- †*Pandoravenator*
- †*Sarcosaurus*
- †*Spinostropheus*

common interaction with Ceratosauria throughout much of the 20th century was the disputation of its existence, performed by the likes of Romer, Lapparent, Lavocat, Colbert, and Charig amongst others.

Ceratosauria's fortune changed in 1986 when Jacques Gauthier, in an attempt to clarify the evolution of birds, grouped the majority of theropods into either Ceratosauria or Tetanurae. In Ceratosauria, he placed the ceratosaurs and coelophysoids.[5] Gauthier's paper brought Ceratosauria's use back in vogue, and by the early 1990s, Abelisauridae had also been included under Ceratosauria. The triumvirate model of ceratosaurs, coelophysoids, and abelisaurids would go unchallenged until the early 2000s. Beginning at the turn of the millennium, a large number of paleontologists began excluding coelophysoids from Ceratosauria. This view is now widely held thanks to several similarities between Ceratosauria and Tetanurae not found in coelophysoids.

Phylogeny


Size comparison of several abelisaurids

Most paleontologists have postulated that Ceratosauria split off from other theropods in the Late Triassic or earliest Jurassic. Despite this, no ceratosaurs have been discovered prior to the Early Jurassic, and even in the Middle Jurassic, species are sparse. Many scientists, such as Carrano and Sampson, have postulated the lack of specimens is due to a poor fossil record, rather than an indictment on the abundance of ceratosaurs at the time. A similarly large gap of specimens exist in the lower Cretaceous, particularly for Abelisauridae. More recent discoveries have resulted in varying phylogenetic results concerning the relationships between *Elaphrosaurus* and the derived Cretaceous noasaurids. The precise relationship between *Ceratosaurus* and the abelisaurids is also not clearly resolved.[6][7][2]

Currently, most paleontologists agree that Ceratosauria contains a slightly more exclusive clade, **Neoceratosauria**, which contains the groups **Ceratosauridae** and **Abelisauroidea**, with some variance as to which taxa are placed into basal polytomy.[2][8] Abelisauroidea is further divided into the Abelisauridae and Noasauridae, with Abelisauridae, including Carnotaurinae. Recently, Rauhut and Carrano have placed Elaphrosaurinae inside Noasauridae while simultaneously moving the previous noasaurs into Noasaurinae.[8] Into their new Noasauridae, they have uniquely included *Deltadromeus* and *Limusaurus*.

It is difficult to discern possible synapomorphies of Ceratosauridae from autapomorphies of *Ceratosaurus* because the remains of the related *Genyodectes* are so fragmentary; e.g. *Ceratosaurus* is different from other ceratosaurians by the very prominent horn on its snout; *Genyodectes*, however, was not found with a complete skull; whether it had a horn is unknown, so it cannot establish that the horn was a shared derived feature of the group. However, due to the shared similarities between the teeth of the two genera, synapomorphies have been recognized in the teeth. The synapomorphies that do exist include: overlap of the second and third premaxillary alveoli in palatal view, largest crown in subadults/adults higher than six centimeters, subquadrangular mesial denticles at two-thirds of the crown in lateral teeth.[9] Currently the only generally-recognized ceratosaurid species outside the genus *Ceratosaurus* is *Genyodectes* from the

Cretaceous or Patagonia. The taxa *Eoabelisaurus* and *Ostafrikasaurus* are also probable ceratosaurs, but it is unknown if they belong to Ceratosauridae.[10] Delcourt (2018) defined Ceratosauridae as "the most inclusive clade containing *Ceratosaurus nasicornis* but not *Carnotaurus sastrei*".[7]

The holotype of *Limusaurus*, one of the most well-known ceratosaurs

Abelisauroidea is a diverse superfamily of ceratosaurians and the sister taxon of Ceratosauridae. It is typically regarded as a Cretaceous group, though the earliest abelisaurid remains are known from the Middle Jurassic of Argentina (classified as the species *Eoabelisaurus mefi*) and possibly Madagascar (fragmentary remains of an unnamed species). Possible abelisaurid remains (an isolated left tibia, right femur, and right tibia) were also discovered in Late Jurassic Tendaguru Beds in Tanzania.

Abelisauroids flourished in the Southern Hemisphere during the Cretaceous period, but their origins can be traced back to at least the Middle Jurassic, when they had a more global distribution (the earliest known abelisauroid remains come from Australian and South American deposits dated to about 170 million years ago).[4]:109 By the Cretaceous period, abelisauroids had apparently become extinct in Asia and North America, possibly due to competition from tyrannosauroids. However, advanced abelisauroids of the family Abelisauridae persisted in the southern continents until the Cretaceous–Paleogene extinction event 66 million years ago.[11]

In an assessment of the phylogenetic position of *Eoabelisaurus*, the analysis found it as the most basal member of the Abelisauridae. Abelisaurid synapomorphies include the laterally covered lacrimal antorbital fossa, broad cervical prespinal fossae, anteroposteriorly short anterior caudal neural spines, absence of a ventral groove in the anterior caudals, presence of rudimentary centrodiapophyseal laminae in the anterior mid-caudals, reduced distal ginglymus in the manual phalanges, and the presence of a flexor depression in the pedal unguals. Alternative phylogenetic placements of *Eoabelisaurus* are significantly suboptimal, except for a slightly more basal position.[12] Noasaurids had longer arms than their relatives the abelisaurids, whose arms were tiny and diminished. Although by no means as large or specialized as the arms of advanced bird-like theropods, noasaurid arms were nevertheless capable of movement and use, possibly even for hunting in large-clawed genera such as *Noasaurus*. Some genera such as *Limusaurus* did have somewhat reduced arms and hands, but far from the extent that abelisaurids acquired. Noasaurids were also nimble and lightly built, with feet showing adaptations for running such as a long central foot bone. Noasaurids varied in size, from the small *Velocisaurus* under 5 feet (1.5 meters) long, to much larger genera such as *Elaphrosaurus* and *Deltadromeus*, which were more than 20 feet (6.1 meters) in length.[2]

The oldest known ceratosaur currently described is *Saltriovenator zanellai* which is dated to the Early Sinemurian, 199-197 Ma.[13] The origin of Ceratosauria could have been in Northern Pangea where *Saltriovenator*, its close relative *Berberosaurus*, and *Carmelopodus* footprints have been found.

The following family tree illustrates a synthesis of the relationships of the major theropod groups based on various studies conducted in the 2010s and demonstrates the position of Ceratosauria within theropods.[14]

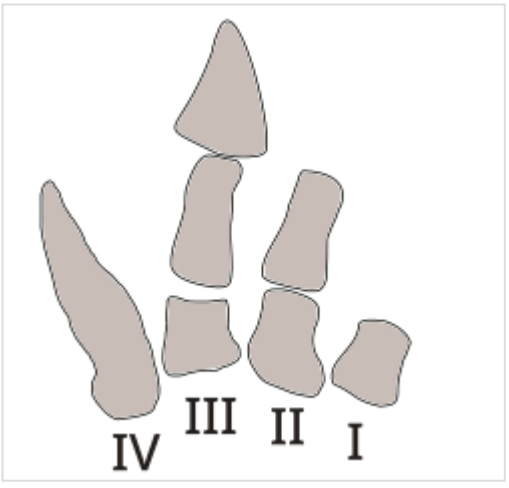
Theropoda

One of the earliest ceratosaurs, *Saltriovenator*, in its environment

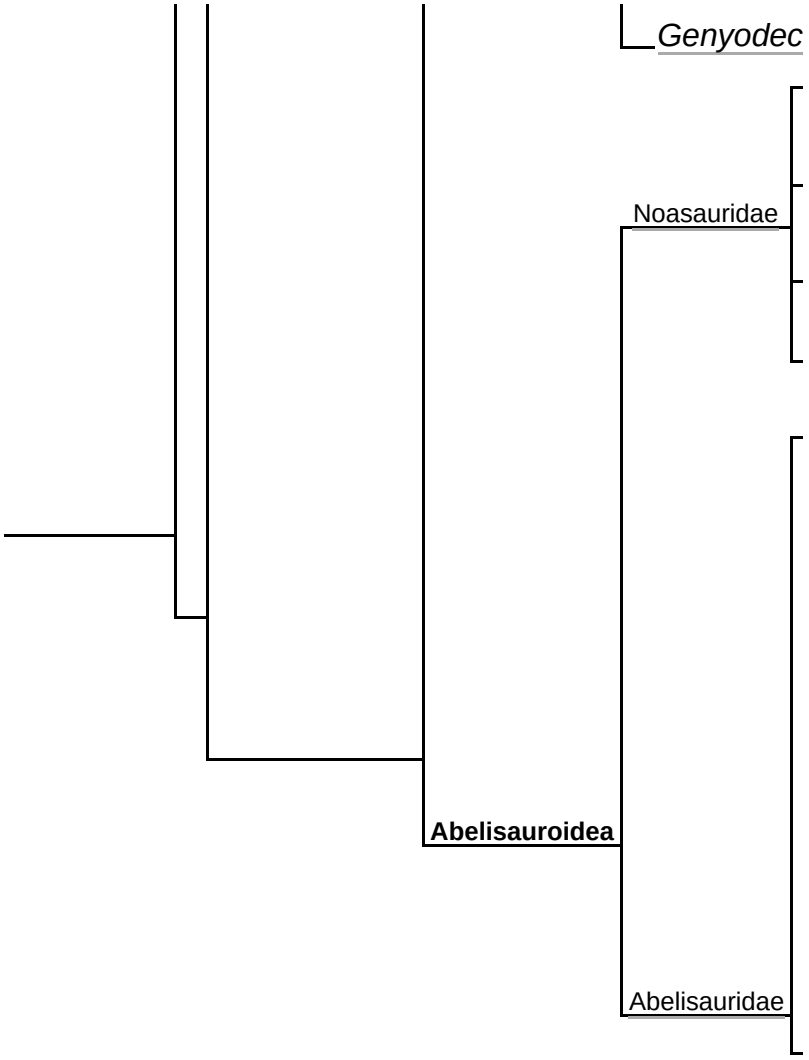
†Herrerasauridae
†*Eoraptor*
†*Eodromaeus*
†*Daemonosaurus*
†*Tawa*
†Coelophysoidea
†Dilophosauridae
†**Ceratosauria**
† Megalosauroide
Neotheropoda
Averostra
Tetanurae
† Allosauroide
Avetheropoda
Coelurosauria

The following cladogram shows the internal relationships within Ceratosauria following an analysis by Diego Pol and Oliver W. M. Rauhut, 2012.[2]

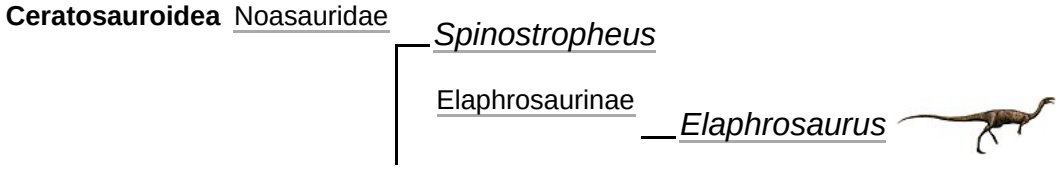
Ceratosauria
Berberosaurus
Deltadromeus
Spinostropheus
Limusaurus
Elaphrosaurus
Neoceratosauria
Ceratosau
Ceratosauridae

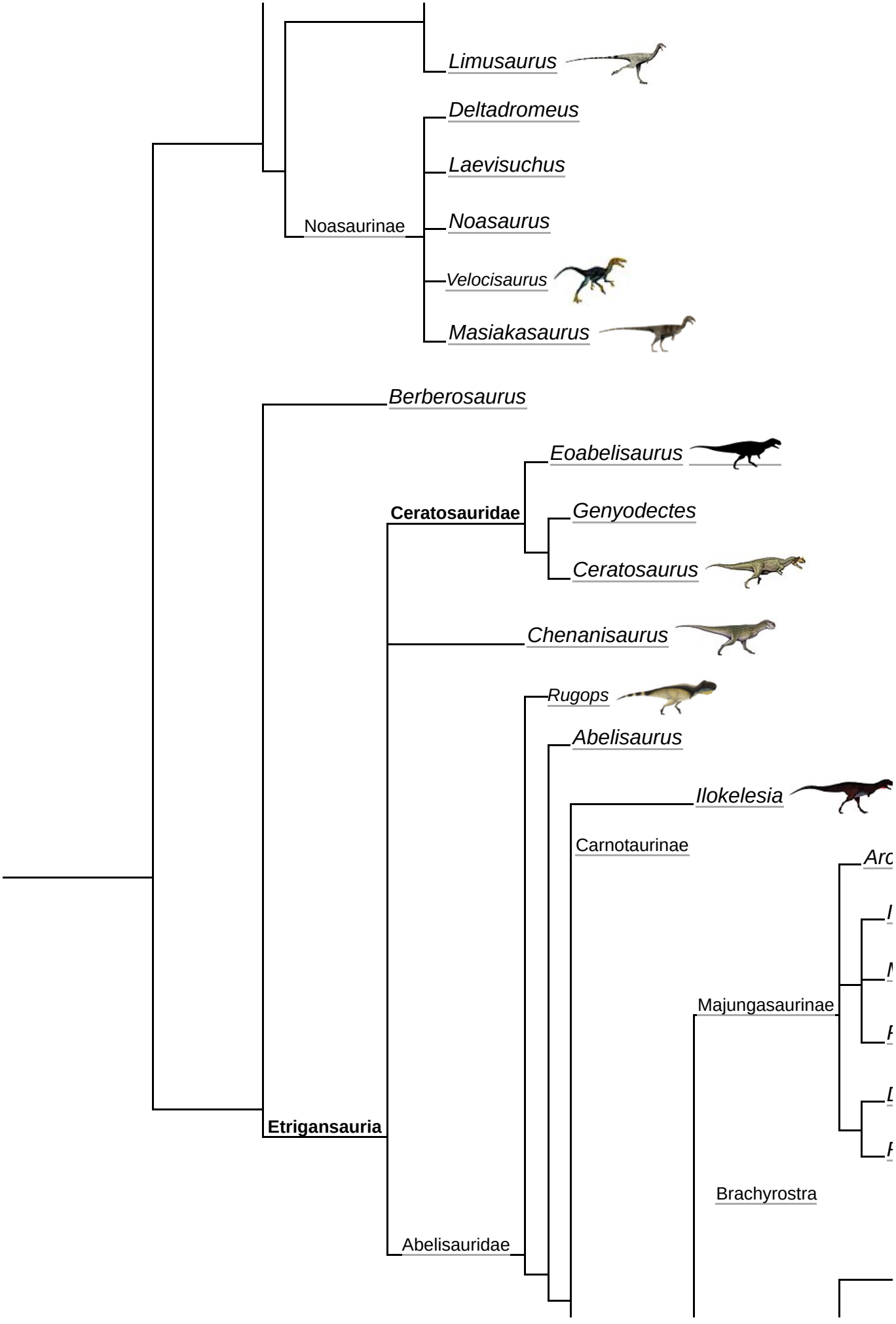


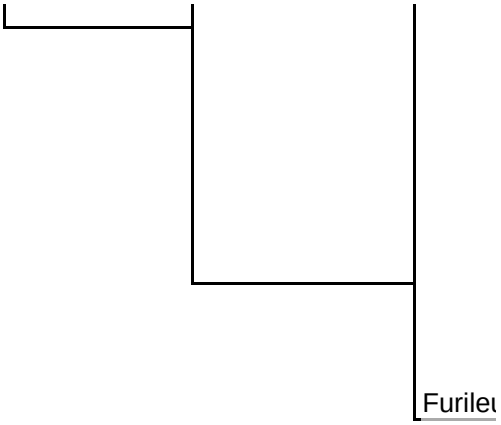
A diagram of the hand bones of *Carnotaurus*, illustrating the four-fingered condition of ceratosaurs, unlike more derived theropods, which only have three fingers



A different conclusion was reached in a 2017 paper on *Limusaurus* ontogeny. Unlike other analyses, Noasauridae was placed more basal than *Ceratosaurus*, with the latter being within Abelisauridae by definition.[15] This was later expanded on in a 2018 paper on ceratosaur paleobiology, which named a new clade Etrigansauria, which contained the families Abelisauridae and Ceratosauridae.[7] The following cladogram is a consensus tree of the latest phylogenies shown in the paper.




Limusaurus
Deltadromeus
Laevisuchus
Noasaurinae
Noasaurus
Velocisaurus
Masiakasaurus
Berberosaurus
Eoabelisaurus
Ceratosauridae
Genyodectes
Ceratosaurus
Chenanisaurus
Rugops
Abelisaurus
Ilokelesia
Carnotaurinae
Majungasaurinae
Brachyrostra
Etrigansauria
Abelisauridae



Paleobiology

Anatomy

The skull of *Masiakasaurus*, exemplifying the unusual tooth structure of some noasaurids

Some of the defining characteristics of Ceratosauria include an increase in height and ornamentation of the skull, as well as a shortening of the forelimbs. Likewise, ceratosaurs fused their ilium, ischium, and pubis together, as well as the astragalus and calcaneum.[4] For less derived members of the group, such as *C. nasicornis*, traits such as raising of the skull and shortening of the forelimbs were not as noticeable. The skull of *C. nasicornis* was rather similar to the basal theropod mold, with a distinguishing nasal crest to go along with lacrimal crests similar to the contemporary *Allosaurus*. *C. nasicornis* had larger teeth than *Allosaurus*, and some paleontologists postulate that it would have had a difficult time attacking larger prey. Abelisaurids, however, carried many of these defining traits to their extremes. Most abelisaurids had largely shortened forelimbs, with *Carnotaurus* having shrunk them further than any large theropod.[16] After analyzing the features of the newly discovered *Rugops primus*, Paul Sereno has postulated that many of these abelisaurid features may lend themselves to scavenging.[17] Despite the huge reduction in size, no taxa in Ceratosauria ever lost a digit or any critical elements of the forelimb. Some joint variation has also been observed in Ceratosauria, and it has been postulated that they may have had better shoulder mobility than other large theropods.[1]

There are two known types of *Ceratosaurus* teeth: one with longitudinal ridges and the other with veined enamel. Both types of teeth have crowns with a teardrop-shaped cross section and carinae running up the middle. The cross section of the tooth's base depends on the position of the tooth in the mouth with front teeth having less symmetric cross sections.

Complete skeletons have been described only for the most advanced abelisaurids (such as *Carnotaurus* and *Aucasaurus*), making the establishment of defining features of the skeleton for the family as a whole more difficult. However, most are known from at least some skull bones, so known shared features come mainly from the skull. Many abelisaurid skull features are shared with carcharodontosaurids.[12] These shared features, along with the fact that Abelisauridae seem to have replaced Carcharodontosauridae in South America, has led to suggestions that the two groups were related. Noasaurids were considered to be distinctive abelisauroids with a peculiar "sickle claw" on the second toe of the foot, convergently developed with that of deinonychosaurians. Among noasaurids, the Argentinean genera *Noasaurus* (Later Cretaceous) and *Ligabueino* (Early Cretaceous) are known from incomplete specimens, including disarticulated non-ungual phalanges and in *Noasaurus*, a claw. A detailed overview of these elements indicates that the supposed raptorial claw of the second pedal digit actually belongs to the first or second finger of the manus, and the putative pedal non-ungual phalanges or both genera also pertain to the manus.[18]

Paleoecology

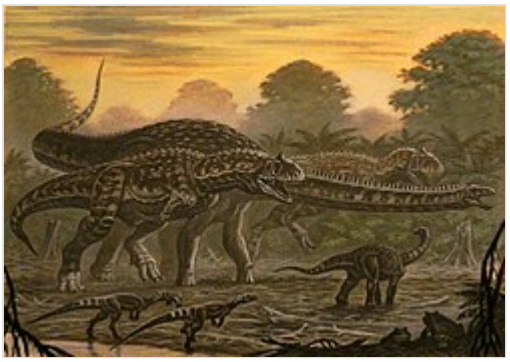
Geography

Ceratosaurs appeared to have had a global population that diverged by the early Jurassic. However, they appear to have largely disappeared from Laurasia in the Cretaceous, with those few specimens that have been discovered having been possibly reintroduced from Gondwana.[19] No confirmed specimens of ceratosaurs in North America during the Cretaceous have been found.

Abelisaurids in particular had great success in Gondwana, particularly in the Cretaceous. Some Gondwanan and Laurasian specimens have recently been found and dated to Late Jurassic, and possibly even the Middle Jurassic, greatly extending the abelisaurid timeline. Some paleontologists have postulated that a large desert may have kept abelisaurids locked in southern Gondwana until the late Jurassic.[2] Whether correlation or causation, it has been largely observed that late Cretaceous ceratosaurs were found less in areas dominated by basal tetanurans (Africa) or coelurosaurs (North America and Asia).

Diet

As with most theropods, ceratosaurs were carnivores—except for some noasaurs like *Limusaurus* and *Berthasaura*, which were omnivores or herbivores with toothless beaks.[20][21] *Ceratosaurus* has been argued to have eaten a large amount of fish and other aquatic creatures, though this has been disputed by many paleontologists.[22] Tooth marks on large animals such as *Allosaurus* indicate that *Ceratosaurus* likely utilized scavenging often.[23] The interesting jaws of the abelisaurids have drawn mixed dietary predictions. One study on *Carnotaurus* found that its bite, thanks to its shortened skull, was suited for hunting small prey, thanks to a quick, but relatively weak bite.[24] On the other hand, other groups of paleontologists have found that the bite of *Carnotaurus* was relatively powerful, and more adept at hunting and wounding large prey.[25]



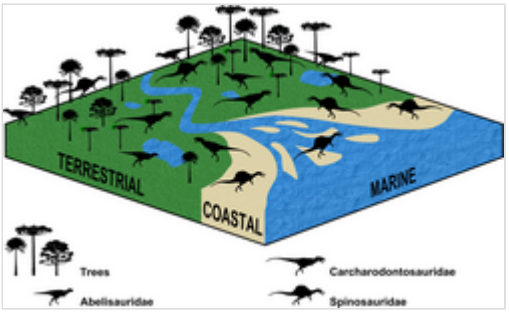
Two *Majungasaurus* (abelisaurids) hunting a sauropod with two *Masiakasaurus* (noasaurids) in the foreground

Others have postulated its skull was built for scavenging. The debate over the eating habits of ceratosaurs is quite active, particularly recently with the increase in abelisaur discoveries. Using three methods, namely a cladistic analysis performed on a dentition-based data matrix, and discriminant and cluster analyses conducted on a large dataset of theropod teeth measurements, three dental morphotypes which are confidently referred to abelisaurid theropods are identifiable. Whether the morphotypes represent different abelisaurid subclades or different positional entities within the jaw of the same abelisaurid species, is unknown. Such an identification, nevertheless, provides additional evidence of abelisaurids feeding on sauropod carcasses.[26]

Studies of *Majungasaurus* indicate that it was a much slower-growing dinosaur than other theropods, taking nearly 20 years to reach adult size. Similar studies on other abelisaurid genera indicate that this slow maturation may have been a common trait to the whole of Abelisauridae. Noasaurines are Late Cretaceous noasaurids known exclusively from southern continents and islands such as South America, Madagascar, and India. Elaphrosaurines were lightly built theropods, with small skulls and long necks and legs. If *Limusaurus* is any indication, adult elaphrosaurines were completely toothless, and their mouths were probably edged with a horny beak. It is likely that *Limusaurus* and other elaphrosaurines were primarily herbivorous as adults, due to mature *Limusaurus* specimens preserving gastroliths and chemical signatures resembling those of herbivorous dinosaurs.

Paleoenvironment

Being found in the Morrison and Tendaguru put the family Ceratosauridae in the presence of other large predators. In North America, it is likely that members of the family such as *C. nasicornis* competed with allosaurids (*A. fragilis*) for food, such as sauropods common to the region at the time. In Africa and Europe members also competed with other large predators for similar food sources. The presence of *C. nasicornis* at the Cleveland-Lloyd Dinosaur Quarry along with the remains of several allosaurids is a good indication of just how close members of this family and other predators coexisted.



A diagram of hypothesized theropod ecologies in Gondwana during the Cretaceous, with ceratosaurian abelisaurs featuring prominently

Most abelisauroid ceratosaurs were found in Madagascar, Asia, or sometimes in Africa. Abelisauridae thrived during the Cretaceous period on the ancient southern supercontinent of Gondwana, and today their fossil remains are found on the modern continents of Africa and South America, as well as on the Indian subcontinent and the island of Madagascar. In Madagascar, *Majungasaurus* was discovered by French paleontologist Charles Depéret. *Majungasaurus* was the most common abelisauroid which we know. In South America, many abelisauroids such as *Skorpiovenator*, *Quilmesaurus*, *Aucasarus*, *Ilokelesia*, and *Pyconemosaurus* are known. *Kurupi itaata* represents the first formally named vertebrate of the Marília Formation (Bauru Group, Bauru Basin) and one of the few theropod records for the Maastrichtian of the Bauru Basin. Its abelisaurid affinities are well established

based on the anatomy of the pelvis and anterior caudal vertebrae; however, closer relationships with other abelisaurids are still unclear. The specimens provide new information on abelisauroids, which are still poorly known in the Brazilian fossil record, and on the distribution of this group of theropod dinosaurs in South America.[27] These discoveries indicate that abelisauroids were the most common large predatory dinosaurs where they lived.[28]

See also

- Gondwana
- Timeline of ceratosaur research

References

1. Carrano, Matthew T.; Sampson, Scott D. (January 1, 2008). "The Phylogeny of Ceratosauria (Dinosauria: Theropoda)". *Journal of Systematic Palaeontology*. **6** (2): 183–236. Bibcode:2008JSPal...6..183C (https://ui.adsabs.harvard.edu/abs/2008JSPal...6..183C). doi:10.1017/S1477201907002246 (https://doi.org/10.1017%2FS1477201907002246). ISSN 1477-2019 (https://search.worldcat.org/issn/1477-2019). S2CID 30068953 (https://api.semanticscholar.org/CorpusID:30068953).
2. Diego Pol & Oliver W. M. Rauhut (2012). "A Middle Jurassic abelisaurid from Patagonia and the early diversification of theropod dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3385738). *Proceedings of the Royal Society B: Biological Sciences*. **279** (1804): 3170–5. doi:10.1098/rspb.2012.0660 (https://doi.org/10.1098%2Frspb.2012.0660). PMC 3385738 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3385738). PMID 22628475 (https://pubmed.ncbi.nlm.nih.gov/22628475).
3. Marsh, O. C. (1884). "On the united metatarsal bones of Ceratosaurus" (https://zenodo.org/record/1450068). *American Journal of Science*. s3-28 (164): 161–162. Bibcode:1884AmJS...28..161M (https://ui.adsabs.harvard.edu/abs/1884AmJS...28..161M). doi:10.2475/ajs.s3-28.164.161 (https://doi.org/10.2475%2Fajs.s3-28.164.161). S2CID 131737380 (https://api.semanticscholar.org/CorpusID:131737380). Archived (https://web.archive.org/web/20200624222814/https://zenodo.org/record/1450068) from the original on June 24, 2020. Retrieved December 3, 2019.
4. Weishampel, David B.; Peter, Dodson; Osmólska, Halszka (2007). *The Dinosauria*. University of California Press. ISBN 978-0-520-24209-8. OCLC 493366196 (https://search.worldcat.org/oclc/493366196).
5. "Saurischian monophyly and the origin of birds" (http://biostor.org/reference/110202). *Memoirs of the California Academy of Sciences*. **8**. 1986. ISSN 0885-4629 (https://search.worldcat.org/issn/0885-4629). Archived (https://web.archive.org/web/20141103042545/http://biostor.org/reference/110202) from the original on November 3, 2014. Retrieved May 31, 2017.
6. Cau, A. (2024). "A Unified Framework for Predatory Dinosaur Macroevolution" (https://www.researchgate.net/publication/379902868). *Bollettino della Società Paleontologica Italiana*. **63** (1): 1–19.
7. Delcourt, Rafael (June 27, 2018). "Ceratosaur palaeobiology: new insights on evolution and ecology of the southern rulers" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6021374). *Scientific Reports*. **8** (1) 9730. Bibcode:2018NatSR...8.9730D (https://ui.adsabs.harvard.edu/abs/2018NatSR...8.9730D). doi:10.1038/s41598-018-28154-x (https://doi.org/10.1038%2Fs41598-018-28154-x). ISSN 2045-2322 (https://search.worldcat.org/issn/2045-2322). PMC 6021374 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6021374). PMID 29950661 (https://pubmed.ncbi.nlm.nih.gov/29950661).

8. Rauhut, Oliver W. M.; Carrano, Matthew T. (November 1, 2016). "The theropod dinosaur Elaphrosaurus bambergi, from the Late Jurassic of Tendaguru, Tanzania". *Zoological Journal of the Linnean Society*. **178** (3): 546–610. doi:10.1111/zoj.12425 (https://doi.org/10.1111%2Fzoj.12425). ISSN 0024-4082 (https://search.worldcat.org/issn/0024-4082).
9. Hendrickx, Christophe; Mateus, Octávio; Araújo, Ricardo; Choiniere, Jonah (November 24, 2019). "The distribution of dental features in non-avian theropod dinosaurs: Taxonomic potential, degree of homoplasy, and major evolutionary trends" (https://palaeo-electronica.org/content/2019/2806-dental-features-in-theropods). *Palaeontologia Electronica*. **22** (3): 1–110. doi:10.26879/820 (https://doi.org/10.26879%2F820). hdl:10362/98436 (https://hdl.handle.net/10362%2F98436). ISSN 1094-8074 (https://search.worldcat.org/issn/1094-8074). Archived (https://web.archive.org/web/20211108131909/https://palaeo-electronica.org/content/2019/2806-dental-features-in-theropods) from the original on November 8, 2021. Retrieved June 17, 2024.
10. Rauhut, Oliver W. M. (2011). "Theropod dinosaurs from the Late Jurassic of Tendaguru (Tanzania)". *Special Papers in Palaeontology*. **86**: 195–239.
11. Martín D. Ezcurra, M.D. and Agnolín, F.L. (2012). "An abelisauroid dinosaur from the Middle Jurassic of Laurasia and its implications on theropod palaeobiogeography and evolution." *Proceedings of the Geologists' Association*, (advance online publication).
12. T Carrano, Matthew. "The appendicular skeleton of *Majungasaurus crenatissimus* (Theropoda: Abelisauridae) from the Late Cretaceous of Madagascar" (https://www.researchgate.net/publication/230808681). *ResearchGate*.
13. Sasso, Cristiano Dal; Maganuco, Simone; Cau, Andrea (December 19, 2018). "The oldest ceratosaurian (Dinosauria: Theropoda), from the Lower Jurassic of Italy, sheds light on the evolution of the three-fingered hand of birds" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6304160). *PeerJ*. **6** e5976. doi:10.7717/peerj.5976 (https://doi.org/10.7717%2Fpeerj.5976). ISSN 2167-8359 (https://search.worldcat.org/issn/2167-8359). PMC 6304160 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6304160). PMID 30588396 (https://pubmed.ncbi.nlm.nih.gov/30588396).
14. Hendrickx, C.; Hartman, S.A.; Mateus, O. (2015). "An overview of non-avian theropod discoveries and classification". *PalArch's Journal of Vertebrate Palaeontology*. **12** (1): 1–73.
15. Wang, S.; Stiegler, J.; Amiot, R.; Wang, X.; Du, G.-H.; Clark, J.M.; Xu, X. (2017). "Extreme Ontogenetic Changes in a Ceratosaurian Theropod" (http://www.cell.com/current-biology/pdf/S0960-9822(16)31269-6.pdf) (PDF). *Current Biology*. **27** (1): 144–148. Bibcode:2017CBio...27..144W (https://ui.adsabs.harvard.edu/abs/2017CBio...27..144W). doi:10.1016/j.cub.2016.10.043 (https://doi.org/10.1016%2Fj.cub.2016.10.043). PMID 28017609 (https://pubmed.ncbi.nlm.nih.gov/28017609).
16. Bonaparte, José; Novas, Fernando; Coria, Rodolfo (1990). "Carnotaurus sastrei Bonaparte, the horned, lightly built carnosaur from the Middle Cretaceous of Patagonia. Contributions in Science" (https://web.archive.org/web/20160629103636/http://www.nhm.org/site/sites/default/files/pdf/contrib_science/CS416.pdf) (PDF). *Natural History Museum of Los Angeles County*. **416**: 1–42. Archived from the original (https://nhm.org/site/sites/default/files/pdf/contrib_science/CS416.pdf) (PDF) on June 29, 2016. Retrieved May 31, 2017.
17. Sereno, Paul C.; Brusatte, Stephen L. (2008). "Basal Abelisaurid and Carcharodontosaurid Theropods from the Lower Cretaceous Elrhaz Formation of Niger" (https://doi.org/10.4202%2Fapp.2008.0102). *Acta Palaeontologica Polonica*. **53** (1): 15–46. doi:10.4202/app.2008.0102 (https://doi.org/10.4202%2Fapp.2008.0102). hdl:20.500.11820/5d55ed2e-52f2-4e4a-9ca1-fd1732f2f964 (https://hdl.handle.net/20.500.11820%2F5d55ed2e-52f2-4e4a-9ca1-fd1732f2f964).
18. Agnolin, Federico; Chiarelli, Pablo. "The position of the claws in Noasauridae (Dinosauria: Abelisauroidea) and its implications for abelisauroid manus evolution" (https://www.researchgate.net/publication/225424245). *ResearchGate*.

19. Tortosa, Thierry; Buffetaut, Eric; Vialle, Nicolas; Dutour, Yves; Turini, Eric; Cheylan, Gilles (2014). "A new abelisaurid dinosaur from the Late Cretaceous of southern France: Palaeobiogeographical implications". *Annales de Paléontologie*. **100** (1): 63–86. Bibcode:2014AnPal.100...63T (https://ui.adsabs.harvard.edu/abs/2014AnPal.100...63T). doi:10.1016/j.annpal.2013.10.003 (https://doi.org/10.1016%2Fj.annpal.2013.10.003).
20. Xu, Xing; Clark, James M.; Mo, Jinyou; Choiniere, Jonah; Forster, Catherine A.; Erickson, Gregory M.; Hone, David W. E.; Sullivan, Corwin; Eberth, David A. (June 18, 2009). "A Jurassic ceratosaur from China helps clarify avian digital homologies" (http://doc.rero.ch/record/209594/files/PAL_E4066.pdf) (PDF). *Nature*. **459** (7249): 940–944. Bibcode:2009Natur.459..940X (https://ui.adsabs.harvard.edu/abs/2009Natur.459..940X). doi:10.1038/nature08124 (https://doi.org/10.1038%2Fnature08124). ISSN 0028-0836 (https://search.worldcat.org/issn/0028-0836). PMID 19536256 (https://pubmed.ncbi.nlm.nih.gov/19536256). S2CID 4358448 (https://api.semanticscholar.org/CorpusID:4358448). Archived (https://web.archive.org/web/20170922012951/http://doc.rero.ch/record/209594/files/PAL_E4066.pdf) (PDF) from the original on September 22, 2017. Retrieved September 6, 2019.
21. de Souza GA, Soares MB, Weinschütz LC, Wilner E, Lopes RT, de Araújo OM, Kellner AW (2021). "The first edentulous ceratosaur from South America" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8602317). *Scientific Reports*. **11** (1) 22281. Bibcode:2021NatSR..1122281D (https://ui.adsabs.harvard.edu/abs/2021NatSR..1122281D). doi:10.1038/s41598-021-01312-4 (https://doi.org/10.1038%2Fs41598-021-01312-4). PMC 8602317 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8602317). PMID 34795306 (https://pubmed.ncbi.nlm.nih.gov/34795306).
22. J., Currie, Philip; B., Koppelhus, Eva; A., Shugar, Martin; L., Wright, Joanna (2004). *Feathered dragons: studies on the transition from dinosaurs to birds*. Indiana University Press. ISBN 978-0-253-34373-4. OCLC 895411495 (https://search.worldcat.org/oclc/895411495).
23. "Prey bone utilization by predatory dinosaurs in the Late Jurassic of North America, with comments on prey bone use by dinosaurs throughout the Mesozoic (PDF Download Available)" (https://www.researchgate.net/publication/268274412). *ResearchGate*. Retrieved May 31, 2017.
24. Mazzetta, Gerardo V.; Cisilino, Adrián P.; Blanco, R. Ernesto; Calvo, Néstor (September 12, 2009). "Cranial mechanics and functional interpretation of the horned carnivorous dinosaur Carnotaurus sastrei". *Journal of Vertebrate Paleontology*. **29** (3): 822–830. Bibcode:2009JVPal..29..822M (https://ui.adsabs.harvard.edu/abs/2009JVPal..29..822M). doi:10.1671/039.029.0313 (https://doi.org/10.1671%2F039.029.0313). hdl:11336/34937 (https://hdl.handle.net/11336%2F34937). ISSN 0272-4634 (https://search.worldcat.org/issn/0272-4634). S2CID 84565615 (https://api.semanticscholar.org/CorpusID:84565615).
25. Kenneth., Carpenter (2005). *The carnivorous dinosaurs*. Indiana University Press. ISBN 978-0-253-34539-4. OCLC 895411496 (https://search.worldcat.org/oclc/895411496).
26. G. Meso, Jorge; Hendrickx, Christophe; A. Baiano, Mattia; I. Canale, Juan; Salgado, Leonardo; Diaz Martinez, Ignacio. "Isolated theropod teeth associated with a sauropod skeleton from the Late Cretaceous Allen Formation of Río Negro, Patagonia, Argentina" (https://app.pan.pl/article/item/app008472020.html). *Acta Paleontologica Polonica*. Archived (https://web.archive.org/web/20210613120900/https://app.pan.pl/article/item/app008472020.html) from the original on June 13, 2021. Retrieved June 30, 2021.
27. A. Farke, Andrew (2013). " "An Abelisauroid Theropod Dinosaur from the Turonian of Madagascar" " (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3630149). *PLOS ONE*. **8** (4) e62047. Bibcode:2013PLoSO...862047F (https://ui.adsabs.harvard.edu/abs/2013PLoSO...862047F). doi:10.1371/journal.pone.0062047 (https://doi.org/10.1371%2Fjournal.pone.0062047). PMC 3630149 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3630149). PMID 23637961 (https://pubmed.ncbi.nlm.nih.gov/23637961).

28. Méndez, Ariel; E Novas, Fernando; Vidoi Iori, Fabiano. "New records of abelisauroid theropods from the Bauru Basin (Upper Cretaceous), Sao Paulo State, Brazil" (https://www.researchgate.net/publication/265381596). *ResearchGate*.

External links

Retrieved from "https://en.wikipedia.org/w/index.php?title=Ceratosauria&oldid=1376847709"

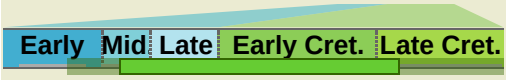
WIKIPEDIA
25 years of the free encyclopedia

Megalosauroidea

Megalosauroidea (meaning 'great/big lizard forms') is a superfamily (or clade) of tetanuran theropod dinosaurs that lived from the Middle Jurassic to the Late Cretaceous period. The group is defined as *Megalosaurus bucklandii* and all taxa sharing a more recent common ancestor with it than with *Allosaurus fragilis* or *Passer domesticus*.[3] Members of the group include *Spinosaurus*, *Megalosaurus*, and *Torvosaurus*. They are possibly paraphyletic in nature with respect to Allosauroidea, which is to say some members of this superfamily might be closer to Allosauroids than to each other, which is undesirable in cladistics.

Megalosauroids

Temporal range: Middle Jurassic - Late Cretaceous,



Probably originated since Toarcian and Possible Campanian-Maastrichtian records[1][2]

Torvosaurus skeletal mount, Museum of Ancient Life

Spinosaurus skeletal mount, National Geographic Museum

Scientific classification

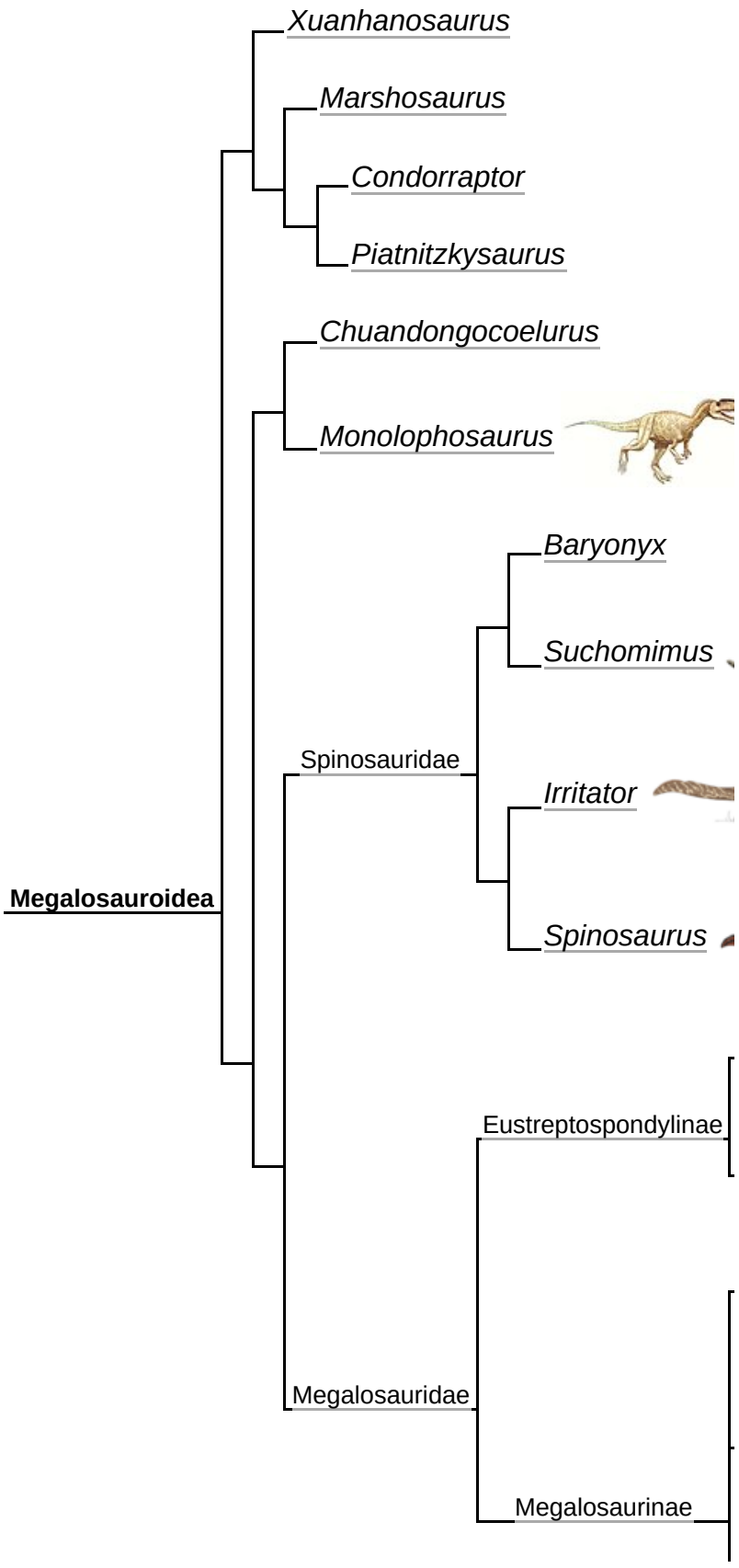
Kingdom:	Animalia
Phylum:	Chordata
Class:	Reptilia
Clade:	Dinosauria
Clade:	Saurischia

Classification

The name **Spinosauroidea** is sometimes used in place of Megalosauroidea. The superfamily Spinosauroidea was named in 1915 by Ernst Stromer. It is a synonym of Megalosauroidea in almost all modern phylogenetic analyses, and it is therefore redundant. Spinosauroidea was defined as a clade in 1998 by Paul Sereno as the node clade containing the common ancestor of *Spinosaurus* and *Torvosaurus* and all its descendants. Thomas Holtz in 2004 defined a branch clade with the same name containing all species closer to *Spinosaurus* than to *Passer domesticus*. The ICZN holds that even clade names (which do not yet have any governing body) should be replaced if having a traditional taxon suffix and being synonyms of ranked taxa at or below the superfamily level. The seniority of Megalosauroidea was not followed in most paleontological literature during the 1990s and early 2000s. A series of papers supporting the validity of *Megalosaurus* as a genus, the relationships of megalosauroids, and the placement of "spinosauroids" among them, published between 2008 and 2010 argued that Megalosauroidea was in fact the valid name for the group.[3]

The classification of megalosauroids follows a study by Benson in 2010. Note that several "wildcard"[3] taxa that are probably megalosauroids were excluded from the final tree, including *Magnosaurus*, *Piveteausaurus* and *Streptospondylus*.[3] Later, *Magnosaurus* and

Streptospondylus were added in the final tree.[4] These are known from remains too fragmentary to be reliably classified.[3] Some of these 'wildcard' taxa, such as *Poekilopleuron* and *Megaraptor*, have been considered possible megalosauroids in the past, but the analysis found that they are more likely allosauroids.[4]



Clade:	Theropoda
Clade:	†Carnosauria (?)
Superfamily:	†Megalosauroidea Huxley, 1889

Type species

†*Megalosaurus bucklandii* Mantell, 1827

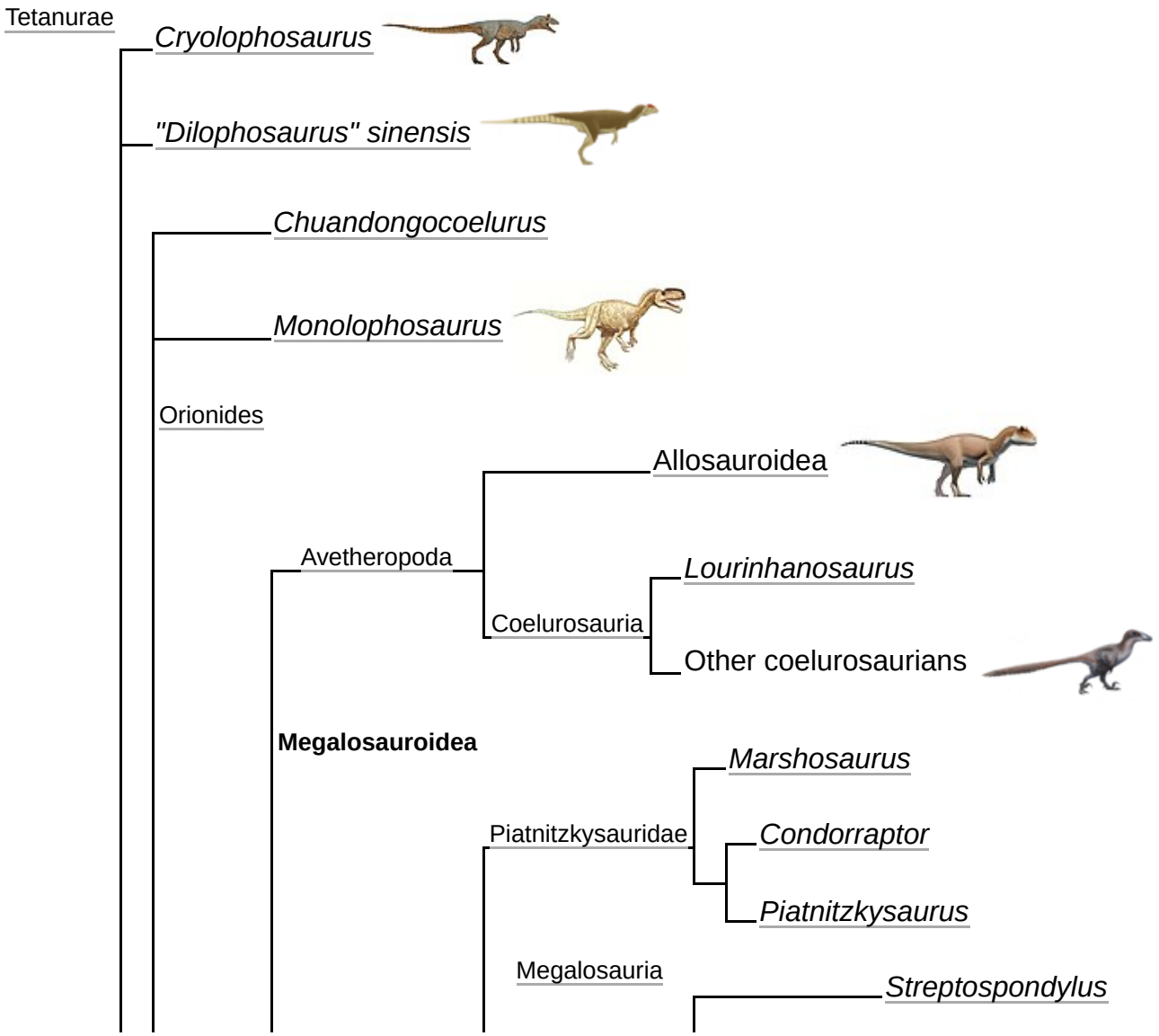
Subgroups

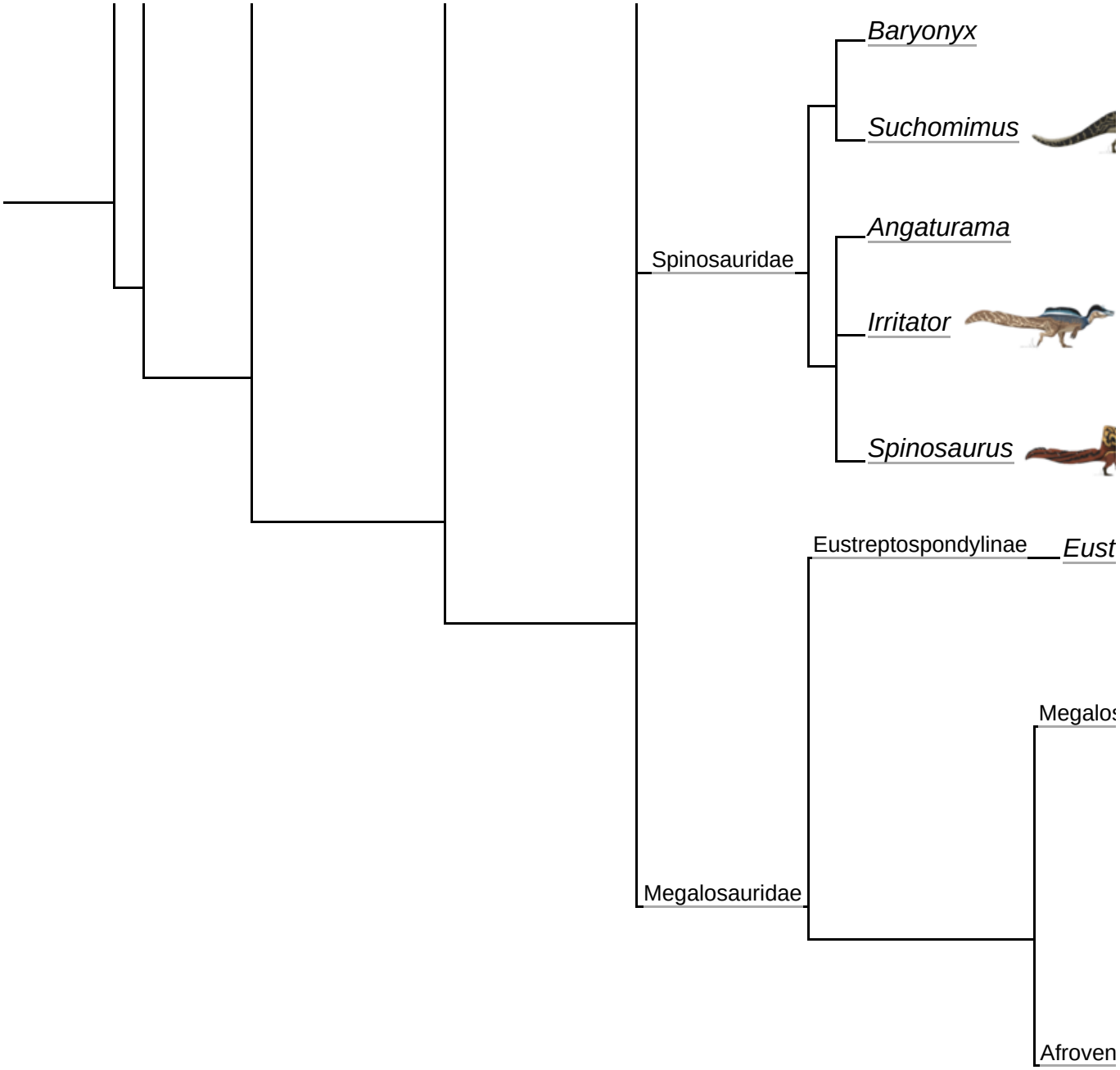
- †*Asfaltovenator*?
- †*Chuandongocoelurus*?
- †*Cruxicheiros*?
- †*Dandakosaurus*?
- †*Gasosaurus*?
- †*Juravenator*?
- †*Kaijiangosaurus*?
- †*Lourinhanosaurus*?
- †*Monolophosaurus*?
- †*Nedcolbertia*?
- †*Sciurumimus*?
- †*Yunyangosaurus*
- †**Piatnitzkysauridae**?
- †**Megalosauria**
  - †*Streptospondylus*
  - †**Megalosauridae**
  - †**Spinosauridae**

Synonyms

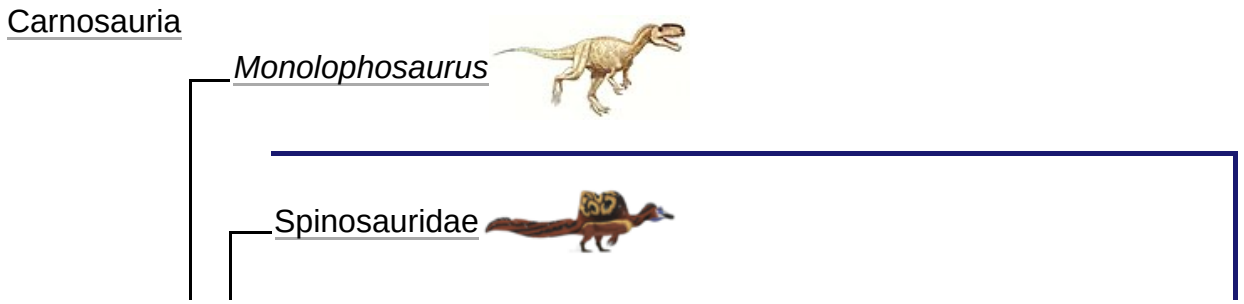
- Spinosauroidea Stromer, 1915
- Torvosauroidea Nopcsa, 1915
- Spinosauria Olshevsky, 1991

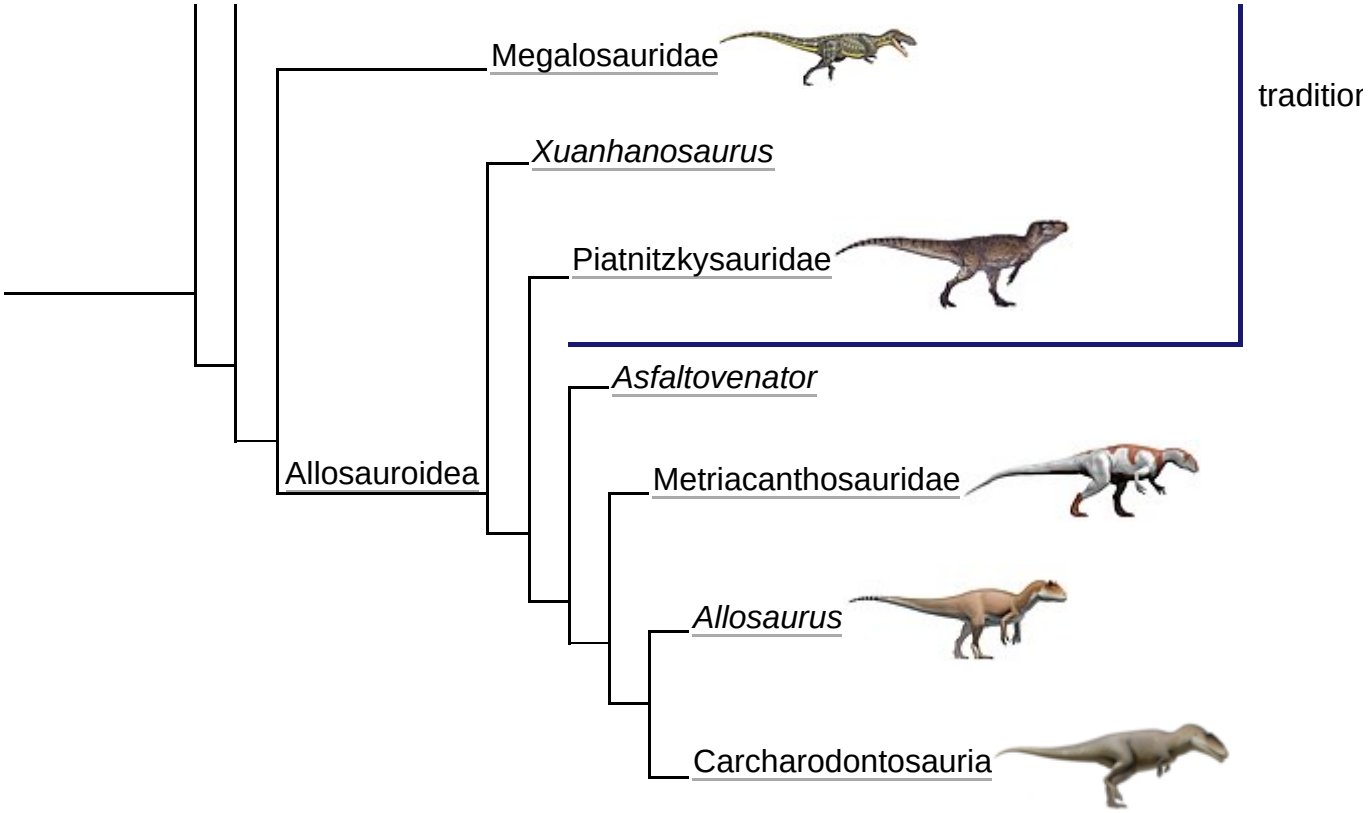
Carrano, Benson & Sampson (2012) performed much larger phylogenetic analysis of the Tetanurae that includes more taxa. They used the clade name **Megalosauria** (Bonaparte, 1850) in their analysis and defined it as the node comprising *Megalosaurus*, *Spinosaurus*, their most recent common ancestor, and all its descendants. Furthermore, a new megalosauroid family Piatnitzkysauridae was named to include all megalosauroids more closely related to *Piatnitzkysaurus* than to either *Spinosaurus* or *Megalosaurus*. Within Megalosauridae a new subfamily was named, Afrovenatorinae, to include all megalosaurids more closely related to *Afrovenator* than to *Megalosaurus*. Unlike Benson *et al.*, 2010, they recovered *Poekilopleuron* as an afrovenatorine, while *Xuanhanosaurus* was recovered as the basalmost metriacanthosaurid. However, the position of these taxa is very unstable, and their exclusion from the analysis gave more resolved and stable cladogram. *Streptospondylus* was also excluded to get more resolved Megalosauridae and Afrovenatorinae. The Chinese tetanuran *Leshansaurus* was included for the first time in a phylogenetic analysis, and was recovered as an afrovenatorine. Both *Chuandongocoelurus* and *Monolophosaurus* were found to be just outside Orionides. The cladogram presented here follows that study.[5]





In 2019, Rauhut and Pol described *Asfaltovenator vialidadi*, a basal allosauroid displaying a mosaic of primitive and derived features seen within Tetanurae. Their phylogenetic analysis found traditional Megalosauroidea to represent a basal grade of carnosaurs, paraphyletic with respect to Allosauroidea.[6]





See also

- Carnosauria
- Orionides

Dinosaurs portal

References

1. Lacerda, M. B. S.; Bittencourt, J. S.; Hutchinson, J. R. (2023). "Macroevolutionary patterns in the pelvis, stylopodium and zeugopodium of megalosauroid theropod dinosaurs and their importance for locomotor function" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10427828). *Royal Society Open Science*. **10** (8). 230481. Bibcode:2023RSOS...1030481L (https://ui.adsabs.harvard.edu/abs/2023RSOS...1030481L). doi:10.1098/rsos.230481 (https://doi.org/10.1098%2Frsos.230481). PMC 10427828 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10427828). PMID 37593714 (https://pubmed.ncbi.nlm.nih.gov/37593714).
2. Olmedo-Romaña, G. J.; Wilson Mantilla, J. A.; Tejada, J. V.; Antoine, P. O.; Burga-Castillo, M. A.; Aliaga-Castillo, A. V.; Varas-Malca, R.; Benites-Palomino, A.; Salas-Gismondi, R. (2025). "Theropod and sauropod dinosaurs from the Campanian-Maastrichtian Bagua Basin of Perú, including the first possible report of Spinosauridae in western South America" (https://www.ameghiniana.org.ar/index.php/ameghiniana/article/view/3627). *Ameghiniana*. **62** (4): 259–287. doi:10.5710/AMGH.13.02.2025.3627 (https://doi.org/10.5710%2FAMGH.13.02.2025.3627).
3. Benson, R.B.J. (2010). "A description of *Megalosaurus bucklandii* (Dinosauria: Theropoda) from the Bathonian of the UK and the relationships of Middle Jurassic theropods" (https://doi.org/10.1111%2Fj.1096-3642.2009.00569.x). *Zoological Journal of the Linnean Society*. **158** (4): 882–935. doi:10.1111/j.1096-3642.2009.00569.x (https://doi.org/10.1111%2Fj.1096-3642.2009.00569.x).
4. Benson, R.B.J., Carrano, M.T and Brusatte, S.L. (2010). "A new clade of archaic large-bodied predatory dinosaurs (Theropoda: Allosauroidea) that survived to the latest

Mesozoic". *Naturwissenschaften*. **97** (1): 71–78. Bibcode:2010NW.....97...71B (https://ui.adsabs.harvard.edu/abs/2010NW.....97...71B). doi:10.1007/s00114-009-0614-x (https://doi.org/10.1007%2Fs00114-009-0614-x). PMID 19826771 (https://pubmed.ncbi.nlm.nih.gov/19826771). S2CID 22646156 (https://api.semanticscholar.org/CorpusID:22646156). Supporting Information (https://doi.org/10.1007%2Fs00114-009-0614-x)

5. Carrano, M. T.; Benson, R. B. J.; Sampson, S. D. (2012). "The phylogeny of Tetanurae (Dinosauria: Theropoda)". *Journal of Systematic Palaeontology*. **10** (2): 211–300. Bibcode:2012JSPal..10..211C (https://ui.adsabs.harvard.edu/abs/2012JSPal..10..211C). doi:10.1080/14772019.2011.630927 (https://doi.org/10.1080%2F14772019.2011.630927). S2CID 85354215 (https://api.semanticscholar.org/CorpusID:85354215).
6. Rauhut, Oliver W. M.; Pol, Diego (2019-12-11). "Probable basal allosauroid from the early Middle Jurassic Cañadón Asfalto Formation of Argentina highlights phylogenetic uncertainty in tetanuran theropod dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6906444). *Scientific Reports*. **9** (1): 18826. Bibcode:2019NatSR...918826R (https://ui.adsabs.harvard.edu/abs/2019NatSR...918826R). doi:10.1038/s41598-019-53672-7 (https://doi.org/10.1038%2Fs41598-019-53672-7). ISSN 2045-2322 (https://search.worldcat.org/issn/2045-2322). PMC 6906444 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6906444). PMID 31827108 (https://pubmed.ncbi.nlm.nih.gov/31827108).

Retrieved from "https://en.wikipedia.org/w/index.php?title=Megalosauroidea&oldid=1368884607"

WIKIPEDIA
25 years of the free encyclopedia

Allosauridae

Allosauridae is an extinct family of medium to large bipedal, carnivorous allosauroid theropod dinosaurs from the Late Jurassic.[1] Allosauridae is a fairly old taxonomic group, having been first named by the American paleontologist Othniel Charles Marsh in 1878.[2]

Description

Allosaurids have a general anatomy typical of other neotheropod dinosaurs, contributing to the difficulty in defining the family's membership. A typical 8m specimen of *Allosaurus fragilis* had a skull of about 0.85m. The premaxilla has five teeth and the maxilla usually around 16. The dentary also typically has 16 teeth. All teeth are serrated and continuously replaced throughout the life of the animal. Allosaurid skulls are characterized by two sets of crests formed by the nasal and lacrimal bones respectively. These crests would have been covered by keratin sheathes.[3] The skull also exhibits features consistent with significant cranial kinesis: a synovial joint between the braincase and the frontals and a loose articulation between the dentary and the angular/surangular.[4] This cranial kinesis would have dampened forces on the bones of the skull and allowed allosaurids to open their mouths to very large angles.

Allosaurids have 28 precaudal vertebrae (9 cervical, 14 dorsal, 5 sacral) and an estimated 45–50 caudal vertebrae.[4] Gastralia and furculae are rarely preserved as fossils but are presumed to occur in all allosaurids.[5] The pubis is highly elongated and extends ventrally to form a pubic foot which like in other large dinosaurs is thought to have been used to support the weight of the body in a resting crouch position.[3]

Like most other theropods, allosaurids have very short forelimbs relative to their hindlimbs with three digits on the hand and four on the foot. The first digit of the hand forms a semi-opposable thumb and digits 4 and 5 are absent. Digits 2–4 of the foot are robust but digit 1 is reduced and does not touch the ground and digit 5 is absent.[6] All distal phalanges

Allosaurids

Temporal range: Bathonian-Tithonian,

An *A. fragilis* skeletal mount, at the Carnegie Museum of Natural History

Scientific classification

Kingdom:	Animalia
Phylum:	Chordata
Class:	Reptilia
Clade:	Dinosauria
Clade:	Saurischia
Clade:	Theropoda
Superfamily:	†Allosauroidea
Family:	†Allosauridae Marsh, 1878

Type species

†*Allosaurus fragilis* Marsh, 1877

Genera[1]

- †*Allosaurus*
- †*Antrodemus*?
- †*Asfaltovenator*?
- †*Epanterias*?
- †*Piveteausaurus*?
- †*Saurophaganax*?

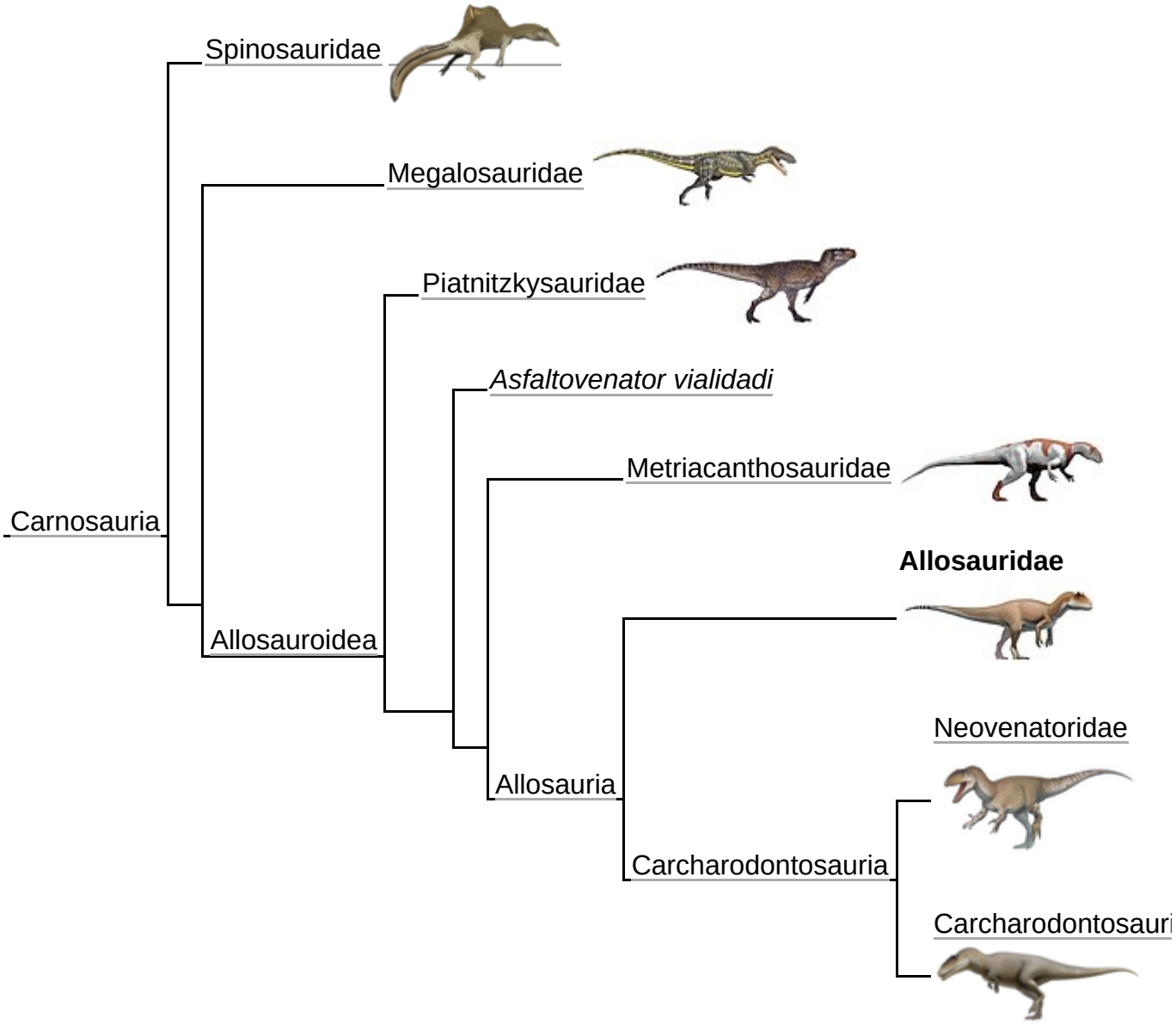
Synonyms

were capped with large claws, those on the hand were especially long and were curved to facilitate raking and grasping of prey items.[3] Phalangeal formulae of the hand and foot are 2-3-4 and 2-3-4-5 respectively.[6]

- Antrodemidae Marsh, 1878
- Labrosauridae Marsh, 1882

Classification

In 2019, Rauhut and Pol described *Asfaltovenator vialidadi*, a basal allosauroid displaying a mosaic of primitive and derived features seen within Tetanurae. Their phylogenetic analysis found traditional Megalosauroidea to represent a basal grade of carnosaurs, paraphyletic with respect to Allosauroidea. They also noted that the definitive taxonomic placement of *Saurophaganax* within Allosauroidea is unstable, being recovered as a sister taxon of Metriacanthosauridae or Allosauria, or even as a basalmost carcharodontosaurian.[7]



In 2024, re-examinations of the specimens attributed to *Saurophaganax* suggested that it is a chimera of multiple dinosaur genera, since some specimens most likely belong to a diplodocid sauropod, while the other referred allosaurid specimens belong to a novel species of *Allosaurus*, named as *Allosaurus anax*.[8]

References

1. Carrano, M. T.; Benson, R. B. J.; Sampson, S. D. (2012). "The phylogeny of Tetanurae (Dinosauria: Theropoda)". *Journal of Systematic Palaeontology*. **10** (2): 211–300. doi:10.1080/14772019.2011.630927 (https://doi.org/10.1080%2F14772019.2011.630927).
2. Marsh, Othniel Charles (1878). "Notice of new dinosaurian reptiles". *American Journal of Science and Arts*. **15**: 241–244.
3. Madsen, James H., Jr. (1993) [1976]. *Allosaurus fragilis: A Revised Osteology*. Utah Geological Survey Bulletin **109** (2nd ed.). Salt Lake City: Utah Geological Survey.
4. Paul, Gregory S. (1988). Predatory Dinosaurs of the World.
5. Chure, Daniel J.; Madsen, James (1996). "On the presence of furculae in some non-maniraptoran theropods". *Journal of Vertebrate Paleontology*. **16** (3): 573–577. doi:10.1080/02724634.1996.10011341 (https://doi.org/10.1080%2F02724634.1996.10011341)
6. Gilmore, Charles W. (1920). Osteology of the Carnivorous Dinosauria in the United States National Museum: With Special Reference to the Genera Antrodemus (Allosaurus) and Ceratosaurus. United States National Museum Bulletin Volume 110.
7. Rauhut, Oliver W. M.; Pol, Diego (December 11, 2019). "Probable basal allosauroid from the early Middle Jurassic Cañadón Asfalto Formation of Argentina highlights phylogenetic uncertainty in tetanuran theropod dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6906444). *Scientific Reports*. **9** (1): 18826. doi:10.1038/s41598-019-53672-7 (https://doi.org/10.1038%2Fs41598-019-53672-7). ISSN 2045-2322 (https://search.worldcat.org/issn/2045-2322). PMC 6906444 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6906444). PMID 31827108 (https://pubmed.ncbi.nlm.nih.gov/31827108). Supplementary information (https://www.nature.com/articles/s41598-019-53672-7#MOESM1)
8. Danison, Andrew; Wedel, Mathew; Barta, Daniel; Woodward, Holly; Flora, Holley; Lee, Andrew; Snively, Eric (December 21, 2024). "Chimerism in specimens referred to *Saurophaganax maximus* reveals a new species of *Allosaurus* (Dinosauria, Theropoda)" (https://journals.library.ualberta.ca/vamp/index.php/VAMP/article/view/29404). *Vertebrate Anatomy Morphology Palaeontology*. **12**. doi:10.18435/vamp29404 (https://doi.org/10.18435%2Fvamp29404). ISSN 2292-1389 (https://search.worldcat.org/issn/2292-1389).

External links

Retrieved from "https://en.wikipedia.org/w/index.php?title=Allosauridae&oldid=1374689861"

WIKIPEDIA 25
25 years of the free encyclopedia

Spinosauridae

Spinosauridae (or **spinosaurids**) is a clade or family of tetanuran theropod dinosaurs comprising ten to seventeen known genera. Spinosaurid fossils have been recovered worldwide, including Africa, Europe, South America, and Asia. Their remains have generally been attributed to the Early to Late Cretaceous.

Spinosaurids were large bipedal carnivores. Their crocodilian-like skulls were long, low, and narrow, bearing conical teeth with reduced or absent serrations. The tips of their upper and lower jaws fanned out into a spoon-shaped structure similar to a rosette, behind which there was a notch in the upper jaw that the expanded tip of the lower jaw fit into. The nostrils of spinosaurids were retracted to a position further back on the head than in most other theropods, and they had bony crests on their heads along the midline of their skulls. Their robust shoulders wielded stocky forelimbs, with three-fingered hands that bore an enlarged claw on the first digit. In many species, the upwards-projecting neural spines of the vertebrae (backbones) were significantly elongated and formed a sail on the animal's back (hence the family's etymology), which supported either a layer of skin or a fatty hump.

The genus *Spinosaurus*, from which the family, one of its subfamilies (Spinosaurinae) and tribe (Spinosaurini) take their names, is among the largest known terrestrial predators from the fossil record, with an estimated length of up to 14 meters (46 ft) and body mass of up to 7.4 metric tons (8.2 short tons).[7] The closely related genus *Sigilmassasaurus* may have reached a similar or greater size, though its taxonomy is disputed. Direct fossil evidence and anatomical adaptations indicate that spinosaurids were at least partially piscivorous (fish-eating), with additional fossil finds indicating they also fed on other dinosaurs and pterosaurs. The osteology of spinosaurid teeth and bones has suggested a semiaquatic lifestyle for some members of this clade. This is further indicated by various anatomical adaptations, such as retracted eyes and nostrils; and the deepening of the tail

Spinosaurids

Temporal range:

Early Cretaceous–Late Cretaceous (Berriasian–Cenomanian), Possible Bathonian record[1]

Montage of four spinosaurids, clockwise from top left: *Baryonyx*, *Irritator*, *Spinosaurus* and *Suchomimus*

Scientific classification

Kingdom:	Animalia
Phylum:	Chordata
Class:	Reptilia
Clade:	Dinosauria
Clade:	Saurischia
Clade:	Theropoda
Clade:	†Carnosauria (?)
Superfamily:	†Megalosauroidea
Family:	†Spinosauridae Stromer, 1915

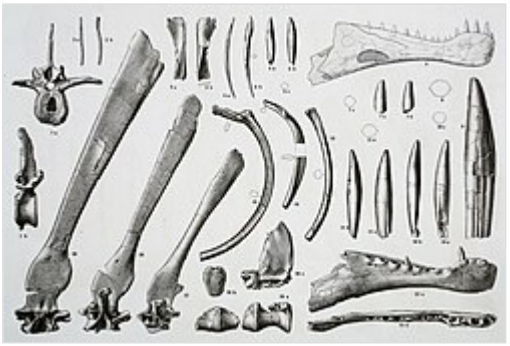
Type species

†*Spinosaurus aegyptiacus*
Stromer, 1915

Subgroups and genera

in some taxa, which has been suggested to have aided in underwater propulsion akin to that of modern crocodilians. Spinosaurs are proposed to be closely related to the megalosaurid theropods of the Jurassic. This is due to both groups sharing many features such an enlarged claw on their first manual ungual and an elongated skull.[8] However, some propose that this group (which is known as the Megalosauroidea) is paraphyletic and that spinosaurs represent either the most basal tetanurans[9] or the basal carnosaurs which are less derived than the megalosaurids.[10] Some have proposed a combination of the two ideas with spinosaurs being in a monophyletic Megalosauroidea inside a more inclusive Carnosauria that is made up of both allosauroids and megalosauroids.[11]

History of discovery



Elements of the holotype specimen of *Spinosaurus aegyptiacus*, as illustrated by Ernst Stromer in 1915

The first spinosaurid fossil, a single conical tooth, was discovered circa 1820 by British paleontologist Gideon Mantell in the Wadhurst Clay Formation.[12] In 1841, naturalist Sir Richard Owen mistakenly assigned it to a crocodilian he named *Suchosaurus* (meaning "crocodile lizard").[13][14] A second species, *S. girardi*, was later named in 1897.[15] However, the spinosaurid nature of *Suchosaurus* was not recognized until a 1998 redescription of *Baryonyx*.[16]

The first fossils referred to a spinosaurid were discovered in 1912 at the Bahariya Formation in Egypt. Consisting of vertebrae, skull fragments, and teeth, these remains became the holotype specimen of the new genus and species *Spinosaurus aegyptiacus* in 1915, when they

- †*Cristatusaurus*
- †*Iberospinus*
- †*Monolophosaurus*?[2]
- †*Ostafrikasaurus*?
- †**Spinosauria** Olshevsky, 1991[3]
  - †**Baryonychinae**
    - †*Baryonyx*
    - †*Cristatusaurus*?
    - †*Eocarcharia*?
    - †*Iberospinus*?
    - †*Protathlitis*?
    - †*Riojavenatrix*
    - †*Suchosaurus*
    - †**Ceratosuchopsini**
      - †*Ceratosuchops*
      - †*Eocarcharia*?
      - †*Riparovenator*
      - †*Suchomimus*
  - †**Spinosaurinae**
    - †*Camarillasaurus*?
    - †*Iberospinus*?
    - †*Ichthyovenator*
    - †*Irritator*
    - †*Siamosaurus*?
    - †*Vallibonavenatrix*
    - †**Spinosaurini**
      - †*Oxalaia**
      - †*Sigilmassasaurus**
      - †*Spinosaurus*

*disputed validity; =*Spinosaurus*?[4][5][6]

Synonyms

- Baryonychidae Charig & Milner, 1986

- Irritatoridae Martill et al., 1996
- Sigilmassasauridae Russell, 1996

were described by German paleontologist Ernst Stromer. The dinosaur's name meant "Egyptian spine lizard", in reference to the unusually long neural spines not seen previously in any other theropod. In April 1944, the holotype of *S. aegyptiacus* was destroyed during an allied bombing raid in World War II.[17][18] In 1934, Stromer referred a partial skeleton also from the Bahariya Formation to a new species of *Spinosaurus;*[19] the specimen has since been alternatively assigned to another African spinosaurid, *Sigilmassasaurus*.[20]

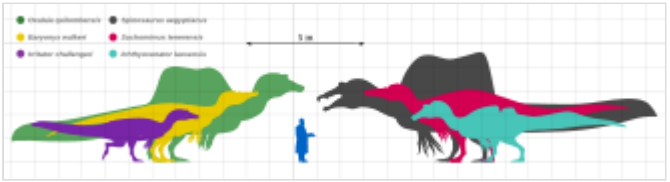
In 1983, a relatively complete skeleton was excavated from the Smokejacks pit in Surrey, England. These remains were described by British paleontologists Alan J. Charig and Angela C. Milner in 1986 as the holotype of a new species, *Baryonyx walkeri*. After the discovery of *Baryonyx*, many new genera have since been described, with the majority from very incomplete remains. However, other finds bear enough fossil material and distinct anatomical features to be assigned with confidence. Paul Sereno and colleagues described *Suchomimus* in 1998, a baryonychine from Niger, on the basis of a partial skeleton found in 1997. In 2004, partial jaw bones were recovered from the Alcântara Formation, these were referred to a new genus of spinosaurine named *Oxalaia* in 2011 by Alexander Kellner.[18]

Fragmentary remains belonging to a large spinosaurid were collected in 2021 from the Vectis Formation on the Isle of Wight. The material lacks distinct characteristics that would prompt the erection of a new species. However, the size of the bones is comparable to the size of *Spinosaurus*, making the "White Rock spinosaurid" one of the largest theropods ever found in Europe in addition to the first theropod identified in the Vectis Formation.[21][22]

In 2024, *Riojavenatrix lacustris* became the fifth spinosaurid species to be named from material found in the Iberian Peninsula, following *Camarillasaurus*, *Iberospinus*, *Protathlitis*, and *Vallibonavenatrix*. The fossils hail from the Early Cretaceous Enciso Group of La Rioja, Spain.[23][24]

In 2026, Paul Sereno and colleagues described *Spinosaurus mirabilis*, a new species of *Spinosaurus* from the Farak Formation of Niger.[25]

Description

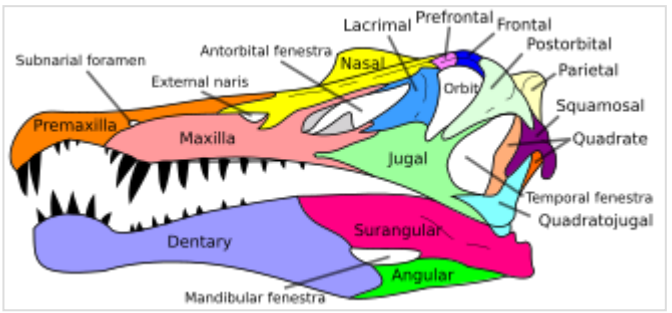


Size comparison of spinosaurid genera (from left to right) *Irritator*, *Baryonyx*, *Oxalaia*, *Spinosaurus*, *Suchomimus*, and *Ichthyovenator* with a human

Although reliable size and weight estimates for most known spinosaurids are hindered by the lack of good material, all known spinosaurids were large animals.[18] The smallest genus known from good material is *Irritator*, which was between 6 and 8 meters (20 and 26 feet) long and around 1 metric ton (1.1 short tons; 0.98 long tons) in weight.[26][27] *Ichthyovenator*, *Baryonyx*, and *Suchomimus* ranged from 7.5 to 11 m (25 to 36 ft) long, and weighed between 1 and 5.2 t (1.1 and 5.7 short tons; 0.98 and 5.12 long tons).[28][27][29] *Oxalaia* may have reached a length of between 12 and 14 m (39 and 46 ft) and a weight of 5 to 7 t (5.5 to 7.7 short tons; 4.9 to 6.9 long tons).[30] The largest known genus is *Spinosaurus*, which was capable of reaching lengths of 14 m (46 ft) and weighed around 7.4 t (8.2 short tons; 7.3 long tons), making it among the largest known theropod dinosaurs and terrestrial predators.[7] The closely allied

Sigilmassasaurus may have grown to a similar or greater length, though its taxonomic relationship with *Spinosaurus* is uncertain.[31] This consistency in large body size among spinosaurids could have evolved as a byproduct of their preference for semiaquatic lifestyles, as without the need to compete with other large theropod dinosaurs for food, they would have been able to grow to massive lengths.[32]

Skull



Annotated skull diagram of *Spinosaurus*

Spinosaurid skulls—similar in many respects to those of crocodilians—were long, low and narrow.[18] As in other theropods, various fenestrae (openings) in the skull aided in reducing its weight. In spinosaurs however, the antorbital fenestrae were greatly reduced, akin to those of crocodilians.[33] The tips of the premaxillae (frontmost snout bones) were expanded in a spoon shape, forming what has been called a "terminal rosette" of enlarged teeth. Behind this expansion, the upper jaw had a notch bearing significantly smaller teeth, into which the also expanded tips of the dentaries (tooth bearing bones of the mandible) fit into, with a notch behind the expansion of the dentary.[18] The maxillae (main upper jaw bones) were long and formed a low branch under the nostrils that connected to the rear of the premaxillae. The teeth at the frontmost part of the maxillae were small, becoming significantly larger soon after and then gradually decreasing in size towards the back of the jaw.[34] Analysis of the teeth of spinosaurids and their comparison to the teeth of tyrannosaurids suggest that the deep roots of spinosaurids helped to better anchor the teeth of these animals and distribute the stress against lateral forces generated during bites in predation and feeding scenarios.[35]

Closeup of the teeth of *Suchomimus*

Despite their highly modified skulls, analysis of the endocasts of *Baryonyx walkeri* and *Ceratosuchops inferodios* reveals spinosaurid brains shared a high degree of similarity with those of other non-maniraptoriform theropods.[36]

Lengthwise atop their skulls ran a thin and shallow sagittal crest that was usually tallest near or above the eyes, either becoming shorter or disappearing entirely towards the front of the head.[18][37][38] *Spinosaurus*'s head crest was comb-shaped and bore distinct vertical grooves,[37] while those of *Baryonyx* and *Suchomimus* looked like small triangular bumps.[39][6] *Irritator*'s median crest stopped above and behind the eyes in a bulbous, flattened shape. However, given that no fully preserved skulls are known for the genus, the complete shape of *Irritator*'s crest is unknown.[34] *Cristatusaurus* and *Suchomimus* (a possible synonym of the former) both had narrow premaxillary crests.[40] *Angaturama* (a possible synonym of *Irritator*) had an unusually tall crest on its premaxillae that nearly overhung the tip of the snout with a small forward protrusion.[38]

Spinosaurid nostrils were set far back on the skull, at least behind the teeth of the premaxillae, instead of at the front of the snout as in most theropods.[18] Those of *Baryonyx* and *Suchomimus* were large and started between the first and fourth maxillary teeth, while *Spinosaurus*'s nostrils were far smaller and

more retracted. *Irritator*'s nostrils were positioned similarly to those of *Baryonyx* and *Suchomimus*, and were between those of *Spinosaurus* and *Suchomimus* in size.[38] Spinosaurids had long secondary palates, bony and rugose structures on the roof of their mouths that are also found in extant crocodilians, but not in most theropod dinosaurs.[33] *Oxalaia* had a particularly elaborate secondary palate, while most spinosaurs had smoother ones.[30] The teeth of spinosaurids were conical, with an oval to circular cross section and either absent or very fine serrations. Their teeth ranged from slightly recurved, such as those of *Baryonyx* and *Suchomimus*, to straight, such as those of *Spinosaurus* and *Siamosaurus*, and the crown was often ornamented with longitudinal grooves or ridges.[38][41]

A 2026 study which analyzed the cranial material referred two controversial spinosaurid taxa (*Cristatusaurus* and *Spinosaurus maroccanus*), in which the data was interpreted as consistent with a higly developed neurovascular system on the premaxilla (tip of the snout) as well as a high tooth replacement rate from a young age.[42]

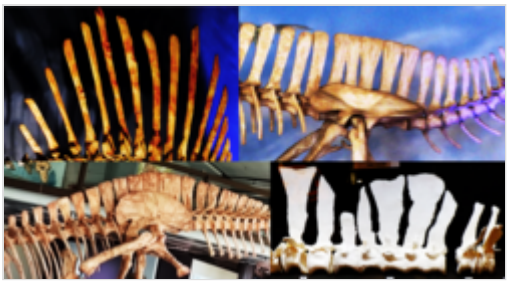
Postcranial skeleton

The coracoid bones of the shoulders in spinosaurids were robust and hook shaped.[6] The arms were relatively large and well-built; the radius (long bone of the forearm) was stout and usually only half as long as the humerus (upper arm bone). *Suchomimus* is the only spinosaur known to have preserved a furcula (wishbone), which shows that spinosaurs had a V-shaped furcula.[43] Spinosaurid hands had three fingers, typical of tetanurans, and wielded an enlarged ungual on the first finger (or "thumb"), which formed the bony core of a keratin claw. In genera like *Baryonyx* and *Suchomimus*, the phalanges (finger bones) were of conventional length for large theropods, and bore hook-shaped, strongly curved hand claws.[18][28] Based on fragmentary material from the forelimbs of *Spinosaurus*, it appears to have had longer, more gracile hands and straighter claws than other spinosaurids.[4]

Hand of spinosaurid specimen MN-4819-V (possibly belonging to *Angaturama*); note the enlarged condition of the first claw

Reconstructed foot bones of *Spinosaurus*; note the straight claws and enlarged hallux (first toe) touching the ground

The hindlimbs of *Suchomimus* and *Baryonyx* were somewhat short and mostly conventional of other megalosauroid theropods.[18][28] *Ichthyovenator*'s hip region was reduced, having the shortest pubis (pubic bone) and ischium (lower and rearmost hip bone) in proportion to the ilium (main hip bone) of any other known theropod.[44] *Spinosaurus* had an even smaller pelvis and hindlimbs in proportion to its body size; its legs composed just over 25 percent of the total body length. Substantially complete spinosaurid foot remains are only known from *Spinosaurus*. Unlike most theropods—which walk on three toes, with the hallux (first toe) being reduced and elevated off the ground—*Spinosaurus* walked on four functional toes, with an enlarged hallux that came in contact with the ground. The unguals of its feet, in contrast with the deeper, smaller and recurved unguals of other theropods, were shallow, long, large in relation to the foot, and had flat bottoms. Based on comparisons with those of modern shorebirds, it is theorized to be probable that the *Spinosaurus*'s feet were webbed.[4]



Reconstructed neural spine sails of four spinosaurids; clockwise from top left: *Spinosaurus*, *Irritator*, *Ichthyovenator*, and *Suchomimus*.

The upward-projecting neural spines of spinosaurid vertebrae (backbones) were very tall, more so than in most theropods. In life, these spines would have been covered in skin or fat tissue and formed a sail down the animal's back, a condition that has also been observed in some carcharodontosaurid and ornithopod dinosaurs.[18][45] The eponymous neural spines of *Spinosaurus* were extremely tall, measuring over 1 m (3 ft 3 in) in height on some of the dorsal (back) vertebrae.[46] *Suchomimus* had a lower, ridge-like sail across the majority of its back, hip, and tail region.[6] *Baryonyx* showed a reduced sail, with a few of the rearmost vertebral spines being somewhat elongated.[39] *Ichthyovenator* had a sinusoidal (wave-like) sail that was separated in two over the hips, with the upper ends of some neural spines being broad and fan-shaped.[44] A neural spine from the holotype of *Vallibonavenatrix* shows a similar morphology to those of *Ichthyovenator*, indicating the presence of a sail in this genus as well.[47] One partial skeleton possibly referable to *Angaturama* also had elongated neural spines on its hip region.[48][49] The presence of a sail in fragmentary taxa like *Sigilmassasaurus* is unknown.[20] In members of the subfamily Spinosaurinae, like *Ichthyovenator* and *Spinosaurus*, the neural spines of the caudal (tail) vertebrae were tall and reclined, accompanied by also elongated chevrons—long, thin bones that form the underside of the tail. This was most pronounced in *Spinosaurus*, in which the spines and chevrons formed a large paddle-like structure, deepening the tail significantly along most of its length.[50][31]

Classification

The family Spinosauridae was named by Stromer in 1915 to include the single genus *Spinosaurus*. The clade was expanded as more close relatives of *Spinosaurus* were uncovered. The first cladistic definition of Spinosauridae was provided by Paul Sereno in 1998 (as "All spinosauroids closer to *Spinosaurus* than to *Torvosaurus*").[6]

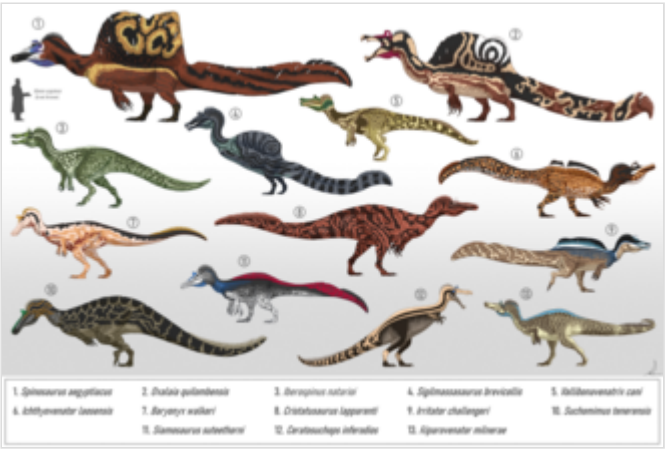
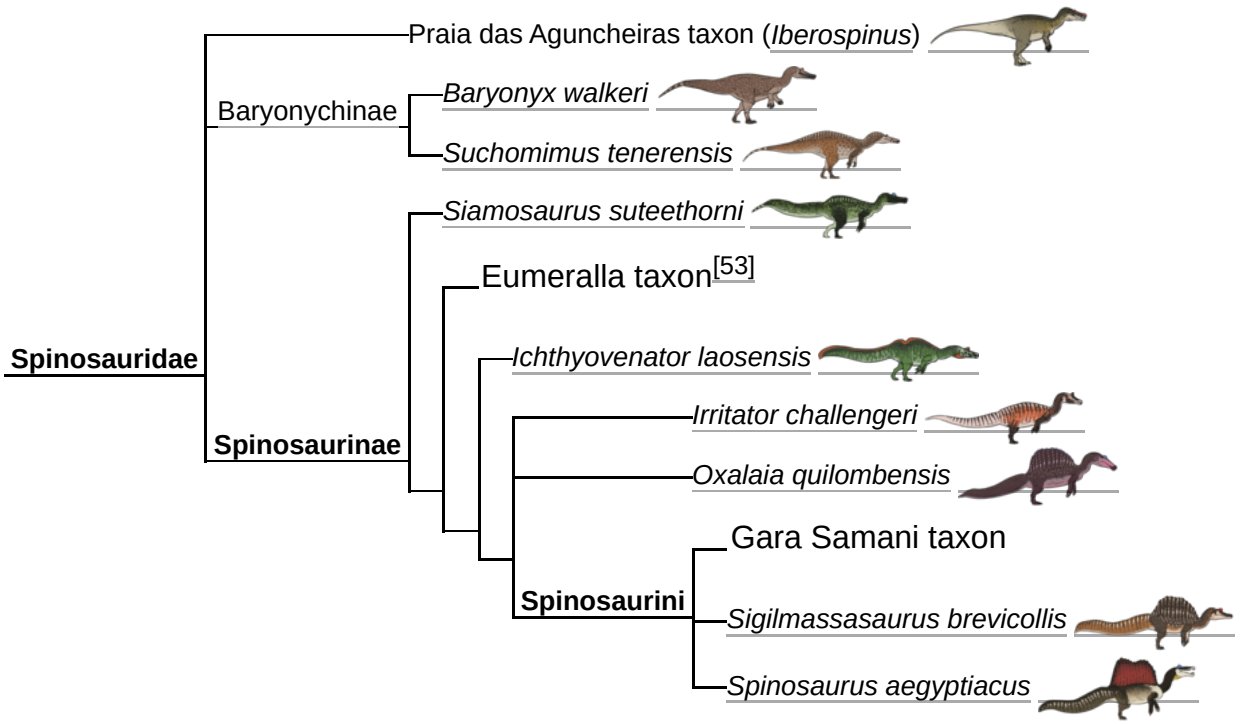


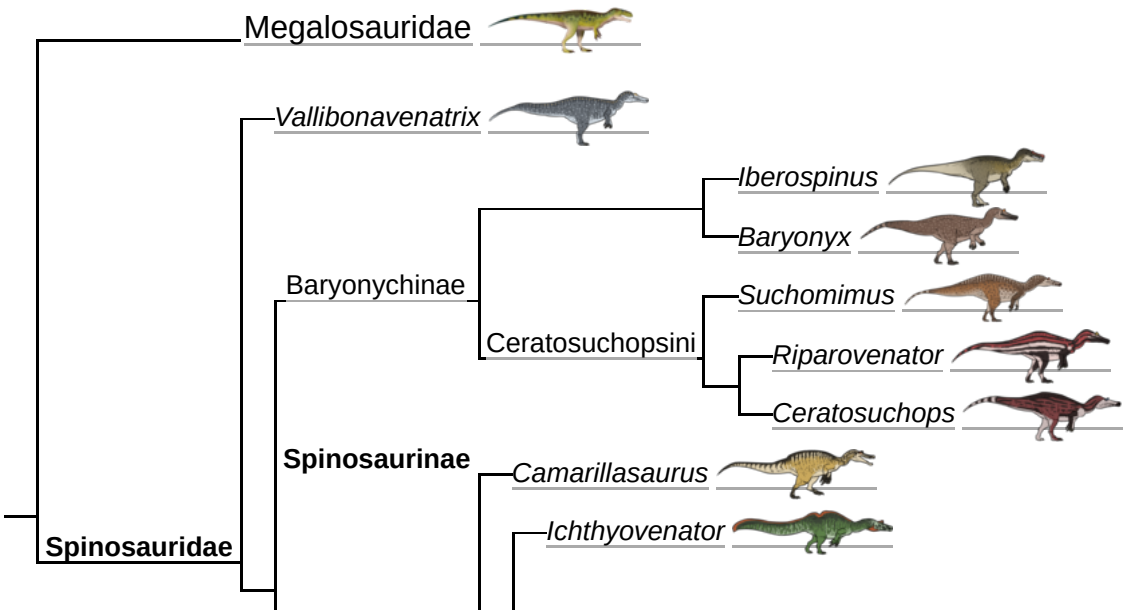
Diagram illustrating various spinosaurids

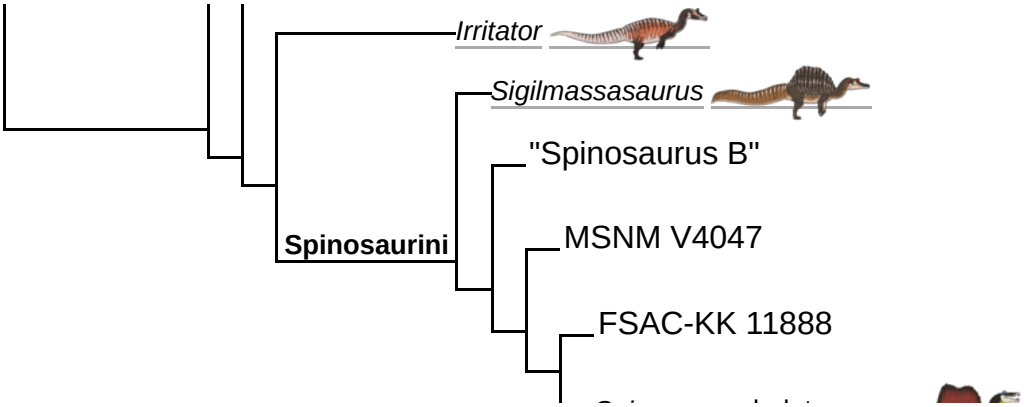
Traditionally, Spinosauridae is divided into two subfamilies: Spinosaurinae, which contains the genera *Icthyovenator*, *Irritator*, *Oxalaia*, *Sigilmassasaurus* and *Spinosaurus*, is marked by unserrated, straight teeth, and external nares which are further back on the skull than in baryonychines,[6][51] and Baryonychinae, which contains the genera *Baryonyx*, *Cristatusaurus*, *Suchosaurus*, *Suchomimus*, *Ceratosuchops*, and *Riparovenator*,[52] which is marked by serrated, slightly curved teeth, smaller size, and more teeth in the lower jaw behind the terminal rosette than in spinosaurines.[6][51] Others, such as *Siamosaurus*, may belong to either Baryonychinae or Spinosaurinae, but are too incompletely known to be assigned with confidence.[52] *Siamosaurus* was classified as a spinosaurine in 2018, but the results are provisional and not entirely conclusive.[31]

The subfamily Spinosaurinae was named by Sereno in 1998, and defined by Thomas Holtz and colleagues in 2004 as all taxa closer to *Spinosaurus aegyptiacus* than to *Baryonyx walkeri*. The subfamily Baryonychinae was named by Charig & Milner in 1986. They erected both the subfamily and the family Baryonychidae for the newly discovered *Baryonyx*, before it was referred to Spinosauridae. Their subfamily was defined by Holtz and colleagues in 2004, as the complementary clade of all taxa closer to *Baryonyx walkeri* than to *Spinosaurus aegyptiacus*. Examinations in 2017 by Marcos Sales and Cesar Schultz suggested that the South American spinosaurids *Angaturama* and *Irritator* may be intermediate between Baronychinae and Spinosaurinae based on their craniodental features and cladistic analysis. A study by Arden et al. (2018) named the tribe Spinosaurini to include *Spinosaurus* and *Sigilmassasaurus*. The results of the 2018 phylogenetic analysis by Arden and colleagues, which included many unnamed taxa, are displayed in the cladogram below:[31]



In 2021, Barker et al. described two new spinosaurid species, *Ceratosuchops inferodios* and *Riparovenator milnerae* as part of a newly proposed Ceratosuchopsini. In the paper, they performed a phylogenetic analysis focused on Spinosauridae. The results of their analysis appear below:[54]





Evolution

Distribution of spinosaurids in Europe and North Africa during the Cretaceous; 1, 3, 4, 5, 6 are *Baryonyx*

Spinosaurids appear to have been widespread from the Barremian to the Cenomanian stages of the Cretaceous period, about 130 to 95 million years ago. Possibly the earliest remains of spinosaurids are known from the Middle Jurassic of Niger and India, the latter of which otherwise has no remains of spinosaurids.[1][55] They shared features such as long, narrow, crocodile-like skulls; sub-circular teeth, with fine to no serrations; the terminal rosette of the snout; and a secondary palate that made them more resistant to torsion. In contrast, the primitive and typical condition for theropods was a tall, narrow snout with blade-like (ziphodont) teeth with serrated carinae.[56] The skull adaptations of spinosaurids converged with those of crocodilians; early members of the latter group had skulls similar to typical theropods, later developing elongated snouts, conical teeth, and secondary palates. These adaptations may have been the result of a dietary change from terrestrial prey to fish. Unlike crocodiles, the post-cranial skeletons of baryonychine spinosaurids do not appear to have aquatic adaptations.[4][56] Sereno and colleagues proposed in 1998 that the large thumb-claw and robust forelimbs of spinosaurids evolved in the Middle Jurassic, before the elongation of the skull and other adaptations related to fish-eating, since the former features are shared with their megalosaurid relatives. They also suggested that the spinosaurines and baryonychines diverged before the Barremian age of the Early Cretaceous.[57]

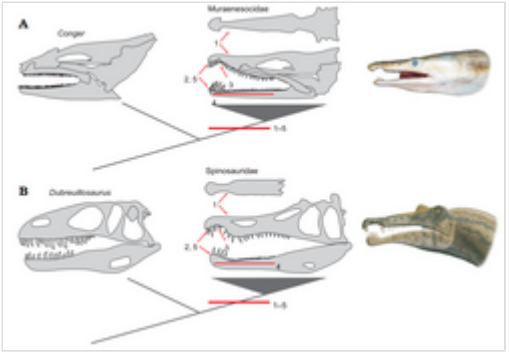
Several theories have been proposed about the biogeography of the spinosaurids. Since *Suchomimus* was more closely related to *Baryonyx* (from Europe) than to *Spinosaurus*—although that genus also lived in Africa—the distribution of spinosaurids cannot be explained as vicariance resulting from continental rifting.[57] Sereno and colleagues[57] proposed that spinosaurids were initially distributed across the supercontinent Pangea, but split with the opening of the Tethys Sea. Spinosaurines would then have evolved in the south (Africa and South America: in Gondwana) and baryonychines in the north (Europe: in Laurasia), with *Suchomimus* the result of a single north-to-south dispersal event.[57] Buffetaut and the Tunisian palaeontologist Mohamed Ouaja also suggested in 2002 that baryonychines could be the ancestors of spinosaurines, which appear to have replaced the former in Africa.[52] Milner suggested in 2003 that spinosaurids originated in Laurasia during the Jurassic, and dispersed via the Iberian land bridge into Gondwana, where they radiated.[58] In 2007, Buffetaut pointed out that palaeogeographical

studies had demonstrated that Iberia was near northern Africa during the Early Cretaceous, which he found to confirm Milner's idea that the Iberian region was a stepping stone between Europe and Africa, which is supported by the presence of baryonychines in Iberia. The direction of the dispersal between Europe and Africa is still unknown,[59] and subsequent discoveries of spinosaurid remains in Asia and possibly Australia indicate that it may have been complex.[60]

In 2016, the Spanish palaeontologist Alejandro Serrano-Martínez and colleagues reported the oldest known spinosaurid fossil, a tooth from the Middle Jurassic of Niger, which they found to suggest that spinosaurids originated in Gondwana, since other known Jurassic spinosaurid teeth are also from Africa, but they found the subsequent dispersal routes unclear.[55] Some later studies instead suggested this tooth belonged to a megalosaurid.[61][62] Candeiro and colleagues suggested in 2017 that spinosaurids of northern Gondwana were replaced by other predators, such as abelisauroids, since no definite spinosaurid fossils are known from after the Cenomanian anywhere in the world. They attributed the disappearance of spinosaurids and other shifts in the fauna of Gondwana to changes in the environment, perhaps caused by transgressions in sea level.[63] Malafaia and colleagues stated in 2020 that *Baryonyx* remains the oldest unquestionable spinosaurid, while acknowledging that older remains had also been tentatively assigned to the group.[64] Barker and colleagues found support for a European origin for spinosaurids in 2021, with an expansion to Asia and Gondwana during the first half of the Early Cretaceous. In contrast to Sereno, these authors suggested there had been at least two dispersal events from Europe to Africa, leading to *Suchomimus* and the African part of Spinosaurinae.[54]

Paleobiology

Diet and feeding



Comparison of a spinosaurid skull with that of *Dubreuillosaurus* and two extant pike conger eels

Spinosaurid teeth resemble those of crocodiles, which are used for piercing and holding prey. Therefore, teeth with small or no serrations, such as in spinosaurids, were not good for cutting or ripping into flesh but instead helped to ensure a strong grip on a struggling prey animal.[65] However, some have argued that spinosaurids converge more with phytosaurs in terms of craniodental anatomy and that phytosaurs are a better analogue for inferring their life modes.[66] Spinosaurid jaws were likened by Romain Vullo and colleagues to those of the pike conger eel, in what they hypothesized was convergent evolution for aquatic feeding. Both kinds of animals have some teeth in the end of the upper and lower jaws that are larger than the others and an area of the upper jaw with smaller teeth, creating a gap into which the enlarged teeth of the lower jaw fit, with the full structure called a terminal rosette.[67]

In the past, spinosaurids have often been considered piscivores (fish-eaters) in the main, based on comparisons of their jaws with those of modern crocodilians.[51] In 2007, British paleontologist Emily J. Rayfield and colleagues conducted biomechanical studies on the skull of *Baryonyx*, which had a long, laterally compressed skull, comparing it to gharial (long, narrow, tubular) and alligator (flat and wide)

Life restoration of *Baryonyx* with a fish in its jaws

skulls. They found that the structure of baryonychine jaws converged on that of gharials, in that the two taxa showed similar response patterns to stress from simulated feeding loads, and did so with and without the presence of a (simulated) secondary palate. The gharial, exemplar of a long, narrow, and tubular snout, is a fish specialist. However, this snout anatomy does not preclude other options for the spinosaurids. The gharial is the most extreme example and a fish specialist; Australian freshwater crocodiles, which have similarly shaped skulls to gharials, also specialize more on fish than sympatric, broad snouted crocodiles and are opportunistic feeders which eat all manner of small aquatic prey, including insects and crustaceans. Thus, spinosaurids' snouts correlate with piscivory; this is consistent with hypotheses of this diet for spinosaurids, in particular baryonychines, but it does not indicate that they were solely piscivorous.[33]

Life restoration of the head of *Spinosaurus*

Further study by Andrew R. Cuff and Rayfield in 2013 on the skulls of *Spinosaurus* and *Baryonyx* did not recover similarities in the skulls of *Baryonyx* and the gharial that the previous study did. *Baryonyx* had, in models where the size difference of the skulls was corrected for, greater resistance to torsion and dorsoventral bending than both *Spinosaurus* and the gharial, while both spinosaurids were inferior to the gharial, alligator, and slender-snouted crocodile in resisting torsion and medio-lateral bending. When the results from the modeling were not scaled according to size, then both spinosaurids performed better than all the crocodilians in resistance to bending and torsion, due to their larger size. Thus, Cuff and Rayfield suggested that the skulls were not efficiently built to deal well with relatively large, struggling prey, but that spinosaurids may overcome prey simply by their size advantage, and not skull build.[68] In 2002, Hans-Dieter Sues and colleagues studied the construction of the spinosaurid skull, and concluded that their mode of feeding was to use extremely quick, powerful strikes to seize small prey items using their jaws, whilst employing the powerful neck muscles in rapid up-and-down motion. Due to the narrow snout, vigorous side-to-side motion of the skull during prey capture is unlikely.[65] Based on the size and positions of their nostrils, Marcos Sales and Cesar Schultz in 2017 suggested that *Spinosaurus* possessed a greater reliance on its sense of smell and had a more piscivorous lifestyle than *Irritator* and baryonychines.[38]

Direct fossil evidence shows that spinosaurids fed on fish as well as a variety of other small to medium-sized animals, including dinosaurs. *Baryonyx* was found with scales of the prehistoric fish *Scheenstia* in its body cavity, and these were abraded, hypothetically by gastric juices. Bones of a young *Iguanodon*, also abraded, were found alongside this specimen. If these represent *Baryonyx*'s meal, the animal was, whether in this case a hunter, or a scavenger, an eater of more diverse fare than fish.[51][65][39] Moreover, there is a documented example of a spinosaurid having eaten a pterosaur, as one *Irritator* tooth was found lodged within the fossil vertebrae of an ornithocheirid pterosaur found in the Romualdo Formation of Brazil. This may represent a predation or a scavenging event.[69][70] A fossil snout referred to *Spinosaurus* was discovered with a vertebra from the sclerorhynchid *Onchopristis* embedded in it.[37] In the Sao Khua Formation of Thailand, isolated tooth crowns from *Siamosaurus* have been found in

Reconstructed *Irritator* skeleton mounted as attacking an anhanguerid pterosaur, National Museum of Rio de Janeiro

association with sauropod remains, indicating possible predation or scavenging.[71] The Portuguese *Iberospinus* fossils were also found associated with isolated *Iguanodon* teeth, and those cases are listed; along with other such associations as support for opportunistic feeding behaviour in spinosaurids.[40]

A 2018 study by Auguste Hassler and colleagues of calcium isotopes in the teeth of North African theropods found that spinosaurids had a mixed diet of fish and herbivorous dinosaurs (such as *Nigersaurus* and *Ouranosaurus*), whereas the other theropods examined (abelisaurids and carcharodontosaurids) mainly fed on herbivorous dinosaurs. This might indicate ecological partitioning between these theropods.[72] Later in the same year, Tito Aureliano and colleagues presented a possible scenario for the food web of Brazilian Romualdo Formation. The researchers proposed that the diet of spinosaurines from this environment may have included—in addition to pterosaurs—terrestrial and aquatic crocodyliforms, juveniles of their own species, turtles, and small to medium-sized dinosaurs. This would have made spinosaurines apex predators within this particular ecosystem.[32]

A 2024 study by D'Amore et al., further vindicates the theory that spinosaurids were similar in niche to generalist or macro-generalist crocodilians. This study likewise suggests their jaws and teeth were well-suited to quick strikes and deep, puncturing bites, but not for slicing flesh or crushing bones. In particular, baryonychine spinosaurids probably did little oral processing of their prey when feeding, but by comparison, spinosaurines were found to be quite capable of processing the meat of relatively large vertebrate prey. None of these findings suggest any spinosaurids from either subfamily were restricted only to fish and small aquatic vertebrates.[73]

Hendrickx (2026) suggested that spinosaurids likely had a generalist dietary preference but probably consumed more aquatic prey than other theropods based on dentition analysis.[74] Pittet (2026) analyzed skull elements referred to the two controversial taxa *Cristatusaurus* and *Spinosaurus maroccanus*, interpreting the analyzed data as consitent with a highly developed neurovascular system of the snout's tip and a high tooth replacement rate from a young age.[42] Buffetaut et al. (2026) reported distinct types of spinosaurid teeth from the Early Cretaceous of eastern Thailand, including a type with no enamel ribbing, interpreted as having possible preference to terrestrial prey, such as turtles.[75]

Forelimb function

The use of the robust forelimbs and giant recurved claws of spinosaurs remains a debated topic. Charig and Milner speculated in 1986 that *Baryonyx* may have crouched by the riverbank and used its claws to gaff fish out of the water, similarly to grizzly bears.[76] In 1987, British biologist Andrew Kitchener argued that with both its crocodile-like snout and enlarged claws, *Baryonyx* seemed to have too many adaptations for piscivory when one would have been enough. Kitchener instead postulated that *Baryonyx* more likely used its arms to scavenge the corpses of large dinosaurs, such as

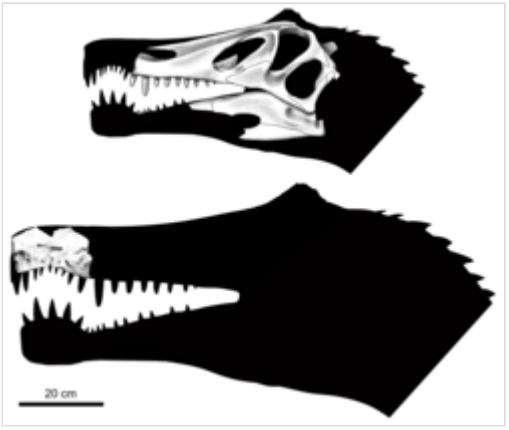
Reconstructed forelimb and hand of *Suchomimus*, Museum of Ancient Life, Utah

Iguanodon, by breaking into the carcass with the large claws, and subsequently probing for viscera with its long snout.[77] In their 1997 article, Charig and Milner rejected this hypothesis, pointing out that in most cases, a carcass would have already been largely emptied out by its initial predators.[39] Later research has also ruled out this sort of specialized scavenging.[18]

In 1986, Charig and Milner suggested that the robust forelimbs and giant thumb claws would have been *Baryonyx*'s primary method of capturing, killing, and tearing apart large prey; whereas its long snout would have been used mostly for fishing.[39] A 2005 study by Canadian paleontologist the François Therrien and colleagues agreed that spinosaur forelimbs were probably used for hunting larger prey items, given that their snouts could not resist the bending stress.[78] In a 2017 review of the family, David Hone and Holtz considered possible functions in digging for water sources or hard to reach prey, as well as burrowing into soil to construct nests.[18] In 2026, Scherer, Steell and Upchurch suggested that baryonychines and other spinosaurids in general had relatively lower skull–forelimb length ratios than other theropod lineages, which are indicative of less forelimb reduction and possibly more reliance on using the forelimbs than the skull for prey capture.[79]

Cranial crests and neural spines

Theropod heads are often decorated with some form of crest, horn, or rugose structure, which in life, would have been extended by keratin.[80] Though there has been little discussion on the head crests of spinosaurs, Hone and Holtz in 2017 considered that their most likely use was for displaying to potential mates or as a means of threatening rivals and other predators.[18] Such has been suggested for theropod cranial structures before, which may have been aided by unusual or bright coloration to provide further visual cues.[80]



Holotype skull specimens of *Irritator challengeri* (top, showing the beginnings of a nasal crest) and *Angaturama limai* (bottom, showing a tall premaxillary crest)

Many theories have been proposed over the years for the use of spinosaurid dorsal sails, such as thermoregulation;[81] to aid in swimming;[82] to store energy or insulate the animal; or for display purposes, such as intimidating rivals and predators, or attracting mates.[83][84] Many elaborate body structures of modern-day animals serve to attract members of the opposite sex during mating. It is possible that the sail of *Spinosaurus* was used for courtship, in a way similar to a peacock's tail. In 1915, Stromer speculated that the size of the neural spines may have differed between males and females.[84] In 2012, French paleontologist Ronan Allain and colleagues suggested considering the high diversity in neural spine elongation observed in theropod dinosaurs, as well as histological research done on the sails of synapsids (stem mammals), the sinusoidal sail of *Ichthyovenator* was likely used for courtship display or recognising members of its own species.[44] In a 2013 blog post, Darren Naish considered the latter function unlikely, favouring the hypothesis of sexual selection for *Ichthyovenator*'s sail because it appears to have evolved on its own, without very close relatives. Naish also notes it is possible similar relatives have not yet been discovered.[85]

In 2015, the German biophysicist Jan Gimsa and colleagues suggested that this feature could also have aided aquatic movement by improving manoeuvrability when submerged, and acted as fulcrum for powerful movements of the neck and tail (similar to those of sailfish or thresher sharks).[86][83]

Ontogeny

Juvenile spinosaurid fossils are somewhat rare. However, an ungual phalanx measuring 21 mm (0.83 in) belonging to a very young *Spinosaurus* indicates that *Spinosaurus*, and probably by extent other spinosaurids, may have developed their semiaquatic adaptations at birth or at a very young age and maintained the adaptations throughout their lives. The specimen, found in 1999 and described by Simone Maganuco, Cristiano Dal Sasso and colleagues in 2018, is believed to have come from a very small juvenile measuring 1.78 m (5.8 ft), making said specimen the smallest known example of a spinosaurid currently described.[87][88]

Paleoecology

Habitat preference

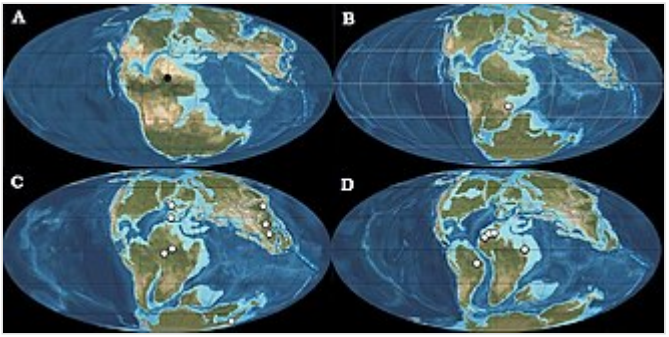
A 2010 publication by Romain Amiot and colleagues found that oxygen isotope ratios of spinosaurid bones indicates semiaquatic lifestyles. Isotope ratios from teeth from *Baryonyx*, *Irritator*, *Siamosaurus*, and *Spinosaurus* were compared with isotopic compositions from contemporaneous theropods, turtles, and crocodilians. The study found that, among theropods, spinosaurid isotope ratios were closer to those of turtles and crocodilians. *Siamosaurus* specimens tended to have the largest difference from the ratios of other theropods, and *Spinosaurus* tended to have the least difference. The authors concluded that spinosaurids, like modern crocodilians and hippopotamuses, spent much of their daily lives in water. The authors also suggested that semiaquatic habits and piscivory in spinosaurids can explain how spinosaurids coexisted with other large theropods: by feeding on different prey items and living in different habitats, the different types of theropods would have been out of direct competition.[89]

In 2016, Sales and colleagues statistically examined the fossil distribution of spinosaurids, abelisaurids, and carcharodontosaurids, and concluded that spinosaurids had the strongest support for association with coastal palaeoenvironments. Spinosaurids also appear to have inhabited inland environments (with their distribution there being comparable to carcharodontosaurids), which indicates they may have been more generalist than usually thought.[90] In 2018, an analysis was conducted on the partial tibia of an indeterminate spinosaurine from the early Albian, the bone was from a sub-adult between 7 and 13 m (22 and 42 ft) in length still growing moderately fast before its death. This specimen (LPP-PV-0042) was found in the Araripe Basin of Brazil and taken to the University of San Carlos for a CT Scan, where it revealed osteosclerosis (high bone density).[32] This condition had previously only been observed in *Spinosaurus*, as a possible way of controlling its buoyancy.[4] The presence of this condition on the leg fragment showed that semi-aquatic adaptations in spinosaurids were already present at least 10 million years before *Spinosaurus aegyptiacus* appeared. According to the phylogenetic bracketing method, this high bone density might have been present in all spinosaurines.[32] In the same year, Canadian palaeontologist Donald M. Henderson published a study of theropod buoyancy (through simulation with 3D models) and found that distantly related theropods floated as well as the tested spinosaurs, and instead supported they would have stayed by the shorelines or shallow water rather than being semi-aquatic.[91]

In 2021, Beevor and colleagues reported the bonebed containing the teeth of *Spinosaurus* and rostral denticles of *Onchopristis*, and suggested this as taphonomic evidence in the Kem Kem group that would support *Spinosaurus* being a semi-aquatic dinosaur.[92] In 2024, Myhrvold and colleagues cited the immediate assumption of spinosaurids being avid divers due to correlations in bone compactness as being subject to errors, such as flawed statistical methods and measurements, as well as sampling bias. Given the amount of variation in specimens and in data collection techniques, they concluded that previous evidence isn't strong enough to put *Spinosaurus* swimming and diving entirely submerged, suggesting that *Spinosaurus* still more likely mostly hung out on shore, akin to wader lifestyle previously interfered.[93]

A 2026 paper by Cau et al. analyzed five spinosaurine skulls and found evidence of the presence of salt glands, which had convergently evolved to be similar to those of semi-aquatic and aquatic birds. This suggests that derived spinosaurines were able to hunt in brackish and saltwater environments. They noted that the evidence of salt glands do not necessarily confirm whether spinosaurines are fully aquatic, diving pursuit predators, or hunted prey like modern wading birds.[94]

Distribution



Generalized locations of spinosaurid fossil discoveries from the Bajocian–Bathonian (A), Tithonian (B), Barremian−Aptian (C), and Albian−Cenomanian (D) marked on maps of those time spans.

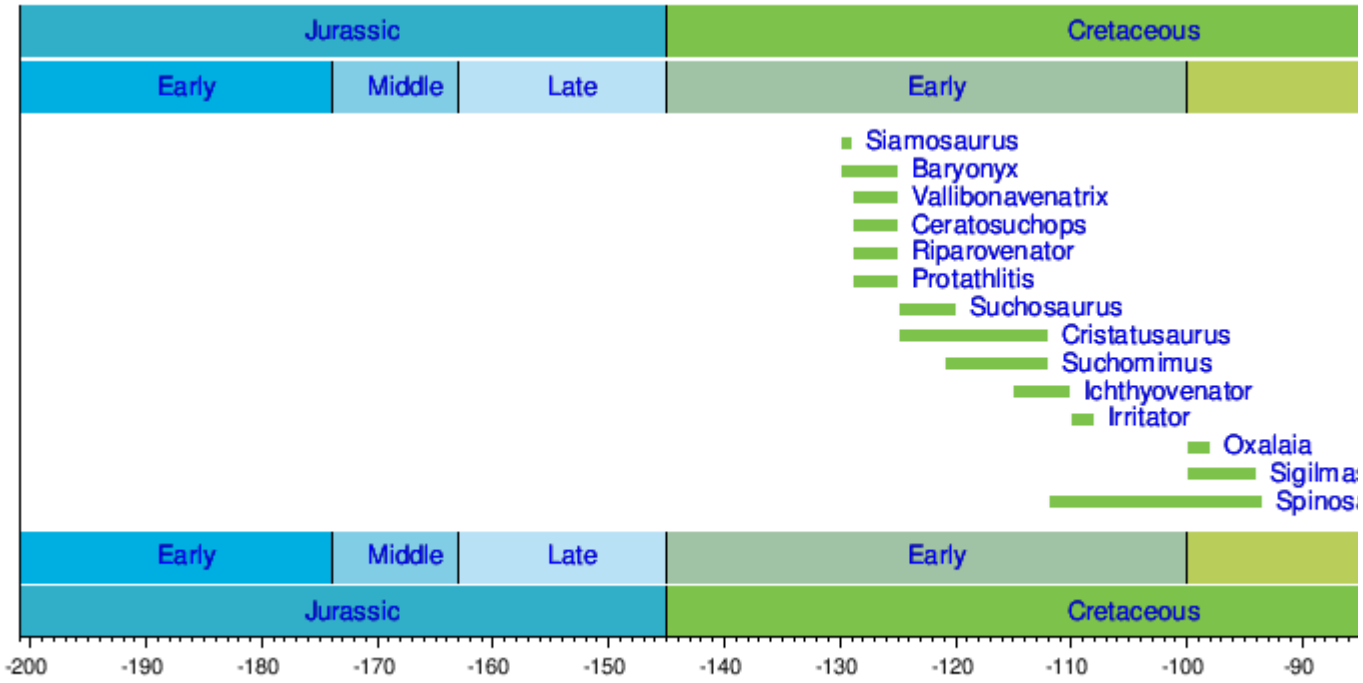
Confirmed spinosaurids have been found on every continent except for North America, Australia and Antarctica, the first of which was *Spinosaurus aegyptiacus*, discovered at the Bahariya Formation in Egypt.[84] Baryonychines were common, such as *Baryonyx*, which lived during the Barremian of England and Spain. *Baryonyx*-like teeth are also found from the earlier Hauterivian and later Aptian sediments of Spain, as well as the Hauterivian of England.[18][60] Baryonychines were represented in Africa, with *Suchomimus tenerensis* and *Cristatusaurus lapparenti* as well as *Baryonyx*-like teeth from the Aptian of Niger.[52][6][95] as well as in Europe, with *Suchosaurus cultridens* and *S. girardi* from the England. *Baryonyx*-like teeth are also reported from the Ashdown Sands of Sussex, in England, and the Burgos Province, in Spain.[60] Other European spinosaurids *Camarillasaurus cirugedae* and *Iberospinus natarioi* are known from the Barremian of Spain and Portugal, respectively.[96][97]

The earliest record of spinosaurines is from Europe, with the Barremian species *Vallibonavenatrix cani* from Spain.[47] Spinosaurines are also present in Albian sediments of Tunisia and Algeria, and in Cenomanian sediments of Egypt and Morocco. In Africa, baryonychines were common in the Aptian, and then replaced by spinosaurines in the Albian and Cenomanian.[52] such as in the Kem Kem beds of Morocco, which housed an ecosystem containing many large coexisting predators.[40][89] A fragment of a spinosaurine lower jaw from the Early Cretaceous was also reported from Tunisia, and referred to *Spinosaurus*.[52] Spinosaurinae's range also extended to South America, particularly Brazil, with the discoveries of *Irritator challengeri*, *Angaturama limai*, and *Oxalaia quilombensis*.[70][30] There was also a fossil tooth in Argentina which has been referred to the Spinosauridae by Leonardo Salgado and colleagues.[98] This referral is doubted by Gengo Tanaka et al., who offers *Hamadasuchus*, a crocodilian, as the most likely animal of origin for these teeth.[99]

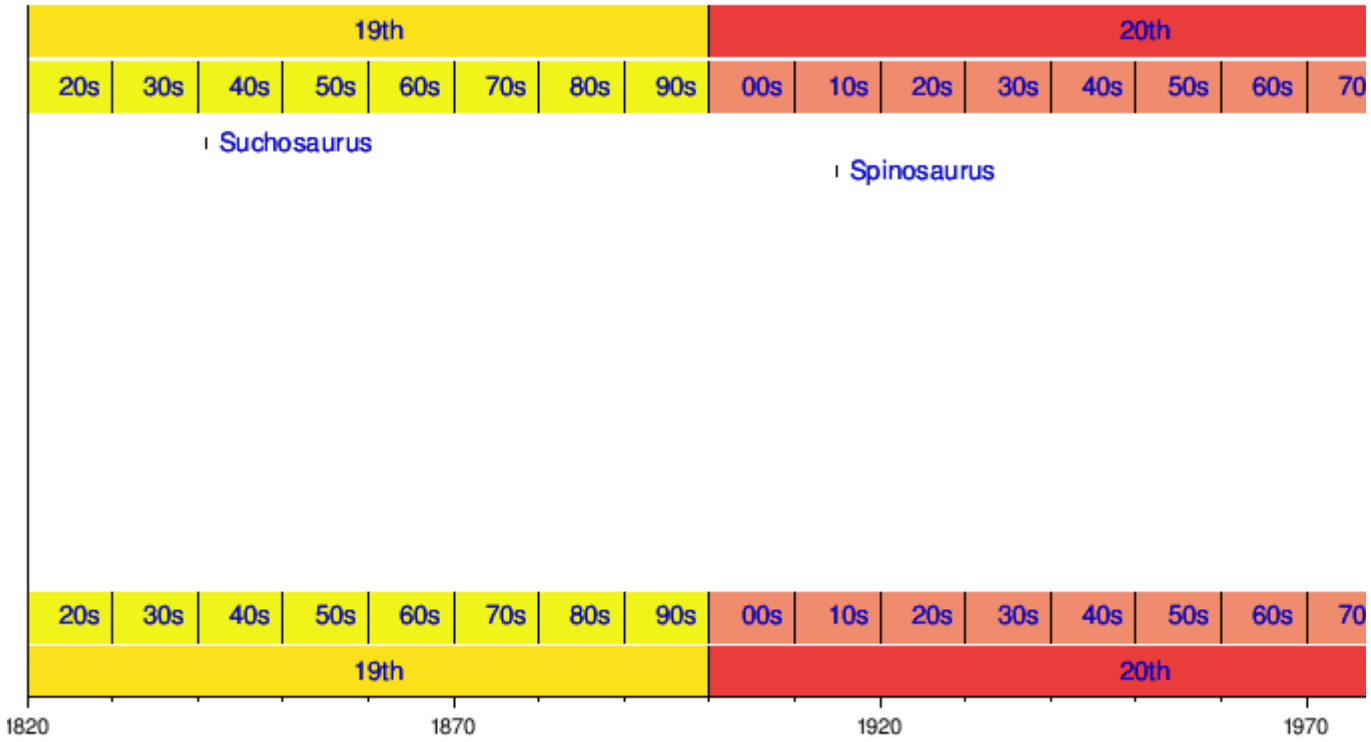
Partial skeletons and numerous fossil teeth indicate spinosaurids were widespread in Asia; three taxa—all spinosaurines—have been named: *Siamosaurus suteethorni* from Thailand, "*Sinopliosaurus*" *fusuiensis* from China, and *Ichthyovenator laosensis* from Laos.[44][52][100] Spinosaurid teeth have been found in Malaysia; they were the first dinosaur remains discovered in the country.[101] Some intermediate specimens extend the known range of spinosaurids past the youngest dates of named taxa. A single theropod tooth attributed to Baryonychinae was found in the mid-Santonian Majiacun Formation of Henan, China,[100] but this tooth lacked spinosaurid synapomorphies,[102][103][104] and it was reclassified as a theropod of uncertain affinities, either an allosauroid or a sister taxon of abelisaurids in 2023.[105] In 2025, Olmedo-Romaña et al. described putative theropod teeth with similarities to spinosaurid teeth from the strata of the Fundo El Triunfo Formation (Peru) interpreted as spanning all or part of the Campanian–Maastrichtian interval, and tentatively classified the studied fossil material as remains of late-surviving members of the family Spinosauridae, as well as the first record of the group from western South America.[106] In the same year, Barker, Naish and Gostling argued that these teeth lack key features of spinosaurid dentition, and that they most likely represent crocodylomorph teeth.[107] At la Cantalera-1, a site in the early Barremian Blesa Formation in Treul, Spain, two types of spinosaurid teeth were found, and they were assigned, tentatively, as indeterminate spinosaurine and baryonychine taxa.[108] An indeterminate spinosaurid was discovered in the Early Cretaceous Eumeralla Formation, Australia.[109] It is known from a single 4 cm long partial cervical vertebra, designated NMV P221081. It is missing most of the neural arch. The specimen is from a juvenile estimated to be about 2 to 3 meters long (6–9 ft). Out of all spinosaurids, it most closely resembles *Baryonyx*.[53] In 2019, it was suggested that the vertebra instead belonged to a megaraptorid theropod, as opposed to a spinosaur.[110]A 2020 publication reported 18 indeterminate spinosaurid teeth from the Aptian Kitadani Formation of Japan.[111]

Indeterminate spinosaurid teeth from the Japanese Kitadani Formation

Timeline of genera



Timeline of genera descriptions



See also

- *Spinosaurus*
- *Oxalaia*
- *Sigilmassasaurus*
- *Baryonyx*

References

1. Sharma, A.; Novas, Fernando E. (2023). "First Jurassic Evidence of a Possible Spinosaurid Pedal Ungual, from the Jaisalmer Basin, India" (https://doi.org/10.54103%2F2039-4942%2F20032). *Rivista Italiana di Paleontologia e Stratigrafia*. **29** (3): 653–670. Bibcode:2023RIPS..12920032S (https://ui.adsabs.harvard.edu/abs/2023RIPS..12920032S). doi:10.54103/2039-4942/20032 (https://doi.org/10.54103%2F2039-4942%2F20032).
2. Rauhut, Oliver W M; Bakirov, Aizek A; Wings, Oliver; Fernandes, Alexandra E; Hübner, Tom R (2024-08-01). "A new theropod dinosaur from the Callovian Balabansai Formation of Kyrgyzstan" (https://doi.org/10.1093%2Fzoolinnean%2Fzlae090). *Zoological Journal of the Linnean Society*. **201** (4) zlae090. doi:10.1093/zoolinnean/zlae090 (https://doi.org/10.1093%2Fzoolinnean%2Fzlae090). ISSN 0024-4082 (https://search.worldcat.org/issn/0024-4082).
3. Olshevsky, G. (1991). "A revision of the parainfraclass Archosauria Cope, 1869, excluding the advanced Crocodylia" (https://web.archive.org/web/20180819050626/http://www.miketaylor.org.uk/tmp/Olshevsky_1991_A_revision_of_the_parainfraclass_Archosauria_Cope_1869_excluding_the_advanced_Crocodylia.pdf) (PDF). *Mesozoic Meanderings*. **2**: 1–196. Archived from the original (http://www.miketaylor.org.uk/tmp/Olshevsky_1991_A_revision_of_the_parainfraclass_Archosauria_Cope_1869_excluding_the_advanced_Crocodylia.pdf) (PDF) on 19 August 2018. Retrieved 14 April 2018.
4. Ibrahim, Nizar; Sereno, Paul C.; Dal Sasso, Cristiano; Maganuco, Simone; Fabri, Matteo; Martill, David M.; Zouhri, Samir; Myhrvold, Nathan; Lurino, Dawid A. (2014). "Semiaquatic adaptations in a giant predatory dinosaur" (https://researchportal.port.ac.uk/portal/en/publications/semiaquatic-adaptations-in-a-giant-predatory-dinosaur(8f11a1ce-3265-4b3b-8c81-6f576856a87f).html). *Science*. **345** (6204): 1613–6. Bibcode:2014Sci...345.1613I (https://ui.adsabs.harvard.edu/abs/2014Sci...345.1613I). doi:10.1126/science.1258750 (https://doi.org/10.1126%2Fscience.1258750). PMID 25213375 (https://pubmed.ncbi.nlm.nih.gov/25213375). S2CID 34421257 (https://api.semanticscholar.org/CorpusID:34421257). Supplementary Information (https://www.science.org/doi/10.1126/science.1258750)
5. Smyth, Robert S.H.; Ibrahim, Nizar; Martill, David M. (October 2020). "Sigilmassasaurus is Spinosaurus: A reappraisal of African spinosaurines". *Cretaceous Research*. **114** 104520. Bibcode:2020CrRes.11404520S (https://ui.adsabs.harvard.edu/abs/2020CrRes.11404520S). doi:10.1016/j.cretres.2020.104520 (https://doi.org/10.1016%2Fj.cretres.2020.104520). S2CID 219487346 (https://api.semanticscholar.org/CorpusID:219487346).
6. Sereno, Paul C.; Beck, Allison L.; Dutheil, Didier B.; Gado, Boubacar; Larsson, Hans C. E.; Lyon, Gabrielle H.; Marcot, Jonathan D.; Rauhut, Oliver W. M.; Sadleir, Rudyard W.; Sidor, Christian A.; Varricchio, David D.; Wilson, Gregory P.; Wilson, Jeffrey A. (13 November 1998). "A Long-Snouted Predatory Dinosaur from Africa and the Evolution of Spinosaurids". *Science*. **282** (5392): 1298–1302. Bibcode:1998Sci...282.1298S (https://ui.adsabs.harvard.edu/abs/1998Sci...282.1298S). CiteSeerX 10.1.1.502.3887 (https://citeseerx.ist.psu.edu/viewdoc/summary?doi=10.1.1.502.3887). doi:10.1126/science.282.5392.1298 (https://doi.org/10.1126%2Fscience.282.5392.1298). PMID 9812890 (https://pubmed.ncbi.nlm.nih.gov/9812890). {{cite journal}}: Cite uses deprecated parameter |citeseerx= (help)

7. Sereno, Paul C.; Myhrvold, Nathan; Henderson, Donald M.; Fish, Frank E.; Vidal, Daniel; Baumgart, Stephanie L.; Keillor, Tyler M.; Formoso, Kiersten K.; Conroy, Lauren L. (2022). "Spinosaurus is not an aquatic dinosaur" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9711522). *eLife*. **11** e80092. doi:10.7554/eLife.80092 (https://doi.org/10.7554%2FeLife.80092). PMC 9711522 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9711522). PMID 36448670 (https://pubmed.ncbi.nlm.nih.gov/36448670).
8. Benson, Rodger (2010). "A description of Megalosaurus bucklandii (Dinosauria: Theropoda) from the Bathonian of the UK and the relationships of Middle Jurassic theropods" (https://doi.org/10.1111%2Fj.1096-3642.2009.00569.x). *Zoological Journal of the Linnean Society*. **158** (4): 882–935. doi:10.1111/j.1096-3642.2009.00569.x (https://doi.org/10.1111%2Fj.1096-3642.2009.00569.x). S2CID 84266680 (https://api.semanticscholar.org/CorpusID:84266680).
9. Holtz, T.R.Jr (1998). "A new phylogeny of the carnivorous dinosaurs" (http://www.mnhn.ul.pt/geologia/gaia/1.pdf) (PDF). *Gaia*. **15**: 5–61. Archived (https://web.archive.org/web/20110719060449/http://www.mnhn.ul.pt/geologia/gaia/1.pdf) (PDF) from the original on 2011-07-19. Retrieved 2008-11-28.
10. Rauhut, Oliver (2019). "Probable basal allosauroid from the early Middle Jurassic Cañadón Asfalto Formation of Argentina highlights phylogenetic uncertainty in tetanuran theropod dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6906444). *Scientific Reports*. **9** (18826) 18826. Bibcode:2019NatSR...918826R (https://ui.adsabs.harvard.edu/abs/2019NatSR...918826R). doi:10.1038/s41598-019-53672-7 (https://doi.org/10.1038%2Fs41598-019-53672-7). PMC 6906444 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6906444). PMID 31827108 (https://pubmed.ncbi.nlm.nih.gov/31827108).
11. Rauhut, Oliver (2012). "Exceptionally preserved juvenile megalosauroid theropod dinosaur with filamentous integument from the Late Jurassic of Germany" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3406838). *PNAS*. **109** (29): 11746–11751. Bibcode:2012PNAS..10911746R (https://ui.adsabs.harvard.edu/abs/2012PNAS..10911746R). doi:10.1073/pnas.1203238109 (https://doi.org/10.1073%2Fpnas.1203238109). PMC 3406838 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3406838). PMID 22753486 (https://pubmed.ncbi.nlm.nih.gov/22753486).
12. Mantell, Gideon Algernon (1822). *The fossils of the South Downs or, Illustrations of the geology of Sussex* (https://www.biodiversitylibrary.org/item/97604). L. Relfe. doi:10.5962/bhl.title.44924 (https://doi.org/10.5962%2Fbhl.title.44924). OCLC 754552732 (https://search.worldcat.org/oclc/754552732).
13. Owen, R. (1840–1845). *Odontography*. London: Hippolyte Baillière, 655 pp, 1–32
14. Owen, R., 1842, *Report on British fossil reptiles. Part II*. Reports of the meetings of the British Association for the Advancement of Science. 11, pp 61-204
15. Sauvage, H. E. (1897–1898). *Vertébrés fossiles du Portugal. Contribution à l'étude des poissons et des reptiles du Jurassique et du Crétacique.* Lisbonne: Direction des Travaux géologiques du Portugal, 46p
16. Milner, A., 2003, "Fish-eating theropods: A short review of the systematics, biology and palaeobiogeography of spinosaurs". In: Huerta Hurtado and Torcida Fernandez-Baldor (eds.). *Actas de las II Jornadas Internacionales sobre Paleontologýa de Dinosaurios y su Entorno (2001)*. pp 129-138
17. Smith, Joshua B.; Lamanna, Matthew C.; Mayr, Helmut; Lacovara, Kenneth J. (March 2006). "New Information Regarding the Holotype of Spinosaurus Aegyptiacus Stromer, 1915". *Journal of Paleontology*. **80** (2): 400–406. doi:10.1666/0022-3360(2006)080[0400:NIRTHO]2.0.CO;2 (https://doi.org/10.1666%2F0022-3360%282006%2980%5B0400%3ANIRTHO%5D2.0.CO%3B2). S2CID 130989487 (https://api.semanticscholar.org/CorpusID:130989487).

18. Hone, David William Elliott; Holtz, Thomas Richard (June 2017). "A Century of Spinosaurs - A Review and Revision of the Spinosauridae with Comments on Their Ecology" (http://qmro.qmul.ac.uk/xmlui/handle/123456789/49404). *Acta Geologica Sinica - English Edition*. **91** (3): 1120–1132. Bibcode:2017AcGlS..91.1120H (https://ui.adsabs.harvard.edu/abs/2017AcGlS..91.1120H). doi:10.1111/1755-6724.13328 (https://doi.org/10.1111%2F1755-6724.13328). S2CID 90952478 (https://api.semanticscholar.org/CorpusID:90952478).
19. Stromer, E. (1934). "Ergebnisse der Forschungsreisen Prof. E. Stromers in den Wüsten Ägyptens. II. Wirbeltier-Reste der Baharije-Stufe (unterstes Cenoman). 13. Dinosauria". *Abhandlungen der Bayerischen Akademie der Wissenschaften Mathematisch-naturwissenschaftliche Abteilung*. Neue Folge (in German). **22**: 1–79.
20. Evers, Serjoscha W.; Rauhut, Oliver W.M.; Milner, Angela C.; McFeeters, Bradley; Allain, Ronan (20 October 2015). "A reappraisal of the morphology and systematic position of the theropod dinosaur Sigilmassasaurus from the 'middle' Cretaceous of Morocco" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4614847). *PeerJ*. **3** e1323. doi:10.7717/peerj.1323 (https://doi.org/10.7717%2Fpeerj.1323). PMC 4614847 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4614847). PMID 26500829 (https://pubmed.ncbi.nlm.nih.gov/26500829).
21. Barker, CT; Lockwood, JAF; Naish, D; Brown, S; Hart, A; Tulloch, E; Gostling, NJ (2022). "A European giant: a large spinosaurid (Dinosauria: Theropoda) from the Vectis Formation (Wealden Group, Early Cretaceous), UK" (https://peerj.com/articles/13543/). *PeerJ*. **10** e13543. doi:10.7717/peerj.13543 (https://doi.org/10.7717%2Fpeerj.13543). PMC 9188774 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9188774). PMID 35702254 (https://pubmed.ncbi.nlm.nih.gov/35702254).
22. Haq, Sana Noor (June 9, 2022). "Scientists unearth remains of one of Europe's biggest predatory dinosaurs" (https://www.cnn.com/2022/06/09/europe/spinosaurid-dinosaur-europe-intl-scn-scli/index.html). *CNN*. Retrieved January 21, 2026.
23. Isasmendi, E; Cuesta, E; Díaz-Martínex, I; Company, J; Sáez-Benito, P; Viera, LI; Torices, A; Pereda-Suberbiola, X (2024). "Increasing the theropod record of Europe: a new basal spinosaurid from the Enciso Group of the Cameros Basin (La Rioja, Spain). Evolutionary implications and palaeobiodiversity" (https://academic.oup.com/zoolinnean/article/202/3/zlad193/7564790?login=false). *Zoological Journal of the Linnean Society*. **202** (3).
24. "Researchers discover a new species of carnivorous dinosaur in La Rioja, Spain" (https://phys.org/news/2024-03-species-carnivorous-dinosaur-la-rioja.html). *Phys.org*. 13 March 2024. Retrieved 25 May 2024.
25. Sereno, Paul C.; Vidal, Daniel; Myhrvold, Nathan P.; Johnson-Ransom, Evan; Ciudad Real, María; Baumgart, Stephanie L.; Sánchez Fontela, Noelia; Green, Todd L.; Saitta, Evan T.; Adamou, Boubé; et al. (February 19, 2026). "Scimitar-crested *Spinosaurus* species from the Sahara caps stepwise spinosaurid radiation". *Science*. **391** (6787): 1–10. doi:10.1126/science.adx5486 (https://doi.org/10.1126%2Fscience.adx5486). ISSN 0036-8075 (https://search.worldcat.org/issn/0036-8075).
26. Dixon, Dougal (2009). *The Ultimate Guide to Dinosaurs* (https://archive.org/details/ultimateguidetod0000dixo). Ticktock Books. ISBN 978-1-84696-988-1.
27. Holtz, Thomas R. Jr. (2011) *Dinosaurs: The Most Complete, Up-to-Date Encyclopedia for Dinosaur Lovers of All Ages,* Winter 2010 Appendix. (http://www.geol.umd.edu/~tholtz/dinoappendix/HoltzappendixWinter2010.pdf)
28. S., Paul, Gregory (2016-10-25). *The Princeton field guide to dinosaurs* (2nd ed.). Princeton, N.J. ISBN 978-1-4008-8314-1. OCLC 954055249 (https://search.worldcat.org/oclc/954055249).
29. Therrien, François; Henderson, Donald M. (12 March 2007). "My theropod is bigger than yours … or not: estimating body size from skull length in theropods". *Journal of Vertebrate Paleontology*. **27** (1): 108–115. doi:10.1671/0272-4634(2007)27[108:mtibty]2.0.co;2 (https://doi.org/10.1671%2F0272-4634%282007%2927%5B108%3Amtibty%5D2.0.co%3B2). S2CID 86025320 (https://api.semanticscholar.org/CorpusID:86025320).

30. Kellner, Alexander WA.; Azevedo, Sergio A.K.; Machado, Elaine B.; Carvalho, Luciana B. de; Henriques, Deise D.R. (March 2011). "A new dinosaur (Theropoda, Spinosauridae) from the Cretaceous (Cenomanian) Alcântara Formation, Cajual Island, Brazil" (https://doi.org/10.1590%2FS0001-37652011000100006). *Anais da Academia Brasileira de Ciências*. **83** (1): 99–108. Bibcode:2011AnABC..83...99K (https://ui.adsabs.harvard.edu/abs/2011AnABC..83...99K). doi:10.1590/S0001-37652011000100006 (https://doi.org/10.1590%2FS0001-37652011000100006). PMID 21437377 (https://pubmed.ncbi.nlm.nih.gov/21437377).
31. Arden, T.M.S.; Klein, C.G.; Zouhri, S.; Longrich, N.R. (2018). "Aquatic adaptation in the skull of carnivorous dinosaurs (Theropoda: Spinosauridae) and the evolution of aquatic habits in *Spinosaurus*". *Cretaceous Research*. **93**: 275–284. Bibcode:2019CrRes..93..275A (https://ui.adsabs.harvard.edu/abs/2019CrRes..93..275A). doi:10.1016/j.cretres.2018.06.013 (https://doi.org/10.1016%2Fj.cretres.2018.06.013). S2CID 134735938 (https://api.semanticscholar.org/CorpusID:134735938).
32. Aureliano, Tito; Ghilardi, Aline M.; Buck, Pedro V.; Fabbri, Matteo; Samathi, Adun; Delcourt, Rafael; Fernandes, Marcelo A.; Sander, Martin (October 2018). "Semi-aquatic adaptations in a spinosaur from the Lower Cretaceous of Brazil" (http://paleorxiv.org/mjt95/). *Cretaceous Research*. **90**: 283–295. Bibcode:2018CrRes..90..283A (https://ui.adsabs.harvard.edu/abs/2018CrRes..90..283A). doi:10.1016/j.cretres.2018.04.024 (https://doi.org/10.1016%2Fj.cretres.2018.04.024). S2CID 134353898 (https://api.semanticscholar.org/CorpusID:134353898).
33. Rayfield, Emily J.; Milner, Angela C.; Xuan, Viet Bui; Young, Philippe G. (12 December 2007). "Functional morphology of spinosaur 'crocodile-mimic' dinosaurs". *Journal of Vertebrate Paleontology*. **27** (4): 892–901. doi:10.1671/0272-4634(2007)27[892:FMOSCD]2.0.CO;2 (https://doi.org/10.1671%2F0272-4634%282007%2927%5B892%3AFMOSCD%5D2.0.CO%3B2). S2CID 85854809 (https://api.semanticscholar.org/CorpusID:85854809).
34. Sues, H. D.; Frey, E.; Martill, D. M.; Scott, D. M. (2002). "Irritator challengeri, a spinosaurid (Dinosauria: Theropoda) from the Lower Cretaceous of Brazil". *Journal of Vertebrate Paleontology*. **22** (3): 535–547. doi:10.1671/0272-4634(2002)022[0535:ICASDT]2.0.CO;2 (https://doi.org/10.1671%2F0272-4634%282002%29022%5B0535%3AICASDT%5D2.0.CO%3B2). S2CID 131050889 (https://api.semanticscholar.org/CorpusID:131050889).
35. Holtz, Thomas R. (2008). "A critical reappraisal of the obligate scavenging hypothesis for Tyrannosaurus rex and other tyrant dinosaurs" (https://books.google.com/books?id=5WH9RnfKco4C&pg=PA371). *Tyrannosaurus Rex, the Tyrant King*. Indiana University Press. pp. 371–396. ISBN 978-0-253-35087-9.
36. Barker, Chris Tijani; Naish, Darren; Trend, Jacob; Michels, Lysanne Veerle; Witmer, Lawrence; Ridgley, Ryan; Rankin, Katy; Clarkin, Claire E.; Schneider, Philipp; Gostling, Neil J. (2023). "Modified skulls but conservative brains? The palaeoneurology and endocranial anatomy of baryonychine dinosaurs (Theropoda: Spinosauridae)" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10184548). *Journal of Anatomy*. **242** (6): 1124–1145. doi:10.1111/joa.13837 (https://doi.org/10.1111%2Fjoa.13837). PMC 10184548 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10184548). PMID 36781174 (https://pubmed.ncbi.nlm.nih.gov/36781174). S2CID 256845477 (https://api.semanticscholar.org/CorpusID:256845477).
37. Sasso, Cristiano Dal; Maganuco, Simone; Buffetaut, Eric; Mendez, Marco A. (30 December 2005). "New information on the skull of the enigmatic theropod Spinosaurus , with remarks on its size and affinities". *Journal of Vertebrate Paleontology*. **25** (4): 888–896. doi:10.1671/0272-4634(2005)025[0888:NIOTSO]2.0.CO;2 (https://doi.org/10.1671%2F0272-4634%282005%29025%5B0888%3ANIOTSO%5D2.0.CO%3B2). S2CID 85702490 (https://api.semanticscholar.org/CorpusID:85702490).

38. Sales, M.A.F.; Schultz, C.L. (2017). "Spinosaur taxonomy and evolution of craniodental features: Evidence from Brazil" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5673194). *PLOS ONE*. **12** (11) e0187070. Bibcode:2017PLoSO..1287070S (https://ui.adsabs.harvard.edu/abs/2017PLoSO..1287070S). doi:10.1371/journal.pone.0187070 (https://doi.org/10.1371%2Fjournal.pone.0187070). PMC 5673194 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5673194). PMID 29107966 (https://pubmed.ncbi.nlm.nih.gov/29107966).
39. Charig, A. J.; Milner, A. C. (1997). "*Baryonyx walkeri*, a fish-eating dinosaur from the Wealden of Surrey" (http://biostor.org/reference/110558). *Bulletin of the Natural History Museum of London*. **53**: 11–70.
40. Hendrickx, Christophe; Mateus, Octávio; Buffetaut, Eric (6 January 2016). "Morphofunctional Analysis of the Quadrate of Spinosauridae (Dinosauria: Theropoda) and the Presence of Spinosaurus and a Second Spinosaurine Taxon in the Cenomanian of North Africa" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4703214). *PLOS ONE*. **11** (1) e0144695. Bibcode:2016PLoSO..1144695H (https://ui.adsabs.harvard.edu/abs/2016PLoSO..1144695H). doi:10.1371/journal.pone.0144695 (https://doi.org/10.1371%2Fjournal.pone.0144695). PMC 4703214 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4703214). PMID 26734729 (https://pubmed.ncbi.nlm.nih.gov/26734729).
41. Bertin, Tor (2010). "A Catalogue of Material and Review of the Spinosauridae" (https://archives.palarch.nl/index.php/jvp/article/view/457). *PalArch's Journal of Vertebrate Palaeontology*. **7** (4): 01–39.
42. Pittet, F. (2026). "Neurovascular system and dental renewal in the rostrum of Spinosauridae: new descriptions and implications on non-olfactive snout sensitivity of dinosaurs" (https://sciencepress.mnhn.fr/en/periodiques/geodiversitas/48/12). *Geodiversitas*. **48** (12): 227–272. doi:10.5252/geodiversitas2026v48a12 (https://doi.org/10.5252%2Fgeodiversitas2026v48a12).
43. Lipkin, Christine; Paul; Sereno, Callistus; Horner, Jack (2007). "The furcula in *Suchomimus tenerensis* and *Tyrannosaurus rex* (Dinosauria: Theropoda: Tetanurae)" (http://publication.plazi.org/id/BA62FFDAFFCCFFB5FFFFFFA2FFDFFFD2). *Journal of Paleontology*. **81** (6): 1523–1527. Bibcode:2007JPal...81.1523L (https://ui.adsabs.harvard.edu/abs/2007JPal...81.1523L). doi:10.1666/06-024.1 (https://doi.org/10.1666%2F06-024.1). S2CID 86234363 (https://api.semanticscholar.org/CorpusID:86234363).
44. Allain, R.; Xaisanavong, T.; Richir, P.; Khentavong, B. (2012). "The first definitive Asian spinosaurid (Dinosauria: Theropoda) from the early cretaceous of Laos". *Naturwissenschaften*. **99** (5): 369–377. Bibcode:2012NW.....99..369A (https://ui.adsabs.harvard.edu/abs/2012NW.....99..369A). doi:10.1007/s00114-012-0911-7 (https://doi.org/10.1007%2Fs00114-012-0911-7). PMID 22528021 (https://pubmed.ncbi.nlm.nih.gov/22528021). S2CID 2647367 (https://api.semanticscholar.org/CorpusID:2647367).
45. Eddy, DR; Clarke, JA (2011). "New Information on the Cranial Anatomy of *Acrocanthosaurus atokensis* and Its Implications for the Phylogeny of Allosauroidea (Dinosauria: Theropoda)" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3061882). *PLOS ONE*. **6** (3) e17932. Bibcode:2011PLoSO...617932E (https://ui.adsabs.harvard.edu/abs/2011PLoSO...617932E). doi:10.1371/journal.pone.0017932 (https://doi.org/10.1371%2Fjournal.pone.0017932). PMC 3061882 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3061882). PMID 21445312 (https://pubmed.ncbi.nlm.nih.gov/21445312).
46. Hecht, Jeff. 1998. "Fish Swam in Fear." New Scientist. November 21. https://www.newscientist.com/article/mg16021610-300-fish-swam-in-fear/.
47. Elisabete Malafaia; José Miguel Gasulla; Fernando Escaso; Iván Narváez; José Luis Sanz; Francisco Ortega (2019). "A new spinosaurid theropod (Dinosauria: Megalosauroidea) from the late Barremian of Vallibona, Spain: Implications for spinosaurid diversity in the Early Cretaceous of the Iberian Peninsula". Cretaceous Research. in press: 104221. doi:10.1016/j.cretres.2019.104221 (https://doi.org/10.1016%2Fj.cretres.2019.104221).

48. *O Estado de S. Paulo* (in Portuguese), 2009-05-14, available at (http://www.estadao.com.br/noticias/vidae,ufrj-expoe-maior-replica-de-dinossauro-carnivoro-no-pais,370960,0.htm); *O Globo*, 2009-05-15, abridgement available at (http://oglobo.globo.com/ciencia/mat/2009/05/14/museu-monta-replica-de-dinossauro-carnivoro-de-6-metros-no-rio-755870546.asp); university's announcement at Moutinho, Sofia (2009-11-05). "Inaugura dia 14 a exposição Dinossauros no Sertão" (https://web.archive.org/web/20110706162356/http://www.ufrj.br/detalha_noticia.php?codnoticia=7486). Archived from the original (http://www.ufrj.br/detalha_noticia.php?codnoticia=7486) on 2011-07-06. Retrieved 2010-01-13.
49. Machado, E. B.; Kellner, A. W. A.; Campos, D. A. (2005). "Preliminary information on a dinosaur (Theropoda, Spinosauridae) pelvis from the Cretaceous Santana Formation (Romualdo Member) Brazil". *Congresso Latino-Americano de Paleontologia de Vertebrados*. 2 (Boletim de resumos): 161–162.
50. Ibrahim, Nizar; Maganuco, Simone; Dal Sasso, Cristiano; Fabbri, Matteo; Auditore, Marco; Bindellini, Gabriele; Martill, David M.; Zouhri, Samir; Mattarelli, Diego A.; Unwin, David M.; Wiemann, Jasmina; Bonadonna, Davide; Amane, Ayoub; Jakubczak, Juliana; Joger, Ulrich; Lauder, George V.; Pierce, Stephanie E. (7 May 2020). "Tail-propelled aquatic locomotion in a theropod dinosaur" (https://doi.org/10.1038%2Fs41586-020-2190-3). *Nature*. **581** (7806): 67–70. Bibcode:2020Natur.581...67I (https://ui.adsabs.harvard.edu/abs/2020Natur.581...67I). doi:10.1038/s41586-020-2190-3 (https://doi.org/10.1038%2Fs41586-020-2190-3). PMID 32376955 (https://pubmed.ncbi.nlm.nih.gov/32376955).
51. Rayfield, E.J (2011). "Structural performance of tetanuran theropod skulls, with emphasis on the Megalosauridae, Spinosauridae and Carcharodontosauridae". *Special Papers in Palaeontology*. **86**: 241–254. OCLC 769169643 (https://search.worldcat.org/oclc/769169643).
52. Buffetaut, Eric; Ouaja, Mohamed (1 September 2002). "A new specimen of *Spinosaurus* (Dinosauria, Theropoda) from the Lower Cretaceous of Tunisia, with remarks on the evolutionary history of the Spinosauridae" (http://doc.rero.ch/record/14728/files/PAL_E1854.pdf) (PDF). *Bulletin de la Société Géologique de France*. **173** (5): 415–421. Bibcode:2002BSGF..173..415B (https://ui.adsabs.harvard.edu/abs/2002BSGF..173..415B). doi:10.2113/173.5.415 (https://doi.org/10.2113%2F173.5.415). hdl:2042/216 (https://hdl.handle.net/2042%2F216).
53. Barrett, Paul M.; Benson, Roger B. J.; Rich, Thomas H.; Vickers-Rich, Patricia (23 December 2011). "First spinosaurid dinosaur from Australia and the cosmopolitanism of Cretaceous dinosaur faunas" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3210678). *Biology Letters*. **7** (6): 933–936. Bibcode:2011BiLet...7..933B (https://ui.adsabs.harvard.edu/abs/2011BiLet...7..933B). doi:10.1098/rsbl.2011.0466 (https://doi.org/10.1098%2Frsbl.2011.0466). PMC 3210678 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3210678). PMID 21693488 (https://pubmed.ncbi.nlm.nih.gov/21693488).
54. Barker, C. T.; Hone, D. W. E.; Naish, D.; Cau, A.; Lockwood, J. A. F.; Foster, B.; Clarkin, C. E.; Schneider, P.; Gostling, N. J. (2021). "New spinosaurids from the Wessex Formation (Early Cretaceous, UK) and the European origins of Spinosauridae" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8481559). *Scientific Reports*. **11** (1): 19340. Bibcode:2021NatSR..1119340B (https://ui.adsabs.harvard.edu/abs/2021NatSR..1119340B). doi:10.1038/s41598-021-97870-8 (https://doi.org/10.1038%2Fs41598-021-97870-8). PMC 8481559 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8481559). PMID 34588472 (https://pubmed.ncbi.nlm.nih.gov/34588472).
55. Serrano-Martínez, A.; Vidal, D.; Sciscio, L.; Ortega, F.; Knoll, F. (2015). "Isolated theropod teeth from the Middle Jurassic of Niger and the early dental evolution of Spinosauridae" (https://doi.org/10.4202%2Fapp.00101.2014). *Acta Palaeontologica Polonica*. doi:10.4202/app.00101.2014 (https://doi.org/10.4202%2Fapp.00101.2014). hdl:10261/152148 (https://hdl.handle.net/10261%2F152148). S2CID 53331040 (https://api.semanticscholar.org/CorpusID:53331040).

56. Holtz Jr., T. R. (1998). "Spinosaurs as crocodile mimics". *Science*. **282** (5392): 1276–1277. Bibcode:1998Sci...282.1276H (https://ui.adsabs.harvard.edu/abs/1998Sci...282.1276H). doi:10.1126/science.282.5392.1276 (https://doi.org/10.1126%2Fscience.282.5392.1276). S2CID 16701711 (https://api.semanticscholar.org/CorpusID:16701711).
57. Sereno, P.C.; Beck, A.L.; Dutheil, D.B.; Gado, B.; Larsson, H.C.E.; Lyon, G.H.; Marcot, J.D.; Rauhut, O.W.M.; Sadleir, R.W.; Sidor, C.A.; Varricchio, D.D.; Wilson, G.P; Wilson, J.A. (1998). "A long-snouted predatory dinosaur from Africa and the evolution of spinosaurids" (https://doi.org/10.1126%2Fscience.282.5392.1298). *Science*. **282** (5392): 1298–1302. Bibcode:1998Sci...282.1298S (https://ui.adsabs.harvard.edu/abs/1998Sci...282.1298S). doi:10.1126/science.282.5392.1298 (https://doi.org/10.1126%2Fscience.282.5392.1298). PMID 9812890 (https://pubmed.ncbi.nlm.nih.gov/9812890).
58. Milner, A. C. (2003). "Fish-eating theropods: A short review of the systematics, biology and palaeobiogeography of spinosaurs". *Actas de las II Jornadas Internacionales Sobre Paleontologýa de Dinosaurios y Su Entorno*: 129–138.
59. Buffetaut, E. (2007). "The spinosaurid dinosaur *Baryonyx* (Saurischia, Theropoda) in the Early Cretaceous of Portugal" (http://doc.rero.ch/record/13845/files/PAL_E769.pdf) (PDF). *Geological Magazine*. **144** (6): 1021–1025. Bibcode:2007GeoM..144.1021B (https://ui.adsabs.harvard.edu/abs/2007GeoM..144.1021B). doi:10.1017/S0016756807003883 (https://doi.org/10.1017%2FS0016756807003883). S2CID 130212901 (https://api.semanticscholar.org/CorpusID:130212901).
60. Mateus, O.; Araújo, R.; Natário, C.; Castanhinha, R. (2011). "A new specimen of the theropod dinosaur *Baryonyx* from the early Cretaceous of Portugal and taxonomic validity of *Suchosaurus*" (http://docentes.fct.unl.pt/sites/default/files/omateus/files/mateus_et_al_2011_a_new_specimen_of_the_theropod_dinosaur_baryonyx_from_the_early_cretaceous_of_portugal_and_taxonomic_validity_of_suchosaurus.pdf) (PDF). *Zootaxa*. 2827. **2827**: 54–68. Bibcode:2011Zoot.28277.1.3M (https://ui.adsabs.harvard.edu/abs/2011Zoot.28277.1.3M). doi:10.11646/zootaxa.2827.1.3 (https://doi.org/10.11646%2Fzootaxa.2827.1.3).
61. Hendrickx, Christophe; Mateus, O.; Araújo, R.; Choiniere, J. (2019). "The distribution of dental features in non-avian theropod dinosaurs: Taxonomic potential, degree of homoplasy, and major evolutionary trends" (https://palaeo-electronica.org/content/2019/2806-dental-features-in-theropods). *Palaeontologia Electronica*. **22** (3): 1–110. doi:10.26879/820 (https://doi.org/10.26879%2F820). hdl:11336/146011 (https://hdl.handle.net/11336%2F146011). ISSN 1094-8074 (https://search.worldcat.org/issn/1094-8074). S2CID 213164229 (https://api.semanticscholar.org/CorpusID:213164229).
62. Soto, M.; Toriño, P.; Perea, D. (2020). "*Ceratosaurus* (Theropoda, Ceratosauria) teeth from the Tacuarembó Formation (Late Jurassic, Uruguay)" (http://adsabs.harvard.edu/abs/2020JSAES.10302781S). *Journal of South American Earth Sciences*. **103** 102781. Bibcode:2020JSAES.10302781S (https://ui.adsabs.harvard.edu/abs/2020JSAES.10302781S). doi:10.1016/j.jsames.2020.102781 (https://doi.org/10.1016%2Fj.jsames.2020.102781). ISSN 0895-9811 (https://search.worldcat.org/issn/0895-9811). S2CID 224842133 (https://api.semanticscholar.org/CorpusID:224842133).
63. Candeiro, C. R. A.; Brusatte, S. L.; Souza, A. L. (2017). "Spinosaurid Dinosaurs from the Early Cretaceous of North Africa and Europe: Fossil Record, Biogeography and Extinction" (https://doi.org/10.11137%2F2017_3_294_302). *Anuário do Instituto de Geociências*. **40** (3): 294–302. doi:10.11137/2017_3_294_302 (https://doi.org/10.11137%2F2017_3_294_302). hdl:20.500.11820/1bc143b5-5bd5-416b-bf10-dd28304d2c9b (https://hdl.handle.net/20.500.11820%2F1bc143b5-5bd5-416b-bf10-dd28304d2c9b).

64. Malafaia, E.; Gasulla, J. M.; Escaso, F.; Narvaéz, I.; Ortega, F. (2020). "An update of the spinosaurid (Dinosauria: Theropoda) fossil record from the Lower Cretaceous of the Iberian Peninsula: distribution, diversity, and evolutionary history". *Journal of Iberian Geology*. **46** (4): 431–444. Bibcode:2020JIbG...46..431M (https://ui.adsabs.harvard.edu/abs/2020JIbG...46..431M). doi:10.1007/s41513-020-00138-9 (https://doi.org/10.1007%2Fs41513-020-00138-9). hdl:20.500.14468/31927 (https://hdl.handle.net/20.500.14468%2F31927). S2CID 222149842 (https://api.semanticscholar.org/CorpusID:222149842).
65. Sues, Hans-Dieter; Frey, Eberhard; Martill, David M.; Scott, Diane M. (19 September 2002). "Irritator challengeri , a spinosaurid (Dinosauria: Theropoda) from the Lower Cretaceous of Brazil". *Journal of Vertebrate Paleontology*. **22** (3): 535–547. doi:10.1671/0272-4634(2002)022[0535:ICASDT]2.0.CO;2 (https://doi.org/10.1671%2F0272-4634%282002%29022%5B0535%3AICASDT%5D2.0.CO%3B2). S2CID 131050889 (https://api.semanticscholar.org/CorpusID:131050889).
66. Yun, Chan-Gyu (23 December 2023). "SPINOSAURS AS PHYTOSAUR MIMICS: A CASE OF CONVERGENT EVOLUTION BETWEEN TWO EXTINCT ARCHOSAURIFORM CLADES" (https://www.researchgate.net/profile/Changyu-Yun/publication/373816152_Spinosaurs_as_phytosaur_mimics_a_case_of_convergent_evolution_between_two_extinct_archosauriform_clades/links/64fe89d9849bbb203b8edb15/Spinosaurs-as-phytosaur-mimics-a-case-of-convergent-evolution-between-two-extinct-archosauriform-clades.pdf) (PDF). *Acta Palaeontologica Romaniae* (20 (1)): 17–29. doi:10.35463/j.apr.2024.01.02 (https://doi.org/10.35463%2Fj.apr.2024.01.02). ISSN 1842-371X (https://search.worldcat.org/issn/1842-371X). Retrieved 8 February 2026 – via ResearchGate.
67. Vullo, Romain; Allain, Ronan; Cavin, Lionel (2016). "Convergent evolution of jaws between spinosaurid dinosaurs and pike conger eels" (https://doi.org/10.4202%2Fapp.00284.2016). *Acta Palaeontologica Polonica*. **61** (4): 825. Bibcode:2016AcPaP..61..825V (https://ui.adsabs.harvard.edu/abs/2016AcPaP..61..825V). doi:10.4202/app.00284.2016 (https://doi.org/10.4202%2Fapp.00284.2016).
68. Cuff, Andrew R.; Rayfield, Emily J. (28 May 2013). "Feeding Mechanics in Spinosaurid Theropods and Extant Crocodilians" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3665537). *PLOS ONE*. **8** (5) e65295. Bibcode:2013PLoSO...865295C (https://ui.adsabs.harvard.edu/abs/2013PLoSO...865295C). doi:10.1371/journal.pone.0065295 (https://doi.org/10.1371%2Fjournal.pone.0065295). PMC 3665537 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3665537). PMID 23724135 (https://pubmed.ncbi.nlm.nih.gov/23724135).
69. Witton, Mark P. (2018). "Pterosaurs in Mesozoic food webs: a review of fossil evidence". *Geological Society, London, Special Publications*. **455** (1): 7–23. Bibcode:2018GSLSP.455....7W (https://ui.adsabs.harvard.edu/abs/2018GSLSP.455....7W). doi:10.1144/SP455.3 (https://doi.org/10.1144%2FSP455.3). S2CID 90573936 (https://api.semanticscholar.org/CorpusID:90573936).
70. Buffetaut, Eric; Martill, David; Escuillié, François (July 2004). "Pterosaurs as part of a spinosaur diet" (https://doi.org/10.1038%2F430033a). *Nature*. **430** (6995): 33. Bibcode:2004Natur.429...33B (https://ui.adsabs.harvard.edu/abs/2004Natur.429...33B). doi:10.1038/430033a (https://doi.org/10.1038%2F430033a). PMID 15229562 (https://pubmed.ncbi.nlm.nih.gov/15229562). S2CID 4398855 (https://api.semanticscholar.org/CorpusID:4398855).
71. Buffetaut, Eric; Suteethorn, Varavudh (June 1999). "The dinosaur fauna of the Sao Khua Formation of Thailand and the beginning of the Cretaceous radiation of dinosaurs in Asia" (http://doc.rero.ch/record/14471/files/PAL_E1664.pdf) (PDF). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **150** (1–2): 13–23. Bibcode:1999PPP...150...13B (https://ui.adsabs.harvard.edu/abs/1999PPP...150...13B). doi:10.1016/S0031-0182(99)00004-8 (https://doi.org/10.1016%2FS0031-0182%2899%2900004-8).

72. Hassler, A.; Martin, J. E.; Amiot, R.; Tacail, T.; Godet, F. Arnaud; Allain, R.; Balter, V. (11 April 2018). "Calcium isotopes offer clues on resource partitioning among Cretaceous predatory dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5904318). *Proceedings of the Royal Society B: Biological Sciences*. **285** (1876) 20180197. doi:10.1098/rspb.2018.0197 (https://doi.org/10.1098%2Frspb.2018.0197). PMC 5904318 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5904318). PMID 29643213 (https://pubmed.ncbi.nlm.nih.gov/29643213).
73. D'Amore, Domenic C.; Johnson-Ransom, Evan; Snively, Eric; Hone, David W. E. (2024-08-28). "Prey size and ecological separation in spinosaurid theropods based on heterodonty and rostrum shape" (https://anatomypubs.onlinelibrary.wiley.com/doi/10.1002/ar.25563). *The Anatomical Record*. **308** (5): 1331–1348. doi:10.1002/ar.25563 (https://doi.org/10.1002%2Far.25563). ISSN 1932-8486 (https://search.worldcat.org/issn/1932-8486). PMID 39205383 (https://pubmed.ncbi.nlm.nih.gov/39205383).
74. Hendrickx, C. (2026). "Dental evolution in non-coelurosaur theropods" (https://doi.org/10.3301%2FIJG.2026.14). *Italian Journal of Geosciences*. **145** (2): 1–34. doi:10.3301/IJG.2026.14 (https://doi.org/10.3301%2FIJG.2026.14).
75. Buffetaut, E.; Suteethorn, S.; Suteethorn, V.; Wongko, K.; Tong, H. (2026). "An Unusual Spinosaurid Tooth Morphotype from the Early Cretaceous of Eastern Thailand" (https://doi.org/10.3390%2Fd18090557). *Diversity*. **18** (9) 557. doi:10.3390/d18090557 (https://doi.org/10.3390%2Fd18090557).
76. Charig, A. J.; Milner, A. C. (1986). "*Baryonyx*, a remarkable new theropod dinosaur". *Nature*. **324** (6095): 359–361. Bibcode:1986Natur.324..359C (https://ui.adsabs.harvard.edu/abs/1986Natur.324..359C). doi:10.1038/324359a0 (https://doi.org/10.1038%2F324359a0). PMID 3785404 (https://pubmed.ncbi.nlm.nih.gov/3785404). S2CID 4343514 (https://api.semanticscholar.org/CorpusID:4343514).
77. Kitchener, Andrew (January 1987). "Function of Claws' claws". *Nature*. **325** (6100): 114. Bibcode:1987Natur.325..114K (https://ui.adsabs.harvard.edu/abs/1987Natur.325..114K). doi:10.1038/325114a0 (https://doi.org/10.1038%2F325114a0). S2CID 4264665 (https://api.semanticscholar.org/CorpusID:4264665).
78. Therrien, F.; Henderson, D.; Ruff, C. (2005). "Bite me – biomechanical models of theropod mandibles and implications for feeding behavior". In Carpenter, K. (ed.). *The Carnivorous Dinosaurs*. Indiana University Press. pp. 179–230. ISBN 978-0-253-34539-4.
79. Scherer, C. R.; Steell, E.; Upchurch, P. (2026). "Drivers and mechanisms of convergent forelimb reduction in non-avian theropod dinosaurs" (https://doi.org/10.1098%2Frspb.2026.0528). *Proceedings of the Royal Society B: Biological Sciences*. **293** (2071) 20260528. doi:10.1098/rspb.2026.0528 (https://doi.org/10.1098%2Frspb.2026.0528).
80. Carpenter, Kenneth (2005). *The Carnivorous Dinosaurs*. Indiana University Press. pp. 285–291. ISBN 978-0-253-34539-4.
81. Halstead, L.B. (1975). *The Evolution and Ecology of the Dinosaurs*. London: Eurobook Limited. pp. 1–116. ISBN 978-0-85654-018-9.
82. Gimsa, Jan; Sleigh, Robert; Gimsa, Ulrike (May 2016). "The riddle of Spinosaurus aegyptiacus' dorsal sail" (https://repository.publisso.de/resource/frl:6402283/data). *Geological Magazine*. **153** (3): 544–547. Bibcode:2016GeoM..153..544G (https://ui.adsabs.harvard.edu/abs/2016GeoM..153..544G). doi:10.1017/S0016756815000801 (https://doi.org/10.1017%2FS0016756815000801). S2CID 51999370 (https://api.semanticscholar.org/CorpusID:51999370).
83. Bailey, J.B. (1997). "Neural spine elongation in dinosaurs: sailbacks or buffalo-backs?". *Journal of Paleontology*. **71** (6): 1124–1146. Bibcode:1997JPal...71.1124B (https://ui.adsabs.harvard.edu/abs/1997JPal...71.1124B). doi:10.1017/S0022336000036076 (https://doi.org/10.1017%2FS0022336000036076). JSTOR 1306608 (https://www.jstor.org/stable/1306608). S2CID 130861276 (https://api.semanticscholar.org/CorpusID:130861276).

84. Stromer, E. (1915). "Ergebnisse der Forschungsreisen Prof. E. Stromers in den Wüsten Ägyptens. II. Wirbeltier-Reste der Baharije-Stufe (unterstes Cenoman). 3. Das Original des Theropoden *Spinosaurus aegyptiacus* nov. gen., nov. spec" (http://www.megaupload.com/?d=3KCCC7LS). *Abhandlungen der Königlich Bayerischen Akademie der Wissenschaften, Mathematisch-physikalische Klasse* (in German). **28** (3): 1–32.
85. Naish, Darren (2013). "Dinosaurs and their exaggerated structures: species recognition aids, or sexual display devices?" (https://blogs.scientificamerican.com/tetrapod-zoology/species-recognition-vs-sexual-selection-in-dinosaurs/). *Scientific American Blog Network*. Retrieved 2020-04-09.
86. Gimsa, J.; Sleigh, R.; Gimsa, U. (2015). "The riddle of *Spinosaurus aegyptiacus* dorsal sail" (https://doi.org/10.1017%2FS0016756815000801). *Geological Magazine*. **153** (3): 544–547. Bibcode:2016GeoM..153..544G (https://ui.adsabs.harvard.edu/abs/2016GeoM..153..544G). doi:10.1017/S0016756815000801 (https://doi.org/10.1017%2FS0016756815000801). S2CID 51999370 (https://api.semanticscholar.org/CorpusID:51999370).
87. "The smallest biggest theropod dinosaur: A new fossil from Africa represents a small juvenile individual of the huge sail-backed Spinosaurus" (https://www.sciencedaily.com/releases/2018/05/180530132947.htm) (Press release). PeerJ. 30 May 2018.
88. Anderson, Natali (31 May 2018). "Paleontologists Find Fossil of Smallest Spinosaurus" (http://www.sci-news.com/paleontology/smallest-spinosaurus-06052.html). *Sci-News.com*.
89. Amiot, R.; Buffetaut, E.; Lécuyer, C.; Wang, X.; Boudad, L.; Ding, Z.; Fourel, F.; Hutt, S.; Martineau, F.; Medeiros, A.; Mo, J.; Simon, L.; Suteethorn, V.; Sweetman, S.; Tong, H.; Zhang, F.; Zhou, Z. (2010). "Oxygen isotope evidence for semi-aquatic habits among spinosaurid theropods". *Geology*. **38** (2): 139–142. Bibcode:2010Geo....38..139A (https://ui.adsabs.harvard.edu/abs/2010Geo....38..139A). doi:10.1130/G30402.1 (https://doi.org/10.1130%2FG30402.1).
90. Sales, M. A. F.; Lacerda, M. B.; Horn, B. L. D.; de Oliveira, I. A. P.; Schultz, C. L.; Bibi, F. (2016). "The "χ" of the Matter: Testing the Relationship between Paleoenvironments and Three Theropod Clades" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4734717). *PLOS ONE*. **11** (2) e0147031. Bibcode:2016PLoSO..1147031S (https://ui.adsabs.harvard.edu/abs/2016PLoSO..1147031S). doi:10.1371/journal.pone.0147031 (https://doi.org/10.1371%2Fjournal.pone.0147031). PMC 4734717 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4734717). PMID 26829315 (https://pubmed.ncbi.nlm.nih.gov/26829315).
91. Henderson, D. M. (2018). "A buoyancy, balance and stability challenge to the hypothesis of a semi-aquatic *Spinosaurus* Stromer, 1915 (Dinosauria: Theropoda)" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6098948). *PeerJ*. **6** e5409. doi:10.7717/peerj.5409 (https://doi.org/10.7717%2Fpeerj.5409). PMC 6098948 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6098948). PMID 30128195 (https://pubmed.ncbi.nlm.nih.gov/30128195).
92. Beevor, Thomas; Quigley, Aaron; Smith, Roy E.; Smyth, Robert S.H.; Ibrahim, Nizar; Zouhri, Samir; Martill, David M. (January 2021). "Taphonomic evidence supports an aquatic lifestyle for Spinosaurus" (https://researchportal.port.ac.uk/portal/en/publications/taphonomic-evidence-supports-an-aquatic-lifestyle-for-spinosaurus(e7fb2358-2ac6-4b6c-9697-225a525e8366).html). *Cretaceous Research*. **117** 104627. Bibcode:2021CrRes.11704627B (https://ui.adsabs.harvard.edu/abs/2021CrRes.11704627B). doi:10.1016/j.cretres.2020.104627 (https://doi.org/10.1016%2Fj.cretres.2020.104627). S2CID 224888268 (https://api.semanticscholar.org/CorpusID:224888268).
93. Myhrvold, Nathan P.; Baumgart, Stephanie L.; Vidal, Daniel; Fish, Frank E.; Henderson, Donald M.; Saitta, Evan T.; Sereno, Paul C. (2024). "Diving dinosaurs? Caveats on the use of bone compactness and pFDA for inferring lifestyle" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10917332). *PLOS ONE*. **19** (3) e0298957. Bibcode:2024PLoSO..1998957M (https://ui.adsabs.harvard.edu/abs/2024PLoSO..1998957M). doi:10.1371/journal.pone.0298957 (https://doi.org/10.1371%2Fjournal.pone.0298957). PMC 10917332 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10917332). PMID 38446841 (https://pubmed.ncbi.nlm.nih.gov/38446841).

94. Cau, Andrea; Gostling, Neil J.; Lacerda, Mauro B. S.; Falasca, Mario; Paterna, Alessandro (2026-05-22). "Avian-like salt glands in Spinosauridae" (https://www.tandfonline.com/doi/full/10.1080/08912963.2026.2669954). *Historical Biology*: 1–13. doi:10.1080/08912963.2026.2669954 (https://doi.org/10.1080%2F08912963.2026.2669954). ISSN 0891-2963 (https://search.worldcat.org/issn/0891-2963).
95. Taquet, Philippe; Russell, Dale A. (September 1998). "New data on spinosaurid dinosaurs from the early cretaceous of the Sahara". *Comptes Rendus de l'Académie des Sciences - Series IIA - Earth and Planetary Science*. **327** (5): 347–353. Bibcode:1998CRASE.327..347T (https://ui.adsabs.harvard.edu/abs/1998CRASE.327..347T). doi:10.1016/S1251-8050(98)80054-2 (https://doi.org/10.1016%2FS1251-8050%2898%2980054-2).
96. Samathi, Adun; Sander, P. Martin; Chanthasit, Phornphen (2 December 2021). "A spinosaurid from Thailand (Sao Khua Formation, Early Cretaceous) and a reassessment of Camarillasaurus cirugedae from the Early Cretaceous of Spain". *Historical Biology*. **33** (12): 3480–3494. Bibcode:2021HBio...33.3480S (https://ui.adsabs.harvard.edu/abs/2021HBio...33.3480S). doi:10.1080/08912963.2021.1874372 (https://doi.org/10.1080%2F08912963.2021.1874372). S2CID 233884025 (https://api.semanticscholar.org/CorpusID:233884025).
97. Mateus, Octávio; Estraviz-López, Darío (16 February 2022). "A new theropod dinosaur from the early cretaceous (Barremian) of Cabo Espichel, Portugal: Implications for spinosaurid evolution" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8849621). *PLOS ONE*. **17** (2) e0262614. Bibcode:2022PLoSO..1762614M (https://ui.adsabs.harvard.edu/abs/2022PLoSO..1762614M). doi:10.1371/journal.pone.0262614 (https://doi.org/10.1371%2Fjournal.pone.0262614). PMC 8849621 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8849621). PMID 35171930 (https://pubmed.ncbi.nlm.nih.gov/35171930).
98. Salgado, Leonardo; Canudo, José I.; Garrido, Alberto C.; Ruiz-Omeñaca, José I.; García, Rodolfo A.; de la Fuente, Marcelo S.; Barco, José L.; Bollati, Raúl (June 2009). "Upper Cretaceous vertebrates from El Anfiteatro area, Río Negro, Patagonia, Argentina". *Cretaceous Research*. **30** (3): 767–784. Bibcode:2009CrRes..30..767S (https://ui.adsabs.harvard.edu/abs/2009CrRes..30..767S). doi:10.1016/j.cretres.2009.01.001 (https://doi.org/10.1016%2Fj.cretres.2009.01.001).
99. Hasegawa, Yoshikazu; Tanaka, Gengo; Takakuwa, Yuji; Koike, Satoshi (2010). "Fine sculptures on a tooth of Spinosaurus (Dinosauria, Theropoda) from Morocco". *Bulletin of Gunma Museum of Natural History* (http://www.gmnh.pref.gunma.jp/wp-content/uploads/bulletin14_2.pdf) (PDF). Vol. 14. pp. 11–20.
100. Hone, David W. E.; Xing, X. U.; De-You, Wang (15 March 2010). "A PROBABLE BARYONYCHINE (THEROPODA: SPINOSAURIDAE) TOOTH FROM THE UPPER CRETACEOUS OF HENAN PROVINCE, CHINA" (http://www.vertpala.ac.cn/EN/Y2010/V48/I1/19). *Vertebrata PalAsiatica*. **48** (1): 19.
101. ResearchSEA (2014). "First discovery of dinosaur fossils in Malaysia" (https://www.sciencedaily.com/releases/2014/02/140224204737.htm). *ScienceDaily*.
102. Katsuhiro, Kubota; Yuji, Takakuwa; Yoshikazu, Hasegawa (2017). "Second discovery of a spinosaurid tooth from the Sebayashi Formation (Lower Cretaceous), Kanna Town, Gunma Prefecture, Japan" (http://www.gmnh.pref.gunma.jp/wp-content/uploads/Second-discovery-of-a-spinosaurid-tooth-from-the-Sebayashi-Formation-Lower-Cretaceous-Kanna-Town-Gunma-Prefecture-Ja1.pdf) (PDF). *Bulletin of the Gunma Museum of Natural History*. **21**: 1–6.
103. Eric Buffetaut; Suravech Suteethorn; Varavudh Suteethorn; Haiyan Tong; Kamonrak Wongko (2019). "Spinosaurid teeth from the Lower Cretaceous of Ko Kut, eastern Thailand" (https://hal.science/hal-03488796v1/file/S0753396919300187.pdf) (PDF). *Annales de Paléontologie*. **105** (3): 239–243. Bibcode:2019AnPal.105..239B (https://ui.adsabs.harvard.edu/abs/2019AnPal.105..239B). doi:10.1016/j.annpal.2019.03.006 (https://doi.org/10.1016%2Fj.annpal.2019.03.006). S2CID 146225359 (https://api.semanticscholar.org/CorpusID:146225359).

104. Puntanon, K.; Samathi, A. (2025). "The occurrence of Spinosauridae (Dinosauria: Theropoda) during the Cretaceous of Asia: Implications for biogeography and distribution" (https://ph03.tci-thaijo.org/index.php/TGJ/article/view/3652). *Thai Geoscience Journal*. **6** (9): 13–28.
105. Barker, Chris T.; Naish, Darren; Gostling, Neil J. (2023). "Isolated tooth reveals hidden spinosaurid dinosaur diversity in the British Wealden Supergroup (Lower Cretaceous)" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10239232). *PeerJ*. **11** e15453. doi:10.7717/peerj.15453 (https://doi.org/10.7717%2Fpeerj.15453). PMC 10239232 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10239232). PMID 37273543 (https://pubmed.ncbi.nlm.nih.gov/37273543).
106. Olmedo-Romaña, G. J.; Wilson Mantilla, J. A.; Tejada, J. V.; Antoine, P. O.; Burga-Castillo, M. A.; Aliaga-Castillo, A. V.; Varas-Malca, R.; Benites-Palomino, A.; Salas-Gismondi, R. (2025). "Theropod and sauropod dinosaurs from the Campanian-Maastrichtian Bagua Basin of Perú, including the first possible report of Spinosauridae in western South America" (https://www.ameghiniana.org.ar/index.php/ameghiniana/article/view/3627). *Ameghiniana*. **62** (4): 259–287. Bibcode:2025Amegh..62.3627O (https://ui.adsabs.harvard.edu/abs/2025Amegh..62.3627O). doi:10.5710/AMGH.13.02.2025.3627 (https://doi.org/10.5710%2FAMGH.13.02.2025.3627).
107. Barker, Chris T.; Naish, D.; Gostling, Neil J. (2025). "Insufficient Evidence for Spinosaurid Survival into the Latest Cretaceous: A Comment on Olmedo-Romaña et al. (2025)" (https://www.ameghiniana.org.ar/index.php/ameghiniana/libraryFiles/downloadPublic/157). *Ameghiniana*. **62** (6): 572–580. doi:10.5710/AMGH.10.12.2025.3673 (https://doi.org/10.5710%2FAMGH.10.12.2025.3673).
108. Alonso, Antonio; Canudo, José Ignacio (17 August 2016). "On the spinosaurid theropod teeth from the early Barremian (Early Cretaceous) Blesa Formation (Spain)". *Historical Biology*. **28** (6): 823–834. Bibcode:2016HBio...28..823A (https://ui.adsabs.harvard.edu/abs/2016HBio...28..823A). doi:10.1080/08912963.2015.1036751 (https://doi.org/10.1080%2F08912963.2015.1036751). S2CID 131023889 (https://api.semanticscholar.org/CorpusID:131023889).
109. "Australian 'Spinosaur' unearthed" (http://www.australiangeographic.com.au/news/2011/06/australian-spinosaur-unearthed). *Australian Geographic*. Retrieved 2018-04-15.
110. Poropat, Stephen F.; White, Matt A.; Vickers-Rich, Patricia; Rich, Thomas H. (2019). "New megaraptorid (Dinosauria: Theropoda) remains from the Lower Cretaceous Eumeralla Formation of Cape Otway, Victoria, Australia" (https://figshare.com/articles/journal_contribution/9963197). *Journal of Vertebrate Paleontology*. **39** (4) e1666273. Bibcode:2019JVPal..39E6273P (https://ui.adsabs.harvard.edu/abs/2019JVPal..39E6273P). doi:10.1080/02724634.2019.1666273 (https://doi.org/10.1080%2F02724634.2019.1666273). S2CID 208603798 (https://api.semanticscholar.org/CorpusID:208603798).
111. Hattori, Soki; Azuma, Yoichi (2020). "Spinosaurid teeth from the Lower Cretaceous Kitadani Formation of the Tetori Group, Fukui, Japan" (https://www.researchgate.net/publication/350343224_Spinosaurid_teeth_from_the_Lower_Cretaceous_Kitadani_Formation_of_the_Tetori_Group_Fukui_Japan). *Memoir of the Fukui Prefectural Dinosaur Museum*. **19** – via ResearchGate.

External links

- Spinosauridae (http://theropoddatabase.com/Megalosauroidea.htm#Spinosauridae) on the Theropod Database

Retrieved from "https://en.wikipedia.org/w/index.php?title=Spinosauridae&oldid=1381152714"

WIKIPEDIA
25 years of the free encyclopedia

Tyrannosauridae

Tyrannosauridae (or **tyrannosaurids**, meaning "tyrant lizards") is a family of coelurosaurian dinosaurs that comprises two subfamilies containing up to fifteen genera, including the eponymous *Tyrannosaurus*. The exact number of genera is controversial, with some experts recognizing as few as three. All of these animals lived near the end of the Cretaceous period and their fossils have been found only in North America and Asia. Well-known genera include *Tyrannosaurus*, *Albertosaurus*, *Gorgosaurus*, *Daspletosaurus*, *Tarbosaurus*, and *Dynamoterror*.

Although descended from smaller ancestors, tyrannosaurids were almost always the largest predators in their respective ecosystems, putting them at the apex of the food chain. The largest species was *Tyrannosaurus rex*, one of the most massive known terrestrial predators of all time, which measured over 13 metres (43 ft) in length[2] and according to most modern estimates up to 8.87 metric tons (9.78 short tons) in weight.[3][4] Tyrannosaurids were bipedal carnivores with massive skulls filled with large teeth. Despite their large size, their legs were (outside of adults of the largest taxa) relatively long and cursorial for swift movement. In contrast, their arms were very small, bearing only two functional digits.

Unlike most other groups of dinosaurs, very complete remains have been discovered for most known tyrannosaurids. This has allowed a variety of research into their biology. Scientific studies have focused on their ontogeny, biomechanics and ecology, among other subjects.

History of discovery

Deinodon teeth, the earliest known tyrannosaurid remains

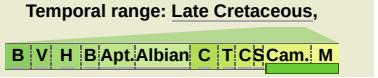
The first remains of tyrannosaurids were uncovered during expeditions led by the Geological Survey of Canada, which located numerous scattered teeth. These distinctive dinosaur teeth were given the name *Deinodon* ("terrible tooth") by Joseph Leidy in 1856. The first good specimens of a tyrannosaurid were found in the Horseshoe Canyon Formation of Alberta, and consisted of nearly complete skulls with partial skeletons. These remains were first studied by Edward Drinker Cope in 1876, who considered them a species of the eastern tyrannosauroid *Dryptosaurus*. In 1905, Henry Fairfield Osborn recognized that the Alberta remains differed considerably from *Dryptosaurus*, and coined a new name for them: *Albertosaurus sarcophagus* ("flesh-eating Alberta lizard").[5] Cope described more tyrannosaur material in 1892, in the form of isolated vertebrae, and gave this animal the name *Manospondylus gigas*. This discovery was mostly overlooked for over a century, and caused controversy in the early 2000s when it was discovered that this material actually belonged to, and had name priority over, *Tyrannosaurus rex*.[6]

In his 1905 paper naming *Albertosaurus*, Osborn described two additional tyrannosaur specimens that had been collected in Montana and Wyoming during a 1902 expedition of the American Museum of Natural History, led by Barnum Brown. Initially, Osborn considered these to be distinct species. The first, he named *Dynamosaurus imperiosus* ("emperor power lizard"), and the second, *Tyrannosaurus rex* ("king tyrant lizard"). A year later, Osborn recognized that these two specimens actually came from the same species. Despite the fact that *Dynamosaurus* had been found first, the name *Tyrannosaurus* had appeared one page earlier in his original article describing both specimens. Therefore, according to the International Code of Zoological Nomenclature (ICZN), the name *Tyrannosaurus* was used.[7]

Barnum Brown went on to collect several more tyrannosaurid specimens from Alberta, including the first to preserve the shortened, two-fingered forelimbs characteristic of the group (which Lawrence Lambe named *Gorgosaurus libratus*, "balanced fierce lizard", in 1914). A second significant find attributed to *Gorgosaurus* was made in 1942, in the form of a well-preserved, though unusually small, complete skull. The specimen waited until after the end of World War II to be studied by Charles W. Gilmore, who named it *Gorgosaurus lancesnis*.[5] This skull was re-studied by Robert T. Bakker, Phil Currie, and Michael Williams in 1988, and assigned to the new genus *Nanotyrannus*.[8] It was also in 1946 that paleontologists from the Soviet Union began expeditions into Mongolia, and uncovered the first tyrannosaur remains from Asia. Evgeny Maleev described new Mongolian species of *Tyrannosaurus* and *Gorgosaurus* in 1955, and one new genus: *Tarbosaurus* ("terrifying lizard"). Subsequent studies, however, showed that all of Maleev's tyrannosaur species were actually one species of *Tarbosaurus* at different stages of growth. A second species of Mongolian tyrannosaurid was found later, described by Sergei Kurzanov in 1976, and given the name *Alioramus remotus* ("remote different branch"), though its status as a true tyrannosaurid and not a more primitive tyrannosaur is still controversial.[9][5]

Tyrannosaurids

Temporal range: Late Cretaceous,



Montage of six tyrannosaurids, clockwise from top left: *Tyrannosaurus*, *Daspletosaurus*, *Tarbosaurus*, *Gorgosaurus*, *Zhuchengtyrannus*, and *Alioramus*

Scientific classification

Kingdom:	Animalia
Phylum:	Chordata
Class:	Reptilia
Clade:	Dinosauria
Clade:	Saurischia
Clade:	Theropoda
Superfamily:	†Tyrannosauroidea
Clade:	†Pantyrannosauria
Clade:	†Eutyrannosauria
Family:	†Tyrannosauridae Osborn, 1906

Type genus

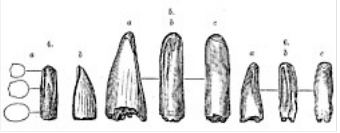
†*Tyrannosaurus*
Osborn, 1905

Subgroups[1]

- †*Aublysodon*
- †*Deinodon*
- †*Raptorex*
- †Albertosaurinae
- †Tyrannosaurinae

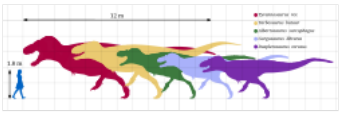
Synonyms

- Deinodontidae Cope, 1866
- Aublysodontidae Nopcsa, 1928
- Shanshanosauridae Dong, 1977



Teeth that have been assigned to *Aublysodon* at various times

Description

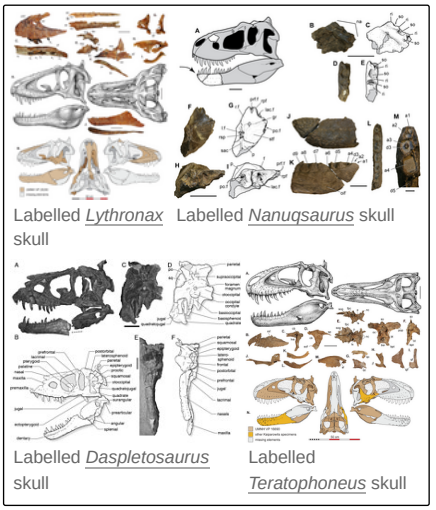


Size comparison of large tyrannosaurid genera *Tyrannosaurus*, *Tarbosaurus*, *Albertosaurus*, *Gorgosaurus* and *Daspletosaurus*

The tyrannosaurids were all large animals, with all species capable of weighing at least 1 metric ton.[10] A single specimen of *Alioramus* of an individual estimated at between 5 and 6 metres (16 and 20 ft) long has been discovered,[9] although it is considered by some experts to be a juvenile.[10][11] *Albertosaurus*, *Gorgosaurus* and *Daspletosaurus* all measured between 8 and 9 metres (26 and 30 ft) long,[12] while *Tarbosaurus* reached lengths of 11 metres (36 ft) from snout to tail.[13] The massive *Tyrannosaurus* reached 13 metres (43 ft) in the largest specimen, RSM P2523.8.[2] Tyrannosaurids are united by numerous synapomorphies, including an interfenestral strut that is less than half the rostrocaudal length of the maxillary fenestra, a lacrimal horn with a distinct apex in lateral view, and a pronounced lateral ridge on the postorbital ramus of the jugal that braces its articulation with the postorbital, among many other derived features.[14]

Skull and dentition

Tyrannosaurid skull anatomy is well understood, as complete skulls are known for all genera but *Alioramus*, which is known only from partial skull remains.[15] *Tyrannosaurus*, *Tarbosaurus*, and *Daspletosaurus* had skulls that exceeded 1 m (3.3 ft) in length.[12] Adult tyrannosaurids had tall, massive skulls, with many bones fused and reinforced for strength. At the same time, hollow chambers within many skull bones and large openings (fenestrae) between those bones helped to reduce skull weight. Many features of tyrannosaurid skulls were also found in their immediate ancestors, including tall premaxillae and fused nasal bones.[10]



Labelled *Lythronax* skull

Labelled *Nanuqsaurus* skull

Labelled *Daspletosaurus* skull

Labelled *Teratophoneus* skull

Tyrannosaurid skulls had many unique characteristics, including fused parietal bones with a prominent sagittal crest, which ran longitudinally along the sagittal suture and separated the two supratemporal fenestrae on the skull roof. Behind these fenestrae, tyrannosaurids had a characteristically tall nuchal crest, which also arose from the parietals but ran along a transverse plane rather than longitudinally. The nuchal crest was especially well-developed in *Tyrannosaurus*, *Tarbosaurus* and *Alioramus*. *Albertosaurus*, *Daspletosaurus* and *Gorgosaurus* had tall crests in front of the eyes on the lacrimal bones, while *Tarbosaurus* and *Tyrannosaurus* had extremely thickened postorbital bones forming crescent-shaped crests behind the eyes. *Alioramus* had a row of six bony crests on top of its snout, arising from the nasal bones; lower crests have been reported on some specimens of *Daspletosaurus* and *Tarbosaurus*, as well as the more basal tyrannosauroid *Appalachiosaurus*.[11][16] The snout and other parts of the skull also sported numerous foramina. According to the 2017 study which described *D. horneri*, scaly integument as well as tactile sensitivity was correlated with the multiple rows of neurovascular foramina seen in crocodilians and tyrannosaurids.[14]

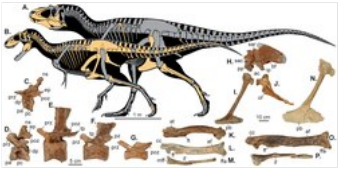
Tyrannosaurus teeth

Tyrannosaurids, like their tyrannosauroid ancestors, were heterodonts, with premaxillary teeth *D*-shaped in cross section and smaller than the rest. Unlike earlier tyrannosauroids and most other theropods, the maxillary and mandibular teeth of mature tyrannosaurids are not blade-like but extremely thickened and often circular in cross-section, with some species having reduced serrations.[10] Tooth counts tend to be consistent within species, and larger species tend to have lower tooth counts than smaller ones. For example, *Alioramus* had 76 to 78 teeth in its jaws, while *Tyrannosaurus* had between 54 and 60.[17]

William Abler observed in 2001 that *Albertosaurus* tooth serrations resemble a crack in the tooth ending in a round void called an ampulla.[18] Tyrannosaurid teeth were used as holdfasts for pulling meat off a body, so when a tyrannosaur would have pulled back on a piece of meat, the tension could cause a purely crack-like serration to spread through the tooth.[18] However, the presence of the ampulla would have distributed these forces over a larger surface area, and lessened the risk of damage to the tooth under strain.[18] The presence of incisions ending in voids has parallels in human engineering. Guitar makers use incisions ending in voids to, as Abler describes, "impart alternating regions of flexibility and rigidity" to the wood they work with.[18] The use of a drill to create an "ampulla" of sorts and prevent the propagation of cracks through material is also used to protect airplane surfaces.[18] Abler demonstrated that a plexiglass bar with incisions called "kerfs" and drilled holes was more than 25% stronger than one with only regularly placed incisions.[18] Unlike tyrannosaurs and other theropods, ancient predators like phytosaurs and *Dimetrodon* had no adaptations to prevent the crack-like serrations of their teeth from spreading when subjected to the forces of feeding.[18]

Postcranial skeleton

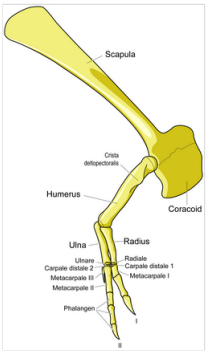
The skull was perched at the end of a thick, *S*-shaped neck, and a long, heavy tail acted as a counterweight to balance out the head and torso, with the center of mass over the hips. Tyrannosaurids are known for their proportionately very small two-fingered forelimbs, although remnants of a vestigial third digit are sometimes found.[10][19] *Tarbosaurus* had the shortest forelimbs compared to its body size, while *Daspletosaurus* had the longest.



Skeletal diagrams of *Lythronax* and *Teratophoneus*

Tyrannosaurids walked exclusively on their hindlimbs, so their leg bones were massive. In contrast to the forelimbs, the hindlimbs were longer compared to body size than almost any other theropods. Juveniles and even some smaller adults, like more basal tyrannosauroids, had longer tibiae than femora, a characteristic of cursorial (fast-running) dinosaurs like ornithomimids. Larger adults had leg proportions characteristic of slower-moving animals, but not to the extent seen in other large theropods like abelisaurids or carnosaurs. The third metatarsals of tyrannosaurids were pinched between the second and fourth metatarsals, forming a structure known as the arctometatarsus.[10] Tyrannosaurids also had large preserved leg muscle attachments and low rotational inertia relative to their body mass, indicating that they could turn more quickly than other large theropods.[20][21]

It is unclear when the arctometatarsus first evolved; it was not present in the earliest tyrannosauroids like *Dilong*,[22] but was found in the later *Appalachiosaurus*.[16] This structure also characterized troodontids, ornithomimids and caenagnathids,[23] but its absence in the earliest tyrannosauroids indicates that it was acquired by convergent evolution.[22]



Labelled diagram of the right arm and shoulder blade of *Gorgosaurus*

Soft tissue reconstruction

A comparative analysis of the teeth suggests that tyrannosaurids, as well as the other large theropods, had lips that protected their teeth from external damage. This anatomical feature gave these animals more visual resemblance to lepidosaurs than to closely related crocodilians.[24]

Classification

The name **Deinodontidae** was coined by Edward Drinker Cope in 1866 for this family,[25] and continued to be used in place of the newer name Tyrannosauridae through the 1960s.[26] The type genus of the Deinodontidae is *Deinodon*, which was named after isolated teeth from Montana.[27] However, in a 1970 review of North American tyrannosaurs, Dale Russell concluded that *Deinodon* was not a valid taxon, and used the name Tyrannosauridae in place of Deinodontidae, stating that this was in accordance with ICZN rules.[12] Therefore, Tyrannosauridae is preferred by modern experts.[5] A petition to officially conserve the family Tyrannosauridae and suppress the family Deinodontidae was submitted to the International Commission on Zoological Nomenclature (ICZN) in 2020,[28] which was approved in 2024.[29]

Tyrannosaurus was named by Henry Fairfield Osborn in 1905, along with the family Tyrannosauridae.[30] The name is derived from the Ancient Greek words τυραννος (*tyrannos*) ('tyrant') and σαυρος (*sauros*) ('lizard'). The very common suffix *-idae* is normally appended to zoological family names and is derived from the Greek suffix *-ιδαι -idai*, which indicates a plural noun.[31]

Taxonomy

Tyrannosauridae is a family in rank-based Linnaean taxonomy, within the superfamily Tyrannosauroidea and the suborder Theropoda.

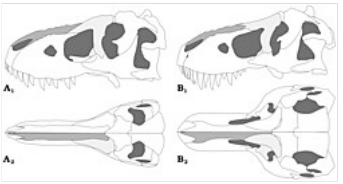


Diagram showing the differences between a generalised *Tarbosaurus* (A) and *Tyrannosaurus* (B) skull

Tyrannosauridae is uncontroversially divided into two subfamilies. Albertosaurinae comprises the North American genera *Albertosaurus* and *Gorgosaurus*, while Tyrannosaurinae includes *Daspletosaurus*, *Teratophoneus*, *Bistahieversor*, *Tarbosaurus*, *Nanuqsaurus*, *Zhuchengtyrannus*, and *Tyrannosaurus* itself.[32] Some authors include the species *Gorgosaurus libratus* in the genus *Albertosaurus* and *Tarbosaurus bataar* in the genus *Tyrannosaurus*,[16][5][33] while others prefer to retain *Gorgosaurus* and *Tarbosaurus* as separate genera.[10][11] Albertosaurines are characterized by more slender builds, lower skulls, and proportionately longer tibiae than tyrannosaurines.[10] In tyrannosaurines, the sagittal crest on the parietals continues forward onto the frontals.[11] In 2014, Lü Junchang *et al.* described the **Alioramini** as a tribe within the Tyrannosauridae containing the genera *Alioramus* and *Qianzhousaurus*. Their phylogenetic analysis indicated that the tribe was located at the base of the Tyrannosaurinae.[34][35] Some authors, such as George Olshevsky and Tracy Ford, have created other subdivisions or tribes for various combinations of tyrannosaurids within the subfamilies.[36][37] However, these have not been phylogenetically defined, and usually consisted of genera that are now considered synonymous with other genera or species.[17]

Additional subfamilies have been named for more fragmentary genera, including Aublysodontinae and Deinodontinae. However, the genera *Aublysodon* and *Deinodon* are usually considered *nomina dubia*, so they and their eponymous subfamilies are usually excluded from taxonomies of tyrannosaurids. An additional tyrannosaurid, *Raptorex*, was initially described as a more primitive tyrannosauroid, but likely represents a juvenile tyrannosaurine similar to *Tarbosaurus*. However, as it is known only from a juvenile specimen, it is also currently considered a *nomen dubium*.[38] However, Thomas Carr maintains its validity and finds it to be similar to tyrannosaurines.[39]

Phylogeny

With the advent of phylogenetic taxonomy in vertebrate paleontology, Tyrannosauridae has been given several explicit definitions. The original was produced by Paul Sereno in 1998, and included all tyrannosauroids closer to Tyrannosaurus than to either *Alectrosaurus*, *Aublysodon* or *Nanotyrannus*.[40]

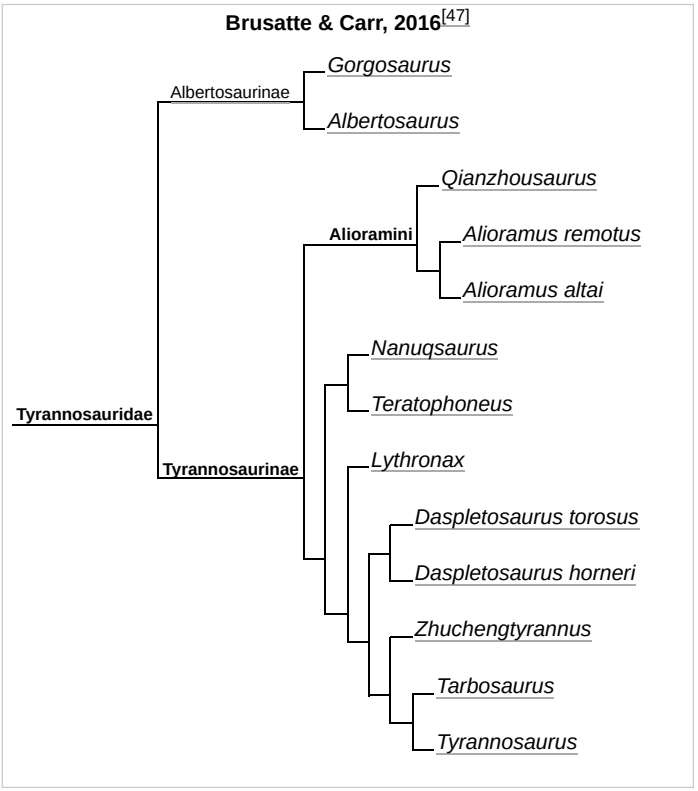
In 2001, Thomas R. Holtz Jr. published a cladistic analysis of the Tyrannosauridae.[41] He concluded that there were two subfamilies: the more primitive Aublysodontinae, characterized by unserrated premaxillary teeth; and the Tyrannosaurinae.[41] The Aublysodontinae included *Aublysodon*, the "Kirtland *Aublysodon*", and *Alectrosaurus*.[41] Holtz also found that *Siamotyrannus* exhibited some of the synapomorphies of the tyrannosauridae, but lay "outside the [family] proper."[41]

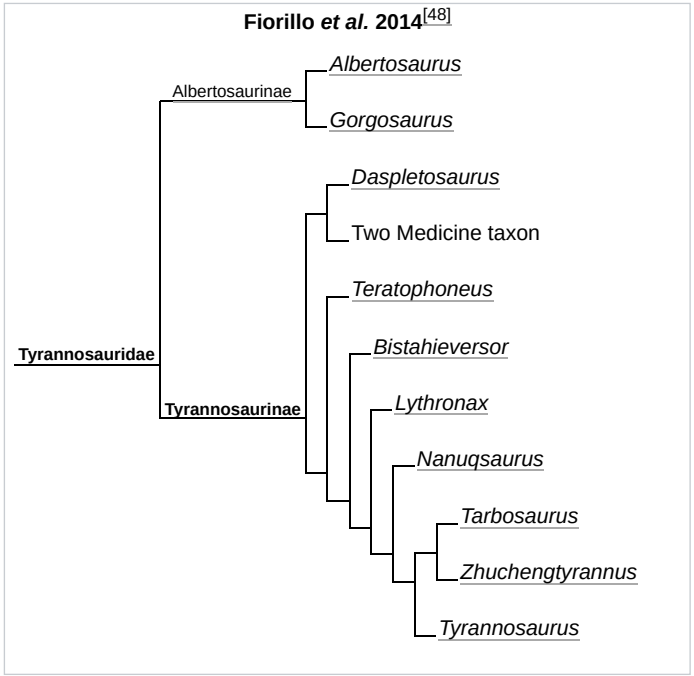
Later in the same paper, he proposed that Tyrannosauridae be defined as "all descendants of the most recent common ancestor of *Tyrannosaurus* and *Aublysodon*".[41] He also criticized definitions previously proposed by other workers, like one proposed by Paul Sereno, that the Tyrannosauridae was "all taxa closer to "Tyrannosaurus" than to *Alectrosaurus*, *Aublysodon*, and *Nanotyrannus*".[41] Holtz observed that since *Nanotyrannus* was probably a misidentified *T. rex* juvenile, Sereno's proposed definition would have the family Tyrannosauridae as a subtaxon of the genus *Tyrannosaurus*.[41] Further, his proposed definition of the subfamily Tyrannosaurinae would also be limited to *Tyrannosaurus*.[41]

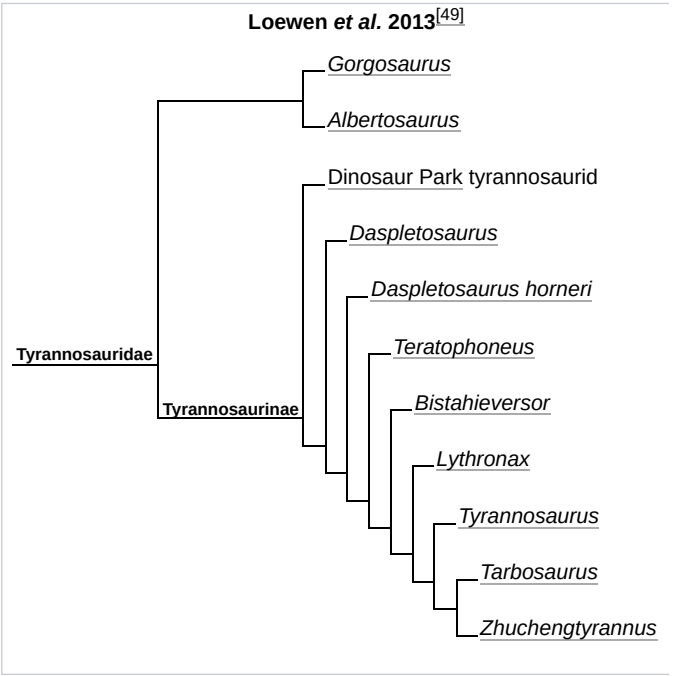
A 2003 attempt by Christopher Brochu included *Albertosaurus*, *Alectrosaurus*, *Alioramus*, *Daspletosaurus*, *Gorgosaurus*, *Tarbosaurus* and *Tyrannosaurus* in the definition.[42] Holtz redefined the clade in 2004 to use all of the above as specifiers except for *Alioramus* and *Alectrosaurus*, which his analysis could not place with certainty. However, in the same paper, Holtz also provided a completely different definition, including all theropods more closely related to *Tyrannosaurus* than to *Eotyrannus*.[10] The most recent definition is that of Sereno in 2005, which defined Tyrannosauridae as the least inclusive clade containing *Albertosaurus*, *Gorgosaurus* and *Tyrannosaurus*.[43]

Cladistic analyses of tyrannosaurid phylogeny often find *Tarbosaurus* and *Tyrannosaurus* to be sister taxa, with *Daspletosaurus* more basal than either. A close relationship between *Tarbosaurus* and *Tyrannosaurus* is supported by numerous skull features, including the pattern of sutures between certain bones, the presence of a crescent-shaped crest on the postorbital bone behind each eye, and a very deep maxilla with a noticeable downward curve on the lower edge, among others.[10][16] An alternative hypothesis was presented in a 2003 study by Phil Currie and colleagues, which found weak support for *Daspletosaurus* as a basal member of a clade also including *Tarbosaurus* and *Alioramus*, both from Asia, based on the absence of a bony prong connecting the nasal and lacrimal bones.[17] *Alioramus* was found to be the closest relative of *Tarbosaurus* in this study, based on a similar pattern of stress distribution in the skull.

A related study also noted a locking mechanism in the lower jaw shared between the two genera.[44] In a separate paper, Currie noted the possibility that *Alioramus* might represent a juvenile *Tarbosaurus*, but stated that the much higher tooth count and more prominent nasal crests in *Alioramus* suggest it is a distinct genus. Similarly, Currie uses the high tooth count of *Nanotyrannus* to suggest that it may be a distinct genus,[11] rather than a juvenile *Tyrannosaurus* as most other experts believe.[10][45] However, the discovery and description of *Qianzhousaurus* has since led many researchers to conclude that *Alioramus* is not a close relation to *Tarbosaurus*, instead belonging to a newly described tribe of tyrannosaurids; the Alioramini. *Qianzhousaurus* further reveals that similar long-snouted tyrannosaurids were widely distributed throughout Asia and would have shared the same environment while avoiding competition with larger and more robust tyrannosaurines by hunting different prey.[46]







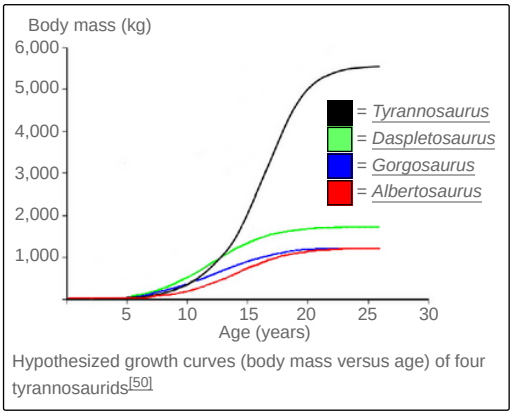
Paleobiology

Growth

Paleontologist Gregory Erickson and colleagues have studied the growth and life history of tyrannosaurids. Analysis of bone histology can determine the age of a specimen when it died. Growth rates can be examined when the age of various individuals are plotted against their size on a graph. Erickson has shown that after a long time as juveniles, tyrannosaurs underwent tremendous growth spurts for about four years midway through their lives. After the rapid growth phase

ended with sexual maturity, growth slowed down considerably in adult animals. A tyrannosaurid growth curve is S-shaped, with the maximum growth rate of individuals around 14 years of age.[50]

Skull of a juvenile *T. rex* formerly classified as *Dinotyrannus megagracilis*



Hypothesized growth curves (body mass versus age) of four tyrannosaurids[50]

The *Stygivenator* holotype (LACM 28471, the "Jordan theropod"), possibly a juvenile *T. rex*, is estimated to have weighed only 29.9 kilograms (66 lb) at only 2 years old, while one of the largest *T. rex* specimens, such as FMNH PR2081 ("Sue"), most likely weighed about 5,654 kg (12,465 lb), estimated to have been 28 years old, an age which may have been close to the maximum for the species.[50] *T. rex* juveniles remained under 1,800 kg (4,000 lb) until approximately 14 years of age, when body size began to increase dramatically. During this rapid growth phase, a young *T. rex* would gain an average of 600 kg (1,300 lb) a year for the next four years. This slowed after 16 years, and at 18 years of age, the curve plateaus again, indicating that growth slowed dramatically.[51] For example, only 600 kg (1,300 lb) separated the 28-year-old "Sue" from a 22-year-old Canadian specimen (RTMP 81.12.1).[50] This sudden change in growth rate may indicate physical maturity, a hypothesis that is supported by the discovery of medullary tissue in the femur of an 18-year-old *T. rex* from Montana (MOR 1125, also known as "B-rex").[52] Medullary tissue is found only in female birds during ovulation, indicating that "B-rex" was of reproductive age.[53]

Other tyrannosaurids exhibit extremely similar growth curves, although with lower growth rates corresponding to their lower adult sizes.[54] Compared to albertosaurines, *Daspletosaurus* showed a faster growth rate during the rapid growth period due to its higher adult weight. The maximum growth rate in *Daspletosaurus* was 180 kilograms (400 lb) per year, based on a mass estimate of 1,800 kg (4,000 lb) in adults. Other authors have suggested higher adult weights for *Daspletosaurus*; this would change the magnitude of the growth rate, but not the overall pattern.[50] The youngest known *Albertosaurus* is a two-year-old discovered in the Dry Island bonebed, which would have weighed about 50 kg (110 lb) and measured slightly more than 2 metres (6.6 ft) in length. The 10-metre (33 ft) specimen from the same quarry is the oldest and largest known, at 28 years of age. The fastest growth rate is estimated to occur around 12–16 years of age, reaching 122 kg (269 lb) per year, based on a 1,300 kg (2,900 lb) adult, which is about a fifth of the rate for *T.-rex*. For *Gorgosaurus*, the calculated maximum growth rate is about 110 kilograms (240 lb) during the rapid growth phase, which is comparable to that of *Albertosaurus*.[50]

In 2021, Funston and colleagues described possible perinatal tyrannosaurid ungual and premaxillary tooth (cf. *Albertosaurus sarcophagus*) from the Horseshoe Canyon Formation (Alberta, Canada) and dentary (cf. *Daspletosaurus horneri*) from the Two Medicine Formation (Montana, United States). They estimated that the ungual and dentary belonged to young tyrannosaurids measuring 1.1 m (3.6 ft) and 71.5 cm (2.35 ft) in total body length respectively. While eggshells have not been found in association with the specimens, the location where the purported neonate tyrannosaurids were uncovered suggests that they used the same nest sites with contemporary species.[55] Longrich et al. (2026) argued that the ungual (UALVP 59599) more closely resembles the morphology (shape) of an ornithomimosaurian ungual, especially that of *Struthiomimus altus*, and re-identified it as an ornithomimid, and that the dentary (MOR 268) belongs to an albertosaurine due to the similarities with *Gorgosaurus*.[56]

In July 2026, Longrich and colleagues described a third metatarsal bone of perinate (hatchling) *Tyrannosaurus rex* and cf. *Gorgosaurus libratus*, RSKM P2416.82 from the Frenchman Formation and TMP 1981.16.475 from the Dinosaur Park Formation respectively. Histological analyses reveal that lines of arrested growth are absent in both specimens, indicative of hatchling or embryonic stage. Synchrotron scans of each specimen reveal the relatively high density of the outer surface and the pattern changes in cortical (outer layer) bone structure, with a layer similar to the hatchling/neonatal line of modern oviparous (egg-laying) and viviparous (giving live birth) taxa that characterize the transition from embryonic to hatchling stage. The scans also detected evidence of Haversian bone remodeling, the continuous process of bone tissue breaking down and rebuilding to form secondary canals (tubular channels connecting bones), which are indicative of hatchling stage and precociality, relative maturity and mobility at birth. The layer lining the medullary cavity of both specimens likely represent endosteal bone deposition, during which new bone matrix is deposited on the inner surface of the cortical bone. The authors also described a partial dentary (FMNH PR 4920) from the Hell Creek Formation and isolated teeth from the Lance Formation as hatchling *T. rex* and interpret their tooth wear as evidence of consuming relatively large vertebrates.[56]

The body length and body mass of the hatchling *T. rex* (RSKM P2416.82) are estimated to be approximately 75 cm (2.46 ft) and 2.5 kg (5.5 lb) respectively, and those of the hatchling cf. *G. libratus* (TMP 1981.16.475) are estimated to be approximately 70 cm (2.3 ft) and 2.4 kg (5.3 lb) respectively. On the basis of these body size estimates, Longrich et al. (2026) suggested that the clutch of the confirmed, reproductively mature female *T. rex* (MOR 1125) would have weighed at least 54–60 kg (119–132 lb) which would possibly correspond to 21-32 eggs. They further stated that larger clutch sizes are also plausible, indicating that tyrannosaurids possibly employed R-selection, a reproductive strategy that maximizes the number of offsprings through large clutch size and minimal parental investment, which would have resulted in rapid maturation of the hatchlings, slower than that of modern birds. It is also likely that the reproduction strategy of tyrannosaurids was an intermediate between R-selection and K-selection, the latter of which would maximize parental investment with fewer number of offsprings and delayed maturation. Given the uniparental care observed in modern crocodilians and basal (early-diverging) birds, most non-avian dinosaurs including *T. rex* were likely uniparental as well. The authors also estimated that the relative size of tyrannosaurid hatchlings and eggs are larger than that of squamates, turtles, crocodilians, pterosaurs, and certain dinosaur clades (protoceratopsids and basal sauropodomorphs), but smaller than that of eumaniraptorans and modern birds, probably weighing less in both relative and absolute size than those of the elephant bird *Aepyornis maximus* and the South Island giant moa.[56]

Fossil footprints from the Wapiti Formation suggest that as tyrannosaurids grew, the feet became wider with thicker toes to support their weight. The broader feet suggest that adult tyrannosaurids were slower-moving than their offspring.[57][58]

Life history

The end of the rapid growth phase suggests the onset of sexual maturity in *Albertosaurus*, although growth continued at a slower rate throughout the animals' lives.[50][54] Sexual maturation while still actively growing appears to be a shared trait among small[59] and large[52] dinosaurs as well as in large mammals, such as humans and elephants.[52] This pattern of relatively early sexual maturation differs strikingly from the pattern in birds, which delay their sexual maturity until after they have finished growing.[52][60]

By tabulating the number of specimens of each age group, Erickson and his colleagues were able to draw conclusions about life history in tyranosauridae populations. Their analysis showed that while juveniles were rare in the fossil record, subadults in the rapid growth phase and adults were far more common. Over half of the known *T. rex* specimens appear to have died within six years of reaching sexual maturity, a pattern that is also seen in other tyrannosaurs and in some large, long-lived birds and mammals today. These species are characterized by high infant mortality rates, followed by relatively low mortality among juveniles. Mortality increases again following sexual maturity, partly due to the stresses of reproduction. While this could be due to preservation or collection biases, Erickson hypothesized that the difference was due to low mortality among juveniles over a certain size, which is also seen in some modern large mammals, like elephants. This low mortality may have resulted from a lack of predation, since tyrannosaurs surpassed all contemporaneous predators in size by the age of two. Paleontologists have not found enough *Daspletosaurus* remains for a similar analysis, but Erickson notes that the same general trend seems to apply.[54]

Nearly complete skeleton of a subadult *Gorgosaurus libratus*, from the Royal Tyrrell Museum of Palaeontology

The tyrannosaurids spent as much as half its life in the juvenile phase before ballooning up to near-maximum size in only a few years.[50] This, along with the lack of predators intermediate in size between adult tyrannosaurids and small theropods, suggests these niches may have been filled by juvenile tyrannosaurids. This is seen in modern Komodo dragons, where hatchlings start off as tree-dwelling insectivores and slowly mature into massive apex predators capable of taking down large vertebrates.[10] However, non-tyrannosaurid large theropods such as *Allosaurus* have also been found to exhibit similar life histories with juveniles filling the niches of cursorial mid-sized carnivores, in spite of living in ecosystems with true mid-sized carnivores present.[61]

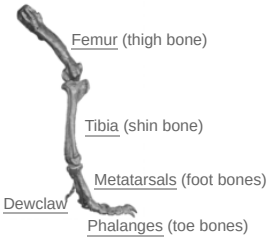
Albertosaurus have been found in aggregations that some have suggested to represent mixed-age packs that filled separate roles during hunting based on age class, with the more cursorial juveniles driving prey towards the adults.[62][63] Similar behavior is practiced occasionally by extant crocodilians.[64]

The discovery of two exceptionally preserved juvenile skulls from *Gorgosaurus* suggests that *Gorgosaurus* underwent the morphological shift from gracile juveniles to robust adults at an earlier age than *Tyrannosaurus*, to which it was compared in a study published by Jared Voris *et al.*, suggests that the ontogenetic changes occurred at roughly 5-7 years of age in *Gorgosaurus*; much earlier than its larger and later relative. However, both tyrannosaur genera underwent these ontogenetic transformations at a similar percent of skull length relative to the large known adult individuals. The study's results likewise indicate that there is a dissociation between body size and cranial development in tyrannosaurs, while simultaneously allowing better identification of juvenile remains that may have been misidentified in museum fossil collections.[65] It is estimated that an ontogenetic dietary shift of *Gorgosaurus* and *Albertosaurus* occurs when the mandibular length reaches 58 cm (1.90 ft), indicating that this is the stage when their bite force increases exponentially and when they begin to pursuit large prey.[66]

Locomotion

Locomotion abilities are best studied for *Tyrannosaurus*, and there are two main issues concerning this: how well it could turn; and what its maximum straight-line speed was likely to have been. *Tyrannosaurus* may have been slow to turn, possibly taking one to two seconds to turn only 45° – an amount that humans, being vertically oriented and tail-less, can spin in a fraction of a second.[67] The cause of the difficulty is rotational inertia, since much of *Tyrannosaurus*'s mass was some distance from its center of gravity, like a human carrying a heavy timber.[68]

Scientists have produced a wide range of maximum speed estimates, mostly around 11 metres per second (25 mph), but a few as low as 5–11 metres per second (11–25 mph), and a few as high as 20 metres per second (45 mph). Researchers have to rely on various estimating techniques because, while there are many tracks of very large theropods walking, so far none have been found of very large theropods running—and this absence *may* indicate that they did not run.[69]



Skeletal anatomy of a *T. rex* right leg

Jack Horner and Don Lessem argued in 1993 that *Tyrannosaurus* was slow and probably could not run (no airborne phase in mid-stride).[70] However, Holtz (1998) concluded that tyrannosaurids and their close relatives were the fastest large theropods.[71] Christiansen (1998) estimated that the leg bones of *Tyrannosaurus* were not significantly stronger than those of elephants, which are relatively limited in their top speed and never actually run (there is no airborne phase), and hence proposed that the dinosaur's maximum speed would have been about 11 metres per second (25 mph), which is about the speed of a human sprinter.[72] Farlow and colleagues (1995) have argued that a 6- to 8-ton *Tyrannosaurus* would have been critically or even fatally injured if it had fallen while moving quickly, since its torso would have slammed into the ground at a deceleration of 6 *g* (six times the acceleration due to gravity, or about 60 metres/s^2) and its tiny arms could not have reduced the impact.[73][74] However, giraffes have been known to gallop at 50 km/h (31 mph), despite the risk that they might break a leg or worse, which can be fatal even in a "safe" environment such as a zoo.[75][76] Thus it is quite possible that *Tyrannosaurus* also moved fast when necessary and had to accept such risks; this scenario has been studied for *Allosaurus* too.[77][78] Most recent research on *Tyrannosaurus* locomotion does not narrow down speeds further than a range from 17 to 40 km/h (11 to 25 mph), i.e. from walking or slow running to moderate-speed running.[69][79][80] A computer model study in 2007 estimated running speeds, based on data taken directly from fossils, and claimed that *T. rex* had a top running speed of 8 metres per second (18 mph).[81][82] (Probably a juvenile individual.[83])

Studies by Eric Snively *et al.*, published in 2019 indicate that tyrannosaurids such as *Tarbosaurus* and *Tyrannosaurus* itself were more manoeuvrable than allosauroids of comparable size due to low rotational inertia compared to their body mass combined with large leg muscles. As a result, it is hypothesized that tyrannosaurids were capable of making relatively quick turns and could likely pivot their bodies more quickly when close to their prey, or that while turning, they could "pirouette" on a single planted foot while the alternating leg was held out in a suspended swing during pursuit. The results of this study potentially could shed light on how agility could have contributed to the success of tyrannosaurid evolution.[20]

Additionally, a 2020 study indicates that tyrannosaurids were exceptionally efficient walkers. Studies by Dececchi *et al.*, compared the leg proportions, body mass, and the gaits of more than 70 species of theropod dinosaurs including tyrannosaurids. The research team then applied a variety of methods to estimate each dinosaur's top speed when running as well as how much energy each dinosaur expended while moving at more relaxed speeds such as when walking. Among smaller to medium-sized species such as dromaeosaurids, longer legs appear to be an adaptation for faster running, in line with previous results by other researchers. But for theropods weighing over 1,000 kg (2,200 lb), top running speed is limited by body size, so longer legs instead were found to have correlated with low-energy walking. The results of the study further indicated that smaller theropods evolved long legs for speed as a means to both aid in hunting and escape from larger predators while larger predatory theropods that evolved long legs did so to reduce the energy costs and increase foraging

efficiency, as they were freed from the demands of predation pressure due to their role as apex predators. Compared to more basal groups of theropods in the study, tyrannosaurids showed a marked increase in foraging efficiency due to reduced energy expenditures during hunting and scavenging. This likely resulted in tyrannosaurs having a reduced need for hunting forays and requiring less food to sustain themselves as a result. Additionally, the research, in conjunction with studies that show tyrannosaurs were more agile than other large-bodied theropods, indicates they were quite well-adapted to a long-distance stalking approach followed by a quick burst of speed to go for the kill. Analogies can be noted between tyrannosaurids and modern wolves as a result, supported by evidence that at least some tyrannosaurids such as *Albertosaurus* were hunting in group settings.[84][85]

Integument

Gorgosaurus restoration with speculative plumage, a trait inferred by phylogenetic bracketing

An ongoing debate in the paleontological community surrounds the extent and nature of tyrannosaurid integumentary covering. Long filamentous structures have been preserved along with skeletal remains of numerous coelurosaurs from the Early Cretaceous Yixian Formation and other nearby geological formations from Liaoning, China.[86] These filaments have usually been interpreted as "protofeathers," homologous with the branched feathers found in birds and some non-avian theropods,[87][88] although other hypotheses have been proposed.[89] A skeleton of *Dilong* was described in 2004 that included the first example of "protofeathers" in a tyrannosauroid. Similarly to down feathers of modern birds, the "protofeathers" found in *Dilong* were branched but not pennaceous, and may have been used for insulation.[22] The discovery and description of the 9-metre (30 ft) feathered tyrannosauroid *Yutyrannus* in 2012 indicates the possibility large tyrannosaurids were also feathered as adults.[90]

Albertosaurus sculpture from Royal Tyrrell Museum depicted with featherless skin

Based on the principle of phylogenetic bracketing, it was predicted that tyrannosaurids might also possess such feathering. However, a study in 2017 published by a team of researchers in *Biology Letters* described tyrannosaurid skin impressions collected in Alberta, Montana, and Mongolia, which came from five genera (*Tyrannosaurus*, *Albertosaurus*, *Gorgosaurus*, *Daspletosaurus* and *Tarbosaurus*).[91] Although the skin impressions are small, they are widely dispersed across the post-cranium, being collectively located on the abdomen, thoracic region, ilium, pelvis, tail, and neck. They show a tight pattern of fine, non-overlapping pebbly scales (which co-author Scott Persons compared to those seen on the flanks of a crocodile[92]) and preserve no hints of feathering. The basic texture is composed of tiny "basement scales" approximately 1 to 2 mm in diameter, with some impressions showing 7 mm "feature scales" interspersed between them. Additional scales can be seen in tyrannosaurid footprints.[93] Studies find that the facial integument of tyrannosaurids had scales on the dentary and maxilla, cornified epidermis and armor-like skin on the subordinate regions.[94][95]

Bell *et al.* performed an ancestral character reconstruction based on what is known about integument distribution in tyrannosauroids. Despite an 89% probability that tyrannosauroids started out with feathers, they determined that scaly tyrannosaurids have a 97% probability of being true. The data "provides compelling evidence of an entirely squamous covering in Tyrannosaurus," the team wrote, although they conceded that plumage may have still been present on the dorsal region where skin impressions haven't been found yet. Bell *et al.* hypothesizes that the scale impressions of tyrannosaurids are possibly reticula which are secondarily derived from feathers though evidence is needed to support this.[91] However, others argue that this is because of taphonomic bias in tyrannosaurids.[96]

It has yet to be determined why such an integumentary change might have occurred. A precedent for feather loss can be seen in other dinosaur groups such as ornithischians, in which filamentous structures were lost, and scales reappeared.[97] Although gigantism has been suggested as a mechanism, Phil R. Bell, who co-authored the study, noted that the feathered *Yutyrannus* overlapped in size with *Gorgosaurus* and *Albertosaurus*. "The problem here is that we have big tyrannosaurs, some with feathers, some without that live in pretty similar climates. So what's the reason for this difference? We really don't know."[98]

Vision

The eye-sockets of *T. rex* faced mainly forwards, giving it good binocular vision

The eye-sockets of *Tyrannosaurus* are positioned so that the eyes would point forward, giving them binocular vision slightly better than that of modern hawks and significantly better than that of falcons. While predatory theropods in general had binocular vision directly in front of their skull, *Tyrannosaurus* had a significantly larger area of overlap, which the authors suggested to be inconsistent with the hypothesis that this genus was an obligate scavenger.[99][100] In contrast, *Tarbosaurus* had a narrower skull more typical of other tyrannosaurids in which the eyes faced primarily sideways.[101] In *Gorgosaurus* specimens, the eye socket was circular rather than oval or keyhole-shaped as in other tyrannosaurid genera.[11] In *Daspletosaurus*, this was a tall oval, somewhere in between the circular shape seen in *Gorgosaurus* and the 'keyhole' shape of *Tyrannosaurus*.[10][11][45]

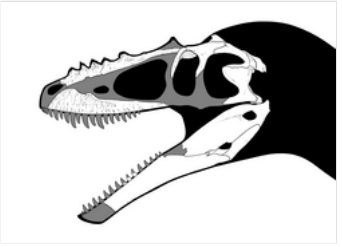
Facial sensitivity

Based on comparisons of bone texture of *Daspletosaurus* with extant crocodilians, a detailed study in 2017 by Thomas D. Carr *et al.* found that tyrannosaurs had large, flat scales on their snouts.[14][102] At the center of these scales were small keratinised patches. In crocodilians, such patches cover bundles of sensory neurons that can detect mechanical, thermal and chemical stimuli.[103][104] They proposed that tyrannosaurs probably also had bundles of sensory neurons under their facial scales and may have used them to identify objects, measure the temperature of their nests and gently pick-up eggs and hatchlings.[14] Similar adaptations are known from spinosaurids and the allosauroid *Neovenator*.

Bony crests

Bony crests are found on the skulls of many theropods, including many tyrannosaurids. *Alioramus*, a possible tyrannosaurid from Mongolia, bears a single row of five prominent bony bumps on the nasal bones; a similar row of much lower bumps is present on the skull of *Appalachiosaurus*, as well as some specimens of *Daspletosaurus*, *Albertosaurus*, and *Tarbosaurus*.[16] In *Albertosaurus*, *Gorgosaurus* and *Daspletosaurus*, there is a prominent horn in front of each eye on the lacrimal bone. The lacrimal horn is absent in *Tarbosaurus* and *Tyrannosaurus*, which instead have a crescent-shaped crest behind each eye on the postorbital bone. These head crests may have been used for display, perhaps for species recognition or courtship behavior.[10]

Thermoregulation



The skull of *Alioramus* with distinct nasal bumps

Tyrannosaurus, like most dinosaurs, was long thought to have an ectothermic ("cold-blooded") reptilian metabolism but was challenged by scientists like Robert T. Bakker and John Ostrom in the early years of the "Dinosaur Renaissance", beginning in the late 1960s.[105][106] *Tyrannosaurus rex* itself was claimed to have been endothermic ("warm-blooded"), implying a very active lifestyle.[107] Since then, several paleontologists have sought to determine the ability of *Tyrannosaurus* to regulate its body temperature. Histological evidence of high growth rates in young *T. rex*, comparable to those of mammals and birds, may support the hypothesis of a high metabolism. Growth curves indicate that, as in mammals and birds, *T. rex* growth was limited mostly to immature animals, rather than the indeterminate growth seen in most other vertebrates.[51] It has been indicated that the temperature difference may have been no more than 4 to 5 °C (7 to 9 °F) between the vertebrae of the torso and the tibia of the lower leg. This small temperature range between the body core and the extremities was claimed by paleontologist Reese Barrick and geochemist William Showers to indicate that *T. rex* maintained a constant internal body temperature (homeothermy) and that it enjoyed a metabolism somewhere between ectothermic reptiles and endothermic mammals.[108] Later they found similar results in *Giganotosaurus* specimens, who lived on a different continent and tens of millions of years earlier in time.[109] Even if *Tyrannosaurus rex* does exhibit evidence of homeothermy, it does not necessarily mean that it was endothermic. Such thermoregulation may also be explained by gigantothermy, as in some living sea turtles.[110][111][112]

Paleoecology

Coexistence of *Daspletosaurus* and *Gorgosaurus*

Skeleton of *Daspletosaurus* (specimen FMNH PR308) at the Field Museum in Chicago

In the Dinosaur Park Formation, *Gorgosaurus* lived alongside a rarer species of the tyrannosaurine *Daspletosaurus*. This is one of the few examples of two tyrannosaur genera coexisting. Similarly sized predators in modern predator guilds are separated into different ecological niches by anatomical, behavioral or geographical differences that limit competition. Niche differentiation between the Dinosaur Park tyrannosaurids is not well understood.[113] In 1970, Dale Russell hypothesized that the more common *Gorgosaurus* actively hunted fleet-footed hadrosaurs, while the rarer and more troublesome ceratopsians and ankylosaurians (horned and heavily armoured dinosaurs) were left to the more heavily built *Daspletosaurus*.[12] However, a specimen of *Daspletosaurus* (OTM 200) from the contemporaneous Two Medicine Formation of Montana preserves the digested remains of a juvenile hadrosaur in its gut region.[114] Unlike some other groups of dinosaurs, neither genus was more common at higher or lower elevations than the other.[113] However, *Gorgosaurus* appears more common in northern formations like the Dinosaur Park, with species of *Daspletosaurus* more abundant to the south. The same pattern is seen in other groups of dinosaurs. Chasmosaurine ceratopsians and hadrosaurine hadrosaurs are also more common in the Two Medicine Formation of Montana and in southwestern North America during the Campanian, while centrosaurines and lambeosaurines dominate in northern latitudes. Holtz has suggested that this pattern indicates shared ecological preferences between tyrannosaurines, chasmosaurines and hadrosaurines. At the end of the later Maastrichtian stage, tyrannosaurines like *Tyrannosaurus rex*, hadrosaurines like *Edmontosaurus* and chasmosaurines like *Triceratops* were widespread throughout western North America, while albertosaurines and centrosaurines became extinct, and lambeosaurines were rare.[10]

Social behavior

There is limited evidence of social behavior among the tyrannosaurids. Researchers reported that a subadult and a juvenile skeleton were found in the same quarry as the "Sue" specimen, which has been used to support the hypothesis that tyrannosaurs may have lived in social groups of some kind.[115] While there is no evidence of gregarious behavior in *Gorgosaurus*,[62][63] there is evidence of some pack behavior for *Albertosaurus* and *Daspletosaurus*.

A young specimen of the Dinosaur Park *Daspletosaurus* species (TMP 94.143.1) shows bite marks on the face that were inflicted by another tyrannosaur. The bite marks are healed over, indicating that the animal survived the bite. A full-grown Dinosaur Park *Daspletosaurus* (TMP 85.62.1) also exhibits tyrannosaur bite marks, showing that attacks to the face were not limited to younger animals. While it is possible that the bites were attributable to other species, intraspecific aggression, including facial biting, is very common among predators. Facial bites are seen in other tyrannosaurs like *Gorgosaurus* and *Tyrannosaurus*, as well as in other theropod genera like *Sinraptor* and *Saurornitholestes*. Darren Tanke and Phil Currie hypothesize that the bites are due to intraspecific competition for territory or resources, or for dominance within a social group.[62]

Full size model diorama of a group of *Albertosaurus*, Royal Tyrrell Museum

Evidence that *Daspletosaurus* lived in social groups comes from a bonebed found in the Two Medicine Formation of Montana. The bonebed includes the remains of three *Daspletosaurus*, including a large adult, a small juvenile, and another individual of intermediate size. At least five hadrosaurs are preserved at the same location. Geologic evidence indicates that the remains were not brought together by river currents but that all of the animals were buried simultaneously at the same location. The hadrosaur remains are scattered and bear many marks from tyrannosaur teeth, indicating that the *Daspletosaurus* were feeding on the hadrosaurs at the time of death. The cause of death is unknown. Currie speculates that the daspletosaurs formed a pack, although this cannot be stated with certainty.[63] Other scientists are skeptical of the evidence for social groups in *Daspletosaurus* and other large theropods;[116] Brian Roach and Daniel Brinkman have suggested that *Daspletosaurus* social interaction would have more closely resembled the modern Komodo dragon, where non-cooperative individuals mob carcasses, frequently attacking and even cannibalizing each other in the process.[117]

The Dry Island bonebed discovered by Barnum Brown and his crew contains the remains of 22 *Albertosaurus*, the most individuals found in one locality of any Cretaceous theropod, and the second-most of any large theropod dinosaur behind the *Allosaurus* assemblage at the Cleveland-Lloyd Dinosaur Quarry in Utah. The group seems to be composed of one very old adult; eight adults between 17 and 23 years old; seven sub-adults undergoing their rapid growth phases at between 12 and 16 years old; and six juveniles between the ages of 2 and 11 years, who had not yet reached the growth phase.[54] The near-absence of herbivore

remains and the similar state of preservation between the many individuals at the *Albertosaurus* bonebed quarry led Phil Currie to conclude that the locality was not a predator trap like the La Brea Tar Pits in California, and that all of the preserved animals died at the same time. Currie claims this as evidence of pack behavior.[118] Other scientists are skeptical, observing that the animals may have been driven together by drought, flood or for other reasons.[54][116][117]

While it generally remains controversial, evidence does exist that supports the theory that at least some tyrannosaurids were social. In British Columbia's Wapiti Formation, a trackway composed of the footprints of three individual tyrannosaurids (named as the ichnogenus *Bellatoripes fredlundi*) was discovered by a local outfitter named Aaron Fredlund and described in the journal PLOS One by Richard McCrea et al. An examination of the trackway found no evidence of one trackway being left long after another had been made, further supporting the hypothesis that three individual tyrannosaurs were traveling together as a group. Further research revealed the animals were traveling at a speed of between 3.9 and 5.2 mph (6.3 and 8.4 km/h) and likely had a hip height of around 7 to 9 feet. As three different genera of tyrannosaurids (*Gorgosaurus*, *Daspletosaurus*, and *Albertosaurus*, respectively) are known from the formation, it is unknown which genus was the maker of the trackway.[119][120][121] Additional evidence in the form of a bone-bed from the Rainbows and Unicorns Quarry in Southern Utah's Kaiparowits Formation described in 2021 attributed to *Teratophoneus* suggests other tyrannosaurids were also social animals. The fossils, consisting of four or possibly five different animals ranging from 4–22 years of age, suggest a mass mortality event, possibly caused by flooding, or less likely by cyanobacterial toxicosis, fire, or drought. The fact that all of the animals preserved seemed to have perished within a short timespan further strengthens the argument for gregarious behavior in tyrannosaurids, with the bone beds of such genera as *Teratophoneus*, *Albertosaurus*, *Tyrannosaurus* and *Daspletosaurus* showcasing suggested social behavior may have been widespread amongst tyrannosauridae in general.[122][123][124]

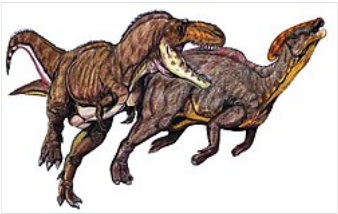
Feeding

Tyrannosaur tooth marks are the most commonly preserved feeding traces of carnivorous dinosaurs.[125] They have been reported from ceratopsians, hadrosaurs and other tyrannosaurs.[125] Tyrannosaurid bones with tooth marks represent about 2% of known fossils with preserved tooth marks.[125] Tyrannosaurid teeth were used as holdfasts for pulling meat off a body, rather than knife-like cutting functions.[126] Tooth wear patterns hint that complex head shaking behaviors may have been involved in tyrannosaur feeding.[126]

Gorgosaurus mounted as attacking a *Zuul*

Speculation on the pack-hunting habits of *Albertosaurus* were made by a few researchers who suggest that the younger members of the pack may have been responsible for driving their prey towards the adults, who were larger and more powerful, but also slower.[118] Juveniles may also have had different lifestyles than adults, filling predator niches between those of the enormous adults and the smaller contemporaneous theropods, the largest of which were two orders of magnitude smaller than an adult *Albertosaurus* in mass.[10] However, as the preservation of behavior in the fossil record is exceedingly rare, these ideas cannot readily be tested. Phil Currie speculates that the *Daspletosaurus* formed packs to hunt, although this cannot be stated with certainty.[63] There is no evidence of such gregarious behavior in *Gorgosaurus*.[62][63]

The debate about whether *Tyrannosaurus* was a predator or a pure scavenger is as old as the debate about its locomotion. Lambe (1917) described a good skeleton of *Tyrannosaurus*'s close relative *Gorgosaurus* and concluded that it and therefore also *Tyrannosaurus* was a pure scavenger, because the *Gorgosaurus*'s teeth showed hardly any wear.[127] This argument is no longer taken seriously, because theropods replaced their teeth quite rapidly. Ever since the first discovery of *Tyrannosaurus* most scientists have agreed that it was a predator, although like modern large predators it would have been happy to scavenge or steal another predator's kill if it had the opportunity.[128][129]

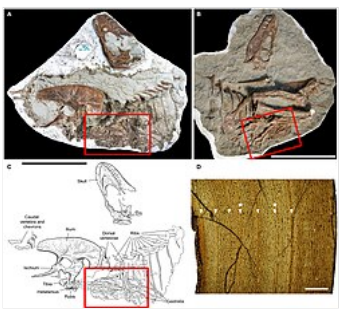


Teratophoneus curriei attacking a *Parasaurolophus cyrtocristatus*

Noted hadrosaur expert Jack Horner is currently the major advocate of the idea that *Tyrannosaurus* was exclusively a scavenger and did not engage in active hunting at all.[70][130][131] Horner has presented several arguments to support the pure scavenger hypothesis. The presence of large olfactory bulbs and olfactory nerves suggests a highly developed sense of smell for sniffing out carcasses over great distances. The teeth could crush bone, and therefore could extract as much food (bone marrow) as possible from carcass remnants, usually the least nutritious parts. At least some of its potential prey could move quickly, while evidence suggests that Tyrannosaurus walked instead of ran.[130][132]

Other evidence suggests hunting behavior in *Tyrannosaurus*. The eye-sockets of tyrannosaurs are positioned so that the eyes would point forward, giving them binocular vision slightly better than that of modern hawks. Tyrannosaur-inflicted damage has been found on skeletons of hadrosaurs and *Triceratops* that seemed to have survived initial attacks.[133][134][135] Some researchers argue that if *Tyrannosaurus* were a scavenger, another dinosaur had to be the top predator in the Amerasian Upper Cretaceous. The top prey were the larger marginocephalians and ornithopods. The other tyrannosaurids share so many characteristics with *Tyrannosaurus* that only small dromaeosaurs remain as feasible top predators. In this light, scavenger hypothesis adherents have suggested that the size and power of tyrannosaurs allowed them to steal kills from smaller predators.[132]

In 2023, a juvenile *Gorgosaurus* with its *in situ* stomach contents containing two *Citipes* juveniles about a year old intact has been reported from the Dinosaur Park Formation. This juvenile would have been 5–7 years old at the time of death, measuring about 4 metres (13 ft) long and weighing around 335 kilograms (739 lb). It is much larger than the two *Citipes* juveniles that weigh about 9–12 kilograms (20–26 lb), contrary to the assumption that tyrannosaurids fed on prey of their size once they reached 16–32 kilograms (35–71 lb), indicating that juvenile tyrannosaurids still consumed much smaller prey after exceeding a certain size threshold. The discovery of this specimen indicated that tyrannosaurids probably did not hunt in multigenerational packs, since its prey size is too small to share with the conspecifics. It is also a direct dietary evidence that reinforces the theory of 'ontogenetic dietary shift' for tyrannosaurids, as previously inferred by ecological modeling and anatomical features among different age groups. Only the remains of the hindlimbs and caudal vertebrae of juvenile *Citipes* were present in the tyrannosaurid's stomach cavity, suggesting that a juvenile *Gorgosaurus* may have had preferential consumption of the muscular hindlimbs.[136]



Gorgosaurus juvenile specimen TMP 2009.12.14, featuring stomach contents comprising remains of *Citipes*

A tyrannosaurid metatarsal from the Campanian-aged Judith River Formation shows bite marks indicative of scavenging from a smaller tyrannosaur also shows that size-asymmetric feeding behaviour between tyrannosaurids took place.[137]

Toothmarks on hadrosaur hind limb bones from the Aguja Formation that were described in 2026 suggest tyrannosaurids may have routinely dismembered carcasses and stripped all meat from less nutritious areas such as the ankles and feet. While scavenging cannot be ruled out, the same feeding behavior has been documented on other fossils in formations where tyrannosaurids are present, suggesting this was typical tyrannosaur feeding behavior.[138]

Cannibalism

Evidence also strongly suggests that tyrannosaurids were at least occasionally cannibalistic. *Tyrannosaurus* itself has strong evidence pointing towards it having been cannibalistic in at least a scavenging capacity based on tooth marks on the foot bones, humerus, and metatarsals of one specimen.[139] Fossils from the Fruitland Formation, Kirtland Formation (both Campanian in age), and Maastichtian-aged Ojo Alamo Formation suggest that cannibalism was present in various tyrannosaurid genera of the San Juan Basin. The evidence gathered from the specimens suggests opportunistic feeding behavior in tyrannosaurids that cannibalized members of their own species.[140] A tyrannosaurid from the Judith River Formation has been discovered with bite traces showing it was fed on by a younger conspecific or a member of a closely related species.[137]

Distribution

The earliest known tyrannosaur remains occurred between 129.4 and 125 million years ago in Ishikawa, Japan near Lat 36.166668 lng 136.633331. Found by Marsh in 1881 (https://paleobiodb.org/classic/displayCollResults?a=basicCollectionSearch&collection_no=32324) and identified by H.F. Osborne in 1906, the age of the remains was determined by H. Matsuoka et al. in 2002.

While earlier tyrannosauroids are found on all three northern continents, with possible remains from Australia, tyrannosaurid fossils (blue dots) are known only from North America and Asia

While earlier tyrannosauroids are found on all three northern continents, tyrannosaurid fossils are known only from North America and Asia. Sometimes fragmentary remains uncovered in the Southern Hemisphere have been reported as "Southern Hemisphere tyrannosaurids," although these seem to have been misidentified abelisaurid fossils.[141]

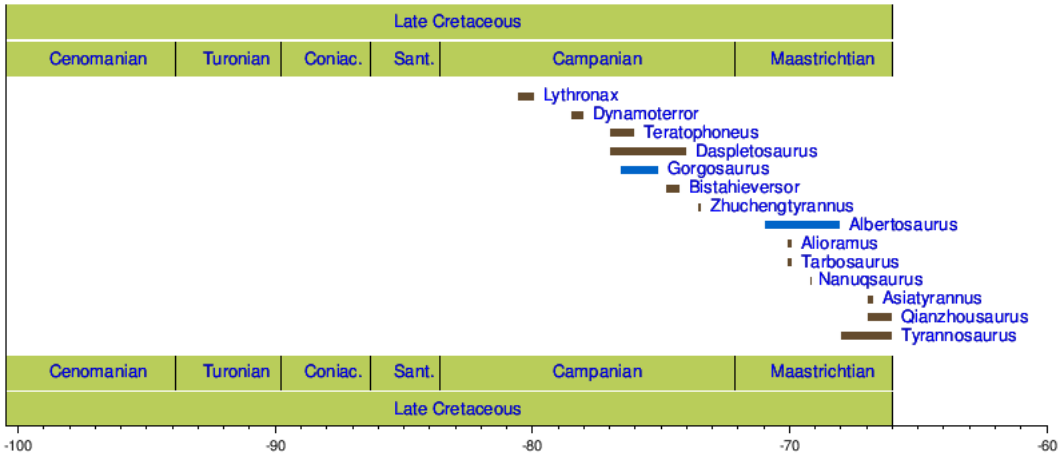
Tyrannosaurid remains have never been recovered from eastern North America, while more basal tyrannosauroids, like *Dryptosaurus* and *Appalachiosaurus*, persisted there until the end of the Cretaceous, indicating that tyrannosaurids must have evolved in or dispersed into western North America after the continent was divided in half by the Western Interior Seaway in the middle of the Cretaceous.[16] Tyrannosaurid fossils have been found in Alaska, which may have provided a route for dispersal between North America and Asia.[142] *Alioramus* and *Tarbosaurus* are found to be related in one cladistic analysis, forming a unique Asian branch of the family.[17] This was later disproven with the discovery of *Qianzhousaurus* and the description of the tyrannosaur family Alioramini. Tyrannosaurid teeth from a large species of unknown variety were discovered in the Nagasaki Peninsula by researchers from the Fukui Prefectural Dinosaur Museum, further expanding the range of the group. The teeth were estimated to be 81 million years old (Campanian Age).[143]

Skeleton cast of *Tarbosaurus bataar*, a tyrannosaurid from Asia

Of the two subfamilies, tyrannosaurines appear to have been more widespread. Albertosaurines are unknown in Asia, which was home to the tyrannosaurines, such as *Tarbosaurus* and *Zhuchengtyrannus*, and *Qianzhousaurus* and *Alioramus* of the Alioramini. Both the Tyrannosaurinae and Albertosaurinae subfamilies were present in the Campanian and early Maastrichtian stages of North America, with tyrannosaurines like *Daspletosaurus* ranging throughout the Western Interior, while the albertosaurines *Albertosaurus* and *Gorgosaurus* are currently known only from the northwestern part of the continent.[23]

By the late Maastrichtian, albertosaurines appear to have gone extinct, while the tyrannosaurine *Tyrannosaurus* roamed from Saskatchewan to Texas. This pattern is mirrored in other North American dinosaur taxa. During the Campanian and early Maastrichtian, lambeosaurine hadrosaurs and centrosaurine ceratopsians are common in the northwest, while hadrosaurines and chasmosaurines were more common to the south. By the end of the Cretaceous, centrosaurines are unknown and lambeosaurines are rare, while hadrosaurines and chasmosaurines were common throughout the Western Interior.[10] A study published in the journal *Scientific Reports* on 2 February 2016, by Steve Brusatte, Thomas Carr *et al*. indicates that during the later Maastrichtian, *Tyrannosaurus* itself might have been partially responsible for the extinction of the other tyrannosaurids in most of western North America. The study indicates that *Tyrannosaurus* might have been an immigrant from Asia as opposed to having evolved in North America (possibly a descendant of the closely related *Tarbosaurus*) that supplanted and outcompeted other tyrannosaurids. This theory is further supported by the fact that few to no other types of tyrannosaurid are found within *Tyrannosaurus'* known range. This could be called into question with the confirmation of Nanotyrannus being a distinct genus and sharing the same range as *Tyrannosaurus*. Likewise, one of the proposals listed by Zanno and Napoli in their description of *Nanotyrannus* suggests that *Nannotyrannus* itself may have originated in Appalachia, suggesting non-tyrannosaurid tyrannosauroids may have been able to coexist with tyrannosaurids by occupying different niches.[144][145]

Timeline of genera



See also

- Timeline of tyrannosaur research

References

1. Holtz, Thomas R. Jr. (2012) *Dinosaurs: The Most Complete, Up-to-Date Encyclopedia for Dinosaur Lovers of All Ages,* Winter 2011 Appendix. (http://www.geol.umd.edu/~tholtz/dinoappendix/HoltzappendixWinter2011.pdf) Archived (https://web.archive.org/web/20170812011954/http://www.geol.umd.edu/~tholtz/dinoappendix/HoltzappendixWinter2011.pdf) 12 August 2017 at the Wayback Machine
2. Article about Scotty on *Osel.cz* (in Czech) (http://www.osel.cz/10883-scotty-ten-nejvetsi-tyranosaurus-ze-vsech.html)
3. Persons, S. W.; Currie, P. J.; Erickson, G. M. (2019). "An Older and Exceptionally Large Adult Specimen of *Tyrannosaurus rex*" (https://doi.org/10.1002%2Far.24118). *The Anatomical Record*. **303** (4): 656–672. doi:10.1002/ar.24118 (https://doi.org/10.1002%2Far.24118). ISSN 1932-8486 (https://search.worldcat.org/issn/1932-8486). PMID 30897281 (https://pubmed.ncbi.nlm.nih.gov/30897281).
4. Lyle, A. (22 March 2019). "Paleontologists identify biggest *Tyrannosaurus rex* ever discovered" (https://www.folio.ca/paleontologists-identify-biggest-tyrannosaurus-rex-ever-discovered/). Folio, University of Alberta. Archived (https://web.archive.org/web/20200804134550/https://www.folio.ca/paleontologists-identify-biggest-tyrannosaurus-rex-ever-discovered/) from the original on 4 August 2020. Retrieved 25 March 2019.
5. Carpenter, Ken. (1992). "Tyrannosaurids (Dinosauria) of Asia and North America". In Mateer, Niall J.; Chen Peiji (eds.). *Aspects of Nonmarine Cretaceous Geology*. Beijing: China Ocean Press. pp. 250–268.
6. Breithaupt, B.H.; Southwell, E.H.; Matthews, N.A. (18 October 2005). "In Celebration of 100 years of *Tyrannosaurus rex*: *Manospondylus gigas*, *Ornithomimus grandis*, and *Dynamosaurus imperiosus*, the Earliest Discoveries of *Tyrannosaurus rex* in the West" (http://gsa.confex.com/gsa/2005AM/finalprogram/abstract_96150.htm). *Abstracts with Programs*. 2005 Salt Lake City Annual Meeting (http://gsa.confex.com/gsa/2005AM/finalprogram/index.html). Vol. 37. Geological Society of America. p. 406. Archived (https://web.archive.org/web/20191022084528/https://gsa.confex.com/gsa/2005AM/finalprogram/abstract_96150.htm) from the original on 22 October 2019. Retrieved 8 October 2008.
7. Breithaup, BH; Southwell EH; Matthews NA (2006). "*Dynamosaurus imperiosus* and the earliest discoveries of *Tyrannosaurus rex* in Wyoming and the West". *New Mexico Museum of Natural History and Science Bulletin*. **35**: 257–258.
8. Bakker R.T; Williams M.; Currie P. (1988). "*Nanotyrannus*, a new genus of pygmy tyrannosaur, from the latest Cretaceous of Montana". *Hunteria*. **1**: 1–30.
9. Kurzanov, Sergei M. "A new carnosaur from the Late Cretaceous of Nogon-Tsav, Mongolia". *The Joint Soviet-Mongolian Paleontological Expedition Transactions* (in Russian). **3**: 93–104.
10. Holtz, Thomas R. (2004). "Tyrannosauroidea". In Weishampel, David B.; Dodson, Peter; Osmólska, Halszka (eds.). *The Dinosauria* (Second ed.). Berkeley: University of California Press. pp. 111–136. ISBN 978-0-520-24209-8.
11. Currie, Philip John (2003). "Cranial anatomy of tyrannosaurids from the Late Cretaceous of Alberta" (http://www.app.pan.pl/archive/published/app48/app48-191.pdf) (PDF). *Acta Palaeontologica Polonica*. **48** (2): 191–226. Archived (https://web.archive.org/web/20180819050324/http://www.app.pan.pl/archive/published/app48/app48-191.pdf) (PDF) from the original on 19 August 2018. Retrieved 12 June 2010.
12. Russell, Dale A. (1970). "Tyrannosaurs from the Late Cretaceous of western Canada". *National Museum of Natural Sciences Publications in Paleontology*. **1**: 1–34.
13. Mortimer, Mickey (21 July 2003). "And the largest Theropod is..." (http://dml.cmnh.org/2003Jul/msg00355.html) The Dinosaur Mailing List. Archived (https://web.archive.org/web/20100325195202/http://dml.cmnh.org/2003Jul/msg00355.html) from the original on 25 March 2010. Retrieved 8 September 2007.
14. Carr, Thomas D.; Varricchio, David J.; Sedlmayr, Jayc C.; Roberts, Eric M.; Moore, Jason R. (30 March 2017). "A new tyrannosaur with evidence for anagenesis and crocodile-like facial sensory system" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5372470). *Scientific Reports*. **7** (1) 44942. Bibcode:2017NatSR...744942C (https://ui.adsabs.harvard.edu/abs/2017NatSR...744942C). doi:10.1038/srep44942 (https://doi.org/10.1038%2Fsrep44942). ISSN 2045-2322 (https://search.worldcat.org/issn/2045-2322). PMC 5372470 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5372470). PMID 28358353 (https://pubmed.ncbi.nlm.nih.gov/28358353).
15. Currie, Philip John (2000). "Theropods from the Cretaceous of Mongolia". *The Age of Dinosaurs in Russia and Mongolia*. Cambridge: Cambridge University Press. pp. 434–455. ISBN 978-0-521-54582-2.
16. Carr, Thomas D.; Williamson, Thomas E.; Schwimmer, David R. (2005). "A new genus and species of tyrannosauroid from the Late Cretaceous (middle Campanian) Demopolis Formation of Alabama". *Journal of Vertebrate Paleontology*. **25** (1): 119–143. doi:10.1671/0272-4634(2005)025[0119:ANGASO]2.0.CO;2 (https://doi.org/10.1671%2F0272-4634%282005%29025%5B0119%3AANGASO%5D2.0.CO%3B2). ISSN 0272-4634 (https://search.worldcat.org/issn/0272-4634). S2CID 86243316 (https://api.semanticscholar.org/CorpusID:86243316).

17. Currie, Philip J.; Hurum, Jørn H; Sabath, Karol. (2003). "Skull structure and evolution in tyrannosaurid phylogeny" (http://www.app.pan.pl/archive/published/app48/app48-227.pdf) (PDF). *Acta Palaeontologica Polonica*. **48** (2): 227–234. Archived (https://web.archive.org/web/20081031093118/http://www.app.pan.pl/archive/published/app48/app48-227.pdf) (PDF) from the original on 31 October 2008. Retrieved 2 October 2010.
18. Abler, W.L. 2001. A kerf-and-drill model of tyrannosaur tooth serrations. p. 84–89. In: *Mesozoic Vertebrate Life*. Ed.s Tanke, D. H., Carpenter, K., Skrepnick, M. W. Indiana University Press.
19. Quinlan, Elizabeth D.; Derstler, Kraig; Miller, Mercedes M. (2007). "Anatomy and function of digit III of the *Tyrannosaurus rex* manus" (http://gsa.confex.com/gsa/2007AM/finalprogram/abstract_132345.htm). *Geological Society of America Annual Meeting — Abstracts with Programs*: 77. Archived (https://web.archive.org/web/20080224015023/http://gsa.confex.com/gsa/2007AM/finalprogram/abstract_132345.htm) from the original on 24 February 2008. Retrieved 15 January 2008. [abstract only]
20. Snively, Eric; O'Brien, Haley; Henderson, Donald M.; Mallison, Heinrich; Surring, Lara A.; Burns, Michael E.; Holtz, Thomas R.; Russell, Anthony P.; Witmer, Lawrence M.; Currie, Philip J.; Hartman, Scott A.; Cotton, John R. (21 February 2019). "Lower rotational inertia and larger leg muscles indicate more rapid turns in tyrannosaurids than in other large theropods" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6387760). *PeerJ*. **7** e6432. Bibcode:2019PeerJ...7e6432S (https://ui.adsabs.harvard.edu/abs/2019PeerJ...7e6432S). doi:10.7717/peerj.6432 (https://doi.org/10.7717%2Fpeerj.6432). ISSN 2167-8359 (https://search.worldcat.org/issn/2167-8359). PMC 6387760 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6387760). PMID 30809441 (https://pubmed.ncbi.nlm.nih.gov/30809441).
21. Hartman, Scott (21 February 2019). "Lower rotational inertia and larger leg muscles indicate more rapid turns in tyrannosaurids than in other large theropods" (https://www.scotthartman.info/publication/snively-2019-lower/). *Scott Hartman - PhD*. Retrieved 29 December 2023.
22. Xu Xing, X; Norell, Mark A.; Kuang Xuewen; Wang Xiaolin; Zhao Qi; Jia Chengkai. (2004). "Basal tyrannosauroids from China and evidence for protofeathers in tyrannosauroids" (http://doc.rero.ch/record/15283/files/PAL_E2582.pdf) (PDF). *Nature*. **431** (7009): 680–684. Bibcode:2004Natur.431..680X (https://ui.adsabs.harvard.edu/abs/2004Natur.431..680X). doi:10.1038/nature02855 (https://doi.org/10.1038%2Fnature02855). PMID 15470426 (https://pubmed.ncbi.nlm.nih.gov/15470426). S2CID 4381777 (https://api.semanticscholar.org/CorpusID:4381777).
23. Holtz, Thomas R. (1994). "The phylogenetic position of the Tyrannosauridae: implications for theropod systematics". *Journal of Paleontology*. **68** (5): 1100–1117. Bibcode:1994JPal...68.1100H (https://ui.adsabs.harvard.edu/abs/1994JPal...68.1100H). doi:10.1017/S0022336000026706 (https://doi.org/10.1017%2FS0022336000026706). JSTOR 1306180 (https://www.jstor.org/stable/1306180). S2CID 129684676 (https://api.semanticscholar.org/CorpusID:129684676).
24. Thomas M. Cullen; Derek William Larson; Mark P Witton; Diane Scott; Tea Maho; Kirstin S. Brink; David C Evans; Robert Reisz (30 March 2023). "Theropod dinosaur facial reconstruction and the importance of soft tissues in paleobiology" (https://doi.org/10.1126%2Fscience.abo7877). *Science*. **379** (6639): 1348–1352. Bibcode:2023Sci...379.1348C (https://ui.adsabs.harvard.edu/abs/2023Sci...379.1348C). doi:10.1126/science.abo7877 (https://doi.org/10.1126%2Fscience.abo7877). PMID 36996202 (https://pubmed.ncbi.nlm.nih.gov/36996202).
25. Cope E.D. (1866). "Discovery of a gigantic dinosaur in the Cretaceous of New Jersey". *Proceedings of the Academy of Natural Sciences of Philadelphia*. **18**: 275–279.
26. Maleev E.A. (1955). "New carnivorous dinosaurs from the Upper Cretaceous Mongolia". *Doklady Akademii Nauk SSSR*. **104** (5): 779–783.
27. Leidy, Joseph (1856). "Notice of remains of extinct reptiles and fishes, discovered by Dr. F.V. Hayden in the badlands of the Judith River, Nebraska Territory". *Proceedings of the Academy of Natural Sciences of Philadelphia*. **8**: 72–73.
28. Yun, C.-G. (2020). "Case 3815 – Tyrannosauridae Osborn, 1906 (Dinosauria, Theropoda): proposed conservation by reversal of precedence with Deinodontidae Cope, 1866 and Dryptosauridae Marsh, 1890". *The Bulletin of Zoological Nomenclature*. **77** (4): 29–34. doi:10.21805/bzn.v77.a008 (https://doi.org/10.21805%2Fbzn.v77.a008).
29. "Opinion 2511 (Case 3815) – Tyrannosauridae Osborn, 1906 (Dinosauria, Theropoda): usage conserved by reversing precedence with Deinodontidae Cope, 1866 and Dryptosauridae Marsh, 1890". *The Bulletin of Zoological Nomenclature*. **81** (1): 156–157. December 2024. doi:10.21805/bzn.v81.a034 (https://doi.org/10.21805%2Fbzn.v81.a034).
30. Osborn, Henry Fairfield (1905). "*Tyrannosaurus* and other Cretaceous carnivorous dinosaurs". *Bulletin of the American Museum of Natural History*. **21** (3): 259–265. doi:10.1111/j.1468-5965.2007.00735_17.x (https://doi.org/10.1111%2Fj.1468-5965.2007.00735_17.x). hdl:2246/1464 (https://hdl.handle.net/2246%2F1464).
31. Liddell, Henry G.; Scott, Robert (1980). *Greek-English Lexicon* (https://archive.org/details/lexicon00lidd) (Abridged ed.). Oxford: Oxford University Press. ISBN 978-0-19-910207-5.
32. Fiorillo, A. R.; Tykoski, R. S. (2014). Dodson, Peter (ed.). "A Diminutive New Tyrannosaur from the Top of the World" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3951350). *PLoS ONE*. **9** (3) e91287. Bibcode:2014PLoSO...991287F (https://ui.adsabs.harvard.edu/abs/2014PLoSO...991287F). doi:10.1371/journal.pone.0091287 (https://doi.org/10.1371%2Fjournal.pone.0091287). PMC 3951350 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3951350). PMID 24621577 (https://pubmed.ncbi.nlm.nih.gov/24621577).
33. Paul, Gregory S. (1988). *Predatory Dinosaurs of the World* (https://archive.org/details/predatorydinosau00paul/page/464). New York: Simon & Schuster. pp. 464pp (https://archive.org/details/predatorydinosau00paul/page/464). ISBN 978-0-671-61946-6.
34. Lü, Junchang; Yi, Laiping; Brusatte, Stephen L.; Yang, Ling; Li, Hua; Chen, Liu (2014). "A new clade of Asian Late Cretaceous long-snouted tyrannosaurids" (https://doi.org/10.1038%2Fncomms4788). *Nature Communications*. **5** 3788. Bibcode:2014NatCo...5.3788L (https://ui.adsabs.harvard.edu/abs/2014NatCo...5.3788L). doi:10.1038/ncomms4788 (https://doi.org/10.1038%2Fncomms4788). PMID 24807588 (https://pubmed.ncbi.nlm.nih.gov/24807588).
35. "Pinocchio rex long-snouted tyrannosaur discovered in Asia" (http://www.cbc.ca/news/technology/pinocchio-rex-long-snouted-tyrannosaur-discovered-in-asia-1.2634651). 7 May 2014. Archived (https://web.archive.org/web/20190202131420/https://www.cbc.ca/news/technology/pinocchio-rex-long-snouted-tyrannosaur-discovered-in-asia-1.2634651) from the original on 2 February 2019. Retrieved 8 May 2014.
36. Olshevsky, George (1995). "The origin and evolution of the tyrannosaurids". *Kyoryugaku Saizensen [Dino Frontline]*. **9–10**: 92–119.
37. Olshevsky G.; Ford T.L. (1995). "The origin and evolution of the Tyrannosauridae, part 2 [in Japanese]". *Dino Frontline*. **6**: 75–99.
38. Fowler, Denver W.; Woodward, H. N.; Freedman, E. A.; Larson, Peter L.; Horner, Jack Robert (2011). "Reanalysis of "Raptorex kriegsteini": A Juvenile Tyrannosaurid Dinosaur from Mongolia" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3126816). *PLOS ONE*. **6** (6) e21376. Bibcode:2011PLoSO...621376F (https://ui.adsabs.harvard.edu/abs/2011PLoSO...621376F). doi:10.1371/journal.pone.0021376 (https://doi.org/10.1371%2Fjournal.pone.0021376). PMC 3126816 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3126816). PMID 21738646 (https://pubmed.ncbi.nlm.nih.gov/21738646).
39. Carr, Thomas D. (25 November 2022). "A reappraisal of tyrannosauroid fossils from the Iren Dabasu Formation (Coniacian–Campanian), Inner Mongolia, People's Republic of China" (https://doi.org/10.1080%2F02724634.2023.2199817). *Journal of Vertebrate Paleontology*. **42** (5) e2199817. Bibcode:2022JVPal..42E9817C (https://ui.adsabs.harvard.edu/abs/2022JVPal..42E9817C). doi:10.1080/02724634.2023.2199817 (https://doi.org/10.1080%2F02724634.2023.2199817). ISSN 0272-4634 (https://search.worldcat.org/issn/0272-4634).
40. Sereno, Paul C. (1998). "A rationale for phylogenetic definitions, with application to the higher-level taxonomy of Dinosauria". *Neues Jahrbuch für Geologie und Paläontologie, Abhandlungen* (in German). **210**: 41–83. doi:10.1127/njgpa/210/1998/41 (https://doi.org/10.1127%2Fnjgpa%2F210%2F1998%2F41).
41. Holtz, TR, Jr. (2001), The phylogeny and taxonomy of the Tyrannosauridae in K Carpenter & D Tanke [eds.], Mesozoic Vertebrate Life. Indiana Univ. Press, pp. 64–83.

42. Brochu, Christopher R. (2003). "Osteology of *Tyrannosaurus rex*: insights from a nearly complete skeleton and high-resolution computed tomographic analysis of the skull" (http://publication.plazi.org/id/6603FFA80B5D0D64FFF6AB533D39FF9A). *Society of Vertebrate Paleontology Memoirs*. **7**: 1–138. doi:10.2307/3889334 (https://doi.org/10.2307%2F3889334). JSTOR 3889334 (https://www.jstor.org/stable/3889334).
43. Sereno, Paul C. (7 November 2005). "Stem Archosauria—TaxonSearch" (https://web.archive.org/web/20070219102659/http://www.taxonsearch.org/dev/file_home.php). Archived from the original (http://www.taxonsearch.org/dev/file_home.php) on 19 February 2007. Retrieved 14 January 2008.
44. Hurum, Jørn H.; Sabath, Karol. (2003). "Giant theropod dinosaurs from Asia and North America: Skulls of *Tarbosaurus bataar* and *Tyrannosaurus rex* compared" (http://www.app.pan.pl/article/item/app48-161.html?pdf=39) (abstract). *Acta Palaeontologica Polonica*. **48** (2): 161–190. Archived (https://web.archive.org/web/20120308104302/http://www.app.pan.pl/article/item/app48-161.html?pdf=39) from the original on 8 March 2012. Retrieved 10 September 2008.
45. Carr, Thomas D. (1999). "Craniofacial ontogeny in Tyrannosauridae (Dinosauria, Coelurosauria)" (https://zenodo.org/record/3371479). *Journal of Vertebrate Paleontology*. **19** (3): 497–520. Bibcode:1999JVPal..19..497C (https://ui.adsabs.harvard.edu/abs/1999JVPal..19..497C). doi:10.1080/02724634.1999.10011161 (https://doi.org/10.1080%2F02724634.1999.10011161). S2CID 83744433 (https://api.semanticscholar.org/CorpusID:83744433). Archived (https://web.archive.org/web/20200220183407/https://zenodo.org/record/3371479) from the original on 20 February 2020. Retrieved 2 December 2019.
46. "Newly found dinosaur is long-nosed cousin of Tyrannosaurus rex" (https://www.sciencedaily.com/releases/2014/05/140507100421.htm). *ScienceDaily*. Archived (https://web.archive.org/web/20180317102219/https://www.sciencedaily.com/releases/2014/05/140507100421.htm) from the original on 17 March 2018. Retrieved 9 March 2018.
47. CITEREFBrusatte_&_Carr2016
48. CITEREFFiorillo_&_Tykoski2014
49. Loewen, M.A.; Irmis, R.B.; Sertich, J.J.W.; Currie, P. J.; Sampson, S. D. (2013). Evans, David C (ed.). "Tyrant Dinosaur Evolution Tracks the Rise and Fall of Late Cretaceous Oceans" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3819173). *PLoS ONE*. **8** (11) e79420. Bibcode:2013PLoSO...879420L (https://ui.adsabs.harvard.edu/abs/2013PLoSO...879420L). doi:10.1371/journal.pone.0079420 (https://doi.org/10.1371%2Fjournal.pone.0079420). PMC 3819173 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3819173). PMID 24223179 (https://pubmed.ncbi.nlm.nih.gov/24223179).
50. Erickson, Gregory M., GM; Makovicky, Peter J.; Currie, Philip J.; Norell, Mark A.; Yerby, Scott A.; Brochu, Christopher A. (2004). "Gigantism and comparative life-history parameters of tyrannosaurid dinosaurs" (http://doc.rero.ch/record/15279/files/PAL_E2578.pdf) (PDF). *Nature*. **430** (7001): 772–775. Bibcode:2004Natur.430..772E (https://ui.adsabs.harvard.edu/abs/2004Natur.430..772E). doi:10.1038/nature02699 (https://doi.org/10.1038%2Fnature02699). PMID 15306807 (https://pubmed.ncbi.nlm.nih.gov/15306807). S2CID 4404887 (https://api.semanticscholar.org/CorpusID:4404887). Archived (https://web.archive.org/web/20200714024211/http://doc.rero.ch/record/15279/files/PAL_E2578.pdf) (PDF) from the original on 14 July 2020. Retrieved 7 January 2023. "Sampled longevities for T. rex ranged from 2 to 28 years" **(Erratum: doi:10.1038/nature16487 (https://doi.org/10.1038%2Fnature16487), PMID 26675726 (https://pubmed.ncbi.nlm.nih.gov/26675726), *Retraction Watch* (http://retractionwatch.com/2016/03/01/high-profile-critic-slams-nature-letters-about-dinosaur-growth-following-corrections/))** and corresponding body mass estimates ranged from 29.9 to 5654 kg
51. Horner, J.R. & Padian, K. (2004). "Age and growth dynamics of Tyrannosaurus rex" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1691809). *Proceedings of the Royal Society of London B*. **271** (1551): 1875–1880. doi:10.1098/rspb.2004.2829 (https://doi.org/10.1098%2Frspb.2004.2829). PMC 1691809 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1691809). PMID 15347508 (https://pubmed.ncbi.nlm.nih.gov/15347508).
52. Lee, Andrew H.; Werning, Sarah (2008). "Sexual maturity in growing dinosaurs does not fit reptilian growth models" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2206579). *Proceedings of the National Academy of Sciences*. **105** (2): 582–587. Bibcode:2008PNAS..105..582L (https://ui.adsabs.harvard.edu/abs/2008PNAS..105..582L). doi:10.1073/pnas.0708903105 (https://doi.org/10.1073%2Fpnas.0708903105). PMC 2206579 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2206579). PMID 18195356 (https://pubmed.ncbi.nlm.nih.gov/18195356).
53. Schweitzer, M.H.; Wittmeyer, J.L.; Horner, J.R. (2005). "Gender-specific reproductive tissue in ratites and *Tyrannosaurus rex*". *Science*. **308** (5727): 1456–1460. Bibcode:2005Sci...308.1456S (https://ui.adsabs.harvard.edu/abs/2005Sci...308.1456S). doi:10.1126/science.1112158 (https://doi.org/10.1126%2Fscience.1112158). PMID 15933198 (https://pubmed.ncbi.nlm.nih.gov/15933198). S2CID 30264554 (https://api.semanticscholar.org/CorpusID:30264554).
54. Erickson, G.M.; Currie, P.J.; Inouye, B.D.; Winn, A.A. (2006). "Tyrannosaur life tables: an example of nonavian dinosaur population biology". *Science*. **313** (5784): 213–217. Bibcode:2006Sci...313..213E (https://ui.adsabs.harvard.edu/abs/2006Sci...313..213E). doi:10.1126/science.1125721 (https://doi.org/10.1126%2Fscience.1125721). PMID 16840697 (https://pubmed.ncbi.nlm.nih.gov/16840697). S2CID 34191607 (https://api.semanticscholar.org/CorpusID:34191607).
55. Funston GF, Powers MJ, Whitebone SA, Brusatte SL, Scannella JB, Horner JR, Currie PJ (2021). "Baby tyrannosaurid bones and teeth from the Late Cretaceous of western North America" (https://www.pure.ed.ac.uk/ws/files/180195224/545._Greg.pdf) (PDF). *Canadian Journal of Earth Sciences*. **58** (9): 756–777. Bibcode:2021CaJES..58..756F (https://ui.adsabs.harvard.edu/abs/2021CaJES..58..756F). doi:10.1139/cjes-2020-0169 (https://doi.org/10.1139%2Fcjes-2020-0169). hdl:20.500.11820/3a870ad5-7258-4fcb-a3b7-238e448c04b4 (https://hdl.handle.net/20.500.11820%2F3a870ad5-7258-4fcb-a3b7-238e448c04b4). S2CID 234053280 (https://api.semanticscholar.org/CorpusID:234053280).
56. Longrich, N. R.; Makovicky, P. J.; Tokaryk, T.; Cooper, D. M. L.; Saitta, E. T.; Erickson, G. M.; Szekely, T.; Snively, E. (2026). "Hatchlings of *Tyrannosaurus rex* and the Evolution of Dinosaur Reproductive Strategies" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC13359715). *Biology*. **15** (13) 1090. doi:10.3390/biology15131090 (https://doi.org/10.3390%2Fbiology15131090). PMC 13359715 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC13359715). PMID 42450637 (https://pubmed.ncbi.nlm.nih.gov/42450637).
57. Enriquez, Nathan James; Campione, Nicolas; Bell, Phil (21 April 2021). "Fat-footed tyrannosaur parents couldn't keep up with their skinnier offspring, fossil footprints reveal" (https://theconversation.com/fat-footed-tyrannosaur-parents-couldnt-keep-up-with-their-skinnier-offspring-fossil-footprints-reveal-154188). *The Conversation*. Archived (https://web.archive.org/web/20210426230718/https://theconversation.com/fat-footed-tyrannosaur-parents-couldnt-keep-up-with-their-skinnier-offspring-fossil-footprints-reveal-154188) from the original on 26 April 2021. Retrieved 28 April 2021.
58. Enriquez, Nathan J.; Campione, Nicolás E.; Brougham, Tom; Fanti, Federico; White, Matt A.; Sissons, Robin L.; Sullivan, Corwin; Vavrek, Matthew J.; Bell, Phil R. (21 April 2021). "Exploring possible ontogenetic trajectories in tyrannosaurids using tracks from the Wapiti Formation (upper Campanian) of Alberta, Canada". *Journal of Vertebrate Paleontology*. **40** (6) e1878201. doi:10.1080/02724634.2021.1878201 (https://doi.org/10.1080%2F02724634.2021.1878201). S2CID 234814620 (https://api.semanticscholar.org/CorpusID:234814620).
59. Erickson, Gregory M.; Curry Rogers, Kristi; Varricchio, David J.; Norell, Mark; Xu, Xing (2007). "Growth patterns in brooding dinosaurs reveals the timing of sexual maturity in non-avian dinosaurs and genesis of the avian condition" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2396186). *Biology Letters*. **3** (5): 558–561. doi:10.1098/rsbl.2007.0254 (https://doi.org/10.1098%2Frsbl.2007.0254). PMC 2396186 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2396186). PMID 17638674 (https://pubmed.ncbi.nlm.nih.gov/17638674).
60. Ricklefs, Robert E. (2007). "Tyrannosaur ageing" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2375931). *Biology Letters*. **3** (2): 214–217. doi:10.1098/rsbl.2006.0597 (https://doi.org/10.1098%2Frsbl.2006.0597). PMC 2375931 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2375931). PMID 17284406 (https://pubmed.ncbi.nlm.nih.gov/17284406).

61. Foster, John R.; Chure, Daniel J. (2006). "Hindlimb allometry in the Late Jurassic theropod dinosaur Allosaurus, with comments on its abundance and distribution". In Foster, John R.; Lucas, Spencer G. (eds.). Paleontology and Geology of the Upper Jurassic Morrison Formation. New Mexico Museum of Natural History and Science Bulletin, 36. Albuquerque, New Mexico: New Mexico Museum of Natural History and Science. pp. 119–122.
62. Tanke, Darren H.; Currie, Philip J. (1998). "Head-biting behavior in theropod dinosaurs: paleopathological evidence" (http://www.mnhn.ul.pt/geologia/gaia/12.pdf) (PDF). *Gaia*. **15**: 167–184. Archived (https://web.archive.org/web/20080227134632/http://www.mnhn.ul.pt/geologia/gaia/12.pdf) (PDF) from the original on 27 February 2008. Retrieved 13 September 2008. [not printed until 2000]
63. Currie, Philip John; Trexler, David; Koppelhus, Eva B.; Wicks, Kelly; Murphy, Nate (2005). "An unusual multi-individual tyrannosaurid bonebed in the Two Medicine Formation (Late Cretaceous, Campanian) of Montana (USA)". In Carpenter, Kenneth (ed.). *The Carnivorous Dinosaurs*. Bloomington: Indiana University Press. pp. 313–324. ISBN 978-0-253-34539-4.
64. Dinets, Vladimir (3 April 2015). "Apparent coordination and collaboration in cooperatively hunting crocodilians". *Ethology Ecology & Evolution*. **27** (2): 244–250. Bibcode:2015EtEcE..27..244D (https://ui.adsabs.harvard.edu/abs/2015EtEcE..27..244D). doi:10.1080/03949370.2014.915432 (https://doi.org/10.1080%2F03949370.2014.915432). ISSN 0394-9370 (https://search.worldcat.org/issn/0394-9370). S2CID 84672219 (https://api.semanticscholar.org/CorpusID:84672219).
65. Voris, Jared T.; Zelenitsky, Darla K.; Therrien, François; Ridgely, Ryan C.; Currie, Philip J.; Witmer, Lawrence M. (2021). "Two exceptionally preserved juvenile specimens of Gorgosaurus libratus (Tyrannosauridae, Albertosaurinae) provide new insight into the timing of ontogenetic changes in tyrannosaurids". *Journal of Vertebrate Paleontology*. **41** (6). e2041651. Bibcode:2021JVPal..41E1651V (https://ui.adsabs.harvard.edu/abs/2021JVPal..41E1651V). doi:10.1080/02724634.2021.2041651 (https://doi.org/10.1080%2F02724634.2021.2041651). hdl:1880/117371 (https://hdl.handle.net/1880%2F117371). ISSN 0272-4634 (https://search.worldcat.org/issn/0272-4634). S2CID 248197540 (https://api.semanticscholar.org/CorpusID:248197540).
66. **Cite error: The named reference `TZVT21` was invoked but never defined (see the help page).**
67. "Tyrannosaurus had poor turning circle" (https://web.archive.org/web/20120507053958/http://www.cosmosmagazine.com/node/1381). Archived from the original on 7 May 2012. Retrieved 25 September 2007.
68. Carrier, D.R.; Walter, R.M.; Lee, D.V. (15 November 2001). "Influence of rotational inertia on turning performance of theropod dinosaurs: clues from humans with increased rotational inertia" (http://jeb.biologists.org/cgi/content/full/204/22/3917). *Journal of Experimental Biology*. **204** (Pt 22): 3917–3926. Bibcode:2001JExpB.204.3917C (https://ui.adsabs.harvard.edu/abs/2001JExpB.204.3917C). doi:10.1242/jeb.204.22.3917 (https://doi.org/10.1242%2Fjeb.204.22.3917). PMID 11807109 (https://pubmed.ncbi.nlm.nih.gov/11807109). Archived (https://web.archive.org/web/20080106180207/http://jeb.biologists.org/cgi/content/full/204/22/3917) from the original on 6 January 2008. Retrieved 12 October 2008.
69. Hutchinson, J.R. (2004). "Biomechanical Modeling and Sensitivity Analysis of Bipedal Running Ability. II. Extinct Taxa" (https://web.archive.org/web/20081031093050/http://www.rvc.ac.uk/AboutUs/Staff/jhutchinson/documents/JRH13.pdf) (PDF). *Journal of Morphology*. **262** (1): 441–461. Bibcode:2004JMorp.262..441H (https://ui.adsabs.harvard.edu/abs/2004JMorp.262..441H). doi:10.1002/jmor.10240 (https://doi.org/10.1002%2Fjmor.10240). PMID 15352202 (https://pubmed.ncbi.nlm.nih.gov/15352202). S2CID 15677774 (https://api.semanticscholar.org/CorpusID:15677774). Archived from the original (http://www.rvc.ac.uk/AboutUs/Staff/jhutchinson/documents/JRH13.pdf) (PDF) on 31 October 2008.
70. Horner, John R.; Don Lessem (1993). *The complete T. rex* (https://archive.org/details/completetrexhows00horn/page/120). New York City: Simon & Schuster. pp. 120 (https://archive.org/details/completetrexhows00horn/page/120). ISBN 978-0-671-74185-3.
71. Holtz, Thomas R. (1 May 1996). "Phylogenetic taxonomy of the Coelurosauria (Dinosauria; Theropoda)" (http://jpaleontol.geoscienceworld.org/cgi/content/abstract/70/3/536). *Journal of Paleontology*. **70** (3): 536–538. Bibcode:1996JPal...70..536H (https://ui.adsabs.harvard.edu/abs/1996JPal...70..536H). doi:10.1017/S0022336000038506 (https://doi.org/10.1017%2FS0022336000038506). S2CID 87599102 (https://api.semanticscholar.org/CorpusID:87599102). Archived (https://web.archive.org/web/20081026053239/http://jpaleontol.geoscienceworld.org/cgi/content/abstract/70/3/536) from the original on 26 October 2008. Retrieved 3 October 2008.
72. Christiansen, P. (1998). "Strength indicator values of theropod long bones, with comments on limb proportions and cursorial potential" (http://www.mnhn.ul.pt/geologia/gaia/19.pdf) (PDF). *Gaia*. **15**: 241–255. ISSN 0871-5424 (https://search.worldcat.org/issn/0871-5424). Archived (https://web.archive.org/web/20081031093050/http://www.mnhn.ul.pt/geologia/gaia/19.pdf) (PDF) from the original on 31 October 2008. Retrieved 12 October 2008.
73. Farlow, James O.; Smith, Matthew B.; Robinson, John M. (1995). "Body mass, bone "strength indicator", and cursorial potential of *Tyrannosaurus rex*" (https://web.archive.org/web/20081023063102/http://www.vertpaleo.org/publications/jvp/15-713-725.cfm). *Journal of Vertebrate Paleontology*. **15** (4): 713–725. Bibcode:1995JVPal..15..713F (https://ui.adsabs.harvard.edu/abs/1995JVPal..15..713F). doi:10.1080/02724634.1995.10011257 (https://doi.org/10.1080%2F02724634.1995.10011257). Archived from the original (http://www.vertpaleo.org/publications/jvp/15-713-725.cfm) on 23 October 2008.
74. "The bigger they come, the harder they fall" (https://www.newscientist.com/article/mg14819982.600-the-bigger-they-come-the-harder-they-fall.html) Archived (https://web.archive.org/web/20150531201705/http://www.newscientist.com/article/mg14819982.600-the-bigger-they-come-the-harder-they-fall.html) 31 May 2015 at the Wayback Machine New Scientist, 7 October 1995, p. 18.
75. "Giraffe" (http://www.wildlifesafari.info/giraffe.html). WildlifeSafari.info. Archived (https://web.archive.org/web/20100612080401/http://www.wildlifesafari.info/giraffe.html) from the original on 12 June 2010. Retrieved 29 April 2006.
76. "The History of Woodland Park Zoo — Chapter 4" (https://web.archive.org/web/20070602141617/http://www.zoo.org/zoo_info/history/chapters/history4.html). Archived from the original (http://www.zoo.org/zoo_info/history/chapters/history4.html) on 2 June 2007. Retrieved 29 April 2006.
77. Alexander, R.M. (7 August 2006). "Dinosaur biomechanics" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1634776). *Proc Biol Sci*. **273** (1596): 1849–1855. Bibcode:2006PBioS.273.1849A (https://ui.adsabs.harvard.edu/abs/2006PBioS.273.1849A). doi:10.1098/rspb.2006.3532 (https://doi.org/10.1098%2Frspb.2006.3532). PMC 1634776 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1634776). PMID 16822743 (https://pubmed.ncbi.nlm.nih.gov/16822743).
78. Hanna, Rebecca R. (2002). "Multiple injury and infection in a sub-adult theropod dinosaur (*Allosaurus fragilis*) with comparisons to allosaur pathology in the Cleveland-Lloyd Dinosaur Quarry collection". *Journal of Vertebrate Paleontology*. **22** (1): 76–90. doi:10.1671/0272-4634(2002)022[0076:MIAIIA]2.0.CO;2 (https://doi.org/10.1671%2F0272-4634%282002%29022%5B0076%3AMIAIIA%5D2.0.CO%3B2). ISSN 0272-4634 (https://search.worldcat.org/issn/0272-4634). S2CID 85654858 (https://api.semanticscholar.org/CorpusID:85654858). catalogues the injuries of the *Allosaurus* known as "Big Al" - at least one was attributed to a fall.
79. Hutchinson, John R.; Garcia, M (28 February 2002). "*Tyrannosaurus* was not a fast runner" (http://researchonline.rvc.ac.uk/id/eprint/1204/). *Nature*. **415** (6875): 1018–1021. Bibcode:2002Natur.415.1018H (https://ui.adsabs.harvard.edu/abs/2002Natur.415.1018H). doi:10.1038/4151018a (https://doi.org/10.1038%2F4151018a). PMID 11875567 (https://pubmed.ncbi.nlm.nih.gov/11875567). S2CID 4389633 (https://api.semanticscholar.org/CorpusID:4389633). Archived (https://web.archive.org/web/20220809012102/https://researchonline.rvc.ac.uk/id/eprint/1204/) from the original on 9 August 2022. Retrieved 7 January 2023.
80. Hajdul, R. (1997). Tendons. *Dinosaur Cards*. Orbis Publishing Ltd. D36044311.

81. Sellers, W.I. & Manning, P.L. (July 2007). "Estimating dinosaur maximum running speeds using evolutionary robotics" (http://dinonews.net/rubriq/articles.php5?action=open&ref=2007_sellers_speed). *Proc. R. Soc. B*. **274** (1626): 2711–6. Bibcode:2007PBioS.274.2711S (https://ui.adsabs.harvard.edu/abs/2007PBioS.274.2711S). doi:10.1098/rspb.2007.0846 (https://doi.org/10.1098%2Frspb.2007.0846). PMC 2279215 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2279215). PMID 17711833 (https://pubmed.ncbi.nlm.nih.gov/17711833). Archived (https://web.archive.org/web/20110207142321/http://dinonews.net/rubriq/articles.php5?action=open&ref=2007_sellers_speed) from the original on 7 February 2011. Retrieved 12 October 2008. This may be a preliminary version of Sellers, W. I., Manning, P.L., Crompton, R.H. and Codd, J.R.,. (2007), "Exploring elastic energy storage effects in bipedal locomotion using evolutionary robotics", *Journal of Biomechanics*, in-review
82. Liz Seward: "T. rex 'would outrun footballer'" (https://news.bbc.co.uk/2/hi/science/nature/6956867.stm) . *BBC News* website, Tuesday, 21 August 2007. The article quotes Dr Bill Sellers, University of Manchester, co-author of a paper published in *Proceedings of the Royal Society B*. Retrieved 22 August 2007.
83. Callison, G.; H. M. Quimby (1984). "Tiny dinosaurs: Are they fully grown?". *Journal of Vertebrate Paleontology*. **3** (4): 200–209. Bibcode:1984JVPal...3..200C (https://ui.adsabs.harvard.edu/abs/1984JVPal...3..200C). doi:10.1080/02724634.1984.10011975 (https://doi.org/10.1080%2F02724634.1984.10011975).
84. Dececchi, T. Alexander; Mloszewska, Aleksandra M.; Holtz, Thomas R.; Habib, Michael B.; Larsson, Hans C. E. (2020). "The fast and the frugal: Divergent locomotory strategies drive limb lengthening in theropod dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7220109). *PLOS ONE*. **15** (5) e0223698. Bibcode:2020PLoSO..1523698D (https://ui.adsabs.harvard.edu/abs/2020PLoSO..1523698D). doi:10.1371/journal.pone.0223698 (https://doi.org/10.1371%2Fjournal.pone.0223698). PMC 7220109 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7220109). PMID 32401793 (https://pubmed.ncbi.nlm.nih.gov/32401793).
85. "T. rex was a champion walker, super-efficient at lower speeds" (https://www.eurekalert.org/pub_releases/2020-05/p-trw051320.php). *EurekAlert!*. Archived (https://web.archive.org/web/20200627052208/https://www.eurekalert.org/pub_releases/2020-05/p-trw051320.php) from the original on 27 June 2020. Retrieved 27 June 2020.
86. Zhou Zhonghe, Z; Barrett, Paul M.; Hilton, Jason. (2003). "An exceptionally preserved Lower Cretaceous ecosystem". *Nature*. **421** (6925): 807–814. Bibcode:2003Natur.421..807Z (https://ui.adsabs.harvard.edu/abs/2003Natur.421..807Z). doi:10.1038/nature01420 (https://doi.org/10.1038%2Fnature01420). PMID 12594504 (https://pubmed.ncbi.nlm.nih.gov/12594504). S2CID 4412725 (https://api.semanticscholar.org/CorpusID:4412725).
87. Chen Peiji, Pei-ji; Dong Zhiming; Zhen Shuonan. (1998). "An exceptionally well-preserved theropod dinosaur from the Yixian Formation of China" (http://doc.rero.ch/record/14898/files/PAL_E2040.pdf) (PDF). *Nature*. **391** (6663): 147–152. Bibcode:1998Natur.391..147C (https://ui.adsabs.harvard.edu/abs/1998Natur.391..147C). doi:10.1038/34356 (https://doi.org/10.1038%2F34356). S2CID 4430927 (https://api.semanticscholar.org/CorpusID:4430927).
88. Xu Xing, X; Zhou Zhonghe; Prum, Richard A. (2003). "Branched integumental structures in Sinornithosaurus and the origin of feathers". *Nature*. **410** (6825): 200–204. Bibcode:2001Natur.410..200X (https://ui.adsabs.harvard.edu/abs/2001Natur.410..200X). doi:10.1038/35065589 (https://doi.org/10.1038%2F35065589). PMID 11242078 (https://pubmed.ncbi.nlm.nih.gov/11242078). S2CID 4426803 (https://api.semanticscholar.org/CorpusID:4426803).
89. Lingham-Soliar, Theagarten; Feduccia, Alan; Wang, Xiaolin (2007). "A new Chinese specimen indicates that 'protofeathers' in the Early Cretaceous theropod dinosaur Sinosauropteryx are degraded collagen fibres" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2270928). *Proceedings of the Royal Society of London. Series B, Biological Sciences*. **274** (1620): 1823–1829. Bibcode:2007PBioS.274.1823L (https://ui.adsabs.harvard.edu/abs/2007PBioS.274.1823L). doi:10.1098/rspb.2007.0352 (https://doi.org/10.1098%2Frspb.2007.0352). PMC 2270928 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2270928). PMID 17521978 (https://pubmed.ncbi.nlm.nih.gov/17521978).
90. "Archived copy" (https://web.archive.org/web/20120417134949/http://www.xinglida.net/pdf/Xu_et_al_2012_Yutyrannus.pdf) (PDF). Archived from the original (http://www.xinglida.net/pdf/Xu_et_al_2012_Yutyrannus.pdf) (PDF) on 17 April 2012. Retrieved 4 April 2012.
91. Bell, Philip R.; Campione, N. E.; Persons, Walter Scott; Currie, Philip John; Larson, Peter L.; Tanke, Darren H.; Bakker, Robert T. (2017). "Tyrannosauroid integument reveals conflicting patterns of gigantism and feather evolution" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5493735). *Biology Letters*. **13** (6) 20170092. doi:10.1098/rsbl.2017.0092 (https://doi.org/10.1098%2Frsbl.2017.0092). PMC 5493735 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5493735). PMID 28592520 (https://pubmed.ncbi.nlm.nih.gov/28592520).
92. http://globalnews.ca/news/3506400/tyrannosaurus-rex-fossil-skin-scales-university-of-alberta/ Archived (https://web.archive.org/web/20170617181635/http://globalnews.ca/news/3506400/tyrannosaurus-rex-fossil-skin-scales-university-of-alberta/) 17 June 2017 at the Wayback Machine What does a T. Rex feel like? U of A paleontologists find out in new fossil discovery
93. Currie, P.J., Badamgarav, D., Koppelhus, E.B. 2003. The First Late Cretaceous Footprints from the Nemegt Locality in the Gobi of Mongolia. Ichnos. Vol.10:1-12.Currie, P.J., Badamgarav, D., Koppelhus, E.B. 2003.
94. Carr, T. D., Varricchio, D. J., Sedlmayr, J. C., Roberts, E. M., & Moore, J. R. (2017). A new tyrannosaur with evidence for anagenesis and crocodile-like facial sensory system. *Scientific Reports*, 7.
95. Witton, Mark; Hone, David (September 2018). "Tyrannosaurid theropods: did they ever smile like crocodiles?" (https://iris.unito.it/bitstream/2318/1703198/1/SVPCA2018AbstractBooklet.pdf) (PDF). *The Annual Symposium of Vertebrate Palaeontology and Comparative Anatomy*. Archived (https://web.archive.org/web/20220716023044/https://iris.unito.it/bitstream/2318/1703198/1/SVPCA2018AbstractBooklet.pdf) (PDF) from the original on 16 July 2022. Retrieved 18 July 2022.
96. Saitta, Evan T.; Fletcher, Ian; Martin, Peter; Pittman, Michael; Kaye, Thomas G.; True, Lawrence D.; Norell, Mark A.; Abbott, Geoffrey D.; Summons, Roger E.; Penkman, Kirsty; Vinther, Jakob (1 November 2018). "Preservation of feather fibers from the Late Cretaceous dinosaur Shuvuuia deserti raises concern about immunohistochemical analyses on fossils" (https://dspace.mit.edu/bitstream/handle/1721.1/125319/Saitta_et_al_feather_fibres_OG_AAM.pdf?sequence=2&isAllowed=y) (PDF). *Organic Geochemistry*. **125**: 142–151. Bibcode:2018OrGeo.125..142S (https://ui.adsabs.harvard.edu/abs/2018OrGeo.125..142S). doi:10.1016/j.orggeochem.2018.09.008 (https://doi.org/10.1016%2Fj.orggeochem.2018.09.008). hdl:1721.1/125319 (https://hdl.handle.net/1721.1%2F125319). ISSN 0146-6380 (https://search.worldcat.org/issn/0146-6380). S2CID 105753275 (https://api.semanticscholar.org/CorpusID:105753275). Archived (https://web.archive.org/web/20220829004119/https://dspace.mit.edu/bitstream/handle/1721.1/125319/Saitta_et_al_feather_fibres_OG_AAM.pdf?sequence=2&isAllowed=y) (PDF) from the original on 29 August 2022. Retrieved 29 August 2022.
97. blogger, Guest. "Ten Things We Don't Know about Tyrannosaurs" (https://blog.thingswedontknow.com/2014/09/ten-things-we-dont-know-about-tyrannosaurs.html). Archived (https://web.archive.org/web/20200930075917/https://blog.thingswedontknow.com/2014/09/ten-things-we-dont-know-about-tyrannosaurs.html) from the original on 30 September 2020. Retrieved 5 January 2021.
98. Guarino, Ben (27 October 2021). "Tyrannosaurus rex had scaly skin and wasn't covered in feathers, a new study says" (https://www.washingtonpost.com/news/speaking-of-science/wp/2017/06/06/tyrannosaurus-rex-had-scaly-skin-and-wasnt-covered-in-feathers-a-new-study-says/). *Washington Post*. ISSN 0190-8286 (https://search.worldcat.org/issn/0190-8286). Archived (https://web.archive.org/web/20170616111207/https://www.washingtonpost.com/news/speaking-of-science/wp/2017/06/06/tyrannosaurus-rex-had-scaly-skin-and-wasnt-covered-in-feathers-a-new-study-says/) from the original on 16 June 2017. Retrieved 7 August 2023.
99. Stevens, Kent A. (June 2006). "Binocular vision in theropod dinosaurs" (https://web.archive.org/web/20120323155306/http://reocities.com/Athens/Bridge/4602/theropod_binocularvision.pdf) (PDF). *Journal of Vertebrate Paleontology*. **26** (2): 321–330. doi:10.1671/0272-4634(2006)26[321:BVITD]2.0.CO;2 (https://doi.org/10.1671%2F0272-4634%282006%2926%5B321%3ABVITD%5D2.0.CO%3B2). ISSN 0272-4634 (https://search.worldcat.org/issn/0272-4634). S2CID 85694979 (https://api.semanticscholar.org/CorpusID:85694979). Archived from the original (http://reocities.com/Athens/Bridge/4602/theropod_binocularvision.pdf) (PDF) on 23 March 2012.

100. Jaffe, Eric (1 July 2006). "Sight for 'Saur Eyes: *T. rex* vision was among nature's best" (http://www.sciencenews.org/view/generic/id/7500/title/Sight_for_Saur_Eyes_%3Ci%3ET._rex%3Ci%3E_vision_was_among_natures_best). *Science News*. **170** (1): 3–4. doi:10.2307/4017288 (https://doi.org/10.2307%2F4017288). JSTOR 4017288 (https://www.jstor.org/stable/4017288). Archived (https://web.archive.org/web/20120929190336/http://www.sciencenews.org/view/generic/id/7500/title/Sight_for_Saur_Eyes_%3Ci%3ET._rex%3Ci%3E_vision_was_among_natures_best) from the original on 29 September 2012. Retrieved 6 October 2008.
101. Saveliev, Sergei V.; Alifanov, Vladimir R. (2005). "A new study of the brain of the predatory dinosaur *Tarbosaurus bataar* (Theropoda, Tyrannosauridae)". *Paleontological Journal*. **41** (3): 281–289. Bibcode:2007PalJ...41..281S (https://ui.adsabs.harvard.edu/abs/2007PalJ...41..281S). doi:10.1134/S0031030107030070 (https://doi.org/10.1134%2FS0031030107030070). S2CID 53529148 (https://api.semanticscholar.org/CorpusID:53529148).
102. Barker, Chris; Naish, Darren; Katsamenis, Orestis; Dyke, Gareth (16 June 2017). "Complex neuroanatomy in the rostrum of the Isle of Wight theropod Neovenator salerii" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5473926). *Scientific Reports*. **7** (1): 3749. Bibcode:2017NatSR...7.3749B (https://ui.adsabs.harvard.edu/abs/2017NatSR...7.3749B). doi:10.1038/s41598-017-03671-3 (https://doi.org/10.1038%2Fs41598-017-03671-3). PMC 5473926 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5473926). PMID 28623335 (https://pubmed.ncbi.nlm.nih.gov/28623335).
103. Leitch, Duncan B.; Catania, Kenneth C. (1 December 2012). "Structure, innervation and response properties of integumentary sensory organs in crocodilians" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4074209). *Journal of Experimental Biology*. **215** (23): 4217–4230. Bibcode:2012JExpB.215.4217L (https://ui.adsabs.harvard.edu/abs/2012JExpB.215.4217L). doi:10.1242/jeb.076836 (https://doi.org/10.1242%2Fjeb.076836). ISSN 0022-0949 (https://search.worldcat.org/issn/0022-0949). PMC 4074209 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4074209). PMID 23136155 (https://pubmed.ncbi.nlm.nih.gov/23136155).
104. Di-Poï, Nicolas; Milinkovitch, Michel C. (2 July 2013). "Crocodylians evolved scattered multi-sensory micro-organs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3711810). *EvoDevo*. **4** (1): 19. doi:10.1186/2041-9139-4-19 (https://doi.org/10.1186%2F2041-9139-4-19). ISSN 2041-9139 (https://search.worldcat.org/issn/2041-9139). PMC 3711810 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3711810). PMID 23819918 (https://pubmed.ncbi.nlm.nih.gov/23819918).
105. Bakker, Robert T. (1968). "The superiority of dinosaurs". *Discovery*. **3** (2): 11–12.
106. Bakker, Robert T. (1972). "Anatomical and ecological evidence of endothermy in dinosaurs". *Nature*. **238** (5359): 81–85. Bibcode:1972Natur.238...81B (https://ui.adsabs.harvard.edu/abs/1972Natur.238...81B). doi:10.1038/238081a0 (https://doi.org/10.1038%2F238081a0). S2CID 4176132 (https://api.semanticscholar.org/CorpusID:4176132).
107. Bakker, Robert T. (1986). *The Dinosaur Heresies* (https://archive.org/details/dinosaurheresies00robe). New York: Kensington Publishing. ISBN 978-0-688-04287-5. OCLC 13699558 (https://search.worldcat.org/oclc/13699558).
108. Barrick, Reese E.; William J. Showers (July 1994). "Thermophysiology of Tyrannosaurus rex: Evidence from Oxygen Isotopes" (https://www.researchgate.net/publication/6089047). *Science*. **265** (5169). New York City: 222–224. Bibcode:1994Sci...265..222B (https://ui.adsabs.harvard.edu/abs/1994Sci...265..222B). doi:10.1126/science.265.5169.222 (https://doi.org/10.1126%2Fscience.265.5169.222). PMID 17750663 (https://pubmed.ncbi.nlm.nih.gov/17750663). S2CID 39392327 (https://api.semanticscholar.org/CorpusID:39392327). Retrieved 6 September 2026 – via ResearchGate.
109. Barrick, Reese E.; William J. Showers (October 1999). "Thermophysiology and biology of *Giganotosaurus*: comparison with *Tyrannosaurus*" (http://palaeo-electronica.org/1999_2/gigan/issue2_99.htm). *Palaeontologia Electronica*. **2** (2). Archived (https://web.archive.org/web/20110517220816/http://palaeo-electronica.org/1999_2/gigan/issue2_99.htm) from the original on 17 May 2011. Retrieved 12 October 2008.
110. Paladino, Frank V.; Spotila, James R.; Dodson, Peter (1999). "A blueprint for giants: modeling the physiology of large dinosaurs". In James O. Farlow; M. K. Brett-Surman (eds.). *The Complete Dinosaur*. Bloomington: Indiana University Press. pp. 491–504. ISBN 978-0-253-21313-6.
111. Chinsamy, Anusuya; Willem J. Hillenius (2004). "Physiology of nonavian dinosaurs". In David B. Weishampel; Peter Dodson; Halszka Osmólska (eds.). *The dinosauria*. Berkeley: University of California Press. pp. 643–659. ISBN 978-0-520-24209-8.
112. Seymour, Roger S. (5 July 2013). "Maximal Aerobic and Anaerobic Power Generation in Large Crocodiles versus Mammals: Implications for Dinosaur Gigantothermy" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3702618). *PLOS ONE*. **8** (7) e69361. Bibcode:2013PLoSO...869361S (https://ui.adsabs.harvard.edu/abs/2013PLoSO...869361S). doi:10.1371/journal.pone.0069361 (https://doi.org/10.1371%2Fjournal.pone.0069361). ISSN 1932-6203 (https://search.worldcat.org/issn/1932-6203). PMC 3702618 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3702618). PMID 23861968 (https://pubmed.ncbi.nlm.nih.gov/23861968).
113. Farlow, James O.; Pianka, Eric R. (2002). "Body size overlap, habitat partitioning and living space requirements of terrestrial vertebrate predators: implications for the paleoecology of large theropod dinosaurs" (http://doc.rero.ch/record/15105/files/PAL_E2372.pdf) (PDF). *Historical Biology*. **16** (1): 21–40. Bibcode:2002HBio...16...21F (https://ui.adsabs.harvard.edu/abs/2002HBio...16...21F). doi:10.1080/0891296031000154687 (https://doi.org/10.1080%2F0891296031000154687). S2CID 18114585 (https://api.semanticscholar.org/CorpusID:18114585).
114. Varricchio, David J. (2001). "Gut contents from a Cretaceous tyrannosaurid: implications for theropod dinosaur digestive tracts" (https://www.researchgate.net/publication/305884983). *Journal of Paleontology*. **75** (2): 401–406. doi:10.1666/0022-3360(2001)075<0401:GCFACT>2.0.CO;2 (https://doi.org/10.1666%2F0022-3360%282001%29075%3C0401%3AGCFACT%3E2.0.CO%3B2). ISSN 0022-3360 (https://search.worldcat.org/issn/0022-3360). S2CID 86113170 (https://api.semanticscholar.org/CorpusID:86113170). Retrieved 6 September 2026 – via ResearchGate.
115. Guinness World Records Ltd. (2003). *2003 Guinness World Records*. pg 90.
116. Eberth, David A.; McCrea, Richard T. (2001). "Were large theropods gregarious?". *Journal of Vertebrate Paleontology*. **21** (Supplement to Number 3): 46A. doi:10.1080/02724634.2001.10010852 (https://doi.org/10.1080%2F02724634.2001.10010852). S2CID 220414868 (https://api.semanticscholar.org/CorpusID:220414868).
117. Roach, Brian T.; Brinkman, Daniel L. (2007). "A reevaluation of cooperative pack hunting and gregariousness in *Deinonychus antirrhopus* and other nonavian theropod dinosaurs". *Bulletin of the Peabody Museum of Natural History*. **48** (1): 103–138. doi:10.3374/0079-032X(2007)48[103:AROCPH]2.0.CO;2 (https://doi.org/10.3374%2F0079-032X%282007%2948%5B103%3AAROCPH%5D2.0.CO%3B2). S2CID 84175628 (https://api.semanticscholar.org/CorpusID:84175628).
118. Currie, Philip John (1998). "Possible evidence of gregarious behavior in tyrannosaurids" (http://www.mnhn.ul.pt/geologia/gaia/21.pdf) (PDF). *Gaia*. **15**: 271–277. Archived (https://web.archive.org/web/20090326062833/http://www.mnhn.ul.pt/geologia/gaia/21.pdf) (PDF) from the original on 26 March 2009. Retrieved 13 September 2008.(not printed until 2000)
119. McCrea, Richard T.; Buckley, Lisa G.; Farlow, James O.; Lockley, Martin G.; Currie, Philip John; Matthews, Neffra A.; Pemberton, S. George (2014). "A 'Terror of Tyrannosaurs': The First Trackways of Tyrannosaurids and Evidence of Gregariousness and Pathology in Tyrannosauridae" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4108409). *PLOS ONE*. **9** (7) e103613. Bibcode:2014PLoSO...9j3613M (https://ui.adsabs.harvard.edu/abs/2014PLoSO...9j3613M). doi:10.1371/journal.pone.0103613 (https://doi.org/10.1371%2Fjournal.pone.0103613). PMC 4108409 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4108409). PMID 25054328 (https://pubmed.ncbi.nlm.nih.gov/25054328).
120. "Tyrannosaurus Tracks Show Dinosaur's Walk" (http://www.popularmechanics.com/science/environment/tyrannosaurus-trackways-reveal-a-dinosaurs-walk-17016027). 23 July 2014. Archived (https://web.archive.org/web/20140930213639/http://www.popularmechanics.com/science/environment/tyrannosaurus-trackways-reveal-a-dinosaurs-walk-17016027) from the original on 30 September 2014. Retrieved 16 January 2015.
121. "Tracks Hint at the Social Life of Tyrant Dinosaurs" (https://web.archive.org/web/20150108013825/http://phenomena.nationalgeographic.com/2014/07/23/tracks-hint-at-the-social-life-of-tyrant-dinosaurs/). 23 July 2014. Archived from the original (http://phenomena.nationalgeographic.com/2014/07/23/tracks-hint-at-the-social-life-of-tyrant-dinosaurs/) on 8 January 2015. Retrieved 16 January 2015.

122. Titus, Alan L.; Knoll, Katja; Sertich, Joseph J. W.; Yamamura, Daigo; Suarez, Celina A.; Glasspool, Ian J.; Ginouves, Jonathan E.; Lukacic, Abigail K.; Roberts, Eric M. (19 April 2021). "Geology and taphonomy of a unique tyrannosaurid bonebed from the upper Campanian Kaiparowits Formation of southern Utah: implications for tyrannosaurid gregariousness" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8061582). *PeerJ*. **9** e11013. Bibcode:2021PeerJ...911013T (https://ui.adsabs.harvard.edu/abs/2021PeerJ...911013T). doi:10.7717/peerj.11013 (https://doi.org/10.7717%2Fpeerj.11013). PMC 8061582 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8061582). PMID 33976955 (https://pubmed.ncbi.nlm.nih.gov/33976955).
123. Eilperin, Juliet (19 April 2021). "Tyrannosaurs likely hunted in packs rather than heading out solo, scientists find" (https://www.washingtonpost.com/climate-environment/2021/04/19/grand-staircase-tyrannosaur/). *The Washington Post*. Retrieved 27 September 2021.
124. "Tyrannosaurus Rex 'hunted in packs' " (https://www.telegraph.co.uk/news/science/dinosaurs/8589113/Tyrannosaurus-Rex-hunted-in-packs.html). 22 June 2011. Archived (https://web.archive.org/web/20210927165533/https://www.telegraph.co.uk/news/science/dinosaurs/8589113/Tyrannosaurus-Rex-hunted-in-packs.html) from the original on 27 September 2021. Retrieved 27 September 2021.
125. Jacobsen, A.R. 2001. Tooth-marked small theropod bone: An extremely rare trace. p. 58-63. In: Mesozoic Vertebrate Life. Ed.s Tanke, D. H., Carpenter, K., Skrepnick, M. W. Indiana University Press.
126. Abler, W.L. 2001. A kerf-and-drill model of tyrannosaur tooth serrations. p. 84-89. In: *Mesozoic Vertebrate Life*. Ed.s Tanke, D. H., Carpenter, K., Skrepnick, M. W. Indiana University Press.
127. Lambe, L. B. (1917). "The Cretaceous theropodous dinosaur *Gorgosaurus*" (https://www.biodiversitylibrary.org/item/252624). *Memoirs of the Geological Survey of Canada*. **100**: 1–84. doi:10.4095/101672 (https://doi.org/10.4095%2F101672). Archived (https://web.archive.org/web/20210909121735/https://www.biodiversitylibrary.org/item/252624) from the original on 9 September 2021. Retrieved 23 September 2021.
128. Farlow, J. O.; Holtz, T. R. (2002). Kowalewski, M.; Kelley, P. H. (eds.). "The Fossil Record of Predation" (https://web.archive.org/web/20081031093048/http://www.yale.edu/ypmip/predation/Chapter_09.pdf) (PDF). The Paleontological Society Papers. pp. 251–266. Archived from the original (http://www.yale.edu/ypmip/predation/Chapter_09.pdf) (PDF) on 31 October 2008.
129. Dorey, M. (1997). Tyrannosaurus. *Dinosaur Cards*. Orbis Publishing Ltd. D36045907.
130. Horner, J.R. (1994). "Steak knives, beady eyes, and tiny little arms (a portrait of *Tyrannosaurus* as a scavenger)". *The Paleontological Society Special Publication*. **7**: 157–164. doi:10.1017/S2475262200009497 (https://doi.org/10.1017%2FS2475262200009497).
131. Amos, J. (31 July 2003). "T. rex goes on trial" (https://news.bbc.co.uk/2/hi/science/nature/3112527.stm). BBC. Archived (https://web.archive.org/web/20120313130020/http://news.bbc.co.uk/1/hi/sci/tech/3112527.stm) from the original on 13 March 2012. Retrieved 12 October 2008.
132. Walters, M., Paker, J. (1995). Dictionary of Prehistoric Life. Claremont Books. ISBN 1-85471-648-4.
133. Erickson G. M.; Olson K. H. (1996). "Bite marks attributable to *Tyrannosaurus rex*: preliminary description and implications" (https://zenodo.org/record/3732430). *Journal of Vertebrate Paleontology*. **16** (1): 175–178. Bibcode:1996JVPal..16..175E (https://ui.adsabs.harvard.edu/abs/1996JVPal..16..175E). doi:10.1080/02724634.1996.10011297 (https://doi.org/10.1080%2F02724634.1996.10011297). Archived (https://web.archive.org/web/20211019153802/https://zenodo.org/record/3732430) from the original on 19 October 2021. Retrieved 12 September 2020.
134. Carpenter, K. (1998). "Evidence of predatory behavior by theropod dinosaurs" (https://web.archive.org/web/20071117132451/http://vertpaleo.org/publications/jvp/15-576-591.cfm). *Gaia*. **15**: 135–144. Archived from the original (http://vertpaleo.org/publications/jvp/15-576-591.cfm) on 17 November 2007. Retrieved 5 December 2007.
135. Fowler, Denver W.; Robert M. Sullivan (2006). "A ceratopsid pelvis with toothmarks from the Upper Cretaceous Kirtland Formation, New Mexico: evidence of late Campanian tyrannosaurid feeding behavior". *New Mexico Museum of Natural History and Science Bulletin*. **35**: 127–130.
136. Therrien, F.; Zelenitsky, D. K.; Tanaka, K.; Voris, J. T.; Erickson, Gregory M.; Currie, Philip John; DeBuhr, C. L.; Kobayashi, Y. (2023). "Exceptionally preserved stomach contents of a young tyrannosaurid reveal an ontogenetic dietary shift in an iconic extinct predator" (https://www.researchgate.net/publication/376355974). *Science Advances*. **9** (49) eadi0505. Bibcode:2023SciA....9I.505T (https://ui.adsabs.harvard.edu/abs/2023SciA....9I.505T). doi:10.1126/sciadv.adi0505 (https://doi.org/10.1126%2Fsciadv.adi0505). PMC 10846869 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10846869). PMID 38064561 (https://pubmed.ncbi.nlm.nih.gov/38064561). Retrieved 6 September 2026 – via ResearchGate.
137. Nielsen, Josephine; Fowler, Denver W.; Wyenberg-Henzler, Taia; Jacobsen, Aase Roland; Pearce, Christof (1 January 2026). "Investigating size-asymmetric feeding among tyrannosaurids using tooth marks on a metatarsal from the Judith River Formation, Montana, USA" (https://doi.org/10.1016%2Fj.eve.2026.100107). *Evolving Earth*. **4** 100107. doi:10.1016/j.eve.2026.100107 (https://doi.org/10.1016%2Fj.eve.2026.100107). ISSN 2950-1172 (https://search.worldcat.org/issn/2950-1172).
138. McElvogue, Bryce M.; Lehman, Thomas M. (21 September 2026). "Tyrannosaur feeding behavior evinced from hadrosaurian dinosaur remains in the Aguja Formation (Upper Cretaceous), Big Bend National Park, Texas, USA" (https://palaeo-electronica.org/content/2026/5962-tyrannosaur-feeding-behavior). *Palaeontologia Electronica*. **29** (3). Society of Vertebrate Paleontology: 1–17. doi:10.26879/1657 (https://doi.org/10.26879%2F1657). ISSN 1094-8074 (https://search.worldcat.org/issn/1094-8074).
139. Longrich, Nicholas R.; Horner, Jack Robert; Erickson, Gregory M.; Currie, Philip John (2010). "Cannibalism in *Tyrannosaurus rex*" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2955550). *PLOS ONE*. **5** (10) e13419. Bibcode:2010PLoSO...513419L (https://ui.adsabs.harvard.edu/abs/2010PLoSO...513419L). doi:10.1371/journal.pone.0013419 (https://doi.org/10.1371%2Fjournal.pone.0013419). PMC 2955550 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2955550). PMID 20976177 (https://pubmed.ncbi.nlm.nih.gov/20976177).
140. "NEW EVIDENCE FOR CANNIBALISM IN TYRANNOSAURID DINOSAURS FROM THE UPPER CRETACEOUS (CAMPANIAN/MAASTRICHTIAN) SAN JUAN BASIN OF NEW MEXICO" (https://www.researchgate.net/publication/348002335). *ResearchGate*. January 2021. pp. 39–56. Archived (https://web.archive.org/web/20211110091047/https://www.researchgate.net/publication/348002335_NEW_EVIDENCE_FOR_CANNIBALISM_IN_TYRANNOSAURID_DINOSAURS_FROM_THE_UPPER_CRETACEOUS_CAMPANIANMAASTRICHTIAN_SAN_JUAN_BASIN_OF_NEW_MEXICO) from the original on 10 November 2021. Retrieved 6 January 2021.
141. Dodson, Peter; Britt, Brooks; Carpenter, Kenneth; Forster, Catherine A.; Gillette, David D.; Norell, Mark A.; Olshevsky, George; Parrish, J. Michael & Weishampel, David B. (1993). "Abelisaurus". *Age of Dinosaurs*. Publications International, LTD. p. 105. ISBN 978-0-7853-0443-2.
142. Fiorillo, Anthony Ricardo; Gangloff, Roland A. (2000). "Theropod teeth from the Prince Creek Formation (Cretaceous) of northern Alaska, with speculations on arctic dinosaur paleoecology" (https://www.tandfonline.com/doi/abs/10.1671/0272-4634%282000%2900%5B0675%3ATTFTPC%5D2.0.CO%3B2). *Journal of Vertebrate Paleontology*. **20** (4): 675–682. doi:10.1671/0272-4634(2000)020[0675:TTFTPC]2.0.CO;2 (https://doi.org/10.1671%2F0272-4634%282000%29020%5B0675%3ATTFTPC%5D2.0.CO%3B2). ISSN 0272-4634 (https://search.worldcat.org/issn/0272-4634). S2CID 130766946 (https://api.semanticscholar.org/CorpusID:130766946). Retrieved 6 September 2026.
143. "These 81-million-year-old teeth are the first bit of evidence of Japan's largest ever dinosaur | Irish Examiner" (http://www.irishexaminer.com/examviral/science-world/these-81-million-year-old-teeth-are-the-first-bit-of-evidence-of-japans-largest-ever-dinosaur-342857.html). 16 July 2015. Archived (https://web.archive.org/web/20160412104057/http://www.irishexaminer.com/examviral/science-world/these-81-million-year-old-teeth-are-the-first-bit-of-evidence-of-japans-largest-ever-dinosaur-342857.html) from the original on 12 April 2016. Retrieved 10 March 2016.
144. Zanno, Lindsay E.; Napoli, James G. (December 2025). "Nanotyrannus and Tyrannosaurus coexisted at the close of the Cretaceous" (https://www.nature.com/articles/s41586-025-09801-6). *Nature*. **648** (8093). Nature Publishing Group: 357–367. Bibcode:2025Natur.648..357Z (https://ui.adsabs.harvard.edu/abs/2025Natur.648..357Z). doi:10.1038/s41586-025-09801-6 (https://doi.org/10.1038%2Fs41586-025-09801-6). ISSN 1476-4687 (https://search.worldcat.org/issn/1476-4687). PMID 41167514 (https://pubmed.ncbi.nlm.nih.gov/41167514).

145. Laura Geggel (29 February 2016). "T. Rex Was Likely an Invasive Species" (https://www.livescience.com/53877-t-rex-was-invasive-species.html). *livescience.com*. Archived (https://web.archive.org/web/20190714170020/https://www.livescience.com/53877-t-rex-was-invasive-species.html) from the original on 14 July 2019. Retrieved 5 January 2021.

- Hieronymus, Tobin; Witmer, Lawrence; Tanke, Darren; Currie, Philip (26 August 2009). "The Facial Integument of Centrosaurine Ceratopsids: Morphological and Histological Correlates of Novel Skin Structures" (https://doi.org/10.1002%2Far.20985). *The Anatomical Record*. **292** (9): 1370–1396. doi:10.1002/ar.20985 (https://doi.org/10.1002%2Far.20985). PMID 19711467 (https://pubmed.ncbi.nlm.nih.gov/19711467).

External links

- Tyrannosauridae (https://web.archive.org/web/20130929074932/http://archosaur.us/theropoddatabase/Tyrannosauroidea.html#Tyrannosauridae) specimen list and discussion at The Theropod Database

Retrieved from "https://en.wikipedia.org/w/index.php?title=Tyrannosauridae&oldid=1379829613"

WIKIPEDIA 25
25 years of the free encyclopedia

Tyrannosaurus

Tyrannosaurus (/tɪˌrænəˈsɔːrəs, taɪ-/)[a] is a genus of large theropod dinosaur. The type species ***Tyrannosaurus rex*** (*rex* meaning 'king' in Latin), often shortened to ***T. rex*** or colloquially **T-Rex**, is one of the best represented theropods. It lived throughout what is now western North America, on what was then an island continent known as Laramidia. *Tyrannosaurus* had a much wider range than other tyrannosaurids. Fossils are found in geological formations which have been dated to the late Maastrichtian age of the late Cretaceous period, 69 to 66 million years ago, with isolated specimens possibly indicating an earlier origin in the middle Campanian age. It was one of the last known members of the tyrannosaurids and among the last non-avian dinosaurs to exist before the Cretaceous–Paleogene extinction event.

Like other tyrannosaurids, *Tyrannosaurus* was a bipedal carnivore with a massive skull balanced by a long, heavy tail. Relative to its large and powerful hind limbs, the forelimbs of *Tyrannosaurus* were short but unusually powerful for their size, and they had two clawed digits. The most complete specimen measures 12.3–12.4 m (40–41 ft) in length, but according to most modern estimates, *Tyrannosaurus* could have exceeded sizes of 13 m (43 ft) in length, 3.7–4 m (12–13 ft) in hip height, and 8.8 t (8.7 long tons; 9.7 short tons) in mass. Although some other theropods might have rivaled or exceeded *Tyrannosaurus* in size, it is still among the largest known land predators, with its estimated bite force being the largest among all terrestrial animals. By far the largest carnivore in its environment, *Tyrannosaurus rex* was most likely an apex predator, preying upon hadrosaurs, juvenile armored herbivores like ceratopsians and ankylosaurs, and possibly sauropods. The question of whether *Tyrannosaurus* was an apex predator or a pure scavenger was among the longest debates in paleontology. Most paleontologists today accept that

Tyrannosaurus

Temporal range: Late Cretaceous

(late Maastrichtian),

Possible middle Campanian record

Reconstruction of the *T. rex* type specimen at the Carnegie Museum of Natural History

Scientific classification

Kingdom:	Animalia
Phylum:	Chordata
Class:	Reptilia
Clade:	Dinosauria
Clade:	Saurischia
Clade:	Theropoda
Superfamily:	†Tyrannosauroidea
Family:	†Tyrannosauridae
Tribe:	†Tyrannosaurini
Genus:	†*Tyrannosaurus* Osborn, 1905

Type species

†***Tyrannosaurus rex***
Osborn, 1905

Other species

Tyrannosaurus was a predator, and the idea of *Tyrannosaurus* being primarily a scavenger has been discredited.[1]

Some specimens of *Tyrannosaurus rex* are nearly complete skeletons. Soft tissue and proteins have been reported in at least one of these specimens. The abundance of fossil material has allowed significant research into many aspects of the animal's biology, including its life history and biomechanics. The feeding habits, physiology, and potential speed of *Tyrannosaurus rex* are a few subjects of debate. Its taxonomy is also controversial. The Asian *Tarbosaurus bataar* is very closely related to *Tyrannosaurus* and has sometimes been seen as a species of this genus. Several North American tyrannosaurids have been synonymized with *Tyrannosaurus*, while some *Tyrannosaurus* specimens have been proposed as distinct species. The validity of these species, such as the more recently discovered ***T. mcraeensis***, is contentious.

Tyrannosaurus has been one of the best-known dinosaurs since the early 20th century. Its fossils have been a popular attraction in museums and it has appeared in media like *Jurassic Park*.

History of research

Earliest finds

Type specimen (AMNH 3982) of *Manospondylus gigas*

A tooth from what is now documented as a *Tyrannosaurus rex* was found in July 1874 upon South Table Mountain (Colorado) by Jarvis Hall student Peter

- †***T. mcraeensis***? Dalman et al., 2024
- See text

Synonyms

Genus synonymy

- *Dinotyrannus*
  Olshevsky, Ford & Yamamoto, 1995
- *Dynamosaurus*
  Osborn, 1905
- *Manospondylus*
  Cope, 1892
- *Stygivenator*?
  Olshevsky, Ford & Yamamoto, 1995

Species synonymy

- *Aublysodon amplus*?
  Marsh, 1892
- *Deinodon amplus*?
  (Marsh, 1892) Hay, 1902
- *Manospondylus amplus*?
  (Marsh, 1892) Olshevsky, 1978
- *Stygivenator amplus*?
  (Marsh, 1892) Olshevsky, Ford & Yamamoto, 1995
- *Tyrannosaurus amplus*?
  (Marsh, 1892) Hay, 1930
- *Aublysodon cristatus*?
  Marsh, 1892
- *Deinodon cristatus*?
  (Marsh, 1892) Hay, 1902
- *Stygivenator cristatus*?
  (Marsh, 1892) Olshevsky, Ford & Yamamoto, 1995
- *Ornithomimus grandis*
  Marsh, 1890
- *Manospondylus gigas*
  Cope, 1892
- *Dynamosaurus imperiosus*
  Osborn, 1905

T. Dotson under the auspices of Arthur Lakes near Golden, Colorado, in sediments belonging to the Denver Formation.[2][3] In the early 1890s, John Bell Hatcher collected postcranial elements in eastern Wyoming. The fossils were believed to be from the large species *Ornithomimus grandis* (now *Deinodon*) but are now considered *T. rex* remains.[4]

In 1892, Edward Drinker Cope found two vertebral fragments of a large dinosaur. Cope believed the fragments belonged to an "agathaumid" (ceratopsid) dinosaur, and named them *Manospondylus gigas*, meaning "giant porous vertebra", in reference to the numerous openings for blood vessels he found in the bone.[4] The *M. gigas* remains were, in 1907, identified by Hatcher as those of a theropod rather than a ceratopsid.[5]

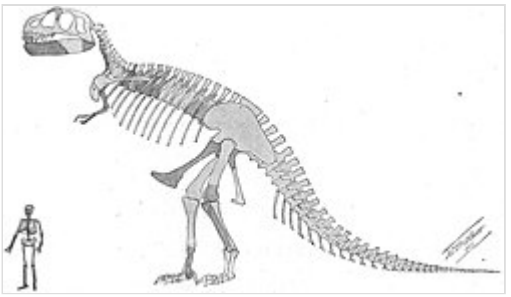
Henry Fairfield Osborn recognized the similarity between *Manospondylus gigas* and *T. rex* as early as 1917, by which time the second vertebra had been lost. Owing to the fragmentary nature of the *Manospondylus* vertebrae, Osborn did not synonymize the two genera, instead considering the older genus indeterminate.[6] In June 2000, the Black Hills Institute (BHI) found around 10% of a *Tyrannosaurus* skeleton (BHI 6248) at a site that might have been the original *M. gigas* locality.[7] While Larson initially considered this specimen as the same animal as the *M. gigas* holotype and suggested that it would take priority over *T. rex*, other researchers including Christopher Brochu remained skeptical that the name *T. rex* would be abandoned.[8] In 2003, Brochu further argued that although both taxa probably represent the same species, it is impossible to demonstrate that the specimen discovered by the Black Hills Institute represents the same animal as *M. gigas* due to the poor preservation of the *M. gigas* holotype; additionally, following the 2000 International Code of Zoological Nomenclature amendment, a name in popular usage like *T. rex* has priority over a name with technical priority but not used as a valid name in formal literature after the year 1899 for over fifty years like *M. gigas*.[9][10]

- *Tyrannosaurus imperiosus*
  (Osborn, 1905) Walker, 1964
- *Tyrannosaurus* "vannus"
  Lawson, 1972 (nomen nudum)
- *Albertosaurus* "megagracilis"
  Paul, 1988a (nomen nudum)
- *Tyrannosaurus* "gigantus"
  Harlan, 1990 (nomen nudum)
- *Dinotyrannus megagracilis*
  Olshevsky, Ford & Yamamoto, 1995
- *Tyrannosaurus* "stanwinstonorum"
  Pickering, 1995 (nomen nudum)
- *Tyrannosaurus* "stanwinstonorus"
  Pickering, 1995 (nomen nudum)
- *Aublysodon molnaris*?
  Paul, 1988a
- *Aublysodon molnari*?
  Paul, 1988a emend Paul, 1989
- *Stygivenator molnari*?
  (Paul, 1988a emend Paul, 1989)
  Olshevsky, Ford & Yamamoto, 1995

Skeleton discovery and naming

Barnum Brown, assistant curator of the American Museum of Natural History, found the first partial skeleton of *T. rex* in eastern Wyoming in 1900. Brown found another partial skeleton in the Hell Creek Formation in Montana in 1902, comprising approximately 34 fossilized bones.[11] Writing at the time Brown said "Quarry No. 1 contains the femur, pubes, humerus, three vertebrae and two undetermined bones of a large Carnivorous Dinosaur not described by Marsh. ... I have never seen anything like it from the Cretaceous."[12] Henry Fairfield Osborn, president of the American Museum of Natural History, named the second skeleton *T. rex* in 1905. The generic name is derived from the Greek words τύραννος

(*tyrannos*, meaning "tyrant") and σαῦρος (*sauros*, meaning "lizard"). Osborn used the Latin word *rex*, meaning "king", for the specific name. The full binomial translates to "tyrant lizard the king" or "King Tyrant Lizard", emphasizing the animal's size and presumed dominance over other species of the time.[11]



Outdated skeletal restoration by William D. Matthew from 1905, published alongside Osborn's description paper

Dynamosaurus imperiosus holotype (NHMUK PV R8020, previously AMNH 5886) on display at the Natural History Museum, London

Osborn named the other specimen *Dynamosaurus imperiosus* in a 1905 paper.[11] In 1906, Osborn recognized that the two skeletons were from the same species and selected *Tyrannosaurus* as the preferred name.[13] In 1941, the *T. rex* type specimen was sold to the Carnegie Museum of Natural History in Pittsburgh, Pennsylvania, for $7,000 (equivalent to $153,224 in 2025).[12] The original *Dynamosaurus* material now resides in the collections of the Natural History Museum, London.[14]

From the 1910s through the end of the 1950s, Barnum's discoveries remained the only specimens of *Tyrannosaurus*, as the Great Depression and wars kept many paleontologists out of the field.[7]

Continued study

Beginning in the 1960s, there was renewed interest in *Tyrannosaurus*, resulting in the recovery of 42 skeletons (5–80% complete by bone count) from western North America.[7] In 1967, Dr. William MacMannis located and recovered the skeleton named "MOR 008", which is 15% complete by bone count and has a reconstructed skull displayed at the Museum of the Rockies. The 1990s saw numerous discoveries, with nearly twice as many finds as in all previous years, including two of the most complete skeletons found to date: Sue and Stan.[7]

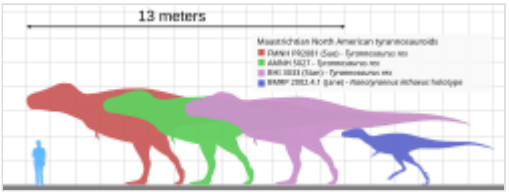
Specimen "Sue", Field Museum of Natural History, Chicago (with some bones reconstructed)

Sue Hendrickson discovered the most complete (approximately 85%) and largest *Tyrannosaurus* skeleton in the Hell Creek Formation on August 12, 1990. The specimen Sue, named after the discoverer, was the object of a legal battle over its ownership. In 1997, the litigation was settled in favor of Maurice Williams, the original land owner. The fossil collection was purchased by the Field Museum of Natural History at auction for $7.6 million, making it the most expensive dinosaur skeleton until the sale of Stan for $31.8 million in 2020.[15] The mounted skeleton opened to the public on May 17, 2000. A study of this specimen's fossilized bones showed that Sue reached full size at age 19 and died at the age of 28, the longest estimated life of any tyrannosaur known.[16]

Replica of "Scotty", the largest known specimen, exhibited in Japan

Another *Tyrannosaurus*, nicknamed Stan (NHMAD 2020.00001, formerly BHI 3033[17]), in honor of amateur paleontologist Stan Sacrison, was recovered from the Hell Creek Formation in 1992. Stan is the second most complete skeleton found, with 199 bones recovered representing 70% of the total.[18] This tyrannosaur also had many bone pathologies, including broken and healed ribs, a broken and healed neck, and a substantial hole in the back of its head, about the size of a *Tyrannosaurus* tooth.[19]

In 1998, 20-year-old Bucky Derflinger noticed a *T. rex* toe exposed above ground, making him the youngest person to discover a *Tyrannosaurus*. The specimen, dubbed Bucky in honor of its discoverer, was a young adult, 3.0 metres (10 ft) tall and 11 metres (35 ft) long. Bucky is the first *Tyrannosaurus* to be found that preserved a furcula (wishbone). Bucky is permanently displayed at The Children's Museum of Indianapolis.[20]



The specimens "Sue", AMNH 5027, "Stan", and "Jane", to scale with a human.

In 2000, crews organized by Jack Horner discovered five *Tyrannosaurus* skeletons near the Fort Peck Reservoir.[21] In 2001, a 50% complete skeleton of a juvenile *Tyrannosaurus* was discovered in the Hell Creek Formation by a crew from the Burpee Museum of Natural History. Dubbed Jane (BMRP 2002.4.1), the find was thought to be the first known skeleton of a pygmy tyrannosaurid, *Nanotyrannus*, but other researchers considered it as the most complete juvenile *T. rex*;[22] currently exhibited at the Burpee Museum of Natural History, Jane is now re-described as the holotype of *N. lethaeus*.[23] In 2002, a skeleton nicknamed "Wyrex", discovered by amateur collectors Dan Wells and Don Wyrick, had 114 bones and was 38% complete. The dig was concluded in 2004 by the Black Hills Institute with the first live online *Tyrannosaurus* excavation.[7]

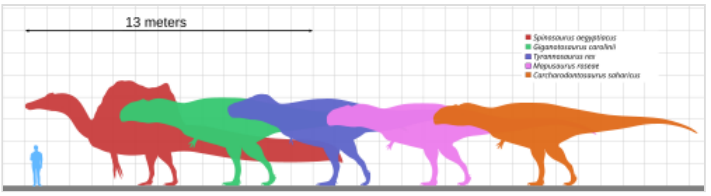
In 2006, Montana State University revealed that it possessed the largest *Tyrannosaurus* skull yet discovered (from a specimen named MOR 008), measuring 5 feet (152 cm) long.[24] Subsequent comparisons indicated that the longest head was 136.5 centimetres (53.7 in) (from specimen LACM 23844) and the widest head was 90.2 centimetres (35.5 in) (from Sue).[25]

Description

Size

T. rex was one of the largest land carnivores of all time. One of its largest and the most complete specimens, Sue (FMNH PR2081), measured 12.3–12.8 m (40–42 ft) long,[26][27][28] was 3.66–3.96 m (12.0–13.0 ft) tall at the hips,[26][29][30] and according to the most recent studies, using a variety of techniques, maximum body masses have been estimated approximately 8.4–9.22 t (8.27–9.07 long tons; 9.26–10.16 short tons).[31][32][33] A specimen nicknamed Scotty (RSM P2523.8), located at the Royal

Saskatchewan Museum, is reported to measure 13 m (43 ft) in length. Using a mass estimation technique that extrapolates from the circumference of the femur, Scotty was estimated as the largest known specimen at 8.87 t (8.73 long tons; 9.78 short tons) in body mass.[31][34]



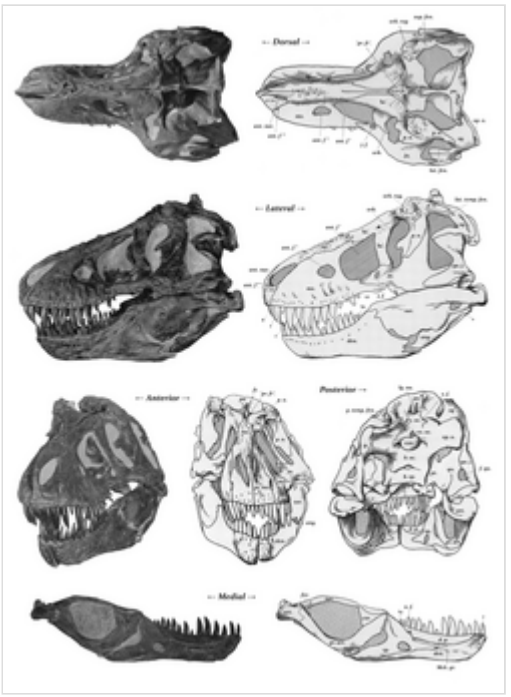
Size (in blue) compared to select giant theropods and a human

Not every adult *Tyrannosaurus* specimen recovered is as big. Historically average adult mass estimates have varied widely over the years, from as low as 4.5 t (4.4 long tons; 5.0 short tons),[35][36] to more than 7.2 t (7.1 long tons; 7.9 short tons),[37] with most modern estimates ranging between 5.4 and 8.0 t (5.3 and 7.9 long tons; 6.0 and 8.8 short tons).[27][38][39][40][41]

A 2024 study found that there was little evidence of size-based sexual dimorphism in *T. rex*.[42]

Skull

The largest known *T. rex* skulls measure up to 1.52 m (5 ft 0 in) in length.[24][26] Large fenestrae (openings) in the skull reduced weight, as in all carnivorous theropods. In other respects *Tyrannosaurus*'s skull was significantly different from those of large non-tyrannosaurid theropods. It was extremely wide at the rear but had a narrow snout, allowing unusually good binocular vision.[43][44] The skull bones were massive and the nasals and some other bones were fused, preventing movement between them; but many were pneumatized (contained a "honeycomb" of tiny air spaces) and thus lighter. These and other skull-strengthening features are part of the tyrannosaurid trend towards an increasingly powerful bite, which easily surpassed that of all non-tyrannosaurids.[45][46][47] The tip of the upper jaw was U-shaped (most non-tyrannosauroid carnivores had V-shaped upper jaws), which increased the amount of tissue and bone a tyrannosaur could rip out with one bite, although it also increased the stresses on the front teeth.[48]


Skull of specimen AMNH 5027 with labelled diagrams in dorsal, lateral, anterior, posterior, and medial (lower jaw) views

The teeth of *T. rex* displayed marked heterodonty (differences in shape).[9][49] The premaxillary teeth, four per side at the front of the upper jaw, were closely packed, *D*-shaped in cross-section, had reinforcing ridges on the rear surface, were incisiform (their tips were chisel-like blades) and curved backwards. The *D*-shaped cross-section, reinforcing ridges and backwards curve reduced the risk that the teeth would snap when *Tyrannosaurus* bit and pulled. The remaining teeth were robust, like "lethal bananas" rather than daggers, more widely spaced and also had reinforcing ridges.[50] Those in the upper jaw, twelve per side in mature individuals,[9] were larger than their counterparts of the lower jaw, except at the rear. The largest found so far is estimated to have been 30.5 cm (12.0 in) long including the root when the animal was alive, making it the largest tooth of any carnivorous dinosaur yet

found.[51] The lower jaw was robust. Its front dentary bone bore thirteen teeth. Behind the tooth row, the lower jaw became notably taller.[9] The upper and lower jaws of *Tyrannosaurus*, like those of many dinosaurs, possessed numerous foramina, or small holes in the bone. Various functions have been proposed for these foramina, such as a crocodile-like sensory system[52] or evidence of extra-oral structures such as scales or potentially lips,[53][54][55] with subsequent research on theropod tooth wear patterns supporting such a proposition.[56]

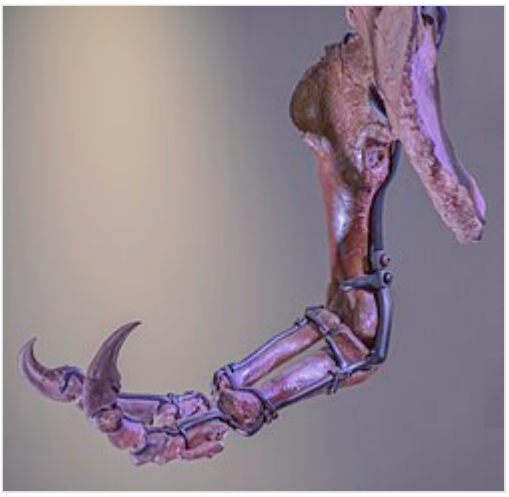
Skeleton

Life restoration showing scaly skin with sparse feathering, and lipped jaws

Skeletal reconstruction of specimen "Sue"

The vertebral column of *Tyrannosaurus* consisted of ten neck vertebrae, thirteen back vertebrae and five sacral vertebrae. The number of tail vertebrae is unknown and could well have varied between individuals but probably numbered at least forty. Sue was mounted with forty-seven of such caudal vertebrae.[9] The neck of *T. rex* formed a natural S-shaped curve like that of other theropods. Compared to these, it was exceptionally short, deep and muscular to support the massive head. The second vertebra, the axis, was especially short. The remaining neck vertebrae were weakly opisthocoelous, i.e. with a convex front of the vertebral body and a concave rear. The vertebral bodies had single pleurocoels, pneumatic depressions created by air sacs, on their sides.[9] The vertebral bodies of the torso were robust but with a narrow waist. Their undersides were keeled. The front sides were concave with a deep vertical trough. They had large pleurocoels. Their neural spines had very rough front and rear sides for the attachment of strong tendons. The sacral vertebrae were fused to each other, both in their vertebral bodies and neural spines. They were pneumatized. They were connected to the pelvis by transverse processes and sacral ribs. The tail was heavy and moderately long, in order to balance the massive head and torso and to provide space for massive locomotor muscles that attached to the thighbones. The thirteenth tail vertebra formed the transition point between the deep tail base and the middle tail that was stiffened by a long front articulation processes. The underside of the trunk was covered by eighteen or nineteen pairs of segmented belly ribs.[9]

The shoulder girdle was longer than the entire forelimb. The shoulder blade had a narrow shaft but was exceptionally expanded at its upper end. It connected via a long forward protrusion to the coracoid, which was rounded. Both shoulder blades were connected by a small furcula. The paired breast bones possibly were made of cartilage only.[9]



Right forelimb of specimen "Sue"

The forelimb or arm was very short. The upper arm bone, the humerus, was short but robust. It had a narrow upper end with an exceptionally rounded head. The lower arm bones, the ulna and radius, were straight elements, much shorter than the humerus. The second metacarpal was longer and wider than the first, whereas normally in theropods the opposite is true.

The forelimbs had only two clawed fingers,[9] along with an additional splint-like small third metacarpal representing the remnant of a third digit.[57]

Pelvic girdle of specimen MOR 555

The pelvis was a large structure. Its upper bone, the ilium, was both very long and high, providing an extensive attachment area for hindlimb muscles. The front pubic bone ended in an enormous pubic boot, longer than the entire shaft of the element. The rear ischium was slender and straight, pointing obliquely to behind and below.[9]

In contrast to the arms, the hindlimbs were among the longest in proportion to body size of any theropod. In the foot, the metatarsus was "arctometatarsalian", meaning that the part of the third metatarsal near the ankle was pinched. The third metatarsal was also exceptionally sinuous.[9] Compensating for the immense bulk of the animal, many bones throughout the skeleton were hollowed, reducing its weight without significant loss of strength.[9]

Classification

Tyrannosaurus is the type genus of the superfamily Tyrannosauroidea, the family Tyrannosauridae, and the subfamily Tyrannosaurinae; in other words it is the standard by which paleontologists decide whether to include other species in the group. Other members of the tyrannosaurine subfamily include the North American *Daspletosaurus* and the Asian *Tarbosaurus*,[22][58] both of which have occasionally been synonymized with *Tyrannosaurus*.[59]

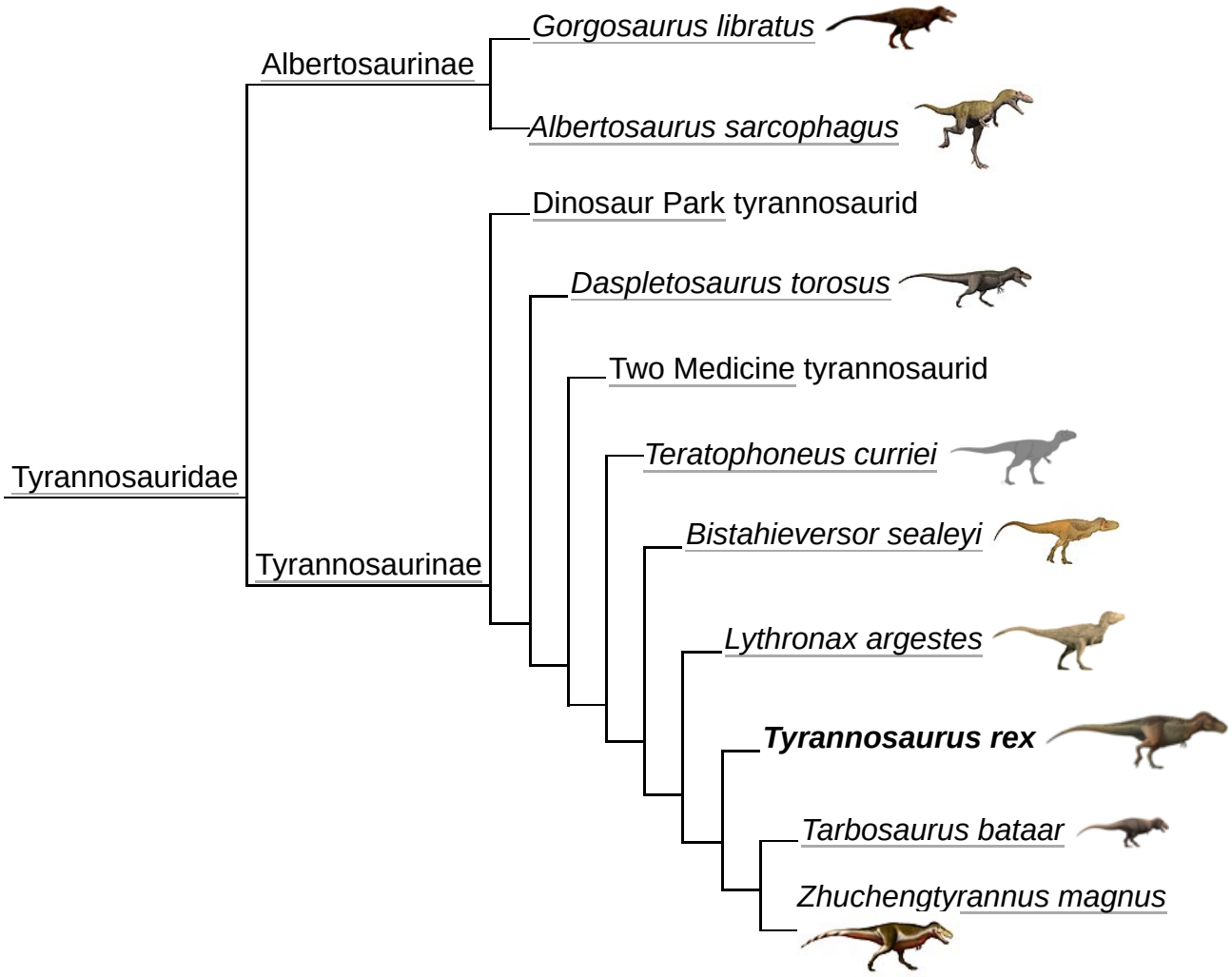
Skull casts of different *Tyrannosaurus* specimens

Tyrannosaurids were once commonly thought to be descendants of earlier large theropods such as megalosaurs and carnosaurs, although more recently they were reclassified with the generally smaller coelurosaurs.[48] The earliest tyrannosaur group were the crested proceratosaurids, while later and more derived members belong to the Pantyrannosauria. Tyrannosaurs started out as small theropods; at least some became larger by the Early Cretaceous. It is likely that tyrannosauroids rose to prominence after the decline in allosauroid and megalosauroid diversity seen during the early stages of the Late Cretaceous.[60]

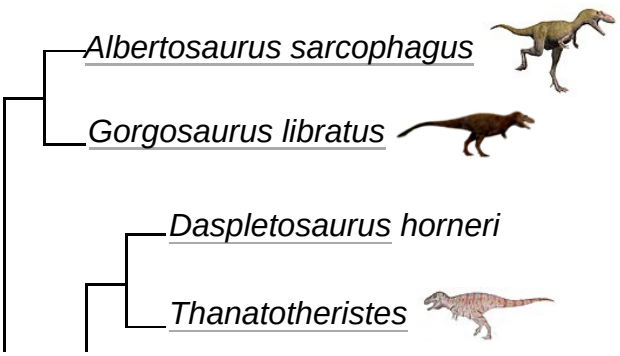
Tyrannosauroids are characterized by their fused nasals and dental arrangement. Pantyrannosaurs are characterized by unique features in their hips as well as an enlarged foramen in the quadrate, a broad postorbital and hourglass shaped nasals. Some of the more derived pantyrannosaurs lack nasal pneumaticity and have a lower humerus to femur ratio with their arms starting to see some reduction. Some pantyrannosaurs started developing an arctometatarsus. Eutyrannosaurs have a rough texture on their nasal bones and their mandibular fenestra is reduced externally. Tyrannosaurids lack kinetic skulls or special crests on their nasal bones, and have a lacrimal with a distinctive process on it. Tyrannosaurids also have an interfenestral strut that is less than half as big as the maxillary fenestra.[61]

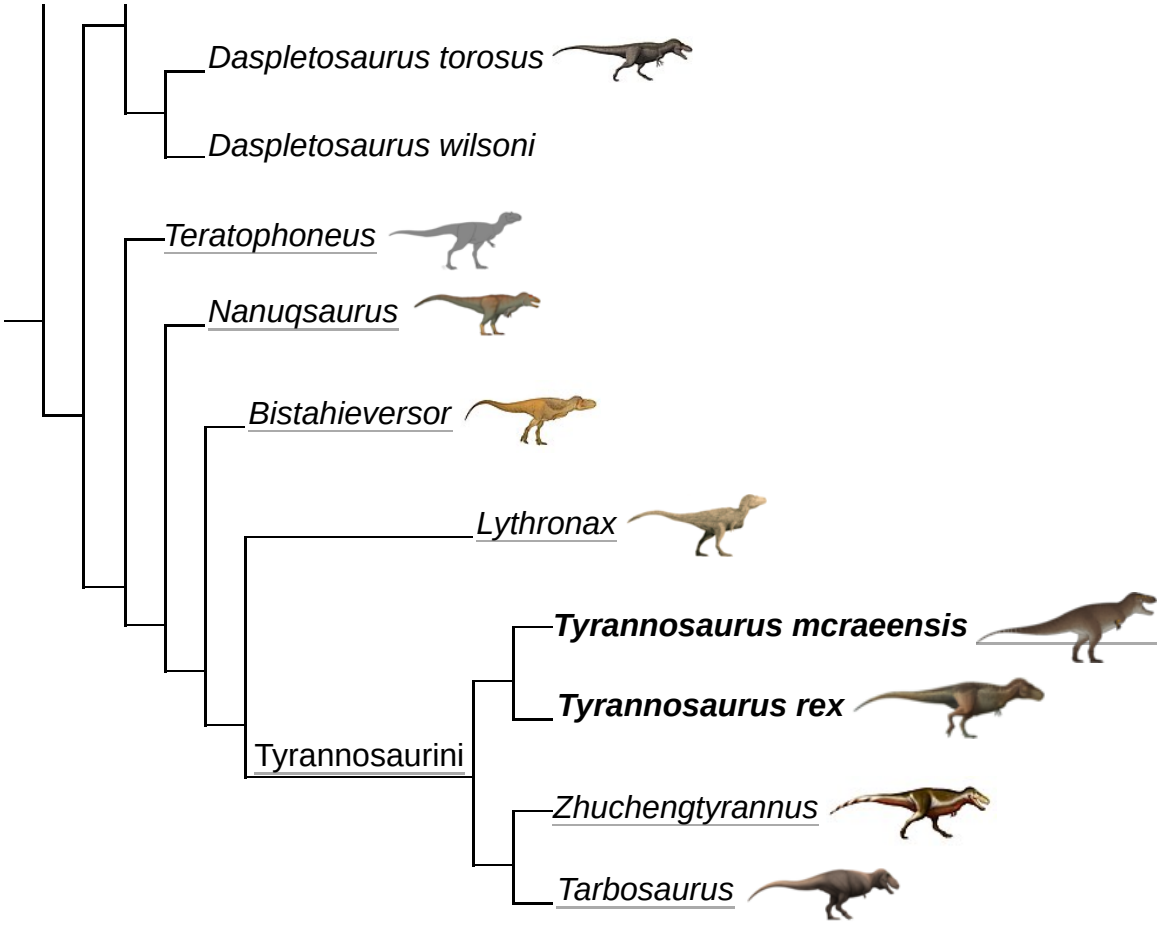
Many phylogenetic analyses have found *Tarbosaurus bataar* to be the sister taxon of *T. rex*.[58] The discovery of the tyrannosaurid *Lythronax* further indicates that *Tarbosaurus* and *Tyrannosaurus* are closely related, forming a clade with fellow Asian tyrannosaurid *Zhuchengtyrannus*, with *Lythronax* being their sister taxon.[62][63] A further study from 2016 by Steve Brusatte, Thomas Carr and colleagues, also indicates that *Tyrannosaurus* may have been an immigrant from Asia, as well as a possible descendant of *Tarbosaurus*.[64]

This cladogram of Tyrannosauridae is based on the phylogenetic analysis conducted by Loewen and colleagues in 2013.[62]



In their 2024 description of *Tyrannosaurus mcraeensis*, Dalman et al. recovered similar results to previous analyses, with *Tyrannosaurus* as the sister taxon to the clade formed by *Tarbosaurus* and *Zhuchengtyrannus*, called the Tyrannosaurini. They also found support for a monophyletic clade containing *Daspletosaurus* and *Thanatotheristes*, typically referred to as the Daspletosaurini.[65][66]





Additional species

In 1955, Soviet paleontologist Evgeny Maleev named a new species, *Tyrannosaurus bataar*, from Mongolia.[67] By 1965, this species was renamed as a distinct genus, *Tarbosaurus bataar*.[68] While most paleontologists continue to maintain the two as distinct genera, some authors such as Thomas Holtz, Kenneth Carpenter, and Thomas Carr argue that the two species are similar enough to be considered members of the same genus, restoring the Mongolian taxon's original binomial name.[48][69][52]

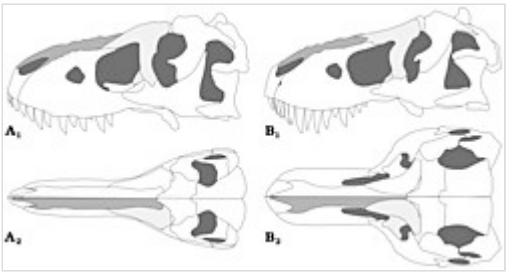


Diagram showing the differences between a generalized *Tarbosaurus* (A) and *Tyrannosaurus rex* (B) skull

Some specimens from the Late Cretaceous deposits of China have been described as new species of *Tyrannosaurus*: *T.* "lanpingensis" based on isolated lateral tooth from the red beds of Yunnan in 1975; *T.* "turpanensis" from the Subashi Formation, Turpan Basin, Xinjiang in 1978; and *T. luanchuanensis* from the Quiba Formation in Henan Province in 1979–1980.[70][71][72] The former two taxa were published without detailed descriptions and are therefore nomina nuda. *T.* "turpanensis" and *T. luanchuanensis* were tentatively listed as junior synonyms of *Tarbosaurus bataar* by Holtz in 2004; Holtz also listed *T.* "lanpingensis," but as a nomen dubium.[58][71][72]

VGI, no. 231/3, a large phalanx bone, assigned to *Tyrannosaurus* sp. by Yarkov in 2000, was found in the Lower Maastrichtian of Bereslavka, Russia. In 2004, Averianov and Yarkov reinterpreted it as a metacarpal or metatarsal that possibly belongs to ceratosaur.[73] In their 2023 overview, Averianov and

Lopatin mention this specimen as well as a single tooth from the same site only as Theropoda indet.[74]

In 2001, various tyrannosaurid teeth and a metatarsal unearthed in a quarry near Zhucheng, China, were assigned by paleontologist Hu Chengzhi to the newly erected species *Tyrannosaurus zhuchengensis*. However, in a nearby site, a right maxilla and left jawbone were assigned to the newly erected tyrannosaurid genus *Zhuchengtyrannus* in 2011. It is possible that *T. zhuchengensis* is synonymous with *Zhuchengtyrannus*. *T. zhuchengensis* is considered to be a *nomen dubium* as the holotype lacks diagnostic features below the level Tyrannosaurinae.[75]

In 2006, a fragmentary tyrannosaurid lacrimal (CM 9401) from the Judith River Formation of Fergus County, Montana, was described as ?*Tyrannosaurus* sp. This isolated right lacrimal was originally collected alongside the holotype specimen of *Deinosuchus rugosus*, a giant crocodilian, and remained undescribed until its re-identification as belonging to a tyrannosaurid theropod in the 1980s by paleontologist Dale Russell. The lacrimal closely resembles those of *Tyrannosaurus rex* in both size and morphology. Notably, it lacks the "lacrimal horn" typically present in earlier tyrannosaurids like *Albertosaurus* and *Gorgosaurus*, instead exhibiting a distinct rugosity along the dorsal surface—consistent with *T. rex* and its Asian relative *Tarbosaurus*. The specimen's size places it within the range of known *T. rex* individuals, suggesting the presence of large tyrannosaurids during the Campanian stage (~75 million years ago), a temporal range earlier than the established Maastrichtian age (~68–66 Ma) for *Tyrannosaurus rex*. However, the exact age and provenance of CM 9401 remain uncertain due to a lack of detailed field documentation.[76]

In 2018, a paper describing tyrannosaurid teeth from the Two Medicine Formation noted a premaxillary tooth (YPM VPPU 023469) had a strong resemblance to the teeth of Sue to the exclusion of any Campanian tyrannosaurid. Additionally, the authors of this paper suggested that CM 9401 also comes from the Two Medicine Formation, as there were preservational similarities between its locality and the Willow Creek anticline where the tooth was found.[77] This would place both specimens in the Flag Butte Member of the Two Medicine Formation, which dates from 77 to 76.3 Ma, far older than any other *Tyrannosaurus* specimen, and directly contemporaneous with *Daspletosaurus.* In 2025, these specimens were used by Charlie Scherer to suggest that the Tyrannosaurini did not evolve directly from *Daspletosaurus*.[78]

Tyrannosaurus imperator and *Tyrannosaurus regina*

In a 2022 study, Gregory S. Paul and colleagues argued that *Tyrannosaurus rex*, as traditionally understood, actually represents three species: the type species *Tyrannosaurus rex*, and two previously unrecognized species: *T. imperator* (meaning "tyrant lizard emperor") and *T. regina* (meaning "tyrant lizard queen"). The holotype of the former (*T. imperator*) is the Sue specimen, and the holotype of the latter (*T. regina*) is Wankel rex. The division into multiple species was primarily based on the high degree of variation in the proportions and robusticity of the femur (and other skeletal elements) across *T. rex* specimens. Presumed robust and gracile morphotypes and the number of small incisiform teeth in the dentary were also used as a line of evidence. Paul and colleagues categorized the species as follows: *T. rex* demonstrates robust anatomy, a moderate ratio of femur length vs circumference, and the possession of a singular slender incisiform dentary tooth, *T. imperator* is robust with a small femur length to circumference ratio and two of the slender teeth, and *T. regina* is more gracile with a high femur ratio and one of the slender teeth.[79]

However, several paleontologists, including Stephen Brusatte, Thomas Carr, Thomas Holtz, David Hone, Jingmai O'Connor, and Lindsay Zanno, criticized the study or expressed skepticism of its conclusions.[80][81][82][83] Holtz and Zanno both remarked that it was plausible that more than one species of *Tyrannosaurus* existed, but felt the new study was insufficient to support the species it proposed. Holtz remarked that, even if *Tyrannosaurus imperator* represented a distinct species from *Tyrannosaurus rex*, it may represent the same species as *Nanotyrannus lancensis* and would need to be called *Tyrannosaurus lancensis*. O'Connor, a curator at the Field Museum, where the *T. imperator* holotype Sue is displayed, regarded the new species as too poorly-supported to justify modifying the exhibit signs. Brusatte, Carr, and O'Connor viewed the distinguishing features proposed between the species as reflecting natural variation within a species. Both Carr and O'Connor expressed concerns about the study's inability to determine which of the proposed species several well-preserved specimens belonged to. Another paleontologist, Philip J. Currie, originally co-authored the study but withdrew from it as he did not want to be involved in naming the new species.[80]

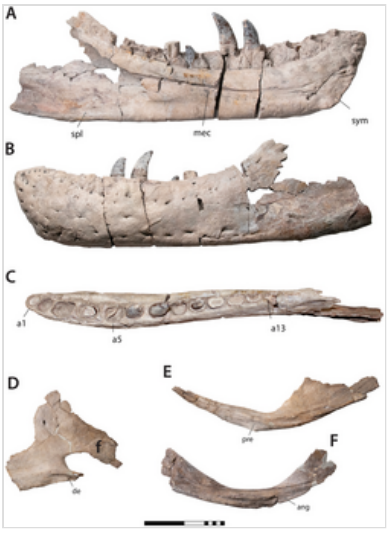
In a paper published in 2025, Paul maintained the validity of *T. imperator* and *T. regina*, arguing that these proposed species could be further distinguished by the shape of their postorbital bosses (bony rugosities behind the eyes).[84]

Tyrannosaurus mcraeensis

Reconstructed skull (top) and speculative life restoration (bottom) of *T. mcraeensis*

In 2024, Dalman and colleagues described the remains of a tyrannosaur discovered in 1983 in the Campanian-early Maastrichtian Hall Lake Formation in New Mexico. Reposited at the New Mexico Museum of Natural History and Science, the fossil material (NMMNH P-3698) consists of the right postorbital, right squamosal, left palatine, and an incomplete maxilla from the skull, the left dentary, right splenial, right prearticular, right angular and right articular from the lower jaws, isolated teeth, and chevrons.[65] Some of the bones were briefly mentioned in 1984 as belonging to *T. rex*,[85] and described in 1986.[86]

Lehman and Carpenter (1990) suggested that NMMNH P-3698 belonged to a new tyrannosaurid genus,[87] while Carr and Williamson (2000) disagreed with their claim.[88] Sullivan and Lucas (2015) argued that there is little evidence to support NMMNH P-3698 as a specimen of *Tyrannosaurus rex*, so they tentatively classified it as cf. *Tyrannosaurus* sp.; they also considered that the McRae tyrannosaur lived before the Lancian (before 67 million years ago) based on its probable association with the *Alamosaurus* fossil.[89]



Mandibular elements of the *T. mcraeensis* holotype

Dalman et al. (2024) proposed the new name *Tyrannosaurus mcraeensis* for the holotype (NMMNH P-3698), referencing the McRae Group, the rock layers to which the Hall Lake Formation belongs. The holotype of *T. mcraeensis* is found in the strata that are around a few million years older than the accepted range of *T. rex*, which existed at the end of the Maastrichtian. The rock layers were initially estimated to date to between 72.7 and 70.9 Ma, correlating to the latest Campanian or earliest Maastrichtian.[65] However, in a 2024 conference abstract, the specific stratigraphic layer which yielded *T. mcraeensis* was estimated to be younger than 69.0 ± 0.4 Ma and older than 66.0 Ma based on the sandstone from the same fossil locality,[90] which would suggest that the age of *T. mcraeensis* is much closer to *T. rex* than previously thought.[78] *T. mcraeensis* was estimated at 12 metres (39 ft) long, which is similar to the size of an adult *T. rex*. The two are distinguished by characters of the skull. Amongst these, the dentary of *T. mcraeensis* is proportionately longer and possesses a less prominent chin, and the lower jaw shallower than that of *T. rex*, suggesting a weaker bite. The teeth are likewise blunter and more laterally compressed, while the post orbital crests are less prominent. Likewise, the skeletal anatomy showcases shared characteristics with *Tarbosaurus* and *Zhuchengtyrannus*.[65][91]

The validity of *T. mcraeensis* was questioned by other researchers in 2025. Morrison and colleagues noted the uncertainty in the previous age estimate and argued that the inclusion of the titanosaurian fossil (cf. *Alamosaurus*) as a biostratigraphic marker of the Hall Lake Formation fauna makes the dating problematic due to the fossil having a huge variability in assumed sedimentation rates. The study also suggested the uncertainty in the morphology of *T. mcraeensis* holotype being reliably outside the known variation of *T. rex*. Regardless of the age and validity of *T. mcraeensis*, the authors concluded that the genus *Tyrannosaurus* still most likely originated from Laramidia and that the ancestors of this genus possibly migrated from Asia.[92] Later that year, Voris and colleagues regarded *T. mcraeensis* as a junior synonym of *T. rex*, agreeing with the problematic stratigraphic data and finding all of the proposed diagnostic characters of *T. mcraeensis* to be in the known *T. rex* specimen range of variation.[93] In his study of the *T. rex* holotype skull, Carr also did not regard *T. mcraeensis* as a valid species, agreeing with the proposed characters of the *T. mcraeensis* holotype being visible throughout the hypodigm of *T. rex*.[94]

Nanotyrannus

Other tyrannosaur fossils found in the same formations as *T. rex* were originally classified as separate taxa, including *Aublysodon* and *Albertosaurus megagracilis*,[59] the latter being named *Dinotyrannus megagracilis* in 1995.[95] These fossils are now universally considered to belong to juvenile *T. rex*.[96] A small but nearly complete skull from Montana, 57.2 centimeters (1.88 ft) long, is an exception. This skull, CMNH 7541, was originally classified as a species of *Gorgosaurus* (*G. lancensis*) by Charles W. Gilmore in 1946.[97] In 1988, the specimen was re-described by Robert T. Bakker, Michael Williams, then the curator of paleontology at the Cleveland Museum of Natural History, and Phil Currie, where the

Cast of CMNH 7541, the holotype of *Nanotyrannus lancensis*, sometimes interpreted as a juvenile *Tyrannosaurus*.

original specimen was housed and is now on display. Their initial research indicated that the skull bones were fused, and that it therefore represented an adult specimen. In light of this, Bakker and colleagues assigned the skull to a new genus named *Nanotyrannus* (meaning "dwarf tyrant", for its apparently small adult size).[98] However, in 1999, a detailed analysis by Thomas Carr revealed the specimen to be a juvenile, leading Carr and many other paleontologists to consider it a juvenile *T. rex* individual.[99][100]

Reconstructed skeleton of "Jane", Burpee Museum of Natural History

In 2001, a more complete juvenile tyrannosaur (nicknamed "Jane", accession number BMRP 2002.4.1), suggested to belong to the same species as the original *Nanotyrannus* specimen, was uncovered. This discovery prompted a conference on tyrannosaurs focused on the issues of the validity of *Nanotyrannus*, held at the Burpee Museum of Natural History in 2005. Several paleontologists who had previously published opinions that *N. lancensis* was a valid species, including Currie and Williams, saw the discovery of "Jane" as a confirmation that *Nanotyrannus* was, in fact, a juvenile *T. rex*.[101][102][103] Peter Larson continued to support the hypothesis that *Nanotyrannus lancensis* was a separate but closely related species, and also argued that *Stygivenator* (LACM 28471), which is generally considered to be a juvenile of *Tyrannosaurus rex*, could be a younger specimen of *Nanotyrannus*.[104][105]

In 2011, news reports about a 2006 discovery of a new, virtually complete theropod specimen found alongside a ceratopsid were made. The specimens were studied by Robert Bakker and Peter Larson on-site, who identified the ceratopsian as *Triceratops* and the theropod as *Nanotyrannus*.[106] It was impossible to determine whether the theropod specimen, nicknamed "Bloody Mary", was distinct from *T. rex*, as the specimen remained in private hands until 2020, when the ownership of the specimen was decided by the Montana Supreme Court to be given to the landowners Mary Anne and Lige Murray.[107] The fossil was acquired by the North Carolina Museum of Natural Sciences in 2020.[108][109] In 2025, paleontologists Lindsay Zanno and James Napoli published an initial description of the "Bloody Mary" specimen as part of an extensive revision of the genus *Nanotyrannus*. They identified this specimen as a skeletally mature individual of *Nanotyrannus* and provided several lines of evidence supporting the validity of this genus. They also described a second species, *N. lethaeus*, based on the Jane specimen due to notable differences between it and *N. lancensis*.[23]

Paleobiology

Life history

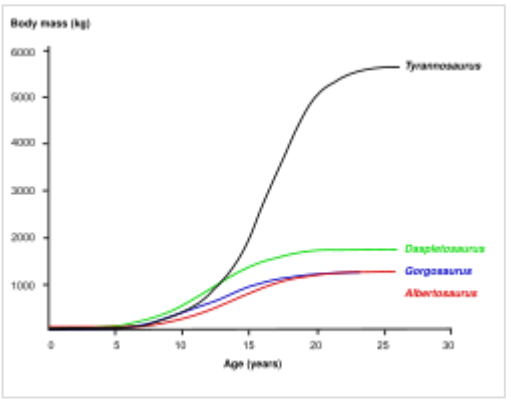
The identification of several specimens as juvenile *T. rex* has allowed scientists to document ontogenetic changes in the species, estimate the lifespan, and determine how quickly the animals would have grown. The *Stygivenator* holotype (LACM 28471, the "Jordan theropod"), possibly a juvenile *T. rex*, is estimated to have weighed only 30 kg (66 lb), while the largest adults, such as FMNH PR2081 (Sue) most likely weighed about 5,650 kg (12,460 lb). Histologic analysis of *T. rex* bones showed LACM 28471 had aged only 2 years when it died. Sue was initially estimated at 28 years old, an age which was at the time

Skull of a juvenile *T. rex* formerly classified as *Dinotyrannus megagracilis*

considered close to the maximum for the species,[38] but the most recent studies estimate Sue's age at 27–33 years.[110] Similarly, Trix (RGM 792.000) is estimated to have been at least 30 years old at time of death.[111]

Histology has also allowed the age of other specimens to be determined. Growth curves can be developed when the ages of different specimens are plotted on a graph along with their mass. A *T. rex* growth curve is S-shaped, with juveniles remaining under 1,800 kg (4,000 lb) until approximately 14 years of age, when body size began to increase dramatically. During this rapid growth phase, a young *T. rex* would gain an average of 600 kg (1,300 lb) a year for the next four years. At 18 years of age, the curve plateaus again, indicating that growth slowed dramatically. For example, only 600 kg (1,300 lb) separated the 28-year-old Sue from a 22-year-old Canadian specimen (RTMP 81.12.1).[38] A 2004 histological study performed by different workers corroborates these results, finding that rapid growth began to slow at around 16 years of age.[112]

A study by Hutchinson and colleagues in 2011 corroborated the previous estimation methods in general, but their estimation of peak growth rates is significantly higher; it found that the "maximum growth rates for T. rex during the exponential stage are 1790 kg/year".[27] Although these results were much higher than previous estimations, the authors noted that these results significantly lowered the great difference between its actual growth rate and the one which would be expected of an animal of its size.[27] The sudden change in growth rate at the end of the growth spurt may indicate physical maturity, a hypothesis which is supported by the discovery of medullary tissue in the femur of a 16 to 20-year-old *T. rex* from Montana (MOR 1125, also known as B-rex). Medullary tissue is found only in female birds during ovulation, indicating that B-rex was of reproductive age.[113] Further study indicates an age of 18 for this specimen.[114] In 2016, it was finally confirmed by Mary Higby Schweitzer and Lindsay Zanno and colleagues that the soft tissue within the femur of MOR 1125 was medullary tissue. This also confirmed the identity of the specimen as a female. The discovery of medullary bone tissue within *Tyrannosaurus* may prove valuable in determining the sex of other dinosaur species in future examinations, as the chemical makeup of medullary tissue is unmistakable.[115] Other tyrannosaurids exhibit extremely similar growth curves, although with lower growth rates corresponding to their lower adult sizes.[116]



A graph showing the hypothesized growth curve, body mass versus age (drawn in black, with other tyrannosaurids for comparison), based on Erickson and colleagues 2004. Studies like this are now known to have been skewed by the inclusion of specimens now considered *Nanotyrannus*

An additional study published in 2020 by Woodward and colleagues, for the journal *Science Advances* indicates that during their growth from juvenile to adult, *Tyrannosaurus* was capable of slowing down its growth to counter environmental factors such as lack of food. Based on BMRP 2002.4.1 and BMRP 2006.4.4 between 13 and 15 years old housed at the Burpee Museum in Illinois then referred to as

juvenile *Tyrannosaurus* specimens, the authors suggested that the rate of maturation for *Tyrannosaurus* was dependent on resource abundance. This study also indicates that in such changing environments, *Tyrannosaurus* was particularly well-suited to an environment that shifted yearly in regards to resource abundance, hinting that other midsize predators might have had difficulty surviving in such harsh conditions and explaining the niche partitioning between juvenile and adult tyrannosaurs. The study further suggested that *Tyrannosaurus* and *Nanotyrannus* are synonymous, due to analysis of the growth rings in the bones of the two specimens studied.[117][118]

In January 2026, Woodward, Myhrvold and Horner performed a comprehensive histological analysis of 17 tyrannosaur specimens, and argued that *Tyrannosaurus* likely experienced a more gradual annual growth rate slope than indicated by earlier studies and evidence of a protracted subadult stage, reaching asymptotic size at around 35–40 years of age. The upper limit of the annual growth rate estimate is approximately 43–53 years. They also found that the growth trajectories of BMRP 2002.4.1 and BMRP 2006.4.4 do not fit with other *Tyrannosaurus* specimens in their growth curve model. While they acknowledged the possibility of these ontogenetically immature specimens representing *Nanotyrannus* as suggested by Zanno and Napoli (2025), they noted that the inconsistencies of these specimens observed in the growth curve do not necessarily bear weight on the proposal that *Nanotyrannus* is a distinct taxon.[119]

In July 2026, Longrich and colleagues described a third metatarsal bone of perinate (hatchling) *Tyrannosaurus rex* and cf. *Gorgosaurus libratus*, RSKM P2416.82 from the Frenchman Formation and TMP 1981.16.475 from the Dinosaur Park Formation respectively. Histological analyses reveal that lines of arrested growth are absent in both specimens, indicative of hatchling or embryonic stage. Synchrotron scans of each specimen reveal the relatively high density of the outer surface and the pattern changes in cortical (outer layer) bone structure, with a layer similar to the hatchling/neonatal line of modern oviparous (egg-laying) and viviparous (giving live birth) taxa that characterize the transition from embryonic to hatchling stage. The scans also detected evidence of Haversian bone remodeling, the continuous process of bone tissue breaking down and rebuilding to form secondary canals (tubular channels connecting bones), which are indicative of hatchling stage and precociality, relative maturity and mobility at birth. The layer lining the medullary cavity of both specimens likely represent endosteal bone deposition, during which new bone matrix is deposited on the inner surface of the cortical bone. The authors also described a partial dentary (FMNH PR 4920) from the Hell Creek Formation and isolated teeth from the Lance Formation as hatchling *T. rex* and interpret their tooth wear as evidence of consuming relatively large vertebrates.[120]

The body length and body mass of the hatchling *T. rex* (RSKM P2416.82) are estimated to be approximately 75 cm (2.46 ft) and 2.5 kg (5.5 lb) respectively, and those of the hatchling cf. *G. libratus* (TMP 1981.16.475) are estimated to be approximately 70 cm (2.3 ft) and 2.4 kg (5.3 lb) respectively. On the basis of these body size estimates, Longrich et al. (2026) suggested that the clutch of the confirmed, reproductively mature female *T. rex* (MOR 1125) would have weighed at least 54–60 kg (119–132 lb) which would possibly correspond to 21-32 eggs. They further stated that larger clutch sizes are also plausible, indicating that tyrannosaurids possibly employed R-selection, a reproductive strategy that maximizes the number of offsprings through large clutch size and minimal parental investment, which would have resulted in rapid maturation of the hatchlings, slower than that of modern birds. It is also likely that the reproduction strategy of tyrannosaurids was an intermediate between R-selection and K-selection, the latter of which would maximize parental investment with fewer number of offsprings and delayed maturation. Given the uniparental care observed in modern crocodilians and basal (early-

diverging) birds, most non-avian dinosaurs including *T. rex* were likely uniparental as well. The authors also estimated that the relative size of tyrannosaurid hatchlings and eggs are larger than that of squamates, turtles, crocodilians, pterosaurs, and certain dinosaur clades (protoceratopsids and basal sauropodomorphs), but smaller than that of eumaniraptorans and modern birds, probably weighing less in both relative and absolute size than those of the elephant bird *Aepyornis maximus* and the South Island giant moa.[120]

Over half of the known *T. rex* specimens appear to have died within six years of reaching sexual maturity, a pattern which is also seen in other tyrannosaurs and in some large, long-lived birds and mammals today. These species are characterized by high infant mortality rates, followed by relatively low mortality among juveniles. Mortality increases again following sexual maturity, partly due to the stresses of reproduction. One study suggests that the rarity of juvenile *T. rex* fossils is due in part to low juvenile mortality rates; the animals were not dying in large numbers at these ages, and thus were not often fossilized. This rarity may also be due to the incompleteness of the fossil record or to the bias of fossil collectors towards larger, more spectacular specimens.[116] In a 2013 lecture, Thomas Holtz Jr. suggested that dinosaurs "lived fast and died young" because they reproduced quickly whereas mammals have long lifespans because they take longer to reproduce.[121] Gregory S. Paul also writes that *Tyrannosaurus* reproduced quickly and died young but attributes their short lifespans to the dangerous lives they lived.[122]

Skin and possible filamentous feathering

Fossilized skin impressions from the tail region of a *Tyrannosaurus rex*, Houston Museum of Natural Science

The discovery of feathered dinosaurs led to debate regarding whether, and to what extent, *Tyrannosaurus* might have been feathered.[123][124] Filamentous structures, which are commonly recognized as the precursors of feathers, have been reported in the small-bodied, basal tyrannosauroid *Dilong paradoxus* from the Early Cretaceous Yixian Formation of China in 2004.[125] Because integumentary impressions of larger tyrannosauroids known at that time showed evidence of scales, the researchers who studied *Dilong* speculated that insulating feathers might have been lost by larger species due to their smaller surface-to-volume ratio.[125] The subsequent discovery of the giant species *Yutyrannus huali*, also from the Yixian, showed that even some large tyrannosauroids had feathers covering much of their bodies, casting doubt on the hypothesis that they were a size-related feature.[126] A 2017 study reviewed known skin impressions of tyrannosaurids, including those of a *Tyrannosaurus* specimen nicknamed "Wyrex" (HMNS 2006.1743.01, formerly known as BHI 6230) which preserves patches of mosaic scales on the tail, hip, and neck.[123] The study concluded that feather covering of large tyrannosaurids such as *Tyrannosaurus* was, if present, limited to the upper side of the trunk.[123]

A conference abstract published in 2016 posited that theropods such as *Tyrannosaurus* had their upper teeth covered in lips, instead of bare teeth as seen in crocodilians. This was based on the presence of enamel, which according to the study needs to remain hydrated, an issue not faced by aquatic animals like crocodilians.[54] However, there has been criticism where it favors the idea for lips, with the 2017 analytical study proposing that tyrannosaurids had large, flat scales on their snouts instead of lips, as

modern crocodiles do.[52][127] But crocodiles possess rather cracked keratinized skin, not flat scales; by observing the hummocky rugosity of tyrannosaurids, and comparing it to extant lizards, researchers have found that tyrannosaurids had squamose scales rather than a crocodillian-like skin.[55][128]

In 2023, Cullen and colleagues supported the idea that theropods like tyrannosaurids had lips based on anatomical patterns, such as those of the foramina on their face and jaws, more similar to those of modern squamates such as monitor lizards or marine iguanas than those of modern crocodilians like alligators. Comparison of the teeth of *Daspletosaurus* and American alligators shows that the enamel of tyrannosaurids had no significant wear and that the teeth of modern crocodilians were eroded on the labial side and were substantially worn. This suggests that it is likely that theropod teeth were kept wet by lips. On the basis of the relationship between hydration and wear resistance, the authors argued that it is unlikely that the teeth of theropods, including tyrannosaurids, would have remained unworn when exposed for a long time, because it would have been hard to maintain hydration. The authors also performed regression analyses to demonstrate the relationship between tooth height and skull length, and found that varanids like the crocodile monitor had substantially greater ratios of tooth height to skull length than those of *Tyrannosaurus*, indicating that the teeth of theropods were not too big to be covered by extraoral tissues when the mouth was closed.[56]

Sexual dimorphism

Skeleton casts mounted in a mating position, Jurassic Museum of Asturias

As the number of known specimens increased, scientists began to analyze the variation between individuals and discovered what appeared to be two distinct body types, or *morphs*, similar to some other theropod species. As one of these morphs was more solidly built, it was termed the 'robust' morph while the other was termed 'gracile'. Several morphological differences associated with the two morphs were used to analyze sexual dimorphism in *T. rex*, with the 'robust' morph usually suggested to be female. For example, the pelvis of several 'robust' specimens seemed to be wider, perhaps to allow the passage of eggs.[129] It was also thought that the 'robust' morphology correlated with a reduced chevron on the first tail vertebra, also ostensibly to allow eggs to pass out of the reproductive tract, as had been erroneously reported for crocodiles.[130]

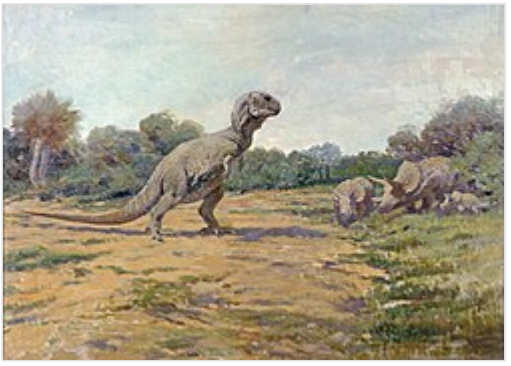
In recent years, evidence for sexual dimorphism has been weakened. A 2005 study reported that previous claims of sexual dimorphism in crocodile chevron anatomy were in error, casting doubt on the existence of similar dimorphism between *T. rex* sexes.[131] A full-sized chevron was discovered on the first tail vertebra of Sue, an extremely robust individual, indicating that this feature could not be used to differentiate the two morphs anyway. As *T. rex* specimens have been found from Saskatchewan to New Mexico, differences between individuals may be indicative of geographic variation rather than sexual dimorphism. The differences could also be age-related, with 'robust' individuals being older animals.[9]

Only a single *Tyrannosaurus* specimen has been conclusively shown to belong to a specific sex. Examination of B-rex demonstrated the preservation of soft tissue within several bones. Some of this tissue has been identified as a medullary tissue, a specialized tissue grown only in modern birds as a source of calcium for the production of eggshell during ovulation. As only female birds lay eggs,

medullary tissue is only found naturally in females, although males are capable of producing it when injected with female reproductive hormones like estrogen. This strongly suggests that B-rex was female and that she died during ovulation.[113] Recent research has shown that medullary tissue is never found in crocodilians, which are thought to be the closest living relatives of dinosaurs. The shared presence of medullary tissue in birds and other theropod dinosaurs is further evidence of the close evolutionary relationship between the two.[132]

Posture

Like many bipedal dinosaurs, *T. rex* was historically depicted as a 'living tripod', with the body at 45 degrees or less from the vertical and the tail dragging along the ground, similar to a kangaroo. This concept dates from Joseph Leidy's 1865 reconstruction of *Hadrosaurus*, the first to depict a dinosaur in a bipedal posture.[133] In 1915, convinced that the creature stood upright, Henry Fairfield Osborn, former president of the American Museum of Natural History, further reinforced the notion in unveiling the first complete *T. rex* skeleton arranged this way. It stood in an upright pose for 77 years, until it was dismantled in 1992.[134]



Outdated reconstruction (by Charles R. Knight), showing upright pose

By 1970, scientists realized this pose was incorrect and could not have been maintained by a living animal, as it would have resulted in the dislocation or weakening of several joints, including the hips and the articulation between the head and the spinal column.[135] The inaccurate AMNH mount inspired similar depictions in many films and paintings (such as Rudolph Zallinger's famous mural *The Age of Reptiles* in Yale University's Peabody Museum of Natural History)[136] until the 1990s, when films such as *Jurassic Park* introduced a more accurate posture to the general public.[137] Modern representations in museums, art, and film show *T. rex* with its body approximately parallel to the ground with the tail extended behind the body to balance the head.[138]

To sit down, *Tyrannosaurus* may have settled its weight backwards and rested its weight on a pubic boot, the wide expansion at the end of the pubis in some dinosaurs. With its weight rested on the pelvis, it may have been free to move the hindlimbs. Getting back up again might have involved some stabilization from the diminutive forelimbs.[139][135] The latter known as Newman's pushup theory has been debated. Nonetheless, *Tyrannosaurus* was probably able to get up if it fell, which only would have required placing the limbs below the center of gravity, with the tail as an effective counterbalance. Healed stress fractures in the forelimbs have been put forward both as evidence that the arms cannot have been very useful[140][141] and as evidence that they were indeed used and acquired wounds,[142] like the rest of the body.

Arms

When *T. rex* was first discovered, the humerus was the only element of the forelimb known.[11] For the initial mounted skeleton as seen by the public in 1915, Osborn substituted longer, three-fingered forelimbs like those of *Allosaurus*.[6] A year earlier, Lawrence Lambe described the short, two-fingered forelimbs of the closely related *Gorgosaurus*.[143] This strongly suggested that *T. rex* had similar forelimbs, but this hypothesis was not confirmed until the first complete *T. rex* forelimbs were identified

The forelimbs might have been used to help *T. rex* rise from a resting pose, as seen in this cast (Bucky specimen)

in 1989, belonging to MOR 555 (the "Wankel rex").[144][145] The remains of Sue also include complete forelimbs.[9] *T. rex* arms are very small relative to overall body size, measuring only 1 meter (3.3 ft) long, and some scholars have labelled them as vestigial. However, the bones show large areas for muscle attachment, indicating considerable strength. This was recognized as early as 1906 by Osborn, who speculated that the forelimbs may have been used to grasp a mate during copulation.[13] Newman (1970) suggested that the forelimbs were used to assist *Tyrannosaurus* in rising from a prone position.[135] Since then, other functions have been proposed, although some scholars find them implausible.[141] Padian (2022) argued that the reduction of the arms in tyrannosaurids did not serve a particular function but was a secondary adaptation, stating that as tyrannosaurids developed larger and more powerful skulls and jaws, the arms got smaller to avoid being bitten or torn by other individuals, particularly during group feedings.[141]

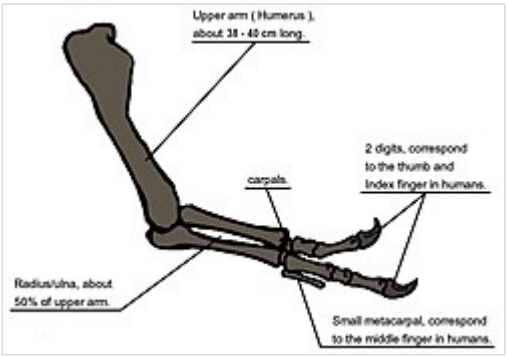


Diagram illustrating arm anatomy

Another possibility is that the forelimbs held struggling prey while it was killed by the tyrannosaur's enormous jaws. This hypothesis may be supported by biomechanical analysis. *T. rex* forelimb bones exhibit extremely thick cortical bone, which has been interpreted as evidence that they were developed to withstand heavy loads. The biceps brachii muscle of an adult *T. rex* was capable of lifting 199 kilograms (439 lb) by itself; other muscles such as the brachialis would work along with the biceps to make elbow flexion even more powerful. The M. biceps muscle of *T. rex* was 3.5 times as powerful as the human equivalent. A *T. rex* forearm had a limited range of motion, with the shoulder and elbow joints allowing only 40 and 45 degrees of motion, respectively. In contrast, the same two joints in *Deinonychus* allow up to 88 and 130 degrees of motion, respectively, while a human arm can rotate 360 degrees at the shoulder and move through 165 degrees at the elbow. The heavy build of the arm bones, strength of the muscles, and limited range of motion may indicate a system evolved to hold fast despite the stresses of a struggling prey animal. In the first detailed scientific description of *Tyrannosaurus* forelimbs, paleontologists Kenneth Carpenter and Matt Smith dismissed notions that the forelimbs were useless or that *Tyrannosaurus* was an obligate scavenger.[146]

The idea that the arms served as weapons when hunting prey have also been proposed by Steven M. Stanley, who suggested that the arms were used for slashing prey, especially by using the claws to rapidly inflict long, deep gashes to its prey.[147] This was dismissed by Padian, who argued that Stanley based his conclusion on incorrectly estimated forelimb size and range of motion.[141]

Thermoregulation

Tyrannosaurus, like most dinosaurs, was long thought to have an ectothermic ("cold-blooded") reptilian metabolism. The idea of dinosaur ectothermy was challenged by scientists like Robert T. Bakker and John Ostrom in the early years of the "dinosaur renaissance", beginning in the late 1960s.[148][149] *T. rex* itself

Restoration showing partial feathering

was claimed to have been endothermic ("warm-blooded"), implying a very active lifestyle.[36] Since then, several paleontologists have sought to determine the ability of *Tyrannosaurus* to regulate its body temperature. Histological evidence of high growth rates in young *T. rex*, comparable to those of mammals and birds, may support the hypothesis of a high metabolism. Growth curves indicate that, as in mammals and birds, *T. rex* growth was limited mostly to immature animals, rather than the indeterminate growth seen in most other vertebrates.[112]

Oxygen isotope ratios in fossilized bone are sometimes used to determine the temperature at which the bone was deposited, as the ratio between certain isotopes correlates with temperature. In one specimen, the isotope ratios in bones from different parts of the body indicated a temperature difference of no more than 4 to 5 °C (7 to 9 °F) between the vertebrae of the torso and the tibia of the lower leg. This small temperature range between the body core and the extremities was claimed by paleontologist Reese Barrick and geochemist William Showers to indicate that *T. rex* maintained a constant internal body temperature (homeothermy) and that it enjoyed a metabolism somewhere between ectothermic reptiles and endothermic mammals.[150] Other scientists have pointed out that the ratio of oxygen isotopes in the fossils today does not necessarily represent the same ratio in the distant past, and may have been altered during or after fossilization (diagenesis).[151] Barrick and Showers have defended their conclusions in subsequent papers, finding similar results in another theropod dinosaur from a different continent and tens of millions of years earlier in time (*Giganotosaurus*).[152] Ornithischian dinosaurs also showed evidence of homeothermy, while varanid lizards from the same formation did not.[153] In 2022, Wiemann and colleagues used a different approach—the spectroscopy of lipoxidation signals, which are byproducts of oxidative phosphorylation and correlate with metabolic rates—to show that various dinosaur genera including *Tyrannosaurus* had endothermic metabolisms, on par with that of modern birds and higher than that of mammals. They also suggested that such a metabolism was ancestrally common to all dinosaurs.[154]

Even if *T. rex* does exhibit evidence of homeothermy, it does not necessarily mean that it was endothermic. Such thermoregulation may also be explained by gigantothermy, as in some living sea turtles.[155][156][157] Similar to contemporary crocodilians, openings (dorsotemporal fenestrae) in the skull roofs of *Tyrannosaurus* may have aided thermoregulation.[158] Isotopic analysis found *T. rex* had a body temperature of 36.3 °C (97.3 °F) ± 2.5 °C, suggesting it had an endothermic metabolism. Its endothermic metabolism suggests it could have inhabited cooler, higher latitude environments.[159]

Soft tissue

In the March 2005 issue of *Science*, Mary Higby Schweitzer of North Carolina State University and colleagues announced the recovery of soft tissue from the marrow cavity of a fossilized leg bone from a *T. rex*. The bone had been intentionally, though reluctantly, broken for shipping and then not preserved in the normal manner, specifically because Schweitzer was hoping to test it for soft tissue.[160] Designated as the Museum of the Rockies specimen 1125, or MOR 1125, the dinosaur was previously excavated from the Hell Creek Formation. Flexible, bifurcating blood vessels and fibrous but elastic bone matrix tissue were recognized. In addition, microstructures resembling blood cells were found inside the matrix and vessels. The structures bear resemblance to ostrich blood cells and vessels. Whether an unknown process, distinct from normal fossilization, preserved the material, or the material is original, the

researchers do not know, and they are careful not to make any claims about preservation.[161] If it is found to be original material, any surviving proteins may be used as a means of indirectly guessing some of the DNA content of the dinosaurs involved, because each protein is typically created by a specific gene. The absence of previous finds may be the result of people assuming preserved tissue was impossible, therefore not looking. Since the first, two more tyrannosaurs and a hadrosaur have also been found to have such tissue-like structures.[160] Research on some of the tissues involved has suggested that birds are closer relatives to tyrannosaurs than other modern animals.[162] The original endogenous chemistry was also found in MOR 1125 based on preservation of elements associated with bone remodeling and redeposition (sulfur, calcium, zinc), which showed that the bone cortices are similar to those of extant birds.[163]

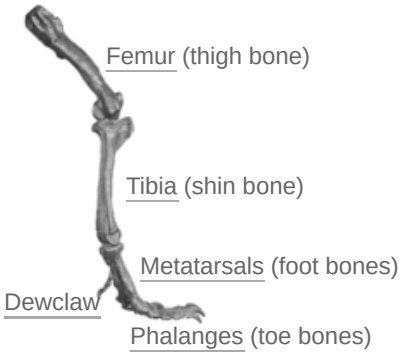
T. rex femur (MOR 1125) from which demineralized matrix and peptides (insets) were obtained

In studies reported in *Science* in April 2007, Asara and colleagues concluded that seven traces of collagen proteins detected in purified *T. rex* bone most closely match those reported in chickens, followed by frogs and newts. The discovery of proteins from a creature tens of millions of years old, along with similar traces the team found in a mastodon bone at least 160,000 years old, upends the conventional view of fossils and may shift paleontologists' focus from bone hunting to biochemistry. Until these finds, most scientists presumed that fossilization replaced all living tissue with inert minerals. Paleontologist Hans Larsson of McGill University in Montreal, who was not part of the studies, called the finds "a milestone", and suggested that dinosaurs could "enter the field of molecular biology and really slingshot paleontology into the modern world".[164]

The presumed soft tissue was called into question by Thomas Kaye of the University of Washington and his co-authors in 2008. They contend that what was really inside the tyrannosaur bone was slimy biofilm created by bacteria that coated the voids once occupied by blood vessels and cells.[165] The researchers found that what previously had been identified as remnants of blood cells, because of the presence of iron, were actually framboids, microscopic mineral spheres bearing iron. They found similar spheres in a variety of other fossils from various periods, including an ammonite. In the ammonite, they found the spheres in a place where the iron they contain could not have had any relationship to the presence of blood.[166] Schweitzer has strongly criticized Kaye's claims and argues that there is no reported evidence that biofilms can produce branching, hollow tubes like those noted in her study.[167] San Antonio, Schweitzer and colleagues published an analysis in 2011 of what parts of the collagen had been recovered, finding that it was the inner parts of the collagen coil that had been preserved, as would have been expected from a long period of protein degradation.[168] Other research challenges the identification of soft tissue as biofilm and confirms finding "branching, vessel-like structures" from within fossilized bone.[169]

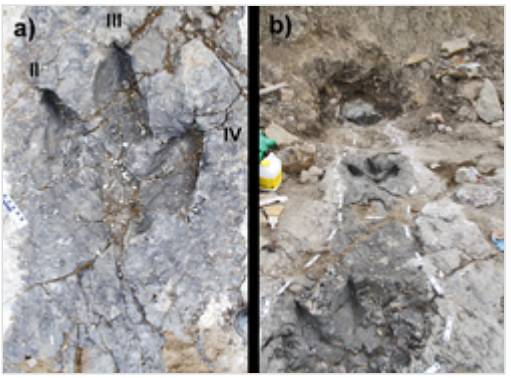
Speed

Scientists have produced a wide range of possible maximum running speeds for *Tyrannosaurus*: mostly around 9 meters per second (32 km/h; 20 mph), but as low as 4.5–6.8 meters per second (16–24 km/h; 10–15 mph) and as high as 20 meters per second (72 km/h; 45 mph). According to John Hutchinson and colleagues, *Tyrannosaurus* was unlikely to run fast .[170]



Skeletal anatomy of a *T. rex* right leg

A 2002 report used a mathematical model (validated by applying it to three living animals: alligators, chickens, and humans; and eight more species, including emus and ostriches[171]) to gauge the leg muscle mass needed for fast running (over 40 km/h or 25 mph).[170] Scientists who think that *Tyrannosaurus* was able to run point out that hollow bones and other features that would have lightened its body may have kept adult weight to a mere 4.5 metric tons (5.0 short tons) or so, or that other animals like ostriches and horses with long, flexible legs are able to achieve high speeds through slower but longer strides.[171] Proposed top speeds exceeded 40 kilometers per hour (25 mph) for *Tyrannosaurus*, but were deemed infeasible because they would require exceptional leg muscles of approximately 40–86% of total body mass. Even moderately fast speeds would have required large leg muscles. If the muscle mass was less, only 18 kilometers per hour (11 mph) for walking or jogging would have been possible.[170] Holtz noted that tyrannosaurids and some closely related groups had significantly longer distal hindlimb components (shin plus foot plus toes) relative to the femur length than most other theropods, and that tyrannosaurids and their close relatives had a tightly interlocked metatarsus (foot bones).[172] The third metatarsal was squeezed between the second and fourth metatarsals to form a single unit called an arctometatarsus. This ankle feature may have helped the animal to run more efficiently.[173] Together, these leg features allowed *Tyrannosaurus* to transmit locomotory forces from the foot to the lower leg more effectively than in earlier theropods.[172]



Only known tyrannosaurid trackway (*Bellatoripes fredlundi*), from the Wapiti Formation, British Columbia

Additionally, a 2020 study indicates that *Tyrannosaurus* and other tyrannosaurids were exceptionally efficient walkers. Studies by Dececchi et al., compared the leg proportions, body mass, and the gaits of more than 70 species of theropod dinosaurs including *Tyrannosaurus* and its relatives. The research team then applied a variety of methods to estimate each dinosaur's top speed when running as well as how much energy each dinosaur expended while moving at more relaxed speeds such as when walking. Among smaller to medium-sized species such as dromaeosaurids, longer legs appear to be an adaptation for faster running, in line with previous results by other researchers. But for theropods weighing over 1,000 kg (2,200 lb), top running speed is limited by body size, so longer legs instead were found to have correlated with low-energy walking. The results further indicate that smaller theropods evolved long legs as a means to both aid in hunting and escape from larger predators while larger theropods that evolved long legs did so to reduce the energy costs and increase foraging efficiency, as they were freed from the demands of predation pressure due to their role as apex predators. Compared to more basal groups of theropods in the study, tyrannosaurs like *Tyrannosaurus* itself showed a marked increase in foraging efficiency due to reduced energy expenditures during hunting or scavenging. This in turn likely resulted in tyrannosaurs having a reduced need for hunting forays and requiring less food to sustain themselves as a result. Additionally, the research, in conjunction with studies that show tyrannosaurs were more agile than other large-bodied theropods, indicates they were quite well-adapted to a long-distance stalking approach

followed by a quick burst of speed to go for the kill. Analogies can be noted between tyrannosaurids and modern wolves as a result, supported by evidence that at least some tyrannosaurids were hunting in group settings.[174][175]

A study published in 2021 by Pasha van Bijlert et al., calculated the preferred walking speed of *Tyrannosaurus*, reporting a speed of 1.28 meters per second (4.6 km/h; 2.9 mph). While walking, animals reduce their energy expenditure by choosing certain step rhythms at which their body parts resonate. The same would have been true for dinosaurs, but previous studies did not fully account for the impact the tail had on their walking speeds. According to the authors, when a dinosaur walked, its tail would slightly sway up and down with each step as a result of the interspinous ligaments suspending the tail. Like rubber bands, these ligaments stored energy when they are stretched due to the swaying of the tail. Using a 3-D model of *Tyrannosaurus* specimen Trix, muscles and ligaments were reconstructed to simulate the tail movements. This results in a rhythmic, energy-efficient walking speed for *Tyrannosaurus* similar to that seen in living animals such as humans, ostriches and giraffes.[176]

A 2017 study estimated the top running speed of *Tyrannosaurus* as 17 mph (27 km/h), speculating that *Tyrannosaurus* exhausted its energy reserves long before reaching top speed, resulting in a parabola-like relationship between size and speed.[177][178] Another 2017 study hypothesized that an adult *Tyrannosaurus* was incapable of running due to high skeletal loads. Using a calculated weight estimate of 7 tons, the model showed that speeds above 11 mph (18 km/h) would have probably shattered the leg bones of *Tyrannosaurus*. The finding may mean that running was also not possible for other giant theropod dinosaurs like *Giganotosaurus*, *Mapusaurus* and *Acrocanthosaurus*.[179] However, studies by Eric Snively and colleagues, published in 2019 indicate that *Tyrannosaurus* and other tyrannosaurids were more maneuverable than allosauroids and other theropods of comparable size due to low rotational inertia compared to their body mass combined with large leg muscles. As a result, it is hypothesized that *Tyrannosaurus* was capable of making relatively quick turns and could likely pivot its body more quickly when close to its prey, or that while turning, the theropod could "pirouette" on a single planted foot while the alternating leg was held out in a suspended swing during a pursuit. The results of this study potentially could shed light on how agility could have contributed to the success of tyrannosaurid evolution.[180] In 2026, Boeye and colleagues analyzed the foot biomechanics of *T. rex* and determined that its foot likely functioned similarly to modern birds. This may have resulted in shorter stride lengths, higher stride frequencies, and a somewhat elevated top speed.[17]

Possible footprints

Two isolated fossilized footprints have been tentatively assigned to *T. rex*. The first was discovered at Philmont Scout Ranch, New Mexico, in 1983 by American geologist Charles Pillmore. Originally thought to belong to a hadrosaurid, examination of the footprint revealed a large 'heel' unknown in ornithopod dinosaur tracks, and traces of what may have been a hallux, the dewclaw-like fourth digit of the tyrannosaur foot. The footprint was published as the ichnogenus *Tyrannosauripus pillmorei* in 1994, by Martin Lockley and Adrian Hunt. Lockley and Hunt suggested that it was very likely the track was made by a *T. rex*, which would

Probable footprint from New Mexico

make it the first known footprint from this species. The track was made in what was once a vegetated wetland mudflat. It measures 83 centimeters (33 in) long by 71 centimeters (28 in) wide.[181]

A second footprint that may have been made by a *Tyrannosaurus* was first reported in 2007 by British paleontologist Phil Manning, from the Hell Creek Formation of Montana. This second track measures 72 centimeters (28 in) long, shorter than the track described by Lockley and Hunt. Whether or not the track was made by *Tyrannosaurus* is unclear, though *Tyrannosaurus* is the only large theropod known to have existed in the Hell Creek Formation.[182][183]

A set of footprints in Glenrock, Wyoming dating to the Maastrichtian stage of the Late Cretaceous and hailing from the Lance Formation were described by Scott Persons, Phil Currie and colleagues in 2016, and are believed to belong to either a juvenile *T. rex* or *Nanotyrannus lancensis*. From measurements and based on the positions of the footprints, the animal was believed to be traveling at a walking speed of around 2.8 to 5 miles per hour and was estimated to have a hip height of 1.56 to 2.06 m (5.1 to 6.8 ft).[184][185][186] A follow-up paper appeared in 2017, increasing the speed estimations by 50–80%.[187]

Rare fossil footprints and trackways found in New Mexico and Wyoming that are assigned to the ichnogenus *Tyrannosauripus* have been attributed to being made by *Tyrannosaurus*, based on the stratigraphic age of the rocks they are preserved in. The first specimen, found in 1994 was described by Lockley and Hunt and consists of a single, large footprint. Another pair of ichnofossils, described in 2021, show a large tyrannosaurid rising from a prone position by rising up using its elbows in conjunction with the pads on their feet to stand. These two unique sets of fossils were found in Ludlow, Colorado and Cimarron, New Mexico.[188] Another ichnofossil described in 2018, perhaps belonging to a juvenile *Tyrannosaurus* or *Nanotyrannus* was uncovered in the Lance Formation of Wyoming. The trackway itself offers a rare glimpse into the walking speed of tyrannosaurids, and the trackmaker is estimated to have been moving at a speed of 4.5–8.0 kilometers per hour (2.8–5.0 mph), significantly faster than previously assumed for estimations of walking speed in tyrannosaurids.[189]

A trackway consisting of four footprints discovered near Marmarth, North Dakota in 2025 and described in 2026 have been attributed to an adult *Tyrannosaurus* based on the shape and size of the tracks, which measure 1 m (3.3 ft). The set of tracks suggests a stride length of 4 m (13 ft) for the animal that made them, with speed estimates of 1.5–2 meters per second (3.4-4.5 mph). The fluvial environment of the upper Hell Creek Formation, from which the footprints were unearthed may explain the significant scarcity of trackways and the uncommon nature of theropod footprints in the formation in general. Three of the footprints are expected to be recovered in autumn of 2026 to be displayed at the Denver Museum of Nature and Science.[190][191][192]

Brain and senses

A study conducted by Lawrence Witmer and Ryan Ridgely of Ohio University found that *Tyrannosaurus* shared the heightened sensory abilities of other coelurosaurs, highlighting relatively rapid and coordinated eye and head movements; an enhanced ability to sense low frequency sounds, which would allow tyrannosaurs to track prey movements from long distances; and an enhanced sense of smell.[193] A study published by Kent Stevens concluded that *Tyrannosaurus* had keen vision. By applying modified perimetry to facial reconstructions of several dinosaurs including *Tyrannosaurus*, the study found that *Tyrannosaurus* had a binocular range of 55 degrees, surpassing that of modern hawks. Stevens estimated that *Tyrannosaurus* had 13 times the visual acuity of a human and surpassed the visual acuity of an eagle,

The eye-sockets faced mainly forwards, giving it good binocular vision (reconstructed skull of the Sue specimen; the original was partly crushed and broken).

which is 3.6 times that of a person. Stevens estimated a limiting far point (that is, the distance at which an object can be seen as separate from the horizon) as far as 6 km (3.7 mi) away, which is greater than the 1.6 km (1 mi) that a human can see.[43][44][194]

Thomas Holtz Jr. would note that high depth perception of *Tyrannosaurus* may have been due to the prey it had to hunt, noting that it had to hunt ceratopsians such as *Triceratops*, ankylosaurs such as *Ankylosaurus*, and hadrosaurs. He would suggest that this made precision more crucial for *Tyrannosaurus* enabling it to, "get in, get that blow in and take it down." In contrast, *Acrocanthosaurus* had limited depth perception because they hunted large sauropods, which were relatively rare during the time of *Tyrannosaurus*.[121]

Though no *Tyrannosaurus* scleral ring has been found, Kenneth Carpenter estimated its size based on that of *Gorgosaurus*. The inferred scleral ring for the Stan specimen is ~7 cm (2.8 in) in diameter with an internal aperture diameter of ~3.5 cm (1.4 in). Based on eye proportions in living reptiles, this implies a pupil diameter of about 2.5 cm (0.98 in), an iris diameter about that of the scleral ring, and an eyeball diameter of 11–12 cm (4.3–4.7 in). Carpenter also estimated an eyeball depth of ~7.7–9.6 cm (3.0–3.8 in). Based on these calculations, the f-number for Stan's eye is 3–3.8; since diurnal animals have f-numbers of 2.1 or higher, this would indicate that *Tyrannosaurus* had poor low-light vision and hunted during the day.[195]

Tyrannosaurus had very large olfactory bulbs and olfactory nerves relative to their brain size, the organs responsible for a heightened sense of smell. This suggests that the sense of smell was highly developed, and implies that tyrannosaurs could detect carcasses by scent alone across great distances. The sense of smell in tyrannosaurs may have been comparable to modern vultures, which use scent to track carcasses for scavenging. Research on the olfactory bulbs has shown that *T. rex* had the most highly developed sense of smell of 21 sampled non-avian dinosaur species.[196]

Cast of the braincase at the Australian Museum, Sydney

Somewhat unusually among theropods, *T. rex* had a very long cochlea. The length of the cochlea is often related to hearing acuity, or at least the importance of hearing in behavior, implying that hearing was a particularly important sense to tyrannosaurs. Specifically, data suggests that *T. rex* heard best in the low-frequency range, and that low-frequency sounds were an important part of tyrannosaur behavior.[193] A 2017 study by Thomas Carr and colleagues found that the snout of tyrannosaurids was highly sensitive, based on a high number of small openings in the facial bones of the related *Daspletosaurus* that contained sensory neurons. The study speculated that tyrannosaurs might have used their sensitive snouts to measure the temperature of their nests and to gently pick up eggs and hatchlings, as seen in modern crocodilians.[52] Another study published in 2021 further suggests that *Tyrannosaurus* had an acute sense of touch, based on neurovascular canals in the front of its jaws, which it could utilize to better detect and consume prey. The

study, published by Kawabe and Hittori et al., suggests that *Tyrannosaurus* could also accurately sense slight differences in material and movement, allowing it to utilize different feeding strategies on different parts of its prey's carcasses depending on the situation. The sensitive neurovascular canals of *Tyrannosaurus* also likely were adapted to performing fine movements and behaviors such as nest building, parental care, and other social behavior such as intraspecific communication. The results of this study also align with results made in studying the related tyrannosaurid *Daspletosaurus horneri* and the allosauroid *Neovenator*, which have similar neurovascular adaptations, suggesting that the faces of theropods were highly sensitive to pressure and touch.[197][198] However, a more recent study reviewing the evolution of the trigeminal canals among sauropsids notes that a much denser network of neurovascular canals in the snout and lower jaw is more commonly encountered in aquatic or semiaquatic taxa (e.g., *Spinosaurus*, *Halszkaraptor*, *Plesiosaurus*), and taxa that developed a rhamphotheca (e.g., *Caenagnathasia*), while the network of canals in *Tyrannosaurus* appears simpler, though still more derived than in most ornithischians, and overall terrestrial taxa such as tyrannosaurids and *Neovenator* may have had average facial sensitivity for non-edentulous terrestrial theropods, although further research is needed. The neurovascular canals in *Tyrannosaurus* may instead have supported soft tissue structures for thermoregulation or social signaling, the latter of which could be confirmed by the fact that the neurovascular network of canals may have changed during ontogeny.[199]

A study by Grant R. Hurlburt, Ryan C. Ridgely and Lawrence Witmer obtained estimates for Encephalization Quotients (EQs), based on reptiles and birds, as well as estimates for the ratio of cerebrum to brain mass. The study concluded that *Tyrannosaurus* had the relatively largest brain of all adult non-avian dinosaurs with the exception of certain small maniraptoriforms (*Bambiraptor*, *Troodon* and *Ornithomimus*). The study found that *Tyrannosaurus*'s relative brain size was still within the range of modern reptiles, being at most 2 standard deviations above the mean of non-avian reptile EQs. The estimates for the ratio of cerebrum mass to brain mass would range from 47.5 to 49.53 percent. According to the study, this is more than the lowest estimates for extant birds (44.6 percent), but still close to the typical ratios of the smallest sexually mature alligators which range from 45.9 to 47.9 percent.[200] Other studies, such as those by Steve Brusatte, indicate the encephalization quotient of *Tyrannosaurus* was similar in range (2.0–2.4) to a chimpanzee (2.2–2.5), though this may be debatable as reptilian and mammalian encephalization quotients are not equivalent.[201]

Social behavior

Mounted skeletons of different age groups (skeleton in lower left based on *Stygivenator*), Natural History Museum of Los Angeles County

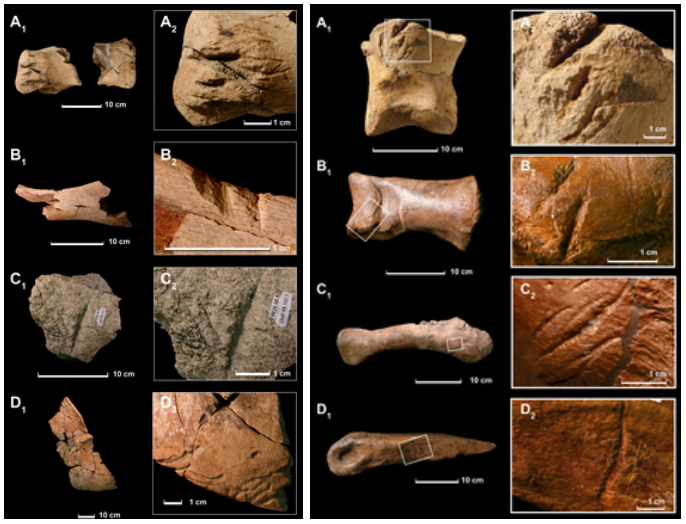
Philip J. Currie suggested that Tyrannosaurus may have been pack hunters, comparing *T. rex* to related species *Tarbosaurus bataar* and *Albertosaurus sarcophagus*, citing fossil evidence that may indicate gregarious (describing animals that travel in herds or packs) behavior.[202] A find in South Dakota where three *T. rex* skeletons were in close proximity may suggest the formation of a pack.[203][204] Cooperative pack hunting may have been an effective strategy for subduing prey with advanced anti-predator adaptations which pose potential lethality such as *Triceratops* and *Ankylosaurus*.[202]

Currie's pack-hunting *T. rex* hypothesis has been criticized for not having been peer-reviewed, but rather was discussed in a television interview and book called *Dino Gangs*.[205] The Currie theory for pack hunting by *T. rex* is based mainly by analogy to a different species, *Tarbosaurus bataar*. Evidence of

gregariousness in *T. bataar* itself has not been peer-reviewed, and to Currie's own admission, can only be interpreted with reference to evidence in other closely related species. According to Currie gregariousness in *Albertosaurus sarcophagus* is supported by the discovery of 26 individuals with varied ages in the Dry Island bonebed. He ruled out the possibility of a predator trap due to the similar preservation state of individuals and the near absence of herbivores.[205][206]

Additional support of tyrannosaurid gregariousness can be found in fossilized trackways from the Upper Cretaceous Wapiti Formation of northeastern British Columbia, Canada, left by three tyrannosaurids traveling in the same direction.[207][208] According to scientists assessing the Dino Gangs program, the evidence for pack hunting in *Tarbosaurus* and *Albertosaurus* is weak and based on group skeletal remains for which alternate explanations may apply (such as drought or a flood forcing dinosaurs to die together in one place).[205] Others researchers have speculated that instead of large theropod social groups, some of these finds represent behavior more akin to Komodo dragon-like mobbing of carcasses, even going as far as to say true pack-hunting behavior may not exist in any non-avian dinosaurs due to its rarity in modern predators.[209] Pathologies of *Tyrannosaurus* specimens have been suggested as evidence of conspecific attack, including "Wyrex" with a hole penetrating its jugal and severe trauma on its tail that shows signs of bone remodeling (not regrowth).[210][211]

Diet and feeding behavior



Tyrannosaurus tooth marks on bones of various herbivorous dinosaurs (left) as well as other *Tyrannosaurus* (right)

Most paleontologists have always accepted that *Tyrannosaurus* was an active predator and, like most large carnivores, would also opportunistically scavenge.[212][213] By far the largest carnivore in its environment, *T. rex* was most likely an apex predator, preying upon hadrosaurs, armored herbivores like ceratopsians and ankylosaurs, and possibly sauropods (such as *Alamosaurus*).[214] Enamel $\delta^{44/42}$Ca values also suggest the possibility that *T. rex* occasionally fed on carcasses of marine reptiles and fish washed up on the shores of the Western Interior Seaway.[215] A study in 2012 by Karl Bates and Peter Falkingham found that *Tyrannosaurus* had the most powerful bite of any terrestrial animal that has ever lived, finding an adult *Tyrannosaurus* could have exerted 35,000 to 57,000 N (7,868 to 12,814 lbf) of force in the back teeth.[216][217][218] Even higher estimates were made by Mason B. Meers in 2003.[46] This allowed it to crush bones during repetitive biting and fully consume the carcasses of large

dinosaurs.[25] Stephan Lautenschlager and colleagues calculated that *Tyrannosaurus* was capable of a maximum jaw gape of around 80 degrees, a necessary adaptation for a wide range of jaw angles to power the creature's strong bite.[219][220]

A discredited fringe hypothesis that was never seriously considered was that *Tyrannosaurus* was an obligate scavenger. The argument originated in a 1917 study by Lambe which argued that large theropods were pure scavengers because *Gorgosaurus* teeth showed hardly any wear.[221] This argument disregarded the fact that theropods rapidly replaced their teeth. Ever since the first discovery of *Tyrannosaurus* most scientists have agreed that it was an active predator, though like modern large predators it would readily scavenge or steal another predator's kill if it had the opportunity.[222][213]

Paleontologist Jack Horner has been a major proponent of the view that *Tyrannosaurus* was not a predator at all but instead was an obligate scavenger.[144][223][224] He has put forward arguments in the popular literature to support the pure scavenger hypothesis:

A *Tyrannosaurus* mounted next to a *Triceratops* at the Los Angeles Natural History Museum

- Tyrannosaur arms are short when compared to other known predators. Horner argues that the arms were too short to make the necessary gripping force to hold on to prey.[224] Other paleontologists such as Thomas Holtz Jr. pointed out that there are plenty of modern-day predators that do not use their forelimbs to hunt such as wolves, hyenas, and secretary birds as well as other extinct animals thought to be predators that would not have used their forelimbs such as phorusrhacids.[225][226]
- Tyrannosaurs had large olfactory bulbs and olfactory nerves (relative to their brain size). These suggest a highly developed sense of smell which could sniff out carcasses over great distances, as modern vultures do. Research on the olfactory bulbs of dinosaurs has shown that *Tyrannosaurus* had the most highly developed sense of smell of 21 sampled dinosaurs.[196] However, good olfactory capabilities are also common in predatory animals, so this fails to provide evidence against predatory behavior.[213]
- Tyrannosaur teeth could crush bone, and therefore could extract as much food (bone marrow) as possible from carcass remnants, usually the least nutritious parts. Karen Chin and colleagues have found bone fragments in coprolites (fossilized feces) that they attribute to tyrannosaurs, but point out that a tyrannosaur's teeth were not well adapted to systematically chewing bone like hyenas do to extract marrow.[227]
- Since at least some of *Tyrannosaurus*'s potential prey could move quickly, evidence that it walked instead of ran could indicate that it was a scavenger.[223] However, most analyses indicate *Tyrannosaurus*, while slower than large modern terrestrial predators due to being far larger, was still faster than large hadrosaurs and ceratopsians.[170][183][213]

Most evidence suggests hunting behavior in *Tyrannosaurus*. The eye sockets of tyrannosaurs are positioned so that the eyes would point forward, giving them binocular vision slightly better than that of modern hawks. It is not obvious why natural selection would have favored this long-term trend if tyrannosaurs had been pure scavengers, which would not have needed the advanced depth perception that stereoscopic vision provides.[43][44] In modern animals, binocular vision is found mainly in predators (though not all predators have well-developed binocular vision).

There is at least one and possibly two fossil specimens of the hadrosaur *Edmontosaurus annectens* with healed caudal injuries caused by *Tyrannosaurus* bites made while the herbivore was alive. One pair of *Edmontosaurus* caudal (tail) vertebrae has the tip of an adult *Tyrannosaurus* tooth embedded in the bone, with evidence of new bone growth that wrapped around the tooth, showing that this individual had also survived a *Tyrannosaurus* attack.[228][229][230][231][232] Another specimen of *Edmontosaurus* (in some newspapers it was misidentified as *Hadrosaurus*) from Montana showed healed supposed tyrannosaur-inflicted damage on its tail vertebrae where some of its caudal neural spines was mutilated;[233] however, in this case the damage is ambiguous and not directly attributable to *Tyrannosaurus*.[228] There is also evidence for an aggressive interaction between a *Triceratops* and a *Tyrannosaurus* in the form of partially healed tyrannosaur tooth marks on a *Triceratops* brow horn and squamosal (a bone of the neck frill); the bitten horn is also broken, with new bone growth after the break. It is not known what the exact nature of the interaction was, though: either animal could have been the aggressor.[234] Since the *Triceratops* wounds healed, it is most likely that the *Triceratops* survived the encounter and managed to overcome the *Tyrannosaurus*. In a battle against a bull *Triceratops*, the *Triceratops* would likely defend itself by inflicting fatal wounds to the *Tyrannosaurus* using its sharp horns.[235] Studies of Sue found a broken and healed fibula and tail vertebrae, scarred facial bones and a tooth from another *Tyrannosaurus* embedded in a neck vertebra, providing evidence for aggressive behavior.[236]

The damage to the tail vertebrae of this *Edmontosaurus annectens* skeleton (on display at the Denver Museum of Nature and Science) indicates that it may have been bitten by a *Tyrannosaurus*

In 1992, William Abler suggested that *Tyrannosaurus* may have had infectious saliva used to kill its prey. Abler observed that the serrations (tiny protuberances) on the cutting edges of the teeth are closely spaced, enclosing little chambers, which he argued trapped pieces of carcass with bacteria, giving *Tyrannosaurus* a deadly, infectious bite, as the Komodo dragon was mistakenly thought to have.[237] Jack Horner and Don Lessem, in a 1993 popular book, questioned Abler's hypothesis, arguing that *Tyrannosaurus*'s tooth serrations as more like cubes in shape than the serrations on a Komodo monitor's teeth, which are rounded.[144]:214–215 The basis for an infectious bite in *Tyrannosaurus* was further undermined when it was found Komodo dragons do not actually have an infectious bite nor hunt by biting prey and tracking it as it weakens.[238]

Tyrannosaurus probably primarily processed carcasses with lateral shakes of the head, like crocodilians. The head was not as maneuverable as the skulls of allosauroids, due to flat joints of the neck vertebrae.[239][213] Studies on hadrosaur vertebrae from the Hell Creek Formation that were punctured by the teeth of what appears to be a late-stage juvenile *Tyrannosaurus* indicate that despite lacking the bone-crushing adaptations of the adults, young individuals were still capable of using the same bone-puncturing feeding technique.[240]

One study of *Tyrannosaurus* specimens with tooth marks in the bones attributable to the same genus was presented as evidence of cannibalism.[241] Tooth marks in the humerus, foot bones and metatarsals may indicate opportunistic scavenging, rather than wounds caused by combat.[241][242] Other tyrannosaurids may also have practiced cannibalism.[241]

Parenting

While there is no direct evidence of *Tyrannosaurus* raising their young (given the rarity of juvenile and nest Tyrannosaur fossils), it has been suggested that like its closest living relatives, birds and crocodiles, *Tyrannosaurus* may have protected and fed its young. Crocodilians and birds are often suggested by some paleontologists to be modern analogues for dinosaur parenting.[243] Direct evidence of parental behavior exists in other dinosaurs such as *Maiasaura peeblesorum*, the first dinosaur to have been discovered to raise its young, as well as more closely related Oviraptorids, the latter suggesting parental behavior in theropods.[244][245][246][247][248] Some very young specimens assigned to *Tyrannosaurus rex* suggest that, although the species may have provided parental care, it was likely limited.[120]

Pathology

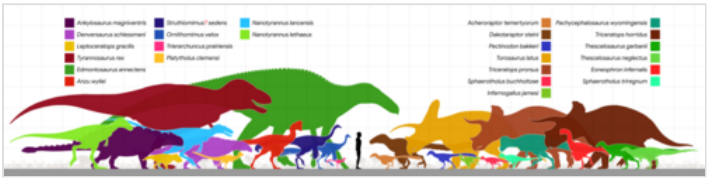
Jaw bones (top) and life restoration showing injuries caused by either parasite infections or bites

In 2001, Bruce Rothschild and others published a study examining evidence for stress fractures and tendon avulsions in theropod dinosaurs and the implications for their behavior. Since stress fractures are caused by repeated trauma rather than singular events they are more likely to be caused by regular behavior than other types of injuries. Of the 81 *Tyrannosaurus* foot bones examined in the study, one was found to have a stress fracture, while none of the 10 hand bones were found to have stress fractures. The researchers found tendon avulsions only among *Tyrannosaurus* and *Allosaurus*. An avulsion injury left a divot on the humerus of Sue the *T. rex*, apparently located at the origin of the deltoid or teres major muscles. The presence of avulsion injuries being limited to the forelimb and shoulder in both *Tyrannosaurus* and *Allosaurus* suggests that theropods may have had a musculature more complex than and functionally different from those of birds. The researchers concluded that Sue's tendon avulsion was probably obtained from struggling prey. The presence of stress fractures and tendon avulsions, in general, provides evidence for a "very active" predation-based diet rather than obligate scavenging.[249]

A 2009 study showed that smooth-edged holes in the skulls of several specimens might have been caused by *Trichomonas*-like parasites that commonly infect birds. According to the study, seriously infected individuals, including "Sue" and MOR 980 ("Peck's Rex"), might therefore have died from starvation

after feeding became increasingly difficult. Previously, these holes had been explained by the bacterious bone infection Actinomycosis or by intraspecific attacks.[250] A subsequent study found that while trichomoniasis has many attributes of the model proposed, the observed sharp margins with little reactive bone shown by the radiographs of *Trichomonas*-infected birds are dissimilar to the reactive bone seen in the affected *T. rex* specimens. Also, trichomoniasis can be very rapidly fatal in birds (14 days or less) albeit in its milder form, and this suggests that if a *Trichomonas*-like protozoan is the culprit, trichomoniasis was less acute in its non-avian dinosaur form during the Late Cretaceous. Finally, the relative size of this type of lesions is much larger in small bird throats, and may not have been enough to choke a *T. rex*.[251] A more recent study examining the pathologies concluded that the osseous alteration observed most closely resembles those around healing human cranial trepanations and healing fractures in the Triassic reptile *Stagonolepis*, in the absence of infection. The possible cause may instead have been intraspecific combat.[252]

Paleoecology



Fauna of Hell Creek (*Tyrannosaurus* in dark red, left).

Tyrannosaurus lived during what is referred to as the Lancian faunal stage (Maastrichtian age) at the end of the Late Cretaceous. *Tyrannosaurus* ranged from Canada in the north to at least New Mexico in the south of Laramidia.[7] Several notable *Tyrannosaurus* remains have been found in the Hell Creek Formation. During the Maastrichtian this area was maritime to subtropical, with a warm and humid climate.[253] The flora consisted mostly of angiosperms, but also included trees like dawn redwood (*Metasequoia*).[254] Archosaurs discovered from this formation include the ceratopsians *Leptoceratops*, *Torosaurus*, and *Triceratops*,[255] the hadrosaurid *Edmontosaurus annectens*,[256] the parksosaurid *Thescelosaurus*,[257] the ankylosaurs *Ankylosaurus* and *Denversaurus*,[258] the pachycephalosaurs *Pachycephalosaurus*, *Platytholus* and *Sphaerotholus*,[259][260] the paravian theropods *Acheroraptor* and *Pectinodon*,[261] other maniraptoran theropods *Anzu*, *Eoneophron*, *Trierarchuncus* and indeterminate ornithomimids,[262][263] the tyrannosauroid *Nanotyrannus*,[23] and the azhdarchid pterosaur *Infernodrakon*.[264]

Tyrannosaurus and other animals of the Hell Creek Formation

Another formation with *Tyrannosaurus* remains is the Lance Formation of Wyoming. This has been interpreted as a bayou environment similar to today's Gulf Coast. The fauna was very similar to Hell Creek, but with *Struthiomimus* replacing its relative *Ornithomimus*. The small ceratopsian *Leptoceratops* also lived in the area.[265]

In its southern range, specifically based on remains discovered from the North Horn Formation of Utah, *Tyrannosaurus rex* lived alongside the titanosaur *Utetitan*, the ceratopsid *Torosaurus* and the indeterminate troodontids and hadrosaurids.[266][267][268] *Tyrannosaurus mcraeensis* from the McRae Group of New Mexico coexisted with the ceratopsid *Sierraceratops* and possibly the titanosaur *Alamosaurus*.[65] Potential remains identified as cf. *Tyrannosaurus* have also been discovered from the Javelina Formation of Texas,[65]

where the remains of the titanosaur (either *Alamosaurus* or *Utetitan*),[268] the ceratopsid *Bravoceratops*, the pterosaurs *Quetzalcoatlus* and *Wellnhopterus*, and possible species of troodontids and hadrosaurids are found.[269][270][271] Its southern range is thought to have been dominated by semi-arid inland plains, following the probable retreat of the Western Interior Seaway as global sea levels fell.[272]

Tyrannosaurus may have also inhabited Mexico's Lomas Coloradas Formation in Sonora. Though skeletal evidence is lacking, six shed and broken teeth from the fossil bed have been thoroughly compared with other theropod genera and appear to be identical to those of *Tyrannosaurus*. If true, the evidence indicates the range of *Tyrannosaurus* was possibly more extensive than previously believed.[273] It is possible that tyrannosaurs were originally Asian species, migrating to North America before the end of the Cretaceous period.[274]

Population estimates

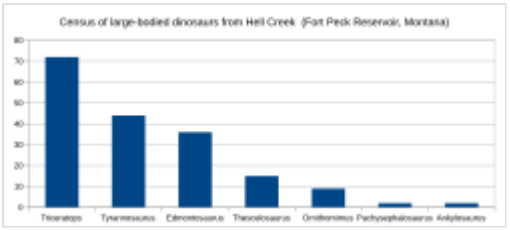


Chart of the time-averaged census for large-bodied dinosaurs from the entire Hell Creek Formation in the study area

According to studies published in 2021 by Charles Marshall et al., the total population of adult *Tyrannosaurus* at any given time was perhaps 20,000 individuals, with computer estimations also suggesting a total population no lower than 1,300 and no higher than 328,000. The authors themselves suggest that the estimate of 20,000 individuals is probably lower than what should be expected, especially when factoring in that disease pandemics could easily wipe out such a small population. Over the span of the genus' existence, it is estimated that there were about 127,000 generations, totalling roughly 2.5 billion animals.[275][276]

In the same paper, it is suggested that in a population of *Tyrannosaurus* adults numbering 20,000, the number of individuals living in an area the size of California could be as high as 3,800 animals, while an area the size of Washington D.C. could support a population of only two adult *Tyrannosaurus*. The study does not take into account the number of juvenile animals present due to their occupation of a different niche than the adults, and thus it is likely the total population was much higher. Simultaneously, studies of living carnivores suggest that some predator populations are higher in density than others of similar weight (such as jaguars and hyenas, which have vastly differing population densities). Lastly, the study suggests that in most cases, only one in 80 million *Tyrannosaurus* would become fossilized, while the chances were likely as high as one in every 16,000 of an individual becoming fossilized in areas that had more dense populations.[275][276]

Meiri (2022) questioned the reliability of the estimates, citing uncertainty in metabolic rate, body size, sex and age-specific survival rates, habitat requirements and range size variability as shortcomings Marshall et al. did not take into account.[277] The authors of the original publication replied that while they agree that their reported uncertainties were probably too small, their framework is flexible enough to accommodate uncerainty in physiology, and that their calculations do not depend on short-term changes but rather on long-term averages. Finally, they remark that they did estimate the range of reasonable survivorship curves and that they did include uncertainty in the time of onset of sexual maturity and in the growth curve by incorporating the uncertainty in the maximum body mass.[278]

Cultural significance

Tyrannosaurus is the most widely known dinosaur in popular culture. Science writer Riley Black states, "In all of prehistory, there is no animal that commands our attention quite like *Tyrannosaurus rex*... Since the time this dinosaur was officially named in 1905, the enormous carnivore has stood as the ultimate dinosaur."[279] Paleontologist David Hone notes the popularity of *Tyrannosaurus* models and skeletal displays in museums and writes that films like *Jurassic Park* and *King Kong* "would not have been the same without it."[280] *T. rex* is the only dinosaur that is commonly known to the general public by its full scientific name (binomial name) and the scientific abbreviation *T. rex* has also come into wide usage.[9]

King Kong fighting *Tyrannosaurus* in the 1933 film

See also

- History of paleontology

Notes

a. lit. 'tyrant lizard'; from Ancient Greek τύραννος *(túrannos)* 'tyrant' and σαῦρος *(saûros)* 'lizard'

References

1. "Time to Slay the T. rex Scavenger "Debate" " (https://www.nationalgeographic.com/science/article/time-to-slay-the-t-rex-scavenger-debate). *Science*. July 16, 2013. Retrieved September 1, 2026.
2. "The Colorado Transcript". July 8, 1874 – via www.coloradohistoricnewspapers.org.
3. Carpenter, K. (November 1, 2002). "Late Cretaceous dinosaurs from the Denver Basin, Colorado" (http://rmg.geoscienceworld.org/cgi/doi/10.2113/gsrocky.37.2.237). *Rocky Mountain Geology*. **37** (2): 237–254. doi:10.2113/gsrocky.37.2.237 (https://doi.org/10.2113%2Fgsrocky.37.2.237). ISSN 1555-7332 (https://search.worldcat.org/issn/1555-7332).
4. Breithaupt, B. H.; Southwell, E. H.; Matthews, N. A. (October 15, 2005). "In Celebration of 100 years of *Tyrannosaurus rex*: *Manospondylus gigas*, *Ornithomimus grandis*, and *Dynamosaurus imperiosus*, the Earliest Discoveries of *Tyrannosaurus rex* in the West" (http://gsa.confex.com/gsa/2005AM/finalprogram/abstract_96150.htm). *Abstracts with Programs; 2005 Salt Lake City Annual Meeting*. **37** (7). Geological Society of America: 406. ISSN 0016-7592 (https://search.worldcat.org/issn/0016-7592). Retrieved October 8, 2008.
5. Hatcher, J. B. (1907). "The Ceratopsia" (https://archive.org/stream/TheCeratopsia/Musgs-1907-hatcherEtAl-theCeratopsiaLkUsaPart1Monograph#page/n111). *Monographs of the United States Geological Survey*. **49**: 113–114. ISSN 0886-7550 (https://search.worldcat.org/issn/0886-7550).

6. Osborn, H. F. (1917). "Skeletal adaptations of *Ornitholestes*, *Struthiomimus*, *Tyrannosaurus*". *Bulletin of the American Museum of Natural History*. **35** (43): 733–771. hdl:2246/1334 (https://hdl.handle.net/2246%2F1334).
7. Larson, N. L. (2008). "One hundred years of *Tyrannosaurus rex*: the skeletons". In Larson, P.; Carpenter, K. (eds.). Tyrannosaurus rex*, The Tyrant King* (https://archive.org/details/tyrannosaurusrex00plar). Bloomington, IN: Indiana University Press. pp. 1 (https://archive.org/details/tyrannosaurusrex00plar/page/n16)–55. ISBN 978-0-253-35087-9.
8. McTavish, B. (July 10, 2000). "The dinosaur formerly known as Tyrannosaurus rex?" (https://www.tampabay.com/archive/2000/07/10/the-dinosaur-formerly-known-as-tyrannosaurus-rex/). Retrieved October 18, 2025.
9. Brochu, C. R. (2003). "Osteology of *Tyrannosaurus rex*: insights from a nearly complete skeleton and high-resolution computed tomographic analysis of the skull" (http://publication.plazi.org/id/6603FFA80B5D0D64FFF6AB533D39FF9A). *Society of Vertebrate Paleontology Memoirs*. **7**: 1–138. doi:10.2307/3889334 (https://doi.org/10.2307%2F3889334). JSTOR 3889334 (https://www.jstor.org/stable/3889334). Archived (https://web.archive.org/web/20250324135253/http://publication.plazi.org/id/6603FFA80B5D0D64FFF6AB533D39FF9A) from the original on March 24, 2025. Retrieved May 24, 2025.
10. "Article 23. Principle of Priority" (https://code.iczn.org/validity-of-names-and-nomenclatural-acts/article-23-principle-of-priority/?frame=1). *code.iczn.org*. Retrieved November 17, 2025.
11. Osborn, H. F. (1905). "*Tyrannosaurus* and other Cretaceous carnivorous dinosaurs". *Bulletin of the AMNH*. **21** (14): 259–265. hdl:2246/1464 (https://hdl.handle.net/2246%2F1464). Retrieved October 6, 2008.
12. Dingus, L.; Norell, M. (May 3, 2010). *Barnum Brown: The Man Who Discovered* Tyrannosaurus rex (https://archive.org/details/barnumbrownmanwh00ding). University of California Press. pp. 90 (https://archive.org/details/barnumbrownmanwh00ding/page/n106), 124. ISBN 978-0-520-94552-4.
13. Osborn, H. F.; Brown, B. (1906). "*Tyrannosaurus*, Upper Cretaceous carnivorous dinosaur". *Bulletin of the AMNH*. **22** (16): 281–296. hdl:2246/1473 (https://hdl.handle.net/2246%2F1473).
14. Breithaupt, B. H.; Southwell, E. H.; Matthews, N. A. (2006). Lucas, S. G.; Sullivan, R. M. (eds.). "*Dynamosaurus imperiosus* and the earliest discoveries of *Tyrannosaurus rex* in Wyoming and the West" (http://econtent.unm.edu/utils/getdownloaditem/collection/bulletins/id/693/filename/694.pdf/mapsto/pdf) (PDF). *New Mexico Museum of Natural History and Science Bulletin*. **35**: 258. Archived (https://web.archive.org/web/20190801022355/http://econtent.unm.edu/utils/getdownloaditem/collection/bulletins/id/693/filename/694.pdf/mapsto/pdf) from the original on August 1, 2019. Retrieved June 25, 2015. "The original skeleton of *Dynamosaurus imperiosus* (AMNH 5866/BM R7995), together with other *T. rex* material (including parts of AMNH 973, 5027, and 5881), were sold to the British Museum of Natural History (now The Natural History Museum) in 1960. This material was used in an interesting 'half-mount' display of this dinosaur in London. Currently the material resides in the research collections."
15. Small, Zachary (October 7, 2020). "T. Rex Skeleton Brings $31.8 Million at Christie's Auction" (https://www.nytimes.com/2020/10/06/arts/design/t-rex-skeleton-brings-31-8-million-at-christies-auction.html). *The New York Times*. Archived (https://web.archive.org/web/20201007175400/https://www.nytimes.com/2020/10/06/arts/design/t-rex-skeleton-brings-31-8-million-at-christies-auction.html) from the original on October 7, 2020. Retrieved May 5, 2021.

16. Erickson, G.; Makovicky, P. J.; Currie, P. J.; Norell, M.; Yerby, S.; Brochu, C. A. (May 26, 2004). "Gigantism and life history parameters of tyrannosaurid dinosaurs" (http://doc.rero.ch/record/15279/files/PAL_E2578.pdf) (PDF). *Nature*. **430** (7001): 772–775. Bibcode:2004Natur.430..772E (https://ui.adsabs.harvard.edu/abs/2004Natur.430..772E). doi:10.1038/nature02699 (https://doi.org/10.1038%2Fnature02699). PMID 15306807 (https://pubmed.ncbi.nlm.nih.gov/15306807). S2CID 4404887 (https://api.semanticscholar.org/CorpusID:4404887). Archived (https://web.archive.org/web/20200714024211/http://doc.rero.ch/record/15279/files/PAL_E2578.pdf) (PDF) from the original on July 14, 2020. Retrieved November 23, 2022. **(Erratum: doi:10.1038/nature16487 (https://doi.org/10.1038%2Fnature16487), PMID 26675726 (https://pubmed.ncbi.nlm.nih.gov/26675726), *Retraction Watch* (http://retractionwatch.com/2016/03/01/high-profile-critic-slams-nature-letters-about-dinosaur-growth-following-corrections/))**
17. Boeye, A. T.; Atkins-Weltman, K. L.; King, J. L.; Swann, Scott (2026). "Evidence of bird-like foot function in *Tyrannosaurus*" (https://royalsocietypublishing.org/rsos/article/13/2/252139/480521/Evidence-of-bird-like-foot-function-in). *Royal Society Open Science*. **13** (2) 252139. Bibcode:2026RSOS...1352139B (https://ui.adsabs.harvard.edu/abs/2026RSOS...1352139B). doi:10.1098/rsos.252139 (https://doi.org/10.1098%2Frsos.252139).
18. "Stan" (https://web.archive.org/web/20100918104233/http://www.museum.manchester.ac.uk/yourvisit/galleries/stan/). *The University of Manchester*. September 18, 2010. Archived from the original (http://www.museum.manchester.ac.uk/yourvisit/galleries/stan/) on September 18, 2010.
19. Fiffer, S. (2000). "Jurassic Farce". Tyrannosaurus *Sue* (https://archive.org/details/tyrannosaurussue00fiff_672). W. H. Freeman and Company, New York. pp. 121 (https://archive.org/details/tyrannosaurussue00fiff_672/page/n139)–122. ISBN 978-0-7167-4017-9.
20. "Meet Bucky The Teenage T. Rex" (https://www.childrensmuseum.org/blog/meet-bucky-the-teenage-t-rex-v2). *Children's Museum of Indianapolis*. July 7, 2014. Archived (https://web.archive.org/web/20141227210245/http://www.childrensmuseum.org/blog/meet-bucky-the-teenage-t-rex-v2) from the original on December 27, 2014. Retrieved December 2, 2019.
21. "Dig pulls up five T. rex specimens" (http://news.bbc.co.uk/2/hi/science/nature/965609.stm). *BBC News*. October 10, 2000. Archived (https://web.archive.org/web/20180819045859/http://news.bbc.co.uk/2/hi/science/nature/965609.stm) from the original on August 19, 2018. Retrieved December 13, 2008.
22. Currie, P. J.; Hurum, J. H.; Sabath, K. (2003). "Skull structure and evolution in tyrannosaurid dinosaurs" (http://www.app.pan.pl/archive/published/app48/app48-227.pdf) (PDF). *Acta Palaeontologica Polonica*. **48** (2): 227–234. Archived (https://web.archive.org/web/20081031093118/http://www.app.pan.pl/archive/published/app48/app48-227.pdf) (PDF) from the original on October 31, 2008. Retrieved October 8, 2008.
23. Zanno, Lindsay E.; Napoli, James G. (October 30, 2025). "*Nanotyrannus* and *Tyrannosaurus* coexisted at the close of the Cretaceous". *Nature*. **648** (8093): 357–367. Bibcode:2025Natur.648..357Z (https://ui.adsabs.harvard.edu/abs/2025Natur.648..357Z). doi:10.1038/s41586-025-09801-6 (https://doi.org/10.1038%2Fs41586-025-09801-6). ISSN 0028-0836 (https://search.worldcat.org/issn/0028-0836). PMID 41167514 (https://pubmed.ncbi.nlm.nih.gov/41167514).
24. "Museum unveils world's largest *T-rex* skull" (https://web.archive.org/web/20060414021235/http://www.montana.edu/cpa/news/nwview.php?article=3607). *Montana State University*. 2006. Archived from the original (http://www.montana.edu/cpa/news/nwview.php?article=3607) on April 14, 2006. Retrieved April 7, 2006.
25. Gignac, P. M.; Erickson, G. M. (2017). "The biomechanics behind extreme osteophagy in *Tyrannosaurus rex*" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5435714). *Scientific Reports*. **7** (1): 2012. Bibcode:2017NatSR...7.2012G (https://ui.adsabs.harvard.edu/abs/2017NatSR...7.2012G). doi:10.1038/s41598-017-02161-w (https://doi.org/10.1038%2Fs41598-017-02161-w). PMC 5435714 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5435714). PMID 28515439 (https://pubmed.ncbi.nlm.nih.gov/28515439).

26. "Sue Fact Sheet" (https://web.archive.org/web/20160818213556/https://www.fieldmuseum.org/sites/default/files/Sue%20Fact%20Sheet.pdf) (PDF). *Sue at the Field Museum*. Field Museum of Natural History. Archived from the original (https://www.fieldmuseum.org/sites/default/files/Sue%20Fact%20Sheet.pdf) (PDF) on August 18, 2016.
27. Hutchinson, J. R.; Bates, K. T.; Molnar, J.; Allen, V.; Makovicky, P. J. (2011). "A Computational Analysis of Limb and Body Dimensions in Tyrannosaurus rex with Implications for Locomotion, Ontogeny, and Growth" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3192160). *PLOS ONE*. **6** (10) e26037. Bibcode:2011PLoSO...626037H (https://ui.adsabs.harvard.edu/abs/2011PLoSO...626037H). doi:10.1371/journal.pone.0026037 (https://doi.org/10.1371%2Fjournal.pone.0026037). PMC 3192160 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3192160). PMID 22022500 (https://pubmed.ncbi.nlm.nih.gov/22022500).
28. Holtz, T. R. (2011). "Dinosaurs: The Most Complete, Up-to-Date Encyclopedia for Dinosaur Lovers of All Ages, Winter 2011 Appendix" (http://www.geol.umd.edu/~tholtz/dinoappendix/HoltzappendixWinter2011.pdf) (PDF). Archived (https://web.archive.org/web/20170812011954/http://www.geol.umd.edu/~tholtz/dinoappendix/HoltzappendixWinter2011.pdf) (PDF) from the original on August 12, 2017. Retrieved January 13, 2012.
29. "How well do you know SUE?" (https://www.fieldmuseum.org/blog/how-well-do-you-know-sue). Field Museum of Natural History. August 11, 2016. Archived (https://web.archive.org/web/20220408042134/https://www.fieldmuseum.org/blog/how-well-do-you-know-sue) from the original on April 8, 2022. Retrieved December 31, 2018.
30. "Sue the T. Rex" (https://www.fieldmuseum.org/blog/sue-t-rex). *Field Museum*. February 5, 2018. Archived (https://web.archive.org/web/20140404170459/http://fieldmuseum.org/sue) from the original on April 4, 2014. Retrieved July 20, 2018.
31. Persons, S. W.; Currie, P. J.; Erickson, G. M. (2019). "An Older and Exceptionally Large Adult Specimen of *Tyrannosaurus rex*" (https://doi.org/10.1002%2Far.24118). *The Anatomical Record*. **303** (4): 656–672. doi:10.1002/ar.24118 (https://doi.org/10.1002%2Far.24118). ISSN 1932-8486 (https://search.worldcat.org/issn/1932-8486). PMID 30897281 (https://pubmed.ncbi.nlm.nih.gov/30897281).
32. Hartman, Scott (July 7, 2013). "Mass estimates: North vs South redux" (http://www.skeletaldrawing.com/home/mass-estimates-north-vs-south-redux772013). Scott Hartman's Skeletal Drawing.com. Archived (https://web.archive.org/web/20131012065922/http://www.skeletaldrawing.com/home/mass-estimates-north-vs-south-redux772013) from the original on October 12, 2013. Retrieved August 24, 2013.
33. Campione, Nicolás E.; Evans, David C.; Brown, Caleb M.; Carrano, Matthew T. (2014). "Body mass estimation in non-avian bipeds using a theoretical conversion to quadruped stylopodial proportions" (https://doi.org/10.1111%2F2041-210X.12226). *Methods in Ecology and Evolution*. **5** (9): 913–923. Bibcode:2014MEcEv...5..913C (https://ui.adsabs.harvard.edu/abs/2014MEcEv...5..913C). doi:10.1111/2041-210X.12226 (https://doi.org/10.1111%2F2041-210X.12226). S2CID 84317234 (https://api.semanticscholar.org/CorpusID:84317234).
34. Lyle, A. (March 22, 2019). "Paleontologists identify biggest *Tyrannosaurus rex* ever discovered" (https://www.folio.ca/paleontologists-identify-biggest-tyrannosaurus-rex-ever-discovered/). Folio, University of Alberta. Archived (https://web.archive.org/web/20200804134550/https://www.folio.ca/paleontologists-identify-biggest-tyrannosaurus-rex-ever-discovered/) from the original on August 4, 2020. Retrieved March 25, 2019.
35. Anderson, J. F.; Hall-Martin, A. J.; Russell, D. (1985). "Long bone circumference and weight in mammals, birds and dinosaurs". *Journal of Zoology*. **207** (1): 53–61. doi:10.1111/j.1469-7998.1985.tb04915.x (https://doi.org/10.1111%2Fj.1469-7998.1985.tb04915.x).
36. Bakker, R. T. (1986). *The Dinosaur Heresies* (https://archive.org/details/dinosaurheresies00robe/page/241). New York: Kensington Publishing. p. 241 (https://archive.org/details/dinosaurheresies00robe/page/241). ISBN 978-0-688-04287-5. OCLC 13699558 (https://search.worldcat.org/oclc/13699558).

37. Henderson, Donald M. (1999). "Estimating the masses and centers of mass of extinct animals by 3-D mathematical slicing". *Paleobiology*. **25** (1): 88–106. doi:10.1666/0094-8373(1999)025<0088:ETMACO>2.3.CO;2 (https://doi.org/10.1666%2F0094-8373%281999%29025%3C0088%3AETMACO%3E2.3.CO%3B2). JSTOR 2665994 (https://www.jstor.org/stable/2665994).
38. Erickson, G. M.; Makovicky, P. J.; Currie, P. J.; Norell, M. A.; Yerby, S. A.; Brochu, C. A. (2004). "Gigantism and comparative life-history parameters of tyrannosaurid dinosaurs" (http://doc.rero.ch/record/15279/files/PAL_E2578.pdf) (PDF). *Nature*. **430** (7001): 772–775. Bibcode:2004Natur.430..772E (https://ui.adsabs.harvard.edu/abs/2004Natur.430..772E). doi:10.1038/nature02699 (https://doi.org/10.1038%2Fnature02699). PMID 15306807 (https://pubmed.ncbi.nlm.nih.gov/15306807). S2CID 4404887 (https://api.semanticscholar.org/CorpusID:4404887). Archived (https://web.archive.org/web/20200714024211/http://doc.rero.ch/record/15279/files/PAL_E2578.pdf) (PDF) from the original on July 14, 2020. Retrieved November 23, 2022. **(Erratum: doi:10.1038/nature16487 (https://doi.org/10.1038%2Fnature16487), PMID 26675726 (https://pubmed.ncbi.nlm.nih.gov/26675726), *Retraction Watch* (http://retractionwatch.com/2016/03/01/high-profile-critic-slams-nature-letters-about-dinosaur-growth-following-corrections/))**
39. Farlow, J. O.; Smith, M. B.; Robinson, J. M. (1995). "Body mass, bone 'strength indicator', and cursorial potential of *Tyrannosaurus rex*" (http://publication.plazi.org/id/0F5BF022FFFAFFDBFFC6B624B6744F00). *Journal of Vertebrate Paleontology*. **15** (4): 713–725. Bibcode:1995JVPal..15..713F (https://ui.adsabs.harvard.edu/abs/1995JVPal..15..713F). doi:10.1080/02724634.1995.10011257 (https://doi.org/10.1080%2F02724634.1995.10011257).
40. Seebacher, Frank (March 26, 2001). "A new method to calculate allometric length-mass relationships of dinosaurs". *Journal of Vertebrate Paleontology*. **21** (1): 51–60. doi:10.1671/0272-4634(2001)021[0051:ANMTCA]2.0.CO;2 (https://doi.org/10.1671%2F0272-4634%282001%29021%5B0051%3AANMTCA%5D2.0.CO%3B2).
41. Christiansen, P.; Fariña, R. A. (2004). "Mass prediction in theropod dinosaurs". *Historical Biology*. **16** (2–4): 85–92. Bibcode:2004HBio...16...85C (https://ui.adsabs.harvard.edu/abs/2004HBio...16...85C). doi:10.1080/08912960412331284313 (https://doi.org/10.1080%2F08912960412331284313). S2CID 84322349 (https://api.semanticscholar.org/CorpusID:84322349).
42. Mallon, Jordan C.; Hone, David W. E. (July 24, 2024). "Estimation of maximum body size in fossil species: A case study using *Tyrannosaurus rex*" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11267449). *Ecology and Evolution*. **14** (7) 11658. Bibcode:2024EcoEv..1411658M (https://ui.adsabs.harvard.edu/abs/2024EcoEv..1411658M). doi:10.1002/ece3.11658 (https://doi.org/10.1002%2Fece3.11658). ISSN 2045-7758 (https://search.worldcat.org/issn/2045-7758). PMC 11267449 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11267449). PMID 39050661 (https://pubmed.ncbi.nlm.nih.gov/39050661).
43. Stevens, Kent A. (June 2006). "Binocular vision in theropod dinosaurs". *Journal of Vertebrate Paleontology*. **26** (2): 321–330. doi:10.1671/0272-4634(2006)26[321:BVITD]2.0.CO;2 (https://doi.org/10.1671%2F0272-4634%282006%2926%5B321%3ABVITD%5D2.0.CO%3B2). S2CID 85694979 (https://api.semanticscholar.org/CorpusID:85694979).
44. Jaffe, E. (July 1, 2006). "Sight for 'Saur Eyes: *T. rex* vision was among nature's best". *Science News*. **170** (1): 3–4. doi:10.2307/4017288 (https://doi.org/10.2307%2F4017288). JSTOR 4017288 (https://www.jstor.org/stable/4017288).
45. Snively, E.; Henderson, D. M.; Phillips, D. S. (2006). "Fused and vaulted nasals of tyrannosaurid dinosaurs: Implications for cranial strength and feeding mechanics" (http://www.app.pan.pl/archive/published/app51/app51-435.pdf) (PDF). *Acta Palaeontologica Polonica*. **51** (3): 435–454. Archived (https://web.archive.org/web/20180819045612/http://www.app.pan.pl/archive/published/app51/app51-435.pdf) (PDF) from the original on August 19, 2018. Retrieved October 8, 2008.

46. Meers, M. B. (August 2003). "Maximum bite force and prey size of *Tyrannosaurus rex* and their relationships to the inference of feeding behavior". *Historical Biology*. **16** (1): 1–12. doi:10.1080/0891296021000050755 (https://doi.org/10.1080%2F0891296021000050755). S2CID 86782853 (https://api.semanticscholar.org/CorpusID:86782853).
47. Erickson, G. M.; Van Kirk, S. D.; Su, J.; Levenston, M. E.; Caler, W. E.; Carter, D. R. (1996). "Bite-force estimation for *Tyrannosaurus rex* from tooth-marked bones" (https://zenodo.org/record/3730962). *Nature*. **382** (6593): 706–708. Bibcode:1996Natur.382..706E (https://ui.adsabs.harvard.edu/abs/1996Natur.382..706E). doi:10.1038/382706a0 (https://doi.org/10.1038%2F382706a0). S2CID 4325859 (https://api.semanticscholar.org/CorpusID:4325859). Archived (https://web.archive.org/web/20200704181008/https://zenodo.org/record/3730962) from the original on July 4, 2020. Retrieved July 4, 2020.
48. Holtz, T. R. (1994). "The Phylogenetic Position of the Tyrannosauridae: Implications for Theropod Systematics". *Journal of Paleontology*. **68** (5): 1100–1117. Bibcode:1994JPal...68.1100H (https://ui.adsabs.harvard.edu/abs/1994JPal...68.1100H). doi:10.1017/S0022336000026706 (https://doi.org/10.1017%2FS0022336000026706). JSTOR 1306180 (https://www.jstor.org/stable/1306180). S2CID 129684676 (https://api.semanticscholar.org/CorpusID:129684676).
49. Smith, J. B. (December 1, 2005). "Heterodonty in *Tyrannosaurus rex*: implications for the taxonomic and systematic utility of theropod dentitions" (http://publication.plazi.org/id/DC6AFFF0D024FFD5FFA3FFD8BE03D62F). *Journal of Vertebrate Paleontology*. **25** (4): 865–887. doi:10.1671/0272-4634(2005)025[0865:HITRIF]2.0.CO;2 (https://doi.org/10.1671%2F0272-4634%282005%29025%5B0865%3AHITRIF%5D2.0.CO%3B2). S2CID 86184190 (https://api.semanticscholar.org/CorpusID:86184190). Archived (https://web.archive.org/web/20250324034130/http://publication.plazi.org/id/DC6AFFF0D024FFD5FFA3FFD8BE03D62F) from the original on March 24, 2025. Retrieved May 24, 2025.
50. Douglas, K.; Young, S. (1998). "The dinosaur detectives" (https://www.newscientist.com/channel/life/dinosaurs/mg15821305.300). *New Scientist*. Archived (https://web.archive.org/web/20080517160559/http://www.newscientist.com/channel/life/dinosaurs/mg15821305.300) from the original on May 17, 2008. Retrieved October 16, 2008. "One palaeontologist memorably described the huge, curved teeth of T. rex as 'lethal bananas' "
51. "Sue's vital statistics" (https://web.archive.org/web/20070929090231/http://www.fieldmuseum.org/SUE/about_vital.asp). *Sue at the Field Museum*. Field Museum of Natural History. Archived from the original (http://www.fieldmuseum.org/sue/about_vital.asp) on September 29, 2007. Retrieved September 15, 2007.
52. Carr, T. D.; Varricchio, D. J.; Sedlmayr, J. C.; Roberts, E. M.; Moore, J. R. (March 30, 2017). "A new tyrannosaur with evidence for anagenesis and crocodile-like facial sensory system" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5372470). *Scientific Reports*. **7** 44942. Bibcode:2017NatSR...744942C (https://ui.adsabs.harvard.edu/abs/2017NatSR...744942C). doi:10.1038/srep44942 (https://doi.org/10.1038%2Fsrep44942). ISSN 2045-2322 (https://search.worldcat.org/issn/2045-2322). PMC 5372470 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5372470). PMID 28358353 (https://pubmed.ncbi.nlm.nih.gov/28358353).
53. Morhardt, Ashley (2009). *Dinosaur smiles: Do the texture and morphology of the premaxilla, maxilla, and dentary bones of sauropsids provide osteological correlates for inferring extra-oral structures reliably in dinosaurs?* (https://www.academia.edu/3871353) (MSc thesis). Western Illinois University. Archived (https://web.archive.org/web/20220210063525/https://www.academia.edu/3871353) from the original on February 10, 2022. Retrieved May 10, 2020.

54. Reisz, R. R.; Larson, D. (2016). "Dental anatomy and skull length to tooth size ratios support the hypothesis that theropod dinosaurs had lips" (https://ejournals.library.ualberta.ca/index.php/VAMP/article/download/28196/20702). *4th Annual Meeting, 2016, Canadian Society of Vertebrate Palaeontology*. ISSN 2292-1389 (https://search.worldcat.org/issn/2292-1389). Archived (https://web.archive.org/web/20170404043155/https://ejournals.library.ualberta.ca/index.php/VAMP/article/download/28196/20702) from the original on April 4, 2017. Retrieved April 3, 2017.
55. "Morphology, taxonomy, and phylogenetic relationships of the Monteviale crocodylians (Oligocene, Italy) p. 67" (https://scholar.google.com/scholar?hl=en&as_sdt=0%2C14&q=MORPHOLOGY%2C+TAXONOMY%2C+AND+PHYLOGENETIC+RELATIONSHIPS+OF+THE+MONTEVIALE+CROCODYLIANS+%28OLIGOCENE%2C+ITALY%29.&btnG=). *The Annual Symposium of Vertebrate Palaeontology and Comparative Anatomy*. 2018. Archived (https://web.archive.org/web/20211110091020/https://scholar.google.com/scholar?hl=en&as_sdt=0%2C14&q=MORPHOLOGY%2C+TAXONOMY%2C+AND+PHYLOGENETIC+RELATIONSHIPS+OF+THE+MONTEVIALE+CROCODYLIANS+%28OLIGOCENE%2C+ITALY%29.&btnG=) from the original on November 10, 2021. Retrieved October 9, 2020.
56. Cullen, Thomas M.; Larson, Derek W.; Witton, Mark P.; Scott, Diane; Maho, Tea; Brink, Kirstin S.; Evans, David C.; Reisz, Robert (March 31, 2023). "Theropod dinosaur facial reconstruction and the importance of soft tissues in paleobiology" (https://doi.org/10.1126%2Fscience.abo7877). *Science*. **379** (6639): 1348–1352. Bibcode:2023Sci...379.1348C (https://ui.adsabs.harvard.edu/abs/2023Sci...379.1348C). doi:10.1126/science.abo7877 (https://doi.org/10.1126%2Fscience.abo7877). ISSN 0036-8075 (https://search.worldcat.org/issn/0036-8075). PMID 36996202 (https://pubmed.ncbi.nlm.nih.gov/36996202). S2CID 257836765 (https://api.semanticscholar.org/CorpusID:257836765).
57. Lipkin, C.; Carpenter, K. (2008). "Looking again at the forelimb of *Tyrannosaurus rex*". In Carpenter, K.; Larson, P. E. (eds.). Tyrannosaurus rex*, the Tyrant King* (https://archive.org/details/tyrannosaurusrex00plar). Bloomington: Indiana University Press. pp. 167 (https://archive.org/details/tyrannosaurusrex00plar/page/n182)–190. ISBN 978-0-253-35087-9.
58. Holtz, T. R. Jr. (2004). "Tyrannosauroidea". In Weishampel, D. B.; Dodson, P.; Osmólska, H. (eds.). *The dinosauria* (https://archive.org/details/dinosauriandedit00weis). Berkeley: University of California Press. pp. 111 (https://archive.org/details/dinosauriandedit00weis/page/n129)–136. ISBN 978-0-520-24209-8.
59. Paul, Gregory S. (1988). *Predatory dinosaurs of the world: a complete illustrated guide* (https://archive.org/details/predatorydinosau00paul/page/228). New York: Simon and Schuster. p. 228 (https://archive.org/details/predatorydinosau00paul/page/228). ISBN 978-0-671-61946-6. OCLC 18350868 (https://search.worldcat.org/oclc/18350868).
60. Li, Daqing; Norell, Mark A.; Gao, Ke-Qin; Smith, Nathan D.; Makovicky, Peter J. (2009). "A longirostrine tyrannosauroid from the Early Cretaceous of China" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2842666). *Proc Biol Sci*. **277** (1679): 183–190. doi:10.1098/rspb.2009.0249 (https://doi.org/10.1098%2Frspb.2009.0249). PMC 2842666 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2842666). PMID 19386654 (https://pubmed.ncbi.nlm.nih.gov/19386654).
61. Brusatte, Stephen L.; Norell, Mark A.; Carr, Thomas D.; Erickson, Gregory M.; Hutchinson, John R.; Balanoff, Amy M.; Bever, Gabe S.; Choiniere, Jonah N.; Makovicky, Peter J.; Xu, Xing (September 17, 2010). "Tyrannosaur Paleobiology: New Research on Ancient Exemplar Organisms" (https://www.research.ed.ac.uk/en/publications/fc52fb23-10e8-466d-a7e9-081260d166c6). *Science*. **329** (5998): 1481–1485. Bibcode:2010Sci...329.1481B (https://ui.adsabs.harvard.edu/abs/2010Sci...329.1481B). doi:10.1126/science.1193304 (https://doi.org/10.1126%2Fscience.1193304). PMID 20847260 (https://pubmed.ncbi.nlm.nih.gov/20847260).

62. Loewen, M. A.; Irmis, R. B.; Sertich, J. J. W.; Currie, P. J.; Sampson, S. D. (2013). Evans, D. C (ed.). "Tyrant Dinosaur Evolution Tracks the Rise and Fall of Late Cretaceous Oceans" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3819173). *PLoS ONE*. **8** (11) e79420. Bibcode:2013PLoSO...879420L (https://ui.adsabs.harvard.edu/abs/2013PLoSO...879420L). doi:10.1371/journal.pone.0079420 (https://doi.org/10.1371%2Fjournal.pone.0079420). PMC 3819173 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3819173). PMID 24223179 (https://pubmed.ncbi.nlm.nih.gov/24223179).
63. Vergano, D. (November 7, 2013). "Newfound "King of Gore" Dinosaur Ruled Before T. Rex" (https://web.archive.org/web/20131108020332/http://news.nationalgeographic.com/news/2013/11/131106-king-gore-tyrannosaurus-dinosaur/). *National Geographic*. Archived from the original (https://news.nationalgeographic.com/news/2013/11/131106-king-gore-tyrannosaurus-dinosaur/) on November 8, 2013. Retrieved November 10, 2017.
64. Geggel, L. (February 29, 2016). "*T. Rex* Was Likely an Invasive Species" (http://www.livescience.com/53877-t-rex-was-invasive-species.html). *Live Science*. Archived (https://web.archive.org/web/20190406221051/https://www.livescience.com/53877-t-rex-was-invasive-species.html) from the original on April 6, 2019. Retrieved November 10, 2017.
65. Dalman, Sebastian G.; Loewen, Mark A.; Pyron, R. Alexander; Jasinski, Steven E.; Malinzak, D. Edward; Lucas, Spencer G.; Fiorillo, Anthony R.; Currie, Philip J.; Longrich, Nicholas R. (January 11, 2024). "A giant tyrannosaur from the Campanian–Maastrichtian of southern North America and the evolution of tyrannosaurid gigantism" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10784284). *Scientific Reports*. **13** (1): 22124. Bibcode:2024NatSR..1422124D (https://ui.adsabs.harvard.edu/abs/2024NatSR..1422124D). doi:10.1038/s41598-023-47011-0 (https://doi.org/10.1038%2Fs41598-023-47011-0). ISSN 2045-2322 (https://search.worldcat.org/issn/2045-2322). PMC 10784284 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10784284). PMID 38212342 (https://pubmed.ncbi.nlm.nih.gov/38212342).
66. Scherer, Charlie Roger; Voiculescu-Holvad, Christian (2024). "Re-analysis of a dataset refutes claims of anagenesis within *Tyrannosaurus*-line tyrannosaurines (Theropoda, Tyrannosauridae)" (https://doi.org/10.1016%2Fj.cretres.2023.105780). *Cretaceous Research*. **155** 105780. Bibcode:2024CrRes.15505780S (https://ui.adsabs.harvard.edu/abs/2024CrRes.15505780S). doi:10.1016/j.cretres.2023.105780 (https://doi.org/10.1016%2Fj.cretres.2023.105780). ISSN 0195-6671 (https://search.worldcat.org/issn/0195-6671).
67. Maleev, E. A. (1955). "(title in Russian)" (http://paleoglot.org/files/Maleev_55a.pdf) [Gigantic carnivorous dinosaurs of Mongolia] (PDF). *Doklady Akademii Nauk SSSR* (in Russian). **104** (4). translated by F. J. Alcock: 634–637. Archived (https://web.archive.org/web/20191211124644/https://paleoglot.org/files/Maleev_55a.pdf) (PDF) from the original on December 11, 2019. Retrieved April 30, 2013.
68. Rozhdestvensky, A. K. (1965). "Growth changes in Asian dinosaurs and some problems of their taxonomy". *Paleontological Journal*. **3**: 95–109.
69. Carpenter, K. (1992). "Tyrannosaurids (Dinosauria) of Asia and North America". In Mateer, N. J.; Chen, P.-j. (eds.). *Aspects of nonmarine Cretaceous geology*. Beijing: China Ocean Press. pp. 250–268. ISBN 978-7-5027-1463-5. OCLC 28260578 (https://search.worldcat.org/oclc/28260578).
70. Hurum, J.H., Sabath, K. (2003). "Giant theropod dinosaurs from Asia and North America: Skulls of *Tarbosaurus bataar* and *Tyrannosaurus rex* compared" (https://web.archive.org/web/20230517131646/https://www.app.pan.pl/archive/published/app48/app48-161.pdf) (PDF). *Acta Palaeontologica Polonica*. **48** (2): 161–190. Archived from the original (https://www.app.pan.pl/archive/published/app48/app48-161.pdf) (PDF) on May 17, 2023.

71. Li, Feng; Bi, Shundong; Pittman, Michael; Brusatte, Stephen L.; Xu, Xing (November 2016). "A new tyrannosaurine specimen (Theropoda: Tyrannosauroidea) with insect borings from the Upper Cretaceous Honglishan Formation of Northwestern China". *Cretaceous Research*. **66**: 155–162. Bibcode:2016CrRes..66..155L (https://ui.adsabs.harvard.edu/abs/2016CrRes..66..155L). doi:10.1016/j.cretres.2016.06.002 (https://doi.org/10.1016%2Fj.cretres.2016.06.002). hdl:20.500.11820/a49b0878-3ba2-4c3e-b4b1-70fe282e43ea (https://hdl.handle.net/20.500.11820%2Fa49b0878-3ba2-4c3e-b4b1-70fe282e43ea).
72. Sebastian G. Dalman, Spencer G. Lucas (January 2018). "Tyrannosaurid dinosaurs (Theropoda: Tyrannosauridae) from the Upper Cretaceous (Early Campanian) Allison Member of the Menefee Formation, New Mexico: Implications for the origin of Tyrannosauridae in North America" (https://www.researchgate.net/publication/328676802). *Fossil Record*. New Mexico Museum of Natural History and Science Bulletin. **6** (79): 99–112.
73. A. O. Averianov, A. A. Yarkov (2004). "Carnivorous dinosaurs (Saurischia, Theropoda) from the Maastrichtian of the Volga-Don Interfluve, Russia" (https://repository.geologyscience.ru/bitstream/handle/123456789/39408/Aver_04.pdf?sequence=1) (PDF). *Paleontological Journal*. **38** (1): 78–82. Archived (https://web.archive.org/web/20240305115625/https://repository.geologyscience.ru/bitstream/handle/123456789/39408/Aver_04.pdf?sequence=1) (PDF) from the original on March 5, 2024.
74. A. O. Averianov, A. V. Lopatin (2023). "Динозавры России: обзор местонахождений [Dinosaurs of Russia: Overview of locations]" (https://new.ras.ru/upload/iblock/a8f/vfw1g7hafto3gx7qbnwgpmuljd0j8ykk.pdf#page=45) (PDF). *Vestnik Rossiiskoi Akademii Nauk* (in Russian). **93** (4): 342–354. doi:10.31857/S0869587323040023 (https://doi.org/10.31857%2FS0869587323040023). Archived (https://web.archive.org/web/20240216183756/https://new.ras.ru/upload/iblock/a8f/vfw1g7hafto3gx7qbnwgpmuljd0j8ykk.pdf#page=45) (PDF) from the original on February 16, 2024.
75. Hone, D. W. E.; Wang, K.; Sullivan, C.; Zhao, X.; Chen, S.; Li, D.; Ji, S.; Ji, Q.; Xu, X. (2011). "A new, large tyrannosaurine theropod from the Upper Cretaceous of China". *Cretaceous Research*. **32** (4): 495–503. Bibcode:2011CrRes..32..495H (https://ui.adsabs.harvard.edu/abs/2011CrRes..32..495H). doi:10.1016/j.cretres.2011.03.005 (https://doi.org/10.1016%2Fj.cretres.2011.03.005).
76. Urban, Michael A.; Lamanna, Matthew C. (December 2006). "Evidence of a Giant Tyrannosaurid (Dinosauria: Theropoda) from the Upper Cretaceous (?Campanian) of Montana". *Annals of Carnegie Museum*. **75** (4): 231–235. doi:10.2992/0097-4463(2006)75[231:EOAGTD]2.0.CO;2 (https://doi.org/10.2992%2F0097-4463%282006%2975%5B231%3AEOAGTD%5D2.0.CO%3B2).
77. Dalman, Sebastian G.; Lucas, Spencer G.; Malinzak, D. Edward (2018). "Tyrannosaurid teeth from the upper Cretaceous (Campanian) Two Medicine Formation of Montana" (https://www.researchgate.net/publication/328676947). *New Mexico Museum of Natural History and Science Bulletin*. **79**: 125–139.
78. Scherer, Charlie Roger (January 11, 2025). "Multiple lines of evidence support anagenesis in Daspletosaurus and cladogenesis in derived tyrannosaurines". *Cretaceous Research*. **169** 106080. Bibcode:2025CrRes.16906080S (https://ui.adsabs.harvard.edu/abs/2025CrRes.16906080S). doi:10.1016/j.cretres.2025.106080 (https://doi.org/10.1016%2Fj.cretres.2025.106080).
79. Paul, Gregory S.; Persons IV, W. Scott; van Raalte, Jay (2022). "The Tyrant Lizard King, Queen and Emperor: Multiple Lines of Morphological and Stratigraphic Evidence Support Subtle Evolution and Probable Speciation Within the North American Genus Tyrannosaurus". *Evolutionary Biology*. **49** (2): 156–179. Bibcode:2022EvBio..49..156P (https://ui.adsabs.harvard.edu/abs/2022EvBio..49..156P). doi:10.1007/s11692-022-09561-5 (https://doi.org/10.1007%2Fs11692-022-09561-5).

80. Elbein, Asher (February 28, 2022). "They Want to Break T. Rex Into 3 Species. Other Paleontologists Aren't Pleased" (https://www.nytimes.com/2022/02/28/science/tyrannosaurus-rex-species.html?). *The New York Times*. Archived (https://web.archive.org/web/20220301014637/https://www.nytimes.com/2022/02/28/science/tyrannosaurus-rex-species.html) from the original on March 1, 2022. Retrieved March 1, 2022.
81. Hunt, Katie (March 1, 2022). "Tyrannosaurus rex may have been misunderstood" (https://edition.cnn.com/2022/02/28/world/t-rex-three-different-dinosaurs-scn/index.html). *CNN*. Archived (https://web.archive.org/web/20220304034107/https://edition.cnn.com/2022/02/28/world/t-rex-three-different-dinosaurs-scn/index.html) from the original on March 4, 2022. Retrieved March 4, 2022.
82. Greshko, Michael (March 1, 2022). "Call to split T. rex into 3 species sparks fierce debate" (https://web.archive.org/web/20220301010250/https://www.nationalgeographic.com/science/article/call-to-split-tyrannosaurus-rex-into-3-species-sparks-fierce-debate). *National Geographic*. Archived from the original (https://www.nationalgeographic.com/science/article/call-to-split-tyrannosaurus-rex-into-3-species-sparks-fierce-debate) on March 1, 2022.
83. Carr, T.D.; Napoli, J.G.; Brusatte, S.L.; Holtz, T.R.; Hone, D.W.E.; Williamson, T.E.; Zanno, L.E. (2022). "Insufficient Evidence for Multiple Species of *Tyrannosaurus* in the Latest Cretaceous of North America: A Comment on "The Tyrant Lizard King, Queen and Emperor: Multiple Lines of Morphological and Stratigraphic Evidence Support Subtle Evolution and Probable Speciation Within the North American Genus *Tyrannosaurus*"" (https://doi.org/10.1007%2Fs11692-022-09573-1). *Evolutionary Biology*. **49** (3): 314–341. Bibcode:2022EvBio..49..327C (https://ui.adsabs.harvard.edu/abs/2022EvBio..49..327C). doi:10.1007/s11692-022-09573-1 (https://doi.org/10.1007%2Fs11692-022-09573-1).
84. Paul, Gregory S. (June 30, 2025). "A presentation of the current data on the exceptionally diverse non-tyrannosaurid eutyrannosaur and Tyrannosaurini genera and species of western North America during the End Cretaceous North American Interchange" (https://doi.org/10.11646%2Fmesozoic.2.2.1). *Mesozoic*. **2** (2): 085–138. doi:10.11646/mesozoic.2.2.1 (https://doi.org/10.11646%2Fmesozoic.2.2.1). ISSN 3021-1867 (https://search.worldcat.org/issn/3021-1867).
85. Lozinsky, Richard P.; Hunt, Adrian P.; Wolberg, Donald L.; Lucas, Spencer G. (1984). "Late Cretaceous (Lancian) dinosaurs from the McRae Formation, Sierra County, New Mexico" (https://geoinfo.nmt.edu/publications/periodicals/nmg/6/n4/nmg_v6_n4_p72.pdf) (PDF). *New Mexico Geology*. **6** (4): 72–77. doi:10.58799/NMG-v6n4.72 (https://doi.org/10.58799%2FNMG-v6n4.72). ISSN 0196-948X (https://search.worldcat.org/issn/0196-948X). S2CID 237011797 (https://api.semanticscholar.org/CorpusID:237011797). Archived (https://web.archive.org/web/20231118063409/http://geoinfo.nmt.edu/publications/periodicals/nmg/6/n4/nmg_v6_n4_p72.pdf) (PDF) from the original on November 18, 2023. Retrieved January 12, 2024.
86. Gillette, David D.; Wolberg, Donald L.; Hunt, Adrian P. (1986). "*Tyrannosaurus rex* from the McRae Formation (Lancian, Upper Cretaceous), Elephant Butte reservoir, Sierra County, New Mexico" (https://nmgs.nmt.edu/publications/guidebooks/downloads/37/37_p0235_p0238.pdf) (PDF). *New Mexico Geological Society Guidebook*. **37**: 235–238. doi:10.56577/FFC-37.235 (https://doi.org/10.56577%2FFFC-37.235). S2CID 251985284 (https://api.semanticscholar.org/CorpusID:251985284). Archived (https://web.archive.org/web/20240112195843/https://nmgs.nmt.edu/publications/guidebooks/downloads/37/37_p0235_p0238.pdf) (PDF) from the original on January 12, 2024. Retrieved January 12, 2024.

87. Lehman, Thomas M.; Carpenter, Kenneth (1990). "A partial skeleton of the tyrannosaurid dinosaur *Aublysodon* from the Upper Cretaceous of New Mexico" (https://zenodo.org/record/1037493). *Journal of Paleontology*. **64** (6): 1026–1032. Bibcode:1990JPal...64.1026L (https://ui.adsabs.harvard.edu/abs/1990JPal...64.1026L). doi:10.1017/S0022336000019843 (https://doi.org/10.1017%2FS0022336000019843). JSTOR 1305741 (https://www.jstor.org/stable/1305741). S2CID 132662000 (https://api.semanticscholar.org/CorpusID:132662000). Archived (https://web.archive.org/web/20230715174200/https://zenodo.org/record/1037493) from the original on July 15, 2023. Retrieved March 17, 2024.
88. Carr, Thomas D.; Williamson, Thomas E. (2000). "A review of Tyrannosauridae (Dinosauria: Coelurosauria) from New Mexico" (http://econtent.unm.edu/cdm/ref/collection/bulletins/id/1015). *Bulletin*. **17**. New Mexico Museum of Natural History and Science: 113–145.
89. Sullivan, Robert M.; Lucas, Spencer G. (2015). "Cretaceous Vertebrates of New Mexico" (https://www.researchgate.net/publication/299592501). *New Mexico Museum of Natural History and Science Bulletin*. **68**.
90. Schantz, E.; Amato, J.M.; Lawton, T.F. *Dating dinosaurs in the Laramide foreland: U-Pb geochronologic constraints on a stratigraphic section containing* Alamosaurus *and the postulated Tyrannosaurus mcraeensis in the Love Ranch Basin* (https://gsa.confex.com/gsa/2024AM/webprogram/Paper405342.html). New Mexico Geological Society of America Abstracts with Programs. Vol. 56. p. 9.
91. Elbein, Asher (January 11, 2024). "New Origin Story for *Tyrannosaurus Rex* Suggested by Fossil - Researchers say the species they named *Tyrannosaurus mcraeensis* predated the dinosaur era's great predator" (https://www.nytimes.com/2024/01/11/science/new-tyrannosaur-species-fossil.html). *The New York Times*. Retrieved January 12, 2024.
92. Morrison, C.; Scherer, C.R.; O'Callaghan, E.V.; Layton, C.; Boisvert, C.; Rolando, M.A.; Durrant, L.; Salas, P.; Allain, S.J.R.; Gascoigne, S.J.L. (2025). "Rise of the king: Gondwanan origins and evolution of megaraptoran dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12055284). *Royal Society Open Science*. **12** (5) 242238. Bibcode:2025RSOS...1242238M (https://ui.adsabs.harvard.edu/abs/2025RSOS...1242238M). doi:10.1098/rsos.242238 (https://doi.org/10.1098%2Frsos.242238). PMC 12055284 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12055284). PMID 40337259 (https://pubmed.ncbi.nlm.nih.gov/40337259).
93. Voris, Jared T.; Zelenitsky, Darla K.; Kobayashi, Yoshitsugu; Modesto, Sean P.; Therrien, François; Tsutsumi, Hiroki; Chinzorig, Tsogtbaatar; Tsogtbaatar, Khishigjav (June 11, 2025). "A new Mongolian tyrannosauroid and the evolution of Eutyrannosauria". *Nature*. **642** (8069): 973–979. Bibcode:2025Natur.642..973V (https://ui.adsabs.harvard.edu/abs/2025Natur.642..973V). doi:10.1038/s41586-025-08964-6 (https://doi.org/10.1038%2Fs41586-025-08964-6). PMID 40500434 (https://pubmed.ncbi.nlm.nih.gov/40500434).
94. Carr, Thomas D. (2025). "Observations on the skull of the type specimen of Tyrannosaurus rex Osborn, 1905". *All Earth*. **37** (1) 2539638: 1–66. Bibcode:2025AEart..3739638C (https://ui.adsabs.harvard.edu/abs/2025AEart..3739638C). doi:10.1080/27669645.2025.2539638 (https://doi.org/10.1080%2F27669645.2025.2539638).
95. Olshevsky, G.; Ford, T.L.; Yamamoto, S. (1995). "The origin and evolution of the tyrannosaurids". *Kyoryugaku Saizensen [Dino Frontline]*. **9–10**: 92–119.
96. Carr, T. D.; Williamson, T. E. (2004). "Diversity of late Maastrichtian Tyrannosauridae (Dinosauria: Theropoda) from western North America" (https://doi.org/10.1111%2Fj.1096-3642.2004.00130.x). *Zoological Journal of the Linnean Society*. **142** (4): 479–523. doi:10.1111/j.1096-3642.2004.00130.x (https://doi.org/10.1111%2Fj.1096-3642.2004.00130.x).
97. Gilmore, C. W. (1946). "A new carnivorous dinosaur from the Lance Formation of Montana". *Smithsonian Miscellaneous Collections*. **106**: 1–19.
98. Bakker, R.T.; Williams, M.; Currie, P.J. (1988). "*Nanotyrannus*, a new genus of pygmy tyrannosaur, from the latest Cretaceous of Montana". *Hunteria*. **1**: 1–30.

99. Carr, T.D. (1999). "Craniofacial ontogeny in Tyrannosauridae (Dinosauria, Coelurosauria)" (https://zenodo.org/record/3371479). *Journal of Vertebrate Paleontology*. **19** (3): 497–520. Bibcode:1999JVPal..19..497C (https://ui.adsabs.harvard.edu/abs/1999JVPal..19..497C). doi:10.1080/02724634.1999.10011161 (https://doi.org/10.1080%2F02724634.1999.10011161). S2CID 83744433 (https://api.semanticscholar.org/CorpusID:83744433). Archived (https://web.archive.org/web/20200220183407/https://zenodo.org/record/3371479) from the original on February 20, 2020. Retrieved December 6, 2019.
100. Tsuihiji, T.; Watabe, M.; Tsogtbaatar, K.; Tsubamoto, T.; Barsbold, R.; Suzuki, S.; Lee, A.H.; Ridgely, R.C.; Kawahara, Y.; Witmer, L.M. (2011). "Cranial osteology of a juvenile specimen of *Tarbosaurus bataar* from the Nemegt Formation (Upper Cretaceous) of Bugin Tsav, Mongolia". *Journal of Vertebrate Paleontology*. **31** (3): 497–517. Bibcode:2011JVPal..31..497T (https://ui.adsabs.harvard.edu/abs/2011JVPal..31..497T). doi:10.1080/02724634.2011.557116 (https://doi.org/10.1080%2F02724634.2011.557116). S2CID 15369707 (https://api.semanticscholar.org/CorpusID:15369707).
101. Currie, P.J. (2003a). "Cranial anatomy of tyrannosaurid dinosaurs from the Late Cretaceous of Alberta, Canada". *Acta Palaeontologica Polonica*. **48**: 191–226.
102. Currie, Henderson, Horner and Williams (2005). "On tyrannosaur teeth, tooth positions and the taxonomic status of *Nanotyrannus lancensis*." In "The origin, systematics, and paleobiology of Tyrannosauridae", a symposium hosted jointly by Burpee Museum of Natural History and Northern Illinois University.
103. Henderson (2005). "Nano No More: The death of the pygmy tyrant." In "The origin, systematics, and paleobiology of Tyrannosauridae", a symposium hosted jointly by Burpee Museum of Natural History and Northern Illinois University.
104. Larson, P. (2013). *The validity of* Nanotyrannus lancensis *(Theropoda, Lancian - Upper Maastrichtian of North America)* (https://vertpaleo.org/wp-content/uploads/2021/03/SVP-2013-merged-book-10-15-2013.pdf) (PDF). Society of Vertebrate Paleontology 73rd Annual Meeting. p. 159. Archived (https://web.archive.org/web/20220303092634/https://vertpaleo.org/wp-content/uploads/2021/03/SVP-2013-merged-book-10-15-2013.pdf) (PDF) from the original on March 3, 2022. Retrieved October 31, 2025.
105. Larson, P.L. (2013). "The case for *Nanotyrannus*". In Parrish, J.M.; Molnar, R.A.; Currie, P.J.; Koppelhus, E.B. (eds.). *Tyrannosaurid Paleobiology*. University of Indiana Press. pp. 15–53. ISBN 978-0-253-00930-2.
106. Black, R. (November 10, 2011). " "Montana's "Dueling Dinosaurs" " (https://www.smithsonianmag.com/science-nature/montanas-dueling-dinosaurs-134669023/). *smithsonianmag.com*. Archived (https://web.archive.org/web/20250714000452/https://www.smithsonianmag.com/science-nature/montanas-dueling-dinosaurs-134669023/) from the original on July 14, 2025. Retrieved October 30, 2025.
107. Cornwall, W. (May 22, 2020). "Court rules 'Dueling Dinos' belong to landowners, in a win for science" (https://www.science.org/content/article/court-rules-dueling-dinos-belong-landowners-win-science). *science.org*. Archived (https://web.archive.org/web/20250405074109/https://www.science.org/content/article/court-rules-dueling-dinos-belong-landowners-win-science) from the original on April 5, 2025. Retrieved October 30, 2025.
108. Pishney, Jon (November 17, 2020). "North Carolina Museum of Natural Sciences to receive the "Dueling Dinosaurs" " (https://naturalsciences.org/calendar/news/north-carolina-museum-of-natural-sciences-to-receive-the-dueling-dinosaurs/). Archived (https://web.archive.org/web/20201118054601/https://naturalsciences.org/calendar/news/north-carolina-museum-of-natural-sciences-to-receive-the-dueling-dinosaurs/) from the original on November 18, 2020. Retrieved October 31, 2025.
109. Pishney, Jon (October 31, 2023). "Dueling Dinosaurs move into the SECU DinoLab | Programs and Events Calendar" (https://naturalsciences.org/calendar/news/dueling-dinosaurs-move-into-the-secu-dinolab/). Archived (https://web.archive.org/web/20240619143936/https://naturalsciences.org/calendar/news/dueling-dinosaurs-move-into-the-secu-dinolab/) from the original on June 19, 2024. Retrieved October 31, 2025.

110. Cullen, Thomas M.; Canale, Juan I.; Apesteguía, Sebastián; Smith, Nathan D.; Hu, Dongyu; Makovicky, Peter J. (November 25, 2020). "Osteohistological analyses reveal diverse strategies of theropod dinosaur body-size evolution" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7739506). *Proceedings of the Royal Society B: Biological Sciences*. **287** (1939) 20202258. doi:10.1098/rspb.2020.2258 (https://doi.org/10.1098%2Frspb.2020.2258). ISSN 0962-8452 (https://search.worldcat.org/issn/0962-8452). PMC 7739506 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7739506). PMID 33234083 (https://pubmed.ncbi.nlm.nih.gov/33234083).
111. "Trix the T. rex is ready for her big Glasgow reveal" (https://www.gla.ac.uk/news/archiveofnews/2019/april/headline_644272_en.html). *www.gla.ac.uk*. Archived (https://web.archive.org/web/20250523204700/https://www.gla.ac.uk/news/archiveofnews/2019/april/headline_644272_en.html) from the original on May 23, 2025. Retrieved May 23, 2025.
112. Horner, J. R.; Padian, K. (2004). "Age and growth dynamics of Tyrannosaurus rex" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1691809). *Proceedings: Biological Sciences*. **271** (1551): 1875–80. doi:10.1098/rspb.2004.2829 (https://doi.org/10.1098%2Frspb.2004.2829). PMC 1691809 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1691809). PMID 15347508 (https://pubmed.ncbi.nlm.nih.gov/15347508).
113. Schweitzer, M. H.; Wittmeyer, J. L.; Horner, J. R. (2005). "Gender-specific reproductive tissue in ratites and *Tyrannosaurus rex*" (http://doc.rero.ch/record/14767/files/PAL_E1895.pdf) (PDF). *Science*. **308** (5727): 1456–60. Bibcode:2005Sci...308.1456S (https://ui.adsabs.harvard.edu/abs/2005Sci...308.1456S). doi:10.1126/science.1112158 (https://doi.org/10.1126%2Fscience.1112158). PMID 15933198 (https://pubmed.ncbi.nlm.nih.gov/15933198). S2CID 30264554 (https://api.semanticscholar.org/CorpusID:30264554). Archived (https://web.archive.org/web/20230312052627/https://doc.rero.ch/record/14767/files/PAL_E1895.pdf) (PDF) from the original on March 12, 2023. Retrieved November 23, 2022.
114. Lee, A. H.; Werning, S. (2008). "Sexual maturity in growing dinosaurs does not fit reptilian growth models" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2206579). *Proceedings of the National Academy of Sciences*. **105** (2): 582–587. Bibcode:2008PNAS..105..582L (https://ui.adsabs.harvard.edu/abs/2008PNAS..105..582L). doi:10.1073/pnas.0708903105 (https://doi.org/10.1073%2Fpnas.0708903105). PMC 2206579 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2206579). PMID 18195356 (https://pubmed.ncbi.nlm.nih.gov/18195356).
115. Schweitzer, M. H.; Zheng, W.; Zanno, L.; Werning, S.; Sugiyama, T. (2016). "Chemistry supports the identification of gender-specific reproductive tissue in Tyrannosaurus rex" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4791554). *Scientific Reports*. **6** 23099. Bibcode:2016NatSR...623099S (https://ui.adsabs.harvard.edu/abs/2016NatSR...623099S). doi:10.1038/srep23099 (https://doi.org/10.1038%2Fsrep23099). PMC 4791554 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4791554). PMID 26975806 (https://pubmed.ncbi.nlm.nih.gov/26975806).
116. Erickson, G. M.; Currie, P. J.; Inouye, B. D.; Winn, A. A. (2006). "Tyrannosaur life tables: an example of nonavian dinosaur population biology". *Science*. **313** (5784): 213–7. Bibcode:2006Sci...313..213E (https://ui.adsabs.harvard.edu/abs/2006Sci...313..213E). doi:10.1126/science.1125721 (https://doi.org/10.1126%2Fscience.1125721). PMID 16840697 (https://pubmed.ncbi.nlm.nih.gov/16840697). S2CID 34191607 (https://api.semanticscholar.org/CorpusID:34191607).
117. Woodward, Holly N; Tremaine, Katie; Williams, Scott A; Zanno, Lindsay E; Horner, John R; Myhrvold, Nathan (January 1, 2020). "Growing up Tyrannosaurus rex: Osteohistology refutes the pygmy "Nanotyrannus" and supports ontogenetic niche partitioning in juvenile Tyrannosaurus" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6938697). *Science Advances*. **6** (1) eaax6250. Bibcode:2020SciA....6.6250W (https://ui.adsabs.harvard.edu/abs/2020SciA....6.6250W). doi:10.1126/sciadv.aax6250 (https://doi.org/10.1126%2Fsciadv.aax6250). PMC 6938697 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6938697). PMID 31911944 (https://pubmed.ncbi.nlm.nih.gov/31911944).

118. Greshko, Michael (January 1, 2020). "These sleek predatory dinosaurs really are teenage T. rex" (https://web.archive.org/web/20200101225622/https://www.nationalgeographic.com/science/2020/01/nanotyrannus-fossils-really-are-teenage-t-rex-dinosaur-growth-rates/). *National Geographic*. Archived from the original (https://www.nationalgeographic.com/science/2020/01/nanotyrannus-fossils-really-are-teenage-t-rex-dinosaur-growth-rates/) on January 1, 2020. Retrieved January 2, 2020.
119. Woodward, H. N.; Myhrvold, N. P.; Horner, J. R. (2026). "Prolonged growth and extended subadult development in the *Tyrannosaurus rex* species complex revealed by expanded histological sampling and statistical modeling" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12811967). *PeerJ*. **14** e20469. Bibcode:2026PeerJ..1420469W (https://ui.adsabs.harvard.edu/abs/2026PeerJ..1420469W). doi:10.7717/peerj.20469 (https://doi.org/10.7717%2Fpeerj.20469). PMC 12811967 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12811967). PMID 41551440 (https://pubmed.ncbi.nlm.nih.gov/41551440).
120. Longrich, Nicholas R.; Makovicky, Peter J.; Tokaryk, Tim; Cooper, David M. L.; Saitta, Evan T.; Erickson, Gregory M.; Szekely, Tamas; Snively, Eric (July 7, 2026). "Hatchlings of *Tyrannosaurus rex* and the Evolution of Dinosaur Reproductive Strategies" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC13359715). *Biology*. **15** (13) 1090. doi:10.3390/biology15131090 (https://doi.org/10.3390%2Fbiology15131090). ISSN 2079-7737 (https://search.worldcat.org/issn/2079-7737). PMC 13359715 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC13359715). PMID 42450637 (https://pubmed.ncbi.nlm.nih.gov/42450637).
121. Holtz, T. R. Jr. (March 19, 2013) [Lecture held March 8, 2013]. *The Life and Times of Tyrannosaurus rex, with Dr. Thomas Holtz* (https://www.youtube.com/watch?v=sqkqkxYGNZc) (Lecture). Seattle, WA: Burke Museum of Natural History and Culture. Archived (https://ghostarchive.org/varchive/youtube/20211211/sqkqkxYGNZc) from the original on December 11, 2021. Retrieved October 12, 2013.
122. Paul, G. S. (2008). "Chapter 18: The Extreme Life Style and Habits of the Gigantic Tyrannosaurid Superpredators of the Cretaceous North America and Asia" (https://books.google.com/books?id=5WH9RnfKco4C&pg=PA307). In Larson, P. L.; Carpenter, K. (eds.). Tyrannosaurus*, The Tyrant King*. Indiana University Press. pp. 307–345. ISBN 978-0-253-35087-9. Retrieved September 14, 2013.
123. Bell, P. R.; Campione, N. E.; Persons IV, W. S.; Currie, P. J.; Larson, P. L.; Tanke, D. H.; Bakker, R. T. (2017). "Tyrannosauroid integument reveals conflicting patterns of gigantism and feather evolution" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5493735). *Biology Letters*. **13** (6) 20170092. doi:10.1098/rsbl.2017.0092 (https://doi.org/10.1098%2Frsbl.2017.0092). PMC 5493735 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5493735). PMID 28592520 (https://pubmed.ncbi.nlm.nih.gov/28592520).
124. Farago, J. (March 7, 2019). "T. Rex Like You Haven't Seen Him: With Feathers" (https://www.nytimes.com/2019/03/07/arts/design/t-rex-exhibition-american-museum-of-natural-history.html). *The New York Times*. Archived (https://web.archive.org/web/20190308002505/https://www.nytimes.com/2019/03/07/arts/design/t-rex-exhibition-american-museum-of-natural-history.html) from the original on March 8, 2019. Retrieved March 7, 2019.
125. Xing, X.; Norell, M. A.; Kuang, X.; Wang, X.; Zhao, Q.; Jia, C. (October 7, 2004). "Basal tyrannosauroids from China and evidence for protofeathers in tyrannosauroids" (http://doc.rero.ch/record/15283/files/PAL_E2582.pdf) (PDF). *Nature*. **431** (7009): 680–684. Bibcode:2004Natur.431..680X (https://ui.adsabs.harvard.edu/abs/2004Natur.431..680X). doi:10.1038/nature02855 (https://doi.org/10.1038%2Fnature02855). PMID 15470426 (https://pubmed.ncbi.nlm.nih.gov/15470426). S2CID 4381777 (https://api.semanticscholar.org/CorpusID:4381777). Archived (https://web.archive.org/web/20230125185242/https://doc.rero.ch/record/15283/files/PAL_E2582.pdf) (PDF) from the original on January 25, 2023. Retrieved January 23, 2023.

126. Xing, X.; Wang, K.; Zhang; Ma, Q.; Xing, L.; Sullivan, C.; Hu, D.; Cheng, S.; Wang, S. (April 5, 2012). "A gigantic feathered dinosaur from the Lower Cretaceous of China" (https://web.archive.org/web/20120417134949/http://www.xinglida.net/pdf/Xu_et_al_2012_Yutyrannus.pdf) (PDF). *Nature*. **484** (7392): 92–95. Bibcode:2012Natur.484...92X (https://ui.adsabs.harvard.edu/abs/2012Natur.484...92X). doi:10.1038/nature10906 (https://doi.org/10.1038%2Fnature10906). PMID 22481363 (https://pubmed.ncbi.nlm.nih.gov/22481363). S2CID 29689629 (https://api.semanticscholar.org/CorpusID:29689629). Archived from the original (http://www.xinglida.net/pdf/Xu_et_al_2012_Yutyrannus.pdf) (PDF) on April 17, 2012.
127. Naish, D. "The Sensitive Face of a Big Predatory Dinosaur" (https://blogs.scientificamerican.com/tetrapod-zoology/the-sensitive-face-of-a-big-predatory-dinosaur/). *Tetrapod Zoology*. Scientific American Blog Network. Archived (https://web.archive.org/web/20181206102230/https://blogs.scientificamerican.com/tetrapod-zoology/the-sensitive-face-of-a-big-predatory-dinosaur/) from the original on December 6, 2018. Retrieved December 5, 2018.
128. Milinkovitch, Michel; Manukyan, Liana; Debry, Adrien; Di-Poi, Nicolas; Martin, Samuel; Singh, Daljit; Lambert, Dominique; Zwicker, Matthias (January 4, 2013). "Crocodile Head Scales Are Not Developmental Units But Emerge from Physical Cracking" (https://doi.org/10.1126%2Fscience.1226265). *Science*. **339** (6115): 78–81. Bibcode:2013Sci...339...78M (https://ui.adsabs.harvard.edu/abs/2013Sci...339...78M). doi:10.1126/science.1226265 (https://doi.org/10.1126%2Fscience.1226265). PMID 23196908 (https://pubmed.ncbi.nlm.nih.gov/23196908). S2CID 6859452 (https://api.semanticscholar.org/CorpusID:6859452).
129. Carpenter, K. (1992). "Variation in *Tyrannosaurus rex*". In Carpenter, K.; Currie, P. J. (eds.). *Dinosaur Systematics: Approaches and Perspectives*. Cambridge: Cambridge University Press. pp. 141–145. ISBN 978-0-521-43810-0.
130. Larson, P. L. (1994). "*Tyrannosaurus* sex". In Rosenberg, G. D.; Wolberg, D. L. (eds.). *Dino Fest*. Vol. 7. The Paleontological Society Special Publications. pp. 139–155.
131. Erickson, G. M.; Kristopher, L. A.; Larson, P. (2005). "Androgynous rex – the utility of chevrons for determining the sex of crocodilians and non-avian dinosaurs" (http://doc.rero.ch/record/14378/files/PAL_E1827.pdf) (PDF). *Zoology (Jena, Germany)*. **108** (4): 277–86. Bibcode:2005Zool..108..277E (https://ui.adsabs.harvard.edu/abs/2005Zool..108..277E). doi:10.1016/j.zool.2005.08.001 (https://doi.org/10.1016%2Fj.zool.2005.08.001). PMID 16351976 (https://pubmed.ncbi.nlm.nih.gov/16351976). Archived (https://web.archive.org/web/20230404163719/https://doc.rero.ch/record/14378/files/PAL_E1827.pdf) (PDF) from the original on April 4, 2023. Retrieved February 2, 2023.
132. Schweitzer, M. H.; Elsey, R. M.; Dacke, C. G.; Horner, J. R.; Lamm, E. T. (2007). "Do egg-laying crocodilian (*Alligator mississippiensis*) archosaurs form medullary bone?". *Bone*. **40** (4): 1152–8. Bibcode:2007Bone...40.1152S (https://ui.adsabs.harvard.edu/abs/2007Bone...40.1152S). doi:10.1016/j.bone.2006.10.029 (https://doi.org/10.1016%2Fj.bone.2006.10.029). PMID 17223615 (https://pubmed.ncbi.nlm.nih.gov/17223615).
133. Leidy, Joseph (1865). "Memoir on the extinct reptiles of the Cretaceous formations of the United States" (https://books.google.com/books?id=H89jMYnptaIC&pg=PA1). *Cretaceous reptiles of the United States* (https://www.biodiversitylibrary.org/item/88605). Smithsonian Contributions to Knowledge. Vol. 14. pp. 1–4. doi:10.5962/bhl.title.39830 (https://doi.org/10.5962%2Fbhl.title.39830).
134. "*Tyrannosaurus*" (https://web.archive.org/web/20081208143829/http://www.amnh.org/exhibitions/expeditions/treasure_fossil/Treasures/Tyrannosaurus/tyrannos.html?dinos). American Museum of Natural History. Archived from the original (http://www.amnh.org/exhibitions/expeditions/treasure_fossil/Treasures/Tyrannosaurus/tyrannos.html?dinos) on December 8, 2008. Retrieved October 16, 2008.
135. Newman, B. H. (1970). "Stance and gait in the flesh-eating *Tyrannosaurus*" (https://zenodo.org/record/3674749). *Biological Journal of the Linnean Society*. **2** (2): 119–123. doi:10.1111/j.1095-8312.1970.tb01707.x (https://doi.org/10.1111%2Fj.1095-8312.1970.tb01707.x). Archived (https://web.archive.org/web/20200804080335/https://zenodo.org/record/3674749) from the original on August 4, 2020. Retrieved March 15, 2020.

136. "The Age of Reptiles Mural" (https://web.archive.org/web/20180929053007/http://peabody.yale.edu/exhibits/age-reptiles-mural). Yale University. 2008. Archived from the original (http://peabody.yale.edu/exhibits/age-reptiles-mural) on September 29, 2018. Retrieved October 16, 2008.
137. Ross, R. M.; Duggan-Haas, D.; Allmon, W. D. (2013). "The Posture of *Tyrannosaurus rex*: Why Do Student Views Lag Behind the Science?". *Journal of Geoscience Education*. **61** (1): 145–160. Bibcode:2013JGeEd..61..145R (https://ui.adsabs.harvard.edu/abs/2013JGeEd..61..145R). doi:10.5408/11-259.1 (https://doi.org/10.5408%2F11-259.1). S2CID 162343784 (https://api.semanticscholar.org/CorpusID:162343784).
138. "*Tyrannosaurus Rex*: Not a tripod anymore" (https://www.aaas.org/tyrannosaurus-rex-not-tripod-anymore). *American Association for the Advancement of Science*. April 2, 2013. Archived (https://web.archive.org/web/20181206053308/https://www.aaas.org/tyrannosaurus-rex-not-tripod-anymore) from the original on December 6, 2018. Retrieved December 5, 2018.
139. Stevens, Kent A. (2011). "Tyrannosaurus rex – "Rex, sit" " (http://ix.cs.uoregon.edu/~kent/paleontology/Tyrannosaurus/index.html). *ix.cs.uoregon.edu*. Archived (https://web.archive.org/web/20201003173931/http://ix.cs.uoregon.edu/~kent/paleontology/Tyrannosaurus/index.html) from the original on October 3, 2020. Retrieved July 26, 2020.
140. "If T. rex fell, how did it get up, given its tiny arms and low center of gravity?" (https://www.scientificamerican.com/article/if-t-rex-fell-how-did-it/). *Scientific American*. Archived (https://web.archive.org/web/20200726053130/https://www.scientificamerican.com/article/if-t-rex-fell-how-did-it/) from the original on July 26, 2020. Retrieved July 26, 2020.
141. Padian K (2022). "Why tyrannosaurid forelimbs were so short: An integrative hypothesis" (https://www.app.pan.pl/archive/published/app67/app009212021.pdf) Archived (https://web.archive.org/web/20220518090424/https://app.pan.pl/archive/published/app67/app009212021.pdf) May 18, 2022, at the Wayback Machine. *Acta Palaeontologica Polonica* **67**(1): p. 63-76
142. Stevens K.A., Larson P, Willis E.D. & Anderson A. "Rex, sit: digital modeling of *Tyrannosaurus rex* at rest". In Larson P & Carpenter K (eds.). *Tyrannosaurus rex, the tyrant king* (Indiana University Press, 2008). p. 192-203
143. Lambe, L. M. (1914). "On a new genus and species of carnivorous dinosaur from the Belly River Formation of Alberta, with a description of the skull of *Stephanosaurus marginatus* from the same horizon". *Ottawa Naturalist*. **27**: 129–135.
144. Horner, J. R.; Lessem, D. (1993). *The complete T. rex* (https://archive.org/details/completetrexhows00horn). New York City: Simon & Schuster. ISBN 978-0-671-74185-3.
145. "A New View of T. Rex | Smithsonian National Museum of Natural History" (https://web.archive.org/web/20200413044033/https://naturalhistory.si.edu/education/teaching-resources/paleontology/new-view-t-rex). April 13, 2020. Archived from the original (https://naturalhistory.si.edu/education/teaching-resources/paleontology/new-view-t-rex) on April 13, 2020. Retrieved April 13, 2020.
146. Carpenter, K.; Smith, M. (2001). "Forelimb Osteology and Biomechanics of *Tyrannosaurus rex*" (https://archive.org/details/mesozoicvertebra0000unse/page/90). In Tanke, D. H.; Carpenter, K. (eds.). *Mesozoic vertebrate life*. Bloomington: Indiana University Press. pp. 90–116 (https://archive.org/details/mesozoicvertebra0000unse/page/90). ISBN 978-0-253-33907-2.
147. Pickrell, J. (November 2, 2017). "T. Rex's Tiny Arms May Have Been Vicious Weapons" (https://web.archive.org/web/20171102164511/https://news.nationalgeographic.com/2017/11/tyrannosaurus-rex-arms-weapons-paleontology-science/). *National Geographic*. Archived from the original (https://news.nationalgeographic.com/2017/11/tyrannosaurus-rex-arms-weapons-paleontology-science/) on November 2, 2017. Retrieved December 10, 2018.

148. Bakker, R. T. (1968). "The superiority of dinosaurs" (https://web.archive.org/web/20060909131058/http://bio.fsu.edu/~amarquez/Evolutionary%20Morphology%20fall%202004/Bakker/Bakker%201968%20-%20Superiority%20of%20DInos.pdf) (PDF). *Discovery*. **3** (2): 11–12. Archived from the original (http://bio.fsu.edu/~amarquez/Evolutionary%20Morphology%20fall%202004/Bakker/Bakker%201968%20-%20Superiority%20of%20DInos.pdf) (PDF) on September 9, 2006. Retrieved October 7, 2008.
149. Bakker, R. T. (1972). "Anatomical and ecological evidence of endothermy in dinosaurs" (https://web.archive.org/web/20060909173036/http://bio.fsu.edu/~amarquez/Evolutionary%20Morphology%20fall%202004/Bakker/14-%20Bakker%201972%20-%20dino%20endothermy.pdf) (PDF). *Nature*. **238** (5359): 81–85. Bibcode:1972Natur.238...81B (https://ui.adsabs.harvard.edu/abs/1972Natur.238...81B). doi:10.1038/238081a0 (https://doi.org/10.1038%2F238081a0). S2CID 4176132 (https://api.semanticscholar.org/CorpusID:4176132). Archived from the original (http://bio.fsu.edu/~amarquez/Evolutionary%20Morphology%20fall%202004/Bakker/14-%20Bakker%201972%20-%20dino%20endothermy.pdf) (PDF) on September 9, 2006. Retrieved October 7, 2008.
150. Barrick, R. E.; Showers, W. J. (1994). "Thermophysiology of *Tyrannosaurus rex*: Evidence from Oxygen Isotopes". *Science*. **265** (5169): 222–224. Bibcode:1994Sci...265..222B (https://ui.adsabs.harvard.edu/abs/1994Sci...265..222B). doi:10.1126/science.265.5169.222 (https://doi.org/10.1126%2Fscience.265.5169.222). PMID 17750663 (https://pubmed.ncbi.nlm.nih.gov/17750663). S2CID 39392327 (https://api.semanticscholar.org/CorpusID:39392327).
151. Trueman, C.; Chenery, C.; Eberth, D. A.; Spiro, B. (2003). "Diagenetic effects on the oxygen isotope composition of bones of dinosaurs and other vertebrates recovered from terrestrial and marine sediments" (http://doc.rero.ch/record/14987/files/PAL_E2137.pdf) (PDF). *Journal of the Geological Society*. **160** (6): 895–901. Bibcode:2003JGSoc.160..895T (https://ui.adsabs.harvard.edu/abs/2003JGSoc.160..895T). doi:10.1144/0016-764903-019 (https://doi.org/10.1144%2F0016-764903-019). S2CID 130658189 (https://api.semanticscholar.org/CorpusID:130658189). Archived (https://web.archive.org/web/20230404175035/https://doc.rero.ch/record/14987/files/PAL_E2137.pdf) (PDF) from the original on April 4, 2023. Retrieved February 2, 2023.
152. Barrick, R. E.; Showers, W. J. (1999). "Thermophysiology and biology of *Giganotosaurus*: comparison with *Tyrannosaurus*" (https://web.archive.org/web/20110517220816/http://palaeo-electronica.org/1999_2/gigan/issue2_99.htm). *Palaeontologia Electronica*. **2** (2). Archived from the original (http://palaeo-electronica.org/1999_2/gigan/issue2_99.htm) on May 17, 2011. Retrieved October 7, 2008.
153. Barrick, R. E.; Stoskopf, M. K.; Showers, W. J. (1999). "Oxygen isotopes in dinosaur bones". In Farlow, J. O.; Brett-Surman, M. K. (eds.). *The Complete Dinosaur*. Bloomington: Indiana University Press. pp. 474–490. ISBN 978-0-253-21313-6.
154. Wiemann, J.; Menéndez, I.; Crawford, J.M.; Fabbri, M.; Gauthier, J.A.; Hull, P.M.; Norell, M.A.; Briggs, D.E.G. (2022). "Fossil biomolecules reveal an avian metabolism in the ancestral dinosaur" (https://resolver.caltech.edu/CaltechAUTHORS:20220531-924660000). *Nature*. **606** (7914): 522–526. Bibcode:2022Natur.606..522W (https://ui.adsabs.harvard.edu/abs/2022Natur.606..522W). doi:10.1038/s41586-022-04770-6 (https://doi.org/10.1038%2Fs41586-022-04770-6). PMID 35614213 (https://pubmed.ncbi.nlm.nih.gov/35614213). S2CID 249064466 (https://api.semanticscholar.org/CorpusID:249064466).
155. Paladino, F. V.; Spotila, J. R.; Dodson, P. (1999). "A blueprint for giants: modeling the physiology of large dinosaurs". In Farlow, J. O.; Brett-Surman, M. K. (eds.). *The Complete Dinosaur*. Bloomington: Indiana University Press. pp. 491–504. ISBN 978-0-253-21313-6.
156. Chinsamy, A.; Hillenius, W. J. (2004). "Physiology of nonavian dinosaurs". In Weishampel, D. B.; Dodson, P.; Osmólska, H. (eds.). *The dinosauria* (https://archive.org/details/dinosauriandedit00weis). Berkeley: University of California Press. pp. 643 (https://archive.org/details/dinosauriandedit00weis/page/n661)–659. ISBN 978-0-520-24209-8.

157. Seymour, R. S. (July 5, 2013). "Maximal Aerobic and Anaerobic Power Generation in Large Crocodiles versus Mammals: Implications for Dinosaur Gigantothermy" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3702618). *PLOS ONE*. **8** (7) e69361. Bibcode:2013PLoSO...869361S (https://ui.adsabs.harvard.edu/abs/2013PLoSO...869361S). doi:10.1371/journal.pone.0069361 (https://doi.org/10.1371%2Fjournal.pone.0069361). ISSN 1932-6203 (https://search.worldcat.org/issn/1932-6203). PMC 3702618 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3702618). PMID 23861968 (https://pubmed.ncbi.nlm.nih.gov/23861968).
158. Holliday, C.M.; Porter, W.R.; Vilet, K.A.; Witmer, L.M. (2019). "The Frontoparietal Fossa and Dorsotemporal Fenestra of Archosaurs and Their Significance for Interpretations of Vascular and Muscular Anatomy in Dinosaurs" (https://doi.org/10.1002%2Far.24218). *The Anatomical Record*. **303** (4): 1060–1074. doi:10.1002/ar.24218 (https://doi.org/10.1002%2Far.24218). PMID 31260177 (https://pubmed.ncbi.nlm.nih.gov/31260177). S2CID 195756776 (https://api.semanticscholar.org/CorpusID:195756776).
159. Flores, Randon J.; Eagle, Robert A.; Trayler, Robin B.; et al. (2026). "The body temperature of Tyrannosaurus rex" (https://www.science.org/doi/10.1126/sciadv.aeb7653?referrer=https%3A%2F%2Fwww.google.com%2F). *Science Advances*. **12** (38). doi:10.1126/sciadv.aeb7653 (https://doi.org/10.1126%2Fsciadv.aeb7653).
160. Fields, H. (2006). "Dinosaur Shocker" (https://web.archive.org/web/20081014011547/http://www.smithsonianmag.com/science-nature/10021606.html). *Smithsonian Magazine*. Archived from the original (http://www.smithsonianmag.com/science-nature/10021606.html) on October 14, 2008. Retrieved October 2, 2008.
161. Schweitzer, M. H.; Wittmeyer, J. L.; Horner, J. R.; Toporski, J. K. (2005). "Soft-tissue vessels and cellular preservation in *Tyrannosaurus rex*". *Science*. **307** (5717): 1952–5. Bibcode:2005Sci...307.1952S (https://ui.adsabs.harvard.edu/abs/2005Sci...307.1952S). doi:10.1126/science.1108397 (https://doi.org/10.1126%2Fscience.1108397). PMID 15790853 (https://pubmed.ncbi.nlm.nih.gov/15790853). S2CID 30456613 (https://api.semanticscholar.org/CorpusID:30456613).
162. Rincon, P. (April 12, 2007). "Protein links T. rex to chickens" (http://news.bbc.co.uk/2/hi/science/nature/6548719.stm). *BBC News*. Archived (https://web.archive.org/web/20180907082950/http://news.bbc.co.uk/2/hi/science/nature/6548719.stm) from the original on September 7, 2018. Retrieved October 2, 2008.
163. Anné, J.; Canoville, A.; Edwards, N. P.; Schweitzer, M. H.; Zanno, L. E. (2023). "Independent Evidence for the Preservation of Endogenous Bone Biochemistry in a Specimen of *Tyrannosaurus rex*" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9953530). *Biology*. **12** (2) 264. doi:10.3390/biology12020264 (https://doi.org/10.3390%2Fbiology12020264). PMC 9953530 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9953530). PMID 36829540 (https://pubmed.ncbi.nlm.nih.gov/36829540).
164. Vergano, D. (April 13, 2007). "Yesterday's T. Rex is today's chicken" (https://www.usatoday.com/tech/science/discoveries/2007-04-12-trex-protein_N.htm). *USA Today*. Archived (https://web.archive.org/web/20120419005821/http://www.usatoday.com/tech/science/discoveries/2007-04-12-trex-protein_N.htm) from the original on April 19, 2012. Retrieved October 8, 2008.
165. Kaye, T. G.; Gaugler, G.; Sawlowicz, Z. (2008). Stepanova, A. (ed.). "Dinosaurian Soft Tissues Interpreted as Bacterial Biofilms" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2483347). *PLOS ONE*. **3** (7) e2808. Bibcode:2008PLoSO...3.2808K (https://ui.adsabs.harvard.edu/abs/2008PLoSO...3.2808K). doi:10.1371/journal.pone.0002808 (https://doi.org/10.1371%2Fjournal.pone.0002808). PMC 2483347 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2483347). PMID 18665236 (https://pubmed.ncbi.nlm.nih.gov/18665236).
166. "New Research Challenges Notion That Dinosaur Soft Tissues Still Survive" (http://newswise.com/articles/view/542898/) (Press release). Newswise. July 24, 2008. Archived (https://web.archive.org/web/20170730134938/http://newswise.com/articles/view/542898/) from the original on July 30, 2017. Retrieved October 8, 2008.

167. "Researchers Debate: Is It Preserved Dinosaur Tissue, or Bacterial Slime?" (https://web.archive.org/web/20080807005940/http://blogs.discovermagazine.com/80beats/2008/07/30/researchers-debate-is-it-preserved-dinosaur-tissue-or-bacterial-slime/) (Press release). Discover. July 30, 2008. Archived from the original (https://blogs.discovermagazine.com/80beats/2008/07/30/researchers-debate-is-it-preserved-dinosaur-tissue-or-bacterial-slime/) on August 7, 2008. Retrieved September 4, 2008.
168. San Antonio, J. D.; Schweitzer, M. H.; Jensen, S. T.; Kalluri, R.; Buckley, M.; Orgel, J. P. R. O. (2011). Van Veen, H. W. (ed.). "Dinosaur Peptides Suggest Mechanisms of Protein Survival" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3110760). *PLOS ONE*. **6** (6) e20381. Bibcode:2011PLoSO...620381S (https://ui.adsabs.harvard.edu/abs/2011PLoSO...620381S). doi:10.1371/journal.pone.0020381 (https://doi.org/10.1371%2Fjournal.pone.0020381). PMC 3110760 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3110760). PMID 21687667 (https://pubmed.ncbi.nlm.nih.gov/21687667).
169. Peterson, J. E.; Lenczewski, M. E.; Scherer, R. P. (October 12, 2010). "Influence of Microbial Biofilms on the Preservation of Primary Soft Tissue in Fossil and Extant Archosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2953520). *PLOS ONE*. **5** (10) e13334. Bibcode:2010PLoSO...513334P (https://ui.adsabs.harvard.edu/abs/2010PLoSO...513334P). doi:10.1371/journal.pone.0013334 (https://doi.org/10.1371%2Fjournal.pone.0013334). PMC 2953520 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2953520). PMID 20967227 (https://pubmed.ncbi.nlm.nih.gov/20967227). "[T]he interpretation of preserved organic remains as microbial biofilm [is] highly unlikely"
170. Hutchinson, J. R.; Garcia, M. (2002). "*Tyrannosaurus* was not a fast runner". *Nature*. **415** (6875): 1018–21. Bibcode:2002Natur.415.1018H (https://ui.adsabs.harvard.edu/abs/2002Natur.415.1018H). doi:10.1038/4151018a (https://doi.org/10.1038%2F4151018a). PMID 11875567 (https://pubmed.ncbi.nlm.nih.gov/11875567). S2CID 4389633 (https://api.semanticscholar.org/CorpusID:4389633).
171. Hutchinson, J. R. (2004). "Biomechanical Modeling and Sensitivity Analysis of Bipedal Running Ability. II. Extinct Taxa" (https://web.archive.org/web/20081031093050/http://www.rvc.ac.uk/AboutUs/Staff/jhutchinson/documents/JRH13.pdf) (PDF). *Journal of Morphology*. **262** (1): 441–461. Bibcode:2004JMorp.262..441H (https://ui.adsabs.harvard.edu/abs/2004JMorp.262..441H). doi:10.1002/jmor.10240 (https://doi.org/10.1002%2Fjmor.10240). PMID 15352202 (https://pubmed.ncbi.nlm.nih.gov/15352202). S2CID 15677774 (https://api.semanticscholar.org/CorpusID:15677774). Archived from the original (http://www.rvc.ac.uk/AboutUs/Staff/jhutchinson/documents/JRH13.pdf) (PDF) on October 31, 2008.
172. Holtz, T. R. (May 1, 1996). "Phylogenetic taxonomy of the Coelurosauria (Dinosauria; Theropoda)". *Journal of Paleontology*. **70** (3): 536–538. Bibcode:1996JPal...70..536H (https://ui.adsabs.harvard.edu/abs/1996JPal...70..536H). doi:10.1017/S0022336000038506 (https://doi.org/10.1017%2FS0022336000038506).
173. Benton, M. (2014). *Vertebrate Palaeontology* (https://books.google.com/books?id=VThUUUtM8A4C&q=tyrannosaurus+metatarsals+squeezing&pg=PA193) (4th ed.). John Wiley & Sons. p. 193. ISBN 978-1-118-40755-4.
174. Dececchi, T. Alexander; Mloszewska, Aleksandra M.; Holtz, Thomas R. Jr.; Habib, Michael B.; Larsson, Hans C. E. (May 13, 2020). "The fast and the frugal: Divergent locomotory strategies drive limb lengthening in theropod dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7220109). *PLOS ONE*. **15** (5) e0223698. Bibcode:2020PLoSO..1523698D (https://ui.adsabs.harvard.edu/abs/2020PLoSO..1523698D). doi:10.1371/journal.pone.0223698 (https://doi.org/10.1371%2Fjournal.pone.0223698). PMC 7220109 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7220109). PMID 32401793 (https://pubmed.ncbi.nlm.nih.gov/32401793).
175. "T. rex was a champion walker, super-efficient at lower speeds" (https://www.eurekalert.org/news-releases/500578). *EurekAlert!*. May 13, 2020. Archived (https://web.archive.org/web/20230816101705/https://www.eurekalert.org/news-releases/500578) from the original on August 16, 2023. Retrieved August 16, 2023.

176. van Bijlert, P. A.; van Soest, A. J. K.; Schulp, A. S. (2021). "Natural Frequency Method: estimating the preferred walking speed of *Tyrannosaurus rex* based on tail natural frequency" (https://phys.org/news/2021-04-dinosaur-biomechanical-tyrannosaurus-rex-gait.html). *Royal Society Open Science*. **8** (4) 201441. Bibcode:2021RSOS....801441V (https://ui.adsabs.harvard.edu/abs/2021RSOS....801441V). doi:10.1098/rsos.201441 (https://doi.org/10.1098%2Frsos.201441). PMC 8059583 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8059583). PMID 33996115 (https://pubmed.ncbi.nlm.nih.gov/33996115). S2CID 233312053 (https://api.semanticscholar.org/CorpusID:233312053). Archived (https://web.archive.org/web/20210422224520/https://phys.org/news/2021-04-dinosaur-biomechanical-tyrannosaurus-rex-gait.html) from the original on April 22, 2021. Retrieved April 22, 2021.
177. "Why *Tyrannosaurus* was a slow runner and why the largest are not always the fastest" (https://www.sciencedaily.com/releases/2017/07/170717115657.htm). *ScienceDaily*. July 17, 2017. Archived (https://web.archive.org/web/20190402111657/https://www.sciencedaily.com/releases/2017/07/170717115657.htm) from the original on April 2, 2019. Retrieved November 10, 2017.
178. Hirt, M. R.; Jetz, W.; Rall, B. C.; Brose, U. (2017). "A general scaling law reveals why the largest animals are not the fastest". *Nature Ecology & Evolution*. **1** (8): 1116–1122. Bibcode:2017NatEE...1.1116H (https://ui.adsabs.harvard.edu/abs/2017NatEE...1.1116H). doi:10.1038/s41559-017-0241-4 (https://doi.org/10.1038%2Fs41559-017-0241-4). PMID 29046579 (https://pubmed.ncbi.nlm.nih.gov/29046579). S2CID 425473 (https://api.semanticscholar.org/CorpusID:425473).
179. Sellers, W. I.; Pond, S. B.; Brassey, C. A.; Manning, P. L.; Bates, K. T. (July 18, 2017). "Investigating the running abilities of *Tyrannosaurus rex* using stress-constrained multibody dynamic analysis" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5518979). *PeerJ*. **5** e3420. Bibcode:2017PeerJ...5e3420S (https://ui.adsabs.harvard.edu/abs/2017PeerJ...5e3420S). doi:10.7717/peerj.3420 (https://doi.org/10.7717%2Fpeerj.3420). ISSN 2167-8359 (https://search.worldcat.org/issn/2167-8359). PMC 5518979 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5518979). PMID 28740745 (https://pubmed.ncbi.nlm.nih.gov/28740745).
180. Cotton, J. R.; Hartman, S. A.; Currie, P. J.; Witmer, L. M.; Russell, A. P.; Holtz, T. R. Jr.; Burns, M. E.; Surring, L. A.; Mallison, H.; Henderson, D. M.; O'Brien, H.; Snively, E. (February 21, 2019). "Lower rotational inertia and larger leg muscles indicate more rapid turns in tyrannosaurids than in other large theropods" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6387760). *PeerJ*. **7** e6432. Bibcode:2019PeerJ...7e6432S (https://ui.adsabs.harvard.edu/abs/2019PeerJ...7e6432S). doi:10.7717/peerj.6432 (https://doi.org/10.7717%2Fpeerj.6432). PMC 6387760 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6387760). PMID 30809441 (https://pubmed.ncbi.nlm.nih.gov/30809441).
181. Lockley, M. G.; Hunt, A. P. (1994). "A track of the giant theropod dinosaur *Tyrannosaurus* from close to the Cretaceous/Tertiary boundary, northern New Mexico". *Ichnos*. **3** (3): 213–218. Bibcode:1994Ichno...3..213L (https://ui.adsabs.harvard.edu/abs/1994Ichno...3..213L). doi:10.1080/10420949409386390 (https://doi.org/10.1080%2F10420949409386390).
182. "A Probable Tyrannosaurid Track From the Hell Creek Formation (Upper Cretaceous), Montana, United States" (https://web.archive.org/web/20071214014855/http://www.nhm.ac.uk/about-us/news/2007/october/news_12515.html). *National Museum of History News*. 2007. Archived from the original (http://www.nhm.ac.uk/about-us/news/2007/october/news_12515.html) on December 14, 2007. Retrieved December 18, 2007.
183. Manning, P. L.; Ott, C.; Falkingham, P. L. (2009). "The first tyrannosaurid track from the Hell Creek Formation (Late Cretaceous), Montana, U.S.A". *PALAIOS*. **23** (10): 645–647. Bibcode:2008Palai..23..645M (https://ui.adsabs.harvard.edu/abs/2008Palai..23..645M). doi:10.2110/palo.2008.p08-030r (https://doi.org/10.2110%2Fpalo.2008.p08-030r). S2CID 129985735 (https://api.semanticscholar.org/CorpusID:129985735).

184. Smith, S. D.; Persons, W. S.; Xing, L. (2016). "A 'Tyrannosaur' trackway at Glenrock, Lance Formation (Maastrichtian), Wyoming". *Cretaceous Research*. **61** (1): 1–4. Bibcode:2016CrRes..61....1S (https://ui.adsabs.harvard.edu/abs/2016CrRes..61....1S). doi:10.1016/j.cretres.2015.12.020 (https://doi.org/10.1016%2Fj.cretres.2015.12.020).
185. Perkins, S. (2016). "You could probably have outrun a *T. rex*". *Palaeontology*. doi:10.1126/science.aae0270 (https://doi.org/10.1126%2Fscience.aae0270).
186. Walton, T. (2016). "Forget all you know from Jurassic Park: For speed, *T. rex* beats *velociraptor*s" (https://www.usatoday.com/story/news/2016/01/27/tyrannosaurs-faster-than-velociraptors/79423372/). *USA Today*. Archived (https://web.archive.org/web/20180819045903/https://www.usatoday.com/story/news/2016/01/27/tyrannosaurs-faster-than-velociraptors/79423372/) from the original on August 19, 2018. Retrieved March 13, 2016.
187. Ruiz, J. (2017). "Comments on "A tyrannosaur trackway at Glenrock, Lance Formation (Maastrichtian), Wyoming" (Smith et al., *Cretaceous Research*, v. 61, pp. 1–4, 2016)". *Cretaceous Research*. **82**: 81–82. doi:10.1016/j.cretres.2017.05.033 (https://doi.org/10.1016%2Fj.cretres.2017.05.033).
188. Caneer, T.; Molkestad, T.; Lucas, S.G. (2021). "TRACKS IN THE UPPER CRETACEOUS OF THE RATON BASIN POSSIBLY SHOW TYRANNOSAURID RISING FROM A PRONE POSITION" (https://www.researchgate.net/publication/348002331). *New Mexico Museum of Natural History and Science*: 29–37.
189. Smith, Sean D.; Persons, W. Scott; Xing, Lida (June 2016). "A tyrannosaur trackway at Glenrock, Lance Formation (Maastrichtian), Wyoming". *Cretaceous Research*. **61**: 1–4. Bibcode:2016CrRes..61....1S (https://ui.adsabs.harvard.edu/abs/2016CrRes..61....1S). doi:10.1016/j.cretres.2015.12.020 (https://doi.org/10.1016%2Fj.cretres.2015.12.020).
190. Falkingham, Peter L.; Hups, Kent M.; Tamez-Galvan, Evan M.; Bercovici, Antoine D.; Lyson, Tyler R. (September 8, 2026). "A large tridactyl trackway from the Upper Cretaceous Hell Creek Formation of North Dakota, likely attributable to an adult Tyrannosaurus rex" (https://www.tandfonline.com/doi/full/10.1080/02724634.2026.2706201). *Journal of Vertebrate Paleontology* e2706201. Taylor & Francis. doi:10.1080/02724634.2026.2706201 (https://doi.org/10.1080%2F02724634.2026.2706201). ISSN 0272-4634 (https://search.worldcat.org/issn/0272-4634).
191. Lazaro, Enrico de (September 9, 2026). "Paleontologists Discover First-Ever Adult Tyrannosaurus rex Trackway | Sci.News" (https://www.sci.news/paleontology/adult-tyrannosaurus-rex-trackway-15056.html). *Sci.News: Breaking Science News*. Retrieved September 10, 2026.
192. "A 66.5-million-year-old trackway captures adult T. rex at walking pace" (https://web.archive.org/web/20260909184411/https://phys.org/news/2026-09-million-year-trackway-captures-adult.html). *Phys.org*. September 9, 2026. Archived from the original (https://phys.org/news/2026-09-million-year-trackway-captures-adult.html) on September 9, 2026. Retrieved September 10, 2026.
193. Witmer, L. M.; Ridgely, R. C. (2009). "New Insights into the Brain, Braincase, and Ear Region of Tyrannosaurs (Dinosauria, Theropoda), with Implications for Sensory Organization and Behavior" (https://doi.org/10.1002%2Far.20983). *The Anatomical Record*. **292** (9): 1266–1296. doi:10.1002/ar.20983 (https://doi.org/10.1002%2Far.20983). PMID 19711459 (https://pubmed.ncbi.nlm.nih.gov/19711459). S2CID 17978731 (https://api.semanticscholar.org/CorpusID:17978731).
194. Stevens, K. A. (April 1, 2011). "The Binocular Vision of Theropod Dinosaurs" (http://ix.cs.uoregon.edu/~kent/paleontology/binocularVision/). Archived (https://web.archive.org/web/20180819050655/http://ix.cs.uoregon.edu/~kent/paleontology/binocularVision/) from the original on August 19, 2018. Retrieved July 19, 2013.
195. Carpenter, K. (2013). "A Closer Look at the Scavenging versus Predation by *Tyrannosaurus rex*". In Parrish, M. J.; Molnar, R. E.; Currie, P. J.; Koppelhus, E. B. (eds.). *Tyrannosaurid Paleobiology*. Life of the Past. Bloomington (Ind.): Indiana University Press. pp. 265–278. ISBN 978-0-253-00930-2.

196. "*T. Rex* brain study reveals a refined 'nose' " (http://www.canada.com/calgaryherald/story.html?id=3641f27e-ca2e-44e8-a56b-f9a0b4aef4b5). *Calgary Herald*. October 28, 2008. Archived (https://web.archive.org/web/20081206065850/http://www.canada.com/calgaryherald/story.html?id=3641f27e-ca2e-44e8-a56b-f9a0b4aef4b5) from the original on December 6, 2008. Retrieved October 29, 2008.
197. Kawabe, Soichiro; Hattori, Soki (2021). "Complex neurovascular system in the dentary of Tyrannosaurus" (https://doi.org/10.1080%2F08912963.2021.1965137). *Historical Biology*. **34** (7): 1137–1145. Bibcode:2022HBio...34.1137K (https://ui.adsabs.harvard.edu/abs/2022HBio...34.1137K). doi:10.1080/08912963.2021.1965137 (https://doi.org/10.1080%2F08912963.2021.1965137).
198. "T. rex's jaw had sensors that made it an even more fearsome predator" (https://phys.org/news/2021-08-rex-jaw-sensors-fearsome-predator.html). *phys.org*. Archived (https://web.archive.org/web/20210823232627/https://phys.org/news/2021-08-rex-jaw-sensors-fearsome-predator.html) from the original on August 23, 2021. Retrieved August 23, 2021.
199. Benoit, Florian Bouabdellah, Emily Lessner, and Julien (January 20, 2022). "The rostral neurovascular system of Tyrannosaurus rex" (https://palaeo-electronica.org/content/2022/3518-t-rex-trigeminal-canals). *Palaeontologia Electronica*. **25** (1): 1–20. doi:10.26879/1178 (https://doi.org/10.26879%2F1178). ISSN 1094-8074 (https://search.worldcat.org/issn/1094-8074). S2CID 246204236 (https://api.semanticscholar.org/CorpusID:246204236). Archived (https://web.archive.org/web/20220316214524/https://palaeo-electronica.org/content/2022/3518-t-rex-trigeminal-canals) from the original on March 16, 2022. Retrieved January 22, 2022.
200. Hurlburt, G. S.; Ridgely, R. C.; Witmer, L. M. (July 5, 2013). "Relative size of brain and cerebrum in Tyrannosaurid dinosaurs: an analysis using brain-endocast quantitative relationships in extant alligators" (https://www.researchgate.net/publication/256536375). In Parrish, M. J.; Molnar, R. E.; Currie, P. J.; Koppelhus, E. B. (eds.). *Tyrannosaurid Paleobiology*. Indiana University Press. pp. 134–154. ISBN 978-0-253-00947-0. Retrieved October 20, 2013.
201. Brusatten, Steve (2018). *The Rise and Fall of the Dinosaurs*. New York, New York: HarperCollins Publishers. p. 219. ISBN 978-0-06-249043-8.
202. "Dino Gangs" (https://web.archive.org/web/20120119125039/http://www.discoveryuk.com/the-loop/dino-gangs/). *Discovery Channel*. June 22, 2011. Archived from the original (http://www.discoveryuk.com/the-loop/dino-gangs/) on January 19, 2012. Retrieved January 19, 2012.
203. Collins, N. (June 22, 2011). "Tyrannosaurus Rex 'hunted in packs' " (https://www.telegraph.co.uk/news/science/dinosaurs/8589113/Tyrannosaurus-Rex-hunted-in-packs.html). *The Telegraph*. Archived (https://ghostarchive.org/archive/20220110/https://www.telegraph.co.uk/news/science/dinosaurs/8589113/Tyrannosaurus-Rex-hunted-in-packs.html) from the original on January 10, 2022. Retrieved March 23, 2014.
204. Wallis, P. (June 11, 2012). "Op-Ed: T. Rex pack hunters? Scary, but likely to be true" (http://digitaljournal.com/article/326451). *Digitaljournal.com*. Archived (https://web.archive.org/web/20210416141803/http://www.digitaljournal.com/article/326451) from the original on April 16, 2021. Retrieved December 23, 2015.
205. Switek, B. (July 25, 2011). "A bunch of bones doesn't make a gang of bloodthirsty tyrannosaurs" (https://www.theguardian.com/science/blog/2011/jul/25/bunch-bones-gang-bloodthirsty-tyrannosaurs). *The Guardian*. Archived (https://web.archive.org/web/20181221011020/https://www.theguardian.com/science/blog/2011/jul/25/bunch-bones-gang-bloodthirsty-tyrannosaurs) from the original on December 21, 2018. Retrieved June 21, 2015.
206. Currie, Philip J. (1998). "Possible evidence of gregarious behaviour in tyrannosaurids" (https://web.archive.org/web/20090326062833/http://www.mnhn.ul.pt/geologia/gaia/21.pdf) (PDF). *Gaia*. **15**: 271–277. Archived from the original (http://www.mnhn.ul.pt/geologia/gaia/21.pdf) (PDF) on March 26, 2009. Retrieved May 3, 2009. (not printed until 2000)

207. Sample, I. (July 23, 2014). "Researchers find first sign that tyrannosaurs hunted in packs" (https://www.theguardian.com/science/2014/jul/23/tyrannosaurs-hunted-packs-tracks-canada). *The Guardian*. Archived (https://web.archive.org/web/20181215141823/https://www.theguardian.com/science/2014/jul/23/tyrannosaurs-hunted-packs-tracks-canada) from the original on December 15, 2018. Retrieved July 28, 2014.
208. McCrea, R. T. (2014). "A 'Terror of Tyrannosaurs': The First Trackways of Tyrannosaurids and Evidence of Gregariousness and Pathology in Tyrannosauridae" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4108409). *PLOS ONE*. **9** (7) e103613. Bibcode:2014PLoSO...9j3613M (https://ui.adsabs.harvard.edu/abs/2014PLoSO...9j3613M). doi:10.1371/journal.pone.0103613 (https://doi.org/10.1371%2Fjournal.pone.0103613). PMC 4108409 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4108409). PMID 25054328 (https://pubmed.ncbi.nlm.nih.gov/25054328).
209. Roach, Brian T.; Brinkman, Daniel T. (2007). "A reevaluation of cooperative pack hunting and gregariousness in *Deinonychus antirrhopus* and other nonavian theropod dinosaurs". *Bulletin of the Peabody Museum of Natural History*. **48** (1): 103–138. doi:10.3374/0079-032X(2007)48[103:AROCPH]2.0.CO;2 (https://doi.org/10.3374%2F0079-032X%282007%2948%5B103%3AAROCPH%5D2.0.CO%3B2). S2CID 84175628 (https://api.semanticscholar.org/CorpusID:84175628).
210. Rothschild, B.M. (2013). "Clawing Their Way to the Top: Tyrannosaurid Pathology and Lifestyle". In Parrish, M. J.; Molnar, R. E.; Currie, P. J.; Koppelhus, E. B. (eds.). *Tyrannosaurid Paleobiology*. Life of the Past. Bloomington (Ind.): Indiana University Press. pp. 211–222. ISBN 978-0-253-00930-2.
211. Anné, J.; Whitney, M.; Brocklehurst, R.; Donnelly, K.; Rothschild, B. (2023). "Unusual lesions seen in the caudals of the hadrosaur, *Edmontosaurus annectens*". *The Anatomical Record*. **306** (3): 594–606. doi:10.1002/ar.25078 (https://doi.org/10.1002%2Far.25078). PMID 36089756 (https://pubmed.ncbi.nlm.nih.gov/36089756).
212. "Time to Slay the *T. rex* Scavenger 'Debate' " (https://web.archive.org/web/20180712030717/https://www.nationalgeographic.com/science/phenomena/2013/07/16/time-to-slay-the-t-rex-scavenger-debate/). *National Geographic*. July 16, 2013. Archived from the original (https://www.nationalgeographic.com/science/phenomena/2013/07/16/time-to-slay-the-t-rex-scavenger-debate/) on July 12, 2018.
213. Witton, Mark Paul (2025). *King Tyrant: A Natural History of Tyrannosaurus Rex*. Princeton University Press. ISBN 978-0-691-24558-4.
214. Black, Riley (April 13, 2012). "When *Tyrannosaurus* Chomped Sauropods". *Smithsonian Magazine*. **25**. Smithsonian Media: 469. doi:10.1671/0272-4634(2005)025[0469:TRFTUC]2.0.CO;2 (https://doi.org/10.1671%2F0272-4634%282005%29025%5B0469%3ATRFTUC%5D2.0.CO%3B2).
215. Martin, Jeremy E.; Hassler, Auguste; Montagnac, Gilles; Therrien, François; Balter, Vincent (February 10, 2022). "The stability of dinosaur communities before the Cretaceous–Paleogene (K–Pg) boundary: A perspective from southern Alberta using calcium isotopes as a dietary proxy". *Geological Society of America Bulletin*. **134** (9–10): 2548–2560. Bibcode:2022GSAB..134.2548M (https://ui.adsabs.harvard.edu/abs/2022GSAB..134.2548M). doi:10.1130/B36222.1 (https://doi.org/10.1130%2FB36222.1). hdl:2164/20498 (https://hdl.handle.net/2164%2F20498).
216. Black, Riley (2012). "The *Tyrannosaurus rex*s Dangerous and Deadly Bite" (https://www.smithsonianmag.com/science-nature/the-tyrannosaurus-rexs-dangerous-and-deadly-bite-37252918/). *Smithsonian Magazine*. Smithsonian Institution. Archived (https://web.archive.org/web/20140513065820/https://www.smithsonianmag.com/science-nature/the-tyrannosaurus-rexs-dangerous-and-deadly-bite-37252918/) from the original on May 13, 2014. Retrieved December 20, 2019.

217. Bates, K. T.; Falkingham, P. L. (February 29, 2012). "Estimating maximum bite performance in *Tyrannosaurus rex* using multi-body dynamics" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3391458). *Biology Letters*. **8** (4): 660–664. Bibcode:2012BiLet...8..660B (https://ui.adsabs.harvard.edu/abs/2012BiLet...8..660B). doi:10.1098/rsbl.2012.0056 (https://doi.org/10.1098%2Frsbl.2012.0056). PMC 3391458 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3391458). PMID 22378742 (https://pubmed.ncbi.nlm.nih.gov/22378742).
218. Scully, C. (2002). *Oxford Handbook of Applied Dental Sciences* (https://archive.org/details/oxfordhandbookof00cris). Oxford University Press. p. 156 (https://archive.org/details/oxfordhandbookof00cris/page/156). ISBN 978-0-19-851096-3.
219. Lautenschlager, Stephan (November 4, 2015). "Estimating cranial musculoskeletal constraints in theropod dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4680622). *Royal Society Open Science*. **2** (11) 150495. Bibcode:2015RSOS....250495L (https://ui.adsabs.harvard.edu/abs/2015RSOS....250495L). doi:10.1098/rsos.150495 (https://doi.org/10.1098%2Frsos.150495). PMC 4680622 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4680622). PMID 26716007 (https://pubmed.ncbi.nlm.nih.gov/26716007).
220. "The better to eat you with? How dinosaurs' jaws influenced diet" (https://web.archive.org/web/20151104162130/https://www.sciencedaily.com/releases/2015/11/151103213705.htm). *ScienceDaily* (Press release). University of Bristol. November 4, 2015. Archived from the original (https://www.sciencedaily.com/releases/2015/11/151103213705.htm) on November 4, 2015. Retrieved September 14, 2018.
221. Lambe, L. B. (1917). "The Cretaceous theropodous dinosaur *Gorgosaurus*" (https://www.biodiversitylibrary.org/item/252624). *Memoirs of the Geological Survey of Canada*. **100**: 1–84. doi:10.4095/101672 (https://doi.org/10.4095%2F101672). Archived (https://web.archive.org/web/20210909121735/https://www.biodiversitylibrary.org/item/252624) from the original on September 9, 2021. Retrieved August 11, 2021.
222. Farlow, J. O.; Holtz (2002). "The fossil record of predation in dinosaurs" (https://web.archive.org/web/20081031093048/http://www.yale.edu/ypmip/predation/Chapter_09.pdf) (PDF). In Kowalewski, M.; Kelley, P. H. (eds.). *The Fossil Record of Predation*. The Paleontological Society Papers. Vol. 8. T. R. Jr. pp. 251–266. Archived from the original (http://www.yale.edu/ypmip/predation/Chapter_09.pdf) (PDF) on October 31, 2008.
223. Horner, J. R. (1994). "Steak knives, beady eyes, and tiny little arms (a portrait of *Tyrannosaurus* as a scavenger)". *The Paleontological Society Special Publication*. **7**: 157–164. doi:10.1017/S2475262200009497 (https://doi.org/10.1017%2FS2475262200009497).
224. Amos, J. (July 31, 2003). "Science/Nature: *T. rex* goes on trial" (http://news.bbc.co.uk/1/hi/sci/tech/3112527.stm). *BBC News*. Archived (https://web.archive.org/web/20120313130020/http://news.bbc.co.uk/1/hi/sci/tech/3112527.stm) from the original on March 13, 2012. Retrieved December 23, 2015.
225. Farlow, James O.; Holtz, Thomas R. (October 2002). "The Fossil Record of Predation in Dinosaurs". *The Paleontological Society Papers*. **8**: 251–266. doi:10.1017/S108933260000111X (https://doi.org/10.1017%2FS108933260000111X).
226. Holtz, Thomas R. (2008). "A critical re-appraisal of the obligate scavenging hypothesis for Tyrannosaurus rex and other tyrant dinosaurs" (https://books.google.com/books?id=5WH9RnfKco4C&pg=PA371). In Larson, Peter L.; Carpenter, Kenneth (eds.). *Tyrannosaurus Rex, the Tyrant King*. Indiana University Press. pp. 371–394. ISBN 978-0-253-35087-9.
227. Chin, K.; Tokaryk, T. T.; Erickson, G. M.; Calk, L. C. (June 18, 1998). "A king-sized theropod coprolite" (https://zenodo.org/record/3943146). *Nature*. **393** (6686): 680–682. Bibcode:1998Natur.393..680C (https://ui.adsabs.harvard.edu/abs/1998Natur.393..680C). doi:10.1038/31461 (https://doi.org/10.1038%2F31461). Archived (https://web.archive.org/web/20201003143612/https://zenodo.org/record/3943146) from the original on October 3, 2020. Retrieved August 12, 2020. Summary at Monastersky, R. (June 20, 1998). "Getting the scoop from the poop of *T. rex*". *Science News*. **153** (25): 391. doi:10.2307/4010364 (https://doi.org/10.2307%2F4010364). JSTOR 4010364 (https://www.jstor.org/stable/4010364).

228. Robert, A. DePalma II; Burnham, David A.; Martin, Larry D.; Rothschild, Bruce M.; Larson, Peter L. (2013). "Physical evidence of predatory behavior in *Tyrannosaurus rex*" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3732924). *Proceedings of the National Academy of Sciences*. **110** (31): 12560–12564. Bibcode:2013PNAS..11012560D (https://ui.adsabs.harvard.edu/abs/2013PNAS..11012560D). doi:10.1073/pnas.1216534110 (https://doi.org/10.1073%2Fpnas.1216534110). PMC 3732924 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3732924). PMID 23858435 (https://pubmed.ncbi.nlm.nih.gov/23858435).
229. Subbaraman, Nidhi(2013, July 16). Crunch! T. rex tooth found in dino tailbone. (http://www.nbcnews.com/science/crunch-t-rex-tooth-found-dino-tailbone-6C10640610,) Retrieved on July 16, 2013
230. "Tyrannosaurus rex tooth found in dinosaur's tail bone 'proves' it was hunter, not scavenger" (https://web.archive.org/web/20130716155946/http://www.news.com.au/technology/sci-tech/tyrannosaurus-rex-tooth-found-in-dinosaurs-tail-bone-proves-it-was-hunter-not-scavenger/story-fn5fsgyc-1226679904234). *news.com.au*. July 16, 2013. Archived from the original (http://www.news.com.au/technology/sci-tech/tyrannosaurus-rex-tooth-found-in-dinosaurs-tail-bone-proves-it-was-hunter-not-scavenger/story-fn5fsgyc-1226679904234) on July 16, 2013. Retrieved July 16, 2013.
231. DePalma, II; Robert, A.; Burnham, David A.; Martin, Larry D.; Rotchild Bruce, M.; Larson, Peter L. (2013). "Physical evidence of predatory behavior in Tyrannosaurus rex" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3732924). *PNAS*. **110** (31): 12560–12564. Bibcode:2013PNAS..11012560D (https://ui.adsabs.harvard.edu/abs/2013PNAS..11012560D). doi:10.1073/pnas.1216534110 (https://doi.org/10.1073%2Fpnas.1216534110). PMC 3732924 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3732924). PMID 23858435 (https://pubmed.ncbi.nlm.nih.gov/23858435).
232. Hotz, Robert Lee(July 16, 2013). Fossil Proves T. Rex Wasn't Just a Scavenger. Retrieved on July 16, 2013, from https://www.wsj.com/articles/SB10001424127887323664204578607911233796672
233. Carpenter, K. (1998). "Evidence of predatory behavior by theropod dinosaurs" (http://www.arca.museus.ul.pt/ArcaSite/obj/gaia/MNHNL-0000778-MG-DOC-web.PDF) (PDF). *Gaia*. **15**: 135–144. Archived (https://web.archive.org/web/20130926134333/http://www.arca.museus.ul.pt/ArcaSite/obj/gaia/MNHNL-0000778-MG-DOC-web.PDF) (PDF) from the original on September 26, 2013. Retrieved March 13, 2014.
234. Happ, J.; Carpenter, K. (2008). "An analysis of predator–prey behavior in a head-to-head encounter between *Tyrannosaurus rex* and *Triceratops*". In Carpenter, K.; Larson, P. E. (eds.). *Tyrannosaurus rex, the Tyrant King (Life of the Past)*. Bloomington: Indiana University Press. pp. 355–368. ISBN 978-0-253-35087-9.
235. Dodson, P. (1996). *The Horned Dinosaurs* (https://archive.org/details/horneddinosaursn00dods). Princeton University Press. p. 19 (https://archive.org/details/horneddinosaursn00dods/page/n32).
236. Tanke, D. H.; Currie, P. J. (1998). "Head-biting behavior in theropod dinosaurs: paleopathological evidence" (https://web.archive.org/web/20080227134632/http://www.mnhn.ul.pt/geologia/gaia/12.pdf) (PDF). *Gaia* (15): 167–184. Archived from the original (http://www.mnhn.ul.pt/geologia/gaia/12.pdf) (PDF) on February 27, 2008.
237. Abler, W. L. (1992). "The serrated teeth of tyrannosaurid dinosaurs, and biting structures in other animals". *Paleobiology*. **18** (2): 161–183. Bibcode:1992Pbio...18..161A (https://ui.adsabs.harvard.edu/abs/1992Pbio...18..161A). doi:10.1017/S0094837300013956 (https://doi.org/10.1017%2FS0094837300013956).
238. Goldstein, E. J. C.; Tyrrell, K. L.; Citron, D. M.; Cox, C. R.; Recchio, I. M.; Okimoto, B.; Bryja, J.; Fry, B. G. (June 1, 2013). "Anaerobic and aerobic bacteriology of the saliva and gingiva from 16 captive Komodo dragons (*Varanus komodoensis*): new implications for the 'bacteria as venom' model". *Journal of Zoo and Wildlife Medicine*. **44** (2): 262–272. doi:10.1638/2012-0022R.1 (https://doi.org/10.1638%2F2012-0022R.1). PMID 23805543 (https://pubmed.ncbi.nlm.nih.gov/23805543).

239. Snively, E.; Cotton, J. R.; Ridgely, R.; Witmer, L. M. (2013). "Multibody dynamics model of head and neck function in *Allosaurus* (Dinosauria, Theropoda)" (https://doi.org/10.26879%2F338). *Palaeontologia Electronica*. **16** (2): 338. Bibcode:2013PalEl..16..338S (https://ui.adsabs.harvard.edu/abs/2013PalEl..16..338S). doi:10.26879/338 (https://doi.org/10.26879%2F338).
240. Peterson, J. E.; Daus, K. N. (March 4, 2019). "Feeding traces attributable to juvenile *Tyrannosaurus rex* offer insight into ontogenetic dietary trends" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6404657). *PeerJ*. **7** e6573. Bibcode:2019PeerJ...7e6573P (https://ui.adsabs.harvard.edu/abs/2019PeerJ...7e6573P). doi:10.7717/peerj.6573 (https://doi.org/10.7717%2Fpeerj.6573). PMC 6404657 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6404657). PMID 30863686 (https://pubmed.ncbi.nlm.nih.gov/30863686).
241. Longrich, N. R.; Horner, J. R.; Erickson, G. M.; Currie, P. J. (2010). "Cannibalism in *Tyrannosaurus rex*" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2955550). *PLOS ONE*. **5** (10) e13419. Bibcode:2010PLoSO...513419L (https://ui.adsabs.harvard.edu/abs/2010PLoSO...513419L). doi:10.1371/journal.pone.0013419 (https://doi.org/10.1371%2Fjournal.pone.0013419). PMC 2955550 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2955550). PMID 20976177 (https://pubmed.ncbi.nlm.nih.gov/20976177).
242. Perkins, S. (October 29, 2015). "Tyrannosaurs were probably cannibals (Comment)" (https://www.science.org/content/article/tyrannosaurs-were-probably-cannibals). *Science*. Archived (https://web.archive.org/web/20221005112604/https://www.science.org/content/article/tyrannosaurs-were-probably-cannibals) from the original on October 5, 2022. Retrieved November 2, 2015.
243. Farlow, James Orville (1989). *Paleobiology of the Dinosaurs* (https://books.google.com/books?id=jXU6atz3RTYC&dq=dinosaur+parenting+behavior&pg=PA21). Geological Society of America. ISBN 978-0-8137-2238-2.
244. "Maiasaura," Dodson, et al. (1994); pages 116-117
245. Horner, J.R.; Makela, R. (1979). "Nest of juveniles provides evidence of family structure among dinosaurs". *Nature*. **282** (5736): 296–298. Bibcode:1979Natur.282..296H (https://ui.adsabs.harvard.edu/abs/1979Natur.282..296H). doi:10.1038/282296a0 (https://doi.org/10.1038%2F282296a0). S2CID 4370793 (https://api.semanticscholar.org/CorpusID:4370793).
246. " "The Best of all Mothers" Maiasaura peeblesorum" (http://bioweb.uwlax.edu/bio203/s2014/fischbac_sama/interactions.htm). *bioweb.uwlax.edu/*. University of Wisconsin-La Crosse. Archived (https://web.archive.org/web/20220616015020/http://bioweb.uwlax.edu/bio203/s2014/fischbac_sama/interactions.htm) from the original on June 16, 2022. Retrieved March 22, 2021.
247. Norell, Mark A.; Clark, James M.; Chiappe, Luis M.; Dashzeveg, Demberelyin (December 28, 1995). "A nesting dinosaur". *Nature*. **378** (6559): 774–776. Bibcode:1995Natur.378..774N (https://ui.adsabs.harvard.edu/abs/1995Natur.378..774N). doi:10.1038/378774a0 (https://doi.org/10.1038%2F378774a0).
248. Watanabe, Myrna E. (March 1, 2009). "Evolving Ideas on the Origins of Parental Care". *BioScience*. **59** (3): 272. Bibcode:2009BiSci..59..272W (https://ui.adsabs.harvard.edu/abs/2009BiSci..59..272W). doi:10.1525/bio.2009.59.3.17 (https://doi.org/10.1525%2Fbio.2009.59.3.17).
249. Rothschild, B.; Tanke, D. H.; Ford, T. L. (2001). "Theropod stress fractures and tendon avulsions as a clue to activity". In Tanke, D. H.; Carpenter, K. (eds.). *Mesozoic Vertebrate Life*. Indiana University Press. pp. 331–336.
250. Wolff, E. D. S.; Salisbury, S. W.; Horner, J. R.; Varricchi, D. J. (2009). Hansen, D. M. (ed.). "Common Avian Infection Plagued the Tyrant Dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2748709). *PLOS ONE*. **4** (9) e7288. Bibcode:2009PLoSO...4.7288W (https://ui.adsabs.harvard.edu/abs/2009PLoSO...4.7288W). doi:10.1371/journal.pone.0007288 (https://doi.org/10.1371%2Fjournal.pone.0007288). PMC 2748709 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2748709). PMID 19789646 (https://pubmed.ncbi.nlm.nih.gov/19789646).

251. Rega, E. (2012). "Disease in Dinosaurs". In Brett-Surman, M.; Holtz, T.; Farlow, J. (eds.). *The Complete Dinosaur*. Bloomington: Indiana University Press.
252. Rothschild, Bruce; O'Connor, Jingmai; Lozado, María Cecilia (December 2022). "Closer examination does not support infection as cause for enigmatic Tyrannosaurus rex mandibular pathologies". *Cretaceous Research*. **140** 105353. Bibcode:2022CrRes.14005353R (https://ui.adsabs.harvard.edu/abs/2022CrRes.14005353R). doi:10.1016/j.cretres.2022.105353 (https://doi.org/10.1016%2Fj.cretres.2022.105353).
253. Arens, Nan Crystal; Allen, Sarah E. (2014), "A florule from the base of the Hell Creek Formation in the type area of eastern Montana: Implications for vegetation and climate" (https://pubs.geoscienceworld.org/books/book/668/chapter/3807605), *Through the End of the Cretaceous in the Type Locality of the Hell Creek Formation in Montana and Adjacent Areas*, Geological Society of America, doi:10.1130/2014.2503(06) (https://doi.org/10.1130%2F2014.2503%2806%29), ISBN 978-0-8137-2503-1, retrieved March 12, 2026
254. Arens, Nan Crystal; Allen, Sarah E. (January 1, 2014). "A florule from the base of the Hell Creek Formation in the type area of eastern Montana: Implications for vegetation and climate". *Through the End of the Cretaceous in the Type Locality of the Hell Creek Formation in Montana and Adjacent Areas*. doi:10.1130/2014.2503(06) (https://doi.org/10.1130%2F2014.2503%2806%29). ISBN 978-0-8137-2503-1.
255. Ott, C.J. (2006). "Cranial Anatomy and Biogeography of the First *Leptoceratops gracilis* (Dinosauria: Ornithischia) Specimens from the Hell Creek Formation, Southeast Montana". In Carpenter, K. (ed.). *Horns and Beaks: Ceratopsian and Ornithopod Dinosaurs*. Indiana University Press. pp. 213–234. ISBN 0-253-34817-X.
256. Campione, N.E.; Evans, D.C. (2011). "Cranial Growth and Variation in Edmontosaurs (Dinosauria: Hadrosauridae): Implications for Latest Cretaceous Megaherbivore Diversity in North America" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3182183). *PLOS ONE*. **6** (9) e25186. Bibcode:2011PLoSO...625186C (https://ui.adsabs.harvard.edu/abs/2011PLoSO...625186C). doi:10.1371/journal.pone.0025186 (https://doi.org/10.1371%2Fjournal.pone.0025186). PMC 3182183 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3182183). PMID 21969872 (https://pubmed.ncbi.nlm.nih.gov/21969872).
257. Weishampel, D.B.; Barrett, P.M.; Coria, R.A.; Le Loeufff, J.; Xu, X.; Zhao, X.; Sahni, A.; Gomani, E.P.M.; Noto, C.R. (2004). "Dinosaur Distribution". In Weishampel, D.B.; Osmólska, H.; Dodson, P. (eds.). *The Dinosauria* (2nd ed.). University of California Press. pp. 517–606. ISBN 0-520-24209-2.
258. Bruns, M.E. (2015). *Intraspecific variation in the armoured dinosaurs (Dinosauria: Ankylosauria)* (PhD thesis). University of Alberta Library. doi:10.7939/R39K46485 (https://doi.org/10.7939%2FR39K46485).
259. Woodruff, D. Cary; Schott, Ryan K.; Evans, David C. (November 15, 2023). "Two new species of small-bodied pachycephalosaurine (Dinosauria, Marginocephalia) from the uppermost Cretaceous of North America suggest hidden diversity in well-sampled formations". *Papers in Palaeontology*. **9** (6) e1535. Bibcode:2023PPal....9E1535W (https://ui.adsabs.harvard.edu/abs/2023PPal....9E1535W). doi:10.1002/spp2.1535 (https://doi.org/10.1002%2Fspp2.1535). ISSN 2056-2799 (https://search.worldcat.org/issn/2056-2799).
260. Horner, John R.; Goodwin, Mark B.; Evans, David C. (April 14, 2023). "A new pachycephalosaurid from the Hell Creek Formation, Garfield County, Montana, U.S.A.". *Journal of Vertebrate Paleontology*. **42** (4) e2190369. Bibcode:2022JVPal..42E0369H (https://ui.adsabs.harvard.edu/abs/2022JVPal..42E0369H). doi:10.1080/02724634.2023.2190369 (https://doi.org/10.1080%2F02724634.2023.2190369). ISSN 0272-4634 (https://search.worldcat.org/issn/0272-4634).

261. Williamson, T.E.; Brusatte, S.L. (2014). "Small Theropod Teeth from the Late Cretaceous of the San Juan Basin, Northwestern New Mexico and Their Implications for Understanding Latest Cretaceous Dinosaur Evolution" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3977837). *PLOS ONE*. **9** (4) e93190. Bibcode:2014PLoSO...993190W (https://ui.adsabs.harvard.edu/abs/2014PLoSO...993190W). doi:10.1371/journal.pone.0093190 (https://doi.org/10.1371%2Fjournal.pone.0093190). PMC 3977837 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3977837). PMID 24709990 (https://pubmed.ncbi.nlm.nih.gov/24709990).
262. Atkins-Weltman, K. L.; Simon, D. J.; Woodward, H. N.; Funston, G. F.; Snively, E. (2024). "A new oviraptorosaur (Dinosauria: Theropoda) from the end-Maastrichtian Hell Creek Formation of North America" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10807829). *PLOS ONE*. **19** (1) e0294901. Bibcode:2024PLoSO..1994901A (https://ui.adsabs.harvard.edu/abs/2024PLoSO..1994901A). doi:10.1371/journal.pone.0294901 (https://doi.org/10.1371%2Fjournal.pone.0294901). PMC 10807829 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10807829). PMID 38266012 (https://pubmed.ncbi.nlm.nih.gov/38266012).
263. Denver W. Fowler; John P. Wilson; Elizabeth A. Freedman Fowler; Christopher R. Noto; Daniel Anduza; John R. Horner (2020). "*Trierarchuncus prairiensis* gen. et sp. nov., the last alvarezsaurid: Hell Creek Formation (uppermost Maastrichtian), Montana" (https://doi.org/10.1016%2Fj.cretres.2020.104560). *Cretaceous Research*. **116** 104560. Bibcode:2020CrRes.11604560F (https://ui.adsabs.harvard.edu/abs/2020CrRes.11604560F). doi:10.1016/j.cretres.2020.104560 (https://doi.org/10.1016%2Fj.cretres.2020.104560). S2CID 225630913 (https://api.semanticscholar.org/CorpusID:225630913).
264. Thomas, Henry N.; Hone, David W. E.; Gomes, Timothy; Peterson, Joseph E. (February 28, 2025). "*Infernodrakon hastacollis* gen. et sp. nov., a new azhdarchid pterosaur from the Hell Creek Formation of Montana, and the pterosaur diversity of Maastrichtian North America". *Journal of Vertebrate Paleontology*. **44** (4) e2442476. Bibcode:2024JVPal..4442476T (https://ui.adsabs.harvard.edu/abs/2024JVPal..4442476T). doi:10.1080/02724634.2024.2442476 (https://doi.org/10.1080%2F02724634.2024.2442476). ISSN 0272-4634 (https://search.worldcat.org/issn/0272-4634).
265. Derstler, K. (1994). "Dinosaurs of the Lance Formation in eastern Wyoming". In Nelson, G. E. (ed.). *The Dinosaurs of Wyoming*. Wyoming Geological Association Guidebook, 44th Annual Field Conference. Wyoming Geological Association. pp. 127–146.
266. Sampson, Scott D.; Loewon, Mark A. (June 27, 2005). "*Tyrannosaurus rex* from the Upper Cretaceous (Maastrichtian) North Horn Formation of Utah: Biogeographic and Paleoecologic Implications". *Journal of Vertebrate Paleontology*. **25** (2): 469–472. doi:10.1671/0272-4634(2005)025[0469:TRFTUC]2.0.CO;2 (https://doi.org/10.1671%2F0272-4634%282005%29025%5B0469%3ATRFTUC%5D2.0.CO%3B2). JSTOR 4524461 (https://www.jstor.org/stable/4524461). S2CID 131583311 (https://api.semanticscholar.org/CorpusID:131583311).
267. Cifelli, Richard L.; Nydam, Randall L.; Eaton, Jeffrey G.; Gardner, James D.; Kirkland, James I. (1999). "Vertebrate faunas of the North Horn Formation (Upper Cretaceous–Lower Paleocene), Emery and Sanpete Counties, Utah". In Gillette, David D. (ed.). *Vertebrate Paleontology in Utah*. Salt Lake City: Utah Geological Survey. pp. 377–388. ISBN 1-55791-634-9.
268. Paul, Gregory S. (October 7, 2025). "Stratigraphic and anatomical evidence for multiple titanosaurid dinosaur taxa in the Late Cretaceous (Campanian-Maastrichtian) of southwestern North America" (https://giw.utahgeology.org/giw/index.php/GIW/article/view/156). *Geology of the Intermountain West*. **12**: 201–220. doi:10.31711/giw.v12.pp201-220 (https://doi.org/10.31711%2Fgiw.v12.pp201-220). Archived (https://web.archive.org/web/20251010051116/https://giw.utahgeology.org/giw/index.php/GIW/article/view/156) from the original on October 10, 2025. Retrieved October 9, 2025.

269. Wick, Steven L.; Lehman, Thomas M. (July 1, 2013). "A new ceratopsian dinosaur from the Javelina Formation (Maastrichtian) of West Texas and implications for chasmosaurine phylogeny". *Naturwissenschaften*. **100** (7): 667–682. Bibcode:2013NW....100..667W (https://ui.adsabs.harvard.edu/abs/2013NW....100..667W). doi:10.1007/s00114-013-1063-0 (https://doi.org/10.1007%2Fs00114-013-1063-0). PMID 23728202 (https://pubmed.ncbi.nlm.nih.gov/23728202). S2CID 16048008 (https://api.semanticscholar.org/CorpusID:16048008).
270. Andres, B.; Langston, W. Jr. (2021). "Morphology and taxonomy of *Quetzalcoatlus* Lawson 1975 (Pterodactyloidea: Azhdarchoidea)" (https://doi.org/10.1080%2F02724634.2021.1907587). *Journal of Vertebrate Paleontology*. **41** (sup1): 142. Bibcode:2021JVPal..41S..46A (https://ui.adsabs.harvard.edu/abs/2021JVPal..41S..46A). doi:10.1080/02724634.2021.1907587 (https://doi.org/10.1080%2F02724634.2021.1907587). ISSN 0272-4634 (https://search.worldcat.org/issn/0272-4634). S2CID 245125409 (https://api.semanticscholar.org/CorpusID:245125409).
271. Tweet, J.S.; Santucci, V.L. (2018). "An Inventory of Non-Avian Dinosaurs from National Park Service Areas" (http://npshistory.com/publications/paleontology/nmmnhs-79-703.pdf) (PDF). *New Mexico Museum of Natural History and Science Bulletin*. **79**: 703–730. Archived (https://web.archive.org/web/20241102020100/https://npshistory.com/publications/paleontology/nmmnhs-79-703.pdf) (PDF) from the original on November 2, 2024. Retrieved June 9, 2024.
272. Jasinski, S. E.; Sullivan, R. M.; Lucas, S. G. (2011). "Taxonomic composition of the Alamo Wash local fauna from the Upper Cretaceous Ojo Alamo Formation (Naashoibito Member) San Juan Basin, New Mexico". *Bulletin*. **53**: 216–271.
273. Serrano-Brañas, C. I.; Torres-Rodrígueza, E.; Luna, P. C. R.; González, I.; González-León, C. (2014). "Tyrannosaurid teeth from the Lomas Coloradas Formation, Cabullona Group (Upper Cretaceous) Sonora, México". *Cretaceous Research*. **49**: 163–171. Bibcode:2014CrRes..49..163S (https://ui.adsabs.harvard.edu/abs/2014CrRes..49..163S). doi:10.1016/j.cretres.2014.02.018 (https://doi.org/10.1016%2Fj.cretres.2014.02.018).
274. Brusatte, S. L.; Carr, T. D. (2016). "The phylogeny and evolutionary history of tyrannosauroid dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4735739). *Scientific Reports*. **6** 20252. Bibcode:2016NatSR...620252B (https://ui.adsabs.harvard.edu/abs/2016NatSR...620252B). doi:10.1038/srep20252 (https://doi.org/10.1038%2Fsrep20252). PMC 4735739 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4735739). PMID 26830019 (https://pubmed.ncbi.nlm.nih.gov/26830019).
275. Chang, Kenneth (April 15, 2021). "How Many Tyrannosaurus Rexes Ever Lived on Earth? Here's a New Clue" (https://ghostarchive.org/archive/20211228/https://www.nytimes.com/2021/04/15/science/tyrannosaurus-rex-population.html). *The New York Times*. Archived from the original (https://www.nytimes.com/2021/04/15/science/tyrannosaurus-rex-population.html) on December 28, 2021.
276. Marshall, Charles R.; Latorre, Daniel V.; Wilson, Connor J.; Frank, Tanner M.; Magoulick, Katherine M.; Zimmt, Joshua B.; Poust, Ashley W. (April 16, 2021). "Absolute abundance and preservation rate of Tyrannosaurus rex" (https://doi.org/10.1126%2Fscience.abc8300). *Science*. **372** (6539): 284–287. Bibcode:2021Sci...372..284M (https://ui.adsabs.harvard.edu/abs/2021Sci...372..284M). doi:10.1126/science.abc8300 (https://doi.org/10.1126%2Fscience.abc8300). PMID 33859033 (https://pubmed.ncbi.nlm.nih.gov/33859033).
277. Meiri, Shai (June 4, 2022). "Population sizes of T. rex cannot be precisely estimated" (https://doi.org/10.21425%2FF5FBG53781). *Frontiers of Biogeography*. **14** (2). doi:10.21425/F5FBG53781 (https://doi.org/10.21425%2FF5FBG53781).

278. Marshall, Charles R.; Latorre, Daniel V.; Wilson, Connor J.; Frank, Tanner M.; Magoulick, Katherine M.; Zimmt, Joshua P.; Poust, Ashley W. (June 4, 2022). "With what precision can the population size of Tyrannosaurus rex be estimated? A reply to Meiri" (https://doi.org/10.21425%2FF5FBG55042). *Frontiers of Biogeography*. **14** (2). doi:10.21425/F5FBG55042 (https://doi.org/10.21425%2FF5FBG55042). Text was copied from this source, which is available under a Creative Commons Attribution 4.0 International License (https://creativecommons.org/licenses/by/4.0/) Archived (https://web.archive.org/web/20171016050101/https://creativecommons.org/licenses/by/4.0/) October 16, 2017, at the Wayback Machine.
279. Riley Black (2022). *The Last Days of the Dinosaurs: An Asteroid, Extinction, and the Beginning of Our World* (https://books.google.com/books?id=6CtBEAAAQBAJ&q=%22in.all.of.prehistory%22&pg=PA1). New York: St Martin's Press. p. 219. ISBN 978-1-250-27105-1.
280. Hone, David (2016). *The Tyrannosaur Chronicles: The Biology of the Tyrant Dinosaurs*. Bloomsbury Publishing. p. 23. ISBN 978-1-4729-1127-8.

Further reading

- Farlow, J. O.; Gatesy, S. M.; Holtz, T. R. Jr.; Hutchinson, J. R.; Robinson, J. M. (2000). "Theropod Locomotion" (https://doi.org/10.1093%2Ficb%2F40.4.640). *American Zoologist*. **40** (4): 640–663. doi:10.1093/icb/40.4.640 (https://doi.org/10.1093%2Ficb%2F40.4.640). JSTOR 3884284 (https://www.jstor.org/stable/3884284).

External links

- The University of Edinburgh Lecture Dr Stephen Brusatte – Tyrannosaur Discoveries Feb 20, 2015 (https://www.youtube.com/watch?v=hVJmPmb_LWY)
- 28 species in the tyrannosaur family tree, when and where they lived (http://www.livescience.com/53877-t-rex-was-invasive-species.html) Stephen Brusatte Thomas Carr 2016
- Australia's answer to T-Rex (https://onesearch.slq.qld.gov.au/permalink/61SLQ_INST/dls06p/alma99274923402061), State Library of Queensland
- American Museum of Natural History (http://www.amnh.org/exhibitions/permanent-exhibitions/fossil-halls/hall-of-saurischian-dinosaurs/tyrannosaurus-rex)

Retrieved from "https://en.wikipedia.org/w/index.php?title=Tyrannosaurus&oldid=1381719568"

WIKIPEDIA
25 years of the free encyclopedia

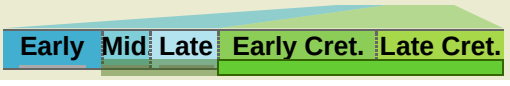
Dromaeosauridae

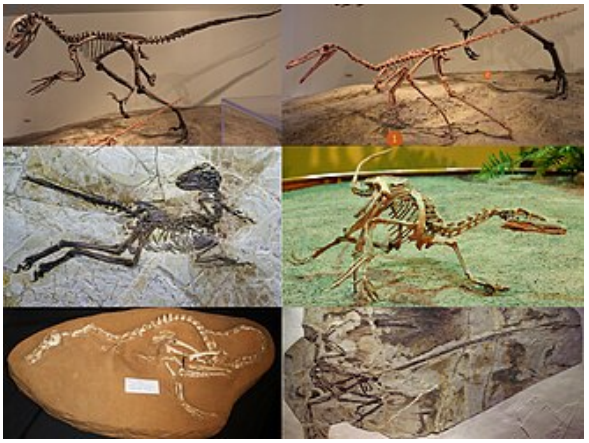
Dromaeosauridae (/ˌdroʊmi.əˈsɔːrɪdiː/) is a family of feathered coelurosaurian theropod dinosaurs. They were generally small to medium-sized feathered carnivores that flourished in the Cretaceous period. The name Dromaeosauridae means 'running lizards', from Greek δρομαῖος (*dromaîos*), meaning 'running at full speed', 'swift', and σαῦρος (*saûros*), meaning 'lizard'. In informal usage, they are often called **raptors**[6] (after *Velociraptor*), a term popularized by the film *Jurassic Park*; several genera include the term "raptor" directly in their name, and popular culture has come to emphasize their bird-like appearance and speculated bird-like behavior.

Definitive dromaeosaurid fossils have been found in North America, Europe and Asia. Some paravian fossils found in other continents have been traditionally regarded as dromaeosaurids, but have recently been reinterpreted as a unique family Unenlagiidae outside Dromaeosauridae, with some authors considering them as a separate lineage of Avialae.[7] The earliest body fossils are known from the Early Cretaceous (145–140 million years ago), and they survived until the end of the Cretaceous (Maastrichtian stage, 66 ma), existing until the Cretaceous–Paleogene extinction event. The presence of dromaeosaurids as early as the Middle Jurassic has been suggested by the discovery of isolated fossil teeth, though no dromaeosaurid body fossils have been found from this period.[8][9]

Dromaeosaurids

Temporal range:





A collection of dromaeosaurid fossil skeletons. Clockwise from upper left: *Deinonychus antirrhopus* (a heavily built eudromaeosaur), *Buitreraptor gonzalezorum* (a long-snouted unenlagiine), *Velociraptor mongoliensis* (a small velociraptorine), *Microraptor gui* (a winged microraptorian), *Halszkaraptor escuilliei* (a semiaquatic halszkaraptorine), *Zhenyuanlong suni* (a long-winged dromaeosaurid)

Scientific classification

Kingdom:	Animalia
Phylum:	Chordata
Class:	Reptilia
Clade:	Dinosauria
Clade:	Saurischia
Clade:	Theropoda
Clade:	Paraves
Family:	†Dromaeosauridae Matthew & Brown, 1922

Description

Technical diagnosis

Dromaeosaurids are diagnosed by the following features: short T-shaped frontals that form the rostral boundary of the supratemporal fenestra; a caudolateral overhanging shelf of the squamosal; a lateral process of the quadrate that contacts the quadratojugal; raised,

stalked, parapophyses on the dorsal vertebrae, a modified pedal digit II; chevrons and prezygapophysis of the caudal vertebrae elongate and spanning several vertebrae; the presence of a subglenoid fossa on the coracoid.[10]

Size and general build

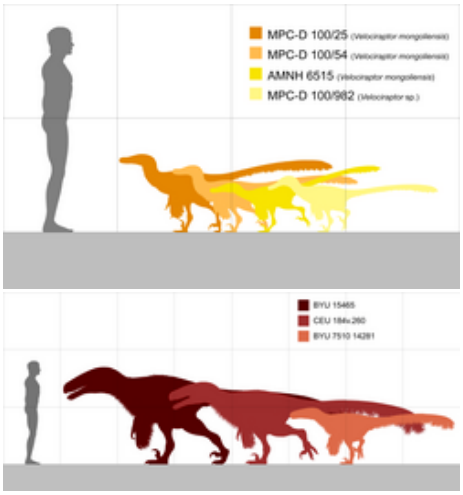
Dromaeosaurids were small to medium-sized dinosaurs, ranging from 1.5–2.07 metres (4.9–6.8 ft) in length (in the case of *Velociraptor*) to approaching or over 6 m (20 ft) (in *Utahraptor*, *Dakotaraptor* and *Achillobator*).[11][12] Large size appears to have evolved at least twice among dromaeosaurids; once among the dromaeosaurines *Utahraptor* and *Achillobator*, and again among the unenlagiines (*Austroraptor*, which measured 5–6 m (16–20 ft) long). A possible third lineage of giant dromaeosaurids is represented by isolated teeth found on the Isle of Wight, England. The teeth belong to an animal the size of the dromaeosaurine *Utahraptor*, but they appear to belong to velociraptorines, judging by the shape of the teeth.[13][14]

The distinctive dromaeosaurid body plan helped to rekindle theories that dinosaurs may have been active, fast, and closely related to birds. Robert Bakker's illustration for John Ostrom's 1969 monograph,[15] showing the dromaeosaurid *Deinonychus* in a fast run, is among the most influential paleontological reconstructions in history.[16] The dromaeosaurid body plan includes a relatively large skull, serrated teeth, narrow snout (an exception being the derived dromaeosaurines), and forward-facing eyes which indicate some degree of binocular vision.[17]

Type species
†*Dromaeosaurus albertensis* Matthew & Brown, 1922
Subgroups[1]

▪ †*Hesperonychus*?[2]
▪ †*Ornithodesmus*
▪ †*Pamparaptor*?[3]
▪ †*Pyroraptor*
▪ †*Variraptor*?[4]
▪ †**Serraraptoria** Wang & Pei 2024
  ▪ †*Daurlong*
  ▪ †*Shanag*?[5]
  ▪ †*Tianyuraptor*
  ▪ †*Zhenyuanlong*
  ▪ †Eudromaeosauria
  ▪ †Microraptoria
▪ †**Unenlagiinia**? Brum et al., 2021
  ▪ †Halszkaraptorinae?
  ▪ †Unenlagiinae

Synonyms
▪ Itemiridae Kurzanov, 1976 ▪ Unenlagiidae? Agnolin & Novas, 2011



Velociraptor (top) and *Utahraptor* (bottom) specimens compared to a 1.8 m (5 ft 11 in) tall human

Life restoration of *Pyroraptor*

Dromaeosaurids, like most other theropods, had a moderately long S-curved neck, and their trunk was relatively short and deep. Like other maniraptorans, they had long arms that could be folded against the body in some species, and relatively large hands with three long fingers (the middle finger being the longest and the first finger being the shortest) ending in large claws.[10] The dromaeosaurid hip structure featured a characteristically large pubic boot projecting beneath the base of the tail. Dromaeosaurid feet bore a large, recurved claw on the second toe. Their tails were slender, with long, low, vertebrae lacking transverse process and neural spines after the 14th caudal vertebra.[10] Ossified uncinate processes of ribs have been identified in several dromaeosaurids.[18][19][20]

Foot

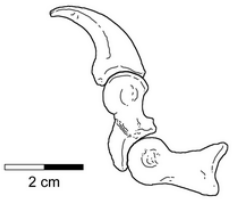


Diagram of the sickle claw and digit of *Adasaurus*

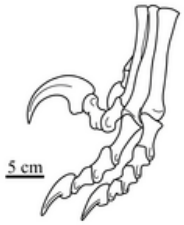


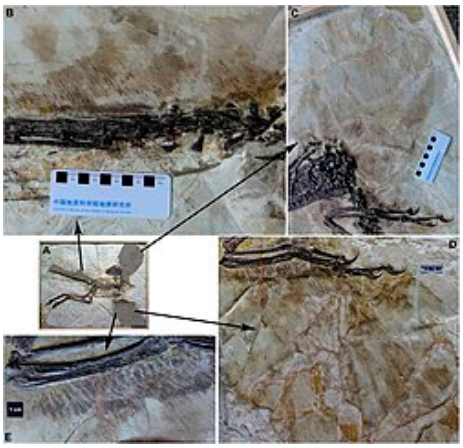
Diagram of the foot of *Deinonychus*

Like other theropods, dromaeosaurids were bipedal; that is, they walked on their hind legs. However, whereas most theropods walked with three toes contacting the ground, fossilized footprint tracks confirm that many early paravian groups, including the dromaeosaurids, held the second toe off the ground in a hyperextended position, with only the third and fourth toes bearing the weight of the animal. This is called functional didactyly.[21] The enlarged second toe bore an unusually large, curved, *falciform* (sickle-shaped, alt. *drepanoid*) claw (held off the ground or 'retracted' when walking), which is thought to have been used in capturing prey and climbing trees (see "Claw function" below). This claw was especially blade-like in the large-bodied predatory eudromaeosaurs.[22] One possible dromaeosaurid species, *Balaur bondoc*, also possessed a first toe which was highly modified in parallel with the second. Both the first and second toes on each foot of *B. bondoc* were also held retracted and bore enlarged, sickle-shaped claws. Similar claws can be found on present day seriema.[23]

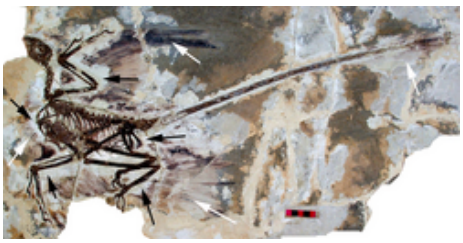
Tail

Dromaeosaurids had long tails. Most of the tail vertebrae bore bony, rod-like extensions (called prezygapophyses), as well as bony tendons in some species. In his study of *Deinonychus*, Ostrom proposed that these features stiffened the tail so that it could only flex at the base, and the whole tail would then move as a single, rigid, lever.[15] However, one well-preserved specimen of *Velociraptor mongoliensis* (IGM 100/986) has an articulated tail skeleton that is curved horizontally in a long S-shape. This suggests that, in life, the tail could bend from side to side with a substantial degree of flexibility.[24] It has been proposed that this tail was used as a stabilizer or counterweight while running or in the air;[24] in *Microraptor*, an elongate diamond-shaped fan of feathers is preserved on the end of the tail. This may have been used as an aerodynamic stabilizer and rudder during gliding or powered flight (see "Flight and gliding" below).[25]

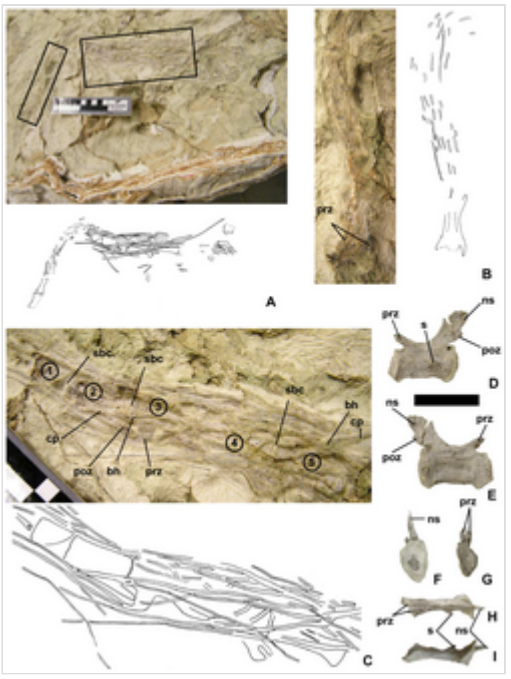
Feathers



Preserved feather traces in a fossil *Zhenyuanlong suni*



Microraptor gui fossil with impressions of feathered wings



Tail of an indeterminate dromaeosaurid (UMNH VP 20209), featuring long prezygapophyses and fossilized bony tendons

There is a large body of evidence showing that dromaeosaurids were covered in feathers. Some dromaeosaurid fossils preserve long, pennaceous feathers on the hands and arms (*remiges*) and tail (*rectrices*), as well as shorter, down-like feathers covering the body.[26][27] Other fossils, which do not preserve actual impressions of feathers, still preserve the associated bumps on the forearm bones where long wing feathers would have attached in life.[28] Overall, this feather pattern looks very much like *Archaeopteryx*.[26]

The first known dromaeosaurid with definitive evidence of feathers was *Sinornithosaurus*, reported from China by Xu et al. in 1999.[27] Many other dromaeosaurid fossils have been found with feathers covering their bodies, some with fully developed feathered wings. *Microraptor* even shows evidence of a second pair of wings on the hind legs.[26] While direct feather impressions are only possible in fine-grained sediments, some fossils found in coarser rocks show evidence of feathers by the presence of quill knobs, the attachment points for wing feathers possessed by some birds. The dromaeosaurids *Rahonavis* and *Velociraptor* have both been found with quill knobs, showing that these forms had feathers despite no impressions having been found. In light of this, it is most likely that even the larger ground-dwelling dromaeosaurids bore feathers, since even flightless birds today retain most of their plumage, and relatively large dromaeosaurids, like *Velociraptor*, are known to have retained pennaceous feathers.[28][29] Though some scientists had suggested that the larger dromaeosaurids lost some or all of their insulatory covering, the discovery of feathers in *Velociraptor* specimens has been cited as evidence that all members of the family retained feathers.[28][30]

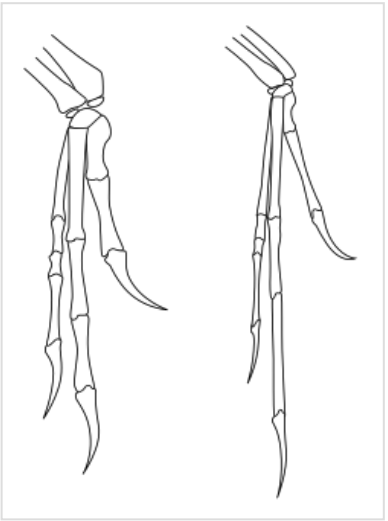
More recently, the discovery of *Zhenyuanlong* established the presence of a full feathered coat in relatively large dromaeosaurids. Additionally, the animal displays proportionally large, aerodynamic wing feathers, as well as a tail-spanning fan, both of which are unexpected traits that may offer an understanding of the integument of large dromaeosaurids.[31] *Dakotaraptor* is an even larger

dromaeosaurid genus with evidence of feathers, albeit indirectly in the form of quill knobs,[32] though the taxon is considered a chimaera by other researchers as even the dinosaurian elements with supposed traits diagnostic for dromaeosaurids are also referrable to caenagnathids and ornithomimosaurs.[33][34]

Classification

Relationship with birds

Dromaeosaurids share many features with early birds (clade Avialae or Aves). The precise nature of their relationship to birds has undergone a great deal of study, and hypotheses about that relationship have changed as large amounts of new evidence became available. As late as 2001, Mark Norell and colleagues analyzed a large survey of coelurosaur fossils and produced the tentative result that dromaeosaurids were most closely related to birds, with troodontids as a more distant outgroup. They even suggested that Dromaeosauridae could be paraphyletic relative to Avialae.[35] In 2002, Hwang and colleagues utilized the work of Norell et al., including new characters and better fossil evidence, to determine that birds (avialans) were better thought of as cousins to the dromaeosaurids and troodontids.[11] The consensus of paleontologists is that there is not yet enough evidence to determine whether any dromaeosaurids could fly or glide, or whether they evolved from ancestors that could.[36]



Comparison of the forelimbs of *Deinonychus* (left) and *Archaeopteryx* (right), one of many skeletal similarities between avians and dromaeosaurids

Alternative theories and flightlessness

Dromaeosaurids are so bird-like that they have led some researchers to argue that they would be better classified as birds. First, since they had feathers, dromaeosaurids (along with many other coelurosaurian theropod dinosaurs) are "birds" under traditional definitions of the word "bird", or "Aves", that are based on the possession of feathers. However, other scientists, such as Lawrence Witmer, have argued that calling a theropod like *Caudipteryx* a bird because it has feathers may stretch the word past any useful meaning.[37]

At least two schools of researchers have proposed that dromaeosaurids may actually be descended from flying ancestors. Hypotheses involving a flying ancestor for dromaeosaurids are sometimes called "Birds Came First" (BCF). George Olshevsky is usually credited as the first author of BCF.[38] In his own work, Gregory S. Paul pointed out numerous features of the dromaeosaurid skeleton that he interpreted as evidence that the entire group had evolved from flying, dinosaurian ancestors, perhaps an animal like *Archaeopteryx*. In that case, the larger dromaeosaurids were secondarily flightless, like the modern ostrich.[29] In 1988, Paul suggested that dromaeosaurids may actually be more closely related to modern birds than to *Archaeopteryx*. By 2002, however, Paul placed dromaeosaurids and *Archaeopteryx* as the closest relatives to one another.[39]

Fossil cast of an extensively feathered *Sinornithosaurus* specimen

In 2002, Hwang et al. found that *Microraptor* was the most primitive dromaeosaurid.[11] Xu and colleagues in 2003 cited the basal position of *Microraptor*, along with feather and wing features, as evidence that the ancestral dromaeosaurid could glide. In that case the larger dromaeosaurids would be secondarily terrestrial—having lost the ability to glide later in their evolutionary history.[26]

Also in 2002, Steven Czerkas described *Cryptovolans*, though it is a probable junior synonym of *Microraptor*. He reconstructed the fossil inaccurately with only two wings and thus argued that dromaeosaurids were powered fliers, rather than passive gliders. He later issued a revised reconstruction in agreement with that of *Microraptor*[40]

The Thermopolis specimen of *Archaeopteryx*, which showed that it also had a hyperextendible second toe[41]

In 2005, Mayr and Peters described the anatomy of a very well preserved specimen of *Archaeopteryx*, and determined that its anatomy was more like non-avian theropods than previously understood. Specifically, they found that *Archaeopteryx* had a primitive palatine, unreversed hallux, and hyper-extendable second toe. Their phylogenetic analysis produced the controversial result that *Confuciusornis* was closer to *Microraptor* than to *Archaeopteryx*, making the Avialae a paraphyletic taxon. They also suggested that the ancestral paravian was able to fly or glide, and that the dromaeosaurids and troodontids were secondarily flightless (or had lost the ability to glide).[42][43] Corfe and Butler criticized this work on methodological grounds.[44]

A challenge to all of these alternative scenarios came when Turner and colleagues in 2007 described a new dromaeosaurid, *Mahakala*, which they found to be the most basal and most primitive member of the Dromaeosauridae, more primitive than *Microraptor*. *Mahakala* had short arms and no ability to glide. Turner et al. also inferred that flight evolved only in the Avialae, and these two points suggested that the ancestral dromaeosaurid could not glide or fly. Based on this cladistic analysis, *Mahakala* suggests that the ancestral condition for dromaeosaurids is non-volant.[45] However, in 2012, an expanded and revised study incorporating the most recent dromaeosaurid finds recovered the *Archaeopteryx*-like *Xiaotingia* as the most primitive member of the clade Dromaeosauridae, which appears to suggest the earliest members of the clade may have been capable of flight.[46]

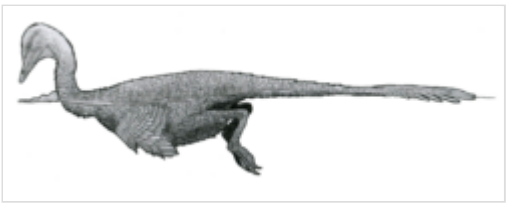
Taxonomy

The authorship of the family Dromaeosauridae is credited to William Diller Matthew and Barnum Brown, who erected it as a subfamily (Dromaeosaurinae) of the family Deinodontidae in 1922, containing only the new genus *Dromaeosaurus*.[47]

The subfamilies of Dromaeosauridae frequently shift in content based on new analysis, but typically consist of the following groups. A number of dromaeosaurids have not been assigned to any particular subfamily, often because they are too poorly preserved to be placed confidently in phylogenetic analysis (see section *Phylogeny* below) or are indeterminate, being assigned to different groups depending on the methodology employed in different papers. The most basal known subfamily of dromaeosaurids is Halszkaraptorinae, a group of bizarre creatures with long fingers and necks, a large number of small teeth, and possible semiaquatic habits.[48] Another enigmatic group, Unenlagiinae, is the most poorly supported subfamily of dromaeosaurids and it is possible that some or all of its members belong outside of Dromaeosauridae.[49][50] The larger, ground-dwelling members like *Buitreraptor* and *Unenlagia* show strong flight adaptations, although they were probably too large to 'take off'. One possible member of this group, *Rahonavis*, is very small, with well-developed wings that show evidence of quill knobs (the attachment points for flight feathers) and it is very likely that it could fly. The next most primitive clade of dromaeosaurids is the Microraptoria. This group includes many of the smallest dromaeosaurids, which show adaptations for living in trees. All known dromaeosaurid skin impressions hail from this group and all show an extensive covering of feathers and well-developed wings. Like the unenlagiines, some species may have been capable of active flight. The most advanced subgroup of dromaeosaurids, Eudromaeosauria, includes stocky and short-legged genera which were likely ambush hunters. This group includes Velociraptorinae, Dromaeosaurinae, and in some studies a third group: Saurornitholestinae. The subfamily Velociraptorinae has traditionally included *Velociraptor*, *Deinonychus*, and *Saurornitholestes*, and while the discovery of *Tsaagan* lent support to this grouping, the inclusion of *Deinonychus*, *Saurornitholestes*, and a few other genera is still uncertain. The Dromaeosaurinae is usually found to consist of medium to giant-sized species, with generally box-shaped skulls (the other subfamilies generally have narrower snouts).[1]

The following classification of the various genera of dromaeosaurids follows the table provided in Holtz, 2011 unless otherwise noted.[1]

Luanchuanraptor, a dromaeosaurid of uncertain placement



Halszkaraptor, a halszkaraptorine

- **Family Dromaeosauridae**
  - *Nuthetes*
  - *Pamparaptor*
  - *Variraptor*
  - *Pyroraptor*
  - *Zhenyuanlong*
  - *Daurlong*
  - **Subfamily Halszkaraptorinae**[48]
    - *Halszkaraptor*
    - *Mahakala*
    - *Hulsanpes*
    - *Natovenator*
  - **Subfamily Unenlagiinae**
    - *Ornithodesmus*
    - *Austroraptor*
    - *Rahonavis*
    - *Unenlagia*

- *Buitreraptor*
- *Neuquenraptor*
- *Unquillosaurus*
- *Ypupiara*
- *Diuqin*
- *Kank*

Austroraptor, an unenlagiine

- **Subfamily Microraptorinae**
  - *Shanag*
  - *Tianyuraptor*
  - *Graciliraptor*
  - *Changyuraptor*
  - *Hesperonychus*
  - *Microraptor*
  - *Sinornithosaurus*
  - *Wulong*
  - *Zhongjianosaurus*
  - *Jian*

Changyuraptor, a microraptorian

- **Node Eudromaeosauria**[22]
  - *Deinonychus*
  - *Dineobellator*
  - *Vectiraptor*
  - **Subfamily Saurornitholestinae**
    - *Bambiraptor*
    - *Saurornitholestes*
    - *Atrociraptor*[22]
    - *Acheroraptor*?
  - **Subfamily Velociraptorinae**
    - *Luanchuanraptor*?
    - *Linheraptor*?
    - *Velociraptor*
    - *Tsaagan*?
    - *Adasaurus*?
    - *Shri*
    - *Kansaignathus*
    - *Kuru*
  - **Subfamily Dromaeosaurinae**
    - *Achillobator*?
    - *Itemirus*?
    - *Dromaeosaurus*
    - *Dakotaraptor*?
    - *Dromaeosauroides*?
    - *Utahraptor*?

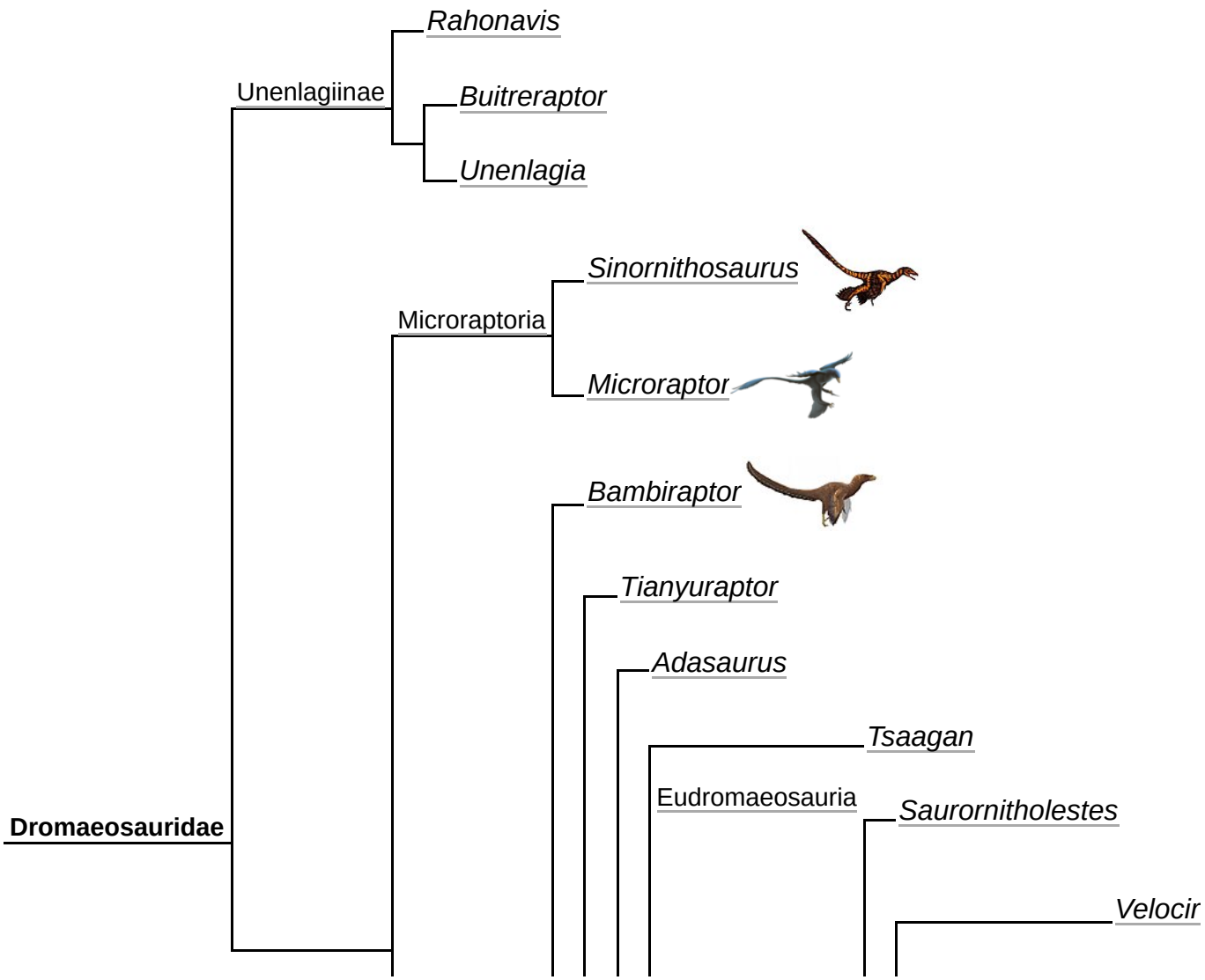
Dromaeosaurus, a eudromaeosaur

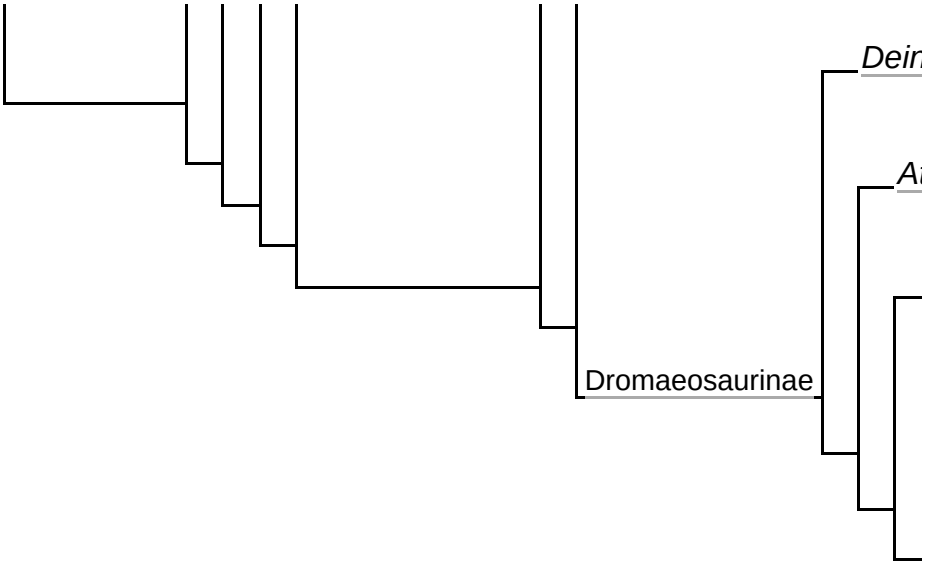
- *Yurgovuchia*?[46]

Phylogeny

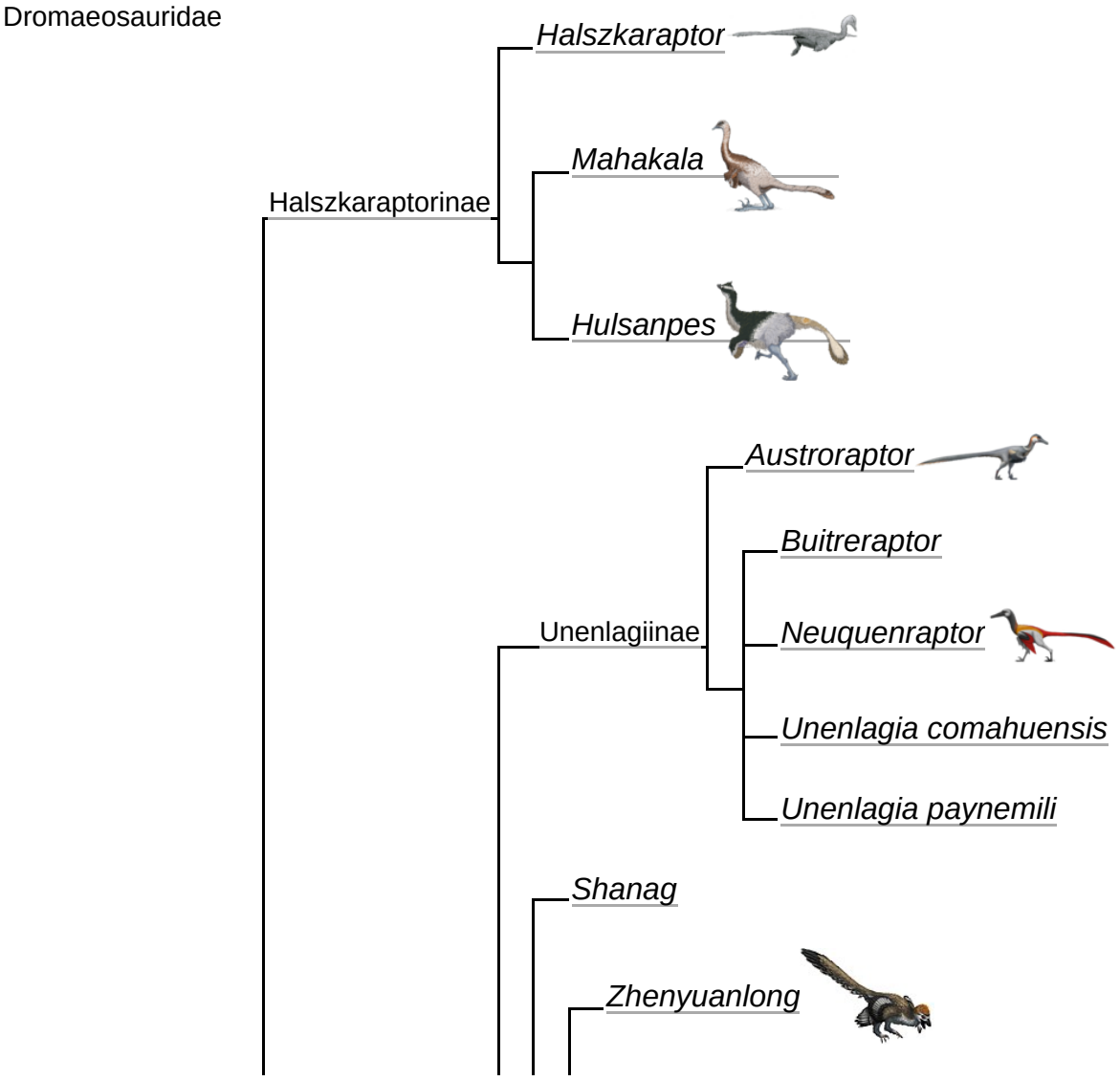
Dromaeosauridae was first defined as a clade by Paul Sereno in 1998, as the most inclusive natural group containing *Dromaeosaurus* but not *Troodon*, *Ornithomimus* or *Passer*. The various "subfamilies" have also been re-defined as clades, usually defined as all species closer to the groups namesake than to *Dromaeosaurus* or any namesakes of other sub-clades (for example, Makovicky defined the clade Unenlagiinae as all dromaeosaurids closer to *Unenlagia* than to *Velociraptor*). The Microraptoria is the only dromaeosaurid sub-clade not converted from a subfamily. Senter and colleagues expressly coined the name without the subfamily suffix *-inae* to avoid perceived issues with erecting a traditional family-group taxon, should the group be found to lie outside dromaeosauridae proper.[51] Sereno offered a revised definition of the sub-group containing *Microraptor* to ensure that it would fall within Dromaeosauridae, and erected the subfamily Microraptorinae, attributing it to Senter et al., though this usage has only appeared on his online TaxonSearch database and has not been formally published.[52] The extensive cladistic analysis conducted by Turner et al. (2012) further supported the monophyly of Dromaeosauridae.[53]

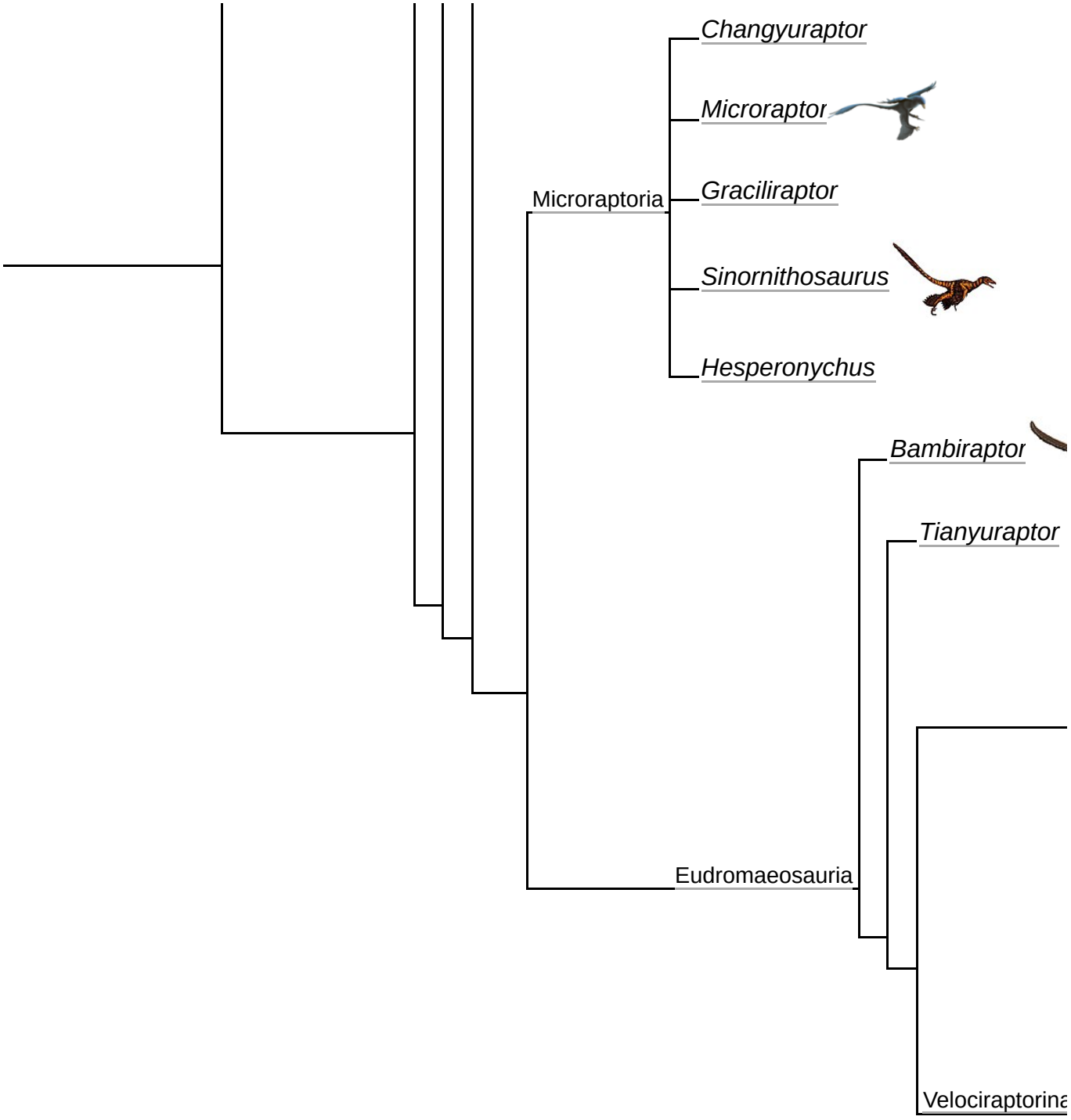
The cladogram below follows a 2015 analysis by DePalma et al. using updated data from the Theropod Working Group.[32]





Another cladogram constructed below follows the phylogenetic analysis conducted in 2017 by Cau et al. using the updated data from the Theropod Working Group in their description of *Halszkaraptor*.[48]




Changyuraptor
Microraptor
Microraptoria
Graciliraptor
Sinornithosaurus
Hesperonychus
Bambiraptor
Tianyuraptor
Eudromaeosauria
Velociraptorina

Paleobiology

Senses

Comparisons between the scleral rings of several dromaeosaurids (*Microraptor*, *Sinornithosaurus*, and *Velociraptor*) and modern birds and reptiles indicate that some dromaeosaurids (including *Microraptor* and *Velociraptor*) may have been nocturnal predators, while *Sinornithosaurus* is inferred to be cathemeral (active throughout the day at short intervals).[54] However, the discovery of iridescent plumage in *Microraptor* has cast doubt on the inference of nocturnality in this genus, as no modern birds that have iridescent plumage are known to be nocturnal.[55]

Studies of the olfactory bulbs of dromaeosaurids reveal that they had similar olfactory ratios for their size to other non-avian theropods and modern birds with an acute sense of smell, such as tyrannosaurids and the turkey vulture, probably reflecting the importance of the olfactory sense in the daily activities of dromaeosaurids such as finding food.[56][57]

Feeding

Dromaeosaurid feeding was discovered to be typical of coelurosaurian theropods, with a characteristic "puncture and pull" feeding method similar to that of modern Komodo dragons.[58] Studies of wear patterns on the teeth of dromaeosaurids by Angelica Torices et al. indicate that dromaeosaurid teeth share similar wear patterns to those seen in the Tyrannosauridae and Troodontidae. However, microwear on the teeth indicated that dromaeosaurids likely preferred larger prey items than the troodontids they often shared their environment with. Such dietary differentiations likely allowed them to inhabit the same environment. The same study also indicated that dromaeosaurids such as *Dromaeosaurus* and *Saurornitholestes* (two dromaeosaurids analyzed in the study) likely included bone in their diet and were better adapted to handle struggling prey while troodontids, equipped with weaker jaws, preyed on softer animals and prey items such as invertebrates and carrion.[59]

Claw function

There is currently disagreement about the function of the enlarged "sickle claw" on the second toe. When John Ostrom described it for *Deinonychus* in 1969, he interpreted the claw as a blade-like slashing weapon, much like the canines of some saber-toothed cats, used with powerful kicks to cut into prey. Adams (1987) suggested that the talon was used to disembowel large ceratopsian dinosaurs.[60] The interpretation of the sickle claw as a killing weapon applied to all dromaeosaurids. However, Manning et al. argued that the claw instead served as a hook, reconstructing the keratinous sheath with an elliptical cross section, instead of the previously inferred inverted teardrop shape.[61] In Manning's interpretation, the second toe claw would be used as a climbing aid when subduing bigger prey and also as a stabbing weapon.

Ostrom compared *Deinonychus* to the ostrich and cassowary. He noted that the bird species can inflict serious injury with the large claw on the second toe.[15] The cassowary has claws up to 125 millimetres (4.9 in) long.[62] Ostrom cited Gilliard (1958) in saying that they can sever an arm or disembowel a man.[63] Kofron (1999 and 2003) studied 241 documented cassowary attacks and found that one human

Life restoration of *Deinonychus* preying on a *Zephyrosaurus* using the sickle claw for prey restraint

Velociraptor restraining an oviraptorosaur with its sickle claws

and two dogs had been killed, but no evidence that cassowaries can disembowel or dismember other animals.[64][65] Cassowaries use their claws to defend themselves, to attack threatening animals, and in agonistic displays such as the Bowed Threat Display.[62] The seriema also has an enlarged second toe claw, and uses it to tear apart small prey items for swallowing.[66]

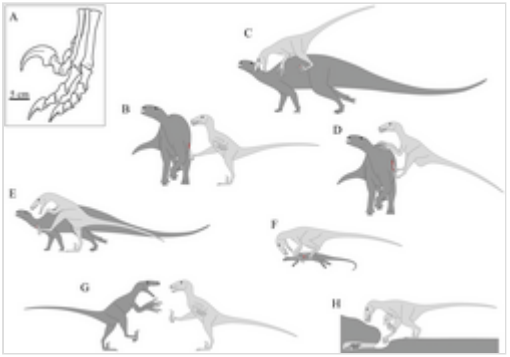
Phillip Manning and colleagues (2009) attempted to test the function of the sickle claw and similarly shaped claws on the forelimbs. They analyzed the bio-mechanics of how stresses and strains would be distributed along the claws and into the limbs, using X-ray imaging to create a three-dimensional contour map of a forelimb claw from *Velociraptor*. For comparison, they analyzed the construction of a claw from a modern predatory bird, the eagle owl. They found that, based on the way that stress was conducted along the claw, they were ideal for climbing. The scientists found that the sharpened tip of the claw was a puncturing and gripping instrument, while the curved and expanded claw base helped transfer stress loads evenly. The Manning team also compared the curvature of the dromaeosaurid "sickle claw" on the foot with curvature in modern birds and mammals. Previous studies had shown that the amount of curvature in a claw corresponded to what lifestyle the animal has: animals with strongly curved claws of a certain shape tend to be climbers, while straighter claws indicate ground-dwelling lifestyles. The sickle claws of the dromaeosaurid *Deinonychus* have a curvature of 160 degrees, well within the range of climbing animals. The forelimb claws they studied also fell within the climbing range of curvature.[67]

Paleontologist Peter Mackovicky commented on the Manning team's study, stating that small, primitive dromaeosaurids (such as *Microraptor*) were likely to have been tree-climbers, but that climbing did not explain why later, gigantic dromaeosaurids such as *Achillobator* retained highly curved claws when they were too large to have climbed trees. Mackovicky speculated that giant dromaeosaurids may have adapted the claw to be used exclusively for latching on to prey.[68]

In 2009 Phil Senter published a study on dromaeosaurid toes and showed that their range of motion was compatible with the excavation of tough insect nests. Senter suggested that small dromaeosaurids such as *Rahonavis* and *Buitreraptor* were small enough to be partial insectivores, while larger genera such as *Deinonychus* and *Neuquenraptor* could have used this ability to catch vertebrate prey residing in insect nests. However, Senter did not test whether the strong curvature of dromaeosaurid claws was also conducive to such activities.[69]

Restoration of a *Saurornitholestes* digging a multituberculate out of a burrow, a function with weak support

In 2011, Denver Fowler and colleagues suggested a new method by which dromaeosaurids may have taken smaller prey. This model, known as the "raptor prey restraint" (RPR) model of predation, proposes that dromaeosaurids killed their prey in a manner very similar to extant accipitrid birds of prey: by leaping onto their quarry, pinning it under their body weight, and gripping it tightly with the large, sickle-shaped claws. Like accipitrids, the dromaeosaurid would then begin to feed on the animal while still alive, until it eventually died from blood loss and organ failure. This proposal is based primarily on comparisons between the morphology and proportions of the feet and legs of dromaeosaurids to several groups of extant birds of prey with known predatory behaviors. Fowler found that the feet and legs of dromaeosaurids most closely resemble those of eagles and hawks, especially in terms of having an enlarged second claw and a similar range of grasping motion. The short metatarsus and foot strength, however, would have been more similar to that of owls. The RPR method of predation would be consistent with other aspects of dromaeosaurid anatomy, such as their unusual dentition and arm morphology. The arms, which could exert a lot of force but were likely covered in long feathers, may have been used as flapping stabilizers for balance while atop a struggling prey animal, along with the stiff counterbalancing tail. Dromaeosaurid jaws, thought by Fowler and colleagues to be comparatively weak, would have been useful for eating prey alive but not as useful for quick, forceful dispatch of the prey. These predatory adaptations working together may also have implications for the origin of flapping in paravians.[70][71]



Proposed scenarios for the sickle claw function, with C, D and F as the more supported behaviors

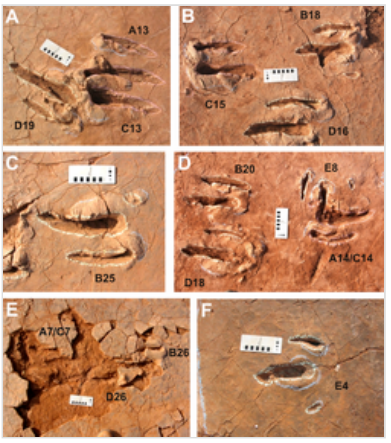
In 2019, Peter Bishop reconstructed the leg skeleton and musculature of *Deinonychus* by using three-dimensional models of muscles, tendons, and bones. With the addition of mathematical models and equations, Bishop simulated the conditions that would provide maximum force at the tip of the sickle claw and therefore the most likely function. Among the proposed modes of the sickle claw use are: kicking to cut, slash or disembowel prey; for gripping onto the flanks of prey; piercing aided by body weight; to attack vital areas of the prey; to restrain prey; intra- or interspecific competition; and digging out prey from hideouts. The results obtained by Bishop showed that a crouching posture increased the claw forces, however, these forces remained relatively weak indicating that the claws were not strong enough to be used in slashing strikes. Rather than being used for slashing, the sickle claws were more likely to be useful in flexed leg angles such as restraining prey and stabbing prey at close quarters. These results are consistent with the Fighting Dinosaurs specimen, which preserves a *Velociraptor* and *Protoceratops* locked in combat, with the former gripping onto the other with its claws in a non-extended leg posture. Despite the obtained results, Bishop considered that the capabilities of the sickle claw could have varied within taxa given that among dromaeosaurids, *Adasaurus* had an unusually smaller sickle claw that retained the characteristic ginglymoid—a structure divided in two parts—and hyperextensible articular surface of the penultimate phalange. He could neither confirm nor disregard that the pedal digit II could have loss or retain its functionally.[72]

A 2020 study by Gianechini et al., also indicates that velociraptorines, dromaeosaurines and other eudromaeosaurs in Laurasia differed greatly in their locomotive and killing techniques from the unenlagiine dromaeosaurids of Gondwana. The shorter second phalanx in the second digit of the foot allowed for increased force to be generated by that digit, which, combined with a shorter and wider metatarsus, and a noticeable marked hinge-like morphology of the articular surfaces of metatarsals and

phalanges, possibly allowed eudromaeosaurs to exert a greater gripping strength than unenlagiines, allowing for more efficient subduing and killing of large prey. In comparison, the unenlagiine dromaeosaurids had a longer and slender subarctometatarsus, and less well-marked hinge joints, a trait that possibly gave them greater cursorial capacities and allowed for greater speed. Additionally, the longer second phalanx of the second digit allowed unenlagiines fast movements of their feet's second digits to hunt smaller and more elusive types of prey. These differences in locomotor and predatory specializations may have been a key feature that influenced the evolutionary pathways that shaped both groups of dromaeosaurs in the northern and southern hemispheres.[73]

Group behavior

Deinonychus fossils have been uncovered in small groups near the remains of the herbivore *Tenontosaurus*, a larger ornithischian dinosaur. This had been interpreted as evidence that these dromaeosaurids hunted in coordinated packs like some modern mammals.[75] However, not all paleontologists found the evidence conclusive, and a subsequent study published in 2007 by Roach and Brinkman suggests that the *Deinonychus* may have actually displayed a disorganized mobbing behavior. Modern diapsids, including birds and crocodiles (the closest relatives of dromaeosaurids), display minimal long-term cooperative hunting (except the aplomado falcon and Harris's hawk); instead, they are usually solitary hunters, either joining forces time to time to increase hunting success (as crocodilians sometimes do), or are drawn to previously killed carcasses, where conflict often occurs between individuals of the same species. For example, in situations where groups of Komodo dragons are eating together, the largest individuals eat first and might attack smaller Komodo dragons that attempt to feed; if the smaller animal dies, it is usually cannibalized. When this information is applied to the sites containing putative pack-hunting behavior in dromaeosaurids, it appears somewhat consistent with a Komodo dragon-like feeding strategy. *Deinonychus* skeletal remains found at these sites are from subadults, with missing parts that may have been eaten by other *Deinonychus*, which a study by Roach et al. presented as evidence against the idea that the animals cooperated in the hunt.[76] A 2020 study done by Frederickson and colleagues found the dietary preferences between juvenile and adult *Deinonychus* to be different. Suggesting that parental feeding ended before the young were large enough to sustain a typical adult diet. This would indicate that the genus did not exhibit mammal-like pack hunting. Despite this, they considered gregariousness to be possible in *Deinonychus*. The Komodo dragon lifestyle was also criticized, due to the lack of spatial distribution of juveniles and adults, suggesting a reduced cannibalistic lifestyle.[77]



Tracks of the ichnogenus *Paravipus didactyloides*, interpreted as representing two individuals that were moving in the same direction[74]

In 2001, multiple *Utahraptor* specimens ranging in age from fully grown adult to tiny three-foot-long baby were found at a site considered by some to be a quicksand predator trap. Some consider this as evidence of family hunting behaviour; however, the full sandstone block is yet to be opened and researchers are unsure as to whether or not the animals died at the same time.[78] Frederickson and colleagues suggests this was a possible sign of gregariousness in *Utahraptor* and dromaeosaurids exhibiting post nestling care.[77]

In 2007, scientists described the first known extensive dromaeosaurid trackway, in Shandong, China. In addition to confirming the hypothesis that the sickle claw was held retracted off the ground, the trackway (made by a large, *Achillobator*-sized species) showed evidence of six individuals of about equal size moving together along a shoreline. The individuals were spaced about one meter apart, traveling in the same direction and walking at a fairly slow pace. The authors of the paper describing these footprints interpreted the trackways as evidence that some species of dromaeosaurids lived in groups. While the trackways clearly do not represent hunting behavior, the idea that groups of dromaeosaurids may have hunted together, according to the authors, could not be ruled out.[21]

Flying and gliding

Restoration of *Microraptor*, featuring an aerodynamic pose

The forearms of dromaeosaurids appear well adapted to resisting the torsional and bending stresses associated with flapping and gliding,[79] and the ability to fly or glide has been suggested for at least five dromaeosaurid species. The first, *Rahonavis ostromi* (originally classified as avian bird, but found to be a dromaeosaurid in later studies[17][80]) may have been capable of powered flight, as indicated by its long forelimbs with evidence of quill knob attachments for long sturdy flight feathers.[81] The forelimbs of *Rahonavis* were more powerfully built than *Archaeopteryx*, and show evidence that they bore strong ligament attachments necessary for flapping flight. Luis Chiappe concluded that, given these adaptations, *Rahonavis* could probably fly but would have been more clumsy in the air than modern birds.[82]

Another species of dromaeosaurid, *Microraptor gui*, may have been capable of gliding using its well-developed wings on both the fore and hind limbs. A 2005 study by Sankar Chatterjee suggested that the wings of *Microraptor* functioned like a split-level "biplane", and that it likely employed a *phugoid* style of gliding, in which it would launch from a perch and swoop downward in a U-shaped curve, then lift again to land on another tree, with the tail and hind wings helping to control its position and speed. Chatterjee also found that *Microraptor* had the basic requirements to sustain level powered flight in addition to gliding.[25]

Changyuraptor yangi is a close relative of *Microraptor gui*, also thought to be a glider or flyer based on the presence of four wings and similar limb proportions. However, it is a considerably larger animal, around the size of a wild turkey, being among the largest known flying Mesozoic paravians.

Another dromaeosaurid species, *Deinonychus antirrhopus*, may display partial flight capacities. The young of this species bore longer arms and more robust pectoral girdles than adults, and which were similar to those seen in other flapping theropods, implying that they may have been capable of flight when young and then lost the ability as they grew.[83]

The possibility that *Sinornithosaurus millenii* was capable of gliding or even powered flight has also been brought up several times,[84][85] though no further studies have occurred.

Zhenyuanlong preserves wing feathers that are aerodynamically shaped, with particularly bird-like coverts as opposed to the longer, wider-spanning coverts of forms like *Archaeopteryx* and *Anchiornis*, as well as fused sternal plates. Due to its size and short arms it is unlikely that *Zhenyuanlong* was capable of

powered flight (though the importance of biomechanical modelling in this regard is stressed[31]), but it may suggest a relatively close descendance from flying ancestors, or even some capacity for gliding or wing-assisted incline running.

Paleopathology

In 2001, Bruce Rothschild and others published a study examining evidence for stress fractures and tendon avulsions in theropod dinosaurs and the implications for their behavior. Since stress fractures are caused by repeated trauma rather than singular events they are more likely to be caused by regular behavior than other types of injuries. The researchers found lesions like those caused by stress fractures on a dromaeosaurid hand claw, one of only two such claw lesions discovered in the course of the study. Stress fractures in the hands have special behavioral significance compared to those found in the feet, since stress fractures in the feet can be obtained while running or during migration. Hand injuries, by contrast, are more likely to be obtained while in contact with struggling prey.[86]

Swimming

At least one dromaeosaurid group, Halszkaraptorinae, whose members are halszkaraptorines, are most likely to have been specialised for aquatic or semiaquatic habits, having developed limb proportions, tooth morphology, and rib cage akin to those of diving birds.[48][87][88]

Life restoration of halszkaraptorine *Natovenator*, depicting swimming behavior

Fishing habits have been proposed for unenlagiines, including comparisons to attributed semi-aquatic spinosaurids,[89] but any aquatic propulsion mechanisms have not been discussed so far.

Reproduction

Restoration of a brooding *Deinonychus*

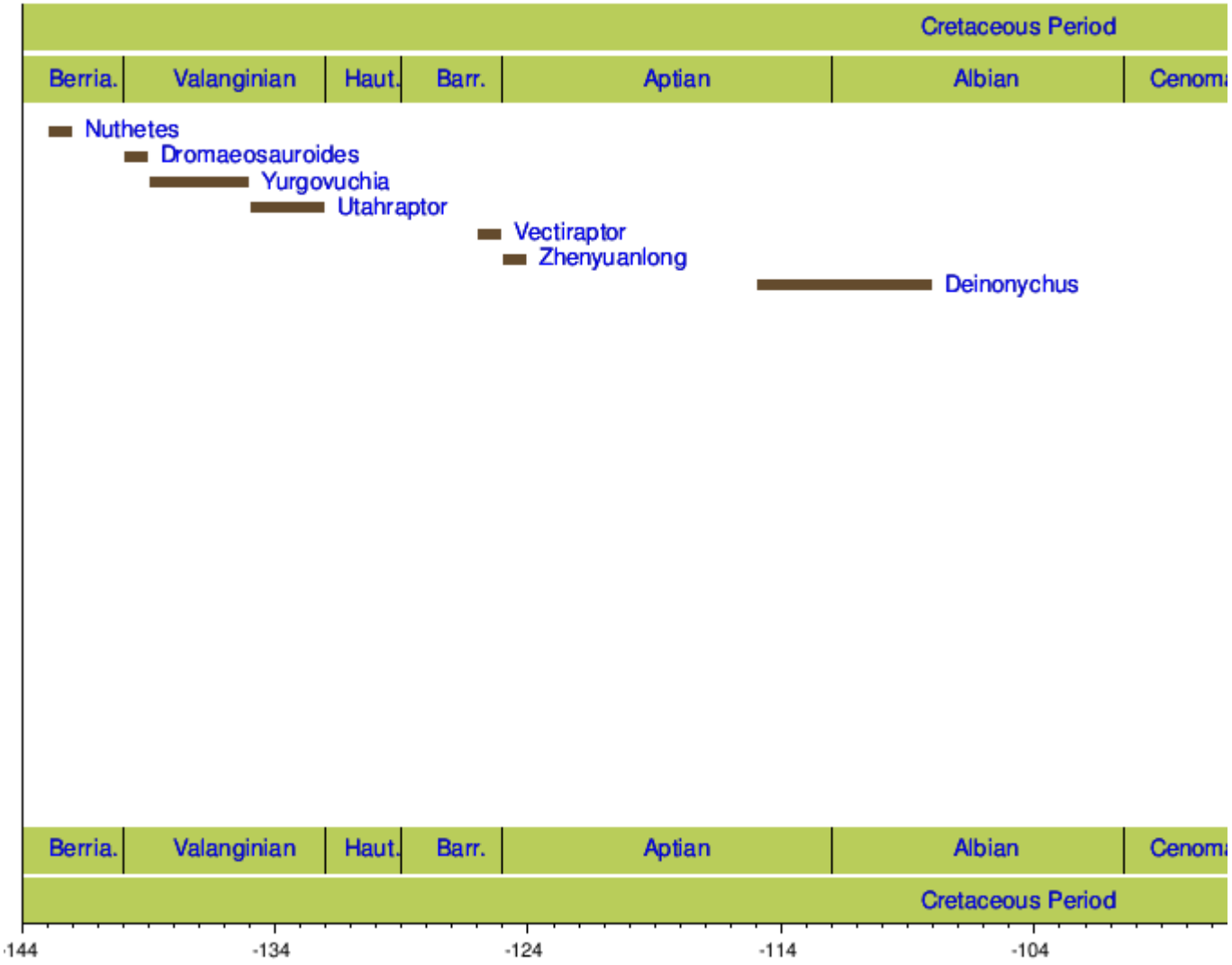
In 2006, Grellet-Tinner and Makovicky reported an egg associated with a specimen of *Deinonychus*. The egg shares similarities with oviraptorid eggs, and the authors interpreted the association as potentially indicative of brooding.[90] A study published in November 2018 by Norell, Yang and Wiemann et al., indicates that *Deinonychus* laid blue eggs, likely to camouflage them as well as creating open nests. Other dromaeosaurids may have done the same, and it is theorized that they and other maniraptoran dinosaurs may have been an origin point for laying colored eggs and creating open nests as many birds do today.[91][92][93]

In popular culture

Velociraptor, a dromaeosaurid, gained much attention after it was featured prominently in the 1993 Steven Spielberg film *Jurassic Park*. However, the dimensions of the *Velociraptor* in the film are much larger than the largest members of that genus. Robert Bakker recalled that Spielberg had been disappointed with the dimensions of *Velociraptor* and so upsized it.[94] Gregory S. Paul, in his 1988 book

Predatory Dinosaurs of the World, also considered *Deinonychus antirrhopus* a species of *Velociraptor*, and so rechristened the species *Velociraptor antirrhopus*.[39] This taxonomic opinion has not been widely followed.[10][95][96]

Timeline of dromaeosaurid genera



See also

- Timeline of dromaeosaurid research
- Avialae

Dinosaurs portal

References

1. Holtz, Thomas R. Jr. (2012) *Dinosaurs: The Most Complete, Up-to-Date Encyclopedia for Dinosaur Lovers of All Ages,* Winter 2011 Appendix. (http://www.geol.umd.edu/~tholtz/dinoappendix/HoltzappendixWinter2011.pdf) Archived (https://web.archive.org/web/20170812011954/http://www.geol.umd.edu/~tholtz/dinoappendix/HoltzappendixWinter2011.pdf) 2017-08-12 at the Wayback Machine

2. Hartman, Scott; Mortimer, Mickey; Wahl, William R.; Lomax, Dean R.; Lippincott, Jessica; Lovelace, David M. (2019-07-10). "A new paravian dinosaur from the Late Jurassic of North America supports a late acquisition of avian flight" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6626525). *PeerJ*. **7** e7247. doi:10.7717/peerj.7247 (https://doi.org/10.7717%2Fpeerj.7247). ISSN 2167-8359 (https://search.worldcat.org/issn/2167-8359). PMC 6626525 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6626525). PMID 31333906 (https://pubmed.ncbi.nlm.nih.gov/31333906).
3. Porfiri, Juan D.; Baiano, Mattia A.; dos Santos, Domenica D.; Gianechini, Federico A.; Pittman, Michael; Lamanna, Matthew C. (2024-06-14). "*Diuqin lechiguanae* gen. et sp. nov., a new unenlagiine (Theropoda: Paraves) from the Bajo de la Carpa Formation (Neuquén Group, Upper Cretaceous) of Neuquén Province, Patagonia, Argentina" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11177497). *BMC Ecology and Evolution*. **24** (1): 77. Bibcode:2024BMCEE..24...77P (https://ui.adsabs.harvard.edu/abs/2024BMCEE..24...77P). doi:10.1186/s12862-024-02247-w (https://doi.org/10.1186%2Fs12862-024-02247-w). ISSN 2730-7182 (https://search.worldcat.org/issn/2730-7182). PMC 11177497 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11177497). PMID 38872101 (https://pubmed.ncbi.nlm.nih.gov/38872101).
4. Allain, Ronan; Pereda-Suberbiola, Xabier (2003). "Dinosaurs of France". *Comptes Rendus Palevol*. **2** (1): 27–44. Bibcode:2003CRPal...2...27A (https://ui.adsabs.harvard.edu/abs/2003CRPal...2...27A). doi:10.1016/S1631-0683(03)00002-2 (https://doi.org/10.1016%2FS1631-0683%2803%2900002-2).
5. Agnolin, Federico L.; Motta, Matias J.; Brissón Egli, Federico; Lo Coco, Gastón; Novas, Fernando E. (2019-02-12). "Paravian Phylogeny and the Dinosaur-Bird Transition: An Overview" (https://doi.org/10.3389%2Ffeart.2018.00252). *Frontiers in Earth Science*. **6**. doi:10.3389/feart.2018.00252 (https://doi.org/10.3389%2Ffeart.2018.00252). hdl:11336/130197 (https://hdl.handle.net/11336%2F130197). ISSN 2296-6463 (https://search.worldcat.org/issn/2296-6463).
6. Acorn, J. (2007). *Deep Alberta: Fossil Facts and Dinosaur Digs* (https://archive.org/details/deepalbertafossi00acor/page/13). University of Alberta Press. pp. 13 (https://archive.org/details/deepalbertafossi00acor/page/13). doi:10.1515/9780888648518 (https://doi.org/10.1515%2F9780888648518). ISBN 978-0-88864-481-7.
7. Motta, Matías J.; Agnolína, Federico L.; Eglia, Federico Brissón; Rozadillaa, Sebastián; Novas, Fernando E. (22 Aug 2025). "Phylogenetic relationships of Unenlagiidae among Paraves (Dinosauria)" (https://www.tandfonline.com/doi/full/10.1080/14772019.2025.2529608). *Journal of Systematic Palaeontology*. **23** (1). doi:10.1080/14772019.2025.2529608 (https://doi.org/10.1080%2F14772019.2025.2529608). hdl:11336/283613 (https://hdl.handle.net/11336%2F283613). Retrieved 22 August 2025.
8. Metcalf, S.J.; Vaughan, R.F.; Benton, M.J.; Cole, J.; Simms, M.J.; Dartnall, D.L. (January 1992). "A new Bathonian (Middle Jurassic) microvertebrate site, within the Chipping Norton Limestone Formation at Hornsleasow Quarry, Gloucestershire". *Proceedings of the Geologists' Association*. **103** (4): 321–342. Bibcode:1992PrGA..103..321M (https://ui.adsabs.harvard.edu/abs/1992PrGA..103..321M). doi:10.1016/S0016-7878(08)80129-0 (https://doi.org/10.1016%2FS0016-7878%2808%2980129-0).
9. Wills, S.; Underwood, C. J.; Barrett, P. M. (2023). "Machine learning confirms new records of maniraptoran theropods in Middle Jurassic UK microvertebrate faunas" (https://doi.org/10.1002%2Fspp2.1487). *Papers in Palaeontology*. **9** (2). e1487. Bibcode:2023PPal....9E1487W (https://ui.adsabs.harvard.edu/abs/2023PPal....9E1487W). doi:10.1002/spp2.1487 (https://doi.org/10.1002%2Fspp2.1487).
10. Norell, M.A.; Makovicky, P.J. (2004). "Dromaeosauridae". In Weishampel, D.B.; Dodson, P.; Osmólska, H. (eds.). *The Dinosauria* (2nd ed.). Berkeley: University of California Press. pp. 196–210. ISBN 978-0-520-24209-8.

11. Hwang, S.H.; Norell, M.A.; Ji, Q.; Gao, K. (2002). "New Specimens of *Microraptor zhaoianus* (Theropoda: Dromaeosauridae) from Northeastern China". *American Museum Novitates* (3381): 44pp. hdl:2246/2870 (https://hdl.handle.net/2246%2F2870).
12. Perle, A.; Norell, M.A.; Clark, J. (1999). "A new maniraptoran theropod – *Achillobator giganticus* (Dromaeosauridae) – from the Upper Cretaceous of Burkhant, Mongolia". *Contributions of the Mongolian-American Paleontological Project*. **101**: 1–105.
13. Naish, D. Hutt, and Martill, D.M. (2001). "Saurischian dinosaurs: theropods." in Martill, D.M. and Naish, D. (eds). *Dinosaurs of the Isle of Wight*. The Palaeontological Association, Field Guides to Fossils. 10, 242–309.
14. "Dino hunts net rare raptor teeth" (https://news.bbc.co.uk/2/hi/science/nature/3659783.stm). *BBC News*. 26 April 2004. Archived (https://web.archive.org/web/20160417043325/http://news.bbc.co.uk/2/hi/science/nature/3659783.stm) from the original on 17 April 2016. Retrieved 7 May 2016.
15. Ostrom, J. H. (1969). "Osteology of *Deinonychus antirrhopus*, an unusual theropod from the Lower Cretaceous of Montana" (https://web.archive.org/web/20100627151832/http://www.peabody.yale.edu/scipubs/bulletins_postillas/ypmB30_1969.pdf) (PDF). *Peabody Museum of Natural History Bulletin*. **30**: 1–165. Archived from the original (http://peabody.yale.edu/scipubs/bulletins_postillas/ypmB30_1969.pdf) (PDF) on 2010-06-27. Retrieved 2010-12-18.
16. Paul, Greg (January 2006). "Drawing Dinosaurs" (http://www.geotimes.org/jan06/feature_drawingdinos.html). *Geotimes*. Archived (https://web.archive.org/web/20080502162556/http://www.geotimes.org/jan06/feature_drawingdinos.html) from the original on 2008-05-02. Retrieved 2007-12-18.
17. Makovicky, Peter J.; Apesteguía, Sebastián; Agnolín, Federico L. (October 2005). "The earliest dromaeosaurid theropod from South America". *Nature*. **437** (7061): 1007–1011. Bibcode:2005Natur.437.1007M (https://ui.adsabs.harvard.edu/abs/2005Natur.437.1007M). doi:10.1038/nature03996 (https://doi.org/10.1038%2Fnature03996). PMID 16222297 (https://pubmed.ncbi.nlm.nih.gov/16222297). S2CID 27078534 (https://api.semanticscholar.org/CorpusID:27078534).
18. Agnolin, Federico; Novas, Fernando E. (2013). *Avian Ancestors* (http://link.springer.com/10.1007/978-94-007-5637-3). SpringerBriefs in Earth System Sciences. Dordrecht: Springer Netherlands. doi:10.1007/978-94-007-5637-3 (https://doi.org/10.1007%2F978-94-007-5637-3). ISBN 978-94-007-5636-6. S2CID 199493087 (https://api.semanticscholar.org/CorpusID:199493087).
19. Codd, Jonathan R; Manning, Phillip L; Norell, Mark A; Perry, Steven F (2008-01-22). "Avian-like breathing mechanics in maniraptoran dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2596187). *Proceedings of the Royal Society B: Biological Sciences*. **275** (1631): 157–161. doi:10.1098/rspb.2007.1233 (https://doi.org/10.1098%2Frspb.2007.1233). ISSN 0962-8452 (https://search.worldcat.org/issn/0962-8452). PMC 2596187 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2596187). PMID 17986432 (https://pubmed.ncbi.nlm.nih.gov/17986432).
20. Paul, Gregory S. (1988). *Predatory dinosaurs of the world: a complete illustrated guide*. New York. ISBN 0-671-61946-2. OCLC 18350868 (https://search.worldcat.org/oclc/18350868).
21. Li, Rihui; Lockley, Martin G.; Makovicky, Peter J.; Matsukawa, Masaki; Norell, Mark A.; Harris, Jerald D.; Liu, Mingwei (19 October 2007). "Behavioral and faunal implications of Early Cretaceous deinonychosaur trackways from China". *Naturwissenschaften*. **95** (3): 185–191. Bibcode:2008NW.....95..185L (https://ui.adsabs.harvard.edu/abs/2008NW.....95..185L). doi:10.1007/s00114-007-0310-7 (https://doi.org/10.1007%2Fs00114-007-0310-7). PMID 17952398 (https://pubmed.ncbi.nlm.nih.gov/17952398). S2CID 16380823 (https://api.semanticscholar.org/CorpusID:16380823).

22. Longrich, Nicholas R.; Currie, Philip J. (31 March 2009). "A microraptorine (Dinosauria–Dromaeosauridae) from the Late Cretaceous of North America" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2664043). *Proceedings of the National Academy of Sciences of the United States of America*. **106** (13): 5002–5007. Bibcode:2009PNAS..106.5002L (https://ui.adsabs.harvard.edu/abs/2009PNAS..106.5002L). doi:10.1073/pnas.0811664106 (https://doi.org/10.1073%2Fpnas.0811664106). PMC 2664043 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2664043). PMID 19289829 (https://pubmed.ncbi.nlm.nih.gov/19289829).
23. Csiki, Zoltán; Vremir, Mátyás; Brusatte, Stephen L.; Norell, Mark A. (31 August 2010). "An aberrant island-dwelling theropod dinosaur from the Late Cretaceous of Romania" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2932599). *Proceedings of the National Academy of Sciences of the United States of America*. **107** (35): 15357–15361. Bibcode:2010PNAS..10715357C (https://ui.adsabs.harvard.edu/abs/2010PNAS..10715357C). doi:10.1073/pnas.1006970107 (https://doi.org/10.1073%2Fpnas.1006970107). PMC 2932599 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2932599). PMID 20805514 (https://pubmed.ncbi.nlm.nih.gov/20805514).
24. Norell, Mark A.; Makovicky, Peter J. (1999). "Important features of the dromaeosaurid skeleton II: information from newly collected specimens of *Velociraptor mongoliensis*". *American Museum Novitates* (3282): 1–45. hdl:2246/3025 (https://hdl.handle.net/2246%2F3025).
25. Chatterjee, Sankar; Templin, R. Jack (30 January 2007). "Biplane wing planform and flight performance of the feathered dinosaur Microraptor gui" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1780066). *Proceedings of the National Academy of Sciences of the United States of America*. **104** (5): 1576–1580. Bibcode:2007PNAS..104.1576C (https://ui.adsabs.harvard.edu/abs/2007PNAS..104.1576C). doi:10.1073/pnas.0609975104 (https://doi.org/10.1073%2Fpnas.0609975104). PMC 1780066 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1780066). PMID 17242354 (https://pubmed.ncbi.nlm.nih.gov/17242354).
26. Xu, Xing; Zhou, Zhonghe; Wang, Xiaolin; Kuang, Xuewen; Zhang, Fucheng; Du, Xiangke (January 2003). "Four-winged dinosaurs from China" (http://doc.rero.ch/record/15275/files/PAL_E2574.pdf) (PDF). *Nature*. **421** (6921): 335–340. Bibcode:2003Natur.421..335X (https://ui.adsabs.harvard.edu/abs/2003Natur.421..335X). doi:10.1038/nature01342 (https://doi.org/10.1038%2Fnature01342). PMID 12540892 (https://pubmed.ncbi.nlm.nih.gov/12540892). S2CID 1160118 (https://api.semanticscholar.org/CorpusID:1160118). Archived (https://web.archive.org/web/20200602051219/http://doc.rero.ch/record/15275/files/PAL_E2574.pdf) (PDF) from the original on 2020-06-02. Retrieved 2023-01-03.
27. Xu, Xing; Wang, Xiao-Lin; Wu, Xiao-Chun (September 1999). "A dromaeosaurid dinosaur with a filamentous integument from the Yixian Formation of China". *Nature*. **401** (6750): 262–266. Bibcode:1999Natur.401..262X (https://ui.adsabs.harvard.edu/abs/1999Natur.401..262X). doi:10.1038/45769 (https://doi.org/10.1038%2F45769). S2CID 4430574 (https://api.semanticscholar.org/CorpusID:4430574).
28. Turner, A. H.; Makovicky, P. J.; Norell, M. A. (21 September 2007). "Feather Quill Knobs in the Dinosaur Velociraptor" (https://doi.org/10.1126%2Fscience.1145076). *Science*. **317** (5845): 1721. Bibcode:2007Sci...317.1721T (https://ui.adsabs.harvard.edu/abs/2007Sci...317.1721T). doi:10.1126/science.1145076 (https://doi.org/10.1126%2Fscience.1145076). PMID 17885130 (https://pubmed.ncbi.nlm.nih.gov/17885130).
29. Paul, Gregory S. (2002). *Dinosaurs of the Air: The Evolution and Loss of Flight in Dinosaurs and Birds.* Baltimore: Johns Hopkins University Press. 472 pp.
30. Prum, Richard O.; Brush, Alan H. (September 2002). "The Evolutionary Origin And Diversification Of Feathers". *The Quarterly Review of Biology*. **77** (3): 261–295. doi:10.1086/341993 (https://doi.org/10.1086%2F341993). PMID 12365352 (https://pubmed.ncbi.nlm.nih.gov/12365352). S2CID 6344830 (https://api.semanticscholar.org/CorpusID:6344830).

31. Lü, Junchang; Brusatte, Stephen L. (16 July 2015). "A large, short-armed, winged dromaeosaurid (Dinosauria: Theropoda) from the Early Cretaceous of China and its implications for feather evolution" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4504142). *Scientific Reports*. **5** (1) 11775. Bibcode:2015NatSR...511775L (https://ui.adsabs.harvard.edu/abs/2015NatSR...511775L). doi:10.1038/srep11775 (https://doi.org/10.1038%2Fsrep11775). PMC 4504142 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4504142). PMID 26181289 (https://pubmed.ncbi.nlm.nih.gov/26181289).
32. "The first giant raptor (Theropoda: Dromaeosauridae) from the Hell Creek Formation" (https://doi.org/10.17161%2Fpaleo.1808.18764). *Paleontological Contributions*. 30 October 2015. doi:10.17161/paleo.1808.18764 (https://doi.org/10.17161%2Fpaleo.1808.18764). hdl:1808/18764 (https://hdl.handle.net/1808%2F18764).
33. Jasinski, Steven E.; Sullivan, Robert M.; Dodson, Peter (2020-03-26). "New Dromaeosaurid Dinosaur (Theropoda, Dromaeosauridae) from New Mexico and Biodiversity of Dromaeosaurids at the end of the Cretaceous" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7099077). *Scientific Reports*. **10** (1): 5105. Bibcode:2020NatSR..10.5105J (https://ui.adsabs.harvard.edu/abs/2020NatSR..10.5105J). doi:10.1038/s41598-020-61480-7 (https://doi.org/10.1038%2Fs41598-020-61480-7). ISSN 2045-2322 (https://search.worldcat.org/issn/2045-2322). PMC 7099077 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7099077). PMID 32218481 (https://pubmed.ncbi.nlm.nih.gov/32218481).
34. Cau, A. (2024). "A Unified Framework for Predatory Dinosaur Macroevolution" (https://web.archive.org/web/20240427205522/https://www.paleoitalia.it/wp-content/uploads/2024/04/Cau_2024_BSPI_ONLINE.pdf) (PDF). *Bollettino della Società Paleontologica Italiana*. **63** (1). doi:10.4435/BSPI.2024.08 (https://doi.org/10.4435%2FBSPI.2024.08) (inactive 12 July 2025). Archived from the original (https://www.paleoitalia.it/wp-content/uploads/2024/04/Cau_2024_BSPI_ONLINE.pdf) (PDF) on April 27, 2024.
35. Norell, M. Clark, J.M., Makovicky, P.J. (2001). "Phylogenetic relationships among coelurosaurian theropods. (http://www.peabody.yale.edu/scipubs/pdfs/Norell.pdf) Archived (https://web.archive.org/web/20070713232508/http://www.peabody.yale.edu/scipubs/pdfs/Norell.pdf) 2007-07-13 at the Wayback Machine" *New Perspectives on the Origin and Evolution of Birds: Proceedings of the International Symposium in Honor of John H. Ostrom*, Yale Peabody Museum: 49–67
36. Padian, Kevin; Dial, Kenneth P. (2005). "Origin of flight: Could 'four-winged' dinosaurs fly?" (https://doi.org/10.1038%2Fnature04354). *Nature*. **438** (7066): E3, discussion E3–4. Bibcode:2005Natur.438E...3P (https://ui.adsabs.harvard.edu/abs/2005Natur.438E...3P). doi:10.1038/nature04354 (https://doi.org/10.1038%2Fnature04354). PMID 16292258 (https://pubmed.ncbi.nlm.nih.gov/16292258). S2CID 4314459 (https://api.semanticscholar.org/CorpusID:4314459).
37. Witmer, L.M. (2005) "The Debate on Avian Ancestry; Phylogeny, Function and Fossils." In "Mesozoic Birds: Above the Heads of Dinosaurs", pp. 3–30. ISBN 0-520-20094-2
38. Olshevsky, George. (1994). "The birds first? A theory to fit the facts — evolution of reptiles into birds". *Omni*, June 1994. Volume 16 No. 9
39. Paul, Gregory S. (1988). *Predatory Dinosaurs of the World.* New York: Simon and Schuster. 464 pp.
40. Czerkas, S.A., Zhang, D., Li, J., and Li, Y. (2002). "Flying Dromaeosaurs", in Czerkas, S.J. (ed.): *Feathered Dinosaurs and the Origin of Flight: The Dinosaur Museum Journal 1.* Blanding: The Dinosaur Museum, 16–26.
41. National Geographic News- *Earliest Bird Had Feet Like Dinosaur, Fossil Shows* (https://web.archive.org/web/20051205022112/http://news.nationalgeographic.com/news/2005/12/1201_051201_archaeopteryx_2.html) – Nicholas Bakalar, December 1, 2005, Page 2. Retrieved 2006-10-18.

42. Mayr, G.; Pohl, B; Peters, DS (2 December 2005). "A Well-Preserved Archaeopteryx Specimen with Theropod Features" (http://doc.rero.ch/record/15488/files/PAL_E2876.pdf) (PDF). *Science*. **310** (5753): 1483–1486. Bibcode:2005Sci...310.1483M (https://ui.adsabs.harvard.edu/abs/2005Sci...310.1483M). doi:10.1126/science.1120331 (https://doi.org/10.1126%2Fscience.1120331). PMID 16322455 (https://pubmed.ncbi.nlm.nih.gov/16322455). S2CID 28611454 (https://api.semanticscholar.org/CorpusID:28611454). Archived (https://web.archive.org/web/20221225090505/https://doc.rero.ch/record/15488/files/PAL_E2876.pdf) (PDF) from the original on 25 December 2022. Retrieved 3 January 2023.
43. Mayr, G. (1 September 2006). "Response to Comment on 'A Well-Preserved Archaeopteryx Specimen with Theropod Features' " (https://doi.org/10.1126%2Fscience.1130964). *Science*. **313** (5791): 1238c. Bibcode:2006Sci...313.1238M (https://ui.adsabs.harvard.edu/abs/2006Sci...313.1238M). doi:10.1126/science.1130964 (https://doi.org/10.1126%2Fscience.1130964).
44. Corfe, I. J.; Butler, RJ (1 September 2006). "Comment on 'A Well-Preserved Archaeopteryx Specimen with Theropod Features' ". *Science*. **313** (5791): 1238b. Bibcode:2006Sci...313.1238C (https://ui.adsabs.harvard.edu/abs/2006Sci...313.1238C). doi:10.1126/science.1130800 (https://doi.org/10.1126%2Fscience.1130800). PMID 16946054 (https://pubmed.ncbi.nlm.nih.gov/16946054). S2CID 19859320 (https://api.semanticscholar.org/CorpusID:19859320).
45. Turner, Alan H.; Pol, Diego; Clarke, Julia A.; Erickson, Gregory M.; Norell, Mark (2007). "A basal dromaeosaurid and size evolution preceding avian flight" (https://doi.org/10.1126%2Fscience.1144066). *Science*. **317** (5843): 1378–1381. Bibcode:2007Sci...317.1378T (https://ui.adsabs.harvard.edu/abs/2007Sci...317.1378T). doi:10.1126/science.1144066 (https://doi.org/10.1126%2Fscience.1144066). PMID 17823350 (https://pubmed.ncbi.nlm.nih.gov/17823350).
46. Senter, P.; Kirkland, J. I.; Deblieux, D. D.; Madsen, S.; Toth, N. (2012). Dodson, Peter (ed.). "New Dromaeosaurids (Dinosauria: Theropoda) from the Lower Cretaceous of Utah, and the Evolution of the Dromaeosaurid Tail" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3352940). *PLOS ONE*. **7** (5) e36790. Bibcode:2012PLoSO...736790S (https://ui.adsabs.harvard.edu/abs/2012PLoSO...736790S). doi:10.1371/journal.pone.0036790 (https://doi.org/10.1371%2Fjournal.pone.0036790). PMC 3352940 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3352940). PMID 22615813 (https://pubmed.ncbi.nlm.nih.gov/22615813).
47. Matthew, W. D.; Brown, B. (1922). "The family Deinodontidae, with notice of a new genus from the Cretaceous of Alberta". *Bulletin of the American Museum of Natural History*. **46**: 367–385. hdl:2246/1300 (https://hdl.handle.net/2246%2F1300).
48. Cau, A.; Beyrand, V.; Voeten, D. F. A. E.; Fernandez, V.; Tafforeau, P.; Stein, K.; Barsbold, R.; Tsogtbaatar, K.; Currie, P. J.; Godefroit, P. (2017). "Synchrotron scanning reveals amphibious ecomorphology in a new clade of bird-like dinosaurs" (https://www.researchgate.net/publication/321609878). *Nature*. **552** (7685): 395–399. Bibcode:2017Natur.552..395C (https://ui.adsabs.harvard.edu/abs/2017Natur.552..395C). doi:10.1038/nature24679 (https://doi.org/10.1038%2Fnature24679). PMID 29211712 (https://pubmed.ncbi.nlm.nih.gov/29211712). S2CID 4471941 (https://api.semanticscholar.org/CorpusID:4471941). Archived (https://web.archive.org/web/20210318014252/https://www.researchgate.net/publication/321609878) from the original on 2021-03-18. Retrieved 2022-12-05.
49. Turner, Alan H.; Hwang, Sunny H.; Norell, Mark A. (2007). "A Small Derived Theropod from Öösh, Early Cretaceous, Baykhangor Mongolia" (https://www.biodiversitylibrary.org/bibliography/169129). *American Museum Novitates* (3557): 1. doi:10.1206/0003-0082(2007)3557[1:ASDTFS]2.0.CO;2 (https://doi.org/10.1206%2F0003-0082%282007%293557%5B1%3AASDTFS%5D2.0.CO%3B2). hdl:2246/5845 (https://hdl.handle.net/2246%2F5845). S2CID 31096081 (https://api.semanticscholar.org/CorpusID:31096081).

50. Agnolin, Federico L.; Novas, Fernando E. (March 2011). "Unenlagiid theropods: are they members of the Dromaeosauridae (Theropoda, Maniraptora)?" (https://doi.org/10.1590%2Fs0001-37652011000100008). *Anais da Academia Brasileira de Ciências*. **83** (1): 117–162. doi:10.1590/s0001-37652011000100008 (https://doi.org/10.1590%2Fs0001-37652011000100008). PMID 21437379 (https://pubmed.ncbi.nlm.nih.gov/21437379).
51. Senter, Phil; Barsbold, R.; Britt, Brooks B.; Burnham, David B. (2004). "Systematics and evolution of Dromaeosauridae (Dinosauria, Theropoda)". *Bulletin of the Gunma Museum of Natural History*. **8**: 1–20.
52. Sereno, P. C. 2005. Stem Archosauria—TaxonSearch (http://www.taxonsearch.org/Archive/stem-archosauria-1.0.php) Archived (https://web.archive.org/web/20090115093329/http://www.taxonsearch.org/Archive/stem-archosauria-1.0.php) 2009-01-15 at the Wayback Machine, version 1.0, November 7, 2005
53. Turner, Alan H.; Makovicky, Peter J.; Norell, Mark A. (17 August 2012). "A Review of Dromaeosaurid Systematics and Paravian Phylogeny" (https://zenodo.org/record/5399588). *Bulletin of the American Museum of Natural History*. **371**: 1–206. doi:10.1206/748.1 (https://doi.org/10.1206%2F748.1). hdl:2246/6352 (https://hdl.handle.net/2246%2F6352). S2CID 83572446 (https://api.semanticscholar.org/CorpusID:83572446). Archived (https://web.archive.org/web/20220103090644/https://zenodo.org/record/5399588) from the original on 3 January 2022. Retrieved 22 November 2021.
54. Schmitz, L.; Motani, R. (14 April 2011). "Nocturnality in Dinosaurs Inferred from Scleral Ring and Orbit Morphology". *Science*. **332** (6030): 705–708. Bibcode:2011Sci...332..705S (https://ui.adsabs.harvard.edu/abs/2011Sci...332..705S). doi:10.1126/science.1200043 (https://doi.org/10.1126%2Fscience.1200043). PMID 21493820 (https://pubmed.ncbi.nlm.nih.gov/21493820). S2CID 33253407 (https://api.semanticscholar.org/CorpusID:33253407).
55. Li, Quanguo; Gao, K.-Q.; Meng, Q.; Clarke, J. A.; Shawkey, M. D.; d'Alba, L.; Pei, R.; Ellison, M.; Norell, M. A.; Vinther, J. (9 March 2012). "Reconstruction of *Microraptor* and the Evolution of Iridescent Plumage". *Science*. **335** (6073): 1215–1219. Bibcode:2012Sci...335.1215L (https://ui.adsabs.harvard.edu/abs/2012Sci...335.1215L). doi:10.1126/science.1213780 (https://doi.org/10.1126%2Fscience.1213780). PMID 22403389 (https://pubmed.ncbi.nlm.nih.gov/22403389). S2CID 206537426 (https://api.semanticscholar.org/CorpusID:206537426).
56. Zelenitsky, Darla K; Therrien, François; Kobayashi, Yoshitsugu (28 October 2008). "Olfactory acuity in theropods: palaeobiological and evolutionary implications" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2660930). *Proceedings of the Royal Society B: Biological Sciences*. **276** (1657): 667–673. doi:10.1098/rspb.2008.1075 (https://doi.org/10.1098%2Frspb.2008.1075). PMC 2660930 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2660930). PMID 18957367 (https://pubmed.ncbi.nlm.nih.gov/18957367).
57. Zelenitsky, Darla K.; Therrien, François; Ridgely, Ryan C.; McGee, Amanda R.; Witmer, Lawrence M. (13 April 2011). "Evolution of olfaction in non-avian theropod dinosaurs and birds" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3203493). *Proceedings of the Royal Society B: Biological Sciences*. **278** (1725): 3625–3634. doi:10.1098/rspb.2011.0238 (https://doi.org/10.1098%2Frspb.2011.0238). PMC 3203493 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3203493). PMID 21490022 (https://pubmed.ncbi.nlm.nih.gov/21490022).
58. Batista, Filipa; Castro, Lígia; Moita, Patrícia; Maggia, Bruno; Ortega, Francisco; Malafaia, Elisabete (3 October 2015). "Microwear structures and surface analysis on isolated theropod teeth from the Upper Jurassic Andrés fossil site, Pombal, Portugal" (https://www.scup.com/doi/full/10.18261/let.58.4.2). *Lethaia*. **58** (4): 1–16. doi:10.18261/let.58.4.2 (https://doi.org/10.18261%2Flet.58.4.2). hdl:10362/191197 (https://hdl.handle.net/10362%2F191197). ISSN 0024-1164 (https://search.worldcat.org/issn/0024-1164). Retrieved 18 March 2026 – via Scandinavian University Press.

59. Torices, A.; Wilkinson, R.; Arbour, V.M.; Ruiz-Omeñaca, J.I.; Currie, P.J. (2018). "Puncture-and-Pull Biomechanics in the Teeth of Predatory Coelurosaurian Dinosaurs" (https://doi.org/10.1016%2Fj.cub.2018.03.042). *Current Biology*. **28** (9): 1467–1474. Bibcode:2018CBio...28E1467T (https://ui.adsabs.harvard.edu/abs/2018CBio...28E1467T). doi:10.1016/j.cub.2018.03.042 (https://doi.org/10.1016%2Fj.cub.2018.03.042). PMID 29706515 (https://pubmed.ncbi.nlm.nih.gov/29706515).
60. Adams, Dawn (1987). "The bigger they are, the harder they fall: Implications of ischial curvature in ceratopsian dinosaurs". In Currie, Philip J.; Koster, E. (eds.). *Fourth symposium on mesozoic terrestrial ecosystems*. Drumheller, Canada: Tyrrell Museum. pp. 1–6.
61. Manning, Phillip L; Payne, David; Pennicott, John; Barrett, Paul M; Ennos, Roland A (11 October 2005). "Dinosaur killer claws or climbing crampons?" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1617199). *Biology Letters*. **2** (1): 110–112. doi:10.1098/rsbl.2005.0395 (https://doi.org/10.1098%2Frsbl.2005.0395). PMC 1617199 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1617199). PMID 17148340 (https://pubmed.ncbi.nlm.nih.gov/17148340).
62. Davies, S.J.J.F. (2002) "Ratites and Tinamous" Oxford University Press. New York, USA
63. Gilliard, Ernest Thomas (1958). *Living birds of the world*. Doubleday. OCLC 681791956 (https://search.worldcat.org/oclc/681791956).
64. Kofron, Christopher P. (1999). "Attacks to humans and domestic animals by the southern cassowary (Casuarius casuarius johnsonii) in Queensland, Australia". *Journal of Zoology*. **249** (4): 375–381. doi:10.1111/j.1469-7998.1999.tb01206.x (https://doi.org/10.1111%2Fj.1469-7998.1999.tb01206.x).
65. Kofron, Christopher P. (2003). "Case histories of attacks by the southern cassowary in Queensland". *Memoirs of the Queensland Museum*. **49** (1): 335–338.
66. Redford, Kent H.; Peters, Gustav (1986). "Notes on the Biology and Song of the Red-Legged Seriema (Cariama cristata)". *Journal of Field Ornithology*. **57** (4): 261–269. JSTOR 4513154 (https://www.jstor.org/stable/4513154).
67. Manning, Phillip L.; Margetts, Lee; Johnson, Mark R.; Withers, Philip J.; Sellers, William I.; Falkingham, Peter L.; Mummery, Paul M.; Barrett, Paul M.; Raymont, David R. (September 2009). "Biomechanics of Dromaeosaurid Dinosaur Claws: Application of X-Ray Microtomography, Nanoindentation, and Finite Element Analysis" (https://doi.org/10.1002%2Far.20986). *The Anatomical Record: Advances in Integrative Anatomy and Evolutionary Biology*. **292** (9): 1397–1405. doi:10.1002/ar.20986 (https://doi.org/10.1002%2Far.20986). PMID 19711472 (https://pubmed.ncbi.nlm.nih.gov/19711472).
68. "Velociraptor's 'killing' claws were for climbing" (https://www.newscientist.com/article/mg20327254.100-velociraptors-killing-claws-were-for-climbing.html). *New Scientist*. **203** (2725): 10. September 2009. doi:10.1016/s0262-4079(09)62381-2 (https://doi.org/10.1016%2Fs0262-4079%2809%2962381-2). Retrieved 2009-09-15.
69. Senter, Phil (2009). "Pedal function in deinonychosaurs (Dinosauria: Theropoda): a comparative study" (http://www.gmnh.pref.gunma.jp/wp-content/uploads/bulletin13_1.pdf) (PDF). *Bulletin of the Gunma Museum of Natural History*. **13**: 1–14. Archived (https://web.archive.org/web/20201101064707/http://www.gmnh.pref.gunma.jp/wp-content/uploads/bulletin13_1.pdf) (PDF) from the original on 2020-11-01. Retrieved 2020-05-09.
70. Fowler, Denver W.; Freedman, Elizabeth A.; Scannella, John B.; Kambic, Robert E. (14 December 2011). "The Predatory Ecology of Deinonychus and the Origin of Flapping in Birds" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3237572). *PLOS ONE*. **6** (12) e28964. Bibcode:2011PLoSO...628964F (https://ui.adsabs.harvard.edu/abs/2011PLoSO...628964F). doi:10.1371/journal.pone.0028964 (https://doi.org/10.1371%2Fjournal.pone.0028964). PMC 3237572 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3237572). PMID 22194962 (https://pubmed.ncbi.nlm.nih.gov/22194962).

71. Choi, Charles (14 December 2011). "Velociraptors' Killer Claws Helped Them Eat Prey Alive" (http://www.livescience.com/17485-velociraptors-killer-claws.html). *LiveScience*. Archived (https://web.archive.org/web/20120703184251/http://www.livescience.com/17485-velociraptors-killer-claws.html) from the original on 3 July 2012. Retrieved 30 December 2011.
72. Bishop, P. J. (2019). "Testing the function of dromaeosaurid (Dinosauria, Theropoda) 'sickle claws' through musculoskeletal modelling and optimization" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6717499). *PeerJ*. **7** e7577. doi:10.7717/peerj.7577 (https://doi.org/10.7717%2Fpeerj.7577). PMC 6717499 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6717499). PMID 31523517 (https://pubmed.ncbi.nlm.nih.gov/31523517).
73. Gianechini, Federico A.; Ercoli, Marcos D.; Díaz-Martínez, Ignacio (5 February 2020). "Differential locomotor and predatory strategies of Gondwanan and derived Laurasian dromaeosaurids (Dinosauria, Theropoda, Paraves): Inferences from morphometric and comparative anatomical studies" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7163733). *Journal of Anatomy*. **236** (5): 772–797. doi:10.1111/joa.13153 (https://doi.org/10.1111%2Fjoa.13153). PMC 7163733 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7163733). PMID 32023660 (https://pubmed.ncbi.nlm.nih.gov/32023660).
74. Mudroch, A.; Richter, U.; Joger, U.; Kosma, R.; Idé, O.; Maga, A. (2011). "Didactyl tracks of paravian theropods (Maniraptora) from the ?Middle Jurassic of Africa" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3038851). *PLOS ONE*. **6** (2): e1462. Bibcode:2011PLoSO...614642M (https://ui.adsabs.harvard.edu/abs/2011PLoSO...614642M). doi:10.1371/journal.pone.0014642 (https://doi.org/10.1371%2Fjournal.pone.0014642). PMC 3038851 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3038851). PMID 21339816 (https://pubmed.ncbi.nlm.nih.gov/21339816). Figure 4.
75. Maxwell, W. Desmond; Ostrom, John H. (27 December 1995). "Taphonomy and paleobiological implications of *Tenontosaurus-Deinonychus* associations". *Journal of Vertebrate Paleontology*. **15** (4): 707–712. Bibcode:1995JVPal..15..707M (https://ui.adsabs.harvard.edu/abs/1995JVPal..15..707M). doi:10.1080/02724634.1995.10011256 (https://doi.org/10.1080%2F02724634.1995.10011256).
76. Roach, Brian T.; Brinkman, Daniel L. (April 2007). "A Reevaluation of Cooperative Pack Hunting and Gregariousness in *Deinonychus antirrhopus* and Other Nonavian Theropod Dinosaurs". *Bulletin of the Peabody Museum of Natural History*. **48** (1): 103–138. doi:10.3374/0079-032X(2007)48[103:AROCPH]2.0.CO;2 (https://doi.org/10.3374%2F0079-032X%282007%2948%5B103%3AAROCPH%5D2.0.CO%3B2). S2CID 84175628 (https://api.semanticscholar.org/CorpusID:84175628).
77. Frederickson, J. A.; Engel, M. H.; Cifelli, R. L. (15 August 2020). "Ontogenetic dietary shifts in Deinonychus antirrhopus (Theropoda; Dromaeosauridae): Insights into the ecology and social behavior of raptorial dinosaurs through stable isotope analysis". *Palaeogeography, Palaeoclimatology, Palaeoecology*. **552** 109780. Bibcode:2020PPP...55209780F (https://ui.adsabs.harvard.edu/abs/2020PPP...55209780F). doi:10.1016/j.palaeo.2020.109780 (https://doi.org/10.1016%2Fj.palaeo.2020.109780). S2CID 219059665 (https://api.semanticscholar.org/CorpusID:219059665).
78. Switek, Brian (7 January 2015). "Utah's Dinosaur 'Death Trap' Reveals Trove of Giant Predators" (https://web.archive.org/web/20191130104345/https://www.nationalgeographic.com/news/2015/1/150106-utahraptor-death-trap-fossil/). *National Geographic News*. Archived from the original (https://www.nationalgeographic.com/news/2015/1/150106-utahraptor-death-trap-fossil/) on November 30, 2019.

79. Hutson, Joel D.; Hutson, Kelda N. (25 September 2015). "An examination of forearm bone mobility in Alligator mississippiensis (Daudin, 1802) and Struthio camelus Linnaeus, 1758 reveals that Archaeopteryx and dromaeosaurs shared an adaptation for gliding and/or flapping" (https://bioone.org/journals/geodiversitas/volume-37/issue-3/g2015n3a3/An-examination-of-forearm-bone-mobility-in-Alligator-mississippiensis-Daudin/10.5252/g2015n3a3.short). *Geodiversitas*. **37** (3): 325–344. doi:10.5252/g2015n3a3 (https://doi.org/10.5252%2Fg2015n3a3). S2CID 55278850 (https://api.semanticscholar.org/CorpusID:55278850). Archived (https://web.archive.org/web/20220925230348/https://bioone.org/journals/geodiversitas/volume-37/issue-3/g2015n3a3/An-examination-of-forearm-bone-mobility-in-Alligator-mississippiensis-Daudin/10.5252/g2015n3a3.short) from the original on 25 September 2022. Retrieved 2 December 2022.
80. Norell, Mark A.; Clark, James M.; Turner, Alan H.; Makovicky, Peter J.; Barsbold, Rinchen; Rowe, Timothy (2006). "A New Dromaeosaurid Theropod from Ukhaa Tolgod (Ömnögov, Mongolia)" (https://www.biodiversitylibrary.org/bibliography/169121). *American Museum Novitates* (3545): 1. doi:10.1206/0003-0082(2006)3545[1:ANDTFU]2.0.CO;2 (https://doi.org/10.1206%2F0003-0082%282006%293545%5B1%3AANDTFU%5D2.0.CO%3B2). hdl:2246/5823 (https://hdl.handle.net/2246%2F5823). Archived (https://web.archive.org/web/20220111134014/https://www.biodiversitylibrary.org/bibliography/169121) from the original on 2022-01-11. Retrieved 2021-06-15.
81. Forster, C. A.; Sampson, SD; Chiappe, LM; Krause, DW (20 March 1998). "The Theropod Ancestry of Birds: New Evidence from the Late Cretaceous of Madagascar". *Science*. **279** (5358): 1915–1919. Bibcode:1998Sci...279.1915F (https://ui.adsabs.harvard.edu/abs/1998Sci...279.1915F). doi:10.1126/science.279.5358.1915 (https://doi.org/10.1126%2Fscience.279.5358.1915). PMID 9506938 (https://pubmed.ncbi.nlm.nih.gov/9506938).
82. Chiappe, L.M. (2007-02-02). *Glorified Dinosaurs: The Origin and Early Evolution of Birds*. Sydney: UNSW Press. ISBN 978-0-471-24723-4.
83. Parsons, William L.; Parsons, Kristen M.; Dodson, Peter (15 April 2015). "Morphological Variations within the Ontogeny of Deinonychus antirrhopus (Theropoda, Dromaeosauridae)" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4398413). *PLOS ONE*. **10** (4) e0121476. Bibcode:2015PLoSO..1021476P (https://ui.adsabs.harvard.edu/abs/2015PLoSO..1021476P). doi:10.1371/journal.pone.0121476 (https://doi.org/10.1371%2Fjournal.pone.0121476). PMC 4398413 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4398413). PMID 25875499 (https://pubmed.ncbi.nlm.nih.gov/25875499).
84. Chatterjee, Sankar; Templin, R. J. (2004). "Feathered Coelurosaurs from China: New Light on the Arboreal Origin of Avian Flight" (https://books.google.com/books?id=L-ZFPTKroVoC&pg=PA251). In Currie, Philip J.; Koppelhus, Eva B.; Shugar, Martin A.; Wright, Joanna L. (eds.). *Feathered Dragons: Studies on the Transition from Dinosaurs to Birds*. Indiana University Press. pp. 251–281. ISBN 978-0-253-34373-4.
85. Naish, Darren (2012). *Planet Dinosaur: The Next Generation of Killer Giants* (https://archive.org/details/planetdinosaurne00scot/page/192). Firefly Books. p. 192 (https://archive.org/details/planetdinosaurne00scot/page/192). ISBN 978-1-77085-049-1.
86. Rothschild, B.; Tanke, D. H.; Ford, T. L. (2001). "Theropod stress fractures and tendon avulsions as a clue to activity". In Tanke, D. H.; Carpenter, K. (eds.). *Mesozoic Vertebrate Life*. Indiana University Press. pp. 331–336.
87. Cau, A. (2020). "The body plan of Halszkaraptor escuilliei (Dinosauria, Theropoda) is not a transitional form along the evolution of dromaeosaurid hypercarnivory" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7047864). *PeerJ*. **8** e8672. doi:10.7717/peerj.8672 (https://doi.org/10.7717%2Fpeerj.8672). PMC 7047864 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7047864). PMID 32140312 (https://pubmed.ncbi.nlm.nih.gov/32140312).

88. Lee, S.; Lee, Y.-N.; Currie, P. J.; Sissons, R.; Park, J.-Y.; Kim, S.-H.; Barsbold, R.; Tsogtbaatar, K. (2022). "A non-avian dinosaur with a streamlined body exhibits potential adaptations for swimming" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9715538). *Communications Biology*. **5** (1185): 1185. doi:10.1038/s42003-022-04119-9 (https://doi.org/10.1038%2Fs42003-022-04119-9). ISSN 2399-3642 (https://search.worldcat.org/issn/2399-3642). PMC 9715538 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9715538). PMID 36456823 (https://pubmed.ncbi.nlm.nih.gov/36456823).
89. Currie, Philip J.; Carabajal, Ariana Paulina (December 2012). "A New Specimen of *Austroraptor cabazai* Novas, Pol, Canale, Porfiri and Calvo, 2008 (Dinosauria, Theropoda, Unenlagiidae) from the Latest Cretaceous (Maastrichtian) of Río Negro, Argentina". *Ameghiniana*. **49** (4): 662–667. doi:10.5710/AMGH.30.8.2012.574 (https://doi.org/10.5710%2FAMGH.30.8.2012.574). hdl:11336/9090 (https://hdl.handle.net/11336%2F9090). S2CID 129058582 (https://api.semanticscholar.org/CorpusID:129058582).
90. Grellet-Tinner, Gerald; Makovicky, Peter (1 June 2006). "A possible egg of the dromaeosaur *Deinonychus antirrhopus*: phylogenetic and biological implications". *Canadian Journal of Earth Sciences*. **43** (6): 705–719. Bibcode:2006CaJES..43..705G (https://ui.adsabs.harvard.edu/abs/2006CaJES..43..705G). doi:10.1139/E06-033 (https://doi.org/10.1139%2FE06-033).
91. Wiemann, Jasmina; Yang, Tzu-Ruei; Norell, Mark A. (31 October 2018). "Dinosaur egg colour had a single evolutionary origin". *Nature*. **563** (7732): 555–558. Bibcode:2018Natur.563..555W (https://ui.adsabs.harvard.edu/abs/2018Natur.563..555W). doi:10.1038/s41586-018-0646-5 (https://doi.org/10.1038%2Fs41586-018-0646-5). PMID 30464264 (https://pubmed.ncbi.nlm.nih.gov/30464264). S2CID 53188171 (https://api.semanticscholar.org/CorpusID:53188171).
92. "Dinosaur Egg Color Had a Single Evolutionary Origin" (https://scitechdaily.com/dinosaur-egg-color-had-a-single-evolutionary-origin/). November 2018. Archived (https://web.archive.org/web/20200223145815/https://scitechdaily.com/dinosaur-egg-color-had-a-single-evolutionary-origin/) from the original on 2020-02-23. Retrieved 2020-02-23.
93. "Dinosaurs put all colored bird eggs in one basket, evolutionarily speaking" (https://news.yale.edu/2018/10/31/dinosaurs-put-all-colored-bird-eggs-one-basket-evolutionarily-speaking). 2018-10-31. Archived (https://web.archive.org/web/20200510173246/https://news.yale.edu/2018/10/31/dinosaurs-put-all-colored-bird-eggs-one-basket-evolutionarily-speaking) from the original on 2020-05-10. Retrieved 2020-02-23.
94. Bakker, Robert T. (1995). *Raptor Red*. New York: Bantam Books. p. 4. ISBN 978-0-553-57561-3.
95. Pérez-Moreno, B.P.; Sanz, J. L.; Sudre, J.; Sigé, B. (1994). "A theropod dinosaur from the Lower Cretaceous of southern France". *Dinosaurs and Other Fossil Reptiles of Europe, Second Georges Cuvier Symposium, Montbéliard; Revue de Paléobiologie, Volume Spécial*. **7**: 173–188.
96. Currie, P. J. (1995). "New information on the anatomy and relationships of *Dromaeosaurus albertensis* (Dinosauria: Theropoda)". *Journal of Vertebrate Paleontology*. **15** (3): 576–591. Bibcode:1995JVPal..15..576C (https://ui.adsabs.harvard.edu/abs/1995JVPal..15..576C). doi:10.1080/02724634.1995.10011250 (https://doi.org/10.1080%2F02724634.1995.10011250). (abstract (https://www.vertpaleo.org/publications/jvp/15-576-591.cfm) Archived (https://web.archive.org/web/20070927204229/https://www.vertpaleo.org/publications/jvp/15-576-591.cfm) 2007-09-27 at the Wayback Machine)

External links

- The Dromaeosauridae: The Raptors! (http://www.ucmp.berkeley.edu/diapsids/saurischia/dromaeosauridae.html), from the University of California Berkeley Museum of Paleontology.

- Dinosaurs (http://www.ageofdinosaur.com/index.htm) – Complete and free online edition of the book "Dinosaurs" as written by W. D. Matthew (cited in this article with authorship of the family Dromaeosauridae), and former Curator of Vertebrate Paleontology at the American Museum of Natural History in New York; Originally published in 1915
- Dromaeosauridae (http://www.dinosaur-world.com/feathered_dinosaurs/dromaeosauridae.htm), Dinosaur-world reference with in-depth description and pictures of many dromaeosauridae dinosaurs

---

Retrieved from "https://en.wikipedia.org/w/index.php?title=Dromaeosauridae&oldid=1381260871"

WIKIPEDIA
25 years of the free encyclopedia

Velociraptor

Velociraptor (/vəˌlɒsɪˈræptər, vəˈlɒsɪræptər/;[2] lit. 'swift thief') is a genus of small dromaeosaurid dinosaurs that lived in Asia during the Late Cretaceous epoch, about 75 million to 71 million years ago (Mya). Two species are currently recognized, although others have been assigned in the past. The type species is ***V. mongoliensis***, named and described in 1924. Fossils of this species have been discovered in the Djadochta Formation, Mongolia. A second species, ***V. osmolskae***, was named in 2008 for skull material from the Bayan Mandahu Formation, China. A possible record is known from the Nemegt Formation.[1]

Smaller than other dromaeosaurids like *Deinonychus* and *Achillobator*, *Velociraptor* was about 1.5–2.07 m (4.9–6.8 ft) long with a body mass of around 14.1–19.7 kg (31–43 lb). It nevertheless shared many of the same anatomical features. It was a bipedal, feathered carnivore with a long tail and an enlarged sickle-shaped claw on each hindfoot, which is thought to have been used to tackle and restrain prey. *Velociraptor* can be distinguished from other dromaeosaurids by its long and low skull, with an upturned snout.

Velociraptor (commonly referred to as "raptor") is one of the dinosaur genera most familiar to the general public due to its prominent role in the *Jurassic Park* films. In reality, *Velociraptor* was roughly the size of a turkey, considerably smaller than the approximately 2 m (6.6 ft) long and 90 kg (200 lb) reptiles seen in the novels and films (which were based on members of the related genus *Deinonychus*). Today, *Velociraptor* is well known to paleontologists, with over a dozen described fossil skeletons. One particularly famous specimen preserves a *Velociraptor* locked in combat with a *Protoceratops*.

History of discovery

During an American Museum of Natural History expedition to the Flaming Cliffs (Bayn Dzak or Bayanzag) of the Djadochta Formation, Gobi Desert, on 11 August 1923, Peter Kaisen discovered the first *Velociraptor* fossil known to

Velociraptor

Temporal range: Late Cretaceous, ~

Possible later Maastrichtian record[1]

Mounted *V. mongoliensis* cast at Royal Belgian Institute of Natural Sciences

Scientific classification

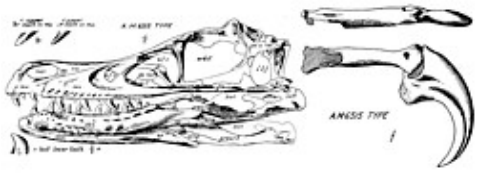
Kingdom:	Animalia
Phylum:	Chordata
Class:	Reptilia
Clade:	Dinosauria
Clade:	Saurischia
Clade:	Theropoda
Family:	†Dromaeosauridae
Clade:	†Eudromaeosauria
Subfamily:	†Velociraptorinae
Genus:	†*Velociraptor* Osborn, 1924

Type species

†***Velociraptor mongoliensis***
Osborn, 1924

Other species

The Flaming Cliffs, Mongolia in 2018



Line diagram of *V. mongoliensis* holotype skull and associated manual ungual

- †***V.? osmolskae***
  Godefroit et al., 2008

science—a crushed but complete skull, associated with one manual claw and adjoining phalanges (AMNH 6515). In 1924, museum president Henry Fairfield Osborn designated the skull and part of the manus as the type specimen of his new genus, *Velociraptor*. This name is derived from the Latin words *velox* ('swift') and *raptor* ('robber' or 'plunderer') and refers to the animal's cursorial nature and carnivorous diet. Osborn named the type species *V. mongoliensis* after its country of origin.[3] Earlier that year, Osborn had informally mentioned the animal in a popular press article, under the name "Ovoraptor djadochtari" (not to be confused with the similarly named *Oviraptor*),[4] eventually changed into *V. mongoliensis* during its formal description.[3]

While North American teams were shut out of communist Mongolia during the Cold War, expeditions by Soviet and Polish scientists, in collaboration with Mongolian colleagues, recovered several more specimens of *Velociraptor*. The most famous is part of the "Fighting Dinosaurs" specimen (MPC-D 100/25; formerly IGM, GIN, or GI SPS), discovered by a Polish-Mongolian team in 1971. The fossil preserves a *Velociraptor* in battle against a *Protoceratops*.[5][6][7] It is considered a national treasure of Mongolia, and in 2000 it was loaned to the American Museum of Natural History in New York City for a temporary exhibition.[8]

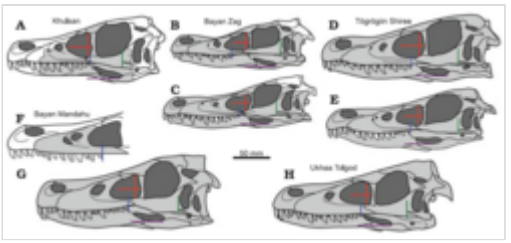
Between 1988 and 1990, a joint Chinese-Canadian team discovered *Velociraptor* remains in northern China.[9] American scientists returned to Mongolia in 1990, and a joint Mongolian-American expedition to the Gobi, led by the American Museum of Natural History and the Mongolian Academy of Sciences, turned up several well-preserved skeletons.[10][11] One such specimen, MPC-D 100/980, was nicknamed "Ichabodcraniosaurus" by Norell's team because the fairly complete specimen was found without its skull (an allusion to the Washington Irving character Ichabod Crane).[12] While Norell and Makovicky provisionally considered it a specimen of *Velociraptor mongoliensis*,[10] it was named as a new species *Shri devi* in 2021.[13]

In 1999, Rinchen Barsbold and Halszka Osmólska reported a juvenile *Velociraptor* specimen (GIN or IGM 100/2000), represented by a complete skeleton including the skull of a young individual. It was found at the Tugriken Shireh locality of the Djadochta Formation during the context of the Mongolian-Japanese Palaeontological Expeditions. The coauthors stated that detailed descriptions of this and other specimens would be published at a later date.[14]

Additional species

Maxillae and a lacrimal (the main tooth-bearing bones of the upper jaw, and the bone that forms the anterior margin of the eye socket, respectively) recovered from the Bayan Mandahu Formation in 1999 by the Sino-Belgian Dinosaur Expeditions were found to pertain to *Velociraptor*, but not to the type species *V. mongoliensis*. Pascal Godefroit and colleagues named these bones *V. osmolskae* (for Polish paleontologist Halszka Osmólska) in 2008.[15] However, the 2013 study noted that while "the elongate

shape of the maxilla in *V. osmolskae* is similar to that of *V. mongoliensis*," phylogenetic analysis found it to be closer to *Linheraptor*, making the genus paraphyletic; thus, *V. osmolskae* might not actually belong to the genus *Velociraptor* and requires reassessment.[16]



Velociraptorine skulls, B, D, E are *V. mongoliensis*, C is *V* sp., and F is *V. osmolskae* (known parts in gray)

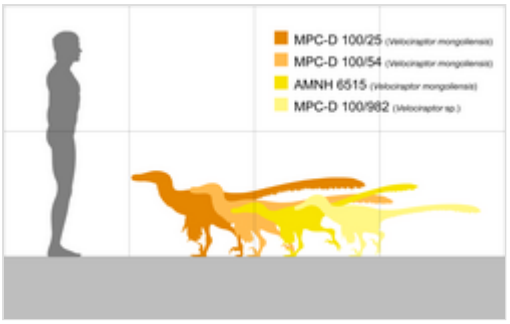
Paleontologists Mark A. Norell and Peter J. Makovicky in 1997 described new and well preserved specimens of *V. mongoliensis*, namely MPC-D 100/985 collected from the Tugrik Shireh locality in 1993, and MPC-D 100/986 collected in 1993 from the Chimney Buttes locality. The team briefly mentioned another specimen, MPC-D 100/982, which by the time of this publication remained undescribed.[11] In 1999 Norell and Makovicky provided more insights into the anatomy of *Velociraptor* with additional specimens. Among these, MPC-D 100/982 was partially described and figured, and referred to *V. mongoliensis* mainly based on cranial similarities with the holotype skull, although they stated that differences were present between the pelvic region of this specimen and other *Velociraptor* specimens. This relatively well-preserved specimen including the skull was discovered and collected in 1995 at the Bayn Dzak locality (specifically at the "Volcano" sub-locality).[10] Martin Kundrát in a 2004 abstract compared the neurocranium of MPC-D 100/982 to another *Velociraptor* specimen, MPC-D 100/976. He concluded that the overall morphology of the former was more derived (advanced) than the latter, suggesting that they could represent distinct taxa.[17]

Velociraptor specimen MPC-D 100/982, possibly a new species

Mark J. Powers in his 2020 master thesis fully described MPC-D 100/982, which he concluded to represent a new and third species of *Velociraptor*. This species, which he considered distinct, was stated to mainly differ from other *Velociraptor* species in having a shallow maxilla morphology.[18] Powers and colleagues also in 2020 used morphometric analyses to compare several dromaeosaurid maxillae, and found the maxilla of MPC-D 100/982 to strongly differ from specimens referred to *Velociraptor*. They indicated that this specimen, based on these results, represents a different species.[19] In 2021 Powers with team used Principal Component Analysis to separate dromaeosaurid maxillae, most notably finding that MPC-D 100/982 falls outside the instraspecific variability of *V. mongoliensis*, arguing for a distinct species. They considered that both *V. mongoliensis* and this new species were ecologically separated based on their skull anatomy.[20] The team in another 2021 abstract reinforced again the species-level separation, noting that additional differences can be found in the hindlimbs.[21]

Description

Velociraptor was a small to medium-sized dromaeosaurid, with adults measuring between 1.5–2.07 m (4.9–6.8 ft) long, approximately 0.5 m (1.6 ft) high at the hips,[22][23] and weighing about 14.1–19.7 kg (31–43 lb).[24][25] Some estimates suggest that *Velociraptor* may have been larger, possibly reaching lengths of about 2.5 m (8 ft) and weighing up to 28 kg (60 lb).[26]



Velociraptor specimens compared in size to a 1.8 m (5 ft 11 in) tall human

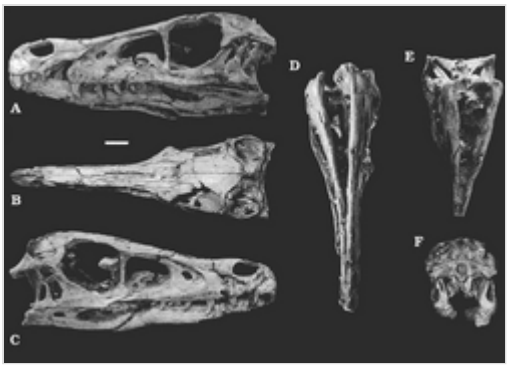
Prominent quill knobs—attachment site of "wing" feathers and direct indicator of a feather covering—have been reported from the ulna of a single *Velociraptor* specimen (IGM 100/981), which represents an animal of estimated 1.5 m (4.9 ft) long and 15 kg (33 lb) in weight. The spacing of 6 preserved knobs suggests that 8 additional knobs may have been present, giving a total of 14 quill knobs that developed large secondaries ("wing" feathers stemming from the forearm).[27] However, the specimen number has been corrected to IGM 100/3503 and its referral to *Velociraptor* may require reevaluation, pending further study.[28] Nevertheless, there is strong phylogenetic evidence from other dromaeosaurid relatives that indicates the presence of feathers in *Velociraptor*, including dromaeosaurids such as *Daurlong*,[29] *Microraptor*,[30] or *Zhenyuanlong*.[31]

Skull

Life restoration

The skull of *Velociraptor* was rather elongated and grew up to 23 cm (9.1 in) long. It was uniquely up-curved at the snout region, concave on the upper surface, and convex on the lower surface. The snout, which occupied about 60% of the entire skull length, was notably narrow and mainly formed by the nasal, premaxilla, and maxilla bones. The premaxilla was the anteriormost bone in the skull, and it was longer than taller. While its posterior end joined the nasal, the main body of the premaxilla touched the maxilla. The maxilla was nearly triangular in shape and the largest element of the snout. On its center or main body, there was a depression developing a small oval to circular-shaped hole, called maxillary fenestra. Though in front of this fenestra were two small openings, referred to as promaxillary fenestrae. The posterior border of the maxilla formed (predominantly) the antorbital fenestra, one of the several large holes in the skull. Both premaxilla and maxilla had several alveoli (tooth sockets) on their bottom surfaces. Above the maxilla and making contact with the premaxilla, there was the nasal bone. It was a thin/narrow and elongated bone contributing to the top surface of the snout. Together, both premaxilla and nasal bones gave form to the naris or narial fenestra (nostril opening), which was relatively large and circular. The posterior end of the nasal was joined by the frontal and lacrimal bones.[32][14]

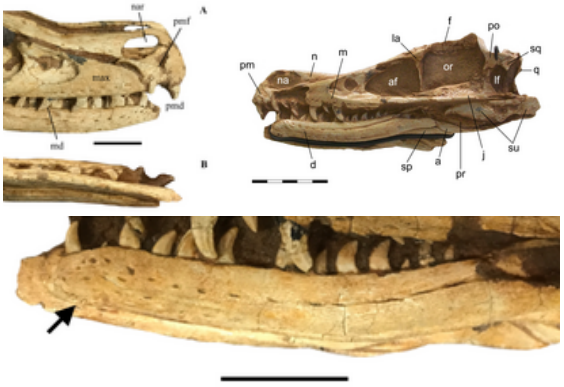
The back or posterior region of the skull was built by the frontal, lacrimal, postorbital, jugal, parietal, quadrate, and quadratojugal bones. The frontal was large element, having a vaguely rectangular shape when seen from above. On its posterior end, this bone was in contact with the parietal, and such elements were the main bodies of the skull roof. The lacrimal was a T-shaped bone and its main body was thin and delicated. Its lower end meet the jugal (often called cheek bone), which was a large, sub-triangular-shaped element. Its lower border was notably straight/horizontal. The postorbital was located just above the jugal: a stocky and strongly T-shaped bone. As a whole, the orbit or orbital fenestra (eye socket)—formed by the lacrimal, jugal, frontal, and postorbital—was large and near circular in shape, being longer than taller. When seen from above, a pair of large and markedly rounded holes were present near the rear of the skull (the temporal fenestrae), whose main components were the postorbital and squamosal. Behind



Skull of MPC-D 100/25 (Fighting Dinosaurs individual), in lateral (A-C), top (B), bottom (D-E), and posterior (E) views

the jugal, an inverted T-shaped bone (also seen in other dromaeosaurids), known as the quadratojugal, was developed. While the upper end of the quadratojugal joined the squamosal, an irregularly-shaped element, its inner side meet the quadrate. The latter was of great importance for the articulation with the lower jaw. The posteriormost bone was the occipital bone and its projection the occipital condyle: a rounded and bulbous protuberance that meet the first vertebra of the neck.[32][14]

The lower jaw of *Velociraptor* comprised mainly the dentary, splenial, angular, surangular, and articular bones. The dentary was a very long, weakly curved, and narrow element that developed several alveoli on its top surface. On its posterior end, it meet the surangular. It had a small hole near its posterior end, called surangular foramen or fenestra. Both bones were the largest elements of the lower jaw of *Velociraptor*, contributing to virtually its entire length. Below them were the smaller splenial and angular, closely articulated to each other. The articular, located on the inner side of the surangular, was a small element that joined the quadrate of the upper skull, enabling the articulation with the lower jaw. An elongated, near oval-shaped hole was developed in the center of the lower jaw (the mandibular fenestra), and it was produced by the joint of the dentary, surangular, and angular bones.[32][14]

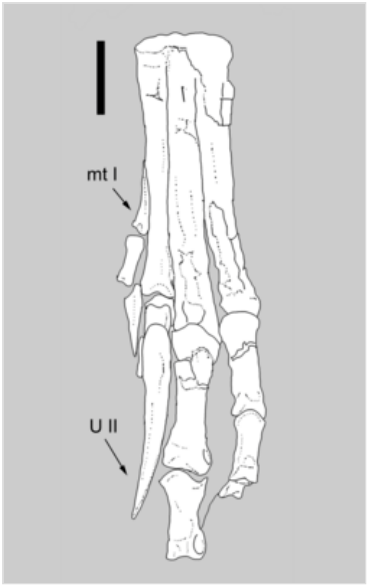


V. mongoliensis holotype skull (right), rostrum (left), and dentary (bottom)

The teeth of *Velociraptor* were fairly homodont (equal in shape) and had several denticles (serrations), each more strongly serrated on the back edge than the front. The premaxilla had 4 alveoli (meaning that 4 teeth were developed), and the maxilla had 11 alveoli. At the dentary, between 14–15 alveoli were present. All teeth present at the premaxilla were poorly curved, and the two first teeth were the longest, with the second having a characteristic large size. The maxillary teeth were more slender, recurved, and most notably, the lower end was strongly more serrated than the upper one.[32][14]

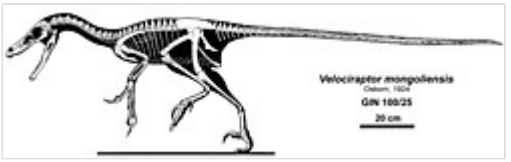
Postcranial skeleton

The arm of *Velociraptor* was formed by the humerus (upper arm bone), radius and ulna (forearm bones), and manus (hand). *Velociraptor*, like other dromaeosaurids, had a large manus with three elongated digits (fingers), which ended up in strongly curved unguals (claw bones) that were similar in construction and flexibility to the wing bones of modern birds. The second digit was the longest of the three digits present, while the first was shortest. The structure of the carpal (wrist) bones prevented pronation of the wrist and forced the manus to be held with the palmar surface facing inward (medially), not downward. The pes (foot) anatomy of *Velociraptor* consisted of the metatarsus—a large element composed of three metatarsals of which the first one was extremely reduced in size—and four digits that developed large



Line diagram of the pes (foot) of *V. mongoliensis* (MPC-D 100/985)

unguals. The first digit, as in other theropods, was a small dewclaw. The second digit, for which *Velociraptor* is most famous, was highly modified and held retracted off the ground, which caused *Velociraptor* and other dromaeosaurids to walk on only their third and fourth digits. It bore a relatively large, sickle-shaped claw, typical of dromaeosaurid and troodontid dinosaurs. This enlarged claw, which could grow to over 6.5 cm (2.6 in) long around its outer edge, was most likely a predatory device used to restrain struggling prey.[11][10]



Skeletal reconstruction of *V. mongoliensis* (MPC-D 100/25)

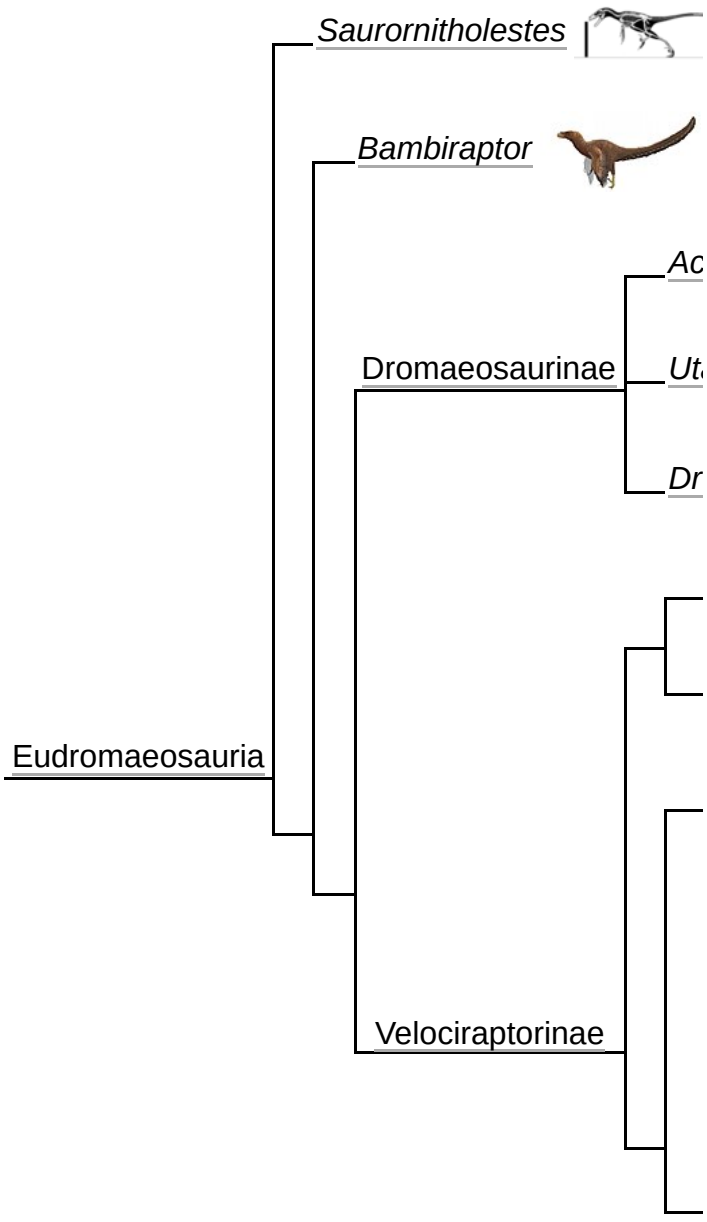
As in other dromaeosaurs, *Velociraptor* tails had prezygapophyses (long bony projections) on the upper surfaces of the vertebrae, as well as ossified tendons underneath. The prezygapophyses began on the tenth tail (caudal) vertebra and extended forward to brace four to ten additional vertebrae, depending on position in the tail. These were once thought to fully stiffen the tail, forcing the entire tail to act as a single rod-like unit. However, at least one specimen has preserved a series of intact tail vertebrae curved sideways into an *S*-shape, suggesting that there was considerably more horizontal flexibility than once thought.[11][10][33]

Classification

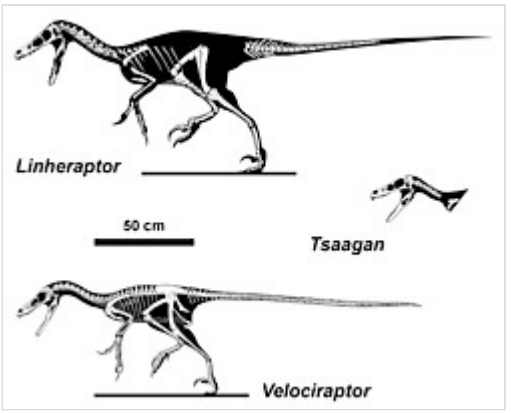
Velociraptor is a member of the group Eudromaeosauria, a derived sub-group of the larger family Dromaeosauridae. It is often placed within its own subfamily, Velociraptorinae. In phylogenetic taxonomy, Velociraptorinae is usually defined as "all dromaeosaurs more closely related to *Velociraptor* than to *Dromaeosaurus*." However, dromaeosaurid classification is highly variable. Originally, the subfamily Velociraptorinae was erected solely to contain *Velociraptor*.[5] Other analyses have often included other genera, usually *Deinonychus* and *Saurornitholestes*,[34] and more recently *Tsaagan*.[35] Several studies published during the 2010s, including expanded versions of the analyses that found support for Velociraptorinae, have failed to resolve it as a distinct group, but rather have suggested it is a paraphyletic grade which gave rise to the Dromaeosaurinae.[36][37]

When first described in 1924, *Velociraptor* was placed in the family Megalosauridae, as was the case with most carnivorous dinosaurs at the time (Megalosauridae, like *Megalosaurus*, functioned as a sort of 'wastebin' taxon, where many unrelated species were grouped together).[3] As dinosaur discoveries multiplied, *Velociraptor* was later recognized as a dromaeosaurid. All dromaeosaurids have also been referred to the family Archaeopterygidae by at least one author (which would, in effect, make *Velociraptor* a flightless bird).[38] In the past, other dromaeosaurid species, including *Deinonychus antirrhopus* and *Saurornitholestes langstoni*, have sometimes been classified in the genus *Velociraptor*. Since *Velociraptor* was the first to be named, these species were renamed *Velociraptor antirrhopus* and *V. langstoni*.[22] As of 2008, the only currently recognized species of *Velociraptor* are *V. mongoliensis*[14][38][39] and *V. osmolskae*.[15] However, several studies have found "*V.*" *osmolskae* to be distantly related to *V. mongoliensis*.[40][41]

Below are the results for the Eudromaeosauria phylogeny based on the phylogenetic analysis conducted by James G. Napoli and team in 2021 during the description of *Kuru*, showing the position of *Velociraptor*:[28]



Size of *Velociraptor* (2) compared with other dromaeosaurs



Comparison of some members of Velociraptorinae, featuring *Linheraptor*, *Tsaagan* and *Velociraptor*

Paleobiology

Feathers

In 2007 Alan H. Turner and colleagues reported the presence of six quill knobs in the ulna of a referred *Velociraptor* specimen (IGM 100/981) from the Ukhaa Tolgod locality of the Djadochta Formation. Turner and colleagues interpreted the presence of feathers on *Velociraptor* as evidence against the idea that the larger, flightless maniraptorans lost their feathers secondarily due to larger body size. Furthermore, they noted that quill knobs are almost never found in flightless bird species today, and that their presence in *Velociraptor* (presumed to have been flightless due to its relatively large size and short forelimbs) is evidence that the ancestors of dromaeosaurids could fly, making *Velociraptor* and other large members of this family secondarily flightless, though it is possible the large wing feathers inferred in the ancestors of *Velociraptor* had a purpose other than flight. The feathers of the flightless *Velociraptor* may have been used for display, for covering their nests while brooding, or for added speed and thrust when running up inclined slopes.[27]

Because of the presence of another dromaeosaurid in Ukhaa Tolgod, *Tsaagan*, Napoli and team have noted that the referral of this specimen to *Velociraptor* is currently subject to reexamination.[28]

Senses

Examinations of the endocranium of *Velociraptor* indicate that it was able to detect and hear a wide range of sound frequencies (2,368–3,965 Hz) and could track prey with ease as a result. The endocranium examinations also further cemented the theory that the dromaeosaur was an agile, swift predator. Fossil evidence suggesting *Velociraptor* scavenged also indicates that it was an opportunistic and actively predatory animal, feeding on carrion during times of drought or famine, if in poor health, or depending on the animal's age.[42]

Feeding

In 2020, Powers and colleagues re-examined the maxillae of several eudromaeosaur taxa concluding that most Asian and North American eudromaeosaurs were separated by snout morphology and ecological strategies. They found the maxilla to be a reliable reference when inferring the shape of the premaxilla and overall snout. For instance, most Asian species have elongated snouts based on the maxilla (namely velociraptorines), indicating a selective feeding in *Velociraptor* and relatives, such as picking up small, fast prey. In contrast, most North American eudromaeosaurs, mostly dromaeosaurines, feature a robust and deep maxillar morphology. However, the large dromaeosurine *Achillobator* is a unique exception to Asian taxa with its deep maxilla.[19]

Isolated tooth of ZPAL MgD-I/97a

Manabu Sakamoto in 2022 performed a Bayesian phylogenetic predictive modelling framework for estimating jaw muscle parameters and bite forces of several extinct archosaurs, based on skull widths and phylogenetic relationships between groups. Among

studied taxa, *Velociraptor* was scored with a bite force of 304 N, which was lower than that of other dromaeosaurids such as *Dromaeosaurus* (885 N) or *Deinonychus* (706 N).[43]

Predatory behavior

The "Fighting Dinosaurs" specimen of *V. mongoliensis* and *Protoceratops andrewsi* and restoration of same

The "Fighting Dinosaurs" specimen, found in 1971, preserves a *Velociraptor mongoliensis* and *Protoceratops andrewsi* in combat and provides direct evidence of predatory behavior. When originally reported, it was hypothesized that the two animals drowned.[7] However, as the animals were preserved in ancient sand dune deposits, it is now thought that the animals were buried in sand, either from a collapsing dune or in a sandstorm. Burial must have been extremely rapid, judging from the lifelike poses in which the animals were preserved. Parts of the *Protoceratops* are missing, which has been seen as evidence of scavenging by other animals.[44] Comparisons between the scleral rings of *Velociraptor*, *Protoceratops*, and modern birds and reptiles indicates that *Velociraptor* may have been nocturnal, while *Protoceratops* may have been cathemeral, active throughout the day during short intervals, suggesting that the fight may have occurred at twilight or during low-light conditions.[45]

Size comparison of the Fighting Dinosaurs

The distinctive claw, on the second digit of dromaeosaurids, has traditionally been depicted as a slashing weapon; its assumed use being to cut and disembowel prey.[46] In the "Fighting Dinosaurs" specimen, the *Velociraptor* lies underneath, with one of its sickle claws apparently embedded in the throat of its prey, while the beak of *Protoceratops* is clamped down upon the right forelimb of its attacker. This suggests *Velociraptor* may have used its sickle claw to pierce vital organs of the throat, such as the jugular vein, carotid artery, or trachea (windpipe), rather than slashing the abdomen. The inside edge of the claw was rounded and not unusually sharp, which may have precluded any sort of cutting or slashing action, although only the bony core of the claw is preserved. The thick abdominal wall of skin and muscle of large prey species would have been difficult to slash without a specialized cutting surface.[44] The slashing hypothesis was tested during a 2005 BBC documentary, *The Truth About Killer Dinosaurs*. The producers of the program created an artificial *Velociraptor* leg with a sickle claw and used a pork belly to simulate the dinosaur's prey. Though the sickle claw did penetrate the abdominal wall, it was unable to tear it open, indicating that the claw was not used to disembowel prey.[47]

Remains of *Deinonychus*, a closely related dromaeosaurid, have commonly been found in aggregations of several individuals. *Deinonychus* has also been found in association with the large ornithopod *Tenontosaurus*, which has been cited as evidence of cooperative (pack) hunting.[48][49] However, the only solid evidence for social behavior of any kind among dromaeosaurids comes from a Chinese trackway which shows six individuals of a large species moving as a group.[50] Although many isolated fossils of

Velociraptor have been found in Mongolia, none were closely associated with other individuals.[39] Therefore, while *Velociraptor* is commonly depicted as a pack hunter, as in *Jurassic Park*, there is only limited fossil evidence to support this theory for dromaeosaurids in general and none specific to *Velociraptor* itself. Dromeosaur footprints in China suggest that a few other raptor genera may have hunted in packs, but there have been no conclusive examples of pack behavior found.[51][52]

V. mongoliensis restraining an oviraptorosaur with its sickle claws

In 2011, Denver Fowler and colleagues suggested a new method by which dromaeosaurs like *Velociraptor* and similar dromaeosaurs may have captured and restrained prey. This model, known as the "raptor prey restraint" (RPR) model of predation, proposes that dromaeosaurs killed their prey in a manner very similar to extant accipitrid birds of prey: by leaping onto their quarry, pinning it under their body weight, and gripping it tightly with the large, sickle-shaped claws. These researchers proposed that, like accipitrids, the dromaeosaur would then begin to feed on the animal while it was still alive, and prey death would eventually result from blood loss and organ failure. This proposal is based primarily on comparisons between the morphology and proportions of the feet and legs of dromaeosaurs to several groups of extant birds of prey with known predatory behaviors. Fowler found that the feet and legs of dromaeosaurs most closely resemble those of eagles and hawks, especially in terms of having an enlarged second claw and a similar range of grasping motion. The short metatarsus and foot strength, however, would have been more similar to that of owls. The RPR method of predation would be consistent with other aspects of *Velociraptor*'s anatomy, such as their unusual jaw and arm morphology. The arms, which could exert a lot of force but were likely covered in long feathers, may have been used as flapping stabilizers for balance while atop a struggling prey animal, along with the stiff counterbalancing tail. The jaws, thought by Fowler and colleagues to be comparatively weak, would have been useful for row saw motion bites like the modern day Komodo dragon, which also has a weak bite, to finish off its prey if the kicks were not powerful enough. These predatory adaptations working together may also have implications for the origin of flapping in paravians.[33]

Scavenging behavior

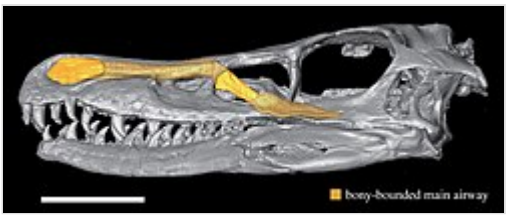
In 2010, Hone and colleagues published a paper on their 2008 discovery of shed teeth of what they believed to be a *Velociraptor* near a tooth-marked jaw bone of what they believed to be a *Protoceratops* in the Bayan Mandahu Formation. The authors concluded that the find represented "late-stage carcass consumption by *Velociraptor*" as the predator would have eaten other parts of a freshly killed *Protoceratops* before biting in the jaw area. The evidence was seen as supporting the inference from the "Fighting Dinosaurs" fossil that *Protoceratops* was part of the diet of *Velociraptor*.[53]

In 2012, Hone and colleagues published a paper that described a *Velociraptor* specimen with a long bone of an azhdarchid pterosaur in its gut. This was interpreted as showing scavenging behaviour.[54]

In a 2024 study by Tse, Miller, and Pittman et al., focusing on the skull morphology and bite forces of various dromaeosaurids, it was discovered that *Velociraptor* had high bite force resistance compared to other dromaeosaurids such as *Dromaeosaurus* itself and *Deinonychus*, the latter of which was much

larger. It is theorized by the authors that high bite force resistance was an adaptation towards obtaining food through scavenging more often than through active predation in *Velociraptor*.[55]

Metabolism



3D scan and nasal cavity reconstruction of *V. mongoliensis* skull MPC-D 100/54

Velociraptor was warm-blooded to some degree, as it required a significant amount of energy to hunt. Modern animals that possess feathery or furry coats, like *Velociraptor* did, tend to be warm-blooded, since these coverings function as insulation. However, bone growth rates in dromaeosaurids and some early birds suggest a more moderate metabolism, compared with most modern warm-blooded mammals and birds. The kiwi is similar to dromaeosaurids in anatomy, feather type, bone structure and even the narrow anatomy of the nasal passages (usually a key indicator of metabolism). The kiwi is a highly active, if specialized, flightless bird, with a stable body temperature and a fairly low resting metabolic rate, making it a good model for the metabolism of primitive birds and dromaeosaurids.[38]

In 2023, Seishiro Tada and team examined the nasal cavities of ectotherm (cold-blooded) or endotherm (warm-blooded) species, in order to evaluate the thermoregulatory physiology of non-avian dinosaurs compared to these groups. They found that the size of the nasal cavity relative to the head size of extant endotherms is larger than those of extant ectotherms, and among taxa, *Velociraptor* was recovered below the extant endotherms level by reconstructing its nasal respiratory cavity. Tada with team suggested that *Velociraptor* and most other non-avian dinosaurs may not have possessed a fully or well-developed nasal thermoregulation apparatus as modern endothermic animals do.[56]

Paleopathology

Norell with colleagues in 1995 reported one *V. mongoliensis* skull bearing two parallel rows of small punctures on its frontal bones that, upon closer examination, match the spacing and size of *Velociraptor* teeth. They suggested that the wound was likely inflicted by another *Velociraptor* during a fight within the species. Because its bone structure shows no sign of healing near the bite wounds and the overall specimen was not scavenged, this individual was likely killed by this fatal wound.[57] In 2001 Molnar and team noted that this specimen is MPC-D 100/976 hailing from the Tugrik Shireh locality, which has also yielded the Fighting Dinosaurs specimen.[58]

In 2012 David Hone and team reported another injured *Velociraptor* specimen (MPC-D 100/54, roughly a sub-adult individual) found with the bones of an azhdarchid pterosaur within its stomach cavity, was carrying or recovering from an injury sustained to one broken rib. From evidence on the pterosaur bones, which were devoid of pitting or deformations from digestion, the *Velociraptor* died shortly after, possibly from the earlier injury. Nevertheless, the team noted that this broken ribs shows signs of bone healing.[54]

Paleoenvironment

Bayan Mandahu Formation

In both Bayan Mandahu and Djadochta formations many of the same genera were present, though they varied at the species level. These differences in species composition may be due a natural barrier separating the two formations, which are relatively close to each other geographically.[15] However, given the lack of any known barrier which would cause the specific faunal compositions found in these areas, it is more likely that those differences indicate a slight time difference.[59]

Restoration of related *Linheraptor* in paleoenvironment

V. osmolskae lived alongside the ankylosaurid *Pinacosaurus mephistocephalus*; alvarezsaurid *Linhenykus*; closely related dromaeosaurid *Linheraptor*; oviraptorids *Machairasaurus* and *Wulatelong*; protoceratopsids *Bagaceratops* and *Protoceratops hellenikorhinus*; and troodontids *Linhevenator*, *Papiliovenator*, and *Philovenator*.[59] Sediments across the formation indicate a similar depositional environment to that of the Djadochta Formation.[60]

Djadochta Formation

Restoration of *V. mongoliensis* in the arid Djadochta settings

Known specimens of *Velociraptor mongoliensis* have been recovered from the Djadochta Formation (also spelled Djadokhta), in the Mongolian province of Ömnögovi. This geological formation is estimated to date back to the Campanian stage (between 75 million and 71 million years ago) of the Late Cretaceous epoch.[61] The abundant sediments—sands, sandstones, or caliche—of the Djadochta Formation were deposited by eolian (wind) processes in arid settings with fields of sand dunes and only intermittent streams, as indicated by very sparse fluvial (river-deposited) sedimentation, under a semi-arid climate.[62][61][63]

The Djadochta Formation is separated into a lower Bayn Dzak Member and upper Turgrugyin Member. *V. mongoliensis* is known from both members, represented by numerous specimens.[61] The Bayn Dzak Member (mainly Bayn Dzak locality) has yielded the oviraptorid *Oviraptor*; ankylosaurid *Pinacosaurus grangeri*; protoceratopsid *Protoceratops andrewsi*; and troodontid *Saurornithoides*.[3][61] The younger Turgrugyin Member (mainly Tugriken Shireh locality) has produced the bird *Elsornis*; dromaeosaurid *Mahakala*: ornithomimid *Aepyornithomimus*; and protoceratopsid *Protoceratops andrewsi*.[23][64]

V. mongoliensis has been found at many of the most famous and prolific Djadochta localities. The type specimen was discovered at the Flaming Cliffs site (sublocality of the larger Bayn Dzak locality/region),[3] while the "Fighting Dinosaurs" were found at the Tugrik Shire locality (also known as

Tugrugeen Shireh and many other spellings).[6] The latter is notorious for its exceptional *in situ* fossil preservation. Based on deposits (such as structureless sandstones), it has been concluded that a large number of specimens were buried alive during powerful sand-bearing events, common to these paleoenvironments.[65]

Cultural significance

The "Dinosaur Input Device" *Velociraptor* used for creating some of the CGI effects in *Jurassic Park* (left), and the design from the *Jurassic World* trilogy, Hong Kong (right)

Velociraptor is commonly perceived as a vicious and cunning killer thanks to their portrayal in the 1990 novel *Jurassic Park* by Michael Crichton and its 1993 film adaptation, directed by Steven Spielberg. The "raptors" portrayed in *Jurassic Park* were actually modeled after the closely related dromaeosaurid *Deinonychus*. Paleontologists in both the novel and film excavate a skeleton in Montana, far from the central Asian range of *Velociraptor* but characteristic of the *Deinonychus* range.[66] Crichton met with the discoverer of *Deinonychus*, John Ostrom, several times at Yale University to discuss details of the animal's possible range of behaviors and appearance. Crichton at one point apologetically told Ostrom that he had decided to use the name *Velociraptor* in place of *Deinonychus* because the former name was "more dramatic." According to Ostrom, Crichton stated that the *Velociraptor* of the novel was based on *Deinonychus* in almost every detail, and that only the name had been changed. The *Jurassic Park* filmmakers also requested all of Ostrom's published papers on *Deinonychus* during production.[67] They portrayed the animals with the size, proportions, and snout shape of *Deinonychus* rather than *Velociraptor*.[68][69]

Production on *Jurassic Park* began before the discovery of the large dromaeosaurid *Utahraptor* was made public in 1991, but as Jody Duncan wrote about this discovery: "Later, after we had designed and built the raptor, there was a discovery of a raptor skeleton in Utah, which they labeled 'super-slasher.' They had uncovered the largest Velociraptor to date and it measured five-and-a-half-feet tall, just like ours. So we designed it, we built it, and then they discovered it. That still boggles my mind."[68] Spielberg's name was briefly considered for naming of the new dinosaur in exchange for funding of field work, but no agreement was reached.[70]

Jurassic Park and its sequel *The Lost World: Jurassic Park* were released before the discovery that dromaeosaurs had feathers, so the *Velociraptor* in both films were depicted as scaled and featherless. For *Jurassic Park III*, the male *Velociraptor* was given quill-like structures along the back of the head and neck, but these structures do not resemble the feathers that *Velociraptor* would have had in reality due to reasons of continuity.[71] The *Jurassic World* sequel trilogy ignored the feathers of *Velociraptor*, adhering

to the designs from *Jurassic Park*.[72] However, the dromaeosaur *Pyroraptor* was feathered for *Jurassic World Dominion*, along with other changes such as stiffening the tail to account for ossified tendons and de-pronating the hands.[73]

See also

- Fighting Dinosaurs
- Timeline of dromaeosaurid research

Dinosaurs portal

Paleontology portal

References

1. Currie, Philip J., ed. (2004). *Feathered dragons: studies on the transition from dinosaurs to birds*. Life of the past. Bloomington: Indiana University Press. ISBN 978-0-253-34373-4.
2. Wells, John C. (2008). *Longman Pronunciation Dictionary* (3 rd ed.). Harlow: Pearson Education. ISBN 978-1-4058-8118-0.
3. Osborn, H. F. (1924). "Three new Theropoda, Protoceratops zone, central Mongolia" (https://digitallibrary.amnh.org/items/25d800cd-76d7-4ced-9dae-39e46f746def). *American Museum Novitates* (144): 1–12. hdl:2246/3223 (https://hdl.handle.net/2246%2F3223). OCLC 40272928 (https://search.worldcat.org/oclc/40272928).
4. Osborn, H. F. (1924). "The discovery of an unknown continent" (https://archive.org/details/naturalhistory2416newy/page/132/mode/2up). *Natural History*. **24** (2): 133–149.
5. Barsbold, Rinchen (1983). "Carnivorous dinosaurs from the Cretaceous of Mongolia" (https://www.geokniga.org/bookfiles/geokniga-hishchnye-dinozavry-mela-mongolii.pdf) (PDF). *Transactions of the Joint Soviet-Mongolian Paleontological Expedition* (in Russian). **19**: 5–119. Archived (https://web.archive.org/web/20210831103445/https://www.geokniga.org/bookfiles/geokniga-hishchnye-dinozavry-mela-mongolii.pdf) (PDF) from the original on 31 August 2021. Retrieved 16 September 2021.
6. Kielan-Jaworowska, Zofia; Barsbold, Rinchen (1972). "Narrative of the Polish-Mongolian Paleontological Expeditions" (http://www.palaeontologia.pan.pl/Archive/1972-27_5-13_1-2.pdf) (PDF). *Paleontologica Polonica*. **27**: 11. Archived (https://web.archive.org/web/20231104085352/http://www.palaeontologia.pan.pl/Archive/1972-27_5-13_1-2.pdf) (PDF) from the original on 4 November 2023. Retrieved 29 April 2022.
7. Barsbold, Rinchen (1974). "Saurornithoididae, a new family of theropod dinosaurs from Central Asia and North America" (http://www.palaeontologia.pan.pl/Archive/1974_30_5-22_1-4.pdf) (PDF). *Paleontologica Polonica*. **30**: 5–22. Archived (https://web.archive.org/web/20150924063432/http://www.palaeontologia.pan.pl/Archive/1974_30_5-22_1-4.pdf) (PDF) from the original on 24 September 2015. Retrieved 16 September 2021.
8. American Museum of Natural History (c. 2000). "Fighting Dinosaurs: New Discoveries from Mongolia: Exhibition Highlights" (https://web.archive.org/web/20101123223227/http://www.amnh.org/exhibitions/fightingdinos/ex-fd.php). Archived from the original (http://www.amnh.org/exhibitions/fightingdinos/ex-fd.php) on 23 November 2010. Retrieved 20 August 2010.
9. Jerzykiewicz, Tomasz; Currie, Philip J.; Eberth, David A.; Johnston, P.A.; Koster, E.H.; Zheng, J.-J. (1993). "Djadokhta Formation correlative strata in Chinese Inner Mongolia: an overview of the stratigraphy, sedimentary geology, and paleontology and comparisons with the type locality in the pre-Altai Gobi". *Canadian Journal of Earth Sciences*. **30** (10): 2180–2195. Bibcode:1993CaJES..30.2180J (https://ui.adsabs.harvard.edu/abs/1993CaJES..30.2180J). doi:10.1139/e93-190 (https://doi.org/10.1139%2Fe93-190).

10. Norell, Mark A.; Makovicky, Peter J. (1999). "Important features of the dromaeosaurid skeleton II: information from newly collected specimens of *Velociraptor mongoliensis*". *American Museum Novitates* (3282): 1–45. hdl:2246/3025 (https://hdl.handle.net/2246%2F3025).
11. Norell, Mark A.; Makovicky, Peter J. (1997). "Important features of the dromaeosaur skeleton: information from a new specimen". *American Museum Novitates* (3215): 1–28. hdl:2246/3557 (https://hdl.handle.net/2246%2F3557).
12. Novacek, Michael J. (1996). *Dinosaurs of the Flaming Cliffs*. New York: Anchor Books. ISBN 0-385-47774-0.
13. Turner, A.H.; Montanari, S.; Norell, M.A. (2021). "A New Dromaeosaurid from the Late Cretaceous Khulsan Locality of Mongolia" (https://www.biodiversitylibrary.org/item/290018). *American Museum Novitates* (3965): 1–48. doi:10.1206/3965.1 (https://doi.org/10.1206%2F3965.1). S2CID 231597229 (https://api.semanticscholar.org/CorpusID:231597229).
14. Barsbold, R.; Osmólska, H. (1999). "The skull of Velociraptor (Theropoda) from the Late Cretaceous of Mongolia" (https://www.app.pan.pl/archive/published/app44/app44-189.pdf) (PDF). *Acta Palaeontologica Polonica*. **44** (2): 189–219. Archived (https://web.archive.org/web/20220714025149/https://www.app.pan.pl/archive/published/app44/app44-189.pdf) (PDF) from the original on 14 July 2022. Retrieved 28 July 2022.
15. Godefroit, Pascal; Currie, Philip J.; Li, Hong; Shang, Chang Yong; Dong, Zhi-ming (2008). "A new species of *Velociraptor* (Dinosauria: Dromaeosauridae) from the Upper Cretaceous of northern China". *Journal of Vertebrate Paleontology*. **28** (2): 432–438. doi:10.1671/0272-4634(2008)28[432:ANSOVD]2.0.CO;2 (https://doi.org/10.1671%2F0272-4634%282008%2928%5B432%3AANSOVD%5D2.0.CO%3B2). S2CID 129414074 (https://api.semanticscholar.org/CorpusID:129414074).
16. Evans, D.C.; Larson, D.W.; Currie, P.J. (2013). "A new dromaeosaurid (Dinosauria: Theropoda) with Asian affinities from the latest Cretaceous of North America". *Naturwissenschaften*. **100** (11): 1041–9. Bibcode:2013NW....100.1041E (https://ui.adsabs.harvard.edu/abs/2013NW....100.1041E). doi:10.1007/s00114-013-1107-5 (https://doi.org/10.1007%2Fs00114-013-1107-5). PMID 24248432 (https://pubmed.ncbi.nlm.nih.gov/24248432). S2CID 14978813 (https://api.semanticscholar.org/CorpusID:14978813).
17. Kundrát, M. (2004). "Two Morphotypes of the Velociraptor Neurocranium" (https://oceanrep.geomar.de/id/eprint/913/1/2004-Journal_of_Morphology.pdf) (PDF). *Journal of Morphology* (Conference Abstract). **260** (3): 305. doi:10.1002/jmor.10224 (https://doi.org/10.1002%2Fjmor.10224). PMID 15124236 (https://pubmed.ncbi.nlm.nih.gov/15124236). S2CID 221869849 (https://api.semanticscholar.org/CorpusID:221869849).
18. Powers, M. J. (2020). *The Evolution of Snout Shape in Eudromaeosaurians and its Ecological Significance* (https://era.library.ualberta.ca/items/c9c289c0-7cc4-42a0-ac53-9ffebf111d16/view/3c913f54-988b-40b8-8db2-f7ad9dee46e5/Powers_Mark_J_202006_MSc.pdf) (PDF) (Master Thesis). Department of Biological Sciences, University of Alberta. doi:10.7939/r3-hz8e-5n76 (https://doi.org/10.7939%2Fr3-hz8e-5n76). Archived (https://web.archive.org/web/20221206171944/https://era.library.ualberta.ca/items/c9c289c0-7cc4-42a0-ac53-9ffebf111d16/view/3c913f54-988b-40b8-8db2-f7ad9dee46e5/Powers_Mark_J_202006_MSc.pdf) (PDF) from the original on 6 December 2022. Retrieved 14 June 2022.
19. Powers, M. A.; Sullivan, C.; Currie, P. J. (2020). "Re-examining ratio based premaxillary and maxillary characters in Eudromaeosauria (Dinosauria: Theropoda): Divergent trends in snout morphology between Asian and North American taxa". *Palaeogeography, Palaeoclimatology, Palaeoecology*. **547** (109704) 109704. Bibcode:2020PPP...54709704P (https://ui.adsabs.harvard.edu/abs/2020PPP...54709704P). doi:10.1016/j.palaeo.2020.109704 (https://doi.org/10.1016%2Fj.palaeo.2020.109704). S2CID 216499705 (https://api.semanticscholar.org/CorpusID:216499705).

20. Powers, M. J.; Norell, M A.; Currie, P. J. (2021). "New shallow snouted species of Velociraptor sheds light on intraspecific variation in Velociraptor mongoliensis and possible niche partitioning between species" (https://journals.library.ualberta.ca/vamp/index.php/VAMP/article/view/29374/21426/78423). *Vertebrate Anatomy Morphology Palaeontology*. CSVP Online Abstracts: 31. Archived (https://web.archive.org/web/20221008180709/https://journals.library.ualberta.ca/vamp/index.php/VAMP/article/view/29374/21426/78423) from the original on 8 October 2022. Retrieved 14 June 2022.
21. Powers, M. J.; Norell, M A.; Currie, P. J. (2021). *Examination of morphological variation across Velociraptor mongoliensis specimens reveals a new species with possible ecomorphological variation in snout dimensions* (https://vertpaleo.org/wp-content/uploads/2021/10/SVP_2021_VirtualBook_final.pdf) (PDF). Virtual Meeting Conference. The Society of Vertebrate Paleontology. p. 211. Archived (https://web.archive.org/web/20211019192436/https://vertpaleo.org/wp-content/uploads/2021/10/SVP_2021_VirtualBook_final.pdf) (PDF) from the original on 19 October 2021. Retrieved 14 June 2022.
22. Paul, G. S. (1988). "Birdlike Avetheropods" (https://archive.org/details/predatorydinosau00paul/page/370). *Predatory Dinosaurs of the World: A Complete Illustrated Guide*. New York: Simon & Schuster. p. 370. ISBN 978-0-671-61946-6.
23. Turner, A.H.; Pol, D.; Clarke, J.A.; Erickson, G.M.; Norell, M.A. (2007). "A Basal Dromaeosaurid and Size Evolution Preceding Avian Flight" (https://doi.org/10.1126%2Fscience.1144066). *Science*. **317** (5843): 1378–1381. Bibcode:2007Sci...317.1378T (https://ui.adsabs.harvard.edu/abs/2007Sci...317.1378T). doi:10.1126/science.1144066 (https://doi.org/10.1126%2Fscience.1144066). PMID 17823350 (https://pubmed.ncbi.nlm.nih.gov/17823350). Supporting Online Material (https://www.science.org/doi/10.1126/science.1144066) Archived (https://web.archive.org/web/20211219095915/https://www.science.org/doi/10.1126/science.1144066) 19 December 2021 at the Wayback Machine
24. Campione, N. E.; Evans, D. C.; Brown, C. M.; Carrano, M. T. (2014). "Body mass estimation in non-avian bipeds using a theoretical conversion to quadruped stylopodial proportions" (https://doi.org/10.1111%2F2041-210X.12226). *Methods in Ecology and Evolution*. **5** (9): 913–923. Bibcode:2014MEcEv...5..913C (https://ui.adsabs.harvard.edu/abs/2014MEcEv...5..913C). doi:10.1111/2041-210X.12226 (https://doi.org/10.1111%2F2041-210X.12226). hdl:10088/25281 (https://hdl.handle.net/10088%2F25281).
25. Campione, N. E.; Evans, D. C. (2020). "The accuracy and precision of body mass estimation in non-avian dinosaurs" (https://doi.org/10.1111%2Fbrv.12638). *Biological Reviews*. **95** (6): 1759–1797. doi:10.1111/brv.12638 (https://doi.org/10.1111%2Fbrv.12638). PMID 32869488 (https://pubmed.ncbi.nlm.nih.gov/32869488). S2CID 221404013 (https://api.semanticscholar.org/CorpusID:221404013).Supporting Information (https://onlinelibrary.wiley.com/doi/abs/10.1111/brv.12638#SupportingInformation) Archived (https://web.archive.org/web/20201110235036/https://onlinelibrary.wiley.com/doi/abs/10.1111/brv.12638#SupportingInformation) 10 November 2020 at the Wayback Machine
26. Paul, Gregory S. (2024). *The Princeton Field Guide to Dinosaurs Third Edition*. Princeton Field Guides (3 ed.). Princeton: Princeton University Press. ISBN 978-0-691-23157-0.
27. Turner, A. H.; Makovicky, P. J.; Norell, M. A. (2007). "Feather Quill Knobs in the Dinosaur Velociraptor" (http://evolucionuader.yolasite.com/resources/Lecturas_complementarias/Articulos_Cient%C3%ADficos/Turner%20et%20al%202007%20-%20Dinosaurios%20con%20plumas..pdf) (PDF). *Science*. **317** (5845): 1721. Bibcode:2007Sci...317.1721T (https://ui.adsabs.harvard.edu/abs/2007Sci...317.1721T). doi:10.1126/science.1145076 (https://doi.org/10.1126%2Fscience.1145076). PMID 17885130 (https://pubmed.ncbi.nlm.nih.gov/17885130). S2CID 11610649 (https://api.semanticscholar.org/CorpusID:11610649). Archived (https://web.archive.org/web/20221130033412/http://evolucionuader.yolasite.com/resources/Lecturas_complementarias/Articulos_Cient%C3%ADficos/Turner%20et%20al%202007%20-%20Dinosaurios%20con%20plumas..pdf) (PDF) from the original on 30 November 2022. Retrieved 1 November 2022.

28. Napoli, J. G.; Ruebenstahl, A. A.; Bhullar, B.-A. S.; Turner, A. H.; Norell, M. A. (2021). "A New Dromaeosaurid (Dinosauria: Coelurosauria) from Khulsan, Central Mongolia" (https://digitallibrary.amnh.org/bitstream/handle/2246/7286/3982.pdf?sequence=1&isAllowed=y) (PDF). *American Museum Novitates* (3982): 1–47. doi:10.1206/3982.1 (https://doi.org/10.1206%2F3982.1). hdl:2246/7286 (https://hdl.handle.net/2246%2F7286). ISSN 0003-0082 (https://search.worldcat.org/issn/0003-0082). S2CID 243849373 (https://api.semanticscholar.org/CorpusID:243849373). Archived (https://web.archive.org/web/20211109144341/https://digitallibrary.amnh.org/bitstream/handle/2246/7286/3982.pdf?sequence=1&isAllowed=y) (PDF) from the original on 9 November 2021. Retrieved 28 August 2022.
29. Wang, X.; Cau, A.; Guo, B.; Ma, F.; Qing, G.; Liu, Y. (2022). "Intestinal preservation in a birdlike dinosaur supports conservatism in digestive canal evolution among theropods" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9675785). *Scientific Reports*. **12** (19965): 19965. Bibcode:2022NatSR..1219965W (https://ui.adsabs.harvard.edu/abs/2022NatSR..1219965W). doi:10.1038/s41598-022-24602-x (https://doi.org/10.1038%2Fs41598-022-24602-x). PMC 9675785 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9675785). PMID 36402874 (https://pubmed.ncbi.nlm.nih.gov/36402874).
30. Hone, D. W. E.; Tischlinger, H.; Xu, X.; Zhang, F. (2010). "The Extent of the Preserved Feathers on the Four-Winged Dinosaur Microraptor gui under Ultraviolet Light" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2821398). *PLOS ONE*. **5** (2) e9223. Bibcode:2010PLoSO...5.9223H (https://ui.adsabs.harvard.edu/abs/2010PLoSO...5.9223H). doi:10.1371/journal.pone.0009223 (https://doi.org/10.1371%2Fjournal.pone.0009223). PMC 2821398 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2821398). PMID 20169153 (https://pubmed.ncbi.nlm.nih.gov/20169153).
31. Lü, J.; Brusatte, S.L. (2015). "A large, short-armed, winged dromaeosaurid (Dinosauria: Theropoda) from the Early Cretaceous of China and its implications for feather evolution" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4504142). *Scientific Reports*. **11775** (5) 11775. Bibcode:2015NatSR...511775L (https://ui.adsabs.harvard.edu/abs/2015NatSR...511775L). doi:10.1038/srep11775 (https://doi.org/10.1038%2Fsrep11775). PMC 4504142 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4504142). PMID 26181289 (https://pubmed.ncbi.nlm.nih.gov/26181289).
32. Sues, H.-D. (1977). "The skull of Velociraptor mongoliensis, a small Cretaceous theropod dinosaur from Mongolia". *Paläontologische Zeitschrift*. **51** (3–4): 173–184. Bibcode:1977PalZ...51..173S (https://ui.adsabs.harvard.edu/abs/1977PalZ...51..173S). doi:10.1007/BF02986567 (https://doi.org/10.1007%2FBF02986567). S2CID 128478862 (https://api.semanticscholar.org/CorpusID:128478862).
33. Fowler, D.W.; Freedman, E.A.; Scannella, J.B.; Kambic, R.E. (2011). "The Predatory Ecology of *Deinonychus* and the Origin of Flapping in Birds" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3237572). *PLOS ONE*. **6** (12) e28964. Bibcode:2011PLoSO...628964F (https://ui.adsabs.harvard.edu/abs/2011PLoSO...628964F). doi:10.1371/journal.pone.0028964 (https://doi.org/10.1371%2Fjournal.pone.0028964). PMC 3237572 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3237572). PMID 22194962 (https://pubmed.ncbi.nlm.nih.gov/22194962).
34. Currie, Philip J. (1995). "New information on the anatomy and relationships of *Dromaeosaurus albertensis* (Dinosauria: Theropoda)" (https://web.archive.org/web/20071117132451/http://vertpaleo.org/publications/jvp/15-576-591.cfm). *Journal of Vertebrate Paleontology*. **15** (3): 576–591. Bibcode:1995JVPal..15..576C (https://ui.adsabs.harvard.edu/abs/1995JVPal..15..576C). doi:10.1080/02724634.1995.10011250 (https://doi.org/10.1080%2F02724634.1995.10011250). Archived from the original (http://vertpaleo.org/publications/jvp/15-576-591.cfm) on 17 November 2007.

35. Norell, M. A.; Clark, J. M.; Turner, A. H.; Makovicky, P. J.; Barsbold, R.; Rowe, T. (2006). "A New Dromaeosaurid Theropod from Ukhaa Tolgod (Ömnögov, Mongolia)" (https://www.researchgate.net/publication/232678611). *American Museum Novitates* (3545): 1–51. doi:10.1206/0003-0082(2006)3545[1:ANDTFU]2.0.CO;2 (https://doi.org/10.1206%2F0003-0082%282006%293545%5B1%3AANDTFU%5D2.0.CO%3B2). hdl:2246/5823 (https://hdl.handle.net/2246%2F5823).
36. Godefroit, Pascal; Cau, Andrea; Hu, Dong-Yu; Escuillié, François; Wu, Wenhao; Dyke, Gareth (2013). "A Jurassic avialan dinosaur from China resolves the early phylogenetic history of birds". *Nature*. **498** (7454): 359–362. Bibcode:2013Natur.498..359G (https://ui.adsabs.harvard.edu/abs/2013Natur.498..359G). doi:10.1038/nature12168 (https://doi.org/10.1038%2Fnature12168). PMID 23719374 (https://pubmed.ncbi.nlm.nih.gov/23719374). S2CID 4364892 (https://api.semanticscholar.org/CorpusID:4364892).
37. DePalma, Robert A.; Burnham, David A.; Martin, Larry D.; Larson, Peter L.; Bakker, Robert T. (2015). "The First Giant Raptor (Theropoda: Dromaeosauridae) from the Hell Creek Formation" (https://kuscholarworks.ku.edu/handle/1808/18764). *Paleontological Contributions* (14). doi:10.17161/paleo.1808.18764 (https://doi.org/10.17161%2Fpaleo.1808.18764). hdl:1808/18764 (https://hdl.handle.net/1808%2F18764). Archived (https://web.archive.org/web/20151106223159/https://kuscholarworks.ku.edu/handle/1808/18764) from the original on 6 November 2015. Retrieved 3 November 2015.
38. Paul, Gregory S. (2002). *Dinosaurs of the Air: The Evolution and Loss of Flight in Dinosaurs and Birds* (https://archive.org/details/dinosaursofairev0000paul). Baltimore: Johns Hopkins University Press. ISBN 978-0-8018-6763-7.
39. Norell, Mark A.; Makovicky, Peter J. (2004). "Dromaeosauridae". In Weishampel, David B.; Dodson, Peter; Osmólska, Halszka (eds.). *The Dinosauria* (Second ed.). Berkeley: University of California Press. pp. 196–209. ISBN 978-0-520-24209-8.
40. Currie, P. J.; Evans, D. C. (2019). "Cranial Anatomy of New Specimens of Saurornitholestes langstoni (Dinosauria, Theropoda, Dromaeosauridae) from the Dinosaur Park Formation (Campanian) of Alberta" (https://doi.org/10.1002%2Far.24241). *The Anatomical Record*. **303** (4): 691–715. doi:10.1002/ar.24241 (https://doi.org/10.1002%2Far.24241). PMID 31497925 (https://pubmed.ncbi.nlm.nih.gov/31497925). S2CID 202002676 (https://api.semanticscholar.org/CorpusID:202002676).
41. Jasinski, S. E.; Sullivan, R. M.; Dodson, P. (2020). "New Dromaeosaurid Dinosaur (Theropoda, Dromaeosauridae) from New Mexico and Biodiversity of Dromaeosaurids at the end of the Cretaceous" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7099077). *Scientific Reports*. **10** (1): 5105. Bibcode:2020NatSR..10.5105J (https://ui.adsabs.harvard.edu/abs/2020NatSR..10.5105J). doi:10.1038/s41598-020-61480-7 (https://doi.org/10.1038%2Fs41598-020-61480-7). ISSN 2045-2322 (https://search.worldcat.org/issn/2045-2322). PMC 7099077 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7099077). PMID 32218481 (https://pubmed.ncbi.nlm.nih.gov/32218481). Material was copied from this source, which is available under a Creative Commons Attribution 4.0 International License (https://creativecommons.org/licenses/by/4.0/) Archived (https://web.archive.org/web/20171016050101/https://creativecommons.org/licenses/by/4.0/) 16 October 2017 at the Wayback Machine.
42. King, J. Logan; Sipla, Justin S.; Georgi, Justin A.; Balanoff, Amy M.; Neenan, James M. (2020). "The endocranium and trophic ecology of Velociraptor mongoliensis" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7542195). *Journal of Anatomy*. **237** (5): 861–869. doi:10.1111/joa.13253 (https://doi.org/10.1111%2Fjoa.13253). PMC 7542195 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7542195). PMID 32648601 (https://pubmed.ncbi.nlm.nih.gov/32648601).
43. Sakamoto, M. (2022). "Estimating bite force in extinct dinosaurs using phylogenetically predicted physiological cross-sectional areas of jaw adductor muscles" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9285543). *PeerJ*. **10** e13731. doi:10.7717/peerj.13731 (https://doi.org/10.7717%2Fpeerj.13731). PMC 9285543 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9285543). PMID 35846881 (https://pubmed.ncbi.nlm.nih.gov/35846881).

44. Carpenter, Kenneth (1998). "Evidence of predatory behavior by theropod dinosaurs" (https://web.archive.org/web/20110719060017/http://www.mnhn.ul.pt/geologia/gaia/9.pdf) (PDF). *Gaia*. **15**: 135–144. Archived from the original (http://www.mnhn.ul.pt/geologia/gaia/9.pdf) (PDF) on 19 July 2011.
45. Schmitz, L.; Motani, R. (2011). "Nocturnality in Dinosaurs Inferred from Scleral Ring and Orbit Morphology". *Science*. **332** (6030): 705–8. Bibcode:2011Sci...332..705S (https://ui.adsabs.harvard.edu/abs/2011Sci...332..705S). doi:10.1126/science.1200043 (https://doi.org/10.1126%2Fscience.1200043). PMID 21493820 (https://pubmed.ncbi.nlm.nih.gov/21493820). S2CID 33253407 (https://api.semanticscholar.org/CorpusID:33253407).
46. Ostrom, John H. (1969). *Osteology of* Deinonychus antirrhopus*, an unusual theropod from the Lower Cretaceous of Montana*. Vol. 30. Bulletin of the Peabody Museum of Natural History. pp. 1–165. doi:10.2307/j.ctvqc6gzx (https://doi.org/10.2307%2Fj.ctvqc6gzx). S2CID 210733704 (https://api.semanticscholar.org/CorpusID:210733704).
47. Manning, P. L.; Payne, D.; Pennicott, J.; Barrett, P. M.; Ennos, R. A. (2006). "Dinosaur killer claws or climbing crampons?" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1617199). *Biology Letters*. **2** (1): 110–112. doi:10.1098/rsbl.2005.0395 (https://doi.org/10.1098%2Frsbl.2005.0395). PMC 1617199 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1617199). PMID 17148340 (https://pubmed.ncbi.nlm.nih.gov/17148340).
48. Maxwell, W. Desmond; Ostrom, John H. (1995). "Taphonomy and paleobiological implications of Tenontosaurus-Deinonychus associations" (https://web.archive.org/web/20070927204328/http://www.vertpaleo.org/publications/jvp/15-707-712.cfm). *Journal of Vertebrate Paleontology*. **15** (4): 707–712. Bibcode:1995JVPal..15..707M (https://ui.adsabs.harvard.edu/abs/1995JVPal..15..707M). doi:10.1080/02724634.1995.10011256 (https://doi.org/10.1080%2F02724634.1995.10011256). Archived from the original (http://vertpaleo.org/publications/jvp/15-707-712.cfm) on 27 September 2007.
49. Brinkman, Daniel L.; Cifelli, Richard L.; Czaplewski, Nicholas J. (1998). "First occurrence of *Deinonychus antirrhopus* (Dinosauria: Theropoda) in the Antlers Formation (Lower Cretaceous: Aptian-Albian) of Oklahoma" (http://www.ogs.ou.edu/pubsscanned/BULLETINS/Bulletin146.pdf) (PDF). *Oklahoma Geological Survey Bulletin*. **146**: 1–27. Archived (https://web.archive.org/web/20180819045405/http://www.ogs.ou.edu/pubsscanned/BULLETINS/Bulletin146.pdf) (PDF) from the original on 19 August 2018. Retrieved 21 August 2010.
50. Li, Rihui; Lockley, M.G.; Makovicky, P.J.; Matsukawa, M.; Norell, M.A.; Harris, J.D.; Liu, M. (2007). "Behavioral and faunal implications of Early Cretaceous deinonychosaur trackways from China". *Die Naturwissenschaften*. **95** (3): 185–191. Bibcode:2008NW.....95..185L (https://ui.adsabs.harvard.edu/abs/2008NW.....95..185L). doi:10.1007/s00114-007-0310-7 (https://doi.org/10.1007%2Fs00114-007-0310-7). PMID 17952398 (https://pubmed.ncbi.nlm.nih.gov/17952398). S2CID 16380823 (https://api.semanticscholar.org/CorpusID:16380823).
51. Switek, Brian (29 March 2011). "Dinosaurs behaving badly: Did velociraptors hunt in packs?" (https://www.theguardian.com/science/blog/2011/mar/29/dinosaurs-behaviour-raptors-pack-hunters). *The Guardian*. Archived (https://web.archive.org/web/20230525041612/https://www.theguardian.com/science/blog/2011/mar/29/dinosaurs-behaviour-raptors-pack-hunters) from the original on 25 May 2023. Retrieved 17 September 2020.
52. Long, John, and Schouten, Peter. (2008). *Feathered Dinosaurs: The Origin of Birds*. Oxford and New York: Oxford University Press. ISBN 978-0-19-537266-3, p. 21.
53. Hone, David; Choiniere, Jonah; Sullivan, Corwin; Xu, Xing; Pittman, Michael; Tan, Qingwei (2010). "New evidence for a trophic relationship between the dinosaurs *Velociraptor* and *Protoceratops*". *Palaeogeography, Palaeoclimatology, Palaeoecology*. **291** (3–4): 488–492. Bibcode:2010PPP...291..488H (https://ui.adsabs.harvard.edu/abs/2010PPP...291..488H). doi:10.1016/j.palaeo.2010.03.028 (https://doi.org/10.1016%2Fj.palaeo.2010.03.028).
54. Hone, D.; Tsuihiji, T.; Watabe, M.; Tsogtbaatr, K. (2012). "Pterosaurs as a food source for small dromaeosaurs". *Palaeogeography, Palaeoclimatology, Palaeoecology*. 331–332: 27. Bibcode:2012PPP...331...27H (https://ui.adsabs.harvard.edu/abs/2012PPP...331...27H). doi:10.1016/j.palaeo.2012.02.021 (https://doi.org/10.1016%2Fj.palaeo.2012.02.021).

55. Tse, Yuen Ting; Miller, Case Vincent; Pittman, Michael (2024). "Morphological disparity and structural performance of the dromaeosaurid skull informs ecology and evolutionary history" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11020771). *BMC Ecology and Evolution*. **24** (1): 39. Bibcode:2024BMCEE..24...39T (https://ui.adsabs.harvard.edu/abs/2024BMCEE..24...39T). doi:10.1186/s12862-024-02222-5 (https://doi.org/10.1186%2Fs12862-024-02222-5). PMC 11020771 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11020771). PMID 38622512 (https://pubmed.ncbi.nlm.nih.gov/38622512).
56. Tada, S.; Tsuihiji, T.; Matsumoto, R.; Hanai, T.; Iwami, Y.; Tomita, N.; Sato, H.; Tsogtbaatar, K. (2023). "Evolutionary process toward avian-like cephalic thermoregulation system in Theropoda elucidated based on nasal structures" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10090882). *Royal Society Open Science*. **10** (4). 220997. Bibcode:2023RSOS...1020997T (https://ui.adsabs.harvard.edu/abs/2023RSOS...1020997T). doi:10.1098/rsos.220997 (https://doi.org/10.1098%2Frsos.220997). PMC 10090882 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10090882). PMID 37063996 (https://pubmed.ncbi.nlm.nih.gov/37063996).
57. Norell, M. A.; Gaffney, E. S.; Dingus, L. (1995). *Discovering Dinosaurs In the American Museum of Natural History* (https://archive.org/details/discoveringdinos00nore). Knopf Inc. p. 43. ISBN 978-0-520-22501-5.
58. Molnar, R.E. (2001). "Theropod Paleopathology: A Literature Survey". In Carpenter, Kenneth; Skrepnick, Michael William; Tanke, Darren H (eds.). *Mesozoic Vertebrate Life*. Indiana Univ. Press. pp. 337–363. ISBN 978-0-253-33907-2. OCLC 248649755 (https://search.worldcat.org/oclc/248649755).
59. Nicholas R. Longrich; Philip J. Currie; Dong Zhi-Ming (2010). "A new oviraptorid (Dinosauria: Theropoda) from the Upper Cretaceous of Bayan Mandahu, Inner Mongolia" (https://doi.org/10.1111%2Fj.1475-4983.2010.00968.x). *Palaeontology*. **53** (5): 945–960. Bibcode:2010Palgy..53..945L (https://ui.adsabs.harvard.edu/abs/2010Palgy..53..945L). doi:10.1111/j.1475-4983.2010.00968.x (https://doi.org/10.1111%2Fj.1475-4983.2010.00968.x).
60. Eberth, D. A. (1993). "Depositional environments and facies transitions of dinosaur-bearing Upper Cretaceous redbeds at Bayan Mandahu (Inner Mongolia, People's Republic of China)". *Canadian Journal of Earth Sciences*. **30** (10): 2196–2213. Bibcode:1993CaJES..30.2196E (https://ui.adsabs.harvard.edu/abs/1993CaJES..30.2196E). doi:10.1139/e93-191 (https://doi.org/10.1139%2Fe93-191).
61. Dashzeveg, D.; Dingus, L.; Loope, D. B.; Swisher III, C. C.; Dulam, T.; Sweeney, M. R. (2005). "New Stratigraphic Subdivision, Depositional Environment, and Age Estimate for the Upper Cretaceous Djadokhta Formation, Southern Ulan Nur Basin, Mongolia" (http://digitallibrary.amnh.org/bitstream/handle/2246/5667/N3498.pdf?sequence=1&isAllowed=y) (PDF). *American Museum Novitates* (3498): 1–31. doi:10.1206/0003-0082(2005)498[0001:NSSDEA]2.0.CO;2 (https://doi.org/10.1206%2F0003-0082%282005%29498%5B0001%3ANSSDEA%5D2.0.CO%3B2). hdl:2246/5667 (https://hdl.handle.net/2246%2F5667). S2CID 55836458 (https://api.semanticscholar.org/CorpusID:55836458). Archived (https://web.archive.org/web/20220101224659/https://digitallibrary.amnh.org/bitstream/handle/2246/5667/N3498.pdf?sequence=1&isAllowed=y) (PDF) from the original on 1 January 2022. Retrieved 28 July 2022.
62. Jerzykiewicz, T. (1997). "Djadokhta Formation". In Currie, P. J.; Padian, K. (eds.). *Encyclopedia of Dinosaurs* (https://archive.org/details/encyclopediadino00curr_075). San Diego: Academic Press. pp. 188 (https://archive.org/details/encyclopediadino00curr_075/page/n218)−191. ISBN 978-0-12-226810-6.

63. Dingus, L.; Loope, D. B.; Dashzeveg, D.; Swisher III, C. C.; Minjin, C.; Novacek, M. J.; Norell, M. A. (2008). "The Geology of Ukhaa Tolgod (Djadokhta Formation, Upper Cretaceous, Nemegt Basin, Mongolia)" (https://digitalcommons.unl.edu/cgi/viewcontent.cgi?article=1218&context=geosciencefacpub). *American Museum Novitates* (3616): 1–40. doi:10.1206/442.1 (https://doi.org/10.1206%2F442.1). hdl:2246/5916 (https://hdl.handle.net/2246%2F5916). S2CID 129735494 (https://api.semanticscholar.org/CorpusID:129735494). Archived (https://web.archive.org/web/20220805093444/https://digitalcommons.unl.edu/cgi/viewcontent.cgi?article=1218&context=geosciencefacpub) from the original on 5 August 2022. Retrieved 27 September 2023.
64. Chinzorig, T.; Kobayashi, Y.; Tsogtbaatar, K.; Currie, P. J.; Watabe, M.; Barsbold, R. (2017). "First Ornithomimid (Theropoda, Ornithomimosauria) from the Upper Cretaceous Djadokhta Formation of Tögrögiin Shiree, Mongolia" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5517598). *Scientific Reports*. **7** (5835): 5835. Bibcode:2017NatSR...7.5835C (https://ui.adsabs.harvard.edu/abs/2017NatSR...7.5835C). doi:10.1038/s41598-017-05272-6 (https://doi.org/10.1038%2Fs41598-017-05272-6). PMC 5517598 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5517598). PMID 28724887 (https://pubmed.ncbi.nlm.nih.gov/28724887).
65. Saneyoshi, M.; Watabe, M.; Suzuki, S.; Tsogtbaatar, K. (2011). "Trace fossils on dinosaur bones from Upper Cretaceous eolian deposits in Mongolia: Taphonomic interpretation of paleoecosystems in ancient desert environments" (https://www.academia.edu/27857932). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **311** (1–2): 38–47. Bibcode:2011PPP...311...38S (https://ui.adsabs.harvard.edu/abs/2011PPP...311...38S). doi:10.1016/j.palaeo.2011.07.024 (https://doi.org/10.1016%2Fj.palaeo.2011.07.024). Archived (https://web.archive.org/web/20221008180700/https://www.academia.edu/27857932) from the original on 8 October 2022. Retrieved 28 July 2022.
66. Crichton, M. (1990). *Jurassic Park* (https://archive.org/details/jurassicparknove00cric/page/117). New York: Alfred A. Knopf. pp. 117 (https://archive.org/details/jurassicparknove00cric/page/117). ISBN 978-0-394-58816-2.
67. Cummings, M. "Yale's legacy in *Jurassic World* (http://news.yale.edu/2015/06/18/yale-s-legacy-jurassic-world) Archived (https://web.archive.org/web/20170706231545/http://news.yale.edu/2015/06/18/yale-s-legacy-jurassic-world) 6 July 2017 at the Wayback Machine." *Yale News*, 18 June 2015.
68. Duncan, J. (2006). *The Winston Effect*. London: Titan Books. p. 175. ISBN 978-1-84576-365-7.
69. Bakker, R.T. (1995). *Raptor Red*. New York: Bantam Books. p. 4. ISBN 978-0-553-57561-3.
70. Adams, B. (15 June 1993). "Director Loses Utahraptor Name Game" (https://www.deseret.com/1993/6/15/19051844/director-loses-utahraptor-name-game). *Deseret News*. Archived (https://web.archive.org/web/20220822053459/https://www.deseret.com/1993/6/15/19051844/director-loses-utahraptor-name-game) from the original on 22 August 2022. Retrieved 21 August 2022.
71. Dhar, M. (17 June 2013). "T. Rex at 20: How 'Jurassic Park' science has evolved" (https://www.washingtonpost.com/national/health-science/t-rex-at-20-how-jurassic-park-science-has-evolved/2013/06/17/1701595c-d1f9-11e2-8cbe-1bcbee06f8f8_story.html). *The Washington Post*. Archived (https://web.archive.org/web/20231230175837/https://www.washingtonpost.com/national/health-science/t-rex-at-20-how-jurassic-park-science-has-evolved/2013/06/17/1701595c-d1f9-11e2-8cbe-1bcbee06f8f8_story.html) from the original on 30 December 2023. Retrieved 21 August 2022.
72. Polo, S. (10 June 2015). "Jurassic World explains its featherless dinos while poking fun at blockbusters" (https://www.polygon.com/2015/6/10/8760275/jurassic-world-dinosaurs-feathers). *Polygon*. Archived (https://web.archive.org/web/20221011135009/http://www.polygon.com/2015/6/10/8760275/jurassic-world-dinosaurs-feathers) from the original on 11 October 2022. Retrieved 21 August 2022.

73. Katwala, A. (10 June 2022). "The Real Story Behind *Jurassic World Dominion*'s Dino Feathers" (https://www.wired.com/story/jurassic-world-dominion-dinosaur-feathers/). *Wired*. Archived (https://web.archive.org/web/20220612111833/https://www.wired.com/story/jurassic-world-dominion-dinosaur-feathers/) from the original on 12 June 2022. Retrieved 21 August 2022.

External links

- Media related to *Velociraptor* at Wikimedia Commons
- Data related to *Velociraptor* at Wikispecies
- Quotations related to *Velociraptor* at Wikiquote
- Wikijunior Dinosaurs/Velociraptor at Wikibooks
- 3D skull model of *Velociraptor mongoliensis* (https://sketchfab.com/3d-models/velociraptor-mongoliensis-skull-4a1551d6fedc47ccbc82bdb1a346f814) at Sketchfab
- Skeletal reconstruction of *Velociraptor mongoliensis* (https://www.skeletaldrawing.com/theropods/velociraptor) at Dr. Scott Hartman's Skeletal Drawing
- Videos about the Fighting Dinosaurs (https://web.archive.org/web/20060621140840/http://www.amnh.org/exhibitions/fightingdinos/videos.html) at American Museum of Natural History (Wayback Machine)

Retrieved from "https://en.wikipedia.org/w/index.php?title=Velociraptor&oldid=1371049913"

WIKIPEDIA
25 years of the free encyclopedia

Troodontidae

Troodontidae /troʊ.əˈdɒntɪdiː/ is a clade of bird-like theropod dinosaurs from the Late Jurassic to Late Cretaceous. During most of the 20th century, troodontid fossils were few and incomplete and they have therefore been allied, at various times, with many dinosaurian lineages. More recent fossil discoveries of complete and articulated specimens (including specimens which preserve feathers, eggs, embryos, and complete juveniles), have helped to increase understanding about this group. Anatomical studies, particularly studies of the most primitive troodontids, like *Sinovenator*, demonstrate striking anatomical similarities with *Archaeopteryx* and primitive dromaeosaurids, and demonstrate that they are relatives comprising a clade called Paraves.

Evolution

The oldest definitive troodontid known is *Hesperornithoides* from the Late Jurassic of Wyoming.[1] The slightly older *Koparion* of Utah is only represented by a single tooth, and small maniraptoran teeth from the Middle Jurassic of England were identified as those of indeterminate troodontids in 2023.[2] Over the Cretaceous, troodontids radiated throughout western North America, Asia, and Europe, suggesting a mostly Laurasian distribution for the group. However, two diagnostic teeth from the latest Cretaceous (Maastrichtian) Kallamedu Formation of southern India were identified as troodontid teeth, suggesting that troodontids either also inhabited Gondwana or managed to disperse to India from elsewhere prior to its separation as an island continent.[3][4] The potential Gondwanan occurrence of troodontids is supported by the existence of Middle Jurassic remains, which suggest that they originated prior to the breakup of Pangaea. However, due to the lack of other remains from the region, it has been suggested that the existence of Gondwanan troodontids should be regarded as provisional.[5]

Troodontids

Temporal range:

Collection of four troodontids, clockwise from top left: *Mei*, unnamed Alaskan troodontid, *Jianianhualong*, *Sinovenator*.

Scientific classification

Kingdom:	Animalia
Phylum:	Chordata
Class:	Reptilia
Clade:	Dinosauria
Clade:	Saurischia
Clade:	Theropoda
Clade:	Paraves
Family:	†Troodontidae Gilmore, 1924

Type species

†*Troodon formosus*
Leidy, 1856

Clade

- †*Albertavenator*

Description

Troodontids are a group of small, bird-like, gracile maniraptorans. All troodontids have unique features of the skull, such as large numbers of closely spaced teeth in the lower jaw. Troodontids have sickle-claws and raptorial hands, and some of the highest non-avian encephalization quotients, suggesting that they were behaviourally advanced and had keen senses.[6] They had unusually long legs compared to other theropods, with a large, curved claw on their retractable second toes, similar to the "sickle-claw" of the dromaeosaurids. However, the sickle-claws of troodontids were not as large or recurved as in dromaeosaurids, and in some instances could not be held off the ground and "retracted" to the same degree. In at least one troodontid, *Borogovia*, the second toe could not be held far off the ground at all and the claw was straight, not curved or sickle-like.

Skull of the troodontid *Saurornithoides mongoliensis*.

Troodontids had unusually large brains among dinosaurs, comparable to those of living flightless birds. Their eyes were also large, and pointed forward, indicating that they had good binocular vision. The ears of troodontids were also unusual among theropods, having enlarged middle ear cavities, indicating acute hearing ability. The placement of this cavity near the eardrum may have aided in the detection of low-frequency sounds.[7] In some troodontids, ears were also asymmetrical, with one ear placed higher on the skull than the other, a feature shared only with some owls. The specialization of the ears may indicate that troodontids hunted in a manner similar to owls, using their hearing to locate small prey.[8]

- †*Almas*
- †*Archaeornithoides*?
- †*Daliansaurus*
- †*Harenadraco*
- †*Hesperornithoides*
- †*Jianianhualong*
- †*Jinfengopteryx*
- †*Koparion*?
- †*Liaoningvenator*
- †*Mei*
- †*Papiliovenator*
- †*Paronychodon*?
- †*Polyodontosaurus*
- †*Sinornithoides*
- †*Sinovenator*
- †*Sinusonasus*
- †*Talos*
- †*Tamarro*
- †*Tochisaurus*
- †*Xixiasaurus*
- †**Troodontinae** Gilmore, 1924
  van der Reest & Currie, 2017
  - †*Borogovia*?
  - †*Byronosaurus*?
  - †*Dinevenator*
  - †*Gobivenator*
  - †*Hypnovenator*
  - †*Latenivenatrix*
  - †*Linhevenator*
  - †*Pectinodon*
  - †*Philovenator*
  - †*Saurornithoides*
  - †*Stenonychosaurus*
  - †*Troodon*
  - †*Urbacodon*

Diet

- †*Xenovenator*
- †*Zanabazar*

Synonyms

- Saurornithoididae Barsbold, 1974

Although most paleontologists believe that they were predatory carnivores, the many small, coarsely serrated teeth, large denticle size, and U-shaped jaws of some species (particularly *Troodon*) suggest that some species may have been omnivorous or herbivorous. Some suggest that the large denticle size is reminiscent of the teeth of extant iguanine lizards.[9][10] In contrast, a few species, such as *Byronosaurus*, had large numbers of needle-like teeth, which seem best-suited for picking up small prey, such as birds, lizards and small mammals. Other morphological characteristics of the teeth, such as the detailed form of the denticles and the presence of blood grooves, also seem to indicate carnivory.[11] Dental microwear also suggests carnivory, as some troodontids have a "puncture and pull" wear pattern on their teeth.[12] Analyses of barium/calcium and strontium/calcium ratios, which are higher in carnivores due to bioaccumulation, found low ratios in teeth of *Stenonychosaurus*, suggesting that it had a diet ranging from mixed to plant-dominant omnivory.[13] Though little is known directly about the predatory behavior of troodontids, Fowler and colleagues theorize that the longer legs and smaller sickle claws (as compared to dromaeosaurids) indicate a more cursorial lifestyle, though the study indicates that troodontids were still likely to have used the unguals for prey manipulation. The proportions of the metatarsals, tarsals and unguals of troodontids appear indicative of their having nimbler, but weaker feet, perhaps better adapted for capturing and subduing smaller prey. This suggests an ecological separation from the slower but more powerful Dromaeosauridae.[14]

Classification

Troodontid fossils were among the first dinosaur remains described. Initially, Leidy (1856) assumed they were lacertilian (lizards), but, by 1924, they were referred to Dinosauria by Gilmore, who suggested that they were ornithischians and allied them with the pachycephalosaurian *Stegoceras* in a Troodontidae. It was not until 1945 that C.M. Sternberg recognized Troodontidae as a theropod family. Since 1969, Troodontidae has typically been allied with Dromaeosauridae, in a clade (natural group) known as Deinonychosauria, but this was by no means a consensus. Holtz (in 1994) erected the clade Bullatosauria, uniting Ornithomimosauria (the "ostrich-dinosaurs") and Troodontidae, on the basis of characteristics including, among others, an inflated braincase (*parabasisphenoid*) and a long, low opening in the upper jaw (the *maxillary fenestra*). Features of the pelvis also suggested they were less advanced than dromaeosaurids. New discoveries of primitive troodontids from China (such as *Sinovenator* and *Mei*), however, display strong similarities between Troodontidae, Dromaeosauridae and the primitive bird *Archaeopteryx*, and most paleontologists, including Holtz, now consider troodontids to be much more closely related to birds than they are to ornithomimosaurs, causing the clade Bullatosauria to be abandoned.

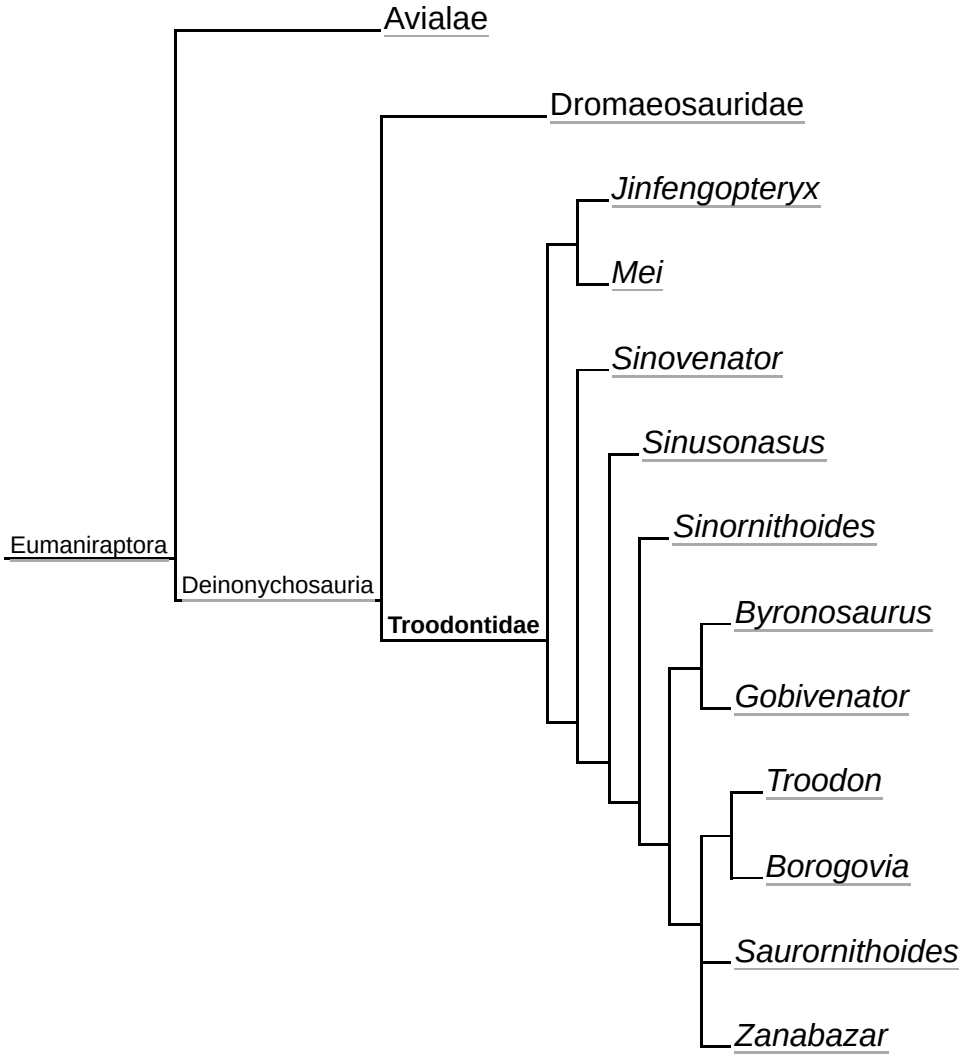
One study of theropod systematics by members of the Theropod Working Group has uncovered striking similarities among the most basal dromaeosaurids, troodontids, and *Archaeopteryx*. This clade is together called Paraves by Novas and Pol.[15] The extensive cladistic analysis conducted by Turner et al., (2012) supported the monophyly of Troodontidae.[16]

Taxonomy

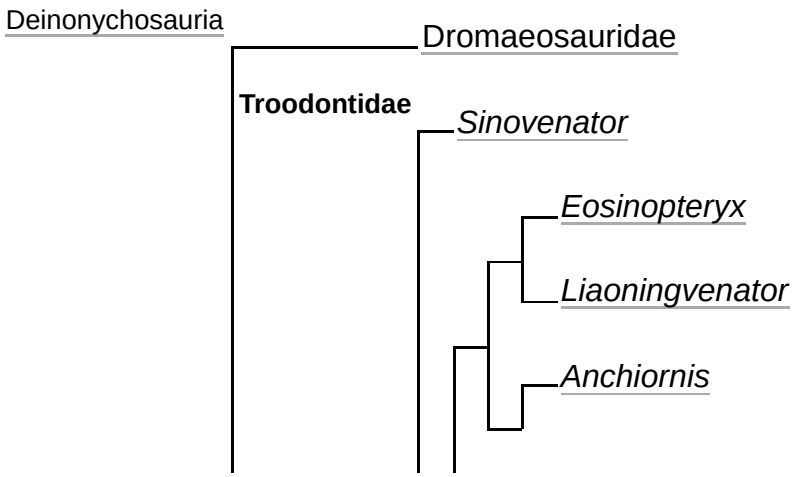
- **Family Troodontidae**
  - †*Almas*
  - †*Jianianhualong*
  - †*Liaoningvenator*
  - †*Sinornithoides*
  - †*Talos*
  - †*Tochisaurus*
  - †*Xixiasaurus*
  - Subfamily Jinfengopteryginae
    - †*Jinfengopteryx*
    - †*Tamarro*
    - IGM 100/1128
  - Subfamily Sinovenatorinae
    - †*Daliansaurus*
    - †*Mei*
    - †*Sinovenator*
    - †*Sinusonasus*
  - Subfamily Troodontinae
    - †*Borogovia*
    - †*Byronosaurus*?
    - †*Gobivenator*
    - †*Hypnovenator*
    - †*Linhevenator*
    - †*Pectinodon*
    - †*Philovenator*
    - †*Urbacodon*
    - Tribe Saurornithoidini
      - †*Saurornithoides*
      - †*Zanabazar*
    - Tribe Troodontini
      - †*Albertavenator*
      - †*Dinevenator*
      - †*Latenivenatrix*
      - †*Stenonychosaurus*
      - †*Troodon*
      - †*Xenovenator*

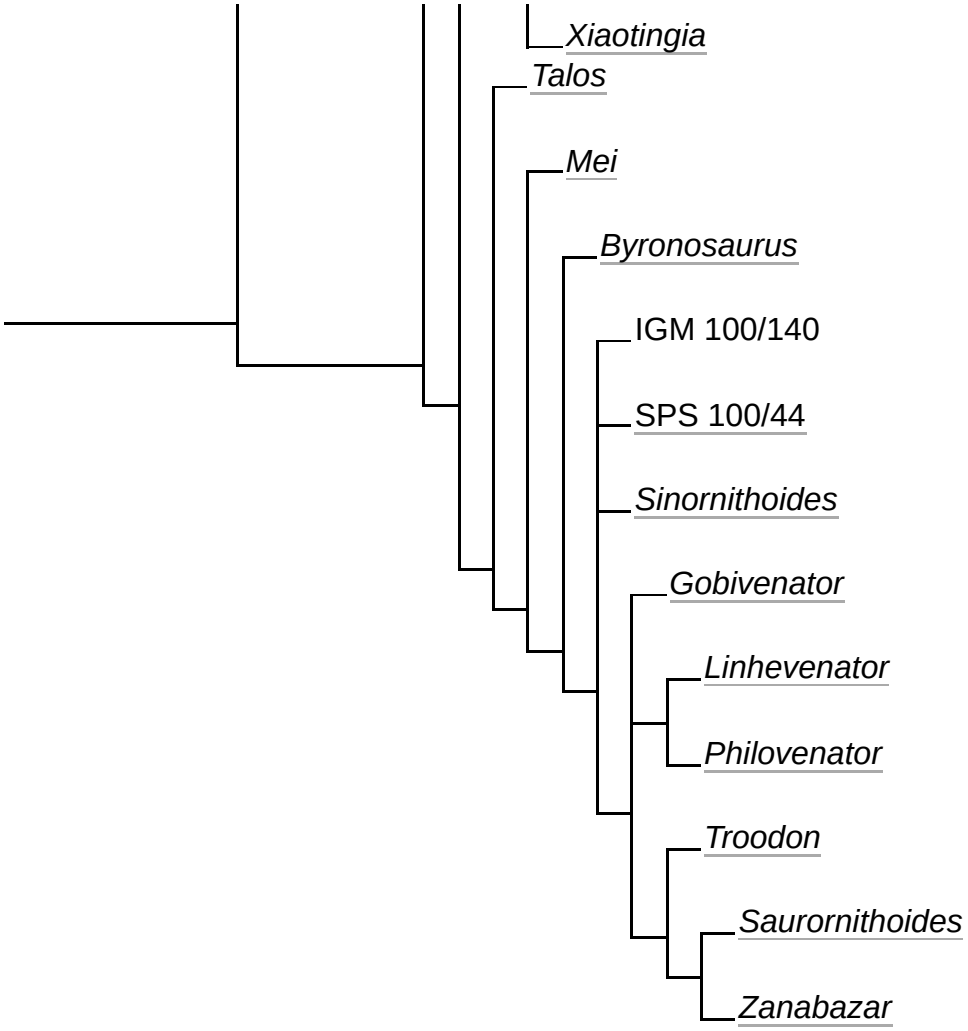
Phylogeny

There are multiple possibilities for the genera included in Troodontidae as well as how they are related. Very primitive species, such as *Anchiornis huxleyi*, have alternately been found to be early troodontids, early members of the closely related group Avialae, or more primitive paravians by various studies. The cladogram below follows the results of a study by Lefèvre et al. 2017.[17]

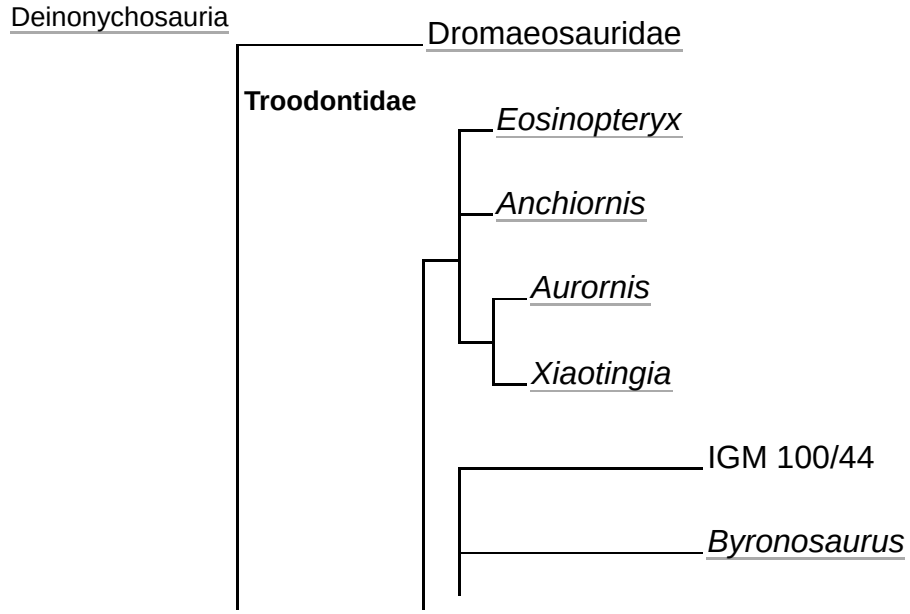


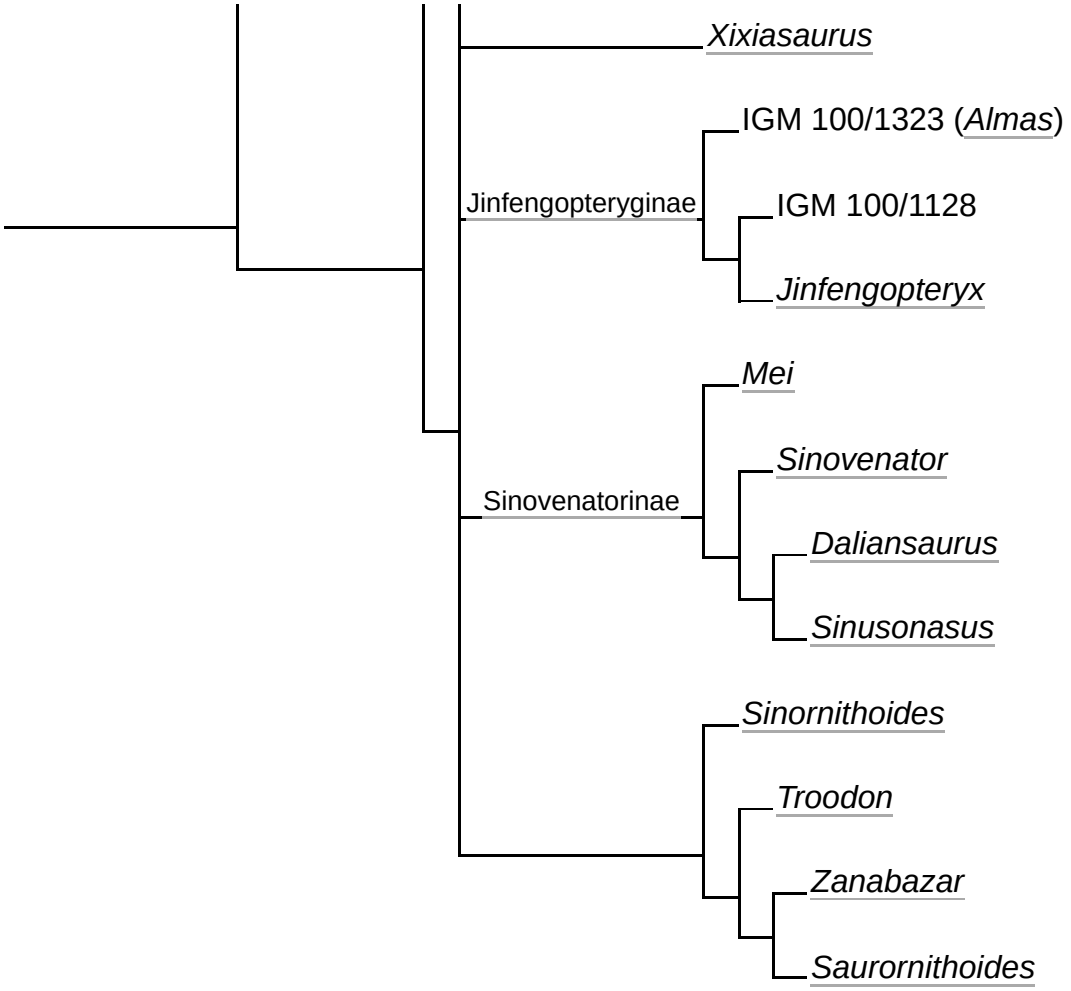
Shen et al. (2017a) explored troodontid phylogeny using a modified version of the Tsuihiji et al. (2014) analysis.[18] It was in turn based on data published by Gao et al. (2012), a slightly modified version of the Xu et al. (2011) analysis,[19] focusing on advanced troodontids. A simplified version is shown below.[20]



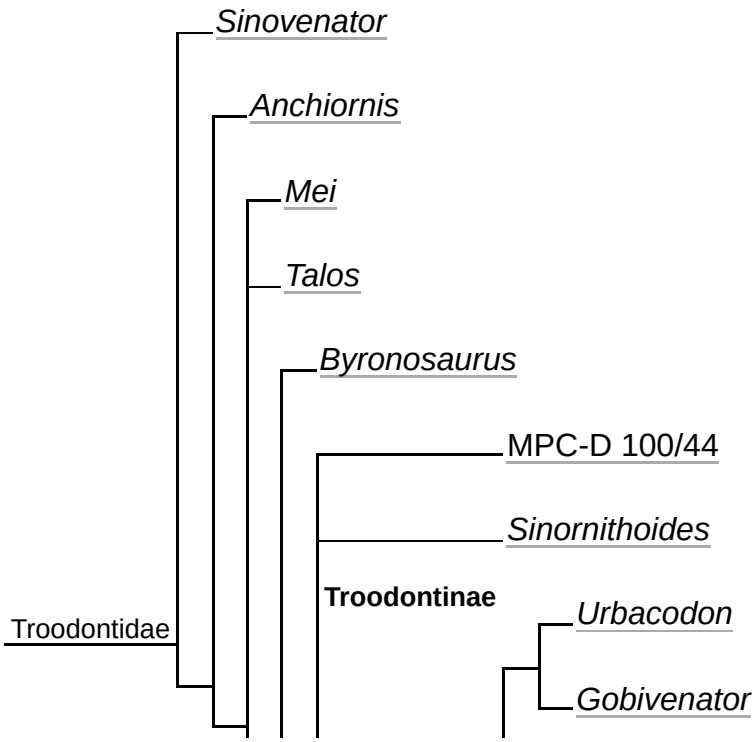


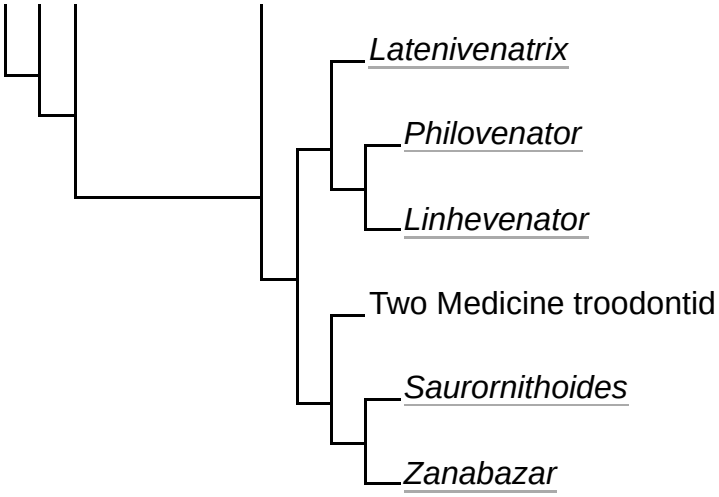
In 2014, Brusatte, Lloyd, Wang and Norell published an analysis on Coelurosauria, based on data from Turner et al. (2012) who named a third subfamily of troodontids, Jinfengopteryginae.[16] Their analysis included more basal troodontid species but failed to resolve many of their interrelationships, resulting in large "polytomies" (sets of species where the branching order in the family tree is uncertain).[21] An updated version of the Brusatte et al. analysis was provided by Shen *et at.* (2017b), who included more taxa and recovered greater resolution. Shen *et at.* named a fourth subfamily of troodontids, the Sinovenatorinae. A simplified version of their analysis is shown below.[22]





Troodontinae is a subfamily of troodontid dinosaurs. The subfamily was first used in 2017 for the group of troodontids descended from the last common ancestor of *Gobivenator mongoliensis* and *Zanabazar junior,* but has been redefined to be the least inclusive clade containing *Saurornithoides mongoliensis* and *Troodon formosus*, utilizing the type species of the clade.[23][17][1] Below is a cladogram of the Troodontinae as published by Aaron van der Reest and Phil Currie, in 2017.[23]





Paleobiology

Prismatoolithus levis eggs, which are considered to have been laid by a troodontid

Many troodontid nests, including eggs that contain fossilized embryos, have been described. Hypotheses about troodontid reproduction have been developed from this evidence (see *Troodon*). A few troodont fossils, including specimens of *Mei* and *Sinornithoides*, demonstrate that these animals roosted like birds, with their heads tucked under their forelimbs.[24] These fossils, as well as numerous skeletal similarities to birds and related feathered dinosaurs, support the idea that troodontids probably bore a bird-like feathered coat. The discovery of fully feathered, primitive troodontids, such as *Jianianhualong*, lend support to this.

In 2004, Mark Norell and colleagues described two partial troodontid skulls (specimen numbers IGM 100/972 and IGM 100/974) found in a nest of oviraptorid eggs in the Djadokhta Formation of Mongolia. The nest is quite certainly that of an oviraptorosaur, since an oviraptorid embryo is still preserved inside one of the eggs. The two partial troodontid skulls were first described by Norell et al. (1994) as dromaeosaurids, but reassigned to the troodontid *Byronosaurus* after further study.[9][25] The troodontids were either hatchlings or embryos, and fragments of eggshell are adhered to them although it seems to be oviraptorid eggshell. The presence of tiny troodontids in an oviraptorid nest is an enigma. Hypotheses explaining how they came to be there include that they were the prey of the adult oviraptorid, that they were there to prey on oviraptorid hatchlings, or that some troodontids may have been nest parasites.[26]

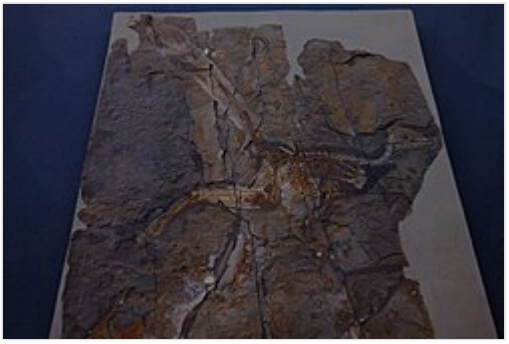
Feeding

Troodontid feeding was discovered to be typical of coelurosaurian theropods, with a characteristic "puncture and pull" feeding method seen also in such theropods as the dromaeosauridae and tyrannosauridae. Studies of wear patterns on the teeth of dromaeosaurids by Angelica Torices et al. indicate that dromaeosaurid teeth share similar wear patterns to those seen in the aforementioned groups. However, micro wear on the teeth indicated that dromaeosaurids likely preferred larger prey items than the troodontids with which they often shared their environment. Such differences in dietary preferences

likely allowed them to inhabit the same ecosystems. The same study also indicated that dromaeosaurids such as *Dromaeosaurus* and *Saurornitholestes* (two dromaeosaurids analyzed in the study) likely included bone in their diet and were better adapted to handle struggling prey, while troodontids, equipped with weaker jaws, preyed on softer-bodied animals and prey items such as invertebrates and carrion that either were immobile or could likely be swallowed whole.[27][28] Ba/Ca and Sr/Ca ratios found in the enamel of Oldman Formation troodontids indicate that they were mixed-feeding to plant-dominant omnivores.[29]

Flight

Compared to most other paravians, troodontids are unspecialised for aerial locomotion. However, *Jinfengopteryx* ranks closely with non-avian theropods known to engage in powered flight like *Microraptor* and *Rahonavis*.[30]



Holotype specimen of *Jinfengopteryx* featuring elongated feathers

Bird evolution

Troodontids are important in research into the origin of birds because they share many anatomical characters with early birds. Crucially, the substantially complete *Hesperornithoides* ("Lori") is a troodontid from the Late Jurassic Morrison Formation, close to the time of *Archaeopteryx*. The discovery of Jurassic troodonts is positive physical evidence that derived deinonychosaurs were present before the time that avians arose. This fact strongly invalidates the "temporal paradox" cited by the few remaining opponents of the idea that birds are closely related to dinosaurs.[31]

Palaeopathology

A variety of palaeopathologies have been diagnosed in troodontid metatarsals from Alberta. These include a superficial cortical fracture that caused a chronic stable callus and a minor avulsion or chip fracture arising from stress at the posterolateral insertion site of the gastrocnemius. A third specimen revealed a pathology that impacted the foot more broadly, resulting in chronic callus formation throughout the shaft, in addition to chronic inflammation at the distal joint of the fourth metatarsal and the deformation of the second metatarsal.[32]

See also

- Timeline of troodontid research
- Evolution of birds

Dinosaurs portal

References

1. Hartman, Scott; Mortimer, Mickey; Wahl, William R.; Lomax, Dean R.; Lippincott, Jessica; Lovelace, David M. (2019-07-10). "A new paravian dinosaur from the Late Jurassic of North America supports a late acquisition of avian flight" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6626525). *PeerJ*. **7** e7247. Bibcode:2019PeerJ...7e7247H (https://ui.adsabs.harvard.

edu/abs/2019PeerJ...7e7247H). doi:10.7717/peerj.7247 (https://doi.org/10.7717%2Fpeerj.7247). ISSN 2167-8359 (https://search.worldcat.org/issn/2167-8359). PMC 6626525 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6626525). PMID 31333906 (https://pubmed.ncbi.nlm.nih.gov/31333906).

2. Wills, Simon; Underwood, Charlie J.; Barrett, Paul M. (2023). Mannion, Philip (ed.). "Machine learning confirms new records of maniraptoran theropods in Middle Jurassic UK microvertebrate faunas" (https://doi.org/10.1002%2Fspp2.1487). *Papers in Palaeontology*. **9** (2) e1487. Bibcode:2023PPal....9E1487W (https://ui.adsabs.harvard.edu/abs/2023PPal....9E1487W). doi:10.1002/spp2.1487 (https://doi.org/10.1002%2Fspp2.1487). ISSN 2056-2799 (https://search.worldcat.org/issn/2056-2799).
3. Goswami, A.; Prasad, G. V. R.; Verma, O.; Flynn, J. J.; Benson, Roger B. J. (16 April 2013). "A troodontid dinosaur from the latest Cretaceous of India" (https://doi.org/10.1038%2Fncomms2716). *Nature Communications*. **4** (1): 1703. Bibcode:2013NatCo...4.1703G (https://ui.adsabs.harvard.edu/abs/2013NatCo...4.1703G). doi:10.1038/ncomms2716 (https://doi.org/10.1038%2Fncomms2716). ISSN 2041-1723 (https://search.worldcat.org/issn/2041-1723). PMID 23591870 (https://pubmed.ncbi.nlm.nih.gov/23591870).
4. Dhiman, H.; Parvez, F.; Goswami, A.; Prasad, G. V. R. (2026). "A second theropod tooth from the Cauvery Basin reaffirms the presence of troodontid dinosaurs in the Cretaceous of India". *The Anatomical Record*. doi:10.1002/ar.70280 (https://doi.org/10.1002%2Far.70280).
5. Ding, A.; Pittman, M.; Upchurch, P.; O'Connor, J.; Field, D. J.; Xu, X. (2020-08-21). "The Biogeography of Coelurosaurian Theropods and Its Impact on Their Evolutionary History" (http://digitallibrary.amnh.org/handle/2246/7237). *BULLETIN OF THE AMERICAN MUSEUM OF NATURAL HISTORY*. hdl:2246/7237 (https://hdl.handle.net/2246%2F7237). Retrieved 2023-12-14.
6. Junchang Lü; Li Xu; Yongqing Liu; Xingliao Zhang; Songhai Jia & Qiang Ji (2010). "A new troodontid (Theropoda: Troodontidae) from the Late Cretaceous of central China, and the radiation of Asian troodontids" (https://www.app.pan.pl/archive/published/app55/app20090047.pdf) (PDF). *Acta Palaeontologica Polonica*. **55** (3): 381–388. doi:10.4202/app.2009.0047 (https://doi.org/10.4202%2Fapp.2009.0047). S2CID 55220748 (https://api.semanticscholar.org/CorpusID:55220748).
7. Currie, P. J. (1985). "Cranial anatomy of *Stenonychosaurus inequalis* (Saurischia, Theropoda) and its bearing on the origin of birds" (http://doc.rero.ch/record/14596/files/PAL_E2272.pdf) (PDF). *Canadian Journal of Earth Sciences*. **22** (11): 1643–1658. Bibcode:1985CaJES..22.1643C (https://ui.adsabs.harvard.edu/abs/1985CaJES..22.1643C). doi:10.1139/e85-173 (https://doi.org/10.1139%2Fe85-173).
8. Castanhinha, R.; Mateus, O. (2006). "On the left-right asymmetry in dinosaurs". *Journal of Vertebrate Paleontology*. **26** (Supp. 3): 48A. doi:10.1080/02724634.2006.10010069 (https://doi.org/10.1080%2F02724634.2006.10010069). S2CID 220413406 (https://api.semanticscholar.org/CorpusID:220413406).
9. Mackovicky, Peter J.; Norell, Mark A. (2004). "Troodontidae". In Weishampel, David B.; Dodson, Peter; Osmólska, Halszka (eds.). *The Dinosauria* (2nd ed.). Berkeley: University of California Press. pp. 184–195. ISBN 978-0-520-24209-8.
10. Holtz, T.R. Jr.; Brinkman, D.L.; Chandler, C.L. (1998). "Denticle morphometrics and a possibly omnivorous feeding habit for the theropod dinosaur *Troodon*" (https://www.geol.umd.edu/~tholtz/gaiatroo.pdf) (PDF). *Gaia*. **15**: 159–166.
11. Currie, PJ; Dong, Z (2001). "New information on Cretaceous troodontids (Dinosauria, Theropoda) from the People's Republic of China" (http://doc.rero.ch/record/14316/files/PAL_E1483.pdf) (PDF). *Canadian Journal of Earth Sciences*. **38** (12): 1753–1766. Bibcode:2001CaJES..38.1753C (https://ui.adsabs.harvard.edu/abs/2001CaJES..38.1753C). doi:10.1139/e01-065 (https://doi.org/10.1139%2Fe01-065).

12. Fan, Yui Chi; Miller, Case Vincent; Pittman, Michael (13 February 2026). "Diet of bird-like troodontid dinosaurs: synthesis of a contentious clade" (https://onlinelibrary.wiley.com/doi/full/10.1002/brv.70145). *Biological Reviews*. doi:10.1002/brv.70145 (https://doi.org/10.1002%2Fbrv.70145). ISSN 1464-7931 (https://search.worldcat.org/issn/1464-7931). Retrieved 14 March 2026 – via Wiley Online Library.
13. Cullen, Thomas M.; Cousens, Brian L. (15 November 2023). "New biogeochemical insights into Mesozoic terrestrial paleoecology and evidence for omnivory in troodontid dinosaurs". *Geological Society of America Bulletin*. **136** (7–8): 2689–2701. doi:10.1130/b37077.1 (https://doi.org/10.1130%2Fb37077.1). ISSN 0016-7606 (https://search.worldcat.org/issn/0016-7606). S2CID 265241862 (https://api.semanticscholar.org/CorpusID:265241862).
14. Fowler, D.W.; Freedman, E.A.; Scannella, J.B.; Kambic, R.E. (2011). "The Predatory Ecology of *Deinonychus* and the Origin of Flapping in Birds" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3237572). *PLOS ONE*. **6** (12) e28964. Bibcode:2011PLoSO...628964F (https://ui.adsabs.harvard.edu/abs/2011PLoSO...628964F). doi:10.1371/journal.pone.0028964 (https://doi.org/10.1371%2Fjournal.pone.0028964). PMC 3237572 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3237572). PMID 22194962 (https://pubmed.ncbi.nlm.nih.gov/22194962).
15. Novas, Fernando E.; Pol, Diego (2005). "New evidence on deinonychosaurian dinosaurs from the Late Cretaceous of Patagonia" (http://doc.rero.ch/record/15096/files/PAL_E2361.pdf) (PDF). *Nature*. **433** (7028): 858–861. Bibcode:2005Natur.433..858N (https://ui.adsabs.harvard.edu/abs/2005Natur.433..858N). doi:10.1038/nature03285 (https://doi.org/10.1038%2Fnature03285). PMID 15729340 (https://pubmed.ncbi.nlm.nih.gov/15729340). S2CID 4413316 (https://api.semanticscholar.org/CorpusID:4413316).
16. Turner, A. H.; Makovicky, P. J.; Norell, M. A. (2012). "A Review of Dromaeosaurid Systematics and Paravian Phylogeny" (http://digitallibrary.amnh.org/bitstream/2246/6352/5/B371-cover.pdf) (PDF). *Bulletin of the American Museum of Natural History*. **371**: 1–206. doi:10.1206/748.1 (https://doi.org/10.1206%2F748.1). S2CID 83572446 (https://api.semanticscholar.org/CorpusID:83572446).
17. Lefèvre, Ulysse; Cau, Andrea; Cincotta, Aude; Hu, Dongyu; Chinsamy, Anusuya; Escuillié, François; Godefroit, Pascal (2017). "A new Jurassic theropod from China documents a transitional step in the macrostructure of feathers". *The Science of Nature*. **104** (9–10): 74. Bibcode:2017SciNa.104...74L (https://ui.adsabs.harvard.edu/abs/2017SciNa.104...74L). doi:10.1007/s00114-017-1496-y (https://doi.org/10.1007%2Fs00114-017-1496-y). PMID 28831510 (https://pubmed.ncbi.nlm.nih.gov/28831510). S2CID 32780661 (https://api.semanticscholar.org/CorpusID:32780661).
18. Tsuihiji, T.; Barsbold, R.; Watabe, M.; Tsogtbaatar, K.; Chinzorig, T.; Fujiyama, Y.; Suzuki, S. (2014). "An exquisitely preserved troodontid theropod with new information on the palatal structure from the Upper Cretaceous of Mongolia". *Naturwissenschaften*. **101** (2): 131–142. Bibcode:2014NW....101..131T (https://ui.adsabs.harvard.edu/abs/2014NW....101..131T). doi:10.1007/s00114-014-1143-9 (https://doi.org/10.1007%2Fs00114-014-1143-9). PMID 24441791 (https://pubmed.ncbi.nlm.nih.gov/24441791). S2CID 13920021 (https://api.semanticscholar.org/CorpusID:13920021).
19. Gao, C.; Morschhauser, E. M.; Varricchio, D. J.; Liu, J.; Zhao, B. (2012). Farke, Andrew A (ed.). "A Second Soundly Sleeping Dragon: New Anatomical Details of the Chinese Troodontid *Mei long* with Implications for Phylogeny and Taphonomy" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3459897). *PLOS ONE*. **7** (9) e45203. Bibcode:2012PLoSO...745203G (https://ui.adsabs.harvard.edu/abs/2012PLoSO...745203G). doi:10.1371/journal.pone.0045203 (https://doi.org/10.1371%2Fjournal.pone.0045203). PMC 3459897 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3459897). PMID 23028847 (https://pubmed.ncbi.nlm.nih.gov/23028847).

20. Cai-zhi Shen; Bo Zhao; Chun-ling Gao; Jun-chang Lü; Martin Kundrát (2017). "A New Troodontid Dinosaur (*Liaoningvenator curriei* gen. et sp. nov.) from the Early Cretaceous Yixian Formation in Western Liaoning Province". *Acta Geoscientica Sinica*. **38** (3): 359–371. doi:10.3975/cagsb.2017.03.06 (https://doi.org/10.3975%2Fcagsb.2017.03.06).
21. Brusatte, Stephen Louis; Lloyd, G. T.; Wang, S. C.; Norell, Mark A. (2014). "Gradual Assembly of Avian Body Plan Culminated in Rapid Rates of Evolution across the Dinosaur-Bird Transition" (https://doi.org/10.1016%2Fj.cub.2014.08.034). *Current Biology*. **24** (20): 2386–92. Bibcode:2014CBio...24.2386B (https://ui.adsabs.harvard.edu/abs/2014CBio...24.2386B). doi:10.1016/j.cub.2014.08.034 (https://doi.org/10.1016%2Fj.cub.2014.08.034). PMID 25264248 (https://pubmed.ncbi.nlm.nih.gov/25264248). S2CID 8879023 (https://api.semanticscholar.org/CorpusID:8879023).
22. Caizhi Shen; Junchang Lü; Sizhao Liu; Martin Kundrát; Stephen L. Brusatte; Hailong Gao (2017). "A new troodontid dinosaur from the Lower Cretaceous Yixian Formation of Liaoning Province, China" (http://www.geojournals.cn/dzxben/ch/reader/view_abstract.aspx?file_no=2017endzxb03001&flag=1). *Acta Geologica Sinica (English Edition)*. **91** (3): 763–780. Bibcode:2017AcGlS..91..763S (https://ui.adsabs.harvard.edu/abs/2017AcGlS..91..763S). doi:10.1111/1755-6724.13307 (https://doi.org/10.1111%2F1755-6724.13307). hdl:20.500.11820/dc010682-fce0-4db4-bef6-7b2b29f5be8a (https://hdl.handle.net/20.500.11820%2Fdc010682-fce0-4db4-bef6-7b2b29f5be8a). S2CID 129939153 (https://api.semanticscholar.org/CorpusID:129939153).
23. van der Reest, A.J.; Currie, P.J. (2017). "Troodontids (Theropoda) from the Dinosaur Park Formation, Alberta, with a description of a unique new taxon: implications for deinonychosaur diversity in North America". *Canadian Journal of Earth Sciences*. **54** (9): 919–935. Bibcode:2017CaJES..54..919V (https://ui.adsabs.harvard.edu/abs/2017CaJES..54..919V). doi:10.1139/cjes-2017-0031 (https://doi.org/10.1139%2Fcjes-2017-0031). hdl:1807/78296 (https://hdl.handle.net/1807%2F78296).
24. Xu; Norell (2004). "A new troodontid dinosaur from China with avian-like sleeping posture" (http://doc.rero.ch/record/15284/files/PAL_E2583.pdf) (PDF). *Nature*. **431** (7010): 838–841. Bibcode:2004Natur.431..838X (https://ui.adsabs.harvard.edu/abs/2004Natur.431..838X). doi:10.1038/nature02898 (https://doi.org/10.1038%2Fnature02898). PMID 15483610 (https://pubmed.ncbi.nlm.nih.gov/15483610). S2CID 4362745 (https://api.semanticscholar.org/CorpusID:4362745).
25. Bever, G.S.; Norell, M.A. (2009). "The perinate skull of *Byronosaurus* (Troodontidae) with observations on the cranial ontogeny of paravian theropods" (https://www.biodiversitylibrary.org/item/280787). *American Museum Novitates* (3657): 51. doi:10.1206/650.1 (https://doi.org/10.1206%2F650.1).
26. Norell, Mark A.; Clark, James M.; Dashzeveg, Demberelyin; Barsbold, Rhinchen; Chiappe, Luis M.; Davidson, Amy R.; McKenna, Malcolm C.; Perle, Altangerel; Novacek, Michael J. (November 4, 1994). "A theropod dinosaur embryo and the affinities of the Flaming Cliffs dinosaur eggs". *Science*. **266** (5186): 779–782. Bibcode:1994Sci...266..779N (https://ui.adsabs.harvard.edu/abs/1994Sci...266..779N). doi:10.1126/science.266.5186.779 (https://doi.org/10.1126%2Fscience.266.5186.779). PMID 17730398 (https://pubmed.ncbi.nlm.nih.gov/17730398). S2CID 22333224 (https://api.semanticscholar.org/CorpusID:22333224).
27. "Dinosaurs' tooth wear sheds light on their predatory lives" (https://www.sciencedaily.com/releases/2018/04/180426130043.htm). *ScienceDaily*. Retrieved 4 February 2019.
28. Guarino, Ben (27 April 2018). "Scratches on dinosaur teeth reveal their fierce, efficient eating habits - The Washington Post" (https://www.washingtonpost.com/news/speaking-of-science/wp/2018/04/27/scratches-on-dinosaur-teeth-reveal-their-fierce-efficient-eating-habits/). *The Washington Post*.

29. Cullen, Thomas M.; Cousens, Brian L. (15 November 2023). "New biogeochemical insights into Mesozoic terrestrial paleoecology and evidence for omnivory in troodontid dinosaurs" (https://pubs.geoscienceworld.org/gsabulletin/article/136/7-8/2689/629495/New-biogeochemical-insights-into-Mesozoic). *Geological Society of America Bulletin*. **136** (7–8): 2689–2701. doi:10.1130/B37077.1 (https://doi.org/10.1130%2FB37077.1). ISSN 0016-7606 (https://search.worldcat.org/issn/0016-7606). Retrieved 21 February 2025 – via GeoScienceWorld.
30. Pei, R.; Pittman, M.; Goloboff, P.A.; Dececchi, T.A.; Habib, M.B.; Kaye, T.G.; Larsson, H.C.E.; Norell, M.A.; Brusatte, S.L.; Xu, X. (18 April 2020). "Powered flight potential approached by wide range of close avian relatives but achieved selectively". bioRxiv 10.1101/2020.04.17.046169 (https://doi.org/10.1101%2F2020.04.17.046169).
31. Hu, D.; Hou, L.; Zhang, L.; Xu, X. (2009). "A pre-*Archaeopteryx* troodontid theropod from China with long feathers on the metatarsus". *Nature*. **461** (7264): 640–643. Bibcode:2009Natur.461..640H (https://ui.adsabs.harvard.edu/abs/2009Natur.461..640H). doi:10.1038/nature08322 (https://doi.org/10.1038%2Fnature08322). PMID 19794491 (https://pubmed.ncbi.nlm.nih.gov/19794491). S2CID 205218015 (https://api.semanticscholar.org/CorpusID:205218015).
32. Garros, Christiana W.; Powers, Mark J.; Dyer, Aaron D.; Currie, Philip John (19 April 2025). "Osteohistological analysis of metatarsals reveals new information on pathology and life history of troodontids from the Campanian Dinosaur Park Formation, Alberta, Canada" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12397072). *Journal of Anatomy*. **247** (3–4): 736–764. doi:10.1111/joa.14262 (https://doi.org/10.1111%2Fjoa.14262). ISSN 0021-8782 (https://search.worldcat.org/issn/0021-8782). PMC 12397072 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12397072). PMID 40252006 (https://pubmed.ncbi.nlm.nih.gov/40252006).

External links

Retrieved from "https://en.wikipedia.org/w/index.php?title=Troodontidae&oldid=1378431387"

WIKIPEDIA
25 years of the free encyclopedia

Sauropoda

Sauropoda (/sɔːˈrɒpədə/), whose members are called **sauropods** (/ˈsɔːrəpɒdz/;[1][2] from *sauro-* + *-pod*; lit. 'lizard-footed'), is a clade of saurischian ('lizard-hipped') dinosaurs. Sauropods had very long necks, long tails, small heads (relative to the rest of their body), and four thick, pillar-like legs. They are notable for the enormous sizes attained by some species, and the group includes the largest animals to have ever lived on land. Well-known genera include *Alamosaurus*, *Apatosaurus*, *Argentinosaurus*, *Brachiosaurus*, *Brontosaurus*, *Camarasaurus*, *Diplodocus*, *Dreadnoughtus*, and *Mamenchisaurus*.[3][4]

The oldest known unequivocal sauropod dinosaurs are known from the Early Jurassic.[5] *Isanosaurus* and *Antetonitrus* were originally described as Triassic sauropods,[6][7] but their age, and in the case of *Antetonitrus* also its sauropod status, were subsequently questioned.[8][5][9] Sauropod-like sauropodomorph tracks from the Fleming Fjord Formation (Greenland) might, however, indicate the occurrence of the group in the Late Triassic.[5] By the Late Jurassic (150 million years ago), sauropods had become widespread (especially the diplodocids and brachiosaurids). By the Late Cretaceous, one group of sauropods, the titanosaurs, had replaced all others and had a near-global distribution. However, as with all other non-avian dinosaurs alive at the time, the titanosaurs died out in the Cretaceous–Paleogene extinction event. Fossilised remains of sauropods have been found on every continent, including Antarctica.[10][11][12][13]

The name Sauropoda was coined by Othniel Charles Marsh in 1878, and is derived from Ancient Greek, meaning "lizard foot".[14]

Sauropods

Temporal range:

Early Jurassic – Late Cretaceous (Sinemurian – Maastrichtian, (probable Late Triassic record)

Six sauropods (top left to bottom right): *Patagotitan*, *Barosaurus*, *Giraffatitan*, *Omeisaurus*, *Shunosaurus*, and *Amargasaurus*

Scientific classification

Kingdom:	Animalia
Phylum:	Chordata
Class:	Reptilia
Clade:	Dinosauria
Clade:	Saurischia

Sauropods are one of the most recognizable groups of dinosaurs, and have become a fixture in popular culture due to their enormousness.

Complete sauropod fossil finds are extremely rare. Many species, especially the largest, are known only from isolated and disarticulated bones. Many near-complete specimens lack heads, tail tips and limbs.

Description

Sauropods were herbivorous (plant-eating), usually quite long-necked[15] quadrupeds (four-legged), often with spatulate (spatula-shaped: broad at the tip, narrow at the neck) teeth. They had relatively tiny heads, massive bodies, and most had long tails. Their hind legs were thick, straight, and powerful, ending in club-like feet with five toes, though only the inner three (or in some cases four) bore claws. Their forelimbs were rather more slender and typically ended in pillar-like hands built for supporting weight; often only the thumb bore a claw. Many illustrations of sauropods in the flesh miss these facts, inaccurately depicting sauropods with hooves capping the claw-less digits of the feet, or more than three claws or hooves on the hands. The proximal caudal vertebrae are extremely diagnostic for sauropods.[16]

Size

The sauropods' most defining characteristic was their size. Even the dwarf sauropods (perhaps 5 to 6 metres (16 to 20 ft) long) were counted among the largest animals in their ecosystem. Their only real competitors in terms of size are the rorquals, such as the blue whale. But, unlike whales, sauropods were primarily terrestrial animals.

Their body structure did not vary as much as other dinosaurs, perhaps due to size constraints, but they displayed ample variety. Some, like the diplodocids, possessed tremendously long tails,

Clade:	†Sauropodomorpha
Clade:	†Sauropodiformes
Clade:	†Sauropoda Marsh, 1878

Subgroups

- †*Amygdalodon*
- †*Blikanasaurus*
- †*Chinshakiangosaurus*
- †*Gongxianosaurus*
- †*Isanosaurus*?
- †*Kotasaurus*
- †*Pulanesaura*?
- †*Volkheimeria*
- †**Lessemsauridae**?
- †**Gravisauria** Allain & Aquesbi, 2008
  - †*Barapasaurus*
  - †*Ohmdenosaurus*
  - †*Rhoetosaurus*
  - †*Spinophorosaurus*
  - †*Tazoudasaurus*
  - †*Vulcanodon*
  - †*Zizhongosaurus*
  - †**Eusauropoda**

Sauropoda *incertae sedis*

- †*Algoasaurus*
- †*Archaeodontosaurus*
- †*Asiatosaurus*
- †*Bothriospondylus*
- †*Ferganasaurus*
- †*Gigantosaurus*
- †*Protognathosaurus*
- †*Qinlingosaurus*
- †*Sanpasaurus*
- †*Ultrasaurus*

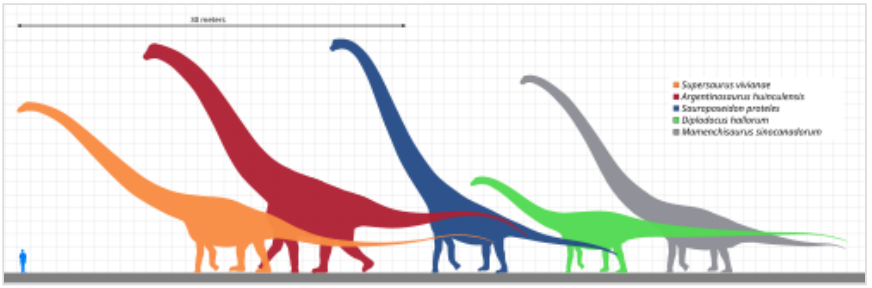
which they may have been able to crack like a whip as a signal or to deter or injure predators,[17] or to make sonic booms.[18][19] *Supersaurus*, at 33 to 34 metres (108 to 112 ft) long,[20] was the longest sauropod known from reasonably complete remains, but others, like the old record holder, *Diplodocus*, were also extremely long. The holotype (and now lost) vertebra of *Amphicoelias fragillimus* (now *Maraapunisaurus*) may have come from an animal 58 metres (190 ft) long;[21] its vertebral column would have been substantially longer than that of the blue whale. However, research published in 2015 speculated that the size estimates of *A. fragillimus* may have been highly exaggerated.[22] The longest dinosaur known from reasonable fossils material is probably *Argentinosaurus huinculensis* with length estimates of 35 to 36 metres (115 to 118 ft).[23][24]

Possible sauropods

- †*Schleitheimia*
- †*Tuebingosaurus*
- †*Yizhousaurus*

Synonyms

- **Opisthocoelia** Owen, 1860
- **Cetiosauria** Seeley, 1870
- **Diplodocia** Tornier, 1913



Size comparison of selected giant sauropod dinosaurs

The heaviest known sauropods include *Argentinosaurus* (65 to 80 metric tons (72 to 88 short tons)[25][23][24]), *Mamenchisaurus sinocanadorum* (60 to 80 metric tons or 66 to 88 short tons[24]), and *Patagotitan* with *Puertasaurus* (50 to 55 metric tons (49 to 54 long tons; 55 to 61 short tons)[23][24]). Additional finds indicate a number of species likely reached or exceeded weights of 40 tons.[26] The largest land animal alive today, the bush elephant, weighs no more than 10.4 metric tons (11.5 short tons).[27]

Among the smallest sauropods were the primitive *Ohmdenosaurus* (4 metres or 13 feet long), the dwarf titanosaur *Magyarosaurus* (6 metres or 20 feet long), and the dwarf brachiosaurid *Europasaurus*, which was 6.2 meters (20 ft) long as a fully-grown adult.[28] Its small stature was probably the result of insular dwarfism occurring in a population of sauropods isolated on an island of the late Jurassic in what is now the Langenberg area of northern Germany.[29][30] The diplodocoid sauropod *Brachytrachelopan* was the shortest member of its group because of its unusually short neck. Unlike other sauropods, whose necks could grow to up to four times the length of their backs, the neck of *Brachytrachelopan* was shorter than its backbone.

Limbs and feet

As massive quadrupeds, sauropods developed specialized "graviportal" (weight-bearing) limbs. The hind feet were broad, and retained three claws in most species.[31] Particularly unusual compared with other animals were the highly modified front feet (*manus*). The front feet of sauropods were very dissimilar from those of modern large quadrupeds, such as elephants. Rather than splaying out to the sides to create a wide foot as in elephants, the manus bones of sauropods were arranged in fully vertical columns, with

Hand (left) and hindfoot (right) of a titanosaur, showing typical anatomy of derived sauropods

extremely reduced finger bones (though it is not clear if the most primitive sauropods, such as *Vulcanodon* and *Barapasaurus*, had such forefeet).[32] The front feet were so modified in eusauropods that individual digits would not have been visible in life.

The arrangement of the forefoot bone (*metacarpal*) columns in eusauropods was semi-circular, so sauropod forefoot prints are horseshoe-shaped. Unlike elephants, print evidence shows that sauropods lacked any fleshy padding to back the front feet, making them concave.[32] The only claw visible in most sauropods was the distinctive thumb claw (associated with digit I). Almost all sauropods had such a claw, though what purpose it served is unknown. The claw was largest (as well as tall and laterally flattened) in diplodocids, and very small in brachiosaurids, some of which seem to have lost the claw entirely based on trackway evidence.[33] Titanosaurs may have lost the thumb claw completely (with the exception of early forms, such as *Janenschia*).

Titanosaurs were most unusual among sauropods, as, across their history as a clade, they lost not just the external claw but also completely lost the digits of the front foot. Advanced titanosaurs had no digits or digit bones, and walked only on horseshoe-shaped "stumps" made up of the columnar metacarpal bones.[34]

Print evidence from Portugal shows that, in at least some sauropods (probably brachiosaurids), the bottom and sides of the forefoot column was likely covered in small, spiny scales, which left score marks in the prints.[35] In titanosaurs, the ends of the metacarpal bones that contacted the ground were unusually broad and squared-off, and some specimens preserve the remains of soft tissue covering this area, suggesting that the front feet were rimmed with some kind of padding in these species.[34]

Matthew Bonnan[36][37] has shown that sauropod dinosaur long bones grew isometrically: that is, there was little to no change in shape as juvenile sauropods became gigantic adults. Bonnan suggested that this odd scaling pattern (most vertebrates show significant shape changes in long bones associated with increasing weight support) might be related to a stilt-walker principle (suggested by amateur scientist Jim Schmidt) in which the long legs of adult sauropods allowed them to easily cover great distances without changing their overall mechanics.

Air sacs

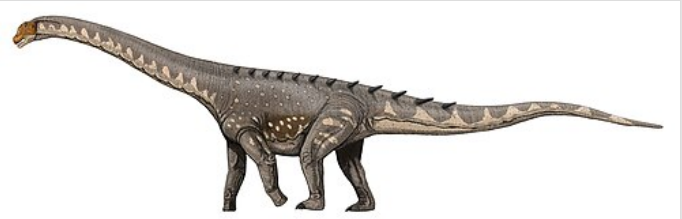
Along with other saurischian dinosaurs (such as theropods, including birds), sauropods had a system of air sacs, evidenced by indentations and hollow cavities in most of their vertebrae that had been invaded by them. Pneumatic, hollow bones are a characteristic feature of all sauropods.[38] These air spaces

reduced the overall weight of the massive necks that the sauropods had, and the air-sac system in general, allowing for a single-direction airflow through stiff lungs, made it possible for the sauropods to get enough oxygen.[39] This adaptation would have advantaged sauropods particularly in the relatively low oxygen conditions of the Jurassic and Early Cretaceous.[40]

The bird-like hollowing of sauropod bones was recognized early in the study of these animals, and, in fact, at least one sauropod specimen found in the 19th century (*Ornithopsis*) was originally misidentified as a pterosaur (a flying reptile) because of this.[41]

Armor

Some sauropods had armor. There were genera with small clubs on their tails, a prominent example being *Shunosaurus*, and several titanosaurs, such as *Saltasaurus* and *Ampelosaurus*, had small bony osteoderms covering portions of their bodies.



Reconstruction of *Ampelosaurus*

Teeth

A study by Michael D'Emic and his colleagues from Stony Brook University found that sauropods evolved high tooth replacement rates to keep up with their large appetites. The study suggested that *Nigersaurus*, for example, replaced each tooth every 14 days, *Camarasaurus* replaced each tooth every 62 days, and *Diplodocus* replaced each tooth once every 35 days.[42] The scientists found qualities of the tooth affected how long it took for a new tooth to grow. *Camarasaurus*'s teeth took longer to grow than those for *Diplodocus* because they were larger.[43]

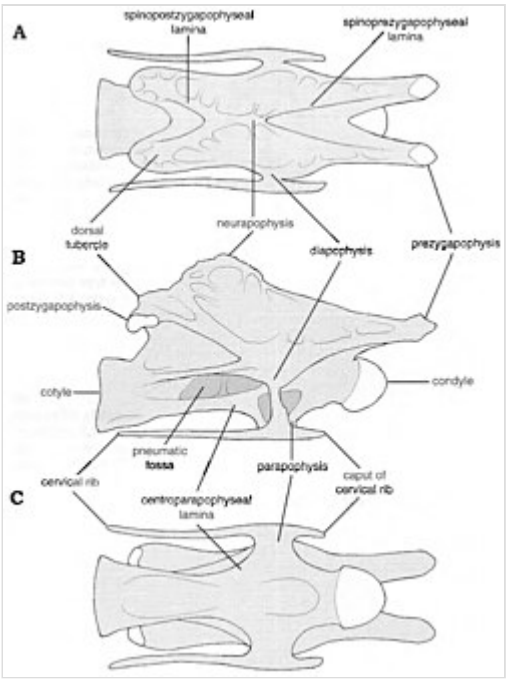
It was also noted by D'Emic and his team that the differences between the teeth of the sauropods also indicated a difference in diet. *Diplodocus* ate plants low to the ground and *Camarasaurus* browsed leaves from top and middle branches. According to the scientists, the specializing of their diets helped the different herbivorous dinosaurs to coexist.[42][43]

Necks

Sauropod necks have been found at over 15 metres (49 ft) in length, a full six times longer than the world record giraffe neck.[39] Enabling this were a number of essential physiological features. The dinosaurs' overall large body size and quadrupedal stance provided a stable base to support the neck, and the head was evolved to be very small and light, losing the ability to orally process food. By reducing their heads to simple harvesting tools that got the plants into the body, the sauropods needed less power to lift their heads, and thus were able to develop necks with less dense muscle and connective tissue. This drastically reduced the overall mass of the neck, enabling further elongation.

Sauropods also had a great number of adaptations in their skeletal structure. Some sauropods had as many as 19 cervical vertebrae, whereas almost all mammals are limited to only seven. Additionally, each vertebra was extremely long and had a number of empty spaces in them which would have been filled only with air. An air-sac system connected to the spaces not only lightened the long necks, but effectively increased the airflow through the trachea, helping the creatures to breathe in enough air. By evolving

vertebrae consisting of 60% air, the sauropods were able to minimize the amount of dense, heavy bone without sacrificing the ability to take sufficiently large breaths to fuel the entire body with oxygen.[39] According to Kent Stevens, computer-modeled reconstructions of the skeletons made from the vertebrae indicate that sauropod necks were capable of sweeping out large feeding areas without needing to move their bodies, but were unable to be retracted to a position much above the shoulders for exploring the area or reaching higher.[44]



Stylized cervical vertebra of a sauropod illustrating terminology

Another proposed function of the sauropods' long necks was essentially a radiator to deal with the extreme amount of heat produced from their large body mass. Considering that the metabolism would have been doing an immense amount of work, it would certainly have generated a large amount of heat as well, and elimination of this excess heat would have been essential for survival.[45] It has also been proposed that the long necks would have cooled the veins and arteries going to the brain, avoiding excessively heated blood from reaching the head. It was in fact found that the increase in metabolic rate resulting from the sauropods' necks was slightly more than compensated for by the extra surface area from which heat could dissipate.[46]

Palaeobiology

Ecology

Dental microwear texture analysis (DMTA) performed on a titanosauriform sauropod from the Turonian-aged Tamagawa Formation suggests that the sauropod fed on plant material that was softer than insect exoskeletons or mollusc shells, with the diet likely consisting of ferns and gymnosperms. The DMTA results also suggested that sauropods likely masticated more energetically than present-day lepidosaurs do.[47]

When sauropods were first discovered, their immense size led many scientists to compare them with modern-day whales. Most studies in the 19th and early 20th centuries concluded that sauropods were too large to have supported their weight on land, and therefore that they must have been mainly aquatic. Most life restorations of sauropods in art through the first three quarters of the 20th century depicted them fully or partially immersed in water.[48] This early notion was cast in doubt beginning in the 1950s, when a study by Kermack (1951) demonstrated that, if the animal were submerged in several metres of water, the pressure would be enough to fatally collapse the lungs and airway.[49] However, this and other early studies of sauropod ecology were flawed in that they ignored a substantial body of evidence that the bodies of sauropods were heavily permeated with air sacs. In 1878, paleontologist E.D. Cope had even referred to these structures as "floats".

Beginning in the 1970s, the effects of sauropod air sacs on their supposed aquatic lifestyle began to be explored. Paleontologists such as Coombs and Bakker used this, as well as evidence from sedimentology and biomechanics, to show that sauropods were primarily terrestrial animals. In 2004, D.M. Henderson noted that, due to their extensive system of air sacs, sauropods would have been buoyant and would not have been able to submerge their torsos completely below the surface of the water; in other words, they would float, and would not have been in danger of lung collapse due to water pressure when swimming.[48]

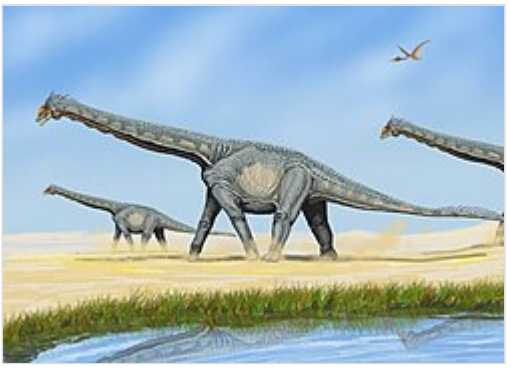
Evidence for swimming in sauropods comes from fossil trackways that have occasionally been found to preserve only the forefeet (manus) impressions. Henderson showed that such trackways can be explained by sauropods with long forelimbs (such as macronarians) floating in relatively shallow water deep enough to keep the shorter hind legs free of the bottom, and using the front limbs to punt forward.[48] However, due to their body proportions, floating sauropods would also have been very unstable and maladapted for extended periods in the water. This mode of aquatic locomotion, combined with its instability, led Henderson to refer to sauropods in water as "tipsy punters".[48]

While sauropods could therefore not have been aquatic as historically depicted, there is evidence that they preferred wet and coastal habitats. Sauropod footprints are commonly found following coastlines or crossing floodplains, and sauropod fossils are often found in wet environments or intermingled with fossils of marine organisms.[48] A good example of this would be the massive Jurassic sauropod trackways found in lagoon deposits on Scotland's Isle of Skye.[50] Studies published in 2021 suggest sauropods could not inhabit polar regions. This study suggests they were largely confined to tropical areas and had metabolisms that were very different to those of other dinosaurs, perhaps intermediate between mammals and reptiles.[51] New studies published by Taia Wyenberg-henzler in 2022 suggest that sauropods in North America declined due to undetermined reasons in regards to their niches and distribution during the end of the Jurassic and into the latest Cretaceous. Why this is remains unclear, but some similarities in feeding niches between iguanodontians, hadrosauroids, and sauropods have been suggested and may have resulted in some competition. However, this cannot fully explain the full decline in distribution of sauropods, as competitive exclusion would have resulted in a much more rapid decline than what is shown in the fossil record. Moreover, it must be determined as to whether sauropod declines in North America was the result of a change in preferred flora that sauropods ate, climate, or other factors. It is also suggested in this same study that iguanodontians and hadrosauroids took advantage of recently vacated niches left by a decline in sauropod diversity during the late Jurassic and the Cretaceous in North America.[52]

Herding and parental care

Many lines of fossil evidence, from both bone beds and trackways, indicate that sauropods were gregarious animals that formed herds. However, the makeup of the herds varied between species. Some bone beds, for example a site from the Middle Jurassic of Argentina, appear to show herds made up of individuals of various age groups, mixing juveniles and adults. However, a number of other fossil sites and trackways indicate that many sauropod species travelled in herds segregated by age, with juveniles forming herds separate from adults. Such segregated herding strategies have been found in species such as *Alamosaurus*, *Bellusaurus* and some diplodocids.[53]

In a review of the evidence for various herd types, Myers and Fiorillo attempted to explain why sauropods appear to have often formed segregated herds. Studies of microscopic tooth wear show that juvenile sauropods had diets that differed from their adult counterparts, so herding together would not have been as productive as herding separately, where individual herd members could forage in a coordinated way. The vast size difference between juveniles and adults may also have played a part in the different feeding and herding strategies.[53]

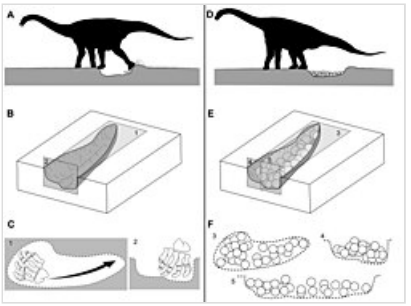


Some sauropods, such as *Alamosaurus sanjuanensis*, formed herds segregated by age.

Cast of Toni, a juvenile *Brachiosaurus* (restored as a diplodocid)

Since the segregation of juveniles and adults must have taken place soon after hatching, and combined with the fact that sauropod hatchlings were most likely precocial, Myers and Fiorillo concluded that species with age-segregated herds would not have exhibited much parental care.[53] On the other hand, scientists who have studied age-mixed sauropod herds suggested that these species may have cared for their young for an extended period before reaching adulthood.[54] A 2014 study suggested that the time from laying the egg to the time of the hatching was likely to have been between 65 and 82 days.[55] Exactly how segregated versus age-mixed herding varied across different groups of sauropods is unknown. Further examples of gregarious behavior will need to be discovered from more sauropod species to detect possible distribution patterns.[53]

Multiple nesting sites discovered in Argentina and India contain 30-400 clutches of fossilized eggs that were found preserved, providing evidence of sauropod maternal care. Researchers suggest that sauropods might have settled in nesting grounds close to volcanic activity for geothermal incubation, in which the mothers keep their eggs warm. This behavior is similar to modern birds and reptiles who follow the same method.[56][57][58]



The illustration of titanosaurs nesting their egg

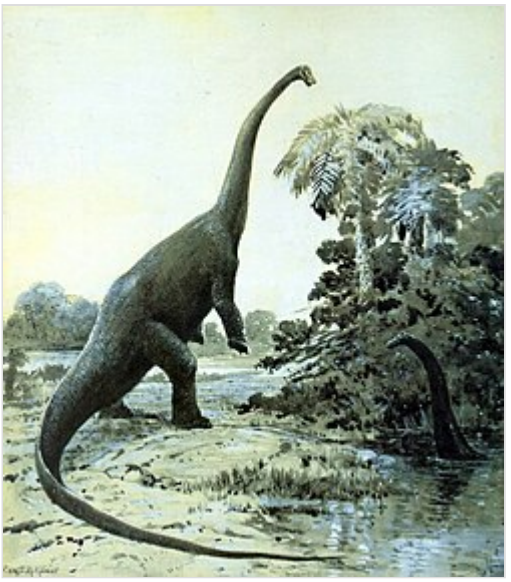
Rearing stance

Since early in the history of their study, scientists, such as Osborn, have speculated that sauropods could rear up on their hind legs, using the tail as the third 'leg' of a tripod.[59] A skeletal mount depicting the diplodocid *Barosaurus lentus* rearing up on its hind legs at the American Museum of Natural History is one illustration of this hypothesis. In a 2005 paper, Rothschild and Molnar reasoned that if sauropods had adopted a bipedal posture at times, there would be evidence of stress fractures in the forelimb 'hands'. However, none were found after they examined a large number of sauropod skeletons.[60]

Heinrich Mallison (in 2009) was the first to study the physical potential for various sauropods to rear into a tripodal stance. Mallison found that some characters previously linked to rearing adaptations were actually unrelated (such as the wide-set hip bones of titanosaurs) or would have hindered rearing. For

Mounted skeleton of *Barosaurus lentus*, depicted in a rearing tripodal stance

example, titanosaurs had an unusually flexible backbone, which would have decreased stability in a tripodal posture and would have put more strain on the muscles. Likewise, it is unlikely that brachiosaurids could rear up onto the hind legs, as their center of gravity was much farther forward than other sauropods, which would cause such a stance to be unstable.[61]



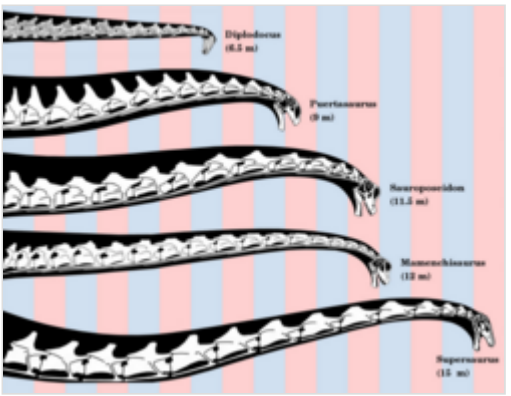
Restoration of a rearing *Diplodocus* by Charles R. Knight, 1911

Diplodocids, on the other hand, appear to have been well adapted for rearing up into a tripodal stance. Diplodocids had a center of mass directly over the hips, giving them greater balance on two legs. Diplodocids also had the most mobile necks of sauropods, a well-muscled pelvic girdle, and tail vertebrae with a specialised shape that would allow the tail to bear weight at the point it touched the ground. Mallison concluded that diplodocids were better adapted to rearing than elephants, which do so occasionally in the wild. He also argues that stress fractures in the wild do not occur from everyday behaviour,[61] such as feeding-related activities (contra Rothschild and Molnar).[60]

Head and neck posture

There is little agreement over how sauropods held their heads and necks, and the postures they could achieve in life.

Whether sauropods' long necks could be used for browsing high trees has been questioned based on calculations suggesting that just pumping blood up to the head in such a posture[62] for long would have used some half of its energy intake.[63] Further, to move blood to such a height—dismissing posited auxiliary hearts in the neck[64]—would require a heart 15 times as large as of a similar-sized whale.[65]



Reconstruction of selected sauropod necks, showing posture and length

The above have been used to argue that the long neck must instead have been held more or less horizontally, presumed to enable feeding on plants over a wide area with less need to move about, yielding a large energy saving for such a large animal. Reconstructions of the necks of *Diplodocus* and *Apatosaurus* have therefore often portrayed them in near-horizontal, so-called "neutral, undeflected posture".[66]

However, research on living animals demonstrates that almost all extant tetrapods hold the base of their necks sharply flexed when alert, showing that any inference from bones about habitual "neutral postures"[66] is deeply unreliable.[67][68] Meanwhile, computer modeling of ostrich necks has raised doubts over the flexibility needed for stationary grazing.[69][70][71]

Trackways and locomotion

Sauropod footprints

Sauropod tracks near Rovereto, Italy

Sauropod trackways and other fossil footprints (known as "ichnites") are known from abundant evidence present on most continents. Ichnites have helped support other biological hypotheses about sauropods, including general fore and hind foot anatomy (see *Limbs and feet* above). Generally, prints from the forefeet are much smaller than the hind feet, and often crescent-shaped. Occasionally ichnites preserve traces of the claws, and help confirm which sauropod groups lost claws or even digits on their forefeet.[72]

Sauropod tracks at Serras de Aire e Candeeiros Natural Park, Portugal

Sauropod tracks from the Villar del Arzobispo Formation of early Berriasian age in Spain support the gregarious behaviour of the group. The tracks are possibly more similar to *Sauropodichnus giganteus* than any other ichnogenera, although they have been suggested to be from a basal titanosauriform. The tracks are wide-gauge, and the grouping as close to *Sauropodichnus* is also supported by the manus-to-pes distance, the morphology of the manus being kidney bean-shaped, and the morphology of the pes being subtriangular. It cannot be identified whether the footprints of the herd were caused by juveniles or adults, because of the lack of previous trackway individual age identification.[73]

A sauropod trackway

Generally, sauropod trackways are divided into three categories based on the distance between opposite limbs: narrow gauge, medium gauge, and wide gauge. The gauge of the trackway can help determine how wide-set the limbs of various sauropods were and how this may have impacted the way they walked.[72] A 2004 study by Day and colleagues found that a general pattern could be found among groups of advanced sauropods, with each sauropod family being characterised by certain trackway gauges. They found that most sauropods other than titanosaurs had narrow-gauge limbs, with strong impressions of the large thumb claw on the forefeet. Medium gauge trackways with claw impressions on the forefeet probably belong to brachiosaurids and other primitive titanosauriformes, which were evolving wider-set limbs but retained their claws. Primitive true titanosaurs also

retained their forefoot claw but had evolved fully wide gauge limbs. Wide gauge limbs were retained by advanced titanosaurs, trackways from which show a wide gauge and lack of any claws or digits on the forefeet.[74]

Occasionally, only trackways from the forefeet are found. Falkingham *et al.*[75] used computer modelling to show that this could be due to the properties of the substrate. These need to be just right to preserve tracks.[76] Differences in hind limb and fore limb surface area, and therefore contact pressure with the substrate, may sometimes lead to only the forefeet trackways being preserved.

Biomechanics and speed

Reconstructed skeleton used to estimate speed, Museo Municipal Carmen Funes, Plaza Huincul, Argentina

In a study published in PLoS ONE on October 30, 2013, by Bill Sellers, Rodolfo Coria, Lee Margetts *et al.*, *Argentinosaurus* was digitally reconstructed to test its locomotion for the first time. Before the study, the most common way of estimating speed was through studying bone histology and ichnology. Commonly, studies about sauropod bone histology and speed focus on the postcranial skeleton, which holds many unique features, such as an enlarged process on the ulna, a wide lobe on the ilia, an inward-slanting top third of the femur, and an extremely ovoid femur shaft. Those features are useful when attempting to explain trackway patterns of graviportal animals. When studying ichnology to calculate sauropod speed, there are a few problems, such as only providing estimates for certain gaits because of preservation bias, and being subject to many more accuracy problems.[77]

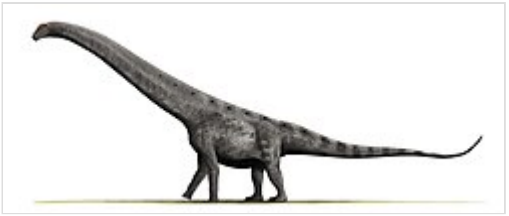
Most likely walking gait of *Argentinosaurus*

To estimate the gait and speed of *Argentinosaurus*, the study performed a musculoskeletal analysis. The only previous musculoskeletal analyses were conducted on hominoids, terror birds, and other dinosaurs. Before they could conduct the analysis, the team had to create a digital skeleton of the animal in question, show where there would be muscle layering, locate the muscles and joints, and finally find the muscle properties before finding the gait and speed. The results of the biomechanics study revealed that *Argentinosaurus* was mechanically competent at a top speed of 2 m/s (5 mph) given the great weight of the animal and the strain that its joints were capable of bearing.[78] The results further revealed that much larger terrestrial vertebrates might be possible, but would require significant body remodeling and possible sufficient behavioral change to prevent joint collapse.[77]

Body size

Sauropods were gigantic descendants of surprisingly small ancestors. Basal dinosauriformes, such as *Pseudolagosuchus* and *Marasuchus* from the Middle Triassic of Argentina, weighed approximately 1 kg (2.2 lb) or less. These evolved into saurischia, which saw a rapid increase of bauplan size, although more primitive members like *Eoraptor*, *Panphagia*, *Pantydraco*, *Saturnalia* and *Guaibasaurus* still retained a

moderate size, possibly under 10 kg (22 lb). Even with these small, primitive forms, there is a notable size increase among sauropodomorphs, although scanty remains of this period make interpretation conjectural. There is one definite example of a small derived sauropodomorph: *Anchisaurus*, under 50 kg (110 lb), even though it is closer to the sauropods than *Plateosaurus* and *Riojasaurus*, which were upwards of 1 t (0.98 long tons; 1.1 short tons) in weight.[45]



Life restoration of *Argentinosaurus*, one of the largest sauropod dinosaurs

Evolving from sauropodomorphs, the sauropods were huge. Their giant size probably resulted from an increased growth rate made possible by tachymetabolic endothermy, a trait which evolved in sauropodomorphs. Once branched into sauropods, sauropodomorphs continued steadily to grow larger, with smaller sauropods, like the Early Jurassic *Barapasaurus* and *Kotasaurus*, evolving into even larger forms like the Middle Jurassic *Mamenchisaurus* and *Patagosaurus*. Responding to the growth of sauropods, their theropod predators grew also, as shown by an *Allosaurus*-sized coelophysoid from Germany.[45]

Size in Neosauropoda

Neosauropoda is quite plausibly the clade of dinosaurs with the largest body sizes ever to have existed. The few exceptions of smaller size are hypothesized to be caused by island dwarfism, or other ecological pressures, although there is a trend in some Titanosauria towards a smaller size. The titanosaurs, however, were some of the largest sauropods ever. Other than titanosaurs, diplodocoids also reached truly gigantic sizes. Meanwhile, a clade of diplodocoids, called Dicraeosauridae, are identified by a small to medium body size. No sauropods were very small, however, for even "dwarf" sauropods are larger than 500 kg (1,100 lb), a size reached by only about 10% of all mammalian species.[45]

Independent gigantism

Although in general, sauropods were large, a gigantic size (40 t (39 long tons; 44 short tons) or more) was reached independently at multiple times in their evolution. Many gigantic forms existed in the Late Jurassic (specifically Kimmeridgian), such as the turiasaur *Turiasaurus*, the mamenchisaurids *Mamenchisaurus* and *Xinjiangtitan*, the diplodocoids *Diplodocus*, *Apatosaurus*, *Supersaurus* and *Barosaurus*, the camarasaurid *Camarasaurus*, and the brachiosaurids *Brachiosaurus* and *Giraffatitan*. Through the Early to Late Cretaceous, the giants *Sauroposeidon*, *Paralititan*, *Argentinosaurus*, *Puertasaurus*, *Antarctosaurus*, *Dreadnoughtus*, *Notocolossus*, *Futalognkosaurus*, *Patagotitan* and *Alamosaurus* lived, with all possibly being titanosaurs. One sparsely known possible giant is *Huanghetitan ruyangensis*, only known from 3 m (9.8 ft) long ribs. These giant species lived in the Late Jurassic to the Late Cretaceous, appearing independently over a time span of 85 million years.[45]

Dwarfism in sauropods

Two well-known island dwarf species of sauropods are the Cretaceous *Magyarosaurus* (at one point its identity as a dwarf was challenged) and the Jurassic *Europasaurus*, both from Europe. Even though these sauropods are small, the only way to prove they are true dwarfs is through a study of their bone histology. A study by Martin Sander and colleagues in 2006 examined eleven individuals of *Europasaurus holgeri* using bone histology and demonstrated that the small island species evolved through a decrease in the growth rate of long bones as compared to rates of growth in ancestral species on the mainland.[79] Two

other possible dwarfs are *Rapetosaurus*, which existed on the island of Madagascar, an isolated island in the Cretaceous, and *Ampelosaurus*, a titanosaur that lived on the Iberian peninsula of southern Spain and France. *Amanzia* from Switzerland might also be a dwarf, but this has yet to be proven.[45] One of the most extreme cases of island dwarfism is found in *Europasaurus*, a relative of the much larger *Camarasaurus* and *Brachiosaurus*: it was only about 6.2 m (20 ft) long, an identifying trait of the species. As for all dwarf species, their reduced growth rate led to their small size.[28][45] Another taxon of tiny sauropods, the saltasaurid titanosaur *Ibirania*, 5.7 metres (19 feet) long, lived a non-insular context in Upper Cretaceous Brazil, and is an example of nanism resultant from other ecological pressures.[80]

Paleopathology and paleoparasitology

Sauropods are rarely known for preserved injuries or signs of illnesses, but more recent discoveries show they could suffer from such pathologies. A diplodocid specimen from the Morrison Formation referred to as "Dolly" was described in 2022 with evidence of a severe respiratory infection.[81][82] Sauropod ribs from Yunyang County, Chongqing, in southwest China show evidence of rib breakage by way of traumatic fracture, bone infection, and osteosclerosis.[83] A sauropod tibia exhibiting initial fracture has been described from the Middle Jurassic of Yunyang County in southwestern China.[84]

Ibirania, a nanoid titanosaur fossil from Brazil, suggests that individuals of various genera were susceptible to diseases such as osteomyelitis and parasite infestations. The specimen hails from the late cretaceous São José do Rio Preto Formation, Bauru Basin, and was described in the journal *Cretaceous Research* by Aureliano et al. (2021).[85] Examination of the titanosaur's bones revealed what appear to be parasitic blood worms similar to the prehistoric *Paleoleishmania* but are 10-100 times larger, that seemed to have caused the osteomyelitis. The fossil is the first known instance of an aggressive case of osteomyelitis being caused by blood worms in an extinct animal.[86][87][88]

History of discovery

The first scraps of fossil remains now recognized as sauropods all came from England and were originally interpreted in a variety of different ways. Their relationship to other dinosaurs was not recognized until well after their initial discovery.

The first sauropod fossil to be scientifically described was a single tooth known by the non-Linnaean descriptor *Rutellum implicatum*.[89] This fossil was described by Edward Lhuyd in 1699, but was not recognized as a giant prehistoric reptile at the time.[90] Dinosaurs would not be recognized as a group until over a century later.

The first reconstruction of a sauropod, the skeleton of *Camarasaurus supremus*. By John A. Ryder, 1877.

Richard Owen published the first modern scientific descriptions of sauropods in 1841, in a book and a paper naming *Cardiodon* and *Cetiosaurus*. *Cardiodon* was known only from two unusual, heart-shaped teeth (from which it got its name), which could not be identified beyond the fact that they came from a previously unknown large reptile. *Cetiosaurus* was known from slightly better, but still scrappy remains. Owen thought at the time that *Cetiosaurus* was a

giant marine reptile related to modern crocodiles, hence its name, which means "whale lizard". A year later, when Owen coined the name Dinosauria, he did not include *Cetiosaurus* and *Cardiodon* in that group.[91]

Modern reconstructed *Camarasaurus* skeleton

In 1850, Gideon Mantell recognized the dinosaurian nature of several bones assigned to *Cetiosaurus* by Owen. Mantell noticed that the leg bones contained a medullary cavity, a characteristic of land animals. He assigned these specimens to the new genus *Pelorosaurus*, and grouped it together with the dinosaurs. However, Mantell still did not recognize the relationship to *Cetiosaurus*.[41]

The next sauropod find to be described and misidentified as something other than a dinosaur were a set of hip vertebrae described by Harry Seeley in 1870. Seeley found that the vertebrae were very lightly constructed for their size and contained openings for air sacs (*pneumatization*). Such air sacs were at the time known only in birds and pterosaurs, and Seeley considered the vertebrae to come from a pterosaur. He named the new genus *Ornithopsis*, or "bird face" because of this.[41]

When more complete specimens of *Cetiosaurus* were described by Phillips in 1871, he finally recognized the animal as a dinosaur related to *Pelorosaurus*.[92] However, it was not until the description of new, nearly complete sauropod skeletons from the United States (representing *Apatosaurus* and *Camarasaurus*) later that year that a complete picture of sauropods emerged. An approximate reconstruction of a complete sauropod skeleton was produced by artist John A. Ryder, hired by paleontologist E.D. Cope, based on the remains of *Camarasaurus*, though many features were still inaccurate or incomplete according to later finds and biomechanical studies.[93] Also in 1877, Richard Lydekker named another relative of *Cetiosaurus*, *Titanosaurus*, based on an isolated vertebra.[41]

In 1878, the most complete sauropod yet was found and described by Othniel Charles Marsh, who named it *Diplodocus*. With this find, Marsh also created a new group to contain *Diplodocus*, *Cetiosaurus*, and their increasing roster of relatives to differentiate them from the other major groups of dinosaurs. Marsh named this group Sauropoda, or "lizard feet".[41]

Several macronarian sauropods; from left to right, *Camarasaurus*, *Brachiosaurus*, *Giraffatitan*, and *Euhelopus*.

Classification

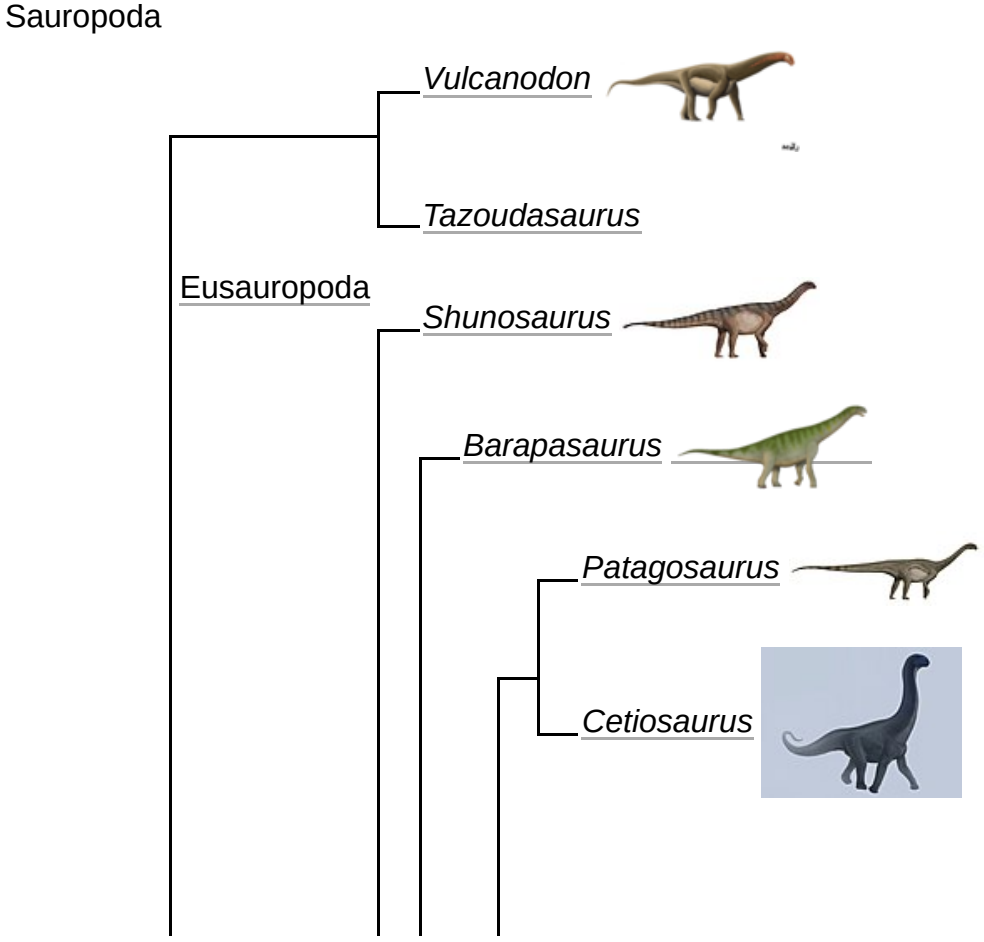
The first phylogenetic definition of Sauropoda was published in 1997 by Salgado and colleagues. They defined the clade as a node-based taxon, containing "the most recent common ancestor of *Vulcanodon karibaensis* and Eusauropoda and all of its descendants".[94] Later, several stem-based definitions were proposed, including one by Yates (2007), who defined Sauropoda as "the most inclusive clade that includes *Saltasaurus loricatus* but not *Melanorosaurus readi*".[95][96]

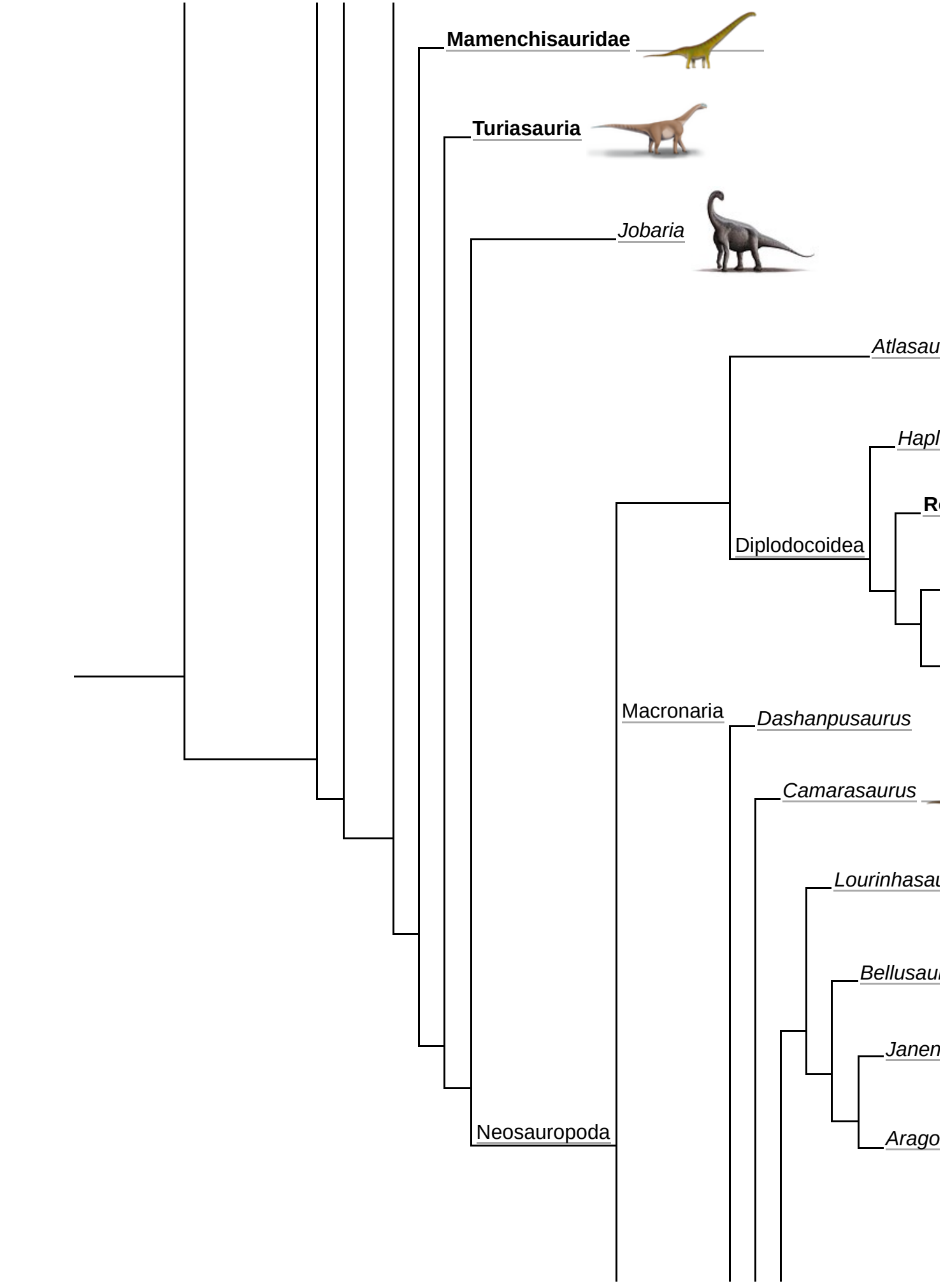
Proponents of this definition also use the clade name **Gravisauria**, defined as the most recent ancestor of *Tazoudasaurus naimi* and *Saltasaurus loricatus* and all of its descendants[97] for the clade equivalent to Sauropoda as defined by Salgado *et al.*[98] The clade Gravisauria was appointed by the French paleontologist Ronan Allain and Moroccan paleontologist Najat Aquesbi in 2008 when a cladistic analysis of the dinosaur found by Allain, *Tazoudasaurus*, as the outcome was that the family Vulcanodontidae. The group includes *Tazoudasaurus* and *Vulcanodon*, and the sister taxon Eusauropoda, but also certain species such as *Antetonitrus*, *Gongxianosaurus* and *Isanosaurus* that do not belong in Vulcanodontidae but to an even more basic position occupied in Sauropoda. It made sense to have Sauropoda compared to this, more derived group that included Vulcanodontidae and Eusauropoda in a definition: defined as the group formed by the last common ancestor of *Tazoudasaurus* and *Saltasaurus* (Bonaparte and Powell, 1980) and all its descendants. Aquesbi mentioned two synapomorphies, shared derived characteristics of Gravisauria: the vertebrae are wider side to side than front to rear and possession of asymmetrical condyles femoris at the bottom of the femur. Those were previously not thought to be Eusauropoda synapomorphies but Allian found these properties also on *Tazoudasaurus*.[99]

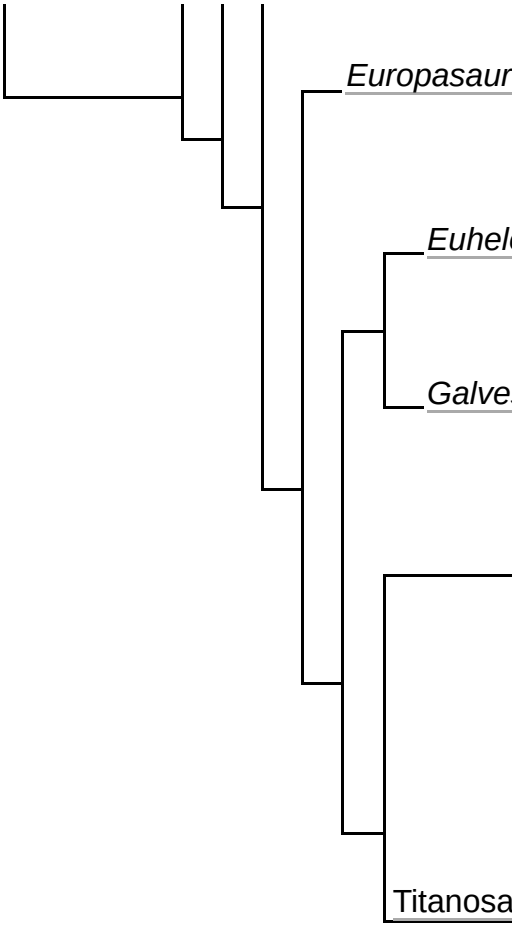
Gravisauria split off in the Early Jurassic, around the Pliensbachian and Toarcian, 183 million years ago, and Aquesbi thought that this was part of a much larger revolution in the fauna, which includes the disappearance of Prosauropoda, Coelophysoidea and basal Thyreophora, which they attributed to a worldwide mass extinction.[99]

The phylogenetic relationships of the sauropods have largely stabilised in recent years, though there are still some uncertainties, such as the placement of *Euhelopus*, *Haplocanthosaurus*, *Jobaria* and Nemegtosauridae.

Cladogram after an analysis presented by Ren and colleagues, 2023:[100]




Mamenchisauridae
Turiasauria
Jobaria
Atlasau
Hapl
R
Diplodocoidea
Macronaria
Dashanpusaurus
Camarasaurus
Lourinhasau
Bellusau
Janen
Arago
Neosauropoda



See also

- List of sauropod species
- Wadadam Fossil Park

Dinosaurs portal

References

1. "sauropod" (https://www.merriam-webster.com/dictionary/sauropod). *Merriam-Webster.com Dictionary*. Merriam-Webster. OCLC 1032680871 (https://search.worldcat.org/oclc/1032680871).
2. "sauropod" (https://www.dictionary.com/browse/sauropod). *Dictionary.com Unabridged* (Online). n.d.
3. Tschopp, E.; Mateus, O.; Benson, R. B. J. (2015). "A specimen-level phylogenetic analysis and taxonomic revision of Diplodocidae (Dinosauria, Sauropoda)" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4393826). *PeerJ*. **3** e857. Bibcode:2015PeerJ...3.e857T (https://ui.adsabs.harvard.edu/abs/2015PeerJ...3.e857T). doi:10.7717/peerj.857 (https://doi.org/10.7717%2Fpeerj.857). PMC 4393826 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4393826). PMID 25870766 (https://pubmed.ncbi.nlm.nih.gov/25870766).
4. blogs.scientificamerican.com tetrapod-zoology 2015-04-24 That Brontosaurus Thing (http://blogs.scientificamerican.com/tetrapod-zoology/2015/04/24/that-brontosaurus-thing/?WT.mc_id=SA_Facebook)

5. Jens N. Lallensack; Hendrik Klein; Jesper Milàn; et al. (2017). "Sauropodomorph dinosaur trackways from the Fleming Fjord Formation of East Greenland: Evidence for Late Triassic sauropods" (https://doi.org/10.4202%2Fapp.00374.2017). *Acta Palaeontologica Polonica*. **62** (4): 833–843. doi:10.4202/app.00374.2017 (https://doi.org/10.4202%2Fapp.00374.2017). hdl:10362/33146 (https://hdl.handle.net/10362%2F33146).
6. Eric Buffetaut; Varavudh Suteethorn; Gilles Cuny; et al. (2000). "The earliest known sauropod dinosaur". *Nature*. **407** (6800): 72–74. Bibcode:2000Natur.407...72B (https://ui.adsabs.harvard.edu/abs/2000Natur.407...72B). doi:10.1038/35024060 (https://doi.org/10.1038%2F35024060). PMID 10993074 (https://pubmed.ncbi.nlm.nih.gov/10993074). S2CID 4387776 (https://api.semanticscholar.org/CorpusID:4387776).
7. Adam M. Yates; James W. Kitching (2003). "The earliest known sauropod dinosaur and the first steps towards sauropod locomotion" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1691423). *Proceedings of the Royal Society B: Biological Sciences*. **270** (1525): 1753–1758. doi:10.1098/rspb.2003.2417 (https://doi.org/10.1098%2Frspb.2003.2417). PMC 1691423 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1691423). PMID 12965005 (https://pubmed.ncbi.nlm.nih.gov/12965005).
8. Blair W. McPhee; Adam M. Yates; Jonah N. Choiniere; Fernando Abdala (2014). "The complete anatomy and phylogenetic relationships of *Antetonitrus ingenipes* (Sauropodiformes, Dinosauria): implications for the origins of Sauropoda". *Zoological Journal of the Linnean Society*. **171** (1): 151–205. doi:10.1111/zoj.12127 (https://doi.org/10.1111%2Fzoj.12127). S2CID 82631097 (https://api.semanticscholar.org/CorpusID:82631097).
9. Blair W. Mcphee; Emese M. Bordy; Lara Sciscio; Jonah N. Choiniere (2017). "The sauropodomorph biostratigraphy of the Elliot Formation of southern Africa: Tracking the evolution of Sauropodomorpha across the Triassic–Jurassic boundary" (https://doi.org/10.4202%2Fapp.00377.2017). *Acta Palaeontologica Polonica*. **62** (3): 441–465. Bibcode:2017AcPaP..62.2017M (https://ui.adsabs.harvard.edu/abs/2017AcPaP..62.2017M). doi:10.4202/app.00377.2017 (https://doi.org/10.4202%2Fapp.00377.2017). hdl:11427/35124 (https://hdl.handle.net/11427%2F35124).
10. Fernando E. Novas (2009). *The Age of Dinosaurs in South America* (https://books.google.com/books?id=dXYi6j7QQ-YC&pg=PA181). Indiana University Press. p. 181. ISBN 978-0-253-35289-7.
11. *Oklahoma Geology Notes* (https://books.google.com/books?id=1gv0AAAAMAAJ&q=%22remains%20of%20sauropods%22). Oklahoma Geological Survey. 2003. p. 40.
12. Beau Riffenburgh (2007). *Encyclopedia of the Antarctic* (https://books.google.com/books?id=fRJtB2MNdJMC&pg=PA415). Taylor & Francis. p. 415. ISBN 978-0-415-97024-2.
13. J. J. Alistair Crame; Geological Society of London (1989). *Origins and Evolution of the Antarctic Biota* (https://books.google.com/books?id=oLwTAQAAIAAJ&q=%22sauropods%22%22antarctica%22). Geological Society. p. 132. ISBN 978-0-903317-44-3.
14. Marsh, O.C. (1878). "Principal characters of American Jurassic dinosaurs. Part I" ". *American Journal of Science and Arts*. **16** (95): 411–416. doi:10.2475/ajs.s3-16.95.411 (https://doi.org/10.2475%2Fajs.s3-16.95.411). hdl:2027/hvd.32044107172876 (https://hdl.handle.net/2027%2Fhvd.32044107172876). S2CID 219245525 (https://api.semanticscholar.org/CorpusID:219245525).

15. Michael P. Taylor; Mathew J. Wedel (2013). "Why sauropods had long necks; and why giraffes have short necks" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3628838). *PeerJ*. **1** e36. arXiv:1209.5439 (https://arxiv.org/abs/1209.5439). Bibcode:2013PeerJ...1..e36T (https://ui.adsabs.harvard.edu/abs/2013PeerJ...1..e36T). doi:10.7717/peerj.36 (https://doi.org/10.7717%2Fpeerj.36). PMC 3628838 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3628838). PMID 23638372 (https://pubmed.ncbi.nlm.nih.gov/23638372). "The necks of the sauropod dinosaurs were by far the longest of any animal..."
16. Tidwell, V., Carpenter, K. & Meyer, S. 2001. New Titanosauriform (Sauropoda) from the Poison Strip Member of the Cedar Mountain Formation (Lower Cretaceous), Utah. In: Mesozoic Vertebrate Life. D. H. Tanke & K. Carpenter (eds.). Indiana University Press, Eds. D.H. Tanke & K. Carpenter. Indiana University Press. 139-165.
17. Bakker, Robert (1994). "The Bite of the Bronto". *Earth*. **3** (6): 26–33.
18. Peterson, Ivars (March 2000). "Whips and Dinosaur Tails" (https://web.archive.org/web/20070714005117/http://www.sciencenews.org/articles/20000318/mathtrek.asp). Science News. Archived from the original (http://www.sciencenews.org/articles/20000318/mathtrek.asp) on 2007-07-14. Retrieved 2007-07-07.
19. Myhrvold, Nathan P.; Currie, Philip John (1997). "Supersonic sauropods? Tail dynamics in the diplodocids" (http://doc.rero.ch/record/15579/files/PAL_E1438.pdf) (PDF). *Paleobiology*. **23** (4): 393–409. Bibcode:1997Pbio...23..393M (https://ui.adsabs.harvard.edu/abs/1997Pbio...23..393M). doi:10.1017/S0094837300019801 (https://doi.org/10.1017%2FS0094837300019801). ISSN 0094-8373 (https://search.worldcat.org/issn/0094-8373). S2CID 83696153 (https://api.semanticscholar.org/CorpusID:83696153).
20. Lovelace, David M.; Hartman, Scott A.; Wahl, William R. (2007). "Morphology of a specimen of *Supersaurus* (Dinosauria, Sauropoda) from the Morrison Formation of Wyoming, and a re-evaluation of diplodocid phylogeny". *Arquivos do Museu Nacional*. **65** (4): 527–544.
21. Carpenter, K. (2006). "Biggest of the big: a critical re-evaluation of the mega-sauropod *Amphicoelias fragillimus*". *New Mexico Museum of Natural History and Science Bulletin*. **36**: 131–138. S2CID 56215581 (https://api.semanticscholar.org/CorpusID:56215581).
22. Cary Woodruff & John R Foster (July 15, 2015). "The fragile legacy of Amphicoelias fragillimus (Dinosauria: Sauropoda; Morrison Formation - Latest Jurassic)" (https://web.archive.org/web/20210831105454/https://peerj.com/preprints/838.pdf) (PDF). *PeerJ PrePrints*. **3** e1037. Archived from the original (https://peerj.com/preprints/838.pdf) (PDF) on August 31, 2021. Retrieved August 21, 2016.
23. Molina-Perez & Larramendi (2020). *Dinosaur Facts and Figures: The Sauropods and Other Sauropodomorphs*. New Jersey: Princeton University Press. pp. 42–267. Bibcode:2020dffs.book.....M (https://ui.adsabs.harvard.edu/abs/2020dffs.book.....M).
24. Paul, Gregory S. (2019). "Determining the largest known land animal: A critical comparison of differing methods for restoring the volume and mass of extinct animals" (http://www.gspauldino.com/Titanomass.pdf) (PDF). *Annals of the Carnegie Museum*. **85** (4): 335–358. Bibcode:2019AnCM...85..335P (https://ui.adsabs.harvard.edu/abs/2019AnCM...85..335P). doi:10.2992/007.085.0403 (https://doi.org/10.2992%2F007.085.0403). S2CID 210840060 (https://api.semanticscholar.org/CorpusID:210840060).

25. Mazzetta, G.V.; Christiansen, P.; Fariña, R.A. (2004). "Giants and Bizarres: Body Size of Some Southern South American Cretaceous Dinosaurs" (http://www.miketaylor.org.uk/tmp/papers/Mazzetta-et-al_04_SA-dino-body-size.pdf) (PDF). *Historical Biology*. **16** (2–4): 71–83. Bibcode:2004HBio...16...71M (https://ui.adsabs.harvard.edu/abs/2004HBio...16...71M). CiteSeerX 10.1.1.694.1650 (https://citeseerx.ist.psu.edu/viewdoc/summary?doi=10.1.1.694.1650). doi:10.1080/08912960410001715132 (https://doi.org/10.1080%2F08912960410001715132). S2CID 56028251 (https://api.semanticscholar.org/CorpusID:56028251). Retrieved 7 July 2024. {{cite journal}}: Cite uses deprecated parameter |citeseerx= (help)
26. Otero, Alejandro; Carballido, José L.; Salgado, Leonardo; Canudo, José Ignacio; Garrido, Alberto C. (12 January 2021). "Report of a giant titanosaur sauropod from the Upper Cretaceous of Neuquén Province, Argentina" (https://www.sciencedirect.com/science/article/abs/pii/S019566712100001X). *Cretaceous Research*. **122** 104754. Bibcode:2021CrRes.12204754O (https://ui.adsabs.harvard.edu/abs/2021CrRes.12204754O). doi:10.1016/j.cretres.2021.104754 (https://doi.org/10.1016%2Fj.cretres.2021.104754). hdl:11336/165354 (https://hdl.handle.net/11336%2F165354). S2CID 233582290 (https://api.semanticscholar.org/CorpusID:233582290).
27. Larramendi, A. (2015). "Shoulder height, body mass and shape of proboscideans" (https://web.archive.org/web/20160304202441/http://www.app.pan.pl/archive/published/app60/app001362014_acc.pdf) (PDF). *Acta Palaeontologica Polonica*. **60**. doi:10.4202/app.00136.2014 (https://doi.org/10.4202%2Fapp.00136.2014). S2CID 2092950 (https://api.semanticscholar.org/CorpusID:2092950). Archived from the original (http://www.app.pan.pl/archive/published/app60/app001362014_acc.pdf) (PDF) on 2016-03-04. Retrieved 2016-04-11.
28. Martin Sander, P.; Mateus, Octávio; Laven, Thomas; Knötschke, Nils (2006). "Bone histology indicates insular dwarfism in a new Late Jurassic sauropod dinosaur". *Nature*. **441** (7094): 739–41. Bibcode:2006Natur.441..739M (https://ui.adsabs.harvard.edu/abs/2006Natur.441..739M). doi:10.1038/nature04633 (https://doi.org/10.1038%2Fnature04633). PMID 16760975 (https://pubmed.ncbi.nlm.nih.gov/16760975). S2CID 4361820 (https://api.semanticscholar.org/CorpusID:4361820).
29. Nicole Klein (2011). *Biology of the Sauropod Dinosaurs: Understanding the Life of Giants* (https://books.google.com/books?id=1GOYYwqctxMC&pg=PA73). Indiana University Press. p. 73. ISBN 978-0-253-35508-9.
30. Reitner, Joachim; Yang, Qun; Wang, Yongdong; Reich, Mike (6 September 2013). *Palaeobiology and Geobiology of Fossil Lagerstätten through Earth History: A Joint Conference of the "Paläontologische Gesellschaft" and the "Palaeontological Society of China", Göttingen, Germany, September 23-27, 2013* (https://books.google.com/books?id=PuVcAgAAQBAJ&pg=PA21). Universitätsverlag Göttingen. p. 21. ISBN 978-3-86395-135-1.
31. Bonnan, M.F. 2005. Pes anatomy in sauropod dinosaurs: implications for functional morphology, evolution, and phylogeny; pp. 346-380 in K. Carpenter and V. Tidwell (eds.), Thunder-Lizards: The Sauropodomorph Dinosaurs. Bloomington, IN: Indiana University Press.
32. Bonnan, Matthew F. (2003). "The evolution of manus shape in sauropod dinosaurs: Implications for functional morphology, forelimb orientation, and phylogeny" (http://doc.rero.ch/record/15172/files/PAL_E2447.pdf) (PDF). *Journal of Vertebrate Paleontology*. **23** (3): 595–613. Bibcode:2003JVPal..23..595B (https://ui.adsabs.harvard.edu/abs/2003JVPal..23..595B). doi:10.1671/A1108 (https://doi.org/10.1671%2FA1108). S2CID 85667519 (https://api.semanticscholar.org/CorpusID:85667519).
33. Upchurch, P. (1994). "Manus claw function in sauropod dinosaurs". *Gaia*. **10**: 161–171.

34. Apesteguía, S. (2005). "Evolution of the titanosaur metacarpus". Pp. 321-345 in Tidwell, V. and Carpenter, K. (eds.) *Thunder-Lizards: The Sauropodomorph Dinosaurs*. Indianapolis: Indiana University Press.
35. Milàn, J.; Christiansen, P.; Mateus, O. (2005). "A three-dimensionally preserved sauropod manus impression from the Upper Jurassic of Portugal: implications for sauropod manus shape and locomotor mechanics". *Kaupia*. **14**: 47–52.
36. Bonnan, M. F. (2004). "Morphometric Analysis of Humerus and Femur Shape in Morrison Sauropods: Implications for Functional Morphology and Paleobiology" (http://doc.rero.ch/record/14932/files/PAL_E2077.pdf) (PDF). *Paleobiology*. **30** (3): 444–470. Bibcode:2004Pbio...30..444B (https://ui.adsabs.harvard.edu/abs/2004Pbio...30..444B). doi:10.1666/0094-8373(2004)030<0444:maohaf>2.0.co;2 (https://doi.org/10.1666%2F0094-8373%282004%29030%3C0444%3Amaohaf%3E2.0.co%3B2). JSTOR 4096900 (https://www.jstor.org/stable/4096900). S2CID 86258781 (https://api.semanticscholar.org/CorpusID:86258781).
37. Bonnan, Matthew F. (2007). "Linear and Geometric Morphometric Analysis of Long Bone Scaling Patterns in Jurassic Neosauropod Dinosaurs: Their Functional and Paleobiological Implications" (https://doi.org/10.1002%2Far.20578). *The Anatomical Record*. **290** (9): 1089–1111. doi:10.1002/ar.20578 (https://doi.org/10.1002%2Far.20578). PMID 17721981 (https://pubmed.ncbi.nlm.nih.gov/17721981). S2CID 41222371 (https://api.semanticscholar.org/CorpusID:41222371).
38. Wedel, M.J. (2009). "Evidence for bird-like air sacs in Saurischian dinosaurs (https://web.archive.org/web/20151115102051/http://sauroposeidon.net/Wedel_2009_air-sacs.pdf)". (pdf) *Journal of Experimental Zoology*, **311A**: 18pp.
39. Taylor, M.P.; Wedel, M.J. (2013). "Why sauropods had long necks; and why giraffes have short necks" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3628838). *PeerJ*. **1** e36. Bibcode:2013PeerJ...1..e36T (https://ui.adsabs.harvard.edu/abs/2013PeerJ...1..e36T). doi:10.7717/peerj.36 (https://doi.org/10.7717%2Fpeerj.36). PMC 3628838 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3628838). PMID 23638372 (https://pubmed.ncbi.nlm.nih.gov/23638372).
40. Ward, Peter Douglas (2006). "The Jurassic: Dinosaur Hegemony in a Low-Oxygen World". *Out of Thin Air: Dinosaurs, Birds, and Earth's Ancient Atmosphere*. Washington, D. C.: Joseph Henry Press. pp. 199–222. ISBN 0-309-10061-5.
41. Taylor, M.P. (2010). "Sauropod dinosaur research: a historical review". In Richard Moody, Eric Buffetaut, David M. Martill and Darren Naish (eds.), *Dinosaurs (and other extinct saurians): a historical perspective*. HTML abstract (http://www.miketaylor.org.uk/dino/pubs/dahp2008/abstract.html).
42. D'Emic, Michael D.; Whitlock, John A.; Smith, Kathlyn M.; Fisher, Daniel C.; Wilson, Jeffrey A. (17 July 2013). "Evolution of High Tooth Replacement Rates in Sauropod Dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3714237). *PLOS ONE*. **8** (7) e69235. Bibcode:2013PLoSO...869235D (https://ui.adsabs.harvard.edu/abs/2013PLoSO...869235D). doi:10.1371/journal.pone.0069235 (https://doi.org/10.1371%2Fjournal.pone.0069235). PMC 3714237 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3714237). PMID 23874921 (https://pubmed.ncbi.nlm.nih.gov/23874921).
43. Barber, Elizabeth (2004-06-09). "No toothbrush required: Dinosaurs replaced their smile every month" (https://www.csmonitor.com/Science/2013/0806/No-toothbrush-required-Dinosaurs-replaced-their-smile-every-month). Christian Science Monitor. Retrieved 2013-08-14.

44. Stevens, K.A. (2013). "The Articulation of Sauropod Necks: Methodology and Mythology" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3812995). *PLOS ONE*. **8** (10) e78572. Bibcode:2013PLoSO...878572S (https://ui.adsabs.harvard.edu/abs/2013PLoSO...878572S). doi:10.1371/journal.pone.0078572 (https://doi.org/10.1371%2Fjournal.pone.0078572). PMC 3812995 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3812995). PMID 24205266 (https://pubmed.ncbi.nlm.nih.gov/24205266).
45. Sander, P. Martin; Christian, Andreas; Clauss, Marcus; Fechner, Regina; Gee, Carole T.; Griebeler, Eva-Maria; Gunga, Hanns-Christian; Hummel, Jürgen; Mallison, Heinrich; Perry, Steven F.; et al. (2011). "Biology of the sauropod dinosaurs: the evolution of gigantism" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3045712). *Biological Reviews*. **86** (1): 117–155. Bibcode:2011BioRv..86..117S (https://ui.adsabs.harvard.edu/abs/2011BioRv..86..117S). doi:10.1111/j.1469-185X.2010.00137.x (https://doi.org/10.1111%2Fj.1469-185X.2010.00137.x). ISSN 1464-7931 (https://search.worldcat.org/issn/1464-7931). PMC 3045712 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3045712). PMID 21251189 (https://pubmed.ncbi.nlm.nih.gov/21251189).
46. Henderson, D.M. (2013). "Sauropod Necks: Are They Really for Heat Loss?" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3812985). *PLOS ONE*. **8** (10) e77108. Bibcode:2013PLoSO...877108H (https://ui.adsabs.harvard.edu/abs/2013PLoSO...877108H). doi:10.1371/journal.pone.0077108 (https://doi.org/10.1371%2Fjournal.pone.0077108). PMC 3812985 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3812985). PMID 24204747 (https://pubmed.ncbi.nlm.nih.gov/24204747).
47. Sakaki, Homare; Winkler, Daniela E.; Kubo, Tai; Hirayama, Ren; Uno, Hikaru; Miyata, Shinya; Endo, Hideki; Sasaki, Kazuhisa; Takisawa, Toshio; Kubo, Mugino O. (August 2022). "Non-occlusal dental microwear texture analysis of a titanosauriform sauropod dinosaur from the Upper Cretaceous (Turonian) Tamagawa Formation, northeastern Japan" (https://doi.org/10.1016%2Fj.cretres.2022.105218). *Cretaceous Research*. **136** 105218. Bibcode:2022CrRes.13605218S (https://ui.adsabs.harvard.edu/abs/2022CrRes.13605218S). doi:10.1016/j.cretres.2022.105218 (https://doi.org/10.1016%2Fj.cretres.2022.105218).
48. Henderson, D.M. (2004). "Tipsy punters: sauropod dinosaur pneumaticity, buoyancy and aquatic habits" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1810024). *Proceedings of the Royal Society of London B*. **271** (Suppl 4): S180–S183. Bibcode:2004PBioS.271.0136H (https://ui.adsabs.harvard.edu/abs/2004PBioS.271.0136H). doi:10.1098/rsbl.2003.0136 (https://doi.org/10.1098%2Frsbl.2003.0136). PMC 1810024 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1810024). PMID 15252977 (https://pubmed.ncbi.nlm.nih.gov/15252977).
49. Kermack, K.A. (1951). "A note on the habits of sauropods". *Ann. Mag. Nat. Hist*. **4** (44): 830–832. doi:10.1080/00222935108654213 (https://doi.org/10.1080%2F00222935108654213).
50. "Giant wading sauropod discovery made on Isle of Skye (Wired UK)" (https://www.wired.co.uk/news/archive/2015-12/02/sauropods-isle-of-skye-dinosaur-discovery). *Wired UK*. 2015-12-02. Retrieved 2016-03-22.
51. "Sauropod dinosaurs were restricted to warmer regions of Earth" (https://phys.org/news/2021-12-sauropod-dinosaurs-restricted-warmer-regions.amp).
52. Wyenberg-Henzler T. 2022. Ecomorphospace occupation of large herbivorous dinosaurs from Late Jurassic through to Late Cretaceous time in North America. PeerJ 10:e13174 https://doi.org/10.7717/peerj.13174

53. Myers, T.S.; Fiorillo, A.R. (2009). "Evidence for gregarious behavior and age segregation in sauropod dinosaurs" (http://doc.rero.ch/record/16633/files/PAL_E962.pdf) (PDF). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **274** (1–2): 96–104. Bibcode:2009PPP...274...96M (https://ui.adsabs.harvard.edu/abs/2009PPP...274...96M). doi:10.1016/j.palaeo.2009.01.002 (https://doi.org/10.1016%2Fj.palaeo.2009.01.002).
54. Coria, R.A. (1994). "On a monospecific assemblage of sauropod dinosaurs from Patagonia: implications for gregarious behavior". *GAIA*. **10**: 209–213.
55. Ruxton, Graeme D.; Birchard, Geoffrey F.; Deeming, D. Charles (2014). "Incubation time as an important influence on egg production and distribution into clutches for sauropod dinosaurs" (https://figshare.com/articles/journal_contribution/24336253). *Paleobiology*. **40** (3): 323–330. Bibcode:2014Pbio...40..323R (https://ui.adsabs.harvard.edu/abs/2014Pbio...40..323R). doi:10.1666/13028 (https://doi.org/10.1666%2F13028). S2CID 84437615 (https://api.semanticscholar.org/CorpusID:84437615).
56. A new Argentinean nesting site showing neosauropod dinosaur reproduction in a Cretaceous hydrothermal environment, Nature Communications, Volume: 1, Article number: 32, DOI: doi:10.1038/ncomms1031
57. H. Dhiman et al.. 2023. New Late Cretaceous titanosaur sauropod dinosaur egg clutches from lower Narmada Valley, India: Palaeobiology and taphonomy. PLoS ONE 18 (1): e0278242; doi: 10.1371/journal.pone.0278242
58. Tanaka, Kohei & Zelenitsky, Darla & Therrien, François & Kobayashi, Yoshitsugu. (2018). Nest substrate reflects incubation style in extant archosaurs with implications for dinosaur nesting habits. Scientific Reports. 8. 10.1038/s41598-018-21386-x.
59. Osborn, H. F. (1899). "A Skeleton of Diplodocus, Recently Mounted in the American Museum" (https://zenodo.org/record/1447822). *Science*. **10** (259): 870–4. Bibcode:1899Sci....10..870F (https://ui.adsabs.harvard.edu/abs/1899Sci....10..870F). doi:10.1126/science.10.259.870 (https://doi.org/10.1126%2Fscience.10.259.870). PMID 17788971 (https://pubmed.ncbi.nlm.nih.gov/17788971).
60. Rothschild, B.M. & Molnar, R.E. (2005). "Sauropod Stress Fractures as Clues to Activity". In Carpenter, K. & Tidswell, V. (eds.). *Thunder Lizards: The Sauropodomorph Dinosaurs*. Indiana University Press. pp. 381–391. ISBN 978-0-253-34542-4.
61. Mallison, H. (2009). "Rearing for food? Kinetic/dynamic modeling of bipedal/tripodal poses in sauropod dinosaurs". P. 63 in Godefroit, P. and Lambert, O. (eds), *Tribute to Charles Darwin and Bernissart Iguanodons: New Perspectives on Vertebrate Evolution and Early Cretaceous Ecosystems*. Brussels.
62. Bujor, Mara (2009-05-29). "Did sauropods walk with their necks upright?" (http://www.zmescience.com/research/did-sauropods-walk-with-their-necks-upright/). ZME Science.
63. Seymour, RS (June 2009). "Raising the sauropod neck: it costs more to get less" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2679936). *Biol. Lett*. **5** (3): 317–9. doi:10.1098/rsbl.2009.0096 (https://doi.org/10.1098%2Frsbl.2009.0096). PMC 2679936 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2679936). PMID 19364714 (https://pubmed.ncbi.nlm.nih.gov/19364714).
64. Choy, DS; Altman, P (1992-08-29). "The cardiovascular system of barosaurus: an educated guess" (https://doi.org/10.1016%2F0140-6736%2892%2991722-k). *Lancet*. **340** (8818): 534–6. doi:10.1016/0140-6736(92)91722-k (https://doi.org/10.1016%2F0140-6736%2892%2991722-k). PMID 1354287 (https://pubmed.ncbi.nlm.nih.gov/1354287). S2CID 7378155 (https://api.semanticscholar.org/CorpusID:7378155).

65. Seymour, RS; Lillywhite, HB (September 2000). "Hearts, neck posture and metabolic intensity of sauropod dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1690760). *Proc. Biol. Sci*. **267** (1455): 1883–7. Bibcode:2000PBioS.267.1883S (https://ui.adsabs.harvard.edu/abs/2000PBioS.267.1883S). doi:10.1098/rspb.2000.1225 (https://doi.org/10.1098%2Frspb.2000.1225). PMC 1690760 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1690760). PMID 11052540 (https://pubmed.ncbi.nlm.nih.gov/11052540).
66. Stevens, K.A.; Parrish, J.M. (1999). "Neck posture and feeding habits of two Jurassic sauropod dinosaurs". *Science*. **284** (5415): 798–800. Bibcode:1999Sci...284..798S (https://ui.adsabs.harvard.edu/abs/1999Sci...284..798S). doi:10.1126/science.284.5415.798 (https://doi.org/10.1126%2Fscience.284.5415.798). PMID 10221910 (https://pubmed.ncbi.nlm.nih.gov/10221910).
67. Taylor, M.P., Wedel, M.J., and Naish, D. (2009). "Head and neck posture in sauropod dinosaurs inferred from extant animals (http://www.app.pan.pl/archive/published/app54/app54-213.pdf)". *Acta Palaeontologica Polonica* 54 (2), 2009: 213-220 abstract (http://www.app.pan.pl/article/item/app54-213.html)
68. Museums and TV have dinosaurs' posture all wrong, claim scientists (https://www.theguardian.com/science/2009/may/27/dinosaurs-sauropods-posture-heads-upright-necks). Guardian, 27 May 2009
69. Cobley MJ; Rayfield EJ; Barrett PM (14 August 2013). "Inter-Vertebral Flexibility of the Ostrich Neck: Implications for Estimating Sauropod Neck Flexibility" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3743800). *PLOS ONE*. **8** (8) e72187. Bibcode:2013PLoSO...872187C (https://ui.adsabs.harvard.edu/abs/2013PLoSO...872187C). doi:10.1371/journal.pone.0072187 (https://doi.org/10.1371%2Fjournal.pone.0072187). PMC 3743800 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3743800). PMID 23967284 (https://pubmed.ncbi.nlm.nih.gov/23967284).
70. n.d (2013-08-14). "Ostrich Necks Reveal Sauropod Movements, Food Habits" (https://www.sciencedaily.com/releases/2013/08/130814191912.htm). Science Daily. Retrieved 2013-08-16.
71. Ghose, Tia (2013-08-15). "Ouch! Long-necked dinosaurs may actually have had stiff necks" (http://www.nbcnews.com/science/ouch-long-necked-dinosaurs-may-actually-have-had-stiff-necks-6C10931266). NBC News Live Science. Retrieved 2013-08-16.
72. Riga, B.J.G.; Calvo, J.O. (2009). "A new wide-gauge sauropod track site from the Late Cretaceous of Mendoza, Neuquen Basin, Argentina" (http://doc.rero.ch/record/208893/files/PAL_E3994.pdf) (PDF). *Palaeontology*. **52** (3): 631–640. Bibcode:2009Palgy..52..631G (https://ui.adsabs.harvard.edu/abs/2009Palgy..52..631G). doi:10.1111/j.1475-4983.2009.00869.x (https://doi.org/10.1111%2Fj.1475-4983.2009.00869.x).
73. Castanera, D.; Barco, J. L.; Díaz-Martínez, I.; Gascón, J. S. H.; Pérez-Lorente, F. L.; Canudo, J. I. (2011). "New evidence of a herd of titanosauriform sauropods from the lower Berriasian of the Iberian range (Spain)". *Palaeogeography, Palaeoclimatology, Palaeoecology*. **310** (3–4): 227–237. Bibcode:2011PPP...310..227C (https://ui.adsabs.harvard.edu/abs/2011PPP...310..227C). doi:10.1016/j.palaeo.2011.07.015 (https://doi.org/10.1016%2Fj.palaeo.2011.07.015).
74. Day, J.J.; Norman, D.B.; Gale, A.S.; Upchurch, P.; Powell, H.P. (2004). "A Middle Jurassic dinosaur trackway site from Oxfordshire, UK" (https://doi.org/10.1111%2Fj.0031-0239.2004.00366.x). *Palaeontology*. **47** (2): 319–348. Bibcode:2004Palgy..47..319D (https://ui.adsabs.harvard.edu/abs/2004Palgy..47..319D). doi:10.1111/j.0031-0239.2004.00366.x (https://doi.org/10.1111%2Fj.0031-0239.2004.00366.x).

75. Falkingham, P. L.; Bates, K. T.; Margetts, L.; Manning, P. L. (2011-02-23). "Simulating sauropod manus-only trackway formation using finite-element analysis" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3030862). *Biology Letters*. **7** (1): 142–145. Bibcode:2011BiLet...7..142F (https://ui.adsabs.harvard.edu/abs/2011BiLet...7..142F). doi:10.1098/rsbl.2010.0403 (https://doi.org/10.1098%2Frsbl.2010.0403). ISSN 1744-9561 (https://search.worldcat.org/issn/1744-9561). PMC 3030862 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3030862). PMID 20591856 (https://pubmed.ncbi.nlm.nih.gov/20591856).
76. Falkingham, P. L.; Bates, K. T.; Margetts, L.; Manning, P. L. (2011-08-07). "The 'Goldilocks' effect: preservation bias in vertebrate track assemblages" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3119880). *Journal of the Royal Society Interface*. **8** (61): 1142–1154. Bibcode:2011JRSI....8.1142F (https://ui.adsabs.harvard.edu/abs/2011JRSI....8.1142F). doi:10.1098/rsif.2010.0634 (https://doi.org/10.1098%2Frsif.2010.0634). ISSN 1742-5689 (https://search.worldcat.org/issn/1742-5689). PMC 3119880 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3119880). PMID 21233145 (https://pubmed.ncbi.nlm.nih.gov/21233145).
77. Sellers, W. I.; Margetts, L.; Coria, R. A. B.; Manning, P. L. (2013). Carrier, David (ed.). "March of the Titans: The Locomotor Capabilities of Sauropod Dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3864407). *PLOS ONE*. **8** (10) e78733. Bibcode:2013PLoSO...878733S (https://ui.adsabs.harvard.edu/abs/2013PLoSO...878733S). doi:10.1371/journal.pone.0078733 (https://doi.org/10.1371%2Fjournal.pone.0078733). PMC 3864407 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3864407). PMID 24348896 (https://pubmed.ncbi.nlm.nih.gov/24348896).
78. Szabo, John (2011). *Dinosaurs*. University of Akron: McGraw Hill. p. 35. ISBN 978-1-121-09332-4.
79. Brian K. Hall; Benedikt Hallgrímsson (1 June 2011). *Strickberger's Evolution* (https://books.google.com/books?id=jrDD3cyA09kC&pg=PA446). Jones & Bartlett Publishers. p. 446. ISBN 978-1-4496-4722-3.
80. Navarro, Bruno A.; Ghilardi, Aline M.; Aureliano, Tito; Díaz, Verónica Díez; Bandeira, Kamila L. N.; Cattaruzzi, André G. S.; Iori, Fabiano V.; Martine, Ariel M.; Carvalho, Alberto B.; Anelli, Luiz E.; Fernandes, Marcelo A.; Zaher, Hussam (2022-09-15). "A New Nanoid Titanosaur (Dinosauria: Sauropoda) from the Upper Cretaceous of Brazil" (https://bioone.org/journals/ameghiniana/volume-59/issue-5/AMGH.25.08.2022.3477/A-New-Nanoid-Titanosaur-Dinosauria--Sauropoda-from-the-Upper/10.5710/AMGH.25.08.2022.3477.full). *Ameghiniana*. **59** (5): 477. Bibcode:2022Amegh..59..477N (https://ui.adsabs.harvard.edu/abs/2022Amegh..59..477N). doi:10.5710/AMGH.25.08.2022.3477 (https://doi.org/10.5710%2FAMGH.25.08.2022.3477). ISSN 0002-7014 (https://search.worldcat.org/issn/0002-7014). S2CID 251875979 (https://api.semanticscholar.org/CorpusID:251875979).
81. "Achoo! Respiratory illness gave young 'Dolly' the dinosaur flu-like symptoms" (https://www.livescience.com/dinosaur-respiratory-infection-first-evidence-sauropod). *Live Science*. 10 February 2022.
82. "Discovery of what ailed Dolly the dinosaur is a first, researchers say" (https://www.cnn.com/2022/02/10/world/dinosaur-respiratory-infection-scn/index.html). *CNN*. 10 February 2022.
83. Tan, Chao; Yu, Hai-Dong; Ren, Xin-Xin; Dai, Hui; Ma, Qing-Yu; Xiong, Can; Zhao, Zhi-Qiang; You, Hai-Lu (2022). "Pathological ribs in sauropod dinosaurs from the Middle Jurassic of Yunyang, Chongqing, Southwestern China" (https://www.tandfonline.com/doi/abs/10.1080/08912963.2022.2045979?journalCode=ghbi20#.Yh0NWORCkb0.facebook). *Historical Biology*. **35** (4): 475–482. doi:10.1080/08912963.2022.2045979 (https://doi.org/10.1080%2F08912963.2022.2045979). S2CID 247172509 (https://api.semanticscholar.org/CorpusID:247172509).

84. Tan, Chao; Wang, Ping; Xiong, Can; Wei, Zhao-Ying; Tu, Yan; Ma, Qing-Yu; Ren, Xin-Xin; You, Hai-Lu (17 September 2024). "A sauropod tibia with initial fracture from the Middle Jurassic in Yunyang, Chongqing, southwest China" (https://www.tandfonline.com/doi/full/10.1080/08912963.2024.2403598). *Historical Biology*. **37** (9): 2131–2138. doi:10.1080/08912963.2024.2403598 (https://doi.org/10.1080%2F08912963.2024.2403598). ISSN 0891-2963 (https://search.worldcat.org/issn/0891-2963). Retrieved 27 September 2024 – via Taylor and Francis Online.
85. Aureliano, Tito; Nascimento, Carolina S. I.; Fernandes, Marcelo A.; Ricardi-Branco, Fresia; Ghilardi, Aline M. (2021-02-01). "Blood parasites and acute osteomyelitis in a non-avian dinosaur (Sauropoda, Titanosauria) from the Upper Cretaceous Adamantina Formation, Bauru Basin, Southeast Brazil" (https://www.sciencedirect.com/science/article/pii/S0195667120303591). *Cretaceous Research*. **118** 104672. Bibcode:2021CrRes.11804672A (https://ui.adsabs.harvard.edu/abs/2021CrRes.11804672A). doi:10.1016/j.cretres.2020.104672 (https://doi.org/10.1016%2Fj.cretres.2020.104672). ISSN 0195-6671 (https://search.worldcat.org/issn/0195-6671). S2CID 225134198 (https://api.semanticscholar.org/CorpusID:225134198).
86. Aureliano, Tito; Nascimento, Carolina S.I.; Fernandes, Marcelo A.; Ricardi-Branco, Fresia; Ghilardi, Aline M. (February 2021). "Blood parasites and acute osteomyelitis in a non-avian dinosaur (Sauropoda, Titanosauria) from the Upper Cretaceous Adamantina Formation, Bauru Basin, Southeast Brazil". *Cretaceous Research*. **118** 104672. Bibcode:2021CrRes.11804672A (https://ui.adsabs.harvard.edu/abs/2021CrRes.11804672A). doi:10.1016/j.cretres.2020.104672 (https://doi.org/10.1016%2Fj.cretres.2020.104672). S2CID 225134198 (https://api.semanticscholar.org/CorpusID:225134198).
87. Baraniuk, Chris (January 2021). "Gruesome 'Blood Worms' Invaded a Dinosaur's Leg Bone, Fossil Suggests" (https://www.scientificamerican.com/article/gruesome-blood-worms-invaded-a-dinosaurs-leg-bone-fossil-suggests/). *Scientific American*.
88. "Cretaceous Titanosaur Suffered from Blood Parasites and Severe Bone Inflammation | Paleontology | Sci-News.com" (http://www.sci-news.com/paleontology/cretaceous-titanosaur-blood-parasites-osteomyelitis-09108.html). *Breaking Science News | Sci-News.com*.
89. Delair, J.B.; Sarjeant, W.A.S. (2002). "The earliest discoveries of dinosaurs: the records re-examined". *Proceedings of the Geologists' Association*. **113** (3): 185–197. Bibcode:2002PrGA..113..185D (https://ui.adsabs.harvard.edu/abs/2002PrGA..113..185D). doi:10.1016/S0016-7878(02)80022-0 (https://doi.org/10.1016%2FS0016-7878%2802%2980022-0).
90. Lhuyd, E. (1699). *Lithophylacii Britannici Ichnographia, sive lapidium aliorumque fossilium Britannicorum singulari figura insignium*. Gleditsch and Weidmann: London.
91. Owen, R. (1842). "Report on British Fossil Reptiles". Part II. Report of the British Association for the Advancement of Science, Plymouth, England.
92. Phillips, J. (1871). *Geology of Oxford and the Valley of the Thames.* Oxford: Clarendon Press, 523 pp.
93. Osborn, H. F.; Mook, C. C. (1919-01-01). "Camarasaurus, Amphicoelias, and other sauropods of Cope" (https://pubs.geoscienceworld.org/gsabulletin/article/30/1/379-388/2870). *Geological Society of America Bulletin*. **30** (1): 379–388. Bibcode:1919GSAB...30..379O (https://ui.adsabs.harvard.edu/abs/1919GSAB...30..379O). doi:10.1130/GSAB-30-379 (https://doi.org/10.1130%2FGSAB-30-379). ISSN 0016-7606 (https://search.worldcat.org/issn/0016-7606).

94. Salgado, L.; Coria, R. A.; Calvo, J. O. (1997). "Evolution of titanosaurid sauropods. 1: Phylogenetic analysis based on the postcranial evidence" (https://www.researchgate.net/publication/283797197). *Ameghiniana*. **34** (1): 3–32. ISSN 1851-8044 (https://search.worldcat.org/issn/1851-8044).
95. Peyre de Fabrègues, C.; Allain, R.; Barriel, V. (2015). "Root causes of phylogenetic incongruence observed within basal sauropodomorph interrelationships: Sauropodomorph Interrelationships" (https://doi.org/10.1111%2Fzoj.12290). *Zoological Journal of the Linnean Society*. **175** (3): 569–586. doi:10.1111/zoj.12290 (https://doi.org/10.1111%2Fzoj.12290). S2CID 83180197 (https://api.semanticscholar.org/CorpusID:83180197).
96. Yates, A. M. (2007). "Solving a dinosaurian puzzle: the identity of *Aliwalia rex* Galton". *Historical Biology*. **19** (1): 93–123. Bibcode:2007HBio...19...93Y (https://ui.adsabs.harvard.edu/abs/2007HBio...19...93Y). doi:10.1080/08912960600866953 (https://doi.org/10.1080%2F08912960600866953). S2CID 85202575 (https://api.semanticscholar.org/CorpusID:85202575).
97. Allain, R.; Aquesbi, N. (2008). "Anatomy and phylogenetic relationships of *Tazoudasaurus naimi* (Dinosauria, Sauropoda) from the late Early Jurassic of Morocco" (https://sciencepress.mnhn.fr/en/periodiques/geodiversitas/30/2/anatomie-et-relations-phylogenetiques-de-tazoudasaurus-naimi-dinosauria-sauropoda-de-la-fin-du-jurassique-inferieur-du-maroc). *Geodiversitas*. **30** (2): 345–424.
98. Pol, D.; Otero, A.; Apaldetti, C.; Martínez, R. N. (2021). "Triassic sauropodomorph dinosaurs from South America: The origin and diversification of dinosaur dominated herbivorous faunas". *Journal of South American Earth Sciences*. **107** 103145. Bibcode:2021JSAES.10703145P (https://ui.adsabs.harvard.edu/abs/2021JSAES.10703145P). doi:10.1016/j.jsames.2020.103145 (https://doi.org/10.1016%2Fj.jsames.2020.103145). hdl:11336/164993 (https://hdl.handle.net/11336%2F164993). S2CID 233579282 (https://api.semanticscholar.org/CorpusID:233579282).
99. Allain, R. and Aquesbi, N. (2008). "Anatomy and phylogenetic relationships of *Tazoudasaurus naimi* (Dinosauria, Sauropoda) from the late Early Jurassic of Morocco." *Geodiversitas*, **30**(2): 345-424.
100. Ren, Xin-Xin; Jiang, Shan; Wang, Xu-Ri; Peng, Guang-Zhao; Ye, Yong; Jia, Lei; You, Hai-Lu (January 2023). "Re-examination of Dashanpusaurus dongi (Sauropoda: Macronaria) supports an early Middle Jurassic global distribution of neosauropod dinosaurs" (https://linkinghub.elsevier.com/retrieve/pii/S0031018222004898). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **610** 111318. Bibcode:2023PPP...61011318R (https://ui.adsabs.harvard.edu/abs/2023PPP...61011318R). doi:10.1016/j.palaeo.2022.111318 (https://doi.org/10.1016%2Fj.palaeo.2022.111318).

External links

- Media related to Sauropoda at Wikimedia Commons
- Strauss, Bob (2008). "Sauropods: The biggest dinosaurs that ever lived" (https://web.archive.org/web/20170202025630/http://dinosaurs.about.com/od/typesofdinosaurs/a/sauropods.htm). *about.com*. Types of dinosaurs. Archived from the original (http://dinosaurs.about.com/od/typesofdinosaurs/a/sauropods.htm) on 2017-02-02. Retrieved 2008-07-07.
- Rogers, K.C.; Wilson, J.A. (2005). *The Sauropods: Evolution and Paleobiology*. Berkeley, CA: University of California Press. ISBN 0-520-24623-3.
- Upchurch, P.; Barrett, P.M.; Dodson, P. (2004). "Sauropoda". In Weishampel, D.; Dodson, P.; Osmólska, H. (eds.). *The Dinosauria* (2nd ed.). Berkeley, CA: University of California Press. pp. 259–322.

Retrieved from "https://en.wikipedia.org/w/index.php?title=Sauropoda&oldid=1381001708"

4

Ornithischia (Kuş Kalçalılar)

Zırhlı, boynuzlu, plakalı ve ördek gagalı dinozorlar

- Ornithischia
- Stegosauria
- Ankylosauria
- Ceratopsidae
- Triceratops
- Hadrosauridae
- Maiasaura

WIKIPEDIA
25 years of the free encyclopedia

Ornithischia

Ornithischia (/ˌɔːrnəˈθɪskiə/) is an extinct clade of mainly herbivorous dinosaurs characterized by a pelvic structure superficially similar to that of birds.[6] The name *Ornithischia*, or "bird-hipped", reflects this similarity and is derived from the Greek stem *ornith-* (ὀρνιθ-), meaning "bird", and *ischion* (ἴσχιον),[a] meaning "hip".[7] However, as birds are theropod dinosaurs, they are only distantly related to this group.[6]

Ornithischians with well-known anatomical adaptations include the ceratopsians or "horn-faced" dinosaurs (e.g. *Triceratops*), the pachycephalosaurs or "thick-headed" dinosaurs, the armored dinosaurs (Thyreophora) such as stegosaurs and ankylosaurs, and the ornithopods.[8][6] There is strong evidence that certain groups of ornithischians lived in herds,[6][9] often segregated by age group, with juveniles forming their own flocks separate from adults.[10] Some were at least partially covered in filamentous (hair- or feather-like) pelts, and there is much debate over whether these filaments found in specimens of *Tianyulong*, *Psittacosaurus*,[11] and *Kulindadromeus* may have been primitive feathers.[12]

Description

Ornithischia is a very large and diverse group of dinosaurs, with members known from all continents, habitats, and a very large range of sizes. They are primarily herbivorous browsers or grazers, but some members may have also been opportunistic omnivores.[13] Ornithischians have multiple features of the skull, teeth, and skeleton in common, including especially the presence of a predentary and palpebral, an increased number of sacral vertebrae, the absence of gastralia, and an opisthopubic pubis.[14] Early ornithischians ranged around 1–2 m (3.3–6.6 ft) in length, with them increasing in size over time so that the largest armoured ornithischians were around 7.5 m (25 ft) and 9 t (8.9 long tons; 9.9 short tons), the largest horned ornithischians were around 8.5 m (28 ft) and 11 t (11 long tons; 12 short tons), and the largest crested ornithischians were around 15 m (49 ft) and 13.5 t (13.3 long tons; 14.9 short tons).[13]

Early ornithischians were relatively small dinosaurs, averaging about 1–2 meters in body length, with a triangular skull that had large circular orbits on the sides. This suggests that early ornithischians had relatively huge eyes that faced laterally. Early ornithischians possessed forelimbs that were considerably shorter than their hindlimbs. Small forelimbs such as those present in early ornithischians would not have been useful for locomotion, and it is evident that early ornithischians were bipedal dinosaurs. The entire skeleton was lightly built, with a largely fenestrated skull and a very stout neck and trunk. The tail is nearly half of the dinosaurs' overall

Ornithischia

Temporal range:

Early Jurassic–Late Cretaceous, (Possible Triassic record)

A collection of ornithischian skeletons (clockwise from top left): *Triceratops* (Ceratopsia), *Heterodontosaurus* (Heterodontosauridae), *Edmontosaurus* (Ornithopoda), *Stegoceras* (Pachycephalosauria), *Europelta* (Ankylosauria), and *Stegosaurus* (Stegosauria).

Scientific classification

Kingdom:	Animalia
Phylum:	Chordata
Class:	Reptilia
Clade:	Dinosauria
Clade:	†Ornithischia Seeley, 1888[1]

Subgroups[2]

- †*Chilesaurus*?
- †*Pisanosaurus*?
- †Silesauridae? (paraphyletic?)
- †**Saphornithischia** Madzia et al., 2021[3]
  - †*Archaeocursor*
  - †*Eocursor*
  - †*Laquintasaura*?

length; their long tails presumably acted as a counterbalance, and as a compensating mechanism for shifts in the creature's center of gravity. The hindlimbs of early ornithischians show that their tibia was considerably longer than their femur, a feature that suggests that early ornithischians were adapted for bipedality, and were fast runners.[15]

Skulls

Skulls of several early diverging ornithischians

Much of the knowledge of early ornithischian anatomy comes from *Lesothosaurus*, which is a taxon known from multiple skulls and skeletons from the Early Jurassic of Lesotho. The rear of its skull is box-like, while the snout tapers to a point. The nasal opening is small, the antorbital fossa that opens from the side of the skull into the palate is large, shallow and triangular, the orbit is large and round and has a palpebral creating a brow, and the lower jaw has a large mandibular fenestra.[16]

The skulls of *Emausaurus* and *Scelidosaurus*, two early members of the armoured group Thyreophora, show similarities in the box-like skull that tapers to the front. The antorbital fossa is smaller and forming an elongate oval in both taxa, and the palpebral which is elongate and slender in *Lesothosaurus* is widened in *Emausaurus* and completely incorporated into the skull as a flat bone in *Scelidosaurus*.[16][17] Skulls in members of the thyreophoran group Stegosauria are much longer and lower, with the width at the back being greater than the height in *Stegosaurus*. The snout and lower jaw are long and deep, and in some genera the premaxilla does not have any teeth. As in *Scelidosaurus*, the palpebral forms the top border of the orbit as a flat brow bone, but the antorbital fossa is reduced to the point of absence in some genera.[18]

Ankylosaurs, the other group of armoured ornithischians, have very robust, immobile skulls, with three significant features that separate them from other groups. The antorbital fossa, supratemporal fenestra and mandibular fenestra are all closed, the sutures separating skull bones are almost completely obliterated by surface texturing, and there is bony armour above the orbits, and at the top and bottom corners of the back of the skull. Teeth are sometimes absent from the premaxilla, and both the upper and lower jaws have deeply inset teeth creating large cheeks. Ankylosaurs also have very extensive and complicated network of sinuses, formed by bone growth in the palate.[19]

The skulls are known from many early ornithopods and some heterodontosaurids, showing similar general features. Skulls are relatively tall with shorter snouts, but the snout is elongated in some later taxa like *Thescelosaurus*. The orbit and antorbital fossa are large, but the nasal opening is small, and while teeth are present in the premaxilla, there is a toothless

- †*Lesothosaurus*
- †Heterodontosauridae
- †**Genasauria** Sereno, 1986[4]
  - †*Serendipaceratops*
  - †Thyreophora
  - †Neornithischia

Dubious ornithischians

- †*Alocodon*
- †*Fabrosaurus*
- †*Ferganocephale*
- †*Gongbusaurus*?
- †*Laosaurus*
- †*Notoceratops*
- †*Syngonosaurus*
- †*Tichosteus*
- †*Trimucrodon*

Synonyms[5]

- Orthopoda Cope, 1866
- Therosauria Haeckel, 1866
- Ornithocephala Bunzel, 1871
- Prionodontia Owen, 1874
- Omosauria Seeley, 1888
- Predentata Marsh, 1894
- Postpubici Jaekel, 1910
- Praedentata Jaekel, 1910
- Phytophagi Jaekel, 1914
- Avipelviens Kuhn, 1964
- Praedenta Glut, 1997
- Onnithischia Buccholz, 2002

front tip that likely formed a keratinous beak. The premaxillary teeth and the first lower tooth in *Heterodontosaurus* are enlarged into sizeable canines.[20] In later ornithopods, the skulls are more elongate and sometimes fully rectangular, with a very large nasal opening, and a thin, elongate palpebral that can extend the entire way across the orbit. Teeth are almost always absent from the premaxilla, the antorbital fossa is reduced and round to slit-like, the tip of the snout is sometimes flared to form a broad beak.[21] Members of the ornithopod family Hadrosauridae show further adaptations, including the formation of dental batteries where teeth are continuously replaced, and in many genera the development of prominent cranial crests formed by multiple different bones of the skull.[22]

Pachycephalosauria, at one time thought to be close to ornithopods and now known to be related instead to ceratopsians, show a unique skull anatomy that is unlike any other ornithischian. The bones of the top of the skull are thickened and in many taxa expanded significantly to form round bony domes as the top of the head, as well as possessing small nodes or elongate spikes along the back edge of the skull. Many taxa are only known from these thick skull domes, which are fused from the frontal and parietal bones. As in many other ornithischians, the snout is short and tapering, the nasal opening is small, the antorbital fossa is sometimes absent, and there are premaxillary teeth, though only three. The two palpebrals are also incorporated into the skull roof as in thyreophorans, rather than free.[23]

Ceratopsians, the sister group to pachycephalosaurs, also display many cranial adaptations, most importantly the evolution of a bone called the rostral that forms the top beak opposite the predentary. The jugal bones flare to the sides to create a pentagonal skull seen from above, the nasal opening is closer to the top of the snout than the teeth, and while the snout tapers in some taxa, it is very deep and short in *Psittacosaurus*. The ceratopsian palpebral is generally triangular, and the back edge of the skull roof forms a flat frill that is enlarged in more derived ceratopsians.[24] The ceratopsian family Ceratopsidae progresses on these features with the addition of horns above each orbit and on the top of the snout, as well as substantial elongation of the frill and in many genera the development of two large parietal fenestrae forming holes in the frill. The skull and frill elongation makes the skulls of *Torosaurus* and *Pentaceratops* the largest of any known terrestrial vertebrate, at over 2 m (6.6 ft) long.[25]

"Bird-hip"

The ornithischian pelvis was "opisthopubic", meaning that the pubis pointed down and backwards (posterior), parallel with the ischium, while in the saurischian pelvis, the pelvis jutted forwards as in lizards.[6][8] This resulted in a four-pronged pelvic structure. In contrast to this, the saurischian pelvis was "propubic", meaning the pubis pointed toward the head (anterior), as in ancestral reptiles (Figure 1b).[6]

Pelvis of *Edmontosaurus*, showing the characteristic ornithischian shape

The opisthopubic pelvis independently evolved at least three times in dinosaurs (in ornithischians, birds and therizinosauroids).[26] Some argue that the opisthopubic pelvis evolved a fourth time, in the clade Dromaeosauridae, but this is controversial, as other authors argue that dromaeosaurids are mesopubic.[26] It has also been argued that the opisthopubic condition is basal to maniraptorans (including among others birds, therizinosauroids and dromaeosaurids), with some clades having later experienced a reversal to the propubic condition.[27]

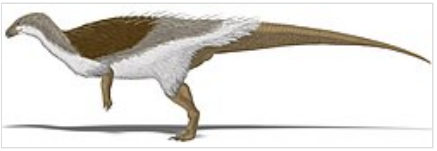
Integument

Cast of preserved *Corythosaurus* skin, showing small arrays of basement scales interspersed with large feature scales

Exceptionally preserved specimens of ornithischians preserved integument, the external tissues scuh as skin, scales, and filaments. Many ornithischians are known to have been covered in scales, similar to those of other dinosaurs. Unlike the overlapping scales of modern lizards, typical ornithischian scales are mosaics of non-overlapping "basement scales", small and of polygonal shape.[28][29][30] In many species, larger feature scales would be interspersed between these basement scales.[29][30][31] An especially abundant record of these scales is known from hadrosaurid mummies, where distinct patterns of scalation can be observed between species.[29][32][33] Other large ornithischians such as ceratopsids, ankylosaurs, iguanodontids and stegosaurs are also known

from traces or preserved scales.[28][30][34][35] Several other types of scalation are found on localized parts of the body in specific taxa. For some examples, hadrosaurs possess "midline feature scales" running along the top of the animal, *Kulindadromeus* and *Haolong* possess large overlapping scales over the tail, and reticulate scales have been documented on hands and feet of ceratopsids and *Kulindadromeus*.[29][31][35][36] In *Psittacosaurus* and *Borealopelta*, the melanosomes of especially well preserved scales have been used to approximate their colour in life.[37][38][39] Some portions of the body would have instead been covered in hard keratin like that found on the claws and beaks of modern birds and reptiles. Examples of these include beaks, the nails on the hands and feet, the horns of horned dinosaurs, the plates of stegosaurs, and the armour of ankylosaurs. The osteoderms found in armored dinosaurs would have been embedded externally within the skin, and covered much of the animal.[28][32][34]

Despite similarities in external scalation, differences in skin thickness can be observed between ornithischians. The small ceratopsian *Psittacosaurus* had skin 1 centimetre (0.39 in) thick, close to the skin thickness of modern rhinos and elephants, whereas large hadrosaurs had especially thin skin only 0.3 centimetres (0.12 in) thick unlike mammalian megafauna.[39]

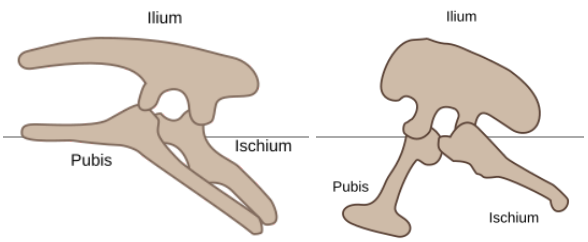


Some ornithischians were covered in a mixture of scales, filaments, and quill-like structures; here, *Thescelosaurus* is depicted with integument inferred from other species

Ornithischians are also known to have possessed quill, spike, and filament-like structures with a disputed connection to the feathers of theropods (including modern birds). Due to being more rarely preserved, their distribution across the group remains more unclear. A specimen of *Psittacosaurus*, otherwise covered in typical scalation, preserves a series of long upward-pointing bristles along the tail of the animal. Likely tubular in life, they are rooted in the skin in clusters.[30][40][41] Extensive filaments are known from the heterodontosaurid *Tianyulong* and the neornithischian *Kulindadromeus*.[12][36][42] Though similar to the bristles of *Psittacosaurus* in being tubular and unbranched, those of *Tianyulong* are denser and preserved under the neck, over the back, and especially long ones over base of the tail. These are similar to the shorter unbranched filaments seen in theropods such as *Sinosauropteryx*. In *Kulindadromeus*, more complex filaments are known.[42] In *Kulindadromeus*, similar unbranched structures are known from across the head and torso. On the limbs, several small filaments branch out of basal plates (which may or may not be scales), similar to downy feathers which lack barbules. Similarly clustered ribbon-like feathers occur on the lower leg. Thusfar, these constitute the only complex filament structures known in ornithischians. Feathers are lacking from the ends of the limbs as well as most of the tail, which are instead scaled, excepting long bristle-like structures emerging from scales along the bottom of the tail.[12][36] The hadrosauroid dinosaur *Haolong* preserves small, thin "cutaneous spikes" of various sizes across the body. Unlike feathers and mammalian hair, these hollow structures were composed of living cells. These were present along the neck and torso and coexisted with basement scales, whilst the tail was strictly scaly.[35] Whether all, some, or none of these structures share an evolutionary origin with true feathers continues to be a matter of debate. Some favor an independent origin, with the earliest dinosaurs being scaled, while other authors favor filaments being present in the earliest dinosaurs or even in the common ancestors of pterosaurs and dinosaurs which subsequently evolved into distinct structures in various lineages.[43][44][45]

Classification

History



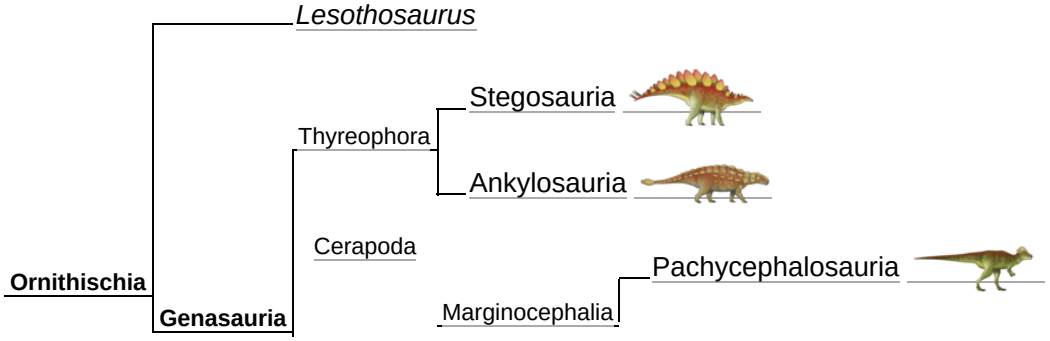
The general pelvis of Ornithischia (left) and Saurischia (right) illustrating the differences highlighted by Seeley (1888) as justifying their independent evolution

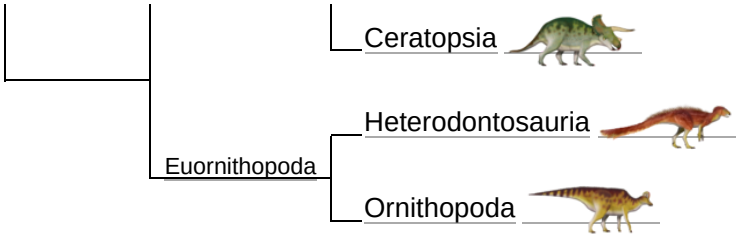
The first recognition of an herbivorous group of dinosaurs was named Orthopoda in 1866 by Edward Drinker Cope,[46] a name that is now recognized as a synonym of Ornithischia.[47] Discussions on the taxonomy of dinosaurs by Othniel Charles Marsh identified two major groups of herbivorous dinosaurs, Ornithopoda and Stegosauria,[48] containing genera from a broad geographic and stratigraphic distribution.[14] While often these groups were placed within Dinosauria, Harry Govier Seeley suggested instead in 1888 that ornithopods and stegosaurs, which shared many features in the skull, limbs, and hip, were unrelated to other dinosaurs, and so he proposed that Dinosauria was an unnatural grouping of two independently-evolved suborders, Saurischia and Ornithischia. It is from the anatomy of the hip that Seeley chose the name Ornithischia, referencing the bird-like anatomy of the ischium bone.[1] Many researchers did not follow the division of Seeley at first, with Marsh naming the group Predentata to unite ornithopods, stegosaurs, and Ceratopsia within Dinosauria,[49] but with additional work and new discoveries the unnatural nature of Dinosauria came to be accepted, and the names Seeley proposed found common use.[50] After further decades, in 1974 Robert T. Bakker and Peter M. Galton provided new evidence in support of the grouping of ornithischians and saurischians together within a natural Dinosauria,[51] which has been supported since.[14]

The first cladistic studies on Ornithischia were published simultaneously in 1984 by David B. Norman, Andrew R. Milner, and Paul C. Sereno. These studies differed somewhat in their results, but found that *Iguanodon* was closer to hadrosaurs than other ornithopods, followed by *Dryosaurus*, *Hypsilophodon* and then *Lesothosaurus* and its relatives.[52][53][54][55][14] While the study of Norman placed ceratopsians between *Hypsilophodon* and more derived ornithopods,[53] the study of Sereno placed ceratopsians with ankylosaurs and stegosaurs.[55] It has since been recognized by that ceratopsians are closer to ornithopods than the armoured ankylosaurs and stegosaurs,[4] but the relationships of some groups are still in states of change, with some more consistent results than others.[56][3] An early study that looked at the relationships within Ornithischia with greater detail was that of Sereno in 1986, who provided features that supported the evolution of all ornithischian groups and shared similarities with earlier studies. Sereno found that *Lesothosaurus* was the most primitive ornithischian, with all other ornithischians united within the clade Genasauria, which has two subgroups. The first subgroup, Thyreophora, unites ankylosaurs and stegosaurs along with more primitive taxa like *Scelidosaurus*, while the second subgroup, Cerapoda, contained ornithopods, ceratopsians, pachycephalosaurs, and small primitive forms.[4] One group of the small primitive forms considered to be cerapodans by Sereno, Heterodontosauridae, has since been found to be a group of very early ornithischians of similar evolutionary status as *Lesothosaurus*,[56] although this result is not definitive.[57]

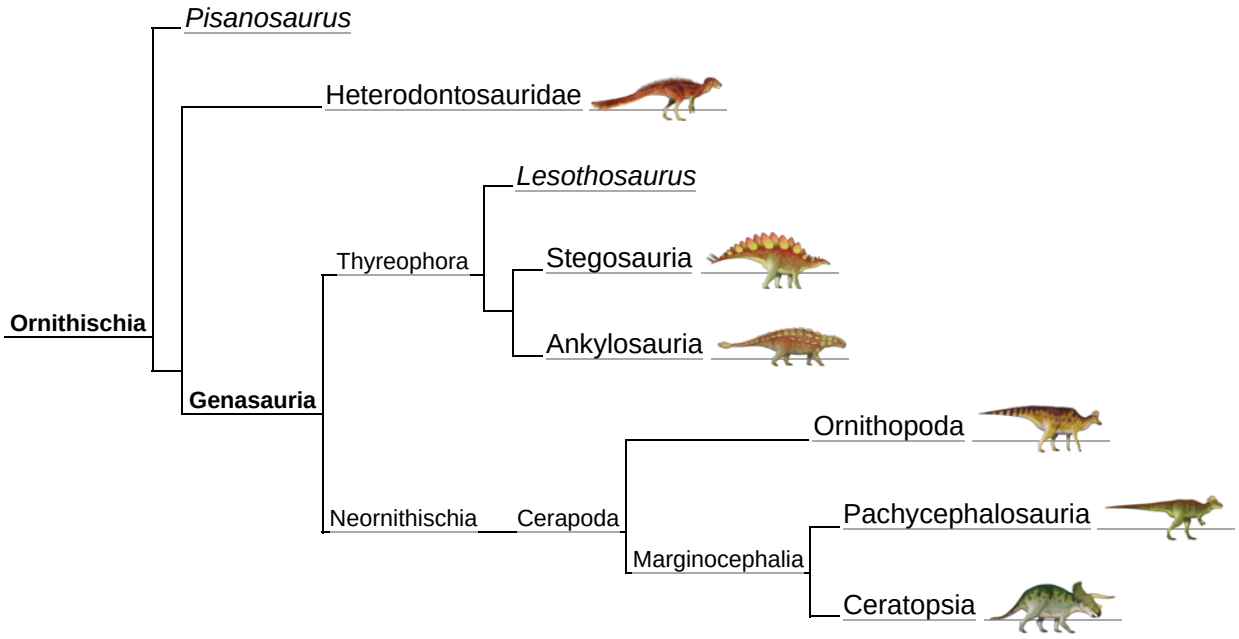
The first large-scale numerical analysis of the phylogenetics of Ornithischia was published in 2008 by Richard J. Butler and colleagues, including many primitive ornithischians and members from all of the major subgroups, to test some of the hypotheses given previously about ornithischian evolution and the relationships of the groups. Thyreophora was found to be a supported group, as well as the clade of pachycephalosaurs and ceratopsians that Sereno named Marginocephalia in 1986. Some taxa considered earlier to be ornithopods, like heterodontosaurids, *Agilisaurus*, *Hexinlusaurus* and *Othnielia*, were instead found to be outside of both Ornithopoda and Ceratopsia, but still closer to those two groups than thyreophorans. The early Argentinian taxon *Pisanosaurus* was found to be the most primitive ornithischian, but while overall results agreed with earlier studies and showed some stability, areas of the evolutionary tree were found to be problematic, and with potential for later change.[56] In 2021, a new phylogenetic study was published authored by Paul-Emile Dieudonné and colleagues that instead found Heterodontosauridae to nest alongside Pachycephalosauria within Marginocephalia, changing the early evolution of ornithopods considerably, and showing that the evolution of ornithischians was far from definitive.[57] Below are the cladograms of Sereno, Butler and colleagues, and Dieudonné and colleagues, restricted to the major clades of Ornithischia, Heterodontosauridae, *Lesothosaurus* and *Pisanosaurus*.

Sereno, 1986[4]

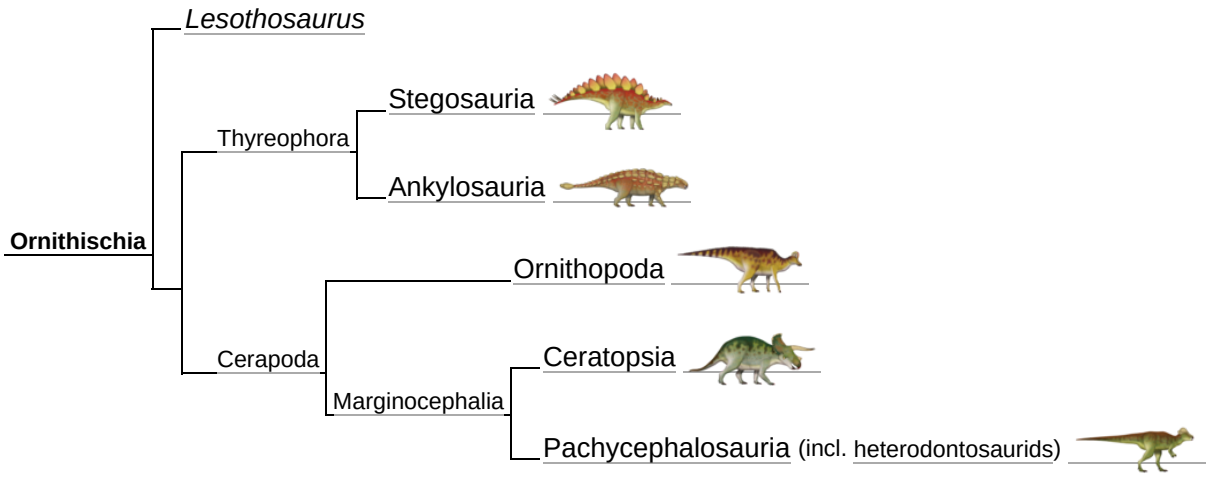




Butler et al., 2008[56]



Dieudonné et al., 2021[57]

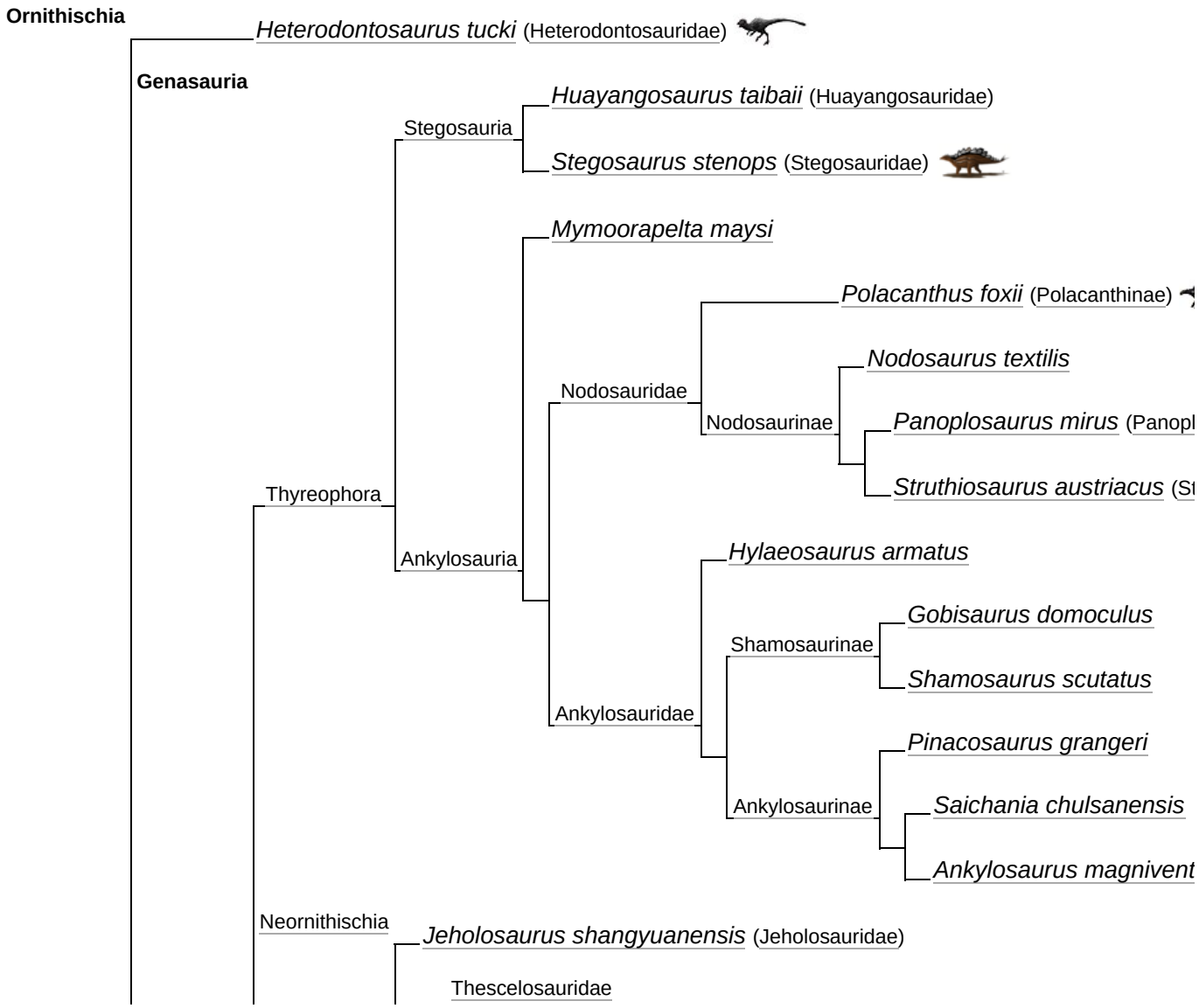


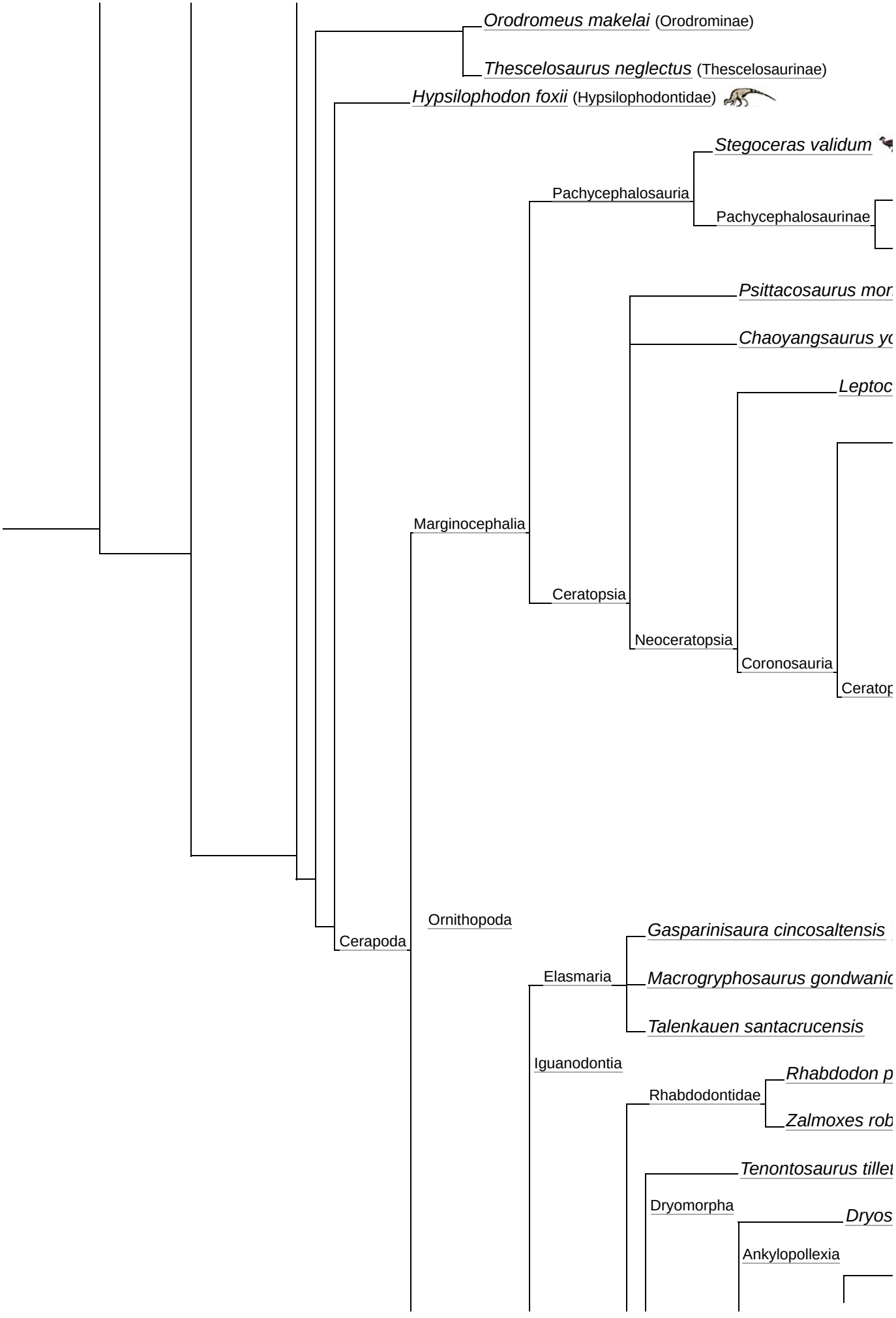
Subgroups

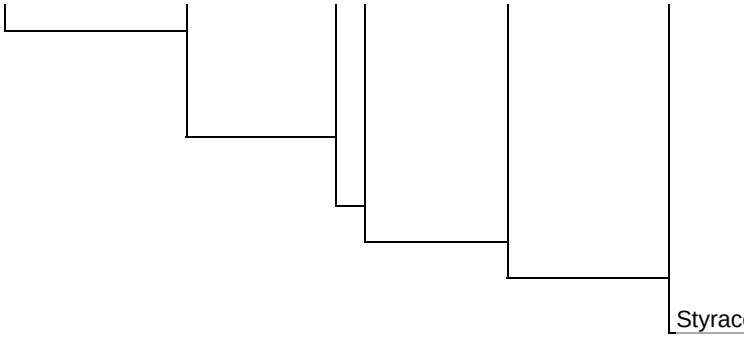
When Ornithischia was first named, Seeley united the orders Ornithopoda and Stegosauria of Marsh's taxonomy within the new group.[1] Ceratopsia was then recognized as a unique group related to ornithopods and stegosaurs by Marsh by 1894, with each of the three suborders still being recognized as distinct groups today.[14][49] Ceratopsians are recognized as group that grew in diversity later in the Cretaceous after evolving in the Late Jurassic, encompassing a diverse array of bodyforms from the small, bipedal *Psittacosaurus* up to the very large, quadrupedal, horned and frilled ceratopsids like *Torosaurus*, which has the longest skull of any terrestrial vertebrate.[24][25] Ornithopods, which range from the Early Jurassic in some studies until the end of the Cretaceous with continuous diversity, are generally bipedal and unarmoured, though some later

groups like Hadrosauridae evolved complex dental anatomy in the form of batteries of teeth.[20][22] Stegosaurs are comparatively limited, restricted to a primarily Jurassic group of moderate to large, quadrupedal herbivores with two rows of vertical plates ornamenting their spine, which possibly did not go extinct until the Late Cretaceous, though at the time of Marsh Stegosauria was used for all armored and quadrupedal taxa, many of which are now separated into Ankylosauria.[18][19] Ankylosaurs were only recognized as a distinct group from stegosaurs in the 1920s despite many members being known for decades before, with the group now encompassing a broad array of heavy, quadrupedal ornithischians with extensive armour covering their body and skull.[19] The fifth recognized major subgroup of ornithischians is Pachycephalosauria,[14] which was first named in 1974 after being confused for a long time with the theropod *Troodon* on account of their similarly omnivorous and unique teeth.[58][23] Pachycephalosaurians are unique for their tall, thickened skulls and small, bipedal bauplan, suggesting that their domes were for sexual display or combat in the form of head-butting or flank-butting.[23] Some taxa, particularly those at one point grouped together in the ornithopod family Hypsilophodontidae, are now recognized to not fall within any of the major ornithischian groups, and either be outside Genasauria, or on the basal stem of Neornithischia outside Cerapoda.[56]

Following the publication of the PhyloCode to provide rules and regulations on the use of taxonomic names for groups, the internal classification of Ornithischia was revised by Daniel Madzia and colleagues in 2021 to provide a framework of definitions and taxa for other studies to follow and modify from. They names the new clade Saphornithischia to unite heterodontosaurids with more derived ornithischians to encompass the concept of the well-supported clear ornithischians, as the origins of the group and the relationships of primitive taxa like *Pisanosaurus* and members of Silesauridae may sometimes be found to be ornithischians outside this core grouping. Madzia and colleagues also provided a composite cladogram of Ornithischia to illustrate the consensus of internal divisions, which can be seen below. Ornithischia has been defined as all taxa closer to *Iguanodon* than *Allosaurus* or *Camarasaurus*. Genasauria has been defined as the smallest clade containing *Ankylosaurus*, *Iguanodon*, *Stegosaurus*, and *Triceratops*.[3]




Orodromeus makelai (Orodrominae)
Thescelosaurus neglectus (Thescelosaurinae)
Hypsilophodon foxii (Hypsilophodontidae)
Stegoceras validum
Pachycephalosauria
Pachycephalosaurinae
Psittacosaurus mor
Chaoyangsaurus yo
Leptoc
Marginocephalia
Ceratopsia
Neoceratopsia
Coronosauria
Ceratop
Ornithopoda
Cerapoda
Gasparinisaura cincosaltensis
Elasmaria
Macrogryphosaurus gondwanic
Talenkauen santacrucensis
Iguanodontia
Rhabdodon p
Rhabdodontidae
Zalmoxes rob
Tenontosaurus tillet
Dryomorpha
Dryos
Ankylopollexia



Multiple taxa within Ornithischia fall around the origin of the group, or cannot be classified definitively. *Lesothosaurus* and *Laquintasaura* have been found as basal thyreophorans or basal ornithischians, *Chilesaurus* is either a theropod or a basal ornithischian, *Pisanosaurus* has been found as a basal ornithischian or a non-ornithischian silesaurid, *Eocursor* has been a basal ornithischian or a basal member of Neornithischia, *Serendipaceratops* cannot be classified beyond Ornithischia as it is either an ankylosaur or a ceratopsian, and *Alocodon*, *Fabrosaurus*, *Ferganocephale*, *Gongbusaurus*, *Taveirosaurus*, *Trimucrodon* and *Xiaosaurus* are dubious ornithischians of uncertain basal classification.[3][59][60][16][61] Depending on the phylogenetic results, Silesauridae could either be a clade within Ornithischia, its members could form an evolutionary gradient, or some members found form a clade while others are part of a gradient.[59][62]

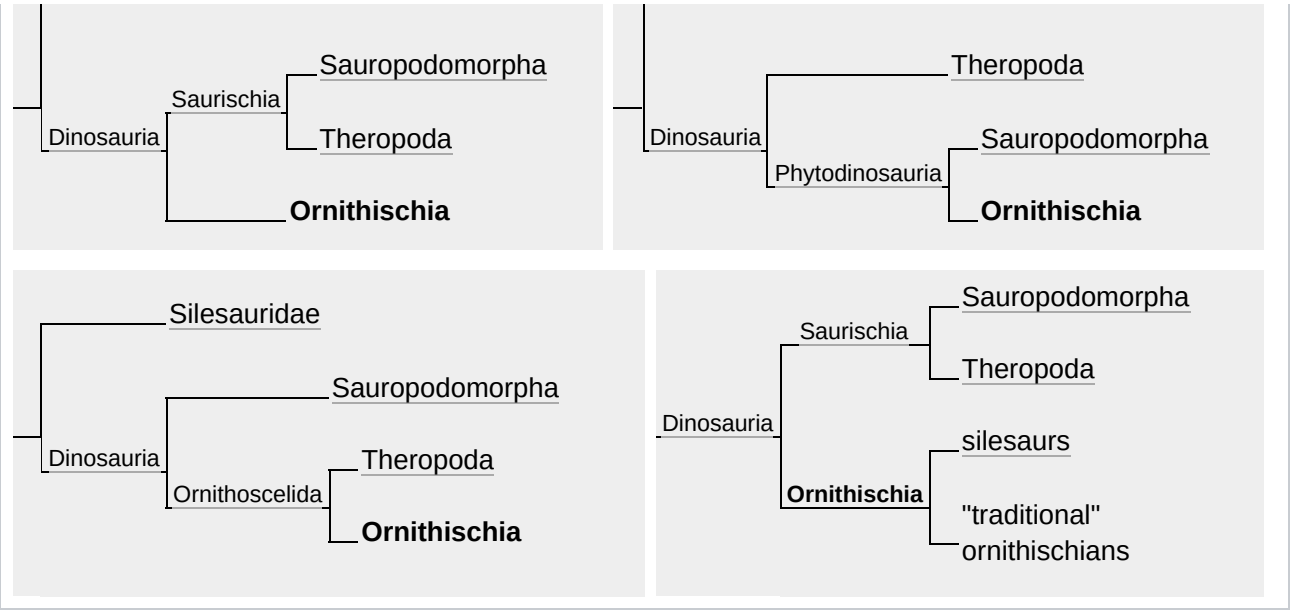
Evolution

Skull of *Revueltosaurus*, a genus originally considered an early ornithischian based on its teeth[63]

For a long time, the only understanding of the origins of Ornithischia came from *Lesothosaurus* and *Pisanosaurus*, which together represented the best-known Early Jurassic and Triassic ornithischians respectively. Many suggestions of taxa and specimens that could be referred to Ornithischia from the Triassic were based on teeth and jaw bones, as they showed similar adaptations for herbivory. The genera *Revueltosaurus*, *Galtonia*, *Pekinosaurus*, *Tecovasaurus*, *Lucianosaurus*, *Protecovasaurus*, *Crosbysaurus*, and *Azendohsaurus* were all at one time considered to be Triassic ornithischians with only their teeth known, but are now recognized to be completely unrelated.[64] The only early ornithischians that were considered to be diagnostic in a 2004 review by Norman and colleagues were *Lesothosaurus*, *Pisanosaurus* and *Technosaurus*, limiting the early ornithischian record to only two Triassic genera from Argentina and the United States and one Early Jurassic genus from South Africa, with all the tooth taxa being considered undiagnostic.[16] Referrals of isolated teeth to Ornithischia based on herbivorous features began to be extensively questioned by William G. Parker and colleagues in 2005 after the discovery of skull and skeleton material clearly from *Revueltosaurus* showing that the "ornithischian-like" teeth were from an animal more closely related to crocodiles than birds, and there were multiple occurrences of herbivory throughout Triassic reptiles.[64][63] Removing the list of Triassic tooth taxa from Ornithischia, the early diversity of the group was substantially reduced, especially in comparison to the known Triassic diversity of theropods and sauropodomorphs. If *Pisanosaurus* represented the earliest ornithischian, there would be at least a 20 million year gap in the evolution of Ornithischia until *Lesothosaurus* and heterodontosaurids. It is possible that the limited early record of ornithischians is due to them inhabiting environments that were less conducive to fossilization, or that the phylogenetics of the group were incorrect and that early ornithischians were already known but identified as members of other groups.[64]

Possible evolutionary relationships of Ornithischia[59]

Silesauridae	Silesauridae

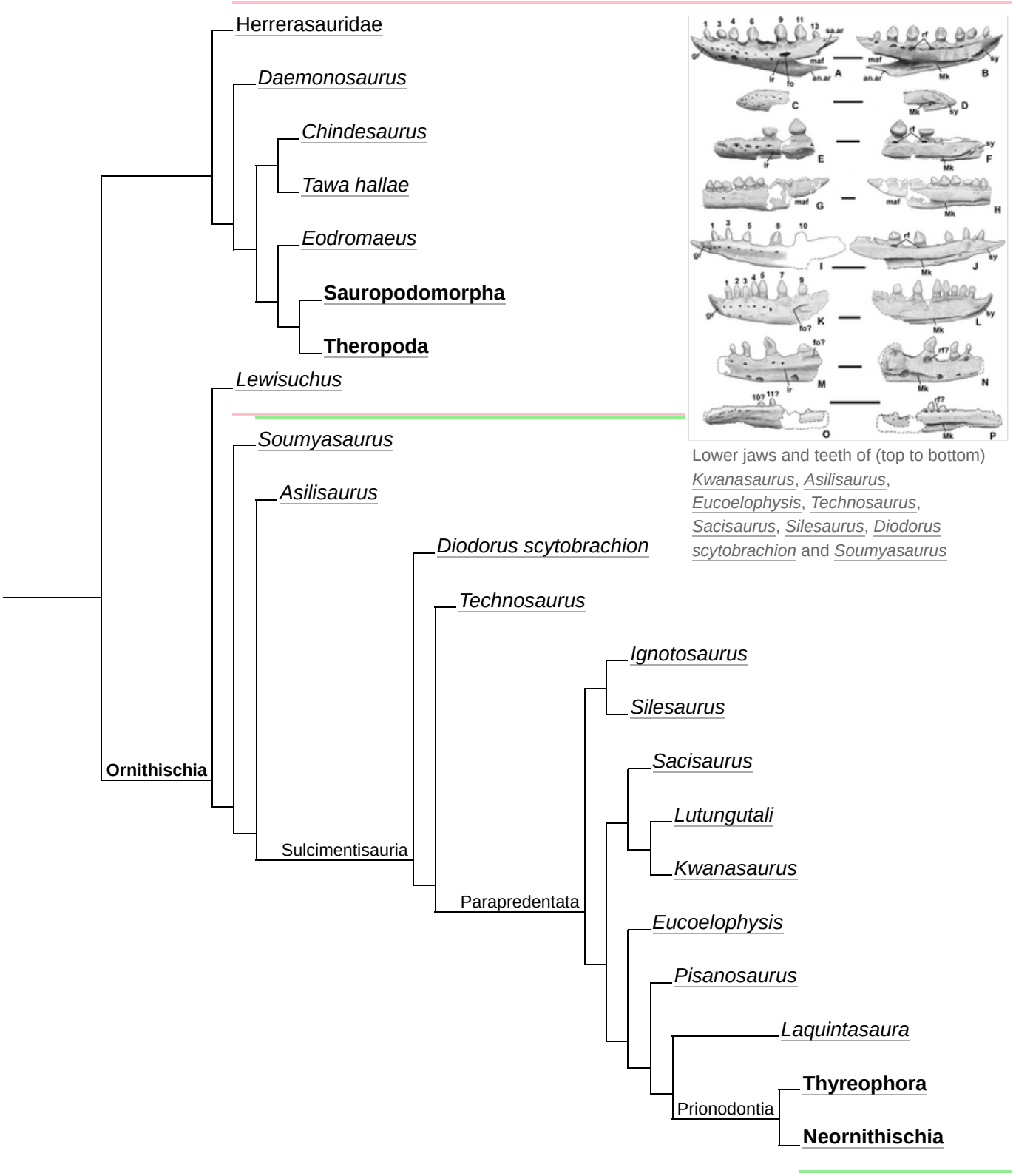


First noted in the 2003 naming of the early taxon *Silesaurus*, some taxa generally considered non-dinosaurs show similarities to ornithischians in the teeth and jaw anatomy.[65][66] These basal taxa, which were then grouped within Silesauridae and commonly as the sister group to Dinosauria, may instead be the earliest ornithischians. They show adaptations for the evolution of herbivory, and can fill in the gap in early evolution of ornithischians that were otherwise only clearly known since the beginning of the Jurassic. This hypothesis has found support in multiple different phylogenetic analyses,[67][68] but the results are not yet accepted as definitive enough to contradict other possible evolutionary strategies of dinosaurs. Alternatively, and more in line with earlier studies on dinosaur evolution, silesaurids may be the sister taxa to the Saurischia-Ornithischia split, or even other arrangements of the three main dinosaur groups Ornithischia, Sauropodomorpha, and Theropoda.[59] The 2017 phylogenetic study of Matthew G. Baron and colleagues suggested that instead of a Saurischia-Ornithischia split, ornithischians were instead closest to theropods in the clade Ornithoscelida, with sauropodomorphs being outside the grouping. Under this case, the omnivory in the earliest sauropodomorphs and ornithischians would be the ancestral condition for dinosaurs, along with the grasping abilities seen in the earliest ornithischians and theropods.[69] While Ornithoscelida is a possible hypothesis for the evolution of dinosaurs and the close relationships of Ornithischia, follow-up studies have not found it statistically more likely than the traditional dichotomy of Ornithischia and Saurischia, or the third alternative, Phytodinosauria, where ornithischians and sauropodomorphs are closer to each other than theropods.[70][71][72]

Along with *Pisanosaurus*, which was supported as the earliest ornithischian for a time before being considered just as likely to be a silesaur rather than an ornithischian, an additional problematic taxon is *Chilesaurus* from the Late Jurassic of Chile. While it was originally named as a derived theropod with unique anatomy, it was found in studies based on Baron and colleagues results to instead be either the basalmost ornithischian, or a sauropodomorph. As the earliest ornithischian, *Chilesaurus* tied multiple details of ornithischian and theropod anatomy together supporting their union in Ornithoscelida, though when it is not the basalmost ornithischian, a traditional Saurischia is recovered. The problematic nature of *Chilesaurus* requires further revisiting of its anatomy, but the details of vertebral air pockets, pelvis shape, and hand support it as a theropod.[59] *Daemonosaurus*, typically a theropod or close relative of herrerasaurs, has also been found as the basalmost ornithischian at times when Ornithoscelida is recovered, but it does not share any unique features with ornithischians and redescribing its anatomy found it fairly confidently to be a basal dinosaur not related closely to Ornithischia.[73]

The phylogenetic analysis of Norman and colleagues in 2022 recovered the members of Silesauridae as forming an ancestral grade within Ornithischia even with the inclusion of *Chilesaurus*, supporting the earlier results of Müller and Garcia and their evolutionary trends for early ornithischian anatomy. Norman and colleagues used Prionodontia over both Saphornithischia and Genasauria, since all were recovered as encompassing the same node.[59] The earliest ornithischians under this reconstruction were faunivorous, as seen by *Lewisuchus*, which has typical teeth like theropods. Serrations on teeth become larger for taxa more derived than *Asilisaurus*, the development of a cingulum in teeth is seen in *Technosaurus* and later ornithischians, the lower jaw becomes more elongate in taxa above *Silesaurus*, and core ornithischians are united by the pubic bone angling backwards, and the modification of the ankle joint.[68]

Dinosauria **Saurischia**



Lower jaws and teeth of (top to bottom) *Kwanasaurus*, *Asilisaurus*, *Eucoelophysis*, *Technosaurus*, *Sacisaurus*, *Silesaurus*, *Diodorus scytobrachion* and *Soumyasaurus*

Palaeoecology

Ornithischians shifted from bipedal to quadrupedal posture at least three times in their evolutionary history and it has been shown primitive members may have been capable of both forms of movement.[74]

Most ornithischians were herbivorous.[6] In fact, most of the unifying characters of Ornithischia are thought to be related to this herbivory.[6] For example, the shift to an opisthopubic pelvis is thought to be related to the development of a large stomach or stomachs and gut which would allow ornithischians to more effectively digest plant matter.[6] The smallest

Ornithischians, such as *Zuul*, would have been preyed on by theropods like *Gorgosaurus*

known ornithischian is *Fruitadens haagarorum*.[75] The largest *Fruitadens* individuals reached just 65–75 cm. Previously, only carnivorous, saurischian theropods were known to reach such small sizes.[75] At the other end of the spectrum, the largest known ornithischians reach about 15 meters (smaller than the largest saurischians).[76]

However, not all ornithischians were strictly herbivorous. Some groups, like the heterodontosaurids, were likely omnivores.[77] At least one species of ankylosaurian, *Liaoningosaurus paradoxus*, appears to have been at least partially carnivorous, with hooked claws, fork-like teeth, and stomach contents suggesting that it may have fed on fish.[78] The members of Genasauria were primarily herbivores.[79] Genasaurians most often had their head at the level of one meter, which suggests they were feeding primarily on "ground-level plants such as ferns, cycads, and other herbaceous gymnosperms."[80]

There is strong evidence that some ornithischians lived in herds.[6][9] This evidence consists of multiple bone beds where large numbers of individuals of the same species and of different age groups died simultaneously.[6][9]

See also

- Dinosaurs portal

Notes

a. plural *ischia*

References

1. Seeley, H.G. (1888). "On the classification of the fossil animals commonly named Dinosauria". *Proceedings of the Royal Society of London*. **43** (258–265): 165–171. doi:10.1098/rspl.1887.0117 (https://doi.org/10.1098%2Frspl.1887.0117).
2. Fonseca, A.O.; Reid, I.J.; Venner, A.; Duncan, R.J.; Garcia, M.S.; Müller, R.T. (2024). "A comprehensive phylogenetic analysis on early ornithischian evolution". *Journal of Systematic Palaeontology*. **22** (1) 2346577. Bibcode:2024JSPal..2246577F (https://ui.adsabs.harvard.edu/abs/2024JSPal..2246577F). doi:10.1080/14772019.2024.2346577 (https://doi.org/10.1080%2F14772019.2024.2346577).
3. Madzia, D.; Arbour, V.M.; Boyd, C.A.; Farke, A.A.; Cruzado-Caballero, P.; Evans, D.C. (2021). "The phylogenetic nomenclature of ornithischian dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8667728). *PeerJ*. **9** e12362. doi:10.7717/peerj.12362 (https://doi.org/10.7717%2Fpeerj.12362). PMC 8667728 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8667728). PMID 34966571 (https://pubmed.ncbi.nlm.nih.gov/34966571).
4. Sereno, P.C. (1986). "Phylogeny of the Bird-Hipped Dinosaurs (Ornithischia)". *National Geographic Research*. **2** (2): 234–256.
5. Reid, I.J.; Fonseca, A.O.; Madzia, D. (2026). "The evolution of ornithischian nomenclature". *Zoological Journal of the Linnean Society*. **207** (3) zlag111. doi:10.1093/zoolinnean/zlag111 (https://doi.org/10.1093%2Fzoolinnean%2Fzlag111).
6. Fastovsky, David E.; Weishampel, David B. (2012). *Dinosaurs: A Concise Natural History*. Cambridge: Cambridge University Press. ISBN 978-1-107-27646-8.
7. Colbert, Edwin H. (Edwin Harris); Knight, Charles Robert (1951). *The dinosaur book: the ruling reptiles and their relatives* (https://archive.org/details/bookruli00colb/page/152/mode/2up). New York: McGraw-Hill. p. 152.
8. "Prehistoric Earth". *The Ultimate Visual Family Dictionary*. New Delhi: DK Pub. 2012. p. 82-83. ISBN 978-0-1434-1954-9.

9. Qi, Zhao; Barrett, Paul M.; Eberth, David A. (2007-09-01). "Social Behaviour and Mass Mortality in the Basal Ceratopsian Dinosaur Psittacosaurus (early Cretaceous, People's Republic of China)" (http://doc.rero.ch/record/16127/files/PAL_E2504.pdf) (PDF). *Palaeontology*. **50** (5): 1023–1029. Bibcode:2007Palgy..50.1023Q (https://ui.adsabs.harvard.edu/abs/2007Palgy..50.1023Q). doi:10.1111/j.1475-4983.2007.00709.x (https://doi.org/10.1111%2Fj.1475-4983.2007.00709.x). ISSN 1475-4983 (https://search.worldcat.org/issn/1475-4983). S2CID 128781816 (https://api.semanticscholar.org/CorpusID:128781816).
10. Zhao, Q. (2013). "Juvenile-only clusters and behaviour of the Early Cretaceous dinosaur *Psittacosaurus*" (https://doi.org/10.4202%2Fapp.2012.0128). *Acta Palaeontologica Polonica*. doi:10.4202/app.2012.0128 (https://doi.org/10.4202%2Fapp.2012.0128).
11. Mayr, Gerald; Peters, Stefan D.; Plodowski, Gerhard; Vogel, Olaf (2002-08-01). "Bristle-like integumentary structures at the tail of the horned dinosaur Psittacosaurus". *Naturwissenschaften*. **89** (8): 361–365. Bibcode:2002NW.....89..361M (https://ui.adsabs.harvard.edu/abs/2002NW.....89..361M). doi:10.1007/s00114-002-0339-6 (https://doi.org/10.1007%2Fs00114-002-0339-6). ISSN 0028-1042 (https://search.worldcat.org/issn/0028-1042). PMID 12435037 (https://pubmed.ncbi.nlm.nih.gov/12435037). S2CID 17781405 (https://api.semanticscholar.org/CorpusID:17781405).
12. Godefroit, P.; Sinitsa, S.M.; Dhouailly, D.; Bolotsky, Y.L.; Sizov, A.V.; McNamara, M.E.; Benton, M.J.; Spagna, P. (2014). "A Jurassic ornithischian dinosaur from Siberia with both feathers and scales" (https://web.archive.org/web/20190209232112/http://palaeo.gly.bris.ac.uk/Benton/reprints/2014Kulinda.pdf) (PDF). *Science*. **345** (6195): 451–455. Bibcode:2014Sci...345..451G (https://ui.adsabs.harvard.edu/abs/2014Sci...345..451G). doi:10.1126/science.1253351 (https://doi.org/10.1126%2Fscience.1253351). hdl:1983/a7ae6dfb-55bf-4ca4-bd8b-a5ea5f323103 (https://hdl.handle.net/1983%2Fa7ae6dfb-55bf-4ca4-bd8b-a5ea5f323103). PMID 25061209 (https://pubmed.ncbi.nlm.nih.gov/25061209). S2CID 206556907 (https://api.semanticscholar.org/CorpusID:206556907). Archived from the original (http://palaeo.gly.bris.ac.uk/Benton/reprints/2014Kulinda.pdf) (PDF) on 2019-02-09. Retrieved 2016-08-28.
13. Paul, G.S. (2024). *The Princeton Field Guide to Dinosaurs* (3rd ed.). Princeton University Press. pp. 261–373. ISBN 978-0-691-23157-0.
14. Weishampel, D.B. (2004). "Ornithischia". In Weishampel, D.B.; Osmólska, H.; Dodson, P. (eds.). *The Dinosauria* (2nd ed.). University of California Press. pp. 323–324. ISBN 0-520-24209-2.
15. Colbert, E. H. (1981). A primitive ornithischian dinosaur from the Kayenta Formation of Arizona. *Museum Northern Arizona Bull*. **53**, 1-61
16. Norman, D.B.; Witmer, L.M.; Weishampel, D.B. (2004). "Basal Ornithischia". In Weishampel, D.B.; Osmólska, H.; Dodson, P. (eds.). *The Dinosauria* (2nd ed.). University of California Press. pp. 325–334. ISBN 0-520-24209-2.
17. Norman, D.B.; Witmer, L.M.; Weishampel, D.B. (2004). "Basal Thyreophora". In Weishampel, D.B.; Osmólska, H.; Dodson, P. (eds.). *The Dinosauria* (2nd ed.). University of California Press. pp. 335–342. ISBN 0-520-24209-2.
18. Galton, P.M.; Upchurch, P.C. (2004). "Stegosauria". In Weishampel, D.B.; Osmólska, H.; Dodson, P. (eds.). *The Dinosauria* (2nd ed.). University of California Press. pp. 343–362. ISBN 0-520-24209-2.
19. Vickaryous, M.K.; Maryańska, T.; Weishampel, D.B. (2004). "Ankylosauria". In Weishampel, D.B.; Osmólska, H.; Dodson, P. (eds.). *The Dinosauria* (2nd ed.). University of California Press. pp. 363–392. ISBN 0-520-24209-2.
20. Norman, D.B.; Sues, H.-D.; Witmer, L.M.; Coria, R.A. (2004). "Basal Ornithopoda". In Weishampel, D.B.; Osmólska, H.; Dodson, P. (eds.). *The Dinosauria* (2nd ed.). University of California Press. pp. 393–412. ISBN 0-520-24209-2.
21. Norman, D.B. (2004). "Basal Iguanodontia". In Weishampel, D.B.; Osmólska, H.; Dodson, P. (eds.). *The Dinosauria* (2nd ed.). University of California Press. pp. 413–437. ISBN 0-520-24209-2.
22. Horner, J.R.; Weishampel, D.B.; Forster, C.A. (2004). "Hadrosauridae". In Weishampel, D.B.; Osmólska, H.; Dodson, P. (eds.). *The Dinosauria* (2nd ed.). University of California Press. pp. 438–463. ISBN 0-520-24209-2.
23. Maryańska, T.; Chapman, R.E.; Weishampel, D.B. (2004). "Pachycephalosauria". In Weishampel, D.B.; Osmólska, H.; Dodson, P. (eds.). *The Dinosauria* (2nd ed.). University of California Press. pp. 464–477. ISBN 0-520-24209-2.
24. You, H.; Dodson, P.; Weishampel, D.B. (2004). "Basal Ceratopsia". In Weishampel, D.B.; Osmólska, H.; Dodson, P. (eds.). *The Dinosauria* (2nd ed.). University of California Press. pp. 478–494. ISBN 0-520-24209-2.
25. Dodson, P.; Forster, C.A.; Sampson, S.D. (2004). "Ceratopsidae". In Weishampel, D.B.; Osmólska, H.; Dodson, P. (eds.). *The Dinosauria* (2nd ed.). University of California Press. pp. 494–513. ISBN 0-520-24209-2.
26. Currie, Philip J.; Padian, Kevin (1997-10-06). *Encyclopedia of Dinosaurs* (https://books.google.com/books?id=7t9M5TsmjOUC). Academic Press. pp. 537–538. ISBN 978-0-08-049474-6.

27. Holtz, T.R. and Osmólska, H. (2004). "Saurischia." In Weishampel, Dodson and Osmólska (eds.), *The Dinosauria*, second edition. Berkeley: University of California Press.
28. Arbour, Victoria M.; Burns, Michael E.; Bell, Phil R.; Currie, Phillip J. (2014). "Epidermal and Dermal Integumentary Structures of Ankylosaurian Dinosaurs". *Journal of Morphology*. **275** (1): 39–50. doi:10.1002/jmor.20194 (https://doi.org/10.1002%2Fjmor.20194). PMID 24105904 (https://pubmed.ncbi.nlm.nih.gov/24105904).
29. Bell, P.R. (2014). "A Review of Hadrosaurid Skin Impressions". In Eberth, D.A.; Evans, D.C. (eds.). *Hadrosaurs*. Indiana University Press. pp. 572–590. ISBN 978-0-253-01385-9.
30. Bell, Phil R.; Hendrickx, Christophe; Pittman, Michael; Kaye, Thomas G.; Mayr, Gerald (August 12, 2022). "The exquisitely preserved integument of Psittacosaurus and the scaly skin of ceratopsian dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9374759). *Communications Biology*. **5** (1): 809. doi:10.1038/s42003-022-03749-3 (https://doi.org/10.1038%2Fs42003-022-03749-3). ISSN 2399-3642 (https://search.worldcat.org/issn/2399-3642). PMC 9374759 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9374759). PMID 35962036 (https://pubmed.ncbi.nlm.nih.gov/35962036).
31. Enriquez, N. J.; Campione, N. E.; Hendrickx, C.; Bell, P. R. (2025). "Epidermal scale growth, allometry and function in non-avian dinosaurs and extant reptiles" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12265030). *Journal of Anatomy*. **247** (2): 250–283. doi:10.1111/joa.14247 (https://doi.org/10.1111%2Fjoa.14247). PMC 12265030 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12265030). PMID 40102911 (https://pubmed.ncbi.nlm.nih.gov/40102911).
32. Bell, P.R. (2012). "Standardized Terminology and Potential Taxonomic Utility for Hadrosaurid Skin Impressions: A Case Study for *Saurolophus* from Canada and Mongolia" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3272031). *PLOS ONE*. **7** (2) e31295. Bibcode:2012PLoSO...731295B (https://ui.adsabs.harvard.edu/abs/2012PLoSO...731295B). doi:10.1371/journal.pone.0031295 (https://doi.org/10.1371%2Fjournal.pone.0031295). PMC 3272031 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3272031). PMID 22319623 (https://pubmed.ncbi.nlm.nih.gov/22319623).
33. Sereno, Paul C.; Saitta, Evan T.; Vidal, Daniel; Myhrvold, Nathan; Real, María Ciudad; Baumgart, Stephanie L.; Bop, Lauren L.; Keillor, Tyler M.; Eriksen, Marcus; Derstler, Kraig (2025-12-18). "Duck-billed dinosaur fleshy midline and hooves reveal terrestrial clay-template "mummification" " (https://www.science.org/doi/10.1126/science.adw3536). *Science*. **391** (6780) eadw3536. doi:10.1126/science.adw3536 (https://doi.org/10.1126%2Fscience.adw3536).
34. Christiansen, N.A.; Tschopp, E. (2010). "Exceptional stegosaur integument impressions from the Upper Jurassic Morrison Formation of Wyoming" (https://doi.org/10.1007%2Fs00015-010-0026-0). *Swiss Journal of Geosciences*. **103** (2): 163–171. Bibcode:2010SwJG..103..163C (https://ui.adsabs.harvard.edu/abs/2010SwJG..103..163C). doi:10.1007/s00015-010-0026-0 (https://doi.org/10.1007%2Fs00015-010-0026-0). S2CID 129246092 (https://api.semanticscholar.org/CorpusID:129246092).
35. Huang, J.; Wu, W.; Mao, L.; Bertozzo, F.; Dhouailly, D.; Robin, N.; Pittman, M.; Kaye, T. G.; Manucci, F.; He, X.; Wang, X.; Godefroit, P. (2026). "Cellular-level preservation of cutaneous spikes in an Early Cretaceous iguanodontian dinosaur". *Nature Ecology & Evolution*: 1–8. doi:10.1038/s41559-025-02960-9 (https://doi.org/10.1038%2Fs41559-025-02960-9).
36. Godefroit, P.; Sinitsa, S.M.; Cincotta, A.; McNamara, M.E.; Reshetova, S.A.; Dhouailly, D. (2020). "Integumentary Structures in Kulindadromeus zabaikalicus, a Basal Neornithischian Dinosaur from the Jurassic of Siberia". In Foth, C.; Rauhut, O. (eds.). *The Evolution of Feathers: From Their Origin to the Present*. Fascinating Life Sciences. Cham: Springer International Publishing. pp. 47–65. doi:10.1007/978-3-030-27223-4_4 (https://doi.org/10.1007%2F978-3-030-27223-4_4). ISBN 978-3-030-27222-7. S2CID 212669998 (https://api.semanticscholar.org/CorpusID:212669998).
37. Vinther, Jakob; Nicholls, Robert; Lautenschlager, Stephan; Pittman, Michael; Kaye, Thomas G.; Rayfield, Emily; Mayr, Gerald; Cuthill, Innes C. (2016). "3D Camouflage in an Ornithischian Dinosaur" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5049543). *Current Biology*. **26** (18): 2456–2462. Bibcode:2016CBio...26.2456V (https://ui.adsabs.harvard.edu/abs/2016CBio...26.2456V). doi:10.1016/j.cub.2016.06.065 (https://doi.org/10.1016%2Fj.cub.2016.06.065). PMC 5049543 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5049543). PMID 27641767 (https://pubmed.ncbi.nlm.nih.gov/27641767).
38. Brown, C.M.; Henderson, D.M.; Vinther, J.; Fletcher, I.; Sistiaga, A.; Herrera, J.; Summons, R.E. (2017). "An Exceptionally Preserved Three-Dimensional Armored Dinosaur Reveals Insights into Coloration and Cretaceous Predator-Prey Dynamics" (https://doi.org/10.1016%2Fj.cub.2017.06.071). *Current Biology*. **27** (16): 2514–2521.e3. Bibcode:2017CBio...27E2514B (https://ui.adsabs.harvard.edu/abs/2017CBio...27E2514B). doi:10.1016/j.cub.2017.06.071 (https://doi.org/10.1016%2Fj.cub.2017.06.071). hdl:1983/281b9117-11bf-495e-8b98-974d3f550fb2 (https://hdl.handle.net/1983%2F281b9117-11bf-495e-8b98-974d3f550fb2). PMID 28781051 (https://pubmed.ncbi.nlm.nih.gov/28781051).
39. Matteo Fabbri; Jasmina Wiemann; Fabio Manucci; Derek E. G. Briggs (2020). "Three-dimensional soft tissue preservation revealed in the skin of a non-avian dinosaur" (https://doi.org/10.1111%2Fpala.12470). *Palaeontology*. **63** (2): 185–193. doi:10.1111/pala.12470 (https://doi.org/10.1111%2Fpala.12470).

40. Mayr, G.; Peters, S. D.; Plodowski, G.; Vogel, O. (2002). "Bristle-like integumentary structures at the tail of the horned dinosaur *Psittacosaurus*" (http://www.miketaylor.org.uk/tmp/papers/gmayr43.pdf) (PDF). *Naturwissenschaften*. **89** (8): 361–365. Bibcode:2002NW.....89..361M (https://ui.adsabs.harvard.edu/abs/2002NW.....89..361M). doi:10.1007/s00114-002-0339-6 (https://doi.org/10.1007%2Fs00114-002-0339-6). PMID 12435037 (https://pubmed.ncbi.nlm.nih.gov/12435037). S2CID 17781405 (https://api.semanticscholar.org/CorpusID:17781405).
41. Mayr, Gerald; Pittman, Michael; Saitta, Evan; Kaye, Thomas G.; Vinther, Jakob; Benson, Roger (2016). "Structure and homology of *Psittacosaurus* tail bristles" (https://research-information.bristol.ac.uk/en/publications/structure-and-homology-of-psittacosaurus-tail-bristles(029c668f-08b9-45f6-a0c5-30ce9256e593).html). *Palaeontology*. **59** (6): 793–802. Bibcode:2016Palgy..59..793M (https://ui.adsabs.harvard.edu/abs/2016Palgy..59..793M). doi:10.1111/pala.12257 (https://doi.org/10.1111%2Fpala.12257). hdl:1983/029c668f-08b9-45f6-a0c5-30ce9256e593 (https://hdl.handle.net/1983%2F029c668f-08b9-45f6-a0c5-30ce9256e593). S2CID 89156313 (https://api.semanticscholar.org/CorpusID:89156313).
42. Zheng, Xiao-Ting; You, Hai-Lu; Xu, Xing; Dong, Zhi-Ming (19 March 2009). "An Early Cretaceous heterodontosaurid dinosaur with filamentous integumentary structures". *Nature*. **458** (7236): 333–336. Bibcode:2009Natur.458..333Z (https://ui.adsabs.harvard.edu/abs/2009Natur.458..333Z). doi:10.1038/nature07856 (https://doi.org/10.1038%2Fnature07856). PMID 19295609 (https://pubmed.ncbi.nlm.nih.gov/19295609). S2CID 4423110 (https://api.semanticscholar.org/CorpusID:4423110).
43. Barrett, Paul M.; Evans, David C.; Campione, Nicolás E. (2015-06-30). "Evolution of dinosaur epidermal structures" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4528472). *Biology Letters*. **11** (6) 20150229. doi:10.1098/rsbl.2015.0229 (https://doi.org/10.1098%2Frsbl.2015.0229). PMC 4528472 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4528472). PMID 26041865 (https://pubmed.ncbi.nlm.nih.gov/26041865).
44. Xu, Xing (2020). "Filamentous Integuments in Nonavialan Theropods and Their Kin: Advances and Future Perspectives for Understanding the Evolution of Feathers". *The Evolution of Feathers*. Fascinating Life Sciences. pp. 67–78. doi:10.1007/978-3-030-27223-4_5 (https://doi.org/10.1007%2F978-3-030-27223-4_5). ISBN 978-3-030-27222-7. S2CID 216384668 (https://api.semanticscholar.org/CorpusID:216384668).
45. Xu, X.; Barrett, P. M. (2025). "The origin and early evolution of feathers: implications, uncertainties and future prospects" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11837858). *Biology Letters*. **21** (2). 20240517. doi:10.1098/rsbl.2024.0517 (https://doi.org/10.1098%2Frsbl.2024.0517). PMC 11837858 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11837858). PMID 39969251 (https://pubmed.ncbi.nlm.nih.gov/39969251).
46. Cope, E.D. (1866). "The anomalous relations existing between the tibia and fibula in certain of the Dinosauria". *Proceedings of the Academy of Natural Sciences of Philadelphia*. **18**: 316–317.
47. Kuhn, O. (1946). "Das System der fossilen und rezenten Amphibien und Reptilien". *Bericht der Naturforschenden Gesellschaft in Bamberg*. **29**: 49–67.
48. Marsh, O.C. (1881). "Principal characters of the American Jurassic Dinosaurs". *American Journal of Science*. **21** (3): 417–423.
49. Marsh, O.C. (1894). "The typical Ornithopoda of the American Jurassic" (https://zenodo.org/record/2161731). *American Journal of Science*. **48** (283): 85–90. Bibcode:1894AmJS...48...85M (https://ui.adsabs.harvard.edu/abs/1894AmJS...48...85M). doi:10.2475/ajs.s3-48.283.85 (https://doi.org/10.2475%2Fajs.s3-48.283.85).
50. Romer, A.S. (1966). "Dinosaurs". *Vertebrate Paleontology* (3rd ed.). University of Chicago Press. pp. 148–163. ISBN 0-7167-1822-7.
51. Bakker, R.T.; Galton, P.M. (1974). "Dinosaur Monophyly and a New Class of Vertebrates". *Nature*. **248** (5444): 168–172. Bibcode:1974Natur.248..168B (https://ui.adsabs.harvard.edu/abs/1974Natur.248..168B). doi:10.1038/248168a0 (https://doi.org/10.1038%2F248168a0).
52. Milner, A.R.; Norman, D.B. (1984). "The biogeography of advanced ornithopod dinosaurs (Archosauria: Ornithischia) - a cladistic-vicariance model". In Reif, W.E.; Westphal, F. (eds.). *Third Symposium on Mesozoic Terrestrial Ecosystems, Short Papers*. ATTEMPTO Verlag. pp. 145–150.
53. Norman, D.B. (1984). "A systematic reappraisal of the reptile order Ornithischia". In Reif, W.E.; Westphal, F. (eds.). *Third Symposium on Mesozoic Terrestrial Ecosystems, Short Papers*. ATTEMPTO Verlag. pp. 157–162.
54. Norman, D.B. (1984). "On the Cranial Morphology and Evolution of Ornithopod Dinosaurs". In Ferguson, M.W.J. (ed.). *The Structure, Development, and Evolution of Reptiles*. Vol. 52. Symposia of the Zoological Society of London. pp. 521–547.
55. Sereno, P.C. (1984). "Thy phylogeny of the Ornithischia : a reappraisal". In Reif, W.E.; Westphal, F. (eds.). *Third Symposium on Mesozoic Terrestrial Ecosystems, Short Papers*. ATTEMPTO Verlag. pp. 219–226.
56. Butler, R.J.; Upchurch, P.; Norman, D.B. (2008). "The phylogeny of the ornithischian dinosaurs". *Journal of Systematic Palaeontology*. **6** (1): 1–40. Bibcode:2008JSPal...6....1B (https://ui.adsabs.harvard.edu/abs/2008JSPal...6....1B). doi:10.1017/S1477201907002271 (https://doi.org/10.1017%2FS1477201907002271).

57. Dieudonné, P.-E.; Cruzado-Caballero, P.; Godefroit, P.; Tortosa, T. (2021). "A new phylogeny of cerapodan dinosaurs" (https://doi.org/10.1080%2F08912963.2020.1793979). *Historical Biology*. **33** (10): 2335–2355. Bibcode:2021HBio...33.2335D (https://ui.adsabs.harvard.edu/abs/2021HBio...33.2335D). doi:10.1080/08912963.2020.1793979 (https://doi.org/10.1080%2F08912963.2020.1793979). hdl:11336/147083 (https://hdl.handle.net/11336%2F147083).
58. Sternberg, C.M. (1945). "Pachycephalosauridae Proposed for Dome-Headed Dinosaurs, *Stegoceras lambei*, n. sp., Described". *Journal of Paleontology*. **19** (5): 534–538.
59. Norman, D.B.; Baron, M.G.; Garcia, M.S.; Müller, R.T. (2022). "Taxonomic, palaeobiological and evolutionary implications of a phylogenetic hypothesis for Ornithischia (Archosauria: Dinosauria)". *Zoological Journal of the Linnean Society*. **196** (4): 1273–1309. doi:10.1093/zoolinnean/zlac062 (https://doi.org/10.1093%2Fzoolinnean%2Fzlac062).
60. Poropat, S.F.; Bell, P.R.; Hart, L.J.; Salisbury, S.W.; Kear, B.P. (2023). "An annotated checklist of Australian Mesozoic tetrapods". *Alcheringa: An Australasian Journal of Palaeontology*. **47** (2): 129–205. Bibcode:2023Alch...47..129P (https://ui.adsabs.harvard.edu/abs/2023Alch...47..129P). doi:10.1080/03115518.2023.2228367 (https://doi.org/10.1080%2F03115518.2023.2228367). hdl:20.500.11937/96166 (https://hdl.handle.net/20.500.11937%2F96166).
61. Sullivan, R.M. (2006). "A taxonomic review of the Pachycephalosauridae (Dinosauria: Ornithischia)". *New Mexico Museum of Natural History and Science Bulletin*. **35**: 347–366.
62. Müller, R.T.; Garcia, M.S. (2023). "A new silesaurid from Carnian beds of Brazil flls a gap in the radiation of avian line archosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10090097). *Scientific Reports*. **13** (1): 4981. Bibcode:2023NatSR..13.4981M (https://ui.adsabs.harvard.edu/abs/2023NatSR..13.4981M). doi:10.1038/s41598-023-32057-x (https://doi.org/10.1038%2Fs41598-023-32057-x). PMC 10090097 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10090097). PMID 37041170 (https://pubmed.ncbi.nlm.nih.gov/37041170).
63. Parker, W.G.; Irmis, R.B.; Nesbitt, S.J.; Martz, J.W.; Browne, L.S. (2005). "The Late Triassic pseudosuchian *Revueltosaurus callenderi* and its implications for the diversity of early ornithischian dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1564089). *Proceedings of the Royal Society B*. **272** (1566): 963–969. doi:10.1098/rspb.2004.3047 (https://doi.org/10.1098%2Frspb.2004.3047). PMC 1564089 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1564089). PMID 16024353 (https://pubmed.ncbi.nlm.nih.gov/16024353).
64. Irmis, R.B.; Parker, W.G.; Nesbitt, S.J.; Liu, J. (2007). "Early ornithischian dinosaurs: the Triassic record". *Historical Biology*. **19** (1): 3–22. Bibcode:2007HBio...19....3I (https://ui.adsabs.harvard.edu/abs/2007HBio...19....3I). doi:10.1080/08912960600719988 (https://doi.org/10.1080%2F08912960600719988).
65. Dzik, J. (2003). "A beaked herbivorous archosaur with dinosaur affinities from the early Late Triassic of Poland". *Journal of Vertebrate Paleontology*. **23** (3): 556–574. Bibcode:2003JVPal..23..556D (https://ui.adsabs.harvard.edu/abs/2003JVPal..23..556D). doi:10.1671/A1097 (https://doi.org/10.1671%2FA1097).
66. Ferigolo, J.; Langer, M.C. (2007). "A Late Triassic dinosauriform from south Brazil and the origin of the ornithischian predentary bone". *Historical Biology*. **19** (1): 23–33. Bibcode:2007HBio...19...23F (https://ui.adsabs.harvard.edu/abs/2007HBio...19...23F). doi:10.1080/08912960600845767 (https://doi.org/10.1080%2F08912960600845767).
67. Cabreira, S.F.; Kellner, A.W.A.; Dias-da-Silva, S.; da Silva, L.R.; Bronzati, M.; Marsola, J.C.A.; Müller, R.T.; Bittencourt, J.S.; Batista, B.J.A.; Raugust, T.; Carrilho, R.; Brodt, A.; Langer, M.C. (2016). "A Unique Late Triassic Dinosauromorph Assemblage Reveals Dinosaur Ancestral Anatomy and Diet" (https://doi.org/10.1016%2Fj.cub.2016.09.040). *Current Biology*. **26** (22): 3090–3095. Bibcode:2016CBio...26.3090C (https://ui.adsabs.harvard.edu/abs/2016CBio...26.3090C). doi:10.1016/j.cub.2016.09.040 (https://doi.org/10.1016%2Fj.cub.2016.09.040). PMID 27839975 (https://pubmed.ncbi.nlm.nih.gov/27839975).
68. Müller, R.T.; Garcia, M.S. (2020). "A paraphyletic 'Silesauridae' as an alternative hypothesis for the initial radiation of ornithischian dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7480155). *Biology Letters*. **16** (8): 1–5. doi:10.1098/rsbl.2020.0417 (https://doi.org/10.1098%2Frsbl.2020.0417). PMC 7480155 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7480155). PMID 32842895 (https://pubmed.ncbi.nlm.nih.gov/32842895).
69. Baron, M.G.; Norman, D.B.; Barrett, P.M. (2017). "A new hypothesis of dinosaur relationships and early dinosaur evolution". *Nature*. **543** (7646): 501–506. Bibcode:2017Natur.543..501B (https://ui.adsabs.harvard.edu/abs/2017Natur.543..501B). doi:10.1038/nature21700 (https://doi.org/10.1038%2Fnature21700). PMID 28332513 (https://pubmed.ncbi.nlm.nih.gov/28332513).
70. Langer, M.C.; Ezcurra, M.D.; Rauhut, O.W.M.; Benton, M.J.; Knoll, F.; McPhee, B.W.; Novas, F.E.; Pol, D.; Brusatte, S.L. (2017). "Untangling the dinosaur family tree". *Nature*. **551** (7678): E1–E3. Bibcode:2017Natur.551E...1L (https://ui.adsabs.harvard.edu/abs/2017Natur.551E...1L). doi:10.1038/nature24011 (https://doi.org/10.1038%2Fnature24011). hdl:1983/d088dae2-c7fa-4d41-9fa2-aeebbfcd2fa3 (https://hdl.handle.net/1983%2Fd088dae2-c7fa-4d41-9fa2-aeebbfcd2fa3). PMID 29094688 (https://pubmed.ncbi.nlm.nih.gov/29094688).

71. Baron, M.G.; Norman, D.B.; Barrett, P.M. (2017). "Baron et al. reply". *Nature*. **551** (7678): E4–E5. Bibcode:2017Natur.551E...4B (https://ui.adsabs.harvard.edu/abs/2017Natur.551E...4B). doi:10.1038/nature24012 (https://doi.org/10.1038%2Fnature24012). PMID 29094705 (https://pubmed.ncbi.nlm.nih.gov/29094705).
72. Parry, L.A.; Baron, M.G.; Vinther, J. (2017). "Multiple optimality criteria support Ornithoscelida" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5666269). *Royal Society Open Science*. **4** (10) 170833. Bibcode:2017RSOS....470833P (https://ui.adsabs.harvard.edu/abs/2017RSOS....470833P). doi:10.1098/rsos.170833 (https://doi.org/10.1098%2Frsos.170833). PMC 5666269 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5666269). PMID 29134086 (https://pubmed.ncbi.nlm.nih.gov/29134086).
73. Nesbitt, S.J.; Sues, H.-D. (2021). "The osteology of the early-diverging dinosaur *Daemonosaurus chauliodus* (Archosauria: Dinosauria) from the *Coelophysis* Quarry (Triassic: Rhaetian) of New Mexico and its relationships to other early dinosaurs". *Zoological Journal of the Linnean Society*. **191** (1): 150–179. doi:10.1093/zoolinnean/zlaa080 (https://doi.org/10.1093%2Fzoolinnean%2Fzlaa080).
74. Jeffrey A. Wilson; Claudia A. Marsicano; Roger M. H. Smith (6 October 2009). "Dynamic Locomotor Capabilities Revealed by Early Dinosaur Trackmakers from Southern Africa" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2752196). *PLOS ONE*. **4** (10) e7331. Bibcode:2009PLoSO...4.7331W (https://ui.adsabs.harvard.edu/abs/2009PLoSO...4.7331W). doi:10.1371/journal.pone.0007331 (https://doi.org/10.1371%2Fjournal.pone.0007331). PMC 2752196 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2752196). PMID 19806213 (https://pubmed.ncbi.nlm.nih.gov/19806213).
75. Butler, Richard J.; Galton, Peter M.; Porro, Laura B.; Chiappe, Luis M.; Henderson, Donald M.; Erickson, Gregory M. (2010-02-07). "Lower limits of ornithischian dinosaur body size inferred from a new Upper Jurassic heterodontosaurid from North America" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2842649). *Proceedings of the Royal Society of London B: Biological Sciences*. **277** (1680): 375–381. doi:10.1098/rspb.2009.1494 (https://doi.org/10.1098%2Frspb.2009.1494). ISSN 0962-8452 (https://search.worldcat.org/issn/0962-8452). PMC 2842649 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2842649). PMID 19846460 (https://pubmed.ncbi.nlm.nih.gov/19846460).
76. Yannan, Ji; Xuri, Wang; Yongqing, Liu; Qiang, Ji (2011-02-01). "Systematics, Behavior and Living Environment of Shantungosaurus Giganteus (Dinosauria: Hadrosauridae)". *Acta Geologica Sinica - English Edition*. **85** (1): 58–65. Bibcode:2011AcGlS..85...58J (https://ui.adsabs.harvard.edu/abs/2011AcGlS..85...58J). doi:10.1111/j.1755-6724.2011.00378.x (https://doi.org/10.1111%2Fj.1755-6724.2011.00378.x). ISSN 1755-6724 (https://search.worldcat.org/issn/1755-6724). S2CID 85351874 (https://api.semanticscholar.org/CorpusID:85351874).
77. Barrett, P. M.; Rayfield, E. J. (2006). "Ecological and evolutionary implications of dinosaur feeding behaviour" (http://doc.rero.ch/record/16151/files/PAL_E2427.pdf) (PDF). *Trends in Ecology & Evolution*. **21** (4): 217–224. doi:10.1016/j.tree.2006.01.002 (https://doi.org/10.1016%2Fj.tree.2006.01.002). PMID 16701088 (https://pubmed.ncbi.nlm.nih.gov/16701088).
78. Ji, Q.; Wu, X.; Cheng, Y.; Ten, F.; Wang, X.; Ji, Y. (2016). "Fish-hunting ankylosaurs (Dinosauria, Ornithischia) from the Cretaceous of China". *Journal of Geology*. **40**: 2.
79. Barrett, P. M., & Rayfield, E. J. (2006). Ecological and evolutionary implications of dinosaur feeding behaviour. Trends in Ecology & Evolution, 21(4), 217–224.
80. Fastovsky, D. E., & Weishampel, D. B. (2012). Dinosaurs: a concise natural history (2nd ed). Cambridge; New York: Cambridge University Press.

External links

- Ornithischia (https://web.archive.org/web/20081201083750/http://www.palaeos.com/Vertebrates/Units/320Ornithischia/100.html#Ornithischia), from Palæos. (cladogram, characteristics)

Retrieved from "https://en.wikipedia.org/w/index.php?title=Ornithischia&oldid=1381917624"

WIKIPEDIA
25 years of the free encyclopedia

Stegosauria

Stegosauria is a group of herbivorous ornithischian dinosaurs that lived during the Jurassic and early Cretaceous periods. Stegosaurian fossils have been found mostly in the Northern Hemisphere (North America, Europe and Asia), Africa and South America. Although their geographical origins are unclear, the earliest unequivocal stegosaurians are known from the Middle Jurassic (Bajocian-Bathonian stages), including *Adratiklit*, *Bashanosaurus*, *Isaberrysaura* and *Thyreosaurus*.

Stegosaurians belong to a clade of armored dinosaurs known as Thyreophora. Originally, they did not differ much from more basal (early-diverging) members of that group, being small, low-slung, running animals protected by armored scutes. An early evolutionary innovation was the development of spikes as defensive weapons. Later species were larger and developed long hindlimbs that no longer allowed them to run. This increased the importance of active defence by the thagomizer, which could ward off even large predators because the tail was in a higher position, pointing horizontally to the rear from the broad pelvis. Stegosaurs had complex arrays of spikes and plates running along their backs, hips and tails.

Stegosauria includes two families, the **Huayangosauridae** and the more diverse **Stegosauridae**. All species were quadrupedal herbivores with characteristic dorsal dermal plates. These large, thin, erect plates are thought to be aligned parasagittally from the neck to near the end of the tail. The end of the tail has pairs of spikes, sometimes referred to as a thagomizer.[6][7] It may be that this is the only scientific term derived from a joke (in this case a *The Far Side* comic). Although defense, thermo-regulation, and display have been theorized to be the possible functions of these dorsal plates, a study of the ontogenetic histology of the plates and spikes suggests that the plates serve different functions at different stages of the stegosaurids' life histories. The terminal spikes in the tail are thought to have been used in old adults, at least, as a weapon for defence.[8] However, the function of stegosaurid plates and spikes, at different life stages, still remains a matter of great debate.

The first stegosaurian finds in the early 19th century were fragmentary. Better fossil material, of the genus *Dacentrurus*, was discovered in 1874 in England. Soon after, in 1877, the first nearly complete skeleton was discovered in the United States. Professor Othniel Charles Marsh that year classified such specimens in the new genus *Stegosaurus*, from which the group acquired its name, and which is still by far the most

Stegosaurs

Temporal range: Middle Jurassic–Early Cretaceous **Possible wider range based on ichnofossils and fragmentary elements**[1][2][3]

Four stegosaurs. Clockwise from top left: *Gigantspinosaurus*, *Hesperosaurus*, *Kentrosaurus*, and *Stegosaurus*

Scientific classification

Kingdom:	Animalia
Phylum:	Chordata
Class:	Reptilia
Clade:	Dinosauria
Clade:	†Ornithischia
Clade:	†Thyreophora
Clade:	†Eurypoda
Clade:	†Stegosauria Marsh, 1877

Subgroups[4][5]

- †*Craterosaurus*
- †*Lexovisaurus*
- †*Monkonosaurus*
- †*Mongolostegus*
- †*Regnosaurus*?
- †**Huayangosauridae** Dong, Tang, & Zhou, 1982
  - †*Baiyinosaurus*

famous stegosaurian. During the latter half of the twentieth century, many important Chinese finds were made, representing about half of the presently known diversity of stegosaurians.

History of research

The first known discovery of a possible stegosaurian was probably made in the early nineteenth century in England. It consisted of a lower jaw fragment and was in 1848 named *Regnosaurus*. In 1845, in the area of the present state of South Africa, remains were discovered that much later would be named *Paranthodon*. In 1874, other remains from England were named *Craterosaurus*. All three taxa were based on fragmentary material and were not recognised as possible stegosaurians until the twentieth century. They gave no reason to suspect the existence of a new, distinctive group of dinosaurs.

Stegosaurus bones illustrated by Arthur Lakes in 1879

In 1874, extensive remains of what was clearly a large herbivore equipped with spikes were uncovered in England; the first partial stegosaurian skeleton known.[9] They were named *Omosaurus* by Richard Owen in 1875. Later, this name was shown to be preoccupied by the phytosaur *Omosaurus* and the stegosaurian was renamed *Dacentrurus*. Other English nineteenth-century and early twentieth-century finds would be assigned to *Omosaurus*; later they would, together with French fossils, be partly renamed *Lexovisaurus* and *Loricatosaurus*.

- †*Bashanosaurus*
- †*Chungkingosaurus*?
- †*Gigantspinosaurus*
- †*Huayangosaurus*
- †*Isaberrysaura*

- †**Stegosauridae** Marsh, 1880
  - †*Paranthodon*
  - †*Tuojiangosaurus*
  - †**Neostegosauria** Sánchez-Fenollosa & Cobos, 2025
    - †*Kentrosaurus*
    - †**Dacentrurinae** Mateus et al., 2009
      - †*Adratiklit*
      - †*Alcovasaurus*
      - †*Dacentrurus*
      - †*Miragaia*
      - †*Thyreosaurus*
    - †**Stegosaurinae** Nopcsa, 1917
      - †*Chialingosaurus*
      - †*Hesperosaurus*
      - †*Hypsirophus*
      - †*Jiangjunosaurus*
      - †*Loricatosaurus*
      - †*Stegosaurus*
      - †*Wuerhosaurus*
      - †*Yanbeilong*
      - †*Yingshanosaurus*

In 1877, Arthur Lakes, a fossil hunter working for Professor Othniel Charles Marsh in Wyoming, excavated a fossil that Marsh named *Stegosaurus* the same year. At first, Marsh still entertained some incorrect notions about its morphology. He assumed that the plates formed a flat skin cover—hence the name, meaning "roof saurian"—and that the animal was bipedal with the spikes sticking out sideways from the rear of the skull. A succession of additional discoveries from the Como Bluff sites allowed a quick update of the presumed build. In 1882, Marsh was able to publish the first skeletal reconstruction of a stegosaur. Hereby, stegosaurians became much better known to the general public. The American finds at the time represented the bulk of known stegosaurian fossils, with about twenty skeletons collected.[9]

The next important discovery was made when a German expedition to the Tendaguru, then part of German East Africa, from 1909 to 1912 excavated over a thousand bones of *Kentrosaurus*. The finds increased the known variability of the group, *Kentrosaurus* being rather small and having long rows of spikes on the hip and tail.

An early life restoration of a *Stegosaurus* from 1910

From the 1950s onwards, the geology of China was systematically surveyed in detail, and infrastructural works led to a vast increase of excavation in the PRC. This resulted in a new wave of Chinese stegosaurian discoveries, starting with *Chialingosaurus* in 1957. Chinese finds of the 1970s and 1980s included *Wuerhosaurus*, *Tuojiangosaurus*, *Chungkingosaurus*, *Huayangosaurus*, *Yingshanosaurus* and *Gigantspinosaurus*. This increased the age range of stegosaurian materials of good fossil preservation, as they represented the first relatively complete skeletons from the Middle Jurassic and the Early Cretaceous. Especially important was *Huayangosaurus*, which provided unique information about the early evolution of the group. In 2007, *Jiangjunosaurus* was reported, the first Chinese stegosaur named since 1994.

Towards the end of the twentieth century, the so-called dinosaur renaissance took place in which a vast increase in scientific attention was given to the Dinosauria. During the 1990s, European and North-American sites became productive again, with fossils such as *Miragaia* having been found in the Lourinhã Formation in Portugal, and a number of relatively complete *Hesperosaurus* skeletons having been excavated in Wyoming. Apart from the fossils per se, important new insights have been gained by applying the analytic method of cladistics, which allowed for precise calculations of interrelationships and the construction of stegosaurian phylogenetic trees.

Description

Stegosaurids are distinguished from other stegosaurians in that the former have lost the plesiomorphic pre-maxillary teeth and lateral scute rows along the trunk.[10] Furthermore, stegosaurids have long, narrow skulls and longer hindlimbs compared to their forelimbs.[7] However, these two features are not diagnostic of Stegosauridae because they may also be present in non-stegosaurid stegosaurians.[1]

Skull

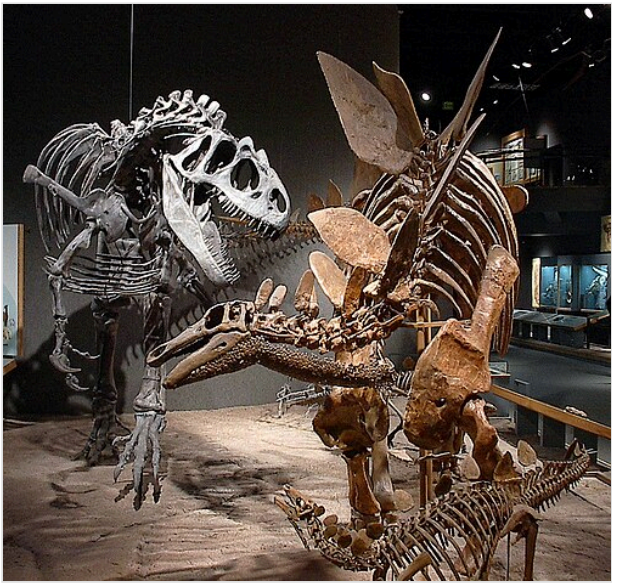
Stegosaurians had characteristic small, long, flat, narrow heads and a horn-covered beak or rhamphotheca,[1] which covered the front of the snout (two premaxillaries) and lower jaw (a single predentary) bones. Similar structures are seen in turtles and birds. Apart from *Huayangosaurus*, stegosaurians subsequently lost all premaxillary teeth within the upper beak. *Huayangosaurus* still had seven per side. The upper and lower jaws are equipped with rows of small teeth. Later species have a vertical bone plate covering the outer side of the lower jaw teeth. The structure of the upper jaw, with a low ridge above and running parallel to the tooth row, indicates the presence of a fleshy cheek. In stegosaurians, the typical archosaurian skull opening, the antorbital fenestra in front of the eye socket, is small, sometimes reduced to a narrow horizontal slit. In general, stegosaurids have proportionally long, low, and narrow snouts with a deep mandible, compared to that of *Huayangosaurus*. Stegosaurids also lack premaxillary teeth.[10]

Postcranial skeleton

All stegosaurians are quadrupedal, with hoof-like toes on all four limbs. All stegosaurians after *Huayangosaurus* have forelimbs much shorter than their hindlimbs. Their hindlimbs are long and straight, designed to carry the weight of the animal while stepping. The condyles of the lower thighbone are short from the front to the rear. This would have limited the supported rotation of the knee joint, making running impossible. *Huayangosaurus* had a thighbone like a running animal. The upper leg was always longer than the lower leg.

Huayangosaurus had relatively long and slender arms. The forelimbs of later forms are very robust, with a massive humerus and ulna. The wrist bones were reinforced by a fusion into two blocks, an ulnar and a radial. The front feet of stegosaurians are commonly depicted in art and in museum displays with fingers splayed out and slanted downward. However, in this position, most bones in the hand would be disarticulated. In reality, the hand bones of stegosaurians

were arranged into vertical columns, with the main fingers, oriented outwards, forming a tube-like structure. This is similar to the hands of sauropod dinosaurs, and is also supported by evidence from stegosaurian footprints and fossils found in a lifelike pose.[11]



Stegosaurus mount showing the high neck posture, the throat ossicles and the robust shoulder girdle and forelimbs

The long hindlimbs elevated the tail base, such that the tail pointed out behind the animal almost horizontally from that high position. While walking, the tail would not have sloped downwards, as this would have impeded the function of the tail base retractor muscles, to pull the thighbones backwards. However, it has been suggested by Robert Thomas Bakker that stegosaurians could rear on their hind legs to reach higher layers of plants, the tail then being used as a "third leg". The mobility of the tail was increased by a reduction or absence of ossified tendons, which, in many ornithischians, stiffen the hip region. *Huayangosaurus* still possessed them. In species that had short forelimbs, the relatively short torso towards the front curved strongly downwards. The dorsal vertebrae typically were very high, with very tall neural arches and transverse processes pointing obliquely upwards to almost the level of the neural spine top. Stegosaurian back vertebrae can easily be identified by this unique configuration. The tall neural arches often house deep neural canals; enlarged canals in the sacral vertebrae have given rise to the incorrect notion of a "second brain". Despite the downwards curvature of the rump, the neck base was not very low, and the head was held a considerable distance off the ground. The neck was flexible and moderately long. *Huayangosaurus* still had the probably original number of nine cervical vertebrae; *Miragaia* has an elongated neck with seventeen.[12]

The stegosaurian shoulder girdle was very robust. In *Huayangosaurus*, the acromion, a process on the lower front edge of the shoulderblade, was moderately developed; the coracoid was about as wide as the lower end of the scapula, with which it formed the shoulder joint. Later forms tend to have a strongly expanded acromion, while the coracoid, largely attached to the acromion, no longer extends to the rear lower corner of the scapula.

The stegosaurian pelvis was originally moderately large, as shown by *Huayangosaurus*. Later species, however, convergent to the Ankylosauria developed very broad pelves, in which the iliac bones formed wide horizontal plates with flaring front blades to allow for an enormous belly-gut. The ilia were attached to the sacral vertebrae via a sacral yoke formed by fused sacral ribs. *Huayangosaurus* still had rather long and obliquely oriented ischia and pubic bones. In more derived species, these became more horizontal and shorter to the rear, while the front prepubic process lengthened.

Armor and ornamention

Like all Thyreophora, stegosaurians were protected by bony scutes that were not part of the skeleton proper but skin ossifications instead: the so-called osteoderms. *Huayangosaurus* had several types. On its neck, back, and tail were two rows of paired small vertical plates and spikes. The very tail end bore a small club. Each flank had a row of smaller osteoderms, culminating in a long shoulder spine in front, curving to the rear. Later forms show very variable configurations, combining plates of various shapes and sizes on the neck and front torso with spikes more to the rear of the animal. They seem to have lost the tail club and the flank rows are apparently absent also, with the exception of the shoulder spine, still shown by *Kentrosaurus* and extremely developed, as its name indicates, in *Gigantspinosaurus*. As far as is known, all forms possessed some sort of thagomizer, though these are rarely preserved articulated, allowing for the establishment of the exact arrangement. A fossil of *Chungkingosaurus* sp. has been reported with three pairs of spikes pointing outwards and a fourth pair pointing to the rear.[13] The most derived species, like *Stegosaurus*, *Hesperosaurus* and *Wuerhosaurus*, have very large and flat back plates. Stegosaurid plates have a thick base and

central portion, but are transversely thin elsewhere. The plates become remarkably large and thin in *Stegosaurus*. They are found in varying sizes along the dorsum, with the central region of the back usually having the largest and tallest plates. The arrangement of these parasagittal dorsal plates has been intensely debated in the past. Discoverer Othniel Charles Marsh suggested a single median row of plates running post-cranially along the longitudinal axis[14] and Lull argued in favour of bilaterally paired arrangement throughout the series.[15] Current scientific consensus lies in the arrangement proposed by Gilmore: two parasagittal rows of staggered alternates, after the discovery of an almost complete skeleton preserved in this manner in rock.[16] Furthermore, no two plates share the same size and shape, making the possibility of bilaterally paired rows even less likely. Plates are usually found with distinct vascular grooves on their lateral surfaces, suggesting the presence of a circulatory network. Stegosaurids also have osteoderms on the throat in the form of small, depressed ossicles and two pairs of elongated spike-like tail spines.[1] With *Stegosaurus* fossils, ossicles have also been found in the throat region, bony skin discs that protected the lower neck.[17]

Many stegosaurs, including *Gigantspinosaurus* and *Kentrosaurus*, have been discovered with parascapular spines, or spines emerging from the shoulder region, which project posteriorly out of the lower part of the shoulder plates. These spines are long, rounded, and comma-shaped in lateral view and have an enlarged base.[18] Stegosaurids also lack lateral scute rows that run longitudinally on either side of the trunk in *Huayangosaurus* and ankylosaurs, indicating yet another secondary loss of a plesiomorphic character.[10] However, the absence of lateral scutes as well as pre-maxillary teeth mentioned above are not specifically diagnostic of stegosaurids, since these features are also present in some other stegosaurians, whose phylogenetic relationships are unclear.[10][18]

The discovery of an impression of the skin covering the dorsal plates has implications for all possible functions of stegosaurian plates. Christiansen and Tschopp (2010)[19] found that the skin was smooth with long, parallel, shallow grooves indicating a keratinous structure covering the plates. The addition of beta-keratin, a strong protein, would indeed allow the plates to bear more weight, suggesting they may have been used for active defense. A keratinous covering would also allow greater surface area for the plates to be used as mating display structures, which could be potentially coloured like the beaks of modern birds. At the same time, this finding implies that the use of plates for thermo-regulation may be less likely because the keratinous covering would make heat transfer from the bone highly ineffective.[19]

Classification

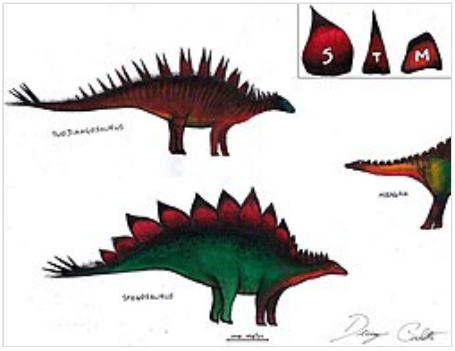
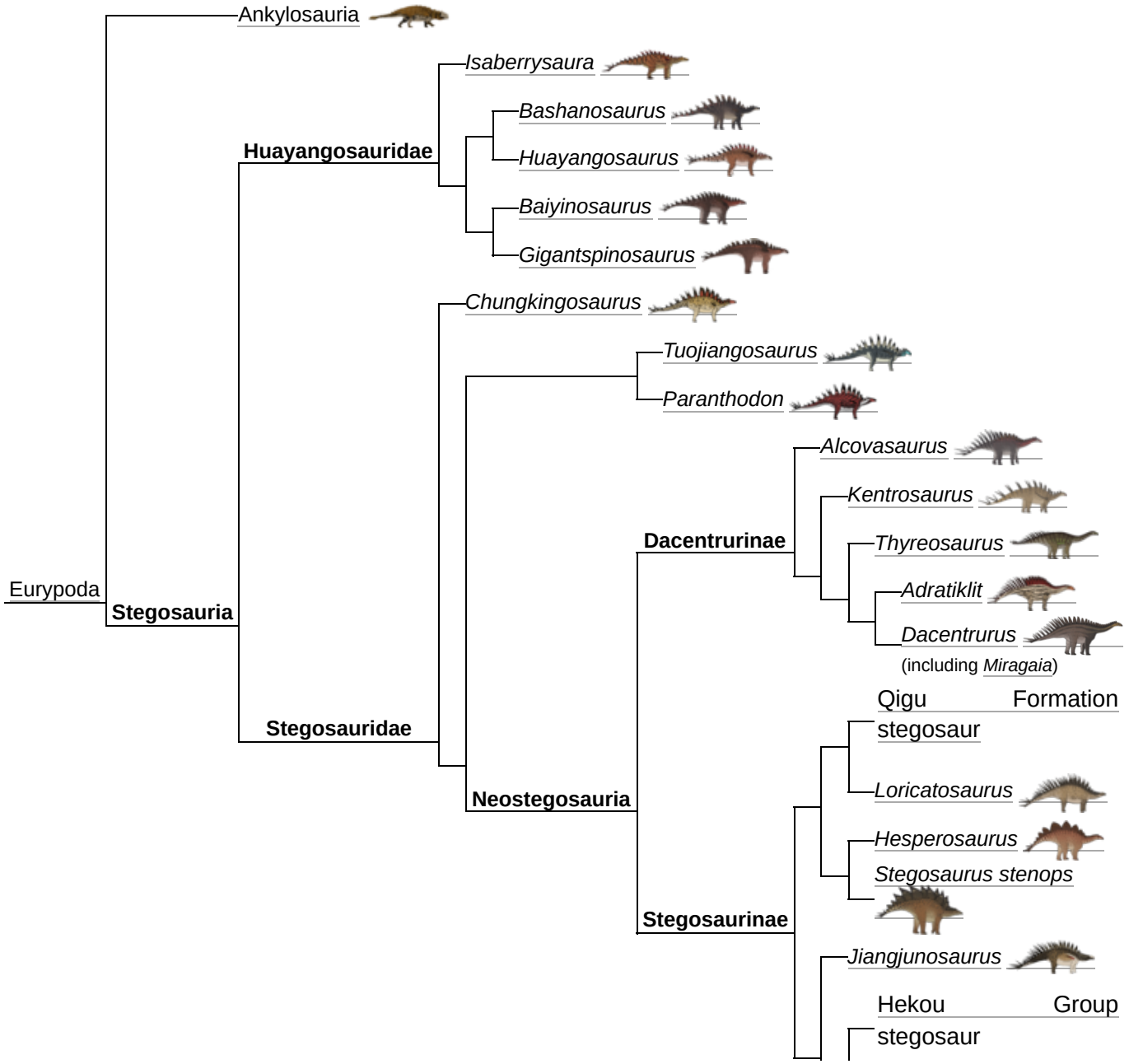


Illustration comparing three stegosaurs

In 1877, Othniel Marsh discovered and named *Stegosaurus armatus*, from which the name of the family 'Stegosauridae' was erected in 1880.[9] In comparison to basal stegosaurians, notable synapomorphies of Stegosauridae include a large antitrochanter (supracetabular process) in the ilium, a long prepubic process, and long femur relative to the length of the humerus.[20] Furthermore, stegosaurid sacral ribs are T-shaped in parasagittal cross-section[1] and the dorsal vertebrae have an elongated neural arch.[9] The first exact clade definition of Stegosauria was given by Peter Malcolm Galton in 1997: all thyreophoran Ornithischia more closely related to *Stegosaurus* than to *Ankylosaurus*.[21] This definition was formalized in the *PhyloCode* by Daniel Madzia and colleagues in 2021 as "the largest clade containing *Stegosaurus stenops*, but not *Ankylosaurus magniventris*".[4] Thus defined, the Stegosauria are by definition the sister group of the Ankylosauria within the Eurypoda. The vast majority of stegosaurian dinosaurs thus far recovered belong to the Stegosauridae, which lived in the later part of the Jurassic and early Cretaceous, and which were defined by Paul Sereno as all stegosaurians more closely related to *Stegosaurus* than to *Huayangosaurus*.[22] This definition was also formalized in the *PhyloCode* by Daniel Madzia and colleagues in 2021 as "the largest clade containing *Stegosaurus stenops*, but not *Huayangosaurus taibaii*".[4] They include per definition the well-known *Stegosaurus*. This group is widespread, with members across the Northern Hemisphere, Africa and South America.[23][24]

Huayangosauridae (derived from *Huayangosaurus*, "Huayang reptile") is a family of stegosaurian dinosaurs from the Jurassic of China.[25] The group is defined as all taxa closer to the namesake genus *Huayangosaurus* than *Stegosaurus*, and was originally named as the family Huayangosaurinae by Dong Zhiming and colleagues in the description of *Huayangosaurus*.[25][4] Huayangosaurinae was originally differentiated by the remaining taxa within Stegosauridae by the presence of teeth in the premaxilla, an antorbital fenestra, and a mandibular fenestra. Huayangosaurinae, known from the Middle Jurassic of the Shaximiao Formation, was proposed to be intermediate between Scelidosaurinae and Stegosaurinae, suggesting that the origins of stegosaurs lay in Asia.[25] Following phylogenetic analyses, Huayangosauridae was expanded to also include the taxon *Chungkingosaurus*, known from specimens from younger Late Jurassic deposits of the Shaximiao Formation.[18] Huayangosauridae is either the sister taxon to all other stegosaurs,[18][9] or close to the origin of the clade, with taxa like *Gigantspinosaurus* or *Isaberrysaura* outside the Stegosauridae-Huayangosauridae split.[4][9] Huayangosauridae was formally defined in 2021 by Daniel Madzia and colleagues, who used the previous definitions of all taxa closer to *Huayangosaurus taibaii* than *Stegosaurus stenops*.[4]

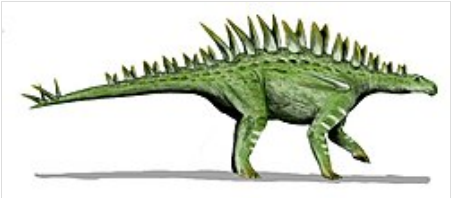
In 2017, Raven and Maidment published a comprehensive phylogenetic framework including most valid stegosaurian genera. Several subsequent publications expanded and corrected this matrix based on novel taxa and revised anatomical interpretations. In 2025, Sánchez-Fenollosa & Cobos compiled these variations and other observations into an updated and expanded dataset. The authors further coined the name **Neostegosauria** for the clade comprising the Dacentrurinae (including *Kentrosaurus*) and the Stegosaurinae. These results are displayed in the cladogram below:[5]



Undescribed species

To date, several genera from China bearing names have been proposed but not formally described, including "Changdusaurus".[26] Until formal descriptions are published, these genera are regarded as *nomina nuda*.

Evolutionary history



Huayangosaurus is among the oldest known stegosaurians for which good material is known, giving an impression of the build of the earliest members of the group

Like the spikes and shields of ankylosaurs, the bony plates and spines of stegosaurians evolved from the low-keeled osteoderms characteristic of basal thyreophorans.[27] One such described genus, *Scelidosaurus*, is proposed to be morphologically close to the last common ancestor of the clade uniting stegosaurians and ankylosaurians, the Eurypoda.[28] Galton (2019) interpreted plates of an armored dinosaur from the Lower Jurassic (Sinemurian-Pliensbachian) Lower Kota Formation of India as fossils of a member of Ankylosauria; the author argued that this finding indicates a probable early Early Jurassic origin for both Ankylosauria and its sister group Stegosauria.[29] Footprints attributed to the ichnotaxon *Deltapodus brodricki* from the Middle Jurassic (Aalenian) of England represent the oldest probable record of stegosaurians reported so far.[2] Outside that, there are assigned fossils to stegosauria from the Toarcian: the specimen "IVPP V.219", a chimaera with bones of the sauropod *Sanpasaurus* is known from the Maanshan Member of the Ziliujing Formation.[30] The earliest possible trackways of stegosaurians are discovered from the Hettangian-aged deposits of France, indicating a possibly earlier origin.[1] The perhaps most basal known stegosaurian, the four-metre-long *Huayangosaurus*, is still close to *Scelidosaurus* in build, with a higher and shorter skull, a short neck, a low torso, long slender forelimbs, short hindlimbs, large condyles on the thighbone, a narrow pelvis, long ischial and pubic shafts, and a relatively long tail. Its small tail club might be a eurypodan synapomorphy. *Huayangosaurus* lived during the Bathonian stage of the Middle Jurassic, about 166 million years ago.

A few million years later, during the Callovian-Oxfordian, from China much larger species are known, with long, "graviportal" (adapted for moving only in a slow manner on land due to a high body weight) hindlimbs: *Chungkingosaurus*, *Chialingosaurus*, *Tuojiangosaurus* and *Gigantspinosaurus*. Most of these are considered members of the derived Stegosauridae. *Lexovisaurus* and *Loricatosaurus*, stegosaurid finds from England and France of approximately equivalent age to the Chinese specimens, are likely the same taxon. During the Late Jurassic, stegosaurids seem to have experienced their greatest radiation. In Europe, *Dacentrurus* and the closely related *Miragaia* were present, while they are represented in Asia by *Jiangjunosaurus*. While older finds had been limited to the northern continents, in this phase, Gondwana was also colonised as shown by *Kentrosaurus* living in Africa. The most derived Jurassic stegosaurians are known from North-America: *Stegosaurus* (perhaps several species thereof) and the somewhat older *Hesperosaurus*. *Stegosaurus* was quite large (some specimens indicate a length of at least seven metres), had high plates, no shoulder spine, and a short, deep rump.

From the Early Cretaceous, far fewer finds are known, and it seems that the group had declined in diversity. Some fragmentary fossils have been described, such as *Craterosaurus* from England, *Paranthodon* from South Africa, and indeterminate remains from the Late Jurassic Cañadón Calcáreo Formation of Argentina[31]. Up until recently, the only substantial discoveries were those of *Wuerhosaurus* from Northern China, the exact age of which is highly uncertain[32] More recent discoveries from Asia however would later begin to fill out the Early Cretaceous diversity of the group. Indeterminate stegosaurs are known from the Early Cretaceous of Siberia, including the Ilek Formation[33]

and Batylykh Formation.[34] The youngest known definitive remains of stegosaurs are those of *Mongolostegus* from Mongolia, a stegosaurine from the Hekou Group of China, and *Yanbeilong* of the Zuoyun Formation of China, all of which date to the Aptian-Albian.[35][36][37]

It has often been suggested that the decline in stegosaur diversity was part of a Jurassic-Cretaceous transition, where angiosperms became the dominant plants, causing a faunal turnover where new groups of herbivores evolved.[38] Although in general the case for such a causal relation is poorly supported by the data, stegosaurians are an exception in that their decline coincides with that of the Cycadophyta.[39]

Though Late Cretaceous stegosaurian fossils have been reported, these have mostly turned out to be misidentified. A well-known example is *Dravidosaurus*, known from Coniacian fossils found in India. Though originally thought to be stegosaurian, in 1991 these badly-eroded fossils were suggested to instead have been based on plesiosaurian pelvis and hindlimb material,[40] and none of the fossils are demonstrably stegosaurian.[41] The reinterpretation of *Dravidosaurus* as a plesiosaur wasn't accepted by Galton and Upchurch (2004), who stated that the skull and plates of *Dravidosaurus* are certainly not plesiosaurian, and noted the need to redescribe the fossil material of *Dravidosaurus*.[1] A purported stegosaurian dermal plate was reported from the latest Cretaceous (Maastrichtian) Kallamedu Formation (southern India); however, Galton & Ayyasami (2017) interpreted the specimen as a bone of a sauropod dinosaur. Nevertheless, the authors considered the survival of stegosaurians into the Maastrichtian to be possible, noting the presence of the stegosaurian ichnotaxon *Deltapodus* in the Maastrichtian Lameta Formation (western India).[3] Furthermore, Ayyasami announced that he was in the process of working on new undescribed and likely stegosaurian bones from the original discovery site of the *Dravidosaurus* fossils.[3]

Paleobiology

Plate function

In an ontogenetic histological analysis of *Stegosaurus* plates and spikes, Hayashi et al. (2012)[8] examined their structure and function through juveniles to old adults. They found that throughout the ontogeny, the dorsal osteoderms are composed of dense ossified collagen fibres in both the cortical and cancellous sections of the bone, suggesting that plates and spikes are formed from the direct mineralization of already existing fibrous networks in the skin. However, the many structural features, seen in the spikes and plates of old adult specimens, are acquired at different stages of development. Extensive vascular networks form in the plates during the change from juveniles to young adults and persist in old adults, but spikes acquire a thick cortex with a large axial vascular channel only in old adults. Hayashi et al. argue that the formation of nourishing vascular networks in young adults supported the growth of large plates. This would have enhanced the size of the animal, which may have helped attract mates and deter rivals.[8] Furthermore, the presence of the vascular networks in the plates of the young adult indicates a secondary use of the plates as a thermoregulatory device for heat loss, much like the elephant ear, toucan bill, or alligator osteoderms. The thickening of the cortical section of the bone and the compaction of bone in the terminal tail-spikes in old adults suggest that they were used as defence weapons, but not until an ontogenetically late stage. The development of the large axial channel in old adults from small canals in young adults facilitated the further enlargement of the spikes by increasing the amount of nourishment supplied. On the other hand, plates do not show a similar degree of bone compaction or cortical thickening, indicating they would not be capable of taking much weight from above. This suggests they were not as important as spikes in active defense.[8]

The protective nature of dorsal plates has also been questioned in the past; Davitashvili (1961) noted that the narrow dorsal location of the plates still left the sides vulnerable. Since the pattern of plates and spines varies between species, he suggested it could be important for intraspecific recognition and as a display for sexual selection.[1] This is corroborated by Spassov's (1982) observations that the plates are arranged for maximum visible effect when viewed laterally during non-aggressive agonistic behaviour, as opposed to from a head-on aggressive stance.[42]

Trace fossils

Stegosaurian tracks were first recognized in 1996 from a hindprint-only trackway discovered at the Cleveland-Lloyd quarry, which is located near Price, Utah.[43] Two years later, a new ichnogenus called *Stegopodus* was erected for another set of stegosaurian tracks which were found near Arches National Park, also in Utah.[43] Unlike the first, this trackway preserved traces of the forefeet. Fossil remains indicate that stegosaurians have five digits on the forefeet and three weight-bearing digits on the hind feet.[43] From this, scientists were able to predict the appearance of stegosaurian tracks in 1990, six years in advance of the first actual discovery of Morrison stegosaurian tracks.[43] More trackways have been found since the erection of *Stegopodus*. None, however, have preserved traces of the front feet, and stegosaurian traces remain rare.[43]

Deltapodus is an ichnogenus attributed as stegosaurian prints, and are known across Europe,[44] North Africa,[45] and China.[46] One *Deltapodus* footprint measures less than 6 cm in length and represents the smallest known stegosaurian track.[46][47] Some tracks preserve exquisite scaly skin pattern.[44]

Australia's 'Dinosaur Coast' in Broome, Western Australia, includes tracks of several different thyreophoran track-makers. Of these, the ichnogenus *Garbina* (a Nyulnyulan word for 'shield') and *Luluichnus* (honours the late Paddy Roe, OAM who went by the name 'Lulu') have been considered registered by stegosaurs.[48] *Garbina* includes the largest stegosaur tracks measuring 80 cm in length. Trackway data show *Garbina* track-makers were capable of bipedal and quadrupedal progression, suggesting an adaptation to facultative bipedalism amongst some stegosaurs.

While has no body fossil evidence currently known for stegosaurs, handprints from underground coal mines near Oakey, Queensland, resembling *Garbina* tracks suggests their occurrence in this country from at least the Middle to Upper Jurassic (Callovian–Tithonian).[49] A single plaster cast of one of these handprints is in the collections of the Queensland Museum.

Tail spikes

There has been debate about whether the spikes were used simply for display, as posited by Gilmore in 1914,[16] or used as a weapon. Robert Bakker noted that it is likely that the stegosaur tail was much more flexible than those of other ornithischian dinosaurs because it lacked ossified tendons, thus lending credence to the idea of the tail as a weapon. He also observed that *Stegosaurus* could have maneuvered its rear easily by keeping its large hindlimbs stationary and pushing off with its very powerfully muscled but short forelimbs, allowing it to swivel deftly to deal with an attack.[50] In 2010, analysis of a digitized model of *Kentrosaurus aethiopicus* showed that the tail could bring the thagomizer around to the sides of the dinosaur, possibly striking an attacker beside it.[51]

Mounted skeleton reconstruction of the stegosaur (*Tuojiangosaurus*) with a mid-sized theropod (*Monolophosaurus*), Field Museum in Chicago

In 2001, a study of tail spikes by McWhinney et al.,[52] showed a high incidence of trauma-related damage. This also supports the theory that the spikes were used in combat. There is also evidence for *Stegosaurus* defending itself from *Allosaurus* predation, in the form of two specimens of the latter (a tail vertebra and a pubis) with partially healed puncture wounds that fit *Stegosaurus* tail spikes.[53] *Stegosaurus stenops* had four dermal spikes, each about 60–90 cm (2–3 ft) long. Discoveries of articulated stegosaur armor show that, at least in some species, these spikes protruded horizontally from the tail, not vertically as is often depicted. Initially, Marsh described *S. armatus* as having eight spikes in its tail, unlike *S. stenops*. However, recent research re-examined this and concluded this species also had four.[54][55]

Posture

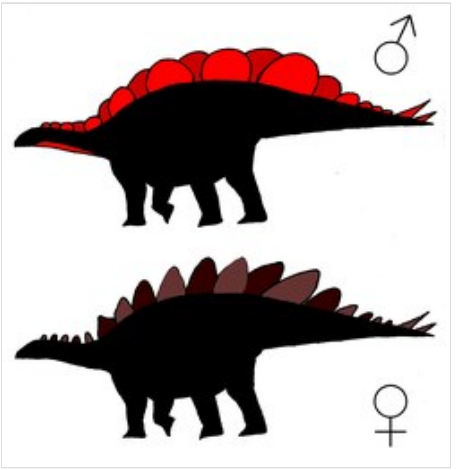
A digital articulation and manipulation of digital scans of specimen material of *Kentrosaurus* inferred that stegosaurids may have used an erect limb posture, like that of most mammals, for habitual locomotion while using a sprawled crocodilian pose for defensive behavior. The sprawled pose would allow them to tolerate the large lateral forces used in swinging the spiked tail against predators as a clubbing device.[56]

Sexual dimorphism

There have been several findings of possible sexual dimorphism in stegosaurids. Saitta (2015)[57] presents evidence of two morphs of *Hesperosaurus* dorsal plates, with one morph having a wide, oval plate with a surface area 45% larger than the narrow, tall morph. Considering that dorsal plates most likely functioned as display structures and that the wide oval shape allowed a broad continuous display, Saitta assigns the wider morph with a larger surface area as male.

Kevin Padian, a paleontologist at the University of California, Berkeley, remarked that Saitta had misidentified features in his specimen's bone tissue sections and said "there's no evidence the animal has stopped growing". Paidan also expressed ethical concerns about the use of private specimens in the study.[58]

Kentrosaurus, *Dacentrurus*, and *Stegosaurus* are also suggested to have exhibited dimorphism in the form of three extra sacral ribs in the females.[1]



Hypothetical sexual dimorphism of *Hesperosaurus mjosi*, including variation in plate shape and coloration

Feeding

In order to explore the feeding habits of stegosaurids, Reichel (2010)[59] created a 3-D model of *Stegosaurus* teeth using the software ZBrush. The model finds that the bite forces of *Stegosaurus* were significantly weaker than those of Labradors, wolves, and humans. The finding suggests that these dinosaurs would be capable of breaking smaller branches and leaves with their teeth, but would not be able to bite through a thick object (12 mm or more in diameter). Parrish et al.'s (2004)[60] description of Jurassic flora in the stegosaurid-rich Morrison Formation supports this finding. The flora during this time period was dominated by seasonal small, fast-growing herbaceous plants, which stegosaurids could consume easily if Reichel's reconstruction is accurate.[59]

Mallison (2010)[56] suggested that *Kentrosaurus* may have used a tripodal stance on their hindlimbs and tail to double the foraging height from the general low browsing height under one metre for stegosaurids. This challenged the view that stegosaurs are primarily low vegetation feeders because of their small heads, short necks, and short forelimbs, since the tripodal stance would also give them access to young trees and high bushes.

Another piece of evidence suggesting that some stegosaurids may have consumed more than just low vegetation was the discovery of the long-necked stegosaurid *Miragaia longicollum*. This dinosaur's neck has at least 17 cervical vertebrae achieved through the transformation of thoracic vertebrae into cervical vertebrae and possible lengthening of the centrum. This is more than most sauropod dinosaurs, which also achieved the elongation of the neck through similar mechanisms and had access to fodder higher off the ground.[12]

Evidence from Yakutia suggests that Early Cretaceous stegosaurs living in high latitude environments were capable of palinal jaw motion and exhibited high rates of tooth replacement and short tooth formation time.[61]

References

1. Peter M. Galton; Paul Upchurch (2004). "Stegosauria". In David B. Weishampel; Peter Dodson; Halszka Osmólska (eds.). *The Dinosauria* (2nd ed.). Berkeley: University of California Press. pp. 343–362. ISBN 978-0-520-24209-8.
2. Peter M. Galton (2017). "Purported earliest bones of a plated dinosaur (Ornithischia: Stegosauria): a "dermal tail spine" and a centrum from the Aalenian-Bajocian (Middle Jurassic) of England, with comments on other early thyreophorans" (https://zenodo.org/record/835532). *Neues Jahrbuch für Geologie und Paläontologie - Abhandlungen*. **285** (1): 1–10. Bibcode:2017NJGPA.285....1G (https://ui.adsabs.harvard.edu/abs/2017NJGPA.285....1G). doi:10.1127/njgpa/2017/0667 (https://doi.org/10.1127%2Fnjgpa%2F2017%2F0667). S2CID 134361050 (https://api.semanticscholar.org/CorpusID:134361050).
3. Peter M. Galton; Krishnan Ayyasami (2017). "Purported latest bone of a plated dinosaur (Ornithischia: Stegosauria), a "dermal plate" from the Maastrichtian (Upper Cretaceous) of southern India". *Neues Jahrbuch für Geologie und Paläontologie - Abhandlungen*. **285** (1): 91–96. Bibcode:2017NJGPA.285...91G (https://ui.adsabs.harvard.edu/abs/2017NJGPA.285...91G). doi:10.1127/njgpa/2017/0671 (https://doi.org/10.1127%2Fnjgpa%2F2017%2F0671).
4. Madzia, D.; Arbour, V.M.; Boyd, C.A.; Farke, A.A.; Cruzado-Caballero, P.; Evans, D.C. (2021). "The phylogenetic nomenclature of ornithischian dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8667728). *PeerJ*. **9** e12362. doi:10.7717/peerj.12362 (https://doi.org/10.7717%2Fpeerj.12362). PMC 8667728 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8667728). PMID 34966571 (https://pubmed.ncbi.nlm.nih.gov/34966571).
5. Sánchez-Fenollosa, Sergio; Cobos, Alberto (2025). "New insights into the phylogeny and skull evolution of stegosaurian dinosaurs: An extraordinary cranium from the European Late Jurassic (Dinosauria: Stegosauria)" (https://doi.org/10.3897%2Fvz.75.e146618). *Vertebrate Zoology*. **75**: 147–171. doi:10.3897/vz.75.e146618 (https://doi.org/10.3897%2Fvz.75.e146618).
6. Billon-Bruyat, Jean-Paul; Marty, Daniel (2010-09-04). "Preface: Symposium on Stegosauria proceedings" (https://doi.org/10.1007%2Fs00015-010-0027-z). *Swiss Journal of Geosciences*. **103** (2): 139–141. doi:10.1007/s00015-010-0027-z (https://doi.org/10.1007%2Fs00015-010-0027-z). ISSN 1661-8726 (https://search.worldcat.org/issn/1661-8726).
7. Sereno, Paul C. (1999-06-25). "The Evolution of Dinosaurs". *Science*. **284** (5423): 2137–2147. doi:10.1126/science.284.5423.2137 (https://doi.org/10.1126%2Fscience.284.5423.2137). ISSN 0036-8075 (https://search.worldcat.org/issn/0036-8075). PMID 10381873 (https://pubmed.ncbi.nlm.nih.gov/10381873).
8. Hayashi, Shoji; Carpenter, Kenneth; Watabe, Mahito; McWHINNEY, Lorrie A. (2012-01-01). "Ontogenetic histology of Stegosaurus plates and spikes" (https://doi.org/10.1111%2Fj.1475-4983.2011.01122.x). *Palaeontology*. **55** (1): 145–161. Bibcode:2012Palgy..55..145H (https://ui.adsabs.harvard.edu/abs/2012Palgy..55..145H). doi:10.1111/j.1475-4983.2011.01122.x (https://doi.org/10.1111%2Fj.1475-4983.2011.01122.x). ISSN 1475-4983 (https://search.worldcat.org/issn/1475-4983).
9. Maidment, S.C.R. (2010). "Stegosauria: a historical review of the body fossil record and phylogenetic relationships". *Swiss Journal of Geosciences*. **103** (Supplement 2, Symposium on Stegosauria): 199–210. Bibcode:2010SwJG..103..199M (https://ui.adsabs.harvard.edu/abs/2010SwJG..103..199M). doi:10.1007/s00015-010-0023-3 (https://doi.org/10.1007%2Fs00015-010-0023-3). S2CID 84415016 (https://api.semanticscholar.org/CorpusID:84415016).
10. Sereno, Paul C., and Dong Zhimin. "The Skull of the Basal Stegosaur Huayangosaurus Taibaii and a Cladistic Diagnosis of Stegosauria." *Journal of Vertebrate Paleontology* 12, no. 3 (1992): 318-43. JSTOR 4523456 (https://www.jstor.org/stable/4523456).
11. Senter, P. (2010). "Evidence for a sauropod-like metacarpal configuration in stegosaurian dinosaurs" (http://doc.rero.ch/record/32124/files/PAL_E2887.pdf) (PDF). *Acta Palaeontologica Polonica*. **55** (3): 427–432. doi:10.4202/app.2009.1105 (https://doi.org/10.4202%2Fapp.2009.1105). S2CID 53328847 (https://api.semanticscholar.org/CorpusID:53328847).
12. Mateus, Octávio; Maidment, Susannah C.R.; Christiansen, Nicolai A. (2009). "A new long-necked 'sauropod-mimic' stegosaur and the evolution of the plated dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2674496). *Proceedings of the Royal Society B: Biological Sciences*. **276** (1663): 1815–21. doi:10.1098/rspb.2008.1909 (https://doi.org/10.1098%2Frspb.2008.1909). PMC 2674496 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2674496). PMID 19324778 (https://pubmed.ncbi.nlm.nih.gov/19324778).
13. Z. Dong, S. Zhou, and Y. Zhang, 1983, "[Dinosaurs from the Jurassic of Sichuan]". *Palaeontologia Sinica, New Series C*, **162**(23): 1–136
14. Marsh, O. C. (1891). Restoration of Stegosaurus. American Journal of Science, 3rd series, 42, 179–182.
15. Lull, R. S. (1910a). Stegosaurus ungulatus Marsh, recently mounted at the Peabody Museum of Yale University. American Journal of Science, 4th series, 30, 361–377

16. Gilmore, C. W. (1914). "Osteology of the armored Dinosauria in the United States National Museum, with special reference to the genus *Stegosaurus*" (https://archive.org/details/osteologyofarmor00gilm). *Series: Smithsonian Institution. United States National Museum. Bulletin 89* (89). Government Printing Office, Washington.
17. Fastovsky D. E.; Weishampel D.B. (2005). "Stegosauria:Hot Plates". In Fastovsky D. E.; Weishampel D. B. (eds.). *The Evolution and Extinction of the Dinosaurs* (2nd ed.). Cambridge University Press. pp. 107–130. ISBN 978-0-521-81172-9.
18. Maidment, S.C.R.; Norman, D.B.; Barrett, P.M.; Upchurch, P. (2008). "Systematics and phylogeny of Stegosauria (Dinosauria: Ornithischia)". *Journal of Systematic Palaeontology*. **6** (4): 367–407. Bibcode:2008JSPal...6..367M (https://ui.adsabs.harvard.edu/abs/2008JSPal...6..367M). doi:10.1017/S1477201908002459 (https://doi.org/10.1017%2FS1477201908002459). S2CID 85673680 (https://api.semanticscholar.org/CorpusID:85673680).
19. Christiansen, Nicolai A.; Tschopp, Emanuel (2010-09-07). "Exceptional stegosaur integument impressions from the Upper Jurassic Morrison Formation of Wyoming" (https://doi.org/10.1007%2Fs00015-010-0026-0). *Swiss Journal of Geosciences*. **103** (2): 163–171. Bibcode:2010SwJG..103..163C (https://ui.adsabs.harvard.edu/abs/2010SwJG..103..163C). doi:10.1007/s00015-010-0026-0 (https://doi.org/10.1007%2Fs00015-010-0026-0). ISSN 1661-8726 (https://search.worldcat.org/issn/1661-8726). S2CID 129246092 (https://api.semanticscholar.org/CorpusID:129246092).
20. David E. Fastovsky, David B. Weishampel. The Evolution and Extinction of the Dinosaurs. p. 123
21. Galton, P.M., 1997, "Stegosauria", pp. 701-703 in: P.J. Currie and K. Padian (eds.), *Encyclopedia of Dinosaurs*, Academic Press, San Diego
22. Sereno, P.C., 1998, "A rationale for phylogenetic definitions, with application to the higher-level taxonomy of Dinosauria", *Neues Jahrbuch für Geologie und Paläontologie, Abhandlungen* **210**: 41-83
23. Pereda-Suberbiola, Xabier; Galton, Peter M.; Mallison, Heinrich; Novas, Fernando (2013). "A plated dinosaur (Ornithischia, Stegosauria) from the Early Cretaceous of Argentina, South America: an evaluation". *Alcheringa: An Australasian Journal of Palaeontology*. **37** (1): 65–78. Bibcode:2013Alch...37...65P (https://ui.adsabs.harvard.edu/abs/2013Alch...37...65P). doi:10.1080/03115518.2012.702531 (https://doi.org/10.1080%2F03115518.2012.702531). hdl:11336/67876 (https://hdl.handle.net/11336%2F67876). S2CID 128632458 (https://api.semanticscholar.org/CorpusID:128632458).
24. Rauhut, Oliver W. M.; Carballido, José Luis; Pol, Diego (2020-12-10). "First osteological record of a stegosaur (Dinosauria, Ornithischia) from the Upper Jurassic of South America" (https://www.tandfonline.com/doi/full/10.1080/02724634.2020.1862133). *Journal of Vertebrate Paleontology*. **40** (6) e1862133. Bibcode:2020JVPal..40E2133R (https://ui.adsabs.harvard.edu/abs/2020JVPal..40E2133R). doi:10.1080/02724634.2020.1862133 (https://doi.org/10.1080%2F02724634.2020.1862133). ISSN 0272-4634 (https://search.worldcat.org/issn/0272-4634). S2CID 234161169 (https://api.semanticscholar.org/CorpusID:234161169).
25. Dong, Z.; Tang, Z.; Zhou, S.W. (1982). "四川自贡大山铺蜀龙动物群简报1.剑龙" (http://www.ivpp.cas.cn/cbw/gjzdwxb/xbwzxz/200904/P020100311296596082254.pdf) [Note on the new Mid-Jurassic stegosaur from Sichuan Basin, China] (PDF). *Vertebrata PalAsiatica* (in Chinese (China)). **20** (1): 83–87.
26. Maidment, Susannah C.R.; Guangbiao Wei (2006). "A review of the Late Jurassic stegosaurs (Dinosauria, Stegosauria) from the People's Republic of China" (http://geolmag.geoscienceworld.org/cgi/content/abstract/143/5/621). *Geological Magazine*. **143** (5): 621–634. Bibcode:2006GeoM..143..621M (https://ui.adsabs.harvard.edu/abs/2006GeoM..143..621M). doi:10.1017/S0016756806002500 (https://doi.org/10.1017%2FS0016756806002500). S2CID 83661067 (https://api.semanticscholar.org/CorpusID:83661067).
27. Norman, David (2001). "*Scelidosaurus*, the earliest complete dinosaur" in *The Armored Dinosaurs*, pp 3-24. Bloomington: Indiana University Press. ISBN 0-253-33964-2.
28. Galton, Peter (1997). "21: *Stegosaurs*" (https://books.google.com/books?id=FOViD-lDPy0C&pg=PA291). In James O. Farlow; M. K. Brett-Surman (eds.). *The Complete Dinosaur*. Bloomington: Indiana University Press. ISBN 978-0-253-21313-6.
29. Peter M. Galton (2019). "Earliest record of an ankylosaurian dinosaur (Ornithischia: Thyreophora): Dermal armor from Lower Kota Formation (Lower Jurassic) of India". *Neues Jahrbuch für Geologie und Paläontologie - Abhandlungen*. **291** (2): 205–219. Bibcode:2019NJGPA.291..205G (https://ui.adsabs.harvard.edu/abs/2019NJGPA.291..205G). doi:10.1127/njgpa/2019/0800 (https://doi.org/10.1127%2Fnjgpa%2F2019%2F0800). S2CID 134302379 (https://api.semanticscholar.org/CorpusID:134302379).
30. "PBDB" (https://paleobiodb.org/classic/basicCollectionSearch?collection_no=140254).

31. Rauhut, Oliver W. M.; Carballido, José Luis; Pol, Diego (2020-12-10). "First osteological record of a stegosaur (Dinosauria, Ornithischia) from the Upper Jurassic of South America" (https://www.tandfonline.com/doi/full/10.1080/02724634.2020.1862133). *Journal of Vertebrate Paleontology*. **40** (6) e1862133. doi:10.1080/02724634.2020.1862133 (https://doi.org/10.1080%2F02724634.2020.1862133). ISSN 0272-4634 (https://search.worldcat.org/issn/0272-4634).
32. Holtz, Thomas R. Jr.; Rey, Luis V. (2007). *Dinosaurs: the most complete, up-to-date encyclopedia for dinosaur lovers of all ages* (https://archive.org/details/dinosaursmostcom00holt). New York: Random House. ISBN 978-0-375-82419-7.
33. Leshchinskiy, S. V.; Fayngerts, A. V.; Ivantsov, S. V. (2019-10-20). "Bol'shoi Ilek as the ilek formation stratotype of the Lower Cretaceous, and the new dinosaur and mammoth faunae site in the south-east of Western Siberia" (https://journals.eco-vector.com/0869-5652/article/view/17790). *Доклады Академии Наук СССР* (in Russian). **488** (5): 513–516. doi:10.31857/S0869-56524885513-516 (https://doi.org/10.31857%2FS0869-56524885513-516). ISSN 0869-5652 (https://search.worldcat.org/issn/0869-5652).
34. Skutschas, Pavel P.; Gvozdkova, Vera A.; Averianov, Alexander O.; Lopatin, Alexey V.; Martin, Thomas; Schellhorn, Rico; Kolosov, Petr N.; Markova, Valentina D.; Kolchanov, Veniamin V.; Grigoriev, Dmitry V.; Kuzmin, Ivan T. (2021-03-17). Pardo-Pérez, Judith (ed.). "Wear patterns and dental functioning in an Early Cretaceous stegosaur from Yakutia, Eastern Russia" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7968641). *PLOS ONE*. **16** (3) e0248163. Bibcode:2021PLoSO..1648163S (https://ui.adsabs.harvard.edu/abs/2021PLoSO..1648163S). doi:10.1371/journal.pone.0248163 (https://doi.org/10.1371%2Fjournal.pone.0248163). ISSN 1932-6203 (https://search.worldcat.org/issn/1932-6203). PMC 7968641 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7968641). PMID 33730093 (https://pubmed.ncbi.nlm.nih.gov/33730093).
35. Tumanova, T. A.; Alifanov, V. R. (2018-12-01). "First Record of Stegosaur (Ornithischia, Dinosauria) from the Aptian–Albian of Mongolia". *Paleontological Journal*. **52** (14): 1771–1779. Bibcode:2018PalJ...52.1771T (https://ui.adsabs.harvard.edu/abs/2018PalJ...52.1771T). doi:10.1134/S0031030118140186 (https://doi.org/10.1134%2FS0031030118140186). ISSN 1555-6174 (https://search.worldcat.org/issn/1555-6174). S2CID 91559457 (https://api.semanticscholar.org/CorpusID:91559457).
36. Li, Ning; Li, Daqing; Peng, Guangzhao; You, Hailu (2024). "The first stegosaurian dinosaur from Gansu Province, China" (https://linkinghub.elsevier.com/retrieve/pii/S0195667124000259). *Cretaceous Research*. **158** (in press) 105852. Bibcode:2024CrRes.15805852L (https://ui.adsabs.harvard.edu/abs/2024CrRes.15805852L). doi:10.1016/j.cretres.2024.105852 (https://doi.org/10.1016%2Fj.cretres.2024.105852).
37. Jia, Lei; Li, Ning; Dong, Liyang; Shi, Jianru; Kang, Zhishuai; Wang, Suozhu; Xu, Shichao; You, Hailu (2024-01-31). "A new stegosaur from the late Early Cretaceous of Zuoyun, Shanxi Province, China". *Historical Biology*. **37** (2): 420–429. doi:10.1080/08912963.2024.2308214 (https://doi.org/10.1080%2F08912963.2024.2308214). ISSN 0891-2963 (https://search.worldcat.org/issn/0891-2963).
38. Bakker, R.T., 1998, "Dinosaur mid-life crisis: the Jurassic-Cretaceous transition in Wyoming and Colorado", In: S.G. Lucas, J.I. Kirkland, & J.W. Estep (eds.) *Lower and Middle Cretaceous Terrestrial Ecosystems; New Mexico Museum of Natural History and Science Bulletin*, **14**: 67-77
39. Butler, R.J., Barrett, P.M., Kenrick, P. and Penn, M.G., 2009, "Diversity patterns amongst herbivorous dinosaurs and plants during the Cretaceous: implications for hypotheses of dinosaur/angiosperm co-evolution", *Journal of Evolutionary Biology*, **22**: 446–459
40. Chatterjee, S., and Rudra, D.K., 1996, "KT events in India: impact, rifting, volcanism and dinosaur extinction," in: Novas & Molnar, eds., *Proceedings of the Gondwanan Dinosaur Symposium, Brisbane, Memoirs of the Queensland Museum*, **39**(3): iv + 489–731 : 489-532
41. Wilson, J. A., Barrett, P. M., & Carrano, M. T. (2011). An associated partial skeleton of *Jainosaurus cf. septentrionalis* (Dinosauria: Sauropoda) from the Late Cretaceous of Chhota Simla, central India. *Palaeontology*, 54(5), 981-998.
42. Spassov, N. B. (1982). The *bizarre* dorsal plates of stegosaurs: ethological approach. Comptes rendus de l'academie bulgare des Sciences, 35, 367–370.
43. "Walk and Don't Look Back: The Footprints; Stegosaurs" in Foster, J. (2007). Jurassic West: The Dinosaurs of the Morrison Formation and Their World. Indiana University Press. pg. 238
44. Mateus, Octávio; Milàn, Jesper; Romano, Michael; Whyte, Martin A. (September 2011). "New Finds of Stegosaur Tracks from the Upper Jurassic Lourinhã Formation, Portugal" (http://www.app.pan.pl/article/item/app20090055.html). *Acta Palaeontologica Polonica*. **56** (3): 651–658. doi:10.4202/app.2009.0055 (https://doi.org/10.4202%2Fapp.2009.0055). ISSN 0567-7920 (https://search.worldcat.org/issn/0567-7920). S2CID 55869900 (https://api.semanticscholar.org/CorpusID:55869900).

45. BELVEDERE, MATTEO; MIETTO, PAOLO (January 2010). "First evidence of stegosaurianDeltapodusfootprints in North Africa (Iouaridène Formation, Upper Jurassic, Morocco)" (https://doi.org/10.1111%2Fj.1475-4983.2009.00928.x). *Palaeontology*. **53** (1): 233–240. Bibcode:2010Palgy..53..233B (https://ui.adsabs.harvard.edu/abs/2010Palgy..53..233B). doi:10.1111/j.1475-4983.2009.00928.x (https://doi.org/10.1111%2Fj.1475-4983.2009.00928.x). ISSN 0031-0239 (https://search.worldcat.org/issn/0031-0239). S2CID 140670546 (https://api.semanticscholar.org/CorpusID:140670546).
46. Xing, Lida; Lockley, Martin G.; Persons, W. Scott; Klein, Hendrik; Romilio, Anthony; Wang, Donghao; Wang, Miaoyan (2021-02-28). "Stegosaur Track Assemblage from Xinjiang, China, Featuring the Smallest Known Stegosaur Record" (https://pubs.geoscienceworld.org/sepm/palaios/article-abstract/36/2/68/595155/STEGOSAUR-TRACK-ASSEMBLAGE-FROM-XINJIANG-CHINA). *PALAIOS*. **36** (2): 68–76. Bibcode:2021Palai..36...68X (https://ui.adsabs.harvard.edu/abs/2021Palai..36...68X). doi:10.2110/palo.2020.036 (https://doi.org/10.2110%2Fpalo.2020.036). ISSN 0883-1351 (https://search.worldcat.org/issn/0883-1351). S2CID 233129489 (https://api.semanticscholar.org/CorpusID:233129489).
47. Queensl, The University of; Lucia, Australia Brisbane St; Gatton, QLD 4072 +61 7 3365 1111 Other Campuses: UQ; Maps, UQ Herston; Queensl, Directions © 2021 The University of. "Tiny cat-sized stegosaur leaves its mark" (https://www.uq.edu.au/news/article/2021/04/tiny-cat-sized-stegosaur-leaves-its-mark). *UQ News*. Retrieved 2021-04-25.
48. Salisbury, Steven W.; Romilio, Anthony; Herne, Matthew C.; Tucker, Ryan T.; Nair, Jay P. (2016-12-12). "The Dinosaurian Ichnofauna of the Lower Cretaceous (Valanginian–Barremian) Broome Sandstone of the Walmadany Area (James Price Point), Dampier Peninsula, Western Australia" (https://doi.org/10.1080%2F02724634.2016.1269539). *Journal of Vertebrate Paleontology*. **36** (sup1): 1–152. Bibcode:2016JVPal..36S...1S (https://ui.adsabs.harvard.edu/abs/2016JVPal..36S...1S). doi:10.1080/02724634.2016.1269539 (https://doi.org/10.1080%2F02724634.2016.1269539). hdl:1959.11/58119 (https://hdl.handle.net/1959.11%2F58119). ISSN 0272-4634 (https://search.worldcat.org/issn/0272-4634).
49. Romilio, Anthony; Salisbury, Steven W.; Jannel, Andréas (2020-06-12). "Footprints of large theropod dinosaurs in the Middle–UpperJurassic (lower Callovian–lower Tithonian) Walloon Coal Measures of southern Queensland, Australia". *Historical Biology*. **33** (10): 2135–2146. doi:10.1080/08912963.2020.1772252 (https://doi.org/10.1080%2F08912963.2020.1772252). ISSN 0891-2963 (https://search.worldcat.org/issn/0891-2963). S2CID 225692077 (https://api.semanticscholar.org/CorpusID:225692077).
50. Bakker, R. T. (1986). *The Dinosaur Heresies* (https://archive.org/details/dinosaurheresies0000bakk). New York: William Morrow. ISBN 978-0-688-04287-5.
51. Naish, Darren (2010). "Heinrich's digital Kentrosaurus: the SJG stegosaur special, part II" (https://web.archive.org/web/20110109004732/http://scienceblogs.com/tetrapodzoology/2011/01/heinrichs_digital_kentrosaurus.php). *Tetrapod Zoology*. Archived from the original (http://scienceblogs.com/tetrapodzoology/2011/01/heinrichs_digital_kentrosaurus.php) on January 9, 2011. Retrieved January 19, 2011.
52. McWhinney, L. A.; Rothschild, B. M. & Carpenter, K. (2001). "Posttraumatic Chronic Osteomyelitis in *Stegosaurus* dermal spikes". In Carpenter, Kenneth (ed.). *The Armored Dinosaurs*. Indiana University Press. pp. 141–56. ISBN 978-0-253-33964-5.
53. Carpenter, Kenneth; Sanders, Frank; McWhinney, Lorrie A. & Wood, Lowell (2005). "Evidence for predator-prey relationships: Examples for *Allosaurus* and *Stegosaurus*". In Carpenter, Kenneth (ed.). *The Carnivorous Dinosaurs* (https://www.researchgate.net/publication/314890308_Evidence_for_Predator-_Prey_Relationships_Examples_for_Allosaurus_and_Stegosaurus). Bloomington and Indianapolis: Indiana University Press. ISBN 978-0-253-34539-4.
54. Marsh, O. C (1877). "A new order of extinct Reptilia (Stegosauria) from the Jurassic of the Rocky Mountains" (https://zenodo.org/record/1450038). *American Journal of Science*. **14** (84): 513–14. Bibcode:1877AmJS...14..513M (https://ui.adsabs.harvard.edu/abs/1877AmJS...14..513M). doi:10.2475/ajs.s3-14.84.513 (https://doi.org/10.2475%2Fajs.s3-14.84.513). S2CID 130078453 (https://api.semanticscholar.org/CorpusID:130078453).
55. Carpenter, K. & Galton, P. M. (2001). "Othniel Charles Marsh and the Eight-Spiked *Stegosaurus*". In Carpenter, Kenneth (ed.). *The Armored Dinosaurs*. Indiana University Press. pp. 76–102. ISBN 978-0-253-33964-5.

56. Mallison, Heinrich (2010-09-07). "CAD assessment of the posture and range of motion of Kentrosaurus aethiopicus Hennig 1915" (https://doi.org/10.1007%2Fs00015-010-0024-2). *Swiss Journal of Geosciences*. **103** (2): 211–233. Bibcode:2010SwJG..103..211M (https://ui.adsabs.harvard.edu/abs/2010SwJG..103..211M). doi:10.1007/s00015-010-0024-2 (https://doi.org/10.1007%2Fs00015-010-0024-2). ISSN 1661-8726 (https://search.worldcat.org/issn/1661-8726). S2CID 132746786 (https://api.semanticscholar.org/CorpusID:132746786).
57. Saitta, Evan Thomas (2015-04-22). "Evidence for Sexual Dimorphism in the Plated Dinosaur Stegosaurus mjosi (Ornithischia, Stegosauria) from the Morrison Formation (Upper Jurassic) of Western USA" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4406738). *PLOS ONE*. **10** (4) e0123503. Bibcode:2015PLoSO..1023503S (https://ui.adsabs.harvard.edu/abs/2015PLoSO..1023503S). doi:10.1371/journal.pone.0123503 (https://doi.org/10.1371%2Fjournal.pone.0123503). ISSN 1932-6203 (https://search.worldcat.org/issn/1932-6203). PMC 4406738 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4406738). PMID 25901727 (https://pubmed.ncbi.nlm.nih.gov/25901727).
58. "Dino 'sexing' study slammed by critics" (https://www.science.org/content/article/dino-sexing-study-slammed-critics). *Science* | *AAAS*. 2015-04-22. Retrieved 2016-06-04.
59. Reichel, Miriam (2010-08-31). "A model for the bite mechanics in the herbivorous dinosaur Stegosaurus (Ornithischia, Stegosauridae)" (https://doi.org/10.1007%2Fs00015-010-0025-1). *Swiss Journal of Geosciences*. **103** (2): 235–240. Bibcode:2010SwJG..103..235R (https://ui.adsabs.harvard.edu/abs/2010SwJG..103..235R). doi:10.1007/s00015-010-0025-1 (https://doi.org/10.1007%2Fs00015-010-0025-1). ISSN 1661-8726 (https://search.worldcat.org/issn/1661-8726). S2CID 84869720 (https://api.semanticscholar.org/CorpusID:84869720).
60. Parrish, Judith Totman; Peterson, Fred; Turner, Christine E (2004-05-15). "Jurassic "savannah"—plant taphonomy and climate of the Morrison Formation (Upper Jurassic, Western USA)" (http://doc.rero.ch/record/14574/files/PAL_E1788.pdf) (PDF). *Sedimentary Geology*. Reconstruction of the Extinct Ecosystem of the Upper Jurassic Morrison Formation. **167** (3–4): 137–162. Bibcode:2004SedG..167..137P (https://ui.adsabs.harvard.edu/abs/2004SedG..167..137P). doi:10.1016/j.sedgeo.2004.01.004 (https://doi.org/10.1016%2Fj.sedgeo.2004.01.004).
61. Skutschas, Pavel P.; Gvozdkova, Vera A.; Averianov, Alexander O.; Lopatin, Alexey V.; Martin, Thomas; Schellhorn, Rico; Kolosov, Petr N.; Markova, Valentina D.; Kolchanov, Veniamin V.; Grigoriev, Dmitry V.; Kuzmin, Ivan T.; Vitenko, Dmitry D. (17 March 2021). Pardo-Pérez, Judith (ed.). "Wear patterns and dental functioning in an Early Cretaceous stegosaur from Yakutia, Eastern Russia" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7968641). *PLOS ONE*. **16** (3) e0248163. Bibcode:2021PLoSO..1648163S (https://ui.adsabs.harvard.edu/abs/2021PLoSO..1648163S). doi:10.1371/journal.pone.0248163 (https://doi.org/10.1371%2Fjournal.pone.0248163). ISSN 1932-6203 (https://search.worldcat.org/issn/1932-6203). PMC 7968641 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7968641). PMID 33730093 (https://pubmed.ncbi.nlm.nih.gov/33730093).

External links

- Stegosauria on Palaeos.com (http://palaeos.com/vertebrates/ornithischia/stegosauria.html)
- https://web.archive.org/web/20080727011652/http://www.kheper.net/evolution/dinosauria/Stegosauria.htm

Retrieved from "https://en.wikipedia.org/w/index.php?title=Stegosauria&oldid=1374690025"

WIKIPEDIA
25 years of the free encyclopedia

Ankylosauria

Ankylosauria is a group of herbivorous dinosaurs that originate of the clade Ornithischia. It includes the great majority of dinosaurs with armor in the form of bony osteoderms, similar to turtles. Ankylosaurs were bulky quadrupeds, with short, powerful limbs. They are known to have first appeared in North Africa during the Middle Jurassic, and persisted until the end of the Late Cretaceous. The two main families of ankylosaurians, Nodosauridae and Ankylosauridae primarily originated from the Northern Hemisphere (North America, Europe and Asia). The more basal Parankylosauria originated from southern Gondwana (South America, Australia and Antarctica) during the Cretaceous period, though it has been suggested that this group is a separate lineage of thyreophorans outside both Ankylosauria and Stegosauria.[4]

Ankylosauria was first named by Henry Fairfield Osborn in 1923.[5] In the Linnaean classification system, the group is usually considered either a suborder or an infraorder. It is contained within the group Thyreophora, which also includes the stegosaurs, armored dinosaurs known for their combination of plates and spikes.

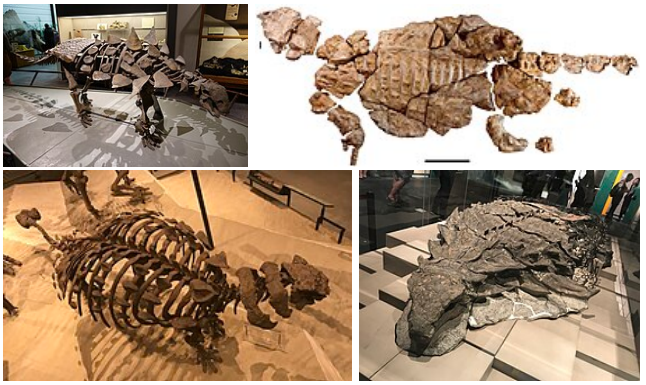
Etymology

The name of this group of dinosaurs is associated with a number of anatomical features in which small and large bony shields fused together, completely covering their back and sides. On the skull these shields fused with the underlying bones, and the dorsal ribs fit snugly to the vertebrae. The Latin name Ankylosauria is derived from the Greek ἀγκύλος [*ankylos*] "curved" or "bent" with the anatomical meaning "hard" or "fused", and σαῦρος [*sauros*] "lizard".[6]

Ankylosaurs

Temporal range:

Middle Jurassic—Late Cretaceous,



Four ankylosaurs. Clockwise from top left: *Gastonia* (a polacanthine), *Kunbarrasaurus* (a parankylosaur), *Borealopelta* (a nodosaurid), and *Akainacephalus* (an ankylosaurid)

Scientific classification

Kingdom:	Animalia
Phylum:	Chordata
Class:	Reptilia
Clade:	Dinosauria
Clade:	†Ornithischia
Clade:	†Thyreophora
Clade:	†Eurypoda
Clade:	†Ankylosauria Osborn, 1923

Subgroups

- †Parankylosauria?
- †**Euankylosauria** Soto-Acuña et al., 2021[1]
  - †Ankylosauridae
  - †Nodosauridae

In the 1908 description of the genus *Ankylosaurus*, Barnum Brown described the family Ankylosauridae as a group of representatives with a "rigid spine", but noted the wide, curved shape of the ribs, suggesting a "strongly curved" back[7] (an error based on the alleged similarity to stegosaurs and glyptodonts, as ankylosaurs have flat backs). Therefore, "rigid lizard" and "curved lizard" could be additional meanings applied to the name of ankylosaurs.[6]

Classification

Mount of ankylosaurid *Zuul* using its tail club to fight off *Gorgosaurus*

Genera of uncertain affinity[2]

- †*Cryptosaurus*
- †*Dracopelta*
- †*Jakapil*?
- †*Minmi*
- †*Mymoorapelta*
- †*Sarcolestes*
- †*Serendipaceratops*?
- †*Sinankylosaurus*?
- †*Spicomellus*
- †*Stegosaurides*?
- †*Tianchisaurus*
- †*Vectipelta*
- †*Zhejiangosaurus*

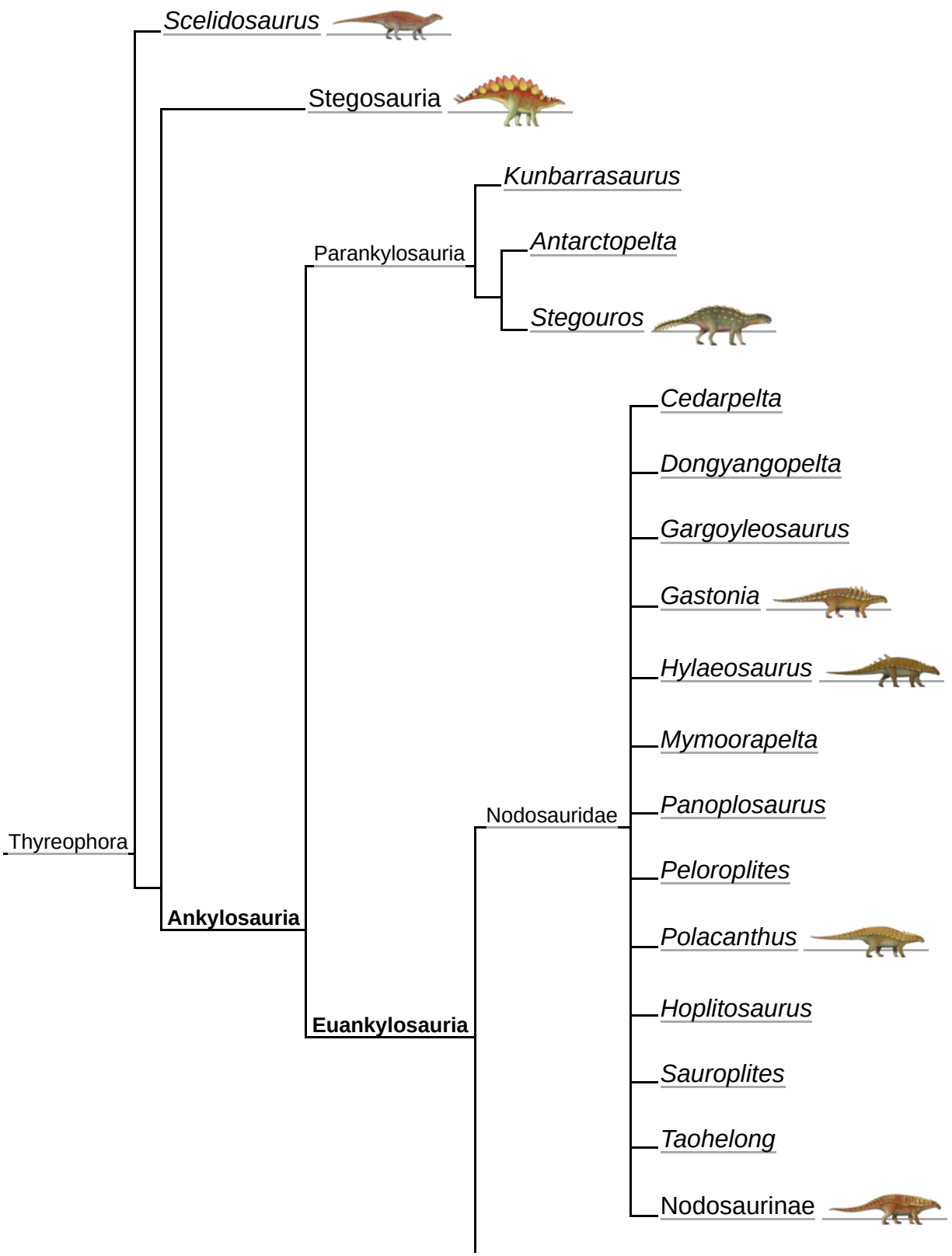
Synonyms[3]

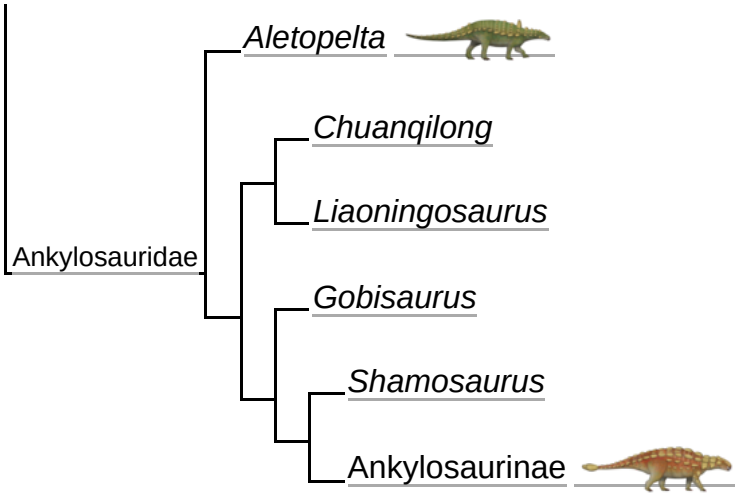
- Ancylosauria Huene, 1914 (*lapsus calami*)
- Ankylosauroidea Norman, 1984
- Ankylosauromorpha Carpenter, 2001
- Apraedentalia Huene, 1948
- Apraedentalidae Huene, 1956
- Nodosauria Brown & Schlaikjer, 1943
- Oligosacralosauroidea Zhao, 1983
- Polysacralosauroidea Zhao, 1983

Ankylosauria and Stegosauria together form the two major subgroups of Thyreophora, a group of armoured dinosaurs distinct from ornithopods and marginocephalians.[8] Historically used for forms lacking large vertical plates, Kenneth Carpenter proposed in 1997 the first informal definition of the group, as all ornithischians closer to *Ankylosaurus* than *Stegosaurus*.[9] This definition was further refined by Paul Sereno in 2005 to specify *Ankylosaurus magniventris* and *Stegosaurus stenops*, the type species of both genera, a definition that was followed by Madzia and colleagues in 2021 when the group name and definition was formalized following the PhyloCode.[8]

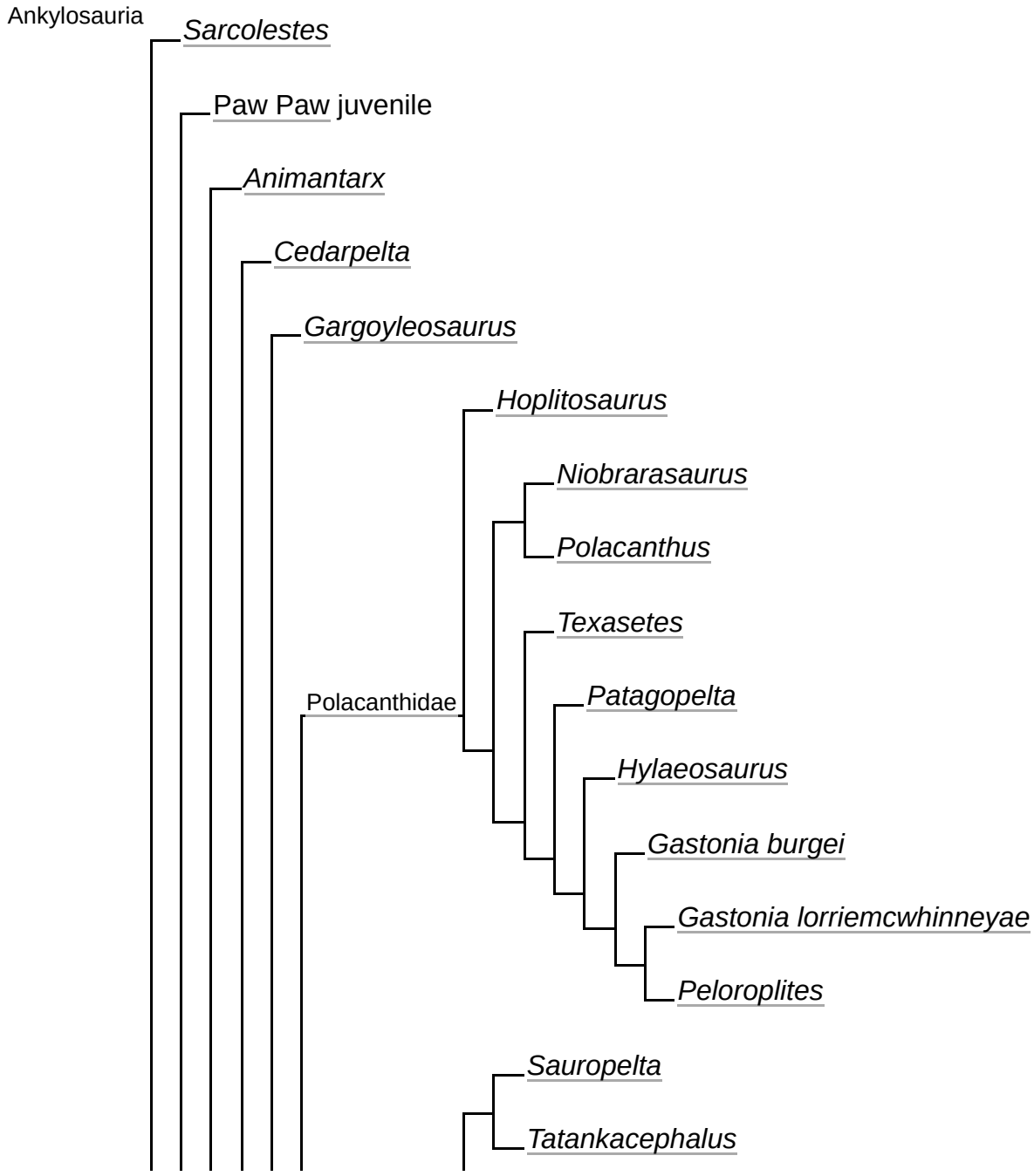
Phylogenetic and morphological studies have differed on the inclusion of certain early taxa into Ankylosauria, especially the armoured Early Jurassic form *Scelidosaurus*. As some analyses, like that of Carpenter from 2001 or David B. Norman in 2021 find *Scelidosaurus* and possibly other early forms like *Emausaurus* and *Scutellosaurus* to fall closer to *Ankylosaurus* than *Stegosaurus*, Carpenter and later Norman suggested redefining Ankylosauria to limit it to the two subclades Nodosauridae and Ankylosauridae, creating the new clade Ankylosauromorpha for all taxa closer to *Ankylosaurus* than *Stegosaurus*.[8][10][11] However, as historically even these primitive forms were considered ankylosaurs if they were more derived than *Stegosaurus*, Madzia and colleagues considered a redefinition of

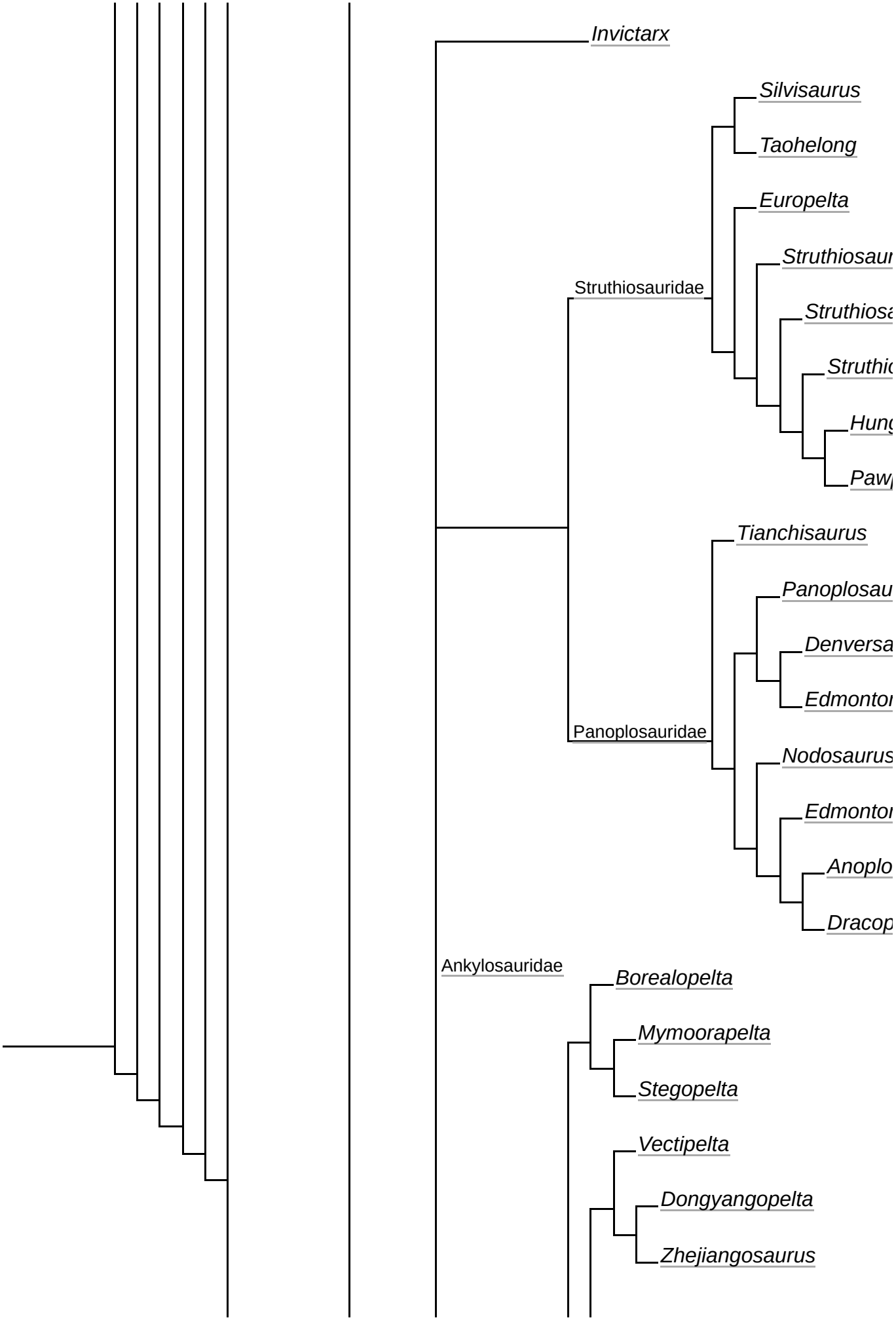
Ankylosauria to be undesirable, instead preferring to abandon Ankylosauromorpha as a same-definition junior synonym of Ankylosauria. The clade **Euankylosauria** was named by Soto-Acuña and colleagues in 2021 in their description of a unique basal ankylosaur *Stegouros* to represent the grouping uniting ankylosaurids and nodosaurids, to the exclusion of their newly discovered clade Parankylosauria.[1] This clade is formally defined in the *PhyloCode* as "the largest clade containing *Ankylosaurus magniventris* and *Nodosaurus textilis*, but not *Stegouros elengassen*".[12]

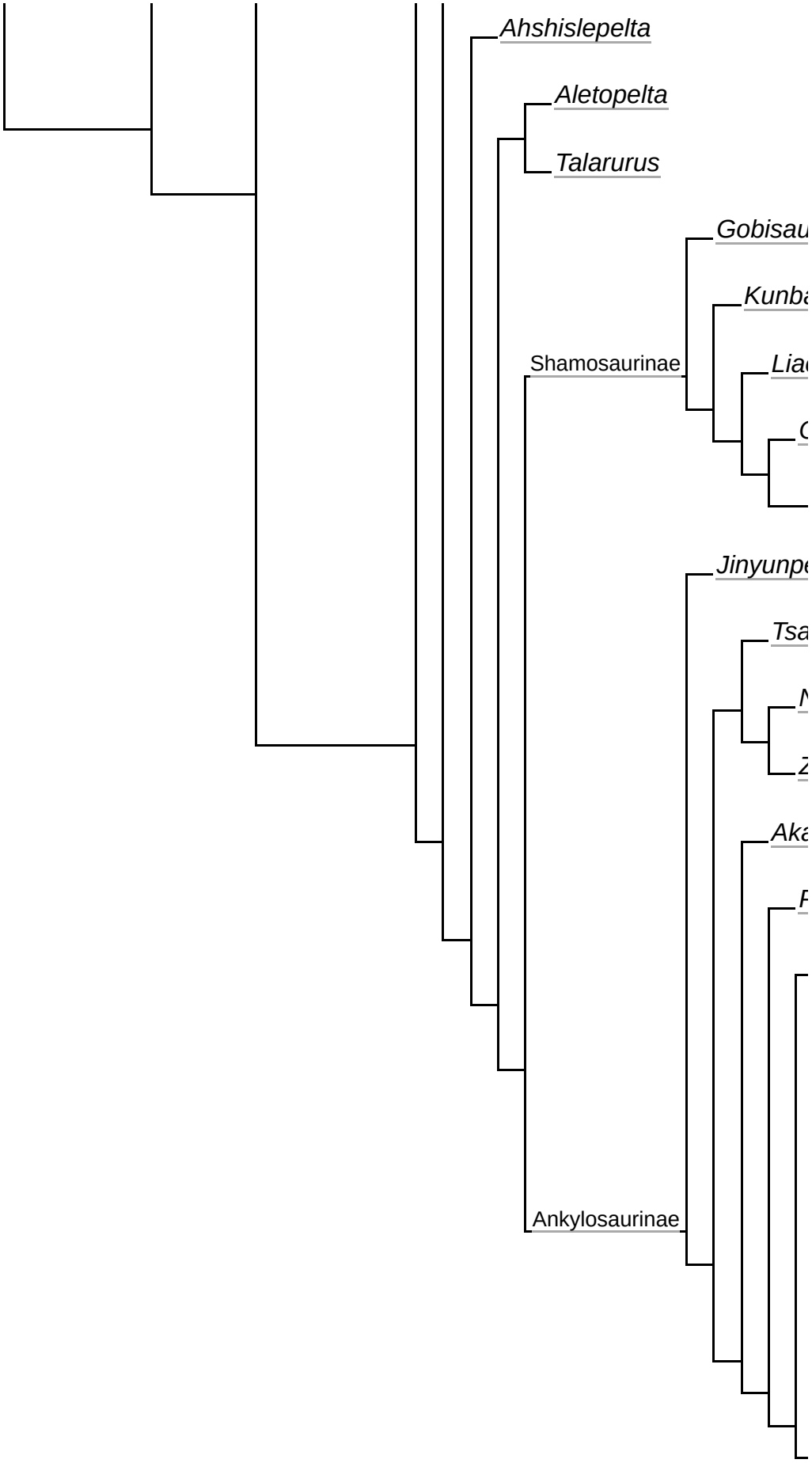




A 2023 review of Thyreophora rejects the traditional Ankylosauridae-Nodosauridae split, instead finding "nodosaurids" to be referrable to three separate families: Panoplosauridae, Polacanthidae, and Struthiosauridae.[13]




Invictarx
Silvisaurus
Taohelong
Europelta
Struthiosaur
Struthiosauridae
Struthiosa
Struthio
Hung
Pawp
Tianchisaurus
Panoplosau
Denversa
Edmonton
Panoplosauridae
Nodosaurus
Edmonton
Anoplo
Dracop
Ankylosauridae
Borealopelta
Mymoorapelta
Stegopelta
Vectipelta
Dongyangopelta
Zhejiangosaurus

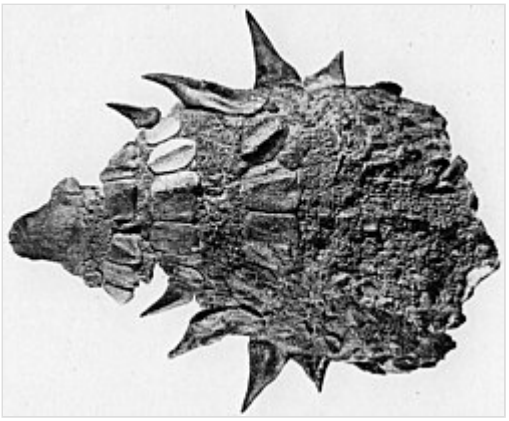

Ahshislepelta
Aletopelta
Talarurus
Shamosaurinae
Ankylosaurinae

Evolution

The origin of ankylosaurs is often misunderstood, and only a few specimens from the Middle Jurassic are known.[14] The ancestry of ankylosaurs has long been sought among stegosaurs, the closest group to ankylosaurs compared to other dinosaurs. Currently, ankylosaurs are a close group of stegosaurs within the Eurypoda clade.[15] They are united by the presence of osteoderms in the skin, the narrow triangular skull of stegosaurs is similar to that of nodosaurids, and some similarity is found in the structure of the palate.[16] Ankylosaurs were definitely present in Africa by the Bathonian age as evidenced by *Spicomellus* in Morocco.[17] Although a possible Early Jurassic (Sinemurian-Pliensbachian) origin has been suggested by Peter Galton in 2019 based on potential ankylosaur remains found in the Kota Formation,[18] Spencer G. Lucas suggested in 2025 that this fossil record might instead be much younger than previously thought due to the age of the formation most likely extending to the Middle Jurassic.[19] An incomplete radius and ulna from the Bajocian-aged strata in Isle of Skye, Scotland are known, the exact affiliation of which to ankylosaurs or stegosaurs is not established.[20] Most likely, ankylosaurs followed a different evolutionary path from stegosaurs, although it is unknown when and how they split off. In the latter, the osteoderms become raised, and the lateral protection disappears.[21] Ankylosaurs evolved towards the development of osteoderms on the surface of the skull, increased armor and further consolidation of the carapace,[22] which suggests that the ancestor of the carapace consisted of separate non-fused osteoderms.[16]

Paleobiology

Possible neonate-sized ankylosaur fossils have been documented in the scientific literature.[23]



Armour of the nodosaurid *Edmontonia*

Armor

All ankylosaurians had armor over much of their bodies from head to tail, mostly scutes and nodules, with large spines in some cases. The scutes, or plates, are rectangular to oval objects organized in transverse (side to side) rows, often with keels. Smaller nodules and plates filled in the open spaces between large plates. The two or three solid, bony ring/band around the neck have three or four scutes which can be

adjacent to or fused to each other; in nodosaurids, this ring usually comes from several scutes adjoining against each other, while ankylosaurids usually have the scutes completely or partially fused to it.[24][25]

Diet and feeding

Ankylosaurs were built low to the ground, typically one foot off the ground surface. They had small, triangular teeth that were loosely packed, similar to stegosaurs. The large hyoid bones left in skeletons indicates that they had long, flexible tongues. They also had a large, side secondary palate. This means that they could breathe while chewing, unlike crocodiles. Their expanded gut region suggests the use of fermentation to digest their food, using symbiotic bacteria and gut flora. Their diet likely consisted of ferns, cycads, and angiosperms. Mallon et al. (2013) examined herbivore coexistence on the island continent of Laramidia during the Late Cretaceous. It was concluded that ankylosaurs were generally restricted to feeding on vegetation at, or below, the height of 1 meter.[26]

Vocalization

In February 2023, scientists reported that, based on analysis of a fossilized larynx from the ankylosaur *Pinacosaurus grangeri*, ankylosaur vocalizations may have been bird-like in the sense that sounds may have been produced low in the trachea (though not by a syrinx as in birds) and modified higher up.[27][28]

See also

- Timeline of ankylosaur research

References

1. Soto-Acuña, Sergio; Vargas, Alexander; Kaluza, Jonatan; Leppe, Marcelo; Botelho, Joao; Palma-Liberona, José; Gutstein, Carolina; Fernández, Roy; Ortiz, Hector; Milla, Verónica; Aravena, Bárbara; Manríquez, Leslie M. E.; Alarcón-Muñoz, Jhonatan; Pino, Juan Pablo; Trevisan, Cristine; Mansilla, Héctor; Hinojosa, Luis Felipe; Muñoz-Walther, Vicente; Rubilar-Rogers, David (2021). "Bizarre tail weaponry in a transitional ankylosaur from subantarctic Chile" (https://www.researchsquare.com/article/rs-821192/v1). *Nature*. **600** (7888): 259–263. Bibcode:2021Natur.600..259S (https://ui.adsabs.harvard.edu/abs/2021Natur.600..259S). doi:10.1038/s41586-021-04147-1 (https://doi.org/10.1038%2Fs41586-021-04147-1). PMID 34853468 (https://pubmed.ncbi.nlm.nih.gov/34853468). S2CID 244799975 (https://api.semanticscholar.org/CorpusID:244799975).
2. Fonseca, A. O.; Reid, I. J.; Venner, A.; Duncan, R. J.; Garcia, M. S.; Müller, R. T. (2024). "A comprehensive phylogenetic analysis on early ornithischian evolution". *Journal of Systematic Palaeontology*. **22** (1). 2346577. Bibcode:2024JSPal..2246577F (https://ui.adsabs.harvard.edu/abs/2024JSPal..2246577F). doi:10.1080/14772019.2024.2346577 (https://doi.org/10.1080%2F14772019.2024.2346577).
3. Reid, I.J.; Fonseca, A.O.; Madzia, D. (2026). "The evolution of ornithischian nomenclature". *Zoological Journal of the Linnean Society*. **207** (3) zlag111. doi:10.1093/zoolinnean/zlag111 (https://doi.org/10.1093%2Fzoolinnean%2Fzlag111).

4. Agnolín, Federico L.; Rozadilla, Sebastián; Marsà, Jordi García; Álvarez Nogueira, Rodrigo; Miner, Santiago; Álvarez-Herrera, Gerardo; Novas, Fernando E.; Pol, Diego (2026-01-13). "New remains of the armored dinosaur *Patagopelta cristata* Riguetti et al. 2022 (Ornithischia, Parankylosauria) from the Late Cretaceous of Patagonia, Argentina". *Historical Biology*: 1–64. doi:10.1080/08912963.2025.2583504 (https://doi.org/10.1080%2F08912963.2025.2583504). ISSN 0891-2963 (https://search.worldcat.org/issn/0891-2963).
5. Osborn, H. F. (1923). "Two Lower Cretaceous dinosaurs of Mongolia." *American Museum Novitates*, **95**: 1–10.[1]
6. Ben Creisler, *Dinosauria Translation and Pronunciation Guide.* (http://www.dinosauria.com/dml/names/dinoj.htm) Archived (https://web.archive.org/web/20111111112017/http://www.dinosauria.com/dml/names/dinoj.htm) 2011-11-11 at the Wayback Machine
7. Brown, B. 1908. The Ankylosauridae, a new family of armored dinosaurs from the Upper Cretaceous. American Museum of Natural History Bulletin 24: 187—201.
8. Madzia, D.; Arbour, V.M.; Boyd, C.A.; Farke, A.A.; Cruzado-Caballero, P.; Evans, D.C. (2021). "The phylogenetic nomenclature of ornithischian dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8667728). *PeerJ*. **9** e12362. doi:10.7717/peerj.12362 (https://doi.org/10.7717%2Fpeerj.12362). PMC 8667728 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8667728). PMID 34966571 (https://pubmed.ncbi.nlm.nih.gov/34966571).
9. Carpenter, K., 1997, "Ankylosauria" pp. 16–20 in: P.J. Currie and K. Padian (eds.), *Encyclopedia of Dinosaurs*, Academic Press, San Diego
10. Carpenter, K. (2001). "Phylogenetic Analysis of the Ankylosauria". In Carpenter, Kenneth (ed.). *The Armored Dinosaurs* (https://books.google.com/books?id=04tQ5_qJN8MC). Indiana University Press. p. 455. ISBN 0-253-33964-2.
11. Norman, D.B. (2021). "*Scelidosaurus harrisonii* (Dinosauria: Ornithischia) from the Early Jurassic of Dorset, England: biology and phylogenetic relationships". *Zoological Journal of the Linnean Society*. **191** (1): 1–86. doi:10.1093/zoolinnean/zlaa061 (https://doi.org/10.1093%2Fzoolinnean%2Fzlaa061).
12. Fonseca, A.O.; Reid, I.J.; Venner, A.; Duncan, R.J.; Garcia, M.S.; Müller, R.T. (2024). "A comprehensive phylogenetic analysis on early ornithischian evolution". *Journal of Systematic Palaeontology*. **22** (1) 2346577. Bibcode:2024JSPal..2246577F (https://ui.adsabs.harvard.edu/abs/2024JSPal..2246577F). doi:10.1080/14772019.2024.2346577 (https://doi.org/10.1080%2F14772019.2024.2346577).
13. Raven, T. J.; Barrett, P. M.; Joyce, C. B.; Maidment, S. C. R. (2023). "The phylogenetic relationships and evolutionary history of the armoured dinosaurs (Ornithischia: Thyreophora)". *Journal of Systematic Palaeontology*. **21** (1). 2205433. Bibcode:2023JSPal..2105433R (https://ui.adsabs.harvard.edu/abs/2023JSPal..2105433R). doi:10.1080/14772019.2023.2205433 (https://doi.org/10.1080%2F14772019.2023.2205433). S2CID 258802937 (https://api.semanticscholar.org/CorpusID:258802937).
14. Arbour, V. M., Currie, P. J. 2016. Systematics, phylogeny and palaeobiogeography of the ankylosaurid dinosaurs. Journal of Systematic Palaeontology 14 (5): 385—444.
15. Norman, D. B., Witmer, L. M. and Weishampel, D. B. 2004. Basal Thyreophora. Pp. 335—342. In D. B. Weishampel, P. Dodson, and H. Osmolska (eds.), The Dinosauria. Berkeley: University of California Press.
16. T.A. Tumanova, 1987, "Pantsirnyye dinozavry Mongolii", Trudy Sovmestnaya Sovetsko-Mongol'skaya Paleontologicheskaya Ekspeditsiya 32: 1-80.

17. Maidment, Susannah C. R.; Strachan, Sarah J.; Ouarhache, Driss; Scheyer, Torsten M.; Brown, Emily E.; Fernandez, Vincent; Johanson, Zerina; Raven, Thomas J.; Barrett, Paul M. (2021-09-23). "Bizarre dermal armour suggests the first African ankylosaur" (https://www.nature.com/articles/s41559-021-01553-6). *Nature Ecology & Evolution*. **5** (12): 1576–1581. Bibcode:2021NatEE...5.1576M (https://ui.adsabs.harvard.edu/abs/2021NatEE...5.1576M). doi:10.1038/s41559-021-01553-6 (https://doi.org/10.1038%2Fs41559-021-01553-6). ISSN 2397-334X (https://search.worldcat.org/issn/2397-334X). PMID 34556830 (https://pubmed.ncbi.nlm.nih.gov/34556830). S2CID 237616095 (https://api.semanticscholar.org/CorpusID:237616095).
18. Galton, Peter M. (2019-02-28). "Earliest record of an ankylosaurian dinosaur (Ornithischia: Thyreophora): Dermal armor from Lower Kota Formation (Lower Jurassic) of India". *Neues Jahrbuch für Geologie und Paläontologie - Abhandlungen*. **291** (2): 205–219. Bibcode:2019NJGPA.291..205G (https://ui.adsabs.harvard.edu/abs/2019NJGPA.291..205G). doi:10.1127/njgpa/2019/0800 (https://doi.org/10.1127%2Fnjgpa%2F2019%2F0800). ISSN 0077-7749 (https://search.worldcat.org/issn/0077-7749). S2CID 134302379 (https://api.semanticscholar.org/CorpusID:134302379).
19. Lucas, S.G. (2025). "Fossil vertebrates, biostratigraphy, biochronology and chronostratigraphy". *Palaeogeography, Palaeoclimatology, Palaeoecology*. **667** 112890. Bibcode:2025PPP...66712890L (https://ui.adsabs.harvard.edu/abs/2025PPP...66712890L). doi:10.1016/j.palaeo.2025.112890 (https://doi.org/10.1016%2Fj.palaeo.2025.112890).
20. Clark, N.D.L. 2001. A thyreophoran dinosaur from the early Bajocian (Middle Jurassic) of the Isle of Skye, Scotland. Scottish Journal of Geology, 37: 19—26.
21. Coombs, W. P. 1978. The families of the ornithischian dinosaur order Ankylosauria. Journal of Paleontology 21: 143—170.
22. Maryanska, T. 1977. Ankylosauridae (Dinosauria) from Mongolia. Palaeontologia Polonica, 37: 85—151.
23. Tanke, D.H. and Brett-Surman, M.K. 2001. Evidence of Hatchling and Nestling-Size Hadrosaurs (Reptilia:Ornithischia) from Dinosaur Provincial Park (Dinosaur Park Formation: Campanian), Alberta, Canada. pp. 206-218. In: Mesozoic Vertebrate Life—New Research Inspired by the Paleontology of Philip J. Currie. Edited by D.H. Tanke and K. Carpenter. Indiana University Press: Bloomington. xviii + 577 pp.
24. Ford, T.L. (2000). "A review of ankylosaur osteoderms from New Mexico and a preliminary review of ankylosaur armor". In Lucas, S.G.; Heckert, A.B. (eds.). *Dinosaurs of New Mexico*. Vol. 17. New Mexico Museum of Natural History and Science Bulletin. pp. 157–176.
25. Coombs Jr., W.P.; Deméré, T.A. (1996). "A Late Cretaceous nodosaurid ankylosaur (Dinosauria: Ornithischia) from marine sediments of coastal California". *Journal of Paleontology*. **70** (2): 311–326. Bibcode:1996JPal...70..311C (https://ui.adsabs.harvard.edu/abs/1996JPal...70..311C). doi:10.1017/S0022336000023404 (https://doi.org/10.1017%2FS0022336000023404).
26. Mallon, Jordan C; David C Evans; Michael J Ryan; Jason S Anderson (2013). "Feeding height stratification among the herbivorous dinosaurs from the Dinosaur Park Formation (upper Campanian) of Alberta, Canada" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3637170). *BMC Ecology*. **13** (1): 14. Bibcode:2013BMCE...13...14M (https://ui.adsabs.harvard.edu/abs/2013BMCE...13...14M). doi:10.1186/1472-6785-13-14 (https://doi.org/10.1186%2F1472-6785-13-14). PMC 3637170 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3637170). PMID 23557203 (https://pubmed.ncbi.nlm.nih.gov/23557203).
27. Wilke, Carolyn (24 February 2023). "What Sounds Did Dinosaurs Make? - A new study of a fossilized ankylosaur suggests it could have uttered birdlike calls" (https://www.nytimes.com/2023/02/24/science/dinosaur-sounds-fossils.html). *The New York Times*. Retrieved 26 February 2023.

28. Yoshida, Junki; Kobayashi, Yoshitsugu; Norell, Mark a. (15 February 2023). "An ankylosaur larynx provides insights for bird-like vocalization in non-avian dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9932143). *Communications Biology*. **152** (1): 152. doi:10.1038/s42003-023-04513-x (https://doi.org/10.1038%2Fs42003-023-04513-x). PMC 9932143 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9932143). PMID 36792659 (https://pubmed.ncbi.nlm.nih.gov/36792659).

Retrieved from "https://en.wikipedia.org/w/index.php?title=Ankylosauria&oldid=1373393860"

WIKIPEDIA
25 years of the free encyclopedia

Ceratopsidae

Ceratopsidae (sometimes spelled **Ceratopidae**) is a family of ceratopsian dinosaurs including *Triceratops*, *Centrosaurus*, and *Styracosaurus*. All known species were quadrupedal herbivores from the Upper Cretaceous. All but one species are known from western North America, which formed the island continent of Laramidia during most of the Late Cretaceous. Ceratopsids are characterized by beaks, rows of shearing teeth in the back of the jaw, elaborate nasal horns, and a thin parietal-squamosal shelf that extends back and up into a frill. The group is divided into two subfamilies—Chasmosaurinae and Centrosaurinae. The chasmosaurines are generally characterized by long, triangular frills and well-developed brow horns. The centrosaurines had well-developed nasal horns or nasal bosses, shorter and more rectangular frills, and elaborate spines on the back of the frill.

These horns and frills show remarkable variation and are the principal means by which the various species have been recognized. Their purpose is not entirely clear. Defense against predators is one possible purpose – although the frills are comparatively fragile in many species – but it is more likely that, as in modern ungulates, they were secondary sexual characteristics used in displays or for intraspecific combat. The massive bosses on the skulls of *Pachyrhinosaurus* and *Achelousaurus* resemble those formed by the base of the horns in modern musk oxen, suggesting that they butted heads. Centrosaurines have frequently been found in massive bone beds with few other species present, suggesting that the animals lived in large herds.

Paleobiology

Behavior

Fossil deposits dominated by large numbers of ceratopsids from individual species suggest that these animals were at least somewhat social.[2] However, the exact nature of ceratopsid social behavior has historically been controversial.[3] In 1997, Lehman argued that the

Ceratopsids

Temporal range: Late Cretaceous, ~

Montage of four ceratopsids. Clockwise from top left: *Titanoceratops*, *Styracosaurus*, *Utahceratops* and *Triceratops*

Scientific classification

Kingdom:	Animalia
Phylum:	Chordata
Class:	Reptilia
Clade:	Dinosauria
Clade:	†Ornithischia
Clade:	†Ceratopsia
Clade:	†Coronosauria
Superfamily:	†Ceratopsoidea
Family:	†Ceratopsidae Marsh, 1888

Subgroups

- †*Ceratops*
- †*Dysganus*
- †*Polyonax*
- †Centrosaurinae

- †Chasmosaurinae

Synonyms

- **Agathaumidae** Cope, 1891
- **Torosauridae** Nopcsa, 1915

aggregations of many individuals preserved in bonebeds originated as local "infestations" and compared them to similar modern occurrences in crocodiles and tortoises.[3] Other authors, such as Scott D. Sampson, interpret these deposits as the remains of large "socially complex" herds.[3]

Modern animals with mating signals as prominent as the horns and frills of ceratopsians tend to form these kinds of large, intricate associations.[4] Sampson found in previous work that the centrosaurine ceratopsids did not achieve fully developed mating signals until nearly fully grown.[5] He finds commonality between the slow growth of mating signals in centrosaurines and the extended adolescence of animals whose social structures are ranked hierarchies founded on age-related differences.[5] In these sorts of groups young males are typically sexually mature for several years before actually beginning to breed, when their mating signals are most fully developed.[6] Females, by contrast do not have such extended adolescence.[6]

Other researchers who support the idea of ceratopsid herding have speculated that these associations were seasonal.[7] This hypothesis portrays ceratopsids as living in small groups near the coasts during the rainy season and inland with the onset of the dry season.[7] Support for the idea that ceratopsids formed herds inland comes from the greater abundance of bonebeds in inland deposits than coastal ones. The migration of ceratopsids away from the coasts may have represented a move to their nesting grounds.[7] Many African herding animals engage in this kind of seasonal herding today.[7] Herds would also have afforded some level of protection from the chief predators of ceratopsids, tyrannosaurids.[8]

Diet

Ceratopsid teeth have a distinctive leaf shape with a primary ridge running down the middle.

Ceratopsids were adapted to processing high-fiber plant material with their highly derived dental batteries and advanced dentition;[9] their skull morphology is consistent with specialisation for feeding on mechanically challenging, low-growing plants.[10] They may have utilized fermentation to break down plant material with a gut microflora.[9] Mallon et al. (2013) examined herbivore coexistence on the island continent of Laramidia, during the Late Cretaceous. It was concluded that ceratopsids were generally restricted to feeding on vegetation at, or below, the height of 1 meter.[11]

Physiology

Ceratopsians probably had the "low mass-specific metabolic rat[e]" typical of large bodied animals.[9]

Sexual dimorphism

According to Scott D. Sampson, if ceratopsids were to have sexual dimorphism modern ecological analogues suggest it would be in their mating signals like horns and frills.[12] No convincing evidence for sexual dimorphism in body size or mating signals is known in ceratopsids, although it was present in the

Ceratopsid skulls at the Natural History Museum of Utah

more primitive ceratopsian *Protoceratops andrewsi*, whose sexes were distinguishable based on frill and nasal prominence size.[12] This is consistent with other known tetrapod groups where midsized animals tended to exhibit markedly more sexual dimorphism than larger ones.[13] However, if there were sexually dimorphic traits, they may have been soft tissue variations like colorations or dewlaps that would not have been preserved as fossils.[13]

Evolution

Map of North America during the Late Cretaceous

Scott D. Sampson has compared the evolution of ceratopsids to that of some mammal groups: both were rapid from a geological perspective and precipitated the simultaneous evolution of large body size, derived feeding structures, and "varied hornlike organs."[3] The earliest ceratopsids, including members of both Centrosaurinae and Chasmosaurinae are known from the early Campanian stage, though the fossil record for early ceratopsids is poor.[14] All but one of the named species of ceratopsid is known from Western North America, which formed the island continent of Laramidia during the Late Cretaceous, separated from the island continent of Appalachia to the east by the Western Interior Seaway. The latitudinal range of ceratopsians across Laramidia extends from Alaska to Mexico. The only named ceratopsid outside of Laramidia is *Sinoceratops*, a centrosaurine from the late Campanian of China.[1] An indeterminate tooth of a ceratopsid is known from Mississippi dating to the late Maastrichtian, a few million years prior to the close of the Cretaceous, indicating that ceratopsids dispersed into eastern North America corresponding to the closure of the Western Interior Seaway at the end of the Cretaceous.[15]

Paleoecology

Size comparison of eight ceratopsids

The chief predators of ceratopsids were tyrannosaurids.[8] $\delta^{44/42}$Ca ratios in tyrannosaurids indicate that ceratopsids were among their most preferred prey.[16]

There is evidence for an aggressive interaction between a *Triceratops* and a *Tyrannosaurus* in the form of partially healed tyrannosaur tooth marks on a *Triceratops* brow horn and squamosal (a bone of the neck frill); the bitten horn is also broken, with new bone growth after the break. It is not known what the exact nature of the interaction was, though: either animal could have been the aggressor.[17] Since the *Triceratops* wounds healed, it is most likely that the *Triceratops* survived the encounter and managed to overcome the

Tyrannosaurus. Paleontologist Peter Dodson estimates that in a battle against a bull *Tyrannosaurus*, the *Triceratops* had the upper hand and would successfully defend itself by inflicting fatal wounds to the *Tyrannosaurus* using its sharp horns.[18]

The Dueling Dinosaurs specimen from the Hell Creek formation shows a *Triceratops* buried in combat with a *Nanotyrannus*.[19][20]

Classification

The clade Ceratopsidae was in 1998 defined by Paul Sereno as the group including the last common ancestor of *Pachyrhinosaurus* and *Triceratops*; and all its descendants.[21] In 2004, Peter Dodson defined it to include *Triceratops*, *Centrosaurus*, and all descendants of their most recent common ancestor.[22] Ceratopsidae was given an official definition in the *PhyloCode* by Daniel Madzia and colleagues in 2021 as "the smallest clade containing *Centrosaurus apertus*, *Ceratops montanus*, *Chasmosaurus belli*, and *Triceratops horridus*".[23] This definition ensures that the type species of Ceratopsidae, *Ceratops montanus*, is included in the clade's definition.[23]

See also

- Timeline of ceratopsian research

Dinosaurs portal

References

- Dodson, P. (1996). *The Horned Dinosaurs*. Princeton University Press, Princeton, New Jersey, pp. xiv-346
- Dodson, P., & Currie, P. J. (1990). "Neoceratopsia." 593–618 *in* Weishampel, D. B., Dodson, P., & Osmólska, H. (eds.), 1990: *The Dinosauria*. University of California Press, Berkeley, Los Angeles, Oxford, 1990 xvi-733.
- Sampson, S. D., 2001, Speculations on the socioecology of Ceratopsid dinosaurs (Ornithischia: Neoceratopsia): In: Mesozoic Vertebrate Life, edited by Tanke, D. H., and Carpenter, K., Indiana University Press, pp. 263–276.

References

1. Dalman, Sebastian G.; Lucas, Spencer G.; Jasinki, Steven G.; Lichtig, Asher J.; Dodson, Peter (2021). "The oldest centrosaurine: a new ceratopsid dinosaur (Dinosauria: Ceratopsidae) from the Allison Member of the Menefee Formation (Upper Cretaceous, early Campanian), northwestern New Mexico, USA" (https://www.researchgate.net/publication/351466021). *PalZ*. **95** (2): 291–335. Bibcode:2021PalZ...95..291D (https://ui.adsabs.harvard.edu/abs/2021PalZ...95..291D). doi:10.1007/s12542-021-00555-w (https://doi.org/10.1007%2Fs12542-021-00555-w). ISSN 0031-0220 (https://search.worldcat.org/issn/0031-0220). S2CID 234351502 (https://api.semanticscholar.org/CorpusID:234351502).
2. "Abstract", Sampson (2001); page 263.
3. "Introduction", Sampson (2001); page 264.
4. "Ceratopsid Socioecology", Sampson (2001); pages 267–268.
5. "Retarded Growth of Mating Signals", Sampson (2001); page 270.

6. "Sociological Correlates in Extant Vertebrates", Sampson (2001); page 265.
7. "Resource Exploitation and Habitat", Sampson (2001); page 269.
8. "Predation Pressure", Sampson (2001); page 272.
9. "Resource Exploitation and Habitat", Sampson (2001); page 268.
10. Mallon, Jordan C.; Anderson, Jason S. (10 July 2013). Butler, Richard J. (ed.). "Skull Ecomorphology of Megaherbivorous Dinosaurs from the Dinosaur Park Formation (Upper Campanian) of Alberta, Canada" (https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0067182). *PLoS ONE*. **8** (7) e67182. doi:10.1371/journal.pone.0067182 (https://doi.org/10.1371%2Fjournal.pone.0067182). ISSN 1932-6203 (https://search.worldcat.org/issn/1932-6203). PMC 3707905 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3707905). PMID 23874409 (https://pubmed.ncbi.nlm.nih.gov/23874409). Retrieved 8 February 2026 – via PLOS ONE.
11. Mallon, Jordan C; David C Evans; Michael J Ryan; Jason S Anderson (2013). "Feeding height stratification among the herbivorous dinosaurs from the Dinosaur Park Formation (upper Campanian) of Alberta, Canada" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3637170). *BMC Ecology*. **13** (1): 14. Bibcode:2013BMCE...13...14M (https://ui.adsabs.harvard.edu/abs/2013BMCE...13...14M). doi:10.1186/1472-6785-13-14 (https://doi.org/10.1186%2F1472-6785-13-14). PMC 3637170 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3637170). PMID 23557203 (https://pubmed.ncbi.nlm.nih.gov/23557203).
12. "Sexual Dimorphism", Sampson (2001); page 269.
13. "Sexual Dimorphism", Sampson (2001); page 270.
14. Brown, Caleb M. (2018-01-16). "Long-horned Ceratopsidae from the Foremost Formation (Campanian) of southern Alberta" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5774296). *PeerJ*. **6** e4265. doi:10.7717/peerj.4265 (https://doi.org/10.7717%2Fpeerj.4265). ISSN 2167-8359 (https://search.worldcat.org/issn/2167-8359). PMC 5774296 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5774296). PMID 29362697 (https://pubmed.ncbi.nlm.nih.gov/29362697).
15. Farke, Andrew A.; Phillips, George E. (23 May 2017). "The first reported ceratopsid dinosaur from eastern North America (Owl Creek Formation, Upper Cretaceous, Mississippi, USA)" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5444368). *PeerJ*. **5** e3342. doi:10.7717/peerj.3342 (https://doi.org/10.7717%2Fpeerj.3342). ISSN 2167-8359 (https://search.worldcat.org/issn/2167-8359). PMC 5444368 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5444368). PMID 28560100 (https://pubmed.ncbi.nlm.nih.gov/28560100).
16. Michailow, Mateusz M.; Lugli, Federico; Cipriani, Anna; Della Giustina, Francesco; Ferretti, Annalisa; Malferrari, Daniele; Fowler, Denver; Fowler, Elizabeth Freedman; Weber, Michael; Tütken, Thomas (1 July 2025). "Combined Ca, Sr isotope and trace element analyses of Late Cretaceous dinosaur teeth: assessing diet versus diagenesis" (https://www.sciencedirect.com/science/article/pii/S001670372500239X). *Geochimica et Cosmochimica Acta*. **400**: 172–189. doi:10.1016/j.gca.2025.05.006 (https://doi.org/10.1016%2Fj.gca.2025.05.006). hdl:11380/1380510 (https://hdl.handle.net/11380%2F1380510). Retrieved 3 October 2025 – via Elsevier Science Direct.
17. Happ, John; Carpenter, Kenneth (2008). "An analysis of predator–prey behavior in a head-to-head encounter between *Tyrannosaurus rex* and *Triceratops*". In Carpenter, Kenneth; Larson, Peter E. (eds.). *Tyrannosaurus rex, the Tyrant King (Life of the Past)*. Bloomington: Indiana University Press. pp. 355–368. ISBN 978-0-253-35087-9.
18. Dodson, Peter, *The Horned Dinosaurs*, Princeton Press. p.19
19. "The Peculiar Case of the Dueling Dinosaurs" (https://missouririvermt.com/blog/the-peculiar-case-of-the-dueling-dinosaurs). *missouririvermt.com*. Retrieved April 20, 2026.
20. "*Nanotyrannus* Confirmed" (https://duelingdinosaurs.org/). *duelingdinosaurs.org*. Retrieved April 20, 2026.

21. Sereno, P. C. (1998). "A rationale for phylogenetic definitions, with application to the higher-level taxonomy of Dinosauria". *Neues Jahrbuch für Geologie und Paläontologie, Abhandlungen*. **210**: 41–83. doi:10.1127/njgpa/210/1998/41 (https://doi.org/10.1127%2Fnjgpa%2F210%2F1998%2F41).
22. Weishampel, David B.; Dodson, Peter; Osmólska, Halszka, eds. (2004). *The Dinosauria* (2nd ed.). Berkeley: University of California Press. ISBN 0-520-24209-2.
23. Madzia, D.; Arbour, V.M.; Boyd, C.A.; Farke, A.A.; Cruzado-Caballero, P.; Evans, D.C. (2021). "The phylogenetic nomenclature of ornithischian dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8667728). *PeerJ*. **9** e12362. doi:10.7717/peerj.12362 (https://doi.org/10.7717%2Fpeerj.12362). PMC 8667728 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8667728). PMID 34966571 (https://pubmed.ncbi.nlm.nih.gov/34966571).

External links

- Media related to Ceratopsidae at Wikimedia Commons
- Data related to Ceratopsidae at Wikispecies

Retrieved from "https://en.wikipedia.org/w/index.php?title=Ceratopsidae&oldid=1367462574"

WIKIPEDIA
25 years of the free encyclopedia

Triceratops

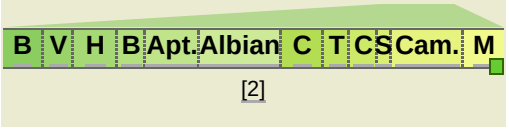
Triceratops (/traɪˈsɛrətɒps/ *try-SERR-ə-tops*;[3] lit. 'three-horned face') is a genus of chasmosaurine ceratopsian dinosaur that lived during the late Maastrichtian age of the Late Cretaceous period, about 69 to 66 million years ago on the island continent of Laramidia,[1] now forming western North America. It was one of the last-known non-avian dinosaurs and lived until the Cretaceous–Paleogene extinction event 66 million years ago. The name *Triceratops*, which means 'three-horned face', is derived from Ancient Greek *trí-* (τρί-), meaning 'three', *kéras* (κέρας), meaning 'horn', and *ṓps* (ὤψ), meaning 'face'.

Bearing a large bony frill, three horns on the skull, and a large, four-legged body, exhibiting convergent evolution with rhinoceroses, *Triceratops* is one of the most recognizable of all dinosaurs and the best-known ceratopsian. It was also one of the largest, measuring around 8–9 m (26–30 ft) long and weighing up to 6–10 t (5.9–9.8 long tons; 6.6–11.0 short tons). It shared the landscape with and was most likely preyed upon by *Tyrannosaurus rex*. The functions of the frills and three distinctive facial horns on its head have inspired countless debates. Traditionally, these have been viewed as defensive weapons against predators. More recent interpretations find it probable that these features were primarily used in species identification, courtship, and dominance display.

Triceratops was traditionally placed within the "short-frilled" ceratopsids, but modern cladistic studies show it to be a member of Chasmosaurinae, which usually have long frills. Two species, ***T. horridus*** and ***T. prorsus***, are considered valid today. Seventeen different species, however, have been named throughout history. Research published in 2010 argued that the contemporaneous *Torosaurus*, a ceratopsid long regarded as a separate genus, represents *Triceratops* in its mature form. This view has been highly disputed, with some morphometric research and the apparent existence of authentic *Torosaurus* subadults favoring the distinctness of the two genera, according to newer studies.

Triceratops

Temporal range: Late Cretaceous



[2]

Skeletal mount of a *T. prorsus* specimen at the Natural History Museum of Los Angeles

Scientific classification

Kingdom:	Animalia
Phylum:	Chordata
Class:	Reptilia
Clade:	Dinosauria
Clade:	†Ornithischia
Clade:	†Ceratopsia
Family:	†Ceratopsidae
Subfamily:	†Chasmosaurinae
Tribe:	†Triceratopsini
Genus:	†*Triceratops* Marsh, 1889b

Type species

†***Ceratops horridus***
Marsh, 1889a

Species

Triceratops has been documented by numerous remains collected since the genus was first described in 1889 by American paleontologist Othniel Charles Marsh. Specimens representing life stages from hatchling to adult have been found. As the archetypal ceratopsian, *Triceratops* has been featured in numerous films, postage stamps, and many other media types.

- †***T. horridus*** (Marsh, 1889a) Marsh 1889b
- †***T. prorsus*** Marsh, 1890

Synonyms

List

- *Agathaumas*? Cope, 1872
- *Polyonax*? Cope, 1874
- *Bison alticornis* Marsh, 1887
- *Ceratops horridus* Marsh, 1889
- *Sterrholophus* Marsh, 1891
- *Claorhynchus*? Cope, 1892
- *Ugrosaurus* Cobabe & Fastovsky, 1987
- *Eotriceratops*? Wu *et al.*, 2007
- *Nedoceratops*? Ukrainsky, 2007
- *Ojoceratops*? Sullivan & Lucas, 2010
- *Tatankaceratops*? Ott & Larson, 2010

Discovery and identification

The first named fossil specimen now attributed to *Triceratops* is a pair of brow horns attached to a skull roof that were found by George Lyman Cannon near Denver, Colorado, in the spring of 1887.[4] This specimen was sent to Othniel Charles Marsh, who believed that the formation from which it came dated from the Pliocene and that the bones belonged to a particularly large and unusual bison, which he named *Bison alticornis*.[4][5] He realized that there were horned dinosaurs by the next year, which saw his publication of the genus *Ceratops* from fragmentary remains.[6]

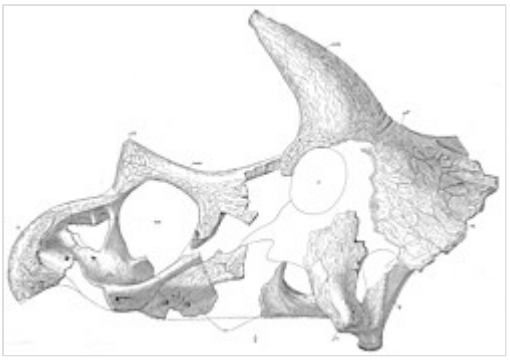
Although not confidently assignable, fossils possibly belonging to *Triceratops* were described as two taxa, *Agathaumas sylvestris* and *Polyonax mortuarius*, in 1872 and 1874, respectively, by Marsh's archrival Edward Drinker Cope.[7][8] *Agathaumas* was named based on a pelvis, several vertebrae, and a few ribs collected by Fielding Bradford Meek and Henry Martyn Bannister near the Green River of southeastern Wyoming from layers coming from the Maastrichtian Lance Formation.[9] Due to the fragmentary nature of the remains, it can only confidently be assigned to Ceratopsidae.[10][11] *Polyonax mortuarius* was collected by Cope himself in 1873 from northeastern Colorado, possibly coming from the Maastrichtian Denver Formation.[12][8] The fossils only consisted of fragmentary horn cores, 3 dorsal vertebrae, and fragmentary limb elements.[8] *Polyonax* has the same issue as *Agathaumas*, with the fragmentary remains non-assignable beyond Ceratopsidae.[13][10]

Illustration of specimen USNM 4739, the horn cores that were erroneously attributed to *Bison alticornis*, the first named specimen of *Triceratops*

The *Triceratops* holotype, YPM 1820, was collected in 1888 from the Lance Formation of Wyoming by fossil hunter John Bell Hatcher, but Marsh initially described this specimen as another species of *Ceratops*.[14] Cowboy Edmund B. Wilson had been startled by the sight of a monstrous skull poking out of the side of a ravine. He tried to recover it by throwing a lasso around one of the horns. When it broke off, the skull tumbling to the bottom of the cleft, Wilson brought the horn to his boss. His boss was rancher and avid fossil collector Charles Arthur

Guernsey, who just so happened to show it to Hatcher. Marsh subsequently ordered Hatcher to locate and salvage the skull.[10] The holotype was first named *Ceratops horridus*. When further preparation uncovered the third nose horn, Marsh changed his mind and gave the piece the new generic name *Triceratops* (lit. 'three horn face'), accepting his *Bison alticornis* as another species of *Ceratops*.[15] It would, however, later be added to *Triceratops*.[16]

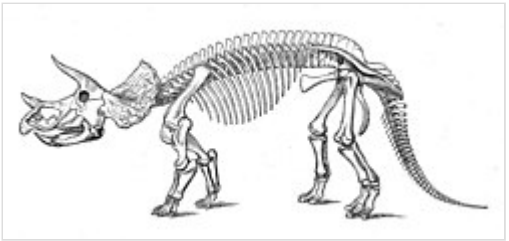
Species



Type specimen YPM 1820 of the type species, *T. horridus*

After *Triceratops* was described, between 1889 and 1891, Hatcher collected another thirty-one of its skulls with great effort. The first species had been named *T. horridus* by Marsh. Its specific name was derived from the Latin word *horridus* meaning "rough" or "rugose", perhaps referring to the type specimen's rough texture, later identified as an aged individual. The additional skulls varied to a lesser or greater degree from the original holotype. This variation is unsurprising, given that *Triceratops* skulls are large three-dimensional objects from individuals of different ages and both sexes that which were subjected to different amounts and directions of pressure during fossilization.[10]

In the first attempt to understand the many species, Richard Swann Lull found two groups, although he did not say how he distinguished them. One group composed of *T. horridus*, *T. prorsus*, and *T. brevicornus* ('the short-horned'). The other composed of *T. elatus* and *T. calicornis*. Two species (*T. serratus* and *T. flabellatus*) stood apart from these groups.[16] By 1933, alongside his revision of the landmark 1907 Hatcher–Marsh–Lull monograph of all known ceratopsians, he retained his two groups and two unaffiliated species, with a third lineage of *T. obtusus* and *T. hatcheri* ('Hatcher's') that was characterized by a very small nasal horn.[11] *T. horridus–T. prorsus–T. brevicornus* was now thought to be the most conservative lineage, with an increase in skull size and a decrease in nasal horn size. *T. elatus–T. calicornis* was defined by having large brow horns and small nasal horns.[11][17] Charles Mortram Sternberg made one modification by adding *T. eurycephalus* ('the wide-headed') and suggesting that it linked the second and third lineages closer together than they were to the *T. horridus* lineage.[18]



1896 skeletal restoration of *T. prorsus* by O. C. Marsh, based on the holotype skull YPM 1822 and referred elements

With time, the idea that the differing skulls might be representative of individual variation within one (or two) species gained popularity. In 1986, John Ostrom and Peter Wellnhofer published a paper in which they proposed that there was only one species, *Triceratops horridus*.[19] Part of their rationale was that there are generally only one or two species of any large animal in a region. To their findings, Thomas Lehman added the old Lull–Sternberg lineages combined with maturity and sexual dimorphism, suggesting that the *T. horridus–T. prorsus–T. brevicornus* lineage was composed of females, the *T. calicornis–T. elatus* lineage was made up of males, and the *T. obtusus–T. hatcheri* lineage was of pathologic old males.[20]

These findings were contested a few years later by paleontologist Catherine Forster, who reanalyzed *Triceratops* material more comprehensively and concluded that the remains fell into two species, *T. horridus* and *T. prorsus*, although the distinctive skull of *T.* ("*Nedoceratops*") *hatcheri* differed enough to warrant a separate genus.[21] She found that *T. horridus* and several other species belonged together and that *T. prorsus* and *T. brevicornus* stood alone. Since there were many more specimens in the first group, she suggested that this meant the two groups were two species. It is still possible to interpret the differences as representing a single species with sexual dimorphism.[10][22]

In 2014, John Scannella and colleagues supported the separation of *T. prorsus* and *T. horridus*, noting that the two species are also separated stratigraphically within the Hell Creek Formation, indicating that they did not live together at the same time.[23]

Valid species

- *T. horridus* (Marsh, 1889) Marsh, 1889 (originally *Ceratops*) (type species)
- *T. prorsus* Marsh, 1890

Cast of the first mounted *T. horridus* skeleton (composite of several specimens, including USNM 2100 and 4842), nicknamed "Hatcher", National Museum of Natural History

Synonyms and doubtful species

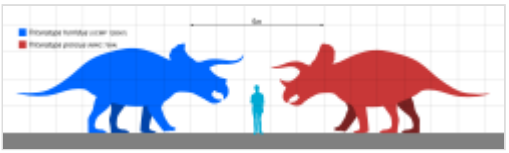
Some of the following species are synonyms, as indicated in parentheses ("=*T. horridus*" or "=*T. prorsus*"). All the others are each considered a *nomen dubium* (lit. 'dubious name') because they are based on remains too poor or incomplete to be distinguished from pre-existing *Triceratops* species.

- *T. albertensis* C. M. Sternberg, 1949
- *T. alticornis* (Marsh 1887) Hatcher, Marsh, and Lull, 1907 [originally *Bison alticornis*, Marsh 1887, and *Ceratops alticornis*, Marsh 1888]
- *T. brevicornus* Hatcher, 1905 (=*T. prorsus*)
- *T. calicornis* Marsh, 1898 (=*T. horridus*)
- *T. elatus* Marsh, 1891 (=*T. horridus*)
- *T. eurycephalus* Schlaikjer, 1935
- *T. flabellatus* Marsh, 1889 (= *Sterrholophus* Marsh, 1891) (=*T. horridus*)
- *T. galeus* Marsh, 1889
- *T. hatcheri* (Hatcher & Lull 1905) Lull, 1933 (contentious; see *Nedoceratops* below)
- *T. ingens* Marsh vide Lull, 1915
- *T. maximus* Brown, 1933
- *T. mortuarius* (Cope, 1874) Kuhn, 1936 (*nomen dubium*; originally *Polyonax mortuarius*)
- *T. obtusus* Marsh, 1898 (=*T. horridus*)
- *T. serratus* Marsh, 1890 (=*T. horridus*)
- *T. sulcatus* Marsh, 1890
- *T. sylvestris* (Cope, 1872) Kuhn, 1936 (*nomen dubium*; originally *Agathaumas sylvestris*)

T. prorsus, Carnegie Museum of Natural History

Description

Size



Size comparison with *T. horridus* in blue and *T. prorsus* in red

Triceratops was a very large animal, measuring around 8–9 m (26–30 ft) in length and weighing up to 6–10 metric tons (6.6–11.0 short tons).[24][25][26] A specimen of *T. horridus* named Kelsey measured 6.7–7.3 meters (22–24 ft) long, has a 2-meter (6.5 ft) skull, stood about 2.3 meters (7.5 ft) tall, and was estimated by the Black Hills Institute to weigh approximately 5.4 metric tons (6.0 short tons).[27][28] A *T. horridus* specimen named Big John was estimated to have been 8.0 meters (26.1 ft) long and 3.0 meters (10 ft) tall.[29]

Skull

Front view of skull with a prominent epoccipital fringe, Houston Museum of Natural Science

Like all chasmosaurines, *Triceratops* had a large skull relative to its body size, among the largest of all land animals. The largest-known skull, specimen MWC 7584 (formerly BYU 12183), is estimated to have been 2.5 meters (8.2 ft) in length when complete[30] and could reach almost a third of the length of the entire animal.[31]

The front of the head was equipped with a large beak in front of its teeth. The core of the top beak was formed by a special rostral bone. Behind it, the premaxillae bones were located, embayed from behind by very large, circular nostrils. In chasmosaurines, the premaxillae met on their midline in a complex bone plate, the rear edge of which was reinforced by the "narial strut". From the base of this strut, a triangular process jutted out into the nostril. *Triceratops* differs from most relatives in that this process was hollowed out on the outer side. Behind the toothless premaxilla, the maxilla bore thirty-six to forty tooth positions, in which three to five teeth per position were vertically stacked. The teeth were closely appressed, forming a "dental battery" curving to the inside. The skull bore a single horn on the snout above the nostrils. In *Triceratops*, the nose horn is sometimes recognisable as a separate ossification, the epinasal.[32]

The skull also featured a pair of supraorbital "brow" horns approximately 1 meter (3.3 ft) long, with one above each eye.[33][23] The jugal bones pointed downward at the rear sides of the skull and were capped by separate epijugals. With *Triceratops*, these were not particularly large and sometimes touched the quadratojugals. The bones of the skull roof were fused and by a folding of the frontal bones, a "double" skull roof was created. In *Triceratops*, some specimens show a fontanelle, an opening in the upper roof layer. The cavity between the layers invaded the bone cores of the brow horns.[32]

At the rear of the skull, the outer squamosal bones and the inner parietal bones grew into a relatively short, bony frill, adorned with epoccipitals in young specimens. These were low triangular processes on the frill edge, representing separate skin ossifications or osteoderms. Typically, with *Triceratops* specimens, there are two epoccipitals present on each parietal bone, with an additional central process on their border. Each squamosal bone had five processes. Most other ceratopsids had large parietal fenestrae, openings in their frills, but those of *Triceratops* were noticeably solid,[34] unless the genus *Torosaurus* represents mature *Triceratops* individuals, which it most likely does not. Under the frill, at the rear of the skull, a huge occipital condyle, up to 106 millimeters (4.2 in) in diameter, connected the head to the neck.[32]

Back of skull, showing rounded joint which connected the head and neck

The lower jaws were elongated and met at their tips in a shared epidentary bone, the core of the toothless lower beak. In the dentary bone, the tooth battery curved to the outside to meet the battery of the upper jaw. At the rear of the lower jaw, the articular bone was exceptionally wide, matching the general width of the jaw joint.[32] *T. horridus* can be distinguished from *T. prorsus* by having a shallower snout.[24]

Postcranial skeleton

Specimen nicknamed "Raymond" that preserves the natural, non-pronated pose of the forelimb

Chasmosaurines showed little variation in their postcranial skeleton.[32] The skeleton of *Triceratops* is markedly robust. Both *Triceratops* species possessed a very sturdy build, with strong limbs, short hands with three hooves each, and short feet with four hooves each.[35] The vertebral column consisted of ten neck, twelve back, ten sacral, and about forty-five tail vertebrae. The front neck vertebrae were fused into a syncervical. Traditionally, this was assumed to have incorporated the first three vertebrae, thus implying that the frontmost atlas was very large and sported a spinous process (the spike-like piece of bone that extends dorsally which combined have a ridge or keel-like appearance). Later interpretations revived an old hypothesis by John Bell Hatcher that, at the very front, a vestige of the real atlas can be observed, the syncervical then consisting of four vertebrae. The vertebral count mentioned is adjusted to this view. In *Triceratops*, the spinous processes of the neck are constant in height and don't gradually slope upwards. Another peculiarity is that the neck ribs only begin to lengthen with the ninth cervical vertebra.[32]

The rather short and high vertebrae of the back were, in its middle region, reinforced by ossified tendons running along the tops of the vertebral arches (the band of bone that encloses the spinal canal). The straight sacrum (the part of the spine at the pelvis) was long and adult individuals show a fusion of all sacral vertebrae. In *Triceratops* the first four and last two sacrals had transverse processes (sideways outgrowths), connecting the vertebral column to the pelvis, that were fused at their distal ends. Sacrals seven and eight had longer processes, causing the sacrum to have an oval profile in top view. On top of

the sacrum, a carapace-like bone plate was formed by a fusion of the spinous processes of the second through fifth vertebrae. *Triceratops* had a large pelvis with a long ilium. The ischium was curved downwards.[32]

Life restorations of *T. horridus* (top) and *T. prorsus* (bottom)

Although certainly quadrupedal, the posture of horned dinosaurs has long been the subject of some debate. Originally, it was believed that the front legs of the animal had to be sprawling at a considerable angle from the thorax in order to better bear the weight of the head.[10] This stance can be seen in paintings by Charles Knight and Rudolph Zallinger. Ichnological evidence in the form of trackways from horned dinosaurs and recent reconstructions of skeletons (both physical and digital) seem to show that *Triceratops* and other ceratopsids maintained an upright stance during normal locomotion, with the elbows flexed to behind and slightly bowed out, in an intermediate state between fully upright and fully sprawling, comparable to the modern rhinoceros.[35][36][37][38] Tracks of a large ceratopsian from the Upper Cretaceous Laramie Formation of Colorado given the name *Ceratopsipes goldenensis* may have been made by *Triceratops* (or the closely related *Torosaurus*).[39]

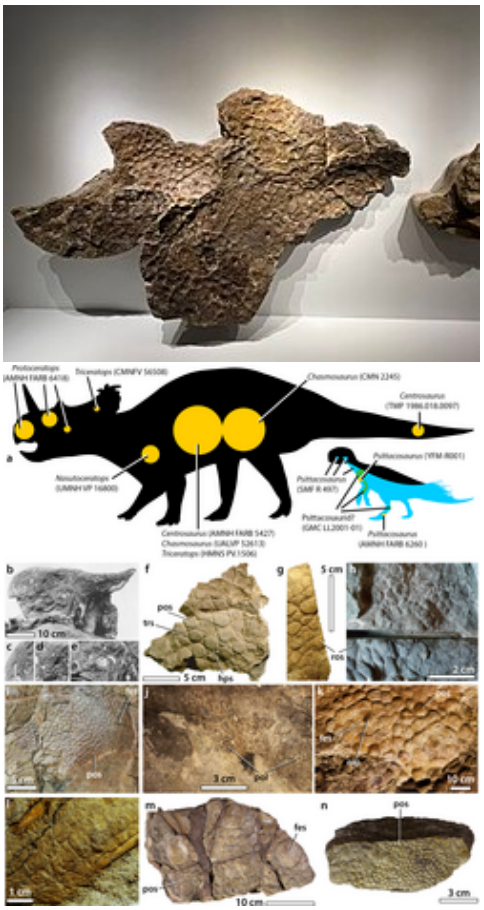
Ceratopsipes goldenensis, possible *Triceratops* track

The hands and forearms of *Triceratops* retained a fairly primitive structure when compared to other quadrupedal dinosaurs, such as thyreophorans and many sauropods. In those two groups, the forelimbs of quadrupedal species were usually rotated so that the hands faced forward with palms backward ("pronated") as the animals walked. *Triceratops*, like other ceratopsians and related quadrupedal ornithopods (together forming the Cerapoda), walked with most of their fingers pointing out and away from the body, the original condition for dinosaurs. This was also retained by bipedal forms, like theropods. In *Triceratops*, the weight of the body was carried by only the first three fingers of the hand, while digits 4 and 5 were vestigial and lacked claws or hooves.[35] The phalangeal formula of the hand is 2-3-4-3-1, meaning that the first or innermost finger of the forelimb has two bones, the next has three, the next has four, etc.[40]

Skin

A *Triceratops* nicknamed "Lane" (HMNS PV.1506) that was unearthed on the Zerbst Ranch in Wyoming in 2002 is preserved with large areas of fossil skin over its trunk−both the nearly complete skeleton and the skin fossil are on display at the Houston Museum of Natural Science. The skin is not preserved as an impression but as a thin layer of clay that formed from a bacterial mat on the desiccated body and recorded the skin in positive relief with exceptional detail, a form of fossilization called "clay templating"

that was only recognized recently based on analysis of two *Edmontosaurus* "mummy" specimens first described in 2025 from the same Wyoming area, dubbed the "mummy zone." Contrary to earlier sources, actual fossilized skin is not preserved.[41][42] The basement scales covering the majority of the body are large in size, the largest known in ceratopsians at up to 90 millimetres (3.5 in) across. Regularly scattered (approximately 15–20 centimetres (5.9–7.9 in) from each other, though in no particular pattern) amongst the basement scales are large feature scales, exceeding 100 millimetres (3.9 in) across, which are hexagonal (six-sided) to heptagonal (seven-sided) in shape. These scales have conical nipple-like projections approximately 1–3 centimetres (0.39–1.18 in) tall rising from their center, with the outer edges of the feature scales being relatively flat. It has been suggested that it is unlikely that the nipple-like projections of the feature scales bore bristles like those known in other ceratopsians like *Psittacosaurus*.[43] A preserved piece of skin from the frill of another specimen (CMN FV 56508) is also known, which consists of small polygonal basement scales.[43]



Fossils of skin from *Triceratops* and other ceratopsians

- top: *Triceratops* fossil skin at the Houston Museum of Natural Science
- bottom: Fossil skin of various ceratopsians (k. is *Triceratops*)

Classification

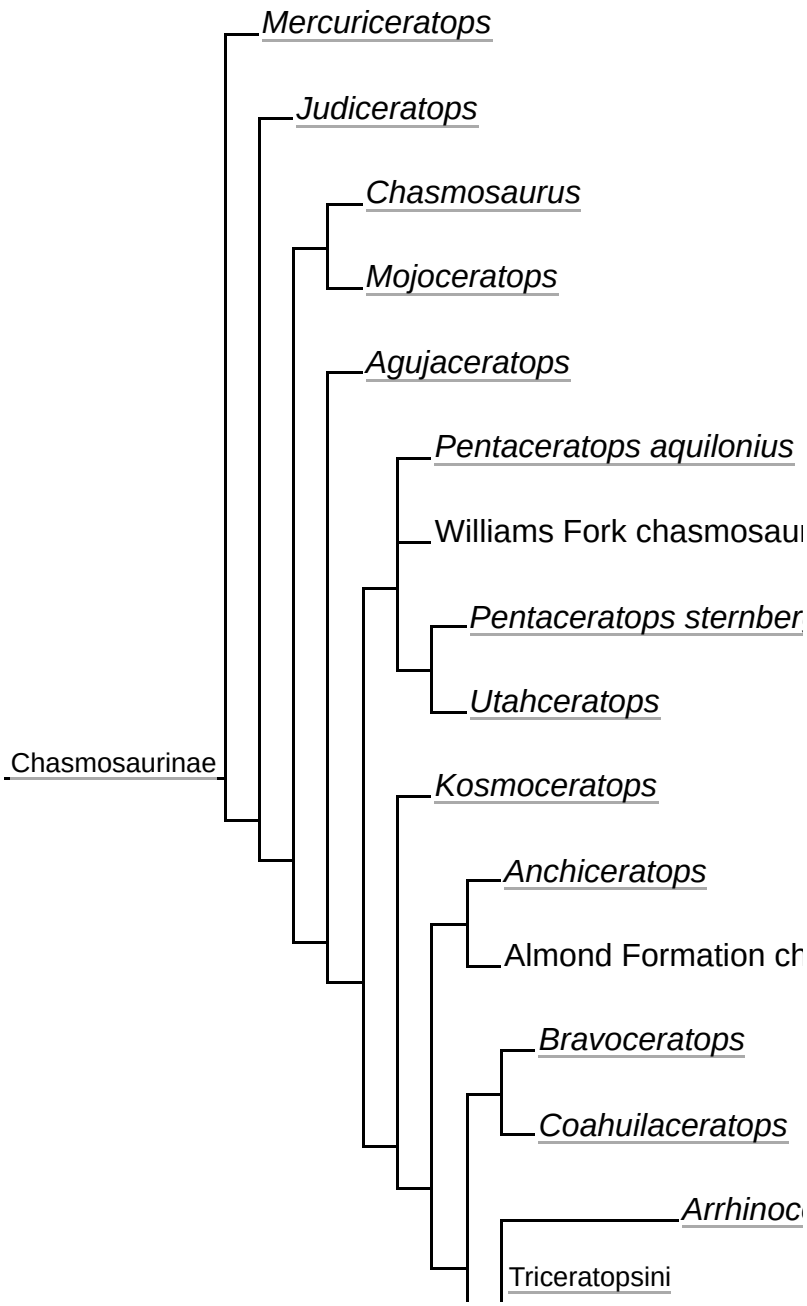
Triceratops is the best-known genus of Ceratopsidae, a family of large, mostly North American ceratopsians. The exact relationship of *Triceratops* among the other ceratopsids has been debated over the years. Confusion stemmed mainly from the combination of a short, solid frill (similar to that of Centrosaurinae), with long brow horns (more akin to Chasmosaurinae).[44] In the first overview of ceratopsians, R. S. Lull hypothesized the existence of two lineages, one of *Monoclonius* and *Centrosaurus* leading to *Triceratops*, the other with *Ceratops* and *Torosaurus*, making *Triceratops* a centrosaurine as the group is understood today.[16] Later revisions supported this view when Lawrence Lambe, in 1915, formally describing the first, short-frilled group as Centrosaurinae (including *Triceratops*), and the second, long-frilled group as Chasmosaurinae.[11][45]

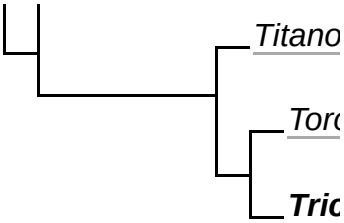
In 1949, Charles Mortram Sternberg was the first to question this position, proposing instead that *Triceratops* was more closely related to *Arrhinoceratops* and *Chasmosaurus* based on skull and horn features, making *Triceratops* a chasmosaurine ("ceratopsine" in his usage) genus.[18] He was largely ignored, with John Ostrom[46] and later David Norman placing *Triceratops* within the Centrosaurinae.[47]

Subsequent discoveries and analyses, however, proved the correctness of Sternberg's view on the position of *Triceratops*, with Thomas Lehman defining both subfamilies in 1990 and diagnosing *Triceratops* as "ceratopsine" on the basis of several morphological features. Apart from the one feature of a shortened frill, *Triceratops* shares no derived traits with centrosaurines.[20] Further research by Peter Dodson, including a 1990 cladistic analysis and a 1993 study using resistant-fit theta-rho analysis, or RFTRA (a morphometric technique which systematically measures similarities in skull shape), reinforces *Triceratops*' placement as a chasmosaurine.[48][49]

The cladogram below follows Longrich (2014), who named a new species of *Pentaceratops*, and included nearly all species of chasmosaurine.[50]

'Horridus', the most complete *Triceratops* fossil known, on display at the Melbourne Museum. Restored and prepared by the team at Pangea Fossils in Victoria British Columbia, Canada.





For many years after its discovery, the deeper evolutionary origins of *Triceratops* and its close relatives remained largely obscure. In 1922, the newly discovered *Protoceratops* was seen as its ancestor by Henry Fairfield Osborn,[10] but many decades passed before additional findings came to light. Recent years have been fruitful for the discovery of several antecedents of *Triceratops*. *Zuniceratops*, the earliest-known ceratopsian with brow horns, was described in the late 1990s, and *Yinlong*, the first known Jurassic ceratopsian, was described in 2006.[51]

Skull of a specimen nicknamed 'Pops' from the Laramie Formation of eastern Colorado, on display at the Weld County Courthouse. Based on the age of the formation, it may be the oldest *Triceratops* known.

These new finds have been vital in illustrating the origins of ceratopsians in general, suggesting an Asian origin in the Jurassic and the appearance of truly horned ceratopsians by the beginning of the Late Cretaceous in North America.[32]

In phylogenetic taxonomy, the genus *Triceratops* has been used as a reference point in the definition of Dinosauria. Dinosaurs have been designated as all descendants of the most recent common ancestor of *Triceratops* and modern birds.[52] Furthermore, Ornithischia has been defined as those dinosaurs more closely related to *Triceratops* than to modern birds.[53]

Paleobiology

A *Triceratops* mounted next to a *Tyrannosaurus* at the Los Angeles Natural History Museum

Although *Triceratops* is commonly portrayed as a herding animal, there is currently little evidence to suggest that they lived in herds. While several other ceratopsians are known from bone beds preserving bones from two to hundreds or even thousands of individuals, there is currently only one documented bonebed dominated by *Triceratops* bones: a site in southeastern Montana with the remains of three juveniles. It may be significant that only juveniles were present.[54] In 2012, a group of three *Triceratops* in relatively complete condition, each of varying sizes from a full-grown adult to a small juvenile, were found near Newcastle, Wyoming. The remains are currently under excavation by paleontologist Peter Larson and a team from the Black Hills Institute. It is believed that the animals were traveling as a family unit, but it remains unknown if the group consists of a mated pair and their offspring, or two females and a juvenile they were caring for. The remains also show signs of predation or scavenging from *Tyrannosaurus*, particularly on the largest specimen, with the bones of the front limbs showing breakage

and puncture wounds from *Tyrannosaurus* teeth.[55] In 2020, Illies and Fowler described the co-ossified distal caudal vertebrae of *Triceratops*. According to them, this pathology could have arisen after one *Triceratops* accidentally stepped on the tail of another member of the herd.[56][57]

Triceratops rib with theropod tooth marks at the middle

For many years, *Triceratops* finds were known only from solitary individuals.[54] These remains are very common. For example, Bruce Erickson, a paleontologist of the Science Museum of Minnesota, has reported having seen 200 specimens of *T. prorsus* in the Hell Creek Formation of Montana.[58] Similarly, Barnum Brown claimed to have seen over 500 skulls in the field.[10]:79 Because *Triceratops* teeth, horn fragments, frill fragments, and other skull fragments are such abundant fossils in the Lancian faunal stage of the late Maastrichtian (Late Cretaceous, 66 mya) of western North America, it is regarded as one of the dominant herbivores of the time, if not the most dominant. In 1986, Robert Bakker estimated it as making up five sixths of the large dinosaur fauna at the end of the Cretaceous.[59] Unlike most animals, skull fossils are far more common than postcranial bones for *Triceratops*, suggesting that the skull had an unusually high preservation potential.[60]

Brain endocast at Minnesota Science Museum

Analysis of the endocranial anatomy of *Triceratops* suggest its sense of smell was poor compared to that of other dinosaurs. Its ears were attuned to low frequency sounds, given the short cochlear lengths recorded in an analysis by Sakagami *et al*,. This same study also suggests that *Triceratops* held its head about 45 degrees to the ground, an angle which would showcase the horns and frill most effectively that simultaneously allowed the animal to take advantage of food through grazing.[61] A 2026 study of the large nasal region suggested that *Triceratops* likely had respiratory turbinates which contained moisture that evaporated as air was inhaled. This released heat and cooled the blood directed to the eyes and brain, which would otherwise overheat within their large, thick skulls. The study also found that in ceratopsians, the lateral nasal nerve supplied the beak rather than the maxillary nerve as in other reptiles. The latter's path was obstructed by the beak and nasal cavity.[62]

A 2022 study by Wiemann and colleagues of various dinosaur genera, including *Triceratops*, suggests that it had an ectothermic (cold blooded) or gigantothermic metabolism, on par with that of modern reptiles. This was uncovered using the spectroscopy of lipoxidation signals, which are byproducts of oxidative phosphorylation and correlate with metabolic rates. They suggested that such metabolisms may have been common for ornithischian dinosaurs in general, with the group evolving towards ectothermy from an ancestor with an endothermic (warm blooded) metabolism.[63] An isotopic analysis study by Rooij and colleagues suggested that *Triceratops* was gigantothermic, if not endothermic due to its large body volume.[64] The very same study conducted by Wiemann et al., revealed that *Triceratops* lived in environments that consisted of floodplains and inland forests.[64]

Dentition and diet

Close up of the jaws and teeth

Triceratops were herbivorous and, because of their low-slung head, their primary food was probably low-growing vegetation, although they may have been able to knock down taller plants with their horns, beak, and sheer bulk.[32][65] The jaws were tipped with a deep, narrow beak, believed to have been better at grasping and plucking than biting.[46][66]

Triceratops teeth were arranged in groups called batteries, which contained 36 to 40 tooth columns in each side of each jaw and 3 to 5 stacked teeth per column, depending on the size of the animal.[32] This gives a range of 432 to 800 teeth, of which only a fraction were in use at any given time (as tooth replacement was continuous throughout the life of the animal).[32] They functioned by shearing in a vertical to near-vertical orientation.[32] Additionally, their teeth wore as they fed, creating fullers that minimised friction as they masticated.[66] The great size and numerous teeth of *Triceratops* suggests that they ate large volumes of fibrous plant material.[66][67][68] Other plants that were a part of its diet included *Populus* plants, Pine plants, *Platanus* plants, *Hazel* plants, and *Taxodium* plants.[69] Some researchers suggest it, along with its cousin *Torosaurus* ate palms and cycads[70][71][72] and others suggest it ate ferns, which then grew in prairies.[73][64] Studies of the isotopes of ceratopsian and hadrosaur teeth revealed that *Triceratops* and *Edmontosaurus* respectively engaged in niche partitioning.[74][75][76]

Functions of the horns and frill

Juvenile and adult skulls—the juvenile skull is about the size of an adult human head

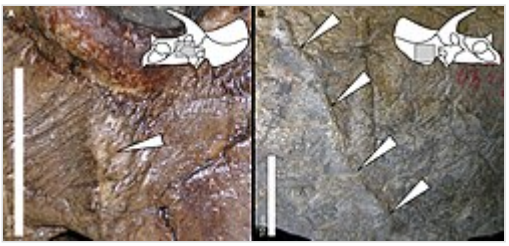
There has been much speculation over the functions of *Triceratops*' head adornments. The two main theories have revolved around use in combat and in courtship display, with the latter now thought to be the most likely primary function.[32]

Early on, Lull postulated that the frills may have served as anchor points for the jaw muscles to aid chewing by allowing increased size and power for the muscles.[77] This has been put forward by other authors over the years, but later studies do not find evidence of large muscle attachments on the frill bones.[78]

Triceratops were long thought to have used their horns and frills in combat with large predators, such as *Tyrannosaurus*, the idea being discussed first by Charles H. Sternberg in 1917 and 70 years later by Robert Bakker.[59][79] There is direct fossil evidence that *Tyrannosaurus* did have aggressive head-on encounters with *Triceratops*, based on partially healed tyrannosaur tooth marks on a *Triceratops* brow horn and squamosal. The bitten horn is also broken, with new bone growth after the break. Which animal was the aggressor, however, is unknown.[80] Paleontologist Peter Dodson estimates that, in a battle against a bull *Tyrannosaurus*, the *Triceratops* had

the upper hand and would successfully defend itself by inflicting fatal wounds to the *Tyrannosaurus* using its sharp horns. *Tyrannosaurus* is also known to have consumed *Triceratops*, as shown by a heavily tooth-scored *Triceratops* ilium and sacrum showing *Tyrannosaurus* feeding traces.[81]

In addition to combat with predators using its horns, *Triceratops* are popularly shown engaging each other in combat with horns locked. While studies show that such activity would be feasible, if unlike that of present-day horned animals,[82] there is disagreement about whether they did so. Although pitting, holes, lesions, and other damage on *Triceratops* skulls (and the skulls of other ceratopsids) are often attributed to horn damage in combat, a 2006 study finds no evidence for horn thrust injuries causing these forms of damage (with there being no evidence of infection or healing). Instead, non-pathological bone resorption, or unknown bone diseases, are suggested as causes.[83] A 2009 study compared incidence rates of skull lesions and periosteal reaction in *Triceratops* and *Centrosaurus*, showing that these were consistent with *Triceratops* using its horns in combat and the frill being adapted as a protective structure, while lower pathology rates in *Centrosaurus* may indicate visual use over physical use of cranial ornamentation or a form of combat focused on the body rather than the head.[84][85][86] The frequency of injury was found to be 14% in *Triceratops*.[87] The researchers also concluded that the damage found on the specimens in the study was often too localized to be caused by bone disease.[88] Histological examination reveals that the frill of *Triceratops* is composed of fibrolamellar bone.[89] This contains fibroblasts that play a critical role in wound healing and is capable of rapidly depositing bone during remodeling.[90][91]



Examples of periosteal reactive bone in selected specimens of *Triceratops*

One skull was found with a hole in the jugal bone, apparently a puncture wound sustained while the animal was alive, as indicated by signs of healing. The hole has a diameter close to that of the distal end of a *Triceratops* horn. This and other apparent healed wounds in the skulls of ceratopsians have been cited as evidence of non-fatal intra-specific competition in these dinosaurs.[92][93] Another specimen, referred to as "Big John", has a similar fenestra to the squamosal caused by what appears to be another *Triceratops* horn and the squamosal bone shows signs of significant healing, further vindicating the hypothesis that this ceratopsian used its horns for intra-specific combat.[94]

The large frill also may have helped to increase body area to regulate body temperature.[95] A similar theory has been proposed regarding the plates of *Stegosaurus*,[96] although this use alone would not account for the bizarre and extravagant variation seen in different members of Ceratopsidae, which would rather support the sexual display theory.[32]

The theory that frills functioned as a sexual display was first proposed by Davitashvili in 1961 and has gained increasing acceptance since.[20][78][97] Evidence that visual display was important, either in courtship or other social behavior, can be seen in the ceratopsians differing markedly in their adornments, making each species highly distinctive. Also, modern living creatures with such displays of horns and adornments use them similarly.[92] A 2006 study of the smallest *Triceratops* skull, ascertained to be that of a juvenile, shows the frill and horns developed at a very early age, predating sexual development. That would suggest they were probably important for visual communication and species recognition in general.[98] However, the use of the exaggerated structures to enable dinosaurs to recognize their own species has been questioned, as no such function exists for such structures in modern species.[99]

Growth and ontogeny

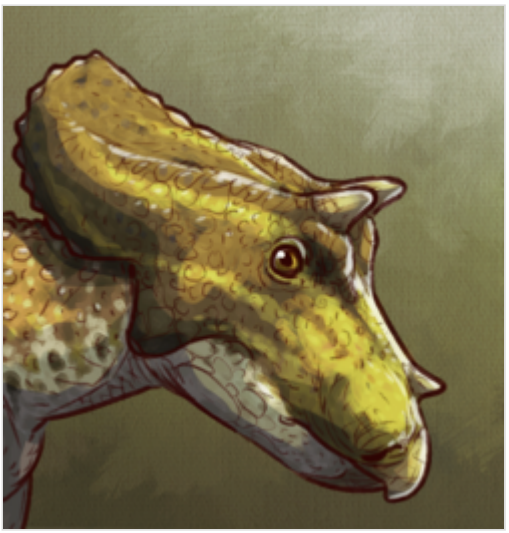
Juvenile *Triceratops* skull found in Montana (cast)

Skull growth series

In 2006, the first extensive ontogenetic study of *Triceratops* was published in the journal *Proceedings of the Royal Society*. The study, by John R. Horner and Mark Goodwin, found that individuals of *Triceratops* could be divided into four general ontogenetic groups: babies, juveniles, subadults, and adults. With a total number of 28 skulls studied, the youngest was only 38 centimeters (15 in) long. Ten of the 28 skulls could be placed in order in a growth series with one representing each age. Each of the four growth stages were found to have identifying features. Multiple ontogenetic trends were discovered, including the size reduction of the epoccipitals, development and reorientation of postorbital horns, and hollowing out of the horns.[100] Big John, one of the *Triceratops*, is estimated to have lived around 60 years when he died.[101]

Torosaurus as growth stage of *Triceratops*

Torosaurus is a ceratopsid genus first identified from a pair of skulls in 1891, two years after the identification of *Triceratops* by Othneil Charles Marsh. The genus *Torosaurus* resembles *Triceratops* in geological age, distribution, anatomy, and size, so it has been recognised as a close relative.[102] Its distinguishing features are an elongated skull and the presence of two ovular fenestrae in the frill. Paleontologists investigating dinosaur ontogeny in Montana's Hell Creek Formation have presented evidence that the two represent a single genus.



Speculative life restoration of a juvenile *Triceratops*

John Scannella, in a paper presented in Bristol at the conference of the Society of Vertebrate Paleontology (September 25, 2009), reclassified *Torosaurus* as especially mature *Triceratops* individuals, perhaps representing a single sex. Horner, Scannella's mentor at Bozeman Campus, Montana State University, noted that ceratopsian skulls consist of metaplastic bone. A characteristic of metaplastic bone is that it lengthens and shortens over time, extending and resorbing to form new shapes. Significant variety is seen even in those skulls already identified as *Triceratops*, Horner said, "where the horn orientation is backwards in juveniles and forward in adults". Approximately 50% of all subadult *Triceratops* skulls have two thin areas in the frill that correspond with the placement of "holes" in *Torosaurus* skulls,

suggesting that holes developed to offset the weight that would otherwise have been added as maturing *Triceratops* individuals grew longer frills.[103] A paper describing these findings in detail was published in July 2010 by Scannella and Horner. It formally argues that *Torosaurus* and the similar contemporary *Nedoceratops* are synonymous with *Triceratops*.[30]

A, *Triceratops prorsus* holotype YPM 1822 and B, *Torosaurus latus* ANSP 15192

The assertion has since ignited much debate. Andrew Farke had, in 2006, stressed that no systematic differences could be found between *Torosaurus* and *Triceratops*, apart from the frill.[102] He nevertheless disputed Scannella's conclusion by arguing in 2011 that the proposed morphological changes required to "age" a *Triceratops* into a *Torosaurus* would be without precedent among ceratopsids. Such changes would include the growth of additional epoccipitals, reversion of bone texture from an adult to immature type and back to adult again, and growth of frill holes at a later stage than usual.[104] A study by Nicholas Longrich and Daniel Field analyzed 35 specimens of both *Triceratops* and *Torosaurus*. The authors concluded that *Triceratops* individuals too old to be considered immature forms are represented in the fossil record, as are *Torosaurus* individuals too young to be considered fully mature adults. The synonymy of *Triceratops* and *Torosaurus* cannot be supported, they said, without more convincing intermediate forms than Scannella and Horner initially produced. Scannella's *Triceratops* specimen with a hole on its frill, they argued, could represent a diseased or malformed individual rather than a transitional stage between an immature *Triceratops* and mature *Torosaurus* form.[105][106]

In 2013, Farke and Leonardo Maiorino published morphometric research, a statistical analysis of the morphospace (shape space) describing the variation of the *Torosaurus*, *Triceratops horridus*, *Triceratops prorsus*, and *Nedoceratops* skulls correlated with maturation. They concluded that *Torosaurus latus* skulls throughout maturation retained a different form from *T. horridus* and *T. prorsus*, the last two species showing an overlapping in their proportions. This is even true when the frill shape is disregarded. *Nedoceratops* proved, except for size, not to be a plausible transitional form between *Torosaurus* and *Triceratops horridus*. Farke and Maiorino admitted that the low number of *Torosaurus* specimens reduced the reliability of these results, but concluded that *Torosaurus* and *Triceratops* were separate taxa, though allowing for the possibility of anagenesis, i.e. the several taxa forming a single chronospecies line of descent, given the lack of good stratigraphic data.[107] The morphometric study was inconclusive on the point of *Torosaurus utahensis*, for which most specimens consist of isolated bones, with its morphospace falling in between *Triceratops* and *Torosaurus latus* and not well separated from either.[107]

Researchers have claimed that distinct juvenile *Torosaurus* have been excavated from a bonebed in the Javelina Formation of Big Bend National Park, basing their identification as *Torosaurus* cf. *utahensis* on their proximity to an adult with a characteristic *Torosaurus* parietal.[108] In 2022, Mallon *et al.* argued that two specimens found in Canada's Frenchman and Scollard Formations, EM P16.1. (at Eastend Historical Museum in Saskatchewan) and UALVP 1646 (at the University of Alberta), are subadults and can be referred to *Torosaurus*, this indicating that it is a valid taxon. The same study also noted that *Torosaurus* indeed lived during the Late Maastrichtian (contemporaneously with *Triceratops*).[109]

Other genera as growth stages of *Triceratops*

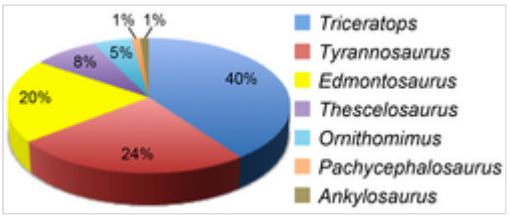
Opinion has varied on the validity of a separate genus for *Nedoceratops*. Scannella and Horner regarded it as an intermediate growth stage between *Triceratops* and *Torosaurus*.[30][110] Farke, in his 2011 redescription of the only known skull, concluded that it was an aged individual of its own valid taxon, *Nedoceratops hatcheri*.[104] Longrich and Fields also did not consider it a transition between *Torosaurus* and *Triceratops*, suggesting that the frill holes were pathological.[106]

Comparisons between the skulls of *Triceratops* and *Nedoceratops*

As described above, Scannella had argued in 2010 that *Nedoceratops* should be considered a synonym of *Triceratops*.[30] Farke (2011) maintained that it represents a valid distinct genus.[104] Longrich agreed with Scannella about *Nedoceratops* and made a further suggestion that the recently described *Ojoceratops* was likewise a synonym. The fossils, he argued, are indistinguishable from the *Triceratops horridus* specimens that were previously attributed to the defunct species *Triceratops serratus*.[111]

Longrich observed that another newly described genus, *Tatankaceratops*, displayed a strange mix of characteristics already found in adult and juvenile *Triceratops*. Rather than representing a distinct genus, *Tatankaceratops* could as easily represent a dwarf *Triceratops* or a *Triceratops* individual with a developmental disorder that caused it to stop growing prematurely.[111]

Paleoecology



Pie chart of the time averaged census for large-bodied dinosaurs from the entire Hell Creek Formation in the study area

Triceratops lived during the Late Cretaceous of western North America, its fossils coming from the Evanston Formation, Scollard Formation, Laramie Formation, Lance Formation, Denver Formation, and Hell Creek Formation.[112] These fossil formations date back to the time of the Cretaceous–Paleogene extinction event, which has been dated to 66 ± 0.07 million years ago.[1] Many animals and plants have been found in these formations, but mostly from the Lance Formation and Hell Creek Formation.[112] *Triceratops* was one of the last ceratopsian genera to appear before the end of the Mesozoic. The related *Torosaurus* and more distantly related diminutive *Leptoceratops* were also present, though their remains have been rarely encountered.[10] Isotopic analysis suggests *Triceratops* specimens from DTB (Darnell *Triceratops* Bonebed) could've inhabited freshwater floodplains, although the it likely wasn't restricted to one environment.[64]

Theropods from these formations include genera of dromaeosaurids, tyrannosaurids, ornithomimids, troodontids,[112] avialans,[113] and caenagnathids.[114] Dromaeosaurids from the Hell Creek Formation are *Acheroraptor* and *Dakotaraptor*. Indeterminate dromaeosaurs are known from other fossil formations. Common teeth previously referred to *Dromaeosaurus* and *Saurornitholestes* were considered to be those of *Acheroraptor*.[115] The tyrannosauroids from the formation are *Nanotyrannus* and *Tyrannosaurus*.[116] Among ornithomimids are the genera *Struthiomimus* and *Ornithomimus*.[112] An undescribed animal named "Orcomimus" could be from the formation.[117] Troodontids are only represented by *Pectinodon* and *Paronychodon* in the Hell Creek Formation with a possible species of *Troodon* from the Lance

Formation. One species of unknown coelurosaur is known from teeth in the Hell Creek and similar formations by a single species, *Richardoestesia*. Oviraptorosaurs from the Hell Creek Formation include *Anzu*.,[114] and avialans include *Avisaurus*,[112] multiple species of *Brodavis*,[118] and several other species of hesperornithoforms, as well as several species of true birds, including *Cimolopteryx*.[113]

Triceratops and other animals of the Hell Creek Formation

Ornithischians are abundant in the Scollard, Laramie, Lance, Denver, and Hell Creek Formation. The main groups of ornithischians are ankylosaurians, ornithopods, ceratopsians, and pachycephalosaurians. Three ankylosaurians are known: *Ankylosaurus*, *Denversaurus*, and possibly a species of *Edmontonia* or an undescribed genus. Multiple genera of ceratopsians are known from the formation other than *Triceratops*. These include the leptoceratopsid *Leptoceratops* and the chasmosaurine ceratopsids *Torosaurus*,[112] *Nedoceratops*, and *Tatankaceratops*.[119] Ornithopods are common in the Hell Creek Formation and are known from several species of the thescelosaurine *Thescelosaurus* and the hadrosaurid *Edmontosaurus*.[112][120] Several pachycephalosaurians have been found in the Hell Creek Formation and in similar formations. Among them are the derived pachycephalosaurids *Stygimoloch*,[112] *Dracorex*,[121] *Pachycephalosaurus*,[112] *Sphaerotholus*, and an undescribed specimen from North Dakota. The first two might be junior synonyms of *Pachycephalosaurus*.

Mammals are plentiful in the Hell Creek Formation. Groups represented include multituberculates, metatherians, and eutherians. The multituberculates represented include *Paracimexomys*,[122] the cimolomyids *Paressonodon*,[123] *Meniscoessus*, *Essonodon*, *Cimolomys*, *Cimolodon*, and *Cimexomys*, and the neoplagiaulacids *Mesodma* and *Neoplagiaulax*. The metatherians are represented by the alphadontids *Alphadon*, *Protalphodon*, and *Turgidodon*, the pediomyids *Pediomys*,[122] *Protolambda*, and *Leptalestes*,[124] the stagodontid *Didelphodon*,[122] the deltatheridiid *Nanocuris*, the herpetotheriid *Nortedelphys*,[123] and the glasbiid *Glasbius*. A few eutherians are known, being represented by *Alostera*,[122] *Protungulatum*,[124] the cimolestids *Cimolestes* and *Batodon*, the gypsonictopsid *Gypsonictops*, and the possible nyctitheriid *Paranyctoides*.[122]

Cultural significance

Since at least the early 20th century, *Triceratops* has been among the most famous dinosaurs and an archetypal ceratopsian.[125] It has been popularly displayed at natural history museums due to the abundance of fossils.[126] A mounted *Triceratops* skeleton was exhibited at the 1900 Paris Exposition, possibly the earliest display of the dinosaur, and was seen by 21,000 visitors during the first two days.[125] Similarly, a model *Triceratops* by American sculptor Louis Paul Jonas was displayed at the 1964 New York World's Fair.[126]

1901 illustration by Charles R. Knight

Triceratops is a staple of dinosaur films including *The Lost World* (1925) and *Jurassic Park* (1993). In literature, they appear as "Gryfs" in Edgar Rice Burroughs's 1921 novel *Tarzan the Terrible*,[125] while the 1956 children's book *The Enormous Egg* by Oliver Butterworth features a *Triceratops* hatching from a chicken egg.[126] After *Tyrannosaurus* and *Stegosaurus*, it has been the most frequently depicted dinosaur on stamps.[125] *Triceratops* is the official state fossil of South Dakota[127] and the official state dinosaur of Wyoming.[128]

In 1928, Charles R. Knight painted a mural incorporating a confrontation between a *Tyrannosaurus* and a *Triceratops* in the Field Museum of Natural History in Chicago, establishing them as enemies in the popular imagination. Paleontologist Robert Bakker said of the imagined rivalry between *Tyrannosaurus* and *Triceratops*, "No matchup between predator and prey has ever been more dramatic. It's somehow fitting that those two massive antagonists lived out their co-evolutionary belligerence through the last days of the last epoch of the Age of Dinosaurs."[129]

See also

- Uncle Beazley

References

1. Husson, D. E.; Galbrun, B.; Laskar, J.; Hinnov, L. A.; Thibault, N.; Gardin, S.; Locklair, R. E. (2011). "Astronomical calibration of the Maastrichtian (Late Cretaceous)". *Earth and Planetary Science Letters*. **305** (3–4): 328–340. Bibcode:2011E&PSL.305..328H (https://ui.adsabs.harvard.edu/abs/2011E&PSL.305..328H). doi:10.1016/j.epsl.2011.03.008 (https://doi.org/10.1016%2Fj.epsl.2011.03.008).
2. Lehman, Thomas M.; Mcdowell, Fred W.; Connelly, James N. (2006). "First Isotopic (U-Pb) Age for the Late Cretaceous Alamosaurus Vertebrate Fauna of West Texas, and Its Significance as a Link between Two Faunal Provinces" (https://www.jstor.org/stable/4524643). *Journal of Vertebrate Paleontology*. **26** (4): 922–928. doi:10.1671/0272-4634(2006)26[922:FIUAFT]2.0.CO;2 (https://doi.org/10.1671%2F0272-4634%282006%2926%5B922%3AFIUAFT%5D2.0.CO%3B2). ISSN 0272-4634 (https://search.worldcat.org/issn/0272-4634). JSTOR 4524643 (https://www.jstor.org/stable/4524643).
3. "Definition of triceratops | Dictionary.com" (https://www.dictionary.com/browse/triceratops). *www.dictionary.com*. Archived (https://web.archive.org/web/20221004023145/https://www.dictionary.com/browse/triceratops) from the original on October 4, 2022. Retrieved September 30, 2022.
4. Carpenter, K. (2006). "*Bison*" *alticornis* and O.C. Marsh's early views on ceratopsians". In Carpenter, K. (ed.). *Horns and Beaks: Ceratopsian and Ornithopod Dinosaurs*. Bloomington and Indianapolis: Indiana University Press. pp. 349–364. ISBN 978-0-253-34817-3.
5. Marsh, O.C. (1887). "Notice of new fossil mammals" (http://ajs.library.cmu.edu/books/pages.cgi?call=AJS_1887_034_1887&layout=vol0/part0/copy0&file=00000332). *American Journal of Science*. **34** (202): 323–331. Bibcode:1887AmJS...34..323M (https://ui.adsabs.harvard.edu/abs/1887AmJS...34..323M). doi:10.2475/ajs.s3-34.202.323 (https://doi.org/10.2475%2Fajs.s3-34.202.323). S2CID 129984410 (https://api.semanticscholar.org/CorpusID:129984410). Archived (https://web.archive.org/web/20180929001329/http://ajs.library.cmu.edu/books/pages.cgi?call=AJS_1887_034_1887&layout=vol0%2Fpart0%2Fcopy0&file=00000332) from the original on September 29, 2018. Retrieved October 19, 2021.

6. Marsh, O.C. (1888). "A new family of horned Dinosauria, from the Cretaceous" (http://ajs.library.cmu.edu/books/pages.cgi?call=AJS_1888_036_1888&layout=vol0/part0/copy0&file=00000493). *American Journal of Science*. **36** (216): 477–478. Bibcode:1888AmJS...36..477M (https://ui.adsabs.harvard.edu/abs/1888AmJS...36..477M). doi:10.2475/ajs.s3-36.216.477 (https://doi.org/10.2475%2Fajs.s3-36.216.477). S2CID 130243398 (https://api.semanticscholar.org/CorpusID:130243398). Archived (https://web.archive.org/web/20200218190455/http://ajs.library.cmu.edu/books/pages.cgi?call=AJS_1888_036_1888&layout=vol0%2Fpart0%2Fcopy0&file=00000493) from the original on February 18, 2020. Retrieved October 19, 2021.
7. Cope, E.D. (1872). "On the existence of Dinosauria in the Transition Beds of Wyoming". Proceedings of the American Philosophical Society. **12**: 481–483.
8. Cope, E.D. (1874). Report on the stratigraphy and Pliocene vertebrate paleontology of northern Colorado. Bulletin of the U.S. Geological and Geographical Survey of the Territories. **9**:9-28.
9. Lull, R. S., & Wright, N. E. (1942). Hadrosaurian dinosaurs of North America(Vol. 40). Geological Society of America.
10. Dodson, P. (1996). *The Horned Dinosaurs* (https://archive.org/details/horneddinosaursn00dods_0). Princeton, New Jersey: Princeton University Press. ISBN 978-0-691-02882-8.
11. Lull, R. S. (1933). "A revision of the Ceratopsia or horned dinosaurs" (https://archive.org/details/revisionofcerato33lull). *Memoirs of the Peabody Museum of Natural History*. **3** (3): 1–175. doi:10.5962/bhl.title.5716 (https://doi.org/10.5962%2Fbhl.title.5716). Retrieved November 20, 2010.
12. "Division of Paleontology" (http://research.amnh.org/paleontology/search.php?action=detail&specimen_id=46569). *research.amnh.org*. Archived (https://web.archive.org/web/20120927015429/http://research.amnh.org/paleontology/search.php?action=detail&specimen_id=46569) from the original on September 27, 2012. Retrieved April 12, 2022.
13. Dodson, P.; Forster, C.A.; Sampson, S.D. (2004). "Ceratopsidae". In Weishampel, David B.; Dodson, Peter; Osmo´lska, Halszka (eds.). *The dinosauria*. Berkeley, Calif.: University of California Press. pp. 494–513. ISBN 978-0-520-94143-4. OCLC 801843269 (https://search.worldcat.org/oclc/801843269)..
14. Marsh, O.C. (1889a). "Notice of new American Dinosauria" (http://ajs.library.cmu.edu/books/pages.cgi?call=AJS_1889_037_1889&layout=vol0/part0/copy0&file=00000339). *American Journal of Science*. **37** (220): 331–336. Bibcode:1889AmJS...37..331M (https://ui.adsabs.harvard.edu/abs/1889AmJS...37..331M). doi:10.2475/ajs.s3-37.220.331 (https://doi.org/10.2475%2Fajs.s3-37.220.331). S2CID 131729220 (https://api.semanticscholar.org/CorpusID:131729220). Archived (https://web.archive.org/web/20180929000453/http://ajs.library.cmu.edu/books/pages.cgi?call=AJS_1889_037_1889&layout=vol0%2Fpart0%2Fcopy0&file=00000339) from the original on September 29, 2018. Retrieved October 19, 2021.
15. Marsh, O.C. (1889b). "Notice of gigantic horned Dinosauria from the Cretaceous" (http://ajs.library.cmu.edu/books/pages.cgi?call=AJS_1889_038_1889&layout=vol0/part0/copy0&file=00000183). *American Journal of Science*. **38** (224): 173–175. Bibcode:1889AmJS...38..173M (https://ui.adsabs.harvard.edu/abs/1889AmJS...38..173M). doi:10.2475/ajs.s3-38.224.173 (https://doi.org/10.2475%2Fajs.s3-38.224.173). S2CID 131187857 (https://api.semanticscholar.org/CorpusID:131187857). Archived (https://web.archive.org/web/20180928235311/http://ajs.library.cmu.edu/books/pages.cgi?call=AJS_1889_038_1889&layout=vol0%2Fpart0%2Fcopy0&file=00000183) from the original on September 28, 2018. Retrieved October 19, 2021.
16. Hatcher, J. B.; Marsh, O. C.; Lull, R. S. (1907). *The Ceratopsia*. Washington, D.C.: Government Printing Office. ISBN 978-0-405-12713-7.
17. Goussard, Florent (2006). "The skull of *Triceratops* in the palaeontology gallery, Muséum national d'Histoire naturelle, Paris" (https://www.researchgate.net/publication/242181069). *Geodiversitas*. **28** (3): 467–476. Archived (https://web.archive.org/web/20211019153736/https://www.researchgate.net/publication/242181069_The_skull_of_Triceratops_in_the_palaeontology_gallery_Museum_National_d%27Histoire_Naturelle_Paris) from the original on October 19, 2021. Retrieved December 22, 2017 – via ResearchGate.

18. Sternberg, C. M. (1949). "The Edmonton fauna and description of a new *Triceratops* from the Upper Edmonton member; phylogeny of the Ceratopsidae". *National Museum of Canada Bulletin*. **113**: 33–46.
19. Ostrom, J. H.; Wellnhofer, P. (1986). "The Munich specimen of *Triceratops* with a revision of the genus". *Zitteliana*. **14**: 111–158.
20. Lehman, T. M. (1990). "The ceratopsian subfamily Chasmosaurinae: sexual dimorphism and systematics". In Carpenter, K.; Currie, P. J. (eds.). *Dinosaur Systematics: Perspectives and Approaches*. Cambridge: Cambridge University Press. pp. 211–229. ISBN 978-0-521-36672-4.
21. Forster, C.A. (1996). "Species resolution in *Triceratops*: cladistic and morphometric approaches". *Journal of Vertebrate Paleontology*. **16** (2): 259–270. Bibcode:1996JVPal..16..259F (https://ui.adsabs.harvard.edu/abs/1996JVPal..16..259F). doi:10.1080/02724634.1996.10011313 (https://doi.org/10.1080%2F02724634.1996.10011313).
22. Lehman, T. M. (1998). "A gigantic skull and skeleton of the horned dinosaur *Pentaceratops sternbergi* from New Mexico". *Journal of Paleontology*. **72** (5): 894–906. Bibcode:1998JPal...72..894L (https://ui.adsabs.harvard.edu/abs/1998JPal...72..894L). doi:10.1017/S0022336000027220 (https://doi.org/10.1017%2FS0022336000027220). JSTOR 1306666 (https://www.jstor.org/stable/1306666). S2CID 132807103 (https://api.semanticscholar.org/CorpusID:132807103).
23. Scannella, John B.; Fowler, Denver W.; Goodwin, Mark B.; Horner, John R. (July 15, 2014). "Evolutionary trends in *Triceratops* from the Hell Creek Formation, Montana" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4104892). *Proceedings of the National Academy of Sciences*. **111** (28): 10245–10250. Bibcode:2014PNAS..11110245S (https://ui.adsabs.harvard.edu/abs/2014PNAS..11110245S). doi:10.1073/pnas.1313334111 (https://doi.org/10.1073%2Fpnas.1313334111). ISSN 0027-8424 (https://search.worldcat.org/issn/0027-8424). PMC 4104892 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4104892). PMID 24982159 (https://pubmed.ncbi.nlm.nih.gov/24982159).
24. Paul, G. S. (2010). *The Princeton Field Guide to Dinosaurs* (https://archive.org/details/princetonfieldgu0000paul/page/265). Princeton University Press. pp. 265–267 (https://archive.org/details/princetonfieldgu0000paul/page/265). ISBN 978-0-691-13720-9.
25. Holtz, Thomas R. Jr. (2011). *Dinosaurs: The Most Complete, Up-to-Date Encyclopedia for Dinosaur Lovers of All Ages* (http://www.geol.umd.edu/~tholtz/dinoappendix/HoltzappendixWinter2011.pdf) (PDF). Archived (https://web.archive.org/web/20170812011954/http://www.geol.umd.edu/~tholtz/dinoappendix/HoltzappendixWinter2011.pdf) (PDF) from the original on August 12, 2017. Retrieved August 2, 2022. "Winter 2011 Appendix"
26. Stein, Walter W. (2019). "TAKING COUNT: A Census of Dinosaur Fossils Recovered From the Hell Creek and Lance Formations (Maastrichtian)" (https://www.aaps-journal.org/pdf/JPS.C.2019.01a.pdf) (PDF). *The Journal of Paleontological Sciences*. **8**: 1–42. Archived (https://web.archive.org/web/20241008104837/https://www.aaps-journal.org/pdf/JPS.C.2019.01a.pdf) (PDF) from the original on October 8, 2024. Retrieved April 21, 2024.
27. "A *Triceratops* Named 'Kelsey' " (https://www.bhigr.com/pages/info/info_klsy.htm). *www.bhigr.com*. Archived (https://web.archive.org/web/20171223102110/https://www.bhigr.com/pages/info/info_klsy.htm) from the original on December 23, 2017. Retrieved December 22, 2017.
28. "Kesley the *Triceratops*" (https://bhigr.com/wp-content/uploads/2022/04/KelseySkeleton_03-copy-scaled.jpg). *www.bhigr.com*.
29. "Big John, largest known triceratops skeleton, sold at auction" (https://www.bbc.com/news/world-europe-58998367). *BBC News*. October 21, 2021. Archived (https://web.archive.org/web/20220112104652/https://www.bbc.com/news/world-europe-58998367) from the original on January 12, 2022. Retrieved December 16, 2025.

30. Scannella, J.; Horner, J.R. (2010). "*Torosaurus* Marsh, 1891, is *Triceratops* Marsh, 1889 (Ceratopsidae: Chasmosaurinae): synonymy through ontogeny". *Journal of Vertebrate Paleontology*. **30** (4): 1157–1168. Bibcode:2010JVPal..30.1157S (https://ui.adsabs.harvard.edu/abs/2010JVPal..30.1157S). doi:10.1080/02724634.2010.483632 (https://doi.org/10.1080%2F02724634.2010.483632). S2CID 86767957 (https://api.semanticscholar.org/CorpusID:86767957).
31. Lambert, D. (1993). *The Ultimate Dinosaur Book* (https://archive.org/details/ultimatedinosaur00lamb/page/152). Dorling Kindersley, New York. pp. 152–167 (https://archive.org/details/ultimatedinosaur00lamb/page/152). ISBN 978-1-56458-304-8.
32. Dodson, P.; Forster, C. A.; Sampson, S. D. (2004). "Ceratopsidae". In Weishampel, D. B.; Dodson, P.; Osmólska, H. (eds.). *The Dinosauria* (second ed.). Berkeley: University of California Press. pp. 494–513. ISBN 978-0-520-24209-8.
33. "Denver museum unveils 7-foot-long, 1,000-pound Triceratops skull" (https://news.google.com/newspapers?nid=894&dat=20031118&id=OCNTAAAAIBAJ&pg=6720,2967799). The Daily Courier. November 18, 2003. Archived (https://web.archive.org/web/20210519173018/https://news.google.com/newspapers?nid=894&dat=20031118&id=OCNTAAAAIBAJ&pg=6720,2967799) from the original on May 19, 2021. Retrieved December 26, 2013.
34. "Making A *Triceratops*. Science Supplies Missing Part! Of Skeleton" (https://news.google.com/newspapers?nid=2249&dat=19011024&id=U5I-AAAAIBAJ&pg=5758,3411357). Boston Evening Transcript. October 24, 1901. Archived (https://web.archive.org/web/20210519173017/https://news.google.com/newspapers?nid=2249&dat=19011024&id=U5I-AAAAIBAJ&pg=5758,3411357) from the original on May 19, 2021. Retrieved December 26, 2013.
35. Fujiwara, Shin-Ichi (December 12, 2009). "A reevaluation of the manus structure in *Triceratops* (Ceratopsia: Ceratopsidae)". *Journal of Vertebrate Paleontology*. **29** (4): 1136–1147. Bibcode:2009JVPal..29.1136F (https://ui.adsabs.harvard.edu/abs/2009JVPal..29.1136F). doi:10.1671/039.029.0406 (https://doi.org/10.1671%2F039.029.0406). ISSN 0272-4634 (https://search.worldcat.org/issn/0272-4634). S2CID 86519018 (https://api.semanticscholar.org/CorpusID:86519018).
36. Christiansen, P.; Paul, G.S. (2001). "Limb bone scaling, limb proportions, and bone strength in neoceratopsian dinosaurs" (http://gspauldino.com/GaiaNeoceratopsian.pdf) (PDF). *Gaia*. **16**: 13–29. Archived (https://web.archive.org/web/20180819045306/http://gspauldino.com/GaiaNeoceratopsian.pdf) (PDF) from the original on August 19, 2018. Retrieved October 29, 2012.
37. Thompson, S.; Holmes, R. (2007). "Forelimb stance and step cycle in *Chasmosaurus irvinensis* (Dinosauria: Neoceratopsia)" (http://palaeo-electronica.org/2007_1/step/index.html). *Palaeontologia Electronica*. **10** (1): 17 p. Archived (https://web.archive.org/web/20181211003615/https://palaeo-electronica.org/2007_1/step/index.html) from the original on December 11, 2018. Retrieved November 20, 2010.
38. Rega, E.; Holmes, R.; Tirabasso, A. (2010). "Habitual locomotor behavior inferred from manual pathology in two Late Cretaceous chasmosaurine ceratopsid dinosaurs, *Chasmosaurus irvinensis* (CMN 41357) and *Chasmosaurus belli* (ROM 843)". In Ryan, Michael J.; Chinnery-Allgeier, Brenda J.; Eberth, David A. (eds.). *New Perspectives on Horned Dinosaurs: The Royal Tyrrell Museum Ceratopsian Symposium*. Bloomington and Indianapolis: Indiana University Press. pp. 340–354. ISBN 978-0-253-35358-0.
39. Tweet, Justin (2025). "Paleontology of Morrison–Golden Fossil Areas National Natural Landmark" (https://www.nps.gov/articles/000/paleontology-of-morrison-golden-fossil-areas-national-natural-landmark.htm). *Park Paleontology News*. **17** (2).
40. Martin, Anthony J. (2006). *Introduction to the study of dinosaurs* (2nd ed.). Malden, MA: Blackwell Pub. ISBN 978-1-4051-3413-2. OCLC 61130756 (https://search.worldcat.org/oclc/61130756).

41. Sereno, Paul C.; Saitta, Evan T.; Vidal, Daniel; Myhrvold, Nathan; Real, María Ciudad; Baumgart, Stephanie L.; Bop, Lauren L.; Keillor, Tyler M.; Eriksen, Marcus; Derstler, Kraig (2026). "Duck-billed dinosaur fleshy midline and hooves reveal terrestrial clay-template 'mummification' ". *Science*. **391** (6780) eadw3536. doi:10.1126/science.adw3536 (https://doi.org/10.1126%2Fscience.adw3536). PMID 41129614 (https://pubmed.ncbi.nlm.nih.gov/41129614).
42. Supplementary information for: Sereno, Paul C.; Saitta, Evan T.; Vidal, Daniel; Myhrvold, Nathan; Real, María Ciudad; Baumgart, Stephanie L.; Bop, Lauren L.; Keillor, Tyler M.; Eriksen, Marcus; Derstler, Kraig (2026). "Duck-billed dinosaur fleshy midline and hooves reveal terrestrial clay-template 'mummification' ". *Science*. **391** (6780) eadw3536. doi:10.1126/science.adw3536 (https://doi.org/10.1126%2Fscience.adw3536). PMID 41129614 (https://pubmed.ncbi.nlm.nih.gov/41129614).
43. Bell, Phil R.; Hendrickx, Christophe; Pittman, Michael; Kaye, Thomas G.; Mayr, Gerald (August 12, 2022). "The exquisitely preserved integument of Psittacosaurus and the scaly skin of ceratopsian dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9374759). *Communications Biology*. **5** (1): 809. doi:10.1038/s42003-022-03749-3 (https://doi.org/10.1038%2Fs42003-022-03749-3). ISSN 2399-3642 (https://search.worldcat.org/issn/2399-3642). PMC 9374759 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9374759). PMID 35962036 (https://pubmed.ncbi.nlm.nih.gov/35962036).
44. "What is special about the *Triceratops*?" (http://dinosaurios.org/triceratops/). Dinosaurios.org. July 24, 2013. Archived (https://web.archive.org/web/20200531214022/http://dinosaurios.org/triceratops/) from the original on May 31, 2020. Retrieved December 26, 2013.
45. Lambe, Lawrence M. (1915). *On Eoceratops canadensis, gen. nov., with remarks on other genera of Cretaceous horned dinosaurs*. Ottawa: Geological Survey of Canada, Government Printing Bureau. ISBN 0-665-82611-7. OCLC 920394016 (https://search.worldcat.org/oclc/920394016).
46. Ostrom, J. H. (1966). "Functional morphology and evolution of the ceratopsian dinosaurs". *Evolution*. **20** (3): 290–308. doi:10.2307/2406631 (https://doi.org/10.2307%2F2406631). JSTOR 2406631 (https://www.jstor.org/stable/2406631). PMID 28562975 (https://pubmed.ncbi.nlm.nih.gov/28562975).
47. Norman, David (1985). *The Illustrated Encyclopaedia of Dinosaurs*. London: Salamander Books. ISBN 978-0-517-46890-6.
48. Dodson, P.; Currie, P. J. (1990). "Neoceratopsia". In Weishampel, D. B.; Dodson, P.; Osmólska, H. (eds.). *The Dinosauria*. Berkeley: University of California Press. pp. 593–618. ISBN 978-0-520-06727-1.
49. Dodson, P. (1993). "Comparative craniology of the Ceratopsia" (http://earth.geology.yale.edu/~ajs/1993/11.1993.07Dodson.pdf) (PDF). *American Journal of Science*. **293**: 200–234. Bibcode:1993AmJS..293..200D (https://ui.adsabs.harvard.edu/abs/1993AmJS..293..200D). doi:10.2475/ajs.293.A.200 (https://doi.org/10.2475%2Fajs.293.A.200). Archived (https://web.archive.org/web/20180819045309/http://earth.geology.yale.edu/~ajs/1993/11.1993.07Dodson.pdf) (PDF) from the original on August 19, 2018. Retrieved January 21, 2007.
50. Longrich, N. R. (2014). "The horned dinosaurs Pentaceratops and Kosmoceratops from the upper Campanian of Alberta and implications for dinosaur biogeography". *Cretaceous Research*. **51**: 292–308. Bibcode:2014CrRes..51..292L (https://ui.adsabs.harvard.edu/abs/2014CrRes..51..292L). doi:10.1016/j.cretres.2014.06.011 (https://doi.org/10.1016%2Fj.cretres.2014.06.011).
51. Xu, X., Forster, C.A., Clark, J.M., and Mo, J. (2006). "A basal ceratopsian with transitional features from the Late Jurassic of northwestern China (https://www.gwu.edu/~newsctr/fossilfind/paper.pdf)." *Proceedings of the Royal Society B: Biological Sciences*, **273**(1598): 2135-2140. doi:10.1098/rspb.2006.3566 (https://doi.org/10.1098%2Frspb.2006.3566)

52. Gauthier, J. A. (1986). "Saurischian monophyly and the origin of birds. The Origin of Birds and the Evolution of Flight, K. Padian (ed.)". *Memoirs of the California Academy of Sciences*. **8**: 1–55.
53. Sereno, P. C. (1998). "A rationale for phylogenetic definitions, with application to the higher-level taxonomy of Dinosauria". *Neues Jahrbuch für Geologie und Paläontologie, Abhandlungen*. **210** (1): 41–83. doi:10.1127/njgpa/210/1998/41 (https://doi.org/10.1127%2Fnjgpa%2F210%2F1998%2F41).
54. Mathews, Joshua C.; Brusatte, Stephen L.; Williams, Scott A.; Henderson, Michael D. (2009). "The first *Triceratops* bonebed and its implications for gregarious behavior". *Journal of Vertebrate Paleontology*. **29** (1): 286–290. Bibcode:2009JVPal..29..286M (https://ui.adsabs.harvard.edu/abs/2009JVPal..29..286M). doi:10.1080/02724634.2009.10010382 (https://doi.org/10.1080%2F02724634.2009.10010382). S2CID 196608646 (https://api.semanticscholar.org/CorpusID:196608646).
55. Smith, Matt (June 4, 2013). "Triceratops trio unearthed in Wyoming – CNN" (http://www.cnn.com/2013/06/03/us/triceratops-found/index.html). *CNN*. Archived (https://web.archive.org/web/20170801233836/http://www.cnn.com/2013/06/03/us/triceratops-found/index.html) from the original on August 1, 2017. Retrieved December 22, 2017.
56. Illies, M. M. Canoy; Fowler, D. W. (2020). "*Triceratops* with a kink: Co-ossification of five distal caudal vertebrae from the Hell Creek Formation of North Dakota". *Cretaceous Research*. **108** 104355. Bibcode:2020CrRes.10804355C (https://ui.adsabs.harvard.edu/abs/2020CrRes.10804355C). doi:10.1016/j.cretres.2019.104355 (https://doi.org/10.1016%2Fj.cretres.2019.104355). S2CID 214215413 (https://api.semanticscholar.org/CorpusID:214215413).
57. Barrera, Nathanial A. (January 9, 2020). "More than old bones: New study sheds light on *Triceratops* behavior and living habits" (https://www.thedickinsonpress.com/incoming/4858206-More-than-old-bones-New-study-sheds-light-on-Triceratops-behavior-and-living-habits). *The Dickinson Press*. Archived (https://web.archive.org/web/20200527200052/https://www.thedickinsonpress.com/incoming/4858206-More-than-old-bones-New-study-sheds-light-on-Triceratops-behavior-and-living-habits) from the original on May 27, 2020. Retrieved March 31, 2020.
58. Erickson, B. R. (1966). "Mounted skeleton of *Triceratops prorsus* in the Science Museum". *Scientific Publications of the Science Museum*. **1**: 1–16.
59. Bakker, R. T. (1986). *The Dinosaur Heresies: New Theories Unlocking The Mystery of the Dinosaurs and Their Extinction* (https://archive.org/details/dinosaurheresies00robe). New York: William Morrow. p. 438 (https://archive.org/details/dinosaurheresies00robe/page/438). ISBN 978-0-14-010055-6.
60. Derstler, K. (1994). "Dinosaurs of the Lance Formation in eastern Wyoming". In Nelson, G. E. (ed.). *The Dinosaurs of Wyoming*. Wyoming Geological Association Guidebook, 44th Annual Field Conference. Wyoming Geological Association. pp. 127–146.
61. Sakagami, Rina; Kawabe, Soichiro (2020). "Endocranial anatomy of the ceratopsid dinosaur *Triceratops* and interpretations of sensory and motor function" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7505063). *PeerJ*. **8** e9888. doi:10.7717/peerj.9888 (https://doi.org/10.7717%2Fpeerj.9888). PMC 7505063 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7505063). PMID 32999761 (https://pubmed.ncbi.nlm.nih.gov/32999761).
62. Tada, S; Tsuihiji, T; Ishikawa, H; Wakimizu, N; Kawabe, S; Sakane, K (2026). "Nasal soft-tissue anatomy of *Triceratops* and other horned dinosaurs". *The Anatomical Record* ar.70150. doi:10.1002/ar.70150 (https://doi.org/10.1002%2Far.70150). PMID 41652936 (https://pubmed.ncbi.nlm.nih.gov/41652936).

63. Wiemann, J.; Menéndez, I.; Crawford, J.M.; Fabbri, M.; Gauthier, J.A.; Hull, P.M.; Norell, M.A.; Briggs, D.E.G. (2022). "Fossil biomolecules reveal an avian metabolism in the ancestral dinosaur" (https://resolver.caltech.edu/CaltechAUTHORS:20220531-924660000). *Nature*. **606** (7914): 522–526. Bibcode:2022Natur.606..522W (https://ui.adsabs.harvard.edu/abs/2022Natur.606..522W). doi:10.1038/s41586-022-04770-6 (https://doi.org/10.1038%2Fs41586-022-04770-6). PMID 35614213 (https://pubmed.ncbi.nlm.nih.gov/35614213). S2CID 249064466 (https://api.semanticscholar.org/CorpusID:249064466).
64. de Rooij, Jimmy; van der Lubbe, Jeroen H. J. L.; Verdegaal, Suzan; Hulscher, Megan; Tooms, Daphne; Kaskes, Pim; Verhage, Oeki; Portanger, Leonie; Schulp, Anne S. (December 1, 2022). "Stable isotope record of *Triceratops* from a mass accumulation (Lance Formation, Wyoming, USA) provides insights into *Triceratops* behaviour and ecology" (https://www.sciencedirect.com/science/article/pii/S003101822200445X). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **607** 111274. Bibcode:2022PPP...60711274D (https://ui.adsabs.harvard.edu/abs/2022PPP...60711274D). doi:10.1016/j.palaeo.2022.111274 (https://doi.org/10.1016%2Fj.palaeo.2022.111274). ISSN 0031-0182 (https://search.worldcat.org/issn/0031-0182). Archived (https://web.archive.org/web/20240226074157/https://www.sciencedirect.com/science/article/pii/S003101822200445X) from the original on February 26, 2024. Retrieved November 22, 2024.
65. Tait, J.; Brown, B. (1928). "How the Ceratopsia carried and used their head". *Transactions of the Royal Society of Canada*. **22**: 13–23.
66. Erickson, Gregory M.; Sidebottom, Mark A.; Kay, David I.; Turner, Kevin T.; Ip, Nathan; Norell, Mark A.; Sawyer, W. Gregory; Krick, Brandon A. (June 5, 2015). "Wear biomechanics in the slicing dentition of the giant horned dinosaur *Triceratops*" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4640618). *Science Advances*. **1** (5) e1500055. Bibcode:2015SciA....1E0055E (https://ui.adsabs.harvard.edu/abs/2015SciA....1E0055E). doi:10.1126/sciadv.1500055 (https://doi.org/10.1126%2Fsciadv.1500055). ISSN 2375-2548 (https://search.worldcat.org/issn/2375-2548). PMC 4640618 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4640618). PMID 26601198 (https://pubmed.ncbi.nlm.nih.gov/26601198).
67. Maiorino, Leonardo; Farke, Andrew A.; Kotsakis, Tassos; Teresi, Luciano; Piras, Paolo (September 11, 2015). "Variation in the shape and mechanical performance of the lower jaws in ceratopsid dinosaurs (Ornithischia, Ceratopsia)" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4609198). *Journal of Anatomy*. **227** (5): 631–646. doi:10.1111/joa.12374 (https://doi.org/10.1111%2Fjoa.12374). ISSN 0021-8782 (https://search.worldcat.org/issn/0021-8782). PMC 4609198 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4609198). PMID 26467240 (https://pubmed.ncbi.nlm.nih.gov/26467240).
68. Carter, Nicholas (January 29, 2019). "A Day in the Life of *Triceratops*" (https://dinomuseum.ca/2019/01/a-day-in-the-life-of-triceratops). *Philip J. Currie Dinosaur Museum*. Retrieved April 15, 2025.
69. "A triceratopsian diet" (https://www.mcgill.ca/redpath/article/triceratopsian-diet). *Redpath Museum*. May 2, 2017. Retrieved April 15, 2025.
70. Ostrom, J. H. (1964). "A functional analysis of jaw mechanics in the dinosaur *Triceratops*" (https://web.archive.org/web/20100626220725/http://www.peabody.yale.edu/scipubs/bulletins_postillas/ypmP088_1964.pdf) (PDF). *Postilla*. **88**: 1–35. Archived from the original (http://www.peabody.yale.edu/scipubs/bulletins_postillas/ypmP088_1964.pdf) (PDF) on June 26, 2010. Retrieved November 20, 2010.
71. Weishampel, D. B. (1984). *Evolution of Jaw Mechanisms in Ornithopod Dinosaurs*. Advances in Anatomy Embryology and Cell Biology. Vol. 87. pp. 1–110. doi:10.1007/978-3-642-69533-9 (https://doi.org/10.1007%2F978-3-642-69533-9). ISBN 978-3-540-13114-4. PMID 6464809 (https://pubmed.ncbi.nlm.nih.gov/6464809). S2CID 12547312 (https://api.semanticscholar.org/CorpusID:12547312).

72. Fricke, Henry C.; Pearson, Dean A. (2008). "Stable isotope evidence for changes in dietary niche partitioning among hadrosaurian and ceratopsian dinosaurs of the Hell Creek Formation, North Dakota". *Paleobiology*. **34** (4): 534–552. Bibcode:2008Pbio...34..534F (https://ui.adsabs.harvard.edu/abs/2008Pbio...34..534F). doi:10.1666/08020.1 (https://doi.org/10.1666%2F08020.1). ISSN 0094-8373 (https://search.worldcat.org/issn/0094-8373).
73. Coe, M. J.; Dilcher, D. L.; Farlow, J. O.; Jarzen, D. M.; Russell, D. A. (1987). "Dinosaurs and land plants". In Friis, E. M.; Chaloner, W. G.; Crane, P. R. (eds.). *The Origins of Angiosperms and their Biological Consequences*. Cambridge University Press. pp. 225–258. ISBN 978-0-521-32357-4.
74. Fricke, Henry C.; Pearson, Dean A. (October 2008). "Stable isotope evidence for changes in dietary niche partitioning among hadrosaurian and ceratopsian dinosaurs of the Hell Creek Formation, North Dakota" (https://www.cambridge.org/core/journals/paleobiology/article/abs/stable-isotope-evidence-for-changes-in-dietary-niche-partitioning-among-hadrosaurian-and-ceratopsian-dinosaurs-of-the-hell-creek-formation-north-dakota/C96E90222D1D728292F0343828FC213B). *Paleobiology*. **34** (4): 534–552. Bibcode:2008Pbio...34..534F (https://ui.adsabs.harvard.edu/abs/2008Pbio...34..534F). doi:10.1666/08020.1 (https://doi.org/10.1666%2F08020.1). ISSN 0094-8373 (https://search.worldcat.org/issn/0094-8373).
75. Sander, P. M.; Gee, C. T.; Hummel, J.; Clauss, Marcus (June 23, 2010). Gee, C. T. (ed.). "Mesozoic plants and dinosaur herbivory" (https://www.zora.uzh.ch/id/eprint/35283/). *Life of the Past*: 331–359. doi:10.5167/uzh-35283 (https://doi.org/10.5167%2Fuzh-35283). ISBN 978-0-253-35456-3.
76. Barrett, Paul M.; Rayfield, Emily J. (April 1, 2006). "Ecological and evolutionary implications of dinosaur feeding behaviour" (https://www.cell.com/trends/ecology-evolution/abstract/S0169-5347(06)00023-1). *Trends in Ecology & Evolution*. **21** (4): 217–224. doi:10.1016/j.tree.2006.01.002 (https://doi.org/10.1016%2Fj.tree.2006.01.002). ISSN 0169-5347 (https://search.worldcat.org/issn/0169-5347). PMID 16701088 (https://pubmed.ncbi.nlm.nih.gov/16701088).
77. Lull, R. S. (1908). "The cranial musculature and the origin of the frill in the ceratopsian dinosaurs" (https://zenodo.org/record/1450156). *American Journal of Science*. **4** (25): 387–399. Bibcode:1908AmJS...25..387L (https://ui.adsabs.harvard.edu/abs/1908AmJS...25..387L). doi:10.2475/ajs.s4-25.149.387 (https://doi.org/10.2475%2Fajs.s4-25.149.387). Archived (https://web.archive.org/web/20191115184638/https://zenodo.org/record/1450156) from the original on November 15, 2019. Retrieved June 12, 2019.
78. Forster, C. A. (1990). *The cranial morphology and systematics of* Triceratops*, with a preliminary analysis of ceratopsian phylogeny* (Ph.D. Dissertation thesis). Philadelphia: University of Pennsylvania. p. 227.
79. Sternberg, C. H. (1917). *Hunting Dinosaurs in the Badlands of the Red Deer River, Alberta, Canada*. Lawrence, Kansas: C. H. Sternberg—The World Company Press. p. 261. doi:10.5962/bhl.title.159811 (https://doi.org/10.5962%2Fbhl.title.159811).
80. Happ, J. (2008). "An analysis of predator-prey behavior in a head-to-head encounter between *Tyrannosaurus rex* and *Triceratops*". In Larson, P.; Carpenter, K. (eds.). *Tyrannosaurus rex, the Tyrant King (Life of the Past)*. Bloomington: Indiana University Press. pp. 355–368. ISBN 978-0-253-35087-9.
81. Erickson, Gregory M.; Olson, Kenneth H. (March 19, 1996). "Bite marks attributable to Tyrannosaurus rex: Preliminary description and implications" (https://zenodo.org/record/3732430). *Journal of Vertebrate Paleontology*. **16** (1): 175–178. Bibcode:1996JVPal..16..175E (https://ui.adsabs.harvard.edu/abs/1996JVPal..16..175E). doi:10.1080/02724634.1996.10011297 (https://doi.org/10.1080%2F02724634.1996.10011297). ISSN 0272-4634 (https://search.worldcat.org/issn/0272-4634). Archived (https://web.archive.org/web/20211019153802/https://zenodo.org/record/3732430) from the original on October 19, 2021. Retrieved June 3, 2020.

82. Farke, A. A. (2004). "Horn Use in *Triceratops* (Dinosauria: Ceratopsidae): Testing Behavioral Hypotheses Using Scale Models" (http://www.nhm.ac.uk/hosted_sites/pe/2004_1/horn/horn.pdf) (PDF). *Palaeo-electronica*. **7** (1): 1–10. Archived (https://web.archive.org/web/20160303204748/http://www.nhm.ac.uk/hosted_sites/pe/2004_1/horn/horn.pdf) (PDF) from the original on March 3, 2016. Retrieved November 20, 2010.
83. Tanke, D. H.; Farke, A. A. (2006). "Bone resorption, bone lesions, and extracranial fenestrae in ceratopsid dinosaurs: a preliminary assessment". In Carpenter, K. (ed.). *Horns and Beaks: Ceratopsian and Ornithopod Dinosaurs*. Bloomington: Indiana University Press. pp. 319–347. ISBN 978-0-253-34817-3.
84. Farke, A.A.; Wolff, E.D.S.; Tanke, D.H.; Sereno, Paul (2009). Sereno, Paul (ed.). "Evidence of Combat in *Triceratops*" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2617760). *PLOS ONE*. **4** (1) e4252. Bibcode:2009PLoSO...4.4252F (https://ui.adsabs.harvard.edu/abs/2009PLoSO...4.4252F). doi:10.1371/journal.pone.0004252 (https://doi.org/10.1371%2Fjournal.pone.0004252). PMC 2617760 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2617760). PMID 19172995 (https://pubmed.ncbi.nlm.nih.gov/19172995).
85. Wall, Michael (January 27, 2009). "Scars Reveal How *Triceratops* Fought" (https://web.archive.org/web/20140820071319/https://www.wired.com/2009/01/dinofight/). *Wired*. ISSN 1059-1028 (https://search.worldcat.org/issn/1059-1028). Archived from the original (https://www.wired.com/2009/01/dinofight/) on August 20, 2014. Retrieved December 13, 2024.
86. Switek, Brian (February 5, 2009). "*Triceratops* v. *Triceratops*" (https://web.archive.org/web/20140416192132/https://www.smithsonianmag.com/science-nature/triceratops-v-triceratops-38373087/). *Smithsonian Magazine*. Archived from the original (https://www.smithsonianmag.com/science-nature/triceratops-v-triceratops-38373087/) on April 16, 2014. Retrieved December 13, 2024.
87. Peterson, JE; Dischler, C; Longrich, NR (2013). "Distributions of Cranial Pathologies Provide Evidence for Head-Butting in Dome-Headed Dinosaurs (Pachycephalosauridae)" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3712952). *PLOS ONE*. **8** (7) e68620. Bibcode:2013PLoSO...868620P (https://ui.adsabs.harvard.edu/abs/2013PLoSO...868620P). doi:10.1371/journal.pone.0068620 (https://doi.org/10.1371%2Fjournal.pone.0068620). PMC 3712952 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3712952). PMID 23874691 (https://pubmed.ncbi.nlm.nih.gov/23874691).
88. Wall, Michael (January 27, 2009). "Scars Reveal How *Triceratops* Fought –" (https://www.wired.com/wiredscience/2009/01/dinofight/). *Wired*. Archived (https://web.archive.org/web/20140112143952/http://www.wired.com/wiredscience/2009/01/dinofight/) from the original on January 12, 2014. Retrieved August 3, 2010.
89. Reid, R.E.H. (1997). "Histology of bones and teeth". In Currie, P. J.; Padian, K. (eds.). *Encyclopedia of Dinosaurs*. San Diego, CA.: Academic Press. pp. 329–339.
90. Horner, JR; Goodwin, MB (2009). "Extreme Cranial Ontogeny in the Upper Cretaceous Dinosaur Pachycephalosaurus" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2762616). *PLOS ONE*. **4** (10) e7626. Bibcode:2009PLoSO...4.7626H (https://ui.adsabs.harvard.edu/abs/2009PLoSO...4.7626H). doi:10.1371/journal.pone.0007626 (https://doi.org/10.1371%2Fjournal.pone.0007626). PMC 2762616 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2762616). PMID 19859556 (https://pubmed.ncbi.nlm.nih.gov/19859556).
91. Horner, JR; Lamm, E (2011). "Ontogeny of the parietal frill of *Triceratops*: a preliminary histological analysis". *Comptes Rendus Palevol*. **10** (5–6): 439–452. doi:10.1016/j.crpv.2011.04.006 (https://doi.org/10.1016%2Fj.crpv.2011.04.006).
92. Farlow, J. O.; Dodson, P. (1975). "The behavioral significance of frill and horn morphology in ceratopsian dinosaurs". *Evolution*. **29** (2): 353–361. doi:10.2307/2407222 (https://doi.org/10.2307%2F2407222). JSTOR 2407222 (https://www.jstor.org/stable/2407222). PMID 28555861 (https://pubmed.ncbi.nlm.nih.gov/28555861).
93. Martin, A. J. (2006). *Introduction to the Study of Dinosaurs* (Second ed.). Oxford: Blackwell Publishing. pp. 299–300. ISBN 978-1-4051-3413-2.

94. D'Anastasio, Ruggero; Cilli, Jacopo; Bacchia, Flavio; Fanti, Federico; Gobbo, Giacomo; Capasso, Luigi (April 7, 2022). "Histological and chemical diagnosis of a combat lesion in *Triceratops*" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8990019). *Scientific Reports*. **12** (1): 3941. Bibcode:2022NatSR..12.3941D (https://ui.adsabs.harvard.edu/abs/2022NatSR..12.3941D). doi:10.1038/s41598-022-08033-2 (https://doi.org/10.1038%2Fs41598-022-08033-2). ISSN 2045-2322 (https://search.worldcat.org/issn/2045-2322). PMC 8990019 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8990019). PMID 35393445 (https://pubmed.ncbi.nlm.nih.gov/35393445).
95. Wheeler, P.E. (1978). "Elaborate CNS cooling structures in large dinosaurs". *Nature*. **275** (5679): 441–443. Bibcode:1978Natur.275..441W (https://ui.adsabs.harvard.edu/abs/1978Natur.275..441W). doi:10.1038/275441a0 (https://doi.org/10.1038%2F275441a0). PMID 692723 (https://pubmed.ncbi.nlm.nih.gov/692723). S2CID 4160470 (https://api.semanticscholar.org/CorpusID:4160470).
96. Farlow, J. O.; Thompson, C. V.; Rosner, D. E. (1976). "Plates of the dinosaur *Stegosaurus*: Forced convection heat loss fins?". *Science*. **192** (4244): 1123–5. Bibcode:1976Sci...192.1123F (https://ui.adsabs.harvard.edu/abs/1976Sci...192.1123F). doi:10.1126/science.192.4244.1123 (https://doi.org/10.1126%2Fscience.192.4244.1123). PMID 17748675 (https://pubmed.ncbi.nlm.nih.gov/17748675). S2CID 44506996 (https://api.semanticscholar.org/CorpusID:44506996).
97. Davitashvili, L. Sh. (1961). *Teoriya Polovogo Otbora (Theory of Sexual Selection)*. Izdatel'stvo Akademii nauk SSSR. p. 538.
98. Goodwin, M.B.; Clemens, W.A.; Horner, J.R. & Padian, K. (2006). "The smallest known *Triceratops* skull: new observations on ceratopsid cranial anatomy and ontogeny" (https://web.archive.org/web/20151105200947/http://www.ucmp.berkeley.edu/people/mbg/Goodwin_et_al_2006.pdf) (PDF). *Journal of Vertebrate Paleontology*. **26** (1): 103–112. doi:10.1671/0272-4634(2006)26[103:TSKTSN]2.0.CO;2 (https://doi.org/10.1671%2F0272-4634%282006%2926%5B103%3ATSKTSN%5D2.0.CO%3B2). ISSN 0272-4634 (https://search.worldcat.org/issn/0272-4634). S2CID 31117040 (https://api.semanticscholar.org/CorpusID:31117040). Archived from the original (http://www.ucmp.berkeley.edu/people/mbg/Goodwin_et_al_2006.pdf) (PDF) on November 5, 2015. Retrieved January 21, 2007.
99. Hone, D. W. E.; Naish, D. (2013). "The 'species recognition hypothesis' does not explain the presence and evolution of exaggerated structures in non-avialan dinosaurs" (https://doi.org/10.1111%2Fjzo.12035). *Journal of Zoology*. **290** (3): 172–180. doi:10.1111/jzo.12035 (https://doi.org/10.1111%2Fjzo.12035).
100. Horner, J.R.; Goodwin, M.B. (2006). "Major cranial changes during *Triceratops* ontogeny" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1635501). *Proceedings of the Royal Society B: Biological Sciences*. **273** (1602): 2757–2761. doi:10.1098/rspb.2006.3643 (https://doi.org/10.1098%2Frspb.2006.3643). PMC 1635501 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1635501). PMID 17015322 (https://pubmed.ncbi.nlm.nih.gov/17015322).
101. Šuligoj, Boris (May 13, 2021). "(Majhen) dinozaver sredi dnevne sobe" (https://www.delo.si/magazin/zanimivosti/majhen-dinozaver-sredi-dnevne-sobe/). *www.delo.si* (in Slovenian). Archived (https://web.archive.org/web/20220112105531/https://www.delo.si/magazin/zanimivosti/majhen-dinozaver-sredi-dnevne-sobe/) from the original on January 12, 2022. Retrieved June 9, 2021.
102. Farke, A. A. (2006). "Cranial osteology and phylogenetic relationships of the chasmosaurine ceratopsid *Torosaurus latus*". In Carpenter, K. (ed.). *Horns and Beaks: Ceratopsian and Ornithopod Dinosaurs*. Bloomington: Indiana University Press. pp. 235–257. ISBN 978-0-253-34817-3.
103. "New Analyses Of Dinosaur Growth May Wipe Out One-third Of Species" (https://www.sciencedaily.com/releases/2009/10/091031002314.htm). *Science News*. ScienceDaily.com. October 31, 2009. Archived (https://web.archive.org/web/20190205024226/https://www.sciencedaily.com/releases/2009/10/091031002314.htm) from the original on February 5, 2019. Retrieved November 3, 2009.

104. Farke, Andrew A. (2011). Claessens, Leon (ed.). "Anatomy and taxonomic status of the chasmosaurine ceratopsid *Nedoceratops hatcheri* from the Upper Cretaceous Lance Formation of Wyoming, U.S.A" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3024410). *PLOS ONE*. **6** (1) e16196. Bibcode:2011PLoSO...616196F (https://ui.adsabs.harvard.edu/abs/2011PLoSO...616196F). doi:10.1371/journal.pone.0016196 (https://doi.org/10.1371%2Fjournal.pone.0016196). PMC 3024410 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3024410). PMID 21283763 (https://pubmed.ncbi.nlm.nih.gov/21283763).
105. Longrich, Nicholas R.; Field, Daniel J. (February 29, 2012). "*Torosaurus* Is Not *Triceratops*: Ontogeny in Chasmosaurine Ceratopsids as a Case Study in Dinosaur Taxonomy" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3290593). *PLOS ONE*. **7** (2) e32623. Bibcode:2012PLoSO...732623L (https://ui.adsabs.harvard.edu/abs/2012PLoSO...732623L). doi:10.1371/journal.pone.0032623 (https://doi.org/10.1371%2Fjournal.pone.0032623). ISSN 1932-6203 (https://search.worldcat.org/issn/1932-6203). PMC 3290593 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3290593). PMID 22393425 (https://pubmed.ncbi.nlm.nih.gov/22393425).
106. Bowdler, Neil (March 1, 2012). "Triceratops and Torosaurus dinosaurs 'two species, not one' " (https://www.bbc.co.uk/news/science-environment-17192624). *BBC News*. Archived (https://web.archive.org/web/20130315215330/http://www.bbc.co.uk/news/science-environment-17192624) from the original on March 15, 2013. Retrieved July 29, 2013.
107. Maiorino L., Farke A.A., Kotsakis T., Piras P.,2013, "Is *Torosaurus Triceratops*? Geometric Morphometric Evidence of Late Maastrichtian Ceratopsid Dinosaurs", *PLoS ONE* **8**(11): e81608. doi:10.1371/journal.pone.0081608
108. Hunt, Rebecca K. and Thomas M. Lehman. 2008. Attributes of the ceratopsian dinosaur *Torosaurus*, and new material from the Javelina Formation (Maastrichtian) of Texas. Journal of Paleontology 82(6): 1127–1138.
109. Mallon, Jordan C; Holmes, Robert B; Bamforth, Emily L; Schumann, Dirk (May 7, 2022). "The record of Torosaurus (Ornithischia: Ceratopsidae) in Canada and its taxonomic implications" (https://doi.org/10.1093%2Fzoolinnean%2Fzlab120). *Zoological Journal of the Linnean Society*. **195** (1): 157–171. doi:10.1093/zoolinnean/zlab120 (https://doi.org/10.1093%2Fzoolinnean%2Fzlab120).
110. Scannella, J. B.; Horner, J. R. (2011). Claessens, Leon (ed.). " 'Nedoceratops': An Example of a Transitional Morphology" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3241274). *PLOS ONE*. **6** (12) e28705. Bibcode:2011PLoSO...628705S (https://ui.adsabs.harvard.edu/abs/2011PLoSO...628705S). doi:10.1371/journal.pone.0028705 (https://doi.org/10.1371%2Fjournal.pone.0028705). PMC 3241274 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3241274). PMID 22194891 (https://pubmed.ncbi.nlm.nih.gov/22194891).
111. Longrich, Nicholas R. (2011). "*Titanoceratops ouranous*, a giant horned dinosaur from the Late Campanian of New Mexico". *Cretaceous Research*. **32** (3): 264–276. Bibcode:2011CrRes..32..264L (https://ui.adsabs.harvard.edu/abs/2011CrRes..32..264L). doi:10.1016/j.cretres.2010.12.007 (https://doi.org/10.1016%2Fj.cretres.2010.12.007).
112. Weishampel, D.B.; Dodson, Peter; Osmólska, H. (2004). *The Dinosauria* (Second ed.). Berkeley: University of California Press. p. 861. ISBN 978-0-520-24209-8.
113. Longrich, N. R.; Tokaryk, T.; Field, D. J. (2011). "Mass extinction of birds at the Cretaceous-Paleogene (K-Pg) boundary" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3174646). *Proceedings of the National Academy of Sciences*. **108** (37): 15253–15257. Bibcode:2011PNAS..10815253L (https://ui.adsabs.harvard.edu/abs/2011PNAS..10815253L). doi:10.1073/pnas.1110395108 (https://doi.org/10.1073%2Fpnas.1110395108). PMC 3174646 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3174646). PMID 21914849 (https://pubmed.ncbi.nlm.nih.gov/21914849).

114. Lamanna, M. C.; Sues, H. D.; Schachner, E. R.; Lyson, T. R. (2014). "A New Large-Bodied Oviraptorosaurian Theropod Dinosaur from the Latest Cretaceous of Western North America" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3960162). *PLOS ONE*. **9** (3) e92022. Bibcode:2014PLoSO...992022L (https://ui.adsabs.harvard.edu/abs/2014PLoSO...992022L). doi:10.1371/journal.pone.0092022 (https://doi.org/10.1371%2Fjournal.pone.0092022). PMC 3960162 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3960162). PMID 24647078 (https://pubmed.ncbi.nlm.nih.gov/24647078).
115. Evans, D. C.; Larson, D. W.; Currie, P. J. (2013). "A new dromaeosaurid (Dinosauria: Theropoda) with Asian affinities from the latest Cretaceous of North America". *Naturwissenschaften*. **100** (11): 1041–1049. Bibcode:2013NW....100.1041E (https://ui.adsabs.harvard.edu/abs/2013NW....100.1041E). doi:10.1007/s00114-013-1107-5 (https://doi.org/10.1007%2Fs00114-013-1107-5). PMID 24248432 (https://pubmed.ncbi.nlm.nih.gov/24248432). S2CID 14978813 (https://api.semanticscholar.org/CorpusID:14978813).
116. Zanno, Lindsay E.; Napoli, James G. (2025). "*Nanotyrannus* and *Tyrannosaurus* coexisted at the close of the Cretaceous". *Nature*. **648** (8093): 357–367. doi:10.1038/s41586-025-09801-6 (https://doi.org/10.1038%2Fs41586-025-09801-6). ISSN 0028-0836 (https://search.worldcat.org/issn/0028-0836). PMID 41167514 (https://pubmed.ncbi.nlm.nih.gov/41167514).
117. Triebold, M. (1997). Wolberg, D.; Stump, E.; Rosenberg, G. (eds.). "The Sandy site: Small dinosaurs from the Hell Creek Formation of South Dakota". *Dinofest International: Proceedings of a Symposium*: 245–248.
118. Martin, L. D.; Kurochkin, E. N.; Tokaryk, T. T. (2012). "A new evolutionary lineage of diving birds from the Late Cretaceous of North America and Asia". *Palaeoworld*. **21**: 59–63. doi:10.1016/j.palwor.2012.02.005 (https://doi.org/10.1016%2Fj.palwor.2012.02.005).
119. Ott, C.J.; Larson, P.L. (2010). "A New, Small Ceratopsian Dinosaur from the Latest Cretaceous Hell Creek Formation, Northwest South Dakota, United States: A Preliminary Description". In Ryan, M.J.; Chinnery-Allgeier, B.J.; Eberth, D.A. (eds.). *New Perspectives on Horned Dinosaurs: The Royal Tyrrell Museum Ceratopsian Symposium*. Bloomington: Indiana University Press. p. 656.
120. Campione, N. S. E.; Evans, D. C. (2011). "Cranial Growth and Variation in Edmontosaurs (Dinosauria: Hadrosauridae): Implications for Latest Cretaceous Megaherbivore Diversity in North America" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3182183). *PLOS ONE*. **6** (9) e25186. Bibcode:2011PLoSO...625186C (https://ui.adsabs.harvard.edu/abs/2011PLoSO...625186C). doi:10.1371/journal.pone.0025186 (https://doi.org/10.1371%2Fjournal.pone.0025186). PMC 3182183 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3182183). PMID 21969872 (https://pubmed.ncbi.nlm.nih.gov/21969872).
121. Bakker, R.T.; Sullivan, R.M.; Porter, V.; Larson, P.; Saulsbury, S.J. (2006). Lucas, S.G.; Sullivan, R.M. (eds.). "*Dracorex hogwartsia*, n. gen., n. sp., a spiked, flat-headed pachycephalosaurid dinosaur from the Upper Cretaceous Hell Creek Formation of South Dakota" (http://econtent.unm.edu/cdm/ref/collection/bulletins/id/710). *Late Cretaceous Vertebrates from the Western Interior*. New Mexico Museum of Natural History and Science Bulletin. **35**: 331–345. Archived (https://web.archive.org/web/20181119203737/http://econtent.unm.edu/cdm/ref/collection/bulletins/id/710) from the original on November 19, 2018. Retrieved January 6, 2015.
122. Kielan-Jaworowska, Zofia; Cifelli, Richard L.; Luo, Zhe-Xi (2004). *Mammals from the Age of Dinosaurs: Origins, Evolution, and Structure*. New York: Columbia University Press. pp. 98–99. ISBN 978-0-231-11918-4.
123. Wilson, G. P. (2013). "Mammals across the K/Pg boundary in northeastern Montana, U.S.A.: Dental morphology and body-size patterns reveal extinction selectivity and immigrant-fueled ecospace filling" (http://www.bioone.org/doi/pdf/10.4202/app.00117.2014). *Paleobiology*. **39** (3): 429–469. Bibcode:2013Pbio...39..429W (https://ui.adsabs.harvard.edu/abs/2013Pbio...39..429W). doi:10.1666/12041 (https://doi.org/10.1666%2F12041). S2CID 36025237 (https://api.semanticscholar.org/CorpusID:36025237).

124. Archibald, J. D.; Zhang, Y.; Harper, T.; Cifelli, R. L. (2011). "Protungulatum, Confirmed Cretaceous Occurrence of an Otherwise Paleocene Eutherian (Placental?) Mammal". *Journal of Mammalian Evolution*. **18** (3): 153–161. doi:10.1007/s10914-011-9162-1 (https://doi.org/10.1007%2Fs10914-011-9162-1). S2CID 16724836 (https://api.semanticscholar.org/CorpusID:16724836).
125. Debus, Allen A. (2014). *Prehistoric Monsters: The Real and Imagined Creatures of the Past That We Love to Fear*. McFarland, Incorporated. pp. 9, 187, 189–191. ISBN 978-0-7864-5815-8.
126. Moore, Randy (2014). *Dinosaurs by the Decades: A Chronology of the Dinosaur in Science and Popular Culture*. ABC-CLIO. pp. 98, 248, 275. ISBN 978-0-313-39365-5.
127. State of South Dakota. "Signs and Symbols of South Dakota..." (https://web.archive.org/web/20080220004837/http://www.state.sd.us/state/sdsym.htm) Archived from the original (http://www.state.sd.us/state/sdsym.htm) on February 20, 2008. Retrieved January 20, 2007.
128. State of Wyoming. "State of Wyoming – General Information" (https://web.archive.org/web/20070210015835/http://wyoming.gov/general/general.asp). Archived from the original (http://wyoming.gov/general/general.asp) on February 10, 2007. Retrieved January 20, 2007.
129. Bakker, R. T. (1986). *The Dinosaur Heresies*. New York: Kensington Publishing. p. 240. On that page, Bakker has his own *T. rex*/*Triceratops* fight.

External links

- LiveScience: Facts about *Triceratops* (http://www.livescience.com/24011-triceratops-facts.html) at LiveScience.com
- Dinosaur Mailing List post on *Triceratops* stance (http://dml.cmnh.org/2002Jul/msg00898.html) Archived (https://web.archive.org/web/20071013213317/http://dml.cmnh.org/2002Jul/msg00898.html) October 13, 2007, at the Wayback Machine
- *Triceratops* For Kids (http://www.enchantedlearning.com/subjects/dinosaurs/dinos/Triceratops.shtml) (a fact sheet about the *Triceratops* with activities for kids)

Retrieved from "https://en.wikipedia.org/w/index.php?title=Triceratops&oldid=1376767985"

WIKIPEDIA
25 years of the free encyclopedia

Hadrosauridae

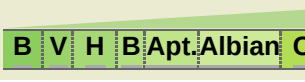
Hadrosauridae (from Ancient Greek ἁδρός *(hadrós)* 'stout, thick' and σαύρα *(saúra)* 'lizard'), is a family of ornithischian dinosaurs informally referred to as **duck-billed dinosaurs**. This group is known as the duck-billed dinosaurs for the flat duck-bill appearance of the bones in their snouts. The ornithopod family, which includes genera such as *Edmontosaurus* and *Parasaurolophus*, was a common group of herbivores during the Late Cretaceous Period.[1] Hadrosaurids are descendants of the Late Jurassic/Early Cretaceous iguanodontian dinosaurs and had a similar body layout. Hadrosaurs were among the most dominant herbivores during the Late Cretaceous in Asia and North America, and during the close of the Cretaceous several lineages dispersed into Europe, Africa, and South America.

Like other ornithischians, hadrosaurids had a predentary bone and a pubic bone which was positioned backwards in the pelvis. Unlike more primitive iguanodonts, the teeth of hadrosaurids are stacked into complex structures known as dental batteries, which acted as effective grinding surfaces. Hadrosauridae is divided into two principal subfamilies: the lambeosaurines (Lambeosaurinae), which had hollow cranial crests or tubes; and the saurolophines (Saurolophinae), identified as hadrosaurines (Hadrosaurinae) in most pre-2010 works, which lacked hollow cranial crests (solid crests were present in some forms). Saurolophines tended to be bulkier than lambeosaurines. Lambeosaurines included the aralosaurins, tsintaosaurins, lambeosaurins and parasaurolophins, while saurolophines included the brachylophosaurins, kritosaurins, saurolophins and edmontosaurins.

Hadrosaurids were facultative bipeds, with the young of some species walking mostly on two legs and the adults walking mostly on four.[2][3]

Hadrosaurids

Temporal range: Late Cretaceous,



Four hadrosaurs. Clockwise from top left: *Maiasaura* (front), *Gryposaurus* (back), *Edmontosaurus* (all saurolophines), *Parasaurolophus*, and *Hypacrosaurus* (both lambeosaurs)

Scientific classification

Kingdom:	Animalia
Phylum:	Chordata
Class:	Reptilia
Clade:	Dinosauria
Clade:	†Ornithischia
Clade:	†Ornithopoda
Clade:	†Hadrosauromorpha
Family:	†Hadrosauridae
	Cope, 1869

Type species

†*Hadrosaurus foulkii*

Leidy, 1858

Subgroups

History of discovery

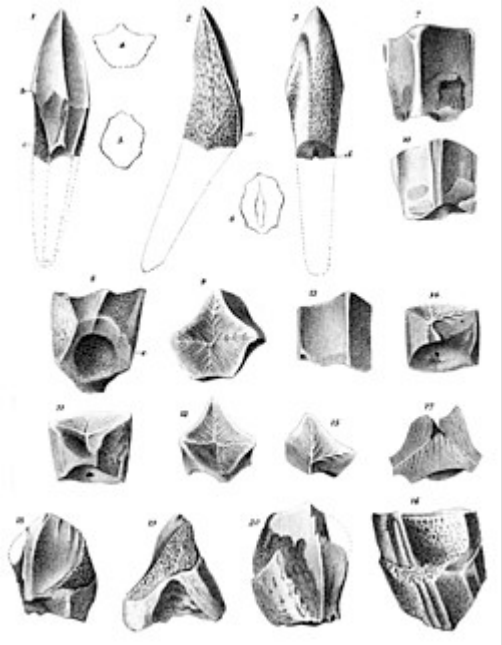


Illustration of *Trachodon mirabilis* teeth

Ferdinand Vandeveer Hayden, during expeditions near the Judith River in 1854 through 1856, discovered the very first dinosaur fossils recognized from North America. These specimens were obtained by Joseph Leidy, who described and named them in 1856; two of the several species named were *Trachodon mirabilis* of the Judith River Formation and *Thespesius occidentalis* of the "Great Lignite Formation". The former was based on a collection of teeth whilst the latter on two caudal centra and a phalanx. Although most of the *Trachodon* teeth turned out to belong to ceratopsids, the holotype and remains of *T. occidentalis* would come to be recognized as the first recognized hadrosaur specimens. Around the same time in Philadelphia, on the other side of the continent, geologist William Parker Foulke was informed of numerous large bones accidentally uncovered by farmer John E. Hopkins some twenty years earlier. Foulke obtained permission to investigate the now scattered fossils in 1858, and these specimens as well were given to Leidy. They were described in the same year as *Hadrosaurus foulkii*, giving a slightly better picture of the form of a hadrosaur. Leidy provided additional description in a 1865 paper.[4] Among his 1858 work Leidy briefly suggested that the animal was likely amphibious in nature; this school of thought about hadrosaurs would come to be dominant for over a century to come.[5]

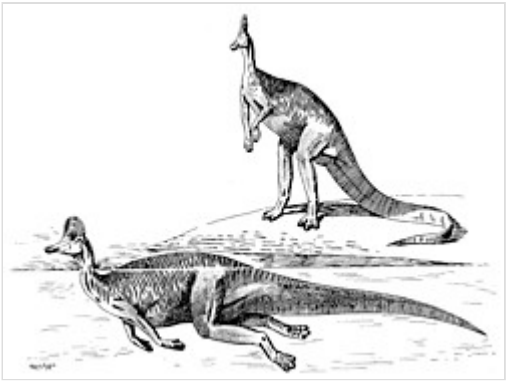
Further discoveries such as "*Hadrosaurus minor*" and "*Ornithotarsus immanis*" would come from the East, and Edward Drinker Cope led an expedition to the Judith River Formation where *Trachodon* was found. Upon the fragments discovered he named seven new species in two genera, as well as assigning material to *Hadrosaurus*.[4] Cope had studied the jaws of hadrosaurs and come to the conclusion that the teeth were fragile and could have been dislodged incredibly easily. As such, he supposed the animals

- †*Aquilarhinus*
- †*Eotrachodon*
- †*Fylax*?
- †*Gobihadros*?
- †*Hadrosaurus*
- †*Lapampasaurus*
- †*Latirhinus*
- †*Lophorhothon*?
- †*Malefica*
- †*Penelopognathus*?
- †*Plesiohadros*?
- †*Sirindhorna*
- †*Yamatosaurus*
- †**Euhadrosauria/Saurolophidae**
  - †Lambeosaurinae
  - †Saurolophinae

Synonyms

- **Trachodontidae** (Lydekker, 1888)
- **Saurolophidae** (Brown, 1914)
- **Lambeosauridae** (Parks, 1923)
- **Cheneosauridae** (Lull & Wright, 1942)
- **Ornithotarsidae** (Cope, 1871)

must have fed largely on soft water plants; he presented this idea to the Philadelphia Academy in 1883, and this idea would come to be very influential on future study.[4][5] Research would continue in the Judith River area for years to come, but the formation never yielded much more than fragmentary remains, and Cope's species as well as *Trachodon* itself would in time be seen as of doubtful validity. The Eastern states, too, would never yield particularly informative specimens. Instead, other sites in the American West would come to provide many very complete specimens that would form the backbone of hadrosaur research. One such specimen was the very complete AMNH 5060 (belonging to *Edmontosaurus annectens*), recovered in 1908 by the fossil collector Charles Hazelius Sternberg and his three sons in Converse County, Wyoming. It was described by Henry Osborn in 1912, who dubbed it the "Dinosaur mummy". This specimen's skin was almost completely preserved in the form of impressions. The skin around its hands, thought to represent webbing, was seen as further bolstering the idea that hadrosaurs were very aquatic animals.[4]



From the mid 19th century through much of the 20th century, hadrosaurs were considered aquatic animals which subsisted on soft water plants

Cope had planned to write a monograph about the group Ornithopoda, but never made much progress towards it before his death. This unrealized endeavor would come to be the inspiration for Richard Swann Lull and Nelda Wright to work on a similar project decades later. Eventually they realized the whole of Ornithopoda was too broad of a scope, until eventually it was narrowed down to specifically North American hadrosaurs. Their monograph, *Hadrosaurian Dinosaurs of North America*, was published in 1942, and looked back at the whole of understanding about the family. It was designed as a definitive work, covering all aspects of their biology and evolution, and as part of it every known species was re-evaluated and many of them redescribed. They agreed with prior authors on the semi-aquatic nature of hadrosaurs, but re-evaluated Cope's idea of weak jaws and found quite the opposite. The teeth were rooted in strong batteries and would be continuously replaced to prevent them getting worn down. Such a system seemed incredibly overbuilt for the job of eating soft Mesozoic plants, and this fact confused the authors. Though they still proposed a diet of water plants, they considered it likely this would be supplemented by occasional forrays into browsing on land plants.[4]

Skeleton of *Maiasaura* posed with a nest; the naming of this genus was one of numerous important developments in the dinosaur renaissance

Twenty years later, in 1964, another very important work would be published, this time by John H. Ostrom. It challenged the idea that hadrosaurs were semi-aquatic animals, which had been held since the work of Leidy back in the 1850s. This new approach was backed using evidence of the environment and climate they lived in, co-existing flora and fauna, physical anatomy, and preserved stomach contents from mummies. Based on evaluation of all this data, Ostrom found the idea that hadrosaurs were adapted for aquatic life incredibly lacking, and instead proposed they were capable terrestrial animals that browsed on plants such as conifers. He remained uncertain, however, as to the purpose of the paddle-like hand Osborn had described, as well as their long and somewhat paddle-like tails. Thus he agreed with the idea that

hadrosaurs would have taken refuge from predators in water.[5] Numerous important studies would follow this; Ostrom's student Peter Dodson published a paper about lambeosaur skull anatomy that made enormous changes to hadrosaur taxonomy in 1975, and Michael K. Brett-Surman conducted a full revision of the group as part of his Graduate studies through the 1970s and 1980s. John R. Horner would also begin to leave his impact on the field, including with the naming of *Maiasaura* in 1979.[6][7][8][9][10][11]

Hadrosaur research experienced a surge in the decade of the 2000s, similar to the research of other dinosaurs. In response to this, the Royal Ontario Museum and the Royal Tyrrell Museum collaborated to arrange the International Hadrosaur Symposium, a professional meeting about ongoing hadrosaur research that was held at the latter institution on September 22 and 23 in 2011. Over fifty presentations were made at the event, thirty-six of which were later incorporated into a book, titled *Hadrosaurs*, published in 2015. The volume was brought together primarily by palaeontologists David A. Eberth and David C. Evans, and featured an afterword from John R. Horner, all of whom also contributed to one or more of the studies published therein.[12] The first chapter of the volume was a study by David B. Weishampel about the rate of ornithopod research over history, and the interest in different aspects of it over that history, using the 2004 volume *The Dinosauria* as the source of data on the amount of works published in each decade. Various periods of high and low activity were found, but the twenty-first century was found to overwhelmingly be the most prolific time, with over two-hundred papers published. The advent of the internet was cited as a likely catalyst for this boom. Hadrosaur research experienced high levels of diversity within the decade, with previously uncommon subjects such as growth, phylogeny, and biogeography experiencing more attention, though the functional morphology of hadrosaurids was found to have declined in study since the dinosaur renaissance.[13]

Distribution

Hadrosaurids likely originated in North America, before shortly dispersing into Asia. During the late Campanian-Maastrichtian, a saurolophine hadrosaurid migrated into South America from North America, giving rise to the clade Austrokritosauria, which is closely related to the tribe Kritosaurini.[14] During the late early Maastrichtian, several lineages of Lambeosaurinae from Asia migrated into the European Ibero-Armorican Island (what is now France and Spain), including Arenysaurini and Tsintaosaurini.[15] One of these lineages later dispersed from Europe into North Africa, as evidenced by *Ajnabia*, a member of Arenysaurini.[16]

Map of various hadrosaur taxa across North America

Classification

The family Hadrosauridae was first used by Edward Drinker Cope in 1869, then containing only *Hadrosaurus*.[17] Since its creation, a major division has been recognized in the group between the hollow-crested subfamily Lambeosaurinae and the subfamily Saurolophinae, historically known as Hadrosaurinae. Both of these have been robustly

supported in all recent literature. Phylogenetic analysis has increased the resolution of hadrosaurid relationships considerably, leading to the widespread usage of tribes (a taxonomic unit below subfamily) to describe the finer relationships within each group of hadrosaurids.[18]

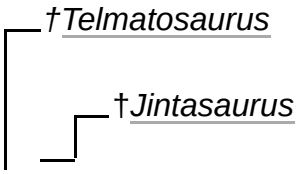
Lambeosaurines have also been traditionally split into Parasaurolophini and Lambeosaurini.[19] These terms entered the formal literature in Evans and Reisz's 2007 redescription of *Lambeosaurus magnicristatus*. Lambeosaurini is defined as all taxa more closely related *Lambeosaurus lambei* than to *Parasaurolophus walkeri*, and Parasaurolophini as all those taxa closer to *P. walkeri* than to *L. lambei*. In recent years Tsintaosaurini and Aralosaurini have also emerged.[20]

The use of the term *Hadrosaurinae* was questioned in a comprehensive study of hadrosaurid relationships by Albert Prieto-Márquez in 2010. Prieto-Márquez noted that, though the name Hadrosaurinae had been used for the clade of mostly crestless hadrosaurids by nearly all previous studies, its type species, *Hadrosaurus foulkii*, has almost always been excluded from the clade that bears its name, in violation of the rules for naming animals set out by the ICZN. Prieto-Márquez defined Hadrosaurinae as just the lineage containing *H. foulkii*, and used the name Saurolophinae instead for the traditional grouping.[18]

Phylogeny

Hadrosauridae was first defined as a clade, by Forster, in a 1997 abstract, as simply "Lambeosaurinae plus Hadrosaurinae and their most recent common ancestor". In 1998, Paul Sereno defined the clade Hadrosauridae as the most inclusive possible group containing *Saurolophus* (a well-known saurolophine) and *Parasaurolophus* (a well-known lambeosaurine), later emending the definition to include *Hadrosaurus*, the type genus of the family. According to Horner et al. (2004), Sereno's definition would place a few other well-known hadrosaurs (such as *Telmatosaurus* and *Bactrosaurus*) outside the family, which led them to define the family to include *Telmatosaurus* by default. Prieto-Marquez reviewed the phylogeny of Hadrosauridae in 2010, including many taxa potentially within the family.[18] The family is now formally defined in the *PhyloCode* as "the smallest clade containing *Hadrosaurus foulkii*, *Lambeosaurus lambei*, and *Saurolophus osborni*".[21] The two main subfamilies of Lambeosaurinae and Saurolophinae belong to the clade **Euhadrosauria** (sometimes called **Saurolophidae**), defined as "the smallest clade containing *Lambeosaurus lambei* and *Saurolophus osborni*, provided it does not include *Hadrosaurus foulkii*".[21] This clade excludes basal hadrosaurids such as *Hadrosaurus* and *Yamatosaurus* but self-destructs if *Hadrosaurus* is descended from the last common ancestor of *Lambeosaurus* and *Saurolophus*.[21]

Below is a cladogram from Prieto-Marquez et al. 2016. This cladogram is a recent modification of the original 2010 analysis, including more characters and taxa. The resulting cladistic tree of their analysis was resolved using Maximum-Parsimony. 61 hadrosauroid species were included, characterized for 273 morphological features: 189 for cranial features and 84 for postcranial features. When characters had multiple states that formed an evolutionary scheme, they were ordered to account for the evolution of one state into the next. The final tree was run through TNT version 1.0.[22]



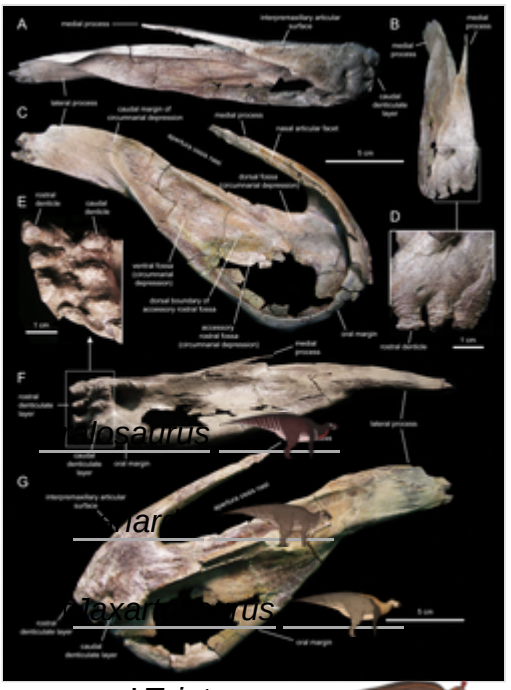
†*Lophorhothon*

†*Claosaurus*

†*Tethyshadros*

†*Hadrosaurus*

†*Eotrachodon*



Premaxillae of †*Eotrachodon*, the taxon named by Prieto-Marquez et al. 2016

†*Tsintaosaurus*

†*Pararhabdodon*

†*Charonosaurus*

†*Parasaurolophus*

†*Lambeosaurus*

†*Corythosaurus*

†*Hypacrosaurus ste*

†Lambeosaurinae

†*Magnapaulia*

Skull of †*Lambeosaurus*, the type taxon of Lambeosaurinae

†*Velafrons*

†*Sahaliyania*

†*Hypa*

†*Olorotitan*

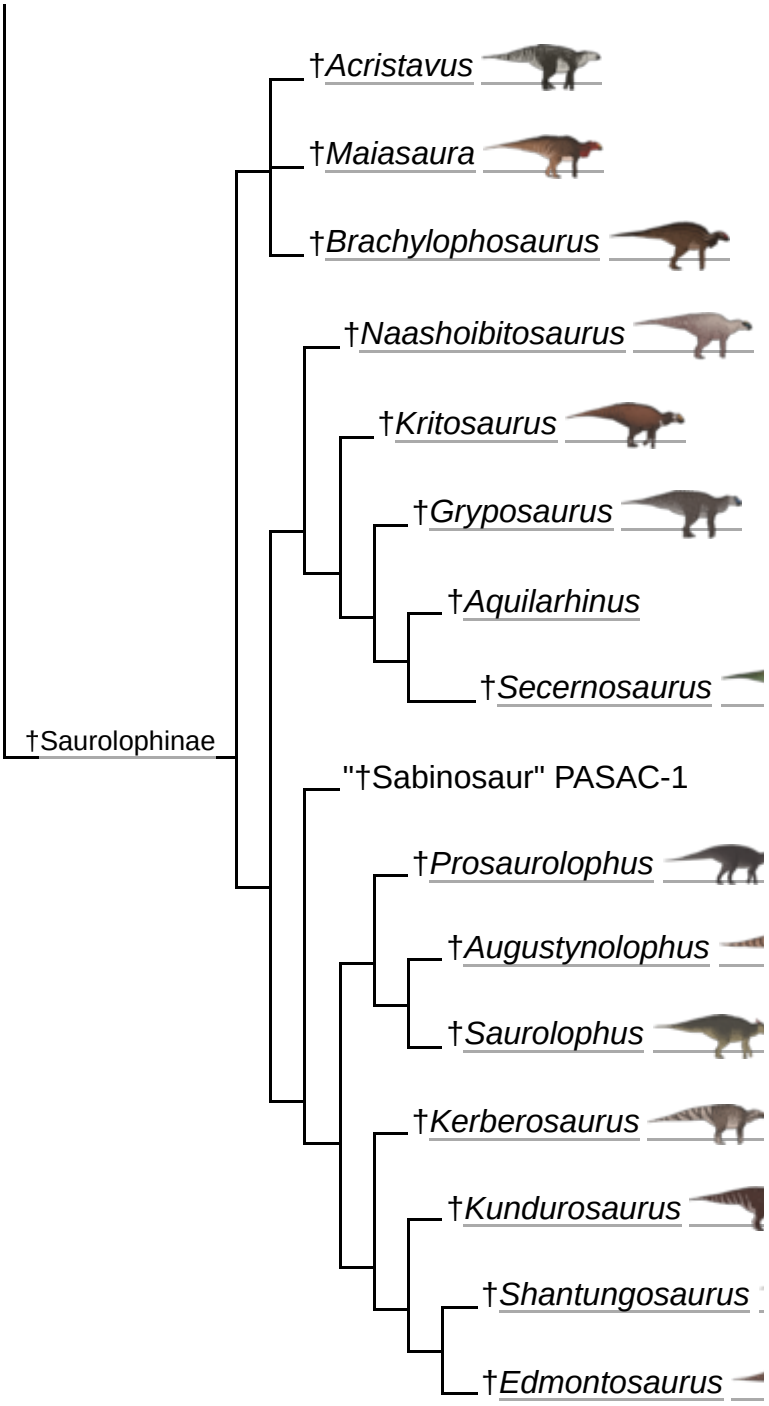
†*Blasisaurus*

Skull of †*Saurolophus*, the type taxon of Saurolophinae

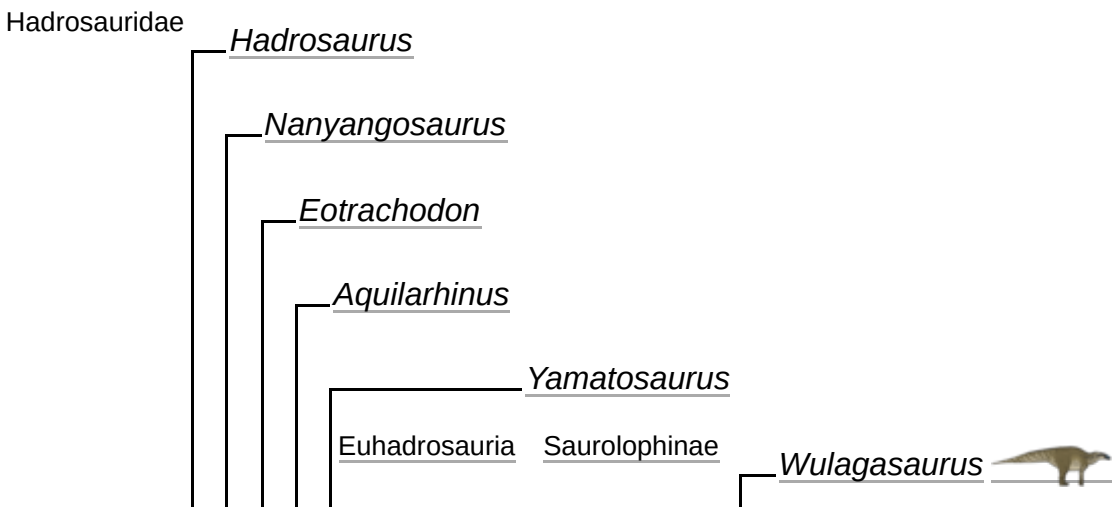
†*Am*

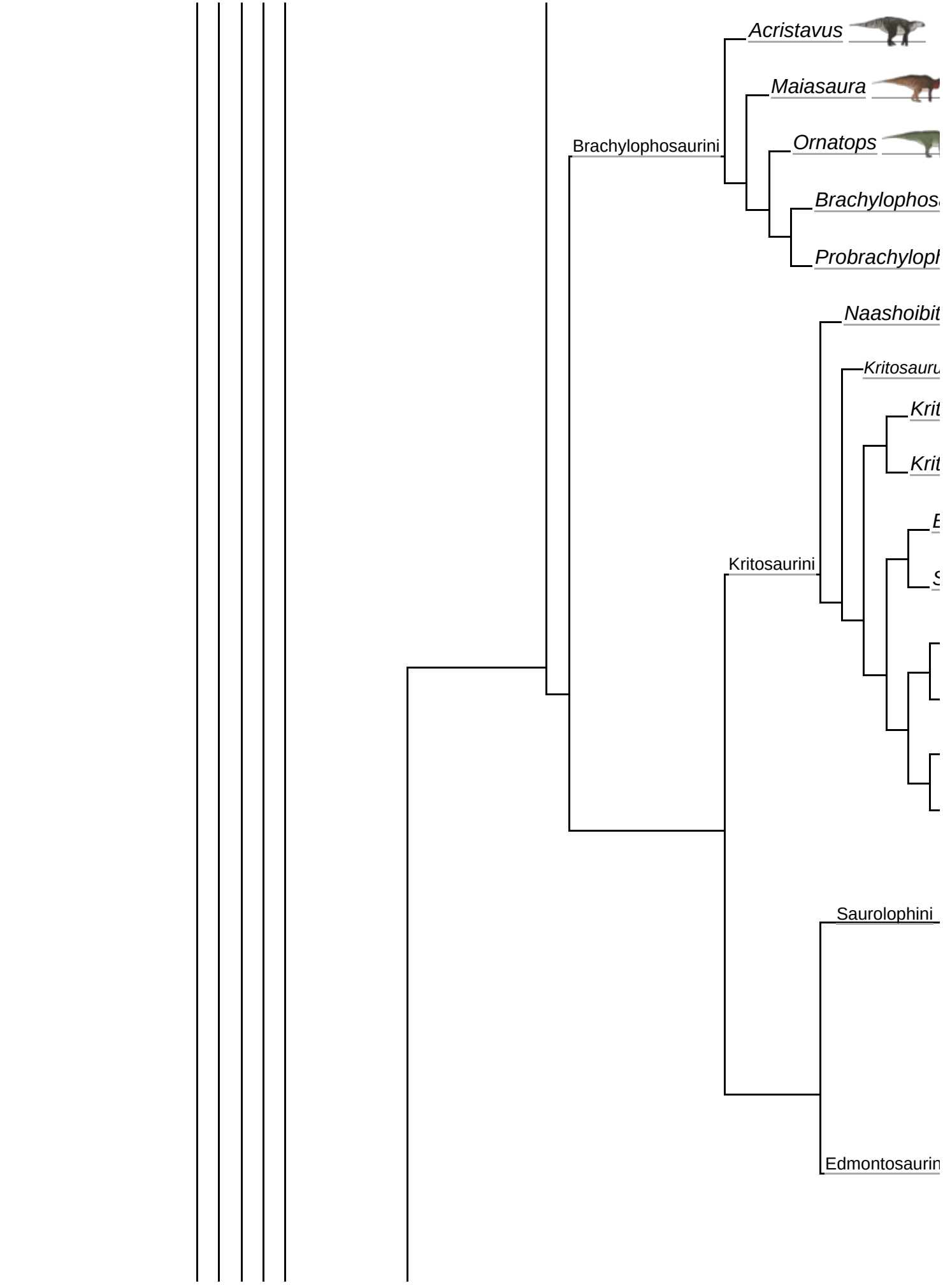
†**Hadrosauridae**

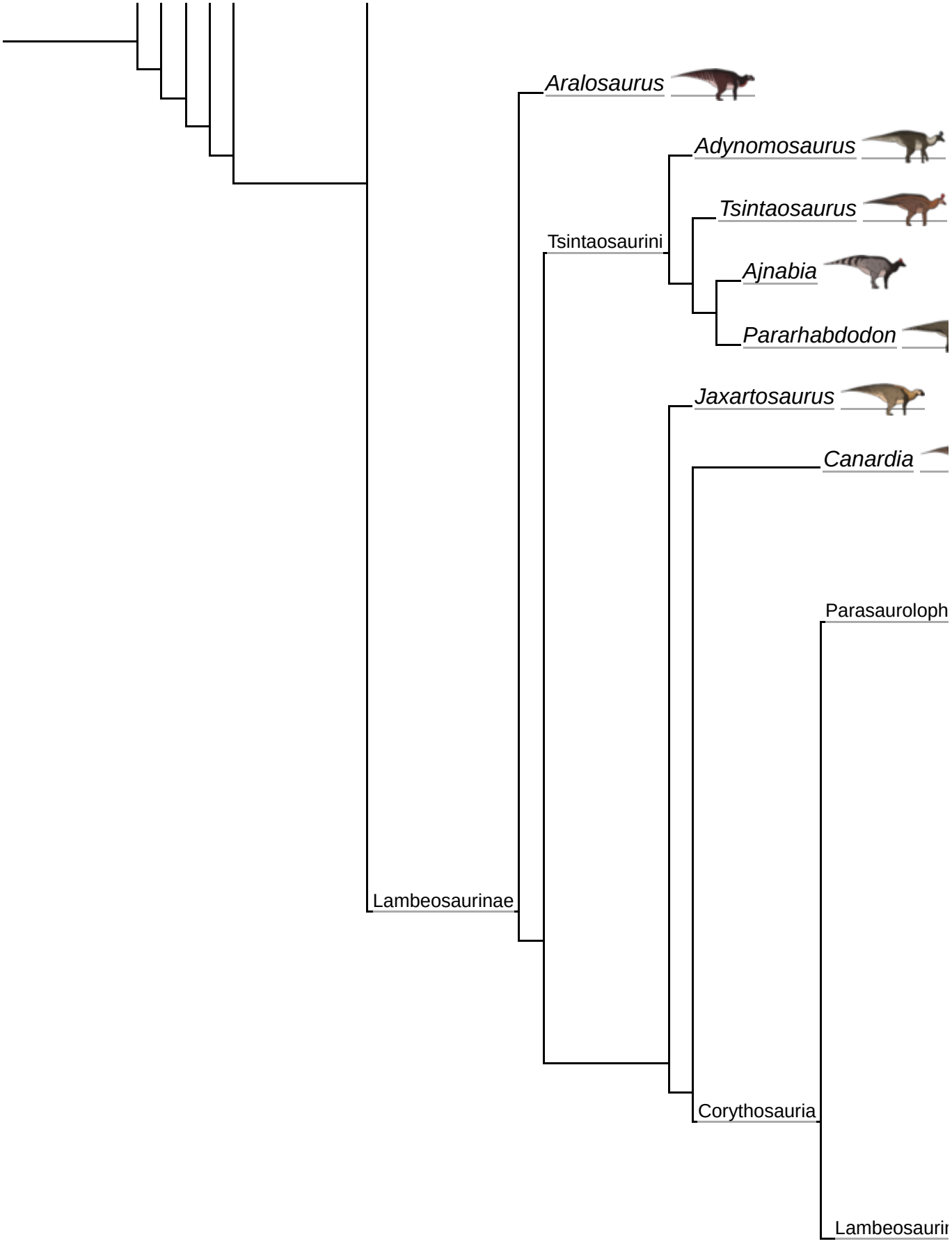
†Saurolophidae



The following cladogram is from Ramírez-Velasco (2022), including most recently named taxa.[23]




Acristavus
Maiasaura
Ornatops
Brachylophosaurini
Brachylophos
Probrachylopl
Naashoibit
Kritosauru
Krit
Krit
Kritosaurini
Saurolophini
Edmontosaurin


Aralosaurus
Adynomosaurus
Tsintaosaurus
Tsintaosaurini
Ajnabia
Pararhabdodon
Jaxartosaurus
Canardia
Parasauroloph
Corythosauria
Lambeosaurinae
Lambeosauri

Anatomy

Edmontosaurus skull, Oxford University Museum of Natural History

The most recognizable aspect of hadrosaur anatomy is the flattened and laterally stretched rostral bones, which gives the distinct duck-bill look. Some members of the hadrosaurs also had massive crests on their heads, probably for display and/or to make noises.[18] In some genera, including *Edmontosaurus*, the whole front of the skull was flat and broadened out to form a beak, which was ideal for clipping leaves and twigs from the forests of Asia, Europe and North America. However, the back of the mouth contained thousands of teeth suitable for grinding food before it was swallowed. This has been hypothesized to have been a crucial factor in the success of this group in the Cretaceous compared to the sauropods.

Skin impressions of multiple hadrosaurs have been found.[24] From these impressions, the hadrosaurs were determined to be scaled, and not feathered like some dinosaurs of other groups.

Hadrosaurs, much like sauropods, are noted for having their manus united in a fleshy, often nail-less pad.[25]

The two major divisions of hadrosaurids are differentiated by their cranial ornamentation. While members of the Lambeosaurinae subfamily have hollow crests that differ depending on species, members of the Saurolophinae (Hadrosaurinae) subfamily have solid crests or none at all. Lambeosaurine crests had air chambers that may have produced a distinct sound and meant that their crests could have been used for both an audio and visual display.

Paleobiology

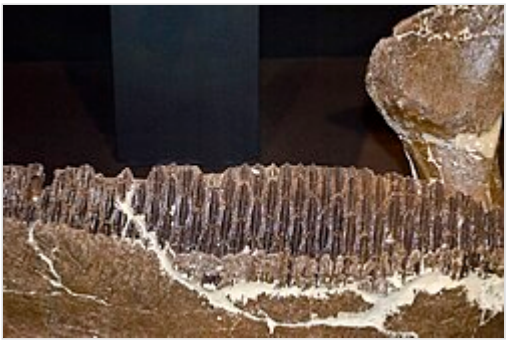
Diet

Early restoration by Charles R. Knight of hadrosaurs as semi-aquatic animals that could only chew soft water plants, a popular idea at the time.

While studying the chewing methods of hadrosaurids in 2009, the paleontologists Vincent Williams, Paul Barrett, and Mark Purnell found that hadrosaurs likely grazed on horsetails and vegetation close to the ground, rather than browsing higher-growing leaves and twigs. This conclusion was based on the evenness of scratches on hadrosaur teeth, which suggested the hadrosaur used the same series of jaw motions over and over again.[26] As a result, the study determined that the hadrosaur diet was probably made of leaves and lacked the bulkier items, such as twigs or stems, that might have required a different chewing method and created different wear patterns.[27] However, Purnell said these conclusions were less secure than the more conclusive evidence regarding the motion of teeth while chewing.[28]

The hypothesis that hadrosaurs were likely grazers rather than browsers appears to contradict previous findings from preserved stomach contents found in the fossilized guts in previous hadrosaur studies.[28] The most recent such finding before the publication of the Purnell study was conducted in 2008, when a team led by University of Colorado at Boulder graduate student Justin S. Tweet found a homogeneous accumulation of millimeter-scale leaf fragments in the gut region of a well-preserved partially grown *Brachylophosaurus*.[29][30] As a result of that finding, Tweet concluded in September 2008 that the animal was likely a browser, not a grazer.[30] In response to such findings, Purnell said that preserved stomach contents are questionable because they do not necessarily represent the usual diet of the animal. The issue remains a subject of debate.[31]

Mallon et al. (2013) examined herbivore coexistence on the island continent of Laramidia, during the Late Cretaceous. It was concluded that hadrosaurids could reach low-growing trees and shrubs that were out of the reach of ceratopsids, ankylosaurs, and other small herbivores. Hadrosaurids were capable of feeding up to a height of 2 m (6 ft 7 in) when standing quadrupedally, and up to a height of 5 m (16 ft) bipedally.[32]



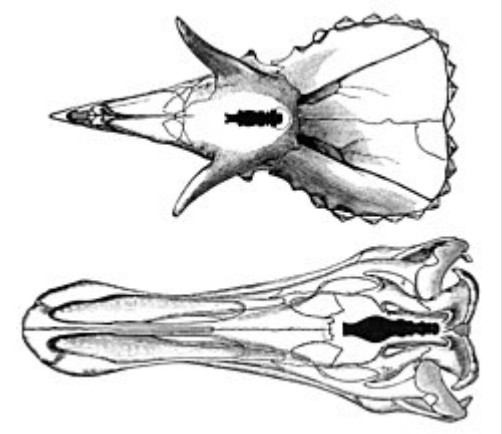
Edmontosaurus dentary with teeth, typical of hadrosauridae

Coprolites (fossilized droppings) of some Late Cretaceous hadrosaurs show that the animals sometimes deliberately ate rotting wood. Wood itself is not nutritious, but decomposing wood would have contained fungi, decomposed wood material and detritus-eating invertebrates, all of which would have been nutritious.[33] Examination of hadrosaur coprolites from the Grand Staircase-Escalante indicates that shellfish such as crustaceans were also an important component of the hadrosaur diet.[34]

It is thought that chewing may have changed throughout life in hadrosaurids. Very young specimens show simple cup shapes occlusion zones, or areas where the teeth contact one another in chewing, whereas in adulthood there is a "dual function" arrangement with two distinct areas of different tooth wear. This change during growth may have helped transition from a diet of softer plants when young to more tough and fibrous ones during adulthood. It is thought the transition between states, characterized by a more

gradual transition from one wear state to another, occurred at different times in the growth of different species; in *Hypacrosaurus stebingeri* it did not occur until a nearly adult stage, whereas in a saurolophine specimen likely less than a year old from the Dinosaur Park Formation the transition had already begun.[35]

Neurology



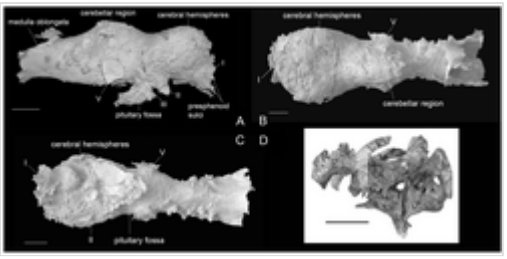
A 1905 diagram showing the small size of an *Edmontosaurus annectens* brain (bottom; alongside that of *Triceratops horridus*, top) commented on in early sources

Hadrosaurs have been noted as having the most complex brains among ornithopods, and indeed among ornithischian dinosaurs as a whole.[36][37][38] The brains of hadrosaurid dinosaurs have been studied as far back at the late 19th century, when Othniel Charles Marsh made an endocast of a specimen then referred to *Claosaurus annectens*; only basic remarks were possible but it was noted that the organ was proportionally small.[39] John Ostrom would give a more informed analysis and review in 1961, pulling on data from *Edmontosaurus regalis*, *E. annectens*, and *Gryposaurus notabilis* (then considered a synonym of *Kritosaurus*). Though still obviously small, Ostrom recognized that the brains may be more significantly developed than expected, but supported the view that dinosaur brains would have only filled some of the endocranial cavity, limiting possibility of analysis.[40] In 1977 James Hopson introduced the use of estimated encephalization quotients to the topic of dinosaur intelligence, finding *Edmontosaurus* to have an EQ of 1.5, above that of other ornithischians including earlier relatives like *Camptosaurus* and *Iguanodon* and similar to that of carnosaurian theropods and modern crocodilians but below that of coelurosaurian theropods. Reasonings suggested for their comparably high intelligence were the need for acute senses in the lack of defensive weapons, and more complex intraspecific behaviours as indicated by their acoustic and visual display structures.[36]

The advent of CT scanning for use in palaeontology has allowed for more widespread application of this without the need for specimen destruction. Modern research using these methods has focused largely on hadrosaurs. In a 2009 study by palaeontologist David C. Evans and colleagues, the brains of lambeosaurine hadrosaur genera *Hypacrosaurus* (adult specimen ROM 702), *Corythosaurus* (juvenile specimen ROM 759 and subadult specimen CMN 34825), and *Lambeosaurus* (juvenile specimen ROM 758) were scanned and compared to each other (on a phylogenetic and ontogenetic level), related taxa, and previous predictions, the first such large-scale look into the neurology of the subfamily. Contra the early works, Evans' studies indicate that only some regions of the hadrosaur brain (the dorsal portion and much of the hindbrain) were loosely correlated to the brain wall, like modern reptiles, with the ventral and lateral regions correlating fairly closely. Also unlike modern reptiles, the brains of the juveniles did not seem to correlate any closer to the brain wall than those of adults. It was cautioned, however, that very young individuals were not included in the study.[37]

As with previous studies, EQ values were investigated, although a wider number range was given to account for uncertainty in brain and body mass. The range for the adult *Hypacrosaurus* was 2.3 to 3.7; the lowest end of this range was still higher than modern reptiles and most non-maniraptoran dinosaurs (nearly all having EQs below two), but fell well short of maniraptorans themselves, which had quotients

higher than four. The size of the cerebral hemispheres was, for the first time, remarked upon. It was found to taking up around 43% of endocranial volume (not considering olfactory bulbs) in ROM 702. This is comparable to their size in saurolophine hadrosaurs, but far larger than in any ornithischians outside of Hadrosauriformes, and all large saurischian dinosaurs; maniraptors *Conchoraptor* and *Archaeopteryx*, an early bird, had very similar proportions. This lends further support to the idea of complex behaviours and relatively high intelligence, for non-avian dinosaurs, in hadrosaurids.[37]



Endocast of an *Amurosaurus* brain in right lateral (A), dorsal (B), and ventral (C) views

Amurosaurus, a close relative of the taxa from the 2009 study, was the subject of a 2013 paper once again looking into a cranial endocast. A nearly identical EQ range of 2.3 to 3.8 was found, and it was again noted this was higher than that of living reptiles, sauropods and other ornithischians, but different EQ estimates for theropods were cited, placing the hadrosaur numbers significantly below even more basal theropods like *Ceratosaurus* (with an EQ range of 3.31 to 5.07) and *Allosaurus* (with a range of 2.4 to 5.24, compared to only 1.6 in the 2009 study);[37][38] more bird-like coelurosaurians theropods such as *Troodon* had stated EQs higher than seven. Additionally, the relative cerebral volume was only 30% in *Amurosaurus*, significantly lower than in *Hypacrosaurus*, closer to that of theropods like *Tyrannosaurus* (with 33%), though still distinctly larger than previously estimated numbers for more primitive iguanodonts like *Lurdusaurus* and *Iguanodon* (both at 19%). This demonstrated a previously unrecognized level of variation in neuro-anatomy within Hadrosauridae.[38]

Reproduction

Eggs of the species *Hypacrosaurus stebingeri*

Neonate sized hadrosaur fossils have been documented in the scientific literature.[41] Tiny hadrosaur footprints have been discovered in the Blackhawk Formation of Utah.[41]

In a 2001 review of hadrosaur eggshell and hatchling material from Alberta's Dinosaur Park Formation, Darren Tanke and M. K. Brett-Surman concluded that hadrosaurs nested in both the ancient upland and lowlands of the formation's depositional environment. The upland nesting grounds may have been preferred by the less common hadrosaurs, like *Brachylophosaurus* and *Parasaurolophus*. However, the authors were unable to determine what specific factors shaped nesting ground choice in the formation's hadrosaurs. They suggested that behavior, diet, soil condition, and competition between dinosaur species all potentially influenced where hadrosaurs nested.[41]

Sub-centimeter fragments of pebbly-textured hadrosaur eggshell have been reported from the Dinosaur Park Formation. This eggshell is similar to the hadrosaur eggshell of Devil's Coulee in southern Alberta as well as that of the Two Medicine and Judith River Formations in Montana, United States. While present, dinosaur eggshell is very rare in the Dinosaur Park Formation and is only found in two different microfossil sites. These sites are distinguished by large numbers of pisidiid clams and other less common shelled invertebrates, like unionid clams and snails. This association is not a coincidence, as the

invertebrate shells would have slowly dissolved and released enough basic calcium carbonate to protect the eggshells from naturally occurring acids that otherwise would have dissolved them and prevented fossilization.[41]

In contrast with eggshell fossils, the remains of very young hadrosaurs are somewhat common. Tanke has observed that an experienced collector could discover multiple juvenile hadrosaur specimens in a single day. The most common remains of young hadrosaurs in the Dinosaur Park Formation are dentaries, bones from limbs and feet, as well as vertebral centra. The material showed little or none of the abrasion that would have resulted from transport, meaning the fossils were buried near their point of origin. Bonebeds 23, 28, 47, and 50 are productive sources of young hadrosaur remains in the formation, especially bonebed 50. The bones of juvenile hadrosaurs and fossil eggshell fragments are not known to have been preserved in association with each other, despite both being present in the formation.[41]

Growth and development

The limbs of the juvenile hadrosaurs are anatomically and proportionally similar to those of adult animals.[41] However, the joints often show "predepositional erosion or concave articular surfaces",[41] which was probably due to the cartilaginous cap covering the ends of the bones.[41] The pelvis of a young hadrosaur was similar to that of an older individual.[41]

Juvenile specimen of the genus *Maiasaura*

The young of some hadrosaurs, namely *Maiasaura*, are suggested to have walked on only their two hind legs, while adults would have walked on all four.[2] As *Maiasaura* aged, the front limbs became more robust in order to take on weight, while the back legs became less robust as they transitioned to walking on all four legs.[2] Bone modeling also seems to support a locomotion shift in *Maiasaura*; outgrowths on the tibiae reinforce the bone in the direction of stresses that would be caused by bipedalism (in a one-year-old individual) and quadrupedalism (in an individual at least four years of age).[42] In contrast, *Edmontosaurus* nestlings appear to have been capable of fully quadrupedal locomotion, and do not show much change in limb proportions through growth.[43] Quadrupedal tracks of a small, presumably juvenile hadrosaur are known from the Cantwell Formation in Alaska.[3]

Daily activity patterns

Comparisons between the scleral rings of several hadrosaur genera (*Corythosaurus*, *Prosaurolophus*, and *Saurolophus*) and modern birds and reptiles suggest that they may have been cathemeral, active throughout the day at short intervals.[44]

Pathology

Spondyloarthropathy has been documented in the spine of a 78-million year old hadrosaurid. Other examples of pathologies in hadrosaurs include healed wounds from predators, such as those found in *Edmontosaurus annectens*, and tumors such as Langerhans cell histiocytosis, hemangiomas, desmoplastic

fibroma, metastatic cancer, and osteoblastomas, found in genera such as *Brachylophosaurus* and *Edmontosaurus*.[45][46] Osteochondrosis is also commonly found in hadrosaurs.[47]

References

1. Case, Judd A.; Martin, James E.; Chaney, Dan S.; Regurero, Marcelo; Marenssi, Sergio A.; Santillana, Sergio M.; Woodburne, Michael O. (25 September 2000). "The first duck-billed dinosaur (family Hadrosauridae) from Antarctica". *Journal of Vertebrate Paleontology*. **20** (3): 612–614. doi:10.1671/0272-4634(2000)020[0612:tfdbdf]2.0.co;2 (https://doi.org/10.1671%2F0272-4634%282000%29020%5B0612%3Atfdbdf%5D2.0.co%3B2). hdl:11336/105444 (https://hdl.handle.net/11336%2F105444). JSTOR 4524132 (https://www.jstor.org/stable/4524132). S2CID 131243139 (https://api.semanticscholar.org/CorpusID:131243139).
2. Dilkes, David W. (2001). "An ontogenetic perspective on locomotion in the Late Cretaceous dinosaur Maiasaura peeblesorum (Ornithischia: Hadrosauridae)". *Canadian Journal of Earth Sciences*. **38** (8): 1205–1227. Bibcode:2001CaJES..38.1205D (https://ui.adsabs.harvard.edu/abs/2001CaJES..38.1205D). doi:10.1139/e01-016 (https://doi.org/10.1139%2Fe01-016).
3. Fiorillo, A.R.; Tykoski, R.S. (2016). "Small hadrosaur manus and pes tracks from the Lower Cantwell Formation (Upper Cretaceous) Denali National Park, Alaska: implications for locomotion in juvenile hadrosaurs" (http://palaios.geoscienceworld.org/content/31/10/479). *PALAIOS*. **31** (10): 479–482. Bibcode:2016Palai..31..479F (https://ui.adsabs.harvard.edu/abs/2016Palai..31..479F). doi:10.2110/palo.2016.049 (https://doi.org/10.2110%2Fpalo.2016.049). S2CID 132975678 (https://api.semanticscholar.org/CorpusID:132975678). Archived (https://web.archive.org/web/20170112201526/http://palaios.geoscienceworld.org/content/31/10/479) from the original on 2017-01-12. Retrieved 2016-10-31.
4. Lull, Richard Swann; and Wright, Nelda E. (1942). *Hadrosaurian Dinosaurs of North America*
5. Ostrom, John H. (1964). "A reconsideration of the paleoecology of the hadrosaurian dinosaurs" (https://doi.org/10.2475%2Fajs.262.8.975). *American Journal of Science*. **262** (8): 975–997. Bibcode:1964AmJS..262..975O (https://ui.adsabs.harvard.edu/abs/1964AmJS..262..975O). doi:10.2475/ajs.262.8.975 (https://doi.org/10.2475%2Fajs.262.8.975).
6. Brett-Surman, Michael K. (1975). *The appendicular anatomy of hadrosaurian dinosaurs*. M.A. thesis. Berkeley: University of California.
7. Brett-Surman, Michael K. (1979). "Phylogeny and paleobiogeography of hadrosaurian dinosaurs". *Nature*. **277** (5697): 560–562. Bibcode:1979Natur.277..560B (https://ui.adsabs.harvard.edu/abs/1979Natur.277..560B). doi:10.1038/277560a0 (https://doi.org/10.1038%2F277560a0). S2CID 4332144 (https://api.semanticscholar.org/CorpusID:4332144).
8. Brett-Surman, Michael K. (1989). *A revision of the Hadrosauridae (Reptilia: Ornithischia) and their evolution during the Campanian and Maastrichtian*. Ph.D. dissertation. Washington, D.C.: George Washington University.
9. Horner, J.R.; Makela, R. (1979). "Nest of juveniles provides evidence of family structure among dinosaurs". *Nature*. **282** (5736): 296–298. Bibcode:1979Natur.282..296H (https://ui.adsabs.harvard.edu/abs/1979Natur.282..296H). doi:10.1038/282296a0 (https://doi.org/10.1038%2F282296a0). S2CID 4370793 (https://api.semanticscholar.org/CorpusID:4370793).

10. Fuentes Buxó, R. 2012. Anàlisi de restes paleontològiques del Maastrichtià superior (Cretaci superior) procedents d'un nou jaciment situat en Les Serretes (Vilamitjana, Conca de Tremp, Espanya). Treball de fi de Master de Paleontologia (UAB-UB-ICP). 129 pp. (https://www.academia.edu/10944379/An%C3%A0lisi_de_restes_paleontol%C3%B2giques_del_Maastrichti%C3%A0_superior_Cretaci_superior_procedents_d_un_nou_jaciment_situat_en_Les_Serretes_Vilamitjana_Conca_de_Tremp_Espanya_) Archived (https://web.archive.org/web/20231129154222/https://www.academia.edu/10944379/An%C3%A0lisi_de_restes_paleontol%C3%B2giques_del_Maastrichti%C3%A0_superior_Cretaci_superior_procedents_d_un_nou_jaciment_situat_en_Les_Serretes_Vilamitjana_Conca_de_Tremp_Espanya_) 2023-11-29 at the Wayback Machine
11. 2014, Fuentes-Buxó, R. & Dalla Vecchia, F.M. 2014. Les Serretes, a new Late Maastrichtian dinosaur site in the eastern Tremp Syncline (Southern Pyrenees, Spain). In: Paleontologia i Evolució. Reconstructing the Terrestrial end-cretaceous paleoenvironments in Europe. pp (https://www.academia.edu/10410968/Les_Serretes_a_new_Late_Maastrichtian_dinosaur_site_in_the_eastern_Tremp_Syncline_Southern_Pyrenees_Spain_) Archived (https://web.archive.org/web/20240121013038/https://www.academia.edu/10410968/Les_Serretes_a_new_Late_Maastrichtian_dinosaur_site_in_the_eastern_Tremp_Syncline_Southern_Pyrenees_Spain_) 2024-01-21 at the Wayback Machine
12. Eberth, David A.; Evans, David C. (2015). "Preface". In Eberth, David A.; Evans, David C. (eds.). *Hadrosaurs*. Bloomington, Indiana: Indiana University Press. pp. xiii–xiv. ISBN 978-0-253-01385-9.
13. Weishampel, David B. (2015). "A history of the study of ornithopods: Where have we been? Where are we now? Where are we going?". In Eberth, David A.; Evans, David C. (eds.). *Hadrosaurs*. Bloomington, Indiana: Indiana University Press. pp. 2–7. ISBN 978-0-253-01385-9.
14. Alarcón-Muñoz, Jhonatan; Vargas, Alexander O.; Püschel, Hans P.; Soto-Acuña, Sergio; Manríquez, Leslie; Leppe, Marcelo; Kaluza, Jonatan; Milla, Verónica; Gutstein, Carolina S.; Palma-Liberona, José; Stinnesbeck, Wolfgang; Frey, Eberhard; Pino, Juan Pablo; Bajor, Dániel; Núñez, Elaine (2023-06-16). "Relict duck-billed dinosaurs survived into the last age of the dinosaurs in subantarctic Chile" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10275600). *Science Advances*. **9** (24) eadg2456. Bibcode:2023SciA....9G2456A (https://ui.adsabs.harvard.edu/abs/2023SciA....9G2456A). doi:10.1126/sciadv.adg2456 (https://doi.org/10.1126%2Fsciadv.adg2456). ISSN 2375-2548 (https://search.worldcat.org/issn/2375-2548). PMC 10275600 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10275600). PMID 37327335 (https://pubmed.ncbi.nlm.nih.gov/37327335).
15. Pérez-Pueyo, Manuel; Cruzado-Caballero, Penélope; Moreno-Azanza, Miguel; Vila, Bernat; Castanera, Diego; Gasca, José Manuel; Puértolas-Pascual, Eduardo; Bádenas, Beatriz; Canudo, José Ignacio (2021-04-02). "The Tetrapod Fossil Record from the Uppermost Maastrichtian of the Ibero-Armorican Island: An Integrative Review Based on the Outcrops of the Western Tremp Syncline (Aragón, Huesca Province, NE Spain)" (https://doi.org/10.3390%2Fgeosciences11040162). *Geosciences*. **11** (4): 162. Bibcode:2021Geosc..11..162P (https://ui.adsabs.harvard.edu/abs/2021Geosc..11..162P). doi:10.3390/geosciences11040162 (https://doi.org/10.3390%2Fgeosciences11040162). hdl:10362/145947 (https://hdl.handle.net/10362%2F145947). ISSN 2076-3263 (https://search.worldcat.org/issn/2076-3263).
16. Longrich, Nicholas R.; Suberbiola, Xabier Pereda; Pyron, R. Alexander; Jalil, Nour-Eddine (April 2021). "The first duckbill dinosaur (Hadrosauridae: Lambeosaurinae) from Africa and the role of oceanic dispersal in dinosaur biogeography" (https://doi.org/10.1016%2Fj.cretres.2020.104678). *Cretaceous Research*. **120** 104678. Bibcode:2021CrRes.12004678L (https://ui.adsabs.harvard.edu/abs/2021CrRes.12004678L). doi:10.1016/j.cretres.2020.104678 (https://doi.org/10.1016%2Fj.cretres.2020.104678). S2CID 228807024 (https://api.semanticscholar.org/CorpusID:228807024).

17. Cope, Edward D. (1969). "Synopsis of the extinct batrachia, reptilia and aves of North America" (https://www.biodiversitylibrary.org/item/123170#page/101/mode/1up). *Transactions of the American Philosophical Society*. **14**. Archived (https://web.archive.org/web/20171112074142/https://www.biodiversitylibrary.org/item/123170#page/101/mode/1up) from the original on 2017-11-12. Retrieved 2019-07-01.
18. Prieto-Márquez, A (2010). "Global phylogeny of Hadrosauridae (Dinosauria: Ornithopoda) using parsimony and Bayesian methods" (https://doi.org/10.1111%2Fj.1096-3642.2009.00617.x). *Zoological Journal of the Linnean Society*. **159** (2): 435–502. doi:10.1111/j.1096-3642.2009.00617.x (https://doi.org/10.1111%2Fj.1096-3642.2009.00617.x).
19. Glut, Donald F. (1997). *Dinosaurs: The Encyclopedia*. Jefferson, North Carolina: McFarland & Co. p. 69. ISBN 0-89950-917-7.
20. Prieto-Márquez, A.; Dalla Vecchia, F. M.; Gaete, R.; Galobart, À. (2013). "Diversity, relationships, and biogeography of the Lambeosaurine dinosaurs from the European archipelago, with description of the new aralosaurin *Canardia garonnensis*" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3724916). *PLOS ONE*. **8** (7) e69835. Bibcode:2013PLoSO...869835P (https://ui.adsabs.harvard.edu/abs/2013PLoSO...869835P). doi:10.1371/journal.pone.0069835 (https://doi.org/10.1371%2Fjournal.pone.0069835). PMC 3724916 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3724916). PMID 23922815 (https://pubmed.ncbi.nlm.nih.gov/23922815).
21. Madzia, D.; Arbour, V.M.; Boyd, C.A.; Farke, A.A.; Cruzado-Caballero, P.; Evans, D.C. (2021). "The phylogenetic nomenclature of ornithischian dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8667728). *PeerJ*. **9** e12362. Bibcode:2021PeerJ...912362M (https://ui.adsabs.harvard.edu/abs/2021PeerJ...912362M). doi:10.7717/peerj.12362 (https://doi.org/10.7717%2Fpeerj.12362). PMC 8667728 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8667728). PMID 34966571 (https://pubmed.ncbi.nlm.nih.gov/34966571).
22. Prieto-Marquez, A.; Erickson, G.M.; Ebersole, J.A. (2016). "A primitive hadrosaurid from southeastern North America and the origin and early evolution of 'duck-billed' dinosaurs" (https://figshare.com/articles/dataset/A_primitive_hadrosaurid_from_southeastern_North_America_and_the_origin_and_early_evolution_of_8216_duck_billed_8217_dinosaurs/1632849). *Journal of Vertebrate Paleontology*. **36** (2) e1054495. Bibcode:2016JVPal..36E4495P (https://ui.adsabs.harvard.edu/abs/2016JVPal..36E4495P). doi:10.1080/02724634.2015.1054495 (https://doi.org/10.1080%2F02724634.2015.1054495). S2CID 86032549 (https://api.semanticscholar.org/CorpusID:86032549). Archived (https://web.archive.org/web/20211125024005/https://figshare.com/articles/dataset/A_primitive_hadrosaurid_from_southeastern_North_America_and_the_origin_and_early_evolution_of_8216_duck_billed_8217_dinosaurs/1632849) from the original on 2021-11-25. Retrieved 2021-08-31.
23. Ramírez-Velasco, Angel Alejandro (2022). "Phylogenetic and biogeography analysis of Mexican hadrosauroids" (https://www.sciencedirect.com/science/article/abs/pii/S0195667122001318). *Cretaceous Research*. **138** 105267. Bibcode:2022CrRes.13805267R (https://ui.adsabs.harvard.edu/abs/2022CrRes.13805267R). doi:10.1016/j.cretres.2022.105267 (https://doi.org/10.1016%2Fj.cretres.2022.105267). S2CID 249559319 (https://api.semanticscholar.org/CorpusID:249559319). Archived (https://web.archive.org/web/20221219141108/https://www.sciencedirect.com/science/article/abs/pii/S0195667122001318) from the original on 2022-12-19. Retrieved 2022-08-02.
24. Bell, P. R. (2012). Farke, Andrew A (ed.). "Standardized Terminology and Potential Taxonomic Utility for Hadrosaurid Skin Impressions: A Case Study for *Saurolophus* from Canada and Mongolia" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3272031). *PLOS ONE*. **7** (2) e31295. Bibcode:2012PLoSO...731295B (https://ui.adsabs.harvard.edu/abs/2012PLoSO...731295B). doi:10.1371/journal.pone.0031295 (https://doi.org/10.1371%2Fjournal.pone.0031295). PMC 3272031 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3272031). PMID 22319623 (https://pubmed.ncbi.nlm.nih.gov/22319623).

25. "Hadrosaur Forelimb Study" (http://palaeo-electronica.org/2009_3/180/discus.htm). Palaeo-electronica.org. Archived (https://web.archive.org/web/20120605021424/http://palaeo-electronica.org/2009_3/180/discus.htm) from the original on 2012-06-05. Retrieved 2013-07-23.
26. Williams, Vincent S.; Barrett, Paul M.; Purnell, Mark A. (2009). "Quantitative analysis of dental microwear in hadrosaurid dinosaurs, and the implications for hypotheses of jaw mechanics and feeding" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2708679). *Proceedings of the National Academy of Sciences*. **106** (27): 11194–11199. Bibcode:2009PNAS..10611194W (https://ui.adsabs.harvard.edu/abs/2009PNAS..10611194W). doi:10.1073/pnas.0812631106 (https://doi.org/10.1073%2Fpnas.0812631106). PMC 2708679 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2708679). PMID 19564603 (https://pubmed.ncbi.nlm.nih.gov/19564603).
27. Bryner, Jeanna (2009-06-29). "Study hints at what and how dinosaurs ate" (https://www.livescience.com/animals/090629-dino-teeth.html). *LiveScience*. Retrieved 2009-06-03.
28. Boyle, Alan (2009-06-29). "How dinosaurs chewed" (https://web.archive.org/web/20090702125027/http://cosmiclog.msnbc.msn.com/archive/2009/06/29/1981788.aspx). *MSNBC*. Archived from the original (https://cosmiclog.msnbc.msn.com/archive/2009/06/29/1981788.aspx) on 2009-07-02. Retrieved 2009-06-03.
29. Tweet, Justin S.; Chin, Karen; Braman, Dennis R.; Murphy, Nate L. (2008). "Probable gut contents within a specimen of *Brachylophosaurus canadensis* (Dinosauria: Hadrosauridae) from the Upper Cretaceous Judith River Formation of Montana". *PALAIOS*. **23** (9): 624–635. Bibcode:2008Palai..23..624T (https://ui.adsabs.harvard.edu/abs/2008Palai..23..624T). doi:10.2110/palo.2007.p07-044r (https://doi.org/10.2110%2Fpalo.2007.p07-044r). S2CID 131393649 (https://api.semanticscholar.org/CorpusID:131393649).
30. Lloyd, Robin (2008-09-25). "Plant-eating dinosaur spills his guts: Fossil suggests hadrosaur's last meal included lots of well-chewed leaves" (https://web.archive.org/web/20160204114241/http://www.nbcnews.com/id/26893497/). *NBC News*. Archived from the original (http://www.nbcnews.com/id/26893497) on February 4, 2016. Retrieved 2009-06-03.
31. This information comes from the aforementioned Alan Boyle source from June 29, 2009. However, this specific information is not included in the body of the article, but rather a response by Boyle to comments in the article. Since the comments were written by Boyle himself, and since they cite information he received specifically from Purnell, they are as legitimate a source of information as the article itself.
32. Mallon, Jordan C.; Evans, David C.; Ryan, Michael J.; Anderson, Jason S. (2013). "Feeding height stratification among the herbivorous dinosaurs from the Dinosaur Park Formation (upper Campanian) of Alberta, Canada" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3637170). *BMC Ecology*. **13** (1): 14. Bibcode:2013BMCE...13...14M (https://ui.adsabs.harvard.edu/abs/2013BMCE...13...14M). doi:10.1186/1472-6785-13-14 (https://doi.org/10.1186%2F1472-6785-13-14). PMC 3637170 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3637170). PMID 23557203 (https://pubmed.ncbi.nlm.nih.gov/23557203).
33. Chin, K. (September 2007). "The Paleobiological Implications of Herbivorous Dinosaur Coprolites from the Upper Cretaceous Two Medicine Formation of Montana: Why Eat Wood?". *PALAIOS*. **22** (5): 554. Bibcode:2007Palai..22..554C (https://ui.adsabs.harvard.edu/abs/2007Palai..22..554C). doi:10.2110/palo.2006.p06-087r (https://doi.org/10.2110%2Fpalo.2006.p06-087r). S2CID 86197149 (https://api.semanticscholar.org/CorpusID:86197149).
34. Chin, Karen; Feldmann, Rodney M.; Tashman, Jessica N. (2017). "Consumption of crustaceans by megaherbivorous dinosaurs: dietary flexibility and dinosaur life history strategies" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5608751). *Scientific Reports*. **7** (1): 11163. Bibcode:2017NatSR...711163C (https://ui.adsabs.harvard.edu/abs/2017NatSR...711163C). doi:10.1038/s41598-017-11538-w (https://doi.org/10.1038%2Fs41598-017-11538-w). PMC 5608751 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5608751). PMID 28935986 (https://pubmed.ncbi.nlm.nih.gov/28935986).

35. Warnock-Juteau, Trystan; Ryan, Michael; Patterson, Tim; Mallon, Jordan (2024). "Computed tomographic investigation of a hatchling skull reveals ontogenetic changes in the dentition and occlusal surface morphology of Hadrosauridae (Dinosauria: Ornithischia)" (https://journals.library.ualberta.ca/vamp/index.php/VAMP/article/view/29395). *Vertebrate Anatomy Morphology Palaeontology*. **11**. doi:10.18435/vamp29395 (https://doi.org/10.18435%2Fvamp29395). Archived (https://web.archive.org/web/20240424192931/https://journals.library.ualberta.ca/vamp/index.php/VAMP/article/view/29395) from the original on 2024-04-24. Retrieved 2024-04-24.
36. Hopson, James A. (1977). "Relative brain size and behavior in archosaurian reptiles". *Annual Review of Ecology and Systematics*. **8** (1): 429–448. Bibcode:1977AnRES...8..429H (https://ui.adsabs.harvard.edu/abs/1977AnRES...8..429H). doi:10.1146/annurev.es.08.110177.002241 (https://doi.org/10.1146%2Fannurev.es.08.110177.002241).
37. Evans, David C.; et al. (2009). "Endocranial Anatomy of Lambeosaurine Hadrosaurids (Dinosauria: Ornithischia): A Sensorineural Perspective on Cranial Crest Function" (https://doi.org/10.1002%2Far.20984). *The Anatomical Record*. **292** (9): 1315–1337. doi:10.1002/ar.20984 (https://doi.org/10.1002%2Far.20984). PMID 19711466 (https://pubmed.ncbi.nlm.nih.gov/19711466). S2CID 15177074 (https://api.semanticscholar.org/CorpusID:15177074).
38. Lauters, Pascaline; et al. (2013). "Cranial Endocast of the Lambeosaurine Hadrosaurid Amurosaurus riabinini from the Amur Region, Russia" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3827337). *PLOS ONE*. **8** (11) e78899. Bibcode:2013PLoSO...878899L (https://ui.adsabs.harvard.edu/abs/2013PLoSO...878899L). doi:10.1371/journal.pone.0078899 (https://doi.org/10.1371%2Fjournal.pone.0078899). PMC 3827337 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3827337). PMID 24236064 (https://pubmed.ncbi.nlm.nih.gov/24236064).
39. Marsh, Othniel Charles (1893). "The skull and brain of Claosaurus" (https://zenodo.org/record/2198923). *American Journal of Science*. **265** (265): 83–86. Bibcode:1893AmJS...45...83M (https://ui.adsabs.harvard.edu/abs/1893AmJS...45...83M). doi:10.2475/ajs.s3-45.265.83 (https://doi.org/10.2475%2Fajs.s3-45.265.83). S2CID 131740074 (https://api.semanticscholar.org/CorpusID:131740074).
40. Ostrom, John H. (1961). "Cranial morphology of the hadrosaurian dinosaurs of North America". *Bulletin of the AMNH*. **122**. hdl:2246/1260 (https://hdl.handle.net/2246%2F1260).
41. Tanke, D.H. and Brett-Surman, M.K. 2001. Evidence of Hatchling and Nestling-Size Hadrosaurs (Reptilia:Ornithischia) from Dinosaur Provincial Park (Dinosaur Park Formation: Campanian), Alberta, Canada. pp. 206-218. In: Mesozoic Vertebrate Life—New Research Inspired by the Paleontology of Philip J. Currie. Edited by D.H. Tanke and K. Carpenter. Indiana University Press: Bloomington. xviii + 577 pp.
42. Cubo, Jorge; Woodward, Holly; Wolff, Ewan; Horner, John R. (2015-07-08). Farke, Andrew A. (ed.). "First Reported Cases of Biomechanically Adaptive Bone Modeling in Non-Avian Dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4495995). *PLOS ONE*. **10** (7) e0131131. Bibcode:2015PLoSO..1031131C (https://ui.adsabs.harvard.edu/abs/2015PLoSO..1031131C). doi:10.1371/journal.pone.0131131 (https://doi.org/10.1371%2Fjournal.pone.0131131). ISSN 1932-6203 (https://search.worldcat.org/issn/1932-6203). PMC 4495995 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4495995). PMID 26153689 (https://pubmed.ncbi.nlm.nih.gov/26153689).
43. Wosik, Mateusz; Goodwin, Mark B.; Evans, David C. (2017-11-02). "A nestling-sized skeleton of Edmontosaurus (Ornithischia, Hadrosauridae) from the Hell Creek Formation of northeastern Montana, U.S.A., with an analysis of ontogenetic limb allometry" (https://www.tandfonline.com/doi/full/10.1080/02724634.2017.1398168). *Journal of Vertebrate Paleontology*. **37** (6) e1398168. Bibcode:2017JVPal..37E8168W (https://ui.adsabs.harvard.edu/abs/2017JVPal..37E8168W). doi:10.1080/02724634.2017.1398168 (https://doi.org/10.1080%2F02724634.2017.1398168). ISSN 0272-4634 (https://search.worldcat.org/issn/0272-4634).

44. Schmitz, L.; Motani, R. (2011). "Nocturnality in Dinosaurs Inferred from Scleral Ring and Orbit Morphology". *Science*. **332** (6030): 705–708. Bibcode:2011Sci...332..705S (https://ui.adsabs.harvard.edu/abs/2011Sci...332..705S). doi:10.1126/science.1200043 (https://doi.org/10.1126%2Fscience.1200043). PMID 21493820 (https://pubmed.ncbi.nlm.nih.gov/21493820). S2CID 33253407 (https://api.semanticscholar.org/CorpusID:33253407).
45. Rothschild, B.M.; Tanke, D.H.; Helbling II, M.; Martin, L.D. (2003). "Epidemiologic study of tumors in dinosaurs" (http://doc.rero.ch/record/13884/files/PAL_E816.pdf) (PDF). *Naturwissenschaften*. **90** (11): 495–500. Bibcode:2003NW.....90..495R (https://ui.adsabs.harvard.edu/abs/2003NW.....90..495R). doi:10.1007/s00114-003-0473-9 (https://doi.org/10.1007%2Fs00114-003-0473-9). PMID 14610645 (https://pubmed.ncbi.nlm.nih.gov/14610645). S2CID 13247222 (https://api.semanticscholar.org/CorpusID:13247222).
46. Carpenter, Kenneth (1998). "Evidence of predatory behavior by theropod dinosaurs" (https://web.archive.org/web/20071117132451/http://vertpaleo.org/publications/jvp/15-576-591.cfm). *Gaia*. **15**: 135–144. Archived from the original (http://vertpaleo.org/publications/jvp/15-576-591.cfm) on 2007-11-17. Retrieved 2009-03-08. [not printed until 2000]
47. Rothschild, Bruce; Tanke, Darren H. (2007). "Osteochondrosis is Late Cretaceous Hadrosauria". In Carpenter Kenneth (ed.). *Horns and Beaks: Ceratopsian and Ornithopod Dinosaurs*. Bloomington and Indianapolis: Indiana University Press. pp. 171–183. ISBN 978-0-253-34817-3.

External links

Dinosaurs portal

- Timeline of hadrosaur research
- UCMP Tree of Life (https://web.archive.org/web/20110604054624/http://www.ucmp.berkeley.edu/diapsids/ornithischia/hadrosauria.html)
- Trachodon (https://web.archive.org/web/20120331070549/http://dinosaurs-on-stamps.info/trachodon.html)
- "Mummified" Dinosaur Discovered In Montana, National Geographic 2002 (https://web.archive.org/web/20021012070200/http://news.nationalgeographic.com/news/2002/10/1010_021010_dinomummy.html)

Retrieved from "https://en.wikipedia.org/w/index.php?title=Hadrosauridae&oldid=1370711933"

WIKIPEDIA 25
25 years of the free encyclopedia

Maiasaura

Maiasaura (from Greek: μαῖα,[a] lit. 'good mother' and Greek: σαύρα, the feminine form of *saurus*, lit. 'lizard') is a large herbivorous saurolophine hadrosaurid ("duck-billed") dinosaur genus that lived in the area currently covered by the state of Montana and the Canadian province of Alberta,[1] in the Upper Cretaceous (mid to late Campanian), from 86.3 to 70.6 million years ago.[2]

Discovered in 1978 during the dinosaur renaissance, the species description of *Maiasaura* first came in 1979 from paleontologists Jack Horner and Bob Makela following the discovery of a site located South of Choteau, Montana. They named the new genus and species ***Maiasaura peeblesorum***, with the genus name *Maiasaura*, ("good mother lizard") referring to Makela and Horner's conclusion that this species took care of their offspring. Collected data at the site indicated that *Maiasaura* fed its young while they were in the nest, the first instance of parental and social behavior in dinosaurs.[3] This was in contradiction with the common idea of the 1970s that dinosaurs laid eggs and abandoned them immediately afterwards, providing no parental care. In 1979, one year after the discovery of the nests and juveniles, *Maiasaura* whole eggs were unearthed, the first discovery of dinosaur whole eggs[4] both on the American continent and in the Western Hemisphere. Further work in the area led to the discovery of more dinosaur eggs, leading to the area being named "Egg Mountain". Computed tomography scans showed embryos at a very early stage of development inside some of the eggs.

Because of these elements, *Maiasaura* is considered one of the most significant discoveries in the history of dinosaur research.[5][6][7]

Maiasaura, whose bones have been found by hundreds in the state of Montana, has been chosen as Montana's state fossil.

Maiasaura

Temporal range: Late Cretaceous (Santonian – Campanian),

Mounted cast, Brussels Natural History Museum

Scientific classification

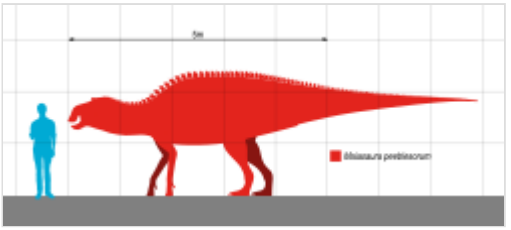
Kingdom:	Animalia
Phylum:	Chordata
Class:	Reptilia
Clade:	Dinosauria
Clade:	†Ornithischia
Clade:	†Ornithopoda
Family:	†Hadrosauridae
Subfamily:	†Saurolophinae
Tribe:	†Brachylophosaurini
Genus:	†*Maiasaura* Horner & Makela, 1979

Type species

†***Maiasaura peeblesorum***
Horner & Makela, 1979

Description

Skull cast, Royal Ontario Museum



Size comparison with human

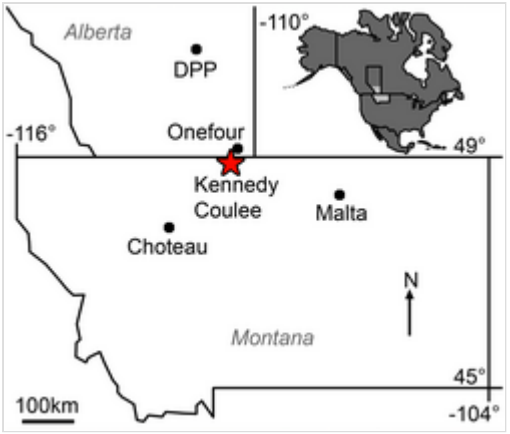
Maiasaura peeblesorum were large, attaining a maximum known length of about 9 metres (30 ft) and an approximate body mass measured at up to 4 metric tons (4.4 short tons).[8] They had a large "duck-billed" mouth structure and rows of hundreds of teeth, typical of hadrosaurids. While hadrosaurids have very similar post-cranial body plans,[9] the distinguishing characteristic of *Maiasaura peeblesorum* is a prominent short, solid crest-like structure situated between the eyes. This crest may have been used in headbutting contests between males during the breeding season.

Maiasaura were herbivorous, terrestrial dinosaurs. They were capable of walking both on two (bipedal) or four (quadrupedal) legs. Studies of the stress patterns of healed bones show that juveniles under four years old walked mainly bipedally, switching to a mainly quadrupedal style of walking when they grew larger.[10] *Maiasaura*, like most other hadrosaurs, possessed little in the way of obvious weaponry, though likely could defend themselves with kicks, stomps, or their muscular tails. It is likely that they primarily resorted to fleeing in the face of danger, using the vast sizes of their herds to be less likely to be targeted. Mass bone beds discovered in the Two Medicine Formation show that herds could be extremely large and comprise as many as 10,000 individuals.[11] Hundreds of specimens have been found throughout all stages of life, allowing for *M. peeblesorum* to be used for understanding how hadrosaurids grew.[12]

Discovery

For years until the 1960s and 1970s, anyone who has traveled through the area south of Choteau, Montana might have come across *Maiasaura* remains, whether such remains have been or have not been attributed to a dinosaurian origin.

The first people who are confirmed as having found *Maiasaura* remains laying on the "Egg Mountain" area (as it is called today), are two homestead families of Bynum, Montana: the Brandvolds and the Trexlers. Marion Kathryn Brandvold (1912–2014, née Nehring), had inherited the "rock shop", Trex Agate Shop, that had been founded in 1937 by her first husband, Clifford "Trex" Trexler (1908–1962). In the years that preceded 1978, she and her second husband, John Brandvold (1937–2020),[13] had been finding small bones and they had been trying to put them together.[14] But in 1978 paleontologist Bob Makela had been told by fellow Berkeley paleontologist Bill Clemens that the State of Montana had called him about a north-western Montana woman who had found bones which she wanted to identify.[15] Makela informed his research partner Jack Horner and both went to the Trex Agate Shop in Bynum. There, Marion Brandvold showed them some dinosaur bones, which the paleontologists could identify,



Map of Brachylophosaurini (Saurolophinae) specimen discoveries in Alberta and Montana. The first remains of *Maiasaura* have been found south of Choteau, visible on the map.

but also two tiny bones that Horner identified as baby hadrosaur bones. Brandvold said she had more at her place, so they followed her to the Brandvolds' house, where she'd been keeping the remains of at least four individuals in a coffee can.[15][16] Two weeks later Brandvold took Makela and Horner to the site in Choteau where she have found the remains. There, bones were visible on the surface, not even needing to dig. During the same Summer of 1978, Makela and Horner excavated the spot and noticed that they were excavating a nest, where in the end they found a total of 15 baby dinosaurs.[15] That was their first excavated nest, but subsequently the site delivered more nests and juveniles, all of them of a till-then unknown species of hadrosaur. The next year in July 1979, during the second summer dig, a lady student named Fran Tannenbaum, under the direction of Jack Horner, discovered the first whole dinosaur egg ever found in North America[17] and in the Western Hemisphere. With time more eggs were discovered and the site of this discovery earned the name of "Egg Mountain", because of the abundance of hadrosaur eggs and eggshell pieces found in it.[18][19]

A skull of *Maiasaura*, specimen PU 22405 (now in the collections of the Yale Peabody Museum of Natural History as YPM PU 22405 following the transfer of the Princeton University vertebrate paleontology collections), was discovered by David Trexler's wife Laurie Trexler in 1979. Trexler partially broke the skull as a consequence of her lack of appropriate excavating tools, but after informing Jack Horner and Robert Makela, both paleontologists managed to extract the whole remaining part of the fossil.[15] It is the specimen that Makela and Horner had used the same year as the holotype of the new species, which they described and named as the type species *Maiasaura peeblesorum*. The generic name *Maiasaura* means 'good mother lizard' and the specific name, *peeblesorum*, honours the families of John and James Peebles, on whose land the finds were made.[20][3]

David Trexler, the second son of Clifford Trexler and Marion Brandvold, grew up in such a context that he ended up by becoming a paleontologist himself. In 1995 he founded in Bynum the Montana Dinosaur Center. That center plays an important role in connexion with the first identified remains of *Maiasaura* because during the years that spanned from 1998 to 2004, David Trexler's mother Marion Brandvold led a legal fight with the Yale and Princeton universities in order to recover the bones of the baby *Maiasauras* that she had found back in the 1970s. In 2004 she obtained a satisfactory issue and, since then, these historically important fossils are preserved at the Montana Dinosaur Center in Bynum.[21]

Over 200 specimens, in all age ranges, have been found.[22] The announcement of the discovery of *Maiasaura* attracted renewed scientific interest to the Two Medicine Formation and many other new kinds of dinosaurs were discovered as a result of the increased attention.[23] Choteau *Maiasaura* remains are found in higher strata than their Two Medicine River counterparts.[24]

The sites that close to Choteau contain *Maiasaura* remains of adults, juveniles, nests and eggs, have been preserved in the rock as if a rapid event had "frozen" the area. It is assumed that such event was the eruption of a volcano located among the nearby Elkhorn mountains. Nests, eggs and animals were

covered by ash and once they fossilised they remained almost perfectly preserved until they were discovered in 1978.[18][19]

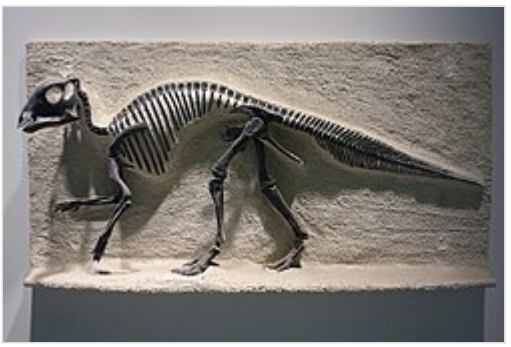
In 1985 The Nature Conservancy purchased the site of Egg Mountain from the James and Marian Peebles family, the original owners of the land, leasing the property to the Peebles for cattle grazing and allowing curator Jack Horner and the Museum of the Rockies to run a paleontology field camp. In 2004, the Museum of the Rockies purchased the property and now the Museum gives "Egg Mountain" the name of "Museum of the Rockies' Beatrice R. Taylor Paleontology Research Site", honouring Beatrice Taylor, whose family donated the money for the purchase.[25]

Classification

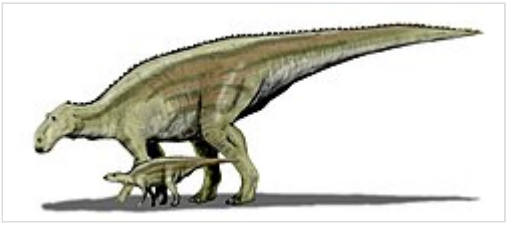
Maiasaura peeblesorum is in the tribe Brachylophosaurini along with these related taxa:

- *Acristavus gagslarsoni*[26]
- *Brachylophosaurus canadensis*[27]
- *Ornatops incantatus*[28]
- *Probrachylophosaurus bergei*[29]

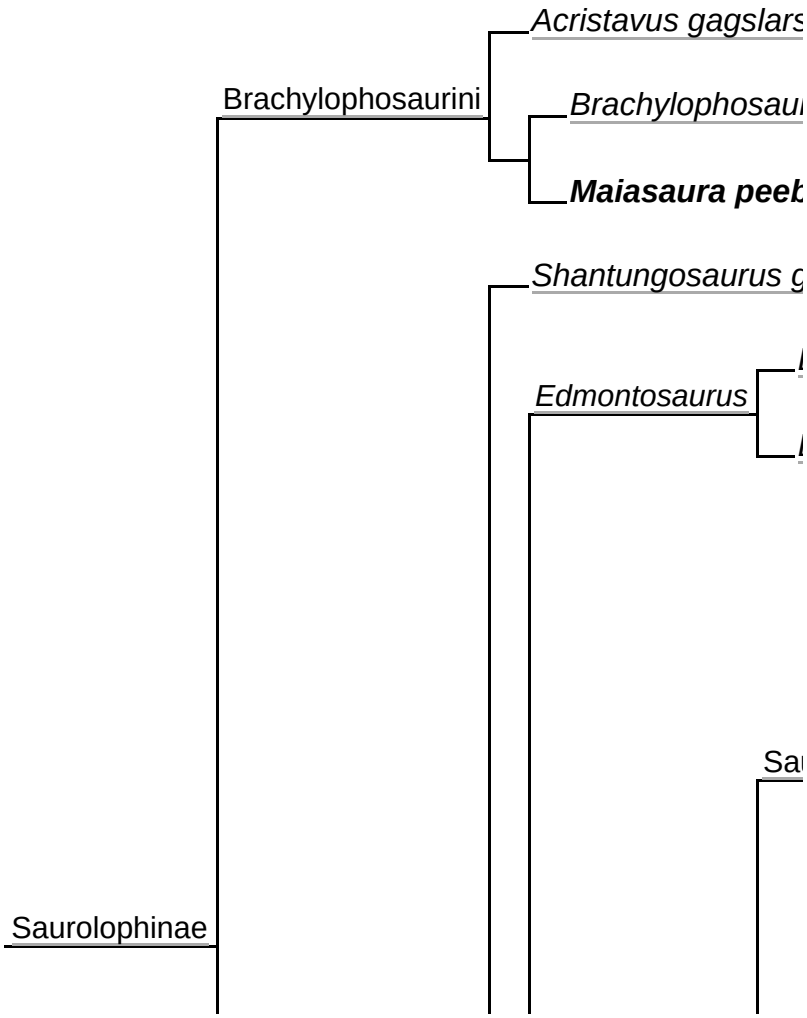
The following cladogram of hadrosaurid relationships was published in 2013 by Albert Prieto-Márquez et al.:[30]

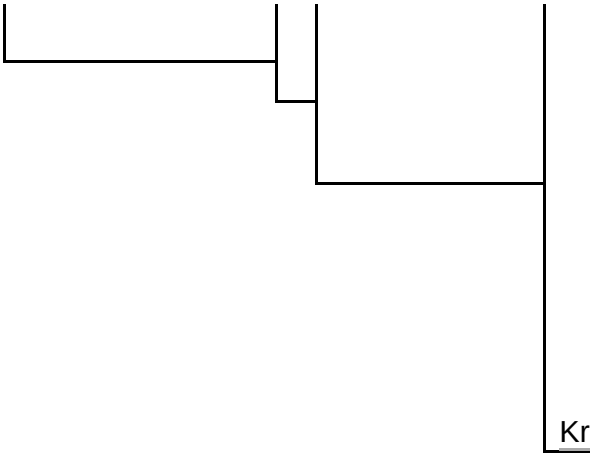


Cast of a juvenile skeleton



Life restorations of an adult and juvenile





Paleobiology

Reconstructed cast by Jack Horner of a *Maiasaura* emerging from its egg

Maiasaura lived in herds and it raised its young in nesting colonies. The nests in the colonies were packed closely together, like those of modern seabirds, with the gap between the nests being around 7 metres (23 ft); less than the length of the adult animal.[31] The nests were made of earth and contained 30 to 40 eggs laid in a circular or spiral pattern. The eggs were about the size of ostrich eggs and are oval shaped with one slightly more pointed end.[32] Fossilized *M. peeblesorum* eggs are black in color and have high, prominent ridges on the outer surface.[32]

The eggs were incubated by the heat resulting from rotting vegetation placed into the nest by the parents, rather than a parent sitting on the nest. Upon hatching, fossils of baby *Maiasaura* show that their legs were not fully developed and thus they were incapable of walking. Fossils also show that their teeth were partly worn, which means that the adults brought food to the nest.[33]

The hatchlings grew from a size of 41 to 147 centimetres (16 to 58 in) long in the span of their first year. At this point, or perhaps after another year, the animal left the nest. This high rate of growth may be evidence of warm bloodedness. The hatchlings had different facial proportions from the adults, with larger eyes and a shorter snout.[33] These features are associated with cuteness, and commonly elicit care from parents in animals dependent on their parents for survival during the early stages of life. Based on metabolic growth rates and age estimations of juvenile specimens both in and outside the nests, it is suggested that infant *Maiasaura* were altricial and left the confines of their nests after 40-75 days (or

roughly a little over 1-2 months old). These studies also were compared to the related lambeosaurine hadrosaur *Hypacrosaurus*, for which neonates are also known. These analyses suggest *Hypacrosaurus* was more precocial than *Maiasaura*, based on its ecological traits.[34]

Studies led by Holly Woodward, Jack Horner, Freedman Fowler et al. have given insight into the life history of *Maiasaura*, resulting in what is perhaps the most detailed life history of any dinosaur known, and to which all others can be compared. From a sample of fifty individual *Maiasaura* tibiae, it was found that Maiasaurs had a mortality rate of about 89.9% in their first year of life. If the animals survived their second year, their mortality rate would drop to 12.7%. The animals would spend their next six years maturing and growing. Sexual maturity was found to occur in their third year, while skeletal maturity was attained at eight years of age. In their eighth year and beyond, the mortality rate for *Maiasaura* would spike back to around 44.4%. The studies that followed also found that Maiasaurs were primarily bipedal as juveniles, and switched to a more quadrupedal stance as they aged. It was also found that *Maiasaura* also included rotting wood in its diet, as well that its environment had a long, dry season prone to drought. The results of the study were published in the journal *Palaeobiology* on September 3, 2015.[35][36]

Reconstruction of a nest with eggs

Baby *Maiasaura* casts on display at Royal Ontario Museum

Diet

A paper from 2007 showed that *Maiasaura* had a diet consisting of fibrous plants, wood, rotting wood, tree bark, leaves, branches, ferns, angiosperms and possibly grasses. This would imply that *Maiasaura* was both a browser and a grazer.[37][38][39] Analysis of its dental wear patterns show that juvenile *M. peeblesorum* exhibited more crush wear than adults, which displayed more shear wear than juveniles. This suggests that adults fed more on tough, fibrous vegetation while juveniles fed more on hard, brittle objects like nuts, seeds, or berries.[40]

Sexual dimorphism

Studies of *Maiasaura* by Saitta et al., suggest that one sex was roughly 45% larger than the other according to the mathematical analysis known as size statistics. However, it cannot be ascertained at this time whether the larger sex was male or female.[41][42]

Palaeoecology

Illustration of a herd of *Maiasaura* walking along a creekbed, as found in the semi-arid Two Medicine Formation fossil bed. This region was characterized by volcanic ash layers and conifer, fern and horsetail vegetation.

Maiasaura is a characteristic fossil of the middle portion (lithofacies 4) of the Two Medicine Formation, dated from about 86.3 to 70.6 million years ago.[2] *Maiasaura* lived alongside the troodontids *Stenonychosaurus* and *Troodon* and the basal ornithopod *Orodromeus*, as well as the dromaeosaurid *Bambiraptor* and the tyrannosaur *Daspletosaurus*.[2] Another species of hadrosaurids, referable to the genus *Hypacrosaurus*, coexisted with *Maiasaura* for some time, as *Hypacrosaurus* remains have been found lower in the Two Medicine Formation than was earlier known.[43] The discovery of an additional hadrosaurid, *Gryposaurus latidens*, in the same range as *Maiasaura* has shown that the border between hypothesized distinct faunas in the upper and middle is less distinct than once thought.[43] There seems to be a major diversification in ornithischian taxa after the appearance of *Maiasaura* within the Two Medicine Formation.[43] The thorough examination of strata found along the Two Medicine River (which exposes the entire upper half of the Two Medicine Formation) indicates that the apparent diversification was a real event rather than a result of preservational biases.[43] While *Maiasaura* has historically been associated with the Two Medicine formation ceratopsid *Einiosaurus* in a single fauna, this is inaccurate, as *Maiasaura* is known exclusively from older strata.[44]

In the Oldman Formation of Alberta, *Maiasaura* lived alongside the ceratopsians *Albertaceratops*, *Anchiceratops*, *Chasmosaurus*, *Coronosaurus*, and *Wendiceratops*, as well as the dromaeosaurids *Dromaeosaurus*, *Saurornitholestes*, and *Hesperonychus*, the tyrannosaurid *Daspletosaurus*, the orodromine thescelosaurid *Albertadromeus*, the pachycephalosaurs *Foraminacephale* and *Hanssuesia*, the ornithomimid *Struthiomimus*, the other hadrosaurids *Brachylophosaurus*, *Corythosaurus*, and *Parasaurolophus*, and the ankylosaurid *Scolosaurus*.[1]

Maiasaura and the popularisation of the "dinosaur renaissance"

Within the first years of the discovery of *Maiasaura*, the importance of such a remarkable find had a durable impact on the public and media contributed to that:

As of 1983, dinosaur renaissance artist Doug Henderson started painting anatomically modern-view life restorations of *Maiasaura* for a children's book authored by Jack Horner, James Gorman and Jeri D. Walton. Finally published in 1985 the book was titled *Maia: A Dinosaur Grows Up* and, via its illustrations, showed to the public *Maiasaura* and other of its contemporary dinosaurs on horizontal-backbone posture, contrary to the till-then traditional, popular and incorrect posture seen in dinosaurs (bipedal or not) that permanently were dragging their tails on the ground.[45]

In 1985 as well, the CBS television documentary *Dinosaur!* showed Jack Horner talking to the camera about the "Egg Mountain" site and its related discoveries. *Dinosaur!* was seminal too showing to the CBS audiences the CT scan image of a *Maiasaura* embryo found inside an egg that had been previously collected at "Egg Mountain".[46]

In 1987, the television documentary film *Digging Dinosaurs*, produced by the television channel WHYY Philadelphia with fund by the William Penn Foundation, showed a fossil dig team led by Bob Makela and Jack Horner, all excavating at a *Maiasaura* fossilised nesting ground in northern Montana. In the film, Horner talks to the camera about his conclusions on how the juveniles hatched and grew, first in the nests, then in company of the adults when joining the herd. He also mentions his interest in the opportunity that this species, found in hundreds of individuals, brings to studies on individual variation within a species of dinosaur.[47]

Horner reappeared on television, this time with extended screen time about the *Maiasaura* discoveries, in another programme, "The Great Dinosaur Hunt", part of *The Infinite Voyage* series of documentaries. *The Great Dinosaur Hunt* first aired on January 4, 1989.[48]

In 1991 two short documentaries produced by Earthtalk Studios: *A Giant Leap for Dinosaurs* and *Dinosaur Hunters*, both directed by Daniel J. Smith, showed Jack Horner at Camp Makela (the scientists' camp that is on site at "Egg Mountain") talking with children and adolescents about the cutting edge of dinosaur research.[49]

See also

- Timeline of hadrosaur research

Dinosaurs portal

Notes

a. *Maĩa* term also related to Maia, one of the seven sisters (Pleiades) and the mother of Hermes by Zeus in Greek mythology.

References

1. McFeeters, Bradley D.; Evans, David C.; Ryan, Michael J.; Maddin, Hillary C. (2021-03-01). "First occurrence of Maiasaura (Dinosauria, Hadrosauridae) from the Upper Cretaceous Oldman Formation of southern Alberta, Canada" (https://cdnsciencepub.com/doi/abs/10.1139/cjes-2019-0207). *Canadian Journal of Earth Sciences*. **58** (3): 286–296. Bibcode:2021CaJES..58..286M (https://ui.adsabs.harvard.edu/abs/2021CaJES..58..286M). doi:10.1139/cjes-2019-0207 (https://doi.org/10.1139%2Fcjes-2019-0207). ISSN 0008-4077 (https://search.worldcat.org/issn/0008-4077). S2CID 233851376 (https://api.semanticscholar.org/CorpusID:233851376).
2. Horner, J. R., Schmitt, J. G., Jackson, F., & Hanna, R. (2001). Bones and rocks of the Upper Cretaceous Two Medicine-Judith River clastic wedge complex, Montana. In Field trip guidebook, Society of Vertebrate Paleontology 61st Annual Meeting: Mesozoic and Cenozoic Paleontology in the Western Plains and Rocky Mountains. Museum of the Rockies Occasional Paper (Vol. 3, pp. 3-14).
3. Horner, J.R.; Makela, R. (1979). "Nest of juveniles provides evidence of family structure among dinosaurs" (https://www.nature.com/articles/282296a0). *Nature*. **282** (5736): 296–298. Bibcode:1979Natur.282..296H (https://ui.adsabs.harvard.edu/abs/1979Natur.282..296H). doi:10.1038/282296a0 (https://doi.org/10.1038%2F282296a0). S2CID 4370793 (https://api.semanticscholar.org/CorpusID:4370793).

4. As of 1913, fragments of dinosaur eggshells had already been discovered, see : Carpenter, Kenneth (1999), *Eggs, Nests, and Baby Dinosaurs: A Look at Dinosaur Reproduction*; Life of the Past. Indiana University Press; ISBN 0-253-33497-7.
5. *www.dinopit*, "The Importance of Maiasaura" (https://www.dinopit.com/maiasaura/)
6. *ebsco.com*, "Maiasaura (https://www.ebsco.com/research-starters/earth-and-atmospheric-sciences/maiasaura)"
7. *montanakids.com* "Maiasaura (https://montanakids.com/history_and_prehistory/dinosaurs/maiasaur.htm)"; August 5, 2020 (archived url (https://web.archive.org/web/20250912165412/https://montanakids.com/history_and_prehistory/dinosaurs/maiasaur.htm): September 12, 2025)
8. Wosik, M.; Chiba, K.; Therrien, F.; Evans, D.C. (2020). "Testing Size–frequency Distributions As a Method of Ontogenetic Aging: A Life-history Assessment of Hadrosaurid Dinosaurs from the Dinosaur Park Formation of Alberta, Canada, with Implications for Hadrosaurid Paleoecology" (https://zenodo.org/record/5001368). *Paleobiology*. **46** (3): 379–404. Bibcode:2020Pbio...46..379W (https://ui.adsabs.harvard.edu/abs/2020Pbio...46..379W). doi:10.1017/pab.2020.2 (https://doi.org/10.1017%2Fpab.2020.2). S2CID 221666530 (https://api.semanticscholar.org/CorpusID:221666530).
9. Brett-Surman, M. K. (February 1979). "Phylogeny and palaeobiogeography of hadrosaurian dinosaurs". *Nature*. **277** (5697): 560–562. Bibcode:1979Natur.277..560B (https://ui.adsabs.harvard.edu/abs/1979Natur.277..560B). doi:10.1038/277560a0 (https://doi.org/10.1038%2F277560a0). ISSN 0028-0836 (https://search.worldcat.org/issn/0028-0836). S2CID 4332144 (https://api.semanticscholar.org/CorpusID:4332144).
10. Cubo, Jorge; Woodward, Holly; Wolff, Ewan; Horner, John R. (2015). "First Reported Cases of Biomechanically Adaptive Bone Modeling in Non-Avian Dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4495995). *PLOS ONE*. **10** (7) e0131131. Bibcode:2015PLoSO..1031131C (https://ui.adsabs.harvard.edu/abs/2015PLoSO..1031131C). doi:10.1371/journal.pone.0131131 (https://doi.org/10.1371%2Fjournal.pone.0131131). PMC 4495995 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4495995). PMID 26153689 (https://pubmed.ncbi.nlm.nih.gov/26153689).
11. Varricchio, David J.; Horner, John R. (1993-05-01). "Hadrosaurid and lambeosaurid bone beds from the Upper Cretaceous Two Medicine Formation of Montana: taphonomic and biologic implications". *Canadian Journal of Earth Sciences*. **30** (5): 997–1006. Bibcode:1993CaJES..30..997V (https://ui.adsabs.harvard.edu/abs/1993CaJES..30..997V). doi:10.1139/e93-083 (https://doi.org/10.1139%2Fe93-083). ISSN 0008-4077 (https://search.worldcat.org/issn/0008-4077).
12. McFeeters, Bradley; Evans, David; Maddin, Hillary (2021). "Ontogeny and variation in the skull roof and braincase of Maiasaura peeblesorum from the Two Medicine Formation of Montana, U.S.A." (http://publication.plazi.org/id/FF82FFBDFFA4FFBDFFEF384F00208708) *Acta Palaeontologica Polonica*. **66**. doi:10.4202/app.00698.2019 (https://doi.org/10.4202%2Fapp.00698.2019). ISSN 0567-7920 (https://search.worldcat.org/issn/0567-7920). S2CID 239729209 (https://api.semanticscholar.org/CorpusID:239729209).
13. John Wallace Brandvold obituary in *frontierfuneralhome.com* (https://www.frontierfuneralhome.com/obituaries/john-brandvold)
14. John R. Horner, *ProQuest, "Dinosaur Behavior" (https://www.proquest.com/openview/e4831391417aa87969b27a0e87c495de/1?pq-origsite=gscholar&cbl=1820941)*, *ProQuest* (*Phi Kappa Phi Journal*); Summer 1998
15. The Accent Podcast YouTube channel, "#24: Jack Horner, Paleontologist: on T-Rex feathers, Building Dinos, Jurassic Park, Dyslexia, etc. (https://www.youtube.com/watch?v=Y8q4PZJ2AWk)", interview with Jack Horner (May 22, 2023)

16. {{Cite web [author=John Ghost |url=https://wheninyourstate.com/montana/how-a-coffee-can-of-baby-bones-cracked-this-150-million-year-montana-mystery/ |title=How a coffee can of baby bones cracked this 150-million-year Montana mystery |website=wheninyourstate.com |date=November 13, 2025 |access-date=2026-05-28 |url-status=live}}
17. Remnick, David J. (August 22, 1979). "Whole Dinosaur Egg Unearthed, 1st in North America" (https://www.washingtonpost.com/archive/politics/1979/08/23/whole-dinosaur-egg-unearthed-1st-in-north-america/b5fab18d-3589-4115-b511-8abcdfdb84f5/). *The Washington Post*.
18. Willie Freimuth, Department of Earth Sciences, Montana State University, "Paleontology - Egg Mountain" (published April 10, 2018 at the Science, Education and Resource Center at Carleton College website (https://serc.carleton.edu/research_education/mt_geoheritage/sites/augusta_choteau/paleontology.html))
19. Montana Department of Transportation, "Elkhorn Mountain Volcanoes and the Boulder Batholith (https://www.hmdb.org/m.asp?m=186551)" (Jefferson City, Jefferson County, Montana, record of *Historical Marker Database*).
20. Browne, Malcolm W. (1980-02-12). "Baby Fossils Lead To New Theories About Dinosaurs; Fossils Lead to New Theories on Dinosaurs Warmblooded Thesis Is Attacked" (https://www.nytimes.com/1980/02/12/archives/baby-fossils-lead-to-new-theories-about-dinosaurs-fossils-lead-to.html). *The New York Times*. ISSN 0362-4331 (https://search.worldcat.org/issn/0362-4331). Retrieved 2026-03-23.
21. *Choteau Acantha* weekly newspaper, "In a victory for the little guy, 91-year-old Marion Brandvold of Bynum has won her fight against Yale and Princeton universities for the return of fossilized baby dinosaur remains that she discovered in the arid badlands west of Choteau in 1978. (https://www.choteauacantha.com/news/article_3231ada3-31a1-5fbc-a168-c65aa0beebdf.html)" (Mar 31, 2004)
22. Horner and Gorman (1988).
23. "Introduction," Trexler (2001); pages 299-300.
24. "Faunal Turnover, Migration, and Evolution," Trexler (2001); page 304.
25. *Montana Living*, "Egg Mountain dinosaur site sold - Dinosaur digs at Egg Mountain get new owners (https://www.montanaliving.com/blogs/outdoors-adventure-in-montana/egg-mountain-dinosaur-site-sold)" (Nov 12, 2004)
26. Gates, Terry A.; Horner, John R.; Hanna, Rebecca R.; Nelson, C. Riley (July 2011). "New unadorned hadrosaurine hadrosaurid (Dinosauria, Ornithopoda) from the Campanian of North America". *Journal of Vertebrate Paleontology*. **31** (4): 798–811. Bibcode:2011JVPal..31..798G (https://ui.adsabs.harvard.edu/abs/2011JVPal..31..798G). doi:10.1080/02724634.2011.577854 (https://doi.org/10.1080%2F02724634.2011.577854). ISSN 0272-4634 (https://search.worldcat.org/issn/0272-4634). S2CID 8878474 (https://api.semanticscholar.org/CorpusID:8878474).
27. Prieto-Marquez, Albert (2005-03-11). "New information on the cranium of*Brachylophosaurus canadensis*(Dinosauria, Hadrosauridae), with a revision of its phylogenetic position". *Journal of Vertebrate Paleontology*. **25** (1): 144–156. doi:10.1671/0272-4634(2005)025[0144:niotco]2.0.co;2 (https://doi.org/10.1671%2F0272-4634%282005%2925%5B0144%3Aniotco%5D2.0.co%3B2). ISSN 0272-4634 (https://search.worldcat.org/issn/0272-4634). S2CID 85767827 (https://api.semanticscholar.org/CorpusID:85767827).
28. McDonald, Andrew T.; Wolfe, Douglas G.; Freedman Fowler, Elizabeth A.; Gates, Terry A. (2021-04-02). "A new brachylophosaurin (Dinosauria: Hadrosauridae) from the Upper Cretaceous Menefee Formation of New Mexico" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8020878). *PeerJ*. **9** e11084. doi:10.7717/peerj.11084 (https://doi.org/10.7717%2Fpeerj.11084). ISSN 2167-8359 (https://search.worldcat.org/issn/2167-8359). PMC 8020878 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8020878). PMID 33859873 (https://pubmed.ncbi.nlm.nih.gov/33859873).

29. Freedman Fowler, Elizabeth A.; Horner, John R. (2015-11-11). "A New Brachylophosaurin Hadrosaur (Dinosauria: Ornithischia) with an Intermediate Nasal Crest from the Campanian Judith River Formation of Northcentral Montana" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4641681). *PLOS ONE*. **10** (11) e0141304. Bibcode:2015PLoSO..1041304F (https://ui.adsabs.harvard.edu/abs/2015PLoSO..1041304F). doi:10.1371/journal.pone.0141304 (https://doi.org/10.1371%2Fjournal.pone.0141304). ISSN 1932-6203 (https://search.worldcat.org/issn/1932-6203). PMC 4641681 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4641681). PMID 26560175 (https://pubmed.ncbi.nlm.nih.gov/26560175).
30. Prieto-Márquez, A.; Wagner, J.R. (2013). "A new species of saurolophine hadrosaurid dinosaur from the Late Cretaceous of the Pacific coast of North America" (http://publication.plazi.org/id/FF97FD39C9534261FFA61F07FFCFBB41). *Acta Palaeontologica Polonica*. **58** (2): 255–268. doi:10.4202/app.2011.0049 (https://doi.org/10.4202%2Fapp.2011.0049).
31. Palmer (1999); page 148.
32. Hirsch, Karl F.; Quinn, Betty (1990-12-20). "Eggs and eggshell fragments from the Upper Cretaceous Two Medicine Formation of Montana". *Journal of Vertebrate Paleontology*. **10** (4): 491–511. Bibcode:1990JVPal..10..491H (https://ui.adsabs.harvard.edu/abs/1990JVPal..10..491H). doi:10.1080/02724634.1990.10011832 (https://doi.org/10.1080%2F02724634.1990.10011832). ISSN 0272-4634 (https://search.worldcat.org/issn/0272-4634).
33. "Maiasaura," Dodson, et al. (1994); pages 116-117.
34. Bert, Hugo; Woodward, Holly; Rinder, Nicolas; Amiot, Romain; Horner, John R.; Lécuyer, Christophe; Sena, Mariana; Cubo, Jorge (2025-07-10). "Neonatal state and degree of necessity for parental care in Maiasaura based on inferred neonatal metabolic rates" (https://hal.science/hal-05156517). *Scientific Reports*. **15** (1): 24827. doi:10.1038/s41598-025-06282-5 (https://doi.org/10.1038%2Fs41598-025-06282-5). ISSN 2045-2322 (https://search.worldcat.org/issn/2045-2322). PMC 12246098 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12246098).
35. "Largest dinosaur population growth study ever shows how Maiasaura lived and died: Decades of research on Montana's state fossil -- the 'good mother lizard' Maiasaura peeblesorum -- has resulted in the most detailed life history of any dinosaur known" (https://www.sciencedaily.com/releases/2015/10/151002191729.htm).
36. Woodward, Holly N.; Freedman Fowler, Elizabeth A.; Farlow, James O.; Horner, John R. (2015). "Maiasaura, a model organism for extinct vertebrate population biology: A large sample statistical assessment of growth dynamics and survivorship". *Paleobiology*. **41** (4): 503–527. Bibcode:2015Pbio...41..503W (https://ui.adsabs.harvard.edu/abs/2015Pbio...41..503W). doi:10.1017/pab.2015.19 (https://doi.org/10.1017%2Fpab.2015.19). S2CID 85902880 (https://api.semanticscholar.org/CorpusID:85902880).
37. Chin, Karen (1 September 2007). "The Paleobiological Implications of Herbivorous Dinosaur Coprolites from the Upper Cretaceous Two Medicine Formation of Montana: Why Eat Wood?" (https://bioone.org/journals/Palaios/volume-22/issue-5/palo.2006.p06-087r/THE-PALEOBIOLOGICAL-IMPLICATIONS-OF-HERBIVOROUS-DINOSAUR-COPROLITES-FROM-THE-UPPER/10.2110/palo.2006.p06-087r.short). *PALAIOS*. **22** (5): 554–566. Bibcode:2007Palai..22..554C (https://ui.adsabs.harvard.edu/abs/2007Palai..22..554C). doi:10.2110/palo.2006.p06-087r (https://doi.org/10.2110%2Fpalo.2006.p06-087r). JSTOR 27670451 (https://www.jstor.org/stable/27670451). S2CID 86197149 (https://api.semanticscholar.org/CorpusID:86197149). Retrieved 29 August 2020.
38. " "The Best of all Mothers" Maiasaura peeblesorum" (https://web.archive.org/web/20220616015020/http://bioweb.uwlax.edu/bio203/s2014/fischbac_sama/interactions.htm). *bioweb.uwlax.edu/*. University of Wisconsin-La Crosse. Archived from the original (http://bioweb.uwlax.edu/bio203/s2014/fischbac_sama/interactions.htm) on 16 June 2022. Retrieved 22 March 2021.

39. Chin, Karen; Gill, Bruce D. (1996). "Dinosaurs, Dung Beetles, and Conifers: Participants in a Cretaceous Food Web". *PALAIOS*. **11** (3): 280–285. Bibcode:1996Palai..11..280C (https://ui.adsabs.harvard.edu/abs/1996Palai..11..280C). doi:10.2307/3515235 (https://doi.org/10.2307%2F3515235). ISSN 0883-1351 (https://search.worldcat.org/issn/0883-1351). JSTOR 3515235 (https://www.jstor.org/stable/3515235) – via JSTOR.
40. Hunter, John P.; Janis, Christine M. (15 May 2026). "Tooth wear in juvenile and adult hadrosaurs: implications for parental care in Maiasaura" (https://www.sciencedirect.com/science/article/pii/S0031018226001707). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **690** 113707. doi:10.1016/j.palaeo.2026.113707 (https://doi.org/10.1016%2Fj.palaeo.2026.113707). Retrieved 25 April 2026 – via Elsevier Science Direct.
41. "Using math to examine the sex differences in dinosaurs" (https://www.sciencedaily.com/releases/2020/08/200826200704.htm).
42. "Statistical analysis reveals differences between dinosaur sexes" (https://www.upi.com/Science_News/2020/08/27/Statistical-analysis-reveals-differences-between-dinosaur-sexes/6841598538634/).
43. "Faunal Turnover, Migration, and Evolution," Trexler (2001); page 306.
44. Sullivan, R. M.; Lucas, S. G. (2006). "The Kirtlandian land-vertebrate "age"–faunal composition, temporal position and biostratigraphic correlation in the nonmarine Upper Cretaceous of western North America". *New Mexico Museum of Natural History and Science Bulletin*. **35**: 7–29.
45. Jack Horner, James Gorman and Jeri D. Walton, *Maia: A Dinosaur Grows Up*, published by Museum of the Rockies, all illustrations by Doug Henderson (1985).
46. "Dinosaur!" (https://www.imdb.com/fr/title/tt0439543/). *IMDB*. Retrieved 30 April 2026.
47. *Digging Dinosaurs*, produced by WHYY-TV Philadelphia (https://archive.org/details/DiggingDinosaurs), aired in 1987 (*archive.org*/*Internet Archive* added date 8 December, 2023)
48. " "The Infinite Voyage" The Great Dinosaur Hunt" (https://www.imdb.com/title/tt0205982/). *IMDB*. Retrieved 30 April 2026.
49. *thisisyellowstone.com*, "Yellowstone Geology (https://thisisyellowstone.com/yellowstone-geology/)", list of geology documentaries by Earthtalk Studios.

Bibliography

- Dodson, Peter & Britt, Brooks & Carpenter, Kenneth & Forster, Catherine A. & Gillette, David D. & Norell, Mark A. & Olshevsky, George & Parrish, J. Michael & Weishampel, David B. *The Age of Dinosaurs*. Publications International, LTD. p. 116-117. ISBN 0-7853-0443-6.
- Horner, Jack and Gorman, James. (1988). *Digging Dinosaurs: The Search that Unraveled the Mystery of Baby Dinosaurs*, Workman Publishing Co.
- Lehman, T. M., 2001, Late Cretaceous dinosaur provinciality: In: Mesozoic Vertebrate Life, edited by Tanke, D. H., and Carpenter, K., Indiana University Press, pp. 310–328.
- Palmer, D., ed. (1999). *The Marshall Illustrated Encyclopedia of Dinosaurs and Prehistoric Animals*. London: Marshall Editions. p. 148. ISBN 1-84028-152-9.
- Trexler, D., 2001, Two Medicine Formation, Montana: geology and fauna: In: Mesozoic Vertebrate Life, edited by Tanke, D. H., and Carpenter, K., Indiana University Press, pp. 298–309.

Retrieved from "https://en.wikipedia.org/w/index.php?title=Maiasaura&oldid=1379507349"

5

Kuşların Kökeni

Tüylü dinozorlar ve dinozorlardan kuşlara geçiş

- Feathered dinosaur
- Archaeopteryx
- Origin of birds

WIKIPEDIA 25
25 years of the free encyclopedia

Feathered dinosaur

A **feathered dinosaur** is any species of dinosaur possessing feathers, including all species of birds. In recent decades, evidence has accumulated that many non-avian dinosaur species also possessed feathers or feather-like structures in some shape or form. The extent to which feathers or feather-like structures were present in dinosaurs as a whole is a subject of ongoing debate and research.

Life restoration of the feathered *Wulong* with colors inferred from preserved melanosomes

It has been suggested that feathers had originally functioned as thermal insulation, as it remains their function in the down feathers of infant birds prior to their eventual modification in birds into structures that support flight.

Since scientific research began on dinosaurs in the early 1800s, they were generally thought to be closely related to modern reptiles such as lizards. The word *dinosaur* itself, coined in 1842 by paleontologist Richard Owen, comes from the Greek for 'terrible lizard'. That view began to shift during the so-called dinosaur renaissance in scientific research in the late 1960s; by the mid-1990s, significant evidence had emerged that dinosaurs were much more closely related to birds, which descended directly from an earlier group of theropod dinosaurs.[1]

Birds are the best-known feathered dinosaurs

Knowledge of the origin of feathers developed as new fossils were discovered throughout the 2000s and the 2010s, and technology enabled scientists to study fossils more closely. Among non-avian dinosaurs, feathers or feather-like integument have been discovered in dozens of genera via direct and indirect fossil evidence.[2] Although the vast majority of feather discoveries have been in coelurosaurian theropods, feather-like integument has also been discovered in at least three ornithischians, suggesting that feathers may have been present on the last common ancestor of the Ornithoscelida, a dinosaur group including both theropods and ornithischians.[3] It is possible that feathers first developed in even earlier archosaurs, in light of the discovery of vaned feathers in pterosaurs.[4][5] Fossil feathers from the dinosaur *Sinosauropteryx* contain traces of beta-proteins (formerly called beta-keratins), confirming that early feathers had a composition similar to that of feathers in modern birds.[6] Crocodilians also possess beta keratin similar to those of birds, which suggests that they evolved from common ancestral genes.[7][8]

History of research

Early

Shortly after the 1859 publication of Charles Darwin's *On the Origin of Species*, the British biologist Thomas Henry Huxley proposed that birds were descendants of dinosaurs. He compared the skeletal structure of *Compsognathus*, a small theropod dinosaur, and the "first bird" *Archaeopteryx lithographica* (both of which were found in the Upper Jurassic Bavarian limestone of Solnhofen). He showed that, apart from its hands and feathers, *Archaeopteryx* was quite similar to *Compsognathus*. Thus *Archaeopteryx* represents a transitional fossil. In 1868, he published *On the Animals which are most nearly intermediate between Birds and Reptiles*, which made that case.[9][10]

The Berlin *Archaeopteryx*

The first restoration of a feathered dinosaur was Huxley's depiction in 1876 of a feathered *Compsognathus*, made to accompany a bird evolution lecture he delivered in New York, in which he speculated that the aforementioned dinosaur might have had feathers.[11]

Dinosaur renaissance

A century later, during the dinosaur renaissance, paleoartists began to create modern restorations of highly active dinosaurs. In 1969, Robert T. Bakker drew a running *Deinonychus*. His student Gregory S. Paul depicted non-avian maniraptoran dinosaurs with feathers and protofeathers, starting in the late 1970s.[12]

Fossil discoveries

The first known specimen of *Archaeopteryx*, on the basis of which the genus was named, was an isolated feather, although whether or not it belongs to *Archaeopteryx* has been controversial.[13][14] One of the earliest discoveries of possible feather impressions by non-avian dinosaurs is a trace fossil (*Fulicopus lyellii*) of the 195–199 million year old Portland Formation in the northeastern United States. Gierlinski (1996, 1997, 1998) and Kundrát (2004) have interpreted traces between two footprints in this fossil as feather impressions from the belly of a squatting dilophosaurid. Although some reviewers have raised questions about the naming and interpretation of this fossil, if correct, that early Jurassic fossil is the oldest known evidence of feathers, almost 30 million years older than the next-oldest-known evidence.[15][16][17]

Cast in Japan of a resting trace from Massachusetts, which was argued to have been made by a theropod like *Dilophosaurus* and to include feather impressions around the belly (arrow), but that has been questioned.

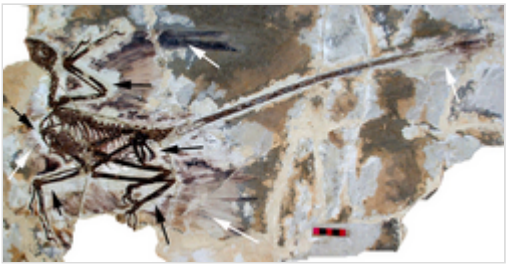
Sinosauropteryx fossil, the first fossil of a definitively non-avialan dinosaur with feathers

The most important discoveries at Liaoning have been a host of feathered dinosaur fossils, with a steady stream of new finds filling in the picture of the dinosaur–bird connection and adding more to theories of the evolutionary development of feathers and flight. Turner *et al*. (2007) reported quill knobs from an ulna of *Velociraptor mongoliensis*, and these are strongly correlated with large and well-developed secondary feathers.[18]

A nesting *Citipati osmolskae* specimen, at the AMNH

Behavioural evidence, in the form of an oviraptorosaur on its nest, showed another link with birds. Its forearms were folded, like those of a bird.[19] Although no feathers were preserved, it is likely that these would have been present to insulate eggs and juveniles.[20]

Not all of the Chinese fossil discoveries proved valid however. In 1999, a supposed fossil of an apparently feathered dinosaur named *Archaeoraptor liaoningensis*, also found in Liaoning, turned out to be a forgery. Comparing the photograph of the specimen with another find, Chinese paleontologist Xu Xing came to the conclusion that it was composed of two portions of different fossil animals. His claim made *National Geographic* review their research and they too came to the same conclusion.[21]



Fossil of *Microraptor gui* includes impressions of feathered wings (see arrows)

In 2011, samples of amber were discovered to contain preserved feathers from 75 to 80 million years ago during the Cretaceous Period, with evidence that they were from both dinosaurs and birds. Initial analysis suggests that some of the feathers were used for insulation, and not flight.[22][23] More complex feathers were revealed to have variations in coloration similar to modern birds, while simpler protofeathers were predominantly dark. Only 11 specimens are currently known. The specimens are too rare to be broken open to study their melanosomes (pigment-bearing organelles), but there are plans for using non-destructive high-resolution X-ray imaging.[24] Melanosomes produce colouration in feathers; as differently-shaped melanosomes produce different colours, subsequent research on melanosomes preserved in feathered dinosaur specimens has led to reconstructions of the life appearance of several dinosaur species. These include *Anchiornis*,[25] *Sinosauropteryx*,[26] *Microraptor*,[27] and *Archaeopteryx*.[14]

In 2016, the discovery was announced of a feathered dinosaur tail preserved in amber that is estimated to be 99 million years old. Lida Xing, a researcher from the China University of Geosciences in Beijing, found the specimen at an amber market in Myanmar. It is the first definitive discovery of dinosaur material in amber.[28][29][30][31]

Current knowledge

Non-avian dinosaur species preserved with evidence of feathers

Several non-avian dinosaurs are now known to have been feathered. Direct evidence of feathers exists for several species. In all examples, the evidence described consists of feather impressions, except those genera inferred to have had feathers based on skeletal or chemical evidence, such as the presence of quill knobs (the anchor points for wing feathers on the forelimb) or a pygostyle (the fused vertebrae at the tail tip which often supports large feathers).[32]

Fossil of *Sinornithosaurus millenii*, the first evidence of feathers in dromaeosaurids

Primitive feather types

Integumentary structures that gave rise to the feathers of birds are seen in the dorsal spines of reptiles and fish. A similar stage in their evolution to the complex coats of birds and mammals can be observed in living reptiles such as iguanas and *Gonocephalus* agamids. Feather structures are thought to have proceeded from simple hollow filaments through several stages of increasing complexity, ending with the large, deeply rooted feathers with strong pens (rachis), barbs and barbules that birds display today.[33]

Cast of a *Caudipteryx* fossil with feather impressions and stomach content

According to Prum's (1999) proposed model, at stage I, the follicle originates with a cylindrical epidermal depression around the base of the feather papilla. The first feather resulted when undifferentiated tubular follicle collar developed out of the old keratinocytes being pushed out. At stage II, the inner, basilar layer of the follicle collar differentiated into longitudinal barb ridges with unbranched keratin filaments, while the thin peripheral layer of the collar became the deciduous sheath, forming a tuft of unbranched barbs with a basal calamus. Stage III consists of two developmental novelties, IIIa and IIIb, as either could have occurred first. Stage IIIa involves helical displacement of barb ridges arising within the collar. The barb ridges on the anterior midline of the follicle fuse together, forming the rachis. The creation of a posterior barb locus follows, giving an indeterminate number of barbs. This resulted in a feather with a symmetrical, primarily branched structure with a rachis and unbranched barbs. In stage IIIb, barbules paired within the peripheral barbule plates of the barb ridges, create branched barbs with rami and barbules. This resulting feather is one with a tuft of branched barbs without a rachis. At stage IV, differentiated distal and proximal barbules produce a closed, pennaceous vane (a contour feather). A closed vane develops when pennulae on the distal barbules form a hooked shape to attach to the simpler proximal barbules of the adjacent barb. Stage V developmental novelties gave rise to additional structural diversity in the closed pennaceous feather. Here, asymmetrical flight feathers, bipinnate plumulaceous feathers, filoplumes, powder down, and bristles evolved.[34]

Some evidence suggests that the original function of simple feathers was insulation. In particular, preserved patches of skin in large, derived, tyrannosauroids show scutes, while those in smaller, more primitive, forms show feathers. This may indicate that the larger forms had complex skins, with both scutes and filaments, or that tyrannosauroids may be like rhinos and elephants, having filaments at birth and then losing them as they developed to maturity.[35] An adult *Tyrannosaurus rex* weighed about as much as an African elephant. If large tyrannosauroids were endotherms, they would have needed to radiate heat efficiently.[36] This is due to the different structural properties of feathers compared to fur.[37]

Fossil cast of a *Sinornithosaurus millenii*

Some evidence also suggests that more derived feather types may have served as insulation. For instance, a study of oviraptorid pennaceous wing feathers and nesting posture suggests that elongated wing feathers evidently may have served to fill gaps between brooding individuals' insulatory body chamber and the outside environment. This "wall" of wing feathers could have shielded eggs from temperature extremes.[38]

Jinfengopteryx elegans fossil

There is an increasing body of evidence that supports the display hypothesis, which states that early feathers were colored and increased reproductive success.[39][40] Coloration could have provided the original adaptation of feathers, implying that all later functions of feathers, such as thermoregulation and flight, were co-opted.[39] This hypothesis has been supported by the discovery of pigmented feathers in multiple species.[41][42][43][44] Supporting the display hypothesis is the fact that fossil feathers have been observed in a ground-dwelling herbivorous dinosaur clade, making it unlikely that feathers functioned as predatory tools or as a means of flight.[45] Additionally, some specimens have iridescent feathers.[46] Pigmented and iridescent feathers may have provided greater attractiveness to mates, providing enhanced reproductive success when compared to non-colored feathers.[47] Current research shows that it is plausible that theropods would have had the visual acuity necessary to see the displays. In a study by Stevens (2006), the binocular field of view for *Velociraptor* has been estimated to be 55 to 60 degrees, which is about that of modern owls. Visual acuity for *Tyrannosaurus* has been predicted to be anywhere from about that of humans to 13 times that of humans.[48] Paleontological and evolutionary developmental studies show that feathers or feather-like structures were converting back to scales.[49][50][51]

The idea that precursors of feathers appeared before they were co-opted for insulation is already stated in Gould and Vrba (1982).[52] The original benefit might have been metabolic. Feathers are largely made of the keratin protein complex, which has disulfide bonds between amino acids that give it stability and elasticity. The metabolism of amino acids containing sulfur can be toxic; however, if the sulfur amino acids are not catabolized as the final products of urea or uric acid but used for the synthesis of keratin instead, the release of hydrogen sulfide is extremely reduced or avoided. For an organism whose

metabolism works at high internal temperatures of 40 °C (104 °F) or greater, it can be extremely important to prevent the excess production of hydrogen sulfide. This hypothesis could be consistent with the need for high metabolic rate of theropod dinosaurs.[53][54]

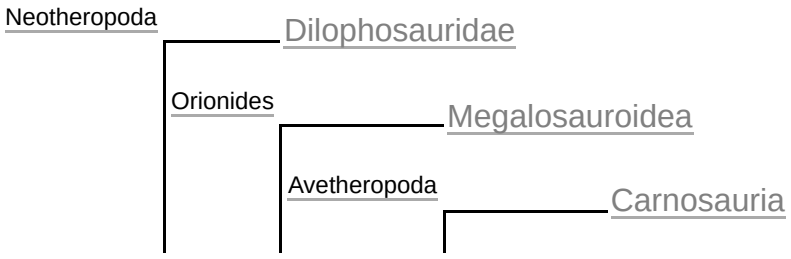
The point is not known with certainty in archosaur phylogeny that the earliest simple "protofeathers" arose, as well as whether they arose once or independently multiple times. Filamentous structures are clearly present in pterosaurs,[55] and long, hollow quills have been reported in specimens of the ornithischian dinosaurs *Psittacosaurus* and *Tianyulong*[56][57] although there has been disagreement.[58][59][60] In 2009, Xu et al. noted that the hollow, unbranched, stiff integumentary structures found on a specimen of *Beipiaosaurus* were strikingly similar to the integumentary structures of *Psittacosaurus* and pterosaurs. They suggested that all of these structures may have been inherited from a common ancestor much earlier in the evolution of archosaurs, possibly in an ornithodire from the Middle Triassic or earlier.[61] More recently, findings in Russia of the basal neornithischian *Kulindadromeus* report that although the lower leg and tail seemed to be scaled, "varied integumentary structures were found directly associated with skeletal elements, supporting the hypothesis that simple filamentous feathers, as well as compound feather-like structures comparable to those in theropods, were widespread amongst the whole dinosaur clade."[62] In contrast, a 2016 study published in the Journal of Geology suggested that the integumentary structures found on *Kulindadromeus* and *Psittacosaurus* may be highly deformed scales rather than filamentous feathers.[58]

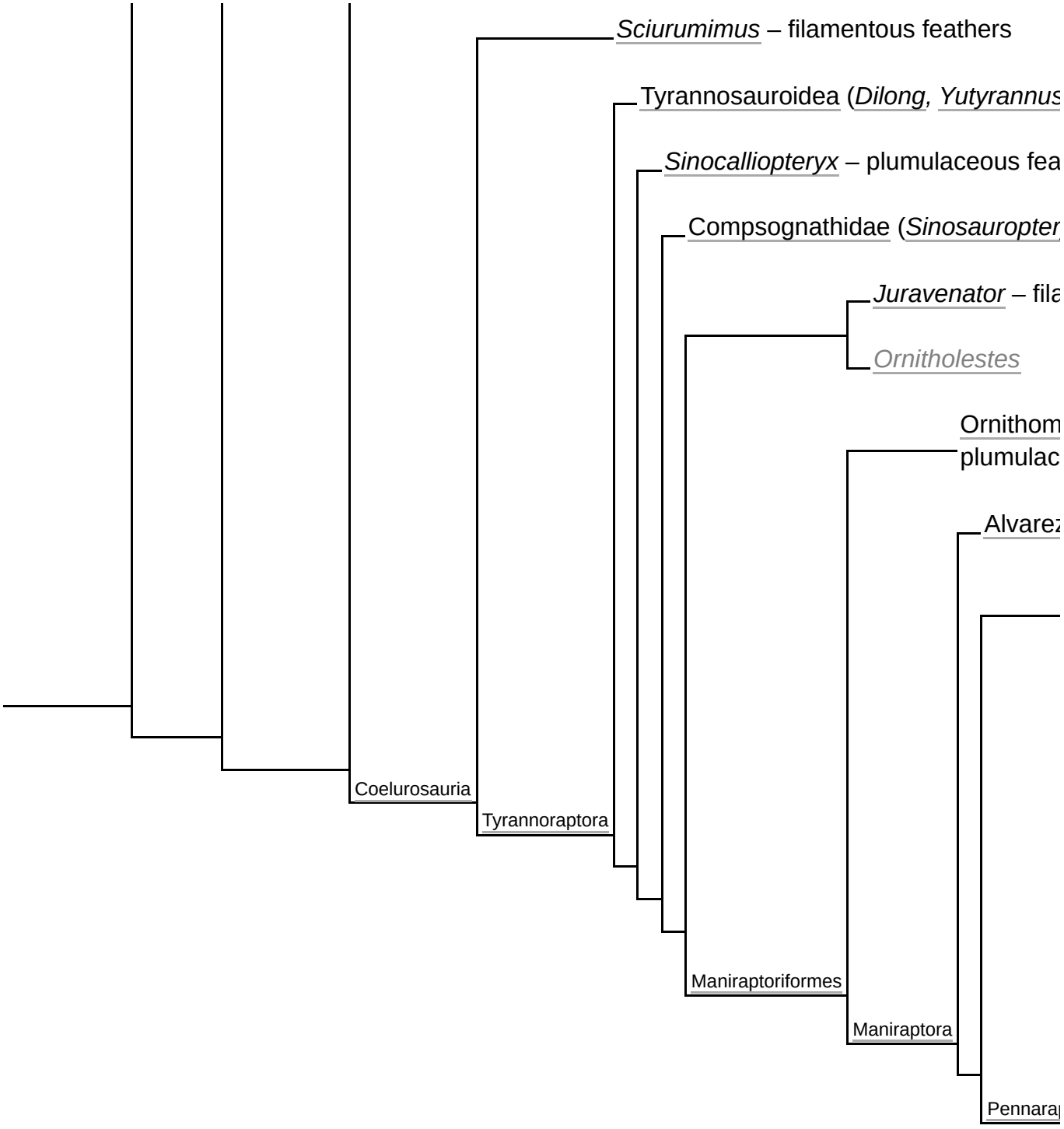
Display feathers are also known from dinosaurs that are very primitive members of the bird lineage, or Avialae. The most primitive example is *Epidexipteryx*, which had a short tail with extremely long, ribbon-like feathers. Oddly enough, the fossil does not preserve wing feathers, suggesting that *Epidexipteryx* was either secondarily flightless, or that display feathers evolved before flight feathers in the bird lineage.[63] Plumaceous feathers are found in nearly all lineages of Theropoda common in the northern hemisphere, and pennaceous feathers are attested as far down the tree as the Ornithomimosauria. The fact that only adult *Ornithomimus* had wing-like structures suggests that pennaceous feathers evolved for mating displays.[64]

Phylogeny and inference of feathers in other dinosaurs

This technique, called phylogenetic bracketing, can also be used to infer the type of feathers a species may have had, since the developmental history of feathers is now reasonably well-known. All feathered species had filamentaceous or plumaceous (downy) feathers, with pennaceous feathers found among the more bird-like groups. The following cladogram is adapted from Godefroit *et al.*, 2013.[65]

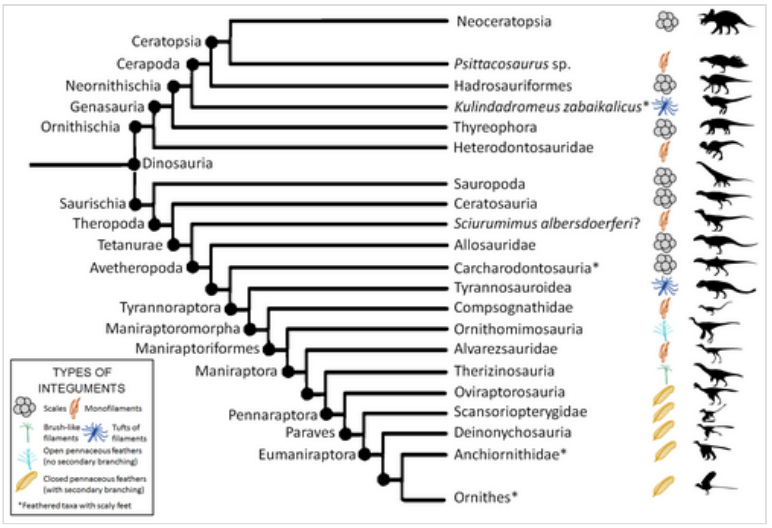
Grey denotes a clade that is not known to contain any feathered specimen at the time of writing, some of which have fossil evidence of scales. The presence or lack of feathered specimens in a given clade does not confirm that all members in a clade have the specified integument, unless corroborated with representative fossil evidence within clade members.



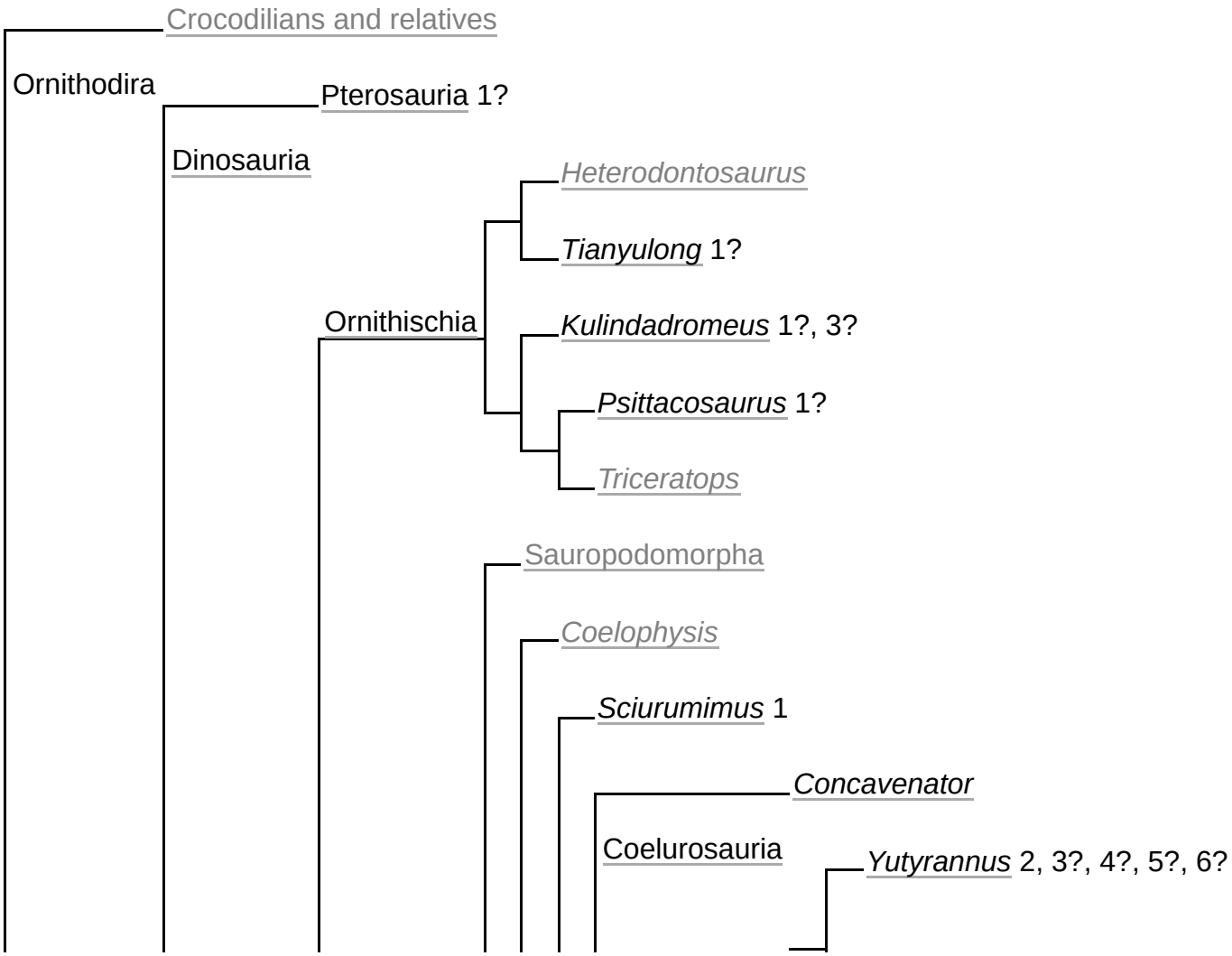

Sciurumimus – filamentous feathers
Tyrannosauroidea (*Dilong*, *Yutyrannus*
Sinocalliopteryx – plumulaceous fea
Compsognathidae (*Sinosauropter*
Juravenator – fila
Ornitholestes
Ornithom
plumulac
Alvarez
Coelurosauria
Tyrannoraptora
Maniraptoriformes
Maniraptora
Pennara

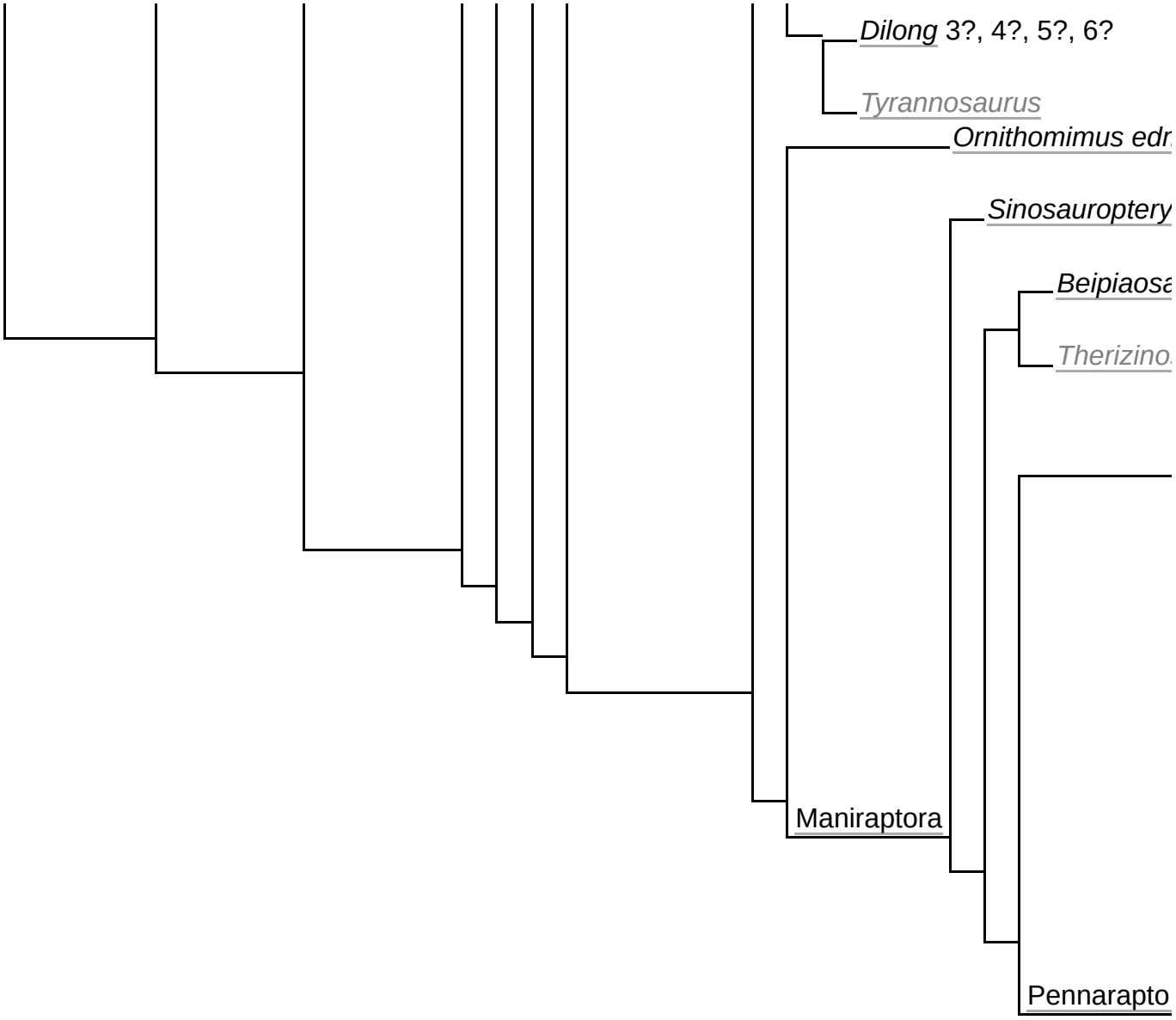
The following cladogram is from Xu (2020).[66]

1. Slender monofilamentous integument
2. Broad monofilamentous integument
3. Basally joining filamentous feather
4. Basally joining shafter filamentous feather
5. Radially branched shafted filamentous feather
6. Bilaterally branched filamentous feather
7. Basally joining branched filamentous feather
8. Basally joining membranous-based filamentous feather
9. Symmetrical open-vaned feather
10. Symmetrical close-vaned feather
11. Asymmetrical close-vaned feather
12. Proximally ribbon-like close-vaned feather
13. Rachis-dominant close-vaned feather



Cladogram showing distribution of feathers in Dinosauria, as of 2019. The groups that are marked with scales did not necessarily lack feathers but simply have never been found with feather impressions.





See also

- Origin of birds – Central question in evolutionary biology of birds
- Dinosaur Discovery Museum

References

1. Brown, Joseph W.; Van Tuinen, M. (2011). "Evolving Perceptions on the Antiquity of the Modern Avian Tree". *Living Dinosaurs*. pp. 306–324. doi:10.1002/9781119990475.ch12 (https://doi.org/10.1002%2F9781119990475.ch12). ISBN 978-1-119-99047-5.
2. Farago, Jason (7 March 2019). "T. Rex Like You Haven't Seen Him: With Feathers" (https://www.nytimes.com/2019/03/07/arts/design/t-rex-exhibition-american-museum-of-natural-history.html). *The New York Times*. Retrieved 7 March 2019.
3. Baron, Matthew G.; Norman, David B.; Barrett, Paul M. (23 March 2017). "A new hypothesis of dinosaur relationships and early dinosaur evolution". *Nature*. **543** (7646): 501–506. Bibcode:2017Natur.543..501B (https://ui.adsabs.harvard.edu/abs/2017Natur.543..501B). doi:10.1038/nature21700 (https://doi.org/10.1038%2Fnature21700). PMID 28332513 (https://pubmed.ncbi.nlm.nih.gov/28332513). S2CID 205254710 (https://api.semanticscholar.org/CorpusID:205254710).
4. Michael Benton, A colourful view of the origin of dinosaur feathers. *Nature* 604, 630-631 (2022)
5. Cincotta, A., Nicolaï, M., Campos, H.B.N. et al. Pterosaur melanosomes support signalling functions for early feathers. Nature 604, 684–688 (2022). https://doi.org/10.1038/s41586-022-04622-3
6. Slater, Tiffany S.; Edwards, Nicholas P.; Webb, Samuel M.; Zhang, Fucheng; McNamara, Maria E. (21 September 2023). "Preservation of corneous β-proteins in Mesozoic feathers" (https://www.nature.com/articles/s41559-023-02177-8). *Nature Ecology & Evolution*. **7** (10): 1706–1713. doi:10.1038/s41559-023-02177-8 (https://doi.org/10.1038%2Fs41559-023-02177-8). ISSN 2397-334X (https://search.worldcat.org/issn/2397-334X). S2CID 262125827 (https://api.semanticscholar.org/CorpusID:262125827).
7. Greenwold, Matthew J.; Sawyer, Roger H. (September 2013). "Molecular evolution and expression of archosaurian β-keratins: Diversification and expansion of archosaurian β-keratins and the origin of feather β-keratins". *Journal of Experimental Zoology Part B: Molecular and Developmental Evolution*. **320** (6): 393–405. doi:10.1002/jez.b.22514 (https://doi.org/10.1002%2Fjez.b.22514). PMID 23744807 (https://pubmed.ncbi.nlm.nih.gov/23744807).
8. Alibardi, L.; Knapp, L. W.; Sawyer, R. H. (1 June 2006). "Beta-keratin localization in developing alligator scales and feathers in relation to the development and evolution of feathers" (https://europepmc.org/article/med/17784647). *Journal of Submicroscopic Cytology and Pathology*. **38** (2–3): 175–192. ISSN 1122-9497 (https://search.worldcat.org/issn/1122-9497). PMID 17784647 (https://pubmed.ncbi.nlm.nih.gov/17784647).
9. Huxley, T.H. (1868). "On the animals which are most nearly intermediate between birds and reptiles" (https://www.biodiversitylibrary.org/page/22247928). *Annals and Magazine of Natural History*. 4th. **2**: 66–75.
10. Foster, Michael; Lankester, E. Ray 1898–1903. *The scientific memoirs of Thomas Henry Huxley*. 4 vols and supplement. London: Macmillan.
11. Huxley, Thomas Henry (1877). "The Hypothesis of Evolution. The Neutral and the Favourable Evidence." (https://archive.org/details/americanaddress01unkngoog). In Huxley, Thomas Henry (ed.). *American Addresses, With A Lecture on Biology*. New York: D. Appleton And Company.
12. "1970s: Dinosaurs Redesigned" (https://web.archive.org/web/20191222045109/https://paleoartistry.webs.com/1970s.htm). *Paleoartistry*. Archived from the original (https://paleoartistry.webs.com/1970s.htm) on 22 December 2019. Retrieved 15 June 2017.

13. Kaye, T. G.; Pittman, M.; Mayr, G.; Schwarz, D.; Xu, X. (2019). "Detection of lost calamus challenges identity of isolated *Archaeopteryx* feather" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6362147). *Scientific Reports*. **9** (1): 1182. Bibcode:2019NatSR...9.1182K (https://ui.adsabs.harvard.edu/abs/2019NatSR...9.1182K). doi:10.1038/s41598-018-37343-7 (https://doi.org/10.1038%2Fs41598-018-37343-7). PMC 6362147 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6362147). PMID 30718905 (https://pubmed.ncbi.nlm.nih.gov/30718905).
14. Carney, R. M.; Tischlinger, H.; Shawkey, M. D. (2020). "Evidence corroborates identity of isolated fossil feather as a wing covert of *Archaeopteryx*" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7528088). *Scientific Reports*. **10** (1): 15593. Bibcode:2020NatSR..1015593C (https://ui.adsabs.harvard.edu/abs/2020NatSR..1015593C). doi:10.1038/s41598-020-65336-y (https://doi.org/10.1038%2Fs41598-020-65336-y). PMC 7528088 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7528088). PMID 32999314 (https://pubmed.ncbi.nlm.nih.gov/32999314).
15. Gierliński, G. (1996). "Feather-like impressions in a theropod resting trace from the Lower Jurassic of Massachusetts". *Museum of Northern Arizona Bulletin*. **60**: 179–184.
16. Kundrát, Martin (15 July 2004). "When did theropods become feathered?-evidence for pre-archaeopteryx feathery appendages". *Journal of Experimental Zoology Part B: Molecular and Developmental Evolution*. **302B** (4): 355–364. doi:10.1002/jez.b.20014 (https://doi.org/10.1002%2Fjez.b.20014). PMID 15287100 (https://pubmed.ncbi.nlm.nih.gov/15287100).
17. **Tetrapod Zoology** (https://scienceblogs.com/tetrapodzoology/2009/09/07/dyzio-feathered-dilophosaur), ScienceBlogs
18. Turner, A. H.; Makovicky, P. J.; Norell, M. A. (21 September 2007). "Feather Quill Knobs in the Dinosaur Velociraptor" (https://doi.org/10.1126%2Fscience.1145076). *Science*. **317** (5845): 1721. Bibcode:2007Sci...317.1721T (https://ui.adsabs.harvard.edu/abs/2007Sci...317.1721T). doi:10.1126/science.1145076 (https://doi.org/10.1126%2Fscience.1145076). PMID 17885130 (https://pubmed.ncbi.nlm.nih.gov/17885130).
19. Norell, M. A.; Clark, J. M.; Chiappe, L. M.; Dashzeveg, D. (1995). "A nesting dinosaur". *Nature*. **378** (6559): 774–776. Bibcode:1995Natur.378..774N (https://ui.adsabs.harvard.edu/abs/1995Natur.378..774N). doi:10.1038/378774a0 (https://doi.org/10.1038%2F378774a0). S2CID 4245228 (https://api.semanticscholar.org/CorpusID:4245228).
20. Hopp, Thomas P.; Orsen, Mark J. (2004). "Dinosaur Brooding Behavior and the Origin of Flight Feathers" (https://books.google.com/books?id=L-ZFPTKroVoC&pg=PA234). In Currie, Philip J.; Koppelhus, Eva B.; Shugar, Martin A.; Wright, Joanna L. (eds.). *Feathered Dragons: Studies on the Transition from Dinosaurs to Birds*. Indiana University Press. pp. 234–250. ISBN 978-0-253-34373-4.
21. "Transcript: The Dinosaur that Fooled the World" (https://www.bbc.co.uk/science/horizon/2001/dinofooltrans.shtml). BBC. Retrieved 22 December 2006.
22. Bock, Walter J. (August 2000). "Explanatory History of the Origin of Feathers1". *American Zoologist*. **40** (4): 478–485. doi:10.1668/0003-1569(2000)040[0478:ehotoo]2.0.co;2 (https://doi.org/10.1668%2F0003-1569%282000%29040%5B0478%3Aehotoo%5D2.0.co%3B2). S2CID 198155047 (https://api.semanticscholar.org/CorpusID:198155047).
23. Chung, Emily (12 September 2011). "Dinosaur feathers found in Alberta amber" (https://www.cbc.ca/news/science/dinosaur-feathers-found-in-alberta-amber-1.1086765). Canadian Broadcasting Corporation. Retrieved 16 September 2011.
24. Switek, Brian (15 September 2011). "Amber inclusions showcase prehistoric feathers" (https://doi.org/10.1038%2Fnews.2011.539). *Nature*. doi:10.1038/news.2011.539 (https://doi.org/10.1038%2Fnews.2011.539).

25. Li, Q.; Gao, K.-Q.; Vinther, J.; Shawkey, M. D.; Clarke, J. A.; D'Alba, L.; Meng, Q.; Briggs, D. E. G.; Prum, R. O. (2010). "Plumage Color Patterns of an Extinct Dinosaur" (http://doc.rero.ch/record/210394/files/PAL_E4402.pdf) (PDF). *Science*. **327** (5971): 1369–1372. Bibcode:2010Sci...327.1369L (https://ui.adsabs.harvard.edu/abs/2010Sci...327.1369L). doi:10.1126/science.1186290 (https://doi.org/10.1126%2Fscience.1186290). PMID 20133521 (https://pubmed.ncbi.nlm.nih.gov/20133521). S2CID 206525132 (https://api.semanticscholar.org/CorpusID:206525132).
26. Smithwick, F. M.; Nicholls, R.; Cuthill, I. C.; Vinther, J. (2017). "Countershading and Stripes in the Theropod Dinosaur Sinosauropteryx Reveal Heterogeneous Habitats in the Early Cretaceous Jehol Biota" (https://doi.org/10.1016%2Fj.cub.2017.09.032). *Current Biology*. **27** (21): 3337–3343.e2. doi:10.1016/j.cub.2017.09.032 (https://doi.org/10.1016%2Fj.cub.2017.09.032). hdl:1983/8ee95b15-5793-42ad-8e57-da6524635349 (https://hdl.handle.net/1983%2F8ee95b15-5793-42ad-8e57-da6524635349). PMID 29107548 (https://pubmed.ncbi.nlm.nih.gov/29107548).
27. Li, Q.; Gao, K.-Q.; Meng, Q.; Clarke, J. A.; Shawkey, M. D.; D'Alba, L.; Pei, R.; Ellison, M.; Norell, M. A.; Vinther, J. (2012). "Reconstruction of *Microraptor* and the Evolution of Iridescent Plumage" (https://web.archive.org/web/20180831072106/https://pdfs.semanticscholar.org/58f0/0bcda8da889ba578e944e91eb8f336e934db.pdf) (PDF). *Science*. **335** (6073): 1215–1219. Bibcode:2012Sci...335.1215L (https://ui.adsabs.harvard.edu/abs/2012Sci...335.1215L). doi:10.1126/science.1213780 (https://doi.org/10.1126%2Fscience.1213780). PMID 22403389 (https://pubmed.ncbi.nlm.nih.gov/22403389). S2CID 206537426 (https://api.semanticscholar.org/CorpusID:206537426). Archived from the original (https://pdfs.semanticscholar.org/58f0/0bcda8da889ba578e944e91eb8f336e934db.pdf) (PDF) on 31 August 2018.
28. St. Fleur, Nicholas (8 December 2016). "That Thing With Feathers Trapped in Amber? It Was a Dinosaur Tail" (https://www.nytimes.com/2016/12/08/science/dinosaur-feathers-amber.html). *The New York Times*. Retrieved 8 December 2016.
29. Romey, Kristin (8 December 2016). "First Dinosaur Tail Found Preserved in Amber" (https://web.archive.org/web/20210225114107/https://www.nationalgeographic.com/animals/article/feathered-dinosaur-tail-amber-theropod-myanmar-burma-cretaceous). *National Geographic Society*. Archived from the original (https://www.nationalgeographic.com/animals/article/feathered-dinosaur-tail-amber-theropod-myanmar-burma-cretaceous) on 25 February 2021. Retrieved 12 December 2016.
30. Rincon, Paul (8 December 2016). " 'Beautiful' dinosaur tail found preserved in amber" (https://www.bbc.com/news/science-environment-38224564). BBC News. Retrieved 8 December 2016.
31. Xing, Lida; McKellar, Ryan C.; Xu, Xing; Li, Gang; Bai, Ming; Persons, W. Scott; Miyashita, Tetsuto; Benton, Michael J.; Zhang, Jianping; Wolfe, Alexander P.; Yi, Qiru; Tseng, Kuowei; Ran, Hao; Currie, Philip J. (December 2016). "A Feathered Dinosaur Tail with Primitive Plumage Trapped in Mid-Cretaceous Amber" (https://doi.org/10.1016%2Fj.cub.2016.10.008). *Current Biology*. **26** (24): 3352–3360. doi:10.1016/j.cub.2016.10.008 (https://doi.org/10.1016%2Fj.cub.2016.10.008). hdl:1983/d3a169c7-b776-4be5-96af-6053c23fa52b (https://hdl.handle.net/1983%2Fd3a169c7-b776-4be5-96af-6053c23fa52b). PMID 27939315 (https://pubmed.ncbi.nlm.nih.gov/27939315).
32. Stephen L. Brusatte; Graeme T. Lloyd; Steve C. Wang; Mark A. Norell (2014). "Gradual assembly of avian body plan culminated in rapid rates of evolution across the dinosaur-bird transition" (https://doi.org/10.1016%2Fj.cub.2014.08.034). *Current Biology*. **24** (20): 2386–2392. doi:10.1016/j.cub.2014.08.034 (https://doi.org/10.1016%2Fj.cub.2014.08.034). PMID 25264248 (https://pubmed.ncbi.nlm.nih.gov/25264248).
33. Prum, R. & Brush A.H.; Brush (2002). "The evolutionary origin and diversification of feathers". *The Quarterly Review of Biology*. **77** (3): 261–295. doi:10.1086/341993 (https://doi.org/10.1086%2F341993). PMID 12365352 (https://pubmed.ncbi.nlm.nih.gov/12365352). S2CID 6344830 (https://api.semanticscholar.org/CorpusID:6344830).

34. Prum, R (1999). "Development and evolutionary origin of feathers". *Journal of Experimental Zoology*. **285** (4): 291–306. doi:10.1002/(SICI)1097-010X(19991215)285:4<291::AID-JEZ1>3.0.CO;2-9 (https://doi.org/10.1002%2F%28SICI%291097-010X%2819991215%29285%3A4%3C291%3A%3AAID-JEZ1%3E3.0.CO%3B2-9). PMID 10578107 (https://pubmed.ncbi.nlm.nih.gov/10578107).
35. Xu, Xing; Norell, Mark A.; Kuang, Xuewen; Wang, Xiaolin; Zhao, Qi; Jia, Chengkai (October 2004). "Basal tyrannosauroids from China and evidence for protofeathers in tyrannosauroids" (http://doc.rero.ch/record/15283/files/PAL_E2582.pdf) (PDF). *Nature*. **431** (7009): 680–684. Bibcode:2004Natur.431..680X (https://ui.adsabs.harvard.edu/abs/2004Natur.431..680X). doi:10.1038/nature02855 (https://doi.org/10.1038%2Fnature02855). PMID 15470426 (https://pubmed.ncbi.nlm.nih.gov/15470426). S2CID 4381777 (https://api.semanticscholar.org/CorpusID:4381777).
36. Norell, M. Xu, X. (2005) "The Varieties of Tyrannosaurs" (https://www.naturalhistorymag.com/htmlsite/master.html?https://www.naturalhistorymag.com/htmlsite/0505/0505_feature.html), Natural History Magazine, May 2005.
37. Dawson, T. J.; Maloney, S. K. (2013). "Fur versus feathers: the different roles of red kangaroo fur and emu feathers in thermoregulation in the Australian arid zone" (https://doi.org/10.1071%2Fam04145). *Australian Mammalogy*. **26** (2): 145. doi:10.1071/am04145 (https://doi.org/10.1071%2Fam04145).
38. Orsen, M. J.; Hopp, T. P. (2004). *Feathered dragons: studies on the transition from dinosaurs to birds*. Indiana University Press. pp. 234–350.
39. Dimond, C. C.; R. J. Cabin; J. S. Brooks (2011). "Feathers, Dinosaurs, and Behavioral Cues: Defining the Visual Display Hypothesis for the Adaptive Function of Feathers in Non-Avian Theropods". *BIOS*. **82** (3): 58–63. doi:10.1893/011.082.0302 (https://doi.org/10.1893%2F011.082.0302). S2CID 98221211 (https://api.semanticscholar.org/CorpusID:98221211).
40. Sumida, S. S.; C. A. Brochu (2000). "Phylogenetic Context for the Origin of Feathers" (https://doi.org/10.1093%2Ficb%2F40.4.486). *American Zoologist*. **40** (4): 485–503. doi:10.1093/icb/40.4.486 (https://doi.org/10.1093%2Ficb%2F40.4.486).
41. Lingham-Soliar, T. (2011). "The evolution of the feather: Sinosauropteryx, a colourful tail". *Journal of Ornithology*. **152** (3): 567–577. doi:10.1007/s10336-010-0620-y (https://doi.org/10.1007%2Fs10336-010-0620-y). S2CID 29827649 (https://api.semanticscholar.org/CorpusID:29827649).
42. Vinther, J.; D. E. G. Briggs; R. O. Prum & V. Saranathan (2008). "The colour of fossil feathers" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2610093). *Biology Letters*. **4** (5): 522–525. doi:10.1098/rsbl.2008.0302 (https://doi.org/10.1098%2Frsbl.2008.0302). PMC 2610093 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2610093). PMID 18611841 (https://pubmed.ncbi.nlm.nih.gov/18611841).
43. Zhang, F. C.; S. L. Kearns; P. J. Orr; M. J. Benton; Z. H. Zhou; D. Johnson; X. Xu; X. L. Wang (2010). "Fossilized melanosomes and the colour of Cretaceous dinosaurs and birds" (http://oro.open.ac.uk/22432/2/41064696.pdf) (PDF). *Nature*. **463** (7284): 1075–1078. Bibcode:2010Natur.463.1075Z (https://ui.adsabs.harvard.edu/abs/2010Natur.463.1075Z). doi:10.1038/nature08740 (https://doi.org/10.1038%2Fnature08740). PMID 20107440 (https://pubmed.ncbi.nlm.nih.gov/20107440). S2CID 205219587 (https://api.semanticscholar.org/CorpusID:205219587).
44. Li, Q.; Gao, K.; Vinther, J.; Shawkey, M. D.; Clarke, J. A.; D'Alba, L.; Meng, Q. (2010). "Plumage Color Patterns of an Extinct Dinosaur" (http://doc.rero.ch/record/210394/files/PAL_E4402.pdf) (PDF). *Science*. **327** (5971): 1369–1372. Bibcode:2010Sci...327.1369L (https://ui.adsabs.harvard.edu/abs/2010Sci...327.1369L). doi:10.1126/science.1186290 (https://doi.org/10.1126%2Fscience.1186290). PMID 20133521 (https://pubmed.ncbi.nlm.nih.gov/20133521). S2CID 206525132 (https://api.semanticscholar.org/CorpusID:206525132).

45. Zelenitsky, Darla K.; Therrien, Francois; Erickson, Gregory M.; DeBuhr, Christopher L.; Kobayashi, Yoshitsugu; Eberth, David A.; Hadfield, Frank (2012). "Feathered Non-Avian Dinosaurs From North America Provide Insight into Wing Origins". *Science*. **338** (6106): 510–514. Bibcode:2012Sci...338..510Z (https://ui.adsabs.harvard.edu/abs/2012Sci...338..510Z). doi:10.1126/science.1225376 (https://doi.org/10.1126%2Fscience.1225376). PMID 23112330 (https://pubmed.ncbi.nlm.nih.gov/23112330). S2CID 2057698 (https://api.semanticscholar.org/CorpusID:2057698).
46. Li, Q. G.; K. Q. Gao; Q. J. Meng; M. D. Shawkey; L. D'Alba; R. Pei; M. Ellison; M. A. Norell; J. Vinther (2012). "Reconstruction of *Microraptor* and the Evolution of Iridescent Plumage". *Science*. **335** (6073): 1215–1219. Bibcode:2012Sci...335.1215L (https://ui.adsabs.harvard.edu/abs/2012Sci...335.1215L). doi:10.1126/science.1213780 (https://doi.org/10.1126%2Fscience.1213780). PMID 22403389 (https://pubmed.ncbi.nlm.nih.gov/22403389). S2CID 206537426 (https://api.semanticscholar.org/CorpusID:206537426).
47. Dimond, C. C.; Cabin, R. J.; Brooks, J. S. (2011). "Feathers, Dinosaurs, and Behavioral Cues: Defining the Visual Display Hypothesis for the Adaptive Function of Feathers in NonAvian Theropods". *BIOS*. **82** (3): 58–63. doi:10.1893/011.082.0302 (https://doi.org/10.1893%2F011.082.0302). S2CID 98221211 (https://api.semanticscholar.org/CorpusID:98221211).
48. Rauhut, Oliver W. M.; Foth, Christian; Tischlinger, Helmut; Norell, Mark A. (17 July 2012). "Exceptionally preserved juvenile megalosauroid theropod dinosaur with filamentous integument from the Late Jurassic of Germany" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3406838). *Proceedings of the National Academy of Sciences of the United States of America*. **109** (29): 11746–11751. Bibcode:2012PNAS..10911746R (https://ui.adsabs.harvard.edu/abs/2012PNAS..10911746R). doi:10.1073/pnas.1203238109 (https://doi.org/10.1073%2Fpnas.1203238109). PMC 3406838 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3406838). PMID 22753486 (https://pubmed.ncbi.nlm.nih.gov/22753486).
49. Dhouailly, Danielle (20 April 2009). "A new scenario for the evolutionary origin of hair, feather, and avian scales" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2736124). *Journal of Anatomy*. **214** (4): 587–606. doi:10.1111/j.1469-7580.2008.01041.x (https://doi.org/10.1111%2Fj.1469-7580.2008.01041.x). PMC 2736124 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2736124). PMID 19422430 (https://pubmed.ncbi.nlm.nih.gov/19422430).
50. Bell, Phil R.; Campione, Nicolás E.; Persons, W. Scott; Currie, Philip J.; Larson, Peter L.; Tanke, Darren H.; Bakker, Robert T. (30 June 2017). "Tyrannosauroid integument reveals conflicting patterns of gigantism and feather evolution" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5493735). *Biology Letters*. **13** (6) 20170092. doi:10.1098/rsbl.2017.0092 (https://doi.org/10.1098%2Frsbl.2017.0092). PMC 5493735 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5493735). PMID 28592520 (https://pubmed.ncbi.nlm.nih.gov/28592520).
51. Zheng, Xiaoting; Zhou, Zhonghe; Wang, Xiaoli; Zhang, Fucheng; Zhang, Xiaomei; Wang, Yan; Wei, Guangjin; Wang, Shuo; Xu, Xing (15 March 2013). "Hind Wings in Basal Birds and the Evolution of Leg Feathers" (https://www.science.org/doi/10.1126/science.1228753). *Science*. **339** (6125): 1309–1312. Bibcode:2013Sci...339.1309Z (https://ui.adsabs.harvard.edu/abs/2013Sci...339.1309Z). doi:10.1126/science.1228753 (https://doi.org/10.1126%2Fscience.1228753). ISSN 0036-8075 (https://search.worldcat.org/issn/0036-8075). PMID 23493711 (https://pubmed.ncbi.nlm.nih.gov/23493711). S2CID 206544531 (https://api.semanticscholar.org/CorpusID:206544531).

52. Gould, Stephen J.; Vrba, Elisabeth S. (1982). "Exaptation: a missing term in the science of form" (https://web.archive.org/web/20220518011405/https://www.insead.edu/sites/default/files/assets/faculty-personal-site/vibha-gaba/documents/Gould%20%26%20Vrba_Exaptation.pdf) (PDF). *Paleobiology*. **8** (1): 4–15. Bibcode:1982Pbio....8....4G (https://ui.adsabs.harvard.edu/abs/1982Pbio....8....4G). doi:10.1017/S0094837300004310 (https://doi.org/10.1017%2FS0094837300004310). S2CID 86436132 (https://api.semanticscholar.org/CorpusID:86436132). Archived from the original (https://www.insead.edu/sites/default/files/assets/faculty-personal-site/vibha-gaba/documents/Gould%20%26%20Vrba_Exaptation.pdf) (PDF) on 18 May 2022. Retrieved 25 March 2022.
53. Reichholf, J. H. (1996). "Die Feder, die Mauser und der Ursprung der Vögel. Ein neure Sicht zur Evolution der Vögel" [The feather, the moult and the origin of the birds. A new perspective on the evolution of birds]. *Archaeopteryx* (in German). **14**: 27–38.
54. Bock, Walter J. (1 August 2000). "Explanatory History of the Origin of Feathers". *American Zoologist*. **40** (4): 478–485. doi:10.1093/icb/40.4.478 (https://doi.org/10.1093%2Ficb%2F40.4.478).
55. Yang, Z.; Jiang, B.; McNamara, M. E.; Kearns, S. L.; Pittman, M.; Kaye, T. G.; Orr, P. J.; Xu, X.; Benton, M. J. (2019). "Pterosaur integumentary structures with complex feather-like branching" (https://research-information.bris.ac.uk/ws/files/184677374/Main_Text_revised_mjb.pdf) (PDF). *Nature Ecology & Evolution*. **3** (1): 24–30. doi:10.1038/s41559-018-0728-7 (https://doi.org/10.1038%2Fs41559-018-0728-7). hdl:1983/1f7893a1-924d-4cb3-a4bf-c4b1592356e9 (https://hdl.handle.net/1983%2F1f7893a1-924d-4cb3-a4bf-c4b1592356e9). PMID 30568282 (https://pubmed.ncbi.nlm.nih.gov/30568282). S2CID 56480710 (https://api.semanticscholar.org/CorpusID:56480710).
56. Mayr, Gerald; Peters, Stefan; Plodowski, Gerhard; Vogel, Olaf (2002). "Bristle-like integumentary structures at the tail of the horned dinosaur *Psittacosaurus*". *Naturwissenschaften*. **89** (8): 361–365. Bibcode:2002NW.....89..361M (https://ui.adsabs.harvard.edu/abs/2002NW.....89..361M). doi:10.1007/s00114-002-0339-6 (https://doi.org/10.1007%2Fs00114-002-0339-6). PMID 12435037 (https://pubmed.ncbi.nlm.nih.gov/12435037). S2CID 17781405 (https://api.semanticscholar.org/CorpusID:17781405).
57. Zheng, Xiao-Ting; You, Hai-Lu; Xu, Xing; Dong, Zhi-Ming (2009). "An Early Cretaceous heterodontosaurid dinosaur with filamentous integumentary structures". *Nature*. **458** (7236): 333–336. Bibcode:2009Natur.458..333Z (https://ui.adsabs.harvard.edu/abs/2009Natur.458..333Z). doi:10.1038/nature07856 (https://doi.org/10.1038%2Fnature07856). PMID 19295609 (https://pubmed.ncbi.nlm.nih.gov/19295609). S2CID 4423110 (https://api.semanticscholar.org/CorpusID:4423110).
58. Qiang, Ji; Xuri, Wang; Yannan, Ji; Ball, Black (2016). "Feathers or scales" (http://www.cqvip.com/qk/96560x/201604/671056951.html). *Journal of Geology* (4): 535–544.
59. Unwin, David (28 September 2020). "No protofeathers on pterosaurs" (https://www.nature.com/articles/s41559-020-01308-9). *Nature Ecology and Evolution*. **4** (12): 1590–1591. doi:10.1038/s41559-020-01308-9 (https://doi.org/10.1038%2Fs41559-020-01308-9). PMID 32989266 (https://pubmed.ncbi.nlm.nih.gov/32989266). S2CID 222168569 (https://api.semanticscholar.org/CorpusID:222168569).
60. Barrett, Paul M.; Evans, David C.; Campione, Nicolás E. (30 June 2015). "Evolution of dinosaur epidermal structures" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4528472). *Biology Letters*. **11** (6) 20150229. doi:10.1098/rsbl.2015.0229 (https://doi.org/10.1098%2Frsbl.2015.0229). PMC 4528472 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4528472). PMID 26041865 (https://pubmed.ncbi.nlm.nih.gov/26041865).

61. Xu, X.; Zheng, X.; You, H. (2009). "A new feather type in a nonavian theropod and the early evolution of feathers" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2630069). *Proceedings of the National Academy of Sciences*. **106** (3): 832–4. Bibcode:2009PNAS..106..832X (https://ui.adsabs.harvard.edu/abs/2009PNAS..106..832X). doi:10.1073/pnas.0810055106 (https://doi.org/10.1073%2Fpnas.0810055106). PMC 2630069 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2630069). PMID 19139401 (https://pubmed.ncbi.nlm.nih.gov/19139401).
62. Godefroit, P; Sinitsa, S; Dhouailly, D; Bolotsky, Y; Sizov, A (2013). *Feather-like structures and scales in a Jurassic neornithischian dinosaur from Siberia* (https://web.archive.org/web/20210301132741/https://www.naturalsciences.be/en/rbins-biblio/12537). Archived from the original (https://www.naturalsciences.be/en/rbins-biblio/12537) on 1 March 2021. Retrieved 6 January 2020.
63. Zhang, Fucheng; Zhou, Zhonghe; Xu, Xing; Wang, Xiaolin; Sullivan, Corwin (2008). "A bizarre Jurassic maniraptoran from China with elongate ribbon-like feathers" (https://www.nature.com/articles/npre.2008.2326.1.pdf) (PDF). *Nature*. **455** (7216): 1105–1108. Bibcode:2008Natur.455.1105Z (https://ui.adsabs.harvard.edu/abs/2008Natur.455.1105Z). doi:10.1038/nature07447 (https://doi.org/10.1038%2Fnature07447). PMID 18948955 (https://pubmed.ncbi.nlm.nih.gov/18948955). S2CID 4362560 (https://api.semanticscholar.org/CorpusID:4362560).
64. Zelenitsky, D. K.; Therrien, F.; Erickson, G. M.; DeBuhr, C. L.; Kobayashi, Y.; Eberth, D. A.; Hadfield, F. (25 October 2012). "Feathered Non-Avian Dinosaurs from North America Provide Insight into Wing Origins". *Science*. **338** (6106): 510–514. Bibcode:2012Sci...338..510Z (https://ui.adsabs.harvard.edu/abs/2012Sci...338..510Z). doi:10.1126/science.1225376 (https://doi.org/10.1126%2Fscience.1225376). PMID 23112330 (https://pubmed.ncbi.nlm.nih.gov/23112330). S2CID 2057698 (https://api.semanticscholar.org/CorpusID:2057698).
65. Godefroit, Pascal; Cau, Andrea; Hu, Dong-Yu; Escuillié, François; Wu, Wenhao; Dyke, Gareth (2013). "A Jurassic avialan dinosaur from China resolves the early phylogenetic history of birds". *Nature*. **498** (7454): 359–362. Bibcode:2013Natur.498..359G (https://ui.adsabs.harvard.edu/abs/2013Natur.498..359G). doi:10.1038/nature12168 (https://doi.org/10.1038%2Fnature12168). PMID 23719374 (https://pubmed.ncbi.nlm.nih.gov/23719374). S2CID 4364892 (https://api.semanticscholar.org/CorpusID:4364892).
66. Xu, Xing (2020), "Filamentous Integuments in Nonavialan Theropods and Their Kin: Advances and Future Perspectives for Understanding the Evolution of Feathers", in Foth, Christian; Rauhut, Oliver W. M. (eds.), *The Evolution of Feathers: From Their Origin to the Present*, Fascinating Life Sciences, Cham: Springer International Publishing, pp. 67–78, doi:10.1007/978-3-030-27223-4_5 (https://doi.org/10.1007%2F978-3-030-27223-4_5), ISBN 978-3-030-27223-4, S2CID 216384668 (https://api.semanticscholar.org/CorpusID:216384668)

External links

- Dinosaur colour and feathers (https://dinocolour.blogs.bristol.ac.uk/), *University of Bristol*, 12 February 2019
- Are Birds Really Dinosaurs? (https://ucmp.berkeley.edu/diapsids/avians.html), DinoBuzz, UC Berkeley.
- Feathered Dinosaurs (https://www.bbc.co.uk/programmes/b099v33p) on *In Our Time* at the BBC

Retrieved from "https://en.wikipedia.org/w/index.php?title=Feathered_dinosaur&oldid=1374680850"

WIKIPEDIA
25 years of the free encyclopedia

Archaeopteryx

Archaeopteryx (/ˌɑːrkiːˈɒptərɪks/ ⓘ; lit. 'ancient wing'), sometimes referred to by its German name, "***Urvogel***" (lit. 'Primeval Bird') is an extinct genus of bird-like dinosaurs. The genus name derives from the Ancient Greek ἀρχαῖος (*archaîos*), meaning 'ancient', and πτέρυξ (*ptérux*), meaning 'feather, wing'. Between the late 19th century and the early 21st century, *Archaeopteryx* was generally accepted by palaeontologists and popular reference books as the oldest known bird (member of the group Avialae).[3] Older potential avialans have since been identified, including *Anchiornis*, *Xiaotingia*, *Aurornis*,[4] and *Baminornis*.[5]

Archaeopteryx lived in the Late Jurassic around 150 million years ago, in what is now southern Germany, during a time when Europe was an archipelago of islands in a shallow warm tropical sea, much closer to the equator than it is now. Similar in size to a Eurasian magpie, with the largest individuals possibly attaining the size of a raven,[6] the largest species of *Archaeopteryx* could grow to about 50 cm (20 in) in length. Despite their small size, broad wings, and inferred ability to fly or glide, *Archaeopteryx* had more in common with other small Mesozoic dinosaurs than with modern birds. In particular, they shared the following features with the dromaeosaurids and troodontids: jaws with sharp teeth, three fingers with claws, a long bony tail, hyperextensible second toes ("killing claw"), feathers (which also suggest warm-bloodedness), and various features of the skeleton.[7][8]

These features make *Archaeopteryx* a clear candidate for a transitional fossil between non-avian dinosaurs and avian dinosaurs (birds).[9][10] Thus, *Archaeopteryx* plays an important role, not only in the study of the origin of birds, but in the study of dinosaurs. It was named from a single feather in 1861,[11] the identity of which has been controversial.[12][13] That same year, the first complete specimen of *Archaeopteryx* was announced. Over the years, twelve more fossils of *Archaeopteryx* have surfaced. Despite variation among these fossils, most experts regard all the remains that have been discovered as belonging to a single species or at least genus, although this is still debated.[14]

Archaeopteryx

Temporal range: Late Jurassic (Tithonian),

The Berlin *Archaeopteryx* specimen (*A. siemensii*)

Scientific classification

Kingdom:	Animalia
Phylum:	Chordata
Class:	Reptilia
Clade:	Dinosauria
Clade:	Saurischia
Clade:	Theropoda
Clade:	Paraves
Family:	†Archaeopterygidae
Genus:	†*Archaeopteryx* Meyer, 1861 (conserved name)

Most of these 14 fossils include impressions of feathers. Because these feathers are of an advanced form (flight feathers), these fossils are evidence that the evolution of feathers began before the Late Jurassic.[15] The type specimen of *Archaeopteryx* was discovered just two years after Charles Darwin published *On the Origin of Species*. *Archaeopteryx* seemed to confirm Darwin's theories and has since become a key piece of evidence for the origin of birds, the transitional fossils debate, and confirmation of evolution. *Archaeopteryx* was long considered to be the beginning of the evolutionary tree of birds. However, in recent years, the discovery of several small, feathered dinosaurs has created a mystery for palaeontologists, raising questions about which animals are the ancestors of modern birds and which are their relatives.[16]

History of discovery

The single feather

Over the years, fourteen body fossil specimens of *Archaeopteryx* have been found. All of the fossils come from the limestone deposits, quarried for centuries, near Solnhofen, Germany. These quarries excavate sediments from the Solnhofen Limestone formation and related units.[17][18] The initial specimen was the first dinosaur to be discovered with feathers.

The initial discovery, a single feather, was unearthed in 1860 or 1861 and described in 1861 by Hermann von Meyer.[19] It is now in the Natural History Museum of Berlin. Though it was the initial holotype, there were indications that it might not have been from the same animal as the body fossils.[11] In 2019 it was reported that laser imaging had revealed the structure of the quill (which had not been visible since some time after the feather was described), and that the feather was inconsistent with the morphology of all other *Archaeopteryx* feathers known, leading to the conclusion that it originated from another dinosaur.[12] This conclusion was challenged in

Type species

†***Archaeopteryx lithographica***

Meyer, 1861 (conserved name)

Other species

- †***A. siemensii***

  Dames, 1897
- †***A. albersdoerferi***

  Kundrat et al., 2018

Synonyms[1][2]

Genus synonymy

- *Griphosaurus*

  Wagner, 1862 (rejected name)
- *Griphornis*

  Woodward, 1862 (rejected name)
- *Archaeornis*

  Petronievics, 1917
- *Jurapteryx*

  Howgate, 1984
- *Wellnhoferia*?

  Elżanowski, 2001
- *Ostromia*?

  Foth & Rauhut, 2017
- *Alcmonavis*?

  Rauhut, Tischlinger & Foth, 2019

Species synonymy

- *Griphosaurus problematicus*

  Wagner, 1862 vide Woodward 1862

  nomen rejectum
- *Griphornis longicaudatus*

  Owen 1862 vide Woodward 1862

  nomen rejectum
- *Griphosaurus longicaudatus*

  (Owen 1862 vide Woodward 1862)

  Owen 1862 vide Brodkorb 1863

  nomen rejectum
- *Archaeopteryx macrura*

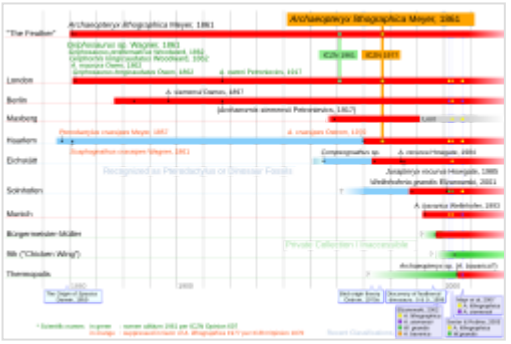
  Owen, 1862 (rejected name)

2020 as being unlikely; the feather was identified on the basis of morphology as most likely having been an upper major primary covert feather.[13]

The first skeleton, known as the **London Specimen** (BMNH 37001),[20] was unearthed in 1861 near Langenaltheim, Germany, and perhaps given to local physician Karl Häberlein in return for medical services. He then sold it for £700 (roughly £83,000 in 2020[21]) to the Natural History Museum in London, where it remains.[17] Missing most of its head and neck, it was described in 1863 by Richard Owen as *Archaeopteryx macrura*, allowing for the possibility it did not belong to the same species as the feather. In the subsequent fourth edition of his *On the Origin of Species*,[22] Charles Darwin said the following about the discovery.[23]

> Until quite recently these authors might have maintained, and some have maintained, that the whole class of birds came suddenly into existence during the Eocene period; but now we know, on the authority of Professor Owen, that a bird certainly lived during the deposition of the upper greensand; and still more recently, that strange bird, the *Archaeopteryx*, with a long lizard-like tail, bearing a pair of feathers on each joint, and with its wings furnished with two free claws, has been discovered in the oolitic slates of Solnhofen. Hardly any recent discovery shows more forcibly than this how little we as yet know of the former inhabitants of the world.

- *Archaeopteryx siemensii*?
  (Dames, 1897)
- *Archaeornis siemensii*
  (Dames, 1897) Peteronievics vide Petroneivics & Woodward 1917
- *Archaeopteryx owenii*
  Petronievics, 1917 (rejected name)
- *Archaeopteryx recurva*
  Howgate, 1984
- *Jurapteryx recurva*
  (Howgate, 1984) Howgate 1985
- *Archaeopteryx bavarica*
  Wellnhofer, 1993
- ?*Wellnhoferia grandis*
  Elżanowski, 2001
- ?*Ostromia crassipes*
  (Meyer, 1857)
- ?*Alcmonavis poeschli*
  Rauhut, Tischlinger & Foth, 2019



Timeline of *Archaeopteryx* discoveries until 2007

The genus name derives from the Ancient Greek ἀρχαῖος (*archaîos*), meaning 'ancient', and πτέρυξ (*ptérux*), meaning 'feather, wing'. Meyer suggested this in his description. At first he referred to a single feather which appeared to resemble a modern bird's remex (wing feather), but he had heard of and been shown a rough sketch of the London specimen, to which he referred as a "*Skelett eines mit ähnlichen Federn bedeckten Tieres*" ('skeleton of an animal covered in similar feathers').

Since then, twelve specimens have been recovered:

The **Berlin Specimen** (HMN 1880/81) was discovered in 1874 or 1875 on the Blumenberg near Eichstätt, Germany, by farmer Jakob Niemeyer. He sold this precious fossil for the money to buy a cow in 1876, to innkeeper Johann Dörr, who again sold it to Ernst Otto Häberlein, the son of K. Häberlein. Placed on sale between 1877 and 1881, with potential buyers including O. C. Marsh of Yale University's Peabody Museum, it eventually was bought for 20,000 Goldmark by the Berlin's Natural History Museum, where it now is displayed. The transaction was financed by Ernst Werner von Siemens, founder

of the company that bears his name.[17] Described in 1884 by Wilhelm Dames, it is the most complete specimen, and the first with a complete head. In 1897 it was named by Dames as a new species, *A. siemensii*; though often considered a synonym of *A. lithographica*, several 21st century studies have concluded that it is a distinct species which includes the Berlin, Munich, and Thermopolis specimens.[24][25]

Consisting of a torso, the **Maxberg Specimen** (S5) was discovered in 1956 near Langenaltheim; it was brought to the attention of the German paleontologist Florian Heller in 1958 and described by him in 1959. The specimen is missing its head and tail, although the rest of the skeleton is mostly intact. Although it was once exhibited at the Maxberg Museum in Solnhofen, it is currently missing. It belonged to Eduard Opitsch, who loaned it to the museum until 1974. After his death in 1991, it was discovered that the specimen was missing and may have been stolen or sold.[26]

Cast of the Maxberg Specimen

The **Haarlem Specimen** (TM 6428/29, also known as the *Teylers Specimen*) was discovered in 1855 near Riedenburg, Germany, and described as a *Pterodactylus crassipes* in 1857 by Meyer. It was reclassified in 1970 by John Ostrom and is currently located at the Teylers Museum in Haarlem, the Netherlands. It was the very first specimen found, but was incorrectly classified at the time. It is also one of the least complete specimens, consisting mostly of limb bones, isolated cervical vertebrae, and ribs. In 2017 it was named as a separate genus *Ostromia*, considered more closely related to *Anchiornis* from China.[27]

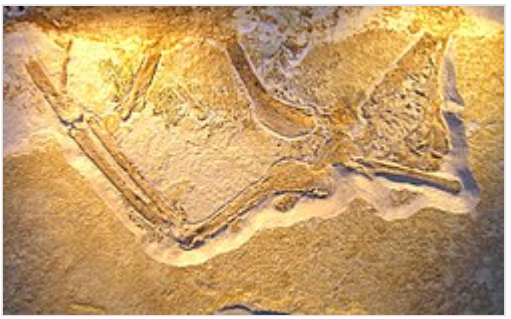
Eichstätt Specimen, once considered a distinct genus, *Jurapteryx*

The **Eichstätt Specimen** (JM 2257) was discovered in 1951 near Workerszell, Germany, and described by Peter Wellnhofer in 1974. Currently located at the Jura Museum in Eichstätt, Germany, it is the smallest known specimen and has the second-best head.[28] In 2007, Phil Senter classified *Jurapteryx*, *A. siemensii* and *A. bavarica* as junior synonyms of *A. lithographica*, suggesting that proposed diagnostic features of these taxa based on proportional and dental differences can be explained by ontogeny.[29]

The **Solnhofen Specimen** ('Unnumbered Specimen') was discovered in the 1970s near Eichstätt, Germany, and described in 1988 by Wellnhofer. Currently located at the Bürgermeister-Müller-Museum in Solnhofen, it originally was classified as *Compsognathus* by an amateur collector, the same mayor Friedrich Müller after which the museum is named. It is the largest specimen known and may belong to a separate genus and species, *Wellnhoferia grandis*. It is missing only portions of the neck, tail, backbone, and head.[30]

The **Munich Specimen** (BSP 1999 I 50, formerly known as the *Solenhofer-Aktien-Verein Specimen*) was discovered on 3 August 1992 near Langenaltheim and described in 1993 by Wellnhofer. It is currently located at the Paläontologisches Museum München in Munich, to which it was sold in 1999 for 1.9 million Deutschmark. What was initially believed to be a bony sternum turned out to be part of the

coracoid,[31] but a cartilaginous sternum may have been present. Only the front of its face is missing. It has been used as the basis for a distinct species, *A. bavarica*,[32] but more recent studies suggest it belongs to *A. siemensii*.[25]



Daiting Specimen, the holotype of *A. albersdoerferi*

An eighth, fragmentary specimen was discovered in 1990 in the younger Mörnsheim Formation at Daiting, Suevia. Therefore, it is known as the **Daiting Specimen**, and had been known since 1996 only from a cast, briefly shown at the Naturkundemuseum in Bamberg. The original was purchased by palaeontologist Raimund Albertsdörfer in 2009.[33] It was on display for the first time with six other original fossils of *Archaeopteryx* at the Munich Mineral Show in October 2009.[34] The Daiting Specimen was subsequently named *Archaeopteryx albersdoerferi* by Kundrat et al. (2018).[35][36] After a lengthy period in a closed private collection, it was moved to the Museum of Evolution at Knuthenborg Safaripark (Denmark) in 2022, where it has since been on display and also been made available for researchers.[37][38]

Bürgermeister-Müller ("chicken wing") Specimen

Another fragmentary fossil was found in 2000. It is in private possession and, since 2004, on loan to the Bürgermeister-Müller Museum in Solnhofen, so it is called the **Bürgermeister-Müller Specimen**; the institute itself officially refers to it as the "Exemplar of the families Ottman & Steil, Solnhofen". As the fragment represents the remains of a single wing of *Archaeopteryx*, it is colloquially known as "chicken wing".[39]

Details of the Wyoming Dinosaur Center *Archaeopteryx* (WDC-CSG-100)

Long in a private collection in Switzerland, the **Thermopolis Specimen** (WDC CSG 100) was discovered in Bavaria and described in 2005 by Mayr, Pohl, and Peters. Donated to the Wyoming Dinosaur Center in Thermopolis, Wyoming, it has the best-preserved head and feet; most of the neck and the lower jaw have not been preserved. The "Thermopolis" specimen was described on 2 December 2005 *Science* journal article as "A well-preserved *Archaeopteryx* specimen with theropod features"; it shows that *Archaeopteryx* lacked a reversed toe—a universal feature of birds—limiting its ability to perch on branches and implying a terrestrial or trunk-climbing lifestyle.[40] This has been interpreted as evidence of theropod ancestry. In 1988, Gregory S. Paul claimed to have found evidence of a hyperextensible second toe,[41] but this was not verified and accepted by other scientists until the Thermopolis specimen was described. "Until now, the feature was thought to belong only to the species' close relatives, the deinonychosaurs."[18] The Thermopolis Specimen was assigned to *Archaeopteryx siemensii* in 2007.[25] The specimen is considered to represent the most complete and best-preserved *Archaeopteryx* remains yet.[25]

The discovery of an eleventh specimen was announced in 2011; it was described in 2014. It is one of the more complete specimens, but is missing much of the skull and one forelimb. It is privately owned and has yet to be given a name.[42][43] Palaeontologists of LMU Munich studied the specimen, which revealed previously unknown features of the plumage, such as feathers on both the upper and lower legs and metatarsus, and the only preserved tail tip.[44][45]

The eleventh specimen

A twelfth specimen had been discovered by an amateur collector in 2010 at the Schamhaupten quarry, but the finding was only announced in February 2014.[46] It was scientifically described in 2018. It represents a complete and mostly articulated skeleton with skull. It is the only specimen lacking preserved feathers. It is from the Painten Formation and somewhat older than the other specimens.[47]

A thirteenth specimen, **SMNK-PAL 10,000**, was published in January 2025, this one from the Mörnsheim Formation. It preserves the right forelimb, shoulder, and fragments of the other limbs, with various features of the shoulder and forelimb resembling *Archaeopteryx* more than any other avialan within the Mörnsheim Formation. However, due to the fragmentary nature of this specimen, it cannot be assigned to a specific species within *Archaeopteryx*.[48]

The existence of a fourteenth specimen (the **Chicago specimen**) was first informally announced in 2024 by the Field Museum in Chicago, US. One of two specimens in an institution outside Europe, the specimen was originally identified in a private collection in Switzerland, and had been acquired by these collectors in 1990, prior to Germany's 2015 ban on exporting *Archaeopteryx* specimens. The specimen was acquired by the Field Museum in 2022, and went on public display in 2024 following two years of preparation.[49][50] In 2025, the paleornithologist Jingmai O'Connor and colleagues officially published a study describing this fourteenth specimen, reporting the first known tertials (specialized inner secondary flight feathers) and other novel features in *Archaeopteryx*.[14]

The Chicago *Archaeopteryx*

Authenticity

Beginning in 1985, an amateur group including astronomer Fred Hoyle and physicist Lee Spetner, published a series of papers claiming that the feathers on the Berlin and London specimens of *Archaeopteryx* were forged.[51][52][53][54] Their claims were repudiated by Alan J. Charig and others at the Natural History Museum in London.[55] Most of their supposed evidence for a forgery was based on unfamiliarity with the processes of lithification; for example, they proposed that, based on the difference in texture associated with the feathers, feather impressions were applied to a thin layer of cement,[52] without realizing that feathers themselves would have caused a textural difference.[55] They also misinterpreted

the fossils, claiming that the tail was forged as one large feather,[52] when visibly this is not the case.[55] In addition, they claimed that the other specimens of *Archaeopteryx* known at the time did not have feathers,[51][52] which is incorrect; the Maxberg and Eichstätt specimens have obvious feathers.[55]

They also expressed disbelief that slabs would split so smoothly, or that one half of a slab containing fossils would have good preservation, but not the counterslab.[51][53] These are common properties of Solnhofen fossils, because the dead animals would fall onto hardened surfaces, which would form a natural plane for the future slabs to split along and would leave the bulk of the fossil on one side and little on the other.[55]

Finally, the motives they suggested for a forgery are not strong, and are contradictory; one is that Richard Owen wanted to forge evidence in support of Charles Darwin's theory of evolution, which is unlikely given Owen's views toward Darwin and his theory. The other is that Owen wanted to set a trap for Darwin, hoping the latter would support the fossils so Owen could discredit him with the forgery; this is unlikely because Owen wrote a detailed paper on the London specimen, so such an action would certainly backfire.[56]

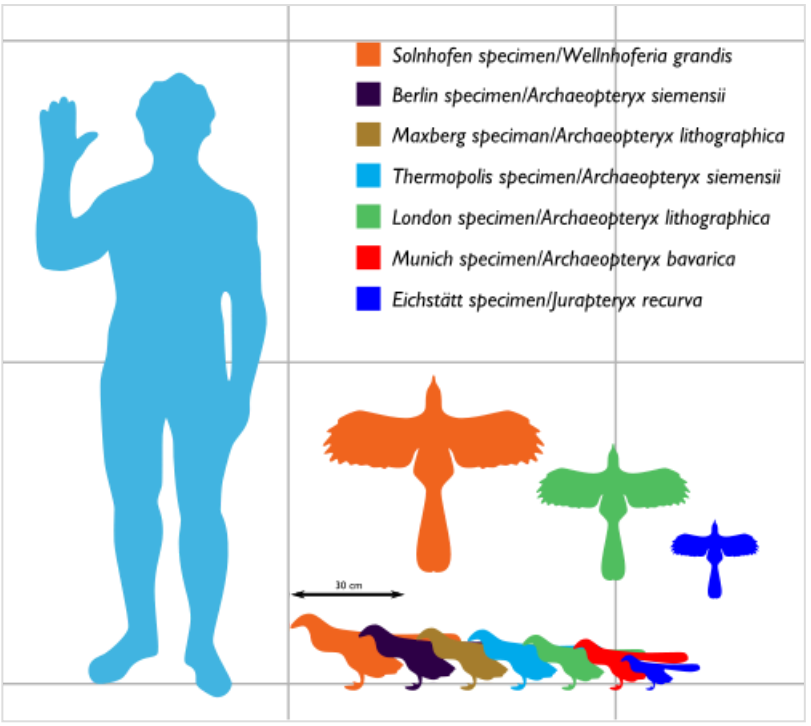
Charig et al. pointed to the presence of hairline cracks in the slabs running through both rock and fossil impressions, and mineral growth over the slabs that had occurred before discovery and preparation, as evidence that the feathers were original.[55] Spetner et al. then attempted to show that the cracks would have propagated naturally through their postulated cement layer,[57] but neglected to account for the fact that the cracks were old and had been filled with calcite, and thus were not able to propagate.[56] They also attempted to show the presence of cement on the London specimen through X-ray spectroscopy, and did find something that was not rock;[57] it was not cement either, and is most probably a fragment of silicone rubber left behind when moulds were made of the specimen.[56] Their suggestions have not been taken seriously by palaeontologists, as their evidence was largely based on misunderstandings of geology, and they never discussed the other feather-bearing specimens, which have increased in number since then. Charig et al. reported a discolouration: a dark band between two layers of limestone – they say it is the product of sedimentation.[55] It is natural for limestone to take on the colour of its surroundings and most limestones are coloured (if not colour banded) to some degree, so the darkness was attributed to such impurities.[58] They also mention that a complete absence of air bubbles in the rock slabs is further proof that the specimen is authentic.[55]

Description

Most of the specimens of *Archaeopteryx* that have been discovered come from the Solnhofen limestone in Bavaria, southern Germany, which is a Lagerstätte, a rare geological formation known for its highly detailed fossils laid down during the early Tithonian stage of the Jurassic period,[59] approximately 150.8–148.5 million years ago.[60]

Archaeopteryx was roughly the size of a raven,[6] with broad wings that were rounded at the ends and a long tail compared to its body length. It could reach up to 50 centimetres (1 ft 8 in) in body length and 70 centimetres (2 ft 4 in) in wingspan, with an estimated mass of 500 to 1,000 grams (18 to 35 oz).[6][61] *Archaeopteryx* feathers, although less documented than its other features, were very similar in structure to modern-day bird feathers.[59] Despite the presence of numerous avian features,[62] *Archaeopteryx* had

many non-avian theropod dinosaur characteristics. Unlike modern birds, *Archaeopteryx* had small teeth,[59] as well as a long bony tail, features which *Archaeopteryx* shared with other dinosaurs of the time.[63]

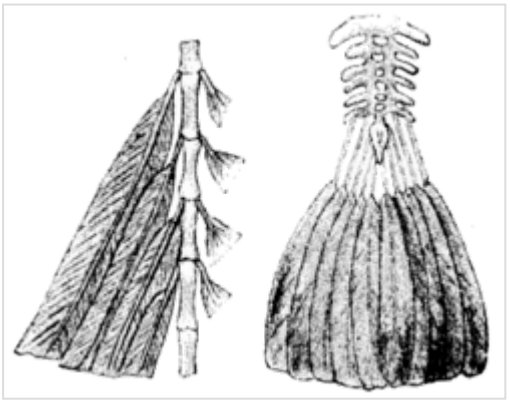


Specimens compared to a human in scale

Because it displays features common to both birds and non-avian dinosaurs, *Archaeopteryx* has often been considered a link between them.[59] In the 1970s, John Ostrom, following Thomas Henry Huxley's lead in 1868, argued that birds evolved within theropod dinosaurs and *Archaeopteryx* was a critical piece of evidence for this argument; it had several avian features, such as a wishbone, flight feathers, wings, and a partially reversed first toe along with dinosaur and theropod features. For instance, it has a long ascending process of the ankle bone, interdental plates, an obturator process of the ischium, and long chevrons in the tail. In particular, Ostrom found that *Archaeopteryx* was remarkably similar to the theropod family Dromaeosauridae.[64][65][66][67]

Archaeopteryx had three separate digits on each fore-leg each ending with a "claw". Few birds have such features. Some birds, such as ducks, swans, Jacanas (*Jacana* sp.), and the hoatzin (*Opisthocomus hoazin*), have them concealed beneath their leg-feathers.[68]

Plumage


Anatomical illustration comparing the "frond-tail" of *Archaeopteryx* with the "fan-tail" of a modern bird

Specimens of *Archaeopteryx* were most notable for their well-developed flight feathers. They were markedly asymmetrical and showed the structure of flight feathers in modern birds, with vanes given stability by a barb-barbule-barbicel arrangement.[69] The tail feathers were less asymmetrical, again in line with the situation in modern birds and also had firm vanes. The thumb did not yet bear a separately movable tuft of stiff feathers.

The body plumage of *Archaeopteryx* is less well-documented and has only been properly researched in the well-preserved Berlin specimen. Thus, as more than one species seems to be involved, the research into the Berlin specimen's feathers does not necessarily hold true for the rest of the species of *Archaeopteryx*. In the Berlin specimen, there are "trousers" of well-developed feathers on the legs; some of these feathers seem to have a basic contour feather structure, but are somewhat decomposed (they lack barbicels as in ratites).[70] In part they are firm and thus capable of supporting flight.[71]

A patch of pennaceous feathers is found running along its back, which was quite similar to the contour feathers of the body plumage of modern birds in being symmetrical and firm, although not as stiff as the flight-related feathers. Apart from that, the feather traces in the Berlin specimen are limited to a sort of "proto-down" not dissimilar to that found in the dinosaur *Sinosauropteryx*: decomposed and fluffy, and possibly even appearing more like fur than feathers in life (although not in their microscopic structure). These occur on the remainder of the body—although some feathers did not fossilize and others were obliterated during preparation, leaving bare patches on specimens—and the lower neck.[70]

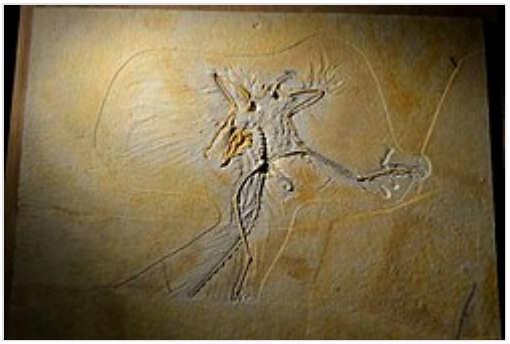
There is no indication of feathering on the upper neck and head. While these conceivably may have been nude, this may still be an artefact of preservation. It appears that most *Archaeopteryx* specimens became embedded in anoxic sediment after drifting some time on their backs in the sea—the head, neck and the tail are generally bent downward, which suggests that the specimens had just started to rot when they were embedded, with tendons and muscle relaxing so that the characteristic shape (death pose) of the fossil specimens was achieved.[72] This would mean that the skin already was softened and loose, which is bolstered by the fact that in some specimens the flight feathers were starting to detach at the point of embedding in the sediment. So it is hypothesized that the pertinent specimens moved along the sea bed in shallow water for some time before burial, the head and upper neck feathers sloughing off, while the more firmly attached tail feathers remained.[24]

Colouration

Artist's restoration illustrating one interpretation of Carney's study[73]

In 2011, graduate student Ryan Carney and colleagues performed the first colour study on an *Archaeopteryx* specimen.[73] Using scanning electron microscopy technology and energy-dispersive X-ray analysis, the team was able to detect the structure of melanosomes in the isolated feather specimen described in 1861. The resultant measurements were then compared to those of 87 modern bird species, and the original colour was calculated with a 95% likelihood to be black. The feather was determined to be black throughout, with heavier pigmentation in the distal tip. The feather studied was most probably a dorsal covert, which would have partly covered the primary feathers on the wings. The study does not mean that *Archaeopteryx* was entirely black, but suggests that it had some black colouration which included the coverts. Carney pointed out that this is consistent with what is known of modern flight characteristics, in that black melanosomes have structural properties that strengthen feathers for flight.[74] In a 2013 study published in the *Journal of Analytical Atomic Spectrometry*, new analyses of *Archaeopteryx*'s feathers revealed that the animal may have had complex light- and dark-coloured plumage, with heavier pigmentation in the distal tips and outer vanes.[75] This analysis of colour distribution was based primarily on the distribution of sulphate within the fossil. An author on the previous *Archaeopteryx* colour study argued against the interpretation of such biomarkers as an indicator of eumelanin in the full *Archaeopteryx* specimen.[76] Carney and other colleagues also argued against the 2013 study's interpretation of the sulphate and trace metals,[77][78] and in a 2020 study published in *Scientific Reports* demonstrated that the isolated covert feather was entirely matte black (as opposed to black and white, or iridescent) and that the remaining "plumage patterns of *Archaeopteryx* remain unknown".[13]

Classification

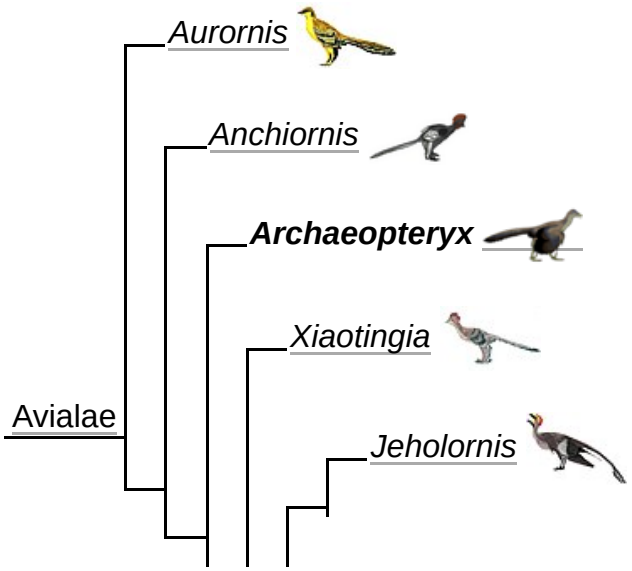


The Thermopolis Specimen

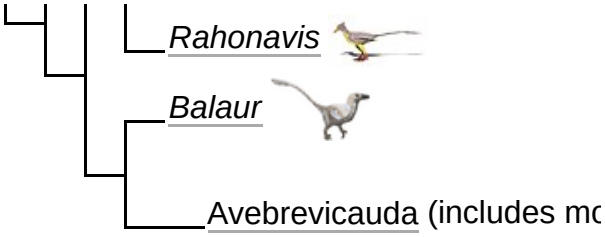
Today, fossils of the genus *Archaeopteryx* are usually assigned to one or two species, *A. lithographica* and *A. siemensii*, but their taxonomic history is complicated. Ten names have been published for the handful of specimens. As interpreted today, the name *A. lithographica* only referred to the single feather described by Meyer. In 1954 Gavin de Beer concluded that the London specimen was the holotype. In 1960, Swinton accordingly proposed that the name *Archaeopteryx lithographica* be placed on the official genera list making the alternative names *Griphosaurus* and *Griphornis* invalid.[79] The ICZN, implicitly accepting De Beer's standpoint, did indeed suppress the plethora of alternative names initially proposed for the first skeleton specimens,[80] which mainly resulted from the acrimonious dispute between Meyer and his opponent Johann Andreas Wagner (whose *Griphosaurus problematicus*—'problematic riddle-lizard'—was a vitriolic sneer at Meyer's *Archaeopteryx*).[81] In addition, in 1977, the Commission ruled that the first species name of the Haarlem specimen, *crassipes*, described by Meyer as a pterosaur before its true nature was realized, was not to be given preference over *lithographica* in instances where scientists considered them to represent the same species.[9][82]

It has been noted that the feather, the first specimen of *Archaeopteryx* described, does not correspond well with the flight-related feathers of *Archaeopteryx*. It certainly is a flight feather of a contemporary species, but its size and proportions indicate that it may belong to another, smaller species of feathered theropod, of which only this feather is known so far.[11] As the feather had been designated the type specimen, the name *Archaeopteryx* should then no longer be applied to the skeletons, thus creating significant nomenclatorial confusion. In 2007, two sets of scientists therefore petitioned the ICZN requesting that the London specimen explicitly be made the type by designating it as the new holotype specimen, or neotype.[83] This suggestion was upheld by the ICZN after four years of debate, and the London specimen was designated the neotype on 3 October 2011.[84]

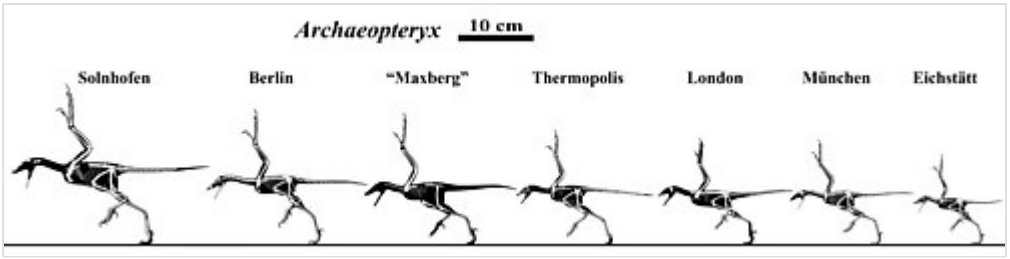
Below is a cladogram published in 2013 by Godefroit et al.[4]



The twelfth specimen



Species



Skeletal restorations of various specimens

It has been argued that all the specimens belong to the same species, *A. lithographica*.[85] Differences do exist among the specimens, and while some researchers regard these as due to the different ages of the specimens, some may be related to actual species diversity. In particular, the Munich, Eichstätt, Solnhofen, and Thermopolis specimens differ from the London, Berlin, and Haarlem specimens in being smaller or much larger, having different finger proportions, having more slender snouts lined with forward-pointing teeth, and the possible presence of a sternum. Due to these differences, most individual specimens have been given their own species name at one point or another. The Berlin specimen has been designated as *Archaeornis siemensii*, the Eichstätt specimen as *Jurapteryx recurva*, the Munich specimen as *Archaeopteryx bavarica*, and the Solnhofen specimen as *Wellnhoferia grandis*.[24]

In 2007, a review of all well-preserved specimens including the then-newly discovered Thermopolis specimen concluded that two distinct species of *Archaeopteryx* could be supported: *A. lithographica* (consisting of at least the London and Solnhofen specimens), and *A. siemensii* (consisting of at least the Berlin, Munich, and Thermopolis specimens). The two species are distinguished primarily by large flexor tubercles on the foot claws in *A. lithographica* (the claws of *A. siemensii* specimens being relatively simple and straight). *A. lithographica* also had a constricted portion of the crown in some teeth and a stouter metatarsus. A supposed additional species, *Wellnhoferia grandis* (based on the Solnhofen specimen), seems to be indistinguishable from *A. lithographica* except in its larger size.[25]

Synonyms

If two names are given, the first denotes the original describer of the "species", the second the author on whom the given name combination is based. As always in zoological nomenclature, putting an author's name in parentheses denotes that the taxon was originally described in a different genus.

- ***Archaeopteryx lithographica*** Meyer, 1861 [conserved name]
  - *Archaeopterix lithographica* Anon., 1861 [*lapsus*]
  - *Griphosaurus problematicus* Wagner, 1862 [rejected name 1961 per ICZN Opinion 607]

- *Griphornis longicaudatus* Owen *vide* Woodward, 1862 [rejected name 1961 per ICZN Opinion 607]
- *Archaeopteryx macrura* Owen, 1862 [rejected name 1961 per ICZN Opinion 607]
- *Archaeopteryx oweni* Petronievics, 1917 [rejected name 1961 per ICZN Opinion 607]
- *Archaeopteryx recurva* Howgate, 1984
- *Jurapteryx recurva* (Howgate, 1984) Howgate, 1985
- *Wellnhoferia grandis* Elżanowski, 2001

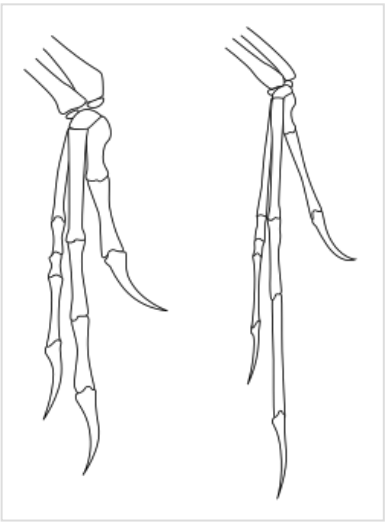
- ***Archaeopteryx siemensii*** Dames, 1897
  - *Archaeornis siemensii* (Dames, 1897) Petronievics, 1917[25]
  - *Archaeopteryx bavarica* Wellnhofer, 1993

"Archaeopteryx" vicensensis (Anon. *fide* Lambrecht, 1933) is a *nomen nudum* for what appears to be an undescribed pterosaur.

The Solnhofen Specimen, by some considered as belonging to the genus *Wellnhoferia*

Phylogenetic position

Modern palaeontology has often classified *Archaeopteryx* as the most primitive bird. However, it is not thought to be a true ancestor of modern birds, but rather a close relative of that ancestor.[86] Nonetheless, *Archaeopteryx* was often used as a model of the true ancestral bird. Several authors have done so.[87] Lowe (1935)[88] and Thulborn (1984)[89] questioned whether *Archaeopteryx* truly was the first bird. They suggested that *Archaeopteryx* was a dinosaur that was no more closely related to birds than were other dinosaur groups. Kurzanov (1987) suggested that *Avimimus* was more likely to be the ancestor of all birds than *Archaeopteryx*.[90] Barsbold (1983)[91] and Zweers and Van den Berge (1997)[92] noted that many maniraptoran lineages are extremely birdlike, and they suggested that different groups of birds may have descended from different dinosaur ancestors.



Comparison of the forelimb of *Archaeopteryx* (right) with that of *Deinonychus* (left)

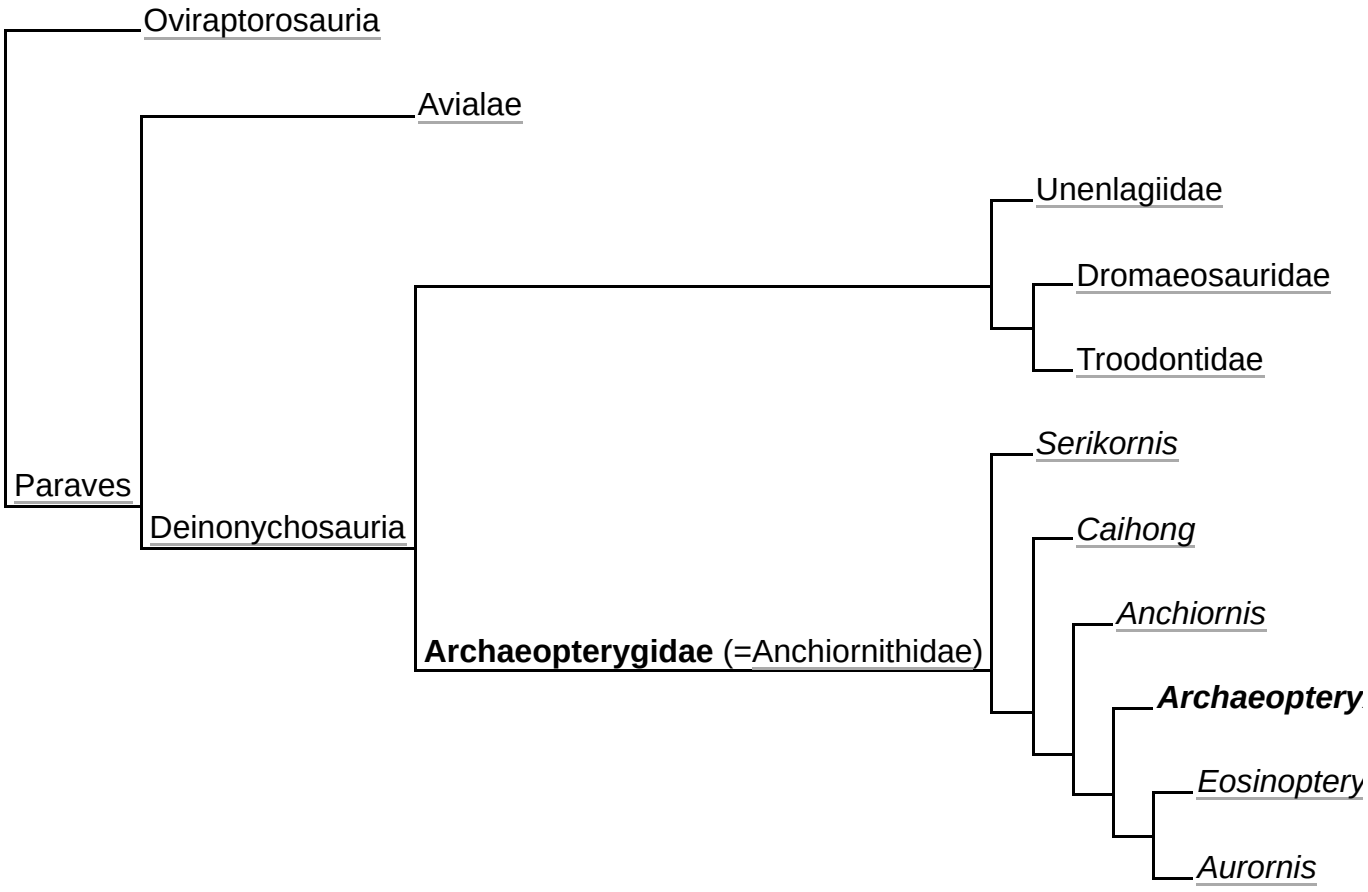
The discovery of the closely related *Xiaotingia* in 2011 led to new phylogenetic analyses that suggested that *Archaeopteryx* is a deinonychosaur rather than an avialan, and therefore, not a "bird" under most common uses of that term.[3] A more thorough analysis was published soon after to test this hypothesis, and failed to arrive at the same result; it found *Archaeopteryx* in its traditional position at the base of *Avialae*, while *Xiaotingia* was recovered as a basal dromaeosaurid or troodontid. The authors of the follow-up study noted that uncertainties still exist, and that it may not be possible to state confidently whether or not *Archaeopteryx* is a member of Avialae or not, barring new and better specimens of relevant species.[93]

Phylogenetic studies conducted by Senter et al. (2012) and Turner, Makovicky, and Norell (2012) also found *Archaeopteryx* to be more closely related to living birds than to dromaeosaurids and troodontids.[94][95] On the other hand, Godefroit et al. (2013) recovered *Archaeopteryx* as more closely related to dromaeosaurids and troodontids in the analysis included in their description of *Eosinopteryx*

brevipenna. The authors used a modified version of the matrix from the study describing *Xiaotingia*, adding *Jinfengopteryx elegans* and *Eosinopteryx brevipenna* to it, as well as adding four additional characters related to the development of the plumage. Unlike the analysis from the description of *Xiaotingia*, the analysis conducted by Godefroit, et al. did not find *Archaeopteryx* to be related particularly closely to *Anchiornis* and *Xiaotingia*, which were recovered as basal troodontids instead.[96]

Agnolín and Novas (2013) found *Archaeopteryx* and (possibly synonymous) *Wellnhoferia* to form a clade sister to the lineage including *Jeholornis* and Pygostylia, with Microraptoria, Unenlagiinae, and the clade containing *Anchiornis* and *Xiaotingia* being successively closer outgroups to the Avialae (defined by the authors as the clade stemming from the last common ancestor of *Archaeopteryx* and Aves).[97] Another phylogenetic study by Godefroit, et al., using a more inclusive matrix than the one from the analysis in the description of *Eosinopteryx brevipenna*, also found *Archaeopteryx* to be a member of Avialae (defined by the authors as the most inclusive clade containing *Passer domesticus*, but not *Dromaeosaurus albertensis* or *Troodon formosus*). *Archaeopteryx* was found to form a grade at the base of Avialae with *Xiaotingia*, *Anchiornis*, and *Aurornis*. Compared to *Archaeopteryx*, *Xiaotingia* was found to be more closely related to extant birds, while both *Anchiornis* and *Aurornis* were found to be more distantly so.[4]

Hu et al. (2018),[98] Wang et al. (2018)[99] and Hartman et al. (2019)[100] found *Archaeopteryx* to have been a deinonychosaur instead of an avialan. More specifically, it and closely related taxa were considered basal deinonychosaurs, with dromaeosaurids and troodontids forming together a parallel lineage within the group. Because Hartman et al. found *Archaeopteryx* isolated in a group of flightless deinonychosaurs (otherwise considered "anchiornithids"), they considered it highly probable that this animal evolved flight independently from bird ancestors (and from *Microraptor* and *Yi*). The following cladogram illustrates their hypothesis regarding the position of *Archaeopteryx*:



The authors, however, found that the *Archaeopteryx* being an avialan was only slightly less likely than this hypothesis, and as likely as Archaeopterygidae and Troodontidae being sister clades.[100]

Palaeobiology

Flight

1880 photo of the Berlin Specimen, showing leg feathers that were removed subsequently, during preparation

As in the wings of modern birds, the flight feathers of *Archaeopteryx* were somewhat asymmetrical and the tail feathers were rather broad. This implies that the wings and tail were used for lift generation, but it is unclear whether *Archaeopteryx* was capable of flapping flight or simply a glider. The lack of a bony breastbone suggests that *Archaeopteryx* was not a very strong flier, but flight muscles might have attached to the thick, boomerang-shaped wishbone, the platelike coracoids, or perhaps, to a cartilaginous sternum. The sideways orientation of the glenoid (shoulder) joint between scapula, coracoid, and humerus—instead of the dorsally angled arrangement found in modern birds—may indicate that *Archaeopteryx* was unable to lift its wings above its back, a requirement for the upstroke found in modern flapping flight. According to a study by Philip Senter in 2006, *Archaeopteryx* was indeed unable to use flapping flight as modern birds do, but it may well have used a downstroke-only flap-assisted gliding technique.[101] However, a more recent study solves this issue by suggesting a different flight stroke configuration for non-avian flying theropods.[39]

Archaeopteryx wings were relatively large, which would have resulted in a low stall speed and reduced turning radius. The short and rounded shape of the wings would have increased drag, but also could have improved its ability to fly through cluttered environments such as trees and brush (similar wing shapes are seen in birds that fly through trees and brush, such as crows and pheasants). The presence of "hind wings", asymmetrical flight feathers stemming from the legs similar to those seen in dromaeosaurids such as *Microraptor*, also would have added to the aerial mobility of *Archaeopteryx*. The first detailed study of the hind wings by Longrich in 2006, suggested that the structures formed up to 12% of the total airfoil. This would have reduced stall speed by up to 6% and turning radius by up to 12%.[71]

The feathers of *Archaeopteryx* were asymmetrical. This has been interpreted as evidence that it was a flyer, because flightless birds tend to have symmetrical feathers. Some scientists, including Thomson and Speakman, have questioned this. They studied more than 70 families of living birds, and found that some flightless types do have a range of asymmetry in their feathers, and that the feathers of *Archaeopteryx* fall into this range.[102] The degree of asymmetry seen in *Archaeopteryx* is more typical for slow flyers than for flightless birds.[103]

In 2010, Robert L. Nudds and Gareth J. Dyke in the journal *Science* published a paper in which they analysed the rachises of the primary feathers of *Confuciusornis* and *Archaeopteryx*. The analysis suggested that the rachises on these two genera were thinner and weaker than those of modern birds

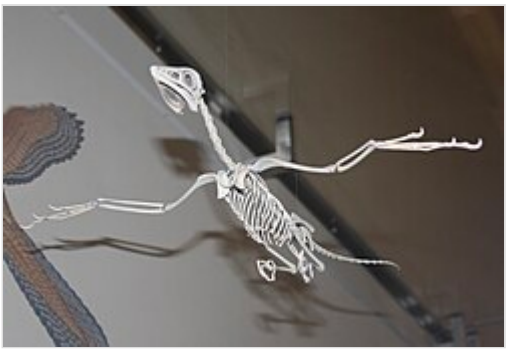
relative to body mass. The authors determined that *Archaeopteryx* and *Confuciusornis*, were unable to use flapping flight.[104] This study was criticized by Philip J. Currie and Luis Chiappe. Chiappe suggested that it is difficult to measure the rachises of fossilized feathers, and Currie speculated that *Archaeopteryx* and *Confuciusornis* must have been able to fly to some degree, as their fossils are preserved in what is believed to have been marine or lake sediments, suggesting that they must have been able to fly over deep water.[105] Gregory Paul also disagreed with the study, arguing in a 2010 response that Nudds and Dyke had overestimated the masses of these early birds, and that more accurate mass estimates allowed powered flight even with relatively narrow rachises. Nudds and Dyke had assumed a mass of 250 g (8.8 oz) for the Munich specimen *Archaeopteryx*, a young juvenile, based on published mass estimates of larger specimens. Paul argued that a more reasonable body mass estimate for the Munich specimen is about 140 g (4.9 oz). Paul also criticized the measurements of the rachises themselves, noting that the feathers in the Munich specimen are poorly preserved. Nudds and Dyke reported a diameter of 0.75 mm (0.03 in) for the longest primary feather, which Paul could not confirm using photographs. Paul measured some of the inner primary feathers, finding rachises 1.25–1.4 mm (0.049–0.055 in) across.[106] Despite these criticisms, Nudds and Dyke stood by their original conclusions. They claimed that Paul's statement, that an adult *Archaeopteryx* would have been a better flyer than the juvenile Munich specimen, was dubious. This, they reasoned, would require an even thicker rachis, evidence for which has not yet been presented.[107] Another possibility is that they had not achieved true flight, but instead used their wings as aids for extra lift while running over water after the fashion of the basilisk lizard, which could explain their presence in lake and marine deposits (see Origin of avian flight).[108][109]

The Munich Specimen

London Specimen of *Archaeopteryx*

In 2004, scientists analysing a detailed CT scan of the braincase of the London *Archaeopteryx* concluded that its brain was significantly larger than that of most dinosaurs, indicating that it possessed the brain size necessary for flying. The overall brain anatomy was reconstructed using the scan. The reconstruction showed that the regions associated with vision took up nearly one-third of the brain. Other well-developed areas involved hearing and muscle coordination.[110] The skull scan also revealed the structure of its inner ear. The structure more closely resembles that of modern birds than the inner ear of non-avian reptiles. These characteristics taken together suggest that *Archaeopteryx* had the keen sense of hearing, balance, spatial perception, and coordination needed to fly.[111] *Archaeopteryx* had a cerebrum-to-brain-volume ratio 78% of the way to modern birds from the condition of non-coelurosaurian dinosaurs such as *Carcharodontosaurus* or *Allosaurus*, which had a crocodile-like anatomy of the brain and inner ear.[112] Newer research shows that while the *Archaeopteryx* brain was more complex than that of more primitive theropods, it had a more generalized brain volume among Maniraptora dinosaurs, even smaller than that of other non-avian dinosaurs in several instances, which indicates the neurological development required for flight was already a common trait in the maniraptoran clade.[113]

Recent studies of flight feather barb geometry reveal that modern birds possess a larger barb angle in the trailing vane of the feather, whereas *Archaeopteryx* lacks this large barb angle, indicating potentially weak flight abilities.[114]



Skeletal reconstruction of *Archaeopteryx* in gliding posture, American Museum of Natural History

Archaeopteryx continues to play an important part in scientific debates about the origin and evolution of birds. Some scientists see it as a semi-arboreal climbing animal, following the idea that birds evolved from tree-dwelling gliders (the "trees down" hypothesis for the evolution of flight proposed by O. C. Marsh). Other scientists see *Archaeopteryx* as running quickly along the ground, supporting the idea that birds evolved flight by running (the "ground up" hypothesis proposed by Samuel Wendell Williston). Still others suggest that *Archaeopteryx* might have been at home both in the trees and on the ground, like modern crows, and this latter view is what currently is considered best supported by morphological characters. Altogether, it appears that the species was not particularly specialized for running on the ground or for perching. A scenario outlined by Elżanowski in 2002 suggested that *Archaeopteryx* used its wings mainly to escape predators by glides punctuated with shallow downstrokes to reach successively higher perches, and alternatively, to cover longer distances (mainly) by gliding down from cliffs or treetops.[24]

In March 2018, scientists reported that *Archaeopteryx* was likely capable of a flight stroke cycle morphologically closer to the grabbing motion of maniraptorans and distinct from that of modern birds.[39][115] This study on *Archaeopteryx*'s bone histology identified biomechanical and physiological adaptations exhibited by modern volant birds that perform intermittent flapping, such as pheasants and other burst flyers.

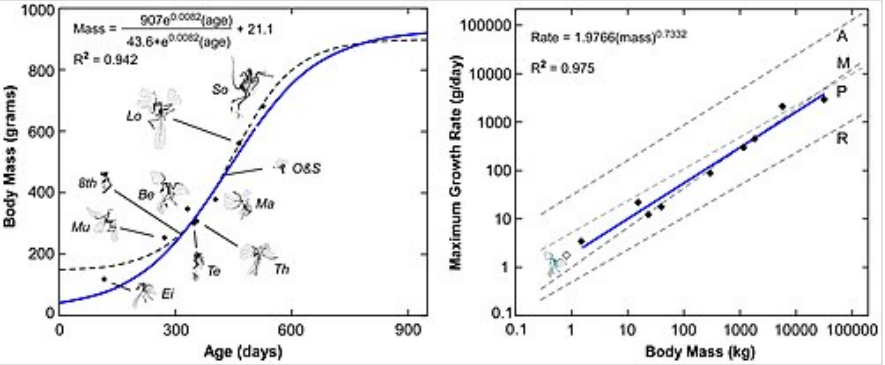
Some researchers suggested that the feather sheaths of *Archaeopteryx* shows a center-out, flight related moulting strategy like modern birds. As it was a weak flier, this would have been extremely advantageous in preserving its maximum flight performance.[116] Kiat and colleagues reinterpreted this purported moulting evidence to be problematic and equivocal at best, and considered that these structures more likely represents the calami traces of the fully grown feathers,[117] though the original authors still remained by their conclusion.[118]

2026 CT-based study of the Chicago *Archaeopteryx* specimen reported avian-like soft-tissue correlates in the feeding apparatus, including structures interpreted as oral papillae and elements of a mobile tongue, suggesting similarities with feeding adaptations seen in modern birds.[119][120]

Growth

An histological study by Erickson, Norell, Zhongue, and others in 2009 estimated that *Archaeopteryx* grew relatively slowly compared to modern birds, presumably because the outermost portions of *Archaeopteryx* bones appear poorly vascularized;[6] in living vertebrates, poorly vascularized bone is correlated with slow growth rate. They also assume that all known skeletons of *Archaeopteryx* come from juvenile specimens. Because the bones of *Archaeopteryx* could not be histologically sectioned in a formal skeletochronological (growth ring) analysis, Erickson and colleagues used bone vascularity (porosity) to estimate bone growth rate. They assumed that poorly vascularized bone grows at similar rates in all birds

and in *Archaeopteryx*. The poorly vascularized bone of *Archaeopteryx* might have grown as slowly as that in a mallard (2.5 micrometres per day) or as fast as that in an ostrich (4.2 micrometres per day). Using this range of bone growth rates, they calculated how long it would take to "grow" each specimen of *Archaeopteryx* to the observed size; it may have taken at least 970 days (there were 375 days in a Late Jurassic year) to reach an adult size of 0.8–1 kg (1.8–2.2 lb). The study also found that the avialans *Jeholornis* and *Sapeornis* grew relatively slowly, as did the dromaeosaurid *Mahakala*. The avialans *Confuciusornis* and *Ichthyornis* grew relatively quickly, following a growth trend similar to that of modern birds.[121] One of the few modern birds that exhibit slow growth is the flightless kiwi, and the authors speculated that *Archaeopteryx* and the kiwi had similar basal metabolic rate.[6]



Growth trends compared with other dinosaurs and birds

Daily activity patterns

Comparisons between the scleral rings of *Archaeopteryx* and modern birds and reptiles indicate that it may have been diurnal, similar to most modern birds.[122]

Palaeoecology

The richness and diversity of the Solnhofen limestones in which all specimens of *Archaeopteryx* have been found have shed light on an ancient Jurassic Bavaria strikingly different from the present day. The latitude was similar to Florida, though the climate was likely to have been drier, as evidenced by fossils of plants with adaptations for arid conditions and a lack of terrestrial sediments characteristic of rivers. Evidence of plants, although scarce, include cycads and conifers while animals found include a large number of insects, small lizards, pterosaurs, and *Compsognathus*.[17]

Restoration of *Archaeopteryx* chasing a juvenile *Compsognathus*

The excellent preservation of *Archaeopteryx* fossils and other terrestrial fossils found at Solnhofen indicates that they did not travel far before becoming preserved.[123] The *Archaeopteryx* specimens found were therefore likely to have lived on the low islands surrounding the Solnhofen lagoon rather than to have been corpses that drifted in from farther away. *Archaeopteryx* skeletons are considerably less numerous in the deposits of Solnhofen than those of pterosaurs, of which seven genera have been found.[124] The pterosaurs included species such as *Rhamphorhynchus* belonging to the Rhamphorhynchidae, the group which dominated the

ecological niche currently occupied by seabirds, and which became extinct at the end of the Jurassic. The pterosaurs, which also included *Pterodactylus*, were common enough that it is unlikely that the specimens found are vagrants from the larger islands 50 km (31 mi) to the north.[125]

Solnhofen lagoon scene with *Archaeopteryx*, *Compsognathus* and *Rhamphorhynchus* by Charles R. Knight for the Field Museum in Chicago

The islands that surrounded the Solnhofen lagoon were low lying, semi-arid, and sub-tropical with a long dry season and little rain.[126] The closest modern analogue for the Solnhofen conditions is said to be Orca Basin in the northern Gulf of Mexico, although it is much deeper than the Solnhofen lagoons.[124] The flora of these islands was adapted to these dry conditions and consisted mostly of low (3 m [10 ft]) shrubs.[125] Contrary to reconstructions of *Archaeopteryx* climbing large trees, these seem to have been mostly absent from the islands; few trunks have been found in the sediments and fossilized tree pollen also is absent.

The lifestyle of *Archaeopteryx* is difficult to reconstruct and there are several theories regarding it. Some researchers suggest that it was primarily adapted to life on the ground,[66] while other researchers suggest that it was principally arboreal on the basis of the curvature of the claws[127] which has since been questioned.[128] The absence of trees does not preclude *Archaeopteryx* from an arboreal lifestyle, as several species of bird live exclusively in low shrubs. Various aspects of the morphology of *Archaeopteryx* point to either an arboreal or ground existence, including the length of its legs and the elongation in its feet; some authorities consider it likely to have been a generalist capable of feeding in both shrubs and open ground, as well as along the shores of the lagoon.[125] It most likely hunted small prey, seizing it with its jaws or claws.[125]

See also

- Dinosaur coloration
- Evolution of birds
- Feathered dinosaur
- Origin of birds
- *Ostromia*
- *Rhamphorhynchus*
- Temporal paradox (paleontology)
- *Xiaotingia*

Paleontology portal

Dinosaurs portal

Birds portal

Evolutionary biology portal

References

1. "Troodontidae Gilmore, 1924" (https://web.archive.org/web/20190403063849/http://theropoddatabase.com/Troodontidae.htm#Troodontidae). *theropoddatabase.com*. Archived from the original (http://theropoddatabase.com/Troodontidae.htm#Troodontidae) on 3 April 2019.
2. O'Connor, J.; Marugán-Lobón, J. (26 March 2026). "Evaluating variation in Solnhofen avialans" (https://doi.org/10.1098%2Frsbl.2025.0601). *Biology Letters*. **22** (3) 20250601. doi:10.1098/rsbl.2025.0601 (https://doi.org/10.1098%2Frsbl.2025.0601).

3. Xu, X; You, H; Du, K; Han, F (28 July 2011). "An *Archaeopteryx*-like theropod from China and the origin of Avialae" (https://web.archive.org/web/20161220071919/http://www.ivpp.ac.cn/qt/papers/201403/P020140314389417822583.pdf) (PDF). *Nature*. **475** (7357): 465–470. doi:10.1038/nature10288 (https://doi.org/10.1038%2Fnature10288). PMID 21796204 (https://pubmed.ncbi.nlm.nih.gov/21796204). S2CID 205225790 (https://api.semanticscholar.org/CorpusID:205225790). Archived from the original (http://www.ivpp.ac.cn/qt/papers/201403/P020140314389417822583.pdf) (PDF) on 20 December 2016. Retrieved 5 November 2016.
4. Godefroit, Pascal; Cau, Andrea; Hu, Dong-Yu; Escuillié, François; Wu, Wenhao; Dyke, Gareth (2013). "A Jurassic avialan dinosaur from China resolves the early phylogenetic history of birds". *Nature*. **498** (7454): 359–362. Bibcode:2013Natur.498..359G (https://ui.adsabs.harvard.edu/abs/2013Natur.498..359G). doi:10.1038/nature12168 (https://doi.org/10.1038%2Fnature12168). PMID 23719374 (https://pubmed.ncbi.nlm.nih.gov/23719374). S2CID 4364892 (https://api.semanticscholar.org/CorpusID:4364892).
5. Chen, Runsheng; Wang, Min; Dong, Liping; Zhou, Guowu; Xu, Xing; Deng, Ke; Xu, Liming; Zhang, Chi; Wang, Linchang; Du, Honggang; Lin, Ganmin; Lin, Min; Zhou, Zhonghe (12 February 2025). "Earliest short-tailed bird from the Late Jurassic of China" (https://www.nature.com/articles/s41586-024-08410-z). *Nature*. **638** (8050): 441–448. Bibcode:2025Natur.638..441C (https://ui.adsabs.harvard.edu/abs/2025Natur.638..441C). doi:10.1038/s41586-024-08410-z (https://doi.org/10.1038%2Fs41586-024-08410-z). ISSN 1476-4687 (https://search.worldcat.org/issn/1476-4687). PMID 39939791 (https://pubmed.ncbi.nlm.nih.gov/39939791).
6. Erickson, Gregory M.; Rauhut, Oliver W. M.; Zhou, Zhonghe; Turner, Alan H.; Inouye, Brian D.; Hu, Dongyu; Norell, Mark A. (2009). Desalle, Robert (ed.). "Was Dinosaurian Physiology Inherited by Birds? Reconciling Slow Growth in *Archaeopteryx*" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2756958). *PLOS ONE*. **4** (10) e7390. Bibcode:2009PLoSO...4.7390E (https://ui.adsabs.harvard.edu/abs/2009PLoSO...4.7390E). doi:10.1371/journal.pone.0007390 (https://doi.org/10.1371%2Fjournal.pone.0007390). PMC 2756958 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2756958). PMID 19816582 (https://pubmed.ncbi.nlm.nih.gov/19816582).
7. Yalden D. W. (1984). "What size was *Archaeopteryx*?". *Zoological Journal of the Linnean Society*. **82** (1–2): 177–188. doi:10.1111/j.1096-3642.1984.tb00541.x (https://doi.org/10.1111%2Fj.1096-3642.1984.tb00541.x).
8. L. M. Chiappe; L. M. Witmer, eds. (2002). *Mesozoic birds: above the heads of dinosaurs* (https://books.google.com/books?id=2MQeh1KCp7sC). Berkeley and Los Angeles, California: University of California Press. p. 151. ISBN 978-0-520-20094-4.
9. *Archaeopteryx*: An Early Bird (http://www.ucmp.berkeley.edu/diapsids/birds/archaeopteryx.html) Archived (https://web.archive.org/web/20200504214042/https://ucmp.berkeley.edu/diapsids/birds/archaeopteryx.html) 4 May 2020 at the Wayback Machine – University of California, Berkeley, Museum of Paleontology. Retrieved 18 October 2006.
10. Ancient birds flew on all fours (https://www.ucalgary.ca/news/september2006/Archaeopteryx-lithographica/) Archived (https://web.archive.org/web/20171203013828/https://www.ucalgary.ca/news/september2006/Archaeopteryx-lithographica/) 3 December 2017 at the Wayback Machine – Nick Longrich, University of Calgary. Discusses how many wings an *Archaeopteryx* had and other questions.
11. Griffiths, P. J. (1996). "The Isolated Archaeopteryx Feather". *Archaeopteryx*. **14**: 1–26.
12. Kaye, T.G.; Pittman, M.; Mayr, G.; Schwarz, D.; Xu, X. (2019). "Detection of lost calamus challenges identity of isolated *Archaeopteryx* feather" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6362147). *Scientific Reports*. **9** (1): 1182. Bibcode:2019NatSR...9.1182K (https://ui.adsabs.harvard.edu/abs/2019NatSR...9.1182K). doi:10.1038/s41598-018-37343-7 (https://doi.org/10.1038%2Fs41598-018-37343-7). PMC 6362147 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6362147). PMID 30718905 (https://pubmed.ncbi.nlm.nih.gov/30718905).

13. Carney, R.M.; Tischlinger, H.; Shawkey, M.D. (2020). "Evidence corroborates identity of isolated fossil feather as a wing covert of *Archaeopteryx*" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7528088). *Scientific Reports*. **10** (1): 15593. Bibcode:2020NatSR..1015593C (https://ui.adsabs.harvard.edu/abs/2020NatSR..1015593C). doi:10.1038/s41598-020-65336-y (https://doi.org/10.1038%2Fs41598-020-65336-y). PMC 7528088 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7528088). PMID 32999314 (https://pubmed.ncbi.nlm.nih.gov/32999314). S2CID 222109311 (https://api.semanticscholar.org/CorpusID:222109311).
14. O'Connor, J.; Clark, A.; Kuo, P.-C.; Kiat, Y.; Fabbri, M.; Shinya, A.; Van Beek, C.; Lu, J.; Wang, M.; Hu, H. (2025). "Chicago *Archaeopteryx* informs on the early evolution of the avian bauplan". *Nature*. **641** (8065): 1201–1207. Bibcode:2025Natur.641.1201O (https://ui.adsabs.harvard.edu/abs/2025Natur.641.1201O). doi:10.1038/s41586-025-08912-4 (https://doi.org/10.1038%2Fs41586-025-08912-4). PMID 40369075 (https://pubmed.ncbi.nlm.nih.gov/40369075).
15. P. Wellnhofer (2004). "The Plumage of *Archaeopteryx*". In Currie PJ, Koppelhus EB, Shugar MA, Wright JL (eds.). *Feathered Dragons*. Indiana University Press. pp. 282–300. ISBN 978-0-253-34373-4.
16. Kaplan, Matt (27 July 2011). "Archaeopteryx no longer first bird" (http://www.nature.com/articles/news.2011.443). *Nature* news.2011.443. doi:10.1038/news.2011.443 (https://doi.org/10.1038%2Fnews.2011.443). ISSN 0028-0836 (https://search.worldcat.org/issn/0028-0836).
17. Chiappe, Luis M. (2007). *Glorified Dinosaurs*. Sydney: UNSW Press. pp. 118–146. ISBN 978-0-471-24723-4.
18. Bakalar, Nicholas (1 December 2005) "Earliest Bird Had Feet Like Dinosaur, Fossil Shows". *National Geographic News*, p. 1 (https://web.archive.org/web/20051204012548/http://news.nationalgeographic.com/news/2005/12/1201_051201_archaeopteryx.html), 2 (https://web.archive.org/web/20051205022112/http://news.nationalgeographic.com/news/2005/12/1201_051201_archaeopteryx_2.html).
19. Meyer, Hermann von (15 August 1861). "Vogel-Federn und Palpipes priscus von Solenhofen" (https://books.google.com/books?id=6RAFAAAAQAAJ&pg=PA561) [Bird feathers and *Palpipes priscus* [a crustacean] from Solenhofen]. *Neues Jahrbuch für Mineralogie, Geognosie, Geologie und Petrefaktenkunde* (in German): 561. *"Aus dem lithographischen Schiefer der Brüche von* Solenhofen *in* Bayern *ist mir in den beiden Gegenplatten eine auf der Ablösungs- oder Spaltungs-Fläche des Gesteins liegende Versteinerung mitgetheilt worden, die mit grosser Deutlichkeit eine Feder erkennen lässt, welche von den Vogel-Federn nicht zu unterscheiden ist."* (From the lithographic slates of the faults of Solenhofen in Bavaria, there has been reported to me a fossil lying on the stone's surface of detachment or cleavage, in both opposing slabs, which can be recognized with great clarity [to be] a feather, which is indistinguishable from a bird's feather.)
20. British Museum of Natural History – 'BMNH 37001' – the type specimen
21. "UK inflation calculator" (https://www.in2013dollars.com/uk/inflation/1861?amount=700). Archived (https://web.archive.org/web/20220117225656/https://www.in2013dollars.com/uk/inflation/1861?amount=700) from the original on 17 January 2022. Retrieved 4 April 2020.
22. Darwin, Charles. *Origin of Species* (http://darwin-online.org.uk/content/frameset?viewtype=text&itemID=F385&pageseq=399) Archived (https://web.archive.org/web/20080507060732/http://darwin-online.org.uk/content/frameset?viewtype=text&itemID=F385&pageseq=399) 7 May 2008 at the Wayback Machine. Chapter 9, p. 367.
23. Darwin, Charles (1859). *On the Origin of Species* (https://web.archive.org/web/20080513170458/http://www.readprint.com/chapter-2217/Charles-Darwin). John Murray. Archived from the original (http://www.readprint.com/chapter-2217/Charles-Darwin) on 13 May 2008. Retrieved 8 June 2008. Please note Darwin's spelling: 'Archeopteryx', not 'Archaeopteryx'.

24. Elżanowski A. (2002). "Archaeopterygidae (Upper Jurassic of Germany)". In Chiappe, L. M.; Witmer, L. M. (eds.). *Mesozoic Birds: Above the Heads of Dinosaurs* (https://archive.org/details/mesozoicbirdsabo00chia_070). Berkeley: University of California Press. pp. 129 (https://archive.org/details/mesozoicbirdsabo00chia_070/page/n139)–159. ISBN 978-0-520-20094-4.
25. Mayr, Gerald; Pohl, Burkhard; Hartman, Scott; Peters, D. Stefan (January 2007). "The tenth skeletal specimen of Archaeopteryx" (https://doi.org/10.1111%2Fj.1096-3642.2006.00245.x). *Zoological Journal of the Linnean Society*. **149** (1): 97–116. doi:10.1111/j.1096-3642.2006.00245.x (https://doi.org/10.1111%2Fj.1096-3642.2006.00245.x).
26. Hecht, Jeff (1 February 2012). "Lost treasures: The Maxberg Archaeopteryx" (https://www.newscientist.com/article/mg21328501-800-lost-treasures-the-maxberg-archaeopteryx). *New Scientist*. Archived (https://web.archive.org/web/20220602140934/https://www.newscientist.com/article/mg21328501-800-lost-treasures-the-maxberg-archaeopteryx/) from the original on 2 June 2022. Retrieved 2 June 2022.
27. Foth, C.; Rauhut, O.W.M. (2017). "Re-evaluation of the Haarlem Archaeopteryx and the radiation of maniraptoran theropod dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5712154). *BMC Evolutionary Biology*. **17** (1): 236. Bibcode:2017BMCEE..17..236F (https://ui.adsabs.harvard.edu/abs/2017BMCEE..17..236F). doi:10.1186/s12862-017-1076-y (https://doi.org/10.1186%2Fs12862-017-1076-y). PMC 5712154 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5712154). PMID 29197327 (https://pubmed.ncbi.nlm.nih.gov/29197327).
28. Howgate, M. E. (September 1984). "The teeth of Archaeopteryx and a reinterpretation of the Eichstätt specimen" (https://academic.oup.com/zoolinnean/article-lookup/doi/10.1111/j.1096-3642.1984.tb00540.x). *Zoological Journal of the Linnean Society*. **82** (1–2): 159–175. doi:10.1111/j.1096-3642.1984.tb00540.x (https://doi.org/10.1111%2Fj.1096-3642.1984.tb00540.x). Archived (https://web.archive.org/web/20220815210439/https://academic.oup.com/zoolinnean/article-lookup/doi/10.1111/j.1096-3642.1984.tb00540.x) from the original on 15 August 2022. Retrieved 21 July 2022.
29. Senter, Phil (2007). "A new look at the phylogeny of coelurosauria (Dlnosauria: Theropoda)". *Journal of Systematic Palaeontology*. **5** (4): 429–463. Bibcode:2007JSPal...5..429S (https://ui.adsabs.harvard.edu/abs/2007JSPal...5..429S). doi:10.1017/S1477201907002143 (https://doi.org/10.1017%2FS1477201907002143). S2CID 83726237 (https://api.semanticscholar.org/CorpusID:83726237).
30. Elżanowski A 2001b. "A new genus and species for the largest specimen of Archaeopteryx (https://www.researchgate.net/publication/228707898_A_new_genus_and_species_for_the_largest_specimen_of_Archaeopteryx) Archived (https://web.archive.org/web/20220909102548/https://www.researchgate.net/publication/228707898_A_new_genus_and_species_for_the_largest_specimen_of_Archaeopteryx) 9 September 2022 at the Wayback Machine". Acta Palaeontologica Polonica 46: 519–532.
31. Wellnhofer, P., & Tischlinger, H. (2004). *Das "Brustbein" von Archaeopteryx bavarica Wellnhofer 1993 – eine Revision*. Archaeopteryx. 22: 3–15. [Article in German].
32. Wellnhofer, Peter. *Archaeopteryx. Der Urvogel Von Solnhofen*. München: Friedrich Pfeil, 2008, p. 54.
33. *Archäologischer Sensationsfund in Daiting*, (in German) Augsburger Allgemeine – Donauwörth edition; published: 28 November 2009; accessed: 23 December 2009.
34. *Sammler und Forscher – ein schwieriges Verhältnis* (https://www.sueddeutsche.de/wissen/archaeopteryx-fossilien-stelldichein-der-urvoegel-1.25046-2) Archived (https://web.archive.org/web/20181026222455/https://www.sueddeutsche.de/wissen/archaeopteryx-fossilien-stelldichein-der-urvoegel-1.25046-2) 26 October 2018 at the Wayback Machine (in German), Sueddeutsche Zeitung; published: 17 May 2010; accessed: 26 October 2018.

35. *Wiedergefundener Archaeopteryx ist wohl neue Art* (http://www.focus.de/wissen/natur/messen-wiedergefundener-archaeopteryx-ist-wohl-neue-art_aid_448677.html) Archived (https://web.archive.org/web/20150924230031/http://www.focus.de/wissen/natur/messen-wiedergefundener-archaeopteryx-ist-wohl-neue-art_aid_448677.html) 24 September 2015 at the Wayback Machine (in German). *Die Zeit*. Accessed: 17 July 2012.
36. Kundrát, Martin; Nudds, John; Kear, Benjamin P.; Lü, Junchang; Ahlberg, Per (2019). "The first specimen of *Archaeopteryx* from the Upper Jurassic Mörnsheim Formation of Germany". *Historical Biology*. **31** (1): 3–63. Bibcode:2019HBio...31....3K (https://ui.adsabs.harvard.edu/abs/2019HBio...31....3K). doi:10.1080/08912963.2018.1518443 (https://doi.org/10.1080%2F08912963.2018.1518443). S2CID 91497638 (https://api.semanticscholar.org/CorpusID:91497638).
37. Frandsen, P.N. (16 November 2022). "Verdens vigtigste fossil kommer til Lolland" (https://folketidende.dk/lolland/verdens-vigtigste-fossil-kommer-til-lolland) (in Danish). Folketidende. Archived (https://web.archive.org/web/20230701100658/https://folketidende.dk/lolland/verdens-vigtigste-fossil-kommer-til-lolland) from the original on 1 July 2023. Retrieved 1 July 2023.
38. "Knuthenborg Safaripark åbner sin største satsning nogensinde" (https://via.ritzau.dk/pressemeddelelse/knuthenborg-safaripark-abner-sin-storste-satsning-nogensinde?publisherId=13560971&releaseId=13664363) (in Danish). Ritzau. 15 November 2022. Archived (https://web.archive.org/web/20230701100658/https://via.ritzau.dk/pressemeddelelse/knuthenborg-safaripark-abner-sin-storste-satsning-nogensinde?publisherId=13560971&releaseId=13664363) from the original on 1 July 2023. Retrieved 1 July 2023.
39. Voeten, Dennis F.A.E.; et al. (2018). "Wing bone geometry reveals active flight in *Archaeopteryx*" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5849612). *Nature Communications*. **9** (1) 923. Bibcode:2018NatCo...9..923V (https://ui.adsabs.harvard.edu/abs/2018NatCo...9..923V). doi:10.1038/s41467-018-03296-8 (https://doi.org/10.1038%2Fs41467-018-03296-8). PMC 5849612 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5849612). PMID 29535376 (https://pubmed.ncbi.nlm.nih.gov/29535376).
40. Mayr, G.; Pohl, B; Peters, DS (2 December 2005). "A Well-Preserved Archaeopteryx Specimen with Theropod Features" (http://doc.rero.ch/record/15488/files/PAL_E2876.pdf) (PDF). *Science*. **310** (5753): 1483–1486. Bibcode:2005Sci...310.1483M (https://ui.adsabs.harvard.edu/abs/2005Sci...310.1483M). doi:10.1126/science.1120331 (https://doi.org/10.1126%2Fscience.1120331). PMID 16322455 (https://pubmed.ncbi.nlm.nih.gov/16322455). S2CID 28611454 (https://api.semanticscholar.org/CorpusID:28611454). Archived (https://web.archive.org/web/20221225090505/https://doc.rero.ch/record/15488/files/PAL_E2876.pdf) (PDF) from the original on 25 December 2022. Retrieved 17 December 2022.
41. Paul, G. S. (1988). *Predatory Dinosaurs of the World, a Complete Illustrated Guide*. Simon and Schuster, New York. 464 p.
42. Switek, Brian (19 October 2011). "Paleontologists Unveil the 11th Archaeopteryx" (http://blogs.smithsonianmag.com/dinosaur/2011/10/paleontologists-unveil-the-11th-archaeopteryx/). *Dinosaur Tracking Blog*. Smithsonian Institution. Archived (https://web.archive.org/web/20111022024203/http://blogs.smithsonianmag.com/dinosaur/2011/10/paleontologists-unveil-the-11th-archaeopteryx/) from the original on 22 October 2011. Retrieved 30 October 2011.
43. Hecht, Jeff (20 October 2011). "Another stunning Archaeopteryx fossil found in Germany" (https://www.newscientist.com/blogs/shortsharpscience/2011/10/stunning-new-archaeopteryx-fos.html). *New Scientist*. Archived (https://web.archive.org/web/20150321065515/http://www.newscientist.com/blogs/shortsharpscience/2011/10/stunning-new-archaeopteryx-fos.html) from the original on 21 March 2015. Retrieved 24 August 2017.

44. Foth, C; Tischlinger, H; Rauhut, O. W (3 July 2014). "New specimen of *Archaeopteryx* provides insights into the evolution of pennaceous feathers". *Nature*. **511** (7507): 79–82. Bibcode:2014Natur.511...79F (https://ui.adsabs.harvard.edu/abs/2014Natur.511...79F). doi:10.1038/nature13467 (https://doi.org/10.1038%2Fnature13467). PMID 24990749 (https://pubmed.ncbi.nlm.nih.gov/24990749). S2CID 4464659 (https://api.semanticscholar.org/CorpusID:4464659).
45. "First show off, then take off" (https://web.archive.org/web/20140707074102/http://www.en.uni-muenchen.de/news/press-services/press-releases/2014/archaeopteryx.html). Ludwig-Maximilians-Universität. 3 July 2014. Archived from the original (http://www.en.uni-muenchen.de/news/press-services/press-releases/2014/archaeopteryx.html) on 7 July 2014.
46. "Schamhaupten: Fossil des Archaeopteryx entdeckt" (https://web.archive.org/web/20141019150803/http://www.donaukurier.de/lokales/kurzmeldungen/riedenburg/Fossil-des-Archaeopteryx-entdeckt;art74375,2880005#plx982820499). *donaukurier.de*. Archived from the original (http://www.donaukurier.de/lokales/kurzmeldungen/riedenburg/Fossil-des-Archaeopteryx-entdeckt;art74375,2880005#plx982820499) on 19 October 2014. Retrieved 18 February 2014.
47. Rauhut, Oliver W.M.; Foth, Christian; Tischlinger, Helmut (2018). "The oldest *Archaeopteryx* (Theropoda: Avialiae): a new specimen from the Kimmeridgian/Tithonian boundary of Schamhaupten, Bavaria" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5788062). *PeerJ*. **6** e4191. Bibcode:2018PeerJ...6e4191R (https://ui.adsabs.harvard.edu/abs/2018PeerJ...6e4191R). doi:10.7717/peerj.4191 (https://doi.org/10.7717%2Fpeerj.4191). PMC 5788062 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5788062). PMID 29383285 (https://pubmed.ncbi.nlm.nih.gov/29383285).
48. Foth, Christian; van de Kamp, Thomas; Tischlinger, Helmut; Kantelis, Theron; Carney, Ryan M.; Zuber, Marcus; Hamann, Elias; Wallaard, Jonathan J. W.; Lenz, Norbert; Rauhut, Oliver W. M.; Frey, Eberhard (3 January 2025). "A new Archaeopteryx from the lower Tithonian Mörnsheim Formation at Mühlheim (Late Jurassic)" (https://doi.org/10.3897%2Ffr.28.e131671). *Fossil Record*. **28** (1): 17–43. Bibcode:2025FossR..28...17F (https://ui.adsabs.harvard.edu/abs/2025FossR..28...17F). doi:10.3897/fr.28.e131671 (https://doi.org/10.3897%2Ffr.28.e131671).
49. "Meet the Chicago Archaeopteryx - Field Museum" (https://www.fieldmuseum.org/exhibition/meet-the-chicago-archaeopteryx). *www.fieldmuseum.org*. Retrieved 6 May 2024.
50. Tribune, Jennifer Day | Chicago (6 May 2024). "Field Museum has a new fossil of an avian dinosaur, unveiled at an event Monday" (https://www.chicagotribune.com/2024/05/06/field-museum-has-a-new-fossil-of-an-avian-dinosaur-unveiled-at-an-event-monday/). *Chicago Tribune*. Archived (https://web.archive.org/web/20240617222229/https://www.chicagotribune.com/2024/05/06/field-museum-has-a-new-fossil-of-an-avian-dinosaur-unveiled-at-an-event-monday/) from the original on 17 June 2024. Retrieved 6 May 2024.
51. Hoyle, F.; Wickramasinghe, N. C.; Watkins, R. S. (1985). "*Archaeopteryx*". *British Journal of Photography*. **132**: 693–694.
52. Watkins, R. S.; Hoyle, F.; Wickrmasinghe, N. C.; Watkins, J.; Rabilizirov, R.; Spetner, L. M. (1985). "*Archaeopteryx* – a photographic study". *British Journal of Photography*. **132**: 264–266.
53. Watkins, R. S.; Hoyle, F.; Wickrmasinghe, N. C.; Watkins, J.; Rabilizirov, R.; Spetner, L. M. (1985). "*Archaeopteryx* – a further comment". *British Journal of Photography*. **132**: 358–359, 367.
54. Watkins, R. S.; Hoyle, F.; Wickrmasinghe, N. C.; Watkins, J.; Rabilizirov, R.; Spetner, L. M. (1985). "*Archaeopteryx* – more evidence". *British Journal of Photography*. **132**: 468–470.
55. Charig, A. J.; Greenaway, F.; Milner, A. C.; Walker, C. A.; Whybrow, P. J. (2 May 1986). "*Archaeopteryx* Is Not a Forgery". *Science*. **232** (4750): 622–626. Bibcode:1986Sci...232..622C (https://ui.adsabs.harvard.edu/abs/1986Sci...232..622C). doi:10.1126/science.232.4750.622 (https://doi.org/10.1126%2Fscience.232.4750.622). PMID 17781413 (https://pubmed.ncbi.nlm.nih.gov/17781413). S2CID 39554239 (https://api.semanticscholar.org/CorpusID:39554239).

56. Nedin, Chris (15 December 2007). "On *Archaeopteryx*, Astronomers, and Forgery" (http://www.talkorigins.org/faqs/archaeopteryx/forgery.html). Archived (https://web.archive.org/web/20070315180204/http://www.talkorigins.org/faqs/archaeopteryx/forgery.html) from the original on 15 March 2007. Retrieved 17 March 2007.
57. Spetner, L. M.; Hoyle, F.; Wickramasinghe, N. C.; Magaritz, M. (1988). "*Archaeopteryx* – more evidence for a forgery". *The British Journal of Photography*. **135**: 14–17.
58. Limestone (mineral) (https://web.archive.org/web/20091028032011/http://encarta.msn.com/encyclopedia_761565838/Limestone_(mineral).html). Encarta.
59. Lambert, David (1993). *The Ultimate Dinosaur Book* (https://archive.org/details/ultimatedinosaur00lamb/page/38). New York: Dorling Kindersley. pp. 38–81 (https://archive.org/details/ultimatedinosaur00lamb/page/38). ISBN 978-1-56458-304-8.
60. Schweigert, G. (2007). "Ammonite biostratigraphy as a tool for dating Upper Jurassic lithographic limestones from South Germany – first results and open questions" (http://doc.rero.ch/record/13825/files/PAL_E722.pdf) (PDF). *Neues Jahrbuch für Geologie und Paläontologie - Abhandlungen*. **245** (1): 117–125. Bibcode:2007NJGPA.245..117S (https://ui.adsabs.harvard.edu/abs/2007NJGPA.245..117S). doi:10.1127/0077-7749/2007/0245-0117 (https://doi.org/10.1127%2F0077-7749%2F2007%2F0245-0117). S2CID 140597349 (https://api.semanticscholar.org/CorpusID:140597349).
61. Paul, Gregory S. (2016). *The Princeton Field Guide to Dinosaurs*. Princeton University Press. p. 144. ISBN 978-1-78684-190-2. OCLC 985402380 (https://search.worldcat.org/oclc/985402380).
62. Holtz, Thomas Jr. (1995). "*Archaeopteryx*s Relationship With Modern Birds" (https://web.archive.org/web/20070209063731/http://dinosauria.com/jdp/archie/archie.htm). *Journal of Dinosaur Paleontology*. Archived from the original (http://www.dinosauria.com/jdp/archie/archie.htm) on 9 February 2007. Retrieved 1 March 2007.
63. Benton, M.J.; Cook, E.; Grigorescu, D.; Popa, E.; Tallódi, E. (1997). "Dinosaurs and other tetrapods in an Early Cretaceous bauxite-filled fissure, northwestern Romania" (http://doc.rero.ch/record/14450/files/PAL_E1638.pdf) (PDF). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **130** (1–4): 275–292. Bibcode:1997PPP...130..275B (https://ui.adsabs.harvard.edu/abs/1997PPP...130..275B). doi:10.1016/S0031-0182(96)00151-4 (https://doi.org/10.1016%2FS0031-0182%2896%2900151-4). Archived (https://web.archive.org/web/20230522234447/https://doc.rero.ch/record/14450/files/PAL_E1638.pdf) (PDF) from the original on 22 May 2023. Retrieved 3 January 2023.
64. Bühler, P.; Bock, W. J. (2002). "Zur Archaeopteryx-Nomenklatur: Missverständnisse und Lösung". *Journal of Ornithology*. **143** (3): 269–286. doi:10.1046/j.1439-0361.2002.02006.x (https://doi.org/10.1046%2Fj.1439-0361.2002.02006.x). [Article in German; English abstract]
65. Olson, Storrs L.; Feduccia, Alan (March 1979). "Flight capability and the pectoral girdle of *Archaeopteryx*". *Nature*. **278** (5701): 247–248. Bibcode:1979Natur.278..247O (https://ui.adsabs.harvard.edu/abs/1979Natur.278..247O). doi:10.1038/278247a0 (https://doi.org/10.1038%2F278247a0). hdl:10088/6524 (https://hdl.handle.net/10088%2F6524). S2CID 4351802 (https://api.semanticscholar.org/CorpusID:4351802).
66. Ostrom, J. H. (1976). "*Archaeopteryx* and the origin of birds" (http://doc.rero.ch/record/161620/files/PAL_E2122.pdf) (PDF). *Biological Journal of the Linnean Society*. **8** (2): 91–182. doi:10.1111/j.1095-8312.1976.tb00244.x (https://doi.org/10.1111%2Fj.1095-8312.1976.tb00244.x). Archived (https://web.archive.org/web/20230126031808/https://doc.rero.ch/record/161620/files/PAL_E2122.pdf) (PDF) from the original on 26 January 2023. Retrieved 21 January 2023.
67. Ostrom, J. H. (1985). "Introduction to Archaeopteryx". In Hecht, M. K. O.; Ostrom, J. H.; Viohl, G.; Wellnhofer, P. (eds.). *The Beginnings of Birds: Proceedings of the International Archaeopteryx Conference*. Eichstätt: Freunde des Jura-Museums Eichstätt. pp. 9–20.
68. Attenborough, David (1998). *The Life of Birds*. BBC Books. ISBN 0-563-38792-0.

69. Feduccia, A.; Tordoff, H. B. (1979). "Feathers of Archaeopteryx: Asymmetric vanes indicate aerodynamic function". *Science*. **203** (4384): 1021–1022. Bibcode:1979Sci...203.1021F (https://ui.adsabs.harvard.edu/abs/1979Sci...203.1021F). doi:10.1126/science.203.4384.1021 (https://doi.org/10.1126%2Fscience.203.4384.1021). PMID 17811125 (https://pubmed.ncbi.nlm.nih.gov/17811125). S2CID 20444096 (https://api.semanticscholar.org/CorpusID:20444096).
70. Christensen, P; Bonde, N. (2004). "Body plumage in Archaeopteryx: a review, and new evidence from the Berlin specimen" (http://dinonews.net/rubriq/biblio.php#plumes_archaeopteryx_christiansen). *Comptes Rendus Palevol*. **3** (2): 99–118. Bibcode:2004CRPal...3...99C (https://ui.adsabs.harvard.edu/abs/2004CRPal...3...99C). doi:10.1016/j.crpv.2003.12.001 (https://doi.org/10.1016%2Fj.crpv.2003.12.001). Archived (https://web.archive.org/web/20070405005210/http://dinonews.net/rubriq/biblio.php#plumes_archaeopteryx_christiansen) from the original on 5 April 2007. Retrieved 1 April 2007.
71. Longrich, N. (2006). "Structure and function of hindlimb feathers in Archaeopteryx lithographica". *Paleobiology*. **32** (3): 417–431. Bibcode:2006Pbio...32..417L (https://ui.adsabs.harvard.edu/abs/2006Pbio...32..417L). doi:10.1666/04014.1 (https://doi.org/10.1666%2F04014.1). S2CID 85832346 (https://api.semanticscholar.org/CorpusID:85832346).
72. Reisdorf, A. G.; Wuttke, M. (2012). "Re-evaluating Moodie's Opisthotonic-Posture Hypothesis in fossil vertebrates. Part I: Reptiles – The taphonomy of the bipedal dinosaurs *Compsognathus longipes* and *Juravenator starki* from the Solnhofen Archipelago (Jurassic, Germany)". *Palaeobiodiversity and Palaeoenvironments*. **92** (1): 119–168. Bibcode:2012PdPe...92..119R (https://ui.adsabs.harvard.edu/abs/2012PdPe...92..119R). doi:10.1007/s12549-011-0068-y (https://doi.org/10.1007%2Fs12549-011-0068-y). S2CID 129785393 (https://api.semanticscholar.org/CorpusID:129785393).
73. Carney, R.; Vinther, Jakob; Shawkey, Matthew D.; d'Alba, Liliana; Ackermann, Jörg (2012). "New evidence on the colour and nature of the isolated Archaeopteryx feather" (https://doi.org/10.1038%2Fncomms1642). *Nature Communications*. **3** 637. Bibcode:2012NatCo...3..637C (https://ui.adsabs.harvard.edu/abs/2012NatCo...3..637C). doi:10.1038/ncomms1642 (https://doi.org/10.1038%2Fncomms1642). PMID 22273675 (https://pubmed.ncbi.nlm.nih.gov/22273675).
74. Switek, Brian (9 November 2011). "Archaeopteryx was robed in black" (https://www.newscientist.com/article/dn21146-archaeopteryx-was-robed-in-black.html). *New Scientist*. Las Vegas. Archived (https://web.archive.org/web/20150426132408/http://www.newscientist.com/article/dn21146-archaeopteryx-was-robed-in-black.html) from the original on 26 April 2015. Retrieved 24 August 2017.
75. Manning, Phillip. L.; Edwards, Nicholas P.; Wogelius, Roy A.; Bergmann, Uwe; Barden, Holly E.; Larson, Peter L.; Schwarz-Wings, Daniela; Egerton, Victoria M.; Sokaras, Dimosthenis; et al. (2013). "Synchrotron-based chemical imaging reveals plumage patterns in a 150 million year old early bird" (https://doi.org/10.1039%2Fc3ja50077b). *Journal of Analytical Atomic Spectrometry*. **28** (7): 1024. doi:10.1039/c3ja50077b (https://doi.org/10.1039%2Fc3ja50077b).
76. Vinther, J (2015). "A guide to the field of palaeo colour: melanin and other pigments can fossilise: reconstructing colour patterns from ancient organisms can give new insights to ecology and behaviour". *BioEssays*. **6** (37): 643–656. Bibcode:2015BiEss..37..643V (https://ui.adsabs.harvard.edu/abs/2015BiEss..37..643V). doi:10.1002/bies.201500018 (https://doi.org/10.1002%2Fbies.201500018). PMID 25854512 (https://pubmed.ncbi.nlm.nih.gov/25854512). S2CID 24966334 (https://api.semanticscholar.org/CorpusID:24966334).
77. Carney, Ryan; Molnar, Julia; Updike, Earl; Brown, William; Jackson, Jessie; Shawkey, Matthew; Lindgren, Johan; Sjövall, Peter; Falkingham, Peter; Gauthier, Jacques (2014). "Archaeopteryx in 4D" (https://web.archive.org/web/20180425135859/http://vertpaleo.org/GlobalPDFS/SVP-2014-Program-and-Abstract-Book-9-18-2014.aspx). *Journal of Vertebrate Paleontology*. **103**. Archived from the original (http://vertpaleo.org/GlobalPDFS/SVP-2014-Program-and-Abstract-Book-9-18-2014.aspx) on 25 April 2018. Retrieved 15 January 2018.

78. "SVP – Press Release – Taking a Deeper Look at "Ancient Wing" " (https://web.archive.org/web/20160419054434/http://vertpaleo.org/Society-News/SVP-Paleo-News/Society-News,-Press-Releases/Press-Release-Taking-a-Deeper-Look-at-Ancient-Wing.aspx). *vertpaleo.org*. Archived from the original (http://vertpaleo.org/Society-News/SVP-Paleo-News/Society-News,-Press-Releases/Press-Release-Taking-a-Deeper-Look-at-Ancient-Wing.aspx) on 19 April 2016. Retrieved 3 April 2016.
79. Swinton, W. E. (1960). "Opinion 1084, Proposed addition of the generic name *Archaeopteryx* VON MEYER, 1861 and the specific name *Lithographica*, VON MEYER, 1861, as published in the binomen *Archaeopteryx Lithographica* to the official lists (Class Aves)" (https://www.biodiversitylibrary.org/page/12220197). *Bulletin of Zoological Nomenclature*. **17** (6–8): 224–226.
80. ICZN (1961). "Opinion 607, *Archaeopteryx* VON MEYER, 1861 (Aves); Addition to the Official list" (https://www.biodiversitylibrary.org/page/12220667). *Bulletin of Zoological Nomenclature*. **18** (4): 260–261. Archived (https://web.archive.org/web/20180915110301/https://www.biodiversitylibrary.org/page/12220667) from the original on 15 September 2018. Retrieved 19 February 2018.
81. Wagner, A. (1861). *Über ein neues, angeblich mit Vogelfedern versehenes Reptil aus dem Solnhofener lithographischen Schiefer. Sitzungberichte der Bayerischen Akademie der Wissenschaften, mathematisch-physikalische Classe* 146–154.
82. ICZN (1977). "Opinion 1070. Conservation of *Archaeopteryx lithographica* VON MEYER 1861 (Aves)" (https://www.biodiversitylibrary.org/page/12226120). *Bulletin of Zoological Nomenclature*. **33**: 165–166.
83. Bock, W. J.; Bühler, P. (2007). "*Archaeopteryx lithographica* von Meyer, 1861 (Aves): proposed conservation of usage by designation of a neotype" (http://iczn.org/content/archaeopteryx-lithographica-von-meyer-1861-aves-proposed-conservation-usage-designation-neot). *Bulletin of Zoological Nomenclature*. **64** (4): 261–262. Archived (https://web.archive.org/web/20120306012804/http://iczn.org/content/archaeopteryx-lithographica-von-meyer-1861-aves-proposed-conservation-usage-designation-neot) from the original on 6 March 2012. Retrieved 8 January 2011.
84. ICZN (2011). "OPINION 2283 (Case 3390) Archaeopteryx lithographica von Meyer, 1861 (Aves): conservation of usage by designation of a neotype" (http://iczn.org/content/opinion-2283-case-3390-archaeopteryx-lithographica-von-meyer-1861-aves-conservation-usage-de). *Bulletin of Zoological Nomenclature*. **68** (3): 230–233. doi:10.21805/bzn.v68i3.a16 (https://doi.org/10.21805%2Fbzn.v68i3.a16). S2CID 160784119 (https://api.semanticscholar.org/CorpusID:160784119). Archived (https://web.archive.org/web/20120318171909/http://iczn.org/content/opinion-2283-case-3390-archaeopteryx-lithographica-von-meyer-1861-aves-conservation-usage-de) from the original on 18 March 2012. Retrieved 7 October 2011.
85. "Archaeopteryx turns out to be singular bird of a feather" (https://www.newscientist.com/article/mg18224432-300-archaeopteryx-turns-out-to-be-singular-bird-of-a-feather). *New Scientist*. 17 April 2004. Archived (https://web.archive.org/web/20200422153325/https://www.newscientist.com/article/mg18224432-300-archaeopteryx-turns-out-to-be-singular-bird-of-a-feather/) from the original on 22 April 2020. Retrieved 26 April 2020.
86. Clarke, Julia A.; Norell, Mark A. (December 2002). "The Morphology and Phylogenetic Position of Apsaravis ukhaana from the Late Cretaceous of Mongolia". *American Museum Novitates* (3387): 1–46. doi:10.1206/0003-0082(2002)387<0001:TMAPPO>2.0.CO;2 (https://doi.org/10.1206%2F0003-0082%282002%29387%3C0001%3ATMAPPO%3E2.0.CO%3B2). hdl:2246/2876 (https://hdl.handle.net/2246%2F2876). S2CID 52971055 (https://api.semanticscholar.org/CorpusID:52971055).
87. Witmer, Lawrence M. (2002). "The debate on avian ancestry". In Witmer, L.; Chiappe, L. (eds.). *Mesozoic Birds: Above the Heads of Dinosaurs* (https://archive.org/details/mesozoicbirdsabo00chia_070). Berkeley: University of California Press. pp. 3 (https://archive.org/details/mesozoicbirdsabo00chia_070/page/n13)–30. ISBN 978-0-520-20094-4.

88. Lowe, P. R. (1935). "On the relationship of the Struthiones to the dinosaurs and to the rest of the avian class, with special reference to the position of *Archaeopteryx*". *Ibis*. **5** (2): 398–432. doi:10.1111/j.1474-919X.1935.tb02979.x (https://doi.org/10.1111%2Fj.1474-919X.1935.tb02979.x).
89. Thulborn, R. A. (1984). "The avian relationships of *Archaeopteryx*, and the origin of birds". *Zoological Journal of the Linnean Society*. **82** (1–2): 119–158. doi:10.1111/j.1096-3642.1984.tb00539.x (https://doi.org/10.1111%2Fj.1096-3642.1984.tb00539.x).
90. Kurzanov, S. M. (1987). "Avimimidae and the problem of the origin of birds". *Transactions of the Joint Soviet-Mongolian Paleontological Expedition*. **31**: 31–94. ISSN 0320-2305 (https://search.worldcat.org/issn/0320-2305).
91. Barsbold, Rhinchen (1983). "Carnivorous dinosaurs from the Cretaceous of Mongolia". *Transactions of the Joint Soviet-Mongolian Paleontological Expedition*. **19**: 5–119. ISSN 0320-2305 (https://search.worldcat.org/issn/0320-2305).
92. Zweers, G. A.; Van den Berge, J. C. (1997). "Evolutionary patterns of avian trophic diversification". *Zoology: Analysis of Complex Systems*. **100**: 25–57. ISSN 0944-2006 (https://search.worldcat.org/issn/0944-2006).
93. Lee, M. S.; Worthy, T. H. (2012). "Likelihood reinstates *Archaeopteryx* as a primitive bird" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3297401). *Biology Letters*. **8** (2): 299–303. Bibcode:2012BiLet...8..299L (https://ui.adsabs.harvard.edu/abs/2012BiLet...8..299L). doi:10.1098/rsbl.2011.0884 (https://doi.org/10.1098%2Frsbl.2011.0884). PMC 3297401 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3297401). PMID 22031726 (https://pubmed.ncbi.nlm.nih.gov/22031726).
94. Senter, Phil; Kirkland, James I.; DeBlieux, Donald D.; Madsen, Scott; Toth, Natalie (2012). Dodson, Peter (ed.). "New Dromaeosaurids (Dinosauria: Theropoda) from the Lower Cretaceous of Utah, and the Evolution of the Dromaeosaurid Tail" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3352940). *PLOS ONE*. **7** (5) e36790. Bibcode:2012PLoSO...736790S (https://ui.adsabs.harvard.edu/abs/2012PLoSO...736790S). doi:10.1371/journal.pone.0036790 (https://doi.org/10.1371%2Fjournal.pone.0036790). PMC 3352940 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3352940). PMID 22615813 (https://pubmed.ncbi.nlm.nih.gov/22615813).
95. Turner, Alan H.; Makovicky, Peter J.; Norell, Mark A. (2012). "A review of dromaeosaurid systematics and paravian phylogeny" (http://digitallibrary.amnh.org/bitstream/2246/6352/5/B371-cover.pdf) (PDF). *Bulletin of the American Museum of Natural History*. **371**: 1–206. doi:10.1206/748.1 (https://doi.org/10.1206%2F748.1). hdl:2246/6352 (https://hdl.handle.net/2246%2F6352). S2CID 83572446 (https://api.semanticscholar.org/CorpusID:83572446). Archived (https://ghostarchive.org/archive/20221009/http://digitallibrary.amnh.org/bitstream/2246/6352/5/B371-cover.pdf) (PDF) from the original on 9 October 2022.
96. Godefroit, P; Demuynck, H; Dyke, G; Hu, D; Escuillié, F; Claeys, P (2013). "Reduced plumage and flight ability of a new Jurassic paravian theropod from China" (https://doi.org/10.1038%2Fncomms2389). *Nature Communications*. **4** 1394. Bibcode:2013NatCo...4.1394G (https://ui.adsabs.harvard.edu/abs/2013NatCo...4.1394G). doi:10.1038/ncomms2389 (https://doi.org/10.1038%2Fncomms2389). PMID 23340434 (https://pubmed.ncbi.nlm.nih.gov/23340434).
97. Agnolin, Federico; Novas, Fernando E (2013). Avian ancestors. A review of the phylogenetic relationships of the theropods Unenlagiidae, Microraptoria, *Anchiornis*, and Scansoriopterygidae. SpringerBriefs in Earth System Sciences. pp. 1–96. doi:10.1007/978-94-007-5637-3 (https://doi.org/10.1007%2F978-94-007-5637-3). ISBN 978-94-007-5636-6. S2CID 199493087 (https://api.semanticscholar.org/CorpusID:199493087).

98. Hu, Dongyu; Clarke, Julia A.; Eliason, Chad M.; Qiu, Rui; Li, Quanguo; Shawkey, Matthew D.; Zhao, Cuilin; D'Alba, Liliana; Jiang, Jinkai; Xu, Xing (15 January 2018). "A bony-crested Jurassic dinosaur with evidence of iridescent plumage highlights complexity in early paravian evolution" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5768872). *Nature Communications*. **9** (1): 217. Bibcode:2018NatCo...9..217H (https://ui.adsabs.harvard.edu/abs/2018NatCo...9..217H). doi:10.1038/s41467-017-02515-y (https://doi.org/10.1038%2Fs41467-017-02515-y). ISSN 2041-1723 (https://search.worldcat.org/issn/2041-1723). PMC 5768872 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5768872). PMID 29335537 (https://pubmed.ncbi.nlm.nih.gov/29335537).
99. Wang, Min; O'Connor, Jingmai K.; Xu, Xing; Zhou, Zhonghe (May 2019). "A new Jurassic scansoriopterygid and the loss of membranous wings in theropod dinosaurs" (http://www.nature.com/articles/s41586-019-1137-z). *Nature*. **569** (7755): 256–259. Bibcode:2019Natur.569..256W (https://ui.adsabs.harvard.edu/abs/2019Natur.569..256W). doi:10.1038/s41586-019-1137-z (https://doi.org/10.1038%2Fs41586-019-1137-z). ISSN 0028-0836 (https://search.worldcat.org/issn/0028-0836). PMID 31068719 (https://pubmed.ncbi.nlm.nih.gov/31068719). S2CID 148571099 (https://api.semanticscholar.org/CorpusID:148571099). Archived (https://web.archive.org/web/20210108062142/https://www.nature.com/articles/s41586-019-1137-z) from the original on 8 January 2021. Retrieved 11 January 2021.
100. Hartman, Scott; Mortimer, Mickey; Wahl, William R.; Lomax, Dean R.; Lippincott, Jessica; Lovelace, David M. (10 July 2019). "A new paravian dinosaur from the Late Jurassic of North America supports a late acquisition of avian flight" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6626525). *PeerJ*. **7** e7247. Bibcode:2019PeerJ...7e7247H (https://ui.adsabs.harvard.edu/abs/2019PeerJ...7e7247H). doi:10.7717/peerj.7247 (https://doi.org/10.7717%2Fpeerj.7247). ISSN 2167-8359 (https://search.worldcat.org/issn/2167-8359). PMC 6626525 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6626525). PMID 31333906 (https://pubmed.ncbi.nlm.nih.gov/31333906).
101. Senter, Phil (2006). "Scapular orientation in theropods and basal birds, and the origin of flapping flight" (https://www.app.pan.pl/article/item/app51-305.html). *Acta Palaeontologica Polonica*. **51** (2): 305–313.
102. Speakman, J. R.; Thomson, S. C. (1994). "Flight capabilities of *Archaeopteryx*" (https://doi.org/10.1038%2F370514a0). *Nature*. **370** (6490): 336–340. Bibcode:1994Natur.370..514S (https://ui.adsabs.harvard.edu/abs/1994Natur.370..514S). doi:10.1038/370514a0 (https://doi.org/10.1038%2F370514a0). PMID 28568098 (https://pubmed.ncbi.nlm.nih.gov/28568098). S2CID 4248184 (https://api.semanticscholar.org/CorpusID:4248184).
103. Norberg, R. A. (1995). "Feather asymmetry in *Archaeopteryx*" (https://doi.org/10.1038%2F374211a0). *Nature*. **374** (6519): 211. Bibcode:1995Natur.374..211M (https://ui.adsabs.harvard.edu/abs/1995Natur.374..211M). doi:10.1038/374211a0 (https://doi.org/10.1038%2F374211a0). S2CID 4352260 (https://api.semanticscholar.org/CorpusID:4352260).
104. Nudds, Robert L.; Dyke, Gareth J. (14 May 2010). "Narrow Primary Feather Rachises in *Confuciusornis* and *Archaeopteryx* Suggest Poor Flight Ability". *Science*. **328** (5980): 887–889. Bibcode:2010Sci...328..887N (https://ui.adsabs.harvard.edu/abs/2010Sci...328..887N). doi:10.1126/science.1188895 (https://doi.org/10.1126%2Fscience.1188895). PMID 20466930 (https://pubmed.ncbi.nlm.nih.gov/20466930). S2CID 12340187 (https://api.semanticscholar.org/CorpusID:12340187).
105. Balter, M. (2010). "Did First Feathers Prevent Early Flight?" (http://news.sciencemag.org/sciencenow/2010/05/did-first-feathers-prevent-early.html) Archived (https://web.archive.org/web/20100518072039/http://news.sciencemag.org/sciencenow/2010/05/did-first-feathers-prevent-early.html) 18 May 2010 at the Wayback Machine *Science Now*, 13 May 2010.

106. Paul, G. S. (15 October 2010). "Comment on 'Narrow Primary Feather Rachises in *Confuciusornis* and *Archaeopteryx* Suggest Poor Flight Ability.' " (https://doi.org/10.1126%2Fscience.1192963). *Science*. **330** (6002): 320. Bibcode:2010Sci...330..320P (https://ui.adsabs.harvard.edu/abs/2010Sci...330..320P). doi:10.1126/science.1192963 (https://doi.org/10.1126%2Fscience.1192963). PMID 20947747 (https://pubmed.ncbi.nlm.nih.gov/20947747).
107. Dyke, G. J.; Nudds, R. L. (15 October 2010). "Response to Comments on "Narrow Primary Feather Rachises in Confuciusornis and Archaeopteryx Suggest Poor Flight Ability" " (http://www.sciencemag.org/cgi/reprint/330/6002/320-d.pdf) (PDF). *Science*. **330** (6002): 320. Bibcode:2010Sci...330..320N (https://ui.adsabs.harvard.edu/abs/2010Sci...330..320N). doi:10.1126/science.1193474 (https://doi.org/10.1126%2Fscience.1193474). S2CID 85044108 (https://api.semanticscholar.org/CorpusID:85044108). Archived (https://ghostarchive.org/archive/20221009/http://www.sciencemag.org/cgi/reprint/330/6002/320-d.pdf) (PDF) from the original on 9 October 2022.
108. Videler, JJ (2005) Avian Flight. Oxford University Press. ISBN 0-19-856603-4 pages 98–117
109. Videler, John J (January 2005). "How Archaeopteryx could run over water" (https://www.researchgate.net/publication/38957284). *ResearchGate*.
110. Witmer, Lawrence M. (4 August 2004). "Inside the oldest bird brain". *Nature*. **430** (7000): 619–620. doi:10.1038/430619a (https://doi.org/10.1038%2F430619a). PMID 15295579 (https://pubmed.ncbi.nlm.nih.gov/15295579). S2CID 1317955 (https://api.semanticscholar.org/CorpusID:1317955).
111. Alonso, Patricio Domínguez; Milner, Angela C.; Ketcham, Richard A.; Cookson, M. John; Rowe, Timothy B. (August 2004). "The avian nature of the brain and inner ear of Archaeopteryx" (http://doc.rero.ch/record/15280/files/PAL_E2579.pdf) (PDF). *Nature*. **430** (7000): 666–669. Bibcode:2004Natur.430..666A (https://ui.adsabs.harvard.edu/abs/2004Natur.430..666A). doi:10.1038/nature02706 (https://doi.org/10.1038%2Fnature02706). PMID 15295597 (https://pubmed.ncbi.nlm.nih.gov/15295597). S2CID 4391019 (https://api.semanticscholar.org/CorpusID:4391019). Archived (https://web.archive.org/web/20230405214607/https://doc.rero.ch/record/15280/files/PAL_E2579.pdf) (PDF) from the original on 5 April 2023. Retrieved 17 December 2022.
112. Larsson, H. C. E. (2001). "Endocranial anatomy of *Carcharodontosaurus saharicus* (Theropoda: Allosauroidea) and its implications for theropod brain evolution". In: Tanke, D. H.; Carpenter, K.; Skrepnick, M. W. (eds.) *Mesozoic Vertebrate Life*. Indiana University Press. pp. 19–33.
113. Balanoff, Amy M.; Bever, Gabe S.; Rowe, Timothy B.; Norell, Mark A. (2013). "Evolutionary origins of the avian brain". *Nature*. **501** (7465): 93–6. Bibcode:2013Natur.501...93B (https://ui.adsabs.harvard.edu/abs/2013Natur.501...93B). doi:10.1038/nature12424 (https://doi.org/10.1038%2Fnature12424). PMID 23903660 (https://pubmed.ncbi.nlm.nih.gov/23903660). S2CID 4451895 (https://api.semanticscholar.org/CorpusID:4451895).
114. Feo, Teresa J.; Field, Daniel J.; Prum, Richard O. (22 March 2015). "Barb geometry of asymmetrical feathers reveals a transitional morphology in the evolution of avian flight" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4345455). *Proceedings of the Royal Society of London B: Biological Sciences*. **282** (1803) 20142864. Bibcode:2015PBioS.28242864F (https://ui.adsabs.harvard.edu/abs/2015PBioS.28242864F). doi:10.1098/rspb.2014.2864 (https://doi.org/10.1098%2Frspb.2014.2864). ISSN 0962-8452 (https://search.worldcat.org/issn/0962-8452). PMC 4345455 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4345455). PMID 25673687 (https://pubmed.ncbi.nlm.nih.gov/25673687).
115. Guarino, Ben (13 March 2018). "This feathery dinosaur probably flew, but not like any bird you know" (https://www.washingtonpost.com/news/speaking-of-science/wp/2018/03/13/this-feathery-dinosaur-probably-flew-but-not-like-any-bird-you-know/). *The Washington Post*. Archived (https://web.archive.org/web/20180626030303/https://www.washingtonpost.com/news/speaking-of-science/wp/2018/03/13/this-feathery-dinosaur-probably-flew-but-not-like-any-bird-you-know/) from the original on 26 June 2018. Retrieved 13 March 2018.

116. Kaye TG, Pittman M, Wahl WR (2020). "*Archaeopteryx* feather sheaths reveal sequential center-out flight-related molting strategy" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7722847). *Communications Biology*. **3** (1) 745. doi:10.1038/s42003-020-01467-2 (https://doi.org/10.1038%2Fs42003-020-01467-2). PMC 7722847 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7722847). PMID 33293660 (https://pubmed.ncbi.nlm.nih.gov/33293660).
117. Kiat Y, Pyle P, Balaban A, O'Connor JK (2021). "Reinterpretation of purported molting evidence in the Thermopolis *Archaeopteryx*" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8257594). *Communications Biology*. **4** (1) 837. doi:10.1038/s42003-021-02349-x (https://doi.org/10.1038%2Fs42003-021-02349-x). PMC 8257594 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8257594). PMID 34226661 (https://pubmed.ncbi.nlm.nih.gov/34226661). S2CID 235738230 (https://api.semanticscholar.org/CorpusID:235738230).
118. Kaye TG, Pittman M (2021). "Reply to: Reinterpretation of purported molting evidence in the Thermopolis *Archaeopteryx*" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8257677). *Communications Biology*. **4** (1) 839. doi:10.1038/s42003-021-02367-9 (https://doi.org/10.1038%2Fs42003-021-02367-9). PMC 8257677 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8257677). PMID 34226634 (https://pubmed.ncbi.nlm.nih.gov/34226634). S2CID 235738222 (https://api.semanticscholar.org/CorpusID:235738222).
119. "Archaeopteryx's Feeding-Related Structures Reflect Elevated Demands of Flight, Scientists Say | Sci.News" (https://www.sci.news/paleontology/archaeopteryx-feeding-structures-flight-14539.html). *Sci.News: Breaking Science News*. 6 February 2026. Retrieved 8 February 2026.
120. O'Connor, Jingmai K.; Clark, Alexander D.; Kuo, Pei-Chen; Wang, Min; Shinya, Akiko; Beek, Constance Van; Chang, Huali (2 February 2026). "Avian features of Archaeopteryx feeding apparatus reflect elevated demands of flight" (https://www.cell.com/the-innovation/abstract/S2666-6758(25)00289-9). *The Innovation*. **7** (2). doi:10.1016/j.xinn.2025.101086 (https://doi.org/10.1016%2Fj.xinn.2025.101086). ISSN 2666-6758 (https://search.worldcat.org/issn/2666-6758). PMC 12881756 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12881756).
121. *EurekAlert!* (8 October 2009), "*Archaeopteryx* was not very bird-like" (http://www.eurekalert.org/pub_releases/2009-10/amon-wn100809.php).
122. Schmitz, L.; Motani, R. (2011). "Nocturnality in Dinosaurs Inferred from Scleral Ring and Orbit Morphology". *Science*. **332** (6030): 705–8. Bibcode:2011Sci...332..705S (https://ui.adsabs.harvard.edu/abs/2011Sci...332..705S). doi:10.1126/science.1200043 (https://doi.org/10.1126%2Fscience.1200043). PMID 21493820 (https://pubmed.ncbi.nlm.nih.gov/21493820). S2CID 33253407 (https://api.semanticscholar.org/CorpusID:33253407).
123. Davis, Paul G.; Briggs, Derek E. G. (February 1998). "The Impact of Decay and Disarticulation on the Preservation of Fossil Birds". *PALAIOS*. **13** (1): 3. Bibcode:1998Palai..13....3D (https://ui.adsabs.harvard.edu/abs/1998Palai..13....3D). doi:10.2307/3515277 (https://doi.org/10.2307%2F3515277). JSTOR 3515277 (https://www.jstor.org/stable/3515277).
124. Bartell, K. W.; Swinburne, N. H. M.; Conway-Morris, S. (1994). *Solnhofen: a study in Mesozoic palaeontology*. Cambridge University Press. ISBN 978-0-521-45830-6.
125. Paul, Gregory S. (2002). *Dinosaurs of the Air: the Evolution and Loss of Flight in Dinosaurs and Birds* (https://archive.org/details/dinosaursofairev0000paul). Baltimore: Johns Hopkins University Press. pp. 159–167. ISBN 978-0-8018-6763-7.
126. Buisonjé, P. H. de (1985). "Climatological conditions during deposition of the Solnhofen limestones". In Hecht, M. K.; Ostrom, J. H.; Viohl, G.; Wellnhofer, P. (eds.). *The beginnings of Birds: Proceedings of the International* Archaeopteryx *Conference, Eichstatt, 1984*. Eichstätt: Freunde des Jura-Museums Eichstätt. pp. 45–65. ISBN 978-3-9801178-0-7.
127. Feduccia, A. (1993). "Evidence from claw geometry indicating arboreal habits of *Archaeopteryx*". *Science*. **259** (5096): 790–793. Bibcode:1993Sci...259..790F (https://ui.adsabs.harvard.edu/abs/1993Sci...259..790F). doi:10.1126/science.259.5096.790 (https://doi.org/10.1126%2Fscience.259.5096.790). PMID 17809342 (https://pubmed.ncbi.nlm.nih.gov/17809342). S2CID 27475082 (https://api.semanticscholar.org/CorpusID:27475082).

128. Pike, A. V. L.; Maitland, D. P. (2004). "Scaling of bird claws". *Journal of Zoology*. **262** (1): 73–81. Bibcode:2004JZoo..262...73P (https://ui.adsabs.harvard.edu/abs/2004JZoo..262...73P). doi:10.1017/s0952836903004382 (https://doi.org/10.1017%2Fs0952836903004382). ISSN 0952-8369 (https://search.worldcat.org/issn/0952-8369).

Further reading

- G. R. de Beer (1954). *Archaeopteryx lithographica: a study based upon the British Museum specimen*. Trustees of the British Museum, London.
- P. Chambers (2002). *Bones of Contention: The Fossil that Shook Science.* John Murray, London. ISBN 0-7195-6059-4.
- A. Feduccia (1996). *The Origin and Evolution of Birds*. Yale University Press, New Haven. ISBN 0-300-06460-8.
- Heilmann, G. (1926). *The Origin of Birds*. Witherby, London.
- T. H. Huxley. (1871). *Manual of the anatomy of vertebrate animals*. London.
- H. von Meyer (1861). *Archaeopterix lithographica (Vogel-Feder) und Pterodactylus von Solenhofen*. *Neues Jahrbuch für Mineralogie, Geognosie, Geologie und Petrefakten-Kunde*. *1861*: 678–679, plate V. [Article in German]. Full text, Google Books (https://books.google.com/books?id=6RAFAAAAQAAJ&pg=PA678).
- P. Shipman (1998). *Taking Wing: Archaeopteryx and the Evolution of Bird Flight*. Weidenfeld & Nicolson, London. ISBN 0-297-84156-4.
- P. Wellnhofer (2008). *Archaeopteryx – Der Urvogel von Solnhofen* (in German). Verlag Friedrich Pfeil, Munich. ISBN 978-3-89937-076-8.

External links

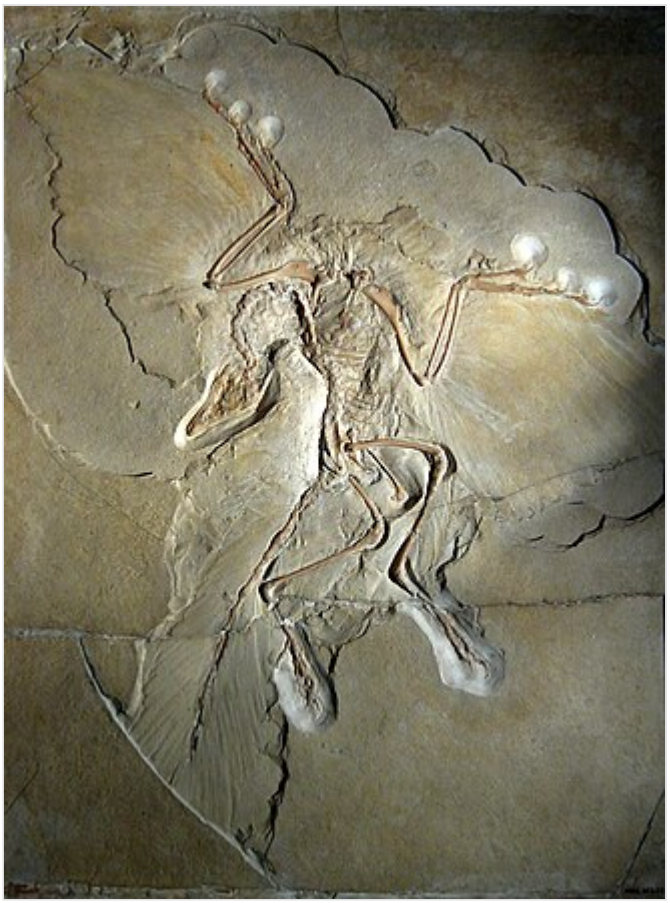
- All About *Archaeopteryx* (http://www.talkorigins.org/faqs/archaeopteryx/info.html), from Talk.Origins.
- Use of SSRL X-ray takes 'transformative glimpse' (https://www.sciencedaily.com/releases/2010/05/100510151348.htm) – A look at chemicals linking birds and dinosaurs
- *Archaeopteryx*: An Early Bird (http://www.ucmp.berkeley.edu/diapsids/birds/archaeopteryx.html) – University of California Museum of Paleontology
- Are Birds Really Dinosaurs? (http://www.ucmp.berkeley.edu/diapsids/avians.html) – University of California Museum of Paleontology
- This Dinosaur Had Feathers and Probably Flew Like a Chicken (https://www.nytimes.com/2025/05/14/science/archaeopteryx-dinosaur-feathers-flight.html). Jingmai O'Connor's analysis of the Chicago Archaeopteryx, 14 May 2025.

Retrieved from "https://en.wikipedia.org/w/index.php?title=Archaeopteryx&oldid=1378563677"

WIKIPEDIA 25
25 years of the free encyclopedia

Origin of birds

The scientific question of which larger group of animals birds evolved within has traditionally been called the "**origin of birds**". The present scientific consensus is that birds are a group of maniraptoran theropod dinosaurs that originated during the Mesozoic era.



The Berlin specimen of *Archaeopteryx lithographica*

A close relationship between birds and dinosaurs was first proposed in the nineteenth century after the discovery of the primitive bird *Archaeopteryx* in Germany. Birds and extinct non-avian dinosaurs share many unique skeletal traits.[1] Moreover, fossils of more than thirty species of non-avian dinosaur with preserved feathers have been collected. There are even very small dinosaurs, such as *Microraptor* and *Anchiornis*, which have long, vaned arm and leg feathers forming wings. The Jurassic basal avialan *Pedopenna* also shows these long foot feathers. Paleontologist Lawrence Witmer concluded in 2009 that this evidence is sufficient to demonstrate that avian evolution went through a four-winged stage.[2] Fossil evidence also demonstrates that birds and dinosaurs shared features such as hollow, pneumatized bones, gastroliths in the digestive system, nest-building, and brooding behaviors.

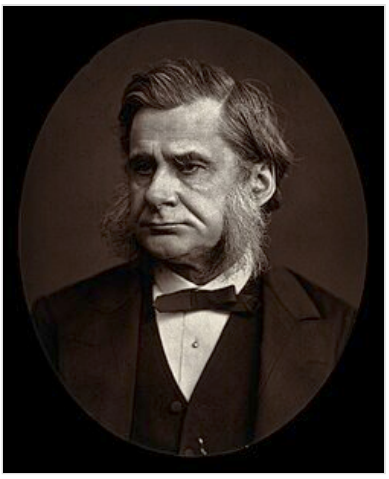
Although the origin of birds has historically been a contentious topic within evolutionary biology, only a few scientists still dispute the dinosaurian origin of birds, suggesting descent from other types of archosaurian reptiles. Within the consensus that supports dinosaurian ancestry, the exact sequence of evolutionary events that gave rise to the early birds within maniraptoran theropods is disputed. The origin of bird flight is a separate but related question for which there are also several proposed answers.

Research history

Huxley, *Archaeopteryx* and early research

Scientific investigation into the origin of birds began shortly after the 1859 publication of Charles Darwin's *On the Origin of Species*.[3] In 1860, a fossilized feather was discovered in Germany's Late Jurassic Solnhofen limestone. Christian Erich Hermann von Meyer described this feather as

Archaeopteryx lithographica the next year.[4] Richard Owen described a nearly complete skeleton in 1863, recognizing it as a bird despite many features reminiscent of reptiles, including clawed forelimbs and a long, bony tail.[5]



Thomas Henry Huxley (1825–1895)

Biologist Thomas Henry Huxley, known as "Darwin's Bulldog" for his tenacious support of the new theory of evolution by means of natural selection, almost immediately seized upon *Archaeopteryx* as a transitional fossil between birds and reptiles. Starting in 1868, and following earlier suggestions by Carl Gegenbaur,[6] and Edward Drinker Cope,[7] Huxley made detailed comparisons of *Archaeopteryx* with various prehistoric reptiles and found that it was most similar to dinosaurs like *Hypsilophodon* and *Compsognathus*.[8][9] The discovery in the late 1870s of the iconic "Berlin specimen" of *Archaeopteryx*, complete with a set of reptilian teeth, provided further evidence. Like Cope, Huxley proposed an evolutionary relationship between birds and dinosaurs. Although Huxley was opposed by the very influential Owen, his conclusions were accepted by many biologists, including Baron Franz Nopcsa,[10] while others, notably Harry Seeley,[11] argued that the similarities were due to convergent evolution.

Heilmann and the thecodont hypothesis

A turning point came in the early twentieth century with the writings of Gerhard Heilmann of Denmark. An artist by trade, Heilmann had a scholarly interest in birds and from 1913 to 1916, expanding on earlier work by Othenio Abel,[12] published the results of his research in several parts, dealing with the anatomy, embryology, behavior, paleontology, and evolution of birds.[13] His work, originally written in Danish as *Vor Nuvaerende Viden om Fuglenes Afstamning*, was compiled, translated into English, and published in 1926 as *The Origin of Birds*.

Heilmann's hypothetical illustration of a pair of fighting 'Proaves' from 1916

Like Huxley, Heilmann compared *Archaeopteryx* and other birds to an exhaustive list of prehistoric reptiles, and also came to the conclusion that theropod dinosaurs like *Compsognathus* were the most similar. However, Heilmann noted that birds had clavicles (collar bones) fused to form a bone called the furcula ("wishbone"), and while clavicles were known in more primitive reptiles, they had not yet been recognized in dinosaurs. Since he was a firm believer in an interpretation of Dollo's law that stated that evolution was not "reversible", Heilmann could not accept that clavicles were lost in dinosaurs and re-evolved in birds. He was therefore forced to rule out dinosaurs as bird ancestors and ascribe all of their similarities to convergent evolution. Heilmann stated that bird ancestors would instead be found among the more primitive "thecodont" grade of reptiles.[14] Heilmann's extremely thorough approach ensured that his book became a

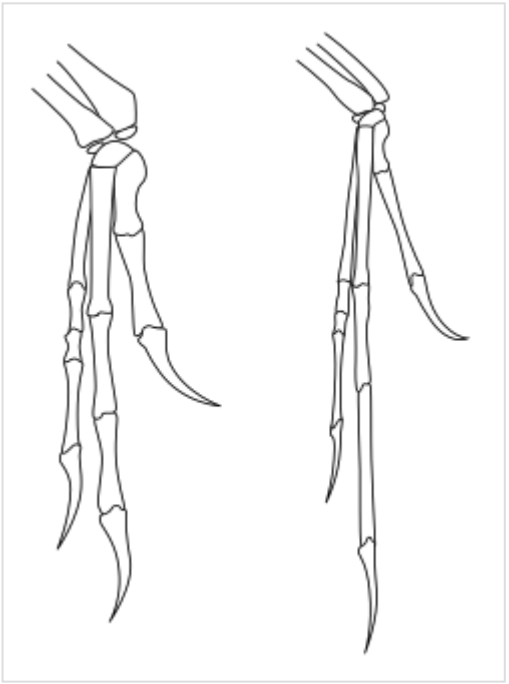
classic in the field, and its conclusions on bird origins, as with most other topics, were accepted by nearly all evolutionary biologists for the next four decades.[15]

Clavicles are relatively delicate bones and therefore in danger of being destroyed or at least damaged beyond recognition. Nevertheless, some fossil theropod clavicles had actually been excavated before Heilmann wrote his book, but these had been misidentified.[16] The absence of clavicles in dinosaurs became the orthodox view despite the discovery of clavicles in the primitive theropod *Segisaurus* in 1936.[17] The next report of clavicles in a dinosaur was in a Russian article in 1983,[18] where Mongolian paleontologist Rinchen Barsbold argued that one or several lineages of theropods may have evolved into birds.[19]

Contrary to what Heilmann believed, paleontologists now accept that clavicles and, in most cases, furculae are a standard feature not just of theropods but of saurischian dinosaurs. Up to late 2007 ossified furculae (i.e. made of bone rather than cartilage) have been found in all types of theropods except the most basal ones, *Eoraptor* and *Herrerasaurus*.[20] The original report of a furcula in the primitive theropod *Segisaurus* (1936) was confirmed by a re-examination in 2005.[21] Joined, furcula-like clavicles have also been found in *Massospondylus*, an Early Jurassic sauropodomorph.[22]

Ostrom, *Deinonychus*, and the dinosaur renaissance

The 'thecodont' hypothesis began to fall out of favor after the 1964 discovery of a new theropod dinosaur in Montana. In 1969, this dinosaur was described and named *Deinonychus* by John Ostrom of Yale University.[23] The next year, Ostrom redescribed a specimen of *Pterodactylus* in the Dutch Teylers Museum as another skeleton of *Archaeopteryx*.[24] The specimen consisted mainly of a single wing and its description made Ostrom aware of the similarities between the wrists of *Archaeopteryx* and *Deinonychus*.[25]



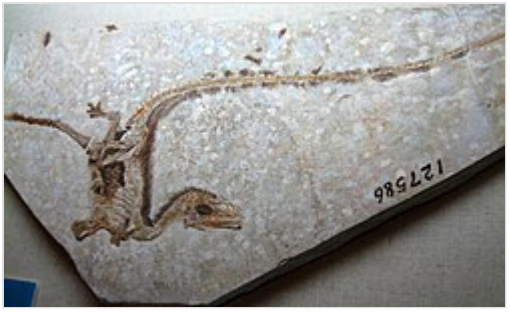
The similarity of the forelimbs of *Deinonychus* (left) and *Archaeopteryx* (right) led John Ostrom to revive the link between dinosaurs and birds.

In 1972, British paleontologist Alick Walker hypothesized that birds arose not from 'thecodonts' but from crocodile ancestors like *Sphenosuchus*.[26] Ostrom's work with both theropods and early birds led him to respond with a series of publications in the mid-1970s in which he laid out the many similarities between birds and theropod dinosaurs, resurrecting the ideas first put forth by Huxley over a century before.[27][28][29] Ostrom's recognition of the dinosaurian ancestry of birds, along with other new ideas about dinosaur metabolism,[30] activity levels, and parental care,[31] began what is known as the dinosaur renaissance, which began in the 1960s and, according to some, continues to this day.[32]

Ostrom's revelations also coincided with the increasing adoption of phylogenetic systematics (cladistics), which began in the 1960s with the work of Willi Hennig.[33] Cladistics is an exact method of arranging species based strictly on their evolutionary relationships, which are calculated by determining the evolutionary tree implying the least number of changes in their anatomical characteristics. In the 1980s,

cladistic methodology was applied to dinosaur phylogeny for the first time by Jacques Gauthier and others, showing unequivocally that birds were a derived group of theropod dinosaurs.[34] Early analyses suggested that dromaeosaurid theropods like *Deinonychus* were particularly closely related to birds, a result that has been corroborated many times since.[35][36]

Feathered dinosaurs in China


Fossil of *Sinosauropteryx prima*

The early 1990s saw the discovery of spectacularly preserved bird fossils in several Early Cretaceous geological formations in the northeastern Chinese province of Liaoning.[37][38] In 1996, Chinese paleontologists described *Sinosauropteryx* as a new genus of bird from the Yixian Formation,[39] but this animal was quickly recognized as a more basal theropod dinosaur closely related to *Compsognathus*. Surprisingly, its body was covered by long filamentous structures. These were dubbed 'protofeathers' and considered homologous with the more advanced feathers of birds,[40] although some scientists disagree with this assessment.[41] Chinese and North American scientists described *Caudipteryx* and *Protarchaeopteryx* soon after. Based on skeletal features, these animals were non-avian dinosaurs, but their remains bore fully formed feathers closely resembling those of birds.[42] "Archaeoraptor", described without peer review in a 1999 issue of *National Geographic*,[43] turned out to be a smuggled forgery,[44] but authentic remains continue to pour out of the Yixian, both legally and illegally. Feathers or "protofeathers" have been found on a wide variety of theropods in the Yixian.[45][46] The morphological gap between non-avian theropods and birds is further closed by the discoveries of extremely bird-like non-avian dinosaurs,[47] as well as non-avian dinosaur-like basal birds.[48]

Digit homology

There is a debate between embryologists and paleontologists whether the hands of theropod dinosaurs and birds are essentially different, based on phalangeal counts—a count of the number of phalanges (finger bones) in the hand.

Embryologists and some paleontologists who oppose the bird-dinosaur link have long numbered the digits of birds II-III-IV based on multiple studies of the development in the egg.[49] This is based on the fact that in most amniotes, the first digit to form in a 5-fingered hand is digit IV, which develops a primary axis. Therefore, embryologists have identified the primary axis in birds as digit IV, and the surviving digits as II-III-IV. The fossils of advanced theropod (Tetanurae) hands appear to have the digits I-II-III (some genera within Avetheropoda also have a reduced digit IV[50]). If this is true, then the II-III-IV development of digits in birds is an indication against theropod (dinosaur) ancestry. However, with no ontogenical (developmental) basis to definitively state which digits are which on a theropod hand (because no non-avian theropods can be observed growing and developing today), the labelling of the theropod hand is not conclusive.

Paleontologists have traditionally identified avian digits as I-II-III. They argue that the digits of birds' number I-II-III, just as those of theropod dinosaurs do, by the conserved phalangeal formula. The phalangeal count for archosaurs is 2-3-4-5-3; many archosaur lineages have a reduced number of digits, but have the same phalangeal formula in the digits that remain. In other words, paleontologists assert that

archosaurs of different lineages tend to lose the same digits when digit loss occurs, from the outside to the inside. The three digits of dromaeosaurs and *Archaeopteryx* have the same phalangeal formula of I-II-III as digits I-II-III of basal archosaurs. Therefore, the lost digits would be V and IV. If this is true, then modern birds would also possess digits I-II-III.[49] Also, one 1999 publication proposed a frame-shift in the digits of the theropod line leading to birds (thus making digit I into digit II, II to III, and so forth).[51][52] However, such frame shifts are rare in amniotes and—to be consistent with the theropod origin of birds—would have had to occur solely in the bird-theropod lineage forelimbs and not the hindlimbs (a condition unknown in any animal).[53] This is called *Lateral Digit Reduction* (LDR) versus *Bilateral Digit Reduction* (BDR) (see also *Limusaurus*).[54]

A small minority, known by the acronym **BAND** (Birds Are Not Dinosaurs),[55] including ornithologists Alan Feduccia and Larry Martin,[56] continues to assert that birds are more closely related to earlier reptiles, such as *Longisquama* or *Euparkeria*, than to dinosaurs.[57][58] Embryological studies of bird developmental biology have raised questions about digit homology in bird and dinosaur forelimbs.[59] However, due to the cogent evidence provided by comparative anatomy and phylogenetics, and the feathered dinosaur fossils from China, the idea that birds are derived dinosaurs remains widely supported among paleontologists.[15]

An alternative to the frame-shift hypothesis is the axis-shift. According to this explanation, the primary limb axis in birds runs through digit III instead of IV.[49][60] This idea is supported by palaeontological observations, which determine the phalangeal formula 2-3-4-1-X for the last common ancestor of ceratosaurs (including *Limusaurus*) and tetanurans (including the tridactyl forms with the phalangeal formula 2-3-4-X-X).[61]

Some later embryological data support the identification of bird digits as I, II, III (as in their theropod ancestors) due to the posterior finger's development outside the *shh*-expressing zone of polarizing activity.[62]

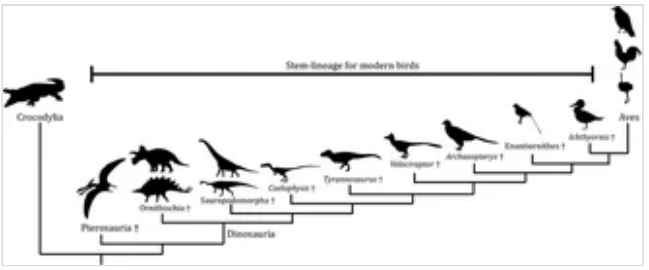
Thermogenic muscle hypothesis

A 2011 publication suggested that selection for the expansion of skeletal muscle, rather than the evolution of flight, was the driving force for the emergence of this clade. Muscles became larger in prospectively endothermic saurians, according to this hypothesis, as a response to the loss of the vertebrate mitochondrial uncoupling protein, UCP1. In mammals, UCP1 functions within brown adipose tissue, which is thermogenic to protect newborns against hypothermia. In modern birds, skeletal muscle serves a similar function and is presumed to have done so in their ancestors. In this view, bipedality and other avian skeletal alterations were side effects of muscle hyperplasia, with further evolutionary modifications of the forelimbs, including adaptations for flight or swimming, and vestigiality, being secondary consequences of two-leggedness.[63][64][65]

Phylogeny

Archaeopteryx has historically been considered the first bird, or *Urvogel*. Although newer fossil discoveries filled the gap between theropods and *Archaeopteryx*, as well as the gap between *Archaeopteryx* and modern birds, phylogenetic taxonomists, in keeping with tradition, almost always use *Archaeopteryx* as a specifier to help define Aves.[67][68] Aves has more rarely been defined as a crown

group consisting only of modern birds.[34] Nearly all palaeontologists regard birds as coelurosaurian theropod dinosaurs.[15] Within Coelurosauria, multiple cladistic analyses have found support for a clade named Maniraptora, consisting of therizinosauroids, oviraptorosaurs, troodontids, dromaeosaurids, and birds.[35][36][69] Of these, dromaeosaurids and troodontids are usually united in the clade Deinonychosauria, which is a sister group to birds (together forming the node-clade Eumaniraptora) within the stem-clade Paraves.[35][70]



Simplified phylogenetic tree showing the relationship between modern birds and other dinosaurs [66]

Other studies have proposed alternative phylogenies, in which certain groups of dinosaurs usually considered non-avian may have evolved from avian ancestors. For example, a 2002 analysis found that oviraptorosaurs were basal avians.[71] Alvarezsaurids, known from Asia and the Americas, have been variously classified as basal maniraptorans,[35][36][72][73] paravians,[69] the sister taxon of ornithomimosaurs,[74] as well as specialized early birds.[75][76] The genus *Rahonavis*, originally described as an early bird,[77] has been identified as a non-avian dromaeosaurid in several studies.[70][78] Dromaeosaurids and troodontids themselves have also been suggested to lie within Aves rather than just outside it.[79][80]

Features linking birds and dinosaurs

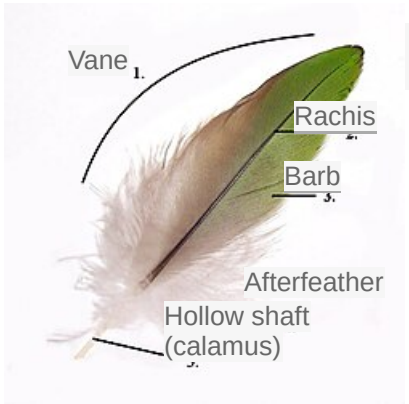
Many anatomical[81] features are shared by birds and other theropod dinosaurs.

Feathers

Archaeopteryx, the first good example of a "feathered dinosaur", was discovered in 1861. The first specimen was found in the Solnhofen limestone in southern Germany, which is a *lagerstätte*, a rare and remarkable geological formation known for its superbly detailed fossils. *Archaeopteryx* is a transitional fossil, with features clearly intermediate between those of non-avian theropod dinosaurs and birds. Discovered just two years after Darwin's seminal *Origin of Species*, its discovery spurred the nascent debate between proponents of evolutionary biology and creationism. This early bird is so dinosaur-like that, without a clear impression of feathers in the surrounding rock, at least one specimen was mistaken for *Compsognathus*.[82]

Since the 1990s, a number of additional feathered dinosaurs have been found, providing even stronger evidence of the close relationship between dinosaurs and modern birds. The first of these were initially described as simple filamentous *protofeathers*, which were reported in dinosaur lineages as primitive as compsognathids and tyrannosauroids.[83] However, feathers indistinguishable from those of modern birds were soon after found in non-avialan dinosaurs as well.[42]

A small minority of researchers have claimed that the simple filamentous "protofeather" structures are simply the result of the decomposition of collagen fiber under the dinosaurs' skin or in fins along their backs, and that species with unquestionable feathers, such as oviraptorosaurs and dromaeosaurs are not dinosaurs, but true birds unrelated to dinosaurs.[84] However, a majority of studies have concluded that



Parts of a feather

feathered dinosaurs are in fact dinosaurs, and that the simpler filaments of unquestionable theropods represent simple feathers. Some researchers have demonstrated the presence of color-bearing melanin in the structures—which would be expected in feathers but not collagen fibers.[85] Others have demonstrated, using studies of modern bird decomposition, that even advanced feathers appear filamentous when subjected to the crushing forces experienced during fossilization, and that the supposed "protofeathers" may have been more complex than previously thought.[86] Detailed examination of the "protofeathers" of *Sinosauropteryx prima* showed that individual feathers consisted of a central quill (*rachis*) with thinner *barbs* branching off from it, similar to but more primitive in structure than modern bird feathers.[87]

The 2022 description of branched feathers in the pterosaur *Tupandactylus* provides strong evidence that "pycnofibers" are not actually a distinct integument unrelated to origin of feathers. The most parsimonious scenario is the presence of feathers in the last common ancestor of pterosaurs and dinosaurs already in the Early Triassic. *Tupandactylus*'s melanosomes indicate visual signalling was an important factor in the evolution of feathers.[88]

Skeleton

Fossil cast of NGMC 91, a probable specimen of *Sinornithosaurus*

Because feathers are often associated with birds, feathered dinosaurs are often touted as the "missing link" between birds and other dinosaurs. However, the multiple skeletal features also shared by the two groups represent the more important proof for paleontologists.

Comparisons of bird and dinosaur skeletons, as well as cladistic analysis, strengthens the case for the link, particularly for a branch of theropods called Maniraptora. Skeletal similarities include the skull, tooth build, neck, uncinate processes on the ribs, an open hip socket,[89] a retroverted long pubis, flexible wrist (semi-lunate carpal), long arms, three-fingered hand, general pectoral girdle, shoulder blade, furcula, and breast bones. Almost all skeletal traits of *Archaeopteryx* can be found in non-avian maniraptorans.

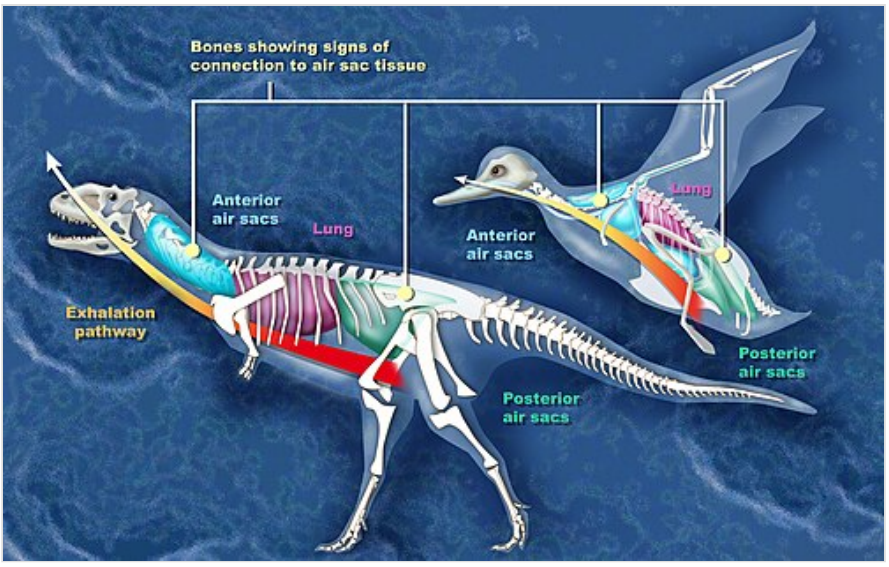
A study comparing embryonic, juvenile and adult archosaur skulls concluded that bird skulls are derived from those of theropod dinosaurs by progenesis, a type of paedomorphic heterochrony, which resulted in retention of juvenile characteristics of their ancestors.[90]

Lungs

Large meat-eating dinosaurs had a complex system of air sacs similar to those found in modern birds, according to an investigation led by Patrick M. O'Connor of Ohio University. In theropod dinosaurs (carnivores that walked on two legs and had birdlike feet) flexible soft tissue air sacs likely pumped air through the stiff lungs, as is the case in birds. "What was once formally considered unique to birds was present in some form in the ancestors of birds", O'Connor said.[91][92]

Heart

Computed tomography (CT) scans conducted in 2000 of the chest cavity of a specimen of the ornithopod *Thescelosaurus* found the apparent remnants of a complex four-chambered heart, much like those found in today's mammals and birds.[93] The idea is controversial within the scientific community, criticised for being bad anatomical science[94] or simply wishful thinking,[95] It is also not very surprising as crocodilians also possess four-chambered hearts.



Comparison between the air sacs of *Majungasaurus* and a bird (duck)

A study published in 2011 applied multiple lines of inquiry to the question of the object's identity, including more advanced CT scanning, histology, X-ray diffraction, X-ray photoelectron spectroscopy, and scanning electron microscopy. From these methods, the authors found that: the object's internal structure does not include chambers but is made up of three unconnected areas of lower density material, and is not comparable to the structure of an ostrich's heart; the "walls" are composed of sedimentary minerals not known to be produced in biological systems, such as goethite, feldspar minerals, quartz, and gypsum, as well as some plant fragments; carbon, nitrogen, and phosphorus, chemical elements important to life, were lacking in their samples; and cardiac cellular structures were absent. There was one possible patch with animal cellular structures. The authors found their data supported identification as a concretion of sand from the burial environment, not the heart, with the possibility that isolated areas of tissues were preserved.[96]

The question of how this find reflects metabolic rate and dinosaur internal anatomy is moot, though, regardless of the object's identity.[96] Both modern crocodilians and birds, the closest living relatives of dinosaurs, have four-chambered hearts (albeit modified in crocodilians), so dinosaurs probably had them as well; the structure is not necessarily tied to metabolic rate.[97]

Sleeping posture

Fossils of the troodonts *Mei* and *Sinornithoides* demonstrate that the dinosaurs slept like certain modern birds, with their heads tucked under their arms.[98] This behavior, which may have helped to keep the head warm, is also characteristic of modern birds.

Reproductive biology

When laying eggs, female birds grow a special type of bone in their limbs. This medullary bone forms as a calcium-rich layer inside the hard outer bone, and is used as a calcium source to make eggshells. The presence of endosteally derived bone tissues lining the interior marrow cavities of portions of a *Tyrannosaurus rex* specimen's hind limb suggested that *T. rex* used similar reproductive strategies, and

revealed that the specimen is female.[99] Further research has found medullary bone in the theropod *Allosaurus* and ornithopod *Tenontosaurus*. Because the line of dinosaurs that includes *Allosaurus* and *Tyrannosaurus* diverged from the line that led to *Tenontosaurus* very early in the evolution of dinosaurs, this suggests that dinosaurs in general produced medullary tissue.[100]

Brooding and care of young

Several *Citipati* specimens have been found resting over the eggs in its nest in a position most reminiscent of brooding.[101]

Numerous dinosaur species, for example *Maiasaura*, have been found in herds mixing both very young and adult individuals, suggesting rich interactions between them.

A dinosaur embryo was found without teeth, which suggests some parental care was required to feed the young dinosaur, possibly the adult dinosaur regurgitated food into the young dinosaur's mouth (*see* altricial). This behaviour is seen in numerous bird species; parent birds regurgitate food into the hatchling's mouth.

A nesting *Citipati osmolskae* specimen, at the American Museum of Natural History in New York City

Gizzard stones

Both birds and dinosaurs use gizzard stones. These stones are swallowed by animals to aid digestion and break down food and hard fibres once they enter the stomach. When found in association with fossils, gizzard stones are called gastroliths.[102] Gizzard stones are also found in some fish (mullets, mud shad, and the gillaroo, a type of trout) and in crocodiles.

Molecular evidence

On several occasions, the extraction of DNA and proteins from Mesozoic dinosaurs fossils has been claimed, allowing for a comparison with birds. Several proteins have putatively been detected in dinosaur fossils,[103] including hemoglobin.[104] In 2023, beta-protein structures were reported from the feathers of the dinosaur *Sinornithosaurus* and the early bird *Confuciusornis*.[105] This confirms that ancient feathers had a composition similar to that of modern birds. Some fossil feathers were reported to have a composition rich in alpha proteins, but fossilization experiments demonstrate that this protein composition is simply an artefact of preservation, because beta-sheet protein structures are readily transformed to alpha-helices during thermal maturation.[105]

In the March 2005 issue of *Science*, Dr. Mary Higby Schweitzer and her team announced the discovery of flexible material resembling actual soft tissue inside a 68-million-year-old *Tyrannosaurus rex* leg bone of specimen MOR 1125 from the Hell Creek Formation in Montana. The seven collagen types obtained from the bone fragments, compared to collagen data from living birds (specifically, a chicken), suggest that older theropods and birds are closely related.[106] The soft tissue allowed a molecular comparison of cellular anatomy and protein sequencing of collagen tissue published in 2007, both of which indicated that *T. rex* and birds are more closely related to each other than either is to *Alligator*.[107][108] A second

molecular study robustly supported the relationship of birds to dinosaurs, though it did not place birds within Theropoda, as expected. This study utilized eight additional collagen sequences extracted from a femur of the "mummified" *Brachylophosaurus canadensis* specimen MOR 2598, a hadrosaur.[109] However, these results have been very controversial. No other peptides of a Mesozoic age have been reported. In 2008, it was suggested that the presumed soft tissue was in fact a bacterial microfilm.[110] In response, it was argued that these very microfilms protected the soft tissue.[111] Another objection was that the results could have been caused by contamination.[112] In 2015, under more controlled conditions safeguarding against contamination, the peptides were still identified.[113] In 2017, a study found that a peptide was present in the bone of the modern ostrich that was identical to that found in the *Tyrannosaurus* and *Brachylophosaurus* specimens, highlighting the danger of a cross-contamination.[114]

The successful extraction of ancient DNA from dinosaur fossils has been reported on two separate occasions, but upon further inspection and peer review, neither of these reports could be confirmed.[115]

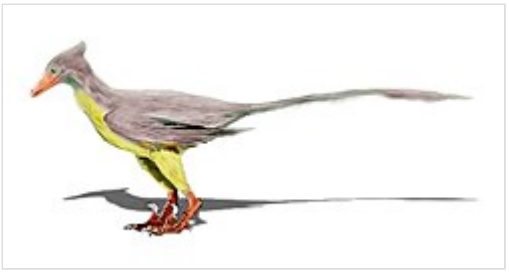
Origin of bird flight

Debates about the origin of bird flight are almost as old as the idea that birds evolved from dinosaurs, which arose soon after the discovery of *Archaeopteryx* in 1862. Two theories have dominated most of the discussion since then: the cursorial ("from the ground up") theory proposes that birds evolved from small, fast predators that ran on the ground; the arboreal ("from the trees down") theory proposes that powered flight evolved from unpowered gliding by arboreal (tree-climbing) animals. A more recent theory, "wing-assisted incline running" (WAIR), is a variant of the cursorial theory and proposes that wings developed their aerodynamic functions as a result of the need to run quickly up very steep slopes such as trees, which would help small feathered dinosaurs escape from predators.

In March 2018, scientists reported that *Archaeopteryx* was likely capable of flight, but in a manner substantially different from that of modern birds.[116][117]

Cursorial ("from the ground up") theory

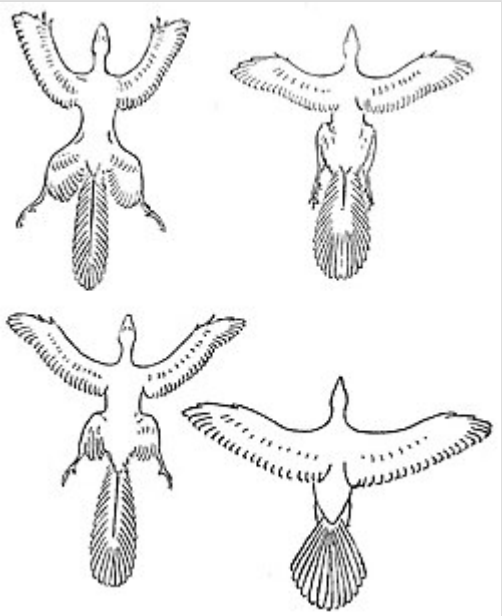
The cursorial theory of the origin of flight was first proposed by Samuel Wendell Williston, and elaborated upon by Baron Nopcsa. This hypothesis proposes that some fast-running animals with long tails used their arms to keep their balance while running. Modern versions of this theory differ in many details from the Williston-Nopcsa version, mainly as a result of discoveries since Nopcsa's time.



Reconstruction of *Rahonavis*, a ground-dwelling feathered dinosaur that some researchers think was well equipped for flight

Nopcsa theorized that increasing the surface area of the outstretched arms could have helped small cursorial predators keep their balance, and that the scales of the forearms elongated, evolving into feathers. The feathers could also have been used to trap insects or other prey. Progressively, the animals leapt for longer distances, helped by their evolving wings. Nopcsa also proposed three stages in the evolution of flight. First, animals

developed passive flight, in which developing wing structures served as a sort of parachute. Second, they achieved active flight by flapping the wings. He used *Archaeopteryx* as an example of this second stage. Finally, birds gained the ability to soar.[118]



Proposed development of flight in a book from 1922: Tetrapteryx, *Archaeopteryx*, Hypothetical Stage, Modern Bird

While some authors had rejected the homology between feathers and scales due to their different proteins,[119] recent studies provide evidence that those structures do share a common origin.[120][121] However, Nopcsa's theory assumes that feathers evolved as part of the evolution of flight, and recent discoveries show that feathers evolved millions of years before flight.[120][88]

Feathers are very common in coelurosaurian dinosaurs (including the early tyrannosauroid *Dilong*).[122] Modern birds are classified as coelurosaurs by nearly all palaeontologists,[123] though not by a few ornithologists.[124][125][126] The modern version of the "from the ground up" hypothesis argues that birds' ancestors were small, feathered, ground-running predatory dinosaurs (rather like roadrunners in their hunting style[127]) that used their forelimbs for balance while pursuing prey, and that the forelimbs and feathers later evolved in ways that provided gliding and then powered flight. The most widely suggested original functions of feathers include thermal insulation and competitive displays, as in modern birds.[128][129]

All of the *Archaeopteryx* fossils come from marine sediments, and it has been suggested that wings may have helped the birds run over water in the manner of the *Jesus Christ Lizard* (common basilisk).[130]

Most recent opposition to the "from the ground up" hypothesis attempt to refute the modern version's assumption that birds are modified coelurosaurian dinosaurs. The criticism is based on embryological analyses that suggest birds' wings are formed from digits 2, 3, and 4, (corresponding to the index, middle, and ring fingers in humans. The first of a bird's three digits forms the alula, which they use to avoid stalling in low-speed flight—for example, when landing). The hands of coelurosaurs, however, are formed by digits 1, 2, and 3 (thumb and first two fingers in humans).[59] However, these embryological analyses were immediately challenged on the embryological grounds that the "hand" often develops differently in clades that have lost some digits in the course of their evolution, and that birds' "hands" do develop from digits 1, 2, and 3.[49][131][132] For more information about this subject, see "Digit homology".

Fowler *et al.* (2011) proposed a model explaining how dromaeosaurids may have hunted. The animal would use its wing as stabilizers while standing on top of its prey eating it alive in the manner of an eagle or a hawk. The authors consider this an important addition to the topic of how flapping movements evolved, arguing they likely precede flight.[133]

Wing-assisted incline running

The wing-assisted incline running (WAIR) hypothesis was prompted by observation of young chukar chicks, and proposes that wings developed their aerodynamic functions as a result of the need to run quickly up very steep slopes such as tree trunks, for example to escape from predators.[134] This makes it a specialized type of cursorial ("from the ground up") theory. Note that in this scenario birds need *downforce* to give their feet increased grip.[135][136] But early birds, including *Archaeopteryx*, lacked the shoulder mechanism by which modern birds' wings produce swift, powerful upstrokes. Since the downforce WAIR depends on is generated by upstrokes, it seems that early birds were incapable of WAIR.[137] Because WAIR is a behavioural trait without osteological specializations, the phylogenetic placement of the flight stroke before the divergence of the Neornithes, the group which contains all extant birds, makes it impossible to determine if WAIR is ancestral to the avian flight stroke or derived from it.[138]

Arboreal ("from the trees down") theory

Most versions of the arboreal hypothesis state that the ancestors of birds were very small dinosaurs that lived in trees, springing from branch to branch. This small dinosaur already had feathers, which were co-opted by evolution to produce longer, stiffer forms that were useful in aerodynamics, eventually producing wings. Wings would have then evolved and become increasingly refined as devices to give the leaper more control, to parachute, to glide, and to fly in stepwise fashion. The arboreal hypothesis also notes that, for arboreal animals, aerodynamics are far more energy efficient, since such animals simply fall to achieve minimum gliding speeds.[139][140]

The four-winged *Microraptor*, a member of the Dromaeosauridae, a group of dinosaurs closely related to birds

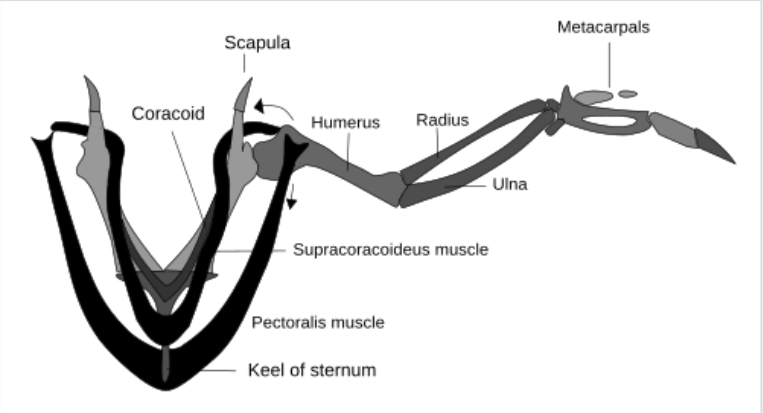
Several small dinosaurs from the Jurassic or Early Cretaceous, all with feathers, have been interpreted as possibly having arboreal and/or aerodynamic adaptations. These include *Scansoriopteryx*, *Epidexipteryx*, *Microraptor*, *Pedopenna*, and *Anchiornis*. *Anchiornis* is particularly important to this subject, as it lived at the beginning of the Late Jurassic, long before *Archaeopteryx*.[141]

Analysis of the proportions of the toe bones of the most primitive birds *Archaeopteryx* and *Confuciusornis*, compared to those of living species, suggest that the early species may have lived both on the ground and in trees.[142]

One study suggested that the earliest birds and their immediate ancestors did not climb trees. This study determined that the amount of toe claw curvature of early birds was more like that seen in modern ground-foraging birds than in perching birds.[143]

Diminished significance of *Archaeopteryx*

Archaeopteryx was the first and for a long time the only known feathered Mesozoic animal. As a result, discussion of the evolution of birds and of bird flight centered on *Archaeopteryx* at least until the mid-1990s.



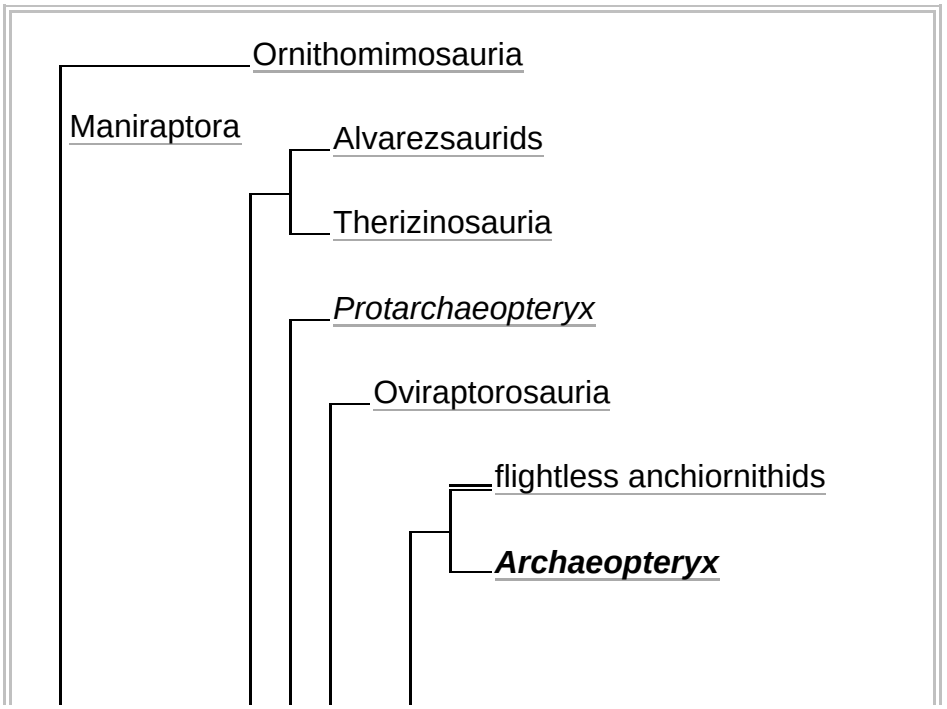
The supracoracoideus works using a pulley-like system to lift the wing while the pectorals provide the powerful downstroke.

There has been debate about whether *Archaeopteryx* could really fly. It appears that *Archaeopteryx* had the brain structures and inner-ear balance sensors that birds use to control their flight.[144] *Archaeopteryx* also had a wing feather arrangement like that of modern birds and similarly asymmetrical flight feathers on its wings and tail. But *Archaeopteryx* lacked the shoulder mechanism by which modern birds' wings produce swift, powerful upstrokes (see diagram above of supracoracoideus pulley); this may mean that it and other early birds were incapable of flapping flight and could only glide.[137]

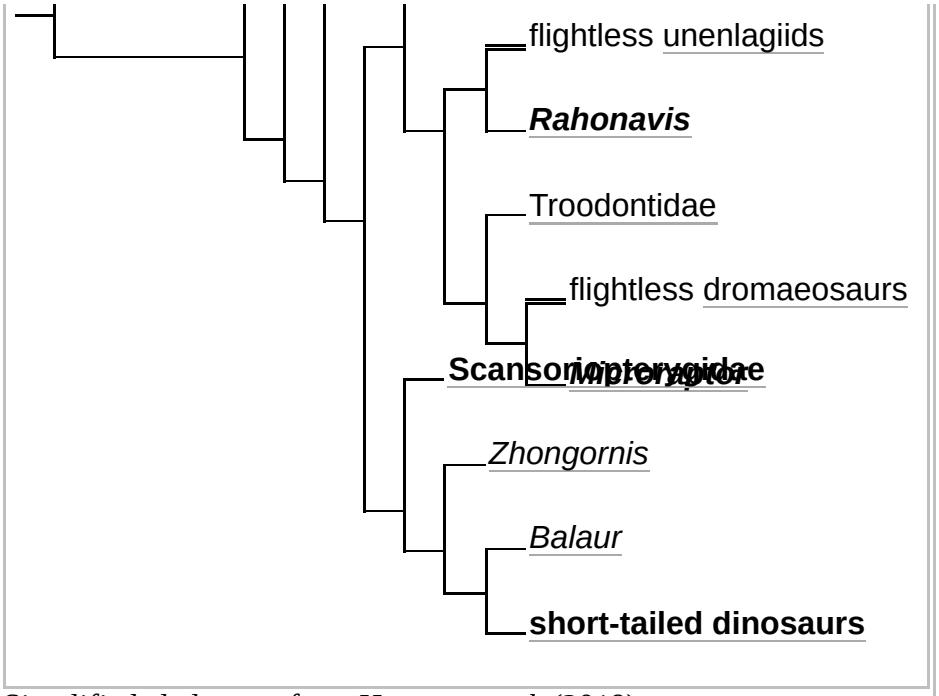
But the discovery since the early 1990s of many feathered dinosaurs means that *Archaeopteryx* is no longer the key figure in the evolution of bird flight. Other small feathered coelurosaurs from the Cretaceous and Late Jurassic show possible precursors of avian flight. These include *Rahonavis*, a ground-runner with a *Velociraptor*-like raised sickle claw on the second toe, that some paleontologists assume to have been better adapted for flight than *Archaeopteryx*,[145] *Scansoriopteryx*, an arboreal dinosaur that may support the "from the trees down" theory,[146] and *Microraptor*, an arboreal dinosaur possibly capable of powered flight but, if so, more like a biplane, as it had well-developed feathers on its legs.[147] As early as 1915, some scientists argued that the evolution of bird flight may have gone through a four-winged (or *tetrapteryx*) stage.[148][149] Hartman *et al*. (2019) found that, because of how basal flying paravians are phylogenetically distributed, flight most likely evolved five times among paravians instead of only once. *Yi*, *Archaeopteryx*, *Rahonavis* and *Microraptor* were thus considered examples of convergent evolution instead of precursors of bird flight.[150]

Secondary flightlessness in dinosaurs

A minority hypothesis, credited to the books *Predatory Dinosaurs of the World* (1988) and *Dinosaurs of the Air* (2002) by scientific illustrator Gregory Paul, suggests that some groups of non-flying carnivorous dinosaurs — especially deinonychosaurs, but perhaps others such as oviraptorosaurs, therizinosaurs, alvarezsaurids and ornithomimosaurs — actually descend from birds or other flighted maniraptorans. Paul also proposed that the ancestors of these groups were



more advanced in their flight adaptations than *Archaeopteryx*. The hypothesis would mean that *Archaeopteryx* is less closely related to extant birds than these dinosaurs are.[151] In 2016, Paul suggested that omnivoropterygid avialans were closely related to oviraptorosaurs and that jeholornithid avialans were closely related to therizinosaurs; he considered them to not be avians but suggested that they shared a flighted ancestor.[152]



Simplified cladogram from Hartman *et al.* (2019)
Groups capable of flight are in bold type.[150]

Mayr *et al.* (2005) analyzed a new, tenth specimen of *Archaeopteryx*, and concluded that *Archaeopteryx* was the sister clade to the Deinonychosauria, but that the more advanced bird *Confuciusornis* was within the Dromaeosauridae.[80][153] This paper, however, excluded all other birds and thus did not sample their character distributions. The paper was criticized by Corfe and Butler (2006) who found the authors could not support their conclusions statistically. Mayr *et al.* agreed that the statistical support for the authors' earlier paper was weak but stated that it is also weak for the alternative scenarios.[154]

Most subsequent cladistic analyses do not support Paul's hypothesis about the position of *Archaeopteryx*. Instead, they indicate that *Archaeopteryx* is closer to birds, within the clade Avialae, than it is to deinonychosaurs or oviraptorosaurs. *Microraptor*, *Pedopenna*, and *Anchiornis* all have winged feet, share many features, and lie close to the base of the clade Paraves. This suggests that the ancestral paravian may have been a four-winged glider.[2] *Deinonychus* may also display partial volancy, with the young being capable of flight or gliding and the adults being flightless.[155] In 2018, a study concluded that the last common ancestor of the Pennaraptora had joint surfaces on the fingers, and between the metatarsus and the wrist, that were optimised to stabilise the hand in flight. This was seen as an indication for secondary flightlessness in heavy basal members of that group.[156] Hartman et al. (2019) found it likely that *Archaeopteryx* was more closely related to dromaeosaurs than to birds, but the distribution of flying taxa among flightless ones suggests multiple origins of flight among maniraptoran theropods instead of only one.[150]

In Euornithes, the earliest unequivocal example of secondary flightlessness is *Patagopteryx*.[157]

See also

Dinosaurs portal

- Bird ichnology

- Dinosaur Discovery Museum
- Evolution of birds
- Feathered dinosaurs
- Flightless birds
- List of extinct birds
- Origin of avian flight
- Temporal paradox (paleontology)

Footnotes

1. Chiappe, Luis M. (2009). "Downsized Dinosaurs: The Evolutionary Transition to Modern Birds" (https://doi.org/10.1007%2Fs12052-009-0133-4). *Evolution: Education and Outreach*. **2** (2): 248–256. doi:10.1007/s12052-009-0133-4 (https://doi.org/10.1007%2Fs12052-009-0133-4).
2. Witmer, LM (2009). "Palaeontology: Feathered dinosaurs in a tangle" (https://doi.org/10.1038%2F461601a). *Nature*. **461** (7264): 601–2. Bibcode:2009Natur.461..601W (https://ui.adsabs.harvard.edu/abs/2009Natur.461..601W). doi:10.1038/461601a (https://doi.org/10.1038%2F461601a). PMID 19794481 (https://pubmed.ncbi.nlm.nih.gov/19794481). S2CID 205049989 (https://api.semanticscholar.org/CorpusID:205049989).
3. Darwin, Charles R. (1859). *On the Origin of Species by Means of Natural Selection, or the Preservation of Favoured Races in the Struggle for Life* (http://darwin-online.org.uk/content/frameset?itemID=F373&viewtype=side&pageseq=16). London: John Murray. p. 502pp. ISBN 978-1-4353-9386-8.
4. von Meyer, C.E. Hermann. (1861). "*Archaeopteryx lithographica* (Vogel-Feder) und *Pterodactylus* von Solnhofen". *Neues Jahrbuch für Mineralogie, Geologie und Paläontologie* (in German). **1861**: 678–679.
5. Owen, Richard. (1863). "On the *Archeopteryx* [sp] of von Meyer, with a description of the fossil remains of a long-tailed species, from the lithographic stone of Solenhofen [sp]" (https://doi.org/10.1098%2Frstl.1863.0003). *Philosophical Transactions of the Royal Society of London*. **153**: 33–47. doi:10.1098/rstl.1863.0003 (https://doi.org/10.1098%2Frstl.1863.0003).
6. Gegenbaur, C. (1863). "Vergleichend-anatomische Bemerkungen über das Fußskelet der Vögel". *Archiv für Anatomie, Physiologie und Wissenschaftliche Medicin*. **1863**: 450–472.
7. Cope, E.D. (1867). "An account of the extinct reptiles which approached the birds". *Proceedings of the Academy of Natural Sciences of Philadelphia*. **19**: 234–235.link (https://archive.org/details/proceedingsofaca19acad)
8. Huxley, T.H. (1868). "On the animals which are most nearly intermediate between birds and reptiles" (https://www.biodiversitylibrary.org/page/22247928). *Annals and Magazine of Natural History*. 4th. **2**: 66–75.
9. Huxley, Thomas H. (1870). "Further Evidence of the Affinity between the Dinosaurian Reptiles and Birds". *Quarterly Journal of the Geological Society of London*. Vol. 26. pp. 12–31. doi:10.1144/GSL.JGS.1870.026.01-02.08 (https://doi.org/10.1144%2FGSL.JGS.1870.026.01-02.08) – via Wikisource.
10. Nopcsa, Franz. (1907). "Ideas on the origin of flight" (https://www.biodiversitylibrary.org/part/72475). *Proceedings of the Zoological Society of London*. **77**: 223–238. doi:10.1111/j.1096-3642.1907.tb01811.x (https://doi.org/10.1111%2Fj.1096-3642.1907.tb01811.x).
11. Seeley, Harry G. (1901). *Dragons of the Air: An Account of Extinct Flying Reptiles* (https://archive.org/details/dragonsairanacc00seelgoog). London: Methuen & Co. p. 239pp.
12. Abel, O (1911). "Die Vorfahren der Vögel und ihre Lebensweise". *Verhandlungen der Zoologisch-Botanischen Gesellschaft in Wien*. **61**: 144–191.

13. Nieuwland, Ilja J.J. (2004). "Gerhard Heilmann and the artist's eye in science, 1912-1927" (https://web.archive.org/web/20160224031948/http://www.palarch.nl/wp-content/ver_2004_3_2.pdf) (PDF). *PalArch's Journal of Vertebrate Palaeontology*. **3** (2). Archived from the original (http://www.palarch.nl/wp-content/ver_2004_3_2.pdf) (PDF) on 2016-02-24. Retrieved 2009-05-21.
14. Heilmann, Gerhard (1926). *The Origin of Birds* (https://archive.org/details/originofbirds0000heil/page/208). London: Witherby. p. 208pp (https://archive.org/details/originofbirds0000heil/page/208). ISBN 978-0-486-22784-9.
15. Padian, Kevin. (2004). "Basal Avialae". In Weishampel, David B.; Dodson, Peter; Osmólska, Halszka (eds.). *The Dinosauria* (Second ed.). Berkeley: University of California Press. pp. 210–231. ISBN 978-0-520-24209-8.
16. For example in 1923, three years before Heilmans's book, Roy Chapman Andrews found a good *Oviraptor* fossil in *Mongolia*, but Henry Fairfield Osborn, who analyzed the fossil in 1924, misidentified the furcula as an interclavicle; described in Paul, G.S. (2002). *Dinosaurs of the Air: The Evolution and Loss of Flight in Dinosaurs and Birds* (https://books.google.com/books?id=OUwXzD3iihAC&q=oviraptor+furcula&pg=PA9). JHU Press. ISBN 978-0-8018-6763-7.
17. Camp, Charles L. (1936). "A new type of small theropod dinosaur from the Navajo Sandstone of Arizona". *Bulletin of the University of California Department of Geological Sciences*. **24**: 39–65.
18. In an *Oviraptor*: Barsbold, R. (1983). "Carnivorous dinosaurs from the Cretaceous of Mongolia". *Trudy Soumestnaya Sovetsko-Mongol'skaya Paleontogicheskaya Ekspeditsiya* (in Russian). **19**: 1–117. See the summary and pictures at "A wish for *Coelophysis*" (https://web.archive.org/web/20080203064406/http://www.hmnh.org/archives/2007/10/11/a-wish-for-coelophysis/). Archived from the original (http://www.hmnh.org/archives/2007/10/11/a-wish-for-coelophysis/) on February 3, 2008.
19. Barsbold, Rinchen. "O ptich'ikh chertakh v stroyenii khishchnykh dinozavrov. ["Avian" features in the morphology of predatory dinosaurs]. Transactions of the Joint Soviet Mongolian Paleontological Expedition 24: 96-103" (http://www.paleoglot.org/files/Barsbold_83b.pdf) (PDF). *Transactions of the Joint Soviet Mongolian Paleontological Expedition 24*.
20. Lipkin, C., Sereno, P.C., and Horner, J.R. (November 2007). "The Furcula In Suchomimus Tenerensis And Tyrannosaurus Rex (Dinosauria: Theropoda: Tetanurae)" (http://jpaleontol.geoscienceworld.org/cgi/content/extract/81/6/1523). *Journal of Paleontology*. **81** (6): 1523–1527. Bibcode:2007JPal...81.1523L (https://ui.adsabs.harvard.edu/abs/2007JPal...81.1523L). doi:10.1666/06-024.1 (https://doi.org/10.1666%2F06-024.1). S2CID 86234363 (https://api.semanticscholar.org/CorpusID:86234363). - full text currently online at "The Furcula in *Suchomimus Tenerensis* and *Tyrannosaurus rex*" (https://web.archive.org/web/20110522043704/http://www.redorbit.com/news/health/1139122/the_furcula_in_suchomimus_tenerensis_and_tyrannosaurus_rex_dinosauria_theropoda/index.html). 2007-11-11. Archived from the original (http://www.redorbit.com/news/health/1139122/the_furcula_in_suchomimus_tenerensis_and_tyrannosaurus_rex_dinosauria_theropoda/index.html) on 2011-05-22. Retrieved 2008-04-17. This lists a large number of theropods in which furculae have been found, as well as describing those of *Suchomimus Tenerensis* and *Tyrannosaurus rex*.
21. Carrano, M.R., Hutchinson, J.R., and Sampson, S.D. (December 2005). "New information on *Segisaurus halli*, a small theropod dinosaur from the Early Jurassic of Arizona" (https://web.archive.org/web/20060512053601/http://www.rvc.ac.uk/AboutUs/Staff/jhutchinson/documents/JH18.pdf) (PDF). *Journal of Vertebrate Paleontology*. **25** (4): 835–849. doi:10.1671/0272-4634(2005)025[0835:NIOSHA]2.0.CO;2 (https://doi.org/10.1671%2F0272-4634%282005%29025%5B0835%3ANIOSHA%5D2.0.CO%3B2). ISSN 0272-4634 (https://search.worldcat.org/issn/0272-4634). S2CID 30039789 (https://api.semanticscholar.org/CorpusID:30039789). Archived from the original (http://www.rvc.ac.uk/AboutUs/Staff/jhutchinson/documents/JH18.pdf) (PDF) on 2006-05-12.

22. Yates, Adam M.; and Vasconcelos, Cecilio C. (2005). "Furcula-like clavicles in the prosauropod dinosaur *Massospondylus*". *Journal of Vertebrate Paleontology*. **25** (2): 466–468. doi:10.1671/0272-4634(2005)025[0466:FCITPD]2.0.CO;2 (https://doi.org/10.1671%2F0272-4634%282005%29025%5B0466%3AFCITPD%5D2.0.CO%3B2). ISSN 0272-4634 (https://search.worldcat.org/issn/0272-4634). S2CID 85679726 (https://api.semanticscholar.org/CorpusID:85679726).
23. Ostrom, John H. (1969). "Osteology of *Deinonychus antirrhopus*, an unusual theropod from the Lower Cretaceous of Montana". *Bulletin of the Peabody Museum of Natural History*. **30**: 1–165.
24. Ostrom, John H. (1970). "*Archaeopteryx*: Notice of a "new" specimen". *Science*. **170** (3957): 537–8. Bibcode:1970Sci...170..537O (https://ui.adsabs.harvard.edu/abs/1970Sci...170..537O). doi:10.1126/science.170.3957.537 (https://doi.org/10.1126%2Fscience.170.3957.537). PMID 17799709 (https://pubmed.ncbi.nlm.nih.gov/17799709). S2CID 46682559 (https://api.semanticscholar.org/CorpusID:46682559).
25. Chambers, Paul (2002). *Bones of Contention: The Archaeopteryx Scandals*. London: John Murray Ltd. pp. 183–184. ISBN 978-0-7195-6054-5.
26. Walker, Alick D. (1972). "New light on the origin of birds and crocodiles". *Nature*. **237** (5353): 257–263. Bibcode:1972Natur.237..257W (https://ui.adsabs.harvard.edu/abs/1972Natur.237..257W). doi:10.1038/237257a0 (https://doi.org/10.1038%2F237257a0). S2CID 4174690 (https://api.semanticscholar.org/CorpusID:4174690).
27. Ostrom, John H. (1973). "The ancestry of birds" (https://doi.org/10.1038%2F242136a0). *Nature*. **242** (5393): 136. Bibcode:1973NPhS..242..136O (https://ui.adsabs.harvard.edu/abs/1973NPhS..242..136O). doi:10.1038/242136a0 (https://doi.org/10.1038%2F242136a0). S2CID 29873831 (https://api.semanticscholar.org/CorpusID:29873831).
28. Ostrom, John H. (1975). "The origin of birds". *Annual Review of Earth and Planetary Sciences*. **3** (1): 55–77. Bibcode:1975AREPS...3...55O (https://ui.adsabs.harvard.edu/abs/1975AREPS...3...55O). doi:10.1146/annurev.ea.03.050175.000415 (https://doi.org/10.1146%2Fannurev.ea.03.050175.000415).
29. Ostrom, John H. (1976). "*Archaeopteryx* and the origin of birds" (http://doc.rero.ch/record/16120/files/PAL_E2122.pdf) (PDF). *Biological Journal of the Linnean Society*. **8** (2): 91–182. doi:10.1111/j.1095-8312.1976.tb00244.x (https://doi.org/10.1111%2Fj.1095-8312.1976.tb00244.x).
30. Bakker, Robert T. (1972). "Anatomical and ecological evidence of endothermy in dinosaurs". *Nature*. **238** (5359): 81–85. Bibcode:1972Natur.238...81B (https://ui.adsabs.harvard.edu/abs/1972Natur.238...81B). doi:10.1038/238081a0 (https://doi.org/10.1038%2F238081a0). S2CID 4176132 (https://api.semanticscholar.org/CorpusID:4176132).
31. Horner, John R.; Makela, Robert (1979). "Nest of juveniles provides evidence of family structure among dinosaurs". *Nature*. **282** (5736): 296–298. Bibcode:1979Natur.282..296H (https://ui.adsabs.harvard.edu/abs/1979Natur.282..296H). doi:10.1038/282296a0 (https://doi.org/10.1038%2F282296a0). S2CID 4370793 (https://api.semanticscholar.org/CorpusID:4370793).
32. Darren Naish, 2021, "Dinosaur Renaissance", in *Dinopedia - A Brief Compendium of Dinosaur Lore*, Princeton University Press
33. Hennig, E.H. Willi (1966). *Phylogenetic Systematics*. translated by Davis, D. Dwight; & Zangerl, Rainer. Urbana: University of Illinois Press. ISBN 978-0-252-06814-0.
34. Gauthier, Jacques. (1986). "Saurischian monophyly and the origin of birds". In Padian, Kevin (ed.). *The Origin of Birds and the Evolution of Flight*. Memoirs of the California Academy of Sciences **8**. pp. 1–55.

35. Senter, Phil (2007). "A new look at the phylogeny of Coelurosauria (Dinosauria: Theropoda)". *Journal of Systematic Palaeontology*. **5** (4): 429–463. Bibcode:2007JSPal...5..429S (https://ui.adsabs.harvard.edu/abs/2007JSPal...5..429S). doi:10.1017/S1477201907002143 (https://doi.org/10.1017%2FS1477201907002143). S2CID 83726237 (https://api.semanticscholar.org/CorpusID:83726237).
36. Turner, Alan H.; Hwang, Sunny; Norell, Mark A. (2007). "A small derived theropod from Öösh, Early Cretaceous, Baykhangor, Mongolia" (https://www.biodiversitylibrary.org/bibliography/169129). *American Museum Novitates* (3557): 1–27. doi:10.1206/0003-0082(2007)3557[1:ASDTFS]2.0.CO;2 (https://doi.org/10.1206%2F0003-0082%282007%293557%5B1%3AASDTFS%5D2.0.CO%3B2). hdl:2246/5845 (https://hdl.handle.net/2246%2F5845). ISSN 0003-0082 (https://search.worldcat.org/issn/0003-0082). S2CID 31096081 (https://api.semanticscholar.org/CorpusID:31096081).
37. Sereno, Paul C.; Rao Chenggang (1992). "Early evolution of avian flight and perching: new evidence from the Lower Cretaceous of China". *Science*. **255** (5046): 845–8. Bibcode:1992Sci...255..845S (https://ui.adsabs.harvard.edu/abs/1992Sci...255..845S). doi:10.1126/science.255.5046.845 (https://doi.org/10.1126%2Fscience.255.5046.845). PMID 17756432 (https://pubmed.ncbi.nlm.nih.gov/17756432). S2CID 41106677 (https://api.semanticscholar.org/CorpusID:41106677).
38. Hou Lian-Hai, Lian-hai; Zhou Zhonghe; Martin, Larry D.; & Feduccia, Alan (1995). "A beaked bird from the Jurassic of China". *Nature*. **377** (6550): 616–618. Bibcode:1995Natur.377..616H (https://ui.adsabs.harvard.edu/abs/1995Natur.377..616H). doi:10.1038/377616a0 (https://doi.org/10.1038%2F377616a0). S2CID 4357707 (https://api.semanticscholar.org/CorpusID:4357707).
39. Ji Qiang; Ji Shu-an (1996). "On the discovery of the earliest bird fossil in China and the origin of birds" (http://www.paleoglot.org/files/Ji&Ji_96.pdf) (PDF). *Chinese Geology*. **233**: 30–33.
40. Chen Pei-ji, Pei-ji; Dong Zhiming; & Zhen Shuo-nan. (1998). "An exceptionally preserved theropod dinosaur from the Yixian Formation of China" (http://doc.rero.ch/record/14898/files/PAL_E2040.pdf) (PDF). *Nature*. **391** (6663): 147–152. Bibcode:1998Natur.391..147C (https://ui.adsabs.harvard.edu/abs/1998Natur.391..147C). doi:10.1038/34356 (https://doi.org/10.1038%2F34356). S2CID 4430927 (https://api.semanticscholar.org/CorpusID:4430927).
41. Lingham-Soliar, Theagarten; Feduccia, Alan; & Wang Xiaolin. (2007). "A new Chinese specimen indicates that 'protofeathers' in the Early Cretaceous theropod dinosaur Sinosauropteryx are degraded collagen fibres" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2270928). *Proceedings of the Royal Society B*. **274** (1620): 1823–9. Bibcode:2007PBioS.274.1823L (https://ui.adsabs.harvard.edu/abs/2007PBioS.274.1823L). doi:10.1098/rspb.2007.0352 (https://doi.org/10.1098%2Frspb.2007.0352). PMC 2270928 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2270928). PMID 17521978 (https://pubmed.ncbi.nlm.nih.gov/17521978).
42. Ji Qiang, Philip J.; Currie, Philip J.; Norell, Mark A.; & Ji Shu-an. (1998). "Two feathered dinosaurs from northeastern China" (http://doc.rero.ch/record/31829/files/PAL_E1436.pdf) (PDF). *Nature*. **393** (6687): 753–761. Bibcode:1998Natur.393..753Q (https://ui.adsabs.harvard.edu/abs/1998Natur.393..753Q). doi:10.1038/31635 (https://doi.org/10.1038%2F31635). S2CID 205001388 (https://api.semanticscholar.org/CorpusID:205001388).
43. Sloan, Christopher P. (1999). "Feathers for *T. rex*?" (https://web.archive.org/web/20121011005616/http://business.highbeam.com/5308/article-1G1-58960652/feathers-t-rex-new-birdlike-fossils-missing-links-dinosaur). *National Geographic*. **196** (5): 98–107. Archived from the original on 2012-10-11 – via HighBeam Research.

44. Monastersky, Richard (2000). "All mixed up over birds and dinosaurs" (https://web.archive.org/web/20081003223042/http://www.sciencenews.org/view/generic/id/94/title/All_mixed_up_over_birds_and_dinosaurs). *Science News*. **157** (3): 38. doi:10.2307/4012298 (https://doi.org/10.2307%2F4012298). JSTOR 4012298 (https://www.jstor.org/stable/4012298). Archived from the original (http://www.sciencenews.org/view/generic/id/94/title/All_mixed_up_over_birds_and_dinosaurs) on 2008-10-03. Retrieved 2008-07-04.
45. Xu Xing, Xing; Tang Zhi-lu; & Wang Xiaolin. (1999). "A therizinosaurid dinosaur with integumentary structures from China". *Nature*. **399** (6734): 350–354. Bibcode:1999Natur.399..350X (https://ui.adsabs.harvard.edu/abs/1999Natur.399..350X). doi:10.1038/20670 (https://doi.org/10.1038%2F20670). S2CID 204993327 (https://api.semanticscholar.org/CorpusID:204993327).
46. Xu Xing, X; Norell, Mark A.; Kuang Xuewen; Wang Xiaolin; Zhao Qi; & Jia Chengkai. (2004). "Basal tyrannosauroids from China and evidence for protofeathers in tyrannosauroids" (http://doc.rero.ch/record/15283/files/PAL_E2582.pdf) (PDF). *Nature*. **431** (7009): 680–4. Bibcode:2004Natur.431..680X (https://ui.adsabs.harvard.edu/abs/2004Natur.431..680X). doi:10.1038/nature02855 (https://doi.org/10.1038%2Fnature02855). PMID 15470426 (https://pubmed.ncbi.nlm.nih.gov/15470426). S2CID 4381777 (https://api.semanticscholar.org/CorpusID:4381777).
47. Xu Xing, X; Zhou Zhonghe; Wang Xiaolin; Kuang Xuewen; Zhang Fucheng; & Du Xiangke (2003). "Four-winged dinosaurs from China" (http://doc.rero.ch/record/15275/files/PAL_E2574.pdf) (PDF). *Nature*. **421** (6921): 335–40. Bibcode:2003Natur.421..335X (https://ui.adsabs.harvard.edu/abs/2003Natur.421..335X). doi:10.1038/nature01342 (https://doi.org/10.1038%2Fnature01342). PMID 12540892 (https://pubmed.ncbi.nlm.nih.gov/12540892). S2CID 1160118 (https://api.semanticscholar.org/CorpusID:1160118).
48. Zhou, Zhonghe; Zhang Fucheng (2002). "A long-tailed, seed-eating bird from the Early Cretaceous of China". *Nature*. **418** (6896): 405–9. Bibcode:2002Natur.418..405Z (https://ui.adsabs.harvard.edu/abs/2002Natur.418..405Z). doi:10.1038/nature00930 (https://doi.org/10.1038%2Fnature00930). PMID 12140555 (https://pubmed.ncbi.nlm.nih.gov/12140555). S2CID 4423299 (https://api.semanticscholar.org/CorpusID:4423299).
49. Chatterjee, Sankar (17 April 1998). "Counting the Fingers of Birds and Dinosaurs" (https://doi.org/10.1126%2Fscience.280.5362.355a). *Science*. **280** (5362): 355a–355. Bibcode:1998Sci...280..355C (https://ui.adsabs.harvard.edu/abs/1998Sci...280..355C). doi:10.1126/science.280.5362.355a (https://doi.org/10.1126%2Fscience.280.5362.355a).
50. University of Maryland department of geology home page, "Theropoda I" on *Avetheropoda* (http://www.geol.umd.edu/~tholtz/G104/10422ther.htm), 14 July 2006.
51. Wagner, G. P.; Gautthier, J. A. (1999). "1,2,3 = 2,3,4: A solution to the problem of the homology of the digits in the avian hand" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC21825). *Proc. Natl. Acad. Sci. U.S.A*. **96** (9): 5111–6. Bibcode:1999PNAS...96.5111W (https://ui.adsabs.harvard.edu/abs/1999PNAS...96.5111W). doi:10.1073/pnas.96.9.5111 (https://doi.org/10.1073%2Fpnas.96.9.5111). PMC 21825 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC21825). PMID 10220427 (https://pubmed.ncbi.nlm.nih.gov/10220427).
52. **Scienceblogs:** *Limusaurus* is awesome (http://scienceblogs.com/tetrapodzoology/2009/06/limusaurus_is_awesome.php) Archived (https://web.archive.org/web/20090622092428/http://scienceblogs.com/tetrapodzoology/2009/06/limusaurus_is_awesome.php) 2009-06-22 at the Wayback Machine.
53. Developmental Biology 8e Online. Chapter 16: Did Birds Evolve From the Dinosaurs? (http://8e.devbio.com/article.php?ch=16&id=161) Archived (https://web.archive.org/web/20080118103234/http://8e.devbio.com/article.php?ch=16&id=161) 2008-01-18 at the Wayback Machine
54. Vargas AO, Wagner GP and Gauthier, JA. 2009. Limusaurus and bird digit identity. Available from Nature Precedings [1]

55. Havstad, Joyce C.; Smith, N. Adam (2019). "Fossils with Feathers and Philosophy of Science" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6701454). *Systematic Biology*. **68** (5): 840–851. doi:10.1093/sysbio/syz010 (https://doi.org/10.1093%2Fsysbio%2Fsyz010). ISSN 1076-836X (https://search.worldcat.org/issn/1076-836X). PMC 6701454 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6701454). PMID 30753719 (https://pubmed.ncbi.nlm.nih.gov/30753719).
56. Darren Naish. 2021. "Birds Are Not Dinosaurs (or BAND)" p 23-27 in *Dinopedia — A Brief Compendium of Dinosaur Lore*. Princeton University Press
57. Martin, Larry D. (2004). "A basal archosaurian origin for birds" (https://web.archive.org/web/20050325192117/http://www.actazool.org/paperdetail.asp?id=1434&volume=50&number=6&bgpage=978&endpage=990&year=2004&month=12). *Acta Zoologica Sinica*. **50** (6): 978–990. Archived from the original on March 25, 2005.
58. Feduccia, Alan; Lingham-Soliar, Theagarten; & Hincliffe, J. Richard. (2005). "Do feathered dinosaurs exist? Testing the hypothesis on neontological and paleontological evidence". *Journal of Morphology*. **266** (2): 125–66. Bibcode:2005JMorp.266..125F (https://ui.adsabs.harvard.edu/abs/2005JMorp.266..125F). doi:10.1002/jmor.10382 (https://doi.org/10.1002%2Fjmor.10382). PMID 16217748 (https://pubmed.ncbi.nlm.nih.gov/16217748). S2CID 15079072 (https://api.semanticscholar.org/CorpusID:15079072).
59. Burke, Ann C.; Feduccia, Alan (1997). "Developmental patterns and the identification of homologies in the avian hand". *Science*. **278** (5338): 666–668. Bibcode:1997Sci...278..666B (https://ui.adsabs.harvard.edu/abs/1997Sci...278..666B). doi:10.1126/science.278.5338.666 (https://doi.org/10.1126%2Fscience.278.5338.666).
60. Xu, Xing; Zhou, Zhonghe; Dudley, Robert; Mackem, Susan; Chuong, Cheng-Ming; Erickson, Gregory M.; Varricchio, David J. (2014). "An integrative approach to understanding bird origins" (https://www.science.org/doi/10.1126/science.1253293). *Science*. **346** (6215) 1253293. doi:10.1126/science.1253293 (https://doi.org/10.1126%2Fscience.1253293). PMID 25504729 (https://pubmed.ncbi.nlm.nih.gov/25504729). S2CID 24228777 (https://api.semanticscholar.org/CorpusID:24228777).
61. Sasso, Cristiano Dal; Maganuco, Simone; Cau, Andrea (2018-12-19). "The oldest ceratosaurian (Dinosauria: Theropoda), from the Lower Jurassic of Italy, sheds light on the evolution of the three-fingered hand of birds" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6304160). *PeerJ*. **6** e5976. Bibcode:2018PeerJ...6e5976D (https://ui.adsabs.harvard.edu/abs/2018PeerJ...6e5976D). doi:10.7717/peerj.5976 (https://doi.org/10.7717%2Fpeerj.5976). ISSN 2167-8359 (https://search.worldcat.org/issn/2167-8359). PMC 6304160 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6304160). PMID 30588396 (https://pubmed.ncbi.nlm.nih.gov/30588396).
62. Tamura, Koji; Nomura, Naoki; Seki, Ryohei; Yonei-Tamura, Sayuri; Yokoyama, Hitoshi (2011). "Embryological Evidence Identifies Wing Digits in Birds as Digits 1, 2, and 3" (https://www.science.org/doi/10.1126/science.1198229). *Science*. **331** (6018): 753–757. Bibcode:2011Sci...331..753T (https://ui.adsabs.harvard.edu/abs/2011Sci...331..753T). doi:10.1126/science.1198229 (https://doi.org/10.1126%2Fscience.1198229). PMID 21311019 (https://pubmed.ncbi.nlm.nih.gov/21311019). S2CID 37952118 (https://api.semanticscholar.org/CorpusID:37952118).
63. Newman SA (2011). "Thermogenesis, muscle hyperplasia, and the origin of birds". *BioEssays*. **33** (9): 653–656. doi:10.1002/bies.201100061 (https://doi.org/10.1002%2Fbies.201100061). PMID 21695679 (https://pubmed.ncbi.nlm.nih.gov/21695679). S2CID 42012034 (https://api.semanticscholar.org/CorpusID:42012034).
64. Newman SA, Mezentseva NV, Badyaev AV (2013). "Gene loss, thermogenesis, and the origin of birds". *Annals of the New York Academy of Sciences*. **1289** (1): 36–47. Bibcode:2013NYASA1289...36N (https://ui.adsabs.harvard.edu/abs/2013NYASA1289...36N). doi:10.1111/nyas.12090 (https://doi.org/10.1111%2Fnyas.12090). PMID 23550607 (https://pubmed.ncbi.nlm.nih.gov/23550607). S2CID 12240405 (https://api.semanticscholar.org/CorpusID:12240405).

65. Mezentseva NV, Kumaratilake JS, Newman SA (2008). "The brown adipocyte differentiation pathway in birds: An evolutionary road not taken" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2375860). *BMC Biology*. **6** (1) 17. doi:10.1186/1741-7007-6-17 (https://doi.org/10.1186%2F1741-7007-6-17). PMC 2375860 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2375860). PMID 18426587 (https://pubmed.ncbi.nlm.nih.gov/18426587).
66. Plotnick, Roy E.; Theodor, Jessica M.; Holtz, Thomas R. (2015-09-24). "Jurassic Pork: What Could a Jewish Time Traveler Eat?" (https://doi.org/10.1186%2Fs12052-015-0047-2). *Evolution: Education and Outreach*. **8** (1): 17. doi:10.1186/s12052-015-0047-2 (https://doi.org/10.1186%2Fs12052-015-0047-2). hdl:1903/27622 (https://hdl.handle.net/1903%2F27622). ISSN 1936-6434 (https://search.worldcat.org/issn/1936-6434). S2CID 16195453 (https://api.semanticscholar.org/CorpusID:16195453).
67. Sereno, Paul C. (1997). "The origin and evolution of dinosaurs" (http://doc.rero.ch/record/15399/files/PAL_E2750.pdf) (PDF). *Annual Review of Earth and Planetary Sciences*. **25** (1): 435–489. Bibcode:1997AREPS..25..435S (https://ui.adsabs.harvard.edu/abs/1997AREPS..25..435S). doi:10.1146/annurev.earth.25.1.435 (https://doi.org/10.1146%2Fannurev.earth.25.1.435).
68. Chiappe, Luis M. (1997). "Aves". In Currie, Philip J.; Padian, Kevin (eds.). *Encyclopedia of Dinosaurs*. San Diego: Academic Press. pp. 45–50. ISBN 978-0-12-226810-6.
69. Holtz, Thomas R.; Osmólska, Halszka. (2004). "Saurischia". In Weishampel, David B.; Dodson, Peter; Osmólska, Halszka (eds.). *The Dinosauria* (Second ed.). Berkeley: University of California Press. pp. 21–24. ISBN 978-0-520-24209-8.
70. Turner, Alan H.; Pol, Diego; Clarke, Julia A.; Erickson, Gregory M.; & Norell, Mark A. (2007). "A basal dromaeosaurid and size evolution preceding avian flight" (https://doi.org/10.1126%2Fscience.1144066). *Science*. **317** (5843): 1378–81. Bibcode:2007Sci...317.1378T (https://ui.adsabs.harvard.edu/abs/2007Sci...317.1378T). doi:10.1126/science.1144066 (https://doi.org/10.1126%2Fscience.1144066). PMID 17823350 (https://pubmed.ncbi.nlm.nih.gov/17823350).
71. Osmólska, Halszka; Maryańska, Teresa; & Wolsan, Mieczysław. (2002). "Avialan status for Oviraptorosauria" (http://app.pan.pl/article/item/app47-097.html). *Acta Palaeontologica Polonica*. **47** (1): 97–116.
72. Martinelli, Agustín G.; Vera, Ezequiel I. (2007). "*Achillesaurus manazzonei*, a new alvarezsaurid theropod (Dinosauria) from the Late Cretaceous Bajo de la Carpa Formation, Río Negro Province, Argentina" (http://www.mapress.com/zootaxa/2007f/z01582p017f.pdf) (PDF). *Zootaxa*. **1582**: 1–17. doi:10.11646/zootaxa.1582.1.1 (https://doi.org/10.11646%2Fzootaxa.1582.1.1).
73. Novas, Fernando E.; Pol, Diego. (2002). "Alvarezsaurid relationships reconsidered". In Chiappe, Luis M.; Witmer, Lawrence M. (eds.). *Mesozoic Birds: Above the Heads of Dinosaurs*. Berkeley: University of California Press. pp. 121–125. ISBN 978-0-520-20094-4.
74. Sereno, Paul C. (1999). "The evolution of dinosaurs". *Science*. **284** (5423): 2137–47. doi:10.1126/science.284.5423.2137 (https://doi.org/10.1126%2Fscience.284.5423.2137). PMID 10381873 (https://pubmed.ncbi.nlm.nih.gov/10381873).
75. Perle, Altangerel; Norell, Mark A.; Chiappe, Luis M.; & Clark, James M. (1993). "Flightless bird from the Cretaceous of Mongolia". *Nature*. **362** (6421): 623–626. Bibcode:1993Natur.362..623A (https://ui.adsabs.harvard.edu/abs/1993Natur.362..623A). doi:10.1038/362623a0 (https://doi.org/10.1038%2F362623a0). S2CID 4252852 (https://api.semanticscholar.org/CorpusID:4252852).
76. Chiappe, Luis M.; Norell, Mark A.; & Clark, James M. (2002). "The Cretaceous, short-armed Alvarezsauridae: *Mononykus* and its kin". In Chiappe, Luis M.; Witmer, Lawrence M. (eds.). *Mesozoic Birds: Above the Heads of Dinosaurs*. Berkeley: University of California Press. pp. 87–119. ISBN 978-0-520-20094-4.

77. Forster, Catherine A.; Sampson, Scott D.; Chiappe, Luis M.; & Krause, David W. (1998). "The theropod ancestry of birds: new evidence from the Late Cretaceous of Madagascar". *Science*. **279** (5358): 1915–9. Bibcode:1998Sci...279.1915F (https://ui.adsabs.harvard.edu/abs/1998Sci...279.1915F). doi:10.1126/science.279.5358.1915 (https://doi.org/10.1126%2Fscience.279.5358.1915). PMID 9506938 (https://pubmed.ncbi.nlm.nih.gov/9506938).
78. Makovicky, Peter J.; Apesteguía, Sebastián; & Agnolín, Federico L. (2005). "The earliest dromaeosaurid theropod from South America". *Nature*. **437** (7061): 1007–11. Bibcode:2005Natur.437.1007M (https://ui.adsabs.harvard.edu/abs/2005Natur.437.1007M). doi:10.1038/nature03996 (https://doi.org/10.1038%2Fnature03996). PMID 16222297 (https://pubmed.ncbi.nlm.nih.gov/16222297). S2CID 27078534 (https://api.semanticscholar.org/CorpusID:27078534).
79. Paul, Gregory S. (2002). *Dinosaurs of the Air: The Evolution and Loss of Flight in Dinosaurs and Birds* (https://archive.org/details/dinosaursofairev0000paul/page/472). Baltimore: Johns Hopkins University Press. p. 472pp (https://archive.org/details/dinosaursofairev0000paul/page/472). ISBN 978-0-8018-6763-7.
80. Mayr, Gerald; Pohl, Burkhard; & Peters, D. Stefan (2005). "A well-preserved *Archaeopteryx* specimen with theropod features" (http://doc.rero.ch/record/15488/files/PAL_E2876.pdf) (PDF). *Science*. **310** (5753): 1483–6. Bibcode:2005Sci...310.1483M (https://ui.adsabs.harvard.edu/abs/2005Sci...310.1483M). doi:10.1126/science.1120331 (https://doi.org/10.1126%2Fscience.1120331). PMID 16322455 (https://pubmed.ncbi.nlm.nih.gov/16322455). S2CID 28611454 (https://api.semanticscholar.org/CorpusID:28611454).
81. Chatterjee, Immoor; L. Immoor (9 September 2005). "The Dinosaurs of the Jurassic Park Movies" (https://web.archive.org/web/20080512005607/http://geolor.com/Jurassic_Park_Movies-Fact_versus_Fiction.htm). Geolor.com. Archived from the original (http://www.geolor.com/Jurassic_Park_Movies-Fact_versus_Fiction.htm) on 12 May 2008. Retrieved June 23, 2007.
82. Wellnhofer, P. (1988). "Ein neuer Exemplar von *Archaeopteryx*". *Archaeopteryx*. **6**: 1–30.
83. Xu X, Norell MA, Kuang X, Wang X, Zhao Q, Jia C (October 2004). "Basal tyrannosauroids from China and evidence for protofeathers in tyrannosauroids" (http://doc.rero.ch/record/15283/files/PAL_E2582.pdf) (PDF). *Nature*. **431** (7009): 680–4. Bibcode:2004Natur.431..680X (https://ui.adsabs.harvard.edu/abs/2004Natur.431..680X). doi:10.1038/nature02855 (https://doi.org/10.1038%2Fnature02855). PMID 15470426 (https://pubmed.ncbi.nlm.nih.gov/15470426). S2CID 4381777 (https://api.semanticscholar.org/CorpusID:4381777).
84. Feduccia, A. (2012). *Riddle of the Feathered Dragons: Hidden Birds of China.* Yale University Press, ISBN 0-300-16435-1, ISBN 978-0-300-16435-0
85. Zhang, F.; Kearns, S.L.; Orr, P.J.; Benton, M.J.; Zhou, Z.; Johnson, D.; Xu, X.; and Wang, X. (2010). "Fossilized melanosomes and the colour of Cretaceous dinosaurs and birds" (http://oro.open.ac.uk/22432/2/41064696.pdf) (PDF). *Nature*. **463** (7284): 1075–1078. Bibcode:2010Natur.463.1075Z (https://ui.adsabs.harvard.edu/abs/2010Natur.463.1075Z). doi:10.1038/nature08740 (https://doi.org/10.1038%2Fnature08740). PMID 20107440 (https://pubmed.ncbi.nlm.nih.gov/20107440). S2CID 205219587 (https://api.semanticscholar.org/CorpusID:205219587).
86. Foth, C (2012). "On the identification of feather structures in stem-line representatives of birds: evidence from fossils and actuopalaeontology". *Paläontologische Zeitschrift*. **86** (1): 91–102. Bibcode:2012PalZ...86...91F (https://ui.adsabs.harvard.edu/abs/2012PalZ...86...91F). doi:10.1007/s12542-011-0111-3 (https://doi.org/10.1007%2Fs12542-011-0111-3). S2CID 86362907 (https://api.semanticscholar.org/CorpusID:86362907).
87. Currie, P.J.; Chen, P.-j. (2001). "Anatomy of *Sinosauropteryx prima* from Liaoning, northeastern China" (http://doc.rero.ch/record/14314/files/PAL_E1480.pdf) (PDF). *Canadian Journal of Earth Sciences*. **38** (1): 705–727. Bibcode:2001CaJES..38.1705C (https://ui.adsabs.harvard.edu/abs/2001CaJES..38.1705C). doi:10.1139/cjes-38-12-1705 (https://doi.org/10.1139%2Fcjes-38-12-1705).

88. Cincotta, Aude; Nicolaï, Michaël; Campos, Hebert Bruno Nascimento; McNamara, Maria; D'Alba, Liliana; Shawkey, Matthew D.; Kischlat, Edio-Ernst; Yans, Johan; Carleer, Robert; Escuillié, François; Godefroit, Pascal (2022). "Pterosaur melanosomes support signalling functions for early feathers" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9046085). *Nature*. **604** (7907): 684–688. Bibcode:2022Natur.604..684C (https://ui.adsabs.harvard.edu/abs/2022Natur.604..684C). doi:10.1038/s41586-022-04622-3 (https://doi.org/10.1038%2Fs41586-022-04622-3). ISSN 1476-4687 (https://search.worldcat.org/issn/1476-4687). PMC 9046085 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9046085). PMID 35444275 (https://pubmed.ncbi.nlm.nih.gov/35444275).
89. Egawa, Shiro (2018). "Morphogenetic mechanism of the acquisition of the dinosaur-type acetabulum" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6227947). *R. Soc. Open Sci*. **5** (10) 180604. Bibcode:2018RSOS....580604E (https://ui.adsabs.harvard.edu/abs/2018RSOS....580604E). doi:10.1098/rsos.180604 (https://doi.org/10.1098%2Frsos.180604). PMC 6227947 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6227947). PMID 30473817 (https://pubmed.ncbi.nlm.nih.gov/30473817).
90. Bhullar, B. A. S.; Marugán-Lobón, J. S.; Racimo, F.; Bever, G. S.; Rowe, T. B.; Norell, M. A.; Abzhanov, A. (2012). "Birds have paedomorphic dinosaur skulls" (http://nrs.harvard.edu/urn-3:HUL.InstRepos:33950782). *Nature*. **487** (7406): 223–226. Bibcode:2012Natur.487..223B (https://ui.adsabs.harvard.edu/abs/2012Natur.487..223B). doi:10.1038/nature11146 (https://doi.org/10.1038%2Fnature11146). PMID 22722850 (https://pubmed.ncbi.nlm.nih.gov/22722850). S2CID 4370675 (https://api.semanticscholar.org/CorpusID:4370675).
91. O'Connor, P.M.; Claessens, L.P.A.M. (2005). "Basic avian pulmonary design and flow-through ventilation in non-avian theropod dinosaurs". *Nature*. **436** (7048): 253–6. Bibcode:2005Natur.436..253O (https://ui.adsabs.harvard.edu/abs/2005Natur.436..253O). doi:10.1038/nature03716 (https://doi.org/10.1038%2Fnature03716). PMID 16015329 (https://pubmed.ncbi.nlm.nih.gov/16015329). S2CID 4390587 (https://api.semanticscholar.org/CorpusID:4390587).
92. Paul C. Sereno, Ricardo N. Martinez, Jeffrey A. Wilson, David J. Varricchio, Oscar A. Alcober, Hans C. E. Larsson (2008). Kemp, Tom (ed.). "Evidence for Avian Intrathoracic Air Sacs in a New Predatory Dinosaur from Argentina" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2553519). *PLOS ONE*. **3** (9) e3303. Bibcode:2008PLoSO...3.3303S (https://ui.adsabs.harvard.edu/abs/2008PLoSO...3.3303S). doi:10.1371/journal.pone.0003303 (https://doi.org/10.1371%2Fjournal.pone.0003303). PMC 2553519 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2553519). PMID 18825273 (https://pubmed.ncbi.nlm.nih.gov/18825273).
93. Fisher, P. E.; Russell, D. A.; Stoskopf, M. K.; Barrick, R. E.; Hammer, M.; Kuzmitz, A. A. (2000). "Cardiovascular evidence for an intermediate or higher metabolic rate in an ornithischian dinosaur". *Science*. **288** (5465): 503–5. Bibcode:2000Sci...288..503F (https://ui.adsabs.harvard.edu/abs/2000Sci...288..503F). doi:10.1126/science.288.5465.503 (https://doi.org/10.1126%2Fscience.288.5465.503). PMID 10775107 (https://pubmed.ncbi.nlm.nih.gov/10775107).
94. Hillenius, W. J.; Ruben, J. A. (2004). "The evolution of endothermy in terrestrial vertebrates: Who? when? why?". *Physiological and Biochemical Zoology*. **77** (6): 1019–42. Bibcode:2004PhyBZ..77.1019H (https://ui.adsabs.harvard.edu/abs/2004PhyBZ..77.1019H). doi:10.1086/425185 (https://doi.org/10.1086%2F425185). PMID 15674773 (https://pubmed.ncbi.nlm.nih.gov/15674773). S2CID 29300018 (https://api.semanticscholar.org/CorpusID:29300018).
95. Rowe, T.; McBride, E. F.; Sereno, P. C.; Russell, D. A.; Fisher, P. E.; Barrick, R. E.; Stoskopf, M. K. (2001). "Dinosaur with a Heart of Stone" (https://doi.org/10.1126%2Fscience.291.5505.783a). *Science*. **291** (5505): 783. doi:10.1126/science.291.5505.783a (https://doi.org/10.1126%2Fscience.291.5505.783a). PMID 11157158 (https://pubmed.ncbi.nlm.nih.gov/11157158).

96. Cleland, Timothy P.; Stoskopf, Michael K.; and Schweitzer, Mary H. (2011). "Histological, chemical, and morphological reexamination of the "heart" of a small Late Cretaceous *Thescelosaurus*". *Naturwissenschaften*. **98** (3): 203–211. Bibcode:2011NW.....98..203C (https://ui.adsabs.harvard.edu/abs/2011NW.....98..203C). doi:10.1007/s00114-010-0760-1 (https://doi.org/10.1007%2Fs00114-010-0760-1). PMID 21279321 (https://pubmed.ncbi.nlm.nih.gov/21279321). S2CID 2408562 (https://api.semanticscholar.org/CorpusID:2408562).
97. Chinsamy, Anusuya; and Hillenius, Willem J. (2004). "Physiology of nonavian dinosaurs". *The Dinosauria*, 2nd. 643–659.
98. Xu, X.; Norell, M.A. (2004). "A new troodontid dinosaur from China with avian-like sleeping posture" (http://doc.rero.ch/record/15284/files/PAL_E2583.pdf) (PDF). *Nature*. **431** (7010): 838–41. Bibcode:2004Natur.431..838X (https://ui.adsabs.harvard.edu/abs/2004Natur.431..838X). doi:10.1038/nature02898 (https://doi.org/10.1038%2Fnature02898). PMID 15483610 (https://pubmed.ncbi.nlm.nih.gov/15483610). S2CID 4362745 (https://api.semanticscholar.org/CorpusID:4362745). See commentary on the article (https://www.theguardian.com/life/news/story/0,12976,1326559,00.html)
99. Schweitzer, M.H.; Wittmeyer, J.L.; and Horner, J.R. (2005). "Gender-specific reproductive tissue in ratites and *Tyrannosaurus rex*" (http://doc.rero.ch/record/14767/files/PAL_E1895.pdf) (PDF). *Science*. **308** (5727): 1456–60. Bibcode:2005Sci...308.1456S (https://ui.adsabs.harvard.edu/abs/2005Sci...308.1456S). doi:10.1126/science.1112158 (https://doi.org/10.1126%2Fscience.1112158). PMID 15933198 (https://pubmed.ncbi.nlm.nih.gov/15933198). S2CID 30264554 (https://api.semanticscholar.org/CorpusID:30264554).
100. Lee, Andrew H.; Werning, Sarah (2008). "Sexual maturity in growing dinosaurs does not fit reptilian growth models" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2206579). *Proceedings of the National Academy of Sciences*. **105** (2): 582–7. Bibcode:2008PNAS..105..582L (https://ui.adsabs.harvard.edu/abs/2008PNAS..105..582L). doi:10.1073/pnas.0708903105 (https://doi.org/10.1073%2Fpnas.0708903105). PMC 2206579 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2206579). PMID 18195356 (https://pubmed.ncbi.nlm.nih.gov/18195356).
101. Norell, M. A., Clark, J. M., Dashzeveg, D., Barsbold, T., Chiappe, L. M., Davidson, A. R., McKenna, M. C. and Novacek, M. J. (November 1994). "A theropod dinosaur embryo and the affinities of the Flaming Cliffs Dinosaur eggs". *Science*. **266** (5186): 779–82. Bibcode:1994Sci...266..779N (https://ui.adsabs.harvard.edu/abs/1994Sci...266..779N). doi:10.1126/science.266.5186.779 (https://doi.org/10.1126%2Fscience.266.5186.779). PMID 17730398 (https://pubmed.ncbi.nlm.nih.gov/17730398). S2CID 22333224 (https://api.semanticscholar.org/CorpusID:22333224).
102. Wings O (2007). "A review of gastrolith function with implications for fossil vertebrates and a revised classification" (http://www.app.pan.pl/article/item/app52-001.html). *Palaeontologica Polonica*. **52** (1): 1–16.
103. Embery, G; Milner, AC; Waddington, RJ; Hall, RC; Langley, MS; Milan, AM (2003). "Identification of proteinaceous material in the bone of the dinosaur Iguanodon". *Connective Tissue Research*. **44** (Suppl 1): 41–6. doi:10.1080/03008200390152070 (https://doi.org/10.1080%2F03008200390152070). PMID 12952172 (https://pubmed.ncbi.nlm.nih.gov/12952172). S2CID 2249126 (https://api.semanticscholar.org/CorpusID:2249126).
104. Schweitzer, MH; Marshall, M; Carron, K; Bohle, DS; Busse, SC; Arnold, EV; Barnard, D; Horner, JR; Starkey, JR (June 1997). "Heme compounds in dinosaur trabecular bone" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC21042). *Proceedings of the National Academy of Sciences of the United States of America*. **94** (12): 6291–6. Bibcode:1997PNAS...94.6291S (https://ui.adsabs.harvard.edu/abs/1997PNAS...94.6291S). doi:10.1073/pnas.94.12.6291 (https://doi.org/10.1073%2Fpnas.94.12.6291). PMC 21042 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC21042). PMID 9177210 (https://pubmed.ncbi.nlm.nih.gov/9177210).

105. Slater, Tiffany S.; Edwards, Nicholas P.; Webb, Samuel M.; Zhang, Fucheng; McNamara, Maria E. (2023-09-21). "Preservation of corneous β-proteins in Mesozoic feathers" (https://www.nature.com/articles/s41559-023-02177-8). *Nature Ecology & Evolution*. **7** (10): 1706–1713. Bibcode:2023NatEE...7.1706S (https://ui.adsabs.harvard.edu/abs/2023NatEE...7.1706S). doi:10.1038/s41559-023-02177-8 (https://doi.org/10.1038%2Fs41559-023-02177-8). ISSN 2397-334X (https://search.worldcat.org/issn/2397-334X). OSTI 2327098 (https://www.osti.gov/biblio/2327098). PMID 37735563 (https://pubmed.ncbi.nlm.nih.gov/37735563). S2CID 262125827 (https://api.semanticscholar.org/CorpusID:262125827).
106. Schweitzer, MH; Wittmeyer, JL; Horner, JR; Toporski, JK (2005). "Soft-Tissue Vessels and Cellular Preservation in Tyrannosaurus rex". *Science*. **307** (5717): 1952–5. Bibcode:2005Sci...307.1952S (https://ui.adsabs.harvard.edu/abs/2005Sci...307.1952S). doi:10.1126/science.1108397 (https://doi.org/10.1126%2Fscience.1108397). PMID 15790853 (https://pubmed.ncbi.nlm.nih.gov/15790853). S2CID 30456613 (https://api.semanticscholar.org/CorpusID:30456613). Also covers the Reproduction Biology paragraph in the Feathered dinosaurs and the bird connection section.
107. Asara, JM; Schweitzer MH, Freimark LM, Phillips M, Cantley LC (2007). "Protein Sequences from Mastodon and Tyrannosaurus Rex Revealed by Mass Spectrometry". *Science*. **316** (5822): 280–5. Bibcode:2007Sci...316..280A (https://ui.adsabs.harvard.edu/abs/2007Sci...316..280A). doi:10.1126/science.1137614 (https://doi.org/10.1126%2Fscience.1137614). PMID 17431180 (https://pubmed.ncbi.nlm.nih.gov/17431180). S2CID 85299309 (https://api.semanticscholar.org/CorpusID:85299309).
108. Organ, CL; Schweitzer, MH; Zheng, W; Freimark, LM; Cantley, LC; Asara, JM (2008). "Molecular phylogenetics of mastodon and *Tyrannosaurus rex*". *Science*. **320** (5875): 499. Bibcode:2008Sci...320..499O (https://ui.adsabs.harvard.edu/abs/2008Sci...320..499O). doi:10.1126/science.1154284 (https://doi.org/10.1126%2Fscience.1154284). PMID 18436782 (https://pubmed.ncbi.nlm.nih.gov/18436782). S2CID 24971064 (https://api.semanticscholar.org/CorpusID:24971064).
109. Schweitzer, M. H.; Zheng W., Organ C. L., Avci R., Suo Z., Freimark L. M., Lebleu V. S., Duncan M. B., Vander Heiden M. G., Neveu J. M., Lane W. S., Cottrell J. S., Horner J. R., Cantley L. C., Kalluri R. & Asara J. M. (2009). "Biomolecular Characterization and Protein Sequences of the Campanian Hadrosaur B. canadensis". *Science*. **324** (5927): 626–31. Bibcode:2009Sci...324..626S (https://ui.adsabs.harvard.edu/abs/2009Sci...324..626S). doi:10.1126/science.1165069 (https://doi.org/10.1126%2Fscience.1165069). PMID 19407199 (https://pubmed.ncbi.nlm.nih.gov/19407199). S2CID 5358680 (https://api.semanticscholar.org/CorpusID:5358680).
110. Kaye, TG; Gaugler, G; Sawlowicz, Z (2008). "Dinosaurian soft tissues interpreted as bacterial biofilms" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2483347). *PLOS ONE*. **3** (7) e2808. Bibcode:2008PLoSO...3.2808K (https://ui.adsabs.harvard.edu/abs/2008PLoSO...3.2808K). doi:10.1371/journal.pone.0002808 (https://doi.org/10.1371%2Fjournal.pone.0002808). PMC 2483347 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2483347). PMID 18665236 (https://pubmed.ncbi.nlm.nih.gov/18665236).
111. Peterson, JE; Lenczewski, ME; Scherer, RP (2010). "Influence of microbial biofilms on the preservation of primary soft tissue in fossil and extant archosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2953520). *PLOS ONE*. **5** (10) e13334. Bibcode:2010PLoSO...513334P (https://ui.adsabs.harvard.edu/abs/2010PLoSO...513334P). doi:10.1371/journal.pone.0013334 (https://doi.org/10.1371%2Fjournal.pone.0013334). PMC 2953520 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2953520). PMID 20967227 (https://pubmed.ncbi.nlm.nih.gov/20967227).
112. Bern, M; Phinney, BS; Goldberg, D (2009). "Reanalysis of *Tyrannosaurus rex* mass spectra" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2738754). *Journal of Proteome Research*. **8** (9): 4328–4332. doi:10.1021/pr900349r (https://doi.org/10.1021%2Fpr900349r). PMC 2738754 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2738754). PMID 19603827 (https://pubmed.ncbi.nlm.nih.gov/19603827).

113. Cleland, TP; et al. (2015). "Mass spectrometry and antibody-based characterization of blood vessels from *Brachylophosaurus canadensis*" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4768904). *Journal of Proteome Research*. **14** (12): 5252–5262. doi:10.1021/acs.jproteome.5b00675 (https://doi.org/10.1021%2Facs.jproteome.5b00675). PMC 4768904 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4768904). PMID 26595531 (https://pubmed.ncbi.nlm.nih.gov/26595531).
114. Buckley, Michael; Warwood, Stacey; van Dongen, Bart; Kitchener, Andrew C.; Manning, Phillip L. (2017). "A fossil protein chimera; difficulties in discriminating dinosaur peptide sequences from modern cross-contamination" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5454271). *Proceedings of the Royal Society B*. **284** (1855) 20170544. Bibcode:2017PBioS.28470544B (https://ui.adsabs.harvard.edu/abs/2017PBioS.28470544B). doi:10.1098/rspb.2017.0544 (https://doi.org/10.1098%2Frspb.2017.0544). PMC 5454271 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5454271). PMID 28566488 (https://pubmed.ncbi.nlm.nih.gov/28566488).
115. Wang, HL; Yan, ZY; Jin, DY (1997). "Reanalysis of published DNA sequence amplified from Cretaceous dinosaur egg fossil" (https://doi.org/10.1093%2Foxfordjournals.molbev.a025796). *Molecular Biology and Evolution*. **14** (5): 589–91. doi:10.1093/oxfordjournals.molbev.a025796 (https://doi.org/10.1093%2Foxfordjournals.molbev.a025796). PMID 9159936 (https://pubmed.ncbi.nlm.nih.gov/9159936).
116. Voeten, Dennis F.A.E.; et al. (13 March 2018). "Wing bone geometry reveals active flight in Archaeopteryx" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5849612). *Nature Communications*. **9** (923) 923. Bibcode:2018NatCo...9..923V (https://ui.adsabs.harvard.edu/abs/2018NatCo...9..923V). doi:10.1038/s41467-018-03296-8 (https://doi.org/10.1038%2Fs41467-018-03296-8). PMC 5849612 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5849612). PMID 29535376 (https://pubmed.ncbi.nlm.nih.gov/29535376).
117. Guarino, Ben (13 March 2018). "This feathery dinosaur probably flew, but not like any bird you know" (https://www.washingtonpost.com/news/speaking-of-science/wp/2018/03/13/this-feathery-dinosaur-probably-flew-but-not-like-any-bird-you-know/). *The Washington Post*. Retrieved 13 March 2018.
118. Terres, John K. (1980). *The Audubon Society Encyclopedia of North American Birds* (https://archive.org/details/audubonsocietyen00terr/page/398). New York, NY: Knopf. pp. 398–401 (https://archive.org/details/audubonsocietyen00terr/page/398). ISBN 978-0-394-46651-4.
119. Poling, J. (1996). "Feathers, scutes and the origin of birds" (https://web.archive.org/web/20080402065705/http://www.dinosauria.com/jdp/archie/scutes.htm). dinosauria.com. Archived from the original (http://www.dinosauria.com/jdp/archie/scutes.htm) on 2008-04-02. Retrieved 2008-03-21.
120. Benton, Michael J.; Dhouailly, Danielle; Jiang, Baoyu; McNamara, Maria (2019). "The Early Origin of Feathers". *Trends in Ecology & Evolution*. **34** (9): 856–869. Bibcode:2019TEcoE..34..856B (https://ui.adsabs.harvard.edu/abs/2019TEcoE..34..856B). doi:10.1016/j.tree.2019.04.018 (https://doi.org/10.1016%2Fj.tree.2019.04.018). hdl:10468/8068 (https://hdl.handle.net/10468%2F8068). ISSN 0169-5347 (https://search.worldcat.org/issn/0169-5347). PMID 31164250 (https://pubmed.ncbi.nlm.nih.gov/31164250). S2CID 174811556 (https://api.semanticscholar.org/CorpusID:174811556).
121. Musser, Jacob M.; Wagner, Günter P.; Prum, Richard O. (2015). "Nuclear β-catenin localization supports homology of feathers, avian scutate scales, and alligator scales in early development: Feather and scale developmental homology" (https://onlinelibrary.wiley.com/doi/10.1111/ede.12123). *Evolution & Development*. **17** (3): 185–194. doi:10.1111/ede.12123 (https://doi.org/10.1111%2Fede.12123). PMID 25963196 (https://pubmed.ncbi.nlm.nih.gov/25963196). S2CID 20485128 (https://api.semanticscholar.org/CorpusID:20485128).

122. Prum, R. & Brush, A.H. (2002). "The evolutionary origin and diversification of feathers" (https://web.archive.org/web/20031015170027/http://www.mcorriss.com/Prum_%26_Brush_2002.pdf) (PDF). *The Quarterly Review of Biology*. **77** (3): 261–95. doi:10.1086/341993 (https://doi.org/10.1086%2F341993). PMID 12365352 (https://pubmed.ncbi.nlm.nih.gov/12365352). S2CID 6344830 (https://api.semanticscholar.org/CorpusID:6344830). Archived from the original on 2003-10-15. Retrieved 2018-09-17.
123. Mayr G, Pohl B, Peters D (2005). "A well-preserved *Archaeopteryx* specimen with theropod features" (http://doc.rero.ch/record/15488/files/PAL_E2876.pdf) (PDF). *Science*. **310** (5753): 1483–6. Bibcode:2005Sci...310.1483M (https://ui.adsabs.harvard.edu/abs/2005Sci...310.1483M). doi:10.1126/science.1120331 (https://doi.org/10.1126%2Fscience.1120331). PMID 16322455 (https://pubmed.ncbi.nlm.nih.gov/16322455). S2CID 28611454 (https://api.semanticscholar.org/CorpusID:28611454).
124. Feduccia, A. (1999). *The Origin and Evolution of Birds* (https://web.archive.org/web/20200602114649/https://yalebooks.yale.edu/). Yale University Press. ISBN 978-0-300-07861-9. Archived from the original (http://yalepress.yale.edu/book.asp?isbn=978-0-300-07861-9) on 2020-06-02. Retrieved 2012-05-10.
125. Feduccia, A. (February 1995). "Explosive Evolution in Tertiary Birds and Mammals" (https://web.archive.org/web/20080320033248/http://taxonomy.zoology.gla.ac.uk/%7Erdmp1c/teaching/L3/tutorials/feduccia/feduccia.html). *Science*. **267** (5198): 637–8. Bibcode:1995Sci...267..637F (https://ui.adsabs.harvard.edu/abs/1995Sci...267..637F). doi:10.1126/science.267.5198.637 (https://doi.org/10.1126%2Fscience.267.5198.637). PMID 17745839 (https://pubmed.ncbi.nlm.nih.gov/17745839). S2CID 42829066 (https://api.semanticscholar.org/CorpusID:42829066). Archived from the original (http://taxonomy.zoology.gla.ac.uk/~rdmp1c/teaching/L3/tutorials/feduccia/feduccia.html) on 2008-03-20.
126. Feduccia, A. (1993).
127. Cretaceous tracks of a bird with a similar lifestyle have been found - Lockley, M.G., Li, R., Harris, J.D., Matsukawa, M., and Liu, M. (August 2007). "Earliest zygodactyl bird feet: evidence from Early Cretaceous roadrunner-like tracks" (http://doc.rero.ch/record/15532/files/PAL_E2941.pdf) (PDF). *Naturwissenschaften*. **94** (8): 657–65. Bibcode:2007NW.....94..657L (https://ui.adsabs.harvard.edu/abs/2007NW.....94..657L). doi:10.1007/s00114-007-0239-x (https://doi.org/10.1007%2Fs00114-007-0239-x). PMID 17387416 (https://pubmed.ncbi.nlm.nih.gov/17387416). S2CID 15821251 (https://api.semanticscholar.org/CorpusID:15821251).
128. Burgers, P. & L. M. Chiappe (1999). "The wing of Archaeopteryx as a primary thrust generator" (http://www.stephenjaygould.org/ctrl/news/file013.html). *Nature*. **399** (6731): 60–62. Bibcode:1999Natur.399...60B (https://ui.adsabs.harvard.edu/abs/1999Natur.399...60B). doi:10.1038/19967 (https://doi.org/10.1038%2F19967). S2CID 4430686 (https://api.semanticscholar.org/CorpusID:4430686).
129. Cowen, R. (1991). *History of Life*. Blackwell Science. ISBN 978-0-7266-0287-0.
130. Videler, J.J. 2005: Avian Flight. Oxford University. Press, Oxford.
131. Vargas, A.O., Fallon, J.F. (October 2004). "Birds have dinosaur wings: The molecular evidence". *Journal of Experimental Zoology Part B: Molecular and Developmental Evolution*. **304B** (1): 86–90. doi:10.1002/jez.b.21023 (https://doi.org/10.1002%2Fjez.b.21023). PMID 15515040 (https://pubmed.ncbi.nlm.nih.gov/15515040).
132. Pennisi, E. (January 2005). "Bird Wings Really Are Like Dinosaurs' Hands" (https://web.archive.org/web/20110727114230/http://www.ncsce.org/PDF_files/shift/Pennisi.pdf) (PDF). *Science*. **307** (5707): 194b. doi:10.1126/science.307.5707.194b (https://doi.org/10.1126%2Fscience.307.5707.194b). PMID 15653478 (https://pubmed.ncbi.nlm.nih.gov/15653478). S2CID 82490156 (https://api.semanticscholar.org/CorpusID:82490156). Archived from the original (http://www.ncsce.org/PDF_files/shift/Pennisi.pdf) (PDF) on 2011-07-27.

133. Fowler, Denver W.; Freedman, Elizabeth A.; Scannella, John B.; Kambic, Robert E. (2011-12-14). "The Predatory Ecology of Deinonychus and the Origin of Flapping in Birds" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3237572). *PLOS ONE*. **6** (12) e28964. Bibcode:2011PLoSO...628964F (https://ui.adsabs.harvard.edu/abs/2011PLoSO...628964F). doi:10.1371/journal.pone.0028964 (https://doi.org/10.1371%2Fjournal.pone.0028964). ISSN 1932-6203 (https://search.worldcat.org/issn/1932-6203). PMC 3237572 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3237572). PMID 22194962 (https://pubmed.ncbi.nlm.nih.gov/22194962).
134. There is a video clip of a very young chick doing this at "Wing assisted incline running and evolution of flight" (https://www.youtube.com/watch?v=MNxt_-f9dmw). *YouTube*. 22 May 2006. Archived (https://ghostarchive.org/varchive/youtube/20211219/MNxt_-f9dmw) from the original on 2021-12-19.
135. Dial, K.P. (2003). "Wing-Assisted Incline Running and the Evolution of Flight" (https://scholarworks.umt.edu/cgi/viewcontent.cgi?article=1048&context=biosci_pubs). *Science*. **299** (5605): 402–4. Bibcode:2003Sci...299..402D (https://ui.adsabs.harvard.edu/abs/2003Sci...299..402D). doi:10.1126/science.1078237 (https://doi.org/10.1126%2Fscience.1078237). PMID 12532020 (https://pubmed.ncbi.nlm.nih.gov/12532020). S2CID 40712093 (https://api.semanticscholar.org/CorpusID:40712093). Summarized in Morelle, Rebecca (24 January 2008). "Secrets of bird flight revealed" (https://news.bbc.co.uk/1/hi/sci/tech/7205086.stm) (Web). *Scientists believe they could be a step closer to solving the mystery of how the first birds took to the air*. BBC News. Retrieved 2008-01-25.
136. Bundle, M.W. & Dial, K.P. (2003). "Mechanics of wing-assisted incline running (WAIR)" (http://dbs.umt.edu/flightlab/pdf/bundle%20and%20dial%20JEB%202003.pdf) (PDF). *The Journal of Experimental Biology*. **206** (Pt 24): 4553–64. Bibcode:2003JExpB.206.4553B (https://ui.adsabs.harvard.edu/abs/2003JExpB.206.4553B). doi:10.1242/jeb.00673 (https://doi.org/10.1242%2Fjeb.00673). PMID 14610039 (https://pubmed.ncbi.nlm.nih.gov/14610039). S2CID 6323207 (https://api.semanticscholar.org/CorpusID:6323207).
137. Senter, P. (2006). "Scapular orientation in theropods and basal birds, and the origin of flapping flight" (http://www.app.pan.pl/article/item/app51-305.html). *Acta Palaeontologica Polonica*. **51** (2): 305–313.
138. Dececchi, T. Alexander; Larsson, Hans C. E. (2011). "Assessing Arboreal Adaptations of Bird Antecedents: Testing the Ecological Setting of the Origin of the Avian Flight Stroke" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3153453). *PLOS ONE*. **6** (8) e22292. Bibcode:2011PLoSO...622292D (https://ui.adsabs.harvard.edu/abs/2011PLoSO...622292D). doi:10.1371/journal.pone.0022292 (https://doi.org/10.1371%2Fjournal.pone.0022292). PMC 3153453 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3153453). PMID 21857918 (https://pubmed.ncbi.nlm.nih.gov/21857918).
139. Chatterjee, Sankar, Templin, R.J. (2004) "Feathered coelurosaurs from China: new light on the arboreal origin of avian flight" pp. 251-281. In Feathered Dragons: Studies on the Transition from Dinosaurs to Birds (P. J. Currie, E. B. Koppelhus, M. A. Shugar, and J. L. Wright (eds.). Indiana University Press, Bloomington.
140. Tarsitano, Samuel F.; Russell, Anthony P.; Horne, Francis; Plummer, Christopher; Millerchip, Karen (2000). "On the Evolution of Feathers from an Aerodynamic and Constructional View Point" (https://doi.org/10.1093%2Ficb%2F40.4.676). *American Zoologist*. **40** (4): 676–686. doi:10.1093/icb/40.4.676 (https://doi.org/10.1093%2Ficb%2F40.4.676).
141. Hu, D.; Hou, L.; Zhang, L. & Xu, X. (2009). "A pre-*Archaeopteryx* troodontid theropod from China with long feathers on the metatarsus". *Nature*. **461** (7264): 640–3. Bibcode:2009Natur.461..640H (https://ui.adsabs.harvard.edu/abs/2009Natur.461..640H). doi:10.1038/nature08322 (https://doi.org/10.1038%2Fnature08322). PMID 19794491 (https://pubmed.ncbi.nlm.nih.gov/19794491). S2CID 205218015 (https://api.semanticscholar.org/CorpusID:205218015).

142. Hopson, James A. "Ecomorphology of avian and nonavian theropod phalangeal proportions:Implications for the arboreal versus terrestrial origin of bird flight" (2001) From New Perspectives on the Origin and Early Evolution of Birds: Proceedings of the International Symposium in Honor of John H. Ostrom. J. Gauthier and L. F. Gall, eds. New Haven: Peabody Mus. Nat. Hist., Yale Univ. ISBN 0-912532-57-2.© 2001 Peabody Museum of Natural History, Yale University. All rights reserved.
143. Glen, C.L. & Bennett, M.B. (November 2007). "Foraging modes of Mesozoic birds and non-avian theropods" (https://doi.org/10.1016%2Fj.cub.2007.09.026). *Current Biology*. **17** (21): R911–2. Bibcode:2007CBio...17.R911G (https://ui.adsabs.harvard.edu/abs/2007CBio...17.R911G). doi:10.1016/j.cub.2007.09.026 (https://doi.org/10.1016%2Fj.cub.2007.09.026). PMID 17983564 (https://pubmed.ncbi.nlm.nih.gov/17983564). S2CID 535424 (https://api.semanticscholar.org/CorpusID:535424).
144. Alonso, P.D., Milner, A.C., Ketcham, R.A., Cokson, M.J. and Rowe, T.B. (August 2004). "The avian nature of the brain and inner ear of Archaeopteryx" (http://doc.rero.ch/record/15280/files/PAL_E2579.pdf) (PDF). *Nature*. **430** (7000): 666–9. Bibcode:2004Natur.430..666A (https://ui.adsabs.harvard.edu/abs/2004Natur.430..666A). doi:10.1038/nature02706 (https://doi.org/10.1038%2Fnature02706). PMID 15295597 (https://pubmed.ncbi.nlm.nih.gov/15295597). S2CID 4391019 (https://api.semanticscholar.org/CorpusID:4391019).
145. Chiappe, L.M. (2007-02-02). *Glorified Dinosaurs: The Origin and Early Evolution of Birds*. Sydney: UNSW Press. ISBN 978-0-471-24723-4.
146. Zhang, F., Zhou, Z., Xu, X. & Wang, X. (2002). "A juvenile coelurosaurian theropod from China indicates arboreal habits". *Naturwissenschaften*. **89** (9): 394–8. Bibcode:2002NW.....89..394Z (https://ui.adsabs.harvard.edu/abs/2002NW.....89..394Z). doi:10.1007/s00114-002-0353-8 (https://doi.org/10.1007%2Fs00114-002-0353-8). PMID 12435090 (https://pubmed.ncbi.nlm.nih.gov/12435090). S2CID 556221 (https://api.semanticscholar.org/CorpusID:556221).
147. Chatterjee, S; Templin, RJ (2007). "Biplane wing planform and flight performance of the feathered dinosaur Microraptor gui" (http://www.pnas.org/cgi/reprint/0609975104v1.pdf) (PDF). *Proceedings of the National Academy of Sciences*. **104** (5): 1576–80. Bibcode:2007PNAS..104.1576C (https://ui.adsabs.harvard.edu/abs/2007PNAS..104.1576C). doi:10.1073/pnas.0609975104 (https://doi.org/10.1073%2Fpnas.0609975104). PMC 1780066 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1780066). PMID 17242354 (https://pubmed.ncbi.nlm.nih.gov/17242354).
148. Beebe, C. W. A. (1915). "Tetrapteryx stage in the ancestry of birds". *Zoologica*. **2**: 38–52.
149. Lucas, Frederic A. (Frederic Augustus); American Museum of Natural History (13 June 2018). *Animals of the past: an account of some of the creatures of the ancient world* (https://archive.org/details/animalsofpastacc00lucauoft). New York : [Amer. Mus. Press] – via Internet Archive.
150. Hartman, Scott; Mortimer, Mickey; Wahl, William R.; Lomax, Dean R.; Lippincott, Jessica; Lovelace, David M. (2019-07-10). "A new paravian dinosaur from the Late Jurassic of North America supports a late acquisition of avian flight" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6626525). *PeerJ*. **7** e7247. Bibcode:2019PeerJ...7e7247H (https://ui.adsabs.harvard.edu/abs/2019PeerJ...7e7247H). doi:10.7717/peerj.7247 (https://doi.org/10.7717%2Fpeerj.7247). ISSN 2167-8359 (https://search.worldcat.org/issn/2167-8359). PMC 6626525 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6626525). PMID 31333906 (https://pubmed.ncbi.nlm.nih.gov/31333906).
151. Paul, G.S. (2002). *Dinosaurs of the Air: The Evolution and Loss of Flight in Dinosaurs and Birds*. Baltimore: Johns Hopkins University Press. p. 257.
152. Paul, G.S. (2016). "Avepods". *The Princeton Field Guide to Dinosaurs* (2nd ed.). Princeton, New Jersey: Princeton University Press. pp. 161, 170. ISBN 978-0-691-16766-4.

153. Mayr, G. (2006). "Response to Comment on *A Well-Preserved Archaeopteryx Specimen with Theropod Features*" (https://doi.org/10.1126%2Fscience.1130964). *Science*. **313** (5791): 1238c. Bibcode:2006Sci...313.1238M (https://ui.adsabs.harvard.edu/abs/2006Sci...313.1238M). doi:10.1126/science.1130964 (https://doi.org/10.1126%2Fscience.1130964).
154. Corfe, I. J.; Butler, RJ (2006). "Comment on *A Well- Preserved Archaeopteryx Specimen With Theropod Features*". *Science*. **313** (5791): 1238b. Bibcode:2006Sci...313.1238C (https://ui.adsabs.harvard.edu/abs/2006Sci...313.1238C). doi:10.1126/science.1130800 (https://doi.org/10.1126%2Fscience.1130800). PMID 16946054 (https://pubmed.ncbi.nlm.nih.gov/16946054). S2CID 19859320 (https://api.semanticscholar.org/CorpusID:19859320).
155. Parsons, William L.; Parsons, Kristen M. (2015). "Morphological Variations within the Ontogeny of Deinonychus antirrhopus (Theropoda, Dromaeosauridae)" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4398413). *PLOS ONE*. **10** (4) e0121476. Bibcode:2015PLoSO..1021476P (https://ui.adsabs.harvard.edu/abs/2015PLoSO..1021476P). doi:10.1371/journal.pone.0121476 (https://doi.org/10.1371%2Fjournal.pone.0121476). PMC 4398413 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4398413). PMID 25875499 (https://pubmed.ncbi.nlm.nih.gov/25875499).
156. Hutson, Joel D.; Hutson, Kelda N. (2018). "Retention of the flight-adapted avian finger-joint complex in the Ostrich helps identify when wings began evolving in dinosaurs" (https://figshare.com/articles/dataset/Retention_of_the_flight-adapted_avian_finger-joint_complex_in_the_Ostrich_helps_identify_when_wings_began_evolving_in_dinosaurs/5840601). *Ostrich: Journal of African Ornithology*. **89** (2): 173–186. Bibcode:2018Ostri..89..173H (https://ui.adsabs.harvard.edu/abs/2018Ostri..89..173H). doi:10.2989/00306525.2017.1422566 (https://doi.org/10.2989%2F00306525.2017.1422566). S2CID 91081348 (https://api.semanticscholar.org/CorpusID:91081348).
157. Chiappe, Luis (2002-01-01), *Osteology of the Flightless Patagopteryx deferrariisi from the Late Cretaceous of Patagonia (Argentina)* (https://www.researchgate.net/publication/281640883), University of California Press, pp. 281–316, ISBN 978-0-520-20094-4, retrieved 2020-07-27

References

- Barsbold, Rinchen (1983): O ptich'ikh chertakh v stroyenii khishchnykh dinozavrov. ["Avian" features in the morphology of predatory dinosaurs]. *Transactions of the Joint Soviet Mongolian Paleontological Expedition* **24**: 96–103. [Original article in Russian.] Translated by W. Robert Welsh, copy provided by Kenneth Carpenter and converted by Matthew Carrano. PDF fulltext (https://web.archive.org/web/20081217051956/http://ravenel.si.edu/paleo/paleoglot/files/Barsbold_83b.pdf)
- Borenstein, Seth (July 31, 2014). "Study traces dinosaur evolution into early birds" (https://web.archive.org/web/20181023110310/http://apnews.excite.com/article/20140731/us-sci-shrinking-dinosaurs-a5c053f221.html). *AP News*. Archived from the original (http://apnews.excite.com/article/20140731/us-sci-shrinking-dinosaurs-a5c053f221.html) on October 23, 2018. Retrieved August 3, 2014.
- Bostwick, Kimberly S (2003). "Bird origins and evolution: data accumulates, scientists integrate, and yet the "debate" still rages". *Cladistics*. **19** (4): 369–371. doi:10.1016/S0748-3007(03)00069-0 (https://doi.org/10.1016%2FS0748-3007%2803%2900069-0) (inactive 22 August 2025).
- Dingus, Lowell & Rowe, Timothy (1997): *The Mistaken Extinction: Dinosaur Evolution and the Origin of Birds.* W. H. Freeman and Company, New York. ISBN 0-7167-2944-X
- Dinosauria On-Line (http://www.dinosauria.com)(1995): . Retrieved 2006-09-30.
- Dinosauria On-Line (http://www.dinosauria.com) (1996): Dinosaurian Synapomorphies Found In *Archaeopteryx* (https://web.archive.org/web/20060812210343/http://www.dinosauria.com/jdp/archie/dinoarch.htm). Retrieved 2006-09-30.

- Heilmann, G. (1926): *The Origin of Birds*. Witherby, London. ISBN 0-486-22784-7 (1972 Dover reprint)
- Mayr, Gerald; Pohl, B.; Peters, D. S. (2005). "A Well-Preserved *Archaeopteryx* Specimen with Theropod Features" (http://doc.rero.ch/record/15488/files/PAL_E2876.pdf) (PDF). *Science*. **310** (5753): 1483–1486. Bibcode:2005Sci...310.1483M (https://ui.adsabs.harvard.edu/abs/2005Sci...310.1483M). doi:10.1126/science.1120331 (https://doi.org/10.1126%2Fscience.1120331). PMID 16322455 (https://pubmed.ncbi.nlm.nih.gov/16322455). S2CID 28611454 (https://api.semanticscholar.org/CorpusID:28611454).
- Olson, Storrs L. (1985): The fossil record of birds. *In:* Farner, D.S.; King, J.R. & Parkes, Kenneth C. (eds.): *Avian Biology* **8**: 79-238. Academic Press, New York.

External links

- 'Dinosaurs Among Us' Retraces an Evolutionary Path from Dinosaurs to Birds (https://www.nytimes.com/2016/03/29/science/dinosaurs-birds-evolution-american-museum-of-natural-history.html), NY Times, March 28, 2016
- DinoBuzz (http://www.ucmp.berkeley.edu/diapsids/avians.html) A popular-level discussion of the dinosaur-bird hypothesis
- *Archaeopteryx* - FAQs (http://www.talkorigins.org/faqs/archaeopteryx.html) from the Usenet newsgroup talk.origins.
- Dinosaurs among us (http://www.amnh.org/exhibitions/dinosaurs-among-us) Article and Video (https://web.archive.org/web/20160406211758/http://news.nationalgeographic.com/2016/04/160405-dinosaurs-feathers-birds-museum-new-york-science/) American Museum of Natural History exhibit of dinosaur evolution leading to birds

Retrieved from "https://en.wikipedia.org/w/index.php?title=Origin_of_birds&oldid=1378548679"

6

Toplu Yok Oluşlar

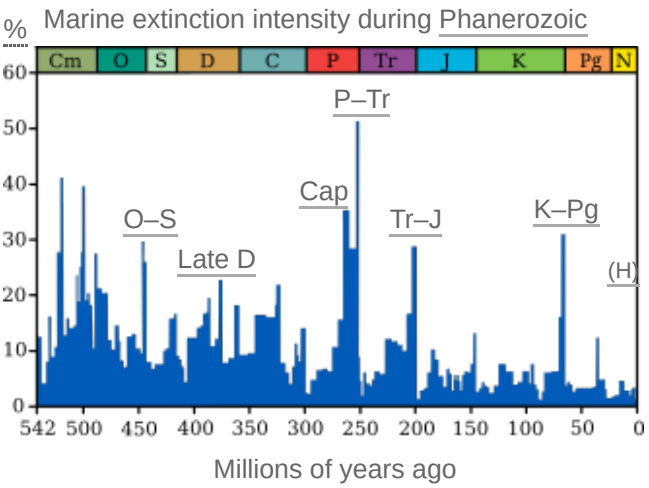
Triyas–Jura ve Kretase–Paleojen yok oluşları ile nedenleri

- Triassic–Jurassic extinction
- Cretaceous–Paleogene extinction event
- Chicxulub crater
- Deccan Traps

WIKIPEDIA
25 years of the free encyclopedia

Triassic–Jurassic extinction

The **Triassic–Jurassic** (**Tr-J**) **extinction event**, often called the **Triassic–Jurassic mass extinction** (**TJME**) or **end-Triassic extinction**, marks the boundary between the Triassic and Jurassic periods, 201.4 million years ago. It represents one of five major extinction events during the Phanerozoic, profoundly affecting life on land and in the oceans.



The blue graph shows the apparent *percentage* (not the absolute number) of marine animal genera becoming extinct during any given time interval. It does not represent all marine species, just those that are readily fossilized. The labels of the traditional "Big Five" extinction events and the more recently recognised Capitanian mass extinction event are clickable links; see Extinction event for more details. *(source and image info)*

In the seas, about 23–34% of marine genera disappeared; corals, bivalves, brachiopods, bryozoans, and radiolarians suffered severe losses of diversity and conodonts were completely wiped out, while marine vertebrates, gastropods, and benthic foraminifera were relatively unaffected. On land, all archosauromorph reptiles other than crocodylomorphs, dinosaurs, and pterosaurs became extinct. Crocodylomorphs, dinosaurs, pterosaurs, and mammals were left largely untouched, allowing them to become the dominant land animals for the next 135 million years. Plants were likewise significantly affected by the crisis, with floral communities undergoing radical ecological restructuring across the extinction event.

The cause of the TJME is generally considered to have been extensive volcanic eruptions in the Central Atlantic magmatic province (CAMP), a large igneous province whose emplacement released large amounts of carbon dioxide into the Earth's atmosphere, causing profound global warming and ocean acidification, and discharged immense quantities of toxic mercury into the environment. Older hypotheses have proposed that gradual changes in climate and sea levels may have been the cause, or perhaps one or more asteroid strikes.

Research history

The earliest research on the TJME was conducted in the mid-20th century, when events in earth history were widely assumed to have been gradual, a paradigm known as uniformitarianism, while comparatively rapid cataclysms as causes of extinction events were dismissed as catastrophism, which had been associated with biblical creationism. Consequently, most researchers believed gradual environmental changes were the best explanation of the extinction; prominent vertebrate palaeontologist Edwin H. Colbert suggested gradual changes in the seasonality of rainfall and eustatic sea level rise that decreased the available land area above sea level were the culprit.[1] In the 1980s, Jack Sepkoski identified the Triassic-Jurassic boundary drop in biodiversity as one of the "Big 5" mass extinction events.[2] After the

discovery that the Cretaceous-Paleogene extinction event was caused by an impact event, the TJME had also been suggested to have been caused by such an impact in the 1980s and 1990s.[3][4] The theory that the TJME was caused by massive volcanism in the Central Atlantic magmatic province (CAMP) first emerged in the 1990s after similar research examining the Permian-Triassic extinction event found it to have been caused by volcanic activity and the emplacement of the CAMP was found to have occurred around the time of the Triassic-Jurassic transition.[5] Despite some early objections,[6] this paradigm remains the scientific consensus in the present day.[7]

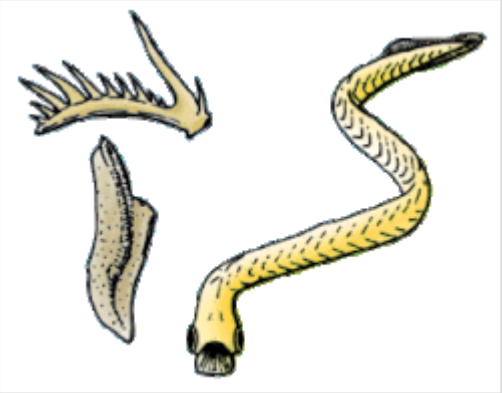
Effects

Marine invertebrates

The Triassic–Jurassic extinction completed the transition from the Palaeozoic evolutionary fauna to the Modern evolutionary fauna that continues to dominate the oceans in the present,[8] a change that began in the aftermath of the end-Guadalupian extinction[9] and continued following the Permian-Triassic extinction event (PTME).[10] Between 23% and 34.1% of marine genera went extinct.[2][11] Plankton diversity dropped suddenly,[12] but it was relatively mildly impacted at the Triassic-Jurassic boundary, although extinction rates among radiolarians rose significantly.[13] Early Hettangian radiolarian communities became depauperate as a result of the TJME and consisted mainly of spumellarians and entactiniids.[14] Benthic foraminifera suffered relatively minor losses of diversity.[15] Some opportunistic foraminifera such as *Triasina hantkeni* increased in abundance as they thrived in oxygen-depleted waters.[16] Ammonites were affected substantially by the Triassic–Jurassic extinction and were nearly wiped out.[17] Ceratitidans, the most prominent group of ammonites in the Triassic, became extinct at the end of the Rhaetian after having their diversity reduced significantly in the Norian, while other ammonite groups such as the Ammonitina, Lytoceratina, and Phylloceratina diversified from the Early Jurassic onward.[18] Bivalves suffered heavy losses, although the extinction was highly selective, with some bivalve clades escaping substantial diversity losses.[19] The Lilliput effect, a term coined to describe a phenomenon wherein organisms shrink in size following a mass extinction, affected megalodontid bivalves,[20] whereas file shell bivalves experienced the Brobdingnag effect, the reverse of the Lilliput effect.[21] There is some evidence of a bivalve cosmopolitanism event during the mass extinction.[22] Additionally, following the TJME, mobile bivalve taxa outnumbered stationary bivalve taxa.[23] Gastropod diversity was barely affected at the Triassic-Jurassic boundary, although gastropods gradually suffered numerous losses over the late Norian and Rhaetian, during the leadup to the TJME.[24] Brachiopods declined in diversity at the end of the Triassic before rediversifying in the Sinemurian and Pliensbachian;[25] the dielasmatoid, athyridoid, and spondylospiroid brachiopods experienced particularly severe declines.[26] Bryozoans, particularly taxa that lived in offshore settings, had already been in decline since the Norian and suffered further losses in the TJME.[27] Ostracods also suffered significant losses,[28] although opportunistic ostracod forms thrived in the eutrophic conditions of the TJME.[29] Conulariids seemingly completely died out at the end of the Triassic.[18] Around 96% of coral genera died out, with integrated corals being especially devastated.[30] Corals practically disappeared from the Tethys Ocean at the end of the Triassic except for its northernmost reaches,[31] resulting in an early Hettangian "coral gap".[32] There is good evidence for a collapse in the reef community, which was likely driven by ocean acidification resulting from CO_2 supplied to the atmosphere by the CAMP eruptions.[33][34]

Most evidence points to a relatively fast recovery from the mass extinction. Benthic ecosystems recovered far more rapidly after the TJME than they did after the PTME.[35] British Early Jurassic benthic marine environments display a relatively rapid recovery that began almost immediately after the end of the mass extinction despite numerous relapses into anoxic conditions during the earliest Jurassic.[36] In the Neuquén Basin, recovery began in the late early Hettangian and lasted until a new biodiversity equilibrium in the late Hettangian.[37] Also despite recurrent anoxic episodes, large bivalves began to reappear shortly after the extinction event.[38] Siliceous sponges dominated the immediate aftermath interval thanks to the enormous influx of silica into the oceans,[39] a consequence of the aerial extent of the CAMP basalts that were exposed to surficial weathering processes.[40][41] In some regions, recovery was slow; in the northern Tethys, carbonate platforms in the TJME's aftermath became dominated by microbial carbonate producers and r-selected calcitic taxa such as *Thaumatoporella parvovesiculifera*, while dasycladacean algae did not reappear until the Sinemurian stage.[42]

Marine vertebrates



Conodonts were a major vertebrate group which died out at the end of the Triassic

Fish did not suffer a mass extinction at the end of the Triassic. The Late Triassic in general did experience a gradual drop in actinopterygiian diversity after an evolutionary explosion in the Middle Triassic. Though this may have been due to falling sea levels or the Carnian Pluvial Event, it may instead be a result of sampling bias considering that Middle Triassic fish have been more extensively studied than Late Triassic fish.[43] Despite the apparent drop in diversity, neopterygiians (which include most modern bony fish) suffered less than more "primitive" actinopterygiians, indicating a biological turnover where modern groups of fish started to supplant earlier groups.[18] Pycnodontiform fish were insignificantly affected.[44] Conodonts, which were prominent index fossils throughout the Paleozoic and Triassic, finally became extinct at the T-J boundary following declining diversity.[18]

Like fish, marine reptiles experienced a substantial drop in diversity between the Middle Triassic and the Jurassic. However, their extinction rate at the Triassic–Jurassic boundary was not elevated. The highest extinction rates experienced by Mesozoic marine reptiles actually occurred at the end of the Ladinian stage, which corresponds to the end of the Middle Triassic. The only marine reptile families which became extinct at or slightly before the Triassic–Jurassic boundary were the placochelyids (the last family of placodonts), making plesiosaurs the only surviving sauropterygians,[45] and giant ichthyosaurs such as shastasaurids.[46] Some authors have argued that the end of the Triassic acted as a genetic "bottleneck" for ichthyosaurs, which never regained the level of anatomical diversity and disparity which they possessed during the Triassic,[47] although analysis of ichthyosaurian and eosauropterygian disparity across the Triassic-Jurassic transition has shown no evidence for such a bottleneck.[48] The high diversity of rhomaelosaurids immediately after the TJME points to a gradual extinction of marine reptiles rather than an abrupt one.[49]

Terrestrial animals

Capitosaurs (such as this *Mastodonsaurus*) were among the major amphibian groups which became extinct at the T–J boundary, though many may have died out earlier.

Terrestrial fauna was affected by the TJME much more severely than marine fauna.[50] One of the earliest pieces of evidence for a Late Triassic extinction was a major turnover in terrestrial tetrapods such as amphibians, reptiles, and synapsids. Edwin H. Colbert drew parallels between the system of extinction and adaptation between the Triassic–Jurassic and Cretaceous–Paleogene boundaries. He recognized how dinosaurs, lepidosaurs (lizards and their relatives), and crocodyliforms (crocodilians and their relatives) filled the niches of more ancient groups of amphibians and reptiles which were extinct by the start of the Jurassic.[1] Olsen (1987) estimated that 42% of all terrestrial tetrapods became extinct at the end of the Triassic, based on his studies of faunal changes in the Newark Supergroup of eastern North America.[3] In contrast to the end-Cretaceous extinction, the TJME substantially affected freshwater ecosystems, and it further differed from the former in that body size did not affect extinction risk.[51] More modern studies have debated whether the turnover in Triassic tetrapods was abrupt at the end of the Triassic, or instead more gradual.[18]

During the Triassic, amphibians were mainly represented by large, crocodile-like members of the order Temnospondyli. Although the earliest lissamphibians (modern amphibians like frogs and salamanders) did appear during the Triassic, they would become more common in the Jurassic while the temnospondyls diminished in diversity past the Triassic–Jurassic boundary.[3] Although the decline of temnospondyls did send shockwaves through freshwater ecosystems, it was probably not as abrupt as some authors have suggested. Brachyopoids, for example, survived until the Cretaceous according to new discoveries in the 1990s. Several temnospondyl groups did become extinct near the end of the Triassic despite earlier abundance, but it is uncertain how close their extinctions were to the end of the Triassic. The last known metoposaurids ("*Apachesaurus*") were from the Redonda Formation, which may have been early Rhaetian or late Norian. *Gerrothorax*, the last known plagiosaurid, has been found in rocks which are probably (but not certainly) Rhaetian, while a capitosaur humerus was found in Rhaetian-age deposits in 2018. Therefore, plagiosaurids and capitosaurs were likely victims of an extinction at the very end of the Triassic, while most other temnospondyls were already extinct.[52]

Terrestrial reptile faunas were dominated by archosauromorphs during the Triassic, particularly phytosaurs and members of Pseudosuchia (the reptile lineage which leads to modern crocodilians). In the Early Jurassic and onwards, dinosaurs and pterosaurs became the most common land reptiles, while small reptiles were mostly represented by lepidosauromorphs (such as lizards and tuatara relatives). Among pseudosuchians, only small crocodylomorphs did not become extinct by the end of the Triassic, with both dominant herbivorous subgroups (such as aetosaurs) and carnivorous ones (rauisuchids) having died out.[3] Phytosaurs, drepanosaurs, trilophosaurids, tanystropheids, and procolophonids, which were other common reptiles in the Late Triassic, had also become extinct by the start of the Jurassic. However, pinpointing the extinction of these different land reptile groups is difficult, as the last stage of the Triassic, the Rhaetian, and the first stage of the Jurassic, the Hettangian, each have few records of large land

Reptile extinction at the end of the Triassic is poorly understood, but phytosaurs (such as this *Redondasaurus*) went from abundant to extinct by the end of the Rhaetian.

animals; some paleontologists have considered only phytosaurs and procolophonids to have become extinct at the Triassic–Jurassic boundary, with other groups having become extinct earlier.[18] However, it is likely that many other groups survived up until the boundary according to British fissure deposits from the Rhaetian. Aetosaurs, kuehneosaurids, drepanosaurs, thecodontosaurids, "saltoposuchids" (like *Terrestrisuchus*), trilophosaurids, and various non-crocodylomorph pseudosuchians are all examples of Rhaetian reptiles which may have become extinct at the Triassic–Jurassic boundary.[53][54][55]

In the TJME's aftermath, dinosaurs experienced a major radiation, filling some of the niches vacated by the victims of the extinction.[56][57] Crocodylomorphs likewise underwent a very rapid and major adaptive radiation,[58] with surviving pseudosuchians being mostly ectothermic as the endothermic pseudosuchians died out during the TJME.[59] Surviving non-mammalian synapsid clades similarly played a role in the post-TJME adaptive radiation during the Early Jurassic.[57]

Herbivorous insects were minimally affected by the TJME; evidence from the Sichuan Basin shows they were overall able to quickly adapt to the floristic turnover by exploiting newly abundant plants.[60] Odonates suffered highly selective losses, and their morphospace was heavily restructured as a result.[61]

Terrestrial plants

The extinction event marks a floral turnover as well, with estimates of the percentage of Rhaetian pre-extinction plants being lost ranging from 17% to 73%.[62] Though spore turnovers are observed across the Triassic-Jurassic boundary, the abruptness of this transition and the relative abundances of given spore types both before and after the boundary are highly variable from one region to another, pointing to a global ecological restructuring rather than a mass extinction of plants.[63] Overall, plants suffered minor diversity losses on a global scale as a result of the extinction, but species turnover rates were high and substantial changes occurred in terms of relative abundance and growth distribution among taxa.[64] Evidence from Central Europe suggests that rather than a sharp, very rapid decline followed by an adaptive radiation, a more gradual turnover in both fossil plants and spores with several intermediate stages is observed over the course of the extinction event.[65] Extinction of plant species can in part be explained by the suspected increased carbon dioxide in the atmosphere as a result of CAMP volcanic activity, which would have created photoinhibition and decreased transpiration levels among species with low photosynthetic plasticity, such as the broad leaved Ginkgoales which declined to near extinction across the Tr–J boundary.[66]

Ferns and other species with dissected leaves displayed greater adaptability to atmosphere conditions of the extinction event,[67] and in some instances were able to proliferate across the boundary and into the Jurassic.[66] The species *Lepidopteris ottonis* evolved a reliance on asexual reproduction amidst the environmental chaos of the TJME.[68] In the Jiyuan Basin of North China, *Classopolis* content increased drastically in concordance with warming, drying, wildfire activity, enrichments in isotopically light carbon, and an overall reduction in floral diversity.[69] In the Sichuan Basin, relatively cool mixed forests

in the late Rhaetian were replaced by hot, arid fernlands during the Triassic–Jurassic transition, which in turn later gave way to a cheirolepid-dominated flora in the Hettangian and Sinemurian.[70] The abundance of ferns in China that were resistant to high levels of aridity increased significantly across the Triassic–Jurassic boundary, though ferns better adapted for moist, humid environments declined, indicating that plants experienced major environmental stress, albeit not an outright mass extinction.[71] In some regions, however, major floral extinctions did occur, with some researchers challenging the hypothesis of there being no significant floral mass extinction on this basis. Evidence from Jameson Land in East Greenland indicates that more than half of plant genera perished in the TJME.[72] In the Newark Supergroup of the United States East Coast, about 60% of the diverse monosaccate and bisaccate pollen assemblages disappear at the Tr–J boundary, indicating a major extinction of plant genera. Early Jurassic pollen assemblages are dominated by *Corollina*, a new genus that took advantage of the empty niches left by the extinction.[73] The site of St. Audrie's Bay displays a shift from diverse gymnosperm-dominated forests to Cheirolepidiaceae-dominated monocultures.[74] The Danish Basin saw 34% of its Rhaetian spore-pollen assemblage, including *Cingulizonates rhaeticus*, *Limbosporites lundbladiae*, *Polypodiisporites polymicroforatus*, and *Ricciisporites tuberculatus*, disappear, with the post-extinction plant community being dominated by pinacean conifers such as *Pinuspollenites minimus* and tree ferns such as *Deltoidospora*, with ginkgos, cycads, cypresses, and corystospermous seed ferns also represented.[75] Swedish assemblages show a low diversity flora consisting mainly of cheirolepids and gingkoaleans was present immediately after the TJME.[76] Along the margins of the European Epicontinental Sea and the European shores of the Tethys, coastal and near-coastal mires fell victim to an abrupt sea level rise. These mires were replaced by a pioneering opportunistic flora after an abrupt sea level fall, although its heyday was short lived and it died out shortly after its rise.[77] The opportunists that established themselves along the Tethyan coastline were primarily spore-producers.[74] In the Eiberg Basin of the Northern Calcareous Alps, there was a very rapid palynomorph turnover.[78] The palynological and palaeobotanical succession in Queensland shows a *Classopolis* bloom after the TJME.[79] Polyploidy may have been an important factor that mitigated a conifer species' risk of going extinct.[80]

Possible causes

Central Atlantic Magmatic Province

The leading and best evidenced explanation for the TJME is massive volcanic eruptions, specifically from the Central Atlantic Magmatic Province (CAMP),[81][82][83] the largest known large igneous province by area, and one of the most voluminous,[84][85] with its flood basalts extending across parts of southwestern Europe,[86][87] northwestern Africa,[88] northeastern South America,[89][90] and southeastern North America.[91][85][92] The coincidence and synchrony of CAMP activity and the TJME is indicated by uranium–lead dating,[93][7] argon-argon dating,[91][87] and palaeomagnetism.[94][95][5] The isotopic composition of fossil soils and marine sediments near the boundary between the Late Triassic and Early Jurassic has been tied to a large negative $\delta^{13}C$ excursion,[96][97][98] with values as low as -2.8%.[99] Carbon isotopes of hydrocarbons (*n*-alkanes) derived from leaf wax and lignin, and total organic carbon from two sections of lake sediments interbedded with the CAMP in eastern North America have shown carbon isotope excursions similar to those found in the mostly marine St. Audrie's Bay section, Somerset, England; the correlation suggests that the TJME began at the same time in marine and terrestrial environments, slightly before the oldest basalts in eastern North America but simultaneous with the

Maximum extent of CAMP volcanism at the Triassic-Jurassic boundary

eruption of the oldest flows in Morocco, with both a critical CO_2 greenhouse and a marine biocalcification crisis.[100] Furthermore, chemostratigraphic analysis in the Junggar Basin has shown that the negative $\delta^{13}C$ excursions associated with CAMP volcanism corresponded in time to biotic turnovers in the palynomorph record, strongly suggesting a causal relationship between the two.[101] Contemporaneous CAMP eruptions, mass extinction, and the carbon isotopic excursions are shown in the same places, making the case for a volcanic cause of a mass extinction.[102][103][104] The observed negative carbon isotope excursion is lower in some sites that correspond to what was then eastern Panthalassa because of the extreme aridity of western Pangaea limiting weathering and erosion there.[105] The negative $\delta^{13}C$ excursion associated with CAMP volcanism lasted for approximately 20,000 to 40,000 years, or about one or two of Earth's axial precession cycles,[106] although the carbon cycle was so disrupted that it did not stabilise until the Sinemurian.[107] Mercury anomalies from deposits in various parts of the world have further bolstered the volcanic cause hypothesis,[108][109] as have anomalies from various platinum-group elements.[110] Nickel enrichments are also observed at the Triassic-Jurassic boundary coevally with light carbon enrichments, providing yet more evidence of massive volcanism.[111]

Some scientists initially rejected the volcanic eruption theory because the Newark Supergroup, a section of rock in eastern North America that records the Triassic–Jurassic boundary, contains no ash-fall horizons and because its oldest basalt flows were estimated to lie around 10 m above the transition zone,[112] which they estimated to have occurred 610 kyr after the TJME.[113] Also among their objections was that the Triassic-Jurassic boundary was poorly defined and the CAMP eruptions poorly constrained temporally.[114] However, updated dating protocol and wider sampling has confirmed that the CAMP eruptions started in Morocco only a few thousand years before the extinction,[7] preceding their onset in Nova Scotia and New Jersey,[115][116][117] and that they continued in several more pulses for the next 600,000 years.[7] Volcanic global warming has also been criticised as an explanation because some estimates have found that the amount of carbon dioxide emitted was only around 250 ppm, not enough to generate a mass extinction.[6] In addition, at some sites, changes in carbon isotope ratios have been attributed to diagenesis and not any primary environmental changes.[118]

Global warming

The flood basalts of the CAMP released gigantic quantities of carbon dioxide,[119] a potent greenhouse gas causing intense global warming.[120] Before the TJME, carbon dioxide levels were around 1,000 ppm as measured by the stomatal index of *Lepidopteris ottonis*, but this quantity jumped to 1,300 ppm at the onset of the extinction event.[121] During the TJME, carbon dioxide concentrations increased fourfold.[122] The record of CAMP degassing shows several distinct pulses of carbon dioxide immediately following each major pulse of magmatism, at least two of which amount to a doubling of atmospheric CO_2.[123] Carbon dioxide was emitted quickly and in enormous quantities compared to other periods of Earth's history, rate of carbon dioxide emissions was one of the most meteoric rises in carbon dioxide levels in Earth's entire history.[124] It is estimated that a single volcanic pulse from the large igneous province would have emitted an amount of carbon dioxide roughly equivalent to projected anthropogenic carbon dioxide emissions for the 21st century.[125] In addition, the flood basalts intruded

through sediments that were rich in organic matter and combusted it,[126][127][128] as evidenced by low $\Delta^{199}Hg$ values showing elevated levels of organic matter-derived mercury in the environment.[129] The degassing of volatiles resulting from volcanic intrusions into organic-rich sediments further enhanced the volcanic warming of the climate.[130][131] Thermogenic carbon release through such contact metamorphism of carbon-rich deposits has been found to be a sensible hypothesis providing a coherent explanation for the magnitude of the negative carbon isotope excursions at the terminus of the Triassic.[132] Global temperatures rose sharply by 3 to 4 °C.[133] In some regions, the temperature rise was as great as 10 °C.[134] Kaolinite-dominated clay mineral spectra reflect the extremely hot and humid greenhouse conditions engendered by the CAMP.[135] Soil erosion occurred as the hydrological cycle was accelerated by the extreme global heat.[136]

The catastrophic dissociation of gas hydrates as a positive feedback resulting from warming, which has been suggested as one possible cause of the PTME, the largest mass extinction of all time,[137] may have exacerbated greenhouse conditions,[138][139] although others suggest that methane hydrate release was temporally mismatched with the TJME and thus not a cause of it.[140][141]

Global cooling

Besides the carbon dioxide-driven long-term global warming, CAMP volcanism had shorter term cooling effects resulting from the emission of sulphur dioxide aerosols.[142][143][7] The extremely voluminous emission of this gas caused sharp rises in Earth's albedo and induced severe volcanic winters.[144] High latitudes had colder climates with evidence of mild glaciation during the extinction interval. Cold periods induced by volcanic ejecta clouding the atmosphere might have favoured endothermic animals, with dinosaurs, pterosaurs, and mammals being more capable at enduring these conditions than large pseudosuchians due to insulation.[145]

Metal poisoning

CAMP volcanism released enormous amounts of toxic mercury.[146][147] The appearance of high rates of mutagenesis of varying severity in fossil spores during the TJME coincides with mercury anomalies and is thus believed by researchers to have been caused by mercury poisoning.[148] $\delta^{202}Hg$ and $\Delta^{199}Hg$ evidence suggests that volcanism caused the mercury loading directly at the Triassic-Jurassic boundary, but that there were later bouts of elevated mercury in the environment during the Early Jurassic caused by eccentricity-forced enhancement of hydrological cycling and erosion that resulted in remobilisation of volcanically injected mercury that had been deposited in wetlands.[149]

Wildfires

The intense, rapid warming is believed to have resulted in increased storminess and lightning activity as a consequence of the more humid climate. The uptick in lightning activity is in turn implicated as a cause of an increase in wildfire activity.[150] Coronene enrichments show an increase in wildfire activity closely temporally connected to biodiversity loss and pulses of acid rain associated with the TJME.[151] Europe became plagued by continental scale wildfires blazing across its fern savannas.[152] The combined presence of charcoal fragments and heightened levels of pyrolytic polycyclic aromatic hydrocarbons in Polish sedimentary facies straddling the Triassic-Jurassic boundary indicates wildfires were extremely commonplace during the earliest Jurassic, immediately after the Triassic-Jurassic transition.[153] Elevated wildfire activity is also known from the Junggar Basin.[154] In the Jiyuan Basin, two distinct pulses of drastically elevated wildfire activity are known: the first mainly affected canopies and occurred amidst

relatively humid conditions while the second predominantly affected ground cover and was associated with aridity.[155] At the Winterswijk quarry in the Netherlands, a surge in wildfire activity has been suggested to correspond to and have caused the sudden decline in coniferous vegetation.[156] Frequent wildfires, combined with increased seismic activity from CAMP emplacement, led to apocalyptic soil degradation.[157]

Anoxia and euxinia

Anoxia was another mechanism of extinction; the end-Triassic extinction was coeval with an uptick in black shale deposition and a pronounced negative $\delta^{238}U$ excursion, indicating a major decrease in marine oxygen availability.[158] Additional evidence for anoxia during the TJME comes from pyrite framboids, which grow in anoxic conditions.[159] Evidence of anoxia has been discovered at the Triassic-Jurassic boundary across the world's oceans; the western Tethys, eastern Tethys, and Panthalassa were all affected by a precipitous drop in seawater oxygen,[160] although at a few sites, the TJME was associated with fully oxygenated waters.[161] Positive $\delta^{15}N$ excursions have also been interpreted as evidence of anoxia concomitant with increased denitrification in marine sediments in the TJME's aftermath.[162]

In northeastern Panthalassa, episodes of anoxia were already occurring during the Rhaetian before the TJME, making its marine ecosystems unstable even before the main crisis began.[163][164][165] This early phase of environmental degradation in eastern Panthalassa may have been caused by an early phase of CAMP activity.[166] Anoxic, reducing conditions were likewise present in western Panthalassa off the coast of what is now Japan for about a million years prior to the TJME.[167] During the TJME, the rapid warming led to the stagnation of ocean circulation in many ocean regions, enabling the development of catastrophic anoxia; in what is now northwestern Europe, shallow seas became salinity stratified, enabling easy development of anoxia.[168] Another factor contributing to anoxia was the increase in continental weathering driven by intense warming that delivered vast quantities of nutrients to the ocean surface and engendered eutrophication;[169][170] this uptick in weathering is evidenced by positive $\delta^{56}Fe$ excursions.[171] A combination of negative $\delta^{66}Zn$ excursions, positive $\delta^{26}Mg$ excursions, and a lack of significant change in $\delta^{65}Cu$ provides further evidence of increased chemical weathering resulting from increased temperature and humidity on land at high latitudes.[172] Increased influx of terrestrial organic matter, in conjunction with reduced salinity, has been directly shown to have enkindled anoxia in the Eiberg Basin.[173] Persistent low $\delta^{238}U$ ratios indicate prolonged global oxygen depletion continued into the Hettangian,[174] with $^{87}Sr/^{86}Sr$ values showing that high influxes of terrestrial nutrients likely continued to eutrophicate the oceans well after the Triassic-Jurassic boundary.[175] The persistence of anoxia into the Hettangian age may have helped delay the recovery of marine life in the extinction's aftermath.[158][176][177]

Euxinia, a form of anoxia defined by not just the absence of dissolved oxygen but high concentrations of hydrogen sulphide, also developed in the oceans, as indicated by findings of increased isorenieratane. The increase in concentration of this substance reveals that populations of green sulphur bacteria, which photosynthesise using hydrogen sulphide instead of water, grew significantly across the Triassic-Jurassic boundary.[178][179] A meteoric shift towards positive sulphur isotope ratios in reduced sulphur species indicates a complete utilisation of sulphate by sulphate reducing bacteria.[180] Off the shores of the Wrangellia Terrane, the onset of photic zone euxinia was preceded by an interval of limited nitrogen availability and increased nitrogen fixation in surface waters while euxinia developed in bottom waters.[181] Recurrent hydrogen sulphide poisoning following the TJME had retarding effects on biotic rediversification.[182][178]

Ocean acidification

Oceanic uptake of volcanogenic carbon and sulphur dioxide would have led to a significant decrease of seawater pH known as ocean acidification, which is discussed as a relevant driver of marine extinction,[183][34][184] acting in conjunction with marine anoxia.[185] Additionally, acidification was enhanced and exacerbated by widespread photic zone euxinia, which caused increased rates of organic matter respiration and carbon dioxide release.[186] Direct evidence of ocean acidification in the former of $\delta^{11}B$ ratios in fossil oysters is known from the TJME interval.[187] Further evidence for ocean acidification as an extinction mechanism comes from the preferential extinction of marine organisms with thick aragonitic skeletons and little biotic control of biocalcification (e.g., corals, hypercalcifying sponges),[188][189] which resulted in a coral reef collapse[33][34] and an early Hettangian "coral gap".[32] The decline of megalodontoid bivalves is also attributed to increased seawater acidity.[190] Extensive fossil remains of malformed calcareous nannoplankton, a common sign of significant drops in pH, have also been extensively reported from the Triassic-Jurassic boundary.[168] Global interruption of carbonate deposition at the Triassic-Jurassic boundary has been cited as additional evidence for catastrophic ocean acidification.[191][183] Upwardly developing aragonite fans in the shallow subseafloor may also reflect decreased pH, these structures being speculated to have precipitated concomitantly with acidification.[192] In some studied sections, the TJME biocalcification crisis is masked by emersion of carbonate platforms induced by marine regression.[193]

Ozone depletion

Research on the role of ozone shield deterioration during the Permian-Triassic mass extinction has suggested that it may have been a factor in the TJME as well.[194][195] A spike in the abundance of unseparated tetrads of *Kraeuselisporites reissingerii* has been interpreted as evidence of increased ultraviolet radiation flux resulting from ozone layer damage caused by volcanic aerosols.[196]

Gradual climate change

The extinctions at the end of the Triassic were initially attributed to gradually changing environments. Within his 1958 study recognizing biological turnover between the Triassic and Jurassic, Edwin H. Colbert's proposal was that this extinction was a result of geological processes decreasing the diversity of land biomes. He considered the Triassic period to be an era of the world experiencing a variety of environments, from towering highlands to arid deserts to tropical marshes. In contrast, the Jurassic period was much more uniform both in climate and elevation due to excursions by shallow seas.[1]

Later studies noted a clear trend towards increased aridification towards the end of the Triassic. Although high-latitude areas like Greenland and Australia actually became wetter, most of the world experienced more drastic changes in climate as indicated by geological evidence. This evidence includes an increase in carbonate and evaporite deposits (which are most abundant in dry climates) and a decrease in coal deposits (which primarily form in humid environments such as coal forests).[18] In addition, the climate may have become much more seasonal, with long droughts interrupted by severe monsoons.[197] The world gradually got warmer over this time as well; from the late Norian to the Rhaetian, mean annual temperatures rose by 7 to 9 °C.[198] The site of Hochalm in Austria preserves evidence of carbon cycle perturbations during the Rhaetian preceding the Triassic-Jurassic boundary, potentially having a role in the ecological crisis.[199]

Sea level fall

Geological formations in Europe and the Middle East seem to indicate a drop in sea levels at the end of the Triassic associated with the TJME.[200][201] Although falling sea levels have sometimes been considered a culprit for marine extinctions, evidence is inconclusive since many sea level drops in geological history are not correlated with increased extinctions. However, there is still some evidence that marine life was affected by secondary processes related to falling sea levels, such as decreased oxygenation (caused by sluggish circulation), or increased acidification. These processes do not seem to have been worldwide, with the sea level fall observed in European sediments believed to be not global but regional,[202] and with even some European sections showing no sign of sea level fall across the Triassic-Jurassic boundary,[203] but they may explain local extinctions in European marine fauna.[18] However, it is not universally accepted that even this local diversity drop was caused by sea level fall.[204] A pronounced sea level change in latest Triassic records from Lake Williston in northeastern British Columbia, which was then the northeastern margin of Panthalassa, resulted in an extinction event of infaunal (sediment-dwelling) bivalves, though not epifaunal ones.[205]

Extraterrestrial impact

Some have hypothesized that an impact from an asteroid or comet caused the Triassic–Jurassic extinction,[11][73] similar to the extraterrestrial object which was the main factor in the Cretaceous–Paleogene extinction about 66 million years ago, as evidenced by the Chicxulub crater in Mexico. However, so far no impact crater of sufficient size has been dated to precisely coincide with the Triassic–Jurassic boundary.[18]

The Manicouagan reservoir in Quebec, a massive crater formed by a Late Triassic impact. Radiometric dating has determined that it is about 13 million years older than the Triassic–Jurassic boundary, and thus an unlikely candidate for a mass extinction.

Nevertheless, the Late Triassic did experience several impacts, including the second-largest confirmed impact in the Mesozoic. The Manicouagan Reservoir in Quebec is one of the most visible large impact craters on Earth, and at 100 km (62 mi) in diameter it is tied with the Eocene Popigai impact structure in Siberia as the fourth-largest impact crater on Earth. Olsen *et al.* (1987) were the first scientists to link the Manicouagan crater to the Triassic–Jurassic extinction, citing its age which at the time was roughly considered to be Late Triassic.[3] More precise radiometric dating by Hodych & Dunning (1992) has shown that the Manicouagan impact occurred about 214 million years ago, about 13 million years before the Triassic–Jurassic boundary. Therefore, it could not have been responsible for an extinction precisely at the Triassic–Jurassic boundary.[206] Nevertheless, the Manicouagan impact did have a widespread effect on the planet; a 214-million-year-old ejecta blanket of shocked quartz has been found in rock layers as far away as England[207] and Japan. There is still a possibility that the Manicouagan impact was responsible for a small extinction midway through the Late Triassic at the Carnian–Norian boundary,[206] although the disputed age of this boundary (and whether an extinction actually occurred in the first place) makes it difficult to correlate the impact with extinction.[207] Onoue *et al.* (2016) alternatively proposed that the Manicouagan impact was responsible

for a marine extinction in the middle of the Norian which affected radiolarians, sponges, conodonts, and Triassic ammonoids. Thus, the Manicouagan impact may have been partially responsible for the gradual decline in the latter two groups which culminated in their extinction at the Triassic–Jurassic boundary.[208] The boundary between the Adamanian and Revueltian land vertebrate faunal zones, which involved extinctions and faunal changes in tetrapods and plants, was possibly also caused by the Manicouagan impact, although discrepancies between magnetochronological and isotopic dating lead to some uncertainty.[209]

Other Triassic craters are closer to the Triassic–Jurassic boundary but also much smaller than the Manicouagan reservoir. The eroded Rochechouart impact structure in France has most recently been dated to 201 ± 2 million years ago,[210] but at 25 km (16 mi) across (possibly up to 50 km (30 mi) across originally), it appears to be too small to have affected the ecosystem,[211] although it has been speculated to have played a role in an alleged much smaller extinction event at the Norian-Rhaetian boundary.[212] The 40 km (25 mi) wide Saint Martin crater in Manitoba has been proposed as a candidate for a possible TJME-causing impact, but it has since been dated to be Carnian.[213] Other putative or confirmed Triassic craters include the 80 km (50 mi) wide Puchezh-Katunki crater in Eastern Russia (though it may be Jurassic in age), the 15 km (9 mi) wide Obolon' crater in Ukraine, and the 9 km (6 mi) wide Red Wing Creek structure in North Dakota. Spray *et al.* (1998) noted an interesting phenomenon, that being how the Manicouagan, Rochechouart, and Saint Martin craters all seem to be at the same latitude, and that the Obolon' and Red Wing craters form parallel arcs with the Rochechouart and Saint Martin craters, respectively. Spray and his colleagues hypothesized that the Triassic experienced a "multiple impact event", a large fragmented asteroid or comet which broke up and impacted the earth in several places at the same time.[4] Such an impact has been observed in the present day, when Comet Shoemaker-Levy 9 broke up and hit Jupiter in 1992. However, the "multiple impact event" hypothesis for Triassic impact craters has not been well-supported; Kent (1998) noted that the Manicouagan and Rochechouart craters were formed in eras of different magnetic polarity,[214] and radiometric dating of the individual craters has shown that the impacts occurred millions of years apart.[18]

Shocked quartz has been found in Rhaetian deposits from the Northern Apennines of Italy, providing possible evidence of an end-Triassic extraterrestrial impact.[215] Certain trace metals indicative of a bolide impact have been found in the late Rhaetian, though not at the Triassic-Jurassic boundary itself; the discoverers of these trace metal anomalies purport that such a bolide impact could only have been an indirect cause of the TJME.[216] The discovery of seismites two to four metres thick coeval with the carbon isotope fluctuations associated with the TJME has been interpreted as evidence of a possible bolide impact, although no definitive link between these seismites and any impact event has been found.[217]

On the other hand, the dissimilarity between the isotopic perturbations characterising the TJME and those characterising the end-Cretaceous mass extinction makes an extraterrestrial impact highly unlikely to have been the cause of the TJME, according to many researchers.[218] Various trace metal ratios, including palladium/iridium, platinum/iridium, and platinum/rhodium, in rocks deposited during the TJME have numerical values very different from what would be expected in an extraterrestrial impact scenario, providing further evidence against this hypothesis.[110] The Triassic-Jurassic boundary furthermore lacks a fern spore spike akin to that observed at the terminus of the Cretaceous, inconsistent with an asteroid impact.[219]

Comparisons to present climate change

The extremely rapid, centuries-long timescale of carbon emissions and global warming caused by pulses of CAMP volcanism has drawn comparisons between the Triassic-Jurassic mass extinction and anthropogenic global warming, currently causing the Holocene extinction.[120] The current rate of carbon dioxide emissions is around 50 gigatonnes per year, hundreds of times faster than during the latest Triassic, although the lack of extremely detailed stratigraphic resolution and pulsed nature of CAMP volcanism means that individual pulses of greenhouse gas emissions likely occurred on comparable timescales to human release of warming gases since the Industrial Revolution.[124] The degassing rate of the first pulse of CAMP volcanism is estimated to have been around half of the rate of modern anthropogenic emissions.[120] Palaeontologists studying the TJME and its impacts warn that a major reduction in humanity's carbon dioxide emissions to slow down climate change is of critical importance for preventing a catastrophe similar to the TJME from befalling the modern biosphere.[124] If human-induced climate change persists as is, predictions can be made as to how various aspects of the biosphere will respond based on records of the TJME. For example, current conditions such the increased carbon dioxide levels, ocean acidification, and ocean deoxygenation create a similar climate to that of the Triassic-Jurassic boundary for marine life, so it is the common assumption that should the trends continue, modern reef-building taxa and skeletal benthic organisms will be preferentially impacted.[220] The end-Triassic reef crisis has been specifically cited as a possible analogue for the fate of present coral reefs should anthropogenic global warming continue.[221]

References

1. Colbert, Edwin H. (15 September 1958). "Tetrapod Extinctions at the End of the Triassic Period" (https://www.pnas.org/content/pnas/44/9/973.full.pdf) (PDF). *Proceedings of the National Academy of Sciences of the United States of America*. **44** (9): 973–977. Bibcode:1958PNAS...44..973C (https://ui.adsabs.harvard.edu/abs/1958PNAS...44..973C). doi:10.1073/pnas.44.9.973 (https://doi.org/10.1073%2Fpnas.44.9.973). ISSN 0027-8424 (https://search.worldcat.org/issn/0027-8424). PMC 528676 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC528676). PMID 16590299 (https://pubmed.ncbi.nlm.nih.gov/16590299).
2. Sepkoski, J. John (1984). "A kinetic model of Phanerozoic taxonomic diversity. III. Post-Paleozoic families and mass extinctions". *Paleobiology*. **10** (2): 246–267. Bibcode:1984Pbio...10..246S (https://ui.adsabs.harvard.edu/abs/1984Pbio...10..246S). doi:10.1017/s0094837300008186 (https://doi.org/10.1017%2Fs0094837300008186). ISSN 0094-8373 (https://search.worldcat.org/issn/0094-8373). S2CID 85595559 (https://api.semanticscholar.org/CorpusID:85595559).
3. Olsen, Paul E.; Shubin, Neil H.; Anders, M. H. (28 August 1987). "New early Jurassic tetrapod assemblages constrain Triassic–Jurassic tetrapod extinction event" (https://www.ldeo.columbia.edu/~polsen/nbcp/olsen.et.al.87.pdf) (PDF). *Science*. **237** (4818): 1025–1029. Bibcode:1987Sci...237.1025O (https://ui.adsabs.harvard.edu/abs/1987Sci...237.1025O). doi:10.1126/science.3616622 (https://doi.org/10.1126%2Fscience.3616622). ISSN 0036-8075 (https://search.worldcat.org/issn/0036-8075). PMID 3616622 (https://pubmed.ncbi.nlm.nih.gov/3616622).

4. Spray, John G.; Kelley, Simon P.; Rowley, David B. (12 March 1998). "Evidence for a late Triassic multiple impact event on Earth" (https://geosci.uchicago.edu/~rowley/Rowley/Publications_files/Nature%201998%20Spray.pdf) (PDF). *Nature*. **392** (6672): 171–173. Bibcode:1998Natur.392..171S (https://ui.adsabs.harvard.edu/abs/1998Natur.392..171S). doi:10.1038/32397 (https://doi.org/10.1038%2F32397). ISSN 1476-4687 (https://search.worldcat.org/issn/1476-4687). S2CID 4413688 (https://api.semanticscholar.org/CorpusID:4413688).
5. Marzoli, Andrea; Renne, Paul R.; Piccirillo, Enzo M.; Ernesto, Marcia; Bellieni, Giuliano; Min, Angelo De (23 April 1999). "Extensive 200-Million-Year-Old Continental Flood Basalts of the Central Atlantic Magmatic Province" (https://www.science.org/doi/10.1126/science.284.5414.616). *Science*. **284** (5414): 616–618. Bibcode:1999Sci...284..616M (https://ui.adsabs.harvard.edu/abs/1999Sci...284..616M). doi:10.1126/science.284.5414.616 (https://doi.org/10.1126%2Fscience.284.5414.616). ISSN 0036-8075 (https://search.worldcat.org/issn/0036-8075). PMID 10213679 (https://pubmed.ncbi.nlm.nih.gov/10213679). Retrieved 31 October 2024.
6. Tanner, L. H.; J. F. Hubert; et al. (7 June 2001). "Stability of atmospheric CO_2 levels across the Triassic/Jurassic boundary". *Nature*. **411** (6838): 675–677. Bibcode:2001Natur.411..675T (https://ui.adsabs.harvard.edu/abs/2001Natur.411..675T). doi:10.1038/35079548 (https://doi.org/10.1038%2F35079548). PMID 11395765 (https://pubmed.ncbi.nlm.nih.gov/11395765). S2CID 4418003 (https://api.semanticscholar.org/CorpusID:4418003).
7. Blackburn, Terrence J.; Olsen, Paul E.; Bowring, Samuel A.; McLean, Noah M.; Kent, Dennis V; Puffer, John; McHone, Greg; Rasbury, Troy; Et-Touhami7, Mohammed (2013). "Zircon U–Pb Geochronology Links the End-Triassic Extinction with the Central Atlantic Magmatic Province" (https://www.personal.kent.edu/~alisonjs/paleo/Blackburn_2013Tr-JExtinctionChronology.pdf) (PDF). *Science*. **340** (6135): 941–945. Bibcode:2013Sci...340..941B (https://ui.adsabs.harvard.edu/abs/2013Sci...340..941B). doi:10.1126/science.1234204 (https://doi.org/10.1126%2Fscience.1234204). PMID 23519213 (https://pubmed.ncbi.nlm.nih.gov/23519213). S2CID 15895416 (https://api.semanticscholar.org/CorpusID:15895416).
8. Schoepfer, Shane D.; Algeo, Thomas J.; Van de Schootbrugge, Bas; Whiteside, Jessica H. (September 2022). "The Triassic–Jurassic transition – A review of environmental change at the dawn of modern life" (https://www.sciencedirect.com/science/article/abs/pii/S0012825222001830). *Earth-Science Reviews*. **232** 104099. Bibcode:2022ESRv..23204099S (https://ui.adsabs.harvard.edu/abs/2022ESRv..23204099S). doi:10.1016/j.earscirev.2022.104099 (https://doi.org/10.1016%2Fj.earscirev.2022.104099). hdl:1874/425545 (https://hdl.handle.net/1874%2F425545). S2CID 250256142 (https://api.semanticscholar.org/CorpusID:250256142). Retrieved 1 February 2023.
9. De la Horra, R.; Galán-Abellán, A. B.; López-Gómez, José; Sheldon, Nathan D.; Barrenechea, J. F.; Luque, F. J.; Arche, A.; Benito, M. I. (August–September 2012). "Paleoecological and paleoenvironmental changes during the continental Middle–Late Permian transition at the SE Iberian Ranges, Spain" (https://www.sciencedirect.com/science/article/abs/pii/S0921818112001221). *Global and Planetary Change*. 94–95: 46–61. Bibcode:2012GPC....94...46D (https://ui.adsabs.harvard.edu/abs/2012GPC....94...46D). doi:10.1016/j.gloplacha.2012.06.008 (https://doi.org/10.1016%2Fj.gloplacha.2012.06.008). hdl:10261/59010 (https://hdl.handle.net/10261%2F59010). Retrieved 15 December 2022.
10. Brayard, Arnaud; Krumenacker, L. J.; Botting, Joseph P.; Jenks, James F.; Bylund, Kevin G.; Fara, Emmanuel; Vennin, Emmanuelle; Olivier, Nicolas; Goudemand, Nicolas; Saucède, Thomas; Charbonnier, Sylvain; Romano, Carlo; Doguzhaeva, Larisa; Thuy, Ben; Hautmann, Michael; Stephen, Daniel A.; Thomazo, Christophe; Escarguel, Gilles (15 February 2017). "Unexpected Early Triassic marine ecosystem and the rise of the Modern evolutionary fauna" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5310825). *Science Advances*. **13** (2) e1602159. Bibcode:2017SciA....3E2159B (https://ui.adsabs.harvard.edu/abs/2017SciA....3E2159B). doi:10.1126/sciadv.1602159 (https://doi.org/10.1126%2Fsciadv.1602159). PMC 5310825 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5310825). PMID 28246643 (https://pubmed.ncbi.nlm.nih.gov/28246643).

11. Ryder, Graham; Fastovsky, David E.; Gartner, Stefan (1996). *The Cretaceous-Tertiary Event and Other Catastrophes in Earth History* (https://books.google.com/books?id=kAup0TOL09gC&pg=PA19). Geological Society of America. p. 19. ISBN 978-0-8137-2307-5.
12. Ward, Peter Douglas; Haggart, J.W.; Carter, E.S.; Wilbur, D.; Tipper, H.W.; Evans, T. (11 May 2001). "Sudden Productivity Collapse Associated with the Triassic-Jurassic Boundary Mass Extinction" (https://www.science.org/doi/10.1126/science.1058574). *Science*. **292** (5519): 1148–1151. Bibcode:2001Sci...292.1148W (https://ui.adsabs.harvard.edu/abs/2001Sci...292.1148W). doi:10.1126/science.1058574 (https://doi.org/10.1126%2Fscience.1058574). PMID 11349146 (https://pubmed.ncbi.nlm.nih.gov/11349146). S2CID 36667702 (https://api.semanticscholar.org/CorpusID:36667702). Retrieved 23 November 2022.
13. Kocsis, Ádám T.; Kiessling, Wolfgang; Pálfy, József (8 April 2016). "Radiolarian biodiversity dynamics through the Triassic and Jurassic: implications for proximate causes of the end-Triassic mass extinction" (https://www.cambridge.org/core/journals/paleobiology/article/abs/radiolarian-biodiversity-dynamics-through-the-triassic-and-jurassic-implications-for-proximate-causes-of-the-endtriassic-mass-extinction/0986175366656DEE63909EF57FCFA87B). *Paleobiology*. **40** (4): 625–639. doi:10.1666/14007 (https://doi.org/10.1666%2F14007). S2CID 129600881 (https://api.semanticscholar.org/CorpusID:129600881). Retrieved 28 May 2023.
14. Longridge, Louise M.; Carter, Elizabeth S.; Smith, Paul L.; Tipper, Howard W. (9 February 2007). "Early Hettangian ammonites and radiolarians from the Queen Charlotte Islands, British Columbia and their bearing on the definition of the Triassic–Jurassic boundary" (https://www.sciencedirect.com/science/article/abs/pii/S0016699512000526). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **244** (1–4): 142–169. Bibcode:2007PPP...244..142L (https://ui.adsabs.harvard.edu/abs/2007PPP...244..142L). doi:10.1016/j.palaeo.2006.06.027 (https://doi.org/10.1016%2Fj.palaeo.2006.06.027). Retrieved 19 February 2025 – via Elsevier Science Direct.
15. Pálfy, József; Demény, Attila; Haas, János; Carter, Elizabeth S.; Görög, Ágnes; Halász, Dóra; Oravecz-Scheffer, Anna; Hetényi, Magdolna; Márton, Emő; Orchard, Michael J.; Ozsvárt, Péter; Vető, István; Zajzon, Norbert (9 February 2007). "Triassic–Jurassic boundary events inferred from integrated stratigraphy of the Csővár section, Hungary" (https://www.sciencedirect.com/science/article/pii/S0031018206004391). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **244** (1–4): 11–33. Bibcode:2007PPP...244...11P (https://ui.adsabs.harvard.edu/abs/2007PPP...244...11P). doi:10.1016/j.palaeo.2006.06.021 (https://doi.org/10.1016%2Fj.palaeo.2006.06.021). Retrieved 19 February 2025 – via Elsevier Science Direct.
16. Ciarapica, Gloria (9 February 2007). "Regional and global changes around the Triassic–Jurassic boundary reflected in the late Norian–Hettangian history of the Apennine basins" (https://www.sciencedirect.com/science/article/pii/S0031018206004408). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **244** (1–4): 34–51. Bibcode:2007PPP...244...34C (https://ui.adsabs.harvard.edu/abs/2007PPP...244...34C). doi:10.1016/j.palaeo.2006.06.022 (https://doi.org/10.1016%2Fj.palaeo.2006.06.022). Retrieved 19 February 2025 – via Elsevier Science Direct.
17. Smith, Paul L.; Longridge, Louise M.; Grey, Melissa; Zhang, Jin; Liang, Bo (4 January 2014). "From near extinction to recovery: Late Triassic to Middle Jurassic ammonoid shell geometry" (https://www.idunn.no/doi/10.1111/let.12058). *Lethaia*. **47** (3): 337–351. Bibcode:2014Letha..47..337S (https://ui.adsabs.harvard.edu/abs/2014Letha..47..337S). doi:10.1111/let.12058 (https://doi.org/10.1111%2Flet.12058). hdl:2429/45186 (https://hdl.handle.net/2429%2F45186). ISSN 0024-1164 (https://search.worldcat.org/issn/0024-1164). Retrieved 28 October 2024.

18. Tanner LH, Lucas SG, Chapman MG (2004). "Assessing the record and causes of Late Triassic extinctions" (https://web.archive.org/web/20071025225841/http://nmnaturalhistory.org/pdf_files/TJB.pdf) (PDF). *Earth-Science Reviews*. **65** (1–2): 103–139. Bibcode:2004ESRv...65..103T (https://ui.adsabs.harvard.edu/abs/2004ESRv...65..103T). doi:10.1016/S0012-8252(03)00082-5 (https://doi.org/10.1016%2FS0012-8252%2803%2900082-5). Archived from the original (http://nmnaturalhistory.org/pdf_files/TJB.pdf) (PDF) on 25 October 2007. Retrieved 22 October 2007.
19. Ros, Sonia; Echevarría, Javier (25 July 2011). "Bivalves and evolutionary resilience: old skills and new strategies to recover from the P/T and T/J extinction events" (http://www.tandfonline.com/doi/abs/10.1080/08912963.2011.578744). *Historical Biology*. **23** (4): 411–429. Bibcode:2011HBio...23..411R (https://ui.adsabs.harvard.edu/abs/2011HBio...23..411R). doi:10.1080/08912963.2011.578744 (https://doi.org/10.1080%2F08912963.2011.578744). hdl:11336/79657 (https://hdl.handle.net/11336%2F79657). ISSN 0891-2963 (https://search.worldcat.org/issn/0891-2963). Retrieved 28 October 2024 – via Taylor and Francis Online.
20. Todaro, Simona; Rigo, Manuel; Randazzo, Vincenzo; Di Stefano, Pietro (June 2018). "The end-Triassic mass extinction: A new correlation between extinction events and δ13C fluctuations from a Triassic-Jurassic peritidal succession in western Sicily" (https://www.sciencedirect.com/science/article/abs/pii/S0037073818300484). *Sedimentary Geology*. **368**: 105–113. Bibcode:2018SedG..368..105T (https://ui.adsabs.harvard.edu/abs/2018SedG..368..105T). doi:10.1016/j.sedgeo.2018.03.008 (https://doi.org/10.1016%2Fj.sedgeo.2018.03.008). S2CID 134941587 (https://api.semanticscholar.org/CorpusID:134941587). Retrieved 27 August 2023.
21. Atkinson, Jed W.; Wignall, Paul Barry; Morton, Jacob D.; Aze, Tracy (9 January 2019). "Body size changes in bivalves of the family Limidae in the aftermath of the end-Triassic mass extinction: the Brobdingnag effect" (https://onlinelibrary.wiley.com/doi/10.1111/pala.12415). *Palaeontology*. **62** (4): 561–582. Bibcode:2019Palgy..62..561A (https://ui.adsabs.harvard.edu/abs/2019Palgy..62..561A). doi:10.1111/pala.12415 (https://doi.org/10.1111%2Fpala.12415). S2CID 134070316 (https://api.semanticscholar.org/CorpusID:134070316). Retrieved 14 January 2023.
22. Yan, Jia; Song, Haijun; Dai, Xu (1 February 2023). "Increased bivalve cosmopolitanism during the mid-Phanerozoic mass extinctions" (https://www.sciencedirect.com/science/article/abs/pii/S0031018222005338). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **611** 111362. Bibcode:2023PPP...61111362Y (https://ui.adsabs.harvard.edu/abs/2023PPP...61111362Y). doi:10.1016/j.palaeo.2022.111362 (https://doi.org/10.1016%2Fj.palaeo.2022.111362). Retrieved 20 February 2023.
23. Abdelhady, Ahmed A.; Ali, Ahmed; Ahmed, Mohamed S.; Elewa, Ashraf M. T. (8 September 2023). "Triassic/Jurassic bivalve biodiversity dynamics: biotic versus abiotic factors" (https://link.springer.com/10.1007/s12517-023-11657-x). *Arabian Journal of Geosciences*. **16** (10): 546. Bibcode:2023ArJG...16..546A (https://ui.adsabs.harvard.edu/abs/2023ArJG...16..546A). doi:10.1007/s12517-023-11657-x (https://doi.org/10.1007%2Fs12517-023-11657-x). ISSN 1866-7511 (https://search.worldcat.org/issn/1866-7511). Retrieved 11 September 2024 – via Springer Link.
24. Hallam, Anthony (2 January 2007). "How catastrophic was the end-Triassic mass extinction?" (https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1502-3931.2002.tb00075.x). *Lethaia*. **35** (2): 147–157. doi:10.1111/j.1502-3931.2002.tb00075.x (https://doi.org/10.1111%2Fj.1502-3931.2002.tb00075.x). Retrieved 28 May 2023.

25. Baeza-Carratalá, José Francisco; Dulai, Alfréd; Sandoval, José (October 2018). "First evidence of brachiopod diversification after the end-Triassic extinction from the pre-Pliensbachian Internal Subbetic platform (South-Iberian Paleomargin)" (https://www.sciencedirect.com/science/article/abs/pii/S0016699518300445). *Geobios*. **51** (5): 367–384. Bibcode:2018Geobi..51..367B (https://ui.adsabs.harvard.edu/abs/2018Geobi..51..367B). doi:10.1016/j.geobios.2018.08.010 (https://doi.org/10.1016%2Fj.geobios.2018.08.010). hdl:10045/81989 (https://hdl.handle.net/10045%2F81989). S2CID 134589701 (https://api.semanticscholar.org/CorpusID:134589701). Retrieved 22 May 2023.
26. Tomašových, Adam; Siblík, Miloš (9 February 2007). "Evaluating compositional turnover of brachiopod communities during the end-Triassic mass extinction (Northern Calcareous Alps): Removal of dominant groups, recovery and community reassembly" (https://www.sciencedirect.com/science/article/abs/pii/S0031018206004482). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **244** (1–4): 170–200. Bibcode:2007PPP...244..170T (https://ui.adsabs.harvard.edu/abs/2007PPP...244..170T). doi:10.1016/j.palaeo.2006.06.028 (https://doi.org/10.1016%2Fj.palaeo.2006.06.028). Retrieved 19 February 2025 – via Elsevier Science Direct.
27. Powers, Catherine M.; Bottjer, David J. (1 November 2007). "Bryozoan paleoecology indicates mid-Phanerozoic extinctions were the product of long-term environmental stress" (https://pubs.geoscienceworld.org/geology/article/35/11/995-998/129752). *Geology*. **35** (11): 995. Bibcode:2007Geo....35..995P (https://ui.adsabs.harvard.edu/abs/2007Geo....35..995P). doi:10.1130/G23858A.1 (https://doi.org/10.1130%2FG23858A.1). ISSN 0091-7613 (https://search.worldcat.org/issn/0091-7613). Retrieved 30 December 2023.
28. Wignall, Paul Barry; Atkinson, Jed W. (September 2020). "A two-phase end-Triassic mass extinction" (https://www.sciencedirect.com/science/article/pii/S0012825220303287). *Earth-Science Reviews*. **208** 103282. Bibcode:2020ESRv..20803282W (https://ui.adsabs.harvard.edu/abs/2020ESRv..20803282W). doi:10.1016/j.earscirev.2020.103282 (https://doi.org/10.1016%2Fj.earscirev.2020.103282). Retrieved 19 February 2025 – via Elsevier Science Direct.
29. Michalík, Jozef; Lintnerová, Otília; Gaździcki, Andrzej; Soták, Ján (9 February 2007). "Record of environmental changes in the Triassic–Jurassic boundary interval in the Zliechov Basin, Western Carpathians" (https://www.sciencedirect.com/science/article/pii/S0031018206004421). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **244** (1–4): 71–88. Bibcode:2007PPP...244...71M (https://ui.adsabs.harvard.edu/abs/2007PPP...244...71M). doi:10.1016/j.palaeo.2006.06.024 (https://doi.org/10.1016%2Fj.palaeo.2006.06.024). Retrieved 19 February 2025 – via Elsevier Science Direct.
30. Stanley Jr., George D.; Shepherd, Hannah M. E.; Robinson, Autumn J. (14 August 2018). "Paleoecological Response of Corals to the End-Triassic Mass Extinction: An Integrational Analysis" (https://link.springer.com/article/10.1007/s12583-018-0793-5). *Journal of Earth Science*. **29** (4): 879–885. Bibcode:2018JEaSc..29..879S (https://ui.adsabs.harvard.edu/abs/2018JEaSc..29..879S). doi:10.1007/s12583-018-0793-5 (https://doi.org/10.1007%2Fs12583-018-0793-5). S2CID 133705370 (https://api.semanticscholar.org/CorpusID:133705370). Retrieved 7 June 2023.
31. Lathuilière, Bernard; Marchal, Denis (12 January 2009). "Extinction, survival and recovery of corals from the Triassic to Middle Jurassic time" (https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1365-3121.2008.00856.x). *Terra Nova*. **21** (1): 57–66. Bibcode:2009TeNov..21...57L (https://ui.adsabs.harvard.edu/abs/2009TeNov..21...57L). doi:10.1111/j.1365-3121.2008.00856.x (https://doi.org/10.1111%2Fj.1365-3121.2008.00856.x). S2CID 128758050 (https://api.semanticscholar.org/CorpusID:128758050). Retrieved 7 June 2023.

32. Martindale, Rowan C.; Berelson, William M.; Corsetti, Frank A.; Bottjer, David J.; West, A. Joshua (15 September 2012). "Constraining carbonate chemistry at a potential ocean acidification event (the Triassic–Jurassic boundary) using the presence of corals and coral reefs in the fossil record" (https://www.sciencedirect.com/science/article/abs/pii/S0031018212003756). *Palaeogeography, Palaeoclimatology, Palaeoecology*. 350–352: 114–123. Bibcode:2012PPP...350..114M (https://ui.adsabs.harvard.edu/abs/2012PPP...350..114M). doi:10.1016/j.palaeo.2012.06.020 (https://doi.org/10.1016%2Fj.palaeo.2012.06.020). Retrieved 7 June 2023.
33. Hönisch, Bärbel; Ridgwell, Andy; Schmidt, Daniela N.; Thomas, Ellen; Gibbs, Samantha J.; Sluijs, Appy; Zeebe, Richard; Kump, Lee; Martindale, Rowan C.; Greene, Sarah E.; Kiessling, Wolfgang (2 March 2012). "The Geological Record of Ocean Acidification" (https://www.science.org/doi/10.1126/science.1208277). *Science*. **335** (6072): 1058–1063. Bibcode:2012Sci...335.1058H (https://ui.adsabs.harvard.edu/abs/2012Sci...335.1058H). doi:10.1126/science.1208277 (https://doi.org/10.1126%2Fscience.1208277). hdl:1874/385704 (https://hdl.handle.net/1874%2F385704). ISSN 0036-8075 (https://search.worldcat.org/issn/0036-8075). PMID 22383840 (https://pubmed.ncbi.nlm.nih.gov/22383840). S2CID 6361097 (https://api.semanticscholar.org/CorpusID:6361097). Retrieved 19 March 2023.
34. Greene, Sarah E.; Martindale, Rowan C.; Ritterbush, Kathleen A.; Bottjer, David J.; Corsetti, Frank A.; Berelson, William M. (1 June 2012). "Recognising ocean acidification in deep time: An evaluation of the evidence for acidification across the Triassic-Jurassic boundary" (http://www.sciencedirect.com/science/article/pii/S0012825212000463). *Earth-Science Reviews*. **113** (1): 72–93. Bibcode:2012ESRv..113...72G (https://ui.adsabs.harvard.edu/abs/2012ESRv..113...72G). doi:10.1016/j.earscirev.2012.03.009 (https://doi.org/10.1016%2Fj.earscirev.2012.03.009). ISSN 0012-8252 (https://search.worldcat.org/issn/0012-8252).
35. Barras, Colin G.; Twitchett, Richard J. (9 February 2007). "Response of the marine infauna to Triassic–Jurassic environmental change: Ichnological data from southern England" (https://www.sciencedirect.com/science/article/pii/S0031018206004500). *Palaeogeography, Palaeoclimatology, Palaeoecology*. Triassic-Jurassic Boundary events: problems, progress, possibilities. **244** (1): 223–241. Bibcode:2007PPP...244..223B (https://ui.adsabs.harvard.edu/abs/2007PPP...244..223B). doi:10.1016/j.palaeo.2006.06.040 (https://doi.org/10.1016%2Fj.palaeo.2006.06.040). ISSN 0031-0182 (https://search.worldcat.org/issn/0031-0182). Retrieved 10 November 2023.
36. Atkinson, J. W.; Wignall, Paul Barry (15 August 2019). "How quick was marine recovery after the end-Triassic mass extinction and what role did anoxia play?" (https://www.sciencedirect.com/science/article/abs/pii/S0031018219302330). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **528**: 99–119. Bibcode:2019PPP...528...99A (https://ui.adsabs.harvard.edu/abs/2019PPP...528...99A). doi:10.1016/j.palaeo.2019.05.011 (https://doi.org/10.1016%2Fj.palaeo.2019.05.011). S2CID 164911938 (https://api.semanticscholar.org/CorpusID:164911938). Retrieved 20 December 2022.
37. Damborenea, Susana E.; Echevarría, Javier; Ros-Franch, Sonia (1 December 2017). "Biotic recovery after the end-Triassic extinction event: Evidence from marine bivalves of the Neuquén Basin, Argentina" (https://www.sciencedirect.com/science/article/abs/pii/S0031018217306946). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **487**: 93–104. Bibcode:2017PPP...487...93D (https://ui.adsabs.harvard.edu/abs/2017PPP...487...93D). doi:10.1016/j.palaeo.2017.08.025 (https://doi.org/10.1016%2Fj.palaeo.2017.08.025). hdl:11336/49626 (https://hdl.handle.net/11336%2F49626). Retrieved 28 May 2023.
38. Opazo, L. Felipe; Twitchett, Richard J. (August 2022). "Bivalve body-size distribution through the Late Triassic mass extinction event" (https://www.cambridge.org/core/journals/paleobiology/article/abs/bivalve-bodysize-distribution-through-the-late-triassic-mass-extinction-event/D395EA5BCADCD7CC2EB903ADD14CC95A). *Paleobiology*. **48** (3): 420–445. Bibcode:2022Pbio...48..420O (https://ui.adsabs.harvard.edu/abs/2022Pbio...48..420O). doi:10.1017/pab.2021.38 (https://doi.org/10.1017%2Fpab.2021.38). ISSN 0094-8373 (https://search.worldcat.org/issn/0094-8373). Retrieved 28 October 2024 – via Cambridge Core.

39. Yager, Joyce A.; West, A. Joshua; Trower, Elizabeth J.; Fischer, Woodward W.; Ritterbush, Kathleen; Rosas, Silvia; Bottjer, David J.; Celestian, Aaron J.; Berelson, William M.; Corsetti, Frank A. (9 January 2025). "Evidence for Low Dissolved Silica in mid-Mesozoic Oceans" (https://ajsonline.org/article/122691-evidence-for-low-dissolved-silica-in-mid-mesozoic-oceans). *American Journal of Science*. **325**. doi:10.2475/001c.122691 (https://doi.org/10.2475%2F001c.122691). ISSN 1945-452X (https://search.worldcat.org/issn/1945-452X). Retrieved 18 February 2025 – via AJS Online.
40. Ritterbrush, Kathleen A.; Bottjer, David J.; Corseti, Frank A.; Rosas, Silvia (1 December 2014). "New evidence on the role of siliceous sponges in ecology and sedimentary facies development in Eastern Panthalassa following the Triassic-Jurassic mass extinction" (https://bioone.org/journals/palaios/volume-29/issue-12/palo.2013.121/NEW-EVIDENCE-ON-THE-ROLE-OF-SILICEOUS-SPONGES-IN-ECOLOGY/10.2110/palo.2013.121.short). *PALAIOS*. **29** (12): 652–668. Bibcode:2014Palai..29..652R (https://ui.adsabs.harvard.edu/abs/2014Palai..29..652R). doi:10.2110/palo.2013.121 (https://doi.org/10.2110%2Fpalo.2013.121). S2CID 140546770 (https://api.semanticscholar.org/CorpusID:140546770). Retrieved 2 April 2023.
41. Ritterbush, Kathleen A.; Rosas, Silvia; Corsetti, Frank A.; Bottjer, David J.; West, A. Joshua (15 February 2015). "Andean sponges reveal long-term benthic ecosystem shifts following the end-Triassic mass extinction" (https://www.sciencedirect.com/science/article/pii/S0031018214005951). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **420**: 193–209. Bibcode:2015PPP...420..193R (https://ui.adsabs.harvard.edu/abs/2015PPP...420..193R). doi:10.1016/j.palaeo.2014.12.002 (https://doi.org/10.1016%2Fj.palaeo.2014.12.002). ISSN 0031-0182 (https://search.worldcat.org/issn/0031-0182). Retrieved 10 November 2023.
42. Montanaro, Andrea; Falzoni, Francesca; Iannace, Alessandro; Parente, Mariano (1 September 2024). "Patterns of extinction and recovery across the Triassic–Jurassic boundary interval in three resilient Southern Tethyan carbonate platforms" (https://doi.org/10.1016%2Fj.palaeo.2024.112335). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **649** 112335. Bibcode:2024PPP...64912335M (https://ui.adsabs.harvard.edu/abs/2024PPP...64912335M). doi:10.1016/j.palaeo.2024.112335 (https://doi.org/10.1016%2Fj.palaeo.2024.112335).
43. Romano, Carlo; Koot, Martha B.; Kogan, Ilja; Brayard, Arnaud; Minikh, Alla V.; Brinkmann, Winand; Bucher, Hugo; Kriwet, Jürgen (27 November 2014). "Permian–Triassic Osteichthyes (bony fishes): diversity dynamics and body size evolution" (https://www.researchgate.net/publication/268810424). *Biological Reviews of the Cambridge Philosophical Society*. **91** (1): 106–147. doi:10.1111/brv.12161 (https://doi.org/10.1111%2Fbrv.12161). ISSN 1469-185X (https://search.worldcat.org/issn/1469-185X). PMID 25431138 (https://pubmed.ncbi.nlm.nih.gov/25431138). S2CID 5332637 (https://api.semanticscholar.org/CorpusID:5332637).
44. Stumpf, Sebastian; Ansorge, Jörg; Pfaff, Cathrin; Kriwet, Jürgen (4 July 2017). "Early Jurassic diversification of pycnodontiform fishes (Actinopterygii, Neopterygii) after the end-Triassic extinction event: evidence from a new genus and species, Grimmenodon aureum" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5646184). *Journal of Vertebrate Paleontology*. **37** (4) e1344679. Bibcode:2017JVPal..37E4679S (https://ui.adsabs.harvard.edu/abs/2017JVPal..37E4679S). doi:10.1080/02724634.2017.1344679 (https://doi.org/10.1080%2F02724634.2017.1344679). ISSN 0272-4634 (https://search.worldcat.org/issn/0272-4634). PMC 5646184 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5646184). PMID 29170576 (https://pubmed.ncbi.nlm.nih.gov/29170576).

45. Fleischle, C. V.; Sander, P. M.; Wintrich, T.; Caspar, K. R. (2019). "Hematological convergence between Mesozoic marine reptiles (Sauropterygia) and extant aquatic amniotes elucidates diving adaptations in plesiosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6873879). *PeerJ*. **7** e8022. Bibcode:2019PeerJ...7e8022F (https://ui.adsabs.harvard.edu/abs/2019PeerJ...7e8022F). doi:10.7717/peerj.8022 (https://doi.org/10.7717%2Fpeerj.8022). PMC 6873879 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6873879). PMID 31763069 (https://pubmed.ncbi.nlm.nih.gov/31763069).
46. Bardet, Nathalie (1 July 1994). "Extinction events among Mesozoic marine reptiles" (http://doc.rero.ch/record/15138/files/PAL_E2412.pdf) (PDF). *Historical Biology*. **7** (4): 313–324. Bibcode:1994HBio....7..313B (https://ui.adsabs.harvard.edu/abs/1994HBio....7..313B). doi:10.1080/10292389409380462 (https://doi.org/10.1080%2F10292389409380462). ISSN 0891-2963 (https://search.worldcat.org/issn/0891-2963).
47. Thorne, Philippa M.; Ruta, Marcello; Benton, Michael J. (17 May 2011). "Resetting the evolution of marine reptiles at the Triassic–Jurassic boundary" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3100925). *Proceedings of the National Academy of Sciences of the United States of America*. **108** (20): 8339–8344. Bibcode:2011PNAS..108.8339T (https://ui.adsabs.harvard.edu/abs/2011PNAS..108.8339T). doi:10.1073/pnas.1018959108 (https://doi.org/10.1073%2Fpnas.1018959108). ISSN 0027-8424 (https://search.worldcat.org/issn/0027-8424). PMC 3100925 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3100925). PMID 21536898 (https://pubmed.ncbi.nlm.nih.gov/21536898).
48. Laboury, Antoine; Stubbs, Thomas L.; Wolniewicz, Andrzej S.; Liu, Jun; Scheyer, Torsten M.; Jones, Marc E H; Fischer, Valentin (22 December 2024). Parins-Fukuchi, Tomomi; Morlon, Hélène (eds.). "Contrasting macroevolutionary patterns in pelagic tetrapods across the Triassic–Jurassic transition" (https://academic.oup.com/evolut/article/79/1/38/7758686). *Evolution*. **79** (1): 38–50. doi:10.1093/evolut/qpae138 (https://doi.org/10.1093%2Fevolut%2Fqpae138). ISSN 0014-3820 (https://search.worldcat.org/issn/0014-3820). PMID 39283731 (https://pubmed.ncbi.nlm.nih.gov/39283731). Retrieved 18 February 2025 – via Oxford Academic.
49. Benson, Roger B. J.; Evans, Mark; Druckenmiller, Patrick S. (16 March 2012). Lalueza-Fox, Carles (ed.). "High Diversity, Low Disparity and Small Body Size in Plesiosaurs (Reptilia, Sauropterygia) from the Triassic–Jurassic Boundary" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3306369). *PLOS ONE*. **7** (3) e31838. Bibcode:2012PLoSO...731838B (https://ui.adsabs.harvard.edu/abs/2012PLoSO...731838B). doi:10.1371/journal.pone.0031838 (https://doi.org/10.1371%2Fjournal.pone.0031838). ISSN 1932-6203 (https://search.worldcat.org/issn/1932-6203). PMC 3306369 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3306369). PMID 22438869 (https://pubmed.ncbi.nlm.nih.gov/22438869).
50. Cribb, Alison T.; Formoso, Kiersten K.; Woolley, C. Henrik; Beech, James; Brophy, Shannon; Byrne, Paul; Cassady, Victoria C.; Godbold, Amanda L.; Larina, Ekaterina; Maxeiner, Philippeter; Wu, Yun-Hsin; Corsetti, Frank A.; Bottjer, David J. (6 December 2023). "Contrasting terrestrial and marine ecospace dynamics after the end-Triassic mass extinction event" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10697803). *Proceedings of the Royal Society B: Biological Sciences*. **290** (2012) 20232232. doi:10.1098/rspb.2023.2232 (https://doi.org/10.1098%2Frspb.2023.2232). ISSN 0962-8452 (https://search.worldcat.org/issn/0962-8452). PMC 10697803 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10697803). PMID 38052241 (https://pubmed.ncbi.nlm.nih.gov/38052241).
51. Buffetaut, Eric (2006). "Continental Vertebrate Extinctions at the Triassic-Jurassic and Cretaceous-Tertiary Boundaries: a Comparison" (https://link.springer.com/chapter/10.1007/3-540-25736-5_11). In Cockell, Charles; Gilmour, Iain; Koeberl, Charles (eds.). *Biological Processes Associated with Impact Events* (https://link.springer.com/book/10.1007/b135965). Impact Studies. Berlin: Springer. pp. 245–256. doi:10.1007/3-540-25736-5_11 (https://doi.org/10.1007%2F3-540-25736-5_11). ISBN 978-3-540-25736-3.

52. Konietzko-Meier, Dorota; Werner, Jennifer D.; Wintrich, Tanja; Martin Sander, P. (31 October 2018). "A large temnospondyl humerus from the Rhaetian (Late Triassic) of Bonenburg (Westphalia, Germany) and its implications for temnospondyl extinction". *Journal of Iberian Geology*. **45** (2): 287–300. doi:10.1007/s41513-018-0092-0 (https://doi.org/10.1007%2Fs41513-018-0092-0). ISSN 1886-7995 (https://search.worldcat.org/issn/1886-7995). S2CID 134049099 (https://api.semanticscholar.org/CorpusID:134049099).
53. Whiteside, D. I.; Marshall, J. E. A. (1 January 2008). "The age, fauna and palaeoenvironment of the Late Triassic fissure deposits of Tytherington, South Gloucestershire, UK" (https://www.researchgate.net/publication/242099762). *Geological Magazine*. **145** (1): 105–147. Bibcode:2008GeoM..145..105W (https://ui.adsabs.harvard.edu/abs/2008GeoM..145..105W). doi:10.1017/S0016756807003925 (https://doi.org/10.1017%2FS0016756807003925). ISSN 0016-7568 (https://search.worldcat.org/issn/0016-7568). S2CID 129614690 (https://api.semanticscholar.org/CorpusID:129614690).
54. Patrick, Erin L.; Whiteside, David I.; Benton, Michael J. (2019). "A new crurotarsan archosaur from the Late Triassic of South Wales" (https://web.archive.org/web/20190830221411/https://cpb-eu-w2.wpmucdn.com/blogs.bristol.ac.uk/dist/6/525/files/2019/08/2019Edgar.pdf) (PDF). *Journal of Vertebrate Paleontology*. **39** (3) e1645147. Bibcode:2019JVPal..39E5147P (https://ui.adsabs.harvard.edu/abs/2019JVPal..39E5147P). doi:10.1080/02724634.2019.1645147 (https://doi.org/10.1080%2F02724634.2019.1645147). S2CID 202848499 (https://api.semanticscholar.org/CorpusID:202848499). Archived from the original (https://cpb-eu-w2.wpmucdn.com/blogs.bristol.ac.uk/dist/6/525/files/2019/08/2019Edgar.pdf) (PDF) on 30 August 2019.
55. Tolchard, Frederick; Nesbitt, Sterling J.; Desojo, Julia B.; Viglietti, Pia; Butler, Richard J.; Choiniere, Jonah N. (1 December 2019). " 'Rauisuchian' material from the lower Elliot Formation of South Africa and Lesotho: Implications for Late Triassic biogeography and biostratigraphy" (http://pure-oai.bham.ac.uk/ws/files/74379812/Tolchard_et_al._in_press.pdf) (PDF). *Journal of African Earth Sciences*. **160** 103610. Bibcode:2019JAfES.16003610T (https://ui.adsabs.harvard.edu/abs/2019JAfES.16003610T). doi:10.1016/j.jafrearsci.2019.103610 (https://doi.org/10.1016%2Fj.jafrearsci.2019.103610). ISSN 1464-343X (https://search.worldcat.org/issn/1464-343X). S2CID 202902771 (https://api.semanticscholar.org/CorpusID:202902771).
56. Chen, Jianbo; Niu, Yi-Ning; Ma, Rongyao; Zhou, Yan-Ling; Liu, Wen-Jie; Wang, Ya-Ming; You, Hai-Lu; Xu, Xing; Shen, Shu-Zhong; Feng, Zhuo (13 January 2026). "Triassic–Jurassic environmental instability on the subtropical eastern Tethyan margin linked to low-latitude dinosaur dispersal" (https://doi.org/10.1038%2Fs43247-025-03083-6). *Communications Earth & Environment*. **7** (1): 91. Bibcode:2026ComEE...7...91C (https://ui.adsabs.harvard.edu/abs/2026ComEE...7...91C). doi:10.1038/s43247-025-03083-6 (https://doi.org/10.1038%2Fs43247-025-03083-6). ISSN 2662-4435 (https://search.worldcat.org/issn/2662-4435).
57. Benton, Michael James (1991). "What really happened in the late Triassic?" (https://www.tandfonline.com/doi/abs/10.1080/10292389109380406). *Historical Biology*. **5** (2–4): 263–278. Bibcode:1991HBio....5..263B (https://ui.adsabs.harvard.edu/abs/1991HBio....5..263B). doi:10.1080/10292389109380406 (https://doi.org/10.1080%2F10292389109380406). Retrieved 15 December 2022.
58. Toljagić, Olja; Butler, Richard J. (23 June 2013). "Triassic–Jurassic mass extinction as trigger for the Mesozoic radiation of crocodylomorphs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3645043). *Biology Letters*. **9** (3): 1–4. Bibcode:2013BiLet...930095T (https://ui.adsabs.harvard.edu/abs/2013BiLet...930095T). doi:10.1098/rsbl.2013.0095 (https://doi.org/10.1098%2Frsbl.2013.0095). PMC 3645043 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3645043). PMID 23536443 (https://pubmed.ncbi.nlm.nih.gov/23536443).

59. Faure-Brac, Mathieu Gabriel (16 December 2024). "Pseudosuchian thermometabolism: A review of the past two decades" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11725720). *The Anatomical Record*. **308** (2): 315–341. doi:10.1002/ar.25609 (https://doi.org/10.1002%2Far.25609). ISSN 1932-8486 (https://search.worldcat.org/issn/1932-8486). PMC 11725720 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11725720). PMID 39682064 (https://pubmed.ncbi.nlm.nih.gov/39682064).
60. Xu, Yuanyuan; Wang, Yongdong; Li, Liqin; Lu, Ning; Zhu, Yanbin; Huang, Zhuanli; McLoughlin, Stephen (9 January 2024). "Plant-insect interactions across the Triassic–Jurassic boundary in the Sichuan Basin, South China" (https://doi.org/10.3389%2Ffevo.2023.1338865). *Frontiers in Ecology and Evolution*. **11** 1338865. Bibcode:2024FrEEv..1138865X (https://ui.adsabs.harvard.edu/abs/2024FrEEv..1138865X). doi:10.3389/fevo.2023.1338865 (https://doi.org/10.3389%2Ffevo.2023.1338865). ISSN 2296-701X (https://search.worldcat.org/issn/2296-701X).
61. Deregnaucourt, Isabelle; Bardin, Jérémie; Villier, Loïc; Julliard, Romain; Béthoux, Olivier (18 August 2023). "Disparification and extinction trade-offs shaped the evolution of Permian to Jurassic Odonata" (https://www.researchgate.net/publication/372525075). *iScience*. **26** (8) 107420. Bibcode:2023iSci...26j7420D (https://ui.adsabs.harvard.edu/abs/2023iSci...26j7420D). doi:10.1016/j.isci.2023.107420 (https://doi.org/10.1016%2Fj.isci.2023.107420). PMC 10424082 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10424082). PMID 37583549 (https://pubmed.ncbi.nlm.nih.gov/37583549). Retrieved 31 October 2024 – via ResearchGate.
62. Lindström, Sofie (1 September 2015). "Palynofloral patterns of terrestrial ecosystem change during the end-Triassic event – a review" (https://pubs.geoscienceworld.org/geolmag/article-abstract/153/2/223/251206/Palynofloral-patterns-of-terrestrial-ecosystem?redirectedFrom=fulltext). *Geological Magazine*. **153** (2): 223–251. doi:10.1017/S0016756815000552 (https://doi.org/10.1017%2FS0016756815000552). S2CID 131410887 (https://api.semanticscholar.org/CorpusID:131410887). Retrieved 28 May 2023.
63. Barbacka, Maria; Pacyna, Grzegorz; Kocsis, Ádam T.; Jarzynka, Agata; Ziaja, Jadwiga; Bodor, Emese (15 August 2017). "Changes in terrestrial floras at the Triassic-Jurassic Boundary in Europe" (https://www.sciencedirect.com/science/article/abs/pii/S0031018216304977). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **480**: 80–93. Bibcode:2017PPP...480...80B (https://ui.adsabs.harvard.edu/abs/2017PPP...480...80B). doi:10.1016/j.palaeo.2017.05.024 (https://doi.org/10.1016%2Fj.palaeo.2017.05.024). Retrieved 12 December 2022.
64. McElwain, Jennifer C.; Popa, Mihai E.; Hesselbo, Stephen P.; Haworth, Matthew; Surlyk, Finn (December 2007). "Macroecological responses of terrestrial vegetation to climatic and atmospheric change across the Triassic/Jurassic boundary in East Greenland". *Paleobiology*. **33** (4): 547–573. Bibcode:2007Pbio...33..547M (https://ui.adsabs.harvard.edu/abs/2007Pbio...33..547M). doi:10.1666/06026.1 (https://doi.org/10.1666%2F06026.1). ISSN 0094-8373 (https://search.worldcat.org/issn/0094-8373). S2CID 129330139 (https://api.semanticscholar.org/CorpusID:129330139).
65. Gravendyck, Julia; Schobben, Martin; Bachelier, Julien B.; Kürschner, Wolfram Michael (November 2020). "Macroecological patterns of the terrestrial vegetation history during the end-Triassic biotic crisis in the central European Basin: A palynological study of the Bonenburg section (NW-Germany) and its supra-regional implications" (https://www.sciencedirect.com/science/article/abs/pii/S0921818120301776). *Global and Planetary Change*. **194** 103286. Bibcode:2020GPC...19403286G (https://ui.adsabs.harvard.edu/abs/2020GPC...194...03286G). doi:10.1016/j.gloplacha.2020.103286 (https://doi.org/10.1016%2Fj.gloplacha.2020.103286). hdl:1874/409017 (https://hdl.handle.net/1874%2F409017). S2CID 225521004 (https://api.semanticscholar.org/CorpusID:225521004). Retrieved 12 December 2022.

66. Yiotis, C.; Evans-Fitz.Gerald, C.; McElwain, J. C. (11 March 2017). "Differences in the photosynthetic plasticity of ferns and Ginkgo grown in experimentally controlled low [O2]: [CO2] atmospheres may explain their contrasting ecological fate across the Triassic–Jurassic mass extinction boundary" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5604595). *Annals of Botany*. **119** (8): 1385–1395. doi:10.1093/aob/mcx018 (https://doi.org/10.1093%2Faob%2Fmcx018). ISSN 0305-7364 (https://search.worldcat.org/issn/0305-7364). PMC 5604595 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5604595). PMID 28334286 (https://pubmed.ncbi.nlm.nih.gov/28334286).
67. Bos, Remco; Lindström, Sofie; van Konijnenburg-van Cittert, Han; Hilgen, Frederik; Hollaar, Teuntje P.; Aalpoel, Hendrik; van der Weijst, Carolien; Sanei, Hamed; Rudra, Arka; Sluijs, Appy; van de Schootbrugge, Bas (1 September 2023). "Triassic-Jurassic vegetation response to carbon cycle perturbations and climate change" (https://doi.org/10.1016%2Fj.gloplacha.2023.104211). *Global and Planetary Change*. **228** 104211. Bibcode:2023GPC...22804211B (https://ui.adsabs.harvard.edu/abs/2023GPC...22804211B). doi:10.1016/j.gloplacha.2023.104211 (https://doi.org/10.1016%2Fj.gloplacha.2023.104211). ISSN 0921-8181 (https://search.worldcat.org/issn/0921-8181).
68. Vajda, Vivi; McLoughlin, Stephen; Slater, Sam M.; Gustafsson, Ola; Rasmusson, Allan G. (1 October 2023). "The 'seed-fern' Lepidopteris mass-produced the abnormal pollen Ricciisporites during the end-Triassic biotic crisis" (https://www.sciencedirect.com/science/article/pii/S0031018223003413). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **627** 111723. Bibcode:2023PPP...62711723V (https://ui.adsabs.harvard.edu/abs/2023PPP...627111723V). doi:10.1016/j.palaeo.2023.111723 (https://doi.org/10.1016%2Fj.palaeo.2023.111723). Retrieved 18 February 2025 – via Elsevier Science Direct.
69. Zhang, Peixin; Lu, Jing; Yang, Minfang; Bond, David P. G.; Greene, Sarah E.; Liu, Le; Zhang, Yuanfu; Wang, Ye; Wang, Ziwei; Li, Shan; Shao, Longyi; Hilton, Jason (28 March 2022). "Volcanically-Induced Environmental and Floral Changes Across the Triassic-Jurassic (T-J) Transition" (https://doi.org/10.3389%2Ffevo.2022.853404). *Frontiers in Ecology and Evolution*. **10** 853404: 1–17. Bibcode:2022FrEEv..1053404Z (https://ui.adsabs.harvard.edu/abs/2022FrEEv..1053404Z). doi:10.3389/fevo.2022.853404 (https://doi.org/10.3389%2Ffevo.2022.853404). ISSN 2296-701X (https://search.worldcat.org/issn/2296-701X).
70. Li, Liqin; Wang, Yongdong; Kürschner, Wolfram M.; Ruhl, Micha; Vajda, Vivi (15 October 2020). "Palaeovegetation and palaeoclimate changes across the Triassic–Jurassic transition in the Sichuan Basin, China" (https://www.sciencedirect.com/science/article/abs/pii/S0031018220303369). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **556** 109891. Bibcode:2020PPP...55609891L (https://ui.adsabs.harvard.edu/abs/2020PPP...55609891L). doi:10.1016/j.palaeo.2020.109891 (https://doi.org/10.1016%2Fj.palaeo.2020.109891). S2CID 225600810 (https://api.semanticscholar.org/CorpusID:225600810). Retrieved 22 May 2023.
71. Zhou, Ning; Xu, Yuanyuan; Li, Liqin; Lu, Ning; An, Pengcheng; Popa, Mihai Emilian; Kürschner, Wolfram Michael; Zhang, Xingliang; Wang, Yongdong (October 2021). "Pattern of vegetation turnover during the end-Triassic mass extinction: Trends of fern communities from South China with global context" (https://doi.org/10.1016%2Fj.gloplacha.2021.103585). *Global and Planetary Change*. **205** 103585. Bibcode:2021GPC...20503585Z (https://ui.adsabs.harvard.edu/abs/2021GPC...20503585Z). doi:10.1016/j.gloplacha.2021.103585 (https://doi.org/10.1016%2Fj.gloplacha.2021.103585).
72. Knetge, Antonietta B.; Barbosa, Catarina; Matthaeus, William J.; Barclay, Richard S.; Glasspool, Ian J.; Gomez, Bernard; Hesselbo, Stephen P.; Popa, Mihai E.; Ruhl, Micha; Sunderlin, David; Surlyk, Finn; McElwain, Jennifer C. (1 December 2025). "Census collection of two fossil plant localities in Jameson Land, East Greenland supports regional ecological turnover and diversity loss at the end-Triassic mass extinction" (https://doi.org/10.1016%2Fj.palaeo.2025.113266). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **679** 113266. Bibcode:2025PPP...67913266K (https://ui.adsabs.harvard.edu/abs/2025PPP...679113266K). doi:10.1016/j.palaeo.2025.113266 (https://doi.org/10.1016%2Fj.palaeo.2025.113266).

73. Fowell, S. J.; Cornet, B.; Olsen, P. E. (1994), "Geologically rapid Late Triassic extinctions: Palynological evidence from the Newark Supergroup", *Geological Society of America Special Papers*, Geological Society of America, pp. 197–206, doi:10.1130/spe288-p197 (https://doi.org/10.1130%2Fspe288-p197), ISBN 978-0-8137-2288-7
74. Bonis, Nina R.; Kürschner, Wolfram M. (2012). "Vegetation history, diversity patterns, and climate change across the Triassic/Jurassic boundary" (https://www.cambridge.org/core/product/identifier/S0094837300000567/type/journal_article). *Paleobiology*. **38** (2): 240–264. Bibcode:2012Pbio...38..240B (https://ui.adsabs.harvard.edu/abs/2012Pbio...38..240B). doi:10.1666/09071.1 (https://doi.org/10.1666%2F09071.1). ISSN 0094-8373 (https://search.worldcat.org/issn/0094-8373). Retrieved 28 March 2024 – via Cambridge Core.
75. Lindström, Sofie; Erlström, Mikael; Piasecki, Stefan; Nielsen, Lars Henrik; Mathiesen, Anders (September 2017). "Palynology and terrestrial ecosystem change of the Middle Triassic to lowermost Jurassic succession of the eastern Danish Basin" (https://linkinghub.elsevier.com/retrieve/pii/S0034666716300318). *Review of Palaeobotany and Palynology*. **244**: 65–95. Bibcode:2017RPaPa.244...65L (https://ui.adsabs.harvard.edu/abs/2017RPaPa.244...65L). doi:10.1016/j.revpalbo.2017.04.007 (https://doi.org/10.1016%2Fj.revpalbo.2017.04.007). Retrieved 28 March 2024 – via Elsevier Science Direct.
76. Quiroz-Cabascango, Daniela; Vajda, Vivi; McLoughlin, Stephen; Niedźwiedzki, Grzegorz (17 June 2026). "Earliest Jurassic plant assemblages from Sweden reveal a low-diversity ginkgoalean and cheirolepid flora dominating the post-extinction landscape" (https://academic.oup.com/aob/article/137/6/1797/8195845). *Annals of Botany*. **137** (6): 1797–1816. doi:10.1093/aob/mcaf143 (https://doi.org/10.1093%2Faob%2Fmcaf143). ISSN 1095-8290 (https://search.worldcat.org/issn/1095-8290). PMC 13275009 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC13275009). PMID 40632901 (https://pubmed.ncbi.nlm.nih.gov/40632901). Retrieved 12 August 2026 – via Oxford Academic.
77. Lindström, Sofie (17 September 2021). "Two-phased Mass Rarity and Extinction in Land Plants During the End-Triassic Climate Crisis" (https://doi.org/10.3389%2Ffeart.2021.780343). *Frontiers in Earth Science*. **9** 780343: 1079. Bibcode:2021FrEaS...9.1079L (https://ui.adsabs.harvard.edu/abs/2021FrEaS...9.1079L). doi:10.3389/feart.2021.780343 (https://doi.org/10.3389%2Ffeart.2021.780343).
78. Bonis, N. R.; Kürschner, W. M.; Krystyn, L. (September 2009). "A detailed palynological study of the Triassic–Jurassic transition in key sections of the Eiberg Basin (Northern Calcareous Alps, Austria)" (https://www.sciencedirect.com/science/article/abs/pii/S0034666709000633). *Review of Palaeobotany and Palynology*. **156** (3–4): 376–400. Bibcode:2009RPaPa.156..376B (https://ui.adsabs.harvard.edu/abs/2009RPaPa.156..376B). doi:10.1016/j.revpalbo.2009.04.003 (https://doi.org/10.1016%2Fj.revpalbo.2009.04.003). Retrieved 28 May 2023.
79. de Jersey, Noel J.; McKellar, John L. (15 January 2013). "The palynology of the Triassic–Jurassic transition in southeastern Queensland, Australia, and correlation with New Zealand" (http://www.tandfonline.com/doi/abs/10.1080/01916122.2012.718609). *Palynology*. **37** (1): 77–114. Bibcode:2013Paly...37...77D (https://ui.adsabs.harvard.edu/abs/2013Paly...37...77D). doi:10.1080/01916122.2012.718609 (https://doi.org/10.1080%2F01916122.2012.718609). ISSN 0191-6122 (https://search.worldcat.org/issn/0191-6122). Retrieved 19 June 2024 – via Taylor and Francis Online.
80. Kürschner, Wolfram M.; Batenburg, Sietske J.; Mander, Luke (7 October 2013). "Aberrant Classopollis pollen reveals evidence for unreduced (2n) pollen in the conifer family Cheirolepidiaceae during the Triassic–Jurassic transition" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3757988). *Proceedings of the Royal Society B: Biological Sciences*. **280** (1768): 1–8. Bibcode:2013PBioS.28031708K (https://ui.adsabs.harvard.edu/abs/2013PBioS.28031708K). doi:10.1098/rspb.2013.1708 (https://doi.org/10.1098%2Frspb.2013.1708). PMC 3757988 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3757988). PMID 23926159 (https://pubmed.ncbi.nlm.nih.gov/23926159).

81. Ernst, Richard E.; Youbi, Nasrrddine (15 July 2017). "How Large Igneous Provinces affect global climate, sometimes cause mass extinctions, and represent natural markers in the geological record" (https://www.sciencedirect.com/science/article/abs/pii/S0031018217302857). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **478**: 30–52. Bibcode:2017PPP...478...30E (https://ui.adsabs.harvard.edu/abs/2017PPP...478...30E). doi:10.1016/j.palaeo.2017.03.014 (https://doi.org/10.1016%2Fj.palaeo.2017.03.014). Retrieved 28 May 2023.
82. Deenen, M. H. L.; Ruhl, M.; Bonis, N. R.; Krijgsman, W.; Kuerschner, W. M.; Reitsma, M.; Van Bergen, M. J. (1 March 2010). "A new chronology for the end-Triassic mass extinction" (https://www.sciencedirect.com/science/article/abs/pii/S0012821X1000021X). *Earth and Planetary Science Letters*. **291** (1–4): 113–125. Bibcode:2010E&PSL.291..113D (https://ui.adsabs.harvard.edu/abs/2010E&PSL.291..113D). doi:10.1016/j.epsl.2010.01.003 (https://doi.org/10.1016%2Fj.epsl.2010.01.003). Retrieved 15 November 2022.
83. Whalen, Lisa; Gazel, Esteban; Vidito, Christopher; Puffer, John; Bizinis, Michael; Henika, William; Caddick, Mark J. (3 September 2015). "Supercontinental inheritance and its influence on supercontinental breakup: The Central Atlantic Magmatic Province and the breakup of Pangea" (https://doi.org/10.1002%2F2015GC005885). *Paleoceanography and Paleoclimatology*. **16** (10): 3532–3554. Bibcode:2015GGG....16.3532W (https://ui.adsabs.harvard.edu/abs/2015GGG....16.3532W). doi:10.1002/2015GC005885 (https://doi.org/10.1002%2F2015GC005885). hdl:10919/71423 (https://hdl.handle.net/10919%2F71423). S2CID 129223849 (https://api.semanticscholar.org/CorpusID:129223849).
84. McHone, J. Gregory (1 January 2003). Hames, W.; Mchone, J. G.; Renne, P.; Ruppel, C. (eds.). *The Central Atlantic Magmatic Province: Insights from Fragments of Pangea, Volume 136* (https://agupubs.onlinelibrary.wiley.com/doi/10.1029/136GM013). American Geophysical Union. p. 241. doi:10.1029/136GM013 (https://doi.org/10.1029%2F136GM013). ISBN 978-1-118-66877-1.
85. Marzen, R. E.; Shillington, D. J.; Lizarralde, D.; Knapp, J. H.; Heffner, D. M.; Davis, J. K.; Harder, S. H. (7 July 2020). "Limited and localized magmatism in the Central Atlantic Magmatic Province" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7341742). *Nature Communications*. **11** (1): 3397. Bibcode:2020NatCo..11.3397M (https://ui.adsabs.harvard.edu/abs/2020NatCo..11.3397M). doi:10.1038/s41467-020-17193-6 (https://doi.org/10.1038%2Fs41467-020-17193-6). PMC 7341742 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7341742). PMID 32636386 (https://pubmed.ncbi.nlm.nih.gov/32636386).
86. Youbi, Nasrrddine; Tavares Martins, Línia; Munhá, José Manuel; Ibouh, Hassan; Madeira, José; Aït Chayeb, El Houssaine; El Boukhari, Abdelmajid (1 January 2003). "The Late Triassic-Early Jurassic Volcanism of Morocco and Portugal in the Framework of the Central Atlantic Magmatic Province: An Overview" (https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/136GM010). In Hames, W.; McHone, J. G.; Renne, Paul R.; Ruppel, C. (eds.). *The Central Atlantic Magmatic Province: Insights from Fragments of Pangea* (https://agupubs.onlinelibrary.wiley.com/doi/book/10.1029/GM136). American Geophysical Union. pp. 179–207. doi:10.1029/136GM010 (https://doi.org/10.1029%2F136GM010). ISBN 978-1-118-66877-1.
87. Verati, Chrystèle; Rapaille, Cédric; Féraud, Gilbert; Marzoli, Andrea; Bertrand, Hervé; Youbi, Nasrrddine (9 February 2007). "40Ar/39Ar ages and duration of the Central Atlantic Magmatic Province volcanism in Morocco and Portugal and its relation to the Triassic–Jurassic boundary" (https://www.sciencedirect.com/science/article/abs/pii/S0031018206004561). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **244** (1–4): 308–325. Bibcode:2007PPP...244..308V (https://ui.adsabs.harvard.edu/abs/2007PPP...244..308V). doi:10.1016/j.palaeo.2006.06.033 (https://doi.org/10.1016%2Fj.palaeo.2006.06.033). Retrieved 28 May 2023.

88. Marzoli, Andrea; Bertrand, Hervé; Youbi, Nasrrddine; Callegaro, Sara; Merle, Renaud; Reisberg, Laurie; Chiaradia, Massimo; Brownlee, Sarah I.; Jourdan, Fred; Zanetti, Alberto; Davies, Joshua H. F. L.; Cuppone, Tiberio; Mahmoudi, Abdelkader; Medina, Fida; Renne, Paul R.; Bellieni, Giuliano; Crivellari, Stefano; El Hachimi, Hind; Bensalah, Mohamed Khalil; Meyzen, Christine M.; Tegner, Christian (19 April 2019). "The Central Atlantic Magmatic Province (CAMP) in Morocco" (https://academic.oup.com/petrology/article/60/5/945/5475177). *Journal of Petrology*. **50** (6): 945–996. doi:10.1093/petrology/egz021 (https://doi.org/10.1093%2Fpetrology%2Fegz021). hdl:11577/3306908 (https://hdl.handle.net/11577%2F3306908). Retrieved 28 May 2023.
89. Rezende, Gabriel L.; Martins, Cristiano Mendel; Nogueira, Afonso C. R.; Domingos, Fabio Garcia; Ribeiro-Filho, Nelson (1 June 2021). "Evidence for the Central Atlantic magmatic province (CAMP) in Precambrian and Phanerozoic sedimentary basins of the southern Amazonian Craton, Brazil" (https://www.sciencedirect.com/science/article/pii/S0895981121000638). *Journal of South American Earth Sciences*. **108** 103216. Bibcode:2021JSAES.10803216R (https://ui.adsabs.harvard.edu/abs/2021JSAES.10803216R). doi:10.1016/j.jsames.2021.103216 (https://doi.org/10.1016%2Fj.jsames.2021.103216). ISSN 0895-9811 (https://search.worldcat.org/issn/0895-9811). S2CID 233565961 (https://api.semanticscholar.org/CorpusID:233565961). Retrieved 12 January 2024 – via Elsevier Science Direct.
90. Marzoli, Andrea; Callegaro, Sara; Dal Corso, Jacopo; Davies, Joshua H. F. L.; Chiaradia, Massimo; Youbi, Nasrrddine; Bertrand, Hervé; Reisberg, Laurie; Merle, Renaud; Jourdan, Fred (16 November 2017). "The Central Atlantic Magmatic Province (CAMP): A Review" (https://link.springer.com/chapter/10.1007/978-3-319-68009-5_4). In Tanner, Lawrence H. (ed.). *The Late Triassic World: Earth in a Time of Transition* (https://link.springer.com/book/10.1007/978-3-319-68009-5). Topics in Geobiology. Vol. 46. Springer Cham. pp. 91–125. doi:10.1007/978-3-319-68009-5_4 (https://doi.org/10.1007%2F978-3-319-68009-5_4). ISBN 978-3-319-68009-5.
91. Hames, W. E.; Renne, Paul R.; Ruppel, C. (1 September 2000). "New evidence for geologically instantaneous emplacement of earliest Jurassic Central Atlantic magmatic province basalts on the North American margin" (https://pubs.geoscienceworld.org/gsa/geology/article-abstract/28/9/859/188925/New-evidence-for-geologically-instantaneous). *Geology*. **28** (9): 859–862. Bibcode:2000Geo....28..859H (https://ui.adsabs.harvard.edu/abs/2000Geo....28..859H). doi:10.1130/0091-7613(2000)28<859:NEFGIE>2.0.CO;2 (https://doi.org/10.1130%2F0091-7613%282000%2928%3C859%3ANEFGIE%3E2.0.CO%3B2). Retrieved 28 May 2023.
92. Goldberg, David S.; Kent, Dennis V.; Olsen, Paul E. (4 January 2010). "Potential on-shore and off-shore reservoirs for CO2 sequestration in Central Atlantic magmatic province basalts" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2824362). *Proceedings of the National Academy of Sciences of the United States of America*. **107** (4): 1327–1332. Bibcode:2010PNAS..107.1327G (https://ui.adsabs.harvard.edu/abs/2010PNAS..107.1327G). doi:10.1073/pnas.0913721107 (https://doi.org/10.1073%2Fpnas.0913721107). PMC 2824362 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2824362). PMID 20080705 (https://pubmed.ncbi.nlm.nih.gov/20080705).
93. Schaltegger, Urs; Guex, Jean; Bartolini, Annachiara; Schoene, Blair; Ovtcharova, Maria (1 March 2008). "Precise U–Pb age constraints for end-Triassic mass extinction, its correlation to volcanism and Hettangian post-extinction recovery" (https://www.sciencedirect.com/science/article/abs/pii/S0012821X07007777). *Earth and Planetary Science Letters*. **166** (1–2): 266–275. Bibcode:2008E&PSL.267..266S (https://ui.adsabs.harvard.edu/abs/2008E&PSL.267..266S). doi:10.1016/j.epsl.2007.11.031 (https://doi.org/10.1016%2Fj.epsl.2007.11.031). Retrieved 30 May 2023.

94. Knight, K. B.; Nomade, S.; Renne, Paul R.; Marzoli, Andrea; Bertrand, Hervé; Youbi, Nasrrddine (30 November 2004). "The Central Atlantic Magmatic Province at the Triassic–Jurassic boundary: paleomagnetic and 40Ar/39Ar evidence from Morocco for brief, episodic volcanism" (https://www.sciencedirect.com/science/article/abs/pii/S0012821X04005667). *Earth and Planetary Science Letters*. **228** (1–2): 143–160. Bibcode:2004E&PSL.228..143K (https://ui.adsabs.harvard.edu/abs/2004E&PSL.228..143K). doi:10.1016/j.epsl.2004.09.022 (https://doi.org/10.1016%2Fj.epsl.2004.09.022). Retrieved 28 May 2023.
95. Marzoli, Andrea; Bertrand, Hervé; Knight, Kim B.; Cirilli, Simonetta; Buratti, Nicoletta; Vérati, Chrystèle; Nomade, Sébastien; Renne, Paul R.; Youbi, Nasrrddine; Martini, Rossana; Allenbach, Karin; Neuwerth, Ralph; Rapaille, Cédric; Zaninetti, Louisette; Bellieni, Giuliano (1 November 2004). "Synchrony of the Central Atlantic magmatic province and the Triassic-Jurassic boundary climatic and biotic crisis" (https://pubs.geoscienceworld.org/gsa/geology/article/32/11/973/103666/Synchrony-of-the-Central-Atlantic-magmatic). *Geology*. **32** (11): 973. Bibcode:2004Geo....32..973M (https://ui.adsabs.harvard.edu/abs/2004Geo....32..973M). doi:10.1130/G20652.1 (https://doi.org/10.1130%2FG20652.1). ISSN 0091-7613 (https://search.worldcat.org/issn/0091-7613) – via GeoScienceWorld.
96. Hu, Fangzhi; Fu, Xiugen; Lin, Li; Song, Chunyan; Wang, Zhongwei; Tian, Kangzhi (January 2020). "Marine Late Triassic-Jurassic carbon-isotope excursion and biological extinction records: New evidence from the Qiangtang Basin, eastern Tethys" (https://www.sciencedirect.com/science/article/abs/pii/S0921818119305788). *Global and Planetary Change*. **185** 103093. Bibcode:2020GPC...18503093H (https://ui.adsabs.harvard.edu/abs/2020GPC...185 03093H). doi:10.1016/j.gloplacha.2019.103093 (https://doi.org/10.1016%2Fj.gloplacha.2019.103093). S2CID 213355203 (https://api.semanticscholar.org/CorpusID:213355203). Retrieved 7 November 2022.
97. Pálfy, József; Demény, Attila; Haas, János; Hetényi, Magdolna; Orchard, Michael J.; Veto, István (1 November 2001). "Carbon isotope anomaly and other geochemical changes at the Triassic-Jurassic boundary from a marine section in Hungary" (https://pubs.geoscienceworld.org/gsa/geology/article-abstract/29/11/1047/197882/Carbon-isotope-anomaly-and-other-geochemical). *Geology*. **29** (11): 1047–1050. Bibcode:2001Geo....29.1047P (https://ui.adsabs.harvard.edu/abs/2001Geo....29.1047P). doi:10.1130/0091-7613(2001)029<1047:CIAAOG>2.0.CO;2 (https://doi.org/10.1130%2F0091-7613%282001%29029%3C1047%3ACIAAOG%3E2.0.CO%3B2). Retrieved 2 April 2023.
98. Hesselbo, Stephen P.; Korte, Christophe; Ullmann, Clemens V.; Ebbesen, Anders L. (April 2020). "Carbon and oxygen isotope records from the southern Eurasian Seaway following the Triassic-Jurassic boundary: Parallel long-term enhanced carbon burial and seawater warming" (https://www.sciencedirect.com/science/article/abs/pii/S0012825219306592). *Earth-Science Reviews*. **203** 103131. Bibcode:2020ESRv..20303131H (https://ui.adsabs.harvard.edu/abs/2020ESRv..20303131H). doi:10.1016/j.earscirev.2020.103131 (https://doi.org/10.1016%2Fj.earscirev.2020.103131). hdl:10871/40906 (https://hdl.handle.net/10871%2F40906). S2CID 213462318 (https://api.semanticscholar.org/CorpusID:213462318). Retrieved 28 May 2023.
99. Al-Suwaidi, Aisha H.; Steuber, Thomas; Suarez, Marina B. (7 July 2016). "The Triassic–Jurassic boundary event from an equatorial carbonate platform (Ghalilah Formation, United Arab Emirates)" (https://www.lyellcollection.org/doi/10.1144/jgs2015-102). *Journal of the Geological Society*. **173** (6): 949–953. Bibcode:2016JGSoc.173..949A (https://ui.adsabs.harvard.edu/abs/2016JGSoc.173..949A). doi:10.1144/jgs2015-102 (https://doi.org/10.1144%2Fjgs2015-102). ISSN 0016-7649 (https://search.worldcat.org/issn/0016-7649) – via Lyell Collection Geological Society Publications.

100. Whiteside, Jessica H.; Olsen, Paul E.; Eglington, Timothy; Brookfield, Michael E.; Sambrotto, Raymond N. (22 March 2010). "Compound-specific carbon isotopes from Earth's largest flood basalt eruptions directly linked to the end-Triassic mass extinction" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2872409). *Proceedings of the National Academy of Sciences of the United States of America*. **107** (15): 6721–6725. Bibcode:2010PNAS..107.6721W (https://ui.adsabs.harvard.edu/abs/2010PNAS..107.6721W). doi:10.1073/pnas.1001706107 (https://doi.org/10.1073%2Fpnas.1001706107). PMC 2872409 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2872409). PMID 20308590 (https://pubmed.ncbi.nlm.nih.gov/20308590).
101. Zhang, Xiaolin (14 February 2024). "Constraining the Triassic–Jurassic boundary carbon cycle perturbations using high-resolution δ 13 C org records from the Haojiagou section in northwestern China" (https://www.lyellcollection.org/doi/10.1144/SP538-2021-116). *Geological Society, London, Special Publications*. **538** (1): 97–113. doi:10.1144/SP538-2021-116 (https://doi.org/10.1144%2FSP538-2021-116). ISSN 0305-8719 (https://search.worldcat.org/issn/0305-8719). Retrieved 19 February 2025 – via Lyell Collection Geological Society Publications.
102. Jeram, Andrew J.; Simms, Michael J.; Hesselbo, Stephen P.; Raine, Robert (December 2021). "Carbon isotopes, ammonites and earthquakes: Key Triassic-Jurassic boundary events in the coastal sections of south-east County Antrim, Northern Ireland, UK" (https://www.sciencedirect.com/science/article/pii/S0016787821000948). *Proceedings of the Geologists' Association*. **132** (6): 702–725. Bibcode:2021PrGA..132..702J (https://ui.adsabs.harvard.edu/abs/2021PrGA..132..702J). doi:10.1016/j.pgeola.2021.10.004 (https://doi.org/10.1016%2Fj.pgeola.2021.10.004). ISSN 0016-7878 (https://search.worldcat.org/issn/0016-7878). S2CID 244698669 (https://api.semanticscholar.org/CorpusID:244698669). Retrieved 10 November 2023.
103. Hesselbo, Stephen P.; Robinson, Stuart A.; Surlyk, Finn; Piasecki, Stefan (1 March 2002). "Terrestrial and marine extinction at the Triassic-Jurassic boundary synchronized with major carbon-cycle perturbation: A link to initiation of massive volcanism?" (https://pubs.geoscienceworld.org/gsa/geology/article-abstract/30/3/251/192348/Terrestrial-and-marine-extinction-at-the-Triassic). *Geology*. **30** (3): 251–254. Bibcode:2002Geo....30..251H (https://ui.adsabs.harvard.edu/abs/2002Geo....30..251H). doi:10.1130/0091-7613(2002)030<0251:TAMEAT>2.0.CO;2 (https://doi.org/10.1130%2F0091-7613%282002%29030%3C0251%3ATAMEAT%3E2.0.CO%3B2). Retrieved 17 April 2023.
104. Lindström, Sofie; Van de Schootbrugge, Bas; Hansen, Katrine H.; Pedersen, Gunver Krarup; Alsen, Peter; Thibault, Nicolas; Dybkjær, Karen; Bjerrum, Christian J.; Nielsen, Lars Henrik (15 July 2017). "A new correlation of Triassic–Jurassic boundary successions in NW Europe, Nevada and Peru, and the Central Atlantic Magmatic Province: A time-line for the end-Triassic mass extinction" (https://www.sciencedirect.com/science/article/abs/pii/S0031018216308926). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **478**: 80–102. Bibcode:2017PPP...478...80L (https://ui.adsabs.harvard.edu/abs/2017PPP...478...80L). doi:10.1016/j.palaeo.2016.12.025 (https://doi.org/10.1016%2Fj.palaeo.2016.12.025). hdl:1874/351998 (https://hdl.handle.net/1874%2F351998). S2CID 133353132 (https://api.semanticscholar.org/CorpusID:133353132). Retrieved 27 August 2023.
105. Ruhl, Micha; Hesselbo, Stephen P.; Al-Suwaidi, A.; Jenkyns, Hugh C.; Damborenea, S. E.; Manceñido, M. O.; Storm, M.; Mather, Tamsin A.; Riccardi, A. C. (September 2020). "On the onset of Central Atlantic Magmatic Province (CAMP) volcanism and environmental and carbon-cycle change at the Triassic–Jurassic transition (Neuquén Basin, Argentina)" (https://www.sciencedirect.com/science/article/abs/pii/S0012825220302750). *Earth-Science Reviews*. **208** 103229. Bibcode:2020ESRv..20803229R (https://ui.adsabs.harvard.edu/abs/2020ESRv..20803229R). doi:10.1016/j.earscirev.2020.103229 (https://doi.org/10.1016%2Fj.earscirev.2020.103229). hdl:10871/121712 (https://hdl.handle.net/10871%2F121712). S2CID 219913748 (https://api.semanticscholar.org/CorpusID:219913748). Retrieved 17 April 2023.

106. Ruhl, Micha; Deenen, M. H. L.; Abels, H. A.; Bonis, N. R.; Krijgsman, W.; Kürschner, W. M. (15 June 2010). "Astronomical constraints on the duration of the early Jurassic Hettangian stage and recovery rates following the end-Triassic mass extinction (St Audrie's Bay/East Quantoxhead, UK)" (https://www.sciencedirect.com/science/article/abs/pii/S0012821X10002426). *Earth and Planetary Science Letters*. **295** (1–2): 262–276. Bibcode:2010E&PSL.295..262R (https://ui.adsabs.harvard.edu/abs/2010E&PSL.295..262R). doi:10.1016/j.epsl.2010.04.008 (https://doi.org/10.1016%2Fj.epsl.2010.04.008). Retrieved 7 June 2023.
107. Van de Schootbrugge, Bas; Payne, Jonathan L.; Tomasovych, A.; Pross, J.; Fiebig, J.; Benbrahim, M.; Föllmi, Karl B.; Quan, T. M. (17 April 2008). "Carbon cycle perturbation and stabilization in the wake of the Triassic-Jurassic boundary mass-extinction event" (https://agupubs.onlinelibrary.wiley.com/doi/full/10.1029/2007GC001914). *Geochemistry, Geophysics, Geosystems*. **9** (4) 2007GC001914: 1–16. Bibcode:2008GGG.....9.4028V (https://ui.adsabs.harvard.edu/abs/2008GGG.....9.4028V). doi:10.1029/2007GC001914 (https://doi.org/10.1029%2F2007GC001914). S2CID 56000418 (https://api.semanticscholar.org/CorpusID:56000418). Retrieved 7 June 2023.
108. Percival, Lawrence M. E.; Ruhl, Micha; Jenkyns, Hugh C.; Mather, Tamsin A.; Whiteside, Jessica H. (19 June 2017). "Mercury evidence for pulsed volcanism during the end-Triassic mass extinction" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5544315). *Proceedings of the National Academy of Sciences of the United States of America*. **114** (30): 7929–7934. Bibcode:2017PNAS..114.7929P (https://ui.adsabs.harvard.edu/abs/2017PNAS..114.7929P). doi:10.1073/pnas.1705378114 (https://doi.org/10.1073%2Fpnas.1705378114). PMC 5544315 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5544315). PMID 28630294 (https://pubmed.ncbi.nlm.nih.gov/28630294).
109. Shen, Jun; Yin, Runsheng; Zhang, Shuang; Algeo, Thomas J.; Bottjer, David J.; Yu, Jianxin; Xu, Guozhen; Penman, Donald; Wang, Yongdong; Li, Liqin; Shi, Xiao; Planavsky, Noah J.; Feng, Qinglai; Xie, Shucheng (13 January 2022). "Intensified continental chemical weathering and carbon-cycle perturbations linked to volcanism during the Triassic–Jurassic transition" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8758789). *Nature Communications*. **13** (1): 299. Bibcode:2022NatCo..13..299S (https://ui.adsabs.harvard.edu/abs/2022NatCo..13..299S). doi:10.1038/s41467-022-27965-x (https://doi.org/10.1038%2Fs41467-022-27965-x). PMC 8758789 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8758789). PMID 35027546 (https://pubmed.ncbi.nlm.nih.gov/35027546). S2CID 256689306 (https://api.semanticscholar.org/CorpusID:256689306).
110. Tegner, Christian; Marzoli, Andrea; McDonald, Iain; Youbi, Nasrrddine; Lindström, Sofie (26 February 2020). "Platinum-group elements link the end-Triassic mass extinction and the Central Atlantic Magmatic Province" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7044291). *Scientific Reports*. **10** (1): 3482. Bibcode:2020NatSR..10.3482T (https://ui.adsabs.harvard.edu/abs/2020NatSR..10.3482T). doi:10.1038/s41598-020-60483-8 (https://doi.org/10.1038%2Fs41598-020-60483-8). PMC 7044291 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7044291). PMID 32103087 (https://pubmed.ncbi.nlm.nih.gov/32103087).
111. Viðarsdóttir, Halla Margrét (2020). "6". *Assessing the biodiversity crisis within the Triassic - Jurassic boundary interval using redox sensitive trace metals and stable carbon isotope geochemistry* (https://lup.lub.lu.se/luur/download?func=downloadFile&recordOId=9026651&fileOId=9026652) (MSc). Lund University. Retrieved 27 August 2023.
112. Fowell, Sarah J.; Olsen, Paul E. (May 1995). "Time calibration of Triassic/Jurassic microfloral turnover, eastern North America—Reply" (https://doi.org/10.1016%2F0040-1951%2894%2900256-9). *Tectonophysics*. **245** (1–2): 96–99. Bibcode:1995Tectp.245...96F (https://ui.adsabs.harvard.edu/abs/1995Tectp.245...96F). doi:10.1016/0040-1951(94)00256-9 (https://doi.org/10.1016%2F0040-1951%2894%2900256-9). ISSN 0040-1951 (https://search.worldcat.org/issn/0040-1951).

113. Whiteside, Jessica H.; Olsen, Paul E.; Kent, Dennis V.; Fowell, Sarah J.; Et-Touhami, Mohammed (9 February 2007). "Synchrony between the Central Atlantic magmatic province and the Triassic–Jurassic mass-extinction event?" (https://www.sciencedirect.com/science/article/pii/S0031018206004585). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **244** (1–4): 345–367. Bibcode:2007PPP...244..345W (https://ui.adsabs.harvard.edu/abs/2007PPP...244..345W). doi:10.1016/j.palaeo.2006.06.035 (https://doi.org/10.1016%2Fj.palaeo.2006.06.035). Retrieved 31 October 2024 – via Elsevier Science Direct.
114. Nomade, S.; Knight, K. B.; Beutel, E.; Renne, P. R.; Verati, C.; Féraud, G.; Marzoli, A.; Youbi, N.; Bertrand, H. (9 February 2007). "Chronology of the Central Atlantic Magmatic Province: Implications for the Central Atlantic rifting processes and the Triassic–Jurassic biotic crisis" (https://www.sciencedirect.com/science/article/pii/S0031018206004573). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **244** (1–4): 326–344. Bibcode:2007PPP...244..326N (https://ui.adsabs.harvard.edu/abs/2007PPP...244..326N). doi:10.1016/j.palaeo.2006.06.034 (https://doi.org/10.1016%2Fj.palaeo.2006.06.034). Retrieved 31 October 2024 – via Elsevier Science Direct.
115. Panfili, Giulia; Cirilli, Simonetta; Dal Corso, Jacopo; Bertrand, Hervé; Medina, Fida; Youbi, Nasrrdine; Marzoli, Andrea (January 2019). "New biostratigraphic constraints show rapid emplacement of the Central Atlantic Magmatic Province (CAMP) during the end-Triassic mass extinction interval" (https://www.sciencedirect.com/science/article/abs/pii/S0921818118303199). *Global and Planetary Change*. **172**: 60–68. Bibcode:2019GPC...172...60P (https://ui.adsabs.harvard.edu/abs/2019GPC...172...60P). doi:10.1016/j.gloplacha.2018.09.009 (https://doi.org/10.1016%2Fj.gloplacha.2018.09.009). S2CID 135154965 (https://api.semanticscholar.org/CorpusID:135154965). Retrieved 29 July 2023.
116. Yager, Joyce A.; West, A. Joshua; Corsetti, Frank A.; Berelson, William M.; Rollins, Nick E.; Rosas, Silvia; Bottjer, David M. (1 September 2017). "Duration of and decoupling between carbon isotope excursions during the end-Triassic mass extinction and Central Atlantic Magmatic Province emplacement" (https://doi.org/10.1016%2Fj.epsl.2017.05.031). *Earth and Planetary Science Letters*. **473**: 227–236. Bibcode:2017E&PSL.473..227Y (https://ui.adsabs.harvard.edu/abs/2017E&PSL.473..227Y). doi:10.1016/j.epsl.2017.05.031 (https://doi.org/10.1016%2Fj.epsl.2017.05.031).
117. Cirilli, Simonetta; Marzoli, A.; Tanner, L.; Bertrand, Hervé; Buratti, N.; Jourdan, F.; Bellieni, G.; Kontak, D.; Renne, P. R. (15 September 2009). "Latest Triassic onset of the Central Atlantic Magmatic Province (CAMP) volcanism in the Fundy Basin (Nova Scotia): New stratigraphic constraints" (https://www.sciencedirect.com/science/article/abs/pii/S0012821X09004324). *Earth and Planetary Science Letters*. **286** (3–4): 514–525. Bibcode:2009E&PSL.286..514C (https://ui.adsabs.harvard.edu/abs/2009E&PSL.286..514C). doi:10.1016/j.epsl.2009.07.021 (https://doi.org/10.1016%2Fj.epsl.2009.07.021). hdl:20.500.11937/17126 (https://hdl.handle.net/20.500.11937%2F17126). Retrieved 29 July 2023.
118. Morante, R.; Hallam, Anthony (1 May 1996). "Organic carbon isotopic record across the Triassic-Jurassic boundary in Austria and its bearing on the cause of the mass extinction" (https://pubs.geoscienceworld.org/gsa/geology/article-abstract/24/5/391/198779/Organic-carbon-isotopic-record-across-the-Triassic). *Geology*. **24** (5): 391–394. Bibcode:1996Geo....24..391M (https://ui.adsabs.harvard.edu/abs/1996Geo....24..391M). doi:10.1130/0091-7613(1996)024<0391:OCIRAT>2.3.CO;2 (https://doi.org/10.1130%2F0091-7613%281996%29024%3C0391%3AOCIRAT%3E2.3.CO%3B2). Retrieved 28 May 2023.
119. Green, Theodore; Renne, Paul R.; Keller, C. Brenhin (12 September 2022). "Continental flood basalts drive Phanerozoic extinctions" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9499591). *Proceedings of the National Academy of Sciences of the United States of America*. **119** (38) e2120441119. Bibcode:2022PNAS..11920441G (https://ui.adsabs.harvard.edu/abs/2022PNAS..11920441G). doi:10.1073/pnas.2120441119 (https://doi.org/10.1073%2Fpnas.2120441119). PMC 9499591 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9499591). PMID 36095185 (https://pubmed.ncbi.nlm.nih.gov/36095185).

120. Capriolo, Manfredo; Mills, Benjamin J. W.; Newton, Robert J.; Corso, Jacobo Dal; Dunhill, Alexander M.; Wignall, Paul Barry; Marzoli, Andrea (February 2022). "Anthropogenic-scale CO2 degassing from the Central Atlantic Magmatic Province as a driver of the end-Triassic mass extinction" (https://doi.org/10.1016%2Fj.gloplacha.2021.103731). *Global and Planetary Change*. **209** 103731. Bibcode:2022GPC...20903731C (https://ui.adsabs.harvard.edu/abs/2022GPC...20903731C). doi:10.1016/j.gloplacha.2021.103731 (https://doi.org/10.1016%2Fj.gloplacha.2021.103731). hdl:10852/91551 (https://hdl.handle.net/10852%2F91551). S2CID 245530815 (https://api.semanticscholar.org/CorpusID:245530815).
121. Slodownik, Miriam; Vajda, Vivi; Steinthorsdottir, Margret (15 February 2021). "Fossil seed fern Lepidopteris ottonis from Sweden records increasing CO2 concentration during the end-Triassic extinction event" (https://doi.org/10.1016%2Fj.palaeo.2020.110157). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **564** 110157. Bibcode:2021PPP...56410157S (https://ui.adsabs.harvard.edu/abs/2021PPP...56410157S). doi:10.1016/j.palaeo.2020.110157 (https://doi.org/10.1016%2Fj.palaeo.2020.110157). S2CID 230527791 (https://api.semanticscholar.org/CorpusID:230527791).
122. Huynh, Tran T.; Poulsen, Christopher J. (25 February 2005). "Rising atmospheric CO2 as a possible trigger for the end-Triassic mass extinction" (https://www.sciencedirect.com/science/article/abs/pii/S0031018204006285). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **217** (3–4): 223–242. Bibcode:2005PPP...217..223H (https://ui.adsabs.harvard.edu/abs/2005PPP...217..223H). doi:10.1016/j.palaeo.2004.12.004 (https://doi.org/10.1016%2Fj.palaeo.2004.12.004). Retrieved 30 May 2023.
123. Schaller, Morgan F.; Wright, James D.; Kent, Dennis V. (18 March 2011). "Atmospheric Pco2 Perturbations Associated with the Central Atlantic Magmatic Province". *Science*. **331** (6023): 1404–1409. Bibcode:2011Sci...331.1404S (https://ui.adsabs.harvard.edu/abs/2011Sci...331.1404S). doi:10.1126/science.1199011 (https://doi.org/10.1126%2Fscience.1199011). ISSN 0036-8075 (https://search.worldcat.org/issn/0036-8075). PMID 21330490 (https://pubmed.ncbi.nlm.nih.gov/21330490). S2CID 206530492 (https://api.semanticscholar.org/CorpusID:206530492).
124. Jiang, Qiang; Jourdan, Fred; Olierook, Hugo K. H.; Merle, Renaud E.; Bourdet, Julien; Fougerouse, Denis; Godel, Belinda; Walker, Alex T. (25 July 2022). "Volume and rate of volcanic CO2 emissions governed the severity of past environmental crises" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9351498). *Proceedings of the National Academy of Sciences of the United States of America*. **119** (31) e2202039119. Bibcode:2022PNAS..11902039J (https://ui.adsabs.harvard.edu/abs/2022PNAS..11902039J). doi:10.1073/pnas.2202039119 (https://doi.org/10.1073%2Fpnas.2202039119). PMC 9351498 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9351498). PMID 35878029 (https://pubmed.ncbi.nlm.nih.gov/35878029). S2CID 251067948 (https://api.semanticscholar.org/CorpusID:251067948).
125. Capriolo, Manfredo; Marzoli, Andrea; Aradi, László E.; Callegaro, Sara; Corso, Jacopo Dal; Newton, Robert J.; Mills, Benjamin J. W.; Wignall, Paul Barry; Bartoli, Omar; Baker, Don R.; Youbi, Nasrrddine; Remusat, Laurent; Spiess, Richard; Szabó, Csaba (7 April 2020). "Deep CO2 in the end-Triassic Central Atlantic Magmatic Province" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7138847). *Nature Communications*. **11** (1): 1670. Bibcode:2020NatCo..11.1670C (https://ui.adsabs.harvard.edu/abs/2020NatCo..11.1670C). doi:10.1038/s41467-020-15325-6 (https://doi.org/10.1038%2Fs41467-020-15325-6). PMC 7138847 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7138847). PMID 32265448 (https://pubmed.ncbi.nlm.nih.gov/32265448). S2CID 215404768 (https://api.semanticscholar.org/CorpusID:215404768).

126. Hua, Xia; Yin, Runsheng; Kemp, David B.; Huang, Chunju; Shen, Jun; Jin, Xin (15 December 2023). "Mercury isotope constraints on the timing and pattern of magmatism during the end-Triassic mass extinction" (https://www.sciencedirect.com/science/article/abs/pii/S0012821X2300451X). *Earth and Planetary Science Letters*. **624** 118438. Bibcode:2023E&PSL.62418438H (https://ui.adsabs.harvard.edu/abs/2023E&PSL.62418438H). doi:10.1016/j.epsl.2023.118438 (https://doi.org/10.1016%2Fj.epsl.2023.118438). Retrieved 18 February 2025 – via Elsevier Science Direct.
127. Lindström, Sofie; Callegaro, Sara; Davies, Joshua; Tegner, Christian; van de Schootbrugge, Bas; Pedersen, Gunver K.; Youbi, Nasrrddine; Sanei, Hamed; Marzoli, Andrea (1 January 2021). "Tracing volcanic emissions from the Central Atlantic Magmatic Province in the sedimentary record" (https://www.sciencedirect.com/science/article/pii/S0012825220304906). *Earth-Science Reviews*. **212** 103444. Bibcode:2021ESRv..21203444L (https://ui.adsabs.harvard.edu/abs/2021ESRv..21203444L). doi:10.1016/j.earscirev.2020.103444 (https://doi.org/10.1016%2Fj.earscirev.2020.103444). hdl:10852/81753 (https://hdl.handle.net/10852%2F81753). ISSN 0012-8252 (https://search.worldcat.org/issn/0012-8252). Retrieved 12 January 2024 – via Elsevier Science Direct.
128. Van de Schootbrugge, Bas; Quan, T. M.; Lindström, S.; Püttmann, W.; Heunisch, C.; Pross, J.; Fiebig, J.; Petschik, R.; Röhling, H.-G.; Richoz, S.; Rosenthal, Y.; Falkowski, P. G. (13 July 2009). "Floral changes across the Triassic/Jurassic boundary linked to flood basalt volcanism" (https://www.nature.com/articles/ngeo577). *Nature Geoscience*. **2** (8): 589–594. Bibcode:2009NatGe...2..589V (https://ui.adsabs.harvard.edu/abs/2009NatGe...2..589V). doi:10.1038/ngeo577 (https://doi.org/10.1038%2Fngeo577). Retrieved 17 April 2023.
129. Shen, Jun; Yin, Runsheng; Algeo, Thomas J.; Svensen, Henrik Hovland; Schoepfer, Shane D. (9 March 2022). "Mercury evidence for combustion of organic-rich sediments during the end-Triassic crisis" (https://www.researchgate.net/publication/359108712). *Nature Communications*. **13** (1): 1307. Bibcode:2022NatCo..13.1307S (https://ui.adsabs.harvard.edu/abs/2022NatCo..13.1307S). doi:10.1038/s41467-022-28891-8 (https://doi.org/10.1038%2Fs41467-022-28891-8). PMC 8907283 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8907283). PMID 35264554 (https://pubmed.ncbi.nlm.nih.gov/35264554). Retrieved 29 March 2023.
130. Davies, J. H. F. L.; Marzoli, Andrea; Bertrand, H.; Youbi, Nasrrddine; Ernesto, M.; Schaltegger, U. (31 May 2017). "End-Triassic mass extinction started by intrusive CAMP activity" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5460029). *Nature Communications*. **8** 15596. Bibcode:2017NatCo...815596D (https://ui.adsabs.harvard.edu/abs/2017NatCo...815596D). doi:10.1038/ncomms15596 (https://doi.org/10.1038%2Fncomms15596). PMC 5460029 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5460029). PMID 28561025 (https://pubmed.ncbi.nlm.nih.gov/28561025). S2CID 13323882 (https://api.semanticscholar.org/CorpusID:13323882).
131. Capriolo, Manfredo; Marzoli, Andrea; Aradi, László E.; Ackerson, Michael R.; Bartoli, Omar; Callegaro, Sara; Dal Corso, Jacopo; Ernesto, Marcia; Gouvêa Vasconcellos, Eleonora M.; De Min, Angelo; Newton, Robert J.; Szabó, Csaba (20 September 2021). "Massive methane fluxing from magma–sediment interaction in the end-Triassic Central Atlantic Magmatic Province" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8452664). *Nature Communications*. **12** (1): 5534. Bibcode:2021NatCo..12.5534C (https://ui.adsabs.harvard.edu/abs/2021NatCo..12.5534C). doi:10.1038/s41467-021-25510-w (https://doi.org/10.1038%2Fs41467-021-25510-w). hdl:11368/2996003 (https://hdl.handle.net/11368%2F2996003). ISSN 2041-1723 (https://search.worldcat.org/issn/2041-1723). PMC 8452664 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8452664). PMID 34545073 (https://pubmed.ncbi.nlm.nih.gov/34545073).

132. Heimdal, Thea H.; Jones, Morgan T.; Svensen, Henrik H. (18 May 2022). "Thermogenic carbon release from the Central Atlantic magmatic province caused major end-Triassic carbon cycle perturbations" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7275695). *Proceedings of the National Academy of Sciences of the United States of America*. **117** (22): 11968–11974. doi:10.1073/pnas.2000095117 (https://doi.org/10.1073%2Fpnas.2000095117). PMC 7275695 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7275695). PMID 32424084 (https://pubmed.ncbi.nlm.nih.gov/32424084).
133. McElwain, J. C.; Beerling, D. J.; Woodward, F. I. (27 August 1999). "Fossil Plants and Global Warming at the Triassic-Jurassic Boundary" (https://www.science.org/doi/10.1126/science.285.5432.1386). *Science*. **285** (5432): 1386–1390. doi:10.1126/science.285.5432.1386 (https://doi.org/10.1126%2Fscience.285.5432.1386). PMID 10464094 (https://pubmed.ncbi.nlm.nih.gov/10464094). Retrieved 15 November 2022.
134. Korte, Christoph; Hesselbo, Stephen P.; Jenkyns, Hugh C.; Rickaby, Rosalind E. M.; Spötl, Christoph (May 2009). "Palaeoenvironmental significance of carbon- and oxygen-isotope stratigraphy of marine Triassic–Jurassic boundary sections in SW Britain" (https://www.researchgate.net/publication/249547523). *Journal of the Geological Society*. **166** (3): 431–445. Bibcode:2009JGSoc.166..431K (https://ui.adsabs.harvard.edu/abs/2009JGSoc.166..431K). doi:10.1144/0016-76492007-177 (https://doi.org/10.1144%2F0016-76492007-177). ISSN 0016-7649 (https://search.worldcat.org/issn/0016-7649). S2CID 128814622 (https://api.semanticscholar.org/CorpusID:128814622). Retrieved 31 October 2023.
135. Pálfy, József; Zajzon, Norbert (15 June 2012). "Environmental changes across the Triassic–Jurassic boundary and coeval volcanism inferred from elemental geochemistry and mineralogy in the Kendlbachgraben section (Northern Calcareous Alps, Austria)" (https://linkinghub.elsevier.com/retrieve/pii/S0012821X12002075). *Earth and Planetary Science Letters*. 335–336: 121–134. Bibcode:2012E&PSL.335..121P (https://ui.adsabs.harvard.edu/abs/2012E&PSL.335..121P). doi:10.1016/j.epsl.2012.01.039 (https://doi.org/10.1016%2Fj.epsl.2012.01.039). Retrieved 19 June 2024 – via Elsevier Science Direct.
136. van de Schootbrugge, Bas; Koutsodendris, Andreas; Taylor, Wilson; Weston, Fabian; Wellman, Charles; Strother, Paul K. (March 2024). "Recognition of an extended record of euglenoid cysts: Implications for the end-Triassic mass extinction" (https://doi.org/10.1016%2Fj.revpalbo.2023.105043). *Review of Palaeobotany and Palynology*. **322** 105043. Bibcode:2024RPaPa.32205043V (https://ui.adsabs.harvard.edu/abs/2024RPaPa.32205043V). doi:10.1016/j.revpalbo.2023.105043 (https://doi.org/10.1016%2Fj.revpalbo.2023.105043).
137. Benton, Michael James; Twitchett, Richard J. (2003). "How to kill (almost) all life: The end-Permian extinction event". *Trends in Ecology & Evolution*. **18** (7): 358–365. doi:10.1016/S0169-5347(03)00093-4 (https://doi.org/10.1016%2FS0169-5347%2803%2900093-4). S2CID 42114053 (https://api.semanticscholar.org/CorpusID:42114053).
138. Ruhl, Micha; Bonis, Nina R.; Reichart, Gert-Jan; Sinninghe Damsté, Jaap S.; Kürschner, Wolfram M. (22 July 2011). "Atmospheric Carbon Injection Linked to End-Triassic Mass Extinction" (https://www.science.org/doi/abs/10.1126/science.1204255). *Science*. **333** (6041): 430–434. Bibcode:2011Sci...333..430R (https://ui.adsabs.harvard.edu/abs/2011Sci...333..430R). doi:10.1126/science.1204255 (https://doi.org/10.1126%2Fscience.1204255). PMID 21778394 (https://pubmed.ncbi.nlm.nih.gov/21778394). S2CID 13537776 (https://api.semanticscholar.org/CorpusID:13537776). Retrieved 9 December 2022.
139. Galli, Maria Teresa; Jadoul, Flavio; Bernasconi, Stefano M.; Weissert, Helmut (1 February 2005). "Anomalies in global carbon cycling and extinction at the Triassic/Jurassic boundary: evidence from a marine C-isotope record" (https://www.sciencedirect.com/science/article/abs/pii/S0031018204005644). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **216** (3–4): 203–214. Bibcode:2005PPP...216..203G (https://ui.adsabs.harvard.edu/abs/2005PPP...216..203G). doi:10.1016/j.palaeo.2004.11.009 (https://doi.org/10.1016%2Fj.palaeo.2004.11.009). Retrieved 9 December 2022.

140. Van de Schootbrugge, Bas; Bachan, Aviv; Suan, Guillaume; Richoz, Sylvain; Payne, Jonathan L. (19 March 2013). "Microbes, mud and methane: cause and consequence of recurrent Early Jurassic anoxia following the end-Triassic mass extinction" (https://doi.org/10.1111%2Fpala.12034). *Palaeontology*. **56** (4): 685–709. Bibcode:2013Palgy..56..685V (https://ui.adsabs.harvard.edu/abs/2013Palgy..56..685V). doi:10.1111/pala.12034 (https://doi.org/10.1111%2Fpala.12034). S2CID 76651746 (https://api.semanticscholar.org/CorpusID:76651746).
141. Lindström, Sofie; Van de Schootbrugge, Bas; Dybkjær, Karen; Pedersen, Gunver Krarup; Fiebig, Jens; Nielsen, Lars Henrik; Richoz, Sylvain (1 June 2012). "No causal link between terrestrial ecosystem change and methane release during the end-Triassic mass extinction" (https://pubs.geoscienceworld.org/gsa/geology/article-abstract/40/6/531/130915/No-causal-link-between-terrestrial-ecosystem?redirectedFrom=fulltext). *Geology*. **40** (6): 531–534. Bibcode:2012Geo....40..531L (https://ui.adsabs.harvard.edu/abs/2012Geo....40..531L). doi:10.1130/G32928.1 (https://doi.org/10.1130%2FG32928.1). Retrieved 27 August 2023.
142. Landwehrs, Jan Philip; Feulner, Georg; Hofmann, Matthias; Petri, Stefan (1 May 2020). "Climatic fluctuations modeled for carbon and sulfur emissions from end-Triassic volcanism" (https://www.sciencedirect.com/science/article/abs/pii/S0012821X20301175). *Earth and Planetary Science Letters*. **537** 116174: 1–11. Bibcode:2020E&PSL.53716174L (https://ui.adsabs.harvard.edu/abs/2020E&PSL.53716174L). doi:10.1016/j.epsl.2020.116174 (https://doi.org/10.1016%2Fj.epsl.2020.116174). S2CID 212982254 (https://api.semanticscholar.org/CorpusID:212982254). Retrieved 29 July 2023.
143. Kaiho, Kunio; Tanaka, Daisuke; Richoz, Sylvain; Jones, David S.; Saito, Ryosuke; Kameyama, Daichi; Ikeda, Masayuki; Takahashi, Satoshi; Aftabuzzaman, Md.; Fujibayashi, Megumu (1 February 2022). "Volcanic temperature changes modulated volatile release and climate fluctuations at the end-Triassic mass extinction" (https://doi.org/10.1016%2Fj.epsl.2021.117364). *Earth and Planetary Science Letters*. **579** 117364. Bibcode:2022E&PSL.57917364K (https://ui.adsabs.harvard.edu/abs/2022E&PSL.57917364K). doi:10.1016/j.epsl.2021.117364 (https://doi.org/10.1016%2Fj.epsl.2021.117364). S2CID 245922701 (https://api.semanticscholar.org/CorpusID:245922701).
144. Kent, Dennis V.; Olsen, Paul E.; Wang, Huapei; Schaller, Morgan F.; Et-Touhami, Mohammed (12 November 2024). "Correlation of sub-centennial-scale pulses of initial Central Atlantic Magmatic Province lavas and the end-Triassic extinctions" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11573653). *Proceedings of the National Academy of Sciences of the United States of America*. **121** (46) e2415486121. Bibcode:2024PNAS..12115486K (https://ui.adsabs.harvard.edu/abs/2024PNAS..12115486K). doi:10.1073/pnas.2415486121 (https://doi.org/10.1073%2Fpnas.2415486121). ISSN 0027-8424 (https://search.worldcat.org/issn/0027-8424). PMC 11573653 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11573653). PMID 39467154 (https://pubmed.ncbi.nlm.nih.gov/39467154).
145. Olsen, Paul; Sha, Jingeng; Fang, Yanan; Chang, Clara; Whiteside, Jessica H.; Kinney, Sean; Sues, Hans-Dieter; Kent, Dennis; Schaller, Morgan; Vajda, Vivi (July 2022). "Arctic ice and the ecological rise of the dinosaurs" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10883366). *Science Advances*. **8** (26) eabo6342. Bibcode:2022SciA....8O6342O (https://ui.adsabs.harvard.edu/abs/2022SciA....8O6342O). doi:10.1126/sciadv.abo6342 (https://doi.org/10.1126%2Fsciadv.abo6342). PMC 10883366 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10883366). PMID 35776799 (https://pubmed.ncbi.nlm.nih.gov/35776799). S2CID 250218588 (https://api.semanticscholar.org/CorpusID:250218588).

146. Thibodeau, Alyson M.; Ritterbush, Kathleen; Yager, Joyce A.; West, A. Joshua; Ibarra, Yadira; Bottjer, David J.; Berelson, William M.; Bergquist, Bridget A.; Corsetti, Frank A. (6 April 2016). "Mercury anomalies and the timing of biotic recovery following the end-Triassic mass extinction" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4823824). *Nature Communications*. **7** (1) 11147. Bibcode:2016NatCo...711147T (https://ui.adsabs.harvard.edu/abs/2016NatCo...711147T). doi:10.1038/ncomms11147 (https://doi.org/10.1038%2Fncomms11147). ISSN 2041-1723 (https://search.worldcat.org/issn/2041-1723). PMC 4823824 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4823824). PMID 27048776 (https://pubmed.ncbi.nlm.nih.gov/27048776).
147. Yager, Joyce A.; West, A. Joshua; Thibodeau, Alyson M.; Corsetti, Frank A.; Rigo, Manuel; Berelson, William M.; Bottjer, David J.; Greene, Sarah E.; Ibarra, Yadira; Jadoul, Flavio; Ritterbush, Kathleen A.; Rollins, Nick; Rosas, Silvia; Di Stefano, Pietro; Sulca, Debbie; Todaro, Simona; Wynn, Peter; Zimmermann, Laura; Bergquist, Bridget A. (December 2021). "Mercury contents and isotope ratios from diverse depositional environments across the Triassic–Jurassic Boundary: Towards a more robust mercury proxy for large igneous province magmatism" (https://linkinghub.elsevier.com/retrieve/pii/S0012825221002762). *Earth-Science Reviews*. **223** 103775. Bibcode:2021ESRv..22303775Y (https://ui.adsabs.harvard.edu/abs/2021ESRv..22303775Y). doi:10.1016/j.earscirev.2021.103775 (https://doi.org/10.1016%2Fj.earscirev.2021.103775). hdl:10447/518179 (https://hdl.handle.net/10447%2F518179). Retrieved 19 June 2024 – via Elsevier Science Direct.
148. Lindström, Sofie; Sanei, Haver; Van de Schootbrugge, Bas; Pedersen, Gunver Krarup; Lesher, Charles E.; Tegner, Christian; Heunisch, Carmen; Dybkjaer, Karen; Outridge, Peter M. (23 October 2019). "Volcanic mercury and mutagenesis in land plants during the end-Triassic mass extinction" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6810405). *Science Advances*. **5** (10) eaaw4018. Bibcode:2019SciA....5.4018L (https://ui.adsabs.harvard.edu/abs/2019SciA....5.4018L). doi:10.1126/sciadv.aaw4018 (https://doi.org/10.1126%2Fsciadv.aaw4018). PMC 6810405 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6810405). PMID 31681836 (https://pubmed.ncbi.nlm.nih.gov/31681836).
149. Bos, Remco; Zheng, Wang; Lindström, Sofie; Sanei, Hamed; Waajen, Irene; Fendley, Isabel M.; Mather, Tamsin A.; Wang, Yang; Rohovec, Jan; Navrátil, Tomáš; Sluijs, Appy; van de Schootbrugge, Bas (27 April 2024). "Climate-forced Hg-remobilization associated with fern mutagenesis in the aftermath of the end-Triassic extinction" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11519498). *Nature Communications*. **15** (1): 3596. Bibcode:2024NatCo..15.3596B (https://ui.adsabs.harvard.edu/abs/2024NatCo..15.3596B). doi:10.1038/s41467-024-47922-0 (https://doi.org/10.1038%2Fs41467-024-47922-0). ISSN 2041-1723 (https://search.worldcat.org/issn/2041-1723). PMC 11519498 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11519498). PMID 38678037 (https://pubmed.ncbi.nlm.nih.gov/38678037).
150. Petersen, Henrik I.; Lindström, Sofie (15 October 2012). "Synchronous Wildfire Activity Rise and Mire Deforestation at the Triassic–Jurassic Boundary" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3471965). *PLOS ONE*. **7** (10) e47236. Bibcode:2012PLoSO...747236P (https://ui.adsabs.harvard.edu/abs/2012PLoSO...747236P). doi:10.1371/journal.pone.0047236 (https://doi.org/10.1371%2Fjournal.pone.0047236). PMC 3471965 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3471965). PMID 23077574 (https://pubmed.ncbi.nlm.nih.gov/23077574).

151. Fang, Linhao; Newton, Robert J.; Zhang, Xiaoyu; Li, Hongjia; Wang, Guangli; Deng, Shenghui; Wignall, Paul Barry; Lu, Yuanzheng; Zhang, Chaokun; Li, Meijun; Wu, Huaichun; He, Tianchen; Xian, Benzhong; Shi, Shengbao; Zhu, Lei; Bottrell, Simon H.; Hesselbo, Stephen P. (15 May 2026). "Pulsed volcanic sulfur emissions linked to the end-Triassic terrestrial crisis" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC13170647). *Science Advances*. **12** (20) eadz6570. Bibcode:2026SciA...12z6570F (https://ui.adsabs.harvard.edu/abs/2026SciA...12z6570F). doi:10.1126/sciadv.adz6570 (https://doi.org/10.1126%2Fsciadv.adz6570). ISSN 2375-2548 (https://search.worldcat.org/issn/2375-2548). PMC 13170647 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC13170647). PMID 42127170 (https://pubmed.ncbi.nlm.nih.gov/42127170).
152. Hollaar, T. P.; Kent, M. S.; Lomax, B. H.; Meredith, W.; Lindström, S. L.; Bos, R.; Looy, Cindy V.; Benca, J. P.; Duijnstee, I. a. P.; Hesselbo, Stephen P.; Richoz, Sylvain; Viðarsdóttir, H. M.; Vandenbroucke, T. R. A.; Vermeer, J.; Kuhlmann, N.; Waajen-Labee, I. M.; Peterse, F.; Nierop, K. G. J.; van de Schootbrugge, Bas (21 July 2026). "Continental-scale fern savannah wildfires during end-Triassic greenhouse warming" (https://www.nature.com/articles/s41561-026-02048-4). *Nature Geoscience*. **19** (8): 973–979. doi:10.1038/s41561-026-02048-4 (https://doi.org/10.1038%2Fs41561-026-02048-4). ISSN 1752-0908 (https://search.worldcat.org/issn/1752-0908). Retrieved 12 August 2026 – via Nature Publishing Group.
153. Marynowski, Leszek; Simoneit, Bernd R. T. (1 December 2009). "Widespread Upper Triassic to Lower Jurassic Wildfire Records from Poland: Evidence from Charcoal and Pyrolytic Polycyclic Aromatic Hydrocarbons" (https://bioone.org/journals/palaios/volume-24/issue-12/palo.2009.p09-044r/WIDESPREAD-UPPER-TRIASSIC-TO-LOWER-JURASSIC-WILDFIRE-RECORDS-FROM-POLAND/10.2110/palo.2009.p09-044r.short). *PALAIOS*. **24** (12): 785–798. Bibcode:2009Palai..24..785M (https://ui.adsabs.harvard.edu/abs/2009Palai..24..785M). doi:10.2110/palo.2009.p09-044r (https://doi.org/10.2110%2Fpalo.2009.p09-044r). S2CID 131470890 (https://api.semanticscholar.org/CorpusID:131470890). Retrieved 29 March 2023.
154. Fang, Yanan; Fang, Linhao; Deng, Shenghui; Lu, Yuanzheng; Wang, Bo; Zhao, Xiangdong; Wang, Yizhe; Zhang, Haichun; Zhang, Xinzhi; Sha, Jingeng (1 September 2021). "Carbon isotope stratigraphy across the Triassic-Jurassic boundary in the high-latitude terrestrial Junggar Basin, NW China" (https://www.sciencedirect.com/science/article/pii/S0031018221003448). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **577** 110559. Bibcode:2021PPP...57710559F (https://ui.adsabs.harvard.edu/abs/2021PPP...57710559F). doi:10.1016/j.palaeo.2021.110559 (https://doi.org/10.1016%2Fj.palaeo.2021.110559). ISSN 0031-0182 (https://search.worldcat.org/issn/0031-0182). Retrieved 12 January 2024 – via Elsevier Science Direct.
155. Zhang, Peixin; Yang, Minfang; Lu, Jing; Jiang, Zhongfeng; Zhou, Kai; Xu, Xiaotao; Wang, Lei; Wu, Li; Zhang, Yuchan; Chen, Huijuan; Zhu, Xuran; Guo, Yanghang; Ye, Huajun; Shao, Longyi; Hilton, Jason (26 January 2024). "Different wildfire types promoted two-step terrestrial plant community change across the Triassic-Jurassic transition" (https://doi.org/10.3389%2Ffevo.2024.1329533). *Frontiers in Ecology and Evolution*. **12** 1329533. Bibcode:2024FrEEv..1229533Z (https://ui.adsabs.harvard.edu/abs/2024FrEEv..1229533Z). doi:10.3389/fevo.2024.1329533 (https://doi.org/10.3389%2Ffevo.2024.1329533). ISSN 2296-701X (https://search.worldcat.org/issn/2296-701X).
156. Bos, Remco; van Zonneveld, Roel-Jan; Reumer, Jelle W.F.; Vis, Geert-Jan; Janssen, Nico; Everwijn, Teun; Sluijs, Appy; van de Schootbrugge, Bas (22 November 2024). "A high-resolution palynological and geochemical study of the end-Triassic mass-extinction based on a new cored succession at Winterswijk (the Netherlands)" (https://doi.org/10.1017%2FS0016756824000323). *Geological Magazine*. **161** e17. Bibcode:2024GeoM..161E..17B (https://ui.adsabs.harvard.edu/abs/2024GeoM..161E..17B). doi:10.1017/S0016756824000323 (https://doi.org/10.1017%2FS0016756824000323). ISSN 0016-7568 (https://search.worldcat.org/issn/0016-7568).

157. Van de Schootbrugge, Bas; Van der Weijst, C. M. H.; Hollaar, T. P.; Vecoli, M.; Strother, P. K.; Kuhlmann, N.; Thein, J.; Visscher, Henk; Van Konijnenburg-van Cittert, H.; Schobben, M. A. N.; Sluijs, Appy; Lindström, Sofie (November 2020). "Catastrophic soil loss associated with end-Triassic deforestation" (https://doi.org/10.1016%2Fj.earscirev.2020.103332). *Earth-Science Reviews*. **210** 103332. Bibcode:2020ESRv..21003332V (https://ui.adsabs.harvard.edu/abs/2020ESRv..21003332V). doi:10.1016/j.earscirev.2020.103332 (https://doi.org/10.1016%2Fj.earscirev.2020.103332). S2CID 225203547 (https://api.semanticscholar.org/CorpusID:225203547).

158. Jost, Adam B.; Bacham, Aviv; Van de Schootbrugge, Bas; Lau, Kimberly V.; Weaver, Karrie L.; Maher, Kate; Payne, Jonathan L. (26 July 2017). "Uranium isotope evidence for an expansion of marine anoxia during the end-Triassic extinction" (https://agupubs.onlinelibrary.wiley.com/doi/10.1002/2017GC006941). *Geochemistry, Geophysics, Geosystems*. **18** (8): 3093–3108. Bibcode:2017GGG....18.3093J (https://ui.adsabs.harvard.edu/abs/2017GGG....18.3093J). doi:10.1002/2017GC006941 (https://doi.org/10.1002%2F2017GC006941). hdl:1874/362214 (https://hdl.handle.net/1874%2F362214). S2CID 133679444 (https://api.semanticscholar.org/CorpusID:133679444). Retrieved 11 March 2023.

159. Hu, Fangzhi; Fu, Xiugen; Wang, Jian; Wei, Hengye; Nie, Ying; Zhang, Jian; Tian, Kangzhi (2 October 2023). "Biological extinction and photic-zone anoxia across the Triassic–Jurassic transition: insights from the Qiangtang Basin, eastern Tethys" (https://www.lyellcollection.org/doi/10.1144/jgs2022-108). *Journal of the Geological Society*. **180** (5) jgs2022-108. Bibcode:2023JGSoc.180..108H (https://ui.adsabs.harvard.edu/abs/2023JGSoc.180..108H). doi:10.1144/jgs2022-108 (https://doi.org/10.1144%2Fjgs2022-108). ISSN 0016-7649 (https://search.worldcat.org/issn/0016-7649). Retrieved 19 February 2025 – via Lyell Collection Geological Society Publications.

160. Tang, Wei; Wang, Jian; Wei, Hengye; Fu, Xiugen; Ke, Puyang (1 August 2023). "Sulfur isotopic evidence for global marine anoxia and low seawater sulfate concentration during the Late Triassic" (https://www.sciencedirect.com/science/article/abs/pii/S1367912023001207). *Journal of Asian Earth Sciences*. **251** 105659. Bibcode:2023JAESc.25105659T (https://ui.adsabs.harvard.edu/abs/2023JAESc.25105659T). doi:10.1016/j.jseaes.2023.105659 (https://doi.org/10.1016%2Fj.jseaes.2023.105659). S2CID 258091074 (https://api.semanticscholar.org/CorpusID:258091074). Retrieved 28 May 2023.

161. Wignall, Paul Barry; Bond, David P. G.; Kuwahara, Kiyoko; Kakuwa, Yoshitaka; Newton, Robert J.; Poulton, Simon W. (March 2010). "An 80 million year oceanic redox history from Permian to Jurassic pelagic sediments of the Mino-Tamba terrane, SW Japan, and the origin of four mass extinctions" (https://www.sciencedirect.com/science/article/abs/pii/S0921818110000287). *Global and Planetary Change*. **71** (1–2): 109–123. Bibcode:2010GPC....71..109W (https://ui.adsabs.harvard.edu/abs/2010GPC....71..109W). doi:10.1016/j.gloplacha.2010.01.022 (https://doi.org/10.1016%2Fj.gloplacha.2010.01.022). Retrieved 7 June 2023.

162. Quan, Tracy M.; Van de Schootbrugge, Bas; Field, M. Paul; Rosenthal, Yair; Falkowski, Paul G. (10 May 2008). "Nitrogen isotope and trace metal analyses from the Mingolsheim core (Germany): Evidence for redox variations across the Triassic-Jurassic boundary" (https://doi.org/10.1029%2F2007GB002981). *Global Biogeochemical Cycles*. **22** (2) 2007GB002981: 1–14. Bibcode:2008GBioC..22.2014Q (https://ui.adsabs.harvard.edu/abs/2008GBioC..22.2014Q). doi:10.1029/2007GB002981 (https://doi.org/10.1029%2F2007GB002981). S2CID 56002825 (https://api.semanticscholar.org/CorpusID:56002825).

163. Larina, Ekaterina; Bottjer, David P.; Corsetti, Frank A.; Zonneveld, John-Paul; Celestian, Aaron J.; Bailey, Jake V. (11 December 2019). "Uppermost Triassic phosphorites from Williston Lake, Canada: link to fluctuating euxinic-anoxic conditions in northeastern Panthalassa before the end-Triassic mass extinction" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6906467). *Scientific Reports*. **9** (1): 18790. Bibcode:2019NatSR...918790L (https://ui.adsabs.harvard.edu/abs/2019NatSR...918790L). doi:10.1038/s41598-019-55162-2 (https://doi.org/10.1038%2Fs41598-019-55162-2). PMC 6906467 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6906467). PMID 31827166 (https://pubmed.ncbi.nlm.nih.gov/31827166).

164. Clement, Annaka M.; Tackett, Lydia S.; Marolt, Samuel (15 March 2024). "Biosediment assemblages reveal disrupted silica cycling and redox conditions throughout the Rhaetian Stage: Evidence for a precursor event to the end-Triassic mass extinction" (https://doi.org/10.1016%2Fj.palaeo.2024.112034). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **638** 112034. Bibcode:2024PPP...63812034C (https://ui.adsabs.harvard.edu/abs/2024PPP...63812034C). doi:10.1016/j.palaeo.2024.112034 (https://doi.org/10.1016%2Fj.palaeo.2024.112034).
165. McCabe, Kayla E.; Marroquín, Selva M.; Caruthers, Andrew H.; Owens, Jeremy D.; Golding, Martyn L.; Reid, Rachel E. B.; Them II, Theodore R.; Trabucho-Alexandre, João P.; Veenma, Yorick P.; Gill, Benjamin C. (1 April 2026). "Deoxygenation in the equatorial Panthalassan Ocean predated the end-Triassic mass extinction" (https://www.nature.com/articles/s43247-026-03362-w). *Communications Earth & Environment*. **7** (1): 460. Bibcode:2026ComEE...7..460M (https://ui.adsabs.harvard.edu/abs/2026ComEE...7..460M). doi:10.1038/s43247-026-03362-w (https://doi.org/10.1038%2Fs43247-026-03362-w). ISSN 2662-4435 (https://search.worldcat.org/issn/2662-4435). Retrieved 12 August 2026 – via Nature Publishing Group.
166. Larina, Ekaterina; Bottjer, David P.; Corsetti, Frank A.; Thibodeau, Alyson M.; Berelson, William M.; West, A. Joshua; Yager, Joyce A. (15 December 2021). "Ecosystem change and carbon cycle perturbation preceded the end-Triassic mass extinction" (https://doi.org/10.1016%2Fj.epsl.2021.117180). *Earth and Planetary Science Letters*. **576** 117180. Bibcode:2021E&PSL.57617180L (https://ui.adsabs.harvard.edu/abs/2021E&PSL.57617180L). doi:10.1016/j.epsl.2021.117180 (https://doi.org/10.1016%2Fj.epsl.2021.117180). S2CID 244179806 (https://api.semanticscholar.org/CorpusID:244179806).
167. Schoepfer, Shane D.; Shen, Jun; Sano, Hiroyoshi; Algeo, Thomas J. (January 2022). "Onset of environmental disturbances in the Panthalassic Ocean over one million years prior to the Triassic-Jurassic boundary mass extinction" (https://linkinghub.elsevier.com/retrieve/pii/S0012825221003718). *Earth-Science Reviews*. **224** 103870. Bibcode:2022ESRv..22403870S (https://ui.adsabs.harvard.edu/abs/2022ESRv..22403870S). doi:10.1016/j.earscirev.2021.103870 (https://doi.org/10.1016%2Fj.earscirev.2021.103870). S2CID 244473296 (https://api.semanticscholar.org/CorpusID:244473296). Retrieved 22 November 2023.
168. Van de Schootbrugge, Bas; Tremolada, F.; Rosenthal, Y.; Bailey, T. R.; Feist-Burkhardt, S.; Brinkhuis, Henk; Pross, J.; Kent, D. V.; Falkowski, P. G. (9 February 2007). "End-Triassic calcification crisis and blooms of organic-walled 'disaster species' " (https://www.sciencedirect.com/science/article/abs/pii/S0031018206004457). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **244** (1–4): 126–141. Bibcode:2007PPP...244..126V (https://ui.adsabs.harvard.edu/abs/2007PPP...244..126V). doi:10.1016/j.palaeo.2006.06.026 (https://doi.org/10.1016%2Fj.palaeo.2006.06.026). Retrieved 30 May 2023.
169. Zhang, Peixin; Yang, Minfang; Lu, Jing; Jiang, Zhongfeng; Zhou, Kai; Xu, Xiaotao; Guo, Yanghang; Chen, Huijuan; Shao, Longyi; Capriolo, Manfredo; Hilton, Jason (August 2026). "Two-step intensification of continental chemical weathering during the Triassic–Jurassic transition linked to CAMP emplacement" (https://www.sciencedirect.com/science/article/pii/S0921818126002626). *Global and Planetary Change*. **263** 105541. Bibcode:2026GPC...26305541Z (https://ui.adsabs.harvard.edu/abs/2026GPC...26305541Z). doi:10.1016/j.gloplacha.2026.105541 (https://doi.org/10.1016%2Fj.gloplacha.2026.105541). Retrieved 12 August 2026 – via Elsevier Science Direct.
170. Li, Jing; Song, Huyue; Du, Yong; Wignall, Paul Barry; Bond, David P. G.; Grasby, Stephen E.; Song, Haijun; Dal Corso, Jacopo; Tian, Li; Chu, Daoliang (5 June 2025). "Spatial and temporal heterogeneity of the marine nitrogen cycle during the end-Triassic mass extinction" (https://www.sciencedirect.com/science/article/pii/S0009254125001421). *Chemical Geology*. **682** 122752. doi:10.1016/j.chemgeo.2025.122752 (https://doi.org/10.1016%2Fj.chemgeo.2025.122752). Retrieved 15 November 2025 – via Elsevier Science Direct.

171. Wan, Ruoqi; Yuan, Chengshuai; Liu, Sheng-Ao; Fang, Linhao; Shen, Jun; Wang, Xiaomei (17 October 2024). "Intensified continental weathering and reductive surface runoff during the Triassic–Jurassic transition" (https://pubs.geoscienceworld.org/geology/article/53/1/13/649334/Intensified-continental-weathering-and-reductive). *Geology*. **53** (1): 13–17. doi:10.1130/G52551.1 (https://doi.org/10.1130%2FG52551.1). ISSN 0091-7613 (https://search.worldcat.org/issn/0091-7613). Retrieved 18 February 2025 – via GeoScienceWorld.
172. Xing, Kai-Chen; Wang, Feng; Teng, Fang-Zhen; Xu, Wen-Liang; Li, Ming; Sun, Yue-Wu; Yang, De-Bin (5 November 2022). "High-latitude climatic response across the Triassic-Jurassic boundary recorded by Mg-Cu-Zn isotopes" (https://www.sciencedirect.com/science/article/pii/S0009254122003795). *Chemical Geology*. **610** 121085. Bibcode:2022ChGeo.61021085X (https://ui.adsabs.harvard.edu/abs/2022ChGeo.61021085X). doi:10.1016/j.chemgeo.2022.121085 (https://doi.org/10.1016%2Fj.chemgeo.2022.121085). Retrieved 19 February 2025 – via Elsevier Science Direct.
173. Bonis, N.R.; Ruhl, M.; Kürschner, W.M. (15 April 2010). "Climate change driven black shale deposition during the end-Triassic in the western Tethys" (https://linkinghub.elsevier.com/retrieve/pii/S0031018209002326). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **290** (1–4): 151–159. Bibcode:2010PPP...290..151B (https://ui.adsabs.harvard.edu/abs/2010PPP...290..151B). doi:10.1016/j.palaeo.2009.06.016 (https://doi.org/10.1016%2Fj.palaeo.2009.06.016). Retrieved 22 November 2023.
174. Somlyay, Anna; Palcsu, László; Kiss, Gabriella Ilona; Clarkson, Matthew O.; Kovács, Emma Blanka; Vallner, Zsolt; Zajzon, Norbert; Pálfy, József (15 July 2023). "Uranium isotope evidence for extensive seafloor anoxia after the end-Triassic mass extinction" (https://www.sciencedirect.com/science/article/pii/S0012821X23002030). *Earth and Planetary Science Letters*. **614** 118190. Bibcode:2023E&PSL.61418190S (https://ui.adsabs.harvard.edu/abs/2023E&PSL.61418190S). doi:10.1016/j.epsl.2023.118190 (https://doi.org/10.1016%2Fj.epsl.2023.118190). hdl:10831/107736 (https://hdl.handle.net/10831%2F107736). Retrieved 18 February 2025 – via Elsevier Science Direct.
175. Heszler, Bernát; Katchinoff, Joachim; Palcsu, László; Horváth, Anikó; Vallner, Zsolt; Kovács, Emma Blanka; Planavsky, Noah J.; Pálfy, József (19 March 2024). "Marine Strontium Isotope Evolution at the Triassic-Jurassic Transition Links Transient Changes in Continental Weathering to Volcanism of the Central Atlantic Magmatic Province" (https://doi.org/10.1029%2F2024GC011464). *Geochemistry, Geophysics, Geosystems*. **25** (3) e2024GC011464. Bibcode:2024GGG....2511464H (https://ui.adsabs.harvard.edu/abs/2024GGG....2511464H). doi:10.1029/2024GC011464 (https://doi.org/10.1029%2F2024GC011464). hdl:10831/113116 (https://hdl.handle.net/10831%2F113116). ISSN 1525-2027 (https://search.worldcat.org/issn/1525-2027).
176. Luo, Genming; Richoz, Sylvain; van de Schootbrugge, Bas; Algeo, Thomas J.; Xie, Shucheng; Ono, Shuhei; Summons, Roger E. (15 June 2018). "Multiple sulfur-isotopic evidence for a shallowly stratified ocean following the Triassic-Jurassic boundary mass extinction" (https://linkinghub.elsevier.com/retrieve/pii/S0016703718302126). *Geochimica et Cosmochimica Acta*. **231**: 73–87. Bibcode:2018GeCoA.231...73L (https://ui.adsabs.harvard.edu/abs/2018GeCoA.231...73L). doi:10.1016/j.gca.2018.04.015 (https://doi.org/10.1016%2Fj.gca.2018.04.015). hdl:1874/366656 (https://hdl.handle.net/1874%2F366656). S2CID 134614697 (https://api.semanticscholar.org/CorpusID:134614697). Retrieved 22 November 2023.
177. Prow-Fleischer, Ashley N.; Lu, Zunli; Blättler, Clara L.; He, Tianchen; Singh, Pulkit; Kemeny, Preston Cosslett; Todes, Jordan P.; Pohl, Alexandre; Bhattacharya, Tripti; van de Schootbrugge, Bas; Wignall, Paul Barry; Todaro, Simona; Payne, Jonathan L. (5 February 2025). "Calcium isotopes support spatial redox gradients on the Tethys European margin across the Triassic-Jurassic boundary" (https://www.sciencedirect.com/science/article/abs/pii/S0009254124006107). *Chemical Geology*. **673** 122530. doi:10.1016/j.chemgeo.2024.122530 (https://doi.org/10.1016%2Fj.chemgeo.2024.122530). Retrieved 18 February 2025 – via Elsevier Science Direct.

178. Richoz, Sylvain; Van de Schootbrugge, Bas; Pross, Jörg; Püttmann, Wilhelm; Quan, Tracy M.; Lindström, Sofie; Heunisch, Carmen; Fiebig, Jens; Maquil, Robert; Schouten, Stefan; Hauzenberger, Christoph A.; Wignall, Paul Barry (12 August 2012). "Hydrogen sulphide poisoning of shallow seas following the end-Triassic extinction" (https://www.nature.com/articles/ngeo1539). *Nature Geoscience*. **5** (1): 662–667. Bibcode:2012NatGe...5..662R (https://ui.adsabs.harvard.edu/abs/2012NatGe...5..662R). doi:10.1038/ngeo1539 (https://doi.org/10.1038%2Fngeo1539). S2CID 128759882 (https://api.semanticscholar.org/CorpusID:128759882). Retrieved 22 May 2023.
179. Jaraula, Caroline M. B.; Grice, Kliti; Twitchett, Richard J.; Böttcher, Michael E.; LeMetayer, Pierre; Dastidar, Apratim G.; Opazo, L. Felipe (1 September 2013). "Elevated pCO2 leading to Late Triassic extinction, persistent photic zone euxinia, and rising sea levels" (https://pubs.geoscienceworld.org/gsa/geology/article-abstract/41/9/955/131334/Elevated-pCO2-leading-to-Late-Triassic-extinction). *Geology*. **41** (9): 955–958. Bibcode:2013Geo....41..955J (https://ui.adsabs.harvard.edu/abs/2013Geo....41..955J). doi:10.1130/G34183.1 (https://doi.org/10.1130%2FG34183.1). Retrieved 30 May 2023.
180. Williford, Kenneth H.; Foriel, Juliet; Ward, Peter D.; Steig, Eric J. (1 September 2009). "Major perturbation in sulfur cycling at the Triassic-Jurassic boundary" (https://pubs.geoscienceworld.org/gsa/geology/article-abstract/37/9/835/30031/Major-perturbation-in-sulfur-cycling-at-the). *Geology*. **37** (9): 835–838. Bibcode:2009Geo....37..835W (https://ui.adsabs.harvard.edu/abs/2009Geo....37..835W). doi:10.1130/G30054A.1 (https://doi.org/10.1130%2FG30054A.1). Retrieved 7 June 2023.
181. Schoepfer, Shane D.; Algeo, Thomas J.; Ward, Peter Douglas; Williford, Kenneth H.; Haggart, James W. (1 October 2016). "Testing the limits in a greenhouse ocean: Did low nitrogen availability limit marine productivity during the end-Triassic mass extinction?" (https://doi.org/10.1016%2Fj.epsl.2016.06.050). *Earth and Planetary Science Letters*. **451**: 138–148. Bibcode:2016E&PSL.451..138S (https://ui.adsabs.harvard.edu/abs/2016E&PSL.451..138S). doi:10.1016/j.epsl.2016.06.050 (https://doi.org/10.1016%2Fj.epsl.2016.06.050). ISSN 0012-821X (https://search.worldcat.org/issn/0012-821X).
182. Beith, Sarah J.; Fox, Calum P.; Marshall, John E. A.; Whiteside, Jessica H. (15 December 2021). "Recurring photic zone euxinia in the northwest Tethys impinged end-Triassic extinction recovery" (https://www.sciencedirect.com/science/article/abs/pii/S003101822100465X). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **584** 110680. Bibcode:2021PPP...58410680B (https://ui.adsabs.harvard.edu/abs/2021PPP...58410680B). doi:10.1016/j.palaeo.2021.110680 (https://doi.org/10.1016%2Fj.palaeo.2021.110680). S2CID 244263152 (https://api.semanticscholar.org/CorpusID:244263152). Retrieved 28 May 2023.
183. Hautmann, Michael (28 July 2004). "Effect of end-Triassic CO2 maximum on carbonate sedimentation and marine mass extinction". *Facies*. **50** (2): 257–261. doi:10.1007/s10347-004-0020-y (https://doi.org/10.1007%2Fs10347-004-0020-y). S2CID 130658467 (https://api.semanticscholar.org/CorpusID:130658467).
184. Ikeda, Masayuki; Hori, Rie S.; Okada, Yuki; Nakada, Ryoichi (15 December 2015). "Volcanism and deep-ocean acidification across the end-Triassic extinction event" (https://www.sciencedirect.com/science/article/pii/S0031018215005568). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **440**: 725–733. Bibcode:2015PPP...440..725I (https://ui.adsabs.harvard.edu/abs/2015PPP...440..725I). doi:10.1016/j.palaeo.2015.09.046 (https://doi.org/10.1016%2Fj.palaeo.2015.09.046). ISSN 0031-0182 (https://search.worldcat.org/issn/0031-0182). Retrieved 12 January 2024 – via Elsevier Science Direct.

185. Fox, Calum P.; Whiteside, Jessica H.; Olsen, Paul E.; Cui, Xingqian; Summons, Roger E.; Idiz, Erdem; Grice, Kliti (5 January 2022). "Two-pronged kill mechanism at the end-Triassic mass extinction" (https://doi.org/10.1130%2FG49560.1). *Geology*. **50** (4): 448–453. Bibcode:2022Geo....50..448F (https://ui.adsabs.harvard.edu/abs/2022Geo....50..448F). doi:10.1130/G49560.1 (https://doi.org/10.1130%2FG49560.1). hdl:20.500.11937/90125 (https://hdl.handle.net/20.500.11937%2F90125). S2CID 245782726 (https://api.semanticscholar.org/CorpusID:245782726).
186. Kasprak, Alex H.; Sepúlveda, Julio; Price-Waldman, Rosalyn; Williford, Kenneth H.; Schoepfer, Shane D.; Haggart, James W.; Ward, Peter D.; Summons, Roger E.; Whiteside, Jessica H. (1 April 2015). "Episodic photic zone euxinia in the northeastern Panthalassic Ocean during the end-Triassic extinction" (https://pubs.geoscienceworld.org/gsa/geology/article-abstract/43/4/307/131839/Episodic-photic-zone-euxinia-in-the-northeastern?redirectedFrom=fulltext). *Geology*. **43** (4): 307–310. Bibcode:2015Geo....43..307K (https://ui.adsabs.harvard.edu/abs/2015Geo....43..307K). doi:10.1130/G36371.1 (https://doi.org/10.1130%2FG36371.1). hdl:1721.1/107847 (https://hdl.handle.net/1721.1%2F107847). S2CID 132681136 (https://api.semanticscholar.org/CorpusID:132681136). Retrieved 10 November 2023.
187. Trudgill, Molly; Rae, James W. B.; Whiteford, Ross; Adloff, Markus; Crumpton-Banks, Jessica; Van Mourik, Michael; Burke, Andrea; Cuperus, Marieke; Corsetti, Frank A.; Doherty, Daniel; Gray, William; Greenop, Rosanna; Hong, Wei-Li; Lepland, Aivo; McIntyre, Andrew; Neiroukh, Noor; Rose, Catherine V.; Ruhl, Micha; Saunders, David; Siri, Magali M. F. R.; Steele, Robert C. J.; Stüeken, Eva E.; West, A. Joshua; Ziegler, Martin; Greene, Sarah E. (14 July 2025). "Pulses of ocean acidification at the Triassic–Jurassic boundary" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12260068). *Nature Communications*. **16** (1): 6471. Bibcode:2025NatCo..16.6471T (https://ui.adsabs.harvard.edu/abs/2025NatCo..16.6471T). doi:10.1038/s41467-025-61344-6 (https://doi.org/10.1038%2Fs41467-025-61344-6). ISSN 2041-1723 (https://search.worldcat.org/issn/2041-1723). PMC 12260068 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12260068). PMID 40659602 (https://pubmed.ncbi.nlm.nih.gov/40659602).
188. Galli, Maria Teresa; Jadoul, Flavio; Bernasconi, Stefano M.; Cirilli, Simonetta; Weissert, Helmut (9 February 2007). "Stratigraphy and palaeoenvironmental analysis of the Triassic–Jurassic transition in the western Southern Alps (Northern Italy)" (https://www.sciencedirect.com/science/article/pii/S003101820600441X). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **244** (1–4): 52–70. Bibcode:2007PPP...244...52G (https://ui.adsabs.harvard.edu/abs/2007PPP...244...52G). doi:10.1016/j.palaeo.2006.06.023 (https://doi.org/10.1016%2Fj.palaeo.2006.06.023). Retrieved 19 February 2025 – via Elsevier Science Direct.
189. Hautmann, Michael; Benton, Michael J.; Tomašových, Adam (1 July 2008). "Catastrophic ocean acidification at the Triassic-Jurassic boundary". *Neues Jahrbuch für Geologie und Paläontologie – Abhandlungen*. **249** (1): 119–127. Bibcode:2008NJGPA.249..119H (https://ui.adsabs.harvard.edu/abs/2008NJGPA.249..119H). doi:10.1127/0077-7749/2008/0249-0119 (https://doi.org/10.1127%2F0077-7749%2F2008%2F0249-0119).
190. Rigo, Manuel; Favero, Marco; Di Stefano, Pietro; Todaro, Simona (15 November 2024). "Organic carbon isotope (δ13Corg) curve and extinction trends across the Triassic/Jurassic boundary at Mt. Sparagio (Italy): A tool for global correlations between peritidal and pelagic successions" (https://doi.org/10.1016%2Fj.palaeo.2024.112440). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **654** 112440. doi:10.1016/j.palaeo.2024.112440 (https://doi.org/10.1016%2Fj.palaeo.2024.112440). hdl:10447/651193 (https://hdl.handle.net/10447%2F651193).
191. Črne, Alenka E.; Weissert, Helmut; Goričan, Špela; Bernasconi, Stefano M. (1 January 2011). "A biocalcification crisis at the Triassic-Jurassic boundary recorded in the Budva Basin (Dinarides, Montenegro)" (https://pubs.geoscienceworld.org/gsa/gsabulletin/article-abstract/123/1-2/40/125647/A-biocalcification-crisis-at-the-Triassic-Jurassic?redirectedFrom=fulltext). *Geological Society of America Bulletin*. **123** (1–2): 40–50. Bibcode:2011GSAB..123...40C (https://ui.adsabs.harvard.edu/abs/2011GSAB..123...40C). doi:10.1130/B30157.1 (https://doi.org/10.1130%2FB30157.1). Retrieved 30 May 2023.

192. Greene, Sarah E.; Bottjer, David J.; Corsetti, Frank A.; Berelson, William M.; Zonneveld, John-Paul (1 November 2012). "A subseafloor carbonate factory across the Triassic-Jurassic transition" (https://pubs.geoscienceworld.org/geology/article-abstract/40/11/1043/130752/A-subseafloor-carbonate-factory-across-the). *Geology*. **40** (11): 1043–1046. Bibcode:2012Geo....40.1043G (https://ui.adsabs.harvard.edu/abs/2012Geo....40.1043G). doi:10.1130/G33205.1 (https://doi.org/10.1130%2FG33205.1). ISSN 0091-7613 (https://search.worldcat.org/issn/0091-7613). Retrieved 19 March 2023.
193. Felber, Roland; Weissert, Helmut J.; Furrer, Heinz; Bontognali, Tomaso R. R. (30 July 2015). "The Triassic–Jurassic boundary in the shallow-water marine carbonates from the western Northern Calcareous Alps (Austria)" (https://doi.org/10.1007%2Fs00015-015-0192-1). *Swiss Journal of Geosciences*. **108** (2–3): 213–224. Bibcode:2015SwJG..108..213F (https://ui.adsabs.harvard.edu/abs/2015SwJG..108..213F). doi:10.1007/s00015-015-0192-1 (https://doi.org/10.1007%2Fs00015-015-0192-1). hdl:20.500.11850/109482 (https://hdl.handle.net/20.500.11850%2F109482). ISSN 1661-8726 (https://search.worldcat.org/issn/1661-8726).
194. Visscher, Henk; Looy, Cindy V.; Collinson, Margaret E.; Brinkhuis, Henk; Cittert, Johanna H. A. van Konijnenburg-van; Kürschner, Wolfram M.; Sephton, Mark A. (31 August 2004). "Environmental mutagenesis during the end-Permian ecological crisis" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC516500). *Proceedings of the National Academy of Sciences of the United States of America*. **101** (35): 12952–12956. Bibcode:2004PNAS..10112952V (https://ui.adsabs.harvard.edu/abs/2004PNAS..10112952V). doi:10.1073/pnas.0404472101 (https://doi.org/10.1073%2Fpnas.0404472101). ISSN 0027-8424 (https://search.worldcat.org/issn/0027-8424). PMC 516500 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC516500). PMID 15282373 (https://pubmed.ncbi.nlm.nih.gov/15282373).
195. Liu, Feng; Peng, Huiping; Marshall, John E. A.; Lomax, Barry H.; Bomfleur, Benjamin; Kent, Matthew S.; Fraser, Wesley T.; Jardine, Phillip E. (6 January 2023). "Dying in the Sun: Direct evidence for elevated UV-B radiation at the end-Permian mass extinction" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9821938). *Science Advances*. **9** (1) eabo6102. Bibcode:2023SciA....9O6102L (https://ui.adsabs.harvard.edu/abs/2023SciA....9O6102L). doi:10.1126/sciadv.abo6102 (https://doi.org/10.1126%2Fsciadv.abo6102). PMC 9821938 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9821938). PMID 36608140 (https://pubmed.ncbi.nlm.nih.gov/36608140).
196. Van de Schootbrugge, Bas; Wignall, Paul Barry (26 October 2015). "A tale of two extinctions: converging end-Permian and end-Triassic scenarios" (https://pubs.geoscienceworld.org/geolmag/article-abstract/153/2/332/251216/A-tale-of-two-extinctions-converging-end-Permian?redirectedFrom=fulltext). *Geological Magazine*. **153** (2): 332–354. doi:10.1017/S0016756815000643 (https://doi.org/10.1017%2FS0016756815000643). hdl:1874/329922 (https://hdl.handle.net/1874%2F329922). S2CID 131750128 (https://api.semanticscholar.org/CorpusID:131750128). Retrieved 26 May 2023.
197. T. Parrish, Judith (1993). "Climate of the Supercontinent Pangea" (https://www.ldeo.columbia.edu/~polsen/cpcp/parrish_93_sm.pdf) (PDF). *The Journal of Geology*. **101** (2): 215–233. Bibcode:1993JG....101..215P (https://ui.adsabs.harvard.edu/abs/1993JG....101..215P). doi:10.1086/648217 (https://doi.org/10.1086%2F648217). JSTOR 30081148 (https://www.jstor.org/stable/30081148). S2CID 128757269 (https://api.semanticscholar.org/CorpusID:128757269).
198. Cleveland, David M.; Nordt, Lee C.; Dworkin, Steven I.; Atchley, Stacy C. (1 November 2008). "Pedogenic carbonate isotopes as evidence for extreme climatic events preceding the Triassic-Jurassic boundary: Implications for the biotic crisis?" (https://pubs.geoscienceworld.org/gsa/gsabulletin/article-abstract/120/11-12/1408/2241/Pedogenic-carbonate-isotopes-as-evidence-for). *Geological Society of America Bulletin*. **120** (11–12): 1408–1415. Bibcode:2008GSAB..120.1408C (https://ui.adsabs.harvard.edu/abs/2008GSAB..120.1408C). doi:10.1130/B26332.1 (https://doi.org/10.1130%2FB26332.1). Retrieved 22 May 2023.

199. Rizzi, Małgorzata; Thibault, Nicolas; Ullmann, Clemens V.; Ruhl, Micha; Olsen, Troels K.; Moreau, Julien; Clémence, Marie-Emilie; Mette, Wolfgang; Korte, Christoph (1 August 2020). "Sedimentology and carbon isotope stratigraphy of the Rhaetian Hochalm section (Late Triassic, Austria)" (https://www.sciencedirect.com/science/article/pii/S0921818120301016). *Global and Planetary Change*. **191** 103210. Bibcode:2020GPC...19103210R (https://ui.adsabs.harvard.edu/abs/2020GPC...19103210R). doi:10.1016/j.gloplacha.2020.103210 (https://doi.org/10.1016%2Fj.gloplacha.2020.103210). hdl:10871/121120 (https://hdl.handle.net/10871%2F121120). ISSN 0921-8181 (https://search.worldcat.org/issn/0921-8181). S2CID 218917014 (https://api.semanticscholar.org/CorpusID:218917014). Retrieved 26 November 2023.
200. Hallam, Anthony (September 1997). "Estimates of the amount and rate of sea-level change across the Rhaetian—Hettangian and Pliensbachian—Toarcian boundaries (latest Triassic to early Jurassic)" (https://www.lyellcollection.org/doi/10.1144/gsjgs.154.5.0773). *Journal of the Geological Society*. **154** (5): 773–779. Bibcode:1997JGSoc.154..773H (https://ui.adsabs.harvard.edu/abs/1997JGSoc.154..773H). doi:10.1144/gsjgs.154.5.0773 (https://doi.org/10.1144%2Fgsjgs.154.5.0773). ISSN 0016-7649 (https://search.worldcat.org/issn/0016-7649). Retrieved 28 October 2024 – via Lyell Collection Geological Society Publications.
201. Urban, Ingrid; Demangel, Isaline; Krystyn, Leopold; Calner, Mikael; Kovács, Zsófia; Gradwohl, Gerit; Lernpeiss, Simon; Maurer, Florian; Richoz, Sylvain (1 June 2023). "Mid-Norian to Hettangian record and time-specific oolites during the end-Triassic Mass Extinction at Wadi Milaha, Musandam Peninsula, United Arab Emirates" (https://doi.org/10.1016%2Fj.jaesx.2023.100138). *Journal of Asian Earth Sciences*. **9** 100138. Bibcode:2023JAESX...900138U (https://ui.adsabs.harvard.edu/abs/2023JAESX...900138U). doi:10.1016/j.jaesx.2023.100138 (https://doi.org/10.1016%2Fj.jaesx.2023.100138).
202. Fox, Calum P.; Cui, Xingqian; Whiteside, Jessica H.; Olsen, Paul E.; Summons, Roger E.; Grice, Kliti (16 November 2020). "Molecular and isotopic evidence reveals the end-Triassic carbon isotope excursion is not from massive exogenous light carbon" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7720136). *Proceedings of the National Academy of Sciences of the United States of America*. **117** (48): 30171–30178. Bibcode:2020PNAS..11730171F (https://ui.adsabs.harvard.edu/abs/2020PNAS..11730171F). doi:10.1073/pnas.1917661117 (https://doi.org/10.1073%2Fpnas.1917661117). PMC 7720136 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7720136). PMID 33199627 (https://pubmed.ncbi.nlm.nih.gov/33199627).
203. Gómez, Juan J.; Goy, Antonio; Barrón, Eduardo (9 February 2007). "Events around the Triassic–Jurassic boundary in northern and eastern Spain: A review" (https://www.sciencedirect.com/science/article/pii/S0031018206004433). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **244** (1–4): 89–110. Bibcode:2007PPP...244...89G (https://ui.adsabs.harvard.edu/abs/2007PPP...244...89G). doi:10.1016/j.palaeo.2006.06.025 (https://doi.org/10.1016%2Fj.palaeo.2006.06.025). Retrieved 19 February 2025 – via Elsevier Science Direct.
204. Hesselbo, Stephen P.; Robinson, Stuart A.; Surlyk, Finn (May 2004). "Sea-level change and facies development across potential Triassic–Jurassic boundary horizons, SW Britain" (https://www.lyellcollection.org/doi/10.1144/0016-764903-033). *Journal of the Geological Society*. **161** (3): 365–379. Bibcode:2004JGSoc.161..365H (https://ui.adsabs.harvard.edu/abs/2004JGSoc.161..365H). doi:10.1144/0016-764903-033 (https://doi.org/10.1144%2F0016-764903-033). ISSN 0016-7649 (https://search.worldcat.org/issn/0016-7649). Retrieved 28 October 2024 – via Lyell Collection Geological Society Publications.
205. Wignall, Paul Barry; Zonneveld, John-Paul; Newton, Robert J.; Amor, K.; Sephton, M. A.; Hartley, S. (27 September 2007). "The end Triassic mass extinction record of Williston Lake, British Columbia" (https://www.sciencedirect.com/science/article/abs/pii/S003101820700329X). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **253** (3–4): 385–406. Bibcode:2007PPP...253..385W (https://ui.adsabs.harvard.edu/abs/2007PPP...253..385W). doi:10.1016/j.palaeo.2007.06.020 (https://doi.org/10.1016%2Fj.palaeo.2007.06.020). Retrieved 28 May 2023.

206. Hodych, J. P.; Dunning, G. R. (1 January 1992). "Did the Manicouagan impact trigger end-of-Triassic mass extinction?" (https://www.researchgate.net/publication/230892382). *Geology*. **20** (1): 51–54. Bibcode:1992Geo....20...51H (https://ui.adsabs.harvard.edu/abs/1992Geo....20...51H). doi:10.1130/0091-7613(1992)020<0051:dtmite>2.3.co;2 (https://doi.org/10.1130%2F0091-7613%281992%29020%3C0051%3Adtmite%3E2.3.co%3B2). ISSN 0091-7613 (https://search.worldcat.org/issn/0091-7613).
207. Racki, Grzegorz (2010). "The Alvarez impact theory of mass extinction; limits to its applicability and the "great expectations syndrome" " (https://www.app.pan.pl/archive/published/app57/app20110058.pdf) (PDF). *Acta Palaeontologica Polonica*. **57** (4): 681–702. doi:10.4202/app.2011.0058 (https://doi.org/10.4202%2Fapp.2011.0058). S2CID 54021858 (https://api.semanticscholar.org/CorpusID:54021858).
208. Onoue, Tetsuji; Sato, Honami; Yamashita, Daisuke; Ikehara, Minoru; Yasukawa, Kazutaka; Fujinaga, Koichiro; Kato, Yasuhiro; Matsuoka, Atsushi (8 July 2016). "Bolide impact triggered the Late Triassic extinction event in equatorial Panthalassa" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4937377). *Scientific Reports*. **6** 29609. Bibcode:2016NatSR...629609O (https://ui.adsabs.harvard.edu/abs/2016NatSR...629609O). doi:10.1038/srep29609 (https://doi.org/10.1038%2Fsrep29609). ISSN 2045-2322 (https://search.worldcat.org/issn/2045-2322). PMC 4937377 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4937377). PMID 27387863 (https://pubmed.ncbi.nlm.nih.gov/27387863).
209. Kent, Dennis V.; Olsen, Paul E.; Lepre, Christopher; Rasmussen, Cornelia; Mundil, Roland; Gehrels, George E.; Giesler, Dominique; Irmis, Randall B.; Geissman, John W.; Parker, William G. (16 October 2019). "Magnetochronology of the entire Chinle Formation (Norian age) in a scientific drill core from Petrified Forest National Park (Arizona, USA) and implications for regional and global correlations in the Late Triassic". *Geochemistry, Geophysics, Geosystems*. **20** (11): 4654–4664. Bibcode:2019GGG....20.4654K (https://ui.adsabs.harvard.edu/abs/2019GGG....20.4654K). doi:10.1029/2019GC008474 (https://doi.org/10.1029%2F2019GC008474). hdl:10150/636323 (https://hdl.handle.net/10150%2F636323). ISSN 1525-2027 (https://search.worldcat.org/issn/1525-2027). S2CID 207980627 (https://api.semanticscholar.org/CorpusID:207980627).
210. Schmieder, M.; Buchner, E.; Schwarz, W. H.; Trieloff, M.; Lambert, P. (5 October 2010). "A Rhaetian $^{40}Ar/^{39}Ar$ age for the Rochechouart impact structure (France) and implications for the latest Triassic sedimentary record" (https://doi.org/10.1111%2Fj.1945-5100.2010.01070.x). *Meteoritics & Planetary Science*. **45** (8): 1225–1242. Bibcode:2010M&PS...45.1225S (https://ui.adsabs.harvard.edu/abs/2010M&PS...45.1225S). doi:10.1111/j.1945-5100.2010.01070.x (https://doi.org/10.1111%2Fj.1945-5100.2010.01070.x). S2CID 129154084 (https://api.semanticscholar.org/CorpusID:129154084).
211. Smith, Roff (16 November 2011). "Dark days of the Triassic: Lost world" (https://doi.org/10.1038%2F479287a). *Nature*. **479** (7373): 287–289. Bibcode:2011Natur.479..287S (https://ui.adsabs.harvard.edu/abs/2011Natur.479..287S). doi:10.1038/479287a (https://doi.org/10.1038%2F479287a). PMID 22094671 (https://pubmed.ncbi.nlm.nih.gov/22094671).
212. Sato, Honami; Ishikawa, Akira; Onoue, Tetsuji; Tomimatsu, Yuki; Rigo, Manuel (30 December 2021). "Sedimentary record of Upper Triassic impact in the Lagonegro Basin, southern Italy: Insights from highly siderophile elements and Re-Os isotope stratigraphy across the Norian/Rhaetian boundary" (https://www.sciencedirect.com/science/article/pii/S0009254121004496). *Chemical Geology*. **586** 120506. Bibcode:2021ChGeo.58620506S (https://ui.adsabs.harvard.edu/abs/2021ChGeo.58620506S). doi:10.1016/j.chemgeo.2021.120506 (https://doi.org/10.1016%2Fj.chemgeo.2021.120506). ISSN 0009-2541 (https://search.worldcat.org/issn/0009-2541). S2CID 239637928 (https://api.semanticscholar.org/CorpusID:239637928). Retrieved 26 November 2023.

213. Schmieder, Martin; Jourdan, Fred; Tohver, Eric; Cloutis, Edward A. (15 November 2014). "40Ar/39Ar age of the Lake Saint Martin impact structure (Canada) – Unchaining the Late Triassic terrestrial impact craters" (https://www.sciencedirect.com/science/article/abs/pii/S0012821X14005445). *Earth and Planetary Science Letters*. **406**: 37–48. Bibcode:2014E&PSL.406...37S (https://ui.adsabs.harvard.edu/abs/2014E&PSL.406...37S). doi:10.1016/j.epsl.2014.08.037 (https://doi.org/10.1016%2Fj.epsl.2014.08.037). Retrieved 30 May 2023.
214. Kent, Dennis V. (10 September 1998). "Impacts on Earth in the Late Triassic" (https://doi.org/10.1038%2F25874). *Nature*. **395** (6698): 126. Bibcode:1998Natur.395..126K (https://ui.adsabs.harvard.edu/abs/1998Natur.395..126K). doi:10.1038/25874 (https://doi.org/10.1038%2F25874). S2CID 4303109 (https://api.semanticscholar.org/CorpusID:4303109).
215. Bice, D. M.; Newton, C. R.; McCauley, S.; Reinerts, P. W.; McRoberts, C. A. (24 January 1992). "Shocked Quartz at the Triassic-Jurassic Boundary in Italy" (https://www.science.org/doi/10.1126/science.255.5043.443). *Science*. **255** (5043): 443–446. Bibcode:1992Sci...255..443B (https://ui.adsabs.harvard.edu/abs/1992Sci...255..443B). doi:10.1126/science.255.5043.443 (https://doi.org/10.1126%2Fscience.255.5043.443). PMID 17842896 (https://pubmed.ncbi.nlm.nih.gov/17842896). S2CID 28314974 (https://api.semanticscholar.org/CorpusID:28314974). Retrieved 30 May 2023.
216. Hori, Rie S.; Fujiki, Toru; Inoue, Eriko; Kimura, Jun-Ichi (9 February 2007). "Platinum group element anomalies and bioevents in the Triassic–Jurassic deep-sea sediments of Panthalassa" (https://www.sciencedirect.com/science/article/abs/pii/S0031018206004615). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **244** (1–4): 391–406. Bibcode:2007PPP...244..391H (https://ui.adsabs.harvard.edu/abs/2007PPP...244..391H). doi:10.1016/j.palaeo.2006.06.038 (https://doi.org/10.1016%2Fj.palaeo.2006.06.038). Retrieved 30 May 2023.
217. Simms, Michael J. (1 June 2003). "Uniquely extensive seismite from the latest Triassic of the United Kingdom: Evidence for bolide impact?" (https://pubs.geoscienceworld.org/gsa/geology/article-abstract/31/6/557/192639/Uniquely-extensive-seismite-from-the-latest?redirectedFrom=fulltext). *Geology*. **31** (6): 557–560. Bibcode:2003Geo....31..557S (https://ui.adsabs.harvard.edu/abs/2003Geo....31..557S). doi:10.1130/0091-7613(2003)031<0557:UESFTL>2.0.CO;2 (https://doi.org/10.1130%2F0091-7613%282003%29031%3C0557%3AUESFTL%3E2.0.CO%3B2). Archived (https://web.archive.org/web/20180602214717/https://pubs.geoscienceworld.org/gsa/geology/article-abstract/31/6/557/192639/uniquely-extensive-seismite-from-the-latest?redirectedFrom=fulltext) from the original on 2 June 2018. Retrieved 31 May 2023.
218. Ward, Peter D.; Garrison, Geoffrey H.; Haggart, James W.; Kring, David A.; Beattie, Michael J. (15 August 2004). "Isotopic evidence bearing on Late Triassic extinction events, Queen Charlotte Islands, British Columbia, and implications for the duration and cause of the Triassic/Jurassic mass extinction" (https://www.sciencedirect.com/science/article/abs/pii/S0012821X04002857). *Earth and Planetary Science Letters*. **224** (3–4): 589–600. Bibcode:2004E&PSL.224..589W (https://ui.adsabs.harvard.edu/abs/2004E&PSL.224..589W). doi:10.1016/j.epsl.2004.04.034 (https://doi.org/10.1016%2Fj.epsl.2004.04.034). Retrieved 23 November 2022.
219. Bonis, Nina R.; Ruhl, Micha; Kürschner, Wolfram R. (1 September 2010). "Milankovitch-scale palynological turnover across the Triassic–Jurassic transition at St. Audrie's Bay, SW UK" (https://pubs.geoscienceworld.org/jgs/article/167/5/877/372819/Milankovitch-scale-palynological-turnover-across). *Journal of the Geological Society*. **167** (5): 877–888. Bibcode:2010JGSoc.167..877B (https://ui.adsabs.harvard.edu/abs/2010JGSoc.167..877B). doi:10.1144/0016-76492009-141 (https://doi.org/10.1144%2F0016-76492009-141). S2CID 128896141 (https://api.semanticscholar.org/CorpusID:128896141). Retrieved 12 December 2022.

220. Singh, Pulkit; Lu, Wanyi; Lu, Zunli; Jost, Adam B.; Lau, Kimberly; Bachan, Aviv; van de Schootbrugge, Bas; Payne, Jonathan L. (March 2023). "Reduction in animal abundance and oxygen availability during and after the end-Triassic mass extinction" (https://onlinelibrary.wiley.com/doi/10.1111/gbi.12533). *Geobiology*. **21** (2): 175–192. Bibcode:2023Gbio...21..175S (https://ui.adsabs.harvard.edu/abs/2023Gbio...21..175S). doi:10.1111/gbi.12533 (https://doi.org/10.1111%2Fgbi.12533). hdl:1874/427044 (https://hdl.handle.net/1874%2F427044). ISSN 1472-4677 (https://search.worldcat.org/issn/1472-4677). PMID 36329603 (https://pubmed.ncbi.nlm.nih.gov/36329603). S2CID 253303479 (https://api.semanticscholar.org/CorpusID:253303479).
221. Pandolfi, John M.; Kiessling, Wolfgang (April 2014). "Gaining insights from past reefs to inform understanding of coral reef response to global climate change" (https://www.sciencedirect.com/science/article/abs/pii/S1877343513001826). *Current Opinion in Environmental Sustainability*. **7**: 52–58. Bibcode:2014COES....7...52P (https://ui.adsabs.harvard.edu/abs/2014COES....7...52P). doi:10.1016/j.cosust.2013.11.020 (https://doi.org/10.1016%2Fj.cosust.2013.11.020). Retrieved 12 December 2022.

Literature

- Hodych, J. P.; G. R. Dunning (1992). "Did the Manicougan impact trigger end-of-Triassic mass extinction?". *Geology*. **20** (1): 51–54. Bibcode:1992Geo....20...51H (https://ui.adsabs.harvard.edu/abs/1992Geo....20...51H). doi:10.1130/0091-7613(1992)020<0051:DTMITE>2.3.CO;2 (https://doi.org/10.1130%2F0091-7613%281992%29020%3C0051%3ADTMITE%3E2.3.CO%3B2).
- McElwain, J. C.; D. J. Beerling; F. I. Woodward (27 August 1999). "Fossil Plants and Global Warming at the Triassic–Jurassic Boundary". *Science*. **285** (5432): 1386–1390. doi:10.1126/science.285.5432.1386 (https://doi.org/10.1126%2Fscience.285.5432.1386). PMID 10464094 (https://pubmed.ncbi.nlm.nih.gov/10464094).
- McHone, J.G. (2003), Volatile emissions of Central Atlantic Magmatic Province basalts: Mass assumptions and environmental consequences, in Hames, W.E. et al., eds., *The Central Atlantic Magmatic Province: Insights from Fragments of Pangea*. American Geophysical Union Monograph 136, p. 241–254.
- Tanner, L.H.; S.G. Lucas; M.G. Chapman (2004). "Assessing the record and causes of Late Triassic extinctions". *Earth-Science Reviews*. **65** (1–2): 103–139. Bibcode:2004ESRv...65..103T (https://ui.adsabs.harvard.edu/abs/2004ESRv...65..103T). doi:10.1016/S0012-8252(03)00082-5 (https://doi.org/10.1016%2FS0012-8252%2803%2900082-5). (https://web.archive.org/web/20060306225005/http://museums.state.nm.us/nmmnh/pdf_files/TJB.pdf)
- Whiteside, Jessica H.; Paul E. Olsen; Timothy Eglinton; Michael E. Brookfield; Raymond N. Sambrotto (22 March 2010). "Compound-specific carbon isotopes from Earth's largest flood basalt eruptions directly linked to the end-Triassic mass extinction" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2872409). *Proceedings of the National Academy of Sciences of the United States of America*. **107** (15): 6721–5. Bibcode:2010PNAS..107.6721W (https://ui.adsabs.harvard.edu/abs/2010PNAS..107.6721W). doi:10.1073/pnas.1001706107 (https://doi.org/10.1073%2Fpnas.1001706107). PMC 2872409 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2872409). PMID 20308590 (https://pubmed.ncbi.nlm.nih.gov/20308590).
- Deenen, M.H.L.; M. Ruhl; N.R. Bonis; W. Krijgsman; W. Kuerscher; M. Reitsma; M.J. van Bergen (2010). "A new chronology for the end-Triassic mass extinction". *Earth and Planetary Science Letters*. **291** (1–4): 113–125. Bibcode:2010E&PSL.291..113D (https://ui.adsabs.harvard.edu/abs/2010E&PSL.291..113D). doi:10.1016/j.epsl.2010.01.003 (https://doi.org/10.1016%2Fj.epsl.2010.01.003).
- Hautmann, M. (2012). "Extinction: End-Triassic Mass Extinction". *eLS*. John Wiley & Sons, Ltd: Chichester. doi:10.1002/9780470015902.a0001655.pub3 (https://doi.org/10.1002%2F9

780470015902.a0001655.pub3). ISBN 978-0-470-01617-6. S2CID 130434497 (https://api.semanticscholar.org/CorpusID:130434497).

- Tetsuji Onoue; Honami Sato; Daisuke Yamashita; Minoru Ikehara; Kazutaka Yasukawa; Koichiro Fujinaga; Yasuhiro Kato; Atsushi Matsuoka (8 July 2016). "Bolide impact triggered the Late Triassic extinction event in equatorial Panthalassa" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4937377). *Scientific Reports*. **6** 29609. Bibcode:2016NatSR...629609O (https://ui.adsabs.harvard.edu/abs/2016NatSR...629609O). doi:10.1038/srep29609 (https://doi.org/10.1038%2Fsrep29609). PMC 4937377 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4937377). PMID 27387863 (https://pubmed.ncbi.nlm.nih.gov/27387863).

External links

- Theories on the Triassic–Jurassic Extinction (http://palaeo.gly.bris.ac.uk/Palaeofiles/Triassic/triextict.htm) Archived (https://web.archive.org/web/20170801065542/http://palaeo.gly.bris.ac.uk/palaeofiles/triassic/triextict.htm) 1 August 2017 at the Wayback Machine
- The Triassic–Jurassic Mass Extinction (https://web.archive.org/web/20090720234639/http://nai.arc.nasa.gov/impact/news_detail.cfm?ID=23&keyword=250)
- 200 million year old mystery (https://www.bbc.co.uk/news/science-environment-15272230) BBC News story, 12-Oct-2011

Retrieved from "https://en.wikipedia.org/w/index.php?title=Triassic–Jurassic_extinction&oldid=1378821292"

WIKIPEDIA
25 years of the free encyclopedia

Cretaceous–Paleogene extinction event

Clockwise from the top:

- Artist's rendering of an asteroid a few kilometers across colliding with the Earth. Such an impact would have released the equivalent energy of several million nuclear weapons detonating simultaneously;
- Badlands near Drumheller, Alberta, where erosion has exposed the K–Pg boundary;
- Complex Cretaceous–Paleogene clay layer (gray) in the Geulhemmergroeve tunnels near Geulhem, The Netherlands (finger is just below the actual Cretaceous–Paleogene boundary);
- Wyoming rock with an intermediate claystone layer that contains 1,000 times more iridium than the upper and lower layers. Picture taken at the San Diego Natural History Museum;
- Rajgad Fort's Citadel, an eroded hill from the Deccan Traps, which are another hypothesized cause of the K–Pg extinction event.

The **Cretaceous–Paleogene** (**K–Pg**) **extinction event**,[a] formerly known as the **Cretaceous–Tertiary** (**K–T**) **extinction event**,[b] was a major mass extinction of three-quarters (75%) of the plant and animal species on Earth[2][3][4] which occurred around 66 million years ago. The event caused the extinction of all the non-avian dinosaurs and most other tetrapods weighing more than 25 kg (55 lb), excepting some ectothermic species such as sea turtles and crocodilians.[5] It marked the ends of both the Cretaceous period and the Mesozoic era, and logged the beginning of the current geological era, the Cenozoic era.

In the geologic record, the K–Pg event is marked by a thin layer of sediment called the K–Pg boundary or K–T boundary, which can be found throughout the world in marine and terrestrial rocks. The boundary clay shows unusually high levels of the metal iridium,[6][7][8] which is more common in asteroids than in the Earth's crust.[9]

It is generally thought that the K–Pg extinction resulted from the impact of a massive asteroid 10 to 15 km (6 to 9 mi) wide,[10][11] creating the Chicxulub impact crater and devastating the global environment primarily through a lingering impact winter which halted photosynthesis in plants and plankton.[12][13] The impact hypothesis, also known as the Alvarez hypothesis, was bolstered by the discovery of the 180 km (112 mi) Chicxulub crater in the Gulf of Mexico's Yucatán Peninsula in the early 1990s.[14] The timing of the ejecta layer, together with the match between fossil-record ecological patterns and modeled environmental disruptions (such as darkness and cooling), supports the conclusion that the Chicxulub impact triggered the mass extinction.[9]

A 2016 drilling project into the Chicxulub peak ring confirmed that it contained granite ejected within minutes from deep in the Earth, and that it contained hardly any gypsum, the usual sulfate-containing sea floor rock in the region. The gypsum would have vaporized and dispersed as an aerosol into the atmosphere, causing longer-term effects on the climate and food chain. In October 2019, researchers proposed a mechanism of the mass extinction, arguing that the Chicxulub asteroid impact event rapidly acidified the oceans and produced long-lasting effects on the climate.[15][16]

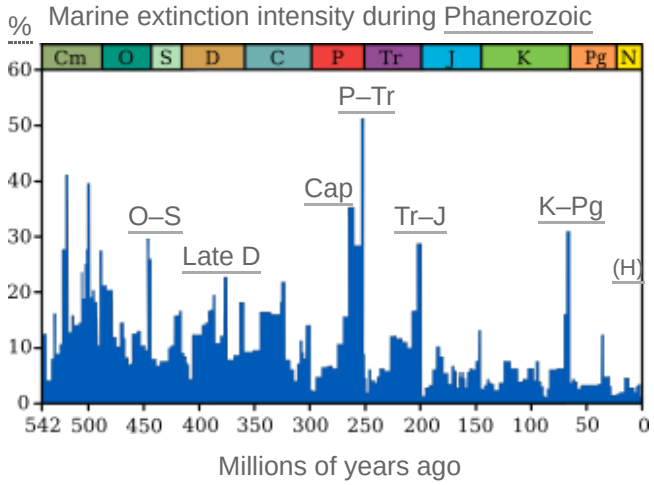
Other proposed causal or contributing factors to the extinction have included the Deccan Traps and other volcanic eruptions,[17][18] climate change, and sea level change. However, scientists in 2020 concluded that climate-modeling of the mass extinction event favored the asteroid impact and not volcanism.[19][20][21]

A wide range of terrestrial species perished in the K–Pg mass extinction, the best-known being the non-avian dinosaurs, along with many mammals, birds,[22] lizards,[23] insects,[24][25] plants, and all of the pterosaurs.[26] The K–Pg mass extinction also killed off plesiosaurs and mosasaurs and devastated teleost fish,[27] sharks, molluscs (especially ammonites and rudists, which became extinct), and many species of plankton in the oceans.[4] However, the extinction also provided evolutionary opportunities. In its wake, many groups underwent remarkable adaptive radiation—sudden and prolific divergence into new forms and species within the disrupted and emptied ecological niches. Mammals in particular diversified in the following Paleogene Period,[28] evolving new forms such as horses, whales, bats, and primates. The surviving group of dinosaurs were avians, a few species of ground and water fowl, which radiated into all modern species of birds.[29] Among the other groups, teleost fish[30] and perhaps lizards[23] also radiated into their modern species.

Extinction patterns

The K–Pg extinction event was global, rapid, and selective, eliminating a vast number of species. Based on marine fossils, it is estimated that 75% or more of all species became extinct.[4]

The event appears to have affected all of the continents at the same time. Non-avian dinosaurs, for example, are seen in the Maastrichtian of North America, Europe, Asia, Africa, South America, and Antarctica, but are not found from the Cenozoic era anywhere in the world.[31] Similarly, fossil pollen



The blue graph shows the apparent *percentage* (not the absolute number) of marine animal genera becoming extinct during any given time interval. It does not represent all marine species, just those that are readily fossilized. The labels of the traditional "Big Five" extinction events and the more recently recognised Capitanian mass extinction event are clickable links; see Extinction event for more details. *(source and image info)*

shows devastation of the plant communities in areas as far apart as New Mexico, Alaska, China, and New Zealand.[26] Nevertheless, high latitudes appear to have been less strongly affected than low latitudes.[32]

Despite the event's severity, there was significant variability in the rate of extinction between and within different clades. Species that depended on photosynthesis declined or became extinct as atmospheric particles blocked sunlight and reduced the solar energy reaching the ground. This plant extinction caused a major reshuffling of the dominant plant groups.[33] Omnivores, insectivores, and carrion-eaters survived the extinction event, perhaps because of the increased availability of their food sources. Neither strictly herbivorous nor strictly carnivorous mammals seem to have survived. Rather, the surviving mammals and birds fed on insects, worms, and snails, which in turn fed on detritus (dead plant and animal matter).[34][35][36]

In stream and lake ecosystems, few animal groups became extinct, including large forms like crocodyliforms and champsosaurs, because such ecosystems rely less directly on food from living plants, and more on detritus washed in from the land, protecting them from this extinction.[37][38] Modern crocodilians can live as scavengers and survive for months without food, and their young are small, grow slowly, and feed largely on invertebrates and dead organisms for their first few years. These characteristics have been linked to crocodilian survival at the end of the Cretaceous period.

Similar, but more complex, patterns have been found in the oceans. Extinction was more severe among animals living in the water column than among animals living on or in the sea floor. Animals in the water column are almost entirely dependent on primary production from living phytoplankton, while animals on the ocean floor always or sometimes feed on detritus.[34] The impact's blockage of sunlight devastated marine food webs which relied on photosynthetic plankton, and “the oceans may have very well returned to a single-celled state that the world had not seen in over half a billion years.”[39] Consequently, coccolithophores—vital to the open ocean ecosystem during the late Cretaceous—were nearly eradicated; however, researchers have theorized that surviving mixotrophic coccolithophores, capable of movement and ingestion of prey particles in addition to photosynthesis, were critical to restoring the algal food web over time.[40] Coccolithophorids and mollusks (including ammonites, rudists, freshwater snails, and mussels), and those organisms whose food chain included these shell builders, became extinct or suffered heavy losses. For example, it is thought that ammonites were the principal prey of mosasaurs, a group of giant marine reptiles that became extinct during the K–Pg event.[41]

The K–Pg extinction had a profound effect on the evolution of life on Earth. The elimination of dominant Cretaceous groups allowed other organisms to take their place, causing a remarkable amount of species diversification during the Paleogene Period.[28] After the K–Pg extinction event, biodiversity required

substantial time to recover, despite the existence of abundant vacant ecological niches.[34] Evidence from the Salamanca Formation suggests that biotic recovery was more rapid in the Southern Hemisphere than in the Northern Hemisphere.[42]

Despite the massive loss of life inferred to have occurred during the extinction, and a number of geologic formations worldwide that span the boundary, only a few fossil sites contain direct evidence of the mass mortality that occurred exactly at the K–Pg boundary. These include the Tanis site of the Hell Creek Formation in North Dakota, United States, which contains a high number of well-preserved fossils that appear to have been buried in a catastrophic flood event likely caused by the impact.[43] Another important site is the Hornerstown Formation in New Jersey, United States, which has a prominent layer at the K–Pg boundary known as the Main Fossiliferous Layer (MFL). It contains a mass accumulation of disarticulated vertebrate remains, likely deposited by a catastrophic impact-related flood.[44]

Microbiota

The K–Pg boundary represents one of the most dramatic turnovers in the fossil record for nannoplankton that formed the calcium deposits for which the Cretaceous is named. The turnover in this group is clearly marked at the species level.[45][46] Statistical analysis of marine losses at this time suggests that the decrease in diversity was caused more by a sharp increase in extinctions than by a decrease in speciation.[47] Major spatial differences existed in calcareous nannoplankton diversity patterns; in the Southern Hemisphere, the extinction was less severe and recovery occurred much faster than in the Northern Hemisphere.[48] Following the extinction, survivor communities dominated for several hundred thousand years. The North Pacific acted as a diversity hotspot from which later nannoplankton communities radiated as they replaced survivor faunas across the globe.[49]

The K–Pg boundary record of dinoflagellates is not so well understood, mainly because only microbial cysts provide a fossil record, and not all dinoflagellate species have cyst-forming stages, which likely causes diversity to be underestimated.[34] Recent studies indicate that there were no major shifts in dinoflagellates through the boundary layer.[50] There were blooms of the taxa *Thoracosphaera operculata* and *Braarudosphaera bigelowii* at the boundary.[51]

Radiolaria have left a geological record since at least the Ordovician times, and their mineral fossil skeletons can be tracked across the K–Pg boundary. There is no evidence of mass extinction of these organisms, and there is support for high productivity of these species in southern high latitudes as a result of cooling temperatures in the early Paleocene.[34] Approximately 46% of diatom species survived the transition from the Cretaceous to the Upper Paleocene, a significant turnover in species but not a catastrophic extinction.[34][52]

The occurrence of planktonic foraminifera across the K–Pg boundary has been studied since the 1930s.[53] Research spurred by the possibility of an impact event at the K–Pg boundary resulted in numerous publications detailing planktonic foraminiferal extinction at the boundary;[34] there is ongoing debate between groups which think the evidence indicates substantial extinction of these species at the K–Pg boundary,[54][55] and those who think the evidence supports a gradual extinction through the boundary.[56][57][58] There is strong evidence that local conditions heavily influenced diversity changes in planktonic foraminifera.[59] Low and mid-latitude communities of planktonic foraminifera experienced high extinction rates, while high latitude faunas were relatively unaffected.[60]

Numerous species of benthic foraminifera became extinct during the event, presumably because they depend on organic debris for nutrients, while biomass in the ocean is thought to have decreased. As the marine microbiota recovered, it is thought that increased speciation of benthic foraminifera resulted from the increase in food sources.[34] In some areas, such as Texas, benthic foraminifera show no sign of any major extinction event, however.[61] Phytoplankton recovery in the early Paleocene provided the food source to support large benthic foraminiferal assemblages, which are mainly detritus-feeding. Ultimate recovery of the benthic populations occurred over several stages lasting several hundred thousand years into the early Paleocene.[62][63]

Marine invertebrates

There is significant variation in the fossil record as to the extinction rate of marine invertebrates across the K–Pg boundary. The apparent rate is influenced by a lack of fossil records, rather than by extinctions.[34]

Discoscaphites iris, an ammonite from the Owl Creek Formation (Upper Cretaceous), in Owl Creek near Ripley, Mississippi

Ostracods, a class of small crustaceans that were prevalent in the upper Maastrichtian, left fossil deposits in a variety of locations. A review of these fossils shows that ostracod diversity was lower in the Paleocene than any other time in the Cenozoic. Current research cannot ascertain whether the extinctions occurred prior to, or during, the boundary interval.[64][65] Ostracods that were heavily sexually selected were more vulnerable to extinction,[66] and ostracod sexual dimorphism was significantly rarer following the mass extinction.[67]

Among decapods, extinction patterns were highly heterogeneous and cannot be neatly attributed to any particular factor. Decapods that inhabited the Western Interior Seaway were especially hard-hit, while other regions of the world's oceans were refugia that increased chances of survival into the Palaeocene.[68] Among retroplumid crabs, the genus *Costacopluma* was a notable survivor.[69]

Around 60% of late-Cretaceous scleractinian coral genera failed to cross the K–Pg boundary into the Paleocene. Further analysis of the coral extinctions shows that approximately 98% of colonial species, ones that inhabit warm, shallow tropical waters, became extinct. The solitary corals, which generally do not form reefs and inhabit colder and deeper (below the photic zone) areas of the ocean were less impacted by the K–Pg boundary. Colonial coral species rely upon symbiosis with photosynthetic algae, which collapsed due to the events surrounding the K–Pg boundary,[70][71] but the use of data from coral fossils to support K–Pg extinction and subsequent Paleocene recovery, must be weighed against the changes that occurred in coral ecosystems through the K–Pg boundary.[34]

Most species of brachiopods, a small phylum of marine invertebrates, survived the K–Pg extinction event and diversified during the early Paleocene.[34]

The numbers of bivalve genera exhibited significant diminution after the K–Pg boundary. Entire groups of bivalves, including rudists (reef-building clams) and inoceramids (giant relatives of modern scallops), became extinct at the K–Pg boundary,[72][73] with the gradual extinction of most inoceramid bivalves

beginning well before the K–Pg boundary.[74] Deposit feeders were the most common bivalves in the catastrophe's aftermath.[75] Abundance was not a factor that affected whether a bivalve taxon went extinct, according to evidence from North America.[76] Veneroid bivalves developed deeper burrowing habitats as the recovery from the crisis ensued.[77]

Rudist bivalves from the Late Cretaceous of the Omani Mountains, United Arab Emirates. Scale bar is 10 mm.

Except for nautiloids (represented by the modern order Nautilida) and coleoids (which had already diverged into modern octopodes, squids, and cuttlefish) all other species of the molluscan class Cephalopoda became extinct at the K–Pg boundary. These included the ecologically significant belemnoids, as well as the ammonoids, a group of highly diverse, numerous, and widely distributed shelled cephalopods.[78][79] The extinction of belemnites enabled surviving cephalopod clades to fill their niches.[80] Ammonite genera became extinct at or near the K–Pg boundary; there was a smaller and slower extinction of ammonite genera prior to the boundary associated with a late Cretaceous marine regression, and a small, gradual reduction in ammonite diversity occurred throughout the very late Cretaceous.[74] Researchers have pointed out that the reproductive strategy of the surviving nautiloids, which rely upon few and larger eggs, played a role in outsurviving their ammonoid counterparts through the extinction event. The ammonoids utilized a planktonic strategy of reproduction (numerous eggs and planktonic larvae), which would have been devastated by the K–Pg extinction event. Additional research has shown that subsequent to this elimination of ammonoids from the global biota, nautiloids began an evolutionary radiation into shell shapes and complexities theretofore known only from ammonoids.[78][79]

Approximately 35% of echinoderm genera became extinct at the K–Pg boundary, although taxa that thrived in low-latitude, shallow-water environments during the late Cretaceous had the highest extinction rate. Mid-latitude, deep-water echinoderms were much less affected at the K–Pg boundary. The pattern of extinction points to habitat loss, specifically the drowning of carbonate platforms, the shallow-water reefs in existence at that time, by the extinction event.[81] Atelostomatans were affected by the Lilliput effect.[82]

Terrestrial invertebrates

Insect damage to the fossilized leaves of flowering plants from fourteen sites in North America was used as a proxy for insect diversity across the K–Pg boundary and analyzed to determine the rate of extinction. Researchers found that Cretaceous sites, prior to the extinction event, had rich plant and insect-feeding diversity. During the early Paleocene, flora were relatively diverse with little predation from insects, even 1.7 million years after the extinction event.[83][84] Studies of the size of the ichnotaxon *Naktodemasis bowni*, produced by either cicada nymphs or beetle larvae, over the course of the K–Pg transition show that the Lilliput effect occurred in terrestrial invertebrates thanks to the extinction event.[85]

The extinction event produced major changes in Paleogene insect communities. Many groups of ants were present in the Cretaceous, but in the Eocene ants became dominant and diverse, with larger colonies. Butterflies diversified as well, perhaps to take the place of leaf-eating insects wiped out by the

extinction. The advanced mound-building termites, Termitidae, also appear to have risen in importance.[86]

Fish

There are fossil records of jawed fishes across the K–Pg boundary, which provide good evidence of extinction patterns of these classes of marine vertebrates. While the deep-sea realm was able to remain seemingly unaffected, there was an equal loss between the open marine apex predators and the durophagous feeders on the continental shelf. Within cartilaginous fish, approximately 7 out of the 41 families of neoselachians (modern sharks, skates, and rays) disappeared after this event and batoids (skates and rays) lost nearly all the identifiable species, while more than 90% of teleost fish (bony fish) families survived.[87][88]

In the Maastrichtian age, 28 shark families and 13 batoid families thrived, of which 25 and 9, respectively, survived the K–Pg boundary event. Forty-seven of all neoselachian genera cross the K–Pg boundary, with 85% being sharks. Batoids display with 15%, a comparably low survival rate.[87][89] Among elasmobranchs, those species that inhabited higher latitudes and lived pelagic lifestyles were more likely to survive, whereas epibenthic lifestyles and durophagy were strongly associated with the likelihood of perishing during the extinction event.[90]

There is evidence of a mass extinction of bony fishes at a fossil site immediately above the K–Pg boundary layer on Seymour Island near Antarctica, apparently precipitated by the K–Pg extinction event;[91][92] the marine and freshwater environments of fishes mitigated the environmental effects of the extinction event.[93] The result was Patterson's Gap, a period in the earliest part of the Cenozoic of decreased acanthomorph diversity,[94] although acanthomorphs diversified rapidly after the extinction.[95] Teleost fish diversified explosively after the mass extinction, filling the niches left vacant by the extinction. Groups appearing in the Paleocene and Eocene epochs include billfish, tunas, eels, and flatfish.[30]

Amphibians

There is limited evidence for extinction of amphibians at the K–Pg boundary. A study of fossil vertebrates across the K–Pg boundary in Montana concluded that no species of amphibian became extinct.[96] Yet there are several species of Maastrichtian amphibian, not included as part of this study, which are unknown from the Paleocene. These include the frog *Theatonius lancensis*[97] and the albanerpetontid *Albanerpeton galaktion*;[98] therefore, some amphibians do seem to have become extinct at the boundary. The relatively low levels of extinction seen among amphibians probably reflect the low extinction rates seen in freshwater animals.[37] Following the mass extinction, frogs radiated substantially, with 88% of modern anuran diversity being traced back to three lineages of frogs that evolved after the cataclysm.[99]

Reptiles

Choristoderes

The choristoderes (a group of semi-aquatic diapsids of uncertain position) survived across the K–Pg boundary[34] subsequently becoming extinct in the Miocene.[100] The gharial-like choristodere genus *Champsosaurus*' palatal teeth suggest that there were dietary changes among the various species across

the K–Pg event.[101]

Turtles

More than 80% of Cretaceous turtle species passed through the K–Pg boundary. All six turtle families in existence at the end of the Cretaceous survived into the Paleogene and are represented by living species.[102] Analysis of turtle survivorship in the Hell Creek Formation shows a minimum of 75% of turtle species survived.[103] Following the extinction event, turtle diversity exceeded pre-extinction levels in the Danian of North America, although in South America it remained diminished.[104] European turtles likewise recovered rapidly following the mass extinction.[105]

Lepidosauria

The rhynchocephalians, which were a globally distributed and diverse group of lepidosaurians during the early Mesozoic, had begun to decline by the mid-Cretaceous, although they remained successful in the Late Cretaceous of southern South America.[106] They are represented today by a single species, the tuatara (*Sphenodon punctatus*) found in New Zealand.[107] Outside of New Zealand, one rhynchocephalian is known to have crossed the K–Pg boundary, *Kawasphenodon peligrensis*, known from the earliest Paleocene (Danian) of Patagonia.[108]

The order Squamata, comprising lizards and snakes, first diversified during the Jurassic and continued to diversify throughout the Cretaceous.[109] They are currently the most successful and diverse group of living reptiles, with more than 10,000 extant species. The only major group of terrestrial lizards to go extinct at the end of the Cretaceous were the polyglyphanodontians, a diverse group of mainly herbivorous lizards known predominantly from the Northern Hemisphere.[110] The mosasaurs, a diverse group of large predatory marine reptiles, also became extinct. Fossil evidence indicates that squamates generally suffered very heavy losses in the K–Pg event, only recovering 10 million years after it. The extinction of Cretaceous lizards and snakes may have led to the evolution of modern groups such as iguanas, monitor lizards, and boas.[23] The diversification of crown group snakes has been linked to the biotic recovery in the aftermath of the K–Pg extinction event.[111] Pan-Gekkotans weathered the extinction event well, with multiple lineages likely surviving.[112]

Marine reptiles

$\mathbf{\Delta}^{44/42}$Ca values indicate that prior to the mass extinction, marine reptiles at the top of food webs were feeding on only one source of calcium, suggesting their populations exhibited heightened vulnerability to extinctions at the terminus of the Cretaceous.[113] Along with the aforementioned mosasaurs, plesiosaurs, represented by the families Elasmosauridae and Polycotylidae, became extinct during the event.[114][115][116][117] The ichthyosaurs had disappeared from fossil record tens of millions of years prior to the K–Pg extinction event.[118]

Crocodyliforms

Ten families of crocodilians or their close relatives are represented in the Maastrichtian fossil records, of which five died out prior to the K–Pg boundary.[119] Five families have both Maastrichtian and Paleocene fossil representatives. All of the surviving families of crocodyliforms inhabited freshwater and terrestrial environments—except for the Dyrosauridae, which lived in freshwater and marine locations. Approximately 50% of crocodyliform representatives survived across the K–Pg boundary, the only

apparent trend being that no large crocodiles survived.[34] Crocodyliform survivability across the boundary may have resulted from their aquatic niche and ability to burrow, which reduced susceptibility to negative environmental effects at the boundary.[93] Jouve and colleagues suggested in 2008 that juvenile marine crocodyliforms lived in freshwater environments as do modern marine crocodile juveniles, which would have helped them survive where other marine reptiles became extinct; freshwater environments were not so strongly affected by the K–Pg extinction event as marine environments were.[120] Among the terrestrial clade Notosuchia, only the family Sebecidae survived; the exact reasons for this pattern are not known.[121] Sebecids were large terrestrial predators, are known from the Eocene of Europe, and would survive in South America into the Miocene.[122] Tethysuchians radiated explosively after the extinction event.[123]

Pterosaurs

Two families of pterosaurs, Azhdarchidae and Nyctosauridae, were definitely present in the Maastrichtian, and they likely became extinct at the K–Pg boundary. Several other pterosaur lineages may have been present during the Maastrichtian, such as the ornithocheirids, pteranodontids, a possible tapejarid, a possible thalassodromid and a basal toothed taxon of uncertain affinities, though they are represented by fragmentary remains that are difficult to assign to any given group.[124][125] While this was occurring, modern birds were undergoing diversification; traditionally it was thought that they replaced archaic birds and pterosaur groups, possibly due to direct competition, or they simply filled empty niches,[93][126][127] but there is no correlation between pterosaur and avian diversities that are conclusive to a competition hypothesis,[128] and small pterosaurs were present in the Late Cretaceous.[129] At least some niches previously held by birds were reclaimed by pterosaurs prior to the K–Pg event.[130]

Non-avian dinosaurs

Scientists agree that all non-avian dinosaurs became extinct at the K–Pg boundary. There is no evidence that late Maastrichtian non-avian dinosaurs could burrow, swim, or dive, which suggests they were unable to shelter themselves from the worst parts of any environmental stress that occurred at the K–Pg boundary. It is possible that small dinosaurs (other than birds) did survive, but they would have been deprived of food, as herbivorous dinosaurs would have found plant material scarce and carnivores would have quickly found prey in short supply.[93]

Tyrannosaurus was among the dinosaurs living on Earth before the extinction

The growing consensus about the endothermy of dinosaurs (see dinosaur physiology) helps to understand their full extinction in contrast with their close relatives, the crocodilians. Ectothermic ("cold-blooded") crocodiles have very limited needs for food (they can survive several months without eating), while endothermic ("warm-blooded") animals of similar size need much more food to sustain their faster metabolism. Thus, under the circumstances of food chain disruption previously mentioned, non-avian dinosaurs died out,[33] while some crocodilians survived. In this context, the survival of other endothermic animals, such as some birds and mammals, could be due,

among other reasons, to their smaller needs for food, related to their small size at the extinction epoch.[131] Prolonged cold is unlikely to have been a reason for the extinction of non-avian dinosaurs given the adaptations of many dinosaurs to cold environments.[132]

Whether the extinction occurred gradually or suddenly has been debated, with both views having support from the fossil record. Interpretations of the dinosaur fossil record have pointed to both a decline in diversity and no decline in diversity during the last few million years of the Cretaceous. It may be that the quality of the dinosaur fossil record is simply not good enough to permit researchers to distinguish between the options.[133] A highly informative sequence of dinosaur-bearing rocks from the K–Pg boundary is found in western North America, particularly the late Maastrichtian-age Hell Creek Formation of Montana.[134] Comparison with the older Judith River Formation (Montana) and Dinosaur Park Formation (Alberta), which both date from approximately 75 Ma, provides information on the changes in dinosaur populations over the last 10 million years of the Cretaceous. These fossil beds are geographically limited, covering only part of one continent.[133] The middle–late Campanian formations show a greater diversity of dinosaurs than any other single group of rocks. The late Maastrichtian rocks contain the largest members of several major clades: *Tyrannosaurus*, *Ankylosaurus*, *Pachycephalosaurus*, *Triceratops*, and *Torosaurus*, which suggests food was plentiful immediately prior to the extinction.[135] A study of 29 fossil sites in Catalan Pyrenees of Europe in 2010 supports the view that dinosaurs there had great diversity until the asteroid impact, with more than 100 living species.[136] More recent research indicates that this figure is obscured by taphonomic biases and the sparsity of the continental fossil record. The results of this study, which were based on estimated real global biodiversity, showed that between 628 and 1,078 non-avian dinosaur species were alive at the end of the Cretaceous and underwent sudden extinction after the Cretaceous–Paleogene extinction event.[137] Alternatively, interpretation based on the fossil-bearing rocks along the Red Deer River in Alberta, Canada, supports the gradual extinction of non-avian dinosaurs; during the last 10 million years of the Cretaceous layers there, the number of dinosaur species seems to have decreased from about 45 to approximately 12. Other scientists have made the same assessment following their research.[138]

Several researchers support the existence of Paleocene non-avian dinosaurs. Evidence of this existence is based on the discovery of dinosaur remains in the Hell Creek Formation up to 1.3 m (4.3 ft) above and 40,000 years later than the K–Pg boundary.[139] Pollen samples recovered near a fossilized hadrosaur femur recovered in the Ojo Alamo Sandstone at the San Juan River in Colorado, indicate the hadrosaur lived during the Cenozoic, approximately 64.5 Ma (about 1 million years after the K–Pg extinction event). If their existence past the K–Pg boundary can be confirmed, these hadrosaurids would be considered a dead clade walking.[140] The scientific consensus is that these fossils were eroded from their original locations and then re-buried in much later sediments (also known as reworked fossils).[141]

Birds

Most paleontologists regard birds as the only surviving dinosaurs (see Origin of birds). It is thought that all non-avian theropods became extinct, including then-flourishing groups such as enantiornithines and hesperornithiforms.[142] Several analyses of bird fossils show divergence of species prior to the K–Pg boundary, and that duck, chicken, and ratite bird relatives coexisted with non-avian dinosaurs.[143] Large collections of bird fossils representing a range of different species provide definitive evidence for the persistence of archaic birds to within 300,000 years of the K–Pg boundary. The absence of these birds in

the Paleogene is evidence that a mass extinction of archaic birds took place there,[22] although *Qinornis* from China has been suggested to be a more basal member of Ornithurae which survived into the Paleocene.[144]

Only a small fraction of ground and water-dwelling Cretaceous bird species survived the impact, giving rise to today's birds.[22][145] The only bird group known for certain to have survived the K–Pg boundary is the Aves.[22] Avians may have been able to survive the extinction as a result of their abilities to dive, swim, or seek shelter in water and marshlands. Many species of avians can build burrows, or nest in tree holes, or termite nests, all of which provided shelter from the environmental effects at the K–Pg boundary. Long-term survival past the boundary was assured as a result of filling ecological niches left empty by extinction of non-avian dinosaurs.[93] Based on molecular sequencing and fossil dating, many species of birds (the Neoaves group in particular) appeared to radiate after the K–Pg boundary.[29][146] The open niche space and relative scarcity of predators following the K–Pg extinction allowed for adaptive radiation of various avian groups. Ratites, for example, rapidly diversified in the early Paleogene and are believed to have convergently developed flightlessness at least three to six times, often fulfilling the niche space for large herbivores once occupied by non-avian dinosaurs.[29][147][148]

Mammals

Mammalian species began diversifying approximately 30 million years prior to the K–Pg boundary. Diversification of mammals stalled across the boundary.[149] All major Late Cretaceous mammalian lineages, including monotremes (egg-laying mammals), multituberculates, metatherians (which includes modern marsupials), eutherians (which includes modern placentals), meridiolestidans,[150] and gondwanatheres[151] survived the K–Pg extinction event, although they suffered losses. In particular, metatherians largely disappeared from North America,[152] and the Asian deltatheroidans became extinct (aside from the lineage leading to *Gurbanodelta*)[153] In the Hell Creek beds of North America, according to one study, at least half of the ten known multituberculate species and all eleven metatherians species are not found above the boundary.[133] Multituberculates in Europe and North America survived relatively unscathed and quickly bounced back in the Paleocene, but Asian forms were devastated, never again to represent a significant component of mammalian fauna.[154] A recent study indicates that metatherians suffered the heaviest losses at the K–Pg event, followed by multituberculates, while eutherians recovered the quickest.[155] The mammals that survived the extinction were generally small, comparable in size to rats; this small size would have helped them find shelter in protected environments. It is postulated that some early monotremes, marsupials, and placentals were semiaquatic or burrowing, as there are multiple mammalian lineages with such habits today. Any burrowing or semiaquatic mammal would have had additional protection from K–Pg boundary environmental stresses.[93]

After the K–Pg extinction, mammals evolved to fill the niches left vacant by the dinosaurs.[156][157] Some research indicates that mammals did not explosively diversify across the K–Pg boundary, despite the ecological niches made available by the extinction of dinosaurs.[158] Several mammalian orders have been interpreted as diversifying immediately after the K–Pg boundary, including Chiroptera (bats) and Cetartiodactyla (a diverse group that today includes whales and dolphins and even-toed ungulates),[158] although recent research concludes that only marsupial orders diversified soon after the K–Pg boundary.[149] However, morphological diversification rates among eutherians after the extinction event were thrice those of before it.[159] Also significant, within the mammalian genera, new species were approximately 9.1% larger after the K–Pg boundary.[160] After about 700,000 years, some mammals had reached 50 kilos (110 pounds), a 100-fold increase over the weight of those which survived the

extinction.[161] It is thought that body sizes of placental mammalian survivors evolutionarily increased first, allowing them to fill niches after the extinctions, with brain sizes increasing later in the Eocene.[162][163]

Terrestrial plants

Plant fossils illustrate the reduction in plant species across the K–Pg boundary. There is overwhelming evidence of global disruption of plant communities at the K–Pg boundary.[164][33] Extinctions are seen both in studies of fossil pollen, and fossil leaves.[26] In North America, the data suggests massive devastation and mass extinction of plants at the K–Pg boundary sections, although there were substantial megafloral changes before the boundary.[165] In North America, approximately 57% of plant species became extinct. In high southern hemisphere latitudes, such as New Zealand and Antarctica, the mass die-off of flora caused no significant turnover in species, but dramatic and short-term changes in the relative abundance of plant groups.[83][166] European flora was also less affected, most likely due to its distance from the site of the Chicxulub impact.[167] In northern Alaska and the Anadyr-Koryak region of Russia, the flora was minimally impacted.[168][169] Another line of evidence of a major floral extinction is that the divergence rate of subviral pathogens (viroids) of angiosperms sharply decreased, which indicates an enormous reduction in the number of flowering plants.[170] However, phylogenetic evidence shows no mass angiosperm extinction.[171]

Due to the wholesale destruction of plants at the K–Pg boundary, there was a proliferation of saprotrophic organisms, such as fungi, that do not require photosynthesis and use nutrients from decaying vegetation. The dominance of fungal species lasted only a few years while the atmosphere cleared and plenty of organic matter to feed on was present. Once the atmosphere cleared photosynthetic organisms returned – initially ferns and other ground-level plants.[172]

In some regions, the Paleocene recovery of plants began with recolonizations by fern species, represented as a fern spike in the geologic record; this same pattern of fern recolonization was observed after the 1980 Mount St. Helens eruption.[173] Just two species of fern appear to have dominated the landscape for centuries after the event.[174] In the sediments below the K–Pg boundary the dominant plant remains are angiosperm pollen grains, but the boundary layer contains little pollen and is dominated by fern spores.[175] More usual pollen levels gradually resume above the boundary layer. This is reminiscent of areas blighted by modern volcanic eruptions, where the recovery is led by ferns, which are later replaced by larger angiosperm plants.[176] In North American terrestrial sequences, the extinction event is best represented by the marked discrepancy between the rich and relatively abundant late-Maastrichtian pollen record and the post-boundary fern spike.[164]

Polyploidy appears to have enhanced the ability of flowering plants to survive the extinction, probably because the additional copies of the genome such plants possessed allowed them to more readily adapt to the rapidly changing environmental conditions that followed the impact.[177]

Beyond extinction impacts, the event also caused more general changes of flora such as giving rise to neotropical rainforest biomes like the Amazonia, replacing species composition and structure of local forests during ~6 million years of recovery to former levels of plant diversity.[178][179]

Fungi

While it appears that many fungi were wiped out at the K–Pg boundary, there is some evidence that some fungal species thrived in the years after the extinction event. Microfossils from that period indicate a great increase in fungal spores, long before the resumption of plentiful fern spores in the recovery after the impact. Monoporisporites and hypha are almost exclusive microfossils for a short span during and after the iridium boundary. These saprophytes would not need sunlight, allowing them to survive during a period when the atmosphere was likely clogged with dust and sulfur aerosols.[172]

The proliferation of fungi has occurred after several extinction events, including the Permian–Triassic extinction event, the largest known mass extinction in Earth's history, with up to 96% of all species suffering extinction.[180]

Seasonal dating

Hell Creek Formation

Detailed research has investigated the season of the Chicxulub impact. A 1991 study of fossil leaves dated the extinction-associated freezing to early June.[181] A later study shifted the dating to spring season, based on the osteological evidence and stable isotope records of well-preserved bones of acipenseriform fishes. The study noted that "the palaeobotanical identities, taphonomic inferences and stratigraphic assumptions" for the June dating have since all been refuted.[182] Depalma et al. (2021) opted for the spring–summer range,[183] but During et al. (2024) reevaluated and criticized this study based on its lack of primary data, unidentified laboratory for the analyses, insufficient methods for accurate replication and problematic isotopic graphs with irregular data and error bars.[184][185] A study of fossilized fish bones found at Tanis in North Dakota suggests that the Cretaceous-Paleogene mass extinction happened during the Northern Hemisphere spring.[186][187][188][189]

Duration

The extinction's rapidity is a controversial issue because some researchers think the extinction was the result of a sudden event, while others argue that it took place over a long period. The exact length of time is difficult to determine because of the Signor–Lipps effect, where the fossil record is so incomplete that most extinct species probably died out long after the most recent fossil that has been found.[190] Scientists have also found very few continuous beds of fossil-bearing rock that cover a time range from several million years before the K–Pg extinction to several million years after it.[34]

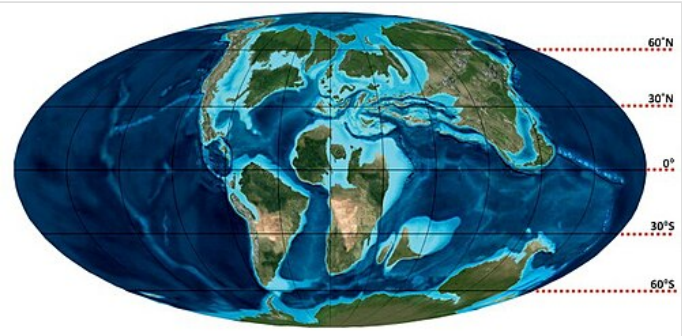
The sedimentation rate and thickness of K–Pg clay from three sites suggest rapid extinction, perhaps over a period of less than 10,000 years.[191] At one site in the Denver Basin of Colorado, after the K–Pg boundary layer was deposited, the fern spike lasted approximately 1,000 years, and no more than 71,000 years; at the same location, the earliest appearance of Cenozoic mammals occurred after approximately 185,000 years, and no more than 570,000 years, "indicating rapid rates of biotic extinction and initial recovery in the Denver Basin during this event."[192] Analysis of the carbon cycle disruptions caused by

the impact constrains them to an interval of just 5,000 years.[193] Models presented at the annual meeting of the American Geophysical Union estimated that the period of global darkness following the Chicxulub impact would have persisted in the Hell Creek Formation nearly 2 years.[194]

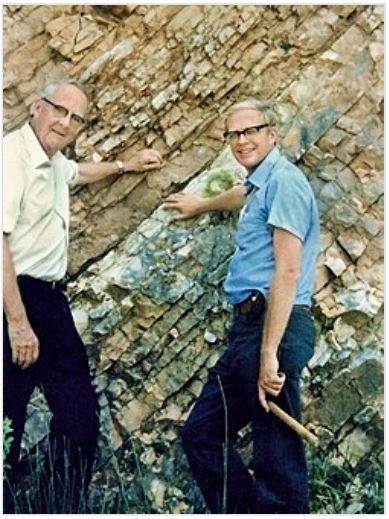
Causes

Chicxulub impact

Evidence for impact



Late Cretaceous global map (Turonian)


Luis, left, and his son Walter Alvarez, right, at the K-T boundary in Gubbio, Italy, 1981

In 1980, a team of researchers consisting of Nobel Prize-winning physicist Luis Alvarez, his son, geologist Walter Alvarez, and chemists Frank Asaro and Helen Michel discovered that sedimentary layers found all over the world at the Cretaceous–Paleogene boundary contain a concentration of iridium many times greater than normal (30, 160, and 20 times in three sections originally studied). Iridium is extremely rare in Earth's crust because it is a siderophile element which mostly sank along with iron into Earth's core during planetary differentiation.[12] Instead, iridium is more common in comets and asteroids.[9] Because of this, the Alvarez team suggested that an asteroid struck the Earth at the time of the K–Pg boundary.[12]

There were earlier speculations on the possibility of an impact event,[195] but this was the first hard evidence,[12] and since then, studies have continued to demonstrate elevated iridium levels in association with the K–Pg boundary.[8][7][6] This hypothesis was viewed as radical when first proposed, but additional evidence soon emerged. The boundary clay was found to be full of minute spherules of rock, crystallized from droplets of molten rock formed by the impact.[196][197] Shocked quartz[c] and other minerals were also identified in the K–Pg boundary.[198][199] The identification of giant tsunami beds along the Gulf Coast and the Caribbean provided more evidence,[200] and suggested that the impact might have occurred nearby, as did the discovery that the K–Pg boundary became thicker in the southern United States, with meter-thick beds of debris occurring in northern New Mexico.[26] A K–Pg boundary "cocktail" of microfossils, lithic fragments, and impact-derived material deposited by gigantic sediment gravity flows was discovered in the Caribbean that served to demarcate the impact.[201]

Further research identified the giant Chicxulub crater, buried under Chicxulub on the coast of Yucatán, as the source of the K–Pg boundary clay. Identified in 1990[14] based on work by geophysicist Glen Penfield in 1978, the crater is oval, with an average diameter of roughly 180 km (110 mi), about the size calculated by the Alvarez team.[202][203] In March 2010, an international panel of 41 scientists reviewed 20 years of scientific literature and endorsed the asteroid hypothesis, specifically the Chicxulub impact, as the cause of the extinction, ruling out other theories such as massive volcanism. They had determined that a 10-to-15-kilometer-wide (6 to 9 mi) asteroid hurtled into Earth at roughly (34,000 mph) into Chicxulub on Mexico's Yucatán Peninsula.[9]

Additional evidence for the impact event is found at the Tanis site in southwestern North Dakota, United States.[204] Tanis is part of the heavily studied Hell Creek Formation, a group of rocks spanning four states in North America renowned for many significant fossil discoveries from the Upper Cretaceous and lower Paleocene.[205] Tanis is an extraordinary and unique site because it appears to record the events from the first minutes until a few hours after the impact of the giant Chicxulub asteroid in extreme detail.[206][207] Amber from the site has been reported to contain microtektites matching those of the Chicxulub impact event.[208] Some researchers question the interpretation of the findings at the site or are skeptical of the team leader, Robert DePalma, who had not yet received his Ph.D. in geology at the time of the discovery and whose commercial activities have been regarded with suspicion.[209] Furthermore, indirect evidence of an asteroid impact as the cause of the mass extinction comes from patterns of turnover in marine plankton.[210]

Some critics of the impact theory have put forward that the impact precedes the mass extinction by about 300,000 years and thus was not its cause.[211][212] However, in a 2013 paper, Paul Renne of the Berkeley Geochronology Center dated the impact at 66.043 ± 0.011 million years ago, based on argon–argon dating. He further posits that the mass extinction occurred within 32,000 years of this date.[213] The dating of hydrothermally altered structures around the crater is consistent with this timeline.[214]

The K–Pg boundary exposure in Trinidad Lake State Park, in the Raton Basin of Colorado, shows an abrupt change from dark- to light-colored rock

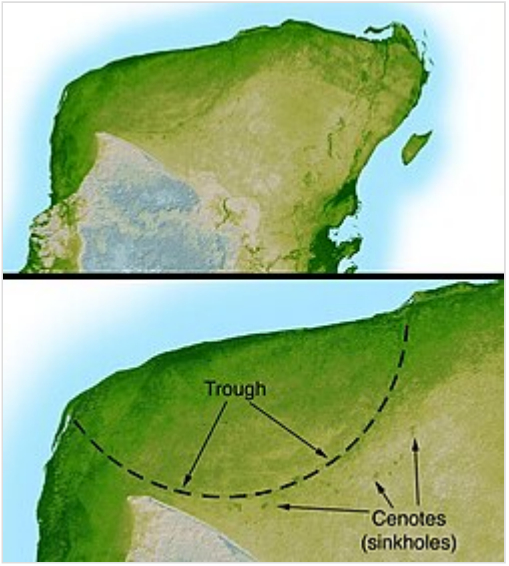
In 2007, it was proposed that the impactor belonged to the Baptistina family of asteroids.[215] This link has been doubted, though not disproved, in part because of a lack of observations of the asteroid and its family.[216] It was reported in 2009 that 298 Baptistina does not share the chemical signature of the K–Pg impactor.[217] Further, a 2011 Wide-field Infrared Survey Explorer (WISE) study of reflected light from the asteroids of the family estimated their break-up at 80 Ma, giving them insufficient time to shift orbits and impact Earth by 66 Ma.[218]

Effects of impact

The collision would have released the same energy as 100 teratonnes of TNT (4.2×10^{23} joules)—more than a billion times the energy of the atomic bombings of Hiroshima and Nagasaki.[9] The Chicxulub impact caused a global catastrophe. Some of the phenomena were brief occurrences immediately following the impact, but there were also long-term geochemical and climatic disruptions that devastated the biosphere.[220][221][222]

Location of Chicxulub crater, Mexico

The scientific consensus is that the asteroid impact at the K–Pg boundary left megatsunami deposits and sediments around the area of the Caribbean Sea and Gulf of Mexico, from the colossal waves created by the impact.[223] These deposits have been identified in the La Popa basin in northeastern Mexico,[224] platform carbonates in northeastern Brazil,[225] in Atlantic deep-sea sediments,[226] and in the form of the thickest-known layer of graded sand deposits, around 100 m (330 ft), in the Chicxulub crater itself, directly above the shocked granite ejecta. The megatsunami has been estimated at more than 100 m (330 ft) tall, as the asteroid fell into relatively shallow seas; in deep seas it would have been 4.6 km (2.9 mi) tall.[227] Fossiliferous sedimentary rocks deposited during the K–Pg impact have been found in the Gulf of Mexico area, including tsunami wash deposits carrying remains of a mangrove-type ecosystem, indicating that water in the Gulf of Mexico sloshed back and forth repeatedly after the impact; dead fish left in these shallow waters were not disturbed by scavengers.[228][229][230][231][232]



Radar topography reveals the 180 km (112 mi)-wide ring of the Chicxulub crater

The re-entry of ejecta into Earth's atmosphere included a brief (hours-long) but intense pulse of infrared radiation, cooking exposed organisms.[93] This is debated, with opponents arguing that local ferocious fires, probably limited to North America, fall short of global firestorms. This is the "Cretaceous–Paleogene firestorm debate". A paper in 2013 by a prominent modeler of nuclear winter suggested that, based on the amount of soot in the global debris layer, the entire terrestrial biosphere might have burned, implying a global soot-cloud blocking out the sun and creating an impact winter effect.[220] If widespread fires occurred this would have exterminated the most vulnerable organisms that survived the period immediately after the impact.[233] Experimental analysis suggests that any impact-induced wildfires were insufficient on their own to cause plant extinctions,[234] and much of the thermal radiation generated by the impact would have been absorbed by the atmosphere and ejecta in the lower atmosphere.[235]

Aside from the hypothesized fire effects on reduction of insolation, the impact would have created a dust cloud that blocked sunlight for up to a year, inhibiting photosynthesis.[236][13][221] The asteroid hit an area of gypsum and anhydrite rock containing a large amount of combustible hydrocarbons and sulfur,[237] much of which was vaporized, thereby injecting sulfuric acid aerosols into the stratosphere, which might have reduced sunlight reaching the Earth's surface by more than 50%.[238] Fine silicate dust also contributed to the intense impact winter,[239] as did soot from wildfires.[240][241][242] The climatic forcing of this impact winter was about 100 times more potent than that of the 1991 eruption of Mount Pinatubo.[243] According to models of the Hell Creek Formation, the onset of global darkness would have reached its maximum in only a few weeks and likely lasted upwards of two years.[194] Freezing temperatures probably lasted for at least three years.[222] At Brazos section, the sea surface temperature dropped as much as 7 °C (13 °F) for decades after the impact.[13] It would take at least ten years for such

aerosols to dissipate, and would account for the extinction of plants and phytoplankton, and subsequently herbivores and their predators. Creatures whose food chains were based on detritus would have a reasonable chance of survival.[131][221] In 2016, a scientific drilling project obtained deep rock-core samples from the peak ring around the Chicxulub impact crater. The discoveries confirmed that the rock comprising the peak ring had been shocked by immense pressure and melted in just minutes from its usual state into its present form. Unlike sea-floor deposits, the peak ring was made of granite originating much deeper in the earth, which had been ejected to the surface by the impact. Gypsum is a sulfate-containing rock usually present in the shallow seabed of the region; it had been almost entirely removed, vaporized into the atmosphere. The impactor was large enough to create a 190-kilometer-wide (120 mi) peak ring, to melt, shock, and eject deep granite, to create colossal water movements, and to eject an immense quantity of vaporized rock and sulfates into the atmosphere, where they would have persisted for several years. This worldwide dispersal of dust and sulfates would have affected climate catastrophically, led to large temperature drops, and devastated the food chain.[244][245]

Artistic impression of the asteroid slamming into tropical, shallow seas of the sulfur-rich Yucatán Peninsula in what is today Southeast Mexico.[219] The aftermath of this immense asteroid collision, which occurred approximately 66 million years ago, is believed to have caused the mass extinction of non-avian dinosaurs and many other species on Earth.[219] The impact spewed hundreds of billions of tons of sulfur into the atmosphere, producing a worldwide blackout, and freezing temperatures which persisted for at least a decade.[219]

The release of large quantities of sulphur aerosols into the atmosphere as a consequence of the impact would also have caused acid rain.[246][238] Oceans acidified as a result.[15][16] This decrease in ocean pH would kill many organisms that grow shells of calcium carbonate.[238] The heating of the atmosphere during the impact itself may have also generated nitric acid rain through the production of nitrogen oxides and their subsequent reaction with water vapour.[247][246]

After the impact winter, the Earth entered a period of global warming as a result of the vaporisation of carbonates into carbon dioxide, whose long residence time in the atmosphere ensured significant warming would occur after more short-lived cooling gases dissipated.[248] Carbon monoxide concentrations also increased and caused particularly devastating global warming because of the consequent increases in tropospheric ozone and methane concentrations.[249] The impact's injection of water vapour into the atmosphere also produced major climatic perturbations.[250]

The end-Cretaceous event is the only mass extinction definitively known to be associated with an impact, and other large extraterrestrial impacts, such as the Manicouagan Reservoir impact, do not coincide with any noticeable extinction events.[251]

Multiple impact event

Other crater-like topographic features have also been proposed as impact craters formed in connection with Cretaceous–Paleogene extinction. This suggests the possibility of near-simultaneous multiple impacts, perhaps from a fragmented asteroidal object similar to the Shoemaker–Levy 9 impact with Jupiter. In addition to the 180 km (110 mi) Chicxulub crater, there is the 24 km (15 mi) Boltysh crater in

Ukraine (65.17 ± 0.64 Ma), the 20 km (12 mi) Silverpit crater in the North Sea (59.5 ± 14.5 Ma) possibly formed by bolide impact, and the controversial and much larger 600 km (370 mi) Shiva crater. Any other craters that might have formed in the Tethys Ocean would since have been obscured by the northward tectonic drift of Africa and India.[252][253][254][255]

The river bed at the Moody Creek Mine, 7 Mile Creek / Waimatuku, Dunollie, New Zealand contains evidence of a devastating event on terrestrial plant communities at the Cretaceous–Paleogene boundary, confirming the severity and global nature of the event[164]

Deccan Traps

The Deccan Traps, which erupted close to the boundary between the Mesozoic and Cenozoic,[256][257][258] have been cited as an alternate explanation for the mass extinction.[259][260] A leading proponent of the Deccan Traps causation is Gerta Keller.[261] Before 2000, arguments that the Deccan Traps flood basalts caused the extinction were usually linked to the view that the extinction was gradual, as the flood basalt events were thought to have started around 68 Mya and lasted more than 2 million years. The most recent evidence shows that the traps erupted over a period of only 800,000 years spanning the K–Pg boundary, and therefore may be responsible for the extinction and the delayed biotic recovery thereafter.[262] Some geophysical models suggest that the Chicxulub impact 66.0 Mya could have triggered some of the largest Deccan eruptions, as well as eruptions at active volcano sites anywhere on Earth.[263][264]

The Deccan Traps could have caused extinction through several mechanisms, including the release of dust and sulfuric aerosols into the air, which might have blocked sunlight and thereby reduced photosynthesis in plants.[265] In addition, the latest Cretaceous saw a rise in global temperatures;[266][267] Deccan Traps volcanism resulted in carbon dioxide emissions that increased the greenhouse effect when the dust and aerosols cleared from the atmosphere.[268][257] Plant fossils record a 250 ppm increase in carbon dioxide concentrations across the K–Pg boundary likely attributable to Deccan Traps activity.[269] The increased carbon dioxide emissions also caused acid rain, evidenced by increased mercury deposition due to increased solubility of mercury compounds in more acidic water.[270]

Evidence for extinctions caused by the Deccan Traps includes the reduction in diversity of marine life when the climate near the K–Pg boundary increased in temperature. The temperature increased about three to four degrees very rapidly between 65.4 and 65.2 million years ago, which is very near the time of the extinction event. Not only did the climate temperature increase, but the water temperature decreased, causing a drastic decrease in marine diversity.[271] Evidence from Tunisia indicates that marine life was deleteriously affected by a major period of increased warmth and humidity linked to a pulse of intense Deccan Traps activity,[272] and that marine extinctions there began before the impact event.[273] Charophyte declines in the Songliao Basin, China before the asteroid impact have been concluded to be connected to climate changes caused by Deccan Traps activity.[274]

In the years when the Deccan Traps hypothesis was linked to a slower extinction, Luis Alvarez (d. 1988) replied that paleontologists were being misled by sparse data. While his assertion was not initially well-received, later intensive field studies of fossil beds lent weight to his claim. Eventually, most paleontologists began to accept the idea that the mass extinctions at the end of the Cretaceous were

largely or at least partly due to a massive Earth impact. Even Walter Alvarez acknowledged that other major changes might have contributed to the extinctions.[275] More recent arguments against the Deccan Traps as an extinction cause include that the timeline of Deccan Traps activity and pulses of climate change has been found by some studies to be asynchronous,[276] that palynological changes do not coincide with intervals of volcanism,[277] and that many sites show climatic stability during the latest Maastrichtian and no sign of major disruptions caused by volcanism.[278] A study published in 2020 concluded that the paleoclimatic data were most coherent with two peaks of degassing caused by the Deccan Traps eruptions, the first more than 200,000 years before the extinction, and a second peak starting around the extinction event and lasting about 355 000 years.[20] The conclusion that the Deccan Traps did not cause the extinction is accepted by the paleontologist Michael Benton.[279] Another study published in the same year argued that the long-term warming caused by the increased volcanism would have increased habitat suitability, not caused an extinction.[21]

Maastrichtian sea-level regression

There is clear evidence that sea levels fell in the final stage of the Cretaceous by more than at any other time in the Mesozoic era. In some Maastrichtian stage rock layers from various parts of the world, the later layers are terrestrial; earlier layers represent shorelines and the earliest layers represent seabeds. These layers do not show the tilting and distortion associated with mountain building, therefore the likeliest explanation is a *regression*, a drop in sea level. There is no direct evidence for the cause of the regression, but the currently accepted explanation is that the mid-ocean ridges became less active and sank under their own weight.[34][280]

A severe regression would have greatly reduced the continental shelf area, the most species-rich part of the sea, and therefore could have been enough to cause a marine mass extinction.[281][282] This change would not have caused the extinction of the ammonites. The regression would also have caused climate changes, partly by disrupting winds and ocean currents and partly by reducing the Earth's albedo and increasing global temperatures.[74] Marine regression also resulted in the loss of epeiric seas, such as the Western Interior Seaway of North America. The loss of these seas greatly altered habitats, removing coastal plains that ten million years before had been host to diverse communities such as are found in rocks of the Dinosaur Park Formation. Another consequence was an expansion of freshwater environments, since continental runoff now had longer distances to travel before reaching oceans. While this change was favorable to freshwater vertebrates, those that prefer marine environments, such as sharks, suffered.[133]

However, sea level fall as a cause of the extinction event is contradicted by other evidence, namely that sections which show no sign of marine regression still show evidence of a major drop in diversity.[283]

Multiple causes

Proponents of multiple causation view the suggested single causes as either too small to produce the vast scale of the extinction, or not likely to produce its observed taxonomic pattern. In a review article, J. David Archibald and David E. Fastovsky discussed a scenario combining three major postulated causes: volcanism, marine regression, and extraterrestrial impact. In this scenario, terrestrial and marine communities were stressed by the changes in, and loss of, habitats. Dinosaurs, as the largest vertebrates, were the first affected by environmental changes, and their diversity declined. At the same time,

particulate materials from volcanism cooled and dried areas of the globe. Then an impact event occurred, causing collapses in photosynthesis-based food chains, both in the already-stressed terrestrial food chains and in the marine food chains.[133]

Artist's rendering of a *Thescelosaurus* shortly after the K–Pg event

Based on studies at Seymour Island in Antarctica, Sierra Petersen and colleagues argue that there were two separate extinction events near the Cretaceous–Paleogene boundary, with one correlating to Deccan Trap volcanism and one correlated with the Chicxulub impact. The team analyzed combined extinction patterns using a new clumped isotope temperature record from a hiatus-free, expanded K–Pg boundary section. They documented a 7.8±3.3 °C warming synchronous with the onset of Deccan Traps volcanism and a second, smaller warming at the time of meteorite impact. They suggested that local warming had been amplified due to the simultaneous disappearance of continental or sea ice. Intra-shell variability indicates a possible reduction in seasonality after Deccan eruptions began, continuing through the meteorite event. Species extinction at Seymour Island occurred in two pulses that coincide with the two observed warming events, directly linking the end-Cretaceous extinction at this site to both volcanic and meteorite events via climate change.[284]

See also

- Climate across Cretaceous–Paleogene boundary – Prehistoric climate changes
- Late Devonian extinction – Second of the big five mass extinctions
- List of possible impact structures on Earth
- Late Ordovician mass extinction – Mass extinction event c. 444 million years ago
- Permian–Triassic extinction event – Earth's most severe extinction event
- Timeline of Cretaceous–Paleogene extinction event research – Research timeline
- Triassic–Jurassic extinction – Mass extinction ending the Triassic period

Explanatory notes

a. The abbreviation is derived from the juxtaposition of **K**, the common abbreviation for the Cretaceous, which in turn originates from the correspondent German term *Kreide*; and **Pg**, which is the abbreviation for the Paleogene.
b. The former designation includes the term 'Tertiary' (abbreviated as **T**), which is now discouraged as a formal geochronological unit by the International Commission on Stratigraphy.[1]
c. Shocked minerals have their internal structure deformed, and are created by intense pressures as in nuclear blasts and meteorite impacts.

Citations

1. Ogg, James G.; Gradstein, F. M.; Gradstein, Felix M. (2004). *A geologic time scale 2004*.

Cambridge, UK: Cambridge University Press. ISBN 978-0-521-78142-8.

2. "International Chronostratigraphic Chart" (https://web.archive.org/web/20140530005940/http://www.stratigraphy.org/index.php/ics-chart-timescale). *stratigraphy.org*. International Commission on Stratigraphy. 2015. Archived from the original (http://www.stratigraphy.org/index.php/ics-chart-timescale) on 30 May 2014. Retrieved 29 April 2015.
3. Fortey, Richard (1999). *Life: A natural history of the first four billion years of life on Earth*. Vintage. pp. 238–260. ISBN 978-0-375-70261-7.
4. Jablonski, D.; Chaloner, W. G. (1994). "Extinctions in the fossil record (and discussion)". *Philosophical Transactions of the Royal Society of London B*. **344** (1307): 11–17. doi:10.1098/rstb.1994.0045 (https://doi.org/10.1098%2Frstb.1994.0045).
5. Muench, David; Muench, Marc; Gilders, Michelle A. (2000). *Primal Forces*. Portland, Oregon: Graphic Arts Center Publishing. p. 20. ISBN 978-1-55868-522-2.
6. Jones, Heather L.; Westerhold, Thomas; Birch, Heather; et al. (18 January 2023). "Stratigraphy of the Cretaceous/Paleogene (K/Pg) boundary at the Global Stratotype Section and Point (GSSP) in El Kef, Tunisia: New insights from the El Kef Coring Project" (https://biblio.vub.ac.be/vubirfiles/105270956/Manuscript_text_revised.docx). *Geological Society of America Bulletin*. **135** (9–10): 2451. Bibcode:2023GSAB..135.2451J (https://ui.adsabs.harvard.edu/abs/2023GSAB..135.2451J). doi:10.1130/B36487.1 (https://doi.org/10.1130%2FB36487.1). S2CID 256021543 (https://api.semanticscholar.org/CorpusID:256021543).
7. Irizarry, Kayla M.; Witts, James T.; Garb, Matthew P.; et al. (15 January 2023). "Faunal and stratigraphic analysis of the basal Cretaceous-Paleogene (K-Pg) boundary event deposits, Brazos River, Texas, USA". *Palaeogeography, Palaeoclimatology, Palaeoecology*. **610** 111334. Bibcode:2023PPP...61011334I (https://ui.adsabs.harvard.edu/abs/2023PPP...61011334I). doi:10.1016/j.palaeo.2022.111334 (https://doi.org/10.1016%2Fj.palaeo.2022.111334). hdl:1983/cd8e64bb-98ce-4940-9a56-3b638da65662 (https://hdl.handle.net/1983%2Fcd8e64bb-98ce-4940-9a56-3b638da65662). S2CID 254345541 (https://api.semanticscholar.org/CorpusID:254345541).
8. Ferreira da Silva, Luiza Carine; Santos, Alessandra; Fauth, Gerson; et al. (April 2023). "High-latitude Cretaceous–Paleogene transition: New paleoenvironmental and paleoclimatic insights from Seymour Island, Antarctica". *Marine Micropaleontology*. **180** 102214. Bibcode:2023MarMP.180j2214F (https://ui.adsabs.harvard.edu/abs/2023MarMP.180j2214F). doi:10.1016/j.marmicro.2023.102214 (https://doi.org/10.1016%2Fj.marmicro.2023.102214). S2CID 256834649 (https://api.semanticscholar.org/CorpusID:256834649).
9. Schulte, Peter; Alegret, L.; Arenillas, I.; et al. (5 March 2010). "The Chicxulub Asteroid Impact and Mass Extinction at the Cretaceous-Paleogene Boundary" (https://web.archive.org/web/20250115020508/https://doc.rero.ch/record/210367/files/PAL_E4389.pdf) (PDF). *Science*. **327** (5970): 1214–1218. Bibcode:2010Sci...327.1214S (https://ui.adsabs.harvard.edu/abs/2010Sci...327.1214S). doi:10.1126/science.1177265 (https://doi.org/10.1126%2Fscience.1177265). PMID 20203042 (https://pubmed.ncbi.nlm.nih.gov/20203042). S2CID 2659741 (https://api.semanticscholar.org/CorpusID:2659741). Archived from the original (http://doc.rero.ch/record/210367/files/PAL_E4389.pdf) (PDF) on 15 January 2025.
10. Sleep, Norman H.; Lowe, Donald R. (9 April 2014). "Scientists reconstruct ancient impact that dwarfs dinosaur-extinction blast" (https://news.agu.org/press-release/scientists-reconstruct-ancient-impact-that-dwarfs-dinosaur-extinction-blast/). *agu.org* (Press release). American Geophysical Union. Archived (https://web.archive.org/web/20170101045924/https://news.agu.org/press-release/scientists-reconstruct-ancient-impact-that-dwarfs-dinosaur-extinction-blast/) from the original on 1 January 2017. Retrieved 30 December 2016.
11. Amos, Jonathan (15 May 2017). "Dinosaur asteroid hit 'worst possible place' " (https://www.bbc.com/news/science-environment-39922998). *BBC News Online*. Archived (https://web.archive.org/web/20180318162335/http://www.bbc.com/news/science-environment-39922998) from the original on 18 March 2018. Retrieved 16 March 2018.

12. Alvarez, Luis W.; Alvarez, Walter; Asaro, F.; Michel, H. V. (1980). "Extraterrestrial cause for the Cretaceous–Tertiary extinction" (https://web.archive.org/web/20190824192051/https://pdfs.semanticscholar.org/f23d/f624d2e7d945277a4c06d6d66008eb0f4242.pdf) (PDF). *Science*. **208** (4448): 1095–1108. Bibcode:1980Sci...208.1095A (https://ui.adsabs.harvard.edu/abs/1980Sci...208.1095A). doi:10.1126/science.208.4448.1095 (https://doi.org/10.1126%2Fscience.208.4448.1095). PMID 17783054 (https://pubmed.ncbi.nlm.nih.gov/17783054). S2CID 16017767 (https://api.semanticscholar.org/CorpusID:16017767). Archived from the original (https://pdfs.semanticscholar.org/f23d/f624d2e7d945277a4c06d6d66008eb0f4242.pdf) (PDF) on 24 August 2019.
13. Vellekoop, J.; Sluijs, A.; Smit, J.; et al. (May 2014). "Rapid short-term cooling following the Chicxulub impact at the Cretaceous-Paleogene boundary" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4040585). *Proceedings of the National Academy of Sciences of the United States of America*. **111** (21): 7537–7541. Bibcode:2014PNAS..111.7537V (https://ui.adsabs.harvard.edu/abs/2014PNAS..111.7537V). doi:10.1073/pnas.1319253111 (https://doi.org/10.1073%2Fpnas.1319253111). PMC 4040585 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4040585). PMID 24821785 (https://pubmed.ncbi.nlm.nih.gov/24821785).
14. Hildebrand, A. R.; Penfield, G. T.; Kring, David A.; et al. (1991). "Chicxulub crater: a possible Cretaceous/Tertiary boundary impact crater on the Yucatán peninsula, Mexico". *Geology*. **19** (9): 867–871. Bibcode:1991Geo....19..867H (https://ui.adsabs.harvard.edu/abs/1991Geo....19..867H). doi:10.1130/0091-7613(1991)019<0867:ccapct>2.3.co;2 (https://doi.org/10.1130%2F0091-7613%281991%29019%3C0867%3Accapct%3E2.3.co%3B2).
15. Joel, Lucas (21 October 2019). "The dinosaur-killing asteroid acidified the ocean in a flash: the Chicxulub event was as damaging to life in the oceans as it was to creatures on land, a study shows" (https://www.nytimes.com/2019/10/21/science/chicxulub-asteroid-ocean-acid.html). *The New York Times*. Archived (https://web.archive.org/web/20191024040435/https://www.nytimes.com/2019/10/21/science/chicxulub-asteroid-ocean-acid.html) from the original on 24 October 2019. Retrieved 24 October 2019.
16. Henehan, Michael J.; Ridgwell, Andy; Thomas, Ellen; Zhang, Shuang; Alegret, Laia; Schmidt, Daniela N.; Rae, James W. B.; Witts, James D.; Landman, Neil H.; Greene, Sarah E.; Huber, Brian T.; Super, James R.; Planavsky, Noah J.; Hull, Pincelli M. (21 October 2019). "Rapid ocean acidification and protracted Earth system recovery followed the end-Cretaceous Chicxulub impact" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6842625). *Proceedings of the National Academy of Sciences of the United States of America*. **116** (45): 22500–22504. Bibcode:2019PNAS..11622500H (https://ui.adsabs.harvard.edu/abs/2019PNAS..11622500H). doi:10.1073/pnas.1905989116 (https://doi.org/10.1073%2Fpnas.1905989116). PMC 6842625 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6842625). PMID 31636204 (https://pubmed.ncbi.nlm.nih.gov/31636204).
17. Keller, Gerta (2012). "The Cretaceous–Tertiary mass extinction, Chicxulub impact, and Deccan volcanism. Earth and life". In Talent, John (ed.). *Earth and Life: Global Biodiversity, Extinction Intervals and Biogeographic Perturbations Through Time* (https://archive.org/details/earthlifeglobalb00tale). Springer. pp. 759 (https://archive.org/details/earthlifeglobalb00tale/page/n774)–793. ISBN 978-90-481-3427-4.
18. Bosker, Bianca (September 2018). "The nastiest feud in science: A Princeton geologist has endured decades of ridicule for arguing that the fifth extinction was caused not by an asteroid but by a series of colossal volcanic eruptions. But she's reopened that debate" (https://www.theatlantic.com/magazine/archive/2018/09/dinosaur-extinction-debate/565769/). *The Atlantic Monthly*. Archived (https://web.archive.org/web/20190221073809/https://www.theatlantic.com/magazine/archive/2018/09/dinosaur-extinction-debate/565769/) from the original on 21 February 2019. Retrieved 30 January 2019.

19. Joel, Lucas (16 January 2020). "Asteroid or Volcano? New Clues to the Dinosaurs' Demise" (https://www.nytimes.com/2020/01/16/science/dinosaurs-extinction-meteorite-volcano.html). *The New York Times*. Archived (https://web.archive.org/web/20250812113047/https://www.nytimes.com/2020/01/16/science/dinosaurs-extinction-meteorite-volcano.html) from the original on 12 August 2025. Retrieved 17 January 2020.
20. Hull, Pincelli M.; Bornemann, André; Penman, Donald E. (17 January 2020). "On impact and volcanism across the Cretaceous-Paleogene boundary" (https://doi.org/10.1126%2Fscience.aay5055). *Science*. **367** (6475): 266–272. Bibcode:2020Sci...367..266H (https://ui.adsabs.harvard.edu/abs/2020Sci...367..266H). doi:10.1126/science.aay5055 (https://doi.org/10.1126%2Fscience.aay5055). hdl:20.500.11820/483a2e77-318f-476a-8fec-33a45fbdc90b (https://hdl.handle.net/20.500.11820%2F483a2e77-318f-476a-8fec-33a45fbdc90b). PMID 31949074 (https://pubmed.ncbi.nlm.nih.gov/31949074). S2CID 210698721 (https://api.semanticscholar.org/CorpusID:210698721).
21. Chiarenza, Alfio Alessandro; Farnsworth, Alexander; Mannion, Philip D.; et al. (21 July 2020). "Asteroid impact, not volcanism, caused the end-Cretaceous dinosaur extinction" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7382232). *Proceedings of the National Academy of Sciences of the United States of America*. **117** (29): 17084–17093. Bibcode:2020PNAS..11717084C (https://ui.adsabs.harvard.edu/abs/2020PNAS..11717084C). doi:10.1073/pnas.2006087117 (https://doi.org/10.1073%2Fpnas.2006087117). ISSN 0027-8424 (https://search.worldcat.org/issn/0027-8424). PMC 7382232 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7382232). PMID 32601204 (https://pubmed.ncbi.nlm.nih.gov/32601204).
22. Longrich, Nicholas R.; Tokaryk, Tim; Field, Daniel J. (2011). "Mass extinction of birds at the Cretaceous–Paleogene (K–Pg) boundary" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3174646). *Proceedings of the National Academy of Sciences of the United States of America*. **108** (37): 15253–15257. Bibcode:2011PNAS..10815253L (https://ui.adsabs.harvard.edu/abs/2011PNAS..10815253L). doi:10.1073/pnas.1110395108 (https://doi.org/10.1073%2Fpnas.1110395108). PMC 3174646 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3174646). PMID 21914849 (https://pubmed.ncbi.nlm.nih.gov/21914849).
23. Longrich, N. R.; Bhullar, B.-A. S.; Gauthier, J. A. (December 2012). "Mass extinction of lizards and snakes at the Cretaceous-Paleogene boundary" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3535637). *Proceedings of the National Academy of Sciences of the United States of America*. **109** (52): 21396–401. Bibcode:2012PNAS..10921396L (https://ui.adsabs.harvard.edu/abs/2012PNAS..10921396L). doi:10.1073/pnas.1211526110 (https://doi.org/10.1073%2Fpnas.1211526110). PMC 3535637 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3535637). PMID 23236177 (https://pubmed.ncbi.nlm.nih.gov/23236177).
24. Labandeira, C. C.; Johnson, K. R.; Lang, P. (2002). "Preliminary assessment of insect herbivory across the Cretaceous-Tertiary boundary: Major extinction and minimum rebound". In Hartman, J.H.; Johnson, K.R.; Nichols, D.J. (eds.). *The Hell Creek formation and the Cretaceous-Tertiary boundary in the northern Great Plains: An integrated continental record of the end of the Cretaceous* (https://books.google.com/books?id=cHvcIeh2f84C). Geological Society of America. pp. 297–327. ISBN 978-0-8137-2361-7.
25. Rehan, Sandra M.; Leys, Remko; Schwarz, Michael P. (2013). "First evidence for a massive extinction event affecting bees close to the K-T boundary" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3806776). *PLOS ONE*. **8** (10) e76683. Bibcode:2013PLoSO...876683R (https://ui.adsabs.harvard.edu/abs/2013PLoSO...876683R). doi:10.1371/journal.pone.0076683 (https://doi.org/10.1371%2Fjournal.pone.0076683). PMC 3806776 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3806776). PMID 24194843 (https://pubmed.ncbi.nlm.nih.gov/24194843).
26. Nichols, D. J.; Johnson, K. R. (2008). *Plants and the K–T Boundary*. Cambridge, UK: Cambridge University Press. Bibcode:2008pktb.book.....N (https://ui.adsabs.harvard.edu/abs/2008pktb.book.....N).

27. Friedman, M. (2009). "Ecomorphological selectivity among marine teleost fishes during the end-Cretaceous extinction" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2664034). *Proceedings of the National Academy of Sciences*. **106** (13). Washington, DC: 5218–5223. Bibcode:2009PNAS..106.5218F (https://ui.adsabs.harvard.edu/abs/2009PNAS..106.5218F). doi:10.1073/pnas.0808468106 (https://doi.org/10.1073%2Fpnas.0808468106). PMC 2664034 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2664034). PMID 19276106 (https://pubmed.ncbi.nlm.nih.gov/19276106).
28. Alroy, John (1999). "The fossil record of North American Mammals: evidence for a Palaeocene evolutionary radiation" (https://doi.org/10.1080%2F106351599260472). *Systematic Biology*. **48** (1): 107–118. doi:10.1080/106351599260472 (https://doi.org/10.1080%2F106351599260472). PMID 12078635 (https://pubmed.ncbi.nlm.nih.gov/12078635).
29. Feduccia, Alan (1995). "Explosive evolution in Tertiary birds and mammals". *Science*. **267** (5198): 637–638. Bibcode:1995Sci...267..637F (https://ui.adsabs.harvard.edu/abs/1995Sci...267..637F). doi:10.1126/science.267.5198.637 (https://doi.org/10.1126%2Fscience.267.5198.637). PMID 17745839 (https://pubmed.ncbi.nlm.nih.gov/17745839). S2CID 42829066 (https://api.semanticscholar.org/CorpusID:42829066).
30. Friedman, M. (2010). "Explosive morphological diversification of spiny-finned teleost fishes in the aftermath of the end-Cretaceous extinction" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2871855). *Proceedings of the Royal Society B*. **277** (1688): 1675–1683. doi:10.1098/rspb.2009.2177 (https://doi.org/10.1098%2Frspb.2009.2177). PMC 2871855 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2871855). PMID 20133356 (https://pubmed.ncbi.nlm.nih.gov/20133356).
31. Weishampel, D. B.; Barrett, P. M. (2004). "Dinosaur distribution". In Weishampel, David B.; Dodson, Peter; Osmólska, Halszka (eds.). *The Dinosauria* (https://archive.org/details/dinosauriandedit00weis) (2nd ed.). Berkeley, CA: University of California Press. pp. 517 (https://archive.org/details/dinosauriandedit00weis/page/n535)–606. ISBN 978-0-520-24209-8. OCLC 441742117 (https://search.worldcat.org/oclc/441742117).
32. Barrera, Enriqueta; Keller, Gerta (1 October 1994). "Productivity across the Cretaceous/Tertiary boundary in high latitudes". *Geological Society of America Bulletin*. **106** (10): 1254–1266. Bibcode:1994GSAB..106.1254B (https://ui.adsabs.harvard.edu/abs/1994GSAB..106.1254B). doi:10.1130/0016-7606(1994)106<1254:PATCTB>2.3.CO;2 (https://doi.org/10.1130%2F0016-7606%281994%29106%3C1254%3APATCTB%3E2.3.CO%3B2).
33. Wilf, P.; Johnson, K.R. (2004). "Land plant extinction at the end of the Cretaceous: A quantitative analysis of the North Dakota megafloral record". *Paleobiology*. **30** (3): 347–368. Bibcode:2004Pbio...30..347W (https://ui.adsabs.harvard.edu/abs/2004Pbio...30..347W). doi:10.1666/0094-8373(2004)030<0347:LPEATE>2.0.CO;2 (https://doi.org/10.1666%2F0094-8373%282004%29030%3C0347%3ALPEATE%3E2.0.CO%3B2). S2CID 33880578 (https://api.semanticscholar.org/CorpusID:33880578).
34. MacLeod, N.; Rawson, P.F.; Forey, P.L.; Banner, F.T.; Boudagher-Fadel, M.K.; Bown, P.R.; Burnett, J.A.; Chambers, P.; Culver, S.; Evans, S.E.; Jeffery, C.; Kaminski, M.A.; Lord, A.R.; Milner, A.C.; Milner, A.R.; Morris, N.; Owen, E.; Rosen, B.R.; Smith, A.B.; Taylor, P.D.; Urquhart, E.; Young, J.R. (1997). "The Cretaceous–Tertiary biotic transition". *Journal of the Geological Society*. **154** (2): 265–292. Bibcode:1997JGSoc.154..265M (https://ui.adsabs.harvard.edu/abs/1997JGSoc.154..265M). doi:10.1144/gsjgs.154.2.0265 (https://doi.org/10.1144%2Fgsjgs.154.2.0265). S2CID 129654916 (https://api.semanticscholar.org/CorpusID:129654916).

35. Sheehan, Peter M.; Hansen, Thor A. (1986). "Detritus feeding as a buffer to extinction at the end of the Cretaceous" (https://web.archive.org/web/20190227164831/http://pdfs.semanticscholar.org/83c8/1affaceb4d7da17e5f8ec0fd91c033adb08f.pdf) (PDF). *Geology*. **14** (10): 868–870. Bibcode:1986Geo....14..868S (https://ui.adsabs.harvard.edu/abs/1986Geo....14..868S). doi:10.1130/0091-7613(1986)14<868:DFAABT>2.0.CO;2 (https://doi.org/10.1130%2F0091-7613%281986%2914%3C868%3ADFAABT%3E2.0.CO%3B2). S2CID 54860261 (https://api.semanticscholar.org/CorpusID:54860261). Archived from the original (http://pdfs.semanticscholar.org/83c8/1affaceb4d7da17e5f8ec0fd91c033adb08f.pdf) (PDF) on 27 February 2019.
36. Aberhan, M.; Weidemeyer, S.; Kieesling, W.; Scasso, R.A.; Medina, F.A. (2007). "Faunal evidence for reduced productivity and uncoordinated recovery in Southern Hemisphere Cretaceous-Paleogene boundary sections". *Geology*. **35** (3): 227–230. Bibcode:2007Geo....35..227A (https://ui.adsabs.harvard.edu/abs/2007Geo....35..227A). doi:10.1130/G23197A.1 (https://doi.org/10.1130%2FG23197A.1).
37. Sheehan, Peter M.; Fastovsky, D. E. (1992). "Major extinctions of land-dwelling vertebrates at the Cretaceous-Tertiary boundary, eastern Montana" (https://digitalcommons.uri.edu/geo_facpubs/108). *Geology*. **20** (6): 556–560. Bibcode:1992Geo....20..556S (https://ui.adsabs.harvard.edu/abs/1992Geo....20..556S). doi:10.1130/0091-7613(1992)020<0556:MEOLDV>2.3.CO;2 (https://doi.org/10.1130%2F0091-7613%281992%29020%3C0556%3AMEOLDV%3E2.3.CO%3B2).
38. García-Girón, Jorge; Chiarenza, Alfio Alessandro; Alahuhta, Janne; DeMar, David G.; Heino, Jani; Mannion, Philip D.; Williamson, Thomas E.; Wilson Mantilla, Gregory P.; Brusatte, Stephen L. (9 December 2022). "Shifts in food webs and niche stability shaped survivorship and extinction at the end-Cretaceous" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9728968). *Science Advances*. **8** (49) eadd5040. Bibcode:2022SciA....8D5040G (https://ui.adsabs.harvard.edu/abs/2022SciA....8D5040G). doi:10.1126/sciadv.add5040 (https://doi.org/10.1126%2Fsciadv.add5040). PMC 9728968 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9728968). PMID 36475805 (https://pubmed.ncbi.nlm.nih.gov/36475805).
39. Black, Riley (2022). *The Last Days of the Dinosaurs: An Asteroid, Extinction, and the Beginning of Our World* (1st ed.). New York: St. Martin's Press. p. 127. ISBN 9781250271044.
40. Gibbs, Samantha J.; Bown, Paul R.; Ward, Ben A.; Alvarez, Sarah A.; Kim, Hojung; Archontikis, Odysseas A.; Sauterey, Boris; Poulton, Alex J.; Wilson, Jamie; Ridgwell, Andy (30 October 2020). "Algal plankton turn to hunting to survive and recover from end-Cretaceous impact darkness" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7608818). *Science Advances*. **6** (44) eabc9123. Bibcode:2020SciA....6.9123G (https://ui.adsabs.harvard.edu/abs/2020SciA....6.9123G). doi:10.1126/sciadv.abc9123 (https://doi.org/10.1126%2Fsciadv.abc9123). PMC 7608818 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7608818). PMID 33127682 (https://pubmed.ncbi.nlm.nih.gov/33127682).
41. Kauffman, E. (2004). "Mosasaur predation on upper Cretaceous nautiloids and ammonites from the United States Pacific Coast" (https://web.archive.org/web/20251225234058/https://doc.rero.ch/record/14992/files/PAL_E2143.pdf) (PDF). *PALAIOS*. **19** (1): 96–100. Bibcode:2004Palai..19...96K (https://ui.adsabs.harvard.edu/abs/2004Palai..19...96K). doi:10.1669/0883-1351(2004)019<0096:MPOUCN>2.0.CO;2 (https://doi.org/10.1669%2F0883-1351%282004%29019%3C0096%3AMPOUCN%3E2.0.CO%3B2). S2CID 130690035 (https://api.semanticscholar.org/CorpusID:130690035). Archived from the original (http://doc.rero.ch/record/14992/files/PAL_E2143.pdf) (PDF) on 25 December 2025.

42. Clyde, William C.; Wilf, Peter; Iglesias, Ari; Slingerland, Rudy L.; Barnum, Timothy; Bijl, Peter K.; Bralower, Timothy J.; Brinkhuis, Henk; Comer, Emily E.; Huber, Brian T.; Ibañez-Mejia, Mauricio; Jicha, Brian R.; Krause, J. Marcelo; Schueth, Jonathan D.; Singer, Bradley S.; Raigemborn, María Sol; Schmitz, Mark D.; Sluijs, Appy; Zamaloa, María del Carmen (1 March 2014). "New age constraints for the Salamanca Formation and lower Río Chico Group in the western San Jorge Basin, Patagonia, Argentina: Implications for Cretaceous-Paleogene extinction recovery and land mammal age correlations" (https://dspace.library.uu.nl/handle/1874/306844). *Geological Society of America Bulletin*. **126** (3–4): 289–306. Bibcode:2014GSAB..126..289C (https://ui.adsabs.harvard.edu/abs/2014GSAB..126..289C). doi:10.1130/B30915.1 (https://doi.org/10.1130%2FB30915.1). hdl:11336/80135 (https://hdl.handle.net/11336%2F80135). S2CID 129962470 (https://api.semanticscholar.org/CorpusID:129962470).
43. During, Melanie A. D.; Smit, Jan; Voeten, Dennis F. A. E.; Berruyer, Camille; Tafforeau, Paul; Sanchez, Sophie; Stein, Koen H. W.; Verdegaal-Warmerdam, Suzan J. A.; van der Lubbe, Jeroen H. J. L. (23 February 2022). "The Mesozoic terminated in boreal spring" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8891016). *Nature*. **603** (7899): 91–94. Bibcode:2022Natur.603...91D (https://ui.adsabs.harvard.edu/abs/2022Natur.603...91D). doi:10.1038/s41586-022-04446-1 (https://doi.org/10.1038%2Fs41586-022-04446-1). PMC 8891016 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8891016). PMID 35197634 (https://pubmed.ncbi.nlm.nih.gov/35197634).
44. Boles, Zachary; Ullmann, Paul; Putnam, Ian; Ford, Mariele; Deckhut, Joseph (12 April 2024). "New vertebrate microfossils expand the diversity of the chondrichthyan and actinopterygian fauna of the Maastrichtian–Danian Hornerstown Formation in New Jersey" (http://publication.plazi.org/id/FFD8FF391304EF00DC40FF91FFE4FFAB). *Acta Palaeontologica Polonica*. doi:10.4202/app.01117.2023 (https://doi.org/10.4202%2Fapp.01117.2023).
45. Pospichal, J. J. (1996). "Calcareous nannofossils and clastic sediments at the Cretaceous–Tertiary boundary, northeastern Mexico". *Geology*. **24** (3): 255–258. Bibcode:1996Geo....24..255P (https://ui.adsabs.harvard.edu/abs/1996Geo....24..255P). doi:10.1130/0091-7613(1996)024<0255:CNACSA>2.3.CO;2 (https://doi.org/10.1130%2F0091-7613%281996%29024%3C0255%3ACNACSA%3E2.3.CO%3B2).
46. Bown, P. (2005). "Selective calcareous nannoplankton survivorship at the Cretaceous–Tertiary boundary". *Geology*. **33** (8): 653–656. Bibcode:2005Geo....33..653B (https://ui.adsabs.harvard.edu/abs/2005Geo....33..653B). doi:10.1130/G21566.1 (https://doi.org/10.1130%2FG21566.1).
47. Bambach, R. K.; Knoll, A. H.; Wang, S. C. (2004). "Origination, extinction, and mass depletions of marine diversity" (http://biology.kenyon.edu/courses/biol241/extinction%20and%20marine%20diversity%20banbach%202004.pdf) (PDF). *Paleobiology*. **30** (4): 522–542. Bibcode:2004Pbio...30..522B (https://ui.adsabs.harvard.edu/abs/2004Pbio...30..522B). doi:10.1666/0094-8373(2004)030<0522:OEAMDO>2.0.CO;2 (https://doi.org/10.1666%2F0094-8373%282004%29030%3C0522%3AOEAMDO%3E2.0.CO%3B2). S2CID 17279135 (https://api.semanticscholar.org/CorpusID:17279135).
48. Jiang, Shijun; Bralower, Timothy J.; Patzkowsky, Mark E.; Kump, Lee R.; Schueth, Jonathan D. (28 February 2010). "Geographic controls on nannoplankton extinction across the Cretaceous/Palaeogene boundary" (https://www.nature.com/articles/ngeo775). *Nature Geoscience*. **3** (4): 280–285. Bibcode:2010NatGe...3..280J (https://ui.adsabs.harvard.edu/abs/2010NatGe...3..280J). doi:10.1038/ngeo775 (https://doi.org/10.1038%2Fngeo775). ISSN 1752-0908 (https://search.worldcat.org/issn/1752-0908). Retrieved 22 August 2024.

49. Schueth, Jonathan D.; Bralower, Timothy J.; Jiang, Shijun; Patzkowsky, Mark E. (September 2015). "The role of regional survivor incumbency in the evolutionary recovery of calcareous nannoplankton from the Cretaceous/Paleogene (K/Pg) mass extinction". *Paleobiology*. **41** (4): 661–679. Bibcode:2015Pbio...41..661S (https://ui.adsabs.harvard.edu/abs/2015Pbio...41..661S). doi:10.1017/pab.2015.28 (https://doi.org/10.1017%2Fpab.2015.28). ISSN 0094-8373 (https://search.worldcat.org/issn/0094-8373).
50. Gedl, P. (2004). "Dinoflagellate cyst record of the deep-sea Cretaceous-Tertiary boundary at Uzgru, Carpathian Mountains, Czech Republic". *Special Publications of the Geological Society of London*. **230** (1): 257–273. Bibcode:2004GSLSP.230..257G (https://ui.adsabs.harvard.edu/abs/2004GSLSP.230..257G). doi:10.1144/GSL.SP.2004.230.01.13 (https://doi.org/10.1144%2FGSL.SP.2004.230.01.13). S2CID 128771186 (https://api.semanticscholar.org/CorpusID:128771186).
51. Tantawy, Abdel Aziz (2011). "Calcareous Nannofossils Across the Cretaceous–Tertiary Boundary at Brazos, Texas, U.S.A.: Extinction and Survivorship, Biostratigraphy, and Paleoecology" (https://pubs.geoscienceworld.org/sepm/books/edited-volume/1197/chapter-abstract/10721260/Calcareous-Nannofossils-Across-the-Cretaceous?redirectedFrom=fulltext). In Keller, Gerta; Adatte, Thierry (eds.). *The End-Cretaceous Mass Extinction and the Chicxulub Impact in Texas* (https://pubs.geoscienceworld.org/sepm/books/edited-volume/1197/The-End-Cretaceous-Mass-Extinction-and-the). Society of Sedimentary Geology. pp. 157–178. doi:10.2110/sepmsp.100.157 (https://doi.org/10.2110%2Fsepmsp.100.157). ISBN 9781565763098.
52. MacLeod, N. (1998). "Impacts and marine invertebrate extinctions". *Special Publications of the Geological Society of London*. **140** (1): 217–246. Bibcode:1998GSLSP.140..217M (https://ui.adsabs.harvard.edu/abs/1998GSLSP.140..217M). doi:10.1144/GSL.SP.1998.140.01.16 (https://doi.org/10.1144%2FGSL.SP.1998.140.01.16). S2CID 129875020 (https://api.semanticscholar.org/CorpusID:129875020).
53. Courtillot, V. (1999). *Evolutionary Catastrophes: The science of mass extinction* (https://books.google.com/books?id=qiegJbafYkUC&pg=PA23). Cambridge, UK: Cambridge University Press. p. 2. ISBN 978-0-521-58392-3.
54. Hansen, T.; Farrand, R.B.; Montgomery, H.A.; Billman, H.G.; Blechschmidt, G. (September 1987). "Sedimentology and extinction patterns across the Cretaceous-Tertiary boundary interval in east Texas". *Cretaceous Research*. **8** (3): 229–252. Bibcode:1987CrRes...8..229H (https://ui.adsabs.harvard.edu/abs/1987CrRes...8..229H). doi:10.1016/0195-6671(87)90023-1 (https://doi.org/10.1016%2F0195-6671%2887%2990023-1).
55. Arenillas, I.; Arz, J. A.; Molina, E.; Dupuis, C. (2000). "An independent test of planktic foraminiferal turnover across the Cretaceous/Paleogene (K/P) boundary at El Kef, Tunisia: Catastrophic mass extinction and possible survivorship". *Micropaleontology*. **46** (1): 31–49. JSTOR 1486024 (https://www.jstor.org/stable/1486024).
56. Keller, Gerta (April 1993). "The Cretaceous-Tertiary boundary transition in the Antarctic Ocean and its global implications". *Marine Micropaleontology*. **21** (1–3): 1–45. Bibcode:1993MarMP..21....1K (https://ui.adsabs.harvard.edu/abs/1993MarMP..21....1K). doi:10.1016/0377-8398(93)90010-U (https://doi.org/10.1016%2F0377-8398%2893%2990010-U).
57. Macleod, Norman; Keller, Gerta (1 November 1991). "How complete are Cretaceous/Tertiary boundary sections? A chronostratigraphic estimate based on graphic correlation". *Geological Society of America Bulletin*. **103** (11): 1439. Bibcode:1991GSAB..103.1439M (https://ui.adsabs.harvard.edu/abs/1991GSAB..103.1439M). doi:10.1130/0016-7606(1991)103<1439:HCACTB>2.3.CO;2 (https://doi.org/10.1130%2F0016-7606%281991%29103%3C1439%3AHCACTB%3E2.3.CO%3B2). ISSN 0016-7606 (https://search.worldcat.org/issn/0016-7606).

58. MacLeod, N. (1996). "Nature of the Cretaceous-Tertiary (K–T) planktonic foraminiferal record: Stratigraphic confidence intervals, Signor–Lipps effect, and patterns of survivorship". In MacLeod, N.; Keller, G. (eds.). *Cretaceous–Tertiary Mass Extinctions: Biotic and environmental changes*. W.W. Norton. pp. 85–138. ISBN 978-0-393-96657-2.
59. Keller, G.; Li, L.; MacLeod, N. (1 January 1996). "The Cretaceous/Tertiary boundary stratotype section at El Kef, Tunisia: how catastrophic was the mass extinction?". *Palaeogeography, Palaeoclimatology, Palaeoecology*. **119** (3): 221–254. Bibcode:1996PPP...119..221K (https://ui.adsabs.harvard.edu/abs/1996PPP...119..221K). doi:10.1016/0031-0182(95)00009-7 (https://doi.org/10.1016%2F0031-0182%2895%2900009-7). ISSN 0031-0182 (https://search.worldcat.org/issn/0031-0182).
60. MacLeod, Norman; Keller, Gerta (Spring 1994). "Comparative biogeographic analysis of planktic foraminiferal survivorship across the Cretaceous/Tertiary (K/T) boundary". *Paleobiology*. **20** (2): 143–177. Bibcode:1994Pbio...20..143M (https://ui.adsabs.harvard.edu/abs/1994Pbio...20..143M). doi:10.1017/S0094837300012653 (https://doi.org/10.1017%2FS0094837300012653). ISSN 0094-8373 (https://search.worldcat.org/issn/0094-8373).
61. Schulte, Peter; Speijer, Robert; Mai, Hartmut; Kontny, Agnes (1 February 2006). "The Cretaceous–Paleogene (K–P) boundary at Brazos, Texas: Sequence stratigraphy, depositional events and the Chicxulub impact" (http://gateway.newisiknowledge.com/gateway/Gateway.cgi?GWVersion=2&SrcAuth=CCC&SrcApp=PRODUCT_NAME&SrcURL=WOS_RETURN_URL&CKEY=SCHU0077060184SP&DestLinkType=FullRecord&DestApp=CCC&SrcDesc=RETURN_ALT_TEXT&SrcAppSID=APP_SID). *Sedimentary Geology*. **184** (1–2): 77–109. Bibcode:2006SedG..184...77S (https://ui.adsabs.harvard.edu/abs/2006SedG..184...77S). doi:10.1016/j.sedgeo.2005.09.021 (https://doi.org/10.1016%2Fj.sedgeo.2005.09.021).
62. Galeotti, S.; Bellagamba, M.; Kaminski, M. A.; Montanari, A. (2002). "Deep-sea benthic foraminiferal recolonisation following a volcaniclastic event in the lower Campanian of the Scaglia Rossa Formation (Umbria-Marche Basin, central Italy)". *Marine Micropaleontology*. **44** (1–2): 57–76. Bibcode:2002MarMP..44...57G (https://ui.adsabs.harvard.edu/abs/2002MarMP..44...57G). doi:10.1016/s0377-8398(01)00037-8 (https://doi.org/10.1016%2Fs0377-8398%2801%2900037-8).
63. Kuhnt, W.; Collins, E. S. (1996). "Cretaceous to Paleogene benthic foraminifers from the Iberia Abyssal Plain". *Scientific Results*. Proceedings of the Ocean Drilling Program. Vol. 149. pp. 203–216. doi:10.2973/odp.proc.sr.149.254.1996 (https://doi.org/10.2973%2Fodp.proc.sr.149.254.1996).
64. Coles, G. P.; Ayress, M. A.; Whatley, R. C. (1990). "A comparison of North Atlantic and 20 Pacific deep-sea Ostracoda". In Whatley, R. C.; Maybury, C. (eds.). *Ostracoda and Global Events*. Chapman & Hall. pp. 287–305. ISBN 978-0-442-31167-4.
65. Brouwers, E. M.; de Deckker, P. (1993). "Late Maastrichtian and Danian Ostracode Faunas from Northern Alaska: Reconstructions of Environment and Paleogeography". *PALAIOS*. **8** (2): 140–154. Bibcode:1993Palai...8..140B (https://ui.adsabs.harvard.edu/abs/1993Palai...8..140B). doi:10.2307/3515168 (https://doi.org/10.2307%2F3515168). JSTOR 3515168 (https://www.jstor.org/stable/3515168).
66. Martins, Maria João Fernandes; Hunt, Gene; Thompson, Carmi Milagros; Lockwood, Rowan; Swaddle, John P.; Puckett, T. Markham (26 August 2020). "Shifts in sexual dimorphism across a mass extinction in ostracods: implications for sexual selection as a factor in extinction risk" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7482269). *Proceedings of the Royal Society B: Biological Sciences*. **287** (1933) 20200730. doi:10.1098/rspb.2020.0730 (https://doi.org/10.1098%2Frspb.2020.0730). ISSN 0962-8452 (https://search.worldcat.org/issn/0962-8452). PMC 7482269 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7482269). PMID 32811315 (https://pubmed.ncbi.nlm.nih.gov/32811315).

67. Samuels-Fair, Maya; Martins, Maria João Fernandes; Lockwood, Rowan; Swaddle, John P.; Hunt, Gene (June 2022). "Temporal shifts in ostracode sexual dimorphism from the Late Cretaceous to the late Eocene of the U.S. Coastal Plain" (https://doi.org/10.1016%2Fj.marmicro.2020.101959). *Marine Micropaleontology*. **174** 101959. Bibcode:2022MarMP.17401959S (https://ui.adsabs.harvard.edu/abs/2022MarMP.17401959S). doi:10.1016/j.marmicro.2020.101959 (https://doi.org/10.1016%2Fj.marmicro.2020.101959). hdl:10400.1/18610 (https://hdl.handle.net/10400.1%2F18610).
68. Schweitzer, Carrie E.; Feldmann, Rodney M. (June 2023). "Selective extinction at the end-Cretaceous and appearance of the modern Decapoda". *Journal of Crustacean Biology*. **43** (2) ruad018. doi:10.1093/jcbiol/ruad018 (https://doi.org/10.1093%2Fjcbiol%2Fruad018). ISSN 0278-0372 (https://search.worldcat.org/issn/0278-0372).
69. Hyžný, Matúš; Perrier, Vincent; Robin, Ninon; Martin, Jeremy E.; Sarr, Raphaël (January 2016). "Costacopluma (Decapoda: Brachyura: Retroplumidae) from the Maastrichtian and Paleocene of Senegal: A survivor of K/Pg events". *Cretaceous Research*. **57**: 142–156. Bibcode:2016CrRes..57..142H (https://ui.adsabs.harvard.edu/abs/2016CrRes..57..142H). doi:10.1016/j.cretres.2015.08.010 (https://doi.org/10.1016%2Fj.cretres.2015.08.010).
70. Vescsei, A.; Moussavian, E. (1997). "Paleocene reefs on the Maiella Platform margin, Italy: An example of the effects of the cretaceous/tertiary boundary events on reefs and carbonate platforms". *Facies*. **36** (1): 123–139. Bibcode:1997Faci...36..123V (https://ui.adsabs.harvard.edu/abs/1997Faci...36..123V). doi:10.1007/BF02536880 (https://doi.org/10.1007%2FBF02536880). S2CID 129296658 (https://api.semanticscholar.org/CorpusID:129296658).
71. Rosen, B. R.; Turnšek, D. (1989). Jell A; Pickett JW (eds.). "Extinction patterns and biogeography of scleractinian corals across the Cretaceous/Tertiary boundary". *Memoir of the Association of Australasian Paleontology*. Proceedings of the Fifth International Symposium on Fossil Cnidaria including Archaeocyatha and Spongiomorphs (8). Brisbane, Queensland: 355–370.
72. Raup, D. M.; Jablonski, D. (1993). "Geography of end-Cretaceous marine bivalve extinctions". *Science*. **260** (5110): 971–973. Bibcode:1993Sci...260..971R (https://ui.adsabs.harvard.edu/abs/1993Sci...260..971R). doi:10.1126/science.11537491 (https://doi.org/10.1126%2Fscience.11537491). PMID 11537491 (https://pubmed.ncbi.nlm.nih.gov/11537491).
73. MacLeod, K. G. (1994). "Extinction of Inoceramid Bivalves in Maastrichtian Strata of the Bay of Biscay Region of France and Spain". *Journal of Paleontology*. **68** (5): 1048–1066. Bibcode:1994JPal...68.1048M (https://ui.adsabs.harvard.edu/abs/1994JPal...68.1048M). doi:10.1017/S0022336000026652 (https://doi.org/10.1017%2FS0022336000026652). S2CID 132641572 (https://api.semanticscholar.org/CorpusID:132641572).
74. Marshall, C. R.; Ward, P. D. (1996). "Sudden and Gradual Molluscan Extinctions in the Latest Cretaceous of Western European Tethys". *Science*. **274** (5291): 1360–1363. Bibcode:1996Sci...274.1360M (https://ui.adsabs.harvard.edu/abs/1996Sci...274.1360M). doi:10.1126/science.274.5291.1360 (https://doi.org/10.1126%2Fscience.274.5291.1360). PMID 8910273 (https://pubmed.ncbi.nlm.nih.gov/8910273). S2CID 1837900 (https://api.semanticscholar.org/CorpusID:1837900).
75. Hansen, Thor A.; Farrell, Benjamin R.; Upshaw, Banks (Spring 1993). "The first 2 million years after the Cretaceous-Tertiary boundary in east Texas: rate and paleoecology of the molluscan recovery". *Paleobiology*. **19** (2): 251–265. Bibcode:1993Pbio...19..251H (https://ui.adsabs.harvard.edu/abs/1993Pbio...19..251H). doi:10.1017/S0094837300015906 (https://doi.org/10.1017%2FS0094837300015906). ISSN 0094-8373 (https://search.worldcat.org/issn/0094-8373).

76. Lockwood, Rowan (4 March 2003). "Abundance not linked to survival across the end-Cretaceous mass extinction: Patterns in North American bivalves" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC151366). *Proceedings of the National Academy of Sciences of the United States of America*. **100** (5): 2478–2482. Bibcode:2003PNAS..100.2478L (https://ui.adsabs.harvard.edu/abs/2003PNAS..100.2478L). doi:10.1073/pnas.0535132100 (https://doi.org/10.1073%2Fpnas.0535132100). ISSN 0027-8424 (https://search.worldcat.org/issn/0027-8424). PMC 151366 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC151366). PMID 12601147 (https://pubmed.ncbi.nlm.nih.gov/12601147).
77. Lockwood, Rowan (Fall 2004). "The K/T event and infaunality: morphological and ecological patterns of extinction and recovery in veneroid bivalves". *Paleobiology*. **30** (4): 507–521. Bibcode:2004Pbio...30..507L (https://ui.adsabs.harvard.edu/abs/2004Pbio...30..507L). doi:10.1666/0094-8373(2004)030<0507:TTEAIM>2.0.CO;2 (https://doi.org/10.1666%2F0094-8373%282004%29030%3C0507%3ATTEAIM%3E2.0.CO%3B2). ISSN 0094-8373 (https://search.worldcat.org/issn/0094-8373).
78. Ward, P. D.; Kennedy, W. J.; MacLeod, K. G.; Mount, J. F. (1991). "Ammonite and inoceramid bivalve extinction patterns in Cretaceous/Tertiary boundary sections of the Biscay region (southwestern France, northern Spain)". *Geology*. **19** (12): 1181–1184. Bibcode:1991Geo....19.1181W (https://ui.adsabs.harvard.edu/abs/1991Geo....19.1181W). doi:10.1130/0091-7613(1991)019<1181:AAIBEP>2.3.CO;2 (https://doi.org/10.1130%2F0091-7613%281991%29019%3C1181%3AAAIBEP%3E2.3.CO%3B2).
79. Harries, P. J.; Johnson, K. R.; Cobban, W. A.; Nichols, D.J. (2002). "Marine Cretaceous-Tertiary boundary section in southwestern South Dakota: Comment and reply". *Geology*. **30** (10): 954–955. Bibcode:2002Geo....30..954H (https://ui.adsabs.harvard.edu/abs/2002Geo....30..954H). doi:10.1130/0091-7613(2002)030<0955:MCTBSI>2.0.CO;2 (https://doi.org/10.1130%2F0091-7613%282002%29030%3C0955%3AMCTBSI%3E2.0.CO%3B2).
80. Iba, Yasuhiro; Mutterlose, Jörg; Tanabe, Kazushige; Sano, Shin-ichi; Misaki, Akihiro; Terabe, Kazunobu (1 May 2011). "Belemnite extinction and the origin of modern cephalopods 35 m.y. prior to the Cretaceous−Paleogene event" (http://pubs.geoscienceworld.org/geology/article/39/5/483/130552/Belemnite-extinction-and-the-origin-of-modern). *Geology*. **39** (5): 483–486. Bibcode:2011Geo....39..483I (https://ui.adsabs.harvard.edu/abs/2011Geo....39..483I). doi:10.1130/G31724.1 (https://doi.org/10.1130%2FG31724.1). ISSN 1943-2682 (https://search.worldcat.org/issn/1943-2682). Retrieved 18 November 2023.
81. Neraudeau, Didier; Thierry, Jacques; Moreau, Pierre (1 January 1997). "Variation in echinoid biodiversity during the Cenomanian-early Turonian transgressive episode in Charentes (France)" (https://pubs.geoscienceworld.org/sgf/bsgf/article-abstract/168/1/51/122879/Variation-in-echinoid-biodiversity-during-the). *Bulletin de la Société Géologique de France*. **168** (1): 51–61.
82. Wiese, Frank; Schlüter, Nils; Zirkel, Jessica; Herrle, Jens O.; Friedrich, Oliver (9 August 2023). Carnevale, Giorgio (ed.). "A 104-Ma record of deep-sea Atelostomata (Holasterioda, Spatangoida, irregular echinoids) – a story of persistence, food availability and a big bang" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10411753). *PLOS ONE*. **18** (8) e0288046. Bibcode:2023PLoSO..1888046W (https://ui.adsabs.harvard.edu/abs/2023PLoSO..1888046W). doi:10.1371/journal.pone.0288046 (https://doi.org/10.1371%2Fjournal.pone.0288046). ISSN 1932-6203 (https://search.worldcat.org/issn/1932-6203). PMC 10411753 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10411753). PMID 37556403 (https://pubmed.ncbi.nlm.nih.gov/37556403).
83. Labandeira, Conrad C.; Johnson, Kirk R.; Wilf, Peter (2002). "Impact of the terminal Cretaceous event on plant–insect associations" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC122319). *Proceedings of the National Academy of Sciences of the United States of America*. **99** (4): 2061–2066. Bibcode:2002PNAS...99.2061L (https://ui.adsabs.harvard.edu/abs/2002PNAS...99.2061L). doi:10.1073/pnas.042492999 (https://doi.org/10.1073%2Fpnas.042492999). PMC 122319 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC122319). PMID 11854501 (https://pubmed.ncbi.nlm.nih.gov/11854501).

84. Wilf, P.; Labandeira, C. C.; Johnson, K. R.; Ellis, B. (2006). "Decoupled plant and insect diversity after the end-Cretaceous extinction". *Science*. **313** (5790): 1112–1115. Bibcode:2006Sci...313.1112W (https://ui.adsabs.harvard.edu/abs/2006Sci...313.1112W). doi:10.1126/science.1129569 (https://doi.org/10.1126%2Fscience.1129569). PMID 16931760 (https://pubmed.ncbi.nlm.nih.gov/16931760). S2CID 52801127 (https://api.semanticscholar.org/CorpusID:52801127).
85. Wiest, Logan A.; Lukens, William E.; Peppe, Daniel J.; Driese, Steven G.; Tubbs, Jack (1 February 2018). "Terrestrial evidence for the Lilliput effect across the Cretaceous-Paleogene (K-Pg) boundary" (http://paleorxiv.org/9x7du/). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **491**: 161–169. Bibcode:2018PPP...491..161W (https://ui.adsabs.harvard.edu/abs/2018PPP...491..161W). doi:10.1016/j.palaeo.2017.12.005 (https://doi.org/10.1016%2Fj.palaeo.2017.12.005). ISSN 0031-0182 (https://search.worldcat.org/issn/0031-0182).
86. Grimaldi, David A. (2007). *Evolution of the Insects*. Cambridge Univ Pr (E). ISBN 978-0-511-12388-7.
87. Kriwet, Jürgen; Benton, Michael J. (2004). "Neoselachian (Chondrichthyes, Elasmobranchii) Diversity across the Cretaceous–Tertiary Boundary". *Palaeogeography, Palaeoclimatology, Palaeoecology*. **214** (3): 181–194. Bibcode:2004PPP...214..181K (https://ui.adsabs.harvard.edu/abs/2004PPP...214..181K). doi:10.1016/j.palaeo.2004.02.049 (https://doi.org/10.1016%2Fj.palaeo.2004.02.049).
88. Patterson, C. (1993). "Osteichthyes: Teleostei". In Benton, M. J. (ed.). *The Fossil Record*. Vol. 2. Springer. pp. 621–656. ISBN 978-0-412-39380-8.
89. Noubhani, Abdelmajid (2010). "The Selachians' faunas of the Moroccan phosphate deposits and the K-T mass extinctions". *Historical Biology*. **22** (1–3): 71–77. Bibcode:2010HBio...22...71N (https://ui.adsabs.harvard.edu/abs/2010HBio...22...71N). doi:10.1080/08912961003707349 (https://doi.org/10.1080%2F08912961003707349). S2CID 129579498 (https://api.semanticscholar.org/CorpusID:129579498).
90. Guinot, Guillaume; Condamine, Fabien L. (23 February 2023). "Global impact and selectivity of the Cretaceous-Paleogene mass extinction among sharks, skates, and rays" (https://hal.science/hal-04262391/file/Manuscrit%20Guinot%20%26%20Condamine%202023.pdf) (PDF). *Science*. **379** (6634): 802–806. Bibcode:2023Sci...379..802G (https://ui.adsabs.harvard.edu/abs/2023Sci...379..802G). doi:10.1126/science.abn2080 (https://doi.org/10.1126%2Fscience.abn2080). PMID 36821692 (https://pubmed.ncbi.nlm.nih.gov/36821692). S2CID 257103123 (https://api.semanticscholar.org/CorpusID:257103123). Archived (https://web.archive.org/web/20260106054933/https://hal.science/hal-04262391/file/Manuscrit%20Guinot%20&%20Condamine%202023.pdf) (PDF) from the original on 6 January 2026.
91. Zinsmeister, W. J. (1 May 1998). "Discovery of fish mortality horizon at the K–T boundary on Seymour Island: Re-evaluation of events at the end of the Cretaceous". *Journal of Paleontology*. **72** (3): 556–571. Bibcode:1998JPal...72..556Z (https://ui.adsabs.harvard.edu/abs/1998JPal...72..556Z). doi:10.1017/S0022336000024331 (https://doi.org/10.1017%2FS0022336000024331). S2CID 132206016 (https://api.semanticscholar.org/CorpusID:132206016).
92. Cione, Alberto L.; Santillana, Sergio; Gouiric-Cavalli, Soledad; Acosta Hospitaleche, Carolina; Gelfo, Javier N.; López, Guillermo M.; Reguero, Marcelo (May 2018). "Before and after the K/Pg extinction in West Antarctica: New marine fish records from Marambio (Seymour) Island" (http://sedici.unlp.edu.ar/handle/10915/147537). *Cretaceous Research*. **85**: 250–265. Bibcode:2018CrRes..85..250C (https://ui.adsabs.harvard.edu/abs/2018CrRes..85..250C). doi:10.1016/j.cretres.2018.01.004 (https://doi.org/10.1016%2Fj.cretres.2018.01.004). hdl:11336/99687 (https://hdl.handle.net/11336%2F99687).

93. Robertson, D. S.; McKenna, M. C.; Toon, O. B.; et al. (2004). "Survival in the first hours of the Cenozoic" (https://repository.geologyscience.ru/bitstream/handle/123456789/38889/Robe_04.pdf) (PDF). *Geological Society of America Bulletin*. **116** (5–6): 760–768. Bibcode:2004GSAB..116..760R (https://ui.adsabs.harvard.edu/abs/2004GSAB..116..760R). doi:10.1130/B25402.1 (https://doi.org/10.1130%2FB25402.1). S2CID 44010682 (https://api.semanticscholar.org/CorpusID:44010682). Archived (https://web.archive.org/web/20190507051856/https://pdfs.semanticscholar.org/8d43/36c3d3e40d27390e37941e4affe0cff84bb2.pdf) (PDF) from the original on 7 May 2019.
94. Friedman, Matt; V. Andrews, James; Saad, Hadeel; El-Sayed, Sanaa (16 June 2023). "The Cretaceous–Paleogene transition in spiny-rayed fishes: surveying "Patterson's Gap" in the acanthomorph skeletal record André Dumont medalist lecture 2018" (https://popups.uliege.be/1374-8505/index.php?id=7048). *Geologica Belgica*. doi:10.20341/gb.2023.002 (https://doi.org/10.20341%2Fgb.2023.002). ISSN 1374-8505 (https://search.worldcat.org/issn/1374-8505). Retrieved 22 August 2024.
95. Alfaro, Michael E.; Faircloth, Brant C.; Harrington, Richard C.; Sorenson, Laurie; Friedman, Matt; Thacker, Christine E.; Oliveros, Carl H.; Černý, David; Near, Thomas J. (12 March 2018). "Explosive diversification of marine fishes at the Cretaceous–Palaeogene boundary" (https://www.nature.com/articles/s41559-018-0494-6). *Nature Ecology & Evolution*. **2** (4): 688–696. Bibcode:2018NatEE...2..688A (https://ui.adsabs.harvard.edu/abs/2018NatEE...2..688A). doi:10.1038/s41559-018-0494-6 (https://doi.org/10.1038%2Fs41559-018-0494-6). ISSN 2397-334X (https://search.worldcat.org/issn/2397-334X). PMID 29531346 (https://pubmed.ncbi.nlm.nih.gov/29531346). Retrieved 22 August 2024.
96. Archibald, J. D.; Bryant, L. J. (1990). "Differential Cretaceous–Tertiary extinction of nonmarine vertebrates; evidence from northeastern Montana". In Sharpton, V.L.; Ward, P.D. (eds.). *Global Catastrophes in Earth History: an Interdisciplinary Conference on Impacts, Volcanism, and Mass Mortality*. Special Paper. Vol. 247. Geological Society of America. pp. 549–562. doi:10.1130/spe247-p549 (https://doi.org/10.1130%2Fspe247-p549). ISBN 978-0-8137-2247-4.
97. Estes, R. (1964). "Fossil vertebrates from the late Cretaceous Lance formation, eastern Wyoming". *University of California Publications, Department of Geological Sciences*. **49**: 1–180.
98. Gardner, J. D. (2000). "Albanerpetontid amphibians from the upper Cretaceous (Campanian and Maastrichtian) of North America". *Geodiversitas*. **22** (3): 349–388.
99. Feng, Yan-Jie; Blackburn, David C.; Liang, Dan; Hillis, David M.; Wake, David B.; Cannatella, David C.; Zhang, Peng (18 July 2017). "Phylogenomics reveals rapid, simultaneous diversification of three major clades of Gondwanan frogs at the Cretaceous–Paleogene boundary" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5530686). *Proceedings of the National Academy of Sciences of the United States of America*. **114** (29) e5864–E5870. Bibcode:2017PNAS..114E5864F (https://ui.adsabs.harvard.edu/abs/2017PNAS..114E5864F). doi:10.1073/pnas.1704632114 (https://doi.org/10.1073%2Fpnas.1704632114). ISSN 0027-8424 (https://search.worldcat.org/issn/0027-8424). PMC 5530686 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5530686). PMID 28673970 (https://pubmed.ncbi.nlm.nih.gov/28673970).
100. Evans, Susan E.; Klembara, Jozef (2005). "A choristoderan reptile (Reptilia: Diapsida) from the Lower Miocene of northwest Bohemia (Czech Republic)". *Journal of Vertebrate Paleontology*. **25** (1): 171–184. doi:10.1671/0272-4634(2005)025[0171:ACRRDF]2.0.CO;2 (https://doi.org/10.1671%2F0272-4634%282005%29025%5B0171%3AACRRDF%5D2.0.CO%3B2). S2CID 84097919 (https://api.semanticscholar.org/CorpusID:84097919).
101. Matsumoto, Ryoko; Evans, Susan E. (November 2015). "Morphology and function of the palatal dentition in Choristodera" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5341546). *Journal of Anatomy*. **228** (3): 414–429. doi:10.1111/joa.12414 (https://doi.org/10.1111%2Fjoa.12414). PMC 5341546 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5341546). PMID 26573112 (https://pubmed.ncbi.nlm.nih.gov/26573112).

102. Novacek, M. J. (1999). "100 million years of land vertebrate evolution: The Cretaceous-early Tertiary transition" (https://www.biodiversitylibrary.org/part/28619). *Annals of the Missouri Botanical Garden*. **86** (2): 230–258. Bibcode:1999AnMBG..86..230N (https://ui.adsabs.harvard.edu/abs/1999AnMBG..86..230N). doi:10.2307/2666178 (https://doi.org/10.2307%2F2666178). JSTOR 2666178 (https://www.jstor.org/stable/2666178).
103. Holroyd, Patricia A.; Wilson, Gregory P.; Hutchinson, J. Howard (2013). "Temporal changes within the latest Cretaceous and early Paleogene turtle faunas of northeastern Montana" (https://books.google.com/books?id=chKDAgAAQBAJ&dq=cretaceous+paleogene+extinction+turtles&pg=PA299). In Wilson, Gregory P.; Clemens, William A.; Horner, John R.; Hartman, Joseph H. (eds.). *Through the End of the Cretaceous in the Type Locality of the Hell Creek Formation in Montana and Adjacent Areas*. Boulder, CO: Geological Society of America. pp. 299–312. ISBN 978-0-8137-2503-1.
104. Cleary, Terri J.; Benson, Roger B. J.; Holroyd, Patricia A.; Barrett, Paul M. (10 May 2020). Mannion, Philip (ed.). "Tracing the patterns of non-marine turtle richness from the Triassic to the Palaeogene: from origin to global spread" (https://research.birmingham.ac.uk/en/publications/1caf4537-e41d-4964-8036-6c808dd09a5e). *Palaeontology*. **63** (5): 753–774. Bibcode:2020Palgy..63..753C (https://ui.adsabs.harvard.edu/abs/2020Palgy..63..753C). doi:10.1111/pala.12486 (https://doi.org/10.1111%2Fpala.12486). ISSN 0031-0239 (https://search.worldcat.org/issn/0031-0239).
105. Pérez-García, Adán (30 January 2020). "Surviving the Cretaceous-Paleogene mass extinction event: A terrestrial stem turtle in the Cenozoic of Laurasia" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6992736). *Scientific Reports*. **10** (1) 1489. Bibcode:2020NatSR..10.1489P (https://ui.adsabs.harvard.edu/abs/2020NatSR..10.1489P). doi:10.1038/s41598-020-58511-8 (https://doi.org/10.1038%2Fs41598-020-58511-8). ISSN 2045-2322 (https://search.worldcat.org/issn/2045-2322). PMC 6992736 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6992736). PMID 32001765 (https://pubmed.ncbi.nlm.nih.gov/32001765).
106. Apesteguía, Sebastián; Novas, Fernando E. (2003). "Large Cretaceous sphenodontian from Patagonia provides insight into lepidosaur evolution in Gondwana". *Nature*. **425** (6958): 609–612. Bibcode:2003Natur.425..609A (https://ui.adsabs.harvard.edu/abs/2003Natur.425..609A). doi:10.1038/nature01995 (https://doi.org/10.1038%2Fnature01995). PMID 14534584 (https://pubmed.ncbi.nlm.nih.gov/14534584). S2CID 4425130 (https://api.semanticscholar.org/CorpusID:4425130).
107. Lutz, D. (2005). *Tuatara: A living fossil*. DIMI Press. ISBN 978-0-931625-43-5.
108. Apesteguía, Sebastián; Gómez, Raúl O.; Rougier, Guillermo W. (7 October 2014). "The youngest South American rhynchocephalian, a survivor of the K/Pg extinction" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4150314). *Proceedings of the Royal Society B: Biological Sciences*. **281** (1792) 20140811. Bibcode:2014PBioS.28140811A (https://ui.adsabs.harvard.edu/abs/2014PBioS.28140811A). doi:10.1098/rspb.2014.0811 (https://doi.org/10.1098%2Frspb.2014.0811). ISSN 0962-8452 (https://search.worldcat.org/issn/0962-8452). PMC 4150314 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4150314). PMID 25143041 (https://pubmed.ncbi.nlm.nih.gov/25143041).
109. Herrera-Flores, Jorge A.; Stubbs, Thomas L.; Benton, Michael J. (March 2021). "Ecomorphological diversification of squamates in the Cretaceous" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8074880). *Royal Society Open Science*. **8** (3). 201961. Bibcode:2021RSOS....801961H (https://ui.adsabs.harvard.edu/abs/2021RSOS....801961H). doi:10.1098/rsos.201961 (https://doi.org/10.1098%2Frsos.201961). ISSN 2054-5703 (https://search.worldcat.org/issn/2054-5703). PMC 8074880 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8074880). PMID 33959350 (https://pubmed.ncbi.nlm.nih.gov/33959350).

110. Xing, Lida; Niu, Kecheng; Evans, Susan E. (January 2023). "A new polyglyphanodontian lizard with a complete lower temporal bar from the Upper Cretaceous of southern China" (https://doi.org/10.1080%2F14772019.2023.2281494). *Journal of Systematic Palaeontology*. **21** (1) 2281494. Bibcode:2023JSPal..2181494X (https://ui.adsabs.harvard.edu/abs/2023JSPal..2181494X). doi:10.1080/14772019.2023.2281494 (https://doi.org/10.1080%2F14772019.2023.2281494). ISSN 1477-2019 (https://search.worldcat.org/issn/1477-2019).
111. Klein, Catherine G.; Pisani, Davide; Field, Daniel J.; Lakin, Rebecca; Wills, Matthew A.; Longrich, Nicholas R. (14 September 2021). "Evolution and dispersal of snakes across the Cretaceous-Paleogene mass extinction" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8440539). *Nature Communications*. **12** (1): 5335. Bibcode:2021NatCo..12.5335K (https://ui.adsabs.harvard.edu/abs/2021NatCo..12.5335K). doi:10.1038/s41467-021-25136-y (https://doi.org/10.1038%2Fs41467-021-25136-y). PMC 8440539 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8440539). PMID 34521829 (https://pubmed.ncbi.nlm.nih.gov/34521829).
112. Čerňanský, Andrej; Daza, Juan; Tabuce, Rodolphe; Saxton, Elizabeth; Vidalenc, Dominique (December 2023). "An early Eocene pan-gekkotan from France could represent an extra squamate group that survived the K-Pg extinction" (https://hal.umontpellier.fr/hal-04295655/document). *Acta Palaeontologica Polonica*. **68**. doi:10.4202/app.01083.2023 (https://doi.org/10.4202%2Fapp.01083.2023). Retrieved 22 August 2024.
113. Martin, Jeremy E.; Vincent, Peggy; Tacail, Théo; Khaldoune, Fatima; Jourani, Essaid; Bardet, Nathalie; Balter, Vincent (5 June 2017). "Calcium Isotopic Evidence for Vulnerable Marine Ecosystem Structure Prior to the K/Pg Extinction" (https://hal.science/hal-01648300). *Current Biology*. **27** (11): 1641–1644.e2. Bibcode:2017CBio...27E1641M (https://ui.adsabs.harvard.edu/abs/2017CBio...27E1641M). doi:10.1016/j.cub.2017.04.043 (https://doi.org/10.1016%2Fj.cub.2017.04.043). hdl:1983/3edf103d-b973-43e7-900b-2f7d39f59d46 (https://hdl.handle.net/1983%2F3edf103d-b973-43e7-900b-2f7d39f59d46). ISSN 0960-9822 (https://search.worldcat.org/issn/0960-9822). PMID 28552352 (https://pubmed.ncbi.nlm.nih.gov/28552352).
114. D'Hondt, Steven (17 August 2005). "Consequences of the Cretaceous/Paleogene Mass Extinction for Marine Ecosystems" (https://digitalcommons.uri.edu/gsofacpubs/1164). *Annual Review of Ecology, Evolution, and Systematics*. **36**: 295–317. doi:10.1146/annurev.ecolsys.35.021103.105715 (https://doi.org/10.1146%2Fannurev.ecolsys.35.021103.105715). S2CID 86073650 (https://api.semanticscholar.org/CorpusID:86073650).
115. Chatterjee, S.; Small, B. J. (1989). "New plesiosaurs from the Upper Cretaceous of Antarctica". *Geological Society, London, Special Publications*. **47** (1): 197–215. Bibcode:1989GSLSP..47..197C (https://ui.adsabs.harvard.edu/abs/1989GSLSP..47..197C). doi:10.1144/GSL.SP.1989.047.01.15 (https://doi.org/10.1144%2FGSL.SP.1989.047.01.15). S2CID 140639013 (https://api.semanticscholar.org/CorpusID:140639013).
116. O'Keefe, F. R. (2001). "A cladistic analysis and taxonomic revision of the Plesiosauria (Reptilia: Sauropterygia)". *Acta Zoologica Fennica*. **213**: 1–63.
117. O'Gorman, José P. (December 2022). "Polycotylidae (Sauropterygia, Plesiosauria) from the La Colonia Formation, Patagonia, Argentina: Phylogenetic affinities of Sulcusuchus erraini and the Late Cretaceous circum-pacific polycotylid diversity". *Cretaceous Research*. **140** 105339. Bibcode:2022CrRes.14005339O (https://ui.adsabs.harvard.edu/abs/2022CrRes.140005339O). doi:10.1016/j.cretres.2022.105339 (https://doi.org/10.1016%2Fj.cretres.2022.105339). S2CID 251749728 (https://api.semanticscholar.org/CorpusID:251749728).
118. Fischer, Valentin; Bardet, Nathalie; Benson, Roger B. J.; Arkhangelsky, Maxim S.; Friedman, Matt (2016). "Extinction of fish-shaped marine reptiles associated with reduced evolutionary rates and global environmental volatility" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4786747). *Nature Communications*. **7** (1) 10825. Bibcode:2016NatCo...710825F (https://ui.adsabs.harvard.edu/abs/2016NatCo...710825F). doi:10.1038/ncomms10825 (https://doi.org/10.1038%2Fncomms10825). PMC 4786747 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4786747). PMID 26953824 (https://pubmed.ncbi.nlm.nih.gov/26953824).

119. Brochu, C. A. (2004). "Calibration age and quartet divergence date estimation". *Evolution*. **58** (6): 1375–1382. Bibcode:2004Evolu..58.1375B (https://ui.adsabs.harvard.edu/abs/2004Evolu..58.1375B). doi:10.1554/03-509 (https://doi.org/10.1554%2F03-509). PMID 15266985 (https://pubmed.ncbi.nlm.nih.gov/15266985). S2CID 198156470 (https://api.semanticscholar.org/CorpusID:198156470).
120. Jouve, S.; Bardet, N.; Jalil, N.-E.; Suberbiola, X. P.; Bouya, B.; Amaghzaz, M. (2008). "The oldest African crocodylian: phylogeny, paleobiogeography, and differential survivorship of marine reptiles through the Cretaceous-Tertiary boundary" (https://web.archive.org/web/20251223034622/http://doc.rero.ch/record/16384/files/PAL_E3543.pdf) (PDF). *Journal of Vertebrate Paleontology*. **28** (2): 409–421. doi:10.1671/0272-4634(2008)28[409:TOACPP]2.0.CO;2 (https://doi.org/10.1671%2F0272-4634%282008%2928%5B409%3ATOACPP%5D2.0.CO%3B2). S2CID 86503283 (https://api.semanticscholar.org/CorpusID:86503283). Archived from the original (http://doc.rero.ch/record/16384/files/PAL_E3543.pdf) (PDF) on 23 December 2025.
121. Aubier, Paul; Jouve, Stéphane; Schnyder, Johann; Cubo, Jorge (20 February 2023). Mannion, Philip (ed.). "Phylogenetic structure of the extinction and biotic factors explaining differential survival of terrestrial notosuchians at the Cretaceous–Palaeogene crisis" (https://doi.org/10.1111%2Fpala.12638). *Palaeontology*. **66** (1) 12638. Bibcode:2023Palgy..6612638A (https://ui.adsabs.harvard.edu/abs/2023Palgy..6612638A). doi:10.1111/pala.12638 (https://doi.org/10.1111%2Fpala.12638). ISSN 0031-0239 (https://search.worldcat.org/issn/0031-0239).
122. Martin, Jeremy E.; Pochat-Cottilloux, Yohan; Laurent, Yves; Perrier, Vincent; Robert, Emmanuel; Antoine, Pierre-Olivier (28 October 2022). "Anatomy and phylogeny of an exceptionally large sebecid (Crocodylomorpha) from the middle Eocene of southern France". *Journal of Vertebrate Paleontology*. **42** (4) e2193828. Bibcode:2022JVPal..42E3828M (https://ui.adsabs.harvard.edu/abs/2022JVPal..42E3828M). doi:10.1080/02724634.2023.2193828 (https://doi.org/10.1080%2F02724634.2023.2193828). ISSN 0272-4634 (https://search.worldcat.org/issn/0272-4634). S2CID 258361595 (https://api.semanticscholar.org/CorpusID:258361595).
123. Forêt, Tom; Aubier, Paul; Jouve, Stéphane; Cubo, Jorge (23 April 2024). "Biotic and abiotic factors and the phylogenetic structure of extinction in the evolution of Tethysuchia" (https://cnrs.hal.science/hal-04558789/document). *Paleobiology*. **50** (2): 285–307. Bibcode:2024Pbio...50..285F (https://ui.adsabs.harvard.edu/abs/2024Pbio...50..285F). doi:10.1017/pab.2024.5 (https://doi.org/10.1017%2Fpab.2024.5). ISSN 0094-8373 (https://search.worldcat.org/issn/0094-8373).
124. Company, J.; Ruiz-Omeñaca, J. I.; Pereda Suberbiola, X. (1999). "A long-necked pterosaur (Pterodactyloidea, Azhdarchidae) from the upper Cretaceous of Valencia, Spain". *Geologie en Mijnbouw*. **78** (3): 319–333. doi:10.1023/A:1003851316054 (https://doi.org/10.1023%2FA%3A1003851316054). hdl:10251/112485 (https://hdl.handle.net/10251%2F112485). S2CID 73638590 (https://api.semanticscholar.org/CorpusID:73638590).
125. Barrett, P. M.; Butler, R. J.; Edwards, N. P.; Milner, A. R. (2008). "Pterosaur distribution in time and space: an atlas" (http://epub.ub.uni-muenchen.de/12007/1/zitteliana_2008_b28_05.pdf) (PDF). *Zitteliana*. **28**: 61–107. Archived (https://web.archive.org/web/20170806150333/https://epub.ub.uni-muenchen.de/12007/1/zitteliana_2008_b28_05.pdf) (PDF) from the original on 6 August 2017. Retrieved 31 August 2015.
126. Slack, K. E.; Jones, C. M.; Ando, T.; Harrison, G. L.; Fordyce, R. E.; Arnason, U.; Penny, D. (2006). "Early Penguin Fossils, Plus Mitochondrial Genomes, Calibrate Avian Evolution" (https://doi.org/10.1093%2Fmolbev%2Fmsj124). *Molecular Biology and Evolution*. **23** (6): 1144–1155. doi:10.1093/molbev/msj124 (https://doi.org/10.1093%2Fmolbev%2Fmsj124). PMID 16533822 (https://pubmed.ncbi.nlm.nih.gov/16533822).

127. Penny, D.; Phillips, M. J. (2004). "The rise of birds and mammals: Are microevolutionary processes sufficient for macroevolution?" (https://web.archive.org/web/20251219155437/http://doc.rero.ch/record/14569/files/PAL_E1781.pdf) (PDF). *Trends in Ecology and Evolution*. **19** (10): 516–522. doi:10.1016/j.tree.2004.07.015 (https://doi.org/10.1016%2Fj.tree.2004.07.015). PMID 16701316 (https://pubmed.ncbi.nlm.nih.gov/16701316). Archived from the original (http://doc.rero.ch/record/14569/files/PAL_E1781.pdf) (PDF) on 19 December 2025.
128. Butler, Richard J.; Barrett, Paul M.; Nowbath, Stephen; Upchurch, Paul (2009). "Estimating the effects of sampling biases on pterosaur diversity patterns: Implications for hypotheses of bird / pterosaur competitive replacement". *Paleobiology*. **35** (3): 432–446. Bibcode:2009Pbio...35..432B (https://ui.adsabs.harvard.edu/abs/2009Pbio...35..432B). doi:10.1666/0094-8373-35.3.432 (https://doi.org/10.1666%2F0094-8373-35.3.432). S2CID 84324007 (https://api.semanticscholar.org/CorpusID:84324007).
129. Prondvai, E.; Bodor, E. R.; Ösi, A. (2014). "Does morphology reflect osteohistology-based ontogeny? A case study of Late Cretaceous pterosaur jaw symphyses from Hungary reveals hidden taxonomic diversity" (http://real.mtak.hu/21860/1/Prondvai_et_al.2014_reposit1_u_110445.946242.pdf) (PDF). *Paleobiology*. **40** (2): 288–321. Bibcode:2014Pbio...40..288P (https://ui.adsabs.harvard.edu/abs/2014Pbio...40..288P). doi:10.1666/13030 (https://doi.org/10.1666%2F13030). hdl:10831/75031 (https://hdl.handle.net/10831%2F75031). S2CID 85673254 (https://api.semanticscholar.org/CorpusID:85673254).
130. Longrich, N. R.; Martill, D. M.; Andres, B. (2018). "Late Maastrichtian pterosaurs from North Africa and mass extinction of Pterosauria at the Cretaceous-Paleogene boundary" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5849296). *PLOS Biology*. **16** (3) e2001663. doi:10.1371/journal.pbio.2001663 (https://doi.org/10.1371%2Fjournal.pbio.2001663). PMC 5849296 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5849296). PMID 29534059 (https://pubmed.ncbi.nlm.nih.gov/29534059).
131. Ocampo, A.; Vajda, V.; Buffetaut, E. (2006). "Unravelling the Cretaceous–Paleogene (K–T) turnover, evidence from flora, fauna and geology in biological processes associated with impact events". In Cockell, C.; Gilmour, I.; Koeberl, C. (eds.). *Biological Processes Associated with Impact Events* (https://archive.org/details/biologicalproces00cock). SpringerLink. pp. 197 (https://archive.org/details/biologicalproces00cock/page/n211)–219. doi:10.1007/3-540-25736-5_9 (https://doi.org/10.1007%2F3-540-25736-5_9). ISBN 978-3-540-25735-6.
132. Buffetaut, Eric (18 November 2004). "Polar dinosaurs and the question of dinosaur extinction: a brief review". *Palaeogeography, Palaeoclimatology, Palaeoecology*. **214** (3): 225–231. Bibcode:2004PPP...214..225B (https://ui.adsabs.harvard.edu/abs/2004PPP...214..225B). doi:10.1016/j.palaeo.2004.02.050 (https://doi.org/10.1016%2Fj.palaeo.2004.02.050).
133. David, Archibald; Fastovsky, David (2004). "Dinosaur extinction" (http://www.bio.sdsu.edu/faculty/archibald/ArchFast.pdf) (PDF). In Weishampel, David B.; Dodson, Peter; Osmólska, Halszka (eds.). *The Dinosauria* (2nd ed.). Berkeley: University of California Press. pp. 672–684. ISBN 978-0-520-24209-8.
134. Fastovsky, David E.; Bercovici, Antoine (January 2016). "The Hell Creek Formation and its contribution to the Cretaceous–Paleogene extinction: A short primer" (https://digitalcommons.uri.edu/geo_facpubs/76). *Cretaceous Research*. **57**: 368–390. Bibcode:2016CrRes..57..368F (https://ui.adsabs.harvard.edu/abs/2016CrRes..57..368F). doi:10.1016/j.cretres.2015.07.007 (https://doi.org/10.1016%2Fj.cretres.2015.07.007).
135. Dodson, Peter (1996). *The Horned Dinosaurs: A Natural History* (https://archive.org/details/horneddinosaursn00dods_0). Princeton, NJ: Princeton University Press. pp. 279–281 (https://archive.org/details/horneddinosaursn00dods_0/page/279). ISBN 978-0-691-05900-6.

136. Rieraa, V.; Marmib, J.; Omsa, O.; Gomez, B. (March 2010). "Orientated plant fragments revealing tidal palaeocurrents in the Fumanya mudflat (Maastrichtian, southern Pyrenees): Insights in palaeogeographic reconstructions". *Palaeogeography, Palaeoclimatology, Palaeoecology*. **288** (1–4): 82–92. Bibcode:2010PPP...288...82R (https://ui.adsabs.harvard.edu/abs/2010PPP...288...82R). doi:10.1016/j.palaeo.2010.01.037 (https://doi.org/10.1016%2Fj.palaeo.2010.01.037).
137. le Loeuff, J. (2012). "Paleobiogeography and biodiversity of Late Maastrichtian dinosaurs: How many dinosaur species became extinct at the Cretaceous-Tertiary boundary?". *Bulletin de la Société Géologique de France*. **183** (6): 547–559. doi:10.2113/gssgfbull.183.6.547 (https://doi.org/10.2113%2Fgssgfbull.183.6.547).
138. Ryan, M. J.; Russell, A. P.; Eberth, D. A.; Currie, P. J. (2001). "The taphonomy of a Centrosaurus (Ornithischia: Ceratopsidae) bone bed from the Dinosaur Park formation (Upper Campanian), Alberta, Canada, with comments on cranial ontogeny". *PALAIOS*. **16** (5): 482–506. Bibcode:2001Palai..16..482R (https://ui.adsabs.harvard.edu/abs/2001Palai..16..482R). doi:10.1669/0883-1351(2001)016<0482:ttoaco>2.0.co;2 (https://doi.org/10.1669%2F0883-1351%282001%29016%3C0482%3Attoaco%3E2.0.co%3B2). S2CID 130116586 (https://api.semanticscholar.org/CorpusID:130116586).
139. Sloan, R. E.; Rigby, K.; van Valen, L. M.; Gabriel, Diane (1986). "Gradual dinosaur extinction and simultaneous ungulate radiation in the Hell Creek formation". *Science*. **232** (4750): 629–633. Bibcode:1986Sci...232..629S (https://ui.adsabs.harvard.edu/abs/1986Sci...232..629S). doi:10.1126/science.232.4750.629 (https://doi.org/10.1126%2Fscience.232.4750.629). PMID 17781415 (https://pubmed.ncbi.nlm.nih.gov/17781415). S2CID 31638639 (https://api.semanticscholar.org/CorpusID:31638639).
140. Fassett, J. E.; Lucas, S. G.; Zielinski, R. A.; Budahn, J. R. (2001). *Compelling new evidence for Paleocene dinosaurs in the Ojo Alamo Sandstone San Juan Basin, New Mexico and Colorado, USA* (http://www.lpi.usra.edu/meetings/impact2000/pdf/3139.pdf) (PDF). International Conference on Catastrophic Events and Mass Extinctions: Impacts and Beyond, 9–12 July 2000. Vol. 1053. Vienna, Austria. pp. 45–46. Bibcode:2001caev.conf.3139F (https://ui.adsabs.harvard.edu/abs/2001caev.conf.3139F). Archived (https://web.archive.org/web/20110605175701/http://www.lpi.usra.edu/meetings/impact2000/pdf/3139.pdf) (PDF) from the original on 5 June 2011. Retrieved 18 May 2007.
141. Sullivan, R. M. (2003). "No Paleocene dinosaurs in the San Juan Basin, New Mexico" (https://web.archive.org/web/20110408155457/http://gsa.confex.com/gsa/2003RM/finalprogram/abstract_47695.htm). *Geological Society of America Abstracts with Programs*. **35** (5): 15. Archived from the original (http://gsa.confex.com/gsa/2003RM/finalprogram/abstract_47695.htm) on 8 April 2011. Retrieved 2 July 2007.
142. Hou, L.; Martin, M.; Zhou, Z.; Feduccia, A. (1996). "Early Adaptive Radiation of Birds: Evidence from Fossils from Northeastern China". *Science*. **274** (5290): 1164–1167. Bibcode:1996Sci...274.1164H (https://ui.adsabs.harvard.edu/abs/1996Sci...274.1164H). doi:10.1126/science.274.5290.1164 (https://doi.org/10.1126%2Fscience.274.5290.1164). PMID 8895459 (https://pubmed.ncbi.nlm.nih.gov/8895459). S2CID 30639866 (https://api.semanticscholar.org/CorpusID:30639866).
143. Clarke, J.A.; Tambussi, C.P.; Noriega, J.I.; Erickson, G.M.; Ketcham, R.A. (2005). "Definitive fossil evidence for the extant avian radiation in the Cretaceous" (https://web.archive.org/web/20230122135139/https://doc.rero.ch/record/15294/files/PAL_E2593.pdf) (PDF). *Nature*. **433** (7023): 305–308. Bibcode:2005Natur.433..305C (https://ui.adsabs.harvard.edu/abs/2005Natur.433..305C). doi:10.1038/nature03150 (https://doi.org/10.1038%2Fnature03150). hdl:11336/80763 (https://hdl.handle.net/11336%2F80763). PMID 15662422 (https://pubmed.ncbi.nlm.nih.gov/15662422). S2CID 4354309 (https://api.semanticscholar.org/CorpusID:4354309). Archived from the original (http://doc.rero.ch/record/15294/files/PAL_E2593.pdf) (PDF) on 22 January 2023.

144. Mayr G., (2007). "The birds from the Paleocene fissure filling of Walbeck (Germany)." *Journal of Vertebrate Paleontology*, **27**(2): 394-408. PDF fulltext (http://www.biologicaldiversity.org/campaigns/global_owl_project/descriptions/Fossil_owls/Berruornis%20halbedelli%20(Mayr)%202007.pdf). doi:10.1671/0272-4634(2007)27[394:TBFTPF]2.0.CO;2 (https://doi.org/10.1671%2F0272-4634%282007%2927%5B394%3ATBFTPF%5D2.0.CO%3B2)
145. "Primitive birds shared dinosaurs' fate" (https://www.sciencedaily.com/releases/2011/09/110919151315.htm). Science Daily. 20 September 2011. Archived (https://web.archive.org/web/20110924053712/http://www.sciencedaily.com/releases/2011/09/110919151315.htm) from the original on 24 September 2011. Retrieved 20 September 2011.
146. Ericson, P. G.; Anderson, C. L.; Britton, T.; et al. (December 2006). "Diversification of Neoaves: integration of molecular sequence data and fossils" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1834003). *Biology Letters*. **2** (4): 543–7. doi:10.1098/rsbl.2006.0523 (https://doi.org/10.1098%2Frsbl.2006.0523). PMC 1834003 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1834003). PMID 17148284 (https://pubmed.ncbi.nlm.nih.gov/17148284).
147. Mitchell, K.J.; Llamas, B.; Soubrier, J.; Rawlence, N. J.; Worthy, T. H.; Wood, J.; Lee, M. S. Y.; Cooper, A. (2014). "Ancient DNA reveals elephant birds and kiwi are sister taxa and clarifies ratite bird evolution". *Science*. **344** (6186): 989–900. Bibcode:2014Sci...344..898M (https://ui.adsabs.harvard.edu/abs/2014Sci...344..898M). doi:10.1126/science.1251981 (https://doi.org/10.1126%2Fscience.1251981). hdl:2328/35953 (https://hdl.handle.net/2328%2F35953). PMID 24855267 (https://pubmed.ncbi.nlm.nih.gov/24855267). S2CID 206555952 (https://api.semanticscholar.org/CorpusID:206555952).
148. Yonezawa, Takahiro; Segawa, Takahiro; Mori, Hiroshi; Campos, Paula F.; Hongoh, Yuichi; Endo, Hideki; Akiyoshi, Ayumi; Kohno, Naoki; Nishida, Shin; Wu, Jiaqi; Jin, Haofei (2017). "Phylogenomics and Morphology of Extinct Paleognaths Reveal the Origin and Evolution of the Ratites" (https://doi.org/10.1016%2Fj.cub.2016.10.029). *Current Biology*. **27** (1): 68–77. Bibcode:2017CBio...27...68Y (https://ui.adsabs.harvard.edu/abs/2017CBio...27...68Y). doi:10.1016/j.cub.2016.10.029 (https://doi.org/10.1016%2Fj.cub.2016.10.029). PMID 27989673 (https://pubmed.ncbi.nlm.nih.gov/27989673).
149. Bininda-Emonds, O. R.; Cardillo M.; Jones, K. E.; MacPhee, R. D.; Beck, R. M.; Grenyer, R.; Price, S. A.; Vos, R. A.; Gittleman, J. L.; Purvis, A. (2007). "The delayed rise of present-day mammals" (https://zotero.org/groups/5435545/items/MLEK8EMP). *Nature*. **446** (7135): 507–512. Bibcode:2007Natur.446..507B (https://ui.adsabs.harvard.edu/abs/2007Natur.446..507B). doi:10.1038/nature05634 (https://doi.org/10.1038%2Fnature05634). PMID 17392779 (https://pubmed.ncbi.nlm.nih.gov/17392779). S2CID 4314965 (https://api.semanticscholar.org/CorpusID:4314965).
150. Gelfo, J. N.; Pascual, R. (2001). "*Peligrotherium tropicalis* (Mammalia, Dryolestida) from the early Paleocene of Patagonia, a survival from a Mesozoic Gondwanan radiation" (https://web.archive.org/web/20120212193611/http://www.mnhn.fr/publication/geodiv/g01n3a4.pdf) (PDF). *Geodiversitas*. **23**: 369–379. Archived from the original (http://www.mnhn.fr/publication/geodiv/g01n3a4.pdf) (PDF) on 12 February 2012.
151. Goin, F. J.; Reguero, M. A.; Pascual, R.; von Koenigswald, W.; Woodburne, M. O.; Case, J. A.; Marenssi, S. A.; Vieytes, C.; Vizcaíno, S. F. (2006). "First gondwanatherian mammal from Antarctica". *Geological Society, London*. Special Publications. **258** (1): 135–144. Bibcode:2006GSLSP.258..135G (https://ui.adsabs.harvard.edu/abs/2006GSLSP.258..135G). doi:10.1144/GSL.SP.2006.258.01.10 (https://doi.org/10.1144%2FGSL.SP.2006.258.01.10). S2CID 129493664 (https://api.semanticscholar.org/CorpusID:129493664).
152. McKenna, M. C.; Bell, S. K. (1997). *Classification of mammals: Above the species level*. Columbia University Press. ISBN 978-0-231-11012-9.

153. Bennett CV; Upchurch P; Goin FJ; Goswami A (6 February 2018). "Deep time diversity of metatherian mammals: implications for evolutionary history and fossil-record quality" (https://www.cambridge.org/core/journals/paleobiology/article/deep-time-diversity-of-metatherian-mammals-implications-for-evolutionary-history-and-fossilrecord-quality/79E914107366FFD593A831E0486E356C). *Paleobiology*. **44** (2). Cambridge University Press: 171–198. doi:10.1017/pab.2017.34 (https://doi.org/10.1017%2Fpab.2017.34). hdl:11336/94590 (https://hdl.handle.net/11336%2F94590).
154. Wood, D. Joseph (2010). *The Extinction of the Multituberculates Outside North America: a Global Approach to Testing the Competition Model* (https://etd.ohiolink.edu/!etd.send_file?accession=osu1275595604&disposition=inline) (M.S.). The Ohio State University. Archived (https://web.archive.org/web/20150408104907/https://etd.ohiolink.edu/!etd.send_file?accession=osu1275595604&disposition=inline) from the original on 8 April 2015. Retrieved 3 April 2015.
155. Pires, Mathias M.; Rankin, Brian D.; Silvestro, Daniele; Quental, Tiago B. (2018). "Diversification dynamics of mammalian clades during the K–Pg mass extinction" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6170748). *Biology Letters*. **14** (9) 20180458. doi:10.1098/rsbl.2018.0458 (https://doi.org/10.1098%2Frsbl.2018.0458). PMC 6170748 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6170748). PMID 30258031 (https://pubmed.ncbi.nlm.nih.gov/30258031).
156. Springer, Mark S.; Foley, Nicole M.; Brady, Peggy L.; Gatesy, John; Murphy, William J. (29 November 2019). "Evolutionary Models for the Diversification of Placental Mammals Across the KPg Boundary" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6896846). *Frontiers in Genetics*. **10** 1241. Bibcode:2019FrGen..10.1241S (https://ui.adsabs.harvard.edu/abs/2019FrGen..10.1241S). doi:10.3389/fgene.2019.01241 (https://doi.org/10.3389%2Ffgene.2019.01241). ISSN 1664-8021 (https://search.worldcat.org/issn/1664-8021). PMC 6896846 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6896846). PMID 31850081 (https://pubmed.ncbi.nlm.nih.gov/31850081).
157. Shupinski, Alex B.; Wagner, Peter J.; Smith, Felisa A.; Lyons, S. Kathleen (3 July 2024). "Unique functional diversity during early Cenozoic mammal radiation of North America" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11286128). *Proceedings of the Royal Society B: Biological Sciences*. **291** (2026) 20240778. doi:10.1098/rspb.2024.0778 (https://doi.org/10.1098%2Frspb.2024.0778). ISSN 1471-2954 (https://search.worldcat.org/issn/1471-2954). PMC 11286128 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11286128). PMID 38955231 (https://pubmed.ncbi.nlm.nih.gov/38955231).
158. Springer, M. S.; Murphy, W. J.; Eizirik, E.; O'Brien, S. J. (2003). "Placental mammal diversification and the Cretaceous–Tertiary boundary" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC298725). *Proceedings of the National Academy of Sciences of the United States of America*. **100** (3): 1056–1061. Bibcode:2003PNAS..100.1056S (https://ui.adsabs.harvard.edu/abs/2003PNAS..100.1056S). doi:10.1073/pnas.0334222100 (https://doi.org/10.1073%2Fpnas.0334222100). PMC 298725 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC298725). PMID 12552136 (https://pubmed.ncbi.nlm.nih.gov/12552136).
159. Halliday, Thomas John Dixon; Upchurch, Paul; Goswami, Anjali (29 June 2016). "Eutherians experienced elevated evolutionary rates in the immediate aftermath of the Cretaceous–Palaeogene mass extinction" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4936024). *Proceedings of the Royal Society B: Biological Sciences*. **283** (1833) 20153026. doi:10.1098/rspb.2015.3026 (https://doi.org/10.1098%2Frspb.2015.3026). ISSN 0962-8452 (https://search.worldcat.org/issn/0962-8452). PMC 4936024 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4936024). PMID 27358361 (https://pubmed.ncbi.nlm.nih.gov/27358361).

160. Alroy, J. (May 1998). "Cope's rule and the dynamics of body mass evolution in North American fossil mammals" (https://web.archive.org/web/20251116123636/https://doc.rero.ch/record/20309/files/PAL_E4068.pdf) (PDF). *Science*. **280** (5364): 731–4. Bibcode:1998Sci...280..731A (https://ui.adsabs.harvard.edu/abs/1998Sci...280..731A). doi:10.1126/science.280.5364.731 (https://doi.org/10.1126%2Fscience.280.5364.731). PMID 9563948 (https://pubmed.ncbi.nlm.nih.gov/9563948). Archived from the original (http://doc.rero.ch/record/20309/files/PAL_E4068.pdf) (PDF) on 16 November 2025.
161. How life blossomed after the dinosaurs died | Science | AAAS (https://www.science.org/content/article/how-life-blossomed-after-dinosaurs-died)
162. "Mammals' bodies outpaced their brains right after the dinosaurs died" (https://www.sciencenews.org/article/mammal-big-bodies-small-brains-dinosaur-extinction-fossils). *Science News*. 31 March 2022. Retrieved 14 May 2022.
163. Bertrand, Ornella C.; Shelley, Sarah L.; Williamson, Thomas E.; Wible, John R.; Chester, Stephen G. B.; Flynn, John J.; Holbrook, Luke T.; Lyson, Tyler R.; Meng, Jin; Miller, Ian M.; Püschel, Hans P.; Smith, Thierry; Spaulding, Michelle; Tseng, Z. Jack; Brusatte, Stephen L. (April 2022). "Brawn before brains in placental mammals after the end-Cretaceous extinction" (https://www.pure.ed.ac.uk/ws/files/251642227/Bertrand_et_al._Science.pdf) (PDF). *Science*. **376** (6588): 80–85. Bibcode:2022Sci...376...80B (https://ui.adsabs.harvard.edu/abs/2022Sci...376...80B). doi:10.1126/science.abl5584 (https://doi.org/10.1126%2Fscience.abl5584). hdl:20.500.11820/d7fb8c6e-886e-4c1d-9977-0cd6406fda20 (https://hdl.handle.net/20.500.11820%2Fd7fb8c6e-886e-4c1d-9977-0cd6406fda20). ISSN 0036-8075 (https://search.worldcat.org/issn/0036-8075). PMID 35357913 (https://pubmed.ncbi.nlm.nih.gov/35357913). S2CID 247853831 (https://api.semanticscholar.org/CorpusID:247853831).
164. Vajda, Vivi; Raine, J. Ian; Hollis, Christopher J. (2001). "Indication of global deforestation at the Cretaceous–Tertiary boundary by New Zealand fern spike". *Science*. **294** (5547): 1700–1702. Bibcode:2001Sci...294.1700V (https://ui.adsabs.harvard.edu/abs/2001Sci...294.1700V). doi:10.1126/science.1064706 (https://doi.org/10.1126%2Fscience.1064706). PMID 11721051 (https://pubmed.ncbi.nlm.nih.gov/11721051). S2CID 40364945 (https://api.semanticscholar.org/CorpusID:40364945).
165. Johnson, K.R.; Hickey, L.J. (1991). "Megafloral change across the Cretaceous Tertiary boundary in the northern Great Plains and Rocky Mountains". In Sharpton, V.I.; Ward, P.D. (eds.). *Global Catastrophes in Earth History: An interdisciplinary conference on impacts, volcanism, and mass mortality*. Geological Society of America. ISBN 978-0-8137-2247-4.
166. Askin, R.A.; Jacobson, S.R. (1996). "Palynological change across the Cretaceous–Tertiary boundary on Seymour Island, Antarctica: environmental and depositional factors". In Keller, G.; MacLeod, N. (eds.). *Cretaceous–Tertiary Mass Extinctions: Biotic and Environmental Changes*. W W Norton. ISBN 978-0-393-96657-2.
167. Wappler, Torsten; Currano, Ellen D.; Wilf, Peter; Rust, Jes; Labandeira, Conrad C. (22 December 2009). "No post-Cretaceous ecosystem depression in European forests? Rich insect-feeding damage on diverse middle Palaeocene plants, Menat, France" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2817104). *Proceedings of the Royal Society B: Biological Sciences*. **276** (1677): 4271–4277. doi:10.1098/rspb.2009.1255 (https://doi.org/10.1098%2Frspb.2009.1255). ISSN 0962-8452 (https://search.worldcat.org/issn/0962-8452). PMC 2817104 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2817104). PMID 19776074 (https://pubmed.ncbi.nlm.nih.gov/19776074).
168. Herman, A. B.; Akhmetiev, M. A.; Kodrul, T. M.; Moiseeva, M. G.; Iakovleva, A. I. (24 February 2009). "Flora development in Northeastern Asia and Northern Alaska during the Cretaceous-Paleogene transitional epoch". *Stratigraphy and Geological Correlation*. **17** (1): 79–97. Bibcode:2009SGC....17...79H (https://ui.adsabs.harvard.edu/abs/2009SGC....17...79H). doi:10.1134/S0869593809010079 (https://doi.org/10.1134%2FS0869593809010079). ISSN 0869-5938 (https://search.worldcat.org/issn/0869-5938).

169. Herman, A. B. (10 December 2013). "Albian-Paleocene flora of the north pacific: Systematic composition, palaeofloristics and phytostratigraphy". *Stratigraphy and Geological Correlation*. **21** (7): 689–747. Bibcode:2013SGC....21..689H (https://ui.adsabs.harvard.edu/abs/2013SGC....21..689H). doi:10.1134/S0869593813070034 (https://doi.org/10.1134%2FS0869593813070034). ISSN 0869-5938 (https://search.worldcat.org/issn/0869-5938).
170. Bajdek, Piotr (10 May 2019). "Divergence rates of subviral pathogens of angiosperms abruptly decreased at the Cretaceous-Paleogene boundary" (https://www.biodiversitylibrary.org/part/424355). *Rethinking Ecology*. **4**: 89–101. Bibcode:2019REco....4...89B (https://ui.adsabs.harvard.edu/abs/2019REco....4...89B). doi:10.3897/rethinkingecology.4.33014 (https://doi.org/10.3897%2Frethinkingecology.4.33014). S2CID 196664424 (https://api.semanticscholar.org/CorpusID:196664424). Retrieved 23 March 2023.
171. Thompson, Jamie B.; Ramírez-Barahona, Santiago (September 2023). "No phylogenetic evidence for angiosperm mass extinction at the Cretaceous–Palaeogene (K-Pg) boundary" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10498348). *Biology Letters*. **19** (9) 20230314. doi:10.1098/rsbl.2023.0314 (https://doi.org/10.1098%2Frsbl.2023.0314). ISSN 1744-957X (https://search.worldcat.org/issn/1744-957X). PMC 10498348 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10498348). PMID 37700701 (https://pubmed.ncbi.nlm.nih.gov/37700701).
172. Vajda, Vivi; McLoughlin, Stephen (5 March 2004). "Fungal Proliferation at the Cretaceous-Tertiary Boundary" (http://urn.kb.se/resolve?urn=urn:nbn:se:nrm:diva-320). *Science*. **303** (5663): 1489. doi:10.1126/science.1093807 (https://doi.org/10.1126%2Fscience.1093807). PMID 15001770 (https://pubmed.ncbi.nlm.nih.gov/15001770). S2CID 44720346 (https://api.semanticscholar.org/CorpusID:44720346).
173. Schultz, P.; d'Hondt, S. (1996). "Cretaceous–Tertiary (Chicxulub) impact angle and its consequences" (https://digitalcommons.uri.edu/gsofacpubs/1183). *Geology*. **24** (11): 963–967. Bibcode:1996Geo....24..963S (https://ui.adsabs.harvard.edu/abs/1996Geo....24..963S). doi:10.1130/0091-7613(1996)024<0963:CTCIAA>2.3.CO;2 (https://doi.org/10.1130%2F0091-7613%281996%29024%3C0963%3ACTCIAA%3E2.3.CO%3B2).
174. Field, Daniel J.; Bercovici, Antoine; Berv, Jacob S.; Dunn, Regan; Fastovsky, David E.; Lyson, Tyler R.; et al. (24 May 2018). "Early evolution of modern birds structured by global forest collapse at the end-Cretaceous mass extinction" (https://digitalcommons.uri.edu/geo_facpubs/75). *Current Biology*. **28** (11): 1825–1831.e2. Bibcode:2018CBio...28E1825F (https://ui.adsabs.harvard.edu/abs/2018CBio...28E1825F). doi:10.1016/j.cub.2018.04.062 (https://doi.org/10.1016%2Fj.cub.2018.04.062). PMID 29804807 (https://pubmed.ncbi.nlm.nih.gov/29804807). S2CID 44075214 (https://api.semanticscholar.org/CorpusID:44075214).
175. "Online guide to the continental Cretaceous–Tertiary boundary in the Raton basin, Colorado and New Mexico" (https://web.archive.org/web/20060925223706/http://esp.cr.usgs.gov/info/kt/stop2b.html). U.S. Geological Survey. 2004. Archived from the original (https://esp.cr.usgs.gov/info/kt/stop2b.html) on 25 September 2006. Retrieved 8 July 2007.
176. Smathers, G.A.; Mueller-Dombois, D. (1974). *Invasion and Recovery of Vegetation after a Volcanic Eruption in Hawaii* (https://web.archive.org/web/20140403124910/http://www.nps.gov/history/history/online_books/science/5/contents.htm). Scientific Monograph. Vol. 5. United States National Park Service. Archived from the original (http://www.nps.gov/history/history/online_books/science/5/contents.htm) on 3 April 2014. Retrieved 9 July 2007.
177. Fawcett, Jeffrey A.; Maere, Steven; Van de Peer, Yves (April 2009). "Plants with double genomes might have had a better chance to survive the Cretaceous-Tertiary extinction event" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2667025). *Proceedings of the National Academy of Sciences of the United States of America*. **106** (14): 5737–5742. Bibcode:2009PNAS..106.5737F (https://ui.adsabs.harvard.edu/abs/2009PNAS..106.5737F). doi:10.1073/pnas.0900906106 (https://doi.org/10.1073%2Fpnas.0900906106). PMC 2667025 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2667025). PMID 19325131 (https://pubmed.ncbi.nlm.nih.gov/19325131).

178. "Dinosaur-killing asteroid strike gave rise to Amazon rainforest" (https://www.bbc.com/news/science-environment-56617409). *BBC News*. 2 April 2021. Retrieved 9 May 2021.
179. Carvalho, Mónica R.; Jaramillo, Carlos; Parra, Felipe de la; et al. (2 April 2021). "Extinction at the end-Cretaceous and the origin of modern Neotropical rainforests" (https://www.researchgate.net/publication/350569900). *Science*. **372** (6537): 63–68. Bibcode:2021Sci...372...63C (https://ui.adsabs.harvard.edu/abs/2021Sci...372...63C). doi:10.1126/science.abf1969 (https://doi.org/10.1126%2Fscience.abf1969). hdl:11336/171234 (https://hdl.handle.net/11336%2F171234). ISSN 0036-8075 (https://search.worldcat.org/issn/0036-8075). PMID 33795451 (https://pubmed.ncbi.nlm.nih.gov/33795451). S2CID 232484243 (https://api.semanticscholar.org/CorpusID:232484243). Retrieved 9 May 2021.
180. Visscher, H.; Brinkhuis, H.; Dilcher, D. L.; Elsik, W. C.; Eshet, Y.; Looy, C. V.; Rampino, M. R.; Traverse, A. (5 March 1996). "The terminal Paleozoic fungal event: evidence of terrestrial ecosystem destabilization and collapse" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC39926). *Proceedings of the National Academy of Sciences of the United States of America*. **93** (5): 2155–2158. Bibcode:1996PNAS...93.2155V (https://ui.adsabs.harvard.edu/abs/1996PNAS...93.2155V). doi:10.1073/pnas.93.5.2155 (https://doi.org/10.1073%2Fpnas.93.5.2155). PMC 39926 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC39926). PMID 11607638 (https://pubmed.ncbi.nlm.nih.gov/11607638).
181. Jack Wolfe (1991). "Palaeobotanical evidence for a June 'impact winter' at the Cretaceous/Tertiary boundary" (https://www.nature.com/articles/352420a0.epdf?sharing_token=SZ2zIDwp4nwPzqW21MBRWdRgN0jAjWel9jnR3ZoTv0M7VLIKoE6x4ag0Nr_AUthe7MxQ_v5tDeeuXolxlXkVi1rOJ-QBzJ-jws0_6AgoDTZ1jGwfolUTVzyctXnuY6AmMFuTIS3OvXWg_NJMTFUMFjCyE1WglpNX1Vg16yxkSt4Mmjas8D9WIxl3VRQ66tWvEI3uj-BYCYv86RvbDAb7-Q%3D%3D). *Nature*. **352** (6334): 420. Bibcode:1991Natur.352..420W (https://ui.adsabs.harvard.edu/abs/1991Natur.352..420W). doi:10.1038/352420a0 (https://doi.org/10.1038%2F352420a0). S2CID 4242454 (https://api.semanticscholar.org/CorpusID:4242454).
182. During, Melanie A. D.; Smit, Jan; Voeten, Dennis F. A. E.; Berruyer, Camille; Tafforeau, Paul; Sanchez, Sophie; Stein, Koen H. W.; Verdegaal-Warmerdam, Suzan J. A.; Van Der Lubbe, Jeroen H. J. L. (23 February 2022). "The Mesozoic terminated in boreal spring" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8891016). *Nature*. **603** (7899): 91–94. Bibcode:2022Natur.603...91D (https://ui.adsabs.harvard.edu/abs/2022Natur.603...91D). doi:10.1038/s41586-022-04446-1 (https://doi.org/10.1038%2Fs41586-022-04446-1). PMC 8891016 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8891016). PMID 35197634 (https://pubmed.ncbi.nlm.nih.gov/35197634).
183. Depalma, Robert A.; Oleinik, Anton A.; Gurche, Loren P.; Burnham, David A.; Klingler, Jeremy J.; McKinney, Curtis J.; Cichocki, Frederick P.; Larson, Peter L.; Egerton, Victoria M.; Wogelius, Roy A.; Edwards, Nicholas P.; Bergmann, Uwe; Manning, Phillip L. (8 December 2021). "Seasonal calibration of the end-cretaceous Chicxulub impact event" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8655067). *Nature*. **11** (1): 23704. Bibcode:2021NatSR..1123704D (https://ui.adsabs.harvard.edu/abs/2021NatSR..1123704D). doi:10.1038/s41598-021-03232-9 (https://doi.org/10.1038%2Fs41598-021-03232-9). PMC 8655067 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8655067). PMID 34880389 (https://pubmed.ncbi.nlm.nih.gov/34880389).
184. During, M. A. D.; Voeten, D. F. A. E.; Van der Lubbe, J. H J. L.; Ahlberg, P. E. (2024). "Calibrations without raw data—A response to "Seasonal calibration of the end-cretaceous Chicxulub impact event" " (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11568822). *PeerJ*. **12** e18519. e18519. Bibcode:2024PeerJ..1218519D (https://ui.adsabs.harvard.edu/abs/2024PeerJ..1218519D). doi:10.7717/peerj.18519 (https://doi.org/10.7717%2Fpeerj.18519). PMC 11568822 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11568822). PMID 39553725 (https://pubmed.ncbi.nlm.nih.gov/39553725).

185. Price, Michael (6 December 2022). "Paleontologist accused of faking data in dino-killing asteroid paper" (https://www.science.org/content/article/paleontologist-accused-faking-data-dino-killing-asteroid-paper). *www.science.org*. Retrieved 7 December 2022.
186. During, Melanie A. D.; Smit, Jan; Voeten, Dennis F. A. E.; et al. (23 February 2022). "The Mesozoic terminated in boreal spring" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8891016). *Nature*. **603** (7899): 91–94. Bibcode:2022Natur.603...91D (https://ui.adsabs.harvard.edu/abs/2022Natur.603...91D). doi:10.1038/s41586-022-04446-1 (https://doi.org/10.1038%2Fs41586-022-04446-1). PMC 8891016 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8891016). PMID 35197634 (https://pubmed.ncbi.nlm.nih.gov/35197634).
187. "Springtime was the season the dinosaurs died, ancient fish fossils suggest" (https://www.science.org/content/article/springtime-was-season-dinosaurs-died-ancient-fish-fossils-suggest). *Science*. 23 February 2022. Retrieved 24 February 2022.
188. Barras, Colin (23 February 2022). "Fossil fish reveal timing of asteroid that killed the dinosaurs" (https://doi.org/10.1038%2Fd41586-022-00511-x). *Nature*. **603** (7899): 17. Bibcode:2022Natur.603...17B (https://ui.adsabs.harvard.edu/abs/2022Natur.603...17B). doi:10.1038/d41586-022-00511-x (https://doi.org/10.1038%2Fd41586-022-00511-x). PMID 35197589 (https://pubmed.ncbi.nlm.nih.gov/35197589). S2CID 247083600 (https://api.semanticscholar.org/CorpusID:247083600).
189. Ouellette, Jennifer (23 February 2022). "An asteroid killed dinosaurs in spring—which might explain why mammals survived – New study sheds light on why species extinction was so selective after the K-Pg impact" (https://arstechnica.com/science/2022/02/fish-fossils-show-asteroid-that-wiped-out-the-dinosaurs-struck-in-the-spring/). *Ars Technica*. Retrieved 26 February 2022.
190. Signor, Philip W. III; Lipps, Jere H. (1982). "Sampling bias, gradual extinction patterns, and catastrophes in the fossil record". In Silver, L.T.; Schultz, Peter H. (eds.). *Geological implications of impacts of large asteroids and comets on the Earth* (https://books.google.com/books?id=efz87W88HawC&pg=291). Boulder, CO: Geological Society of America. pp. 291–296. ISBN 978-0-8137-2190-3. OCLC 4434434112 (https://search.worldcat.org/oclc/4434434112). Special Publication 190. Retrieved 25 October 2015.
191. Mukhopadhyay, Sujoy (2001). "A Short Duration of the Cretaceous-Tertiary Boundary Event: Evidence from Extraterrestrial Helium-3" (https://authors.library.caltech.edu/36545/7/Mukhopad3.sup.pdf) (PDF). *Science*. **291** (5510): 1952–1955. Bibcode:2001Sci...291.1952M (https://ui.adsabs.harvard.edu/abs/2001Sci...291.1952M). doi:10.1126/science.291.5510.1952 (https://doi.org/10.1126%2Fscience.291.5510.1952). PMID 11239153 (https://pubmed.ncbi.nlm.nih.gov/11239153). Archived (https://web.archive.org/web/20230307171536/https://authors.library.caltech.edu/36545/7/Mukhopad3.sup.pdf) (PDF) from the original on 7 March 2023.
192. Clyde, William C.; Ramezani, Jahandar; Johnson, Kirk R.; Bowring, Samuel A.; Jones, Matthew M. (15 October 2016). "Direct high-precision U–Pb geochronology of the end-Cretaceous extinction and calibration of Paleocene astronomical timescales". *Earth and Planetary Science Letters*. **452**: 272–280. Bibcode:2016E&PSL.452..272C (https://ui.adsabs.harvard.edu/abs/2016E&PSL.452..272C). doi:10.1016/j.epsl.2016.07.041 (https://doi.org/10.1016%2Fj.epsl.2016.07.041).
193. Renne, Paul R.; Deino, Alan L.; Hilgen, Frederik J.; Kuiper, Klaudia F.; Mark, Darren F.; Mitchell, William S.; Morgan, Leah E.; Mundil, Roland; Smit, Jan (8 February 2013). "Time Scales of Critical Events Around the Cretaceous-Paleogene Boundary". *Science*. **339** (6120): 684–687. Bibcode:2013Sci...339..684R (https://ui.adsabs.harvard.edu/abs/2013Sci...339..684R). doi:10.1126/science.1230492 (https://doi.org/10.1126%2Fscience.1230492). ISSN 0036-8075 (https://search.worldcat.org/issn/0036-8075). PMID 23393261 (https://pubmed.ncbi.nlm.nih.gov/23393261).
194. Weisberger, Mindy (22 December 2021). "Darkness caused by dino-killing asteroid snuffed out life on Earth in 9 months" (https://www.livescience.com/cretaceous-extinction-darkness). *livescience.com*. Retrieved 17 November 2022.

195. de Laubenfels, M. W. (1956). "Dinosaur extinction: One more hypothesis". *Journal of Paleontology*. **30** (1): 207–218. JSTOR 1300393 (https://www.jstor.org/stable/1300393).
196. Smit, J.; Klaver, J. (1981). "Sanidine spherules at the Cretaceous-Tertiary boundary indicate a large impact event". *Nature*. **292** (5818): 47–49. Bibcode:1981Natur.292...47S (https://ui.adsabs.harvard.edu/abs/1981Natur.292...47S). doi:10.1038/292047a0 (https://doi.org/10.1038%2F292047a0). S2CID 4331801 (https://api.semanticscholar.org/CorpusID:4331801).
197. Olsson, Richard K.; Miller, Kenneth G.; Browning, James V.; Habib, Daniel; Sugarman, Peter J. (1 August 1997). "Ejecta layer at the Cretaceous-Tertiary boundary, Bass River, New Jersey (Ocean Drilling Program Leg 174AX)". *Geology*. **25** (8): 759. Bibcode:1997Geo....25..759O (https://ui.adsabs.harvard.edu/abs/1997Geo....25..759O). doi:10.1130/0091-7613(1997)025<0759:ELATCT>2.3.CO;2 (https://doi.org/10.1130%2F0091-7613%281997%29025%3C0759%3AELATCT%3E2.3.CO%3B2). ISSN 0091-7613 (https://search.worldcat.org/issn/0091-7613).
198. Bohor, B. F.; Foord, E. E.; Modreski, P. J.; Triplehorn, D. M. (1984). "Mineralogic evidence for an impact event at the Cretaceous-Tertiary boundary". *Science*. **224** (4651): 867–9. Bibcode:1984Sci...224..867B (https://ui.adsabs.harvard.edu/abs/1984Sci...224..867B). doi:10.1126/science.224.4651.867 (https://doi.org/10.1126%2Fscience.224.4651.867). PMID 17743194 (https://pubmed.ncbi.nlm.nih.gov/17743194). S2CID 25887801 (https://api.semanticscholar.org/CorpusID:25887801).
199. Bohor, B. F.; Modreski, P. J.; Foord, E. E. (1987). "Shocked quartz in the Cretaceous-Tertiary boundary clays: Evidence for a global distribution" (https://zenodo.org/record/1230978). *Science*. **236** (4802): 705–709. Bibcode:1987Sci...236..705B (https://ui.adsabs.harvard.edu/abs/1987Sci...236..705B). doi:10.1126/science.236.4802.705 (https://doi.org/10.1126%2Fscience.236.4802.705). PMID 17748309 (https://pubmed.ncbi.nlm.nih.gov/17748309). S2CID 31383614 (https://api.semanticscholar.org/CorpusID:31383614).
200. Bourgeois, J.; Hansen, T. A.; Wiberg, P. A.; Kauffman, E. G. (1988). "A tsunami deposit at the Cretaceous-Tertiary boundary in Texas". *Science*. **241** (4865): 567–570. Bibcode:1988Sci...241..567B (https://ui.adsabs.harvard.edu/abs/1988Sci...241..567B). doi:10.1126/science.241.4865.567 (https://doi.org/10.1126%2Fscience.241.4865.567). PMID 17774578 (https://pubmed.ncbi.nlm.nih.gov/17774578). S2CID 7447635 (https://api.semanticscholar.org/CorpusID:7447635).
201. Bralower, Timothy J.; Paull, Charles K.; Mark Leckie, R. (1 April 1998). "The Cretaceous-Tertiary boundary cocktail: Chicxulub impact triggers margin collapse and extensive sediment gravity flows". *Geology*. **26** (4): 331. Bibcode:1998Geo....26..331B (https://ui.adsabs.harvard.edu/abs/1998Geo....26..331B). doi:10.1130/0091-7613(1998)026<0331:TCTBCC>2.3.CO;2 (https://doi.org/10.1130%2F0091-7613%281998%29026%3C0331%3ATCTBCC%3E2.3.CO%3B2). ISSN 0091-7613 (https://search.worldcat.org/issn/0091-7613).
202. Pope, K. O.; Ocampo, A. C.; Kinsland, G. L.; Smith, R. (1996). "Surface expression of the Chicxulub crater". *Geology*. **24** (6): 527–530. Bibcode:1996Geo....24..527P (https://ui.adsabs.harvard.edu/abs/1996Geo....24..527P). doi:10.1130/0091-7613(1996)024<0527:SEOTCC>2.3.CO;2 (https://doi.org/10.1130%2F0091-7613%281996%29024%3C0527%3ASEOTCC%3E2.3.CO%3B2). PMID 11539331 (https://pubmed.ncbi.nlm.nih.gov/11539331).
203. Stöffler, Dieter; Artemieva, Natalya A.; Ivanov, Boris A.; Hecht, Lutz; Kenkmann, Thomas; Schmitt, Ralf Thomas; Tagle, Roald Alberto; Wittmann, Axel (26 January 2010). "Origin and emplacement of the impact formations at Chicxulub, Mexico, as revealed by the ICDP deep drilling at Yaxcopoil-1 and by numerical modeling". *Meteoritics & Planetary Science*. **39** (7): 1035–1067. doi:10.1111/j.1945-5100.2004.tb01128.x (https://doi.org/10.1111%2Fj.1945-5100.2004.tb01128.x). ISSN 1086-9379 (https://search.worldcat.org/issn/1086-9379).

204. Depalma, Robert A.; Smit, Jan; Burnham, David A.; Kuiper, Klaudia; Manning, Phillip L.; Oleinik, Anton; Larson, Peter; Maurrasse, Florentin J.; Vellekoop, Johan; Richards, Mark A.; Gurche, Loren; Alvarez, Walter (2019). "A seismically induced onshore surge deposit at the KPG boundary, North Dakota" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6486721). *Proceedings of the National Academy of Sciences of the United States of America*. **116** (17): 8190–8199. Bibcode:2019PNAS..116.8190D (https://ui.adsabs.harvard.edu/abs/2019PNAS..116.8190D). doi:10.1073/pnas.1817407116 (https://doi.org/10.1073%2Fpnas.1817407116). PMC 6486721 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6486721). PMID 30936306 (https://pubmed.ncbi.nlm.nih.gov/30936306).
205. "National Natural Landmarks – National Natural Landmarks (U.S. National Park Service)" (https://www.nps.gov/subjects/nnlandmarks/site.htm?Site=HECR-MT). *www.nps.gov*. Retrieved 22 March 2019. "Year designated: 1966"
206. Smit, J., *et al.* (2017) Tanis, a mixed marine-continental event deposit at the KPG Boundary in North Dakota caused by a seiche triggered by seismic waves of the Chicxulub Impact (https://gsa.confex.com/gsa/2017AM/webprogram/Paper305713.html) Paper No. 113–15, presented 23 October 2017 at the GSA Annual Meeting, Seattle, Washington, USA.
207. DePalma, R. *et al.* (2017) Life after impact: A remarkable mammal burrow from the Chicxulub aftermath in the Hell Creek Formation, North Dakota (https://gsa.confex.com/gsa/2017AM/webprogram/Paper305627.html) Paper No. 113–16, presented 23 October 2017 at the GSA Annual Meeting, Seattle, Washington, USA.
208. Kaskes, P.; Goderis, S.; Belza, J.; Tack, P.; DePalma, R. A.; Smit, J.; Vincze, Laszlo; Vabgaecje, F.; Claeys, P. (2019). "Caught in amber: Geochemistry and petrography of uniquely preserved Chicxulub microtektites from the Tanis K-Pg site from North Dakota (USA)". *Large Meteorite Impacts VI 2019 (LPI Contrib. No. 2136)* (https://cris.vub.be/ws/portalfiles/portal/48767201/5090_Kaskes_et_al._2019_Abstract_Tanis_spherules_FINAL.pdf) (PDF). Vol. 6. Houston, TX: Lunar and Planetary Institute. pp. 1–2. Retrieved 11 April 2021.
209. Barras, Colin (5 April 2019). "Does fossil site record dino-killing impact?". *Science*. **364** (6435): 10–11. Bibcode:2019Sci...364...10B (https://ui.adsabs.harvard.edu/abs/2019Sci...364...10B). doi:10.1126/science.364.6435.10 (https://doi.org/10.1126%2Fscience.364.6435.10). PMID 30948530 (https://pubmed.ncbi.nlm.nih.gov/30948530). S2CID 96434764 (https://api.semanticscholar.org/CorpusID:96434764).
210. Alegret, Laia; Molina, Eustoquio; Thomas, Ellen (July 2003). "Benthic foraminiferal turnover across the Cretaceous/Paleogene boundary at Agost (southeastern Spain): paleoenvironmental inferences". *Marine Micropaleontology*. **48** (3–4): 251–279. Bibcode:2003MarMP..48..251A (https://ui.adsabs.harvard.edu/abs/2003MarMP..48..251A). doi:10.1016/S0377-8398(03)00022-7 (https://doi.org/10.1016%2FS0377-8398%2803%2900022-7).
211. Keller, G.; Adatte, T.; Stinnesbeck, W.; Rebolledo-Vieyra, M.; Fucugauchi, J. U.; Kramar, U.; Stüben, D. (2004). "Chicxulub impact predates the K–T boundary mass extinction" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC374316). *Proceedings of the National Academy of Sciences of the United States of America*. **101** (11): 3753–3758. Bibcode:2004PNAS..101.3753K (https://ui.adsabs.harvard.edu/abs/2004PNAS..101.3753K). doi:10.1073/pnas.0400396101 (https://doi.org/10.1073%2Fpnas.0400396101). PMC 374316 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC374316). PMID 15004276 (https://pubmed.ncbi.nlm.nih.gov/15004276).
212. Keller, Gerta; Adatte, Thierry; Stinnesbeck, Wolfgang; STüBEN, Doris; Berner, Zsolt; Kramar, Utz; Harting, Markus (26 January 2010). "More evidence that the Chicxulub impact predates the K/T mass extinction". *Meteoritics & Planetary Science*. **39** (7): 1127–1144. doi:10.1111/j.1945-5100.2004.tb01133.x (https://doi.org/10.1111%2Fj.1945-5100.2004.tb01133.x). ISSN 1086-9379 (https://search.worldcat.org/issn/1086-9379).

213. Perlman, David. "Dinosaur extinction battle flares" (http://www.sfgate.com/science/article/Dinosaur-extinction-battle-flares-4261978.php). *sfgate.com*. Archived (https://web.archive.org/web/20130208122400/http://www.sfgate.com/science/article/Dinosaur-extinction-battle-flares-4261978.php) from the original on 8 February 2013. Retrieved 8 February 2013.
214. Timms, Nicholas E.; Kirkland, Christopher L.; Cavosie, Aaron J.; Rae, Auriol S.P.; Rickard, William D.A.; Evans, Noreen J.; Erickson, Timmons M.; Wittmann, Axel; Ferrière, Ludovic; Collins, Gareth S.; Gulick, Sean P.S. (15 July 2020). "Shocked titanite records Chicxulub hydrothermal alteration and impact age" (https://doi.org/10.1016%2Fj.gca.2020.04.031). *Geochimica et Cosmochimica Acta*. **281**: 12–30. Bibcode:2020GeCoA.281...12T (https://ui.adsabs.harvard.edu/abs/2020GeCoA.281...12T). doi:10.1016/j.gca.2020.04.031 (https://doi.org/10.1016%2Fj.gca.2020.04.031). hdl:10044/1/79097 (https://hdl.handle.net/10044%2F1%2F79097).
215. Bottke, W. F.; Vokrouhlický, D.; Nesvorný, D. (September 2007). "An asteroid breakup 160 Myr ago as the probable source of the K/T impactor". *Nature*. **449** (7158): 48–53. Bibcode:2007Natur.449...48B (https://ui.adsabs.harvard.edu/abs/2007Natur.449...48B). doi:10.1038/nature06070 (https://doi.org/10.1038%2Fnature06070). PMID 17805288 (https://pubmed.ncbi.nlm.nih.gov/17805288). S2CID 4322622 (https://api.semanticscholar.org/CorpusID:4322622).
216. Majaess, D. J.; Higgins, D.; Molnar, L. A.; Haegert, M. J.; Lane, D. J.; Turner, D. G.; Nielsen, I. (February 2009). "New constraints on the asteroid 298 Baptistina, the alleged family member of the K/T impactor". *The Journal of the Royal Astronomical Society of Canada*. **103** (1): 7–10. arXiv:0811.0171 (https://arxiv.org/abs/0811.0171). Bibcode:2009JRASC.103....7M (https://ui.adsabs.harvard.edu/abs/2009JRASC.103....7M).
217. Reddy, V.; Emery, J. P.; Gaffey, M. J.; Bottke, W. F.; Cramer, A.; Kelley, M. S. (December 2009). "Composition of 298 Baptistina: Implications for the K/T impactor link" (https://doi.org/10.1111%2Fj.1945-5100.2009.tb02001.x). *Meteoritics & Planetary Science*. **44** (12): 1917–1927. Bibcode:2009M&PS...44.1917R (https://ui.adsabs.harvard.edu/abs/2009M&PS...44.1917R). doi:10.1111/j.1945-5100.2009.tb02001.x (https://doi.org/10.1111%2Fj.1945-5100.2009.tb02001.x). S2CID 39644763 (https://api.semanticscholar.org/CorpusID:39644763).
218. "NASA's WISE raises doubt about asteroid family believed responsible for dinosaur extinction" (https://www.sciencedaily.com/releases/2011/09/110919144042.htm). *ScienceDaily*. 20 September 2011. Archived (https://web.archive.org/web/20110923215602/http://www.sciencedaily.com/releases/2011/09/110919144042.htm) from the original on 23 September 2011. Retrieved 21 September 2011.
219. Osterloff, Emily (2018). "How an asteroid ended the age of the dinosaurs" (https://www.nhm.ac.uk/discover/how-an-asteroid-caused-extinction-of-dinosaurs.html). London: Natural History Museum. Archived (https://web.archive.org/web/20220426174952/https://www.nhm.ac.uk/discover/how-an-asteroid-caused-extinction-of-dinosaurs.html) from the original on 26 April 2022. Retrieved 18 May 2022.
220. Robertson, D. S.; Lewis, W. M.; Sheehan, P. M.; Toon, O. B. (2013). "K/Pg extinction: Re-evaluation of the heat/fire hypothesis" (https://doi.org/10.1002%2Fjgrg.20018). *Journal of Geophysical Research*. **118** (1): 329–336. Bibcode:2013JGRG..118..329R (https://ui.adsabs.harvard.edu/abs/2013JGRG..118..329R). doi:10.1002/jgrg.20018 (https://doi.org/10.1002%2Fjgrg.20018).
221. Pope, K. O.; d'Hondt, S. L.; Marshall. C. R. (1998). "Meteorite impact and the mass extinction of species at the Cretaceous/Tertiary boundary" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC33889). *Proceedings of the National Academy of Sciences of the United States of America*. **95** (19): 11028–11029. Bibcode:1998PNAS...9511028P (https://ui.adsabs.harvard.edu/abs/1998PNAS...9511028P). doi:10.1073/pnas.95.19.11028 (https://doi.org/10.1073%2Fpnas.95.19.11028). PMC 33889 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC33889). PMID 9736679 (https://pubmed.ncbi.nlm.nih.gov/9736679).

222. Brugger, Julia; Feulner, Georg; Petri, Stefan (2016). "Baby, it's cold outside: Climate model simulations of the effects of the asteroid impact at the end of the Cretaceous" (https://publications.pik-potsdam.de/pubman/item/item_21282_1/component/file_21283/7402oa.pdf) (PDF). *Geophysical Research Letters*. **44** (1): 419–427. Bibcode:2017GeoRL..44..419B (https://ui.adsabs.harvard.edu/abs/2017GeoRL..44..419B). doi:10.1002/2016GL072241 (https://doi.org/10.1002%2F2016GL072241). S2CID 53631053 (https://api.semanticscholar.org/CorpusID:53631053).
223. Bourgeois, J. (2009). "Chapter 3. Geologic effects and records of tsunamis" (http://faculty.washington.edu/jbourgeo/BourgeoisTheSeaCh3.pdf) (PDF). In Robinson, A.R.; Bernard, E.N. (eds.). *Tsunamis*. The Sea (Ideas and Observations on Progress in the Study of the Seas). Vol. 15. Boston, MA: Harvard University. ISBN 978-0-674-03173-9. Retrieved 29 March 2012.
224. Lawton, T. F.; Shipley, K. W.; Aschoff, J. L.; Giles, K. A.; Vega, F. J. (2005). "Basinward transport of Chicxulub ejecta by tsunami-induced backflow, La Popa basin, northeastern Mexico, and its implications for distribution of impact-related deposits flanking the Gulf of Mexico". *Geology*. **33** (2): 81–84. Bibcode:2005Geo....33...81L (https://ui.adsabs.harvard.edu/abs/2005Geo....33...81L). doi:10.1130/G21057.1 (https://doi.org/10.1130%2FG21057.1).
225. Albertão, G. A.; P. P. Martins Jr. (1996). "A possible tsunami deposit at the Cretaceous-Tertiary boundary in Pernambuco, northeastern Brazil" (http://www.repositorio.ufop.br/handle/123456789/4089). *Sed. Geol*. **104** (1–4): 189–201. Bibcode:1996SedG..104..189A (https://ui.adsabs.harvard.edu/abs/1996SedG..104..189A). doi:10.1016/0037-0738(95)00128-X (https://doi.org/10.1016%2F0037-0738%2895%2900128-X).
226. Norris, R. D.; Firth, J.; Blusztajn, J. S. & Ravizza, G. (2000). "Mass failure of the North Atlantic margin triggered by the Cretaceous-Paleogene bolide impact". *Geology*. **28** (12): 1119–1122. Bibcode:2000Geo....28.1119N (https://ui.adsabs.harvard.edu/abs/2000Geo....28.1119N). doi:10.1130/0091-7613(2000)28<1119:MFOTNA>2.0.CO;2 (https://doi.org/10.1130%2F0091-7613%282000%2928%3C1119%3AMFOTNA%3E2.0.CO%3B2).
227. Bryant, Edward (June 2014). *Tsunami: The Underrated Hazard* (https://books.google.com/books?id=tOkpBAAAQBAJ&pg=PA178). Springer. p. 178. ISBN 978-3-319-06133-7. Archived (https://web.archive.org/web/20190901055132/https://books.google.com/books?id=tOkpBAAAQBAJ&pg=PA178&lpg=PA17) from the original on 1 September 2019. Retrieved 30 August 2017.
228. Smit, Jan; Montanari, Alessandro; Swinburne, Nicola H.; Alvarez, Walter; Hildebrand, Alan R.; Margolis, Stanley V.; Claeys, Philippe; Lowrie, William; Asaro, Frank (1992). "Tektite-bearing, deep-water clastic unit at the Cretaceous-Tertiary boundary in northeastern Mexico". *Geology*. **20** (2): 99–103. Bibcode:1992Geo....20...99S (https://ui.adsabs.harvard.edu/abs/1992Geo....20...99S). doi:10.1130/0091-7613(1992)020<0099:TBDWCU>2.3.CO;2 (https://doi.org/10.1130%2F0091-7613%281992%29020%3C0099%3ATBDWCU%3E2.3.CO%3B2). PMID 11537752 (https://pubmed.ncbi.nlm.nih.gov/11537752).
229. *Field guide to Cretaceous-tertiary boundary sections in northeastern Mexico* (https://www.lpi.usra.edu/lpi/contribution_docs/LPI-000827.pdf) (PDF). Lunar and Planetary Institute. 1994. Archived (https://web.archive.org/web/20190821130759/https://www.lpi.usra.edu/lpi/contribution_docs/LPI-000827.pdf) (PDF) from the original on 21 August 2019. Retrieved 25 June 2019.
230. Smit, Jan (1999). "The global stratigraphy of the Cretaceous-Tertiary boundary impact ejecta". *Annual Review of Earth and Planetary Sciences*. **27**: 75–113. Bibcode:1999AREPS..27...75S (https://ui.adsabs.harvard.edu/abs/1999AREPS..27...75S). doi:10.1146/annurev.earth.27.1.75 (https://doi.org/10.1146%2Fannurev.earth.27.1.75).
231. Kring, David A. (2007). "The Chicxulub impact event and its environmental consequences at the Cretaceous-Tertiary boundary". *Palaeogeography, Palaeoclimatology, Palaeoecology*. **255** (1–2): 4–21. Bibcode:2007PPP...255....4K (https://ui.adsabs.harvard.edu/abs/2007PPP...255....4K). doi:10.1016/j.palaeo.2007.02.037 (https://doi.org/10.1016%2Fj.palaeo.2007.02.037).

232. "Chicxulub impact event" (https://www.lpi.usra.edu/science/kring/Chicxulub/regional-effects/). *www.lpi.usra.edu*. Archived (https://web.archive.org/web/20190726023401/https://www.lpi.usra.edu/science/kring/Chicxulub/regional-effects/) from the original on 26 July 2019. Retrieved 25 June 2019.
233. Pope, K.O.; Baines, K.H.; Ocampo, A.C.; Ivanov, B. A. (1997). "Energy, volatile production, and climatic effects of the Chicxulub Cretaceous/Tertiary impact" (https://doi.org/10.1029%2F97JE01743). *Journal of Geophysical Research*. **102** (E9): 21645–21664. Bibcode:1997JGR...10221645P (https://ui.adsabs.harvard.edu/abs/1997JGR...10221645P). doi:10.1029/97JE01743 (https://doi.org/10.1029%2F97JE01743). PMID 11541145 (https://pubmed.ncbi.nlm.nih.gov/11541145).
234. Belcher, Claire M.; Hadden, Rory M.; Rein, Guillermo; Morgan, Joanna V.; Artemieva, Natalia; Goldin, Tamara (22 January 2015). "An experimental assessment of the ignition of forest fuels by the thermal pulse generated by the Cretaceous–Palaeogene impact at Chicxulub" (https://www.research.ed.ac.uk/en/publications/a9e74939-2e71-4b99-91a4-ba4cdc65da7b). *Journal of the Geological Society*. **172** (2): 175–185. Bibcode:2015JGSoc.172..175B (https://ui.adsabs.harvard.edu/abs/2015JGSoc.172..175B). doi:10.1144/jgs2014-082 (https://doi.org/10.1144%2Fjgs2014-082). hdl:10044/1/18936 (https://hdl.handle.net/10044%2F1%2F18936). ISSN 0016-7649 (https://search.worldcat.org/issn/0016-7649).
235. Adair, Robert K. (1 June 2010). "Wildfires and animal extinctions at the Cretaceous/Tertiary boundary" (https://pubs.aip.org/ajp/article/78/6/567/1042048/Wildfires-and-animal-extinctions-at-the-Cretaceous). *American Journal of Physics*. **78** (6): 567–573. Bibcode:2010AmJPh..78..567A (https://ui.adsabs.harvard.edu/abs/2010AmJPh..78..567A). doi:10.1119/1.3192770 (https://doi.org/10.1119%2F1.3192770). ISSN 0002-9505 (https://search.worldcat.org/issn/0002-9505). Retrieved 22 August 2024.
236. Morgan, Joanna V.; Bralower, Timothy J.; Brugger, Julia; Wünnemann, Kai (12 April 2022). "The Chicxulub impact and its environmental consequences" (https://www.nature.com/articles/s43017-022-00283-y). *Nature Reviews Earth & Environment*. **3** (5): 338–354. Bibcode:2022NRvEE...3..338M (https://ui.adsabs.harvard.edu/abs/2022NRvEE...3..338M). doi:10.1038/s43017-022-00283-y (https://doi.org/10.1038%2Fs43017-022-00283-y). ISSN 2662-138X (https://search.worldcat.org/issn/2662-138X). Retrieved 22 August 2024.
237. Kaiho, Kunio; Oshima, Naga (2017). "Site of asteroid impact changed the history of life on Earth: The low probability of mass extinction" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5680197). *Scientific Reports*. **7** (1). Article number 14855. Bibcode:2017NatSR...714855K (https://ui.adsabs.harvard.edu/abs/2017NatSR...714855K). doi:10.1038/s41598-017-14199-x (https://doi.org/10.1038%2Fs41598-017-14199-x). PMC 5680197 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5680197). PMID 29123110 (https://pubmed.ncbi.nlm.nih.gov/29123110).
238. Ohno, S.; Kadono, T.; Kurosawa, K.; et al. (2014). "Production of sulphate-rich vapour during the Chicxulub impact and implications for ocean acidification". *Nature Geoscience*. **7** (4): 279–282. Bibcode:2014NatGe...7..279O (https://ui.adsabs.harvard.edu/abs/2014NatGe...7..279O). doi:10.1038/ngeo2095 (https://doi.org/10.1038%2Fngeo2095).
239. Senel, Cem Berk; Kaskes, Pim; Temel, Orkun; Vellekoop, Johan; Goderis, Steven; DePalma, Robert; Prins, Maarten A.; Claeys, Philippe; Karatekin, Özgür (30 October 2023). "Chicxulub impact winter sustained by fine silicate dust". *Nature Geoscience*. **16** (11): 1033–1040. Bibcode:2023NatGe..16.1033S (https://ui.adsabs.harvard.edu/abs/2023NatGe..16.1033S). doi:10.1038/s41561-023-01290-4 (https://doi.org/10.1038%2Fs41561-023-01290-4). hdl:1871.1/d1e86494-4bc1-4237-a9f8-5ebfc269d70a (https://hdl.handle.net/1871.1%2Fd1e86494-4bc1-4237-a9f8-5ebfc269d70a). ISSN 1752-0894 (https://search.worldcat.org/issn/1752-0894). S2CID 264805571 (https://api.semanticscholar.org/CorpusID:264805571).

240. Bardeen, Charles G.; Garcia, Rolando R.; Toon, Owen B.; Conley, Andrew J. (21 August 2017). "On transient climate change at the Cretaceous−Paleogene boundary due to atmospheric soot injections" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5594694). *Proceedings of the National Academy of Sciences of the United States*. **114** (36) e7415–E7424. Bibcode:2017PNAS..114E7415B (https://ui.adsabs.harvard.edu/abs/2017PNAS..114E7415B). doi:10.1073/pnas.1708980114 (https://doi.org/10.1073%2Fpnas.1708980114). ISSN 0027-8424 (https://search.worldcat.org/issn/0027-8424). PMC 5594694 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5594694). PMID 28827324 (https://pubmed.ncbi.nlm.nih.gov/28827324).
241. Lyons, Shelby L.; Karp, Allison T.; Bralower, Timothy J.; Grice, Kliti; Schaefer, Bettina; Gulick, Sean P. S.; Morgan, Joanna V.; Freeman, Katherine H. (28 September 2020). "Organic matter from the Chicxulub crater exacerbated the K–Pg impact winter" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7568312). *Proceedings of the National Academy of Sciences of the United States of America*. **117** (41): 25327–25334. Bibcode:2020PNAS..11725327L (https://ui.adsabs.harvard.edu/abs/2020PNAS..11725327L). doi:10.1073/pnas.2004596117 (https://doi.org/10.1073%2Fpnas.2004596117). ISSN 0027-8424 (https://search.worldcat.org/issn/0027-8424). PMC 7568312 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7568312). PMID 32989138 (https://pubmed.ncbi.nlm.nih.gov/32989138).
242. Kaiho, Kunio; Oshima, Naga; Adachi, Kouji; Adachi, Yukimasa; Mizukami, Takuya; Fujibayashi, Megumu; Saito, Ryosuke (14 July 2016). "Global climate change driven by soot at the K-Pg boundary as the cause of the mass extinction" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4944614). *Scientific Reports*. **6** (1) 28427. Bibcode:2016NatSR...628427K (https://ui.adsabs.harvard.edu/abs/2016NatSR...628427K). doi:10.1038/srep28427 (https://doi.org/10.1038%2Fsrep28427). ISSN 2045-2322 (https://search.worldcat.org/issn/2045-2322). PMC 4944614 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4944614). PMID 27414998 (https://pubmed.ncbi.nlm.nih.gov/27414998).
243. Pierazzo, Elisabetta; Hahmann, Andrea N.; Sloan, Lisa C. (5 July 2004). "Chicxulub and Climate: Radiative Perturbations of Impact-Produced S-Bearing Gases". *Astrobiology*. **3** (1): 99–118. doi:10.1089/153110703321632453 (https://doi.org/10.1089%2F153110703321632453). ISSN 1531-1074 (https://search.worldcat.org/issn/1531-1074). PMID 12804368 (https://pubmed.ncbi.nlm.nih.gov/12804368).
244. Hand, Eric (17 November 2016). "Updated: Drilling of dinosaur-killing impact crater explains buried circular hills". *Science*. doi:10.1126/science.aaf5684 (https://doi.org/10.1126%2Fscience.aaf5684).
245. "Chicxulub crater dinosaur extinction" (https://www.nytimes.com/2016/11/18/science/chicxulub-crater-dinosaur-extinction.html). *The New York Times*. New York, NY. 18 November 2016. Archived (https://web.archive.org/web/20171109221221/https://www.nytimes.com/2016/11/18/science/chicxulub-crater-dinosaur-extinction.html) from the original on 9 November 2017. Retrieved 14 October 2017.
246. Sigurdsson, H.; D'Hondt, S.; Carey, S. (April 1992). "The impact of the Cretaceous/Tertiary bolide on evaporite terrane and generation of major sulfuric acid aerosol" (https://digitalcommons.uri.edu/gsofacpubs/1094). *Earth and Planetary Science Letters*. **109** (3–4): 543–559. Bibcode:1992E&PSL.109..543S (https://ui.adsabs.harvard.edu/abs/1992E&PSL.109..543S). doi:10.1016/0012-821X(92)90113-A (https://doi.org/10.1016%2F0012-821X%2892%2990113-A).
247. Prinn, Ronald G.; Fegley, Bruce (May 1987). "Bolide impacts, acid rain, and biospheric traumas at the Cretaceous-Tertiary boundary". *Earth and Planetary Science Letters*. **83** (1–4): 1–15. Bibcode:1987E&PSL..83....1P (https://ui.adsabs.harvard.edu/abs/1987E&PSL..83....1P). doi:10.1016/0012-821X(87)90046-X (https://doi.org/10.1016%2F0012-821X%2887%2990046-X).

248. Pope, Kevin O.; Baines, Kevin H.; Ocampo, Adriana C.; Ivanov, Boris A. (December 1994). "Impact winter and the Cretaceous/Tertiary extinctions: Results of a Chicxulub asteroid impact model". *Earth and Planetary Science Letters*. **128** (3–4): 719–725. Bibcode:1994E&PSL.128..719P (https://ui.adsabs.harvard.edu/abs/1994E&PSL.128..719P). doi:10.1016/0012-821X(94)90186-4 (https://doi.org/10.1016%2F0012-821X%2894%2990186-4). PMID 11539442 (https://pubmed.ncbi.nlm.nih.gov/11539442).
249. Kawaragi, Ko; Sekine, Yasuhito; Kadono, Toshihiko; Sugita, Seiji; Ohno, Sohsuke; Ishibashi, Ko; Kurosawa, Kosuke; Matsui, Takafumi; Ikeda, Susumu (30 May 2009). "Direct measurements of chemical composition of shock-induced gases from calcite: an intense global warming after the Chicxulub impact due to the indirect greenhouse effect of carbon monoxide". *Earth and Planetary Science Letters*. **282** (1–4): 56–64. Bibcode:2009E&PSL.282...56K (https://ui.adsabs.harvard.edu/abs/2009E&PSL.282...56K). doi:10.1016/j.epsl.2009.02.037 (https://doi.org/10.1016%2Fj.epsl.2009.02.037).
250. Pierazzo, Elisabetta; Kring, David A.; Melosh, H. Jay (25 November 1998). "Hydrocode simulation of the Chicxulub impact event and the production of climatically active gases". *Journal of Geophysical Research: Planets*. **103** (E12): 28607–28625. Bibcode:1998JGR...10328607P (https://ui.adsabs.harvard.edu/abs/1998JGR...10328607P). doi:10.1029/98JE02496 (https://doi.org/10.1029%2F98JE02496). ISSN 0148-0227 (https://search.worldcat.org/issn/0148-0227).
251. Brannen, Peter (2017). *The Ends of the World: Volcanic Apocalypses, Lethal Oceans, and Our Quest to Understand Earth's Past Mass Extinctions*. Harper Collins. p. 336. ISBN 978-0-06-236480-7.
252. Mullen, L. (13 October 2004). "Debating the Dinosaur Extinction" (https://web.archive.org/web/20120625152542/http://www.astrobio.net/exclusive/1243/debating-the-dinosaur-extinction). *Astrobiology Magazine*. Archived from the original (http://www.astrobio.net/exclusive/1243/debating-the-dinosaur-extinction) on 25 June 2012. Retrieved 29 March 2012.
253. Mullen, L. (20 October 2004). "Multiple impacts" (https://web.archive.org/web/20120406162059/http://www.astrobio.net/exclusive/1253/multiple-impacts). *Astrobiology Magazine*. Archived from the original (http://www.astrobio.net/exclusive/1253/multiple-impacts) on 6 April 2012. Retrieved 29 March 2012.
254. Mullen, L. (3 November 2004). "Shiva: Another K–T impact?" (https://web.archive.org/web/20111211091219/http://www.astrobio.net/exclusive/1281/shiva-another-k-t-impact). *Astrobiology Magazine*. Archived from the original (http://www.astrobio.net/exclusive/1281/shiva-another-k-t-impact) on 11 December 2011. Retrieved 29 March 2012.
255. Chatterjee, Sankar (August 1997). "Multiple Impacts at the KT Boundary and the Death of the Dinosaurs" (https://books.google.com/books?id=3IORF1Ei3LIC&pg=PA31). *30th International Geological Congress*. Vol. 26. pp. 31–54. ISBN 978-90-6764-254-5.
256. Kaneoka, Ichiro (January 1980). "40Ar/39Ar dating on volcanic rocks of the Deccan Traps, India". *Earth and Planetary Science Letters*. **46** (2): 233–243. doi:10.1016/0012-821X(80)90009-6 (https://doi.org/10.1016%2F0012-821X%2880%2990009-6).
257. Duncan, R. A.; Pyle, D. G. (30 June 1988). "Rapid eruption of the Deccan flood basalts at the Cretaceous/Tertiary boundary". *Nature*. **333** (6176): 841–843. Bibcode:1988Natur.333..841D (https://ui.adsabs.harvard.edu/abs/1988Natur.333..841D). doi:10.1038/333841a0 (https://doi.org/10.1038%2F333841a0). ISSN 0028-0836 (https://search.worldcat.org/issn/0028-0836). S2CID 4351454 (https://api.semanticscholar.org/CorpusID:4351454).
258. Courtillot, V.; Féraud, G.; Maluski, H.; Vandamme, D.; Moreau, M. G.; Besse, J. (30 June 1988). "Deccan flood basalts and the Cretaceous/Tertiary boundary". *Nature*. **333** (6176): 843–846. Bibcode:1988Natur.333..843C (https://ui.adsabs.harvard.edu/abs/1988Natur.333..843C). doi:10.1038/333843a0 (https://doi.org/10.1038%2F333843a0). ISSN 0028-0836 (https://search.worldcat.org/issn/0028-0836). S2CID 4326163 (https://api.semanticscholar.org/CorpusID:4326163).

259. Courtillot, Vincent; Besse, Jean; Vandamme, Didier; Montigny, Raymond; Jaeger, Jean-Jacques; Cappetta, Henri (November 1986). "Deccan flood basalts at the Cretaceous/Tertiary boundary?". *Earth and Planetary Science Letters*. **80** (3–4): 361–374. Bibcode:1986E&PSL..80..361C (https://ui.adsabs.harvard.edu/abs/1986E&PSL..80..361C). doi:10.1016/0012-821X(86)90118-4 (https://doi.org/10.1016%2F0012-821X%2886%2990118-4).
260. Courtillot, V. (December 1990). "Deccan volcanism at the Cretaceous-Tertiary boundary: past climatic crises as a key to the future?". *Palaeogeography, Palaeoclimatology, Palaeoecology*. **89** (3): 291–299. Bibcode:1990PPP....89..291C (https://ui.adsabs.harvard.edu/abs/1990PPP....89..291C). doi:10.1016/0031-0182(90)90070-N (https://doi.org/10.1016%2F0031-0182%2890%2990070-N).
261. Bosker, Bianca (10 August 2018). "The Nastiest Feud in Science" (https://www.theatlantic.com/magazine/archive/2018/09/dinosaur-extinction-debate/565769/). *The Atlantic*. ISSN 2151-9463 (https://search.worldcat.org/issn/2151-9463). Retrieved 6 July 2026.
262. Keller, G.; Adatte, T.; Gardin, S.; Bartolini, A.; Bajpai, S. (2008). "Main Deccan volcanism phase ends near the K–T boundary: Evidence from the Krishna-Godavari Basin, SE India". *Earth and Planetary Science Letters*. **268** (3–4): 293–311. Bibcode:2008E&PSL.268..293K (https://ui.adsabs.harvard.edu/abs/2008E&PSL.268..293K). doi:10.1016/j.epsl.2008.01.015 (https://doi.org/10.1016%2Fj.epsl.2008.01.015).
263. Renne, P. R.; et al. (2015). "State shift in Deccan volcanism at the Cretaceous-Paleogene boundary, possibly induced by impact" (https://escholarship.org/content/qt5fk799n8/qt5fk799n8.pdf?t=riwyyv) (PDF). *Science*. **350** (6256): 76–78. Bibcode:2015Sci...350...76R (https://ui.adsabs.harvard.edu/abs/2015Sci...350...76R). doi:10.1126/science.aac7549 (https://doi.org/10.1126%2Fscience.aac7549). PMID 26430116 (https://pubmed.ncbi.nlm.nih.gov/26430116).
264. Richards, M. A.; et al. (2015). "Triggering of the largest Deccan eruptions by the Chicxulub impact" (https://escholarship.org/content/qt86f3521g/qt86f3521g.pdf?t=ph1pq1) (PDF). *Geological Society of America Bulletin*. **127** (11–12): 1507–1520. Bibcode:2015GSAB..127.1507R (https://ui.adsabs.harvard.edu/abs/2015GSAB..127.1507R). doi:10.1130/B31167.1 (https://doi.org/10.1130%2FB31167.1). S2CID 3463018 (https://api.semanticscholar.org/CorpusID:3463018).
265. Callegaro, Sara; Baker, Don R.; Renne, Paul R.; Melluso, Leone; Geraki, Kalotina; Whitehouse, Martin J.; De Min, Angelo; Marzoli, Andrea (6 October 2023). "Recurring volcanic winters during the latest Cretaceous: Sulfur and fluorine budgets of Deccan Traps lavas" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10550224). *Science Advances*. **9** (40) eadg8284. Bibcode:2023SciA....9G8284C (https://ui.adsabs.harvard.edu/abs/2023SciA....9G8284C). doi:10.1126/sciadv.adg8284 (https://doi.org/10.1126%2Fsciadv.adg8284). ISSN 2375-2548 (https://search.worldcat.org/issn/2375-2548). PMC 10550224 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10550224). PMID 37792933 (https://pubmed.ncbi.nlm.nih.gov/37792933).
266. Nordt, Lee; Atchley, Stacy; Dworkin, Steve (December 2003). "Terrestrial Evidence for Two Greenhouse Events in the Latest Cretaceous". *Geological Society of America Today*. **13** (12): 4. Bibcode:2003GSAT...13l...4N (https://ui.adsabs.harvard.edu/abs/2003GSAT...13l...4N). doi:10.1130/1052-5173(2003)013<4:TEFTGE>2.0.CO;2 (https://doi.org/10.1130%2F1052-5173%282003%29013%3C4%3ATEFTGE%3E2.0.CO%3B2). ISSN 1052-5173 (https://search.worldcat.org/issn/1052-5173).
267. Berggren, W.A; Aubry, M.-P; van Fossen, M; Kent, D.V; Norris, R.D; Quillévéré, F (1 June 2000). "Integrated Paleocene calcareous plankton magnetobiochronology and stable isotope stratigraphy: DSDP Site 384 (NW Atlantic Ocean)". *Palaeogeography, Palaeoclimatology, Palaeoecology*. **159** (1–2): 1–51. Bibcode:2000PPP...159....1B (https://ui.adsabs.harvard.edu/abs/2000PPP...159....1B). doi:10.1016/S0031-0182(00)00031-6 (https://doi.org/10.1016%2FS0031-0182%2800%2900031-6).

268. Courtillot, Vincent (1990). "A volcanic eruption". *Scientific American*. **263** (4): 85–92. Bibcode:1990SciAm.263d..85C (https://ui.adsabs.harvard.edu/abs/1990SciAm.263d..85C). doi:10.1038/scientificamerican1090-85 (https://doi.org/10.1038%2Fscientificamerican1090-85). PMID 11536474 (https://pubmed.ncbi.nlm.nih.gov/11536474).
269. Milligan, Joseph N.; Royer, Dana L.; Franks, Peter J.; Upchurch, Garland R.; McKee, Melissa L. (7 March 2019). "No Evidence for a Large Atmospheric CO_2 Spike Across the Cretaceous-Paleogene Boundary". *Geophysical Research Letters*. **46** (6): 3462–3472. Bibcode:2019GeoRL..46.3462M (https://ui.adsabs.harvard.edu/abs/2019GeoRL..46.3462M). doi:10.1029/2018GL081215 (https://doi.org/10.1029%2F2018GL081215). ISSN 0094-8276 (https://search.worldcat.org/issn/0094-8276).
270. Sial, A. N.; Lacerda, L. D.; Ferreira, V. P.; Frei, R.; Marquillas, R. A.; Barbosa, J. A.; Gaucher, C.; Windmöller, C. C.; Pereira, N. S. (1 October 2013). "Mercury as a proxy for volcanic activity during extreme environmental turnover: The Cretaceous–Paleogene transition" (http://www.repositorio.ufc.br/handle/riufc/65773). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **387**: 153–164. Bibcode:2013PPP...387..153S (https://ui.adsabs.harvard.edu/abs/2013PPP...387..153S). doi:10.1016/j.palaeo.2013.07.019 (https://doi.org/10.1016%2Fj.palaeo.2013.07.019).
271. Keller, Gerta (July 2001). "The end-cretaceous mass extinction in the marine realm: Year 2000 assessment" (https://web.archive.org/web/20251218105943/https://doc.rero.ch/record/14191/files/PAL_E1154.pdf) (PDF). *Planetary and Space Science*. **49** (8): 817–830. Bibcode:2001P&SS...49..817K (https://ui.adsabs.harvard.edu/abs/2001P&SS...49..817K). doi:10.1016/S0032-0633(01)00032-0 (https://doi.org/10.1016%2FS0032-0633%2801%2900032-0). Archived from the original (http://doc.rero.ch/record/14191/files/PAL_E1154.pdf) (PDF) on 18 December 2025.
272. Adatte, Thierry; Keller, Gerta; Stinnesbeck, Wolfgang (28 February 2002). "Late Cretaceous to early Paleocene climate and sea-level fluctuations: the Tunisian record". *Palaeogeography, Palaeoclimatology, Palaeoecology*. **178** (3–4): 165–196. Bibcode:2002PPP...178..165A (https://ui.adsabs.harvard.edu/abs/2002PPP...178..165A). doi:10.1016/S0031-0182(01)00395-9 (https://doi.org/10.1016%2FS0031-0182%2801%2900395-9).
273. Keller, Gerta (October 1988). "Extinction, survivorship and evolution of planktic foraminifera across the Cretaceous/Tertiary boundary at El Kef, Tunisia". *Marine Micropaleontology*. **13** (3): 239–263. Bibcode:1988MarMP..13..239K (https://ui.adsabs.harvard.edu/abs/1988MarMP..13..239K). doi:10.1016/0377-8398(88)90005-9 (https://doi.org/10.1016%2F0377-8398%2888%2990005-9).
274. Zhang, Laiming; Wang, Chengshan; Wignall, Paul B.; Kluge, Tobias; Wan, Xiaoqiao; Wang, Qian; Gao, Yuan (1 March 2018). "Deccan volcanism caused coupled pCO2 and terrestrial temperature rises, and pre-impact extinctions in northern China". *Geology*. **46** (3): 271–274. Bibcode:2018Geo....46..271Z (https://ui.adsabs.harvard.edu/abs/2018Geo....46..271Z). doi:10.1130/G39992.1 (https://doi.org/10.1130%2FG39992.1). ISSN 0091-7613 (https://search.worldcat.org/issn/0091-7613).
275. Alvarez, W (1997). *T. rex and the Crater of Doom* (https://archive.org/details/trexcraterofdoo000alva). Princeton University Press. pp. 130 (https://archive.org/details/trexcraterofdoo000alva/page/130)–146. ISBN 978-0-691-01630-6.
276. Sprain, Courtney J.; Renne, Paul R.; Vanderkluysen, Loÿc; Pande, Kanchan; Self, Stephen; Mittal, Tushar (22 February 2019). "The eruptive tempo of Deccan volcanism in relation to the Cretaceous-Paleogene boundary" (https://escholarship.org/content/qt8sj8w1rf/qt8sj8w1rf.pdf?t=rhh5bw) (PDF). *Science*. **363** (6429): 866–870. Bibcode:2019Sci...363..866S (https://ui.adsabs.harvard.edu/abs/2019Sci...363..866S). doi:10.1126/science.aav1446 (https://doi.org/10.1126%2Fscience.aav1446). ISSN 0036-8075 (https://search.worldcat.org/issn/0036-8075). PMID 30792301 (https://pubmed.ncbi.nlm.nih.gov/30792301).

277. Cripps, J.A.; Widdowson, M.; Spicer, R.A.; Jolley, D.W. (1 February 2005). "Coastal ecosystem responses to late stage Deccan Trap volcanism: the post K–T boundary (Danian) palynofacies of Mumbai (Bombay), west India" (https://hull-repository.worktribe.com/output/3566589). *Palaeogeography, Palaeoclimatology, Palaeoecology*. **216** (3–4): 303–332. Bibcode:2005PPP...216..303C (https://ui.adsabs.harvard.edu/abs/2005PPP...216..303C). doi:10.1016/j.palaeo.2004.11.007 (https://doi.org/10.1016%2Fj.palaeo.2004.11.007).
278. Dzombak, R.M.; Sheldon, N.D.; Mohabey, D.M.; Samant, B. (September 2020). "Stable climate in India during Deccan volcanism suggests limited influence on K–Pg extinction" (https://doi.org/10.1016%2Fj.gr.2020.04.007). *Gondwana Research*. **85**: 19–31. Bibcode:2020GondR..85...19D (https://ui.adsabs.harvard.edu/abs/2020GondR..85...19D). doi:10.1016/j.gr.2020.04.007 (https://doi.org/10.1016%2Fj.gr.2020.04.007).
279. Benton, Michael (2023). *Extinctions: How Life Survives, Adapts and Evolves*. London, UK: Thames & Hudson. p. 197. ISBN 978-0-500-02546-8.
280. Li, Liangquan; Keller, Gerta (1998). "Abrupt deep-sea warming at the end of the Cretaceous". *Geology*. **26** (11): 995–998. Bibcode:1998Geo....26..995L (https://ui.adsabs.harvard.edu/abs/1998Geo....26..995L). doi:10.1130/0091-7613(1998)026<0995:ADSWAT>2.3.CO;2 (https://doi.org/10.1130%2F0091-7613%281998%29026%3C0995%3AADSWAT%3E2.3.CO%3B2). S2CID 115136793 (https://api.semanticscholar.org/CorpusID:115136793).
281. Keller, Gerta (June 1989). "Extended Cretaceous/Tertiary boundary extinctions and delayed population change in planktonic foraminifera from Brazos River, Texas". *Paleoceanography and Paleoclimatology*. **4** (3): 287–332. Bibcode:1989PalOc...4..287K (https://ui.adsabs.harvard.edu/abs/1989PalOc...4..287K). doi:10.1029/PA004i003p00287 (https://doi.org/10.1029%2FPA004i003p00287). ISSN 0883-8305 (https://search.worldcat.org/issn/0883-8305).
282. Keller, Gerta; Barrera, Enriqueta (1990). "The Cretaceous/Tertiary boundary impact hypothesis and the paleontological record". In Sharpton, Virgil L.; Ward, Peter D. (eds.). *Global Catastrophes in Earth History; An Interdisciplinary Conference on Impacts, Volcanism, and Mass Mortality*. Geological Society of America. doi:10.1130/SPE247-p563 (https://doi.org/10.1130%2FSPE247-p563). ISBN 9780813722474.
283. Jiang, M. J.; Gartner, S. (1986). "Calcareous Nannofossil Succession across the Cretaceous/Tertiary Boundary in East-Central Texas". *Micropaleontology*. **32** (3): 232. Bibcode:1986MiPal..32..232J (https://ui.adsabs.harvard.edu/abs/1986MiPal..32..232J). doi:10.2307/1485619 (https://doi.org/10.2307%2F1485619). JSTOR 1485619 (https://www.jstor.org/stable/1485619).
284. Petersen, Sierra V.; Dutton, Andrea; Lohmann, Kyger C. (2016). "End-Cretaceous extinction in Antarctica linked to both Deccan volcanism and meteorite impact via climate change" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4935969). *Nature Communications*. **7** 12079. Bibcode:2016NatCo...712079P (https://ui.adsabs.harvard.edu/abs/2016NatCo...712079P). doi:10.1038/ncomms12079 (https://doi.org/10.1038%2Fncomms12079). PMC 4935969 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4935969). PMID 27377632 (https://pubmed.ncbi.nlm.nih.gov/27377632).

Further reading

- Cowen, R. (2000). "The K–T extinction" (http://www.ucmp.berkeley.edu/education/events/cowen1b.html). University of California Museum of Paleontology. Retrieved 2 August 2007.
- DePalma, Robert A.; et al. (1 April 2019). "A seismically induced onshore surge deposit at the KPg boundary, North Dakota" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6486721). *PNAS*. **116** (17): 8190–8199. Bibcode:2019PNAS..116.8190D (https://ui.adsabs.harvard.edu/abs/2019PNAS..116.8190D). doi:10.1073/pnas.1817407116 (https://doi.org/10.1073%2Fpnas.1817407116). PMC 6486721 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6486721). PMID 30936306 (https://pubmed.ncbi.nlm.nih.gov/30936306).

- Fortey, Richard (2005). *Earth: An Intimate History* (https://books.google.com/books?id=8i8UzOYJlgUC). New York: Vintage Books. ISBN 978-0-375-70620-2. OCLC 54537112 (https://search.worldcat.org/oclc/54537112).
- Papers and presentations resulting from the 2016 Chicxulub drilling project (https://gsa.confex.com/gsa/2017AM/webprogram/Session43180.html)—The Geological Society of America, GSA Annual Meeting in Seattle, Washington, USA – 2017, Session No. 192
- Kring, D.A. (2005). "Chicxulub impact event: Understanding the K–T boundary" (https://web.archive.org/web/20070629230804/http://www.lpl.arizona.edu/SIC/impact_cratering/Chicxulub/Chicx_title.html). NASA Space Imagery Center. Archived from the original (http://www.lpl.arizona.edu/SIC/impact_cratering/Chicxulub/Chicx_title.html) on 29 June 2007. Retrieved 2 August 2007.
- Preston, Douglas (8 April 2019). "The Day the Dinosaurs Died" (https://www.newyorker.com/magazine/2019/04/08/the-day-the-dinosaurs-died). *The New Yorker*. pp. 52–65. "Robert A. de Palm has found strong evidence that the dinosaurs—and nearly all other life on Earth—were indeed wiped out 66 million years ago by the Chicxulub asteroid"

External links

- What killed the dinosaurs? (https://www.ucmp.berkeley.edu/diapsids/extinction.html) — University of California Museum of Paleontology (1995)
- The Great Chicxulub Debate 2004 (https://www.geolsoc.org.uk/chicxulub) — Geological Society of London

Retrieved from "https://en.wikipedia.org/w/index.php?title=Cretaceous–Paleogene_extinction_event&oldid=1382035744"

WIKIPEDIA
25 years of the free encyclopedia

Chicxulub crater

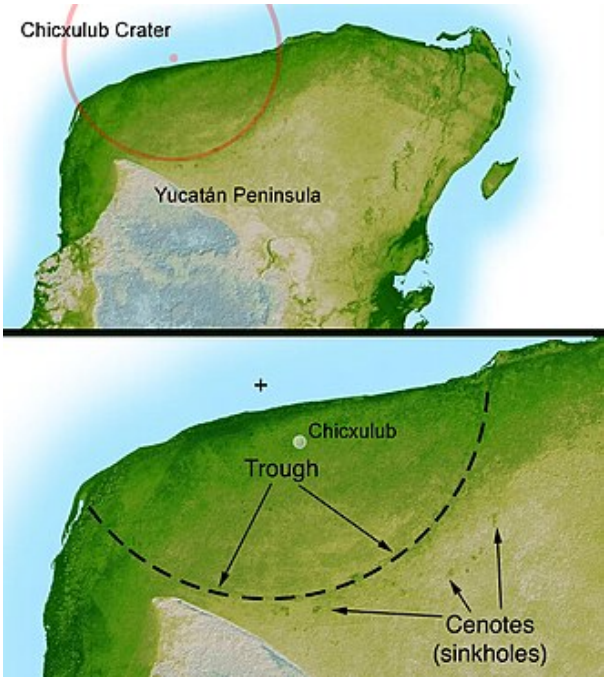
The **Chicxulub crater** is an impact crater buried underneath the Yucatán Peninsula in Mexico. The crater is named after the onshore inland community of Chicxulub Pueblo.[5] It was formed slightly over 66 million years ago when an asteroid, about ten kilometers (six miles) in diameter, struck Earth. The crater is estimated to be two hundred kilometers (120 mi) in diameter and is buried to a depth of about one kilometer (0.6 mi) beneath younger sedimentary rocks.[2]:3 It is one of the largest impact structures on Earth, alongside the much older Sudbury and Vredefort impact structures,[6] and the only one whose peak ring is intact and directly accessible for scientific research.[7]

The crater was discovered by Antonio Camargo and Glen Penfield, geophysicists who had been looking for petroleum in the Yucatán Peninsula during the late 1970s. Penfield was initially unable to obtain evidence that the geological feature was a crater and gave up his search. Later, through contact with Alan R. Hildebrand in 1990, Penfield obtained samples that suggested it was an impact feature. Evidence for the crater's impact origin includes shocked quartz, a gravity anomaly, and tektites in surrounding areas.[5]

The date of the impact coincides with the Cretaceous–Paleogene boundary (commonly known as the K–Pg or K–T boundary). It is now widely accepted that the devastation and climate disruption resulting from the impact was the primary cause of the Cretaceous–Paleogene extinction event, a mass extinction of 75% of plant and animal species on Earth, including all non-avian dinosaurs.[7]

Chicxulub crater

Chicxulub impact structure



Imaging from NASA's Shuttle Radar Topography Mission STS-99 reveals part of the diameter ring of the crater in the form of a shallow circular trough. Numerous cenotes (sinkholes) cluster around the trough marking the inner crater rim.[1]

Location within North America

Discovery

In the late 1970s, geologist Walter Alvarez and his father, Nobel Prize–winning scientist Luis Walter Alvarez, put forth their theory that the Cretaceous–

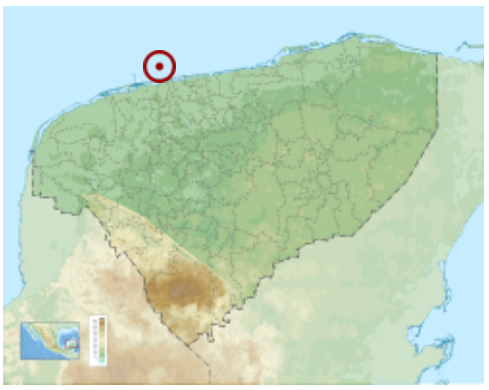
Paleogene extinction was caused by an impact event.[8][9] The main evidence of such an impact was contained in a thin layer of clay present in the Cretaceous–Paleogene boundary (K–Pg boundary) in Gubbio, Italy. The Alvarezes and colleagues reported that it contained an abnormally high concentration of iridium, a chemical element rare on Earth but common in asteroids.[8][10] Iridium levels in this layer were as much as 160 times above the background level.[11] It was hypothesized that the iridium was spread into the atmosphere when the impactor was vaporized and settled across Earth's surface among other material thrown up by the impact, producing the layer of iridium-enriched clay.[12] At the time, there was no consensus on what caused the Cretaceous–Paleogene extinction and the boundary layer, with theories including a nearby supernova, climate change, or a geomagnetic reversal.[11]:1095 The Alvarezes' impact hypothesis was rejected by many paleontologists, who believed that the lack of fossils found close to the K–Pg boundary—the "three-meter problem"—suggested a more gradual die-off of fossil species.[9][13]

The Alvarezes, joined by Frank Asaro and Helen Michel from University of California, Berkeley, published their paper on the iridium anomaly in *Science* in June 1980.[11] Almost simultaneously Jan Smit and Jan Hertogen published their iridium findings from Caravaca, Spain, in *Nature* in May 1980.[14] These papers were followed by other reports of similar iridium spikes at the K–Pg boundary across the globe, and sparked wide interest in the cause of the K–Pg extinction; over 2,000 papers were published in the 1980s on the topic.[13]:82[15] There were no known impact craters that were the right age and size, spurring a search for a suitable candidate.[9] Recognizing the scope of the work, Lee Hunt and Lee Silver organized a cross-discipline meeting in Snowbird, Utah, in 1981. Unknown to them, evidence of the crater they were looking for was being presented the same week, and would be largely missed by the scientific community.[13]:83–84[15]

In 1978, geophysicists Glen Penfield and Antonio Camargo were working for the Mexican state-owned oil company Petróleos Mexicanos (Pemex) as part of

Location within Mexico

Location within Gulf of Mexico



Location within Yucatán (state)

Location	Yucatán, Mexico 21°24′N 89°30′W
Impact crater structure	
Confidence	Confirmed
Diameter	~180–200 km (110–120 mi)
Depth	~1 km (0.6 mi) burial depth[2]:3
Impactor diameter	~10 km (6 mi)[3]
Age	Cretaceous–Paleogene boundary[4]
Exposed	No
Drilled	Yes
Bolide type	Carbonaceous chondrite
Pronunciation	[/ʧiːkʃuːˈluːb/ *cheek-shoo-LOOB*

Spanish: [tʃikʃuˈlub]]

an airborne magnetic survey of the Gulf of Mexico north of the Yucatán Peninsula.[17]:20–21 Penfield's job was to use geophysical data to scout possible locations for oil drilling.[8] In the offshore magnetic data, Penfield noted anomalies whose depth he estimated and mapped. He then obtained onshore gravity data from the 1940s. When the gravity maps and magnetic anomalies were compared, Penfield described a shallow "bullseye", 180 km (110 mi) in diameter, appearing on the otherwise non-magnetic and uniform surroundings—clear evidence to him of an impact feature.[8][5] A decade earlier, the same map had suggested a crater to contractor Robert Baltosser, but Pemex corporate policy prevented him from publicizing his conclusion.[17]:20

Artist's impression of the asteroid slamming into tropical, shallow seas of the sulfur-rich Yucatán Peninsula in what is today Southeast Mexico.[16] The aftermath of the asteroid collision, which occurred approximately 66 million years ago, is believed to have caused the mass extinction of non-avian dinosaurs and many other species on Earth.[16] The impact spewed hundreds of billions of tons of sulfur into the atmosphere, producing a worldwide blackout and freezing temperatures which persisted for at least a decade.[16]

Penfield presented his findings to Pemex, who rejected the crater theory, instead deferring to findings that ascribed the feature to volcanic activity.[5] Pemex disallowed release of specific data, but let Penfield and Camargo present the results at the 1981 Society of Exploration Geophysicists conference.[15] That year's conference was under-attended and their report attracted little attention, as many experts on impact craters and the K–Pg boundary were attending the Snowbird conference instead. Carlos Byars, a *Houston Chronicle* journalist who was familiar with Penfield and had seen the gravitational and magnetic data himself, wrote a front-page story[18] on Penfield and Camargo's claim, but the news did not propagate widely.[17]:23

Although Penfield had plenty of geophysical data sets, he had no rock cores or other physical evidence of an impact.[8] He knew Pemex had drilled exploratory wells in the region. In 1951, one well bored into what was described as a thick layer of andesite about 1.3 kilometers (4,300 ft) down. This layer could have resulted from the intense heat and pressure of an Earth impact, but at the time of the borings it was dismissed as a lava dome—a feature uncharacteristic of the region's geology.[8] Penfield was encouraged by William C. Phinney, curator of lunar rocks at the Johnson Space Center, to find these samples to support his hypothesis.[5] Penfield tried to secure site samples, but was told they had been lost or destroyed. When attempts to return to the drill sites to look for corroborating rocks proved fruitless, Penfield abandoned his search, published his findings and returned to his Pemex work.[8] Seeing the 1980 *Science* paper, Penfield wrote to Walter Alvarez about the Yucatán structure, but received no response.[15]

Alvarez and other scientists continued their search for the crater, although they were searching in oceans based on incorrect analysis of glassy spherules from the K–Pg boundary that suggested the impactor had landed in open water.[13] Unaware of Penfield's discovery, University of Arizona graduate student Alan R. Hildebrand and faculty adviser William V. Boynton looked for a crater near the Brazos River in Texas.[13] Their evidence included greenish-brown clay with surplus iridium, containing shocked quartz grains and small weathered glass beads that looked to be tektites.[2] Thick, jumbled deposits of coarse rock fragments were also present, thought to have been scoured from one place and deposited elsewhere by an impact event. Such deposits occur in many locations but seemed concentrated in the Caribbean Basin at

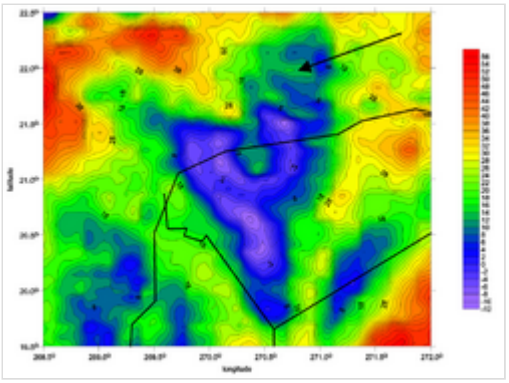
the K–Pg boundary. When Haitian professor Florentine Morás discovered what he thought to be evidence of an ancient volcano on Haiti, Hildebrand suggested it could be a telltale feature of a nearby impact. Tests on samples retrieved from the K–Pg boundary revealed more tektite glass, formed only in the heat of asteroid impacts and high-yield nuclear detonations.[8]

In 1990, Carlos Byars told Hildebrand of Penfield's earlier discovery of a possible impact crater.[19]:50 Hildebrand contacted Penfield and the pair soon secured two drill samples from the Pemex wells, which had been stored in New Orleans for decades.[5] Hildebrand's team tested the samples, which clearly showed shock-metamorphic materials.[8] A team of California researchers surveying satellite images found a cenote (sinkhole) ring centered on the town of Chicxulub Pueblo that matched the one Penfield saw earlier; the cenotes were thought to be caused by subsidence of bolide-weakened lithostratigraphy around the impact crater wall.[20] More recent evidence suggests the crater is 300 km (190 mi) wide, and the 180 km (110 mi) ring observed is an inner wall of the larger crater.[21] Hildebrand, Penfield, Boynton, Camargo, and others published their paper identifying the crater in 1991.[2][13] The crater was named for the nearby town of Chicxulub Pueblo. Penfield also recalled that part of the motivation for the name was "to give the academics and NASA naysayers a challenging time pronouncing it" after years of dismissing its existence.[5]

In March 2010, forty-one experts from many countries reviewed the available evidence: twenty years' worth of data spanning a variety of fields. They concluded that the impact at Chicxulub triggered the mass extinctions at the K–Pg boundary.[9][7] Dissenters, notably Gerta Keller of Princeton University, have proposed an alternate culprit: the eruption of the Deccan Traps in what is now the Indian subcontinent. This period of intense volcanism occurred before and after the Chicxulub impact;[9][22] dissenting studies argue that the worst of the volcanic activity occurred *before* the impact, and the role of the Deccan Traps was instead shaping the evolution of surviving species post-impact.[23] A 2013 study compared isotopes in impact glass from the Chicxulub impact with isotopes in ash from the K–Pg boundary, concluding that they were dated almost exactly the same, and within experimental error.[4]

Impact specifics

A 2013 study published in *Science* estimated the age of the impact as 66,043,000 ± 11,000 years ago (± 43,000 years ago considering systematic error), based on multiple lines of evidence, including argon–argon dating of tektites from Haiti and bentonite horizons overlying the impact horizon in northeastern Montana.[4] This date was supported by a 2015 study based on argon–argon dating of tephra found in lignite beds in the Hell Creek and overlying Fort Union formations in northeastern Montana.[24] A 2018 study based on argon–argon dating of spherules from Gorgonilla Island, Colombia, obtained a slightly different result of 66,051,000 ± 31,000 years ago.[25] The impact has been interpreted to have occurred in the Northern Hemisphere's spring season based on annual isotope curves in sturgeon and paddlefish bones



Free-air gravity anomaly over the Chicxulub structure (coastline and state boundaries shown as black lines). The arrow indicates a possible companion crater.

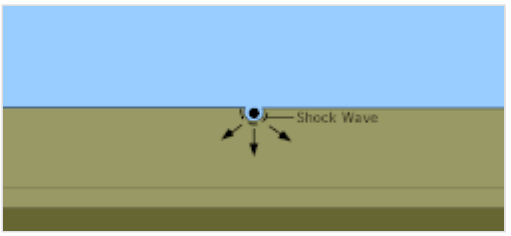
found in an ejecta-bearing sedimentary unit at the Tanis site in southwestern North Dakota. This sedimentary unit is thought to have formed within hours of impact.[26]

The site of the crater at the time of impact was a marine carbonate platform.[27] The water depth at the impact site varied from 100 meters (330 ft) on the western edge of the crater to over 1,200 meters (3,900 ft) on the northeastern edge, with an estimated depth at the center of the impact of approximately 650 meters (2,130 ft).[28] The seafloor rocks consisted of a sequence of Jurassic–Cretaceous marine sediments 3 kilometers (1.9 mi) thick. They were predominantly carbonate rock, including dolomite (35–40% of total sequence) and limestone (25–30%), along with evaporites (anhydrite 25–30%) and minor amounts of shale and sandstone (3–4%) underlain by approximately 35 kilometers (22 mi) of continental crust, composed of igneous crystalline basement including granite.[29]

The impactor was around 10 kilometers (6 mi) in diameter[3]—large enough that, if set at sea level, it would have reached taller than Mount Everest.[13]:9 A 2021 study estimated the impactor had a velocity of 20 kilometers per second (12 mi/s) inclined 45–60° to horizontal, impacting from the northeast.[30]

The estimates of released energy in the literature vary from ~10^{23} joules[31][32][33][34][35] to 10^{24} J[36][20] and even to 10^{25} J,[37] equivalent to tens of millions to billions of megatons of TNT.

Effects



An animation showing the Chicxulub impact and subsequent crater formation

Such amounts of energy generated winds in excess of 1,000 kilometers per hour (600 mph) near the blast's center,[38] and produced a transient cavity 100 kilometers (60 mi) wide and 30 kilometers (20 mi) deep that later collapsed. This formed a crater mainly under the sea and currently covered by about 1,000 meters (3,000 ft) of sediment.[27][39] The impact, expansion of water after filling the crater, and related seismic activity spawned megatsunamis over 100 meters (300 ft) tall, with one simulation suggesting the immediate waves from the impact may have reached up to 1.5 kilometers (0.9 mi) high.[40][41] The waves scoured the sea floor, leaving ripples underneath what is now Louisiana with average wavelengths of 600 meters (2,000 ft) and average wave heights of 16 meters (52 ft), the largest ripples documented.[42][43] Material shifted by subsequent earthquakes and the waves reached to what are now Texas and Florida, and may have disturbed sediments as far as 6,000 kilometers (3,700 mi) from the impact site.[40][44][45] The impact triggered a seismic event with an estimated moment magnitude of 9 to 11 M_w.[32]

A cloud of hot dust, ash and steam would have spread from the crater, with as much as 25 trillion metric tons of excavated material being ejected into the atmosphere by the blast. Some of this material escaped orbit, dispersing throughout the Solar System,[9] while some of it fell back to Earth, vaporizing upon re-entry. The rock heated Earth's surface and ignited wildfires, estimated to have enveloped nearly 70% of the planet's forests. The effect on living creatures even hundreds of kilometers away was immense, and much of present-day Mexico and the United States would have been devastated.[8][9][13]:10–13 Fossil evidence for an instantaneous extinction of diverse animals was found in a soil layer only 10 centimeters (4 in) thick in New Jersey, 2,500 kilometers (1,600 mi) away from the impact site, indicating that death and burial under debris occurred suddenly and quickly over wide distances on nearby land.[39] Field

research from the Hell Creek Formation in North Dakota published in 2019 shows the simultaneous mass extinction of myriad species, combined with geological and atmospheric features that are consistent with the impact event.[9]

Due to the relatively shallow water at the impact site, the rock that was vaporized included sulfur-rich gypsum from the lower part of the Cretaceous sequence, and this was injected into the atmosphere.[39] This global dispersal of dust and sulfates would have led to a sudden and catastrophic effect on the climate worldwide, instigating large temperature drops and devastating the food chain. Researchers stated that the impact not only generated an environmental calamity that extinguished life, but it also induced a vast subsurface hydrothermal system that became an oasis for the recovery of life.[46][47] Using seismic images of the crater in 2008, scientists determined that the impactor landed in deeper water than previously assumed, which may have resulted in increased sulfate aerosols in the atmosphere as a result of more water vapor being available to react with the vaporized anhydrite. This could have made the impact even deadlier by rapidly cooling the climate and generating acid rain.[48]

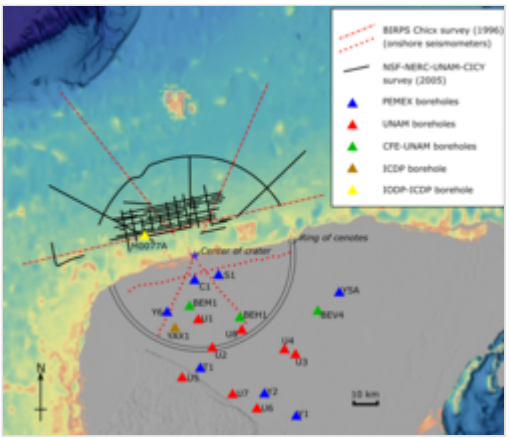
The emission of dust and particles could have covered the entire surface of Earth for several years, possibly up to a decade, creating a harsh environment for biological life. Production of carbon dioxide caused by the destruction of carbonate rocks would have led to a sudden greenhouse effect.[2]:5 The large oil field present at the impact site may also have combusted, creating a large cloud of soot further blocking sunlight.[49] For a decade or longer, sunlight would have been blocked from reaching the surface of Earth by the dust and soot particles in the atmosphere, cooling the surface dramatically. Photosynthesis by plants would also have been interrupted, affecting the entire food chain.[50][51] A model of the event developed by Lomax et al. (2001) suggests that net primary productivity rates may have increased to higher than pre-impact levels over the long term because of the high carbon dioxide concentrations.[52]

A long-term local effect of the impact was the creation of the Yucatán sedimentary basin which "ultimately produced favorable conditions for human settlement in a region where surface water is scarce".[53]

Post-discovery investigations

Geophysical data

Two seismic reflection datasets have been acquired over the offshore parts of the crater since its discovery. Older 2D seismic datasets have also been used that were originally acquired for hydrocarbon exploration. A set of three long-record 2D lines was acquired in October 1996, with a total length of 650 kilometers (400 mi), by the BIRPS group. The longest of the lines, *Chicx-A*, was shot parallel to the coast, while *Chicx-B* and *Chicx-C* were shot NW–SE and SSW–NNE respectively. In addition to the conventional seismic reflection imaging, data was recorded onshore to allow for wide-angle refraction imaging.[54][55]



Location of seismic surveys and boreholes

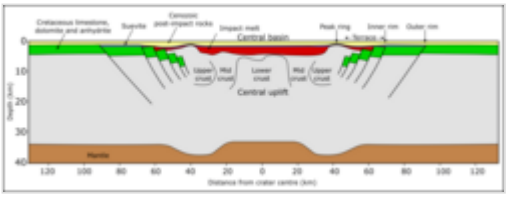
In 2005, another set of profiles was acquired, bringing the total length of the 2D deep-penetration seismic data up to 2,470 kilometers (1,530 mi). This survey also used ocean bottom seismometers and land stations to allow 3D travel time inversion to improve the understanding of the velocity structure of the crater. The data was concentrated around the interpreted offshore peak ring to help identify possible drilling locations. At the same time, gravity data was acquired along 7,638 kilometers (4,746 mi) of profiles. The acquisition was funded by the National Science Foundation (NSF), Natural Environment Research Council (NERC) with logistical assistance from the National Autonomous University of Mexico (UNAM) and the Centro de Investigación Científica de Yucatán (CICY – Yucatán Center for Scientific Investigation).[27][56]

Borehole drilling

Intermittent core samples from hydrocarbon exploration boreholes drilled by Pemex on the Yucatán peninsula have provided some useful data. UNAM drilled a series of eight fully-cored boreholes in 1995, three of which penetrated deep enough to reach the ejecta deposits outside the main crater rim (UNAM-5, 6, and 7). Between 2001 and 2002, a scientific borehole was drilled near the Hacienda Yaxcopoil, known as *Yaxcopoil-1* (or more commonly Yax-1), to a depth of 1,511 meters (4,957 ft) below the surface, as part of the International Continental Scientific Drilling Program. The borehole was cored continuously, passing through 100 meters (330 ft) of impactites. Three fully-cored boreholes were also drilled by the Comisión Federal de Electricidad (Federal Electricity Commission) with UNAM. One of them, (BEV-4), was deep enough to reach the ejecta deposits.[57]

In 2016, a joint United Kingdom–United States team obtained the first offshore core samples from the peak ring in the central zone of the crater with the drilling of the borehole known as M0077A, part of Expedition 364 of the International Ocean Discovery Program. The borehole reached 1,335 meters (4,380 ft) below the seafloor.[58]

Morphology



Schematic cross-section of the Chicxulub impact structure

The form and structure (geomorphology) of the Chicxulub crater is known mainly from geophysical data. It has a well-defined concentric multi-ring structure. The outermost ring was identified using seismic reflection data. It is up to 130 kilometers (81 mi) from the crater center, and is a ring of normal faults, throwing down towards the crater center, marking the outer limit of significant crustal deformation. This makes it one of the three largest impact structures on Earth.[59][60] Moving toward the center, the next ring is the main crater rim, also known as the "inner rim," which correlates with a ring of cenotes onshore and a major circular Bouguer gravity gradient anomaly.[28][61] This ring has a radius that varies between 70 and 85 kilometers (43 and 53 mi).[27] The next inner ring structure is the peak ring. The area between the inner rim and peak ring is described as the "terrace zone", characterized by a series of fault blocks defined by normal faults dipping towards the crater center, sometimes referred to as "slump blocks". The peak ring is about 80 km in diameter and of variable height, 400 to 600 meters (1,300 to 2,000 ft) above the base of the crater in the west and northwest and 200 to 300 meters (660 to 980 ft) in the north, northeast,

and east.[27] The central part of the crater lies above a zone where the mantle was uplifted such that the Mohorovičić discontinuity is shallower by about 1–2 kilometers (0.6–1.2 mi) compared to regional values.[27][60]

The ring structures are best developed to the south, west and northwest, becoming more indistinct towards the north and northeast of the structure. This is interpreted to be a result of variable water depth at the time of impact, with less-well-defined rings resulting from the areas with water depths significantly deeper than 100 meters (300 ft).[28]

Geology

Pre-impact geology

Before the impact, the geology of the Yucatán area, sometimes referred to as the "target rocks", consisted of a sequence of mainly Cretaceous limestones, overlying red beds of uncertain age above an unconformity with the dominantly granitic basement. The basement forms part of the Maya Block and information about its makeup and age in the Yucatán area has come only from drilling results around the Chicxulub crater and the analysis of basement material found as part of the ejecta at more distant K–Pg boundary sites. The Maya block is one of a group of crustal blocks found at the edge of the Gondwana continent. Zircon ages are consistent with the presence of an underlying Grenville age crust, with large amounts of late Ediacaran arc-related igneous rocks, interpreted to have formed in the Pan-African orogeny. Late Paleozoic granitoids (the distinctive "pink granite") were found in the peak ring borehole M0077A, with an estimated age of 326 ± 5 million years ago (Carboniferous). These have an adakitic composition and are interpreted to represent the effects of slab detachment during the Marathon-Ouachita orogeny, part of the collision between Laurentia and Gondwana that created the Pangaea supercontinent.[62]

The center of the crater is near Chicxulub Puerto.

Stela in the main square of Chicxulub Puerto commemorating the impact

Red beds of variable thickness, up to 115 meters (377 ft), overlay the granitic basement, particularly in the southern part of the area. These continental clastic rocks are thought to be of Triassic-to-Jurassic age, although they may extend into the Lower Cretaceous. The lower part of the Lower Cretaceous sequence consists of dolomite with interbedded anhydrite and gypsum, with the upper part being limestone, with dolomite and anhydrite in part. The thickness of the Lower Cretaceous varies from 750 meters (2,460 ft) up to 1,675 meters (5,495 ft) in the boreholes. The Upper Cretaceous sequence is

mainly platform limestone, with marl and interbedded anhydrite. It varies in thickness from 600 to 1,200 meters (2,000 to 3,900 ft). There is evidence for a Cretaceous basin within the Yucatán area that has been named the Yucatán Trough, running approximately south–north, widening northwards, explaining the observed thickness variations.[63]

Impact rocks

The most common observed impact rocks are suevites, found in many of the boreholes drilled around the Chicxulub crater. Most of the suevites were resedimented soon after the impact by the resurgence of oceanic water into the crater. This gave rise to a layer of suevite extending from the inner part of the crater out as far as the outer rim.[64]

Impact melt rocks are thought to fill the central part of the crater, with a maximum thickness of 3 kilometers (1.9 mi). The samples of melt rock that have been studied have overall compositions similar to that of the basement rocks, with some indications of mixing with carbonate source, presumed to be derived from the Cretaceous carbonates. An analysis of melt rocks sampled by the M0077A borehole indicates two types of melt rock, an upper impact melt (UIM), which has a clear carbonate component as shown by its overall chemistry and the presence of rare limestone clasts and a lower impact melt-bearing unit (LIMB) that lacks any carbonate component. The difference between the two impact melts is interpreted to be a result of the upper part of the initial impact melt, represented by the LIMB in the borehole, becoming mixed with materials from the shallow part of the crust either falling back into the crater or being brought back by the resurgence forming the UIM.[65]

The "pink granite", a granitoid rich in alkali feldspar found in the peak ring borehole shows many deformation features that record the extreme strains associated with the formation of the crater and the subsequent development of the peak ring.[46][66] The granitoid has an unusually low density and P-wave velocity compared to typical granitic basement rocks. Study of the core from M0077A shows the following deformation features in apparent order of development: pervasive fracturing along and through grain boundaries, a high density of shear faults, bands of cataclasite and ultra-cataclasite and some ductile shear structures. This deformation sequence is interpreted to result from initial crater formation involving acoustic fluidization followed by shear faulting with the development of cataclasites with fault zones containing impact melts.[67]

The peak ring drilling below the sea floor also discovered evidence of a massive hydrothermal system, which modified approximately 1.4×10^5 km^3 of Earth's crust and lasted for hundreds of thousands of years. These hydrothermal systems may provide support for the impact origin of life hypothesis for the Hadean eon,[68] when the entire surface of Earth was affected by impactors much larger than the Chicxulub impactor.[69]

Post-impact geology

After the immediate effects of the impact had stopped, sedimentation in the Chicxulub area returned to the shallow water platform carbonate depositional environment that characterised it before the impact. The sequence, which dates back as far as the Paleocene, consists of marl and limestone, reaching a thickness of about 1,000 m (3,000 ft).[2]:3 The K–Pg boundary inside the crater is significantly deeper than in the surrounding area.[2]:4

On the Yucatán peninsula, the inner rim of the crater is marked by clusters of cenotes,[70] which are the surface expression of a zone of preferential groundwater flow, moving water from a recharge zone in the south to the coast through a karstic aquifer system.[2]:4[71] From the cenote locations, the karstic aquifer is clearly related to the underlying crater rim,[72] possibly through higher levels of fracturing, caused by differential compaction.[73]

Astronomical origin and type of impactor

There is broad consensus that the Chicxulub impactor was a C-type asteroid with a carbonaceous chondrite-like composition, rather than a comet.[3][74] This type of asteroid originally formed in the outer Solar System, beyond the orbit of Jupiter.[74] In 1998, a meteorite, approximately 2.5 millimeters ($\frac{1}{8}$ in) across, was described from a deep sea sediment core from the North Pacific, from a sediment sequence spanning the Cretaceous–Paleogene boundary (when the site was located in the central Pacific), with the meteorite being found at the base of the K-Pg boundary iridium anomaly within the sediment core. It was suggested to be a fragment of the Chicxulub impactor. Analysis suggested that it best fitted the criteria of the CV, CO and CR groups of carbonaceous chondrites.[75] Ruthenium isotope ratios found in impact layers support a carbonaceous chondrite composition for the impactor.[74] A 2021 paper suggested, based on geochemical evidence including the excess of chromium isotope ^{54}Cr and the ratios of platinum group metals found in marine impact layers, that the impactor matched the characteristics of CM or CR carbonaceous chondrites.[3] A 2026 paper reported that the nickel isotopes present constrained the impactor to the CO group instead.[76]

A 2007 *Nature* report proposed a specific astronomical origin for the Chicxulub asteroid.[50] The authors, William F. Bottke, David Vokrouhlický, and David Nesvorný, argued that a collision in the asteroid belt 160 million years ago between a 170 km (110 mi) diameter parent body and another 60 km (37 mi) diameter body resulted in the Baptistina family of asteroids, the largest surviving member of which is 298 Baptistina. They proposed that the Chicxulub asteroid was also a member of this group.[77] Subsequent evidence has disproven this theory. A 2009 spectrographic analysis revealed that 298 Baptistina has a different composition more typical of an S-type asteroid than the presumed carbonaceous chondrite composition of the Chicxulub impactor.[78] In 2011, data from the Wide-field Infrared Survey Explorer revised the date of the collision which created the Baptistina family to about 80 million years ago, allowing only 15 million years for the process of resonance and collision, which takes many tens of millions of years.[79] In 2010, another hypothesis implicated the newly discovered asteroid 354P/LINEAR, a member of the Flora family, as a possible remnant cohort of the K–Pg impactor.[80] In 2021, a numerical simulation study argued that the impactor likely originated in the outer main part of the asteroid belt.[81]

Some scholars have argued that the impactor was a comet, not an asteroid. Two papers in 1984 proposed it to be a comet originating from the Oort cloud, and it was proposed in 1992 that tidal disruption of comets could potentially increase impact rates.[3] In 2021, Avi Loeb and a colleague suggested in *Scientific Reports* that the impactor was a fragment from a disrupted comet.[82] A rebuttal in *Astronomy & Geophysics* countered that Loeb et al. had ignored that the amount of iridium deposited around the globe, 2.0×10^8 to 2.8×10^8 kg (4.4×10^8 to 6.2×10^8 lb), was too large for a comet of the size implied by the crater, and that they had overestimated likely comet impact rates. They concluded that all available evidence strongly favors an asteroid impactor, effectively ruling out a comet.[3] Ruthenium isotope ratios in impact layers also strongly support an asteroid rather than a comet nature for the impactor.[74]

See also

- Barberton Greenstone Belt
- List of impact structures on Earth
- List of possible impact structures on Earth
- Nadir crater
- Permian–Triassic extinction event
- Tenejapa-Lacandón Formation
- Timeline of Cretaceous–Paleogene extinction event research

References

1. "PIA03379: Shaded Relief with Height as Color, Yucatan Peninsula, Mexico" (https://photojournal.jpl.nasa.gov/catalog/PIA03379). *Shuttle Radar Topography Mission*. NASA. March 6, 2003. Archived (https://web.archive.org/web/20170313212222/http://photojournal.jpl.nasa.gov/catalog/PIA03379) from the original on March 13, 2017. Retrieved October 28, 2010.
2. Hildebrand, Alan R.; Penfield, Glen T.; Kring, David A.; et al. (September 1991). "Chicxulub Crater; a possible Cretaceous/Tertiary boundary impact crater on the Yucatan Peninsula, Mexico". *Geology*. **19** (9): 867–871. Bibcode:1991Geo....19..867H (https://ui.adsabs.harvard.edu/abs/1991Geo....19..867H). doi:10.1130/0091-7613(1991)019<0867:CCAPCT>2.3.CO;2 (https://doi.org/10.1130%2F0091-7613%281991%29019%3C0867%3ACCAPCT%3E2.3.CO%3B2).
3. Desch, Steve; Jackson, Alan; Noviello, Jessica; Anbar, Ariel (June 1, 2021). "The Chicxulub impactor: comet or asteroid?". *Astronomy & Geophysics*. **62** (3): 3.34 – 3.37. arXiv:2105.08768 (https://arxiv.org/abs/2105.08768). doi:10.1093/astrogeo/atab069 (https://doi.org/10.1093%2Fastrogeo%2Fatab069).
4. Renne, Paul R.; Deino, Alan L.; Hilgen, Frederik J.; et al. (February 8, 2013). "Time Scales of Critical Events Around the Cretaceous-Paleogene Boundary". *Science*. **339** (6120): 684–687. Bibcode:2013Sci...339..684R (https://ui.adsabs.harvard.edu/abs/2013Sci...339..684R). doi:10.1126/science.1230492 (https://doi.org/10.1126%2Fscience.1230492). PMID 23393261 (https://pubmed.ncbi.nlm.nih.gov/23393261).
5. Penfield, Glen (2019). "Unlikely Impact The unexpected discovery of the Paleogene-Cretaceous impact crater" (https://www.researchgate.net/publication/337771587). *AAPG Explorer*. **40** (12): 20–23. Archived (https://web.archive.org/web/20251130195517/https://www.aapg.org/news-and-media/details/explorer/articleid/55293/unlikely-impact) from the original on November 30, 2025. Retrieved November 30, 2025.
6. Grieve, Richard; Therriault, Ann (May 2000). "Vredefort, Sudbury, Chicxulub: Three of a Kind?". *Annual Review of Earth and Planetary Sciences*. **28** (1): 305–338. Bibcode:2000AREPS..28..305G (https://ui.adsabs.harvard.edu/abs/2000AREPS..28..305G). doi:10.1146/annurev.earth.28.1.305 (https://doi.org/10.1146%2Fannurev.earth.28.1.305).

7. Schulte, Peter; Alegret, Laia; Arenillas, Ignacio; et al. (March 5, 2010). "The Chicxulub Asteroid Impact and Mass Extinction at the Cretaceous-Paleogene Boundary" (https://lirias.kuleuven.be/handle/123456789/264213). *Science*. **327** (5970): 1214–1218. Bibcode:2010Sci...327.1214S (https://ui.adsabs.harvard.edu/abs/2010Sci...327.1214S). doi:10.1126/science.1177265 (https://doi.org/10.1126%2Fscience.1177265). PMID 20203042 (https://pubmed.ncbi.nlm.nih.gov/20203042).; Rincon, Paul (March 4, 2010). "Dinosaur extinction link to crater confirmed" (https://news.bbc.co.uk/2/hi/science/nature/8550504.stm). BBC. Archived (https://web.archive.org/web/20191031034913/http://news.bbc.co.uk/2/hi/science/nature/8550504.stm) from the original on October 31, 2019. Retrieved March 5, 2010.
8. Bates, Robin (series producer); Chesmar, Terri and Baniewicz, Rich (associate producers); Bakker, Robert T.; Hildebrand, Alan; Melosh, Gene; Moras, Florentine; Penfield, Glen (interviewees) (1992). *The Dinosaurs! Episode 4: "Death of the Dinosaur"* (https://imdb.com/title/tt0103400) (TV-series). PBS Video, WHYY-TV. Archived (https://web.archive.org/web/20240918010428/https://www.imdb.com/title/tt0103400/) from the original on September 18, 2024. Retrieved September 25, 2007.
9. Preston, Douglas (March 29, 2019). "The Day The Dinosaurs Died" (https://www.newyorker.com/magazine/2019/04/08/the-day-the-dinosaurs-died). *The New Yorker*. Archived (https://web.archive.org/web/20190518000523/https://www.newyorker.com/magazine/2019/04/08/the-day-the-dinosaurs-died) from the original on May 18, 2019. Retrieved May 13, 2019.
10. Alvarez, W.; Alvarez, L.W.; Asaro, F.; Michel, H.V. (1979). "Anomalous iridium levels at the Cretaceous/Tertiary boundary at Gubbio, Italy: Negative results of tests for a supernova origin". In Christensen, W.K.; Birkelund, T. (eds.). *Cretaceous/Tertiary Boundary Events Symposium*. Vol. 2. University of Copenhagen, Copenhagen, Denmark. p. 69.; Becker, Luann (March 2002). "Repeated Blows". *Scientific American*. **286** (3): 76–83. Bibcode:2002SciAm.286c..76B (https://ui.adsabs.harvard.edu/abs/2002SciAm.286c..76B). doi:10.1038/scientificamerican0302-76 (https://doi.org/10.1038%2Fscientificamerican0302-76). PMID 11857903 (https://pubmed.ncbi.nlm.nih.gov/11857903).
11. Alvarez, Luis; Alvarez, Walter; Asaro, Frank; Michel, Helen (June 6, 1980). "Extraterrestrial Cause for the Cretaceous-Tertiary Extinction". *Science*. **208** (4408): 1095–1108. Bibcode:1980Sci...208.1095A (https://ui.adsabs.harvard.edu/abs/1980Sci...208.1095A). doi:10.1126/science.208.4448.1095 (https://doi.org/10.1126%2Fscience.208.4448.1095). PMID 17783054 (https://pubmed.ncbi.nlm.nih.gov/17783054).
12. Mayell, Hillary (May 15, 2005). "Asteroid Rained Glass Over Entire Earth, Scientists Say" (https://web.archive.org/web/20160918074556/http://news.nationalgeographic.com/news/2005/04/0415_050418_chicxulub.html). *National Geographic News*. Archived from the original (http://news.nationalgeographic.com/news/2005/04/0415_050418_chicxulub.html) on September 18, 2016. Retrieved October 1, 2007.
13. Alvarez, Walter (2008). *T. Rex and the Crater of Doom*. Princeton University Press. ISBN 978-0-691-13103-0.
14. Smit, Jan; Hertogen, Jan (1980). "An extraterrestrial event at the Cretaceous-tertiary boundary". *Nature*. **285** (5762): 198–200. Bibcode:1980Natur.285..198S (https://ui.adsabs.harvard.edu/abs/1980Natur.285..198S). doi:10.1038/285198a0 (https://doi.org/10.1038%2F285198a0).
15. Weinreb, David B. (March 2002). "Catastrophic Events in the History of Life: Toward a New Understanding of Mass Extinctions in the Fossil Record – Part I" (https://www.jyi.org/2002-march/2017/10/23/catastrophic-events-in-the-history-of-life-toward-a-new-understanding-of-mass-extinctions-in-the-fossil-record-part-i). *Journal of Young Investigators*. **5** (6). Archived (https://web.archive.org/web/20240918010420/https://www.jyi.org/2002-march/2017/10/23/catastrophic-events-in-the-history-of-life-toward-a-new-understanding-of-mass-extinctions-in-the-fossil-record-part-i) from the original on September 18, 2024. Retrieved July 27, 2024.

16. Osterloff, Emily (2018). "How an asteroid ended the age of the dinosaurs" (https://www.nhm.ac.uk/discover/how-an-asteroid-caused-extinction-of-dinosaurs.html). London, England: Natural History Museum. Archived (https://web.archive.org/web/20220426174952/https://www.nhm.ac.uk/discover/how-an-asteroid-caused-extinction-of-dinosaurs.html) from the original on April 26, 2022. Retrieved May 18, 2022.
17. Verschuur, Gerrit L. (1996). *Impact!: The Threat of Comets and Asteroids*. Oxford University Press (U.S.). ISBN 978-0-19-511919-0.
18. Byars, Carlos (December 13, 1981). "Mexican site may be a link to dinosaurs' disappearance" (https://infoweb.newsbank.com/apps/news/openurl?ctx_ver=z39.88-2004&rft_id=info%3Asid/infoweb.newsbank.com&svc_dat=AMNEWS&req_dat=0D1C2A34C3EF45E0&rft_val_format=info%3Aofi/fmt%3Akev%3Amtx%3Actx&rft_dat=document_id%3Aimage%252Fv2%253A14DB39C1C40322B4%2540EANX-NB-1642F1C7C1E6C3CA%25402444952-1642EDDD95D04DD1%25400-1642EDDD95D04DD1%2540). *Houston Chronicle* (Newspaper [digital microfilm]). Vol. 81, no. 61 (Final ed.). pp. 1, 18. Retrieved September 1, 2024 – via NewsBank.
19. Frankel, Charles (1999). *The End of the Dinosaurs: Chicxulub Crater and Mass Extinctions*. Cambridge, England: Cambridge University Press. p. 236 (https://archive.org/details/endofdinosaursch00fran/page/236). ISBN 978-0-521-47447-4.
20. Pope KO; Baines, K.H.; Ocampo, A.C.; Ivanov, B.A. (1997). "Energy, volatile production, and climatic effects of the Chicxulub Cretaceous/Tertiary impact" (https://doi.org/10.1029%2F97JE01743). *Journal of Geophysical Research*. **102** (E9). Washington, D.C.: American Geophysical Union: 245–264. Bibcode:1997JGR...10221645P (https://ui.adsabs.harvard.edu/abs/1997JGR...10221645P). doi:10.1029/97JE01743 (https://doi.org/10.1029%2F97JE01743). PMID 11541145 (https://pubmed.ncbi.nlm.nih.gov/11541145).
21. Sharpton, Vernon L.; Marin, Luis E. (May 1997). "The Cretaceous–Tertiary impact crater and the cosmic projectile that produced it". *Annals of the New York Academy of Sciences*. **822** (1). New York City: Wiley-Blackwell: 353–380. Bibcode:1997NYASA.822..353S (https://ui.adsabs.harvard.edu/abs/1997NYASA.822..353S). doi:10.1111/j.1749-6632.1997.tb48351.x (https://doi.org/10.1111%2Fj.1749-6632.1997.tb48351.x). PMID 11543120 (https://pubmed.ncbi.nlm.nih.gov/11543120).
22. Keller, Gerta; Mateo, Paula; Monkenbusch, Johannes; et al. (November 2020). "Mercury linked to Deccan Traps volcanism, climate change and the end-Cretaceous mass extinction" (https://doi.org/10.1016%2Fj.gloplacha.2020.103312). *Global and Planetary Change*. **194** 103312. Bibcode:2020GPC...19403312K (https://ui.adsabs.harvard.edu/abs/2020GPC...19403312K). doi:10.1016/j.gloplacha.2020.103312 (https://doi.org/10.1016%2Fj.gloplacha.2020.103312).
23. Hull, Pincelli M.; Bornemann, André; Penman, Donald E.; et al. (January 17, 2020). "On impact and volcanism across the Cretaceous-Paleogene boundary" (https://doi.org/10.1126%2Fscience.aay5055). *Science*. **367** (6475): 266–272. Bibcode:2020Sci...367..266H (https://ui.adsabs.harvard.edu/abs/2020Sci...367..266H). doi:10.1126/science.aay5055 (https://doi.org/10.1126%2Fscience.aay5055). hdl:20.500.11820/483a2e77-318f-476a-8fec-33a45fbdc90b (https://hdl.handle.net/20.500.11820%2F483a2e77-318f-476a-8fec-33a45fbdc90b). PMID 31949074 (https://pubmed.ncbi.nlm.nih.gov/31949074).
24. Sprain, C.J.; Renne, P.R.; Wilson, G.P.; Clemens, W.A. (March 1, 2015). "High-resolution chronostratigraphy of the terrestrial Cretaceous-Paleogene transition and recovery interval in the Hell Creek region, Montana". *Geological Society of America Bulletin*. **127** (3–4): 393–409. Bibcode:2015GSAB..127..393S (https://ui.adsabs.harvard.edu/abs/2015GSAB..127..393S). doi:10.1130/B31076.1 (https://doi.org/10.1130%2FB31076.1).
25. Renne, Paul R.; Arenillas, Ignacio; Arz, José A.; et al. (June 2018). "Multi-proxy record of the Chicxulub impact at the Cretaceous-Paleogene boundary from Gorgonilla Island, Colombia" (http://urn.kb.se/resolve?urn=urn:nbn:se:nrm:diva-3235). *Geology*. **46** (6): 547–550. Bibcode:2018Geo....46..547R (https://ui.adsabs.harvard.edu/abs/2018Geo....46..547R). doi:10.1130/G40224.1 (https://doi.org/10.1130%2FG40224.1).

26. During, Melanie A.D.; Smit, Jan; Voeten, Dennis F.A.E.; et al. (February 23, 2022). "The Mesozoic terminated in boreal spring" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8891016). *Nature*. **603** (7899): 91–94. Bibcode:2022Natur.603...91D (https://ui.adsabs.harvard.edu/abs/2022Natur.603...91D). doi:10.1038/s41586-022-04446-1 (https://doi.org/10.1038%2Fs41586-022-04446-1). PMC 8891016 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8891016). PMID 35197634 (https://pubmed.ncbi.nlm.nih.gov/35197634).
27. Gulick, S.P.S.; Christeson, G.L.; Barton, P.J.; et al. (January 2013). "Geophysical characterization of the Chicxulub impact crater" (https://doi.org/10.1002%2Frog.20007). *Reviews of Geophysics*. **51** (1): 31–52. Bibcode:2013RvGeo..51...31G (https://ui.adsabs.harvard.edu/abs/2013RvGeo..51...31G). doi:10.1002/rog.20007 (https://doi.org/10.1002%2Frog.20007).
28. Gulick, Sean P. S.; Barton, Penny J.; Christeson, Gail L.; et al. (February 2008). "Importance of pre-impact crustal structure for the asymmetry of the Chicxulub impact crater". *Nature Geoscience*. **1** (2): 131–135. Bibcode:2008NatGe...1..131G (https://ui.adsabs.harvard.edu/abs/2008NatGe...1..131G). doi:10.1038/ngeo103 (https://doi.org/10.1038%2Fngeo103).
29. Navarro, Karina F.; Urrutia-Fucugauchi, Jaime; Villagran-Muniz, Mayo; et al. (August 2020). "Emission spectra of a simulated Chicxulub impact-vapor plume at the Cretaceous–Paleogene boundary". *Icarus*. **346** 113813. Bibcode:2020Icar..34613813N (https://ui.adsabs.harvard.edu/abs/2020Icar..34613813N). doi:10.1016/j.icarus.2020.113813 (https://doi.org/10.1016%2Fj.icarus.2020.113813).
30. Collins, G. S.; Patel, N.; Davison, T. M.; Rae, A. S. P.; Morgan, J. V.; Gulick, S. P. S. (May 26, 2020). "A steeply-inclined trajectory for the Chicxulub impact" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7251121). *Nature Communications*. **11** (1): 1480. Bibcode:2020NatCo..11.1480C (https://ui.adsabs.harvard.edu/abs/2020NatCo..11.1480C). doi:10.1038/s41467-020-15269-x (https://doi.org/10.1038%2Fs41467-020-15269-x). PMC 7251121 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7251121). PMID 32457325 (https://pubmed.ncbi.nlm.nih.gov/32457325).
31. Morgan, Joanna V.; Bralower, Timothy J.; Brugger, Julia; Wünnemann, Kai (2022). "The Chicxulub impact and its environmental consequences" (https://www.nature.com/articles/s43017-022-00283-y). *Nature Reviews Earth & Environment*. **3** (5): 338–354. Bibcode:2022NRvEE...3..338M (https://ui.adsabs.harvard.edu/abs/2022NRvEE...3..338M). doi:10.1038/s43017-022-00283-y (https://doi.org/10.1038%2Fs43017-022-00283-y). ISSN 2662-138X (https://search.worldcat.org/issn/2662-138X). Archived (https://web.archive.org/web/20260509152551/https://www.nature.com/articles/s43017-022-00283-y) from the original on May 9, 2026. Retrieved May 29, 2026.
32. Richards, Mark A.; Alvarez, Walter; Self, Stephen; Karlstrom, Leif; Renne, Paul R.; Manga, Michael; Sprain, Courtney J.; Smit, Jan; Vanderkluysen, Loÿc; Gibson, Sally A. (November 2015). "Triggering of the largest Deccan eruptions by the Chicxulub impact". *GSA Bulletin*. **127** (11–12): 1507–1520. Bibcode:2015GSAB..127.1507R (https://ui.adsabs.harvard.edu/abs/2015GSAB..127.1507R). doi:10.1130/B31167.1 (https://doi.org/10.1130%2FB31167.1).
33. Kring, David A. (November 2007). "The Chicxulub impact event and its environmental consequences at the Cretaceous–Tertiary boundary". *Palaeogeography, Palaeoclimatology, Palaeoecology*. **255** (1–2): 4–21. Bibcode:2007PPP...255....4K (https://ui.adsabs.harvard.edu/abs/2007PPP...255....4K). doi:10.1016/j.palaeo.2007.02.037 (https://doi.org/10.1016%2Fj.palaeo.2007.02.037).
34. Bralower, Timothy J.; Paull, Charles K.; Mark Leckie, R. (1998). "The Cretaceous-Tertiary boundary cocktail: Chicxulub impact triggers margin collapse and extensive sediment gravity flows". *Geology*. **26** (4): 331. Bibcode:1998Geo....26..331B (https://ui.adsabs.harvard.edu/abs/1998Geo....26..331B). doi:10.1130/0091-7613(1998)026<0331:TCTBCC>2.3.CO;2 (https://doi.org/10.1130%2F0091-7613%281998%29026%3C0331%3ATCTBCC%3E2.3.CO%3B2).

35. Parkos, Devon; Alexeenko, Alina; Kulakhmetov, Marat; Johnson, Brandon C.; Melosh, H. Jay (December 2015). "NO x production and rainout from Chicxulub impact ejecta reentry". *Journal of Geophysical Research: Planets*. **120** (12): 2152–2168. doi:10.1002/2015JE004857 (https://doi.org/10.1002%2F2015JE004857).
36. Pope, Kevin O.; Baines, Kevin H.; Ocampo, Adriana C.; Ivanov, Boris A. (September 25, 1997). "Energy, volatile production, and climatic effects of the Chicxulub Cretaceous/Tertiary impact". *Journal of Geophysical Research: Planets*. **102** (E9): 21645–21664. Bibcode:1997JGR...10221645P (https://ui.adsabs.harvard.edu/abs/1997JGR...10221645P). doi:10.1029/97JE01743 (https://doi.org/10.1029%2F97JE01743). PMID 11541145 (https://pubmed.ncbi.nlm.nih.gov/11541145).
37. Durand-Manterola, Hector Javier; Cordero-Tercero, Guadalupe (March 19, 2014), *Assessments of the energy, mass and size of the Chicxulub Impactor*, arXiv:1403.6391 (https://arxiv.org/abs/1403.6391)
38. "Chicxulub Impact Event: Regional Effects" (https://www.lpi.usra.edu/science/kring/Chicxulub/regional-effects/). *Lunar and Planetary Institute*. Archived (https://web.archive.org/web/20190726023401/https://www.lpi.usra.edu/science/kring/Chicxulub/regional-effects/) from the original on July 26, 2019. Retrieved June 1, 2020.
39. Amos, Jonathan (May 15, 2017). "Dinosaur asteroid hit 'worst possible place' " (https://www.bbc.com/news/science-environment-39922998). Science and Environment. *BBC News*. Archived (https://web.archive.org/web/20180318162335/http://www.bbc.com/news/science-environment-39922998) from the original on March 18, 2018. Retrieved August 19, 2017.
40. "Huge Global Tsunami Followed Dinosaur-Killing Asteroid Impact" (https://eos.org/articles/huge-global-tsunami-followed-dinosaur-killing-asteroid-impact). December 20, 2018. Archived (https://web.archive.org/web/20200711221446/https://eos.org/articles/huge-global-tsunami-followed-dinosaur-killing-asteroid-impact) from the original on July 11, 2020. Retrieved July 11, 2020.
41. Bryant, Edward (June 2014). *Tsunami: The underrated hazard* (https://books.google.com/books?id=tOkpBAAAQBAJ&pg=PA178). Springer. p. 178. ISBN 978-3-319-06133-7.
42. Koumoundouros, Tessa (July 14, 2021). "Fossilized Tsunami 'Megaripples' Reveal The Devastation From The Chicxulub Asteroid" (https://www.sciencealert.com/tsunami-megaripples-from-the-dinosaur-killing-asteroid-impact-discovered-in-louisiana). *ScienceAlert*. Retrieved January 1, 2022.
43. Kinsland, Gary L.; Egedahl, Kaare; Strong, Martell Albert; Ivy, Robert (September 15, 2021). "Chicxulub impact tsunami megaripples in the subsurface of Louisiana: Imaged in petroleum industry seismic data". *Earth and Planetary Science Letters*. **570** 117063. Bibcode:2021E&PSL.57017063K (https://ui.adsabs.harvard.edu/abs/2021E&PSL.57017063K). doi:10.1016/j.epsl.2021.117063 (https://doi.org/10.1016%2Fj.epsl.2021.117063).
44. Palmer, Jane (February 25, 2016). "We Finally Know How Much the Dino-Killing Asteroid Reshaped Earth" (http://www.smithsonianmag.com/science-nature/we-finally-know-how-much-dino-killing-asteroid-reshaped-earth-180958222/). *Smithsonian.com*. Smithsonian Institution. Archived (https://web.archive.org/web/20160228025905/http://www.smithsonianmag.com/science-nature/we-finally-know-how-much-dino-killing-asteroid-reshaped-earth-180958222/) from the original on February 28, 2016. Retrieved February 26, 2016.

45. Goto, Kazuhisa; Tada, Ryuji; Tajika, Eiichi; et al. (2004). "Evidence for ocean water invasion into the Chicxulub crater at the Cretaceous/Tertiary boundary" (https://doi.org/10.1111%2Fj.1945-5100.2004.tb00943.x). *Meteoritics & Planetary Science*. **39** (8): 1233–1247. Bibcode:2004M&PS...39.1233G (https://ui.adsabs.harvard.edu/abs/2004M&PS...39.1233G). doi:10.1111/j.1945-5100.2004.tb00943.x (https://doi.org/10.1111%2Fj.1945-5100.2004.tb00943.x)., Range, Molly M.; Arbic, SAND-Brian K.; Johnson, Brandon C.; et al. (December 14, 2018). "The Chicxulub Impact Produced a Powerful Global Tsunami" (https://agu.confex.com/agu/fm18/meetingapp.cgi/Paper/445502). *AGU Fall Meeting Abstracts*. **2018**. AGU. Bibcode:2018AGUFMPP53B..07R (https://ui.adsabs.harvard.edu/abs/2018AGUFMPP53B..07R). Archived (https://web.archive.org/web/20200715022234/https://agu.confex.com/agu/fm18/meetingapp.cgi/Paper/445502) from the original on July 15, 2020. Retrieved July 11, 2020 – via agu.confex.com., Matsui, Takafumi; Imamura, F.; Tajika, Eiichi; Nakano, Y.; Fujisawa, Y. (2002). "Generation and propagation of a tsunami from the Cretaceous-Tertiary impact event". *Catastrophic events and mass extinctions: Impacts and beyond*. doi:10.1130/0-8137-2356-6.69 (https://doi.org/10.1130%2F0-8137-2356-6.69). ISBN 978-0-8137-2356-3.
46. Kring, David A; Claeys, Philippe; Gulick, Sean P.S.; Morgan, Joanna V.; Collins, Gareth S. (October 10, 2017). "Chicxulub and the Exploration of Large Peak-Ring Impact Craters through Scientific Drilling" (https://web.archive.org/web/20171010155225/https://www.geosociety.org/gsatoday/science/G352A/GSATG352A.pdf) (PDF). *GSA Today*. The Geological Society of America. Archived from the original (https://www.geosociety.org/gsatoday/science/G352A/GSATG352A.pdf) (PDF) on October 10, 2017. Retrieved February 1, 2022.
47. Shaulis, Barry J.; Riller, Ulrich; Cockell, Charles; Coolen, Marco J.L. (2017). "Probing the impact-generated hydrothermal system in the peak ring of the Chicxulub crater and its potential as a habitat" (https://web.archive.org/web/20201026101301/https://www.hou.usra.edu/meetings/lpsc2017/pdf/1212.pdf) (PDF). *Lunar and Planetary Science*. **XLVIII** (1964): 1212. Bibcode:2017LPI....48.1212K (https://ui.adsabs.harvard.edu/abs/2017LPI....48.1212K). Archived from the original (https://www.hou.usra.edu/meetings/lpsc2017/pdf/1212.pdf) (PDF) on October 26, 2020.
48. Airhart, Marc (January 1, 2008). "Seismic Images Show Dinosaur-Killing Meteor Made Bigger Splash" (http://www.jsg.utexas.edu/news/2008/01/seismic-images-show-dinosaur-killing-meteor-made-bigger-splash/). Archived (https://web.archive.org/web/20141220175132/http://www.jsg.utexas.edu/news/2008/01/seismic-images-show-dinosaur-killing-meteor-made-bigger-splash/) from the original on December 20, 2014. Retrieved November 29, 2011.
49. Kaiho, Kunio; Oshima, Naga; Adachi, Kouji; Adachi, Yukimasa; Mizukami, Takuya; Fujibayashi, Megumu; Saito, Ryosuke (July 14, 2016). "Global climate change driven by soot at the K-Pg boundary as the cause of the mass extinction" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4944614). *Scientific Reports*. **6** (1) 28427. Bibcode:2016NatSR...628427K (https://ui.adsabs.harvard.edu/abs/2016NatSR...628427K). doi:10.1038/srep28427 (https://doi.org/10.1038%2Fsrep28427). ISSN 2045-2322 (https://search.worldcat.org/issn/2045-2322). PMC 4944614 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4944614). PMID 27414998 (https://pubmed.ncbi.nlm.nih.gov/27414998).
50. Perlman, David (September 6, 2007). "Scientists say they know where dinosaur-killing asteroid came from" (https://web.archive.org/web/20120404030311/http://www.sfgate.com/cgi-bin/article.cgi?f=/c/a/2007/09/06/MNVFRUVCK.DTL). *San Francisco Chronicle*. Archived from the original (http://www.sfgate.com/cgi-bin/article.cgi?f=/c/a/2007/09/06/MNVFRUVCK.DTL) on April 4, 2012. Retrieved October 3, 2007.

51. Pope KO; Ocampo AC; Kinsland GL; Smith R (1996). "Surface expression of the Chicxulub crater". *Geology*. **24** (6): 527–530. Bibcode:1996Geo....24..527P (https://ui.adsabs.harvard.edu/abs/1996Geo....24..527P). doi:10.1130/0091-7613(1996)024<0527:SEOTCC>2.3.CO;2 (https://doi.org/10.1130%2F0091-7613%281996%29024%3C0527%3ASEOTCC%3E2.3.CO%3B2). PMID 11539331 (https://pubmed.ncbi.nlm.nih.gov/11539331). See also a similar 1998 report by the same group (https://ntrs.nasa.gov/api/citations/19980211540/downloads/19980211540.pdf) Archived (https://web.archive.org/web/20240616152811/https://ntrs.nasa.gov/api/citations/19980211540/downloads/19980211540.pdf) June 16, 2024, at the Wayback Machine.
52. Lomax, B.; Beerling, D.; Upchurch, G. Jr.; Otto-Bliesner, B. (2001). "Rapid (10-yr) recovery of terrestrial productivity in a simulation study of the terminal Cretaceous impact event". *Earth and Planetary Science Letters*. **192** (2): 137–144. Bibcode:2001E&PSL.192..137L (https://ui.adsabs.harvard.edu/abs/2001E&PSL.192..137L). doi:10.1016/S0012-821X(01)00447-2 (https://doi.org/10.1016%2FS0012-821X%2801%2900447-2).
53. Winemiller, Terance L. (2007). *The Chicxulub meteor impact and ancient locational decisions on the Yucatán Peninsula, Mexico: The application of remote sensing, GIS, and GPS in settlement pattern Studies* (http://www.asprs.org/a/publications/proceedings/tampa2007/0080.pdf) (PDF). ASPRS 2007 Annual Conference. Tampa, Florida: American Society for Photogrammetry and Remote Sensing. Archived (https://web.archive.org/web/20170810112923/http://asprs.org/a/publications/proceedings/tampa2007/0080.pdf) (PDF) from the original on August 10, 2017. Retrieved October 2, 2012.
54. Morgan, J.; Warner, M.; Brittan, J.; et al. (1997). "Size and morphology of the Chicxulub impact crater". *Nature*. **390** (6659): 472–476. Bibcode:1997Natur.390..472M (https://ui.adsabs.harvard.edu/abs/1997Natur.390..472M). doi:10.1038/37291 (https://doi.org/10.1038%2F37291).
55. Snyder, D.B.; Hobbs, R.W. (1999). "Deep seismic reflection profiles across the Chicxulub crater" (https://books.google.com/books?id=gV_H9cK94q8C&q=Chix&pg=PA263). In Dressler, B.O.; Sharpton, V.L. (eds.). *Large Meteorite Impacts and Planetary Evolution II*. Special Publication. Vol. 339. Geological Society of America. ISBN 978-0-8137-2339-6.
56. Morgan, J.; Urrutia-Fucugauchi, J.; Gulick, S.; et al. (2005). "Chicxulub Crater Seismic Survey prepares way for future drilling" (https://doi.org/10.1029%2F2005EO360001). *Eos*. **86** (36): 325–328. Bibcode:2005EOSTr..86..325M (https://ui.adsabs.harvard.edu/abs/2005EOSTr..86..325M). doi:10.1029/2005EO360001 (https://doi.org/10.1029%2F2005EO360001).
57. Center for Lunar Science and Exploration (2019). "Classroom Illustrations: Chicxulub Crater" (https://www.lpi.usra.edu/exploration/training/illustrations/chicxulub-crater/). Archived (https://web.archive.org/web/20220126000444/https://www.lpi.usra.edu/exploration/training/illustrations/chicxulub-crater/) from the original on January 26, 2022. Retrieved March 24, 2022.; Urrutia-Fucugauchi, J.; Chavez-Aguirre, J.M.; Pérez-Cruz, L.; De la Rosa, J.L. (2008). "Impact ejecta and carbonate sequence in the eastern sector of the Chicxulub crater". *Comptes Rendus Geoscience*. **340** (12): 801–810. Bibcode:2008CRGeo.340..801U (https://ui.adsabs.harvard.edu/abs/2008CRGeo.340..801U). doi:10.1016/j.crte.2008.09.001 (https://doi.org/10.1016%2Fj.crte.2008.09.001).
58. Amos, Jonathan (April 5, 2016). "Project to drill into 'dinosaur crater' gets under way" (https://www.bbc.com/news/science-environment-35950946). *BBC News*. Archived (https://web.archive.org/web/20160406072841/http://www.bbc.com/news/science-environment-35950946) from the original on April 6, 2016. Retrieved April 5, 2016.; Amos, Jonathan (May 25, 2016). "Chicxulub 'dinosaur' crater drill project declared a success" (https://www.bbc.com/news/science-environment-36377679). *BBC News*. Archived (https://web.archive.org/web/20160528081109/http://www.bbc.com/news/science-environment-36377679) from the original on May 28, 2016. Retrieved May 25, 2016.

59. Morgan, J.; Warner, M.; the Chicxulub Working Group; et al. (1997). "Size and morphology of the Chicxulub impact crater". *Nature*. **390** (6659): 472–476. Bibcode:1997Natur.390..472M (https://ui.adsabs.harvard.edu/abs/1997Natur.390..472M). doi:10.1038/37291 (https://doi.org/10.1038%2F37291).
60. Melosh, J. (2001). "Deep down at Chicxulub". *Nature*. **414** (6866): 861–862. doi:10.1038/414861a (https://doi.org/10.1038%2F414861a). PMID 11780048 (https://pubmed.ncbi.nlm.nih.gov/11780048).
61. Hildebrand, A.; Pilkington, M.; Conors, M.; Ortiz-Aleman, C.; Chavez, R.E. (1995). "Size and structure of the Chicxulub crater revealed by horizontal gravity gradients and cenotes". *Nature*. **376** (6539): 415–417. Bibcode:1995Natur.376..415H (https://ui.adsabs.harvard.edu/abs/1995Natur.376..415H). doi:10.1038/376415a0 (https://doi.org/10.1038%2F376415a0).
62. Zhao, J.; Xiao, L.; Gulick, S.P.S.; et al. (2020). "Geochemistry, geochronology and petrogenesis of Maya Block granitoids and dykes from the Chicxulub Impact Crater, Gulf of México: Implications for the assembly of Pangea" (https://biblio.vub.ac.be/vubirfiles/78237141/Zhao_etal_accepted.pdf) (PDF). *Gondwana Research*. **82**: 128–150. Bibcode:2020GondR..82..128Z (https://ui.adsabs.harvard.edu/abs/2020GondR..82..128Z). doi:10.1016/j.gr.2019.12.003 (https://doi.org/10.1016%2Fj.gr.2019.12.003). Archived (https://web.archive.org/web/20220518033345/https://biblio.vub.ac.be/vubirfiles/78237141/Zhao_etal_accepted.pdf) (PDF) from the original on May 18, 2022. Retrieved May 18, 2022.
63. Guzmán-Hidalgo, E.; Grajales-Nishimura, J.M.; Eberli, G.P.; et al. (2021). "Seismic stratigraphic evidence of a pre-impact basin in the Yucatán Platform: morphology of the Chicxulub crater and K/Pg boundary deposits". *Marine Geology*. **441** 106594. Bibcode:2021MGeol.44106594G (https://ui.adsabs.harvard.edu/abs/2021MGeol.44106594G). doi:10.1016/j.margeo.2021.106594 (https://doi.org/10.1016%2Fj.margeo.2021.106594).
64. Kaskes, P.; de Graaf, S.J.; Feignon, J.-G.; et al. (2022). "Formation of the crater suevite sequence from the Chicxulub peak ring: A petrographic, geochemical, and sedimentological characterization" (https://biblio.vub.ac.be/vubirfiles/72890608/b36020.pdf) (PDF). *GSA Bulletin*. **134** (3–4): 895–927. Bibcode:2022GSAB..134..895K (https://ui.adsabs.harvard.edu/abs/2022GSAB..134..895K). doi:10.1130/B36020.1 (https://doi.org/10.1130%2FB36020.1). Archived (https://web.archive.org/web/20220518033347/https://biblio.vub.ac.be/vubirfiles/72890608/b36020.pdf) (PDF) from the original on May 18, 2022. Retrieved May 18, 2022.
65. de Graaf, S.J.; Kaskes, P.; Déhais, T.; et al. (2022). "New insights into the formation and emplacement of impact melt rocks within the Chicxulub impact structure, following the 2016 IODP-ICDP Expedition 364" (https://research.vu.nl/en/publications/e92c261f-b2e7-45c7-a12a-340e59ac33f2). *GSA Bulletin*. **134** (1–2): 293–315. Bibcode:2022GSAB..134..293D (https://ui.adsabs.harvard.edu/abs/2022GSAB..134..293D). doi:10.1130/B35795.1 (https://doi.org/10.1130%2FB35795.1). hdl:2013/ULB-DIPOT:oai:dipot.ulb.ac.be:2013/324574 (https://hdl.handle.net/2013%2FULB-DIPOT%3Aoai%3Adipot.ulb.ac.be%3A2013%2F324574).
66. St. Fleur, Nicholas (November 17, 2016). "Drilling into the Chicxulub Crater, Ground Zero of the Dinosaur Extinction" (https://www.nytimes.com/2016/11/18/science/chicxulub-crater-dinosaur-extinction.html). *The New York Times*. Archived (https://web.archive.org/web/20161119200501/http://www.nytimes.com/2016/11/18/science/chicxulub-crater-dinosaur-extinction.html?_r=0) from the original on November 19, 2016. Retrieved March 1, 2017.
67. Riller, U.; Poelchau, M.H.; Rae, A.S.P.; et al. (2018). "Rock fluidization during peak-ring formation of large impact structures" (http://eprints.gla.ac.uk/173925/1/173925.pdf) (PDF). *Nature*. **562** (7728): 511–518. Bibcode:2018Natur.562..511R (https://ui.adsabs.harvard.edu/abs/2018Natur.562..511R). doi:10.1038/s41586-018-0607-z (https://doi.org/10.1038%2Fs41586-018-0607-z). PMID 30356184 (https://pubmed.ncbi.nlm.nih.gov/30356184). Archived (https://web.archive.org/web/20220523012518/http://eprints.gla.ac.uk/173925/1/173925.pdf) (PDF) from the original on May 23, 2022. Retrieved May 18, 2022.
68. Kring, David; Tikoo, Sonia M.; Schmieder, Martin; et al. (2020). "Probing the hydrothermal system of the Chicxulub impact crater". *Science Advances*. **6** (22). doi:10.1126/sciadv.aaz3053 (https://doi.org/10.1126%2Fsciadv.aaz3053).

69. Marchi, S.; Bottke, W.F.; Elkins-Tanton, L.T.; et al. (2014). "Widespread mixing and burial of Earth's Hadean crust by asteroid impacts". *Nature*. **511** (7511): 578–582. Bibcode:2014Natur.511..578M (https://ui.adsabs.harvard.edu/abs/2014Natur.511..578M). doi:10.1038/nature13539 (https://doi.org/10.1038%2Fnature13539). PMID 25079556 (https://pubmed.ncbi.nlm.nih.gov/25079556).
70. "Meteor impact site" (https://www.youtube.com/watch?v=dNRTtLLuNM8). *National Geographic* (video). Earth: The biography. July 11, 2008. Archived (https://web.archive.org/web/20151017002003/https://www.youtube.com/watch?v=dNRTtLLuNM8) from the original on October 17, 2015. Retrieved August 19, 2015.
71. Pérez-Ceballos, R.; Canul-Macario, C.; Pacheco-Castro, R.; et al. (2021). "Regional Hydrogeochemical Evolution of Groundwater in the Ring of Cenotes, Yucatán (Mexico): An Inverse Modelling Approach" (https://doi.org/10.3390%2Fw13050614). *Water*. **13** (5): 614. Bibcode:2021Water..13..614P (https://ui.adsabs.harvard.edu/abs/2021Water..13..614P). doi:10.3390/w13050614 (https://doi.org/10.3390%2Fw13050614).
72. Kring, David A. "Discovering the Crater" (https://web.archive.org/web/20071010021337/http://www.lpl.arizona.edu/SIC/impact_cratering/Chicxulub/Discovering_crater.html). *lpl.arizona.edu*. Archived from the original (http://www.lpl.arizona.edu/SIC/impact_cratering/Chicxulub/Discovering_crater.html) on October 10, 2007. Retrieved October 12, 2007.
73. Hildebrand, A.R.; Pilkington, M.; Ortiz-Aleman, C.; et al. (1998). "Mapping Chicxulub crater structure with gravity and seismic reflection data". In Grady, M.M.; Hutchinson, R.; McCall, G.J.H.; Rothery, D.A. (eds.). *Meteorites: Flux with Time and Impact Effects*. Special Publications. Vol. 140. London: Geological Society. p. 160. doi:10.1144/GSL.SP.1998.140.01.12 (https://doi.org/10.1144%2FGSL.SP.1998.140.01.12). ISBN 978-1-86239-017-1.
74. Fischer-Gödde, Mario; Tusch, Jonas; Goderis, Steven; Bragagni, Alessandro; Mohr-Westheide, Tanja; Messling, Nils; Elfers, Bo-Magnus; Schmitz, Birger; Reimold, Wolf U.; Maier, Wolfgang D.; Claeys, Philippe; Koeberl, Christian; Tissot, François L.H.; Bizzarro, Martin; Münker, Carsten (August 16, 2024). "Ruthenium isotopes show the Chicxulub impactor was a carbonaceous-type asteroid" (https://doi.org/10.1126%2Fscience.adk4868). *Science*. **385** (6710): 752–756. Bibcode:2024Sci...385..752F (https://ui.adsabs.harvard.edu/abs/2024Sci...385..752F). doi:10.1126/science.adk4868 (https://doi.org/10.1126%2Fscience.adk4868). PMID 39146402 (https://pubmed.ncbi.nlm.nih.gov/39146402).
75. Kyte, Frank T. (November 1998). "A meteorite from the Cretaceous/Tertiary boundary". *Nature*. **396** (6708): 237–239. Bibcode:1998Natur.396..237K (https://ui.adsabs.harvard.edu/abs/1998Natur.396..237K). doi:10.1038/24322 (https://doi.org/10.1038%2F24322).
76. Makhatadze, Georgy V.; Moynier, Frédéric; Brun, Leslie-Anne; Goderis, Steven; Claeys, Philippe; Koeberl, Christian (July 17, 2026). "The origin of Cretaceous-Palaeogene impactor revealed by nickel isotopes". *Science Advances*. **12** (29). doi:10.1126/sciadv.aef4858 (https://doi.org/10.1126%2Fsciadv.aef4858).
77. Bottke, William F.; Vokrouhlický, David; Nesvorný, David (September 2007). "An asteroid breakup 160 Myr ago as the probable source of the K/T impactor". *Nature*. **449** (7158): 48–53. Bibcode:2007Natur.449...48B (https://ui.adsabs.harvard.edu/abs/2007Natur.449...48B). doi:10.1038/nature06070 (https://doi.org/10.1038%2Fnature06070). PMID 17805288 (https://pubmed.ncbi.nlm.nih.gov/17805288).; Ingham, Richard (September 5, 2007). "Traced: The asteroid breakup that wiped out the dinosaurs" (https://web.archive.org/web/20071114031858/http://afp.google.com/article/ALeqM5jz3TGi2zcsmdYQDxwbdCwq2kanMA). *Google News*. Agence France-Presse. Archived from the original (http://afp.google.com/article/ALeqM5jz3TGi2zcsmdYQDxwbdCwq2kanMA) on November 14, 2007. Retrieved September 27, 2007.

78. Reddy, Vishnu; Emery, Joshua P.; Gaffey, Michael J.; et al. (December 2009). "Composition of 298 Baptistina: Implications for the K/T impactor link" (https://doi.org/10.1111%2Fj.1945-5100.2009.tb02001.x). *Meteoritics & Planetary Science*. **44** (12): 1917–1927. Bibcode:2009M&PS...44.1917R (https://ui.adsabs.harvard.edu/abs/2009M&PS...44.1917R). doi:10.1111/j.1945-5100.2009.tb02001.x (https://doi.org/10.1111%2Fj.1945-5100.2009.tb02001.x).
79. Masiero, Joseph R.; Mainzer, A.K.; Grav, T.; et al. (November 10, 2011). "Main belt asteroids with WISE / NEOWISE. I. Preliminary albedos and diameters". *The Astrophysical Journal*. **741** (2): 68. arXiv:1109.4096 (https://arxiv.org/abs/1109.4096). Bibcode:2011ApJ...741...68M (https://ui.adsabs.harvard.edu/abs/2011ApJ...741...68M). doi:10.1088/0004-637X/741/2/68 (https://doi.org/10.1088%2F0004-637X%2F741%2F2%2F68).
80. "Smashed asteroids may be related to dinosaur killer" (https://www.reuters.com/article/us-space-asteroid-idUSTRE61154120100202). *Reuters*. February 2, 2010. Archived (https://web.archive.org/web/20191004074906/https://www.reuters.com/article/us-space-asteroid-idUSTRE61154120100202) from the original on October 4, 2019. Retrieved July 5, 2021.
81. Nesvorný, David; Bottke, William F.; Marchi, Simone (November 1, 2021). "Dark primitive asteroids account for a large share of K/Pg-scale impacts on the Earth". *Icarus*. **368** 114621. arXiv:2107.03458 (https://arxiv.org/abs/2107.03458). Bibcode:2021Icar..36814621N (https://ui.adsabs.harvard.edu/abs/2021Icar..36814621N). doi:10.1016/j.icarus.2021.114621 (https://doi.org/10.1016%2Fj.icarus.2021.114621).
82. Ferreira, Becky (February 15, 2021). "Where Did the Dinosaur-Killing Impactor Come From? – A new study blames a comet fragment for the death of the dinosaurs 66 million years ago. But most experts maintain that an asteroid caused this cataclysmic event" (https://www.nytimes.com/2021/02/15/science/dinosaur-extinction-kt-comet-asteroid.html). *The New York Times*. Archived (https://web.archive.org/web/20210215200655/https://www.nytimes.com/2021/02/15/science/dinosaur-extinction-kt-comet-asteroid.html) from the original on February 15, 2021. Retrieved February 15, 2021.; Siraj, Amir (February 15, 2021). "Breakup of a long-period comet as the origin of the dinosaur extinction" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7884440). *Scientific Reports*. **11** (3803) 3803. arXiv:2102.06785 (https://arxiv.org/abs/2102.06785). Bibcode:2021NatSR..11.3803S (https://ui.adsabs.harvard.edu/abs/2021NatSR..11.3803S). doi:10.1038/s41598-021-82320-2 (https://doi.org/10.1038%2Fs41598-021-82320-2). PMC 7884440 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7884440). PMID 33589634 (https://pubmed.ncbi.nlm.nih.gov/33589634).

- Kornel, Katherine (September 10, 2019). "A New Timeline of the Day the Dinosaurs Began to Die Out – By drilling into the Chicxulub crater, scientists assembled a record of what happened just after the asteroid impact" (https://www.nytimes.com/2019/09/10/science/chicxulub-asteroid-impact-dinosaurs.html). *The New York Times*. Archived (https://web.archive.org/web/20190925002058/https://www.nytimes.com/2019/09/10/science/chicxulub-asteroid-impact-dinosaurs.html) from the original on September 25, 2019. Retrieved September 25, 2019.

External links

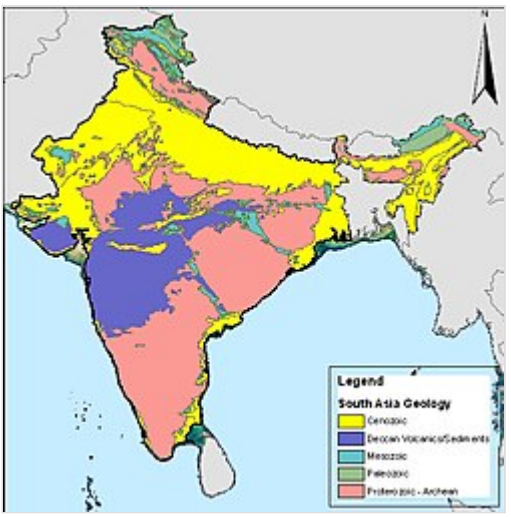
- Chicxulub Crater (http://www.chicxulubcrater.org) Archived (https://web.archive.org/web/20180722035405/http://www.chicxulubcrater.org/) July 22, 2018, at the Wayback Machine
- Chicxulub: Variations in the magnitude of the gravity field at sea level image (http://www.lpi.usra.edu/publications/slidesets/craters/slide_37.html) (Lunar and Planetary Institute, USRA)
- "Doubts on Dinosaurs" (http://www.scientificamerican.com/article.cfm?id=doubts-on-dinosaurs) – *Scientific American* (paywalled)
- Papers and presentations resulting from the 2016 Chicxulub drilling project (https://gsa.confex.com/gsa/2017AM/webprogram/Session43180.html)

Retrieved from "https://en.wikipedia.org/w/index.php?title=Chicxulub_crater&oldid=1379989654"

WIKIPEDIA 25
25 years of the free encyclopedia

Deccan Traps

The **Deccan Traps** are a large igneous province of west-central India (17–24°N, 73–74°E). They are one of the largest volcanic features on Earth, with lava erupted from multiple volcanic centres.[2] The Traps consist of many layers of solidified flood basalt that together are more than about 2 kilometres (1.2 mi) thick, cover an area of about 500,000 square kilometres (200,000 sq mi),[3] and have a volume of about 1 million km^3 (200,000 cu mi).[4] Originally, the Deccan Traps may have covered about 1.5 million km^2 (600,000 sq mi),[5] with a correspondingly larger original volume. This volume overlies the Archean age Indian Shield, which is likely the lithology the province passed through during eruption. The province is commonly divided into four subprovinces: the main Deccan, the Malwa Plateau, the Mandla Lobe and the Saurashtran Plateau.[6]



Geologic map of India, showing the Deccan Traps in blue-purple

The eruptions occurred over a 600,000-to-800,000-year time period between around 66.3 to 65.6 million years ago, spanning the Cretaceous–Paleogene boundary. While some authors initially suggested the eruptions were a major cause of the Cretaceous–Paleogene mass extinction event at roughly the same time,[7] this theory has been rejected as a result of research into the Chicxulub impact, now thought to be the primary cause of the extinction.[8][9][10][11] While some scholars continue to argue the eruptions may have contributed, it is now generally accepted that the Deccan eruptions played a minor role at most or may have even partially negated the effects of the impact.[9][10]

The Western Ghats at Matheran in Maharashtra

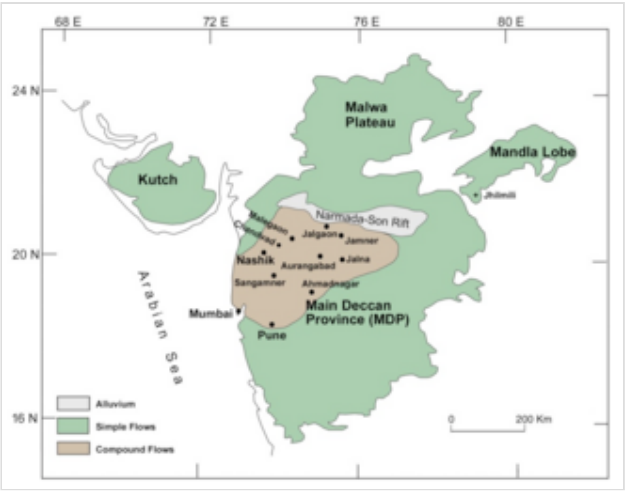
The Deccan Traps are thought to have been produced in major part by the still active Réunion hotspot, responsible for the creation of the modern Mascarene Islands in the Indian Ocean.[12]

Oblique satellite view of the Deccan Traps

Etymology

The term *trap* has been used in geology since 1785–1795 for such rock formations. It is derived from the Swedish word for stairs (*trapp*) and refers to the step-like hills forming the landscape of the region.[13] The name *Deccan* has Sanskrit origins meaning "southern".[6]

History



Map of the Deccan Traps[1]

The Deccan Traps began forming 66.25 million years ago,[5] at the end of the Cretaceous period, although it is possible that some of the oldest material may underlie younger material.[2][6] The bulk of the volcanic eruption occurred at the Western Ghats between 66 and 65 million years ago when lava began to extrude in fissure eruptions.[14] Determining the exact age for Deccan rock is made difficult by a number of limitations, one being that the transition between eruption events may have lasted only a few thousand years and the resolution of dating methods is not sufficient to pinpoint these events. In this way, determining the rate of magma emplacement is also difficult to constrain.[2] This series of eruptions may have lasted for less than 30,000 years.[15]

Deccan Traps at Ajanta Caves

The original area covered by the lava flows is estimated to have been as large as 1.5 million km^2 (0.58 million sq mi), approximately half the size of modern India. The Deccan Traps region was reduced to its current size by erosion and plate tectonics; the present area of directly observable lava flows is around 500,000 km^2 (200,000 sq mi).

The Deccan Traps are segmented into three stratigraphic units: the Upper, Middle and Lower traps. While it was previously interpreted that these groups represented their own key points in the sequence of events in Deccan extrusion, it is now more widely accepted that these horizons relate more closely to paleotopography and distance from the eruption site.[6]

Effect on mass extinctions and climate

The release of volcanic gases, particularly sulfur dioxide, during the formation of the traps may have contributed to climate change. An average drop in temperature of about 2 °C (3.6 °F) was recorded during this period.[16]

Because of its magnitude, some scientists (notably Gerta Keller) have speculated that the gases released during the formation of the Deccan Traps played a major role in the Cretaceous–Paleogene (K–Pg) extinction event (also known as the Cretaceous–Tertiary or K–T extinction).[17] It has been theorized that sudden cooling due to sulfurous volcanic gases released by the formation of the traps and toxic gas emissions may have contributed significantly to the K–Pg mass extinction.[18] However, the current consensus among the scientific community is that the extinction was primarily triggered by the Chicxulub impact event in North America, which will have produced a sunlight-blocking dust cloud that killed much of the plant life and reduced global temperature (this cooling is called an impact winter).[19]

A 2014 study suggested the extinction may have been caused by both the volcanism and the impact event.[20][21] This was followed by a similar study in 2015, both of which consider the hypothesis that the impact exacerbated or induced the Deccan volcanism, since the events occurred approximately at antipodes.[22][23] A 2020 study questioned the idea that the Deccan Traps were a contributory factor at all, suggesting that the Deccan Traps eruptions may have even partially negated the climatic change induced by the impact.[10]

A major criticism of the Deccan Traps as the primary cause of the extinctions is that the extinction event appears to be globally geologically instantaneous and simultaneous in both marine and terrestrial environments, as would be expected from an impact cause, rather than staggered as would be expected from a large igneous province.[10]

A more recent discovery appears to demonstrate the scope of the destruction from the impact alone, however. In a March 2019 article in the Proceedings of the National Academy of Sciences, an international team of twelve scientists revealed the contents of the Tanis fossil site discovered near Bowman, North Dakota, that appeared to show a devastating mass destruction of an ancient lake and its inhabitants at the time of the Chicxulub impact. In the paper, the group reports that the geology of the site is strewn with fossilized trees and remains of fish and other animals. The lead researcher, Robert A. DePalma of the University of Kansas, was quoted in *The New York Times* as stating that "You would be blind to miss the carcasses sticking out... It is impossible to miss when you see the outcrop". Evidence correlating this find to the Chicxulub impact included tektites bearing "the unique chemical signature of other tektites associated with the Chicxulub event" found in the gills of fish fossils and embedded in amber, an iridium-rich top layer that is considered another signature of the event, and an atypical lack of evidence for scavenging, perhaps suggesting that there were few survivors. The exact mechanism of the site's destruction has been debated as either an impact-caused tsunami or lake and river seiche activity triggered by post-impact earthquakes, though there has yet been no firm conclusion upon which researchers have settled.[24][25]

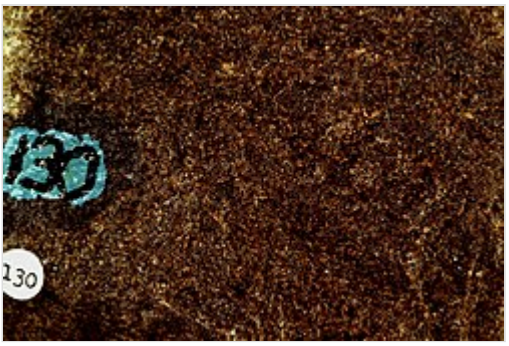
A 2024 study of glycerol dialkyl glycerol tetraether levels in fossilized peat found that the Deccan Traps caused long-term warming of around 3 °C over the course of the final 100,000 years of the Maastrichtian, as well as about 5 °C drop in temperature for less than 10,000 years around 30,000 years prior to the K–Pg boundary (coinciding with the peak of the Poladpur eruptive phase), but by the time of the K–Pg boundary, global temperatures had returned to previous levels. This suggests that the Deccan Traps were not the primary cause of extinction.[11]

Petrology

Within the Deccan Traps, at least 95% of the lavas are tholeiitic basalts.[26] Major mineral constituents are olivine, pyroxenes and plagioclase, as well as certain Fe-Ti-rich oxides. These magmas are <7% MgO. However, many of these minerals are observed as highly altered forms.[2] Other rock types present include alkali basalt, nephelinite, lamprophyre and carbonatite.

Mantle xenoliths have been described from Kachchh (northwestern India) and elsewhere in the western Deccan and contain spinel lherzolite and pyroxenite constituents.[2][27]

While the Deccan traps have been categorized in many different ways including the three different stratigraphic groups, geochemically the province can be split into as many as eleven different formations. Many of the petrologic differences in these units are a product of varying degrees of crustal contamination.[2]



Sample of tholeiitic basalt from the Deccan Traps

Fossils

The Deccan Traps are famous for the beds of fossils that have been found between layers of lava. Particularly well-known species include the frog *Oxyglossus pusillus* (Owen) of the Eocene of India and the toothed frog *Indobatrachus*, an early lineage of modern frogs, which is now placed in the Australian family Myobatrachidae.[28][29] The Infratrappean Beds (Lameta Formation) and Intertrappean Beds also contain fossil freshwater molluscs.[30]

Paleoart of the Deccan trap during the Late Cretaceous

Crystals of epistilbite and calcite in a vug in Deccan Traps basalt lava from Jalgaon District, Maharashtra

Theories of formation

It is postulated that the Deccan Traps eruption was associated with a deep mantle plume. High $^{3}He/^{4}He$ ratios of the main pulse of the eruption are often seen in magmas with mantle plume origin.[31] The area of long-term eruption (the hotspot), known as the Réunion hotspot, is suspected of both causing the Deccan Traps eruption and opening the rift that separated the Mascarene Plateau from India. Regional crustal thinning supports the theory of this rifting event and likely encouraged the rise of the plume in this area.[6] Seafloor spreading at the boundary between the Indian and African Plates subsequently pushed India north over the plume, which now lies under Réunion island in the Indian Ocean, southwest of India. The mantle plume model has, however, been challenged.[32]

Data continues to emerge that supports the plume model. The motion of the Indian tectonic plate and the eruptive history of the Deccan traps show strong correlations. Based on data from marine magnetic profiles, a pulse of unusually rapid plate motion began at the same time as the first pulse of Deccan flood basalts, which is dated at 67 million years ago. The spreading rate rapidly increased and reached a maximum at the same time as the peak basaltic eruptions. The spreading rate then dropped off, with the decrease occurring around 63 million years ago, by which time the main phase of Deccan volcanism ended. This correlation is seen as driven by plume dynamics.[33]

The motions of the Indian and African plates have also been shown to be coupled, the common element being the position of these plates relative to the location of the Réunion plume head. The onset of accelerated motion of India coincides with a large slowing of the rate of counterclockwise rotation of Africa. The close correlations between the plate motions suggest that they were both driven by the force of the Réunion plume.[33]

When comparing the Na_8, Fe_8 and Si_8 contents of the Deccan to other major igneous provinces, the Deccan appears to have undergone the greatest degree of melting suggesting a deep plume origin. Olivine appears to have fractionated at near-Moho depths with additional fractionation of gabbro ~6 km below the surface.[2] Features such as widespread faulting, frequent diking events, high heat flux, and positive gravity anomalies suggest that the extrusive phase of the Deccan Traps is associated with the existence of a triple junction which may have existed during the Late Cretaceous, having been caused by a deep mantle plume. Not all of these diking events are attributed to large-scale contributions to the overall flow volume. It can be difficult, however, to locate the largest dikes as they are often located towards the west coast and are therefore believed to currently reside under water.[6]

Suggested link to impact events

Chicxulub crater

An illustration of the Deccan Trap eruption that may have caused the extinction of the dinosaurs

Although the Deccan Traps began erupting well before the impact, in a 2015 study it was proposed based on argon–argon dating that the impact may have caused an increase in permeability that allowed magma to reach the surface and produced the most voluminous flows, accounting for around 70% of the volume.[34] The combination of the asteroid impact and the resulting increase in eruptive volume may have been responsible for the mass extinctions that occurred at the time that separates the Cretaceous and Paleogene periods, known as the K–Pg boundary.[35][36] However this proposal has been questioned by other authors, who describe the suggestion as being "convenient interpretations based on superficial and cursory observations."[37]

Shiva crater

A geological structure that exists in the sea floor off the west coast of India has been suggested as a possible impact crater, in this context called the Shiva crater. It was also dated approximately 66 million years ago, potentially matching the Deccan traps. The researchers claiming that this feature is an impact crater suggest that the impact may have been the triggering event for the Deccan Traps as well as contributing to the acceleration of the Indian plate in the early Paleogene.[38] However, the current consensus in the Earth science community is that this feature is unlikely to be an actual impact crater.[39][40]

See also

- Columbia River Basalt Group
- Emeishan Traps
- Geology of India
- Krishna Godavari Basin
- Lameta Formation
- List of flood basalt provinces
- List of volcanoes in India
- Siberian Traps
- Verneshot
- Viluy traps

References

1. Götze, Jens; Hofmann, Beda; Machałowski, Tomasz; Tsurkan, Mikhail V.; Jesionowski, Teofil; Ehrlich, Hermann; Kleeberg, Reinhard; Ottens, Berthold (16 June 2020). "Biosignatures in Subsurface Filamentous Fabrics (SFF) from the Deccan Volcanic Province, India" (https://doi.org/10.3390%2Fmin10060540). *Minerals*. **10** (6): 540. Bibcode:2020Mine...10..540G (https://ui.adsabs.harvard.edu/abs/2020Mine...10..540G). doi:10.3390/min10060540 (https://doi.org/10.3390%2Fmin10060540).
2. Sen, Gautam (1 December 2001). "Generation of Deccan Trap magmas" (https://doi.org/10.1007%2FBF02702904). *Journal of Earth System Science*. **110** (4): 409–431. Bibcode:2001InEPS.110..409S (https://ui.adsabs.harvard.edu/abs/2001InEPS.110..409S). doi:10.1007/BF02702904 (https://doi.org/10.1007%2FBF02702904). ISSN 0973-774X (https://search.worldcat.org/issn/0973-774X). S2CID 52107268 (https://api.semanticscholar.org/CorpusID:52107268).
3. Singh, R. N.; Gupta, K. R. (1994). "Workshop yields new insight into volcanism at Deccan Traps, India". *Eos*. **75** (31): 356. Bibcode:1994EOSTr..75..356S (https://ui.adsabs.harvard.edu/abs/1994EOSTr..75..356S). doi:10.1029/94EO01005 (https://doi.org/10.1029%2F94EO01005).
4. Dessert, Céline; Dupréa, Bernard; Françoisa, Louis M.; Schotta, Jacques; Gaillardet, Jérôme; Chakrapani, Govind; Bajpai, Sujit (2001). "Erosion of Deccan Traps determined by river geochemistry: impact on the global climate and the 87Sr/86Sr ratio of seawater". *Earth and Planetary Science Letters*. **188** (3–4): 459–474. Bibcode:2001E&PSL.188..459D (https://ui.adsabs.harvard.edu/abs/2001E&PSL.188..459D). doi:10.1016/S0012-821X(01)00317-X (https://doi.org/10.1016%2FS0012-821X%2801%2900317-X).
5. "What really killed the dinosaurs?" (https://news.mit.edu/2014/volcanic-eruption-dinosaur-extinction-1211) Jennifer Chu, MIT News Office, 11 December 2014
6. Macdougall, J. D. (1988). *Continental Flood Basalts*. Dordrecht: Springer Netherlands. ISBN 978-94-015-7805-9. OCLC 851375252 (https://search.worldcat.org/oclc/851375252).
7. Schoene, Blair; Eddy, Michael P.; Keller, C. Brenhin; Samperton, Kyle M. (16 April 2021). "An evaluation of Deccan Traps eruption rates using geochronologic data" (https://gchron.copernicus.org/articles/3/181/2021/). *Geochronology*. **3** (1): 181–198. Bibcode:2021GeChr...3..181S (https://ui.adsabs.harvard.edu/abs/2021GeChr...3..181S). doi:10.5194/gchron-3-181-2021 (https://doi.org/10.5194%2Fgchron-3-181-2021). ISSN 2628-3719 (https://search.worldcat.org/issn/2628-3719).

8. Morgan, Joanna V.; Bralower, Timothy J.; Brugger, Julia; Wünnemann, Kai (12 April 2022). "The Chicxulub impact and its environmental consequences" (https://www.nature.com/articles/s43017-022-00283-y). *Nature Reviews Earth & Environment*. **3** (5): 338–354. Bibcode:2022NRvEE...3..338M (https://ui.adsabs.harvard.edu/abs/2022NRvEE...3..338M). doi:10.1038/s43017-022-00283-y (https://doi.org/10.1038%2Fs43017-022-00283-y). ISSN 2662-138X (https://search.worldcat.org/issn/2662-138X).
9. Smit, J. (21 June 2022), "The KPg boundary Chicxulub impact-extinction hypothesis: The winding road towards a solid theory" (https://research.vu.nl/ws/portalfiles/portal/222623549/The_KPg_boundary_Chicxulub_impact_extinction_hypothesis.pdf) (PDF), in Koeberl, Christian; Claeys, Philippe; Montanari, Alessandro (eds.), *From the Guajira Desert to the Apennines, and from Mediterranean Microplates to the Mexican Killer Asteroid: Honoring the Career of Walter Alvarez*, Geological Society of America, pp. 391–414, doi:10.1130/2022.2557(19) (https://doi.org/10.1130%2F2022.2557%2819%29), ISBN 978-0-8137-2557-4, retrieved 13 August 2024
10. Chiarenza, Alfio Alessandro; Farnsworth, Alexander; Mannion, Philip D.; Lunt, Daniel J.; Valdes, Paul J.; Morgan, Joanna V.; Allison, Peter A. (21 July 2020). "Asteroid impact, not volcanism, caused the end-Cretaceous dinosaur extinction" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7382232). *Proceedings of the National Academy of Sciences*. **117** (29): 17084–17093. doi:10.1073/pnas.2006087117 (https://doi.org/10.1073%2Fpnas.2006087117). ISSN 0027-8424 (https://search.worldcat.org/issn/0027-8424). PMC 7382232 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7382232). PMID 32601204 (https://pubmed.ncbi.nlm.nih.gov/32601204).
11. O'Connor, Lauren K.; Jerrett, Rhodri M.; Price, Gregory D.; Lyson, Tyler R.; Lengger, Sabine K.; Peterse, Francien; van Dongen, Bart E. (20 December 2024). "Terrestrial evidence for volcanogenic sulfate-driven cooling event ~30 kyr before the Cretaceous–Paleogene mass extinction" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11654674). *Science Advances*. **10** (51) eado5478. doi:10.1126/sciadv.ado5478 (https://doi.org/10.1126%2Fsciadv.ado5478). ISSN 2375-2548 (https://search.worldcat.org/issn/2375-2548). PMC 11654674 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11654674). PMID 39693422 (https://pubmed.ncbi.nlm.nih.gov/39693422).
12. Glišović, Petar; Forte, Alessandro M. (10 February 2017). "On the deep-mantle origin of the Deccan Traps" (https://www.science.org/doi/10.1126/science.aah4390). *Science*. **355** (6325): 613–616. Bibcode:2017Sci...355..613G (https://ui.adsabs.harvard.edu/abs/2017Sci...355..613G). doi:10.1126/science.aah4390 (https://doi.org/10.1126%2Fscience.aah4390). ISSN 0036-8075 (https://search.worldcat.org/issn/0036-8075). PMID 28183974 (https://pubmed.ncbi.nlm.nih.gov/28183974).
13. Trap (https://www.dictionary.com/browse/trap) at dictionary.reference.com
14. Krishnan, M.S. (2006). *Geology of India and Burma* (6th ed.). New Delhi: CBS Publishers and Distributors. ISBN 81-239-0012-0. OCLC 778055464 (https://search.worldcat.org/oclc/778055464).
15. "India's Smoking Gun: Dino-killing Eruptions (https://www.sciencedaily.com/releases/2005/08/050810130729.htm)." *ScienceDaily,* 10 August 2005.
16. Royer, D. L.; Berner, R. A.; Montañez, I. P.; Tabor, N. J.; Beerling, D. J. (2004). "CO_2 as a primary driver of Phanerozoic climate" (https://doi.org/10.1130%2F1052-5173%282004%29014%3C4%3ACAAPDO%3E2.0.CO%3B2). *GSA Today*. **14** (3): 4–10. Bibcode:2004GSAT...14c...4R (https://ui.adsabs.harvard.edu/abs/2004GSAT...14c...4R). doi:10.1130/1052-5173(2004)014<4:CAAPDO>2.0.CO;2 (https://doi.org/10.1130%2F1052-5173%282004%29014%3C4%3ACAAPDO%3E2.0.CO%3B2). ISSN 1052-5173 (https://search.worldcat.org/issn/1052-5173).
17. Courtillot, Vincent (1990). "A Volcanic Eruption". *Scientific American*. **263** (4): 85–92. Bibcode:1990SciAm.263d..85C (https://ui.adsabs.harvard.edu/abs/1990SciAm.263d..85C). doi:10.1038/scientificamerican1090-85 (https://doi.org/10.1038%2Fscientificamerican1090-85). PMID 11536474 (https://pubmed.ncbi.nlm.nih.gov/11536474).

18. Beardsley, Tim (1988). "Star-Struck?". *Scientific American*. **258** (4): 37–40. Bibcode:1988SciAm.258d..37B (https://ui.adsabs.harvard.edu/abs/1988SciAm.258d..37B). doi:10.1038/scientificamerican0488-37b (https://doi.org/10.1038%2Fscientificamerican0488-37b).
19. Schulte, Peter; et al. (5 March 2010). "The Chicxulub Asteroid Impact and Mass Extinction at the Cretaceous-Paleogene Boundary" (http://doc.rero.ch/record/210367/files/PAL_E4389.pdf) (PDF). *Science*. **327** (5970): 1214–1218. Bibcode:2010Sci...327.1214S (https://ui.adsabs.harvard.edu/abs/2010Sci...327.1214S). doi:10.1126/science.1177265 (https://doi.org/10.1126%2Fscience.1177265). ISSN 1095-9203 (https://search.worldcat.org/issn/1095-9203). PMID 20203042 (https://pubmed.ncbi.nlm.nih.gov/20203042). S2CID 2659741 (https://api.semanticscholar.org/CorpusID:2659741). Archived (https://web.archive.org/web/20170921215225/http://doc.rero.ch/record/210367/files/PAL_E4389.pdf) (PDF) from the original on 21 September 2017.
20. Keller, G., *Deccan volcanism, the Chicxulub impact, and the end-Cretaceous mass extinction: Coincidence? Cause and effect?,* in *Volcanism, Impacts, and Mass Extinctions: Causes and Effects,* GSA Special Paper 505, pp. 29–55, 2014 abstract (http://specialpapers.gsapubs.org/content/early/2014/06/10/2014.2505_03.1.abstract) Archived (https://web.archive.org/web/20170618024315/http://specialpapers.gsapubs.org/content/early/2014/06/10/2014.2505_03.1.abstract) 18 June 2017 at the Wayback Machine
21. Schoene, B.; Samperton, K. M.; Eddy, M. P.; Keller, G.; Adatte, T.; Bowring, S. A.; Khadri, S. F. R.; Gertsch, B. (11 December 2014). "U-Pb geochronology of the Deccan Traps and relation to the end-Cretaceous mass extinction". *Science*. **347** (6218): 182–184. Bibcode:2015Sci...347..182S (https://ui.adsabs.harvard.edu/abs/2015Sci...347..182S). doi:10.1126/science.aaa0118 (https://doi.org/10.1126%2Fscience.aaa0118). PMID 25502315 (https://pubmed.ncbi.nlm.nih.gov/25502315). S2CID 206632431 (https://api.semanticscholar.org/CorpusID:206632431).
22. Renne, P. R.; Sprain, C. J.; Richards, M. A.; Self, S.; Vanderkluysen, L.; Pande, K. (2 October 2015). "State shift in Deccan volcanism at the Cretaceous-Paleogene boundary, possibly induced by impact" (https://doi.org/10.1126%2Fscience.aac7549). *Science*. **350** (6256): 76–78. Bibcode:2015Sci...350...76R (https://ui.adsabs.harvard.edu/abs/2015Sci...350...76R). doi:10.1126/science.aac7549 (https://doi.org/10.1126%2Fscience.aac7549). PMID 26430116 (https://pubmed.ncbi.nlm.nih.gov/26430116).
23. Sample, Ian (2 October 2015). "Asteroid that killed dinosaurs also intensified volcanic eruptions - study" (https://www.theguardian.com/science/2015/oct/01/asteroid-that-killed-dinosaurs-also-intensified-volcanic-eruptions-study). *The Guardian*. Retrieved 2 October 2015.
24. "Stunning discovery offers glimpse of minutes following 'dinosaur-killer' Chicxulub impact" (https://web.archive.org/web/20190410034107/https://news.ku.edu/2019/03/29/stunning-discovery-offers-glimpse-minutes-following-%E2%80%98dinosaur-killer%E2%80%99-chicxulub-impact). *news.ku.edu*. 29 March 2019. Archived from the original (https://news.ku.edu/2019/03/29/stunning-discovery-offers-glimpse-minutes-following-%E2%80%98dinosaur-killer%E2%80%99-chicxulub-impact) on 10 April 2019. Retrieved 10 April 2019.
25. Broad, William J. and Chang, Kenneth (29 March 2019). "Fossil Site Reveals Day That Meteor Hit Earth and, Maybe, Wiped Out Dinosaurs" (https://ghostarchive.org/archive/20220101/https://www.nytimes.com/2019/03/29/science/dinosaurs-extinction-asteroid.html). *The New York Times*. Archived from the original (https://www.nytimes.com/2019/03/29/science/dinosaurs-extinction-asteroid.html) on 1 January 2022.
26. Aramaki, S.; Fukuoka, T.; Deshmukh, S. S.; Fujii, T.; Sano, T. (1 December 2001). "Differentiation Processes of Deccan Trap Basalts: Contribution from Geochemistry and Experimental Petrology". *Journal of Petrology*. **42** (12): 2175–2195. doi:10.1093/petrology/42.12.2175 (https://doi.org/10.1093%2Fpetrology%2F42.12.2175). ISSN 0022-3530 (https://search.worldcat.org/issn/0022-3530).

27. Dessai, A.G.; Vaselli, O. (October 1999). "Petrology and geochemistry of xenoliths in lamprophyres from the Deccan Traps: implications for the nature of the deep crust boundary in western India" (https://rruff.info/doclib/MinMag/Volume_63/63-5-703.pdf) (PDF). *Mineralogical Magazine*. **63** (5): 703–722. Bibcode:1999MinM...63..703D (https://ui.adsabs.harvard.edu/abs/1999MinM...63..703D). doi:10.1180/minmag.1999.063.5.08 (https://doi.org/10.1180%2Fminmag.1999.063.5.08). Archived (https://web.archive.org/web/20161215200622/http://rruff.info/doclib/MinMag/Volume_63/63-5-703.pdf) (PDF) from the original on 15 December 2016.
28. Noble, Gladwyn Kingsley (1930). "The Fossil Frogs of the Intertrappean Beds of Bombay, India". *American Museum of Natural History*. **401**: 1930. hdl:2246/3061 (https://hdl.handle.net/2246%2F3061).
29. "Myobatrachinae" (https://web.archive.org/web/20170922052151/http://tolweb.org/Myobatrachinae/16946). Archived from the original (http://tolweb.org/Myobatrachinae/16946) on 22 September 2017. Retrieved 19 June 2011.
30. Hartman, J.H., Mohabey, D.M., Bingle, M., Scholz, H., Bajpai, S., and Sharma, R., 2006, Initial survivorship of nonmarine molluscan faunas in end-Cretaceous Deccan intertrappean strata, India: Geological Society of America (annual meeting, Philadelphia) Abstracts with Programs, v. 38, no. 7, p. 143.
31. Basu, Asish R.; Renne, Paul R.; DasGupta, Deb K.; Teichmann, Friedrich; Poreda, Robert J. (13 August 1993). "Early and Late Alkali Igneous Pulses and a High-3He Plume Origin for the Deccan Flood Basalts" (https://www.science.org/doi/10.1126/science.261.5123.902). *Science*. **261** (5123): 902–906. Bibcode:1993Sci...261..902B (https://ui.adsabs.harvard.edu/abs/1993Sci...261..902B). doi:10.1126/science.261.5123.902 (https://doi.org/10.1126%2Fscience.261.5123.902). PMID 17783739 (https://pubmed.ncbi.nlm.nih.gov/17783739). S2CID 23709446 (https://api.semanticscholar.org/CorpusID:23709446).
32. Sheth, Hetu C. "The Deccan Beyond the Plume Hypothesis (http://www.mantleplumes.org/Deccan.html)." *MantlePlumes.org,* 2006.
33. Cande, S.C.; Stegman, D.R. (2011). "Indian and African plate motions driven by the push force of the Réunion plume head". *Nature*. **475** (7354): 47–52. Bibcode:2011Natur.475...47C (https://ui.adsabs.harvard.edu/abs/2011Natur.475...47C). doi:10.1038/nature10174 (https://doi.org/10.1038%2Fnature10174). PMID 21734702 (https://pubmed.ncbi.nlm.nih.gov/21734702). S2CID 205225348 (https://api.semanticscholar.org/CorpusID:205225348).
34. Richards, Mark A.; Alvarez, Walter; Self, Stephen; Karlstrom, Leif; Renne, Paul R.; Manga, Michael; Sprain, Courtney J.; Smit, Jan; Vanderkluysen, Loÿc; Gibson, Sally A. (2015). "Triggering of the largest Deccan eruptions by the Chicxulub impact" (http://eprints.esc.cam.ac.uk/3289/1/Gibson%20bul%20gsa.pdf) (PDF). *Geological Society of America Bulletin*. **127** (11–12): 1507–1520. Bibcode:2015GSAB..127.1507R (https://ui.adsabs.harvard.edu/abs/2015GSAB..127.1507R). doi:10.1130/B31167.1 (https://doi.org/10.1130%2FB31167.1). S2CID 3463018 (https://api.semanticscholar.org/CorpusID:3463018). Archived (https://web.archive.org/web/20180219045415/http://eprints.esc.cam.ac.uk/3289/1/Gibson%20bul%20gsa.pdf) (PDF) from the original on 19 February 2018.
35. Renne, P. R.; et al. (2015). "State shift in Deccan volcanism at the Cretaceous-Paleogene boundary, possibly induced by impact" (https://doi.org/10.1126%2Fscience.aac7549). *Science*. **350** (6256): 76–78. Bibcode:2015Sci...350...76R (https://ui.adsabs.harvard.edu/abs/2015Sci...350...76R). doi:10.1126/science.aac7549 (https://doi.org/10.1126%2Fscience.aac7549). PMID 26430116 (https://pubmed.ncbi.nlm.nih.gov/26430116).
36. Sprain, Courtney J.; Renne, Paul R.; Vanderkluysen, Loÿc; Pande, Kanchan; Self, Stephen; Mittal, Tushar (21 February 2019). "The eruptive tempo of Deccan volcanism in relation to the Cretaceous-Paleogene boundary" (https://doi.org/10.1126%2Fscience.aav1446). *Science*. **363** (6429): 866–870. Bibcode:2019Sci...363..866S (https://ui.adsabs.harvard.edu/abs/2019Sci...363..866S). doi:10.1126/science.aav1446 (https://doi.org/10.1126%2Fscience.aav1446). PMID 30792301 (https://pubmed.ncbi.nlm.nih.gov/30792301). S2CID 67876911 (https://api.semanticscholar.org/CorpusID:67876911).

37. Dole, Gauri; Patil Pillai, Shilpa; Upasani, Devdutt; Kale, Vivek S. (January 2017). "Triggering of the largest Deccan eruptions by the Chicxulub impact: Comment" (https://pubs.geoscienceworld.org/gsabulletin/article/129/1-2/253-255/185478). *Geological Society of America Bulletin*. **129** (1–2): 253–255. Bibcode:2017GSAB..129..253D (https://ui.adsabs.harvard.edu/abs/2017GSAB..129..253D). doi:10.1130/B31520.1 (https://doi.org/10.1130%2FB31520.1). ISSN 0016-7606 (https://search.worldcat.org/issn/0016-7606).
38. Chatterjee, Sankar. "The Shiva Crater: Implications for Deccan Volcanism, India-Seychelles Rifting, Dinosaur Extinction, and Petroleum Entrapment at the KT Boundary (https://gsa.confex.com/gsa/2003AM/finalprogram/abstract_58126.htm) Archived (https://web.archive.org/web/20161202220808/https://gsa.confex.com/gsa/2003AM/finalprogram/abstract_58126.htm) 2 December 2016 at the Wayback Machine." Paper No. 60-8, Seattle Annual Meeting, November 2003.
39. Mullen, Leslie (2 November 2004). "Shiva: Another K–Pg Impact?" (http://www.spacedaily.com/news/deepimpact-04r.html). *Spacedaily.com*. Retrieved 20 February 2008. - original article at source (https://web.archive.org/web/20110804221725/http://www.astrobio.net/exclusive/1281/shiva-another-k-t-impact)
40. Moskowitz, Clara (18 October 2009). "New Dino-destroying Theory Fuels Hot Debate" (http://www.space.com/scienceastronomy/091018-dinosaur-crater.html). space.com.

External links

- "Animated simulation by the Geodynamics group at the Geological Survey of Norway illustrating the Indian plate moving through the Indian Ocean" (https://web.archive.org/web/20110723122146/http://www.geodynamics.no/indexOld.htm). Archived from the original (http://www.geodynamics.no/indexOld.htm) on 23 July 2011.
- Scientist argues that volcanoes, not meteorite, killed dinosaurs (https://www.nbcnews.com/id/wbna50119631)
- The Deccan Traps/Volcanism Theory (https://hoopermuseum.carleton.ca/saleem/volcanism.htm)

Retrieved from "https://en.wikipedia.org/w/index.php?title=Deccan_Traps&oldid=1376521594"